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DOE FUNDAMENTALS HANDBOOK
NUCLEAR PHYSICS
AND REACTOR THEORY
Volume 1 of 2
U.S. Department of Energy FSC-6910
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ABSTRACT
The Nuclear Physics and Reactor Theory Handbook was developed to assist nuclear facility operating contractors in providing operators, maintenance personnel, and the technical staff with the necessary fundamentals training to ensure a basic understanding of nuclear physics and reactor theory. The handbook includes information on atomic and nuclear physics; neutron characteristics; reactor theory and nuclear parameters; and the theory of reactor operation. This information will provide personnel with a foundation for understanding the scientific principles that are associated with various DOE nuclear facility operations and maintenance.
Key Words: Training Material, Atomic Physics, The Chart of the Nuclides, Radioactivity, Radioactive Decay, Neutron Interaction, Fission, Reactor Theory, Neutron Characteristics, Neutron Life Cycle, Reactor Kinetics
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FOREWORD
The Department of Energy (DOE) Fundamentals Handbooks consist of ten academic subjects, which include Mathematics; Classical Physics; Thermodynamics, Heat Transfer, and Fluid Flow; Instrumentation and Control; Electrical Science; Material Science; Mechanical Science; Chemistry; Engineering Symbology, Prints, and Drawings; and Nuclear Physics and Reactor Theory. The handbooks are provided as an aid to DOE nuclear facility contractors.
These handbooks were first published as Reactor Operator Fundamentals Manuals in 1985 for use by DOE category A reactors. The subject areas, subject matter content, and level of detail of the Reactor Operator Fundamentals Manuals were determined from several sources. DOE Category A reactor training managers determined which materials should be included, and served as a primary reference in the initial development phase. Training guidelines from the commercial nuclear power industry, results of job and task analyses, and independent input from contractors and operations-oriented personnel were all considered and included to some degree in developing the text material and learning objectives.
The DOE Fundamentals Handbooks represent the needs of various DOE nuclear facilities' fundamental training requirements. To increase their applicability to nonreactor nuclear facilities,

Facility Training Coordination Program Steering Committee for review and comment. To update their reactor-specific content, DOE Category A reactor training managers also reviewed and commented on the content. On the basis of feedback from these sources, information that applied to two or more DOE nuclear facilities was considered generic and was included. The final draft of each of the handbooks was then reviewed by these two groups. This approach has resulted in revised modular handbooks that contain sufficient detail such that each facility may adjust the content to fit their specific needs.

Each handbook contains an abstract, a foreword, an overview, learning objectives, and text material, and is divided into modules so that content and order may be modified by individual DOE contractors to suit their specific training needs. Each handbook is supported by a separate examination bank with an answer key.

The DOE Fundamentals Handbooks have been prepared for the Assistant Secretary for Nuclear Energy, Office of Nuclear Safety Policy and Standards, by the DOE Training Coordination Program. This program is managed by EG&G Idaho, Inc.

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NUCLEAR PHYSICS AND REACTOR THEORY

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OVERVIEW

The Department of Energy Fundamentals Handbook entitled Nuclear Physics and Reactor Theory was prepared as an information resource for personnel who are responsible for the operation of the Department's nuclear facilities. Almost all processes that take place in a nuclear facility involves the transfer of some type of energy. A basic understanding of nuclear physics and reactor theory is necessary for DOE nuclear facility operators, maintenance personnel, and the technical staff to safely operate and maintain the facility and facility support systems. The information in this handbook is presented to provide a foundation for applying engineering concepts to the job. This knowledge will help personnel understand the impact that their actions may have on the safe and reliable operation of facility components and systems.

The Nuclear Physics and Reactor Theory handbook consists of four modules that are contained in two volumes. The following is a brief description of the information presented in each module of the handbook.

Volume 1 of 2

Module 1 - Atomic and Nuclear Physics

Introduces concepts of atomic physics including the atomic nature of matter, the chart of the nuclides, radioactivity and radioactive decay, neutron interactions and fission, and the interaction of radiation with matter.

Module 2 - Reactor Theory (Nuclear Parameters)

Provides information on reactor theory and neutron characteristics. Includes topics such as neutron sources, neutron flux, neutron cross sections, reaction rates, neutron moderation, and prompt and delayed neutrons.

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NUCLEAR PHYSICS AND REACTOR THEORY

OVERVIEW (Cont.)

Volume 2 of 2

Module 3 - Reactor Theory (Nuclear Parameters)

Explains the nuclear parameters associated with reactor theory. Topics include the neutron life cycle, reactivity and reactivity coefficients, neutron poisons, and

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Module 4 - Reactor Theory (Reactor Operations)
Introduces the reactor operations aspect of reactor theory. Topics include subcritical multiplication, reactor kinetics, and reactor operation.
The information contained in this handbook is not all-encompassing. An attempt to present the entire subject of nuclear physics and reactor theory would be impractical. However, the Nuclear Physics and Reactor Theory handbook presents enough information to provide the reader with the fundamental knowledge necessary to understand the advanced theoretical concepts presented in other subject areas, and to understand basic system and equipment operation.
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Department of Energy
Fundamentals Handbook
NUCLEAR PHYSICS
AND REACTOR THEORY
Module 1
Atomic and Nuclear Physics
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Atomic and Nuclear Physics DOE-HDBK-1019/1-93 OBJECTIVES
TERMINAL OBJECTIVE
1.0 Given sufficient information, DESCRIBE atoms, including components, structure, and nomenclature.
ENABLING OBJECTIVES
1.1 STATE the characteristics of the following atomic particles, including mass, charge, and location within the atom:
a. Proton
b. Neutron
c. Electron
1.2 DESCRIBE the Bohr model of an atom.
1.3 DEFINE the following terms:
a. Nuclide c. Atomic number
b. Isotope d. Mass number
1.4 Given the standard A
ZX notation for a particular nuclide, DETERMINE the following:
a. Number of protons
b. Number of neutrons
c. Number of electrons
1.5 DESCRIBE the three forces that act on particles within the nucleus and affect the stability of the nucleus.
1.6 DEFINE the following terms:
a. Enriched uranium
b. Depleted uranium
1.7 DEFINE the following terms:
a. Mass defect
b. Binding energy
1.8 Given the atomic mass for a nuclide and the atomic masses of a neutron, proton, and electron, CALCULATE the mass defect and binding energy of the nuclide.

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TERMINAL OBJECTIVE
2.0 Given necessary references, DESCRIBE the various modes of radioactive decay.
ENABLING OBJECTIVES
2.1 DESCRIBE the following processes:
a. Alpha decay d. Electron capture
b. Beta-minus decay e. Internal conversions
c. Beta-plus decay f. Isomeric transitions
2.2 Given a Chart of the Nuclides, WRITE the radioactive decay chain for a nuclide.
2.3 EXPLAIN why one or more gamma rays typically accompany particle emission.
2.4 Given the stability curve on the Chart of the Nuclides, DETERMINE the type of radioactive decay that the nuclides in each region of the chart will typically undergo.
2.5 DEFINE the following terms:
a. Radioactivity d. Radioactive decay constant
b. Curie e. Radioactive half-life
c. Becquerel
2.6 Given the number of atoms and either the half-life or decay constant of a nuclide, CALCULATE the activity.
2.7 Given the initial activity and the decay constant of a nuclide, CALCULATE the activity at any later time.
2.8 CONVERT between the half-life and decay constant for a nuclide.
2.9 Given the Chart of the Nuclides and the original activity, PLOT the radioactive decay curve for a nuclide on either linear or semi-log coordinates.
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2.10 DEFINE the following terms:
a. Radioactive equilibrium
b. Transient radioactive equilibrium
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Atomic and Nuclear Physics DOE-HDBK-1019/1-93 OBJECTIVES
TERMINAL OBJECTIVE
3.0 Without references, DESCRIBE the different nuclear interactions initiated by neutrons.
ENABLING OBJECTIVES
3.1 DESCRIBE the following scattering interactions between a neutron and a nucleus:
a. Elastic scattering
b. Inelastic scattering
3.2 STATE the conservation laws that apply to an elastic collision between a neutron and a nucleus.
3.3 DESCRIBE the following reactions where a neutron is absorbed in a nucleus:
a. Radiative capture
b. Particle ejection
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OBJECTIVES DOE-HDBK-1019/1-93 Atomic and Nuclear Physics
TERMINAL OBJECTIVE
4.0 Without references, DESCRIBE the fission process.
ENABLING OBJECTIVES
4.1 EXPLAIN the fission process using the liquid drop model of a nucleus.
4.2 DEFINE the following terms:
a. Excitation energy
b. Critical energy

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a. Fissile material
b. Fissionable material
c. Fertile material
4.4 DESCRIBE the processes of transmutation, conversion, and breeding.
4.5 DESCRIBE the curve of Binding Energy per Nucleon versus mass number and give a qualitative description of the reasons for its shape.
4.6 EXPLAIN why only the heaviest nuclei are easily fissioned.
4.7 EXPLAIN why uranium-235 fissions with thermal neutrons and uranium-238 fissions only with fast neutrons.
4.8 CHARACTERIZE the fission products in terms of mass groupings and radioactivity.
4.9 Given the nuclides involved and their masses, CALCULATE the energy released from fission.
4.10 Given the curve of Binding Energy per Nucleon versus mass number, CALCULATE the energy released from fission.
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Atomic and Nuclear Physics DOE-HDBK-1019/1-93 OBJECTIVES
TERMINAL OBJECTIVE
5.0 Without references, DESCRIBE how the various types of radiation interact with matter.
ENABLING OBJECTIVES
5.1 DESCRIBE interactions of the following with matter:
a. Alpha particle c. Positron
b. Beta particle d. Neutron
5.2 DESCRIBE the following ways that gamma radiation interacts with matter:
a. Compton scattering
b. Photoelectric effect
c. Pair production
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OBJECTIVES DOE-HDBK-1019/1-93 Atomic and Nuclear Physics
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Atomic and Nuclear Physics DOE-HDBK-1019/1-93 ATOMIC NATURE OF MATTER
ATOMIC NATURE OF MATTER
All matter is composed of atoms. The atom is the smallest amount of matter that retains the properties of an element. Atoms themselves are composed of smaller particles, but these smaller particles no longer have the same properties as the overall element.
EO 1.1 STATE the characteristics of the following atomic particles, including mass, charge, and location within the atom:
a. Proton
b. Neutron
c. Electron
EO 1.2 DESCRIBE the Bohr model of an atom.
EO 1.3 DEFINE the following terms:
a. Nuclide c. Atomic number
b. Isotope d. Mass number
EO 1.4 Given the standard A
ZX notation for a particular nuclide,

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a. Number of protons
b. Number of neutrons
c. Number of electrons
EO 1.5 DESCRIBE the three forces that act on particles within the nucleus and affect the stability of the nucleus.
Structure of Matter
Early Greek philosophers speculated that the earth was made up of different combinations of basic substances, or elements. They considered these basic elements to be earth, air, water, and fire. Modern science shows that the early Greeks held the correct concept that matter consists of a combination of basic elements, but they incorrectly identified the elements.

In 1661 the English chemist Robert Boyle published the modern criterion for an element. He defined an element to be a basic substance that cannot be broken down into any simpler substance after it is isolated from a compound, but can be combined with other elements to form compounds. To date, 105 different elements have been confirmed to exist, and researchers claim to have discovered three additional elements. Of the 105 confirmed elements, 90 exist in nature and 15 are man-made.
Another basic concept of matter that the Greeks debated was whether matter was continuous or discrete. That is, whether matter could be continuously divided and subdivided into ever smaller particles or whether eventually an indivisible particle would be encountered. Democritus in about 450 B.C. argued that substances were ultimately composed of small, indivisible particles that he labeled atoms. He further suggested that different substances were composed of different atoms or combinations of atoms, and that one substance could be converted into another by rearranging the atoms. It was impossible to conclusively prove or disprove this proposal for more than 2000 years.
The modern proof for the atomic nature of matter was first proposed by the English chemist John Dalton in 1803. Dalton stated that each chemical element possesses a particular kind of atom, and any quantity of the element is made up of identical atoms of this kind. What distinguishes one element from another element is the kind of atom of which it consists, and the basic physical difference between kinds of atoms is their weight.
Subatomic Particles
For almost 100 years after Dalton established the atomic nature of atoms, it was considered impossible to divide the atom into even smaller parts. All of the results of chemical experiments during this time indicated that the atom was indivisible. Eventually, experimentation into electricity and radioactivity indicated that particles of matter smaller than the atom did indeed
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exist. In 1906, J. J. Thompson won the Nobel Prize in physics for establishing the existence of electrons. Electrons are negatively-charged particles that have 1/1835 the mass of the hydrogen atom. Soon after the discovery of electrons, protons were discovered. Protons are relatively large particles that have almost the same mass as a hydrogen atom and a positive charge equal in magnitude (but opposite in sign) to that of the electron. The third subatomic particle to be discovered, the neutron, was not found until 1932. The neutron has almost the same mass as the proton, but it is electrically neutral.

Bohr Model of the Atom
The British physicist Ernest Rutherford postulated that the positive charge in an atom is

in orbits around it. Niels Bohr, coupling Rutherford's postulation with the quantum theory introduced by Max Planck, proposed that the atom consists of a dense nucleus of protons surrounded by electrons traveling in discrete orbits at fixed distances from the nucleus. An electron in one of these orbits or shells has a specific or discrete quantity of energy (quantum). When an electron moves from one allowed orbit to another allowed orbit, the energy difference between the two states is emitted or absorbed in the form of a single quantum of radiant energy called a photon. Figure 1 is Bohr's model of the hydrogen atom showing an electron as having just dropped from the third shell to the first shell with the emission of a photon that has an energy = hv. (h = Planck's constant = 6.63 x 10-34 J-s and v = frequency of the photon.) Bohr's theory was the first to successfully account for the discrete energy levels of this radiation as measured in the laboratory. Although Bohr's atomic model is designed specifically to explain the hydrogen atom, his theories apply generally to the structure of all atoms. Additional information on electron shell theory can be found in the Chemistry Fundamentals Handbook.
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Figure 1 Bohr's Model of the Hydrogen Atom
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ATOMIC NATURE OF MATTER DOE-HDBK-1019/1-93 Atomic and Nuclear Physics
Properties of the three subatomic particles are listed in Table 1.
TABLE 1
Properties of Subatomic Particles
Particle Location Charge Mass
Neutron Nucleus none 1.008665 amu
Proton Nucleus +1 1.007277 amu
Electron Shells around nucleus -1 0.0005486 amu
Measuring Units on the Atomic Scale
The size and mass of atoms are so small that the use of normal measuring units, while possible, is often inconvenient. Units of measure have been defined for mass and energy on the atomic scale to make measurements more convenient to express. The unit of measure for mass is the atomic mass unit (amu). One atomic mass unit is equal to 1.66 x 10-24 grams. The reason for this particular value for the atomic mass unit will be discussed in a later chapter. Note from Table 1 that the mass of a neutron and a proton are both about 1 amu. The unit for energy is the electron volt (eV). The electron volt is the amount of energy acquired by a single electron when it falls through a potential difference of one volt. One electron volt is equivalent to 1.602 x 10-19 joules or 1.18 x 10-19 foot-pounds.
Nuclides
The total number of protons in the nucleus of an atom is called the atomic number of the atom and is given the symbol Z. The number of electrons in an electrically-neutral atom is the same as the number of protons in the nucleus. The number of neutrons in a nucleus is known as the neutron number and is given the symbol N. The mass number of the nucleus is the total number of nucleons, that is, protons and neutrons in the nucleus. The mass number is given the symbol A and can be found by the equation Z + N = A.
Each of the chemical elements has a unique atomic number because the atoms of different elements contain a different number of protons. The atomic number of an atom identifies the particular element.
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Atomic and Nuclear Physics DOE-HDBK-1019/1-93 ATOMIC NATURE OF MATTER
Each type of atom that contains a unique combination of
Figure 2 Nomenclature for

protons and neutrons is called a nuclide. Not all combinations of numbers of protons and neutrons are possible, but about 2500 specific nuclides with unique combinations of neutrons and protons have been identified. Each nuclide is denoted by the chemical symbol of the element with the atomic number written as a subscript and the mass number written as a superscript, as shown in Figure 2. Because each element has a unique name, chemical symbol, and atomic number, only one of the three is necessary to identify the element. For this reason nuclides can also be identified by either the chemical name or the chemical symbol followed by the mass number (for example, U-235 or uranium-235). Another common format is to use the abbreviation of the chemical element with the mass number superscripted (for example, 235U). In this handbook the format used in the text will usually be the element's name followed by the mass number. In equations and tables, the format in Figure 2 will usually be used.

Example:

State the name of the element and the number of protons, electrons, and neutrons in the nuclides listed below.

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1
1H
10
5B
14
7N
11
4
4
8Cd
23
9
9
4Pu

Solution:

The name of the element can be found from the Periodic Table (refer to Chemistry Fundamentals Handbook) or the Chart of the Nuclides (to be discussed later). The number of protons and electrons are equal to Z. The number of neutrons is equal to Z - A.

Nuclide Element Protons Electrons Neutrons
1
1H hydrogen 1 1 0
10
5B boron 5 5 5
14

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11
4
4
8Cd cadmium 48 48 66
23
9
9
4Pu plutonium 94 94 145
Isotopes
Isotopes are nuclides that have the same atomic number and are therefore the same element, but differ in the number of neutrons. Most elements have a few stable isotopes and several unstable, radioactive isotopes. For example, oxygen has three stable isotopes that can be found in nature (oxygen-16, oxygen-17, and oxygen-18) and eight radioactive isotopes. Another example is hydrogen, which has two stable isotopes (hydrogen-1 and hydrogen-2) and a single radioactive isotope (hydrogen-3).
The isotopes of hydrogen are unique in that they are each commonly referred to by a unique name instead of the common chemical element name. Hydrogen-1 is almost always referred to as hydrogen, but the term protium is infrequently used also. Hydrogen-2 is commonly called deuterium and symbolized 2
1D. Hydrogen-3 is commonly called tritium and symbolized 31T. This
text will normally use the symbology 2
1H and 31H for deuterium and tritium, respectively.
Atomic and Nuclear Radii
The size of an atom is difficult to define exactly due to the fact that the electron cloud, formed by the electrons moving in their various orbitals, does not have a distinct outer edge. A reasonable measure of atomic size is given by the average distance of the outermost electron from the nucleus. Except for a few of the lightest atoms, the average atomic radii are
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approximately the same for all atoms, about 2 x 10 -8 cm.
Like the atom the nucleus does not have a sharp outer boundary. Experiments have shown that the nucleus is shaped like a sphere with a radius that depends on the atomic mass number of the atom. The relationship between the atomic mass number and the radius of the nucleus is shown in the following equation.
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Atomic and Nuclear Physics DOE-HDBK-1019/1-93 ATOMIC NATURE OF MATTER
r = (1.25 x 10 -13 cm) A1/3
where:
r = radius of the nucleus (cm)
A = atomic mass number (dimensionless)
The values of the nuclear radii for some light, intermediate, and heavy nuclides are shown in Table 2.
TABLE 2
Calculated Values for Nuclear
Radii
Nuclide Radius of Nucleus
1
1H 1.25 x 10 -13 cm
10

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5
2
6
6Fe 4.78 x 10 -13 cm
17
7
8
2Hf 7.01 x 10 -13 cm
23
9
8
2U 7.74 x 10 -13 cm
25
9
2
8Cf 7.89 x 10 -13 cm
From the table, it is clear that the radius of a typical atom (e.g. 2 x 10 -8 cm) is more than 25,000 times larger than the radius of the largest nucleus.
Nuclear Forces
In the Bohr model of the atom, the nucleus consists of positively-charged protons and electrically■neutra as nucleons. One problem that the Bohr model of the atom presented was accounting for an attractive force to overcome the repulsive force between protons.
Two forces present in the nucleus are (1) electrostatic forces between charged particles and (2) gravitational forces between any two objects that have mass. It is possible to calculate the magnitude of the gravitational force and electrostatic force based upon principles from classical physics.
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Fg
G m1 m2
r
2
Fe
K Q1 Q2
r
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2
ATOMIC NATURE OF MATTER DOE-HDBK-1019/1-93 Atomic and Nuclear Physics
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Newton stated that the gravitational force between two bodies is directly proportional to the masses of the two bodies and inversely proportional to the square of the distance between the bodies. This relationship is shown in the equation below.
where:
F = gravitational force (newtons) g
m = mass of first body (kilograms) 1
m = mass of second body (kilograms) 2
G = gravitational constant (6.67 x 10 N-m /kg)
-11 2 2
r = distance between particles (meters)

between the objects, the greater the gravitational force. So even though the masses of nucleons are very small, the fact that the distance between nucleons is extremely short may make the gravitational force significant. It is necessary to calculate the value for the gravitational force and compare it to the value for other forces to determine the significance of the gravitational force in the nucleus. The gravitational force between two protons that are separated by a distance of 10 meters is about 10 newtons. -20 -24

Coulomb's Law can be used to calculate the force between two protons. The electrostatic force is directly proportional to the electrical charges of the two particles and inversely proportional to the square of the distance between the particles. Coulomb's Law is stated as the following equation.

where:

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F = electrostatic force (newtons) e

K = electrostatic constant (9.0 x 10 N-m /C)

9 2 2

Q = charge of first particle (coulombs) 1

Q = charge of second particle (coulombs) 2

r = distance between particles (meters)

Using this equation, the electrostatic force between two protons that are separated by a distance of 10 meters is about 10 newtons. Comparing this result with the calculation of the

-20 12

gravitational force (10 newtons) shows that the gravitational force is so small that it can be

-24

neglected.

Atomic and Nuclear Physics DOE-HDBK-1019/1-93 ATOMIC NATURE OF MATTER

If only the electrostatic and gravitational forces existed in the nucleus, then it would be impossible to have stable nuclei composed of protons and neutrons. The gravitational forces are much too small to hold the nucleons together compared to the electrostatic forces repelling the protons. Since stable atoms of neutrons and protons do exist, there must be another attractive force acting within the nucleus. This force is called the nuclear force.

The nuclear force is a strong attractive force that is independent of charge. It acts equally only between pairs of neutrons, pairs of protons, or a neutron and a proton. The nuclear force has a very short range; it acts only over distances approximately equal to the diameter of the nucleus (10 -13 cm). The attractive nuclear force between all nucleons drops off with distance much faster than the repulsive electrostatic force between protons.

TABLE 3

Forces Acting in the Nucleus

Force	Interaction	Range
Gravitational	Very weak attractive force between all nucleons	Relatively long
Electrostatic	Strong repulsive force between like charged particles (protons)	Relatively long
Nuclear Force	Strong attractive force between all nucleons	Extremely short

In stable atoms, the attractive and repulsive forces in the nucleus balance. If the forces do not balance, the atom cannot be stable, and the nucleus will emit radiation in an attempt to achieve a more stable configuration.

Summary

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Atomic Nature of Matter Summary
Atoms consist of three basic subatomic particles. These particles are the proton, the neutron, and the electron.
Protons are particles that have a positive charge, have about the same mass as a hydrogen atom, and exist in the nucleus of an atom.
Neutrons are particles that have no electrical charge, have about the same mass as a hydrogen atom, and exist in the nucleus of an atom.
Electrons are particles that have a negative charge, have a mass about eighteen hundred times smaller than the mass of a hydrogen atom, and exist in orbital shells around the nucleus of an atom.
The Bohr model of the atom consists of a dense nucleus of protons and neutrons (nucleons) surrounded by electrons traveling in discrete orbits at fixed distances from the nucleus.
Nuclides are atoms that contain a particular number of protons and neutrons.
Isotopes are nuclides that have the same atomic number and are therefore the same element, but differ in the number of neutrons.
Fixture checkpoint marker-0015-xray is recorded on fixture page 0015.
The atomic number of an atom is the number of protons in the nucleus.
The mass number of an atom is the total number of nucleons (protons and neutrons) in the nucleus.
The notation A
ZX is used to identify a specific nuclide. "Z" represents the atomic number, which is equal to the number of protons. "A" represents the mass number, which is equal to the number of nucleons. "X" represents the chemical symbol of the element.
Number of protons = Z
Number of electrons = Z
Number of neutrons = A - Z
The stability of a nucleus is determined by the different forces interacting within it. The electrostatic force is a relatively long-range, strong, repulsive force that acts between the positively charged protons. The nuclear force is a relatively short-range attractive force between all nucleons. The gravitational force the long range, relatively weak attraction between masses, is negligible compared to the other forces.

CHART OF THE NUCLIDES
The Chart of the Nuclides, like the Periodic Table, is a convenient format for presenting a large amount of scientific information in an organized manner.
EO 1.6 DEFINE the following terms:
a. Enriched uranium
b. Depleted uranium
Chart of the Nuclides
A tabulated chart called the Chart of the Nuclides lists the stable and unstable nuclides in addition to pertinent information about each one. Figure 3 shows a small portion of a typical chart. This

and the number of neutrons (N = A - Z) on the horizontal axis.
The completely gray squares indicate stable isotopes. Those in white squares are artificially radioactive, meaning that they are produced by artificial techniques and do not occur naturally. By consulting a complete chart, other types of isotopes can be found, such as naturally occurring radioactive types (but none are found in the region of the chart that is illustrated in Figure 3). Located in the box on the far left of each horizontal row is general information about the element. The box contains the chemical symbol of the element in addition to the average atomic weight of the naturally occurring substance and the average thermal neutron absorption cross section, which will be discussed in a later module. The known isotopes (elements with the same atomic number Z but different mass number A) of each element are listed to the right.

Figure 3 Nuclide Chart for Atomic Numbers 1 to 6

Information for Stable Nuclides
For the stable isotopes, in addition to the symbol and the atomic mass number, the number percentage of each isotope in the naturally occurring element is listed, as well as the thermal neutron activation cross section and the mass in atomic mass units (amu). A typical block for a stable nuclide from the Chart of the Nuclides is shown in Figure 4.
Figure 4 Stable Nuclides
Information for Unstable Nuclides
For unstable isotopes the additional information includes the half life, the mode of decay (for example, β^-, α), the total disintegration energy in MeV (million electron volts), and the mass in
Fixture checkpoint marker-0016-xray is recorded on fixture page 0016.
amu when available. A typical block for an unstable nuclide from the Chart of the Nuclides is shown in Figure 5.
Figure 5 Unstable Nuclides

Neutron - Proton Ratios
Figure 6 shows the distribution of the stable nuclides plotted on the same axes as the Chart of the Nuclides. As the mass numbers become higher, the ratio of neutrons to protons in the nucleus becomes larger. For helium-4 (2 protons and 2 neutrons) and oxygen-16 (8 protons and 8 neutrons) this ratio is unity. For indium-115 (49 protons and 66 neutrons) the ratio of neutrons to protons has increased to 1.35, and for uranium-238 (92 protons and 146 neutrons) the neutron■to-proton
Figure 6 Neutron - Proton Plot of the Stable Nuclides
If a heavy nucleus were to split into two fragments, each fragment would form a nucleus that would have approximately the same neutron-to-proton ratio as the heavy nucleus. This high neutron-to-proton ratio places the fragments below and to the right of the stability curve displayed by Figure 6. The instability caused by this excess of neutrons is generally rectified by successive beta emissions, each of which converts a neutron to a proton and moves the nucleus toward a more stable neutron-to-proton ratio.

■

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The relative abundance of an isotope in nature compared to other isotopes of the same element is relatively constant. The Chart of the Nuclides presents the relative abundance of the naturally occurring isotopes of an element in units of atom percent. Atom percent is the percentage of the atoms of an element that are of a particular isotope. Atom percent is abbreviated as a/o.
For example, if a cup of water contains 8.23 x 10 atoms of oxygen, and the isotopic abundance 24 of oxygen-18 is 0.20%, then there are 1.65 x 10 atoms of oxygen-18 in the cup. 22
The atomic weight for an element is defined as the average atomic weight of the isotopes of the element. The atomic weight for an element can be calculated by summing the products of the isotopic abundance of the isotope with the atomic mass of the isotope.
Example:
Calculate the atomic weight for the element lithium. Lithium-6 has an atom percent abundance of 7.5% and an atomic mass of 6.015122 amu. Lithium-7 has an atomic abundance of 92.5% and an atomic mass of 7.016003 amu.
Solution:
The other common measurement of isotopic abundance is weight percent (w/o). Weight percent is the percent weight of an element that is a particular isotope. For example, if a sample of material contained 100 kg of uranium that was 28 w/o uranium-235, then 28 kg of uranium-235 was present in the sample.
Enriched and Depleted Uranium
Natural uranium mined from the earth contains the isotopes uranium-238, uranium-235 and uranium-234. The majority (99.2745%) of all the atoms in natural uranium are uranium-238. Most of the remaining atoms (0.72%) are uranium-235, and a slight trace (0.0055%) are uranium-234. Although all isotopes of uranium have similar chemical properties, each of the isotopes has significantly different nuclear properties. For reasons that will be discussed in later modules, the isotope uranium-235 is usually the desired material for use in reactors.
A vast amount of equipment and energy are expended in processes that separate the isotopes of uranium (and other elements). The details of these processes are beyond the scope of this module. These processes are called enrichment processes because they selectively increase the proportion of a particular isotope. The enrichment process typically starts with feed material that has the proportion of isotopes that occur naturally. The process results in two types of
CHART OF THE NUCLIDES DOE-HDBK-1019/1-93 Atomic and Nuclear Physics
In the case of uranium, the natural uranium ore is 0.72 a/o uranium-235. The desired outcome
Fixture checkpoint marker-0017-xray is recorded on fixture page 0017.
of the enrichment process is to produce enriched uranium. Enriched uranium is defined as uranium in which the isotope uranium-235 has a concentration greater than its natural value. The enrichment process will also result in the byproduct of depleted uranium. Depleted uranium is defined as uranium in which the isotope uranium-235 has a concentration less than its natural value. Although depleted uranium is referred to as a by-product of the enrichment process, it does have uses in the nuclear field and in commercial and defense industries.
Summary
The important information in this chapter is summarized below.
Chart of the Nuclides Summary
Enriched uranium is uranium in which the isotope uranium-235 has a concentration greater than its natural value of 0.7%.
Depleted uranium is uranium in which the isotope uranium-235 has a concentration less than its natural value of 0.7%.
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Atomic and Nuclear Physics DOE-HDBK-1019/1-93 MASS DEFECT AND BINDING ENERGY

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The separate laws of Conservation of Mass and Conservation of Energy are not applied strictly on the nuclear level. It is possible to convert between mass and energy. Instead of two separate conservation laws, a single conservation law states that the sum of mass and energy is conserved. Mass does not magically appear and disappear at random. A decrease in mass will be accompanied by a corresponding increase in energy and vice versa.

EO 1.7 DEFINE the following terms:

a. Mass defect

b. Binding energy

EO 1.8 Given the atomic mass for a nuclide and the atomic masses of a neutron, proton, and electron, CALCULATE the mass defect and binding energy of the nuclide.

Mass Defect

Careful measurements have shown that the mass of a particular atom is always slightly less than the sum of the masses of the individual neutrons, protons, and electrons of which the atom consists. The difference between the mass of the atom and the sum of the masses of its parts is called the mass defect (Δm). The mass defect can be calculated using Equation (1-1). In calculating the mass defect it is important to use the full accuracy of mass measurements because the difference in mass is small compared to the mass of the atom. Rounding off the masses of atoms and particles to three or four significant digits prior to the calculation will result in a calculated mass defect of zero.

$\Delta m = [Z(mp + me) + (A-Z)mn] - matom$ (1-1)

where:

Δm = mass defect (amu)

mp = mass of a proton (1.007277 amu)

mn = mass of a neutron (1.008665 amu)

me = mass of an electron (0.000548597 amu)

matom = mass of nuclide A

ZX (amu)

Z = atomic number (number of protons)

A = mass number (number of nucleons)

Example:

Calculate the mass defect for lithium-7. The mass of lithium-7 is 7.016003 amu.

Solution:

Δ m Z m

p

me A Z mn matom

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Δ m 3 1.007826 amu 7 3 1.008665 amu 7.016003 amu

Δ m 0.0421335 amu

Binding Energy

The loss in mass, or mass defect, is due to the conversion of mass to binding energy when the nucleus is formed. Binding energy is defined as the amount of energy that must be supplied to a nucleus to completely separate its nuclear particles (nucleons). It can also be understood as the amount of energy that would be released if the nucleus was formed from the separate particles. Binding energy is the energy equivalent of the mass defect. Since the mass defect

the binding energy using a conversion factor derived by the mass-energy relationship from Einstein's Theory of Relativity.
Einstein's famous equation relating mass and energy is E = mc2 where c is the velocity of light (c = 2.998 x 108 m/sec). The energy equivalent of 1 amu can be determined by inserting this quantity of mass into Einstein's equation and applying conversion factors.
E m c 2
1 amu
1.6606 x 10 27 kg
1 amu
2.998 x 108 m
sec
2
1 N
1
kg m
sec2
Fixture checkpoint marker-0019-xray is recorded on fixture page 0019.
1 J
1 N m
1.4924 x 10 10 J 1 MeV
1.6022 x 10 13 J
931.5 MeV
Conversion Factors:
1 amu = 1.6606 x 10 -27 kg
1 newton = 1 kg-m/sec2
1 joule = 1 newton-meter
1 MeV = 1.6022 x 10-13 joules
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Atomic and Nuclear Physics DOE-HDBK-1019/1-93 MASS DEFECT AND BINDING ENERGY
Since 1 amu is equivalent to 931.5 MeV of energy, the binding energy can be calculated using Equation (1-2).
(1-2)
B.E. ∆m
931.5 MeV
1 amu
Example:
Calculate the mass defect and binding energy for uranium-235. One uranium-235 atom has a mass of 235.043924 amu.
Solution:
Step 1: Calculate the mass defect using Equation (1-1).
∆m Z m
p
me A Z mn matom
∆m 92 1.007826 amu 235 92 1.008665 amu 235.043924 amu
∆m 1.91517 amu
Step 2: Use the mass defect and Equation (1-2) to calculate the binding energy.
B.E. ∆m
931.5 MeV

1.91517 amu
931.5 MeV
1 amu
1784 MeV
Energy Levels of Atoms
The electrons that circle the nucleus move in fairly well-defined orbits. Some of these electrons are more tightly bound in the atom than others. For example, only 7.38 eV is required to remove the outermost electron from a lead atom, while 88,000 eV is required to remove the innermost electron. The process of removing an electron from an atom is called ionization, and the energy required to remove the electron is called the ionization energy.
In a neutral atom (number of electrons = Z) it is possible for the electrons to be in a variety of different orbits, each with a different energy level. The state of lowest energy is the one in which the atom is normally found and is called the ground state. When the atom possesses more energy than its ground state energy, it is said to be in an excited state.
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An atom cannot stay in the excited state for an indefinite period of time. An excited atom will eventually transition to either a lower-energy excited state, or directly to its ground state, by emitting a discrete bundle of electromagnetic energy called an x-ray. The energy of the x-ray will be equal to the difference between the energy levels of the atom and will typically range from several eV to 100,000 eV in magnitude.
Energy Levels of the Nucleus
The nucleons in the nucleus of an atom, like the electrons that circle the nucleus, exist in shells
Fixture checkpoint marker-0020-xray is recorded on fixture page 0020.
that correspond to energy states. The energy shells of the nucleus are less defined and less understood than those of the electrons. There is a state of lowest energy (the ground state) and discrete possible excited states for a nucleus. Where the discrete energy states for the electrons of an atom are measured in eV or keV, the energy levels of the nucleus are considerably greater and typically measured in MeV.
A nucleus that is in the excited state will not remain at that energy level for an indefinite period. Like the electrons in an excited atom, the nucleons in an excited nucleus will transition towards their lowest energy configuration and in doing so emit a discrete bundle of electromagnetic radiation called a gamma ray (γ-ray). The only differences between x-rays and γ-rays are their energy levels and whether they are emitted from the electron shell or from the nucleus.
The ground state and the excited states of
Figure 7 Energy Level Diagram - Nickel-60
a nucleus can be depicted in a nuclear energy-level diagram. The nuclear energy-level diagram consists of a stack of horizontal bars, one bar for each of the excited states of the nucleus. The vertical distance between the bar representing an excited state and the bar representing the ground state is proportional to the energy level of the excited state with respect to the ground state. This difference in energy between the ground state and the excited state is called the excitation energy

a nuclide has zero excitation energy. The
bars for the excited states are labeled with
their respective energy levels. Figure 7 is
the energy level diagram for nickel-60.
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Atomic and Nuclear Physics DOE-HDBK-1019/1-93 MASS DEFECT AND BINDING ENERGY
Summary
The important information in this chapter is summarized below.
Mass Defect and Binding Energy Summary
Mass defect is the difference between the mass of the atom and the sum of the
masses of its constituent parts.
Binding energy is the amount of energy that must be supplied to a nucleus to
completely separate its nuclear particles. Binding energy is the energy equivalent
of the mass defect.
Mass defect can be calculated by using the equation below.
$\Delta m = [Z(mp + me) + (A-Z)mn] - matom$
Binding energy can be calculated by multiplying the mass defect by the factor
of 931.5 MeV per amu.
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MODES OF RADIOACTIVE DECAY DOE-HDBK-1019/1-93 Atomic and Nuclear Physics
MODES OF RADIOACTIVE DECAY
Most atoms found in nature are stable and do not emit particles or energy that
change form over time. Some atoms, however, do not have stable nuclei. These
atoms emit radiation in order to achieve a more stable configuration.
EO 2.1 DESCRIBE the following processes:
a. Alpha decay d. Electron capture
b. Beta-minus decay e. Internal conversions
c. Beta-plus decay f. Isomeric transitions
EO 2.2 Given a Chart of the Nuclides, WRITE the radioactive decay
chain for a nuclide.
Fixture checkpoint marker-0021-xray is recorded on fixture page 0021.
EO 2.3 EXPLAIN why one or more gamma rays typically accompany
particle emission.
EO 2.4 Given the stability curve on the Chart of the Nuclides,
DETERMINE the type of radioactive decay that the nuclides in
each region of the chart will typically undergo.
Stability of Nuclei
As mass numbers become larger, the ratio of neutrons to protons in the nucleus becomes larger for the stable nuclei. Non-stable nuclei may have an excess or deficiency of neutrons and undergo a transformation process known as beta (β) decay. Non-stable nuclei can also undergo a variety of other processes such as alpha (α) or neutron (n) decay. As a result of these decay processes, the final nucleus is in a more stable or more tightly bound configuration.
Natural Radioactivity
In 1896, the French physicist Becquerel discovered that crystals of a uranium salt emitted rays that were similar to x-rays in that they were highly penetrating, could affect a photographic plate, and induced electrical conductivity in gases. Becquerel's discovery was followed in 1898 by the identification of two other radioactive elements, polonium and radium, by Pierre and Marie Curie.

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Atomic and Nuclear Physics DOE-HDBK-1019/1-93 MODES OF RADIOACTIVE DECAY
Heavy elements, such as uranium or thorium, and their unstable decay chain elements emit radiation in their naturally occurring state. Uranium and thorium, present since their creation at the beginning of geological time, have an extremely slow rate of decay. All naturally occurring nuclides with atomic numbers greater than 82 are radioactive.
Nuclear Decay
Whenever a nucleus can attain a more stable (i.e., more tightly bound) configuration by emitting radiation, a spontaneous disintegration process known as radioactive decay or nuclear decay may occur. In practice, this "radiation" may be electromagnetic radiation, particles, or both.
Detailed studies of radioactive decay and nuclear reaction processes have led to the formulation of useful conservation principles. The four principles of most interest in this module are discussed below.
Conservation of electric charge implies that charges are neither created nor destroyed. Single positive and negative charges may, however, neutralize each other. It is also possible for a neutral particle to produce one charge of each sign.
Conservation of mass number does not allow a net change in the number of nucleons. However, the conversion of a proton to a neutron and vice versa is allowed.
Conservation of mass and energy implies that the total of the kinetic energy and the energy equivalent of the mass in a system must be conserved in all decays and reactions. Mass can be converted to energy and energy can be converted to mass, but the sum of mass and energy must be constant.
Conservation of momentum is responsible for the distribution of the available kinetic energy among product nuclei, particles, and/or radiation. The total amount is the same before and after the reaction even though it may be distributed differently among entirely different nuclides and/or particles.
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■ ■ ■ ■ ■ ■
■ ■ ■ ■ ■
MODES OF RADIOACTIVE DECAY DOE-HDBK-1019/1-93 Atomic and Nuclear Physics
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Alpha Decay (■)
Alpha decay is the emission of alpha particles (helium nuclei) which may be represented as either $^{4}_{2}He$ or ■. When an unstable nucleus ejects an alpha particle, the atomic number is reduced by 2 and the mass number decreased by 4. An example is uranium-234 which decays by the ejection of an alpha particle accompanied by the emission of a 0.068 MeV gamma.
The combined kinetic energy of the daughter nucleus (Thorium-230) and the ■ particle is designated
Fixture checkpoint marker-0022-xray is recorded on fixture page 0022.
as KE. The sum of the KE and the gamma energy is equal to the difference in mass between the original nucleus (Uranium-234) and the final particles (equivalent to the binding energy released, since ■m = BE). The alpha particle will carry off as much as 98% of the kinetic energy and, in most cases, can be considered to carry off all the kinetic energy.
Beta Decay (■)
Beta decay is the emission of electrons of nuclear rather than orbital origin. These particles are electrons that have been expelled by excited nuclei and may have a charge of either sign.
If both energy and momentum are to be conserved, a third type of particle, the neutrino, ■, must be involved. The neutrino is associated with positive electron emission, and its antiparticle, the antineutrino, , is emitted with a negative electron. These uncharged particles have only the

they pass through all materials with so few interactions that the energy they possess cannot be recovered. The neutrinos and antineutrinos are included here only because they carry a portion of the kinetic energy that would otherwise belong to the beta particle, and therefore, must be considered for energy and momentum to be conserved. They are normally ignored since they are not significant in the context of nuclear reactor applications.

Negative electron emission, represented as e, -1 -1■, or simply as e or ■ , effectively converts a neutro to a proton, thus increasing the atomic number by one and leaving the mass number unchanged. This is a common mode of decay for nuclei with an excess of neutrons, such as fission fragments below and to the right of the neutron-proton stability curve (refer to Figure 6). An example of a typical beta minus-decay reaction is shown below.

■ ■
■
■ ■
■ ■ ■ ■

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Figure 9 Orbital Electron Capture

Positively charged electrons (beta-plus) are known as positrons. Except for sign, they are nearly identical to their negatively charged cousins. When a positron, represented as e, ■, or simply as +1 +1 0 0
e or ■ , is ejected from the nucleus, the atomic number is decreased by one and the mass number + + remains unchanged. A proton has been converted to a neutron. An example of a typical positron (beta-plus) decay is shown below.

Electron Capture (EC, K-capture)

Nuclei having an excess of protons may capture an electron from one of the inner orbits which immediately combines with a proton in the nucleus to form a neutron. This process is called electron capture (EC). The electron is normally captured from the innermost orbit (the K-shell), and, consequently, this process is sometimes called K-capture. The following example depicts electron capture.

A neutrino is formed at the same time that the neutron is formed, and energy carried off by it serves to conserve momentum. Any energy that is available due to the atomic mass of the product being appreciably less than that of the parent will appear as gamma radiation. Also, there will always be characteristic x-rays given off when an electron from one of the higher energy shells moves in to fill the vacancy in the K-shell.

Electron capture is shown graphically in Figure 8.

Electron capture and positron emission result in the production of the same daughter product, and

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For positron emission to occur, however, the mass of the daughter product must be less than the mass of the parent by an amount equal to at least twice the mass of an electron. This mass difference between the parent and daughter is necessary to account for two items present in the parent but not in the daughter. One item is the positron ejected from the nucleus of the parent. The other item is that the daughter product has one less orbital electron than the parent. If this requirement is not met, then orbital electron capture takes place exclusively.

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Gamma Emission (■)

Gamma radiation is a high-energy electromagnetic radiation that originates in the nucleus. It is emitted in the form of photons, discrete bundles of energy that have both wave and particle properties. Often a daughter nuclide is left in an excited state after a radioactive parent nucleus undergoes a transformation by alpha decay, beta decay, or electron capture. The nucleus will drop to the ground state by the emission of gamma radiation.

Internal Conversion

The usual method for an excited nucleus to go from the excited state to the ground state is by emission of gamma radiation. However, in some cases the gamma ray (photon) emerges from the nucleus only to interact with one of the innermost orbital electrons and, as a result, the energy of the Fixture checkpoint marker-0024-xray is recorded on fixture page 0024.
photon is transferred to the electron. The gamma ray is then said to have undergone internal conversion. The conversion electron is ejected from the atom with kinetic energy equal to the gamma energy minus the binding energy of the orbital electron. An orbital electron then drops to a lower energy state to fill the vacancy, and this is accompanied by the emission of characteristic x-rays.

Isomers and Isomeric Transition

Isomeric transition commonly occurs immediately after particle emission; however, the nucleus may remain in an excited state for a measurable period of time before dropping to the ground state at its own characteristic rate. A nucleus that remains in such an excited state is known as a nuclear isomer because it differs in energy and behavior from other nuclei with the same atomic number and mass number. The decay of an excited nuclear isomer to a lower energy level is called an isomeric transition. It is also possible for the excited isomer to decay by some alternate means, for example, by beta emission.

An example of gamma emission accompanying particle emission is illustrated by the decay of nitrogen-16 below.

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Decay Chains

When an unstable nucleus decays, the resulting daughter nucleus is not necessarily stable. The nucleus resulting from the decay of a parent is often itself unstable, and will undergo an additional decay. This is especially common among the larger nuclides.

It is possible to trace the steps of an unstable atom as it goes through multiple decays trying to achieve stability. The list of the original unstable nuclide, the nuclides that are involved as intermediate steps in the decay, and the final stable nuclide is known as the decay chain. One common method for stating the decay chain is to state each of the nuclides involved in the standard X format. Arrows are used between nuclides to indicate where decays occur, with the type of decay A Z
indicated above the arrow and the half-life below the arrow. The half-life for decay will be discussed in the next chapter.

Example:

Write the decay chains for rubidium-91 and actinium-215. Continue the chains until a stable nuclide or a nuclide with a half-life greater than 1 x 10 years is reached. 6

Solution:

Predicting Type of Decay

Radioactive nuclides tend to decay in a way that results in a daughter nuclide that lies closer to the line of stability. Due to this, it is possible to predict the type of decay that a nuclide will undergo based on its location relative to the line of stability on the Chart of the Nuclides.

Figure 9 illustrates the type of decay nuclides in different regions of the chart will typically undergo. Nuclides that are below and to the right of the line of stability will usually undergo ■ decay. - Nuclides that are above and to the left of the line of stability will usually undergo either ■ decay or + electron capture. Most nuclides that will undergo ■ decay are found in the upper right hand region of the chart. These are general rules that have many exceptions, especially in the region of the heavy nuclides.

Fixture checkpoint marker-0025-xray is recorded on fixture page 0025.

Figure 9 Types of Radioactive Decay Relative to the Line of Stability

Summary

The important information in this chapter is summarized below.

Modes of Radioactive Decay Summary

Alpha decay is the emission of an alpha particle (2 protons and 2 neutrons) from an unstable nucleus. The daughter nuclide has an atomic number 2 less than the parent nuclide and a mass number 4 less than the parent nuclide. The daughter nucleus commonly releases its excitation energy by gamma emission.

Beta-minus decay effectively converts a neutron to a proton and an electron, which is immediately ejected from the nucleus. The daughter nuclide has its atomic number increased by 1 and the same mass number compared to the parent.

Beta-plus decay effectively converts a proton to a neutron and a positron, which is immediately ejected from the nucleus. The daughter nuclide has its atomic number decreased by 1 and the same mass number compared to the parent.

In electron capture, the nucleus absorbs an electron from the innermost orbit.

Internal conversion occurs when a gamma ray, emitted by the nucleus as it goes from the excited state to the ground state, interacts with one of the innermost electrons of the same atom. The electron is ejected from the atom.
An isomeric transition is the decay of an excited nucleus to a lower-energy level by the emission of a gamma ray.
Decay chains can be found by tracing the steps an unstable atom goes through as it tries to achieve stability.
Many modes of radioactive decay result in a daughter nuclide that has an energy level above the ground state. This excitation energy is usually released immediately in the form of a gamma ray.
The type of decay that a nuclide will typically undergo can be determined by its relationship to the line of stability on the Chart of the Nuclides. Nuclides that lie below and to the right of the line of stability will typically beta minus decay. Nuclides above and to the left of the line will typically either beta plus decay or electron capture. Most alpha emitters are found in the upper, right-hand corner of the chart.
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RADIOACTIVITY DOE-HDBK-1019/1-93 Atomic and Nuclear Physics
RADIOACTIVITY
The rate at which a sample of radioactive material decays is not constant. As individual atoms of the material decay, there are fewer of those types of atoms remaining. Since the rate of decay is directly proportional to the number of atoms, the rate of decay will decrease as the number of atoms decreases.
EO 2.5 DEFINE the following terms:
a. Radioactivity d. Radioactive decay constant
b. Curie e. Radioactive half-life
c. Becquerel
EO 2.6 Given the number of atoms and either the half-life or decay constant of a nuclide, CALCULATE the activity.
EO 2.7 Given the initial activity and the decay constant of a nuclide, CALCULATE the activity at any later time.
EO 2.8 CONVERT between the half-life and decay constant for a nuclide.
EO 2.9 Given the Chart of the Nuclides and the original activity, PLOT the radioactive decay curve for a nuclide on either linear or
Fixture checkpoint marker-0026-xray is recorded on fixture page 0026.
semi-log coordinates.
EO 2.10 DEFINE the following terms:
a. Radioactive equilibrium
b. Transient radioactive equilibrium
Radioactive Decay Rates
Radioactivity is the property of certain nuclides of spontaneously emitting particles or gamma radiation. The decay of radioactive nuclides occurs in a random manner, and the precise time at which a single nucleus will decay cannot be determined. However, the average behavior of a very large sample can be predicted accurately by using statistical methods. These studies have revealed that there is a certain probability that in a given time interval a certain fraction of the nuclei within a sample of a particular nuclide will decay. This probability per unit time that an atom of a nuclide will decay is known as the radioactive decay constant, λ. The units for the

constant units can also be expressed as second-1, minute-1, hour-1, and year-1.
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■■
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The activity (A) of a sample is the rate of decay of that sample. This rate of decay is usually measured in the number of disintegrations that occur per second. For a sample containing millions of atoms, the activity is the product of the decay constant and the number of atoms present in the sample.
The relationship between the activity, number of atoms, and decay constant is shown in Equation (1-3).
A = ■ N (1-3)
Fixture checkpoint marker-0027-xray is recorded on fixture page 0027.
where:
A = Activity of the nuclide (disintegrations/second)
■ = decay constant of the nuclide (second)-1
N = Number of atoms of the nuclide in the sample
Since ■ is a constant, the activity and the number of atoms are always proportional.
Units of Measurement for Radioactivity
Two common units to measure the activity of a substance are the curie (Ci) and becquerel (Bq). A curie is a unit of measure of the rate of radioactive decay equal to 3.7 x 10 disintegrations 10 per second. This is approximately equivalent to the number of disintegrations that one gram of radium-226 will undergo in one second. A becquerel is a more fundamental unit of measure of radioactive decay that is equal to 1 disintegration per second. Currently, the curie is more widely used in the United States, but usage of the becquerel can be expected to broaden as the metric system slowly comes into wider use. The conversion between curies and becquerels is shown below.
1 curie = 3.7 x 10 becquerels 10
Variation of Radioactivity Over Time
The rate at which a given radionuclide sample decays is stated in Equation (1-3) as being equal to the product of the number of atoms and the decay constant. From this basic relationship it is possible to use calculus to derive an expression which can be used to calculate how the number of atoms present will change over time. The derivation is beyond the scope of this text, but Equation (1-4) is the useful result.
(1-4)
where:
N = number of atoms present at time t
N = number of atoms initially present o
■ = decay constant (time)-1
t = time
A Ao
e
t
A Aoe
t
A
Ao
e

ln A
Ao
t
t
ln
A
Ao
t
1/2
ln
1
2
t
1/2
ln 2 0.693
RADIOACTIVITY DOE-HDBK-1019/1-93 Atomic and Nuclear Physics
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Since the activity and the number of atoms are always proportional, they may be used interchangeably to describe any given radionuclide population. Therefore, the following is true.
Fixture checkpoint marker-0028-xray is recorded on fixture page 0028.
(1-5)
where:
A = activity present at time t
A = activity initially present o
= decay constant (time)
-1
t = time
Radioactive Half-Life
One of the most useful terms for estimating how quickly a nuclide will decay is the radioactive half-life. The radioactive half-life is defined as the amount of time required for the activity to decrease to one-half of its original value. A relationship between the half-life and decay constant can be developed from Equation (1-5). The half-life can be calculated by solving Equation (1-5) for the time, t, when the current activity, A, equals one-half the initial activity A .
o
First, solve Equation (1-5) for t.
If A is equal to one-half of Ao, then A/Ao is equal to one-half. Substituting this in the equation above yields an expression for t .
1/2
(1-6)
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The basic features of decay of a radionuclide sample are shown by the graph in Figure 10.
Figure 10 Radioactive Decay as a Function of Time in Units of Half-Life
Assuming an initial number of atoms No, the population, and consequently, the activity may be noted to decrease by one-half of this value in a time of one half-life. Additional decreases occur so that whenever one half-life elapses, the number of atoms drops to one-half of what its value was at the beginning of that time interval. After five half-lives have elapsed, only 1/32, or 3.1%, of the original number of atoms remains. After seven half-lives, only 1/128, or 0.78%, of the atoms remains. The number of atoms existing after 5 to 7 half-lives can usually be

information on calculating the number of atoms contained within a sample.
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■
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Example:
A sample of material contains 20 micrograms of californium-252.
Californium-252 has a half-life of 2.638 years.
Calculate:
(a) The number of californium-252 atoms initially present
(b) The activity of the californium-252 in curies
(c) The number of californium-252 atoms that will remain in 12 years
(d) The time it will take for the activity to reach 0.001 curies
Solution:
(a) The number of atoms of californium-252 can be determined as below.
(b) First, use Equation (1-6) to calculate the decay constant.
Use this value for the decay constant in Equation (1-3) to determine the activity.
N No
e
t
4.78 x 1016 e
(0.263/yr) (12 yr)
2.04 x 1015
A Aoe
t
Fixture checkpoint marker-0029-xray is recorded on fixture page 0029.
A
Ao
e
t
ln A
Ao
t
t
ln
A
Ao
t
ln
0.001 Ci
0.0108 Ci
0.263 year 1
t 9.05 years
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(c) The number of californium atoms that will remain in 12 years can be calculated from Equation (1-4).

from Equation (1-5). First, solve Equation (1-5) for time.
Inserting the appropriate values in the right side of this equation will result in the required time.
Plotting Radioactive Decay
It is useful to plot the activity of a nuclide as it changes over time. Plots of this type can be used to determine when the activity will fall below a certain level. This plot is usually done showing activity on either a linear or a logarithmic scale. The decay of the activity of a single nuclide on a logarithmic scale will plot as a straight line because the decay is exponential.
t
1/2
0.693
0.693
t
1/2
0.693
7.13 seconds
0.0972 second 1
A Aoe
t
RADIOACTIVITY DOE-HDBK-1019/1-93 Atomic and Nuclear Physics
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Example:
Plot the radioactive decay curve for nitrogen-16 over a period of 100 seconds. The initial activity is 142 curies and the half-life of nitrogen-16 is 7.13 seconds. Plot the curve on both linear rectangular coordinates and on a semi-log scale.
Solution:
First, use Equation (1-6) to calculate the decay constant corresponding to a half-life of 7.13 seconds.
Use the decay constant determined above to calculate the activity at various times using Equation (1-5).
Time Activity
0 seconds 142 Ci
20 seconds 20.3 Ci
Fixture checkpoint marker-0030-xray is recorded on fixture page 0030.
40 seconds 2.91 Ci
60 seconds 0.416 Ci
80 seconds 0.0596 Ci
100 seconds 0.00853 Ci
■■
■■
■■
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Figure 11 Linear and Semi-log Plots of Nitrogen-16 Decay
Plotting the data points calculated above on both linear and semilog scales results in the graphs shown in Figure 11.
If a substance contains more than one radioactive nuclide, the total activity is the sum of the individual activities of each nuclide. As an example, consider a sample of material that

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1 x 10 atoms of manganese-54 that has a half-life of 312.2 days (■ = 2.57 x 10 sec), and 1 6 -8 -1
x 10 atoms of cobalt-60 that has a half-life of 1925 days (■ = 4.17 x 10 sec). 6 -9 -1
The initial activity of each of the nuclides would be the product of the number of atoms and the decay constant.

Figure 12 Combined Decay of Iron-56, Manganese-54, and Cobalt-60

Plotting the manner in which the activities of each of the three nuclides decay over time demonstrates that initially the activity of the shortest-lived nuclide (iron-59) dominates the total activity, then manganese-54 dominates. After almost all of the iron and manganese have

Fixture checkpoint marker-0031-xray is recorded on fixture page 0031.

decayed away, the only contributor to activity will be the cobalt-60. A plot of this combined decay is shown in Figure 12.

Radioactive Equilibrium

Radioactive equilibrium exists when a radioactive nuclide is decaying at the same rate at which it is being produced. Since the production rate and decay rate are equal, the number of atoms present remains constant over time.

An example of radioactive equilibrium is the concentration of sodium-24 in the coolant circulating through a sodium-cooled nuclear reactor. Assume that the sodium-24 is being produced at a rate of 1 x 10 atoms per second. If the sodium-24 were stable and did not decay, 6 the amount of sodium-24 present after some period of time could be calculated by multiplying the production rate by the amount of time. Plotting the amount of material present would result in the graph in Figure 13.

However, sodium-24 is not stable, and it decays with a half-life of 14.96 hours. If no sodium-24 is present initially and production starts at a rate of 1 x 10 atoms per second, the rate 6 of decay will initially be zero because there is no sodium-24 present to decay. The rate of decay of sodium-24 will increase as the amount of sodium-24 increases.

■
■
■
■

Figure 13 Cumulative Production of Sodium-24 Over Time

The amount of sodium-24 present will initially increase rapidly, then it will increase at a continually decreasing rate until the rate of decay is equal to the rate of production. It is possible to calculate how much sodium-24 will be present at equilibrium by setting the production rate (R) equal to the decay rate (■ N).

where:

R = production rate (atoms/second)
■ = decay constant (second)-1
N = number of atoms

It is possible to calculate the equilibrium value for sodium-24 being produced at a rate of 1 x 10 atoms/second. 6

■ ■ ■■
■
■
■

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Figure 14 Approach of Sodium-24 to Equilibrium

The development of the equation to calculate how the amount of sodium-24 changes over time as it approaches the equilibrium value is beyond the scope of this handbook. However, the equation is presented below.

This equation can be used to calculate the values of the amount of sodium-24 present at different times. As the time increases, the exponential term approaches zero, and the number of atoms present will approach R/■. A plot of the approach of sodium-24 to equilibrium is shown in Figure 14.

Transient Radioactive Equilibrium

Transient radioactive equilibrium occurs when the parent nuclide and the daughter nuclide decay at essentially the same rate.

For transient equilibrium to occur, the parent must have a long half-life when compared to the daughter. An example of this type of compound decay process is barium-140, which decays by beta emission to lanthanum-140, which in turn decays by beta emission to stable cerium-140.

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Figure 15 Transient Equilibrium in the Decay of Barium-140

The decay constant for barium-140 is considerably smaller than the decay constant for lanthanum-140. Remember that the rate of decay of both the parent and daughter can be represented as ■N. Although the decay constant for barium-140 is smaller, the actual rate of decay (■N) is initially larger than that of lanthanum-140 because of the great difference in their initial concentrations. As the concentration of the daughter increases, the rate of decay of the daughter will approach and eventually match the decay rate of the parent. When this occurs, they are said to be in transient equilibrium. A plot of the barium-lanthanum-cerium decay chain reaching transient equilibrium is shown in Figure 15.

Secular equilibrium occurs when the parent has an extremely long half-life. In the long decay chain for a naturally radioactive element, such as thorium-232, where all of the elements in the chain are in secular equilibrium, each of the descendants has built up to an equilibrium amount and all decay at the rate set by the original parent. The only exception is the final stable element on the end of the chain. Its number of atoms is constantly increasing.

Summary

The important information in this chapter is summarized on the following page.

■
■■
■

Radioactivity Summary

& Radioactivity is the decay of unstable atoms by the emission of particles and electromagnetic radiation.

& A curie (Ci) is a unit of radioactivity equal to 3.7 x 10 disintegrations per 10 second.

& A becquerel (Bq) is a unit of radioactivity equal to 1 disintegration per second.

& The radioactive decay constant (■) is the probability per unit time that an atom will decay.

decrease to one-half its original value.
& The activity of a substance can be calculated from the number of atoms and the decay constant based on the equation below.
& The amount of activity remaining after a particular time can be calculated from the equation below.
& The relationship between the decay constant and the half-life is shown below.
& Plots of radioactive decay can be useful to describe the variation of activity over time. If decay is plotted using semi-log scale the plot results in a straight line.
& Radioactive equilibrium exists when the production rate of a material is equal to the removal rate.
& Transient radioactive equilibrium exists when the parent nuclide and the daughter nuclide decay at essentially the same rate. This occurs only when the parent has a long half-life compared to the daughter.
Atomic and Nuclear Physics DOE-HDBK-1019/1-93 NEUTRON INTERACTIONS
NEUTRON INTERACTIONS
Neutrons can cause many different types of interactions. The neutron may simply scatter off the nucleus in two different ways, or it may actually be absorbed into the nucleus. If a neutron is absorbed into the nucleus, it may result in the emission of a gamma ray or a subatomic particle, or it may cause the nucleus to fission.
EO 3.1 DESCRIBE the following scattering interactions between a neutron and a nucleus:
a. Elastic scattering
b. Inelastic scattering
Fixture checkpoint marker-0033-xray is recorded on fixture page 0033.
EO 3.2 STATE the conservation laws that apply to an elastic collision between a neutron and a nucleus.
EO 3.3 DESCRIBE the following reactions where a neutron is absorbed in a nucleus:
a. Radiative capture
b. Particle ejection
Scattering
A neutron scattering reaction occurs when a nucleus, after having been struck by a neutron, emits a single neutron. Despite the fact that the initial and final neutrons do not need to be (and often are not) the same, the net effect of the reaction is as if the projectile neutron had merely "bounced off," or scattered from, the nucleus. The two categories of scattering reactions, elastic and inelastic scattering, are described in the following paragraphs.
Elastic Scattering
In an elastic scattering reaction between a neutron and a target nucleus, there is no energy transferred into nuclear excitation. Momentum and kinetic energy of the "system" are conserved although there is usually some transfer of kinetic energy from the neutron to the target nucleus. The target nucleus gains the amount of kinetic energy that the neutron loses.

mn
vn,i mTvT,i mnvn,f mTvT,f
1
2
m v

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1
2
mn
v
2
n,i
1
2
mT
v
2
T,i
1
2
mn
v
2
n,f
1
2
mT
v
2
T,f
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Figure 16 Elastic Scattering
Figure 16 illustrates the process of elastic scattering of a neutron off a target nucleus. In the elastic scattering reaction, the conservation of momentum and kinetic energy is represented by the equations below.
Conservation of momentum (mv)
Fixture checkpoint marker-0034-xray is recorded on fixture page 0034.
Conservation of kinetic energy
where:
mn = mass of the neutron
m = mass of the target nucleus T
vn,i = initial neutron velocity
vn,f = final neutron velocity
vT,i = initial target velocity
vT,f = final target velocity
Atomic and Nuclear Physics DOE-HDBK-1019/1-93 NEUTRON INTERACTIONS
Elastic scattering of neutrons by nuclei can occur in two ways. The more unusual of the two interactions is the absorption of the neutron, forming a compound nucleus, followed by the re-emission of a neutron in such a way that the total kinetic energy is conserved and the nucleus returns to its ground state. This is known as resonance elastic scattering and is very dependent upon the initial kinetic energy possessed by the neutron. Due to formation of the compound nucleus, it is also referred to as compound elastic scattering. The second, more usual method, is termed potential elastic scattering and can be understood by visualizing the neutrons and

with incident neutrons that have an energy of up to about 1 MeV. In potential scattering, the neutron does not actually touch the nucleus and a compound nucleus is not formed. Instead, the neutron is acted on and scattered by the short range nuclear forces when it approaches close enough to the nucleus.

Inelastic Scattering

In inelastic scattering, the incident neutron is absorbed by the target nucleus, forming a compound nucleus. The compound nucleus will then emit a neutron of lower kinetic energy which leaves the original nucleus in an excited state. The nucleus will usually, by one or more gamma emissions, emit this excess energy to reach its ground state. Figure 17 shows the process of inelastic scattering.

For the nucleus that has reached its ground state, the sum of the kinetic energy of the exit

Figure 17 Inelastic Scattering

neutron, the target nucleus, and the total gamma energy emitted is equal to the initial kinetic energy of the incident neutron.

1
0
n
238
92
U
239
92
U
239
92
U
0
0
1
0
n
10
5
B
11
5
B

Fixture checkpoint marker-0035-xray is recorded on fixture page 0035.

7
3
Li
4
2

Absorption Reactions

Most absorption reactions result in the loss of a neutron coupled with the production of a

is emitted at some later time. Radiative capture, particle ejection, and fission are all categorized as absorption reactions and are briefly described below.

Radiative Capture

In radiative capture the incident neutron enters the target nucleus forming a compound nucleus. The compound nucleus then decays to its ground state by gamma emission. An example of a radiative capture reaction is shown below.

Particle Ejection

In a particle ejection reaction the incident particle enters the target nucleus forming a compound nucleus. The newly formed compound nucleus has been excited to a high enough energy level to cause it to eject a new particle while the incident neutron remains in the nucleus. After the new particle is ejected, the remaining nucleus may or may not exist in an excited state depending upon the mass-energy balance of the reaction. An example of a particle ejection reaction is shown below.

Fission

One of the most important interactions that neutrons can cause is fission, in which the nucleus Fixture checkpoint marker-0036-xray is recorded on fixture page 0036.
that absorbs the neutron actually splits into two similarly sized parts. Fission will be discussed in detail in the next chapter.

Summary

The important information in this chapter is summarized below.

Neutron Interactions Summary

Interactions where a neutron scatters off a target nucleus are either elastic or inelastic. In elastic scattering, kinetic energy and momentum are conserved and no energy is transferred into excitation energy of the target nucleus. In inelastic scattering, some amount of kinetic energy is transferred into excitation energy of the target nucleus.

The conservation principles that apply to an elastic collision are conservation of kinetic energy and conservation of momentum.

Radiative capture is the absorption of a neutron by the target nucleus, resulting in an excited nucleus which subsequently (typically within a small fraction of a second) releases its excitation energy in the form of a gamma ray.

Particle ejection occurs when a neutron is absorbed by a target nucleus, resulting in the formation of a compound nucleus. The compound nucleus immediately ejects a particle (for example, alpha or proton).

1
0
n
235
92
U
236
92
U
140
55
Cs

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37
Rb 3
1
0
n
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NUCLEAR FISSION
Nuclear fission is a process in which an atom splits and releases energy, fission products, and neutrons. The neutrons released by fission can, in turn, cause the fission of other atoms.
EO 4.1 EXPLAIN the fission process using the liquid drop model of a nucleus.
EO 4.2 DEFINE the following terms:
a. Excitation energy (E_{exc})
b. Critical energy (E_{crit})
EO 4.3 DEFINE the following terms:
a. Fissile material
b. Fissionable material
c. Fertile material
EO 4.4 DESCRIBE the processes of transmutation, conversion, and breeding.
Fixture checkpoint marker-0037-xray is recorded on fixture page 0037.
EO 4.5 DESCRIBE the curve of Binding Energy per Nucleon versus mass number and give a qualitative description of the reasons for its shape.
EO 4.6 EXPLAIN why only the heaviest nuclei are easily fissioned.
EO 4.7 EXPLAIN why uranium-235 fissions with thermal neutrons and uranium-238 fissions only with fast neutrons.
Fission
In the fission reaction the incident neutron enters the heavy target nucleus, forming a compound nucleus that is excited to such a high energy level ($E > E_{crit}$) that the nucleus "splits" (E_{exc} (fissions) into two large fragments plus some neutrons. An example of a typical fission reaction is shown below.
A large amount of energy is released in the form of radiation and fragment kinetic energy.
Atomic and Nuclear Physics DOE-HDBK-1019/1-93 NUCLEAR FISSION
Liquid Drop Model of a Nucleus
The nucleus is held together by the attractive nuclear force between nucleons, which was discussed in a previous chapter. The characteristics of the nuclear force are listed below.
(a) very short range, with essentially no effect beyond nuclear dimensions (10^{-13} cm)
(b) stronger than the repulsive electrostatic forces within the nucleus
(c) independent of nucleon pairing, in that the attractive forces between pairs of neutrons are no different than those between pairs of protons or a neutron and a proton
(d) saturable, that is, a nucleon can attract only a few of its nearest neighbors
One theory of fission considers the fissioning of a nucleus similar in some respects to the splitting of a liquid drop. This analogy is justifiable to some extent by the fact that a liquid drop is held

any deformation in the same manner as nuclear forces are assumed to hold the nucleus together. By considering the nucleus as a liquid drop, the fission process can be described.

Referring to Figure 18(A), the nucleus in the ground state is undistorted, and its attractive nuclear forces are greater than the repulsive electrostatic forces between the protons within the nucleus. When an incident particle (in this instance a neutron) is absorbed by the target nucleus, a compound nucleus is formed. The compound nucleus temporarily contains all the charge and mass involved in the reaction and exists in an excited state. The excitation energy added to the compound nucleus is equal to the binding energy contributed by the incident particle plus the kinetic energy possessed by that particle. Figure 18(B) illustrates the excitation energy thus imparted to the compound nucleus, which may cause it to oscillate and become distorted. If the excitation energy is greater than a certain critical energy, the oscillations may cause the compound nucleus to become dumbbell-shaped. When this happens, the attractive nuclear forces (short-range) in the neck area are small due to saturation, while the repulsive electrostatic forces (long-range) are only slightly less than before. When the repulsive electrostatic forces exceed the attractive nuclear forces, nuclear fission occurs, as illustrated in Figure 18(C).

Figure 18 Liquid Drop Model of Fission

Critical Energy

The measure of how far the energy level of a nucleus is above its ground state is called the excitation energy (Eexc). For fission to occur, the excitation energy must be above a particular value for that nuclide. The critical energy (Ecrit) is the minimum excitation energy required for fission to occur.

Fissile Material

A fissile material is composed of nuclides for which fission is possible with neutrons of any energy level. What is especially significant about these nuclides is their ability to be fissioned with zero kinetic energy neutrons (thermal neutrons). Thermal neutrons have very low kinetic energy levels (essentially zero) because they are roughly in equilibrium with the thermal motion of surrounding materials. Therefore, in order to be classified as fissile, a material must be

Fixture checkpoint marker-0038-xray is recorded on fixture page 0038.

capable of fissioning after absorbing a thermal neutron. Consequently, they impart essentially no kinetic energy to the reaction. Fission is possible in these materials with thermal neutrons, since the change in binding energy supplied by the neutron addition alone is high enough to exceed the critical energy. Some examples of fissile nuclides are uranium-235, uranium-233, and plutonium-239.

Fissionable Material

A fissionable material is composed of nuclides for which fission with neutrons is possible. All fissile nuclides fall into this category. However, also included are those nuclides that can be fissioned only with high energy neutrons. The change in binding energy that occurs as the result of neutron absorption results in a nuclear excitation energy level that is less than the required critical energy. Therefore, the additional excitation energy must be supplied by the kinetic energy of the incident neutron. The reason for this difference between fissile and fissionable materials is the so-called odd-even effect for nuclei. It has been observed that nuclei with even numbers of neutrons and/or protons are more stable than those with odd numbers. Therefore, adding a neutron to change a nucleus with an odd number of neutrons to a nucleus with an even number of neutrons produces an appreciably higher binding energy than adding a neutron to a nucleus

energy neutrons to cause fission are thorium-232, uranium-238, and plutonium-240. Table 4 indicates the critical energy (Ecrit) and the binding energy change for an added neutron (BEn) to target nuclei of interest. For fission to be possible, the change in binding energy plus the kinetic energy must equal or exceed the critical energy ($\Delta BE + KE > Ecrit$).

TABLE 4

Critical Energies Compared to Binding Energy of Last Neutron

Target Nucleus | Critical Energy Ecrit | Binding Energy of Last Neutron BEn | BEn - Ecrit

$^{232}_{90}Th$ 7.5 MeV 5.4 MeV -2.1 MeV

$^{238}_{92}U$ 7.0 MeV 5.5 MeV -1.5 MeV

$^{235}_{92}U$ 6.5 MeV 6.8 MeV +0.3 MeV

$^{238}_{92}U$ 6.0 MeV 7.0 MeV +1.0 MeV

$^{240}_{94}Pu$ 5.0 MeV 6.6 MeV +1.6 MeV

Uranium-235 fissions with thermal neutrons because the binding energy released by the absorption of a neutron is greater than the critical energy for fission; therefore uranium-235 is a fissile material. The binding energy released by uranium-238 absorbing a thermal neutron is

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less than the critical energy, so additional energy must be possessed by the neutron for fission to be possible. Consequently, uranium-238 is a fissionable material.

Fertile Material

All of the neutron absorption reactions that do not result in fission lead to the production of new nuclides through the process known as transmutation. These nuclides can, in turn, be transmuted again or may undergo radioactive decay to produce still different nuclides. The nuclides that are produced by this process are referred to as transmutation products. Because several of the fissile nuclides do not exist in nature, they can only be produced by nuclear reactions (transmutation). The target nuclei for such reactions are said to be fertile. Fertile materials are materials that can

by which two fertile nuclides, thorium-232 and uranium-238, produce uranium-233 and plutonium-239, respectively.

Figure 19 Conversion of Fertile Nuclides to Fissile Nuclides

If a reactor contains fertile material in addition to its fissile fuel, some new fuel will be produced as the original fuel is burned up. This is called conversion. Reactors that are specifically designed to produce fissionable fuel are called "breeder" reactors. In such reactors, the amount of fissionable fuel produced is greater than the amount of fuel burnup. If less fuel is produced than used, the process is called conversion, and the reactor is termed a "converter."

Binding Energy Per Nucleon (BE/A)

As the number of particles in a nucleus increases, the total binding energy also increases. The rate of increase, however, is not uniform. This lack of uniformity results in a variation in the

Fixture checkpoint marker-0040-xray is recorded on fixture page 0040.

amount of binding energy associated with each nucleon within the nucleus. This variation in the binding energy per nucleon (BE/A) is easily seen when the average BE/A is plotted versus atomic mass number (A), as shown in Figure 20.

Figure 20 Binding Energy per Nucleon vs. Mass Number

Figure 20 illustrates that as the atomic mass number increases, the binding energy per nucleon decreases for A > 60. The BE/A curve reaches a maximum value of 8.79 MeV at A = 56 and decreases to about 7.6 MeV for A = 238. The general shape of the BE/A curve can be explained using the general properties of nuclear forces. The nucleus is held together by very short-range attractive forces that exist between nucleons. On the other hand, the nucleus is being forced apart by long range repulsive electrostatic (coulomb) forces that exist between all the protons in the nucleus.

As the atomic number and the atomic mass number increase, the repulsive electrostatic forces within the nucleus increase due to the greater number of protons in the heavy elements. To overcome this increased repulsion, the proportion of neutrons in the nucleus must increase to maintain stability. This increase in the neutron-to-proton ratio only partially compensates for the growing proton-proton repulsive force in the heavier, naturally occurring elements. Because the repulsive forces are increasing, less energy must be supplied, on the average, to remove a nucleon from the nucleus. The BE/A has decreased. The BE/A of a nucleus is an indication of its degree of stability. Generally, the more stable nuclides have higher BE/A than the less stable ones. The increase in the BE/A as the atomic mass number decreases from 260 to 60 is the primary reason for the energy liberation in the fission process. In addition, the increase in the BE/A as the atomic mass number increases from 1 to 60 is the reason for the energy liberation in the fusion process, which is the opposite reaction of fission.

The heaviest nuclei require only a small distortion from a spherical shape (small energy addition) for the relatively large coulomb forces forcing the two halves of the nucleus apart to overcome the attractive nuclear forces holding the two halves together. Consequently, the heaviest nuclei are easily fissionable compared to lighter nuclei.

Summary

The important information in this chapter is summarized on the following page.

Nuclear Fission Summary

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In the ground state the nucleus is nearly spherical in shape. After the absorption of a neutron, the nucleus will be in an excited state and start to oscillate and become distorted. If the oscillations cause the nucleus to become shaped like a dumbbell, the repulsive electrostatic forces will overcome the short-range attractive nuclear forces, and the nucleus will split in two.

Excitation energy is the amount of energy a nucleus has above its ground state. Critical energy is the minimum excitation energy that a nucleus must have before it can fission.

Fissile material is material for which fission is possible with neutrons that have zero kinetic energy. Fissionable material is material for which fission caused by neutron absorption is possible provided the kinetic energy added with the binding energy is greater than the critical energy. Fertile material is material that can undergo transmutation to become fissile material.

Transmutation is the process of neutron absorption and subsequent decay, which changes one nuclide to another nuclide. Conversion is the process of transmuting fertile material into fissile material in a reactor, where the amount of fissile material produced is less than the amount of fissile material consumed. Breeding is the same as conversion, except the amount of fissile material produced is more than the amount of fissile material consumed.

The curve of binding energy per nucleon increases quickly through the light

Fixture checkpoint marker-0041-xray is recorded on fixture page 0041.

nuclides and reaches a maximum at a mass number of about 56. The curve decreases slowly for mass numbers greater than 60.

The heaviest nuclei are easily fissionable because they require only a small distortion from the spherical shape to allow the coulomb forces to overcoming the attractive nuclear force, forcing the two halves of the nucleus apart.

Uranium-235 fissions with thermal neutrons because the binding energy released by the absorption of a neutron is greater than the critical energy for fission. The binding energy released by uranium-238 absorbing a neutron is less than the critical energy, so additional kinetic energy must be possessed by the neutron for fission to be possible.

1
0
n
235
92
U
236
92
U
140
55
Cs
93
37
Rb 3
1

n
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ENERGY RELEASE FROM FISSION
Fission of heavy nuclides converts a small amount of mass into an enormous amount of energy. The amount of energy released by fission can be determined based on either the change in mass that occurs during the reaction or by the difference in binding energy per nucleon between the fissile nuclide and the fission products.
EO 4.8 CHARACTERIZE the fission products in terms of mass groupings and radioactivity.
EO 4.9 Given the nuclides involved and their masses, CALCULATE the energy released from fission.
EO 4.10 Given the curve of Binding Energy per nucleon versus mass number, CALCULATE the energy released from fission.
Calculation of Fission Energy
Nuclear fission results in the release of enormous quantities of energy. It is necessary to be able to calculate the amount of energy that will be produced. The logical manner in which to pursue this is to first investigate a typical fission reaction such as the one listed below.
It can be seen that when the compound nucleus splits, it breaks into two fission fragments, rubidium-93, cesium-140, and some neutrons. Both fission products then decay by multiple - emissions as a result of the high neutron-to-proton ratio possessed by these nuclides.
In most cases, the resultant fission fragments have masses that vary widely. Figure 21 gives the percent yield for atomic mass numbers. The most probable pair of fission fragments for the thermal fission of the fuel uranium-235 have masses of about 95 and 140. Note that the vertical axis of the fission yield curve is on a logarithmic scale. Therefore, the formation of fission fragments of mass numbers of about 95 and 140 is highly likely.
Fixture checkpoint marker-0042-xray is recorded on fixture page 0042.
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Figure 21 Uranium-235 Fission Yield vs.
Mass Number
Referring now to the binding energy per nucleon curve (Figure 20), we can estimate the amount of energy released by our "typical" fission by plotting this reaction on the curve and calculating the change in binding energy (ΔBE) between the reactants on the left-hand side of the fission equation and the products on the right-hand side. Plotting the reactant and product nuclides on the curve shows that the total binding energy of the system after fission is greater than the total binding energy of the system before fission. When there is an increase in the total binding energy of a system, the system has become more stable by releasing an amount of energy equal to the increase in total binding energy of the system. Therefore, in the fission process, the energy liberated is equal to the increase in the total binding energy of the system.
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Figure 22 graphically depicts that the binding energy per nucleon for the products (C,
Figure 22 Change in Binding Energy for Typical Fission
rubidium-93 and B, cesium-140) is greater than that for the reactant (A, uranium-235). The total binding energy for a nucleus can be found by multiplying the binding energy per nucleon by the number of nucleons.
TABLE 5

SYNTHETIC FIXTURE PAGE 0043 OF 1200

Binding Energy per Nucleon Curve
Nuclide B.E. per Nucleon
(BE/A)
Mass Number
(A)
Binding Energy
(BE/A) x (A)
9
3
3
7Rb 8.7 MeV 93 809 MeV
14
5
0
5Cs 8.4 MeV 140 1176 MeV
23
9
5
2U 7.6 MeV 235 1786 MeV
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BE BEproducts BEreactants
BERb 93 BECs 140 BEU 235
809 MeV 1176 MeV 1786 MeV
199 MeV
1
0
n
235
92
U
236
92
U
140
Fixture checkpoint marker-0043-xray is recorded on fixture page 0043.
55
Cs
93
37
Rb 3
1
0
n
Mass difference Mass of Reactants Mass of Products
236.052589 amu 235.85208 amu
0.200509 amu
EInst 0.020059 amu
931.5 MeV

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186.8 MeV
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The energy released will be equivalent to the difference in binding energy (BE) between the reactants and the products.
The energy liberation during the fission process can also be explained from the standpoint of the conservation of mass-energy. During the fission process, there is a decrease in the mass of the system. There must, therefore, be energy liberated equal to the energy equivalent of the mass lost in the process. This method is more accurate than the previously illustrated method and is used when actually calculating the energy liberated during the fission process.
Again, referring to the "typical" fission reaction.
Fixture checkpoint marker-0044-xray is recorded on fixture page 0044.
E , the instantaneous energy, is the energy released immediately after the fission process. It Inst is equal to the energy equivalent of the mass lost in the fission process. It can be calculated as shown below.
Mass of the Reactants Mass of the Products
U 235.043924 amu Rb 92.91699 amu 235
92
93
37
n 1.008665 amu Cs 139.90910 amu 1
0
140
55
3 (n) 3.02599 amu 1
0
236.052589 amu 235.85208 amu
This mass difference can be converted to an energy equivalent.
93
37
Rb
93
38
Sr
93
39
Y
93
40
Zr
93
41
Nb
140
55
Cs
140
56

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140
57
La
140
58
Ce
EDecay mRb 93 mNb 93 4 melectron
931.5MeV
amu
92.91699 amu 92.90638 amu 4 0.0005486 amu
931.5 MeV
amu
0.008416 amu
931.5 MeV
amu
7.84 MeV
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Fixture checkpoint marker-0045-xray is recorded on fixture page 0045.
The total energy released per fission will vary from the fission to the next depending on what fission products are formed, but the average total energy released per fission of uranium-235 with a thermal neutron is 200 MeV.
As illustrated in the preceding example, the majority of the energy liberated in the fission process is released immediately after the fission occurs and appears as the kinetic energy of the fission fragments, kinetic energy of the fission neutrons, and instantaneous gamma rays. The remaining energy is released over a period of time after the fission occurs and appears as kinetic energy of the beta, neutrino, and decay gamma rays.
Estimation of Decay Energy
In addition to this instantaneous energy release during the actual fission reaction, there is additional energy released when the fission fragments decay by emission. This additional
-
energy is called decay energy, EDecay. The decay chains for rubidium-93 and cesium-140 are shown below.
The energy released during the decay for each chain will be equivalent to the mass difference between the original fission product and the sum of the final stable nuclide and the beta particles emitted.
The energy released in the decay chain of rubidium-93 is calculated below.
EDecay mRb 93 mNb 93 3 melectron
931.5MeV
amu
139.90910 amu 139.90543 amu 3 0.0005486 amu
931.5 MeV
amu
0.000202 amu
931.5 MeV
amu
1.89 MeV
Atomic and Nuclear Physics DOE-HDBK-1019/1-93 ENERGY RELEASE FROM FISSION

The energy released in the decay chain of cesium-140 is calculated below.
The total decay energy is the sum of the energies of the two chains, or 9.73 MeV.
Distribution of Fission Energy
The average energy distribution for the energy released per fission with a thermal neutron in uranium-235 is shown in Tables 6 and 7.
TABLE 6
Instantaneous Energy from Fission
Kinetic Energy of Fission Products 167 Mev
Energy of Fission Neutrons 5 MeV
Instantaneous Gamma-ray Energy 5 MeV
Capture Gamma-ray Energy 10 MeV
Total Instantaneous Energy 187 MeV
TABLE 7
Delayed Energy from Fission
Beta Particles From Fission Products 7 Mev
Gamma-rays from Fission Products 6 MeV
Neutrinos 10 MeV
Total Delayed Energy 23 MeV
ENERGY RELEASE FROM FISSION DOE-HDBK-1019/1-93 Atomic and Nuclear Physics
Because the 10 MeV of neutrino energy shown in Table 7 is not absorbed in the reactor, the average value of 200 MeV per fission is still accurate. Note in Table 6 that some fission neutrons undergo radiative capture and the resultant gamma ray emission provides an additional 10 MeV of instantaneous energy, which contributes to the total of 187 MeV instantaneous energy.
All of the energy released, with the exception of the neutrino energy, is ultimately transformed
Fixture checkpoint marker-0046-xray is recorded on fixture page 0046.
into heat through a number of processes. The fission fragments, with their high positive charge and kinetic energy, cause ionization directly as they rip orbital electrons from the surrounding atoms. In this ionization process, kinetic energy is transferred to the surrounding atoms of the fuel material, resulting in an increase in temperature. The beta particles and gamma rays also give up their energy through ionization, and the fission neutrons interact and lose their energy through elastic scattering. Of the 200 MeV released per fission, about seven percent (13 MeV) is released at some time after the instant of fission. When a reactor is shut down, fissions essentially cease, but energy is still being released from the decay of fission products. The heat produced by this decay energy is referred to as "decay heat." Although decay energy represents about seven percent of reactor heat production during reactor operation, once the reactor is shut down the decay heat production drops off quickly to a small fraction of its value while operating. The decay heat produced is significant, however, and systems must be provided to keep the reactor cool even after shutdown.
Summary
The important information in this chapter is summarized below.
Energy Release From Fission Summary
Fission products have some general characteristics in common.
They generally decay by β- emission.
The most common mass numbers are grouped near 95 and 140.
The energy released by fission can be calculated based on the difference in mass between the masses of the reactants before fission and the fission fragments and fission neutrons after fission.

change in binding energy per nucleon between the fissile nuclide and the fission products.

INTERACTION OF RADIATION WITH MATTER

Different types of radiation interact with matter in widely different ways. A large, massive, charged alpha particle cannot penetrate a piece of paper and even has a limited range in dry air. A neutrino, at the other extreme, has a low probability of interacting with any matter, even if it passed through the diameter of the earth.

EO 5.1 DESCRIBE interactions of the following with matter:

a. Alpha particle c. Positron
b. Beta particle d. Neutron

EO 5.2 DESCRIBE the following ways that gamma radiation interacts with matter:

a. Photoelectric effect
b. Compton scattering
c. Pair production

Interaction of Radiation With Matter

Radiation can be classified into two general groups, charged and uncharged; therefore, it may be expected that interactions with matter fall into two general types. Charged particles directly ionize the media through which they pass, while uncharged particles and photons can cause ionization only indirectly or by secondary radiation.

A moving charged particle has an electrical field surrounding it, which interacts with the atomic structure of the medium through which it is passing. This interaction decelerates the particle and accelerates electrons in the atoms of the medium. The accelerated electrons may acquire enough energy to escape from the parent atom. This process, whereby radiation "strips" off orbital electrons, is called ionization. Uncharged moving particles have no electrical field, so they can only lose energy and cause ionization by such means as collisions or scattering. A photon can lose energy by the photoelectric effect, Compton effect, or pair production.

Fixture checkpoint marker-0047-xray is recorded on fixture page 0047.

$$I \propto m z^2 K.E.$$

Because ionizing radiation creates ions in pairs, the intensity of ionization or the specific ionization is defined as the number of ion-pairs formed per centimeter of travel in a given material. The amount of ionization produced by a charged particle per unit path length, which is a measure of its ionizing power, is roughly proportional to the particle's mass and the square of its charge as illustrated in the equation below.

where:

I is the ionizing power
m is the mass of the particle

K.E. is its kinetic energy
Since m for an alpha particle is about 7300 times as large as m for a beta particle, and z is twice as great, an alpha will produce much more ionization per unit path length than a beta particle of the same energy. This phenomenon occurs because the larger alpha particle moves slower for a given energy and thus acts on a given electron for a longer time.
Alpha Radiation
Alpha radiation is normally produced from the radioactive decay of heavy nuclides and from certain nuclear reactions. The alpha particle consists of 2 neutrons and 2 protons, so it is essentially the same as the nucleus of a helium atom. Because it has no electrons, the alpha particle has a charge of +2. This positive charge causes the alpha particle to strip electrons from the orbits of atoms in its vicinity. As the alpha particle passes through material, it removes electrons from the orbits of atoms it passes near. Energy is required to remove electrons and the energy of the alpha particle is reduced by each reaction. Eventually the particle will expend its kinetic energy, gain 2 electrons in orbit, and become a helium atom. Because of its strong positive charge and large mass, the alpha particle deposits a large amount of energy in a short distance of travel. This rapid, large deposition of energy limits the penetration of alpha particles. The most energetic alpha particles are stopped by a few centimeters of air or a sheet of paper.
Beta-Minus Radiation
A beta-minus particle is an electron that has been ejected at a high velocity from an unstable nucleus. An electron has a small mass and an electrical charge of -1. Beta particles cause ionization by displacing electrons from atom orbits. The ionization occurs from collisions with orbiting electrons. Each collision removes kinetic energy from the beta particle, causing it to slow down. Eventually the beta particle will be slowed enough to allow it to be captured as an
2 electrons 0.000549 amu
electron
931.5 MeV
amu
1.02 MeV
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orbiting electron in an atom. Although more penetrating than the alpha, the beta is relatively easy to stop and has a low power of penetration. Even the most energetic beta radiation can be stopped by a few millimeters of metal.
Positron Radiation
Positively charged electrons are called positrons. Except for the positive charge, they are identical to beta-minus particles and interact with matter in a similar manner. Positrons are very short-lived, however, and quickly are annihilated by interaction with a negatively charged
Fixture checkpoint marker-0048-xray is recorded on fixture page 0048.
electron, producing two gammas with a combined energy (calculated below) equal to the rest mass of the positive and negative electrons.
Neutron Radiation
Neutrons have no electrical charge. They have nearly the same mass as a proton (a hydrogen atom nucleus). A neutron has hundreds of times more mass than an electron, but 1/4 the mass of an alpha particle. The source of neutrons is primarily nuclear reactions, such as fission, but they may also be produced from the decay of radioactive nuclides. Because of its lack of charge, the neutron is difficult to stop and has a high penetrating power.

scatter, inelastic scatter, and absorption. In elastic scatter, a neutron collides with a nucleus and bounces off. This reaction transmits some of the kinetic energy of the neutron to the nucleus of the atom, resulting in the neutron being slowed, and the atom receives some kinetic energy (motion). This process is sometimes referred to as "the billiard ball effect."

As the mass of the nucleus approaches the mass of the neutron, this reaction becomes more effective in slowing the neutron. Hydrogenous material attenuates neutrons most effectively.

In the inelastic scatter reaction, the same neutron/nucleus collision occurs as in elastic scatter. However, in this reaction, the nucleus receives some internal energy as well as kinetic energy. This slows the neutron, but leaves the nucleus in an excited state. When the nucleus decays to its original energy level, it normally emits a gamma ray.

In the absorption reaction, the neutron is actually absorbed into the nucleus of an atom. The neutron is captured, but the atom is left in an excited state. If the nucleus emits one or more gamma rays to reach a stable level, the process is called radiative capture. This reaction occurs at most neutron energy levels, but is more probable at lower energy levels.

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Gamma Radiation

Gamma radiation is electromagnetic radiation. It is commonly referred to as a gamma ray and is very similar to an x-ray. The difference is that gamma rays are emitted from the nucleus of an atom, and x-rays are produced by orbiting electrons. The x-ray is produced when orbiting electrons move to a lower energy orbit or when fast-moving electrons approaching an atom are deflected and decelerated as they react with the atom's electrical field (called Bremsstrahlung).

The gamma ray is produced by the decay of excited nuclei and by nuclear reactions. Because the gamma ray has no mass and no charge, it is difficult to stop and has a very high penetrating power. A small fraction of the original gamma stream will pass through several feet of concrete or several meters of water.

There are three methods of attenuating gamma rays. The first method is referred to as the photo-electric effect. When a low energy gamma strikes an atom, the total energy of the gamma is expended in ejecting an electron from orbit. The result is ionization of the atom and expulsion of a high energy electron. This reaction is most predominant with low energy gammas interacting in materials with high atomic weight and rarely occurs with gammas having an energy above 1 MeV. Annihilation of the gamma results. Any gamma energy in excess of the binding energy of the electron is carried off by the electron in the form of kinetic energy.

The second method of attenuation of gammas is called Compton scattering. The gamma interacts with an orbital or free electron; however, in this case, the photon loses only a fraction of its energy. The actual energy loss depending on the scattering angle of the gamma. The gamma continues on at lower energy, and the energy difference is absorbed by the electron. This reaction becomes important for gamma energies of about 0.1 MeV and higher.

At higher energy levels, a third method of attenuation is predominant. This method is pair-production. When a high energy gamma passes close enough to a heavy nucleus, the gamma completely disappears, and an electron and a positron are formed. For this reaction to take place, the original gamma must have at least 1.02 MeV energy. Any energy greater than 1.02 MeV becomes kinetic energy shared between the electron and positron. The probability of pair■production incre

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Atomic and Nuclear Physics INTERACTION OF RADIATION WITH MATTER

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An alpha particle deposits a large amount of energy in a short distance of travel due to its large mass and charge.
Beta-minus particles interact with the electrons orbiting the nucleus of atoms, causing ionization by displacing the electrons. The beta particle loses energy with each interaction. After the beta particle loses enough energy, it is captured in the orbital shells of an atom.
Positrons interact with matter much the same way as beta minus particles. After the positron has lost most of its energy by ionizing atoms, it is annihilated by interaction with an electron. The electron-positron pair disappear and are replaced by two gammas, each with the energy equivalent of the mass of an electron (0.51 MeV).
Neutrons interact with matter by elastic scattering, inelastic scattering, or absorption.
Photoelectric effect is where a gamma interacts with an electron orbiting an atom. The entire energy of the gamma is transferred to the electron, and the electron is ejected from its orbit.
In Compton scattering a gamma interacts with an orbital electron, but only part of the gamma energy is transferred to the electron. The electron is ejected from its orbit, and the gamma is scattered off at a lower energy.
In pair-production, a gamma interacts with the electric field of a nucleus and is converted into an electron-positron pair. The gamma must have an energy greater than 1.02 MeV for this to occur.
Summary
Fixture checkpoint marker-0050-xray is recorded on fixture page 0050.
The important information in this chapter is summarized below.
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Department of Energy
Fundamentals Handbook
NUCLEAR PHYSICS
AND REACTOR THEORY
Module 2
Reactor Theory (Neutron Characteristics)
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OBJECTIVES DOE-HDBK-1019/1-93 Reactor Theory (Neutron Characteristics)
TERMINAL OBJECTIVE
1.0 Without references, EXPLAIN how neutron sources produce neutrons.
ENABLING OBJECTIVES
1.1 DEFINE the following terms:
a. Intrinsic neutron source
b. Installed neutron source
1.2 LIST three examples of reactions that produce neutrons in intrinsic neutron sources.
1.3 LIST three examples of reactions that produce neutrons in installed neutron sources.

Reactor Theory (Neutron Characteristics) DOE-HDBK-1019/1-93 OBJECTIVES
TERMINAL OBJECTIVE
2.0 Given the necessary information for calculations, EXPLAIN basic concepts in reactor physics and perform calculations.
ENABLING OBJECTIVES
2.1 DEFINE the following terms:
a. Atom density d. Barn
b. Neutron flux e. Macroscopic cross section
c. Microscopic cross section f. Mean free path
2.2 EXPRESS macroscopic cross section in terms of microscopic cross section.
2.3 DESCRIBE how the absorption cross section of typical nuclides varies with neutron
Fixture checkpoint marker-0052-xray is recorded on fixture page 0052.
energy at energies below the resonance absorption region.
2.4 DESCRIBE the cause of resonance absorption in terms of nuclear energy levels.
2.5 DESCRIBE the energy dependence of resonance absorption peaks for typical light and heavy nuclei.
2.6 EXPRESS mean free path in terms of macroscopic cross section.
2.7 Given the number densities (or total density and component fractions) and microscopic cross sections of components, CALCULATE the macroscopic cross section for a mixture.
2.8 CALCULATE a macroscopic cross section given a material density, atomic mass, and microscopic cross section.
2.9 EXPLAIN neutron shadowing or self-shielding.

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2.11 DESCRIBE the relationship between neutron flux and reactor power.
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OBJECTIVES DOE-HDBK-1019/1-93 Reactor Theory (Neutron Characteristics)
ENABLING OBJECTIVES (Cont.)
2.12 DEFINE the following concepts:
a. Thermalization d. Average logarithmic energy decrement
b. Moderator e. Macroscopic slowing down power
c. Moderating ratio
2.13 LIST three desirable characteristics of a moderator.
2.14 Given an average fractional energy loss per collision, CALCULATE the energy loss after a specified number of collisions.
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Reactor Theory (Neutron Characteristics) DOE-HDBK-1019/1-93 OBJECTIVES
TERMINAL OBJECTIVE
Fixture checkpoint marker-0053-xray is recorded on fixture page 0053.
3.0 Without references, EXPLAIN the production process and effects on fission of prompt and delayed neutrons.
ENABLING OBJECTIVES
3.1 STATE the origin of prompt neutrons and delayed neutrons.
3.2 STATE the approximate fraction of neutrons that are born as delayed neutrons from the fission of the following nuclear fuels:
a. Uranium-235
b. Plutonium-239
3.3 EXPLAIN the mechanism for production of delayed neutrons.
3.4 EXPLAIN prompt and delayed neutron generation times.
3.5 Given prompt and delayed neutron generation times and delayed neutron fraction, CALCULATE the average generation time.
3.6 EXPLAIN the effect of delayed neutrons on reactor control.
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OBJECTIVES DOE-HDBK-1019/1-93 Reactor Theory (Neutron Characteristics)
TERMINAL OBJECTIVE
4.0 Without references, DESCRIBE the neutron energy spectrum for the type of reactor presented in this module.
ENABLING OBJECTIVES
4.1 STATE the average energy at which prompt neutrons are produced.
4.2 DESCRIBE the neutron energy spectrum in the following reactors:
a. Fast reactor
b. Thermal reactor
4.3 EXPLAIN the reason for the particular shape of the fast, intermediate, and slow energy regions of the neutron flux spectrum for a thermal reactor.
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Reactor Theory (Neutron Characteristics) DOE-HDBK-1019/1-93 NEUTRON SOURCES
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NEUTRON SOURCES
Neutrons from a variety of sources are always present in a reactor core. This is true even when the reactor is shut down. Some of these neutrons are produced by naturally occurring (intrinsic) neutron sources, while others may be the result of fabricated (installed) neutron sources that are incorporated into the design of the

are often grouped together and classified as source neutrons.
EO 1.1 DEFINE the following terms:
a. Intrinsic neutron source
b. Installed neutron source
EO 1.2 LIST three examples of reactions that produce neutrons in intrinsic neutron sources.
EO 1.3 LIST three examples of reactions that produce neutrons in installed neutron sources.
Neutron Sources
In addition to neutron-induced fission, neutrons are produced by other reactions. The neutrons produced by reactions other than neutron-induced fission are called source neutrons. Source neutrons are important because they ensure that the neutron population remains high enough to allow a visible indication of neutron level on the most sensitive monitoring instruments while the reactor is shutdown and during the startup sequence. This verifies instrument operability and allows monitoring of neutron population changes. Source neutrons can be classified as either intrinsic or installed neutron sources.
Intrinsic Neutron Sources
Some neutrons will be produced in the materials present in the reactor due to a variety of unavoidable reactions that occur because of the nature of these materials. Intrinsic neutron sources are those neutron-producing reactions that always occur in reactor materials.
■ ■ ■ ■
Fixture checkpoint marker-0054-xray is recorded on fixture page 0054.
■ ■ ■
■ ■
NEUTRON SOURCES DOE-HDBK-1019/1-93 Reactor Theory (Neutron Characteristics)
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A limited number of neutrons will always be present, even in a reactor core that has never been operated, due to spontaneous fission of some heavy nuclides that are present in the fuel. Uranium■238, ur of the rate at which different heavy nuclides produce neutrons by spontaneous fission.
Californium-252 is not an intrinsic neutron source, but will be discussed in the section on installed neutron sources.
TABLE 1
Neutron Production by Spontaneous Fission
Nuclide T (Fission) T (1/2 1/2 ■-decay) neutrons/sec/gram
U 1.8 x 10 years 6.8 x 10 years 8.0 x 10 235
92
17 8 -4
U 8.0 x 10 years 4.5 x 10 years 1.6 x 10 238
92
15 9 -2
Pu 5.5 x 10 years 2.4 x 10 years 3.0 x 10 239
94
5 4 -2
Pu 1.2 x 10 years 6.6 x 10 years 1.0 x 10 240
94
11 3 3
Cf 66.0 years 2.65 years 2.3 x 10 252
98

Another intrinsic neutron source is a reaction involving natural boron and fuel. In some reactors, natural boron is loaded into the reactor core as a neutron absorber to improve reactor control or increase core life-time. Boron-11 (80.1% of natural boron) undergoes a reaction with the alpha particle emitted by the radioactive decay of heavy nuclides in the fuel to yield a neutron as shown below.

The boron-11 must be mixed with, or in very close proximity to, the fuel for this reaction because of the short path length of the alpha particle. For a reactor core with this configuration, this (■,n) reaction is an important source of neutrons for reactor startup.

In a reactor that has been operated, another source of neutrons becomes significant. Neutrons may be produced by the interaction of a gamma ray and a deuterium nucleus. This reaction is commonly referred to as a photoneutron reaction because it is initiated by electromagnetic radiation and results in the production of a neutron. The photoneutron reaction is shown below.

■ ■ ■
■ ■
■ ■ ■
■ ■

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There is an abundant supply of high energy gammas in a reactor that has been operated because many of the fission products are gamma emitters. All water-cooled reactors have some deuterium present in the coolant in the reactor core because a small fraction of natural hydrogen is the isotope deuterium. The atom percentage of deuterium in the water ranges from close to the naturally occurring value (0.015%) for light water reactors to above 90% deuterium for heavy water reactors. Therefore, the required conditions for production of photoneutrons exist.

The supply of gamma rays decreases with time after shutdown as the gamma emitters decay; therefore, the photoneutron production rate also decreases. In a few particular reactors, additional D O (heavy water) may be added to the reactor to increase the production of 2 photoneutrons following a long shutdown period.

Fixture checkpoint marker-0055-xray is recorded on fixture page 0055.

Installed Neutron Sources

Because intrinsic neutron sources can be relatively weak or dependent upon the recent power history of the reactor, many reactors have artificial sources of neutrons installed. These neutron sources ensure that shutdown neutron levels are high enough to be detected by the nuclear instruments at all times. This provides a true picture of reactor conditions and any change in these conditions. An installed neutron source is an assembly placed in or near the reactor for the sole purpose of producing source neutrons.

One strong source of neutrons is the artificial nuclide californium-252, which emits neutrons at the rate of about 2 x 10 neutrons per second per gram as the result of spontaneous fission. 12 Important drawbacks for some applications may be its high cost and its short half-life (2.65 years).

Many installed neutron sources use the (■,n) reaction with beryllium. These sources are composed of a mixture of metallic beryllium (100% beryllium-9) with a small quantity of an alpha particle emitter, such as a compound of radium, polonium, or plutonium. The reaction that occurs is shown below.

The beryllium is intimately (homogeneously) mixed with the alpha emitter and is usually enclosed in a stainless steel capsule.

Another type of installed neutron source that is widely used is a photoneutron source that employs the (■,n) reaction with beryllium. Beryllium is used for photoneutron sources because

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1.66 MeV. Thus, a gamma ray with greater energy than 1.66 MeV can cause neutrons to be ejected by the (■,n) reaction as shown below.
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Neutron Sources Summary
& Intrinsic neutron sources are sources of neutrons from materials that are in the reactor for other purposes such as fuel, burnable poison, or moderator.
& Installed neutron sources are materials or components placed in the reactor specifically for the purpose of producing source neutrons.
& Examples of intrinsic neutron sources are listed below.
Spontaneous fission of heavy nuclides in fuel, such as uranium-238, uranium-235, and plutonium-239, results in fission fragments and free neutrons.
Boron-11 mixed with the fuel undergoes an alpha-neutron reaction and becomes nitrogen-14.
Deuterium present in the reactor coolant undergoes a gamma■neutron reaction and becomes hydrogen-1.
& Examples of installed neutron sources are listed below.
Spontaneous fission of californium-252 results in fission fragments and free neutrons.
Beryllium-9 undergoes an alpha-neutron reaction (alpha from the decay of plutonium, polonium, or radium) and becomes carbon-12.
Beryllium-9 undergoes a gamma-neutron reaction (high energy gamma from decay of antimony-124) and becomes beryllium-8.
Many startup sources of this type use antimony and beryllium because after activation with neutrons the radioactive antimony becomes an emitter of high energy gammas. The photoneutron sources of this type are constructed somewhat differently from the (■,n) types. One design incorporates a capsule of irradiated antimony enclosed in a beryllium sleeve. The entire assembly is then encased in a stainless steel cladding. A large reactor may have several neutron sources of this type installed within the core.
Summary
The important information in this chapter is summarized below.
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NUCLEAR CROSS SECTIONS AND NEUTRON FLUX
Fixture checkpoint marker-0056-xray is recorded on fixture page 0056.
To determine the frequency of neutron interactions, it is necessary to describe the availability of neutrons to cause interaction and the probability of a neutron interacting with material. The availability of neutrons and the probability of interaction are quantified by the neutron flux and nuclear cross section.
EO 2.1 DEFINE the following terms:
a. Atom density d. Barn
b. Neutron flux e. Macroscopic cross section
c. Microscopic cross section f. Mean free path
EO 2.2 EXPRESS macroscopic cross section in terms of microscopic cross section.
EO 2.3 DESCRIBE how the absorption cross section of typical nuclides varies with neutron energy at energies below the resonance

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EO 2.4 DESCRIBE the cause of resonance absorption in terms of nuclear energy levels.
EO 2.5 DESCRIBE the energy dependence of resonance absorption peaks for typical light and heavy nuclei.
EO 2.6 EXPRESS mean free path in terms of macroscopic cross section.
EO 2.7 Given the number densities (or total density and component fractions) and microscopic cross sections of components, CALCULATE the macroscopic cross section for a mixture.
EO 2.8 CALCULATE a macroscopic cross section given a material density, atomic mass, and microscopic cross section.
EO 2.9 EXPLAIN neutron shadowing or self-shielding.
'

Fixture checkpoint marker-0057-xray is recorded on fixture page 0057.

Introduction

Fission neutrons are born with an average energy of about 2 MeV. These fast neutrons interact with the reactor core materials in various absorption and scattering reactions. Collisions that result in scattering are useful in slowing neutrons to thermal energies. Thermal neutrons may be absorbed by fissile nuclei to produce more fissions or be absorbed in fertile material for conversion to fissionable fuel. Absorption of neutrons in structural components, coolant, and other non-fuel material results in the removal of neutrons without fulfilling any useful purpose. To safely and efficiently operate a nuclear reactor it is necessary to predict the probability that a particular absorption or scattering reaction will occur. Once these probabilities are known, if the availability of neutrons can be determined, then the rate at which these nuclear reactions take place can be predicted.

Atom Density

One important property of a material is the atom density. The atom density is the number of atoms of a given type per unit volume of the material. To calculate the atom density of a substance use Equation (2-1).

(2-1)

where:

N = atom density (atoms/cm)3
' = density (g/cm)3
N = Avogadro's number (6.022 x 10 atoms/mole) A
23
M = gram atomic weight

N
NA
M
2.699 g
cm3
6.022 x 1023 atoms
mole
26.9815 g
mole
6.024 x 1022 atoms

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Example:
A block of aluminum has a density of 2.699 g/cm . If the gram atomic weight of
3
aluminum is 26.9815 g, calculate the atom density of the aluminum.
Solution:
Cross Sections
The probability of a neutron interacting with a nucleus for a particular reaction is dependent upon not only the kind of nucleus involved, but also the energy of the neutron. Accordingly, the absorption of a thermal neutron in most materials is much more probable than the absorption of a fast neutron. Also, the probability of interaction will vary depending upon the type of reaction involved.
The probability of a particular reaction occurring between a neutron and a nucleus is called the microscopic cross section ( ) of the nucleus for the particular reaction. This cross section will vary with the energy of the neutron. The microscopic cross section may also be regarded as the effective area the nucleus presents to the neutron for the particular reaction. The larger the effective area, the greater the probability for reaction.
Fixture checkpoint marker-0058-xray is recorded on fixture page 0058.
Because the microscopic cross section is an area, it is expressed in units of area, or square centimeters. A square centimeter is tremendously large in comparison to the effective area of a nucleus, and it has been suggested that a physicist once referred to the measure of a square centimeter as being "as big as a barn" when applied to nuclear processes. The name has persisted and microscopic cross sections are expressed in terms of barns. The relationship between barns and cm is shown below. 2
1 barn = 10-24 cm2
N
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Whether a neutron will interact with a certain volume of material depends not only on the microscopic cross section of the individual nuclei but also on the number of nuclei within that volume. Therefore, it is necessary to define another kind of cross section known as the macroscopic cross section ( ). The macroscopic cross section is the probability of a given reaction occurring per unit travel of the neutron. is related to the microscopic cross section ( ) by the relationship shown below.
(2-2)
where:
= macroscopic cross section (cm)
-1
N = atom density of material (atoms/cm)
3
= microscopic cross-section (cm)
2
The difference between the microscopic and macroscopic cross sections is extremely important and is restated for clarity. The microscopic cross section ( ) represents the effective target area that a single nucleus presents to a bombarding particle. The units are given in barns or cm .

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The macroscopic cross section ( ) represents the effective target area that is presented by all of the nuclei contained in 1 cm of the material. The units are given as 1/cm or cm .
3 -1
A neutron interacts with an atom of the material it enters in two basic ways. It will either interact through a scattering interaction or through an absorption reaction. The probability of a neutron being absorbed by a particular atom is the microscopic cross section for absorption, . The probability of a neutron scattering off of a particular nucleus is the microscopic cross a section for scattering, s
. The sum of the microscopic cross section for absorption and the microscopic cross section for scattering is the total microscopic cross section, . T
= + T a s
Both the absorption and the scattering microscopic cross sections can be further divided. For instance, the scattering cross section is the sum of the elastic scattering cross section (se) and the inelastic scattering cross section ( ). si
= + s se si
The microscopic absorption cross section ( ) includes all reactions except scattering. However, a for most purposes it is sufficient to merely separate it into two categories, fission (f) and capture ( ). Radiative capture of neutrons was described in the Neutron Interactions chapter c of Module 1.
= + a f c
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Figure 1 Typical Neutron Absorption Cross Section vs. Neutron Energy
The variation of absorption cross sections with neutron energy is often complicated. For many elements the absorption cross sections are small, ranging from a fraction of a barn to a few
Fixture checkpoint marker-0059-xray is recorded on fixture page 0059.
barns for slow (or thermal) neutrons.
For a considerable number of nuclides of moderately high (or high) mass numbers, an examination of the variation of the absorption cross section with the energy of the incident neutron reveals the existence of three regions on a curve of absorption cross section versus neutron energy. This cross section is illustrated in Figure 1. First, the cross section decreases steadily with increasing neutron energy in a low energy region, which includes the thermal range (E < 1 eV). In this region the absorption cross section, which is often high, is inversely proportional to the velocity (v). This region is frequently referred to as the "1/v region," because the absorption cross section is proportional to 1/v, which is the reciprocal of neutron velocity. Following the 1/v region, there occurs the "resonance region" in which the cross sections rise sharply to high values called "resonance peaks" for neutrons of certain energies, and then fall again. These energies are called resonance energies and are a result of the affinity of the nucleus for neutrons whose energies closely match its discrete, quantum energy levels. That is, when the binding energy of a neutron plus the kinetic energy of the neutron are exactly equal to the amount required to raise a compound nucleus from its ground state to a quantum level, resonance absorption occurs. The following example problem further illustrates this point.
■
(

state, calculate the kinetic energy a neutron must possess to undergo resonant absorption in uranium-235 at this resonance energy level.
BE = [Mass(U) + Mass(neutron) - Mass(U)] x 931 MeV/amu 235 236
BE = (235.043925 + 1.008665 - 236.045563) x 931 MeV/amu
BE = (0.007025 amu) x 931 MeV/amu = 6.54 MeV
6.8 MeV - 6.54 MeV = 0.26 MeV
The difference between the binding energy and the quantum energy level equals the amount of kinetic energy the neutron must possess. The typical heavy nucleus will have many closely■spaced resonanc complex and have more possible configurations and corresponding energy states. Light nuclei, being less complex, have fewer possible energy states and fewer resonances that are sparsely distributed at higher energy levels.
For higher neutron energies, the absorption cross section steadily decreases as the energy of the neutron increases. This is called the "fast neutron region." In this region the absorption cross sections are usually less than 10 barns.
With the exception of hydrogen, for which the value is fairly large, the elastic scattering cross sections are generally small, for example, 5 barns to 10 barns. This is close to the magnitude of the actual geometric cross sectional area expected for atomic nuclei. In potential scattering, the cross section is essentially constant and independent of neutron energy. Resonance elastic scattering and inelastic scattering exhibit resonance peaks similar to those associated with absorption cross sections. The resonances occur at lower energies for heavy nuclei than for light nuclei. In general, the variations in scattering cross sections are very small when compared to the variations that occur in absorption cross sections.
Mean Free Path
If a neutron has a certain probability of undergoing a particular interaction in one centimeter of travel, then the inverse of this value describes how far the neutron will travel (in the average case) before undergoing an interaction. This average distance traveled by a neutron before interaction is known as the mean free path for that interaction and is represented by the symbol ■. The relationship between the mean free path (■) and the macroscopic cross section (*) is shown below.
(2-3)
'
(
)
Fixture checkpoint marker-0060-xray is recorded on fixture page 0060.
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Reactor Theory (Neutron Characteristics) NUCLEAR CROSS SECTIONS AND NEUTRON FLUX
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Calculation of Macroscopic Cross Section and Mean Free Path
Most materials are composed of several elements, and because most elements are composed of several isotopes, most materials involve many cross sections, one for each isotope involved. Therefore, to include all the isotopes within a given material, it is necessary to determine the macroscopic cross section for each isotope and then sum all the individual macroscopic cross sections. Equation (2-4) can be used to determine the macroscopic cross section for a composite material.
* = N) + N) + N) + ... N) (2-4) 1 1 2 2 3 3 nn
where:
N = the number of nuclei per cm of the n element n
3 th

th
The following example problems illustrate the calculation of the macroscopic cross section for a single element and for combinations of materials.
Example 1:
Find the macroscopic thermal neutron absorption cross section for iron, which has a density of 7.86 g/cm . The microscopic cross section for absorption of iron is 2.56 barns 3 and the gram atomic weight is 55.847 g.
Solution:
Step 1: Using Equation (2-1), calculate the atom density of iron.
Step 2: Use this atom density in Equation (2-2) to calculate the macroscopic
Fixture checkpoint marker-0061-xray is recorded on fixture page 0061.
cross section.
'
'

Example 2:
An alloy is composed of 95% aluminum and 5% silicon (by weight). The density of the alloy is 2.66 g/cm . Properties of aluminum and silicon are shown below. 3
Element Gram Atomic Weight)) a (barns) s (barns)
Aluminum 26.9815 0.23 1.49
Silicon 28.0855 0.16 2.20
1. Calculate the atom densities for the aluminum and silicon.
2. Determine the absorption and scattering macroscopic cross sections for thermal neutrons.
3. Calculate the mean free paths for absorption and scattering.
Solution:
Step 1: The density of the aluminum will be 95% of the total density. Using Equation (2-1) yields the atom densities.
a NAl a,Al NSi a,Si
5.64 x 1022 atoms
cm3
0.23 x 10 24 cm22.85 x 1021 atoms
cm3
0.16 x 10 24 cm2
0.0134 cm 1
s NAl s,Al NSi s,Si
5.64 x 1022 atoms
cm3
1.49 x 10 24 cm22.85 x 1021 atoms
cm3
2.20 x 10 24 cm2
0.0903 cm 1
a
1
a
1

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74.3 cm
s
1
s
1
0.0903 cm 1
11.1 cm
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Step 2: The macroscopic cross sections for absorption and scattering are calculated using Equation (2-4).
Step 3: The mean free paths are calculated by inserting the macroscopic cross sections calculated above into Equation (2-3).
Thus, a neutron must travel an average of 74.3 cm to interact by absorption in this alloy, but it must travel only 11.1 cm to interact by scattering.
)
Fixture checkpoint marker-0062-xray is recorded on fixture page 0062.
)
)
)
(■
(■
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NUCLEAR CROSS SECTIONS AND NEUTRON FLUX Reactor Theory (Neutron Characteristics)
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Effects of Temperature on Cross Section
As discussed, the microscopic absorption cross section varies significantly as neutron energy varies. The microscopic cross sections provided on most charts and tables are measured for a standard neutron velocity of 2200 meters/second, which corresponds to an ambient temperature of 68(F. Therefore, if our material is at a higher temperature, the absorption cross section will be lower than the value for 68(F, and any cross sections which involve absorption (for example,) ,) ,)) must be corrected for the existing temperature. acf
The following formula is used to correct microscopic cross sections for temperature. Although the example illustrates absorption cross section, the same formula may be used to correct capture and fission cross sections.
where:
) = microscopic cross section corrected for temperature
) = microscopic cross section at reference temperature (68(F or 20(C) o
T = reference temperature (68(F) in degrees Rankine ((R) or Kelvin ((K) o
T = temperature for which corrected value is being calculated
NOTE: When using this formula, all temperatures must be converted to (R or (K.
(R = (F + 460
(K = (C + 273
Example:
What is the value of)f for uranium-235 for thermal neutrons at 500(F? Uranium-235 has a) of 583 barns at 68(F. f
Solution:

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Reactor Theory (Neutron Characteristics) NUCLEAR CROSS SECTIONS AND NEUTRON FLUX
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Neutron Flux
Macroscopic cross sections for neutron reactions with materials determine the probability of one neutron undergoing a specific reaction per centimeter of travel through that material. If one wants to determine how many reactions will actually occur, it is necessary to know how many neutrons are traveling through the material and how many centimeters they travel each second. It is convenient to consider the number of neutrons existing in one cubic centimeter at any one instant and the total distance they travel each second while in that cubic centimeter. The number of neutrons existing in a cm of material at any instant is called neutron density and is 3 represented by the symbol n with units of neutrons/cm . The total distance these neutrons can 3 travel each second will be determined by their velocity.
A good way of defining neutron flux (1) is to consider it to be the total path length covered by all neutrons in one cubic centimeter during one second. Mathematically, this is the equation below.
1 = n v (2-5)
where:
1 = neutron flux (neutrons/cm -sec) 2
n = neutron density (neutrons/cm)3
v = neutron velocity (cm/sec)
The term neutron flux in some applications (for example, cross section measurement) is used as parallel beams of neutrons traveling in a single direction. The intensity (I) of a neutron beam is the product of the neutron density times the average neutron velocity. The directional beam intensity is equal to the number of neutrons per unit area and time (neutrons/cm -sec) falling on 2
Fixture checkpoint marker-0063-xray is recorded on fixture page 0063.
a surface perpendicular to the direction of the beam.
One can think of the neutron flux in a reactor as being comprised of many neutron beams traveling in various directions. Then, the neutron flux becomes the scalar sum of these directional flux intensities (added as numbers and not vectors), that is, 1 = I + I + I +...I . 123 n
Since the atoms in a reactor do not interact preferentially with neutrons from any particular direction, all of these directional beams contribute to the total rate of reaction. In reality, at a given point within a reactor, neutrons will be traveling in all directions.
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Self-Shielding
In some locations within the reactor, the flux level may be significantly lower than in other areas due to a phenomenon referred to as neutron shadowing or self-shielding. For example, the interior of a fuel pin or pellet will "see" a lower average flux level than the outer surfaces since an appreciable fraction of the neutrons will have been absorbed and therefore cannot reach the interior of the fuel pin. This is especially important at resonance energies, where the absorption cross sections are large.
Summary
The important information in this chapter is summarized below.
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Reactor Theory (Neutron Characteristics) NUCLEAR CROSS SECTIONS AND NEUTRON FLUX
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Nuclear Cross Section and Neutron Flux Summary

material.
& Microscopic cross section ()) is the probability of a given reaction occurring between a neutron and a nucleus.
& Microscopic cross sections are measured in units of barns, where 1 barn = 10-24 cm .2
& Macroscopic cross section (*) is the probability of a given reaction occurring per unit length of travel of the neutron. The units for macroscopic cross section are cm . -1
& The mean free path (■) is the average distance that a neutron travels in a material between interactions.
& Neutron flux (1) is the total path length traveled by all neutrons in one cubic centimeter of material during one second.
& The macroscopic cross section for a material can be calculated using the equation below.
* = N)
& The absorption cross section for a material usually has three distinct regions. At low neutron energies (<1 eV) the cross section is inversely proportional to the neutron velocity.
& Resonance absorption occurs when the sum of the kinetic energy of the neutron and its binding energy is equal to an allowed nuclear energy level of the nucleus.
& Resonance peaks exist at intermediate energy levels. For higher neutron energies, the absorption cross section steadily decreases as the neutron energy increases.
& The mean free path equals 1/*.
& The macroscopic cross section for a mixture of materials can be calculated using the equation below.
* = N 1 1 2 2 3 3 nn) + N) + N) + ... N)
& Self-shielding is where the local neutron flux is depressed within a material due to neutron absorption near the surface of the material.
1)
REACTION RATES DOE-HDBK-1019/1-93 Reactor Theory (Neutron Characteristics) NP-02 Page 18 Rev. 0
Fixture checkpoint marker-0064-xray is recorded on fixture page 0064.
REACTION RATES
It is possible to determine the rate at which a nuclear reaction will take place based on the neutron flux, cross section for the interaction, and atom density of the target. This relationship illustrates how a change in one of these items affects the reaction rate.
EO 2.10 Given the neutron flux and macroscopic cross section, CALCULATE the reaction rate.
EO 2.11 DESCRIBE the relationship between neutron flux and reactor power.
Reaction Rates
If the total path length of all the neutrons in a cubic centimeter in a second is known, (neutron flux (1)), and if the probability of having an interaction per centimeter path length is also known (macroscopic cross section (*)), multiply them together to get the number of interactions taking place in that cubic centimeter in one second. This value is known as the reaction rate and is denoted by the symbol R. The reaction rate can be calculated by the equation shown below.
R = 1 * (2-6)

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R = reaction rate (reactions/sec)
1 = neutron flux (neutrons/cm -sec) 2
* = macroscopic cross section (cm) -1
Substituting the fact that * = N) into Equation (2-6) yields the equation below.
where:
) = microscopic cross section (cm)2
N = atom density (atoms/cm)3
Rf f
1 x 1013 neutrons
cm2
sec
0.1 cm 1
1 x 1012 fissions
cm3
sec
Rf f
f
Rf
1.29 x 1012 fissions
cm3
sec
3 x 1013 neutrons
cm2
sec
0.043 cm 1
Reactor Theory (Neutron Characteristics) DOE-HDBK-1019/1-93 REACTION RATES
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The reaction rate calculated will depend on which macroscopic cross section is used in the calculation. Normally, the reaction rate of greatest interest is the fission reaction rate.
Example:
If a one cubic centimeter section of a reactor has a macroscopic fission cross section of
0.1 cm , and if the thermal neutron flux is 10 neutrons/cm -sec, what is the fission rate -1 13 2
in that cubic centimeter?
Solution:
In addition to using Equation (2-6) to determine the reaction rate based on the physical properties of material, it is also possible to algebraically manipulate the equation to determine physical properties if the reaction rate is known.
Example:
Fixture checkpoint marker-0065-xray is recorded on fixture page 0065.
A reactor operating at a flux level of 3 x 10 neutrons/cm -sec contains 10 atoms of
13 2 20
uranium-235 per cm . The reaction rate is 1.29 x 10 fission/cm . Calculate and .
3 12 3
f f
Solution:
Step 1: The macroscopic cross section can be determined by solving
Equation (2-6) for and substituting the appropriate values. f
Rf N f

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Rf
N
1.29 x 1012 fissions
cm3
sec
1 x 1020 atoms
cm3
3 x 1013 neutrons
cm2
sec
4.3 x 10 22 cm2 1 barn
1 x 10 24 cm2
430 barns
1 watt 1 erg
1 x 10 7 watt sec
Fixture checkpoint marker-0066-xray is recorded on fixture page 0066.
1 MeV
1.602 x 10 6
erg
1 fission
200 MeV
3.12 x 1010 fissions
second
REACTION RATES DOE-HDBK-1019/1-93 Reactor Theory (Neutron Characteristics)
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Step 2: To find the microscopic cross section, replace f with (N x f) and solve
for .
f
Reactor Power Calculation
Multiplying the reaction rate per unit volume by the total volume of the core results in the total number of reactions occurring in the core per unit time. If the amount of energy involved in each reaction were known, it would be possible to determine the rate of energy release (power) due to a certain reaction.
In a reactor where the average energy per fission is 200 MeV, it is possible to determine the number of fissions per second that are necessary to produce one watt of power using the following conversion factors.
1 fission = 200 MeV
1 MeV = 1.602 x 10 ergs -6
1 erg = 1 x 10 watt-sec -7
This is equivalent to stating that 3.12 x 10 fissions release 1 watt-second of energy. 10
P
th f V
3.12 x 1010 fissions
watt sec
Reactor Theory (Neutron Characteristics) DOE-HDBK-1019/1-93 REACTION RATES
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The power released in a reactor can be calculated based on Equation (2-6). Multiplying the reaction rate by the volume of the reactor results in the total fission rate for the entire reactor.

reactor in units of watts. This relationship is shown mathematically in Equation (2-7) below.
(2-7)
where:
P = power (watts)
= thermal neutron flux (neutrons/cm -sec) th
2
= macroscopic cross section for fission (cm) f
-1
V = volume of core (cm)
3
Relationship Between Neutron Flux and Reactor Power
In an operating reactor the volume of the reactor is constant. Over a relatively short period of time (days or weeks), the number density of the fuel atoms is also relatively constant. Since the atom density and microscopic cross section are constant, the macroscopic cross section must also be constant. Examining Equation (2-7), it is apparent that if the reactor volume and macroscopic cross section are constant, then the reactor power and the neutron flux are directly proportional. This is true for day-to-day operation. The neutron flux for a given power level will increase very slowly over a period of months due to the burnup of the fuel and resulting decrease in atom density and macroscopic cross section.
REACTION RATES DOE-HDBK-1019/1-93 Reactor Theory (Neutron Characteristics)
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Reaction Rates Summary
Fixture checkpoint marker-0067-xray is recorded on fixture page 0067.
& The reaction rate is the number of interactions of a particular type occurring in a cubic centimeter of material in a second.
& The reaction rate can be calculated by the equation below.
R = 1 *
& Over a period of several days, while the atom density of the fuel can be considered constant, the neutron flux is directly proportional to reactor power.
Summary
The important information in this chapter is summarized below.
Reactor Theory (Neutron Characteristics) DOE-HDBK-1019/1-93 NEUTRON MODERATION
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NEUTRON MODERATION
In thermal reactors, the neutrons that cause fission are at a much lower energy than the energy level at which they were born from fission. In this type of reactor, specific materials must be included in the reactor design to reduce the energy level of the neutrons in an efficient manner.
EO 2.12 DEFINE the following concepts:
a. Thermalization d. Average logarithmic energy decrement
b. Moderator e. Macroscopic slowing down power
c. Moderating ratio
EO 2.13 LIST three desirable characteristics of a moderator.
EO 2.14 Given an average fractional energy loss per collision, CALCULATE the energy loss after a specified number of collisions.
Neutron Slowing Down and Thermalization
Fission neutrons are produced at an average energy level of 2 MeV and immediately begin to

a number of collisions with nuclei, the speed of a neutron is reduced to such an extent that it has approximately the same average kinetic energy as the atoms (or molecules) of the medium in which the neutron is undergoing elastic scattering. This energy, which is only a small fraction of an electron volt at ordinary temperatures (0.025 eV at 20(C), is frequently referred to as the thermal energy, since it depends upon the temperature. Neutrons whose energies have been reduced to values in this region (< 1 eV) are designated thermal neutrons. The process of reducing the energy of a neutron to the thermal region by elastic scattering is referred to as thermalization, slowing down, or moderation. The material used for the purpose of thermalizing neutrons is called a moderator. A good moderator reduces the speed of neutrons in a small number of collisions, but does not absorb them to any great extent. Slowing the neutrons in as few collisions as possible is desirable in order to reduce the amount of neutron leakage from the core and also to reduce the number of resonance absorptions in non-fuel materials. Neutron leakage and resonance absorption will be discussed in the next module.

ln Ei

ln Ef

ln

Ei

Ef

2

A

2

3

The ideal moderating material (moderator) should have the following nuclear properties.

large scattering cross section

small absorption cross section

large energy loss per collision

A convenient measure of energy loss per collision is the logarithmic energy decrement. The

Fixture checkpoint marker-0068-xray is recorded on fixture page 0068.

average logarithmic energy decrement is the average decrease per collision in the logarithm of the neutron energy. This quantity is represented by the symbol (Greek letter xi).

(2-8)

where:

= average logarithmic energy decrement

Ei = average initial neutron energy

Ef = average final neutron energy

The symbol is commonly called the average logarithmic energy decrement because of the fact that a neutron loses, on the average, a fixed fraction of its energy per scattering collision. Since the fraction of energy retained by a neutron in a single elastic collision is a constant for a given material, is also a constant. Because it is a constant for each type of material and does not depend upon the initial neutron energy, is a convenient quantity for assessing the moderating ability of a material.

The values for the lighter nuclei are tabulated in a variety of sources. The following commonly used approximation may be used when a tabulated value is not available.

This approximation is relatively accurate for mass numbers (A) greater than 10, but for some low values of A it may be in error by over three percent.

N

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ln
Ehigh
Elow
N
ln
Ehigh
Elow
ln
2 x 106eV
0.025 eV
0.948
19.2 collisions
Reactor Theory (Neutron Characteristics) DOE-HDBK-1019/1-93 NEUTRON MODERATION
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Since represents the average logarithmic energy loss per collision, the total number of collisions necessary for a neutron to lose a given amount of energy may be determined by dividing into the difference of the natural logarithms of the energy range in question. The number of collisions (N) to travel from any energy, E , to any lower energy, E , can be calculated as shown below. high low
Example:
How many collisions are required to slow a neutron from an energy of 2 MeV to a thermal energy of 0.025 eV, using water as the moderator? Water has a value of 0.948 for .
Solution:
Sometimes it is convenient, based upon information known, to work with an average fractional energy loss per collision as opposed to an average logarithmic fraction. If the initial neutron energy level and the average fractional energy loss per collision are known, the final energy level for a given number of collisions may be computed using the following formula.
! (
NEUTRON MODERATION DOE-HDBK-1019/1-93 Reactor Theory (Neutron Characteristics)
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(2-9)
where:
E = initial neutron energy o
E = neutron energy after N collisions N
x = average fractional energy loss per collision
N = number of collisions
Fixture checkpoint marker-0069-xray is recorded on fixture page 0069.
Example:
If the average fractional energy loss per collision in hydrogen is 0.63, what will be the energy of a 2 MeV neutron after (a) 5 collisions? (b) 10 collisions?
Solution:
a)
b)
Macroscopic Slowing Down Power
Although the logarithmic energy decrement is a convenient measure of the ability of a material to slow neutrons, it does not measure all necessary properties of a moderator. A better measure of the capabilities of a material is the macroscopic slowing down power. The macroscopic slowing down power (MSDP) is the product of the logarithmic energy decrement and the

calculate the macroscopic slowing down power.
(2-10)
MR
s
a
Reactor Theory (Neutron Characteristics) DOE-HDBK-1019/1-93 NEUTRON MODERATION
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Moderating Ratio
Macroscopic slowing down power indicates how rapidly a neutron will slow down in the material in question, but it still does not fully explain the effectiveness of the material as a moderator. An element such as boron has a high logarithmic energy decrement and a good slowing down power, but it is a poor moderator because of its high probability of absorbing neutrons.
The most complete measure of the effectiveness of a moderator is the moderating ratio. The Fixture checkpoint marker-0070-xray is recorded on fixture page 0070.
moderating ratio is the ratio of the macroscopic slowing down power to the macroscopic cross section for absorption. The higher the moderating ratio, the more effectively the material performs as a moderator. Equation (2-11) shows how to calculate the moderating ratio of a material.
(2-11)
Moderating properties of different materials are compared in Table 2.
TABLE 2
Moderating Properties of Materials
Material Collisions to Slowing Down
Number of Macroscopic
Thermalize Power
Moderating
Ratio
H2O 0.927 19 1.425 62
D2O 0.510 35 0.177 4830
Helium 0.427 42 9 x 10 51 -6
Beryllium 0.207 86 0.154 126
Boron 0.171 105 0.092 0.00086
Carbon 0.158 114 0.083 216
NEUTRON MODERATION DOE-HDBK-1019/1-93 Reactor Theory (Neutron Characteristics)
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Summary
The important information in this chapter is summarized below.
Neutron Moderation Summary
& Thermalization is the process of reducing the energy level of a neutron from the energy level at which it is produced to an energy level in the thermal range.
& The moderator is the reactor material that is present for the purpose of thermalizing neutrons.
& Moderating ratio is the ratio of the macroscopic slowing down power to the macroscopic cross section for absorption.
& The average logarithmic energy decrement (!) is the average change in the logarithm of neutron energy per collision.
& Macroscopic slowing down power is the product of the average logarithmic energy decrement and the macroscopic cross section for scattering.

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1. large scattering cross section
2. small absorption cross section
3. large energy loss per collision

& The energy loss after a specified number of collisions can be calculated using the equation below.

E = E (1 - x) N o

N

■

■ ■

Reactor Theory (Neutron Characteristics) DOE-HDBK-1019/1-93 PROMPT AND DELAYED NEUTRONS Rev. 0 Page 29 NP-02

PROMPT AND DELAYED NEUTRONS

Not all neutrons are released at the same time following fission. Most neutrons are released virtually instantaneously and are called prompt neutrons. A very small fraction of neutrons are released after the decay of fission products and are called delayed neutrons. Although delayed neutrons are a very small fraction of the total number of neutrons, they play an extremely important role in the control of the reactor.

EO 3.1 STATE the origin of prompt neutrons and delayed neutrons.

EO 3.2 STATE the approximate fraction of neutrons that are born as

Fixture checkpoint marker-0071-xray is recorded on fixture page 0071.

delayed neutrons from the fission of the following nuclear fuels:

a. Uranium-235

b. Plutonium-239

EO 3.3 EXPLAIN the mechanism for production of delayed neutrons.

EO 3.4 EXPLAIN prompt and delayed neutron generation times.

EO 3.5 Given prompt and delayed neutron generation times and delayed neutron fraction, CALCULATE the average generation time.

EO 3.6 EXPLAIN the effect of delayed neutrons on reactor control.

Neutron Classification

The great majority (over 99%) of the neutrons produced in fission are released within about 10-13 seconds of the actual fission event. These are called prompt neutrons. A small portion of fission neutrons are delayed neutrons, which are produced for some time after the fission process has taken place. The delayed neutrons are emitted immediately following the first beta decay of a fission fragment known as a delayed neutron precursor. An example of a delayed neutron precursor is bromine-87, shown below.

PROMPT AND DELAYED NEUTRONS DOE-HDBK-1019/1-93 Reactor Theory (Neutron Characteristics) NP-02 Page 30 Rev. 0

For most applications, it is convenient to combine the known precursors into groups with appropriately averaged properties. These groups vary somewhat depending on the fissile material in use. Table 3 lists the characteristics for the six precursor groups resulting from thermal fission of uranium-235. The fraction of all neutrons that are produced by each of these precursors is called the delayed neutron fraction for that precursor. The total fraction of all neutrons born as delayed neutrons is called the delayed neutron fraction (■). The fraction of delayed neutrons produced varies depending on the predominant fissile nuclide in use. The delayed neutron fractions (■) for the fissile nuclides of most interest are as follows: uranium-233 (0.0026), uranium-235 (0.0065), uranium-238 (0.0148), and plutonium-239 (0.0021).

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Delayed Neutron Precursor Groups
for Thermal Fission in Uranium-235
Group Half-Life Delayed Neutron Average Energy
(sec) Fraction (MeV)
1 55.7 0.00021 0.25
2 22.7 0.00142 0.46
3 6.2 0.00127 0.41
4 2.3 0.00257 0.45
5 0.61 0.00075 0.41
6 0.23 0.00027 -
Total - 0.0065 -
Neutron Generation Time
The neutron generation time is the time required for neutrons from one generation to cause the fissions that produce the next generation of neutrons. The generation time for prompt neutrons (5* - pronounced "ell-star") is the total time from birth to rebirth. Three time intervals are involved: (a) the time it takes a fast neutron to slow down to thermal energy, (b) the time the now thermal neutron exists prior to absorption in fuel, and (c) the time required for a fissionable nucleus to emit a fast neutron after neutron absorption.
Fast neutrons slow to thermal energies or leak out of the reactor in 10 seconds to 10 seconds, -4 -6 depending on the moderator. In water moderated reactors, thermal neutrons tend to exist for about 10 seconds before they are absorbed. Fission and fast neutron production following -4 neutron absorption in a fissionable nucleus occurs in about 10 seconds. Thus, fast reactors -13 have an 5* of about 10 seconds, while thermal reactors have an 5* of about 10 seconds + 10 -6 -6 -4 seconds, which is about 10 seconds to 10 seconds. -4 -5

■ ■ ■ ■
■ ■ ■ ■
■ ■

Fixture checkpoint marker-0072-xray is recorded on fixture page 0072.
Reactor Theory (Neutron Characteristics) DOE-HDBK-1019/1-93 PROMPT AND DELAYED NEUTRONS
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On the other hand, the average generation time for the six delayed neutron groups is the total time from the birth of the fast neutron to the emission of the delayed neutron. Again, three time intervals are involved: (a) the time it takes a fast neutron to slow down to thermal energy, (b) the time the thermal neutron exists prior to absorption, and (c) the average time from neutron absorption to neutron emission by the six precursor groups. The average time for decay of precursors from uranium-235 is 12.5 seconds. The other terms in the delayed neutron generation time are insignificant when compared to this value, and the average delayed neutron generation time becomes ■ 12.5 seconds.
A neutron generation time in the range of 10 seconds to 10 seconds or faster could result in -4 -5 very rapid power excursions, and control would not be possible without the dependence upon delayed neutrons to slow down the rate of the reaction. The average generation time, and hence the rate that power can rise, is determined largely by the delayed neutron generation time. The following equation shows this mathematically.
(2-12)
Example:
Assume a prompt neutron generation time for a particular reactor of 5 x 10 seconds and -5 a delayed neutron generation time of 12.5 seconds. If ■ is 0.0065, calculate the average generation time.

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This example demonstrates the effect delayed neutrons have on the neutron generation time and thus reactor control. If a reactor were to be operated in a sustained chain reaction using only prompt neutrons (■ = 0), the generation time from the previous example would be about 5 x 10 seconds. However, by operating the reactor such that a 0.0065 fraction of neutrons are -5 delayed, the generation life time is extended to 0.0813 seconds, providing time for adequate operator control. Therefore, although only a small fraction of the total neutron population, delayed neutrons are extremely important to the control and maintenance of a sustained fission chain reaction.

Summary

The important information in this chapter is summarized on the following page.

■ ■ ■ ■

PROMPT AND DELAYED NEUTRONS DOE-HDBK-1019/1-93 Reactor Theory (Neutron Characteristics) NP-02 Page 32 Rev. 0

Prompt and Delayed Neutrons Summary

& Prompt neutrons are released directly from fission within 10 seconds of the -13 fission event.

& Delayed neutrons are released from the decay of fission products that are called delayed neutron precursors. Delayed neutron precursors are grouped according to half-life. Half-lives vary from fractions of a second to almost a minute.

& The fraction of neutrons born as delayed neutrons is different for different fuel materials. Following are values for some common fuel materials.

Uranium-235 0.0065

Plutonium-239 0.0021

& Delayed neutrons are produced by a classification of fission products known as delayed neutron precursors. When a delayed neutron precursor undergoes a ■ decay, it results in an excited daughter nucleus which immediately ejects a - neutron. Therefore, these delayed neutrons appear with a half-life of the delayed neutron precursor.

& The delayed neutron generation time is the total time from the birth of the fast neutron to the emission of the delayed neutron in the next generation. Delayed neutron generation times are dominated by the half-life of the delayed neutron precursor. The average delayed neutron generation time is about 12.5 seconds.

& A prompt neutron generation time is the sum of the amount of time it takes a fast neutron to thermalize, the amount of time the neutron exists as a thermal neutron

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before it is absorbed, and the amount of time between a fissionable nuclide absorbing a neutron and fission neutrons being released. Prompt neutron generation time is about 5 x 10 seconds. -5

& The average neutron generation time can be calculated from the prompt and delayed neutron generation times and the delayed neutron fraction using Equation (2-12).

& Delayed neutrons are responsible for the ability to control the rate at which power can rise in a reactor. If only prompt neutrons existed, reactor control would not be possible due to the rapid power changes.

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Figure 2 Prompt Fission Neutron Energy Spectrum for Thermal Fission of Uranium-235

The number of neutrons that exist at a given energy level varies. A plot of either the fraction of neutrons or the neutron flux at a given energy versus the energy level is called a neutron energy spectrum. The neutron energy spectrum varies widely for different types of reactors.

EO 4.1 STATE the average energy at which prompt neutrons are produced.

EO 4.2 DESCRIBE the neutron energy spectrum in the following reactors:

a. Fast reactor

b. Thermal reactor

EO 4.3 EXPLAIN the reason for the particular shape of the fast, intermediate, and slow energy regions of the neutron flux spectrum for a thermal reactor.

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Prompt Neutron Energies

The neutrons produced by fission are high energy neutrons, and almost all fission neutrons have energies between 0.1 MeV and 10 MeV. The neutron energy distribution, or spectrum, may best be described by plotting the fraction of neutrons per MeV as a function of neutron energy, as shown in Figure 2. From this figure it can be seen that the most probable neutron energy is about 0.7 MeV. Also, from this data it can be shown that the average energy of fission neutrons is about 2 MeV. Figure 2 is the neutron energy spectrum for thermal fission in uranium-235. The values will vary slightly for other nuclides.

NEUTRON FLUX SPECTRUM DOE-HDBK-1019/1-93 Reactor Theory (Neutron Characteristics) NP-02 Page 34 Rev. 0

Figure 3 Comparison of Neutron Flux Spectra for Thermal and Fast Breeder Reactor

Thermal and Fast Breeder Reactor Neutron Spectra

The spectrum of neutron energies produced by fission varies significantly from the energy spectrum, or flux, existing in a reactor at a given time. Figure 3 illustrates the difference in neutron flux spectra between a thermal reactor and a fast breeder reactor. The energy distribution of neutrons from fission is essentially the same for both reactors, so the differences in the curve shapes may be attributed to the neutron moderation or slowing down effects. No attempt is made to thermalize or slow down neutrons in the fast breeder reactor (liquid metal cooled); therefore, an insignificant number of neutrons exist in the thermal range. For the thermal reactor (water moderated), the spectrum of neutrons in the fast region (> 0.1 MeV) has

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In the thermal reactor, the flux in the intermediate energy region (1 eV to 0.1 MeV) has approximately a 1/E dependence. That is, if the energy (E) is halved, the flux doubles. This 1/E dependence is caused by the slowing down process, where elastic collisions remove a constant fraction of the neutron energy per collision (on the average), independent of energy; thus, the neutron loses larger amounts of energy per collision at higher energies than at lower energies. The fact that the neutrons lose a constant fraction of energy per collision causes the neutrons to tend to "pile up" at lower energies, that is, a greater number of neutrons exist at the lower energies as a result of this behavior.
vp
2 k T
m
vp
2 k T
m
2 1.38 x 10 16 erg
K
293 K
1.66 x 10 24 g
Fixture checkpoint marker-0075-xray is recorded on fixture page 0075.
2.2 x 105 cm
sec
1 m
100 cm
2200 m
sec
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In the thermal region the neutrons achieve a thermal equilibrium with the atoms of the moderator material. In any given collision they may gain or lose energy, and over successive collisions will gain as much energy as they lose. These thermal neutrons, even at a specific temperature, do not all have the same energy or velocity; there is a distribution of energies, usually referred to as the Maxwell distribution (e.g., Figure 2). The energies of most thermal neutrons lie close to the most probable energy, but there is a spread of neutrons above and below this value.
Most Probable Neutron Velocities
The most probable velocity (vp) of a thermal neutron is determined by the temperature of the medium and can be determined by Equation (2-13) .
(2-13)
where:
v = most probable velocity of neutron (cm/sec) p
k = Boltzman's constant (1.38 x 10 erg/ K) -16
T = absolute temperature in degrees Kelvin (K)
m = mass of neutron (1.66 x 10 grams) -24
Example:
Calculate the most probable velocities for neutrons in thermal equilibrium with their surroundings at the following temperatures. a) 20 C, b) 260 C.
Solution:
a) Calculate the most probable velocity for 20 C using Equation (2-13).
vp

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AND REACTOR THEORY
Volume 1 of 2
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NUCLEAR PHYSICS AND REACTOR THEORY
ABSTRACT
The Nuclear Physics and Reactor Theory Handbook was developed to assist nuclear facility operating contractors in providing operators, maintenance personnel, and the technical staff with the necessary fundamentals training to ensure a basic understanding of nuclear physics and reactor theory. The handbook includes information on atomic and nuclear physics; neutron characteristics; reactor theory and nuclear parameters; and the theory of reactor operation. This information will provide personnel with a foundation for understanding the scientific principles that are associated with various DOE nuclear facility operations and maintenance.
Key Words: Training Material, Atomic Physics, The Chart of the Nuclides, Radioactivity, Radioactive Decay, Neutron Interaction, Fission, Reactor Theory, Neutron Characteristics, Neutron Life Cycle, Reactor Kinetics
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NUCLEAR PHYSICS AND REACTOR THEORY
FOREWORD
The Department of Energy (DOE) Fundamentals Handbooks consist of ten academic subjects, which include Mathematics; Classical Physics; Thermodynamics, Heat Transfer, and Fluid Flow; Instrumentation and Control; Electrical Science; Material Science; Mechanical Science; Chemistry; Engineering Symbology, Prints, and Drawings; and Nuclear Physics and Reactor Theory. The handbooks are provided as an aid to DOE nuclear facility contractors.
These handbooks were first published as Reactor Operator Fundamentals Manuals in 1985 for use by DOE category A reactors. The subject areas, subject matter content, and level of detail of the Reactor Operator Fundamentals Manuals were determined from several sources. DOE Category A reactor training managers determined which materials should be included, and served as a primary reference in the initial development phase. Training guidelines from the commercial nuclear power industry, results of job and task analyses, and independent input from contractors and operations-oriented personnel were all considered and included to some degree in developing the text material and learning objectives.
The DOE Fundamentals Handbooks represent the needs of various DOE nuclear facilities' fundamental training requirements. To increase their applicability to nonreactor nuclear facilities, the Reactor Operator Fundamentals Manual learning objectives were distributed to the Nuclear Facility Training Coordination Program Steering Committee for review and comment. To update their reactor-specific content, DOE Category A reactor training managers also reviewed and

to two or more DOE nuclear facilities was considered generic and was included. The final draft of each of the handbooks was then reviewed by these two groups. This approach has resulted in revised modular handbooks that contain sufficient detail such that each facility may adjust the content to fit their specific needs.

Each handbook contains an abstract, a foreword, an overview, learning objectives, and text material, and is divided into modules so that content and order may be modified by individual DOE contractors to suit their specific training needs. Each handbook is supported by a separate examination bank with an answer key.

The DOE Fundamentals Handbooks have been prepared for the Assistant Secretary for Nuclear Energy, Office of Nuclear Safety Policy and Standards, by the DOE Training Coordination Program. This program is managed by EG&G Idaho, Inc.

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NUCLEAR PHYSICS AND REACTOR THEORY

OVERVIEW

The Department of Energy Fundamentals Handbook entitled Nuclear Physics and Reactor Theory was prepared as an information resource for personnel who are responsible for the operation of the Department's nuclear facilities. Almost all processes that take place in a nuclear facility involves the transfer of some type of energy. A basic understanding of nuclear physics and reactor theory is necessary for DOE nuclear facility operators, maintenance personnel, and the technical staff to safely operate and maintain the facility and facility support systems. The information in this handbook is presented to provide a foundation for applying engineering concepts to the job. This knowledge will help personnel understand the impact that their actions may have on the safe and reliable operation of facility components and systems.

The Nuclear Physics and Reactor Theory handbook consists of four modules that are contained in two volumes. The following is a brief description of the information presented in each module of the handbook.

Volume 1 of 2

Module 1 - Atomic and Nuclear Physics

Introduces concepts of atomic physics including the atomic nature of matter, the chart of the nuclides, radioactivity and radioactive decay, neutron interactions and fission, and the interaction of radiation with matter.

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Module 2 - Reactor Theory (Nuclear Parameters)

Provides information on reactor theory and neutron characteristics. Includes topics such as neutron sources, neutron flux, neutron cross sections, reaction rates, neutron moderation, and prompt and delayed neutrons.

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NUCLEAR PHYSICS AND REACTOR THEORY

OVERVIEW (Cont.)

Volume 2 of 2

Module 3 - Reactor Theory (Nuclear Parameters)

Explains the nuclear parameters associated with reactor theory. Topics include the neutron life cycle, reactivity and reactivity coefficients, neutron poisons, and control rods.

Module 4 - Reactor Theory (Reactor Operations)

Introduces the reactor operations aspect of reactor theory. Topics include

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The information contained in this handbook is not all-encompassing. An attempt to present the entire subject of nuclear physics and reactor theory would be impractical. However, the Nuclear Physics and Reactor Theory handbook presents enough information to provide the reader with the fundamental knowledge necessary to understand the advanced theoretical concepts presented in other subject areas, and to understand basic system and equipment operation.
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TERMINAL OBJECTIVE
1.0 Given sufficient information, DESCRIBE atoms, including components, structure, and nomenclature.
ENABLING OBJECTIVES
1.1 STATE the characteristics of the following atomic particles, including mass, charge, and location within the atom:
a. Proton
b. Neutron
c. Electron
1.2 DESCRIBE the Bohr model of an atom.
1.3 DEFINE the following terms:
a. Nuclide c. Atomic number
b. Isotope d. Mass number
1.4 Given the standard A
ZX notation for a particular nuclide, DETERMINE the following:
Fixture checkpoint marker-0081-xray is recorded on fixture page 0081.
a. Number of protons
b. Number of neutrons
c. Number of electrons
1.5 DESCRIBE the three forces that act on particles within the nucleus and affect the stability of the nucleus.
1.6 DEFINE the following terms:
a. Enriched uranium
b. Depleted uranium
1.7 DEFINE the following terms:
a. Mass defect
b. Binding energy
1.8 Given the atomic mass for a nuclide and the atomic masses of a neutron, proton, and electron, CALCULATE the mass defect and binding energy of the nuclide.

TERMINAL OBJECTIVE
2.0 Given necessary references, DESCRIBE the various modes of radioactive decay.

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2.1 DESCRIBE the following processes:
a. Alpha decay d. Electron capture
b. Beta-minus decay e. Internal conversions
c. Beta-plus decay f. Isomeric transitions
2.2 Given a Chart of the Nuclides, WRITE the radioactive decay chain for a nuclide.
2.3 EXPLAIN why one or more gamma rays typically accompany particle emission.
2.4 Given the stability curve on the Chart of the Nuclides, DETERMINE the type of radioactive decay that the nuclides in each region of the chart will typically undergo.
2.5 DEFINE the following terms:
a. Radioactivity d. Radioactive decay constant
b. Curie e. Radioactive half-life
c. Becquerel
2.6 Given the number of atoms and either the half-life or decay constant of a nuclide, CALCULATE the activity.
2.7 Given the initial activity and the decay constant of a nuclide, CALCULATE the activity at any later time.
2.8 CONVERT between the half-life and decay constant for a nuclide.
2.9 Given the Chart of the Nuclides and the original activity, PLOT the radioactive decay curve for a nuclide on either linear or semi-log coordinates.
2.10 DEFINE the following terms:
a. Radioactive equilibrium
b. Transient radioactive equilibrium
NP-01 Page viii Rev. 0
Atomic and Nuclear Physics DOE-HDBK-1019/1-93 OBJECTIVES
TERMINAL OBJECTIVE
3.0 Without references, DESCRIBE the different nuclear interactions initiated by neutrons.
ENABLING OBJECTIVES
3.1 DESCRIBE the following scattering interactions between a neutron and a nucleus:
a. Elastic scattering
b. Inelastic scattering
3.2 STATE the conservation laws that apply to an elastic collision between a neutron and a nucleus.
3.3 DESCRIBE the following reactions where a neutron is absorbed in a nucleus:
a. Radiative capture
b. Particle ejection
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OBJECTIVES DOE-HDBK-1019/1-93 Atomic and Nuclear Physics
Fixture checkpoint marker-0082-xray is recorded on fixture page 0082.
TERMINAL OBJECTIVE
4.0 Without references, DESCRIBE the fission process.
ENABLING OBJECTIVES
4.1 EXPLAIN the fission process using the liquid drop model of a nucleus.
4.2 DEFINE the following terms:
a. Excitation energy
b. Critical energy
4.3 DEFINE the following terms:
a. Fissile material
b. Fissionable material

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4.4 DESCRIBE the processes of transmutation, conversion, and breeding.
4.5 DESCRIBE the curve of Binding Energy per Nucleon versus mass number and give a qualitative description of the reasons for its shape.
4.6 EXPLAIN why only the heaviest nuclei are easily fissioned.
4.7 EXPLAIN why uranium-235 fissions with thermal neutrons and uranium-238 fissions only with fast neutrons.
4.8 CHARACTERIZE the fission products in terms of mass groupings and radioactivity.
4.9 Given the nuclides involved and their masses, CALCULATE the energy released from fission.
4.10 Given the curve of Binding Energy per Nucleon versus mass number, CALCULATE the energy released from fission.
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Atomic and Nuclear Physics DOE-HDBK-1019/1-93 OBJECTIVES
TERMINAL OBJECTIVE
Fixture checkpoint marker-0083-xray is recorded on fixture page 0083.
5.0 Without references, DESCRIBE how the various types of radiation interact with matter.
ENABLING OBJECTIVES
5.1 DESCRIBE interactions of the following with matter:
a. Alpha particle c. Positron
b. Beta particle d. Neutron
5.2 DESCRIBE the following ways that gamma radiation interacts with matter:
a. Compton scattering
b. Photoelectric effect
c. Pair production
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OBJECTIVES DOE-HDBK-1019/1-93 Atomic and Nuclear Physics
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NP-01 Page xii Rev. 0
Atomic and Nuclear Physics DOE-HDBK-1019/1-93 ATOMIC NATURE OF MATTER
ATOMIC NATURE OF MATTER
All matter is composed of atoms. The atom is the smallest amount of matter that retains the properties of an element. Atoms themselves are composed of smaller particles, but these smaller particles no longer have the same properties as the overall element.
EO 1.1 STATE the characteristics of the following atomic particles, including mass, charge, and location within the atom:
a. Proton
b. Neutron
c. Electron
EO 1.2 DESCRIBE the Bohr model of an atom.
EO 1.3 DEFINE the following terms:
a. Nuclide c. Atomic number
b. Isotope d. Mass number
EO 1.4 Given the standard A ZX notation for a particular nuclide, DETERMINE the following:
a. Number of protons
b. Number of neutrons

SYNTHETIC FIXTURE PAGE 0084 OF 1200

EO 1.5 DESCRIBE the three forces that act on particles within the nucleus and affect the stability of the nucleus.

Structure of Matter

Early Greek philosophers speculated that the earth was made up of different combinations of basic substances, or elements. They considered these basic elements to be earth, air, water, and fire. Modern science shows that the early Greeks held the correct concept that matter consists of a combination of basic elements, but they incorrectly identified the elements.

In 1661 the English chemist Robert Boyle published the modern criterion for an element. He defined an element to be a basic substance that cannot be broken down into any simpler substance after it is isolated from a compound, but can be combined with other elements to form compounds. To date, 105 different elements have been confirmed to exist, and researchers claim to have discovered three additional elements. Of the 105 confirmed elements, 90 exist in nature and 15 are man-made.

Another basic concept of matter that the Greeks debated was whether matter was continuous or discrete. That is, whether matter could be continuously divided and subdivided into ever smaller particles or whether eventually an indivisible particle would be encountered. Democritus in about 450 B.C. argued that substances were ultimately composed of small, indivisible particles that he labeled atoms. He further suggested that different substances were composed of different atoms or combinations of atoms, and that one substance could be converted into another by rearranging

Fixture checkpoint marker-0084-xray is recorded on fixture page 0084.

the atoms. It was impossible to conclusively prove or disprove this proposal for more than 2000 years.

The modern proof for the atomic nature of matter was first proposed by the English chemist John Dalton in 1803. Dalton stated that each chemical element possesses a particular kind of atom, and any quantity of the element is made up of identical atoms of this kind. What distinguishes one element from another element is the kind of atom of which it consists, and the basic physical difference between kinds of atoms is their weight.

Subatomic Particles

For almost 100 years after Dalton established the atomic nature of atoms, it was considered impossible to divide the atom into even smaller parts. All of the results of chemical experiments during this time indicated that the atom was indivisible. Eventually, experimentation into electricity and radioactivity indicated that particles of matter smaller than the atom did indeed exist. In 1906, J. J. Thompson won the Nobel Prize in physics for establishing the existence of electrons. Electrons are negatively-charged particles that have 1/1835 the mass of the hydrogen atom. Soon after the discovery of electrons, protons were discovered. Protons are relatively large particles that have almost the same mass as a hydrogen atom and a positive charge equal in magnitude (but opposite in sign) to that of the electron. The third subatomic particle to be discovered, the neutron, was not found until 1932. The neutron has almost the same mass as the proton, but it is electrically neutral.

Bohr Model of the Atom

The British physicist Ernest Rutherford postulated that the positive charge in an atom is concentrated in a small region called a nucleus at the center of the atom with electrons existing in orbits around it. Niels Bohr, coupling Rutherford's postulation with the quantum theory introduced by Max Planck, proposed that the atom consists of a dense nucleus of protons

electron in one of these orbits or shells has a specific or discrete quantity of energy (quantum). When an electron moves from one allowed orbit to another allowed orbit, the energy difference between the two states is emitted or absorbed in the form of a single quantum of radiant energy called a photon. Figure 1 is Bohr's model of the hydrogen atom showing an electron as having just dropped from the third shell to the first shell with the emission of a photon that has an energy = hv. (h = Planck's constant = 6.63 x 10-34 J-s and v = frequency of the photon.) Bohr's theory was the first to successfully account for the discrete energy levels of this radiation as measured in the laboratory. Although Bohr's atomic model is designed specifically to explain the hydrogen atom, his theories apply generally to the structure of all atoms. Additional information on electron shell theory can be found in the Chemistry Fundamentals Handbook.

Figure 1 Bohr's Model of the Hydrogen Atom

Properties of the three subatomic particles are listed in Table 1.

TABLE 1

Properties of Subatomic Particles

Particle	Location	Charge	Mass
Neutron	Nucleus	none	1.008665 amu
Proton	Nucleus	+1	1.007277 amu
Electron	Shells around nucleus	-1	0.0005486 amu

Measuring Units on the Atomic Scale

The size and mass of atoms are so small that the use of normal measuring units, while possible, is often inconvenient. Units of measure have been defined for mass and energy on the atomic scale to make measurements more convenient to express. The unit of measure for mass is the atomic mass unit (amu). One atomic mass unit is equal to 1.66 x 10-24 grams. The reason for this particular value for the atomic mass unit will be discussed in a later chapter. Note from Table 1 that the mass of a neutron and a proton are both about 1 amu. The unit for energy is the electron volt (eV). The electron volt is the amount of energy acquired by a single electron

Fixture checkpoint marker-0085-xray is recorded on fixture page 0085.

when it falls through a potential difference of one volt. One electron volt is equivalent to 1.602 x 10-19 joules or 1.18 x 10-19 foot-pounds.

Nuclides

The total number of protons in the nucleus of an atom is called the atomic number of the atom and is given the symbol Z. The number of electrons in an electrically-neutral atom is the same as the number of protons in the nucleus. The number of neutrons in a nucleus is known as the neutron number and is given the symbol N. The mass number of the nucleus is the total number of nucleons, that is, protons and neutrons in the nucleus. The mass number is given the symbol A and can be found by the equation Z + N = A.

Each of the chemical elements has a unique atomic number because the atoms of different elements contain a different number of protons. The atomic number of an atom identifies the particular element.

Each type of atom that contains a unique combination of

Figure 2 Nomenclature for Identifying Nuclides

protons and neutrons is called a nuclide. Not all combinations of numbers of protons and neutrons are

combinations of neutrons and protons have been identified. Each nuclide is denoted by the chemical symbol of the element with the atomic number written as a subscript and the mass number written as a superscript, as shown in Figure 2. Because each element has a unique name, chemical symbol, and atomic number, only one of the three is necessary to identify the element. For this reason nuclides can also be identified by either the chemical name or the chemical symbol followed by the mass number (for example, U-235 or uranium-235). Another common format is to use the abbreviation of the chemical element with the mass number superscripted (for example, 235U). In this handbook the format used in the text will usually be the element's name followed by the mass number. In equations and tables, the format in Figure 2 will usually be used.

Example:

State the name of the element and the number of protons, electrons, and neutrons in the nuclides listed below.

1
1H
10
5B
14
7N
11
4
4
8Cd
23
9
9
4Pu

Solution:

The name of the element can be found from the Periodic Table (refer to Chemistry Fixture checkpoint marker-0086-xray is recorded on fixture page 0086. Fundamentals Handbook) or the Chart of the Nuclides (to be discussed later). The number of protons and electrons are equal to Z. The number of neutrons is equal to Z - A.

Nuclide Element Protons Electrons Neutrons
1
1H hydrogen 1 1 0
10
5B boron 5 5 5
14
7N nitrogen 7 7 7
11
4

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8Cd cadmium 48 48 66
23
9
9
4Pu plutonium 94 94 145
Isotopes
Isotopes are nuclides that have the same atomic number and are therefore the same element, but differ in the number of neutrons. Most elements have a few stable isotopes and several unstable, radioactive isotopes. For example, oxygen has three stable isotopes that can be found in nature (oxygen-16, oxygen-17, and oxygen-18) and eight radioactive isotopes. Another example is hydrogen, which has two stable isotopes (hydrogen-1 and hydrogen-2) and a single radioactive isotope (hydrogen-3).
Fixture checkpoint marker-0087-xray is recorded on fixture page 0087.
The isotopes of hydrogen are unique in that they are each commonly referred to by a unique name instead of the common chemical element name. Hydrogen-1 is almost always referred to as hydrogen, but the term protium is infrequently used also. Hydrogen-2 is commonly called deuterium and symbolized 2
1D. Hydrogen-3 is commonly called tritium and symbolized 31T. This
text will normally use the symbology 2
1H and 31H for deuterium and tritium, respectively.
Atomic and Nuclear Radii
The size of an atom is difficult to define exactly due to the fact that the electron cloud, formed by the electrons moving in their various orbitals, does not have a distinct outer edge. A reasonable measure of atomic size is given by the average distance of the outermost electron from the nucleus. Except for a few of the lightest atoms, the average atomic radii are approximately the same for all atoms, about 2 x 10 -8 cm.
Like the atom the nucleus does not have a sharp outer boundary. Experiments have shown that the nucleus is shaped like a sphere with a radius that depends on the atomic mass number of the atom. The relationship between the atomic mass number and the radius of the nucleus is shown in the following equation.
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Atomic and Nuclear Physics DOE-HDBK-1019/1-93 ATOMIC NATURE OF MATTER
r = (1.25 x 10 -13 cm) A1/3
where:
r = radius of the nucleus (cm)
A = atomic mass number (dimensionless)
The values of the nuclear radii for some light, intermediate, and heavy nuclides are shown in Table 2.
TABLE 2
Calculated Values for Nuclear
Radii
Nuclide Radius of Nucleus
1
1H 1.25 x 10 -13 cm
10
5B 2.69 x 10 -13 cm
5
2

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6Fe 4.78 x 10 -13 cm
17
7
8
2Hf 7.01 x 10 -13 cm
23
9
8
2U 7.74 x 10 -13 cm
25
9
2
8Cf 7.89 x 10 -13 cm
From the table, it is clear that the radius of a typical atom (e.g. 2 x 10 -8 cm) is more than 25,000 times larger than the radius of the largest nucleus.
Nuclear Forces
In the Bohr model of the atom, the nucleus consists of positively-charged protons and electrically■neutra as nucleons. One problem that the Bohr model of the atom presented was accounting for an attractive force to overcome the repulsive force between protons.
Fixture checkpoint marker-0088-xray is recorded on fixture page 0088.
Two forces present in the nucleus are (1) electrostatic forces between charged particles and (2) gravitational forces between any two objects that have mass. It is possible to calculate the magnitude of the gravitational force and electrostatic force based upon principles from classical physics.
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Fg
G m1 m2
r
2
Fe
K Q1 Q2
r
2
ATOMIC NATURE OF MATTER DOE-HDBK-1019/1-93 Atomic and Nuclear Physics
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Newton stated that the gravitational force between two bodies is directly proportional to the masses of the two bodies and inversely proportional to the square of the distance between the bodies. This relationship is shown in the equation below.
where:
F = gravitational force (newtons) g
m = mass of first body (kilograms) 1
m = mass of second body (kilograms) 2
G = gravitational constant (6.67 x 10 N-m /kg)
-11 2 2
r = distance between particles (meters)
The equation illustrates that the larger the masses of the objects or the smaller the distance between the objects, the greater the gravitational force. So even though the masses of nucleons are very small, the fact that the distance between nucleons is extremely short may make the

compare it to the value for other forces to determine the significance of the gravitational force in the nucleus. The gravitational force between two protons that are separated by a distance of 10 meters is about 10 newtons. -20 -24

Coulomb's Law can be used to calculate the force between two protons. The electrostatic force is directly proportional to the electrical charges of the two particles and inversely proportional to the square of the distance between the particles. Coulomb's Law is stated as the following equation.

where:

F = electrostatic force (newtons) e

K = electrostatic constant (9.0 x 10 N-m /C)

9 2 2

Q = charge of first particle (coulombs) 1

Q = charge of second particle (coulombs) 2

r = distance between particles (meters)

Using this equation, the electrostatic force between two protons that are separated by a distance of 10 meters is about 10 newtons. Comparing this result with the calculation of the

-20 12

gravitational force (10 newtons) shows that the gravitational force is so small that it can be

-24

neglected.

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If only the electrostatic and gravitational forces existed in the nucleus, then it would be impossible to have stable nuclei composed of protons and neutrons. The gravitational forces are much too small to hold the nucleons together compared to the electrostatic forces repelling the protons. Since stable atoms of neutrons and protons do exist, there must be another attractive force acting within the nucleus. This force is called the nuclear force.

Fixture checkpoint marker-0089-xray is recorded on fixture page 0089.

The nuclear force is a strong attractive force that is independent of charge. It acts equally only between pairs of neutrons, pairs of protons, or a neutron and a proton. The nuclear force has a very short range; it acts only over distances approximately equal to the diameter of the nucleus (10 -13 cm). The attractive nuclear force between all nucleons drops off with distance much faster than the repulsive electrostatic force between protons.

TABLE 3

Forces Acting in the Nucleus

Force	Interaction	Range
Gravitational	Very weak attractive force between all nucleons	Relatively long
Electrostatic	Strong repulsive force between like charged particles (protons)	Relatively long
Nuclear Force	Strong attractive force between all nucleons	Extremely short

In stable atoms, the attractive and repulsive forces in the nucleus balance. If the forces do not balance, the atom cannot be stable, and the nucleus will emit radiation in an attempt to achieve a more stable configuration.

Summary

The important information in this chapter is summarized on the following page.

SYNTHETIC FIXTURE PAGE 0090 OF 1200

Atoms consist of three basic subatomic particles. These particles are the proton, the neutron, and the electron.
Protons are particles that have a positive charge, have about the same mass as a hydrogen atom, and exist in the nucleus of an atom.
Neutrons are particles that have no electrical charge, have about the same mass as a hydrogen atom, and exist in the nucleus of an atom.
Electrons are particles that have a negative charge, have a mass about eighteen hundred times smaller than the mass of a hydrogen atom, and exist in orbital shells around the nucleus of an atom.
The Bohr model of the atom consists of a dense nucleus of protons and neutrons (nucleons) surrounded by electrons traveling in discrete orbits at fixed distances from the nucleus.
Nuclides are atoms that contain a particular number of protons and neutrons.
Isotopes are nuclides that have the same atomic number and are therefore the same element, but differ in the number of neutrons.
The atomic number of an atom is the number of protons in the nucleus.
The mass number of an atom is the total number of nucleons (protons and neutrons) in the nucleus.
The notation A
ZX is used to identify a specific nuclide. "Z" represents the atomic number, which is equal to the number of protons. "A" represents the mass number, which is equal to the number of nucleons. "X" represents the chemical symbol of the element.
Number of protons = Z
Number of electrons = Z
Number of neutrons = A - Z
The stability of a nucleus is determined by the different forces interacting within it. The electrostatic force is a relatively long-range, strong, repulsive force that acts between the positively charged protons. The nuclear force is a relatively short-range attractive force between all nucleons. The gravitational force the long range, relatively weak attraction between masses, is negligible compared to the other forces.
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Fixture checkpoint marker-0090-xray is recorded on fixture page 0090.
Atomic and Nuclear Physics DOE-HDBK-1019/1-93 CHART OF THE NUCLIDES
CHART OF THE NUCLIDES
The Chart of the Nuclides, like the Periodic Table, is a convenient format for presenting a large amount of scientific information in an organized manner.
EO 1.6 DEFINE the following terms:
a. Enriched uranium
b. Depleted uranium
Chart of the Nuclides
A tabulated chart called the Chart of the Nuclides lists the stable and unstable nuclides in addition to pertinent information about each one. Figure 3 shows a small portion of a typical chart. This chart plots a box for each individual nuclide, with the number of protons (Z) on the vertical axis and the number of neutrons (N = A - Z) on the horizontal axis.
The completely gray squares indicate stable isotopes. Those in white squares are artificially

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By consulting a complete chart, other types of isotopes can be found, such as naturally occurring radioactive types (but none are found in the region of the chart that is illustrated in Figure 3). Located in the box on the far left of each horizontal row is general information about the element. The box contains the chemical symbol of the element in addition to the average atomic weight of the naturally occurring substance and the average thermal neutron absorption cross section, which will be discussed in a later module. The known isotopes (elements with the same atomic number Z but different mass number A) of each element are listed to the right.
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CHART OF THE NUCLIDES DOE-HDBK-1019/1-93 Atomic and Nuclear Physics
Figure 3 Nuclide Chart for Atomic Numbers 1 to 6
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Atomic and Nuclear Physics DOE-HDBK-1019/1-93 CHART OF THE NUCLIDES
Information for Stable Nuclides
For the stable isotopes, in addition to the symbol and the atomic mass number, the number percentage of each isotope in the naturally occurring element is listed, as well as the thermal neutron activation cross section and the mass in atomic mass units (amu). A typical block for a stable nuclide from the Chart of the Nuclides is shown in Figure 4.
Figure 4 Stable Nuclides
Information for Unstable Nuclides
For unstable isotopes the additional information includes the half life, the mode of decay (for example, β^-, α), the total disintegration energy in MeV (million electron volts), and the mass in amu when available. A typical block for an unstable nuclide from the Chart of the Nuclides is shown in Figure 5.
Figure 5 Unstable Nuclides
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CHART OF THE NUCLIDES DOE-HDBK-1019/1-93 Atomic and Nuclear Physics
Neutron - Proton Ratios
Figure 6 shows the distribution of the stable nuclides plotted on the same axes as the Chart of the Nuclides. As the mass numbers become higher, the ratio of neutrons to protons in the nucleus becomes larger. For helium-4 (2 protons and 2 neutrons) and oxygen-16 (8 protons and 8 neutrons) this ratio is unity. For indium-115 (49 protons and 66 neutrons) the ratio of neutrons to protons has increased to 1.35, and for uranium-238 (92 protons and 146 neutrons) the neutron■to-proton
Figure 6 Neutron - Proton Plot of the Stable Nuclides
If a heavy nucleus were to split into two fragments, each fragment would form a nucleus that would have approximately the same neutron-to-proton ratio as the heavy nucleus. This high neutron-to-proton ratio places the fragments below and to the right of the stability curve displayed by Figure 6. The instability caused by this excess of neutrons is generally rectified by successive beta emissions, each of which converts a neutron to a proton and moves the nucleus toward a more stable neutron-to-proton ratio.
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■
Atomic and Nuclear Physics DOE-HDBK-1019/1-93 CHART OF THE NUCLIDES
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Natural Abundance of Isotopes
The relative abundance of an isotope in nature compared to other isotopes of the same element is relatively constant. The Chart of the Nuclides presents the relative abundance of the naturally

the atoms of an element that are of a particular isotope. Atom percent is abbreviated as a/o.
For example, if a cup of water contains 8.23 x 10 atoms of oxygen, and the isotopic abundance 24
of oxygen-18 is 0.20%, then there are 1.65 x 10 atoms of oxygen-18 in the cup. 22
The atomic weight for an element is defined as the average atomic weight of the isotopes of the element. The atomic weight for an element can be calculated by summing the products of the isotopic abundance of the isotope with the atomic mass of the isotope.
Example:
Calculate the atomic weight for the element lithium. Lithium-6 has an atom percent abundance of 7.5% and an atomic mass of 6.015122 amu. Lithium-7 has an atomic abundance of 92.5% and an atomic mass of 7.016003 amu.
Solution:
The other common measurement of isotopic abundance is weight percent (w/o). Weight percent is the percent weight of an element that is a particular isotope. For example, if a sample of material contained 100 kg of uranium that was 28 w/o uranium-235, then 28 kg of uranium-235 was present in the sample.
Enriched and Depleted Uranium
Natural uranium mined from the earth contains the isotopes uranium-238, uranium-235 and
Fixture checkpoint marker-0092-xray is recorded on fixture page 0092.
uranium-234. The majority (99.2745%) of all the atoms in natural uranium are uranium-238. Most of the remaining atoms (0.72%) are uranium-235, and a slight trace (0.0055%) are uranium-234. Although all isotopes of uranium have similar chemical properties, each of the isotopes has significantly different nuclear properties. For reasons that will be discussed in later modules, the isotope uranium-235 is usually the desired material for use in reactors.
A vast amount of equipment and energy are expended in processes that separate the isotopes of uranium (and other elements). The details of these processes are beyond the scope of this module. These processes are called enrichment processes because they selectively increase the proportion of a particular isotope. The enrichment process typically starts with feed material that has the proportion of isotopes that occur naturally. The process results in two types of
CHART OF THE NUCLIDES DOE-HDBK-1019/1-93 Atomic and Nuclear Physics
In the case of uranium, the natural uranium ore is 0.72 a/o uranium-235. The desired outcome of the enrichment process is to produce enriched uranium. Enriched uranium is defined as uranium in which the isotope uranium-235 has a concentration greater than its natural value. The enrichment process will also result in the byproduct of depleted uranium. Depleted uranium is defined as uranium in which the isotope uranium-235 has a concentration less than its natural value. Although depleted uranium is referred to as a by-product of the enrichment process, it does have uses in the nuclear field and in commercial and defense industries.
Summary
The important information in this chapter is summarized below.
Chart of the Nuclides Summary
Enriched uranium is uranium in which the isotope uranium-235 has a concentration greater than its natural value of 0.7%.
Depleted uranium is uranium in which the isotope uranium-235 has a concentration less than its natural value of 0.7%.
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Atomic and Nuclear Physics DOE-HDBK-1019/1-93 MASS DEFECT AND BINDING ENERGY
MASS DEFECT AND BINDING ENERGY
The separate laws of Conservation of Mass and Conservation of Energy are not applied strictly on the nuclear level. It is possible to convert between mass and

states that the sum of mass and energy is conserved. Mass does not magically appear and disappear at random. A decrease in mass will be accompanied by a corresponding increase in energy and vice versa.

EO 1.7 DEFINE the following terms:

a. Mass defect

b. Binding energy

EO 1.8 Given the atomic mass for a nuclide and the atomic masses of a neutron, proton, and electron, CALCULATE the mass defect and binding energy of the nuclide.

Mass Defect

Careful measurements have shown that the mass of a particular atom is always slightly less than the sum of the masses of the individual neutrons, protons, and electrons of which the atom consists. The difference between the mass of the atom and the sum of the masses of its parts is called the mass defect (Δm). The mass defect can be calculated using Equation (1-1). In calculating the mass defect it is important to use the full accuracy of mass measurements because the difference in mass is small compared to the mass of the atom. Rounding off the masses of atoms and particles to three or four significant digits prior to the calculation will result in a calculated mass defect of zero.

$\Delta m = [\, Z(mp + me) + (A-Z)mn \,] - matom$ (1-1)

where:

Δm = mass defect (amu)

mp = mass of a proton (1.007277 amu)

mn = mass of a neutron (1.008665 amu)

me = mass of an electron (0.000548597 amu)

Fixture checkpoint marker-0093-xray is recorded on fixture page 0093.

matom = mass of nuclide A

ZX (amu)

Z = atomic number (number of protons)

A = mass number (number of nucleons)

Example:

Calculate the mass defect for lithium-7. The mass of lithium-7 is 7.016003 amu.

Solution:

Δ m Z m

p

me A Z mn matom

Δ m 3 1.007826 amu 7 3 1.008665 amu 7.016003 amu

Δ m 0.0421335 amu

Binding Energy

The loss in mass, or mass defect, is due to the conversion of mass to binding energy when the nucleus is formed. Binding energy is defined as the amount of energy that must be supplied to a nucleus to completely separate its nuclear particles (nucleons). It can also be understood as the amount of energy that would be released if the nucleus was formed from the separate particles. Binding energy is the energy equivalent of the mass defect. Since the mass defect was converted to binding energy (BE) when the nucleus was formed, it is possible to calculate the binding energy using a conversion factor derived by the mass-energy relationship from Einstein's Theory of Relativity.

(c = 2.998 x 108 m/sec). The energy equivalent of 1 amu can be determined by inserting this quantity of mass into Einstein's equation and applying conversion factors.
E m c 2
1 amu
1.6606 x 10 27 kg
1 amu
2.998 x 108 m
sec
2
1 N
1
kg m
sec2
1 J
1 N m
1.4924 x 10 10 J 1 MeV
1.6022 x 10 13 J
931.5 MeV
Conversion Factors:
1 amu = 1.6606 x 10 -27 kg
1 newton = 1 kg-m/sec2
1 joule = 1 newton-meter
1 MeV = 1.6022 x 10-13 joules

Since 1 amu is equivalent to 931.5 MeV of energy, the binding energy can be calculated using Equation (1-2).
(1-2)
B.E. ∆m
931.5 MeV
1 amu
Fixture checkpoint marker-0094-xray is recorded on fixture page 0094.
Example:
Calculate the mass defect and binding energy for uranium-235. One uranium-235 atom has a mass of 235.043924 amu.
Solution:
Step 1: Calculate the mass defect using Equation (1-1).
∆m Z m
p
me A Z mn matom
∆m 92 1.007826 amu 235 92 1.008665 amu 235.043924 amu
∆m 1.91517 amu
Step 2: Use the mass defect and Equation (1-2) to calculate the binding energy.
B.E. ∆m
931.5 MeV
1 amu
1.91517 amu
931.5 MeV

1784 MeV

Energy Levels of Atoms

The electrons that circle the nucleus move in fairly well-defined orbits. Some of these electrons are more tightly bound in the atom than others. For example, only 7.38 eV is required to remove the outermost electron from a lead atom, while 88,000 eV is required to remove the innermost electron. The process of removing an electron from an atom is called ionization, and the energy required to remove the electron is called the ionization energy.

In a neutral atom (number of electrons = Z) it is possible for the electrons to be in a variety of different orbits, each with a different energy level. The state of lowest energy is the one in which the atom is normally found and is called the ground state. When the atom possesses more energy than its ground state energy, it is said to be in an excited state.

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An atom cannot stay in the excited state for an indefinite period of time. An excited atom will eventually transition to either a lower-energy excited state, or directly to its ground state, by emitting a discrete bundle of electromagnetic energy called an x-ray. The energy of the x-ray will be equal to the difference between the energy levels of the atom and will typically range from several eV to 100,000 eV in magnitude.

Energy Levels of the Nucleus

The nucleons in the nucleus of an atom, like the electrons that circle the nucleus, exist in shells that correspond to energy states. The energy shells of the nucleus are less defined and less understood than those of the electrons. There is a state of lowest energy (the ground state) and discrete possible excited states for a nucleus. Where the discrete energy states for the electrons of an atom are measured in eV or keV, the energy levels of the nucleus are considerably greater and typically measured in MeV.

A nucleus that is in the excited state will not remain at that energy level for an indefinite period. Like the electrons in an excited atom, the nucleons in an excited nucleus will transition towards their lowest energy configuration and in doing so emit a discrete bundle of electromagnetic radiation called a gamma ray (γ-ray). The only differences between x-rays and γ-rays are their energy levels and whether they are emitted from the electron shell or from the nucleus.

The ground state and the excited states of

Figure 7 Energy Level Diagram - Nickel-60

a nucleus can be depicted in a nuclear energy-level diagram. The nuclear energy-level diagram consists of a stack of horizontal bars, one bar for each of the excited states of the nucleus. The vertical distance between the bar representing an

Fixture checkpoint marker-0095-xray is recorded on fixture page 0095.

excited state and the bar representing the ground state is proportional to the energy level of the excited state with respect to the ground state. This difference in energy between the ground state and the excited state is called the excitation energy of the excited state. The ground state of a nuclide has zero excitation energy. The bars for the excited states are labeled with

the energy level diagram for nickel-60.
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Atomic and Nuclear Physics DOE-HDBK-1019/1-93 MASS DEFECT AND BINDING ENERGY
Summary
The important information in this chapter is summarized below.
Mass Defect and Binding Energy Summary
Mass defect is the difference between the mass of the atom and the sum of the masses of its constituent parts.
Binding energy is the amount of energy that must be supplied to a nucleus to completely separate its nuclear particles. Binding energy is the energy equivalent of the mass defect.
Mass defect can be calculated by using the equation below.
$\Delta m = [Z(mp + me) + (A-Z)mn] - matom$
Binding energy can be calculated by multiplying the mass defect by the factor of 931.5 MeV per amu.
Fixture checkpoint marker-0096-xray is recorded on fixture page 0096.
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MODES OF RADIOACTIVE DECAY DOE-HDBK-1019/1-93 Atomic and Nuclear Physics
MODES OF RADIOACTIVE DECAY
Most atoms found in nature are stable and do not emit particles or energy that change form over time. Some atoms, however, do not have stable nuclei. These atoms emit radiation in order to achieve a more stable configuration.
EO 2.1 DESCRIBE the following processes:
a. Alpha decay d. Electron capture
b. Beta-minus decay e. Internal conversions
c. Beta-plus decay f. Isomeric transitions
EO 2.2 Given a Chart of the Nuclides, WRITE the radioactive decay chain for a nuclide.
EO 2.3 EXPLAIN why one or more gamma rays typically accompany particle emission.
EO 2.4 Given the stability curve on the Chart of the Nuclides, DETERMINE the type of radioactive decay that the nuclides in each region of the chart will typically undergo.
Stability of Nuclei
As mass numbers become larger, the ratio of neutrons to protons in the nucleus becomes larger for the stable nuclei. Non-stable nuclei may have an excess or deficiency of neutrons and undergo a transformation process known as beta (β) decay. Non-stable nuclei can also undergo a variety of other processes such as alpha (α) or neutron (n) decay. As a result of these decay processes, the final nucleus is in a more stable or more tightly bound configuration.
Natural Radioactivity
In 1896, the French physicist Becquerel discovered that crystals of a uranium salt emitted rays that were similar to x-rays in that they were highly penetrating, could affect a photographic plate, and induced electrical conductivity in gases. Becquerel's discovery was followed in 1898 by the identification of two other radioactive elements, polonium and radium, by Pierre and Marie Curie.
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Atomic and Nuclear Physics DOE-HDBK-1019/1-93 MODES OF RADIOACTIVE DECAY
Heavy elements, such as uranium or thorium, and their unstable decay chain elements emit

at the beginning of geological time, have an extremely slow rate of decay. All naturally occurring nuclides with atomic numbers greater than 82 are radioactive.

Nuclear Decay

Whenever a nucleus can attain a more stable (i.e., more tightly bound) configuration by emitting radiation, a spontaneous disintegration process known as radioactive decay or nuclear decay may occur. In practice, this "radiation" may be electromagnetic radiation, particles, or both.

Detailed studies of radioactive decay and nuclear reaction processes have led to the formulation of useful conservation principles. The four principles of most interest in this module are discussed below.

Conservation of electric charge implies that charges are neither created nor destroyed. Single positive and negative charges may, however, neutralize each other. It is also possible for a neutral particle to produce one charge of each sign.

Conservation of mass number does not allow a net change in the number of nucleons. However, the conversion of a proton to a neutron and vice versa is allowed.

Conservation of mass and energy implies that the total of the kinetic energy and the energy equivalent of the mass in a system must be conserved in all decays and reactions. Mass can be converted to energy and energy can be converted to mass, but the sum of mass and energy must be constant.

Conservation of momentum is responsible for the distribution of the available kinetic energy among product nuclei, particles, and/or radiation. The total amount is the same before and after the reaction even though it may be distributed differently among entirely different nuclides and/or particles.

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■ ■ ■ ■ ■ ■

■ ■ ■ ■ ■

Alpha Decay (■)

Alpha decay is the emission of alpha particles (helium nuclei) which may be represented as either He or ■. When an unstable nucleus ejects an alpha particle, the atomic number is reduced by 2 4 4 2 2

and the mass number decreased by 4. An example is uranium-234 which decays by the ejection of an alpha particle accompanied by the emission of a 0.068 MeV gamma.

The combined kinetic energy of the daughter nucleus (Thorium-230) and the ■ particle is designated as KE. The sum of the KE and the gamma energy is equal to the difference in mass between the original nucleus (Uranium-234) and the final particles (equivalent to the binding energy released, since ■m = BE). The alpha particle will carry off as much as 98% of the kinetic energy and, in most cases, can be considered to carry off all the kinetic energy.

Beta Decay (■)

Beta decay is the emission of electrons of nuclear rather than orbital origin. These particles are electrons that have been expelled by excited nuclei and may have a charge of either sign.

If both energy and momentum are to be conserved, a third type of particle, the neutrino, ■, must be involved. The neutrino is associated with positive electron emission, and its antiparticle, the antineutrino, , is emitted with a negative electron. These uncharged particles have only the weakest interaction with matter, no mass, and travel at the speed of light. For all practical purposes, they pass through all materials with so few interactions that the energy they possess cannot be recovered. The neutrinos and antineutrinos are included here only because they carry a portion of

for energy and momentum to be conserved. They are normally ignored since they are not significant in the context of nuclear reactor applications.

Negative electron emission, represented as e, -1 -1■, or simply as e or ■ , effectively converts a neutro to a proton, thus increasing the atomic number by one and leaving the mass number unchanged. This is a common mode of decay for nuclei with an excess of neutrons, such as fission fragments below and to the right of the neutron-proton stability curve (refer to Figure 6). An example of a typical beta minus-decay reaction is shown below.

■ ■
■
■ ■
■ ■ ■ ■

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Figure 9 Orbital Electron Capture

Positively charged electrons (beta-plus) are known as positrons. Except for sign, they are nearly identical to their negatively charged cousins. When a positron, represented as e, ■, or simply as +1 +1 0 0
e or ■ , is ejected from the nucleus, the atomic number is decreased by one and the mass number + + remains unchanged. A proton has been converted to a neutron. An example of a typical positron (beta-plus) decay is shown below.

Electron Capture (EC, K-capture)

Nuclei having an excess of protons may capture an electron from one of the inner orbits which immediately combines with a proton in the nucleus to form a neutron. This process is called electron capture (EC). The electron is normally captured from the innermost orbit (the K-shell), and, consequently, this process is sometimes called K-capture. The following example depicts electron capture.

A neutrino is formed at the same time that the neutron is formed, and energy carried off by it serves to

Fixture checkpoint marker-0098-xray is recorded on fixture page 0098.

conserve momentum. Any energy that is available due to the atomic mass of the product being appreciably less than that of the parent will appear as gamma radiation. Also, there will always be characteristic x-rays given off when an electron from one of the higher energy shells moves in to fill the vacancy in the K-shell.

Electron capture is shown graphically in Figure 8.

Electron capture and positron emission result in the production of the same daughter product, and they exist as competing processes.

For positron emission to occur, however, the mass of the daughter product must be less than the mass of the parent by an amount equal to at least twice the mass of an electron. This mass difference

in the daughter. One item is the positron ejected from the nucleus of the parent. The other item is that the daughter product has one less orbital electron than the parent. If this requirement is not met, then orbital electron capture takes place exclusively.

■

■ ■ ■ ■

■ ■

Gamma Emission (■)

Gamma radiation is a high-energy electromagnetic radiation that originates in the nucleus. It is emitted in the form of photons, discrete bundles of energy that have both wave and particle properties. Often a daughter nuclide is left in an excited state after a radioactive parent nucleus undergoes a transformation by alpha decay, beta decay, or electron capture. The nucleus will drop to the ground state by the emission of gamma radiation.

Internal Conversion

The usual method for an excited nucleus to go from the excited state to the ground state is by emission of gamma radiation. However, in some cases the gamma ray (photon) emerges from the nucleus only to interact with one of the innermost orbital electrons and, as a result, the energy of the photon is transferred to the electron. The gamma ray is then said to have undergone internal conversion. The conversion electron is ejected from the atom with kinetic energy equal to the gamma energy minus the binding energy of the orbital electron. An orbital electron then drops to a lower energy state to fill the vacancy, and this is accompanied by the emission of characteristic x-rays.

Isomers and Isomeric Transition

Isomeric transition commonly occurs immediately after particle emission; however, the nucleus may remain in an excited state for a measurable period of time before dropping to the ground state at its own characteristic rate. A nucleus that remains in such an excited state is known as a nuclear isomer because it differs in energy and behavior from other nuclei with the same atomic number and mass number. The decay of an excited nuclear isomer to a lower energy level is called an isomeric transition. It is also possible for the excited isomer to decay by some alternate means, for example, by beta emission.

An example of gamma emission accompanying particle emission is illustrated by the decay of nitrogen-16 below.

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■

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When an unstable nucleus decays, the resulting daughter nucleus is not necessarily stable. The nucleus resulting from the decay of a parent is often itself unstable, and will undergo an additional decay. This is especially common among the larger nuclides.
It is possible to trace the steps of an unstable atom as it goes through multiple decays trying to achieve stability. The list of the original unstable nuclide, the nuclides that are involved as intermediate steps in the decay, and the final stable nuclide is known as the decay chain. One common method for stating the decay chain is to state each of the nuclides involved in the standard X format. Arrows are used between nuclides to indicate where decays occur, with the type of decay A Z
indicated above the arrow and the half-life below the arrow. The half-life for decay will be discussed in the next chapter.
Example:
Write the decay chains for rubidium-91 and actinium-215. Continue the chains until a stable
Fixture checkpoint marker-0100-xray is recorded on fixture page 0100.
nuclide or a nuclide with a half-life greater than 1 x 10 years is reached. 6
Solution:
Predicting Type of Decay
Radioactive nuclides tend to decay in a way that results in a daughter nuclide that lies closer to the line of stability. Due to this, it is possible to predict the type of decay that a nuclide will undergo based on its location relative to the line of stability on the Chart of the Nuclides.
Figure 9 illustrates the type of decay nuclides in different regions of the chart will typically undergo. Nuclides that are below and to the right of the line of stability will usually undergo ■ decay. - Nuclides that are above and to the left of the line of stability will usually undergo either ■ decay or + electron capture. Most nuclides that will undergo ■ decay are found in the upper right hand region of the chart. These are general rules that have many exceptions, especially in the region of the heavy nuclides.
MODES OF RADIOACTIVE DECAY DOE-HDBK-1019/1-93 Atomic and Nuclear Physics
Figure 9 Types of Radioactive Decay Relative to the Line of Stability
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Atomic and Nuclear Physics DOE-HDBK-1019/1-93 MODES OF RADIOACTIVE DECAY
Summary
The important information in this chapter is summarized below.
Modes of Radioactive Decay Summary
Alpha decay is the emission of an alpha particle (2 protons and 2 neutrons) from an unstable nucleus. The daughter nuclide has an atomic number 2 less than the parent nuclide and a mass number 4 less than the parent nuclide. The daughter nucleus commonly releases its excitation energy by gamma emission.
Beta-minus decay effectively converts a neutron to a proton and an electron, which is immediately ejected from the nucleus. The daughter nuclide has its atomic number increased by 1 and the same mass number compared to the parent.
Beta-plus decay effectively converts a proton to a neutron and a positron, which is immediately ejected from the nucleus. The daughter nuclide has its atomic number decreased by 1 and the same mass number compared to the parent.
In electron capture, the nucleus absorbs an electron from the innermost orbit. This electron combines with a proton to form a neutron.
Internal conversion occurs when a gamma ray, emitted by the nucleus as it goes from the excited state to the ground state, interacts with one of the innermost

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An isomeric transition is the decay of an excited nucleus to a lower-energy level by the emission of a gamma ray.
Decay chains can be found by tracing the steps an unstable atom goes through as it tries to achieve stability.
Many modes of radioactive decay result in a daughter nuclide that has an energy level above the ground state. This excitation energy is usually released immediately in the form of a gamma ray.
The type of decay that a nuclide will typically undergo can be determined by its relationship to the line of stability on the Chart of the Nuclides. Nuclides that lie below and to the right of the line of stability will typically beta minus decay. Nuclides above and to the left of the line will typically either beta plus decay or electron capture. Most alpha emitters are found in the upper, right-hand corner of the chart.
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RADIOACTIVITY
The rate at which a sample of radioactive material decays is not constant. As individual atoms of the material decay, there are fewer of those types of atoms remaining. Since the rate of decay is directly proportional to the number of atoms, the rate of decay will decrease as the number of atoms decreases.
Fixture checkpoint marker-0101-xray is recorded on fixture page 0101.
EO 2.5 DEFINE the following terms:
a. Radioactivity d. Radioactive decay constant
b. Curie e. Radioactive half-life
c. Becquerel
EO 2.6 Given the number of atoms and either the half-life or decay constant of a nuclide, CALCULATE the activity.
EO 2.7 Given the initial activity and the decay constant of a nuclide, CALCULATE the activity at any later time.
EO 2.8 CONVERT between the half-life and decay constant for a nuclide.
EO 2.9 Given the Chart of the Nuclides and the original activity, PLOT the radioactive decay curve for a nuclide on either linear or semi-log coordinates.
EO 2.10 DEFINE the following terms:
a. Radioactive equilibrium
b. Transient radioactive equilibrium
Radioactive Decay Rates
Radioactivity is the property of certain nuclides of spontaneously emitting particles or gamma radiation. The decay of radioactive nuclides occurs in a random manner, and the precise time at which a single nucleus will decay cannot be determined. However, the average behavior of a very large sample can be predicted accurately by using statistical methods. These studies have revealed that there is a certain probability that in a given time interval a certain fraction of the nuclei within a sample of a particular nuclide will decay. This probability per unit time that an atom of a nuclide will decay is known as the radioactive decay constant, λ. The units for the decay constant are inverse time such as 1/second, 1/minute, 1/hour, or 1/year. These decay constant units can also be expressed as second-1, minute-1, hour-1, and year-1.
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The activity (A) of a sample is the rate of decay of that sample. This rate of decay is usually measured in the number of disintegrations that occur per second. For a sample containing millions of atoms, the activity is the product of the decay constant and the number of atoms present in the sample.
The relationship between the activity, number of atoms, and decay constant is shown in Equation (1-3).
A = ■ N (1-3)
where:
A = Activity of the nuclide (disintegrations/second)
■ = decay constant of the nuclide (second)-1
N = Number of atoms of the nuclide in the sample
Since ■ is a constant, the activity and the number of atoms are always proportional.
Units of Measurement for Radioactivity
Two common units to measure the activity of a substance are the curie (Ci) and becquerel (Bq). A curie is a unit of measure of the rate of radioactive decay equal to 3.7 x 10 disintegrations 10 per second. This is approximately equivalent to the number of disintegrations that one gram of radium-226 will undergo in one second. A becquerel is a more fundamental unit of measure of radioactive decay that is equal to 1 disintegration per second. Currently, the curie is more widely used in the United States, but usage of the becquerel can be expected to broaden as the metric system slowly comes into wider use. The conversion between curies and becquerels is shown below.
1 curie = 3.7 x 10 becquerels 10
Variation of Radioactivity Over Time
The rate at which a given radionuclide sample decays is stated in Equation (1-3) as being equal to the product of the number of atoms and the decay constant. From this basic relationship it
Fixture checkpoint marker-0102-xray is recorded on fixture page 0102.
is possible to use calculus to derive an expression which can be used to calculate how the number of atoms present will change over time. The derivation is beyond the scope of this text, but Equation (1-4) is the useful result.
(1-4)
where:
N = number of atoms present at time t
N = number of atoms initially present o
■ = decay constant (time)-1
t = time
A Ao
e
t
A Aoe
t
A
Ao
e
t
ln A
Ao

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t
ln
A
Ao
t
1/2
ln
1
2
t
1/2
ln 2 0.693
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Since the activity and the number of atoms are always proportional, they may be used interchangeably to describe any given radionuclide population. Therefore, the following is true.
(1-5)
where:
A = activity present at time t
A = activity initially present o
= decay constant (time)
-1
t = time
Radioactive Half-Life
One of the most useful terms for estimating how quickly a nuclide will decay is the radioactive half-life. The radioactive half-life is defined as the amount of time required for the activity to decrease to one-half of its original value. A relationship between the half-life and decay constant can be developed from Equation (1-5). The half-life can be calculated by solving Equation (1-5) for the time, t, when the current activity, A, equals one-half the initial activity A .
o
First, solve Equation (1-5) for t.
If A is equal to one-half of Ao, then A/Ao is equal to one-half. Substituting this in the equation above yields an expression for t .
1/2
Fixture checkpoint marker-0103-xray is recorded on fixture page 0103.
(1-6)
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The basic features of decay of a radionuclide sample are shown by the graph in Figure 10.
Figure 10 Radioactive Decay as a Function of Time in Units of Half-Life
Assuming an initial number of atoms No, the population, and consequently, the activity may be noted to decrease by one-half of this value in a time of one half-life. Additional decreases occur so that whenever one half-life elapses, the number of atoms drops to one-half of what its value was at the beginning of that time interval. After five half-lives have elapsed, only 1/32, or 3.1%, of the original number of atoms remains. After seven half-lives, only 1/128, or 0.78%, of the atoms remains. The number of atoms existing after 5 to 7 half-lives can usually be assumed to be negligible. The Chemistry Fundamentals Handbook contains additional information on calculating the number of atoms contained within a sample.
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■

Example:
A sample of material contains 20 micrograms of californium-252.
Californium-252 has a half-life of 2.638 years.
Calculate:
(a) The number of californium-252 atoms initially present
(b) The activity of the californium-252 in curies
(c) The number of californium-252 atoms that will remain in 12 years
(d) The time it will take for the activity to reach 0.001 curies
Fixture checkpoint marker-0104-xray is recorded on fixture page 0104.
Solution:
(a) The number of atoms of californium-252 can be determined as below.
(b) First, use Equation (1-6) to calculate the decay constant.
Use this value for the decay constant in Equation (1-3) to determine the activity.
N No
e
t
4.78 x 1016 e
(0.263/yr) (12 yr)
2.04 x 1015
A Aoe
t
A
Ao
e
t
ln A
Ao
t
t
ln
A
Ao
t
ln
0.001 Ci
0.0108 Ci
0.263 year 1
t 9.05 years

(c) The number of californium atoms that will remain in 12 years can be calculated from Equation (1-4).
(d) The time that it will take for the activity to reach 0.001 Ci can be determined from Equation (1-5). First, solve Equation (1-5) for time.
Inserting the appropriate values in the right side of this equation will result in the

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Plotting Radioactive Decay
It is useful to plot the activity of a nuclide as it changes over time. Plots of this type can be used to determine when the activity will fall below a certain level. This plot is usually done showing activity on either a linear or a logarithmic scale. The decay of the activity of a single nuclide on a logarithmic scale will plot as a straight line because the decay is exponential.
t
1/2
0.693
0.693
t
1/2
0.693
7.13 seconds
0.0972 second 1
A Aoe
t
RADIOACTIVITY DOE-HDBK-1019/1-93 Atomic and Nuclear Physics
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Fixture checkpoint marker-0105-xray is recorded on fixture page 0105.
Example:
Plot the radioactive decay curve for nitrogen-16 over a period of 100 seconds. The initial activity is 142 curies and the half-life of nitrogen-16 is 7.13 seconds. Plot the curve on both linear rectangular coordinates and on a semi-log scale.
Solution:
First, use Equation (1-6) to calculate the decay constant corresponding to a half-life of 7.13 seconds.
Use the decay constant determined above to calculate the activity at various times using Equation (1-5).
Time Activity
0 seconds 142 Ci
20 seconds 20.3 Ci
40 seconds 2.91 Ci
60 seconds 0.416 Ci
80 seconds 0.0596 Ci
100 seconds 0.00853 Ci
■ ■
■ ■
■ ■
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Figure 11 Linear and Semi-log Plots of Nitrogen-16 Decay
Plotting the data points calculated above on both linear and semilog scales results in the graphs shown in Figure 11.
If a substance contains more than one radioactive nuclide, the total activity is the sum of the individual activities of each nuclide. As an example, consider a sample of material that contained 1 x 10 atoms of iron-59 that has a half-life of 44.51 days (■ = 1.80 x 10 sec), 6 -7 -1 1 x 10 atoms of manganese-54 that has a half-life of 312.2 days (■ = 2.57 x 10 sec), and 1 6 -8 -1 x 10 atoms of cobalt-60 that has a half-life of 1925 days (■ = 4.17 x 10 sec). 6 -9 -1

decay constant.

Figure 12 Combined Decay of Iron-56, Manganese-54, and Cobalt-60

Plotting the manner in which the activities of each of the three nuclides decay over time demonstrates that initially the activity of the shortest-lived nuclide (iron-59) dominates the total activity, then manganese-54 dominates. After almost all of the iron and manganese have decayed away, the only contributor to activity will be the cobalt-60. A plot of this combined decay is shown in Figure 12.

Radioactive Equilibrium

Radioactive equilibrium exists when a radioactive nuclide is decaying at the same rate at which it is being produced. Since the production rate and decay rate are equal, the number of atoms present remains constant over time.

An example of radioactive equilibrium is the concentration of sodium-24 in the coolant circulating through a sodium-cooled nuclear reactor. Assume that the sodium-24 is being produced at a rate of 1 x 10 atoms per second. If the sodium-24 were stable and did not decay, 6 the amount of sodium-24 present after some period of time could be calculated by multiplying the production rate by the amount of time. Plotting the amount of material present would result in the graph in Figure 13.

However, sodium-24 is not stable, and it decays with a half-life of 14.96 hours. If no sodium-24 is present initially and production starts at a rate of 1 x 10 atoms per second, the rate 6 of decay will initially be zero because there is no sodium-24 present to decay. The rate of decay of sodium-24 will increase as the amount of sodium-24 increases.

■
■

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■
■

Figure 13 Cumulative Production of Sodium-24 Over Time

The amount of sodium-24 present will initially increase rapidly, then it will increase at a continually decreasing rate until the rate of decay is equal to the rate of production. It is possible to calculate how much sodium-24 will be present at equilibrium by setting the production rate (R) equal to the decay rate (■ N).

where:

R = production rate (atoms/second)
■ = decay constant (second)-1
N = number of atoms

It is possible to calculate the equilibrium value for sodium-24 being produced at a rate of 1 x 10 atoms/second. 6

■ ■ ■■
■
■
■
■

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The development of the equation to calculate how the amount of sodium-24 changes over time as it approaches the equilibrium value is beyond the scope of this handbook. However, the equation is presented below.

This equation can be used to calculate the values of the amount of sodium-24 present at different times. As the time increases, the exponential term approaches zero, and the number of atoms present will approach R/■. A plot of the approach of sodium-24 to equilibrium is shown in Figure 14.

Transient Radioactive Equilibrium

Transient radioactive equilibrium occurs when the parent nuclide and the daughter nuclide decay at essentially the same rate.

For transient equilibrium to occur, the parent must have a long half-life when compared to the daughter. An example of this type of compound decay process is barium-140, which decays by beta emission to lanthanum-140, which in turn decays by beta emission to stable cerium-140.

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Figure 15 Transient Equilibrium in the Decay of Barium-140

The decay constant for barium-140 is considerably smaller than the decay constant for lanthanum-140. Remember that the rate of decay of both the parent and daughter can be represented as ■N. Although the decay constant for barium-140 is smaller, the actual rate of decay (■N) is initially larger than that of lanthanum-140 because of the great difference in their initial concentrations. As the concentration of the daughter increases, the rate of decay of the daughter will approach and eventually match the decay rate of the parent. When this occurs, they are said to be in transient equilibrium. A plot of the barium-lanthanum-cerium decay chain reaching transient equilibrium is shown in Figure 15.

Secular equilibrium occurs when the parent has an extremely long half-life. In the long decay chain for a naturally radioactive element, such as thorium-232, where all of the elements in the chain are in secular equilibrium, each of the descendants has built up to an equilibrium amount and all decay at the rate set by the original parent. The only exception is the final stable element on the end of the chain. Its number of atoms is constantly increasing.

Summary

The important information in this chapter is summarized on the following page.

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■■

■

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Radioactivity Summary

& Radioactivity is the decay of unstable atoms by the emission of particles and electromagnetic radiation.

& A curie (Ci) is a unit of radioactivity equal to 3.7 x 10 disintegrations per 10 second.

& A becquerel (Bq) is a unit of radioactivity equal to 1 disintegration per second.

& The radioactive decay constant (■) is the probability per unit time that an atom will decay.

& The radioactive half-life is the amount of time required for the activity to decrease to one-half its original value.

& The activity of a substance can be calculated from the number of atoms and the

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& The amount of activity remaining after a particular time can be calculated from the equation below.
& The relationship between the decay constant and the half-life is shown below.
& Plots of radioactive decay can be useful to describe the variation of activity over time. If decay is plotted using semi-log scale the plot results in a straight line.
& Radioactive equilibrium exists when the production rate of a material is equal to the removal rate.
& Transient radioactive equilibrium exists when the parent nuclide and the daughter nuclide decay at essentially the same rate. This occurs only when the parent has a long half-life compared to the daughter.
Atomic and Nuclear Physics DOE-HDBK-1019/1-93 NEUTRON INTERACTIONS
NEUTRON INTERACTIONS
Neutrons can cause many different types of interactions. The neutron may simply scatter off the nucleus in two different ways, or it may actually be absorbed into the nucleus. If a neutron is absorbed into the nucleus, it may result in the emission of a gamma ray or a subatomic particle, or it may cause the nucleus to fission.
EO 3.1 DESCRIBE the following scattering interactions between a neutron and a nucleus:
a. Elastic scattering
b. Inelastic scattering
EO 3.2 STATE the conservation laws that apply to an elastic collision between a neutron and a nucleus.
EO 3.3 DESCRIBE the following reactions where a neutron is absorbed in a nucleus:
a. Radiative capture
b. Particle ejection
Scattering
A neutron scattering reaction occurs when a nucleus, after having been struck by a neutron, emits a single neutron. Despite the fact that the initial and final neutrons do not need to be (and often are not) the same, the net effect of the reaction is as if the projectile neutron had merely "bounced off," or scattered from, the nucleus. The two categories of scattering reactions, elastic and inelastic scattering, are described in the following paragraphs.
Elastic Scattering
In an elastic scattering reaction between a neutron and a target nucleus, there is no energy transferred into nuclear excitation. Momentum and kinetic energy of the "system" are conserved although there is usually some transfer of kinetic energy from the neutron to the target nucleus. The target nucleus gains the amount of kinetic energy that the neutron loses.

Fixture checkpoint marker-0108-xray is recorded on fixture page 0108.
mn
vn,i mTvT,i mnvn,f mTvT,f
1
2
m v
2
1
2

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v
2
n,i
1
2
mT
v
2
T,i
1
2
mn
v
2
n,f
1
Fixture checkpoint marker-0109-xray is recorded on fixture page 0109.
2
mT
v
2
T,f
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Figure 16 Elastic Scattering
Figure 16 illustrates the process of elastic scattering of a neutron off a target nucleus. In the elastic scattering reaction, the conservation of momentum and kinetic energy is represented by the equations below.
Conservation of momentum (mv)
Conservation of kinetic energy
where:
mn = mass of the neutron
m = mass of the target nucleus T
vn,i = initial neutron velocity
vn,f = final neutron velocity
vT,i = initial target velocity
vT,f = final target velocity
Atomic and Nuclear Physics DOE-HDBK-1019/1-93 NEUTRON INTERACTIONS
Elastic scattering of neutrons by nuclei can occur in two ways. The more unusual of the two interactions is the absorption of the neutron, forming a compound nucleus, followed by the re-emission of a neutron in such a way that the total kinetic energy is conserved and the nucleus returns to its ground state. This is known as resonance elastic scattering and is very dependent upon the initial kinetic energy possessed by the neutron. Due to formation of the compound nucleus, it is also referred to as compound elastic scattering. The second, more usual method, is termed potential elastic scattering and can be understood by visualizing the neutrons and nuclei to be much like billiard balls with impenetrable surfaces. Potential scattering takes place with incident neutrons that have an energy of up to about 1 MeV. In potential scattering, the neutron does not actually touch the nucleus and a compound nucleus is not formed. Instead, the

enough to the nucleus.

Inelastic Scattering

In inelastic scattering, the incident neutron is absorbed by the target nucleus, forming a compound nucleus. The compound nucleus will then emit a neutron of lower kinetic energy which leaves the original nucleus in an excited state. The nucleus will usually, by one or more gamma emissions, emit this excess energy to reach its ground state. Figure 17 shows the process of inelastic scattering.

For the nucleus that has reached its ground state, the sum of the kinetic energy of the exit

Figure 17 Inelastic Scattering

neutron, the target nucleus, and the total gamma energy emitted is equal to the initial kinetic energy of the incident neutron.

1
0
n
238
92
U
239
92
U
239
92

Fixture checkpoint marker-0110-xray is recorded on fixture page 0110.

U
0
0
1
0
n
10
5
B
11
5
B
7
3
Li
4
2

Absorption Reactions

Most absorption reactions result in the loss of a neutron coupled with the production of a charged particle or gamma ray. When the product nucleus is radioactive, additional radiation is emitted at some later time. Radiative capture, particle ejection, and fission are all categorized as absorption reactions and are briefly described below.

In radiative capture the incident neutron enters the target nucleus forming a compound nucleus. The compound nucleus then decays to its ground state by gamma emission. An example of a radiative capture reaction is shown below.

Particle Ejection

In a particle ejection reaction the incident particle enters the target nucleus forming a compound nucleus. The newly formed compound nucleus has been excited to a high enough energy level to cause it to eject a new particle while the incident neutron remains in the nucleus. After the new particle is ejected, the remaining nucleus may or may not exist in an excited state depending upon the mass-energy balance of the reaction. An example of a particle ejection reaction is shown below.

Fission

One of the most important interactions that neutrons can cause is fission, in which the nucleus that absorbs the neutron actually splits into two similarly sized parts. Fission will be discussed in detail in the next chapter.

Atomic and Nuclear Physics DOE-HDBK-1019/1-93 NEUTRON INTERACTIONS

Summary

The important information in this chapter is summarized below.

Neutron Interactions Summary

Interactions where a neutron scatters off a target nucleus are either elastic or inelastic. In elastic scattering, kinetic energy and momentum are conserved and no energy is transferred into excitation energy of the target nucleus. In inelastic scattering, some amount of kinetic energy is transferred into excitation energy of the target nucleus.

The conservation principles that apply to an elastic collision are conservation of kinetic energy and conservation of momentum.

Radiative capture is the absorption of a neutron by the target nucleus, resulting in an excited nucleus which subsequently (typically within a small fraction of a second) releases its excitation energy in the form of a gamma ray.

Particle ejection occurs when a neutron is absorbed by a target nucleus, resulting in the formation of a compound nucleus. The compound nucleus immediately

Fixture checkpoint marker-0111-xray is recorded on fixture page 0111.

ejects a particle (for example, alpha or proton).

1
0
n
235
92
U
236
92
U
140
55
Cs
93
37
Rb 3

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0
n
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NUCLEAR FISSION
Nuclear fission is a process in which an atom splits and releases energy, fission products, and neutrons. The neutrons released by fission can, in turn, cause the fission of other atoms.
EO 4.1 EXPLAIN the fission process using the liquid drop model of a nucleus.
EO 4.2 DEFINE the following terms:
a. Excitation energy (E) exc
b. Critical energy (E) crit
EO 4.3 DEFINE the following terms:
a. Fissile material
b. Fissionable material
c. Fertile material
EO 4.4 DESCRIBE the processes of transmutation, conversion, and breeding.
EO 4.5 DESCRIBE the curve of Binding Energy per Nucleon versus mass number and give a qualitative description of the reasons for its shape.
EO 4.6 EXPLAIN why only the heaviest nuclei are easily fissioned.
EO 4.7 EXPLAIN why uranium-235 fissions with thermal neutrons and uranium-238 fissions only with fast neutrons.
Fission
In the fission reaction the incident neutron enters the heavy target nucleus, forming a compound nucleus that is excited to such a high energy level (E > E) that the nucleus "splits" exc crit (fissions) into two large fragments plus some neutrons. An example of a typical fission reaction is shown below.
A large amount of energy is released in the form of radiation and fragment kinetic energy.
Atomic and Nuclear Physics DOE-HDBK-1019/1-93 NUCLEAR FISSION
Liquid Drop Model of a Nucleus
The nucleus is held together by the attractive nuclear force between nucleons, which was discussed in a previous chapter. The characteristics of the nuclear force are listed below.
(a) very short range, with essentially no effect beyond nuclear dimensions (10-13 cm)
Fixture checkpoint marker-0112-xray is recorded on fixture page 0112.
(b) stronger than the repulsive electrostatic forces within the nucleus
(c) independent of nucleon pairing, in that the attractive forces between pairs of neutrons are no different than those between pairs of protons or a neutron and a proton
(d) saturable, that is, a nucleon can attract only a few of its nearest neighbors
One theory of fission considers the fissioning of a nucleus similar in some respects to the splitting of a liquid drop. This analogy is justifiable to some extent by the fact that a liquid drop is held together by molecular forces that tend to make the drop spherical in shape and that try to resist any deformation in the same manner as nuclear forces are assumed to hold the nucleus together. By considering the nucleus as a liquid drop, the fission process can be described.

forces are greater than the repulsive electrostatic forces between the protons within the nucleus. When an incident particle (in this instance a neutron) is absorbed by the target nucleus, a compound nucleus is formed. The compound nucleus temporarily contains all the charge and mass involved in the reaction and exists in an excited state. The excitation energy added to the compound nucleus is equal to the binding energy contributed by the incident particle plus the kinetic energy possessed by that particle. Figure 18(B) illustrates the excitation energy thus imparted to the compound nucleus, which may cause it to oscillate and become distorted. If the excitation energy is greater than a certain critical energy, the oscillations may cause the compound nucleus to become dumbbell-shaped. When this happens, the attractive nuclear forces (short-range) in the neck area are small due to saturation, while the repulsive electrostatic forces (long-range) are only slightly less than before. When the repulsive electrostatic forces exceed the attractive nuclear forces, nuclear fission occurs, as illustrated in Figure 18(C).

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Figure 18 Liquid Drop Model of Fission

Critical Energy

The measure of how far the energy level of a nucleus is above its ground state is called the excitation energy (Eexc). For fission to occur, the excitation energy must be above a particular value for that nuclide. The critical energy (Ecrit) is the minimum excitation energy required for fission to occur.

Fissile Material

A fissile material is composed of nuclides for which fission is possible with neutrons of any energy level. What is especially significant about these nuclides is their ability to be fissioned with zero kinetic energy neutrons (thermal neutrons). Thermal neutrons have very low kinetic energy levels (essentially zero) because they are roughly in equilibrium with the thermal motion of surrounding materials. Therefore, in order to be classified as fissile, a material must be capable of fissioning after absorbing a thermal neutron. Consequently, they impart essentially no kinetic energy to the reaction. Fission is possible in these materials with thermal neutrons, since the change in binding energy supplied by the neutron addition alone is high enough to exceed the critical energy. Some examples of fissile nuclides are uranium-235, uranium-233, and plutonium-239.

Fissionable Material

A fissionable material is composed of nuclides for which fission with neutrons is possible. All fissile nuclides fall into this category. However, also included are those nuclides that can be fissioned only with high energy neutrons. The change in binding energy that occurs as the result of neutron absorption results in a nuclear excitation energy level that is less than the required critical energy. Therefore, the additional excitation energy must be supplied by the kinetic energy of the incident neutron. The reason for this difference between fissile and fissionable materials is the so-called odd-even effect for nuclei. It has been observed that nuclei with even numbers of neutrons and/or protons are more stable than those with odd numbers. Therefore, adding a neutron to change a nucleus with an odd number of neutrons to a nucleus with an even number of neutrons produces an appreciably higher binding energy than adding a neutron to a nucleus already possessing an even number of neutrons. Some examples of nuclides requiring high energy neutrons to cause fission are thorium-232, uranium-238, and plutonium-240. Table 4 indicates the critical energy (Ecrit) and the binding energy change for an added neutron (BEn) to

energy must equal or exceed the critical energy (ΔBE + KE > Ecrit).
TABLE 4
Critical Energies Compared to Binding Energy of Last Neutron
Target
Nucleus
Critical Energy
Ecrit
Binding Energy of
Last Neutron BEn
BEn - Ecrit
23
9
2
0Th 7.5 MeV 5.4 MeV -2.1 MeV
23
9
8
2U 7.0 MeV 5.5 MeV -1.5 MeV
23
9
5
Fixture checkpoint marker-0114-xray is recorded on fixture page 0114.
2U 6.5 MeV 6.8 MeV +0.3 MeV
23
9
3
2U 6.0 MeV 7.0 MeV +1.0 MeV
23
9
9
4Pu 5.0 MeV 6.6 MeV +1.6 MeV
Uranium-235 fissions with thermal neutrons because the binding energy released by the absorption of a neutron is greater than the critical energy for fission; therefore uranium-235 is a fissile material. The binding energy released by uranium-238 absorbing a thermal neutron is less than the critical energy, so additional energy must be possessed by the neutron for fission to be possible. Consequently, uranium-238 is a fissionable material.
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Fertile Material
All of the neutron absorption reactions that do not result in fission lead to the production of new nuclides through the process known as transmutation. These nuclides can, in turn, be transmuted again or may undergo radioactive decay to produce still different nuclides. The nuclides that are produced by this process are referred to as transmutation products. Because several of the fissile nuclides do not exist in nature, they can only be produced by nuclear reactions (transmutation). The target nuclei for such reactions are said to be fertile. Fertile materials are materials that can undergo transmutation to become fissile materials. Figure 19 traces the transmutation mechanism by which two fertile nuclides, thorium-232 and uranium-238, produce uranium-233 and plutonium-239, respectively.

If a reactor contains fertile material in addition to its fissile fuel, some new fuel will be produced as the original fuel is burned up. This is called conversion. Reactors that are specifically designed to produce fissionable fuel are called "breeder" reactors. In such reactors, the amount of fissionable fuel produced is greater than the amount of fuel burnup. If less fuel is produced than used, the process is called conversion, and the reactor is termed a "converter."

Binding Energy Per Nucleon (BE/A)

As the number of particles in a nucleus increases, the total binding energy also increases. The rate of increase, however, is not uniform. This lack of uniformity results in a variation in the amount of binding energy associated with each nucleon within the nucleus. This variation in the binding energy per nucleon (BE/A) is easily seen when the average BE/A is plotted versus atomic mass number (A), as shown in Figure 20.

Figure 20 Binding Energy per Nucleon vs. Mass Number

Figure 20 illustrates that as the atomic mass number increases, the binding energy per nucleon decreases for A > 60. The BE/A curve reaches a maximum value of 8.79 MeV at A = 56 and decreases to about 7.6 MeV for A = 238. The general shape of the BE/A curve can be explained using the general properties of nuclear forces. The nucleus is held together by very short-range attractive forces that exist between nucleons. On the other hand, the nucleus is being forced apart by long range repulsive electrostatic (coulomb) forces that exist between all the protons in the nucleus.

As the atomic number and the atomic mass number increase, the repulsive electrostatic forces within the nucleus increase due to the greater number of protons in the heavy elements. To overcome this increased repulsion, the proportion of neutrons in the nucleus must increase to maintain stability. This increase in the neutron-to-proton ratio only partially compensates for the growing proton-proton repulsive force in the heavier, naturally occurring elements. Because the

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repulsive forces are increasing, less energy must be supplied, on the average, to remove a nucleon from the nucleus. The BE/A has decreased. The BE/A of a nucleus is an indication of its degree of stability. Generally, the more stable nuclides have higher BE/A than the less stable ones. The increase in the BE/A as the atomic mass number decreases from 260 to 60 is the primary reason for the energy liberation in the fission process. In addition, the increase in the BE/A as the atomic mass number increases from 1 to 60 is the reason for the energy liberation in the fusion process, which is the opposite reaction of fission.

The heaviest nuclei require only a small distortion from a spherical shape (small energy addition) for the relatively large coulomb forces forcing the two halves of the nucleus apart to overcome the attractive nuclear forces holding the two halves together. Consequently, the heaviest nuclei are easily fissionable compared to lighter nuclei.

Summary

The important information in this chapter is summarized on the following page.

Nuclear Fission Summary

The fission process can be explained using the liquid drop model of a nucleus.
In the ground state the nucleus is nearly spherical in shape. After the absorption of a neutron, the nucleus will be in an excited state and start to oscillate and

dumbbell, the repulsive electrostatic forces will overcome the short-range attractive nuclear forces, and the nucleus will split in two.
Excitation energy is the amount of energy a nucleus has above its ground state. Critical energy is the minimum excitation energy that a nucleus must have before it can fission.
Fissile material is material for which fission is possible with neutrons that have zero kinetic energy. Fissionable material is material for which fission caused by neutron absorption is possible provided the kinetic energy added with the binding energy is greater than the critical energy. Fertile material is material that can undergo transmutation to become fissile material.
Transmutation is the process of neutron absorption and subsequent decay, which changes one nuclide to another nuclide. Conversion is the process of transmuting fertile material into fissile material in a reactor, where the amount of fissile material produced is less than the amount of fissile material consumed. Breeding is the same as conversion, except the amount of fissile material produced is more than the amount of fissile material consumed.
The curve of binding energy per nucleon increases quickly through the light nuclides and reaches a maximum at a mass number of about 56. The curve decreases slowly for mass numbers greater than 60.
The heaviest nuclei are easily fissionable because they require only a small distortion from the spherical shape to allow the coulomb forces to overcoming the attractive nuclear force, forcing the two halves of the nucleus apart.
Uranium-235 fissions with thermal neutrons because the binding energy released by the absorption of a neutron is greater than the critical energy for fission. The binding energy released by uranium-238 absorbing a neutron is less than the critical energy, so additional kinetic energy must be possessed by the neutron for fission to be possible.
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1
0
n
235
92
U
236
Fixture checkpoint marker-0116-xray is recorded on fixture page 0116.
92
U
140
55
Cs
93
37
Rb 3
1
0
n
ENERGY RELEASE FROM FISSION DOE-HDBK-1019/1-93 Atomic and Nuclear Physics

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ENERGY RELEASE FROM FISSION
Fission of heavy nuclides converts a small amount of mass into an enormous amount of energy. The amount of energy released by fission can be determined based on either the change in mass that occurs during the reaction or by the difference in binding energy per nucleon between the fissile nuclide and the fission products.
EO 4.8 CHARACTERIZE the fission products in terms of mass groupings and radioactivity.
EO 4.9 Given the nuclides involved and their masses, CALCULATE the energy released from fission.
EO 4.10 Given the curve of Binding Energy per nucleon versus mass number, CALCULATE the energy released from fission.
Fixture checkpoint marker-0117-xray is recorded on fixture page 0117.
Calculation of Fission Energy
Nuclear fission results in the release of enormous quantities of energy. It is necessary to be able to calculate the amount of energy that will be produced. The logical manner in which to pursue this is to first investigate a typical fission reaction such as the one listed below.
It can be seen that when the compound nucleus splits, it breaks into two fission fragments, rubidium-93, cesium-140, and some neutrons. Both fission products then decay by multiple - emissions as a result of the high neutron-to-proton ratio possessed by these nuclides.
In most cases, the resultant fission fragments have masses that vary widely. Figure 21 gives the percent yield for atomic mass numbers. The most probable pair of fission fragments for the thermal fission of the fuel uranium-235 have masses of about 95 and 140. Note that the vertical axis of the fission yield curve is on a logarithmic scale. Therefore, the formation of fission fragments of mass numbers of about 95 and 140 is highly likely.
Atomic and Nuclear Physics DOE-HDBK-1019/1-93 ENERGY RELEASE FROM FISSION
Figure 21 Uranium-235 Fission Yield vs.
Mass Number
Referring now to the binding energy per nucleon curve (Figure 20), we can estimate the amount of energy released by our "typical" fission by plotting this reaction on the curve and calculating the change in binding energy (ΔBE) between the reactants on the left-hand side of the fission equation and the products on the right-hand side. Plotting the reactant and product nuclides on the curve shows that the total binding energy of the system after fission is greater than the total binding energy of the system before fission. When there is an increase in the total binding energy of a system, the system has become more stable by releasing an amount of energy equal to the increase in total binding energy of the system. Therefore, in the fission process, the energy liberated is equal to the increase in the total binding energy of the system.
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Figure 22 graphically depicts that the binding energy per nucleon for the products (C,
Figure 22 Change in Binding Energy for Typical Fission
rubidium-93 and B, cesium-140) is greater than that for the reactant (A, uranium-235). The total binding energy for a nucleus can be found by multiplying the binding energy per nucleon by the number of nucleons.
TABLE 5
Binding Energies Calculated from
Binding Energy per Nucleon Curve
Nuclide B.E. per Nucleon

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Mass Number
(A)
Binding Energy
(BE/A) x (A)
9
3
3
7Rb 8.7 MeV 93 809 MeV
14
5
0
5Cs 8.4 MeV 140 1176 MeV
23
9
5
2U 7.6 MeV 235 1786 MeV
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BE BEproducts BEreactants
BERb 93 BECs 140 BEU 235
Fixture checkpoint marker-0118-xray is recorded on fixture page 0118.
809 MeV 1176 MeV 1786 MeV
199 MeV
1
0
n
235
92
U
236
92
U
140
55
Cs
93
37
Rb 3
1
0
n
Mass difference Mass of Reactants Mass of Products
236.052589 amu 235.85208 amu
0.200509 amu
EInst 0.020059 amu
931.5 MeV
amu
186.8 MeV
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The energy released will be equivalent to the difference in binding energy (BE) between the reactants and the products.
The energy liberation during the fission process can also be explained from the standpoint of the conservation of mass-energy. During the fission process, there is a decrease in the mass of the system. There must, therefore, be energy liberated equal to the energy equivalent of the mass lost in the process. This method is more accurate than the previously illustrated method and is used when actually calculating the energy liberated during the fission process.
Again, referring to the "typical" fission reaction.
E , the instantaneous energy, is the energy released immediately after the fission process. It Inst is equal to the energy equivalent of the mass lost in the fission process. It can be calculated as shown below.
Mass of the Reactants Mass of the Products
U 235.043924 amu Rb 92.91699 amu 235
92
93
37
n 1.008665 amu Cs 139.90910 amu 1
0
140
55
3 (n) 3.02599 amu 1
0
236.052589 amu 235.85208 amu
This mass difference can be converted to an energy equivalent.
93
37
Fixture checkpoint marker-0119-xray is recorded on fixture page 0119.
Rb
93
38
Sr
93
39
Y
93
40
Zr
93
41
Nb
140
55
Cs
140
56
Ba
140
57

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140
58
Ce
EDecay mRb 93 mNb 93 4 melectron
931.5MeV
amu
92.91699 amu 92.90638 amu 4 0.0005486 amu
931.5 MeV
amu
0.008416 amu
931.5 MeV
amu
7.84 MeV
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The total energy released per fission will vary from the fission to the next depending on what fission products are formed, but the average total energy released per fission of uranium-235 with a thermal neutron is 200 MeV.
As illustrated in the preceding example, the majority of the energy liberated in the fission process is released immediately after the fission occurs and appears as the kinetic energy of the fission fragments, kinetic energy of the fission neutrons, and instantaneous gamma rays. The remaining energy is released over a period of time after the fission occurs and appears as kinetic energy of the beta, neutrino, and decay gamma rays.
Estimation of Decay Energy
In addition to this instantaneous energy release during the actual fission reaction, there is additional energy released when the fission fragments decay by emission. This additional
-
energy is called decay energy, EDecay. The decay chains for rubidium-93 and cesium-140 are shown below.
The energy released during the decay for each chain will be equivalent to the mass difference between the original fission product and the sum of the final stable nuclide and the beta particles emitted.
The energy released in the decay chain of rubidium-93 is calculated below.
Fixture checkpoint marker-0120-xray is recorded on fixture page 0120.
EDecay mRb 93 mNb 93 3 melectron
931.5MeV
amu
139.90910 amu 139.90543 amu 3 0.0005486 amu
931.5 MeV
amu
0.000202 amu
931.5 MeV
amu
1.89 MeV
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The energy released in the decay chain of cesium-140 is calculated below.
The total decay energy is the sum of the energies of the two chains, or 9.73 MeV.

The average energy distribution for the energy released per fission with a thermal neutron in uranium-235 is shown in Tables 6 and 7.

TABLE 6

Instantaneous Energy from Fission

Kinetic Energy of Fission Products 167 Mev

Energy of Fission Neutrons 5 MeV

Instantaneous Gamma-ray Energy 5 MeV

Capture Gamma-ray Energy 10 MeV

Total Instantaneous Energy 187 MeV

TABLE 7

Fixture checkpoint marker-0121-xray is recorded on fixture page 0121.

Delayed Energy from Fission

Beta Particles From Fission Products 7 Mev

Gamma-rays from Fission Products 6 MeV

Neutrinos 10 MeV

Total Delayed Energy 23 MeV

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Because the 10 MeV of neutrino energy shown in Table 7 is not absorbed in the reactor, the average value of 200 MeV per fission is still accurate. Note in Table 6 that some fission neutrons undergo radiative capture and the resultant gamma ray emission provides an additional 10 MeV of instantaneous energy, which contributes to the total of 187 MeV instantaneous energy.

All of the energy released, with the exception of the neutrino energy, is ultimately transformed into heat through a number of processes. The fission fragments, with their high positive charge and kinetic energy, cause ionization directly as they rip orbital electrons from the surrounding atoms. In this ionization process, kinetic energy is transferred to the surrounding atoms of the fuel material, resulting in an increase in temperature. The beta particles and gamma rays also give up their energy through ionization, and the fission neutrons interact and lose their energy through elastic scattering. Of the 200 MeV released per fission, about seven percent (13 MeV) is released at some time after the instant of fission. When a reactor is shut down, fissions essentially cease, but energy is still being released from the decay of fission products. The heat produced by this decay energy is referred to as "decay heat." Although decay energy represents about seven percent of reactor heat production during reactor operation, once the reactor is shut down the decay heat production drops off quickly to a small fraction of its value while operating. The decay heat produced is significant, however, and systems must be provided to keep the reactor cool even after shutdown.

Summary

The important information in this chapter is summarized below.

Energy Release From Fission Summary

Fission products have some general characteristics in common.

They generally decay by β- emission.

The most common mass numbers are grouped near 95 and 140.

The energy released by fission can be calculated based on the difference in mass between the masses of the reactants before fission and the fission fragments and fission neutrons after fission.

Another method to determine the energy released by fission is based on the change in binding energy per nucleon between the fissile nuclide and the fission products.

INTERACTION OF RADIATION WITH MATTER
Different types of radiation interact with matter in widely different ways. A large, massive, charged alpha particle cannot penetrate a piece of paper and even has a limited range in dry air. A neutrino, at the other extreme, has a low probability of interacting with any matter, even if it passed through the diameter of the earth.
EO 5.1 DESCRIBE interactions of the following with matter:
a. Alpha particle c. Positron
b. Beta particle d. Neutron
EO 5.2 DESCRIBE the following ways that gamma radiation interacts with matter:
a. Photoelectric effect
b. Compton scattering
c. Pair production
Interaction of Radiation With Matter
Fixture checkpoint marker-0122-xray is recorded on fixture page 0122.
Radiation can be classified into two general groups, charged and uncharged; therefore, it may be expected that interactions with matter fall into two general types. Charged particles directly ionize the media through which they pass, while uncharged particles and photons can cause ionization only indirectly or by secondary radiation.
A moving charged particle has an electrical field surrounding it, which interacts with the atomic structure of the medium through which it is passing. This interaction decelerates the particle and accelerates electrons in the atoms of the medium. The accelerated electrons may acquire enough energy to escape from the parent atom. This process, whereby radiation "strips" off orbital electrons, is called ionization. Uncharged moving particles have no electrical field, so they can only lose energy and cause ionization by such means as collisions or scattering. A photon can lose energy by the photoelectric effect, Compton effect, or pair production.

$$I \propto m z^2 K.E.$$

Because ionizing radiation creates ions in pairs, the intensity of ionization or the specific ionization is defined as the number of ion-pairs formed per centimeter of travel in a given material. The amount of ionization produced by a charged particle per unit path length, which is a measure of its ionizing power, is roughly proportional to the particle's mass and the square of its charge as illustrated in the equation below.
where:
I is the ionizing power
m is the mass of the particle
z is the number of unit charges it carries
K.E. is its kinetic energy
Since m for an alpha particle is about 7300 times as large as m for a beta particle, and z is twice

of the same energy. This phenomenon occurs because the larger alpha particle moves slower for a given energy and thus acts on a given electron for a longer time.

Alpha Radiation

Alpha radiation is normally produced from the radioactive decay of heavy nuclides and from certain nuclear reactions. The alpha particle consists of 2 neutrons and 2 protons, so it is essentially the same as the nucleus of a helium atom. Because it has no electrons, the alpha particle has a charge of +2. This positive charge causes the alpha particle to strip electrons from the orbits of atoms in its vicinity. As the alpha particle passes through material, it removes electrons from the orbits of atoms it passes near. Energy is required to remove electrons and the energy of the alpha particle is reduced by each reaction. Eventually the particle will expend its kinetic energy, gain 2 electrons in orbit, and become a helium atom. Because of its strong positive charge and large mass, the alpha particle deposits a large amount of energy in a short distance of travel. This rapid, large deposition of energy limits the penetration of alpha particles. The most energetic alpha particles are stopped by a few centimeters of air or a sheet of paper.

Beta-Minus Radiation

A beta-minus particle is an electron that has been ejected at a high velocity from an unstable nucleus. An electron has a small mass and an electrical charge of -1. Beta particles cause ionization by displacing electrons from atom orbits. The ionization occurs from collisions with orbiting electrons. Each collision removes kinetic energy from the beta particle, causing it to slow down. Eventually the beta particle will be slowed enough to allow it to be captured as an

2 electrons 0.000549 amu

electron

931.5 MeV

Fixture checkpoint marker-0123-xray is recorded on fixture page 0123.

amu

1.02 MeV

orbiting electron in an atom. Although more penetrating than the alpha, the beta is relatively easy to stop and has a low power of penetration. Even the most energetic beta radiation can be stopped by a few millimeters of metal.

Positron Radiation

Positively charged electrons are called positrons. Except for the positive charge, they are identical to beta-minus particles and interact with matter in a similar manner. Positrons are very short-lived, however, and quickly are annihilated by interaction with a negatively charged electron, producing two gammas with a combined energy (calculated below) equal to the rest mass of the positive and negative electrons.

Neutron Radiation

Neutrons have no electrical charge. They have nearly the same mass as a proton (a hydrogen atom nucleus). A neutron has hundreds of times more mass than an electron, but 1/4 the mass of an alpha particle. The source of neutrons is primarily nuclear reactions, such as fission, but they may also be produced from the decay of radioactive nuclides. Because of its lack of charge, the neutron is difficult to stop and has a high penetrating power.

Neutrons are attenuated (reduced in energy and numbers) by three major interactions, elastic scatter, inelastic scatter, and absorption. In elastic scatter, a neutron collides with a nucleus and bounces off. This reaction transmits some of the kinetic energy of the neutron to the nucleus

(motion). This process is sometimes referred to as "the billiard ball effect."
As the mass of the nucleus approaches the mass of the neutron, this reaction becomes more effective in slowing the neutron. Hydrogenous material attenuates neutrons most effectively.
In the inelastic scatter reaction, the same neutron/nucleus collision occurs as in elastic scatter. However, in this reaction, the nucleus receives some internal energy as well as kinetic energy. This slows the neutron, but leaves the nucleus in an excited state. When the nucleus decays to its original energy level, it normally emits a gamma ray.
In the absorption reaction, the neutron is actually absorbed into the nucleus of an atom. The neutron is captured, but the atom is left in an excited state. If the nucleus emits one or more gamma rays to reach a stable level, the process is called radiative capture. This reaction occurs at most neutron energy levels, but is more probable at lower energy levels.
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Gamma Radiation
Gamma radiation is electromagnetic radiation. It is commonly referred to as a gamma ray and is very similar to an x-ray. The difference is that gamma rays are emitted from the nucleus of an atom, and x-rays are produced by orbiting electrons. The x-ray is produced when orbiting electrons move to a lower energy orbit or when fast-moving electrons approaching an atom are deflected and decelerated as they react with the atom's electrical field (called Bremsstrahlung). The gamma ray is produced by the decay of excited nuclei and by nuclear reactions. Because the gamma ray has no mass and no charge, it is difficult to stop and has a very high penetrating power. A small fraction of the original gamma stream will pass through several feet of concrete or several meters of water.
There are three methods of attenuating gamma rays. The first method is referred to as the photo-electric effect. When a low energy gamma strikes an atom, the total energy of the gamma is expended in ejecting an electron from orbit. The result is ionization of the atom and expulsion of a high energy electron. This reaction is most predominant with low energy gammas interacting in materials with high atomic weight and rarely occurs with gammas having an energy above 1 MeV. Annihilation of the gamma results. Any gamma energy in excess of the binding energy of the electron is carried off by the electron in the form of kinetic energy.
The second method of attenuation of gammas is called Compton scattering. The gamma interacts
Fixture checkpoint marker-0124-xray is recorded on fixture page 0124.
with an orbital or free electron; however, in this case, the photon loses only a fraction of its energy. The actual energy loss depending on the scattering angle of the gamma. The gamma continues on at lower energy, and the energy difference is absorbed by the electron. This reaction becomes important for gamma energies of about 0.1 MeV and higher.
At higher energy levels, a third method of attenuation is predominant. This method is pair-production. When a high energy gamma passes close enough to a heavy nucleus, the gamma completely disappears, and an electron and a positron are formed. For this reaction to take place, the original gamma must have at least 1.02 MeV energy. Any energy greater than 1.02 MeV becomes kinetic energy shared between the electron and positron. The probability of pair■production incre
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Atomic and Nuclear Physics INTERACTION OF RADIATION WITH MATTER
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Interaction of Radiation with Matter Summary
An alpha particle deposits a large amount of energy in a short distance of travel due to its large mass and charge.

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causing ionization by displacing the electrons. The beta particle loses energy with each interaction. After the beta particle loses enough energy, it is captured in the orbital shells of an atom.
Positrons interact with matter much the same way as beta minus particles. After the positron has lost most of its energy by ionizing atoms, it is annihilated by interaction with an electron. The electron-positron pair disappear and are replaced by two gammas, each with the energy equivalent of the mass of an electron (0.51 MeV).
Neutrons interact with matter by elastic scattering, inelastic scattering, or absorption.
Photoelectric effect is where a gamma interacts with an electron orbiting an atom. The entire energy of the gamma is transferred to the electron, and the electron is ejected from its orbit.
In Compton scattering a gamma interacts with an orbital electron, but only part of the gamma energy is transferred to the electron. The electron is ejected from its orbit, and the gamma is scattered off at a lower energy.
In pair-production, a gamma interacts with the electric field of a nucleus and is converted into an electron-positron pair. The gamma must have an energy greater than 1.02 MeV for this to occur.
Summary
The important information in this chapter is summarized below.
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Department of Energy
Fundamentals Handbook
NUCLEAR PHYSICS
AND REACTOR THEORY
Module 2
Reactor Theory (Neutron Characteristics)
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OBJECTIVES DOE-HDBK-1019/1-93 Reactor Theory (Neutron Characteristics)
TERMINAL OBJECTIVE
1.0 Without references, EXPLAIN how neutron sources produce neutrons.
ENABLING OBJECTIVES
1.1 DEFINE the following terms:
a. Intrinsic neutron source
b. Installed neutron source
1.2 LIST three examples of reactions that produce neutrons in intrinsic neutron sources.
1.3 LIST three examples of reactions that produce neutrons in installed neutron sources.
Fixture checkpoint marker-0127-xray is recorded on fixture page 0127.

Reactor Theory (Neutron Characteristics) DOE-HDBK-1019/1-93 OBJECTIVES
TERMINAL OBJECTIVE
2.0 Given the necessary information for calculations, EXPLAIN basic concepts in reactor physics and perform calculations.
ENABLING OBJECTIVES
2.1 DEFINE the following terms:
a. Atom density d. Barn
b. Neutron flux e. Macroscopic cross section
c. Microscopic cross section f. Mean free path
2.2 EXPRESS macroscopic cross section in terms of microscopic cross section.
2.3 DESCRIBE how the absorption cross section of typical nuclides varies with neutron energy at energies below the resonance absorption region.
2.4 DESCRIBE the cause of resonance absorption in terms of nuclear energy levels.
2.5 DESCRIBE the energy dependence of resonance absorption peaks for typical light and heavy nuclei.
2.6 EXPRESS mean free path in terms of macroscopic cross section.
2.7 Given the number densities (or total density and component fractions) and microscopic cross sections of components, CALCULATE the macroscopic cross section for a mixture.
2.8 CALCULATE a macroscopic cross section given a material density, atomic mass, and microscopic cross section.
2.9 EXPLAIN neutron shadowing or self-shielding.
2.10 Given the neutron flux and macroscopic cross section, CALCULATE the reaction rate.
2.11 DESCRIBE the relationship between neutron flux and reactor power.

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ENABLING OBJECTIVES (Cont.)
2.12 DEFINE the following concepts:
a. Thermalization d. Average logarithmic energy decrement
b. Moderator e. Macroscopic slowing down power
c. Moderating ratio
2.13 LIST three desirable characteristics of a moderator.
2.14 Given an average fractional energy loss per collision, CALCULATE the energy loss after a specified number of collisions.
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Reactor Theory (Neutron Characteristics) DOE-HDBK-1019/1-93 OBJECTIVES
TERMINAL OBJECTIVE
3.0 Without references, EXPLAIN the production process and effects on fission of prompt and delayed neutrons.
ENABLING OBJECTIVES
3.1 STATE the origin of prompt neutrons and delayed neutrons.
3.2 STATE the approximate fraction of neutrons that are born as delayed neutrons from the fission of the following nuclear fuels:
a. Uranium-235
b. Plutonium-239
3.3 EXPLAIN the mechanism for production of delayed neutrons.
3.4 EXPLAIN prompt and delayed neutron generation times.
3.5 Given prompt and delayed neutron generation times and delayed neutron fraction, CALCULATE the average generation time.
3.6 EXPLAIN the effect of delayed neutrons on reactor control.
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OBJECTIVES DOE-HDBK-1019/1-93 Reactor Theory (Neutron Characteristics)
TERMINAL OBJECTIVE
4.0 Without references, DESCRIBE the neutron energy spectrum for the type of reactor presented in this module.
Fixture checkpoint marker-0128-xray is recorded on fixture page 0128.
ENABLING OBJECTIVES
4.1 STATE the average energy at which prompt neutrons are produced.
4.2 DESCRIBE the neutron energy spectrum in the following reactors:
a. Fast reactor
b. Thermal reactor
4.3 EXPLAIN the reason for the particular shape of the fast, intermediate, and slow energy regions of the neutron flux spectrum for a thermal reactor.
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Reactor Theory (Neutron Characteristics) DOE-HDBK-1019/1-93 NEUTRON SOURCES
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NEUTRON SOURCES
Neutrons from a variety of sources are always present in a reactor core. This is true even when the reactor is shut down. Some of these neutrons are produced by naturally occurring (intrinsic) neutron sources, while others may be the result of fabricated (installed) neutron sources that are incorporated into the design of the reactor. The neutrons produced by sources other than neutron-induced fission are often grouped together and classified as source neutrons.
EO 1.1 DEFINE the following terms:

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b. Installed neutron source
EO 1.2 LIST three examples of reactions that produce neutrons in intrinsic neutron sources.
EO 1.3 LIST three examples of reactions that produce neutrons in installed neutron sources.
Neutron Sources
In addition to neutron-induced fission, neutrons are produced by other reactions. The neutrons produced by reactions other than neutron-induced fission are called source neutrons. Source neutrons are important because they ensure that the neutron population remains high enough to allow a visible indication of neutron level on the most sensitive monitoring instruments while the reactor is shutdown and during the startup sequence. This verifies instrument operability and allows monitoring of neutron population changes. Source neutrons can be classified as either intrinsic or installed neutron sources.
Intrinsic Neutron Sources
Some neutrons will be produced in the materials present in the reactor due to a variety of unavoidable reactions that occur because of the nature of these materials. Intrinsic neutron sources are those neutron-producing reactions that always occur in reactor materials.
■ ■ ■ ■
■ ■ ■
■ ■
NEUTRON SOURCES DOE-HDBK-1019/1-93 Reactor Theory (Neutron Characteristics)
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A limited number of neutrons will always be present, even in a reactor core that has never been operated, due to spontaneous fission of some heavy nuclides that are present in the fuel. Uranium■238, ur of the rate at which different heavy nuclides produce neutrons by spontaneous fission.
Californium-252 is not an intrinsic neutron source, but will be discussed in the section on installed neutron sources.
TABLE 1
Neutron Production by Spontaneous Fission
Nuclide T (Fission) T (1/2 1/2 ■-decay) neutrons/sec/gram
U 1.8 x 10 years 6.8 x 10 years 8.0 x 10 235
92
17 8 -4
U 8.0 x 10 years 4.5 x 10 years 1.6 x 10 238
92
15 9 -2
Fixture checkpoint marker-0129-xray is recorded on fixture page 0129.
Pu 5.5 x 10 years 2.4 x 10 years 3.0 x 10 239
94
5 4 -2
Pu 1.2 x 10 years 6.6 x 10 years 1.0 x 10 240
94
11 3 3
Cf 66.0 years 2.65 years 2.3 x 10 252
98
12
Another intrinsic neutron source is a reaction involving natural boron and fuel. In some reactors, natural boron is loaded into the reactor core as a neutron absorber to improve reactor control or

particle emitted by the radioactive decay of heavy nuclides in the fuel to yield a neutron as shown below.

The boron-11 must be mixed with, or in very close proximity to, the fuel for this reaction because of the short path length of the alpha particle. For a reactor core with this configuration, this (■,n) reaction is an important source of neutrons for reactor startup.

In a reactor that has been operated, another source of neutrons becomes significant. Neutrons may be produced by the interaction of a gamma ray and a deuterium nucleus. This reaction is commonly referred to as a photoneutron reaction because it is initiated by electromagnetic radiation and results in the production of a neutron. The photoneutron reaction is shown below.

■ ■ ■
■ ■
■ ■ ■
■ ■

Fixture checkpoint marker-0130-xray is recorded on fixture page 0130.

There is an abundant supply of high energy gammas in a reactor that has been operated because many of the fission products are gamma emitters. All water-cooled reactors have some deuterium present in the coolant in the reactor core because a small fraction of natural hydrogen is the isotope deuterium. The atom percentage of deuterium in the water ranges from close to the naturally occurring value (0.015%) for light water reactors to above 90% deuterium for heavy water reactors. Therefore, the required conditions for production of photoneutrons exist.

The supply of gamma rays decreases with time after shutdown as the gamma emitters decay; therefore, the photoneutron production rate also decreases. In a few particular reactors, additional D O (heavy water) may be added to the reactor to increase the production of 2 photoneutrons following a long shutdown period.

Installed Neutron Sources

Because intrinsic neutron sources can be relatively weak or dependent upon the recent power history of the reactor, many reactors have artificial sources of neutrons installed. These neutron sources ensure that shutdown neutron levels are high enough to be detected by the nuclear instruments at all times. This provides a true picture of reactor conditions and any change in these conditions. An installed neutron source is an assembly placed in or near the reactor for the sole purpose of producing source neutrons.

One strong source of neutrons is the artificial nuclide californium-252, which emits neutrons at the rate of about 2 x 10 neutrons per second per gram as the result of spontaneous fission. 12 Important drawbacks for some applications may be its high cost and its short half-life (2.65 years).

Many installed neutron sources use the (■,n) reaction with beryllium. These sources are composed of a mixture of metallic beryllium (100% beryllium-9) with a small quantity of an alpha particle emitter, such as a compound of radium, polonium, or plutonium. The reaction that occurs is shown below.

The beryllium is intimately (homogeneously) mixed with the alpha emitter and is usually enclosed in a stainless steel capsule.

Another type of installed neutron source that is widely used is a photoneutron source that employs the (■,n) reaction with beryllium. Beryllium is used for photoneutron sources because its stable isotope beryllium-9 has a weakly attached last neutron with a binding energy of only 1.66 MeV. Thus, a gamma ray with greater energy than 1.66 MeV can cause neutrons to be ejected by the (■,n) reaction as shown below.

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Neutron Sources Summary
& Intrinsic neutron sources are sources of neutrons from materials that are in the reactor for other purposes such as fuel, burnable poison, or moderator.
& Installed neutron sources are materials or components placed in the reactor specifically for the purpose of producing source neutrons.
& Examples of intrinsic neutron sources are listed below.
Spontaneous fission of heavy nuclides in fuel, such as uranium-238, uranium-235, and plutonium-239, results in fission fragments and free neutrons.
Boron-11 mixed with the fuel undergoes an alpha-neutron reaction and becomes nitrogen-14.
Deuterium present in the reactor coolant undergoes a gamma■neutron reaction and becomes hydrogen-1.
& Examples of installed neutron sources are listed below.
Spontaneous fission of californium-252 results in fission fragments and free neutrons.
Beryllium-9 undergoes an alpha-neutron reaction (alpha from the decay of plutonium, polonium, or radium) and becomes carbon-12.
Beryllium-9 undergoes a gamma-neutron reaction (high energy gamma from decay of antimony-124) and becomes beryllium-8.
Fixture checkpoint marker-0131-xray is recorded on fixture page 0131.
Many startup sources of this type use antimony and beryllium because after activation with neutrons the radioactive antimony becomes an emitter of high energy gammas. The photoneutron sources of this type are constructed somewhat differently from the (■,n) types. One design incorporates a capsule of irradiated antimony enclosed in a beryllium sleeve. The entire assembly is then encased in a stainless steel cladding. A large reactor may have several neutron sources of this type installed within the core.
Summary
The important information in this chapter is summarized below.
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Reactor Theory (Neutron Characteristics) NUCLEAR CROSS SECTIONS AND NEUTRON FLUX
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NUCLEAR CROSS SECTIONS AND NEUTRON FLUX
To determine the frequency of neutron interactions, it is necessary to describe the availability of neutrons to cause interaction and the probability of a neutron interacting with material. The availability of neutrons and the probability of interaction are quantified by the neutron flux and nuclear cross section.
EO 2.1 DEFINE the following terms:
a. Atom density d. Barn
b. Neutron flux e. Macroscopic cross section
c. Microscopic cross section f. Mean free path
EO 2.2 EXPRESS macroscopic cross section in terms of microscopic cross section.
EO 2.3 DESCRIBE how the absorption cross section of typical nuclides varies with neutron energy at energies below the resonance absorption region.
EO 2.4 DESCRIBE the cause of resonance absorption in terms of nuclear energy levels.

for typical light and heavy nuclei.
EO 2.6 EXPRESS mean free path in terms of macroscopic cross section.
EO 2.7 Given the number densities (or total density and component fractions) and microscopic cross sections of components, CALCULATE the macroscopic cross section for a mixture.
EO 2.8 CALCULATE a macroscopic cross section given a material density, atomic mass, and microscopic cross section.
EO 2.9 EXPLAIN neutron shadowing or self-shielding.
'

Introduction

Fission neutrons are born with an average energy of about 2 MeV. These fast neutrons interact with the reactor core materials in various absorption and scattering reactions. Collisions that result in scattering are useful in slowing neutrons to thermal energies. Thermal neutrons may be absorbed by fissile nuclei to produce more fissions or be absorbed in fertile material for conversion to fissionable fuel. Absorption of neutrons in structural components, coolant, and other non-fuel material results in the removal of neutrons without fulfilling any useful purpose. To safely and efficiently operate a nuclear reactor it is necessary to predict the probability that a particular absorption or scattering reaction will occur. Once these probabilities are known, if the availability of neutrons can be determined, then the rate at which these nuclear reactions take place can be predicted.

Atom Density

One important property of a material is the atom density. The atom density is the number of atoms of a given type per unit volume of the material. To calculate the atom density of a substance use Equation (2-1).

Fixture checkpoint marker-0132-xray is recorded on fixture page 0132.

(2-1)

where:

N = atom density (atoms/cm)3
' = density (g/cm)3
N = Avogadro's number (6.022 x 10 atoms/mole) A
23
M = gram atomic weight
N
NA
M
2.699 g
cm3
6.022 x 1023 atoms
mole
26.9815 g
mole
6.024 x 1022 atoms
cm3

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Example:
A block of aluminum has a density of 2.699 g/cm . If the gram atomic weight of
3
aluminum is 26.9815 g, calculate the atom density of the aluminum.
Solution:
Cross Sections
The probability of a neutron interacting with a nucleus for a particular reaction is dependent upon not only the kind of nucleus involved, but also the energy of the neutron. Accordingly, the absorption of a thermal neutron in most materials is much more probable than the absorption of a fast neutron. Also, the probability of interaction will vary depending upon the type of reaction involved.
The probability of a particular reaction occurring between a neutron and a nucleus is called the microscopic cross section ( ) of the nucleus for the particular reaction. This cross section will vary with the energy of the neutron. The microscopic cross section may also be regarded as the effective area the nucleus presents to the neutron for the particular reaction. The larger the effective area, the greater the probability for reaction.
Because the microscopic cross section is an area, it is expressed in units of area, or square centimeters. A square centimeter is tremendously large in comparison to the effective area of a nucleus, and it has been suggested that a physicist once referred to the measure of a square centimeter as being "as big as a barn" when applied to nuclear processes. The name has persisted and microscopic cross sections are expressed in terms of barns. The relationship between barns and cm is shown below. 2
1 barn = 10-24 cm2
N
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NUCLEAR CROSS SECTIONS AND NEUTRON FLUX Reactor Theory (Neutron Characteristics)
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Whether a neutron will interact with a certain volume of material depends not only on the microscopic cross section of the individual nuclei but also on the number of nuclei within that volume. Therefore, it is necessary to define another kind of cross section known as the macroscopic cross section ( ). The macroscopic cross section is the probability of a given reaction occurring per unit travel of the neutron. is related to the microscopic cross section ( ) by the relationship shown below.
(2-2)
Fixture checkpoint marker-0133-xray is recorded on fixture page 0133.
where:
= macroscopic cross section (cm)
-1
N = atom density of material (atoms/cm)
3
= microscopic cross-section (cm)
2
The difference between the microscopic and macroscopic cross sections is extremely important and is restated for clarity. The microscopic cross section ( ) represents the effective target area that a single nucleus presents to a bombarding particle. The units are given in barns or cm .
2
The macroscopic cross section ( ) represents the effective target area that is presented by all of the nuclei contained in 1 cm of the material. The units are given as 1/cm or cm .

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A neutron interacts with an atom of the material it enters in two basic ways. It will either interact through a scattering interaction or through an absorption reaction. The probability of a neutron being absorbed by a particular atom is the microscopic cross section for absorption, . The probability of a neutron scattering off of a particular nucleus is the microscopic cross a section for scattering, s
. The sum of the microscopic cross section for absorption and the microscopic cross section for scattering is the total microscopic cross section, . T
= + T a s
Both the absorption and the scattering microscopic cross sections can be further divided. For instance, the scattering cross section is the sum of the elastic scattering cross section (se) and the inelastic scattering cross section ( ). si
Fixture checkpoint marker-0134-xray is recorded on fixture page 0134.
= + s se si
The microscopic absorption cross section ( ) includes all reactions except scattering. However, a for most purposes it is sufficient to merely separate it into two categories, fission (f) and capture ( ). Radiative capture of neutrons was described in the Neutron Interactions chapter c of Module 1.
= + a f c
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Figure 1 Typical Neutron Absorption Cross Section vs. Neutron Energy
The variation of absorption cross sections with neutron energy is often complicated. For many elements the absorption cross sections are small, ranging from a fraction of a barn to a few barns for slow (or thermal) neutrons.
For a considerable number of nuclides of moderately high (or high) mass numbers, an examination of the variation of the absorption cross section with the energy of the incident neutron reveals the existence of three regions on a curve of absorption cross section versus neutron energy. This cross section is illustrated in Figure 1. First, the cross section decreases steadily with increasing neutron energy in a low energy region, which includes the thermal range (E < 1 eV). In this region the absorption cross section, which is often high, is inversely proportional to the velocity (v). This region is frequently referred to as the "1/v region," because the absorption cross section is proportional to 1/v, which is the reciprocal of neutron velocity. Following the 1/v region, there occurs the "resonance region" in which the cross sections rise sharply to high values called "resonance peaks" for neutrons of certain energies, and then fall again. These energies are called resonance energies and are a result of the affinity of the nucleus for neutrons whose energies closely match its discrete, quantum energy levels. That is, when the binding energy of a neutron plus the kinetic energy of the neutron are exactly equal to the amount required to raise a compound nucleus from its ground state to a quantum level, resonance absorption occurs. The following example problem further illustrates this point.
■
(
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Assuming that uranium-236 has a nuclear quantum energy level at 6.8 MeV above its ground state, calculate the kinetic energy a neutron must possess to undergo resonant absorption in uranium-235 at this resonance energy level.

BE = (235.043925 + 1.008665 - 236.045563) x 931 MeV/amu
BE = (0.007025 amu) x 931 MeV/amu = 6.54 MeV
6.8 MeV - 6.54 MeV = 0.26 MeV
The difference between the binding energy and the quantum energy level equals the amount of kinetic energy the neutron must possess. The typical heavy nucleus will have many closely■spaced resonanc complex and have more possible configurations and corresponding energy states. Light nuclei, being less complex, have fewer possible energy states and fewer resonances that are sparsely distributed at higher energy levels.
For higher neutron energies, the absorption cross section steadily decreases as the energy of the neutron increases. This is called the "fast neutron region." In this region the absorption cross sections are usually less than 10 barns.
With the exception of hydrogen, for which the value is fairly large, the elastic scattering cross sections are generally small, for example, 5 barns to 10 barns. This is close to the magnitude of the actual geometric cross sectional area expected for atomic nuclei. In potential scattering, the cross section is essentially constant and independent of neutron energy. Resonance elastic scattering and inelastic scattering exhibit resonance peaks similar to those associated with absorption cross sections. The resonances occur at lower energies for heavy nuclei than for light nuclei. In general, the variations in scattering cross sections are very small when compared to
Fixture checkpoint marker-0135-xray is recorded on fixture page 0135.
the variations that occur in absorption cross sections.
Mean Free Path
If a neutron has a certain probability of undergoing a particular interaction in one centimeter of travel, then the inverse of this value describes how far the neutron will travel (in the average case) before undergoing an interaction. This average distance traveled by a neutron before interaction is known as the mean free path for that interaction and is represented by the symbol ■. The relationship between the mean free path (■) and the macroscopic cross section (*) is shown below.
(2-3)
'
(
)
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Calculation of Macroscopic Cross Section and Mean Free Path
Most materials are composed of several elements, and because most elements are composed of several isotopes, most materials involve many cross sections, one for each isotope involved. Therefore, to include all the isotopes within a given material, it is necessary to determine the macroscopic cross section for each isotope and then sum all the individual macroscopic cross sections. Equation (2-4) can be used to determine the macroscopic cross section for a composite material.
* = N) + N) + N) + ... N) (2-4) 1 1 2 2 3 3 nn
where:
N = the number of nuclei per cm of the n element n
3 th
) = the microscopic cross section of the n element n
th
The following example problems illustrate the calculation of the macroscopic cross section for

Example 1:
Find the macroscopic thermal neutron absorption cross section for iron, which has a density of 7.86 g/cm . The microscopic cross section for absorption of iron is 2.56 barns 3 and the gram atomic weight is 55.847 g.
Solution:
Step 1: Using Equation (2-1), calculate the atom density of iron.
Step 2: Use this atom density in Equation (2-2) to calculate the macroscopic cross section.
'
'

Example 2:
An alloy is composed of 95% aluminum and 5% silicon (by weight). The density of the alloy is 2.66 g/cm . Properties of aluminum and silicon are shown below. 3
Element Gram Atomic Weight)) a (barns) s (barns)
Aluminum 26.9815 0.23 1.49
Silicon 28.0855 0.16 2.20
1. Calculate the atom densities for the aluminum and silicon.
2. Determine the absorption and scattering macroscopic cross sections for thermal neutrons.
3. Calculate the mean free paths for absorption and scattering.
Solution:
Step 1: The density of the aluminum will be 95% of the total density. Using
Fixture checkpoint marker-0136-xray is recorded on fixture page 0136.
Equation (2-1) yields the atom densities.
a NAl a,Al NSi a,Si
5.64 x 1022 atoms
cm3
0.23 x 10 24 cm22.85 x 1021 atoms
cm3
0.16 x 10 24 cm2
0.0134 cm 1
s NAl s,Al NSi s,Si
5.64 x 1022 atoms
cm3
1.49 x 10 24 cm22.85 x 1021 atoms
cm3
2.20 x 10 24 cm2
0.0903 cm 1
a
1
a
1
0.01345 cm 1
74.3 cm
s

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s
1
0.0903 cm 1
11.1 cm
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Step 2: The macroscopic cross sections for absorption and scattering are calculated using Equation (2-4).
Step 3: The mean free paths are calculated by inserting the macroscopic cross sections calculated above into Equation (2-3).
Thus, a neutron must travel an average of 74.3 cm to interact by absorption in this alloy, but it must travel only 11.1 cm to interact by scattering.
)
)
)
)
(■
(■
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Effects of Temperature on Cross Section
As discussed, the microscopic absorption cross section varies significantly as neutron energy varies. The microscopic cross sections provided on most charts and tables are measured for a standard neutron velocity of 2200 meters/second, which corresponds to an ambient temperature of 68(F. Therefore, if our material is at a higher temperature, the absorption cross section will be lower than the value for 68(F, and any cross sections which involve absorption (for example,) ,) ,)) must be corrected for the existing temperature. acf
The following formula is used to correct microscopic cross sections for temperature. Although the example illustrates absorption cross section, the same formula may be used to correct capture and fission cross sections.
Fixture checkpoint marker-0137-xray is recorded on fixture page 0137.
where:
) = microscopic cross section corrected for temperature
) = microscopic cross section at reference temperature (68(F or 20(C) o
T = reference temperature (68(F) in degrees Rankine ((R) or Kelvin ((K) o
T = temperature for which corrected value is being calculated
NOTE: When using this formula, all temperatures must be converted to (R or (K.
(R = (F + 460
(K = (C + 273
Example:
What is the value of)f for uranium-235 for thermal neutrons at 500(F? Uranium-235 has a) of 583 barns at 68(F. f
Solution:
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Macroscopic cross sections for neutron reactions with materials determine the probability of one neutron undergoing a specific reaction per centimeter of travel through that material. If one wants to determine how many reactions will actually occur, it is necessary to know how many neutrons are traveling through the material and how many centimeters they travel each second. It is convenient to consider the number of neutrons existing in one cubic centimeter at any one instant and the total distance they travel each second while in that cubic centimeter. The number of neutrons existing in a cm of material at any instant is called neutron density and is 3 represented by the symbol n with units of neutrons/cm . The total distance these neutrons can 3 travel each second will be determined by their velocity.

A good way of defining neutron flux (1) is to consider it to be the total path length covered by all neutrons in one cubic centimeter during one second. Mathematically, this is the equation below.

1 = n v (2-5)

where:

1 = neutron flux (neutrons/cm -sec) 2

n = neutron density (neutrons/cm)3

v = neutron velocity (cm/sec)

The term neutron flux in some applications (for example, cross section measurement) is used as parallel beams of neutrons traveling in a single direction. The intensity (I) of a neutron beam is the product of the neutron density times the average neutron velocity. The directional beam intensity is equal to the number of neutrons per unit area and time (neutrons/cm -sec) falling on 2 a surface perpendicular to the direction of the beam.

One can think of the neutron flux in a reactor as being comprised of many neutron beams traveling in various directions. Then, the neutron flux becomes the scalar sum of these directional flux intensities (added as numbers and not vectors), that is, 1 = I + I + I +...I . 123 n

Since the atoms in a reactor do not interact preferentially with neutrons from any particular direction, all of these directional beams contribute to the total rate of reaction. In reality, at a given point within a reactor, neutrons will be traveling in all directions.

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Self-Shielding

In some locations within the reactor, the flux level may be significantly lower than in other areas due to a phenomenon referred to as neutron shadowing or self-shielding. For example, the interior of a fuel pin or pellet will "see" a lower average flux level than the outer surfaces since an appreciable fraction of the neutrons will have been absorbed and therefore cannot reach the interior of the fuel pin. This is especially important at resonance energies, where the absorption cross sections are large.

Summary

Fixture checkpoint marker-0138-xray is recorded on fixture page 0138.

The important information in this chapter is summarized below.

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Nuclear Cross Section and Neutron Flux Summary

& Atom density (N) is the number of atoms of a given type per unit volume of material.

& Microscopic cross section ()) is the probability of a given reaction occurring

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& Microscopic cross sections are measured in units of barns, where 1 barn = 10-24 cm .2
& Macroscopic cross section (*) is the probability of a given reaction occurring per unit length of travel of the neutron. The units for macroscopic cross section are cm . -1
& The mean free path (■) is the average distance that a neutron travels in a material between interactions.
& Neutron flux (1) is the total path length traveled by all neutrons in one cubic centimeter of material during one second.
& The macroscopic cross section for a material can be calculated using the equation below.
* = N)
& The absorption cross section for a material usually has three distinct regions. At low neutron energies (<1 eV) the cross section is inversely proportional to the neutron velocity.
& Resonance absorption occurs when the sum of the kinetic energy of the neutron
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and its binding energy is equal to an allowed nuclear energy level of the nucleus.
& Resonance peaks exist at intermediate energy levels. For higher neutron energies, the absorption cross section steadily decreases as the neutron energy increases.
& The mean free path equals 1/*.
& The macroscopic cross section for a mixture of materials can be calculated using the equation below.
* = N 1 1 2 2 3 3 nn) + N) + N) + ... N)
& Self-shielding is where the local neutron flux is depressed within a material due to neutron absorption near the surface of the material.
1)
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REACTION RATES
It is possible to determine the rate at which a nuclear reaction will take place based on the neutron flux, cross section for the interaction, and atom density of the target. This relationship illustrates how a change in one of these items affects the reaction rate.
EO 2.10 Given the neutron flux and macroscopic cross section, CALCULATE the reaction rate.
EO 2.11 DESCRIBE the relationship between neutron flux and reactor power.
Reaction Rates
If the total path length of all the neutrons in a cubic centimeter in a second is known, (neutron flux (1)), and if the probability of having an interaction per centimeter path length is also known (macroscopic cross section (*)), multiply them together to get the number of interactions taking place in that cubic centimeter in one second. This value is known as the reaction rate and is denoted by the symbol R. The reaction rate can be calculated by the equation shown below.
R = 1 * (2-6)
where:
R = reaction rate (reactions/sec)
1 = neutron flux (neutrons/cm -sec) 2

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Substituting the fact that * = N) into Equation (2-6) yields the equation below.
where:
) = microscopic cross section (cm)2
N = atom density (atoms/cm)3
Rf f
1 x 1013 neutrons
cm2
sec
0.1 cm 1
1 x 1012 fissions
cm3
sec
Rf f
f
Rf
1.29 x 1012 fissions
cm3
sec
3 x 1013 neutrons
cm2
sec
0.043 cm 1
Reactor Theory (Neutron Characteristics) DOE-HDBK-1019/1-93 REACTION RATES
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The reaction rate calculated will depend on which macroscopic cross section is used in the calculation. Normally, the reaction rate of greatest interest is the fission reaction rate.
Example:
If a one cubic centimeter section of a reactor has a macroscopic fission cross section of
0.1 cm , and if the thermal neutron flux is 10 neutrons/cm -sec, what is the fission rate -1 13 2
in that cubic centimeter?
Solution:
In addition to using Equation (2-6) to determine the reaction rate based on the physical properties of material, it is also possible to algebraically manipulate the equation to determine physical properties if the reaction rate is known.
Example:
A reactor operating at a flux level of 3 x 10 neutrons/cm -sec contains 10 atoms of
13 2 20
uranium-235 per cm . The reaction rate is 1.29 x 10 fission/cm . Calculate and .
3 12 3
f f
Solution:
Step 1: The macroscopic cross section can be determined by solving
Equation (2-6) for and substituting the appropriate values. f
Rf N f
f
Rf
N

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cm3
sec
1 x 1020 atoms
cm3
3 x 1013 neutrons
cm2
sec
4.3 x 10 22 cm2 1 barn
1 x 10 24 cm2
430 barns
1 watt 1 erg
1 x 10 7 watt sec
1 MeV
1.602 x 10 6
erg
1 fission
200 MeV
3.12 x 1010 fissions
second
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Step 2: To find the microscopic cross section, replace f with (N x f) and solve
for .
f
Reactor Power Calculation
Multiplying the reaction rate per unit volume by the total volume of the core results in the total number of reactions occurring in the core per unit time. If the amount of energy involved in each reaction were known, it would be possible to determine the rate of energy release (power) due to a certain reaction.
In a reactor where the average energy per fission is 200 MeV, it is possible to determine the
Fixture checkpoint marker-0141-xray is recorded on fixture page 0141.
number of fissions per second that are necessary to produce one watt of power using the following conversion factors.
1 fission = 200 MeV
1 MeV = 1.602 x 10 ergs -6
1 erg = 1 x 10 watt-sec -7
This is equivalent to stating that 3.12 x 10 fissions release 1 watt-second of energy. 10
P
th f V
3.12 x 1010 fissions
watt sec
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The power released in a reactor can be calculated based on Equation (2-6). Multiplying the reaction rate by the volume of the reactor results in the total fission rate for the entire reactor. Dividing by the number of fissions per watt-sec results in the power released by fission in the reactor in units of watts. This relationship is shown mathematically in Equation (2-7) below.
(2-7)

P = power (watts)
= thermal neutron flux (neutrons/cm -sec) th
2
= macroscopic cross section for fission (cm) f
-1
V = volume of core (cm)
3
Relationship Between Neutron Flux and Reactor Power
In an operating reactor the volume of the reactor is constant. Over a relatively short period of time (days or weeks), the number density of the fuel atoms is also relatively constant. Since the atom density and microscopic cross section are constant, the macroscopic cross section must also be constant. Examining Equation (2-7), it is apparent that if the reactor volume and macroscopic cross section are constant, then the reactor power and the neutron flux are directly proportional. This is true for day-to-day operation. The neutron flux for a given power level will increase very slowly over a period of months due to the burnup of the fuel and resulting decrease in atom density and macroscopic cross section.
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Reaction Rates Summary
& The reaction rate is the number of interactions of a particular type occurring in a cubic centimeter of material in a second.
& The reaction rate can be calculated by the equation below.
R = 1 *
& Over a period of several days, while the atom density of the fuel can be considered constant, the neutron flux is directly proportional to reactor power.
Summary
The important information in this chapter is summarized below.
Reactor Theory (Neutron Characteristics) DOE-HDBK-1019/1-93 NEUTRON MODERATION
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NEUTRON MODERATION
In thermal reactors, the neutrons that cause fission are at a much lower energy than the energy level at which they were born from fission. In this type of reactor, specific materials must be included in the reactor design to reduce the energy level of the neutrons in an efficient manner.
EO 2.12 DEFINE the following concepts:
a. Thermalization d. Average logarithmic energy decrement
b. Moderator e. Macroscopic slowing down power
Fixture checkpoint marker-0142-xray is recorded on fixture page 0142.
c. Moderating ratio
EO 2.13 LIST three desirable characteristics of a moderator.
EO 2.14 Given an average fractional energy loss per collision, CALCULATE the energy loss after a specified number of collisions.
Neutron Slowing Down and Thermalization
Fission neutrons are produced at an average energy level of 2 MeV and immediately begin to slow down as the result of numerous scattering reactions with a variety of target nuclei. After a number of collisions with nuclei, the speed of a neutron is reduced to such an extent that it has approximately the same average kinetic energy as the atoms (or molecules) of the medium in

of an electron volt at ordinary temperatures (0.025 eV at 20(C), is frequently referred to as the thermal energy, since it depends upon the temperature. Neutrons whose energies have been reduced to values in this region (< 1 eV) are designated thermal neutrons. The process of reducing the energy of a neutron to the thermal region by elastic scattering is referred to as thermalization, slowing down, or moderation. The material used for the purpose of thermalizing neutrons is called a moderator. A good moderator reduces the speed of neutrons in a small number of collisions, but does not absorb them to any great extent. Slowing the neutrons in as few collisions as possible is desirable in order to reduce the amount of neutron leakage from the core and also to reduce the number of resonance absorptions in non-fuel materials. Neutron leakage and resonance absorption will be discussed in the next module.
ln Ei
ln Ef
ln
Ei
Fixture checkpoint marker-0143-xray is recorded on fixture page 0143.
Ef
2
A
2
3
NEUTRON MODERATION DOE-HDBK-1019/1-93 Reactor Theory (Neutron Characteristics)
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The ideal moderating material (moderator) should have the following nuclear properties.
large scattering cross section
small absorption cross section
large energy loss per collision
A convenient measure of energy loss per collision is the logarithmic energy decrement. The average logarithmic energy decrement is the average decrease per collision in the logarithm of the neutron energy. This quantity is represented by the symbol (Greek letter xi).
(2-8)
where:
= average logarithmic energy decrement
Ei = average initial neutron energy
Ef = average final neutron energy
The symbol is commonly called the average logarithmic energy decrement because of the fact that a neutron loses, on the average, a fixed fraction of its energy per scattering collision. Since the fraction of energy retained by a neutron in a single elastic collision is a constant for a given material, is also a constant. Because it is a constant for each type of material and does not depend upon the initial neutron energy, is a convenient quantity for assessing the moderating ability of a material.
The values for the lighter nuclei are tabulated in a variety of sources. The following commonly used approximation may be used when a tabulated value is not available.
This approximation is relatively accurate for mass numbers (A) greater than 10, but for some low values of A it may be in error by over three percent.
N
ln Ehigh ln Elow
ln
Ehigh

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N
ln
Ehigh
Elow
ln
2 x 106eV
0.025 eV
0.948
19.2 collisions
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Since represents the average logarithmic energy loss per collision, the total number of collisions necessary for a neutron to lose a given amount of energy may be determined by dividing into the difference of the natural logarithms of the energy range in question. The number of collisions (N) to travel from any energy, E , to any lower energy, E , can be calculated as shown below. high low
Example:
How many collisions are required to slow a neutron from an energy of 2 MeV to a thermal energy of 0.025 eV, using water as the moderator? Water has a value of 0.948 for .
Solution:
Sometimes it is convenient, based upon information known, to work with an average fractional
Fixture checkpoint marker-0144-xray is recorded on fixture page 0144.
energy loss per collision as opposed to an average logarithmic fraction. If the initial neutron energy level and the average fractional energy loss per collision are known, the final energy level for a given number of collisions may be computed using the following formula.
! (
NEUTRON MODERATION DOE-HDBK-1019/1-93 Reactor Theory (Neutron Characteristics)
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(2-9)
where:
E = initial neutron energy o
E = neutron energy after N collisions N
x = average fractional energy loss per collision
N = number of collisions
Example:
If the average fractional energy loss per collision in hydrogen is 0.63, what will be the energy of a 2 MeV neutron after (a) 5 collisions? (b) 10 collisions?
Solution:
a)
b)
Macroscopic Slowing Down Power
Although the logarithmic energy decrement is a convenient measure of the ability of a material to slow neutrons, it does not measure all necessary properties of a moderator. A better measure of the capabilities of a material is the macroscopic slowing down power. The macroscopic slowing down power (MSDP) is the product of the logarithmic energy decrement and the macroscopic cross section for scattering in the material. Equation (2-10) illustrates how to calculate the macroscopic slowing down power.
(2-10)

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s
a
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Moderating Ratio
Macroscopic slowing down power indicates how rapidly a neutron will slow down in the material in question, but it still does not fully explain the effectiveness of the material as a moderator. An element such as boron has a high logarithmic energy decrement and a good slowing down power, but it is a poor moderator because of its high probability of absorbing neutrons.
The most complete measure of the effectiveness of a moderator is the moderating ratio. The moderating ratio is the ratio of the macroscopic slowing down power to the macroscopic cross section for absorption. The higher the moderating ratio, the more effectively the material performs as a moderator. Equation (2-11) shows how to calculate the moderating ratio of a material.
(2-11)
Moderating properties of different materials are compared in Table 2.
TABLE 2
Moderating Properties of Materials
Material Collisions to Slowing Down
Number of Macroscopic
Thermalize Power
Moderating
Ratio
H2O 0.927 19 1.425 62
D2O 0.510 35 0.177 4830
Helium 0.427 42 9 x 10 51 -6
Beryllium 0.207 86 0.154 126
Boron 0.171 105 0.092 0.00086
Fixture checkpoint marker-0145-xray is recorded on fixture page 0145.
Carbon 0.158 114 0.083 216
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Summary
The important information in this chapter is summarized below.
Neutron Moderation Summary
& Thermalization is the process of reducing the energy level of a neutron from the energy level at which it is produced to an energy level in the thermal range.
& The moderator is the reactor material that is present for the purpose of thermalizing neutrons.
& Moderating ratio is the ratio of the macroscopic slowing down power to the macroscopic cross section for absorption.
& The average logarithmic energy decrement (!) is the average change in the logarithm of neutron energy per collision.
& Macroscopic slowing down power is the product of the average logarithmic energy decrement and the macroscopic cross section for scattering.
& There are three desirable characteristics of a moderator.
1. large scattering cross section
2. small absorption cross section

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& The energy loss after a specified number of collisions can be calculated using the equation below.
E = E (1 - x) N o
N
■
■ ■
Reactor Theory (Neutron Characteristics) DOE-HDBK-1019/1-93 PROMPT AND DELAYED NEUTRONS Rev. 0 Page 29 NP-02
PROMPT AND DELAYED NEUTRONS
Not all neutrons are released at the same time following fission. Most neutrons are released virtually instantaneously and are called prompt neutrons. A very small fraction of neutrons are released after the decay of fission products and are called delayed neutrons. Although delayed neutrons are a very small fraction of the total number of neutrons, they play an extremely important role in the control of the reactor.
EO 3.1 STATE the origin of prompt neutrons and delayed neutrons.
EO 3.2 STATE the approximate fraction of neutrons that are born as delayed neutrons from the fission of the following nuclear fuels:
a. Uranium-235
b. Plutonium-239
EO 3.3 EXPLAIN the mechanism for production of delayed neutrons.
EO 3.4 EXPLAIN prompt and delayed neutron generation times.
EO 3.5 Given prompt and delayed neutron generation times and delayed neutron fraction, CALCULATE the average generation time.
EO 3.6 EXPLAIN the effect of delayed neutrons on reactor control.
Neutron Classification
The great majority (over 99%) of the neutrons produced in fission are released within about 10-13 seconds of the actual fission event. These are called prompt neutrons. A small portion of fission neutrons are delayed neutrons, which are produced for some time after the fission process has taken place. The delayed neutrons are emitted immediately following the first beta decay of a fission fragment known as a delayed neutron precursor. An example of a delayed neutron precursor is bromine-87, shown below.
PROMPT AND DELAYED NEUTRONS DOE-HDBK-1019/1-93 Reactor Theory (Neutron Characteristics) NP-02 Page 30 Rev. 0
Fixture checkpoint marker-0146-xray is recorded on fixture page 0146.
For most applications, it is convenient to combine the known precursors into groups with appropriately averaged properties. These groups vary somewhat depending on the fissile material in use. Table 3 lists the characteristics for the six precursor groups resulting from thermal fission of uranium-235. The fraction of all neutrons that are produced by each of these precursors is called the delayed neutron fraction for that precursor. The total fraction of all neutrons born as delayed neutrons is called the delayed neutron fraction (■). The fraction of delayed neutrons produced varies depending on the predominant fissile nuclide in use. The delayed neutron fractions (■) for the fissile nuclides of most interest are as follows: uranium-233 (0.0026), uranium-235 (0.0065), uranium-238 (0.0148), and plutonium-239 (0.0021).
TABLE 3
Delayed Neutron Precursor Groups
for Thermal Fission in Uranium-235

(sec) Fraction (MeV)
1 55.7 0.00021 0.25
2 22.7 0.00142 0.46
3 6.2 0.00127 0.41
4 2.3 0.00257 0.45
5 0.61 0.00075 0.41
6 0.23 0.00027 -
Total - 0.0065 -
Neutron Generation Time
The neutron generation time is the time required for neutrons from one generation to cause the fissions that produce the next generation of neutrons. The generation time for prompt neutrons (5* - pronounced "ell-star") is the total time from birth to rebirth. Three time intervals are
Fixture checkpoint marker-0147-xray is recorded on fixture page 0147.
involved: (a) the time it takes a fast neutron to slow down to thermal energy, (b) the time the now thermal neutron exists prior to absorption in fuel, and (c) the time required for a fissionable nucleus to emit a fast neutron after neutron absorption.
Fast neutrons slow to thermal energies or leak out of the reactor in 10 seconds to 10 seconds, -4 -6 depending on the moderator. In water moderated reactors, thermal neutrons tend to exist for about 10 seconds before they are absorbed. Fission and fast neutron production following -4 neutron absorption in a fissionable nucleus occurs in about 10 seconds. Thus, fast reactors -13 have an 5* of about 10 seconds, while thermal reactors have an 5* of about 10 seconds + 10 -6 -6 -4 seconds, which is about 10 seconds to 10 seconds. -4 -5

■■■■
■■■■
■■

Reactor Theory (Neutron Characteristics) DOE-HDBK-1019/1-93 PROMPT AND DELAYED NEUTRONS
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On the other hand, the average generation time for the six delayed neutron groups is the total time from the birth of the fast neutron to the emission of the delayed neutron. Again, three time intervals are involved: (a) the time it takes a fast neutron to slow down to thermal energy, (b) the time the thermal neutron exists prior to absorption, and (c) the average time from neutron absorption to neutron emission by the six precursor groups. The average time for decay of precursors from uranium-235 is 12.5 seconds. The other terms in the delayed neutron generation time are insignificant when compared to this value, and the average delayed neutron generation time becomes ■ 12.5 seconds.
A neutron generation time in the range of 10 seconds to 10 seconds or faster could result in -4 -5 very rapid power excursions, and control would not be possible without the dependence upon delayed neutrons to slow down the rate of the reaction. The average generation time, and hence the rate that power can rise, is determined largely by the delayed neutron generation time. The following equation shows this mathematically.
(2-12)
Example:
Assume a prompt neutron generation time for a particular reactor of 5 x 10 seconds and -5 a delayed neutron generation time of 12.5 seconds. If ■ is 0.0065, calculate the average generation time.
Solution:
This example demonstrates the effect delayed neutrons have on the neutron generation time and thus reactor control. If a reactor were to be operated in a sustained chain reaction using only

5 x 10 seconds. However, by operating the reactor such that a 0.0065 fraction of neutrons are -5
delayed, the generation life time is extended to 0.0813 seconds, providing time for adequate
operator control. Therefore, although only a small fraction of the total neutron population,
delayed neutrons are extremely important to the control and maintenance of a sustained fission
chain reaction.

Summary

The important information in this chapter is summarized on the following page.

■ ■ ■ ■

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Prompt and Delayed Neutrons Summary

& Prompt neutrons are released directly from fission within 10 seconds of the -13
fission event.

& Delayed neutrons are released from the decay of fission products that are called
delayed neutron precursors. Delayed neutron precursors are grouped according to
half-life. Half-lives vary from fractions of a second to almost a minute.

& The fraction of neutrons born as delayed neutrons is different for different fuel
materials. Following are values for some common fuel materials.

Uranium-235 0.0065

Fixture checkpoint marker-0148-xray is recorded on fixture page 0148.

Plutonium-239 0.0021

& Delayed neutrons are produced by a classification of fission products known as
delayed neutron precursors. When a delayed neutron precursor undergoes a
■ decay, it results in an excited daughter nucleus which immediately ejects a -
neutron. Therefore, these delayed neutrons appear with a half-life of the delayed
neutron precursor.

& The delayed neutron generation time is the total time from the birth of the fast
neutron to the emission of the delayed neutron in the next generation. Delayed
neutron generation times are dominated by the half-life of the delayed neutron
precursor. The average delayed neutron generation time is about 12.5 seconds.

& A prompt neutron generation time is the sum of the amount of time it takes a fast
neutron to thermalize, the amount of time the neutron exists as a thermal neutron
before it is absorbed, and the amount of time between a fissionable nuclide
absorbing a neutron and fission neutrons being released. Prompt neutron
generation time is about 5 x 10 seconds. -5

& The average neutron generation time can be calculated from the prompt and
delayed neutron generation times and the delayed neutron fraction using
Equation (2-12).

& Delayed neutrons are responsible for the ability to control the rate at which power
can rise in a reactor. If only prompt neutrons existed, reactor control would not be
possible due to the rapid power changes.

Reactor Theory (Neutron Characteristics) DOE-HDBK-1019/1-93 NEUTRON FLUX SPECTRUM
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Figure 2 Prompt Fission Neutron Energy Spectrum for
Thermal Fission of Uranium-235

NEUTRON FLUX SPECTRUM

The number of neutrons that exist at a given energy level varies. A plot of either
the fraction of neutrons or the neutron flux at a given energy versus the energy

widely for different types of reactors.
EO 4.1 STATE the average energy at which prompt neutrons are produced.
EO 4.2 DESCRIBE the neutron energy spectrum in the following reactors:
a. Fast reactor
b. Thermal reactor
EO 4.3 EXPLAIN the reason for the particular shape of the fast, intermediate, and slow energy regions of the neutron flux spectrum for a thermal reactor.
Prompt Neutron Energies
The neutrons produced by fission are high energy neutrons, and almost all fission neutrons have energies between 0.1 MeV and 10 MeV. The neutron energy distribution, or spectrum, may best be described by plotting the fraction of neutrons per MeV as a function of neutron energy, as shown in Figure 2. From this figure it can be seen that the most probable neutron energy is about 0.7 MeV. Also, from this data it can be shown that the average energy of fission neutrons is about
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2 MeV. Figure 2 is the neutron energy spectrum for thermal fission in uranium-235. The values will vary slightly for other nuclides.
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Figure 3 Comparison of Neutron Flux Spectra for Thermal and Fast Breeder Reactor
Thermal and Fast Breeder Reactor Neutron Spectra
The spectrum of neutron energies produced by fission varies significantly from the energy spectrum, or flux, existing in a reactor at a given time. Figure 3 illustrates the difference in neutron flux spectra between a thermal reactor and a fast breeder reactor. The energy distribution of neutrons from fission is essentially the same for both reactors, so the differences in the curve shapes may be attributed to the neutron moderation or slowing down effects.
No attempt is made to thermalize or slow down neutrons in the fast breeder reactor (liquid metal cooled); therefore, an insignificant number of neutrons exist in the thermal range. For the thermal reactor (water moderated), the spectrum of neutrons in the fast region (> 0.1 MeV) has a shape similar to that for the spectrum of neutrons emitted by the fission process.
In the thermal reactor, the flux in the intermediate energy region (1 eV to 0.1 MeV) has approximately a 1/E dependence. That is, if the energy (E) is halved, the flux doubles. This

constant fraction of the neutron energy per collision (on the average), independent of energy; thus, the neutron loses larger amounts of energy per collision at higher energies than at lower energies. The fact that the neutrons lose a constant fraction of energy per collision causes the neutrons to tend to "pile up" at lower energies, that is, a greater number of neutrons exist at the lower energies as a result of this behavior.
vp
2 k T
m
vp
2 k T
m
2 1.38 x 10 16 erg
K
293 K
1.66 x 10 24 g
2.2 x 105 cm
sec
1 m
100 cm
2200 m
sec
Reactor Theory (Neutron Characteristics) DOE-HDBK-1019/1-93 NEUTRON FLUX SPECTRUM
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In the thermal region the neutrons achieve a thermal equilibrium with the atoms of the moderator material. In any given collision they may gain or lose energy, and over successive collisions will gain as much energy as they lose. These thermal neutrons, even at a specific temperature, do not all have the same energy or velocity; there is a distribution of energies, usually referred to as the Maxwell distribution (e.g., Figure 2). The energies of most thermal neutrons lie close to the most probable energy, but there is a spread of neutrons above and below this value.
Most Probable Neutron Velocities
The most probable velocity (vp) of a thermal neutron is determined by the temperature of the medium and can be determined by Equation (2-13) .
(2-13)
Fixture checkpoint marker-0150-xray is recorded on fixture page 0150.
where:
v = most probable velocity of neutron (cm/sec) p
k = Boltzman's constant (1.38 x 10 erg/ K) -16
T = absolute temperature in degrees Kelvin (K)
m = mass of neutron (1.66 x 10 grams) -24
Example:
Calculate the most probable velocities for neutrons in thermal equilibrium with their surroundings at the following temperatures. a) 20 C, b) 260 C.
Solution:
a) Calculate the most probable velocity for 20 C using Equation (2-13).
vp
2 k T
m
2 1.38 x 10 16 erg

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NUCLEAR PHYSICS AND REACTOR THEORY
ABSTRACT
The Nuclear Physics and Reactor Theory Handbook was developed to assist nuclear facility operating contractors in providing operators, maintenance personnel, and the technical staff with the necessary fundamentals training to ensure a basic understanding of nuclear physics and reactor theory. The handbook includes information on atomic and nuclear physics; neutron characteristics; reactor theory and nuclear parameters; and the theory of reactor operation. This information will provide personnel with a foundation for understanding the scientific principles that are associated with various DOE nuclear facility operations and maintenance.
Key Words: Training Material, Atomic Physics, The Chart of the Nuclides, Radioactivity, Radioactive Decay, Neutron Interaction, Fission, Reactor Theory, Neutron Characteristics, Neutron Life Cycle, Reactor Kinetics
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NUCLEAR PHYSICS AND REACTOR THEORY
FOREWORD
The Department of Energy (DOE) Fundamentals Handbooks consist of ten academic subjects, which include Mathematics; Classical Physics; Thermodynamics, Heat Transfer, and Fluid Flow; Instrumentation and Control; Electrical Science; Material Science; Mechanical Science; Chemistry; Engineering Symbology, Prints, and Drawings; and Nuclear Physics and Reactor Theory. The handbooks are provided as an aid to DOE nuclear facility contractors.
These handbooks were first published as Reactor Operator Fundamentals Manuals in 1985 for use by DOE category A reactors. The subject areas, subject matter content, and level of detail of the Reactor Operator Fundamentals Manuals were determined from several sources. DOE Category A reactor training managers determined which materials should be included, and served as a primary reference in the initial development phase. Training guidelines from the commercial nuclear power industry, results of job and task analyses, and independent input from contractors and operations-oriented personnel were all considered and included to some degree in developing the text material and learning objectives.
The DOE Fundamentals Handbooks represent the needs of various DOE nuclear facilities' fundamental training requirements. To increase their applicability to nonreactor nuclear facilities, the Reactor Operator Fundamentals Manual learning objectives were distributed to the Nuclear Facility Training Coordination Program Steering Committee for review and comment. To update their reactor-specific content, DOE Category A reactor training managers also reviewed and commented on the content. On the basis of feedback from these sources, information that applied to two or more DOE nuclear facilities was considered generic and was included. The final draft of each of the handbooks was then reviewed by these two groups. This approach has resulted

content to fit their specific needs.
Each handbook contains an abstract, a foreword, an overview, learning objectives, and text material, and is divided into modules so that content and order may be modified by individual DOE contractors to suit their specific training needs. Each handbook is supported by a separate examination bank with an answer key.
The DOE Fundamentals Handbooks have been prepared for the Assistant Secretary for Nuclear Energy, Office of Nuclear Safety Policy and Standards, by the DOE Training Coordination Program. This program is managed by EG&G Idaho, Inc.
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NUCLEAR PHYSICS AND REACTOR THEORY
OVERVIEW
The Department of Energy Fundamentals Handbook entitled Nuclear Physics and Reactor Theory was prepared as an information resource for personnel who are responsible for the operation of the Department's nuclear facilities. Almost all processes that take place in a nuclear facility involves the transfer of some type of energy. A basic understanding of nuclear physics and reactor theory is necessary for DOE nuclear facility operators, maintenance personnel, and Fixture checkpoint marker-0152-xray is recorded on fixture page 0152.
the technical staff to safely operate and maintain the facility and facility support systems. The information in this handbook is presented to provide a foundation for applying engineering concepts to the job. This knowledge will help personnel understand the impact that their actions may have on the safe and reliable operation of facility components and systems.
The Nuclear Physics and Reactor Theory handbook consists of four modules that are contained in two volumes. The following is a brief description of the information presented in each module of the handbook.
Volume 1 of 2
Module 1 - Atomic and Nuclear Physics
Introduces concepts of atomic physics including the atomic nature of matter, the chart of the nuclides, radioactivity and radioactive decay, neutron interactions and fission, and the interaction of radiation with matter.
Module 2 - Reactor Theory (Nuclear Parameters)
Provides information on reactor theory and neutron characteristics. Includes topics such as neutron sources, neutron flux, neutron cross sections, reaction rates, neutron moderation, and prompt and delayed neutrons.
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NUCLEAR PHYSICS AND REACTOR THEORY
OVERVIEW (Cont.)
Volume 2 of 2
Module 3 - Reactor Theory (Nuclear Parameters)
Explains the nuclear parameters associated with reactor theory. Topics include the neutron life cycle, reactivity and reactivity coefficients, neutron poisons, and control rods.
Module 4 - Reactor Theory (Reactor Operations)
Introduces the reactor operations aspect of reactor theory. Topics include subcritical multiplication, reactor kinetics, and reactor operation.
The information contained in this handbook is not all-encompassing. An attempt to present the entire subject of nuclear physics and reactor theory would be impractical. However,

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reader with the fundamental knowledge necessary to understand the advanced theoretical concepts presented in other subject areas, and to understand basic system and equipment operation.
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Fundamentals Handbook
NUCLEAR PHYSICS
AND REACTOR THEORY
Module 1
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TERMINAL OBJECTIVE
1.0 Given sufficient information, DESCRIBE atoms, including components, structure, and nomenclature.
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ENABLING OBJECTIVES
1.1 STATE the characteristics of the following atomic particles, including mass, charge, and location within the atom:
a. Proton
b. Neutron
c. Electron
1.2 DESCRIBE the Bohr model of an atom.
1.3 DEFINE the following terms:
a. Nuclide c. Atomic number
b. Isotope d. Mass number
1.4 Given the standard A
ZX notation for a particular nuclide, DETERMINE the following:
a. Number of protons
b. Number of neutrons
c. Number of electrons
1.5 DESCRIBE the three forces that act on particles within the nucleus and affect the stability of the nucleus.
1.6 DEFINE the following terms:
a. Enriched uranium
b. Depleted uranium
1.7 DEFINE the following terms:
a. Mass defect
b. Binding energy
1.8 Given the atomic mass for a nuclide and the atomic masses of a neutron, proton, and electron, CALCULATE the mass defect and binding energy of the nuclide.

TERMINAL OBJECTIVE
2.0 Given necessary references, DESCRIBE the various modes of radioactive decay.
ENABLING OBJECTIVES
2.1 DESCRIBE the following processes:
a. Alpha decay d. Electron capture

c. Beta-plus decay f. Isomeric transitions
2.2 Given a Chart of the Nuclides, WRITE the radioactive decay chain for a nuclide.
2.3 EXPLAIN why one or more gamma rays typically accompany particle emission.
2.4 Given the stability curve on the Chart of the Nuclides, DETERMINE the type of radioactive decay that the nuclides in each region of the chart will typically undergo.
2.5 DEFINE the following terms:
a. Radioactivity d. Radioactive decay constant
b. Curie e. Radioactive half-life
c. Becquerel
2.6 Given the number of atoms and either the half-life or decay constant of a nuclide, CALCULATE the activity.
2.7 Given the initial activity and the decay constant of a nuclide, CALCULATE the activity at any later time.
2.8 CONVERT between the half-life and decay constant for a nuclide.
2.9 Given the Chart of the Nuclides and the original activity, PLOT the radioactive decay curve for a nuclide on either linear or semi-log coordinates.
2.10 DEFINE the following terms:
a. Radioactive equilibrium
b. Transient radioactive equilibrium
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Atomic and Nuclear Physics DOE-HDBK-1019/1-93 OBJECTIVES
TERMINAL OBJECTIVE
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3.0 Without references, DESCRIBE the different nuclear interactions initiated by neutrons.
ENABLING OBJECTIVES
3.1 DESCRIBE the following scattering interactions between a neutron and a nucleus:
a. Elastic scattering
b. Inelastic scattering
3.2 STATE the conservation laws that apply to an elastic collision between a neutron and a nucleus.
3.3 DESCRIBE the following reactions where a neutron is absorbed in a nucleus:
a. Radiative capture
b. Particle ejection
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OBJECTIVES DOE-HDBK-1019/1-93 Atomic and Nuclear Physics
TERMINAL OBJECTIVE
4.0 Without references, DESCRIBE the fission process.
ENABLING OBJECTIVES
4.1 EXPLAIN the fission process using the liquid drop model of a nucleus.
4.2 DEFINE the following terms:
a. Excitation energy
b. Critical energy
4.3 DEFINE the following terms:
a. Fissile material
b. Fissionable material
c. Fertile material
4.4 DESCRIBE the processes of transmutation, conversion, and breeding.
4.5 DESCRIBE the curve of Binding Energy per Nucleon versus mass number and give a

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4.6 EXPLAIN why only the heaviest nuclei are easily fissioned.
4.7 EXPLAIN why uranium-235 fissions with thermal neutrons and uranium-238 fissions only with fast neutrons.
4.8 CHARACTERIZE the fission products in terms of mass groupings and radioactivity.
4.9 Given the nuclides involved and their masses, CALCULATE the energy released from fission.
4.10 Given the curve of Binding Energy per Nucleon versus mass number, CALCULATE the energy released from fission.

TERMINAL OBJECTIVE
5.0 Without references, DESCRIBE how the various types of radiation interact with matter.
ENABLING OBJECTIVES
5.1 DESCRIBE interactions of the following with matter:
a. Alpha particle c. Positron
b. Beta particle d. Neutron
5.2 DESCRIBE the following ways that gamma radiation interacts with matter:
a. Compton scattering
b. Photoelectric effect
c. Pair production

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ATOMIC NATURE OF MATTER
All matter is composed of atoms. The atom is the smallest amount of matter that retains the properties of an element. Atoms themselves are composed of smaller particles, but these smaller particles no longer have
Fixture checkpoint marker-0158-xray is recorded on fixture page 0158.
the same properties as the overall element.
EO 1.1 STATE the characteristics of the following atomic particles, including mass, charge, and location within the atom:
a. Proton
b. Neutron
c. Electron
EO 1.2 DESCRIBE the Bohr model of an atom.
EO 1.3 DEFINE the following terms:
a. Nuclide c. Atomic number
b. Isotope d. Mass number
EO 1.4 Given the standard A ZX notation for a particular nuclide, DETERMINE the following:
a. Number of protons
b. Number of neutrons
c. Number of electrons
EO 1.5 DESCRIBE the three forces that act on particles within the nucleus and affect the stability of the nucleus.

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Early Greek philosophers speculated that the earth was made up of different combinations of basic substances, or elements. They considered these basic elements to be earth, air, water, and fire. Modern science shows that the early Greeks held the correct concept that matter consists of a combination of basic elements, but they incorrectly identified the elements.

In 1661 the English chemist Robert Boyle published the modern criterion for an element. He defined an element to be a basic substance that cannot be broken down into any simpler substance after it is isolated from a compound, but can be combined with other elements to form compounds. To date, 105 different elements have been confirmed to exist, and researchers claim to have discovered three additional elements. Of the 105 confirmed elements, 90 exist in nature and 15 are man-made.

Another basic concept of matter that the Greeks debated was whether matter was continuous or discrete. That is, whether matter could be continuously divided and subdivided into ever smaller particles or whether eventually an indivisible particle would be encountered. Democritus in about 450 B.C. argued that substances were ultimately composed of small, indivisible particles that he labeled atoms. He further suggested that different substances were composed of different atoms or combinations of atoms, and that one substance could be converted into another by rearranging the atoms. It was impossible to conclusively prove or disprove this proposal for more than 2000 years.

The modern proof for the atomic nature of matter was first proposed by the English chemist John Dalton in 1803. Dalton stated that each chemical element possesses a particular kind of atom, and any quantity of the element is made up of identical atoms of this kind. What distinguishes one element from another element is the kind of atom of which it consists, and the basic physical difference between kinds of atoms is their weight.

Subatomic Particles

For almost 100 years after Dalton established the atomic nature of atoms, it was considered impossible to divide the atom into even smaller parts. All of the results of chemical experiments during this time indicated that the atom was indivisible. Eventually, experimentation into electricity and radioactivity indicated that particles of matter smaller than the atom did indeed exist. In 1906, J. J. Thompson won the Nobel Prize in physics for establishing the existence of electrons. Electrons are negatively-charged particles that have 1/1835 the mass of the hydrogen atom. Soon after the discovery of electrons, protons were discovered. Protons are relatively large particles that have almost the same mass as a hydrogen atom and a positive charge equal in magnitude (but opposite in sign) to that of the electron. The third subatomic particle to be discovered, the neutron, was not found until 1932. The neutron has almost the same mass as the

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proton, but it is electrically neutral.

Bohr Model of the Atom

The British physicist Ernest Rutherford postulated that the positive charge in an atom is concentrated in a small region called a nucleus at the center of the atom with electrons existing in orbits around it. Niels Bohr, coupling Rutherford's postulation with the quantum theory introduced by Max Planck, proposed that the atom consists of a dense nucleus of protons surrounded by electrons traveling in discrete orbits at fixed distances from the nucleus. An electron in one of these orbits or shells has a specific or discrete quantity of energy (quantum). When an electron moves from one allowed orbit to another allowed orbit, the energy difference

called a photon. Figure 1 is Bohr's model of the hydrogen atom showing an electron as having just dropped from the third shell to the first shell with the emission of a photon that has an energy = hv. (h = Planck's constant = 6.63 x 10-34 J-s and v = frequency of the photon.) Bohr's theory was the first to successfully account for the discrete energy levels of this radiation as measured in the laboratory. Although Bohr's atomic model is designed specifically to explain the hydrogen atom, his theories apply generally to the structure of all atoms. Additional information on electron shell theory can be found in the Chemistry Fundamentals Handbook.
Figure 1 Bohr's Model of the Hydrogen Atom

Properties of the three subatomic particles are listed in Table 1.
TABLE 1
Properties of Subatomic Particles
Fixture checkpoint marker-0160-xray is recorded on fixture page 0160.

Particle	Location	Charge	Mass
Neutron	Nucleus	none	1.008665 amu
Proton	Nucleus	+1	1.007277 amu
Electron	Shells around nucleus	-1	0.0005486 amu

Measuring Units on the Atomic Scale
The size and mass of atoms are so small that the use of normal measuring units, while possible, is often inconvenient. Units of measure have been defined for mass and energy on the atomic scale to make measurements more convenient to express. The unit of measure for mass is the atomic mass unit (amu). One atomic mass unit is equal to 1.66 x 10-24 grams. The reason for this particular value for the atomic mass unit will be discussed in a later chapter. Note from Table 1 that the mass of a neutron and a proton are both about 1 amu. The unit for energy is the electron volt (eV). The electron volt is the amount of energy acquired by a single electron when it falls through a potential difference of one volt. One electron volt is equivalent to 1.602 x 10-19 joules or 1.18 x 10-19 foot-pounds.
Nuclides
The total number of protons in the nucleus of an atom is called the atomic number of the atom and is given the symbol Z. The number of electrons in an electrically-neutral atom is the same as the number of protons in the nucleus. The number of neutrons in a nucleus is known as the neutron number and is given the symbol N. The mass number of the nucleus is the total number of nucleons, that is, protons and neutrons in the nucleus. The mass number is given the symbol A and can be found by the equation Z + N = A.
Each of the chemical elements has a unique atomic number because the atoms of different elements contain a different number of protons. The atomic number of an atom identifies the particular element.

Each type of atom that contains a unique combination of
Figure 2 Nomenclature for
Identifying Nuclides
protons and neutrons is called a nuclide. Not all combinations of numbers of protons and neutrons are possible, but about 2500 specific nuclides with unique combinations of neutrons and protons have been identified. Each nuclide is denoted by the chemical

a subscript and the mass number written as a superscript, as shown in Figure 2. Because each element has a unique name, chemical symbol, and atomic number, only one of the three is necessary to identify the element. For this reason nuclides can also be identified by either the chemical name or the chemical symbol followed by the mass number (for example, U-235 or uranium-235).
Another common format is to use the abbreviation of the chemical element with the mass number superscripted (for example, 235U). In this handbook the format used in the text will usually be the element's name followed by the mass number. In equations and tables, the format in Figure 2 will usually be used.
Example:
State the name of the element and the number of protons, electrons, and neutrons in the nuclides listed below.
1
1H
10
5B
14
7N
Fixture checkpoint marker-0161-xray is recorded on fixture page 0161.
11
4
4
8Cd
23
9
9
4Pu

Solution:
The name of the element can be found from the Periodic Table (refer to Chemistry Fundamentals Handbook) or the Chart of the Nuclides (to be discussed later). The number of protons and electrons are equal to Z. The number of neutrons is equal to Z - A.
Nuclide Element Protons Electrons Neutrons
1
1H hydrogen 1 1 0
10
5B boron 5 5 5
14
7N nitrogen 7 7 7
11
4
4
8Cd cadmium 48 48 66
23

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9
4Pu plutonium 94 94 145
Isotopes
Isotopes are nuclides that have the same atomic number and are therefore the same element, but differ in the number of neutrons. Most elements have a few stable isotopes and several unstable, radioactive isotopes. For example, oxygen has three stable isotopes that can be found in nature (oxygen-16, oxygen-17, and oxygen-18) and eight radioactive isotopes. Another example is hydrogen, which has two stable isotopes (hydrogen-1 and hydrogen-2) and a single radioactive isotope (hydrogen-3).
The isotopes of hydrogen are unique in that they are each commonly referred to by a unique name instead of the common chemical element name. Hydrogen-1 is almost always referred to as hydrogen, but the term protium is infrequently used also. Hydrogen-2 is commonly called deuterium and symbolized 2
1D. Hydrogen-3 is commonly called tritium and symbolized 31T. This
text will normally use the symbology 2
1H and 31H for deuterium and tritium, respectively.
Atomic and Nuclear Radii
The size of an atom is difficult to define exactly due to the fact that the electron cloud, formed by the electrons moving in their various orbitals, does not have a distinct outer edge. A reasonable measure of atomic size is given by the average distance of the outermost electron from the nucleus. Except for a few of the lightest atoms, the average atomic radii are approximately the same for all atoms, about 2 x 10 -8 cm.
Like the atom the nucleus does not have a sharp outer boundary. Experiments have shown that the nucleus is shaped like a sphere with a radius that depends on the atomic mass number of the atom. The relationship between the atomic mass number and the radius of the nucleus is shown in the following equation.
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Fixture checkpoint marker-0162-xray is recorded on fixture page 0162.
Atomic and Nuclear Physics DOE-HDBK-1019/1-93 ATOMIC NATURE OF MATTER
r = (1.25 x 10 -13 cm) A1/3
where:
r = radius of the nucleus (cm)
A = atomic mass number (dimensionless)
The values of the nuclear radii for some light, intermediate, and heavy nuclides are shown in Table 2.
TABLE 2
Calculated Values for Nuclear
Radii
Nuclide Radius of Nucleus
1
1H 1.25 x 10 -13 cm
10
5B 2.69 x 10 -13 cm
5
2
6
6Fe 4.78 x 10 -13 cm
17

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8
2Hf 7.01 x 10 -13 cm
23
9
8
2U 7.74 x 10 -13 cm
25
9
2
8Cf 7.89 x 10 -13 cm
From the table, it is clear that the radius of a typical atom (e.g. 2 x 10 -8 cm) is more than 25,000 times larger than the radius of the largest nucleus.
Nuclear Forces
In the Bohr model of the atom, the nucleus consists of positively-charged protons and electrically■neutra as nucleons. One problem that the Bohr model of the atom presented was accounting for an attractive force to overcome the repulsive force between protons.
Two forces present in the nucleus are (1) electrostatic forces between charged particles and (2) gravitational forces between any two objects that have mass. It is possible to calculate the magnitude of the gravitational force and electrostatic force based upon principles from classical physics.
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Fg
G m1 m2
r
2
Fe
K Q1 Q2
r
2
ATOMIC NATURE OF MATTER DOE-HDBK-1019/1-93 Atomic and Nuclear Physics
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Newton stated that the gravitational force between two bodies is directly proportional to the masses of the two bodies and inversely proportional to the square of the distance between the bodies. This relationship is shown in the equation below.
Fixture checkpoint marker-0163-xray is recorded on fixture page 0163.
where:
F = gravitational force (newtons) g
m = mass of first body (kilograms) 1
m = mass of second body (kilograms) 2
G = gravitational constant (6.67 x 10 N-m /kg)
-11 2 2
r = distance between particles (meters)
The equation illustrates that the larger the masses of the objects or the smaller the distance between the objects, the greater the gravitational force. So even though the masses of nucleons are very small, the fact that the distance between nucleons is extremely short may make the gravitational force significant. It is necessary to calculate the value for the gravitational force and compare it to the value for other forces to determine the significance of the gravitational force in the nucleus. The gravitational force between two protons that are separated by a distance of

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Coulomb's Law can be used to calculate the force between two protons. The electrostatic force is directly proportional to the electrical charges of the two particles and inversely proportional to the square of the distance between the particles. Coulomb's Law is stated as the following equation.
where:
F = electrostatic force (newtons) e
K = electrostatic constant (9.0 x 10 N-m /C)
9 2 2
Q = charge of first particle (coulombs) 1
Q = charge of second particle (coulombs) 2
r = distance between particles (meters)
Fixture checkpoint marker-0164-xray is recorded on fixture page 0164.
Using this equation, the electrostatic force between two protons that are separated by a distance of 10 meters is about 10 newtons. Comparing this result with the calculation of the
-20 12
gravitational force (10 newtons) shows that the gravitational force is so small that it can be
-24
neglected.
Atomic and Nuclear Physics DOE-HDBK-1019/1-93 ATOMIC NATURE OF MATTER
If only the electrostatic and gravitational forces existed in the nucleus, then it would be impossible to have stable nuclei composed of protons and neutrons. The gravitational forces are much too small to hold the nucleons together compared to the electrostatic forces repelling the protons. Since stable atoms of neutrons and protons do exist, there must be another attractive force acting within the nucleus. This force is called the nuclear force.
The nuclear force is a strong attractive force that is independent of charge. It acts equally only between pairs of neutrons, pairs of protons, or a neutron and a proton. The nuclear force has a very short range; it acts only over distances approximately equal to the diameter of the nucleus (10 -13 cm). The attractive nuclear force between all nucleons drops off with distance much faster than the repulsive electrostatic force between protons.
TABLE 3
Forces Acting in the Nucleus
Force Interaction Range
Gravitational Very weak attractive force between all nucleons Relatively long
Electrostatic Strong repulsive force between like charged particles (protons) Relatively long
Nuclear Force Strong attractive force between all nucleons Extremely short
In stable atoms, the attractive and repulsive forces in the nucleus balance. If the forces do not balance, the atom cannot be stable, and the nucleus will emit radiation in an attempt to achieve a more stable configuration.
Summary
The important information in this chapter is summarized on the following page.
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ATOMIC NATURE OF MATTER DOE-HDBK-1019/1-93 Atomic and Nuclear Physics
Atomic Nature of Matter Summary
Atoms consist of three basic subatomic particles. These particles are the proton, the neutron, and the electron.

hydrogen atom, and exist in the nucleus of an atom.
Neutrons are particles that have no electrical charge, have about the same mass as a hydrogen atom, and exist in the nucleus of an atom.
Electrons are particles that have a negative charge, have a mass about eighteen hundred times smaller than the mass of a hydrogen atom, and exist in orbital shells around the nucleus of an atom.
The Bohr model of the atom consists of a dense nucleus of protons and neutrons (nucleons) surrounded by electrons traveling in discrete orbits at fixed distances from the nucleus.
Nuclides are atoms that contain a particular number of protons and neutrons.
Isotopes are nuclides that have the same atomic number and are therefore the same element, but differ in the number of neutrons.
The atomic number of an atom is the number of protons in the nucleus.
The mass number of an atom is the total number of nucleons (protons and neutrons) in the nucleus.
The notation A
ZX is used to identify a specific nuclide. "Z" represents the atomic number, which is equal to the number of protons. "A" represents the mass
Fixture checkpoint marker-0165-xray is recorded on fixture page 0165.
number, which is equal to the number of nucleons. "X" represents the chemical symbol of the element.
Number of protons = Z
Number of electrons = Z
Number of neutrons = A - Z
The stability of a nucleus is determined by the different forces interacting within it. The electrostatic force is a relatively long-range, strong, repulsive force that acts between the positively charged protons. The nuclear force is a relatively short-range attractive force between all nucleons. The gravitational force the long range, relatively weak attraction between masses, is negligible compared to the other forces.
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Atomic and Nuclear Physics DOE-HDBK-1019/1-93 CHART OF THE NUCLIDES
CHART OF THE NUCLIDES
The Chart of the Nuclides, like the Periodic Table, is a convenient format for presenting a large amount of scientific information in an organized manner.
EO 1.6 DEFINE the following terms:
a. Enriched uranium
b. Depleted uranium
Chart of the Nuclides
A tabulated chart called the Chart of the Nuclides lists the stable and unstable nuclides in addition to pertinent information about each one. Figure 3 shows a small portion of a typical chart. This chart plots a box for each individual nuclide, with the number of protons (Z) on the vertical axis and the number of neutrons (N = A - Z) on the horizontal axis.
The completely gray squares indicate stable isotopes. Those in white squares are artificially radioactive, meaning that they are produced by artificial techniques and do not occur naturally. By consulting a complete chart, other types of isotopes can be found, such as naturally occurring radioactive types (but none are found in the region of the chart that is illustrated in Figure 3).

element. The box contains the chemical symbol of the element in addition to the average atomic weight of the naturally occurring substance and the average thermal neutron absorption cross section, which will be discussed in a later module. The known isotopes (elements with the same atomic number Z but different mass number A) of each element are listed to the right.

Figure 3 Nuclide Chart for Atomic Numbers 1 to 6

Information for Stable Nuclides

For the stable isotopes, in addition to the symbol and the atomic mass number, the number percentage of each isotope in the naturally occurring element is listed, as well as the thermal neutron activation cross section and the mass in atomic mass units (amu). A typical block for a stable nuclide from the Chart of the Nuclides is shown in Figure 4.

Figure 4 Stable Nuclides

Information for Unstable Nuclides

For unstable isotopes the additional information includes the half life, the mode of decay (for example, β^-, α), the total disintegration energy in MeV (million electron volts), and the mass in amu when available. A typical block for an unstable nuclide from the Chart of the Nuclides is shown in Figure 5.

Figure 5 Unstable Nuclides

Neutron - Proton Ratios

Fixture checkpoint marker-0166-xray is recorded on fixture page 0166.

Figure 6 shows the distribution of the stable nuclides plotted on the same axes as the Chart of the Nuclides. As the mass numbers become higher, the ratio of neutrons to protons in the nucleus becomes larger. For helium-4 (2 protons and 2 neutrons) and oxygen-16 (8 protons and 8 neutrons) this ratio is unity. For indium-115 (49 protons and 66 neutrons) the ratio of neutrons to protons has increased to 1.35, and for uranium-238 (92 protons and 146 neutrons) the neutron■to-proton

Figure 6 Neutron - Proton Plot of the Stable Nuclides

If a heavy nucleus were to split into two fragments, each fragment would form a nucleus that would have approximately the same neutron-to-proton ratio as the heavy nucleus. This high neutron-to-proton ratio places the fragments below and to the right of the stability curve displayed by Figure 6. The instability caused by this excess of neutrons is generally rectified by successive beta emissions, each of which converts a neutron to a proton and moves the nucleus toward a more stable neutron-to-proton ratio.

■

Natural Abundance of Isotopes

The relative abundance of an isotope in nature compared to other isotopes of the same element is relatively constant. The Chart of the Nuclides presents the relative abundance of the naturally occurring isotopes of an element in units of atom percent. Atom percent is the percentage of the atoms of an element that are of a particular isotope. Atom percent is abbreviated as a/o.

For example, if a cup of water contains 8.23×10^{24} atoms of oxygen, and the isotopic abundance

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The atomic weight for an element is defined as the average atomic weight of the isotopes of the element. The atomic weight for an element can be calculated by summing the products of the isotopic abundance of the isotope with the atomic mass of the isotope.
Example:
Calculate the atomic weight for the element lithium. Lithium-6 has an atom percent abundance of 7.5% and an atomic mass of 6.015122 amu. Lithium-7 has an atomic abundance of 92.5% and an atomic mass of 7.016003 amu.
Solution:
The other common measurement of isotopic abundance is weight percent (w/o). Weight percent is the percent weight of an element that is a particular isotope. For example, if a sample of material contained 100 kg of uranium that was 28 w/o uranium-235, then 28 kg of uranium-235 was present in the sample.
Enriched and Depleted Uranium
Natural uranium mined from the earth contains the isotopes uranium-238, uranium-235 and uranium-234. The majority (99.2745%) of all the atoms in natural uranium are uranium-238. Most of the remaining atoms (0.72%) are uranium-235, and a slight trace (0.0055%) are uranium-234. Although all isotopes of uranium have similar chemical properties, each of the isotopes has significantly different nuclear properties. For reasons that will be discussed in later modules, the isotope uranium-235 is usually the desired material for use in reactors.
A vast amount of equipment and energy are expended in processes that separate the isotopes of uranium (and other elements). The details of these processes are beyond the scope of this module. These processes are called enrichment processes because they selectively increase the proportion of a particular isotope. The enrichment process typically starts with feed material that has the proportion of isotopes that occur naturally. The process results in two types of
CHART OF THE NUCLIDES DOE-HDBK-1019/1-93 Atomic and Nuclear Physics
In the case of uranium, the natural uranium ore is 0.72 a/o uranium-235. The desired outcome of the enrichment process is to produce enriched uranium. Enriched uranium is defined as uranium in which the isotope uranium-235 has a concentration greater than its natural value. The enrichment process will also result in the byproduct of depleted uranium. Depleted uranium is defined as uranium in which the isotope uranium-235 has a concentration less than its natural value. Although depleted uranium is referred to as a by-product of the enrichment process, it does have uses in the nuclear field and in commercial and defense industries.
Fixture checkpoint marker-0167-xray is recorded on fixture page 0167.
Summary
The important information in this chapter is summarized below.
Chart of the Nuclides Summary
Enriched uranium is uranium in which the isotope uranium-235 has a concentration greater than its natural value of 0.7%.
Depleted uranium is uranium in which the isotope uranium-235 has a concentration less than its natural value of 0.7%.
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Atomic and Nuclear Physics DOE-HDBK-1019/1-93 MASS DEFECT AND BINDING ENERGY
MASS DEFECT AND BINDING ENERGY
The separate laws of Conservation of Mass and Conservation of Energy are not applied strictly on the nuclear level. It is possible to convert between mass and energy. Instead of two separate conservation laws, a single conservation law states that the sum of mass and energy is conserved. Mass does not magically appear and disappear at random. A decrease in mass will be accompanied by a

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EO 1.7 DEFINE the following terms:
a. Mass defect
b. Binding energy
EO 1.8 Given the atomic mass for a nuclide and the atomic masses of
a neutron, proton, and electron, CALCULATE the mass defect
and binding energy of the nuclide.
Mass Defect
Careful measurements have shown that the mass of a particular atom is always slightly less than the sum of the masses of the individual neutrons, protons, and electrons of which the atom consists. The difference between the mass of the atom and the sum of the masses of its parts is called the mass defect (∆m). The mass defect can be calculated using Equation (1-1). In calculating the mass defect it is important to use the full accuracy of mass measurements because the difference in mass is small compared to the mass of the atom. Rounding off the masses of atoms and particles to three or four significant digits prior to the calculation will result in a calculated mass defect of zero.
∆m = [Z(mp + me) + (A-Z)mn] - matom (1-1)
where:
∆m = mass defect (amu)
mp = mass of a proton (1.007277 amu)
mn = mass of a neutron (1.008665 amu)
me = mass of an electron (0.000548597 amu)
matom = mass of nuclide A
ZX (amu)
Z = atomic number (number of protons)
A = mass number (number of nucleons)

Example:
Calculate the mass defect for lithium-7. The mass of lithium-7 is 7.016003 amu.
Solution:
∆ m Z m
p
me A Z mn matom
∆ m 3 1.007826 amu 7 3 1.008665 amu 7.016003 amu
∆ m 0.0421335 amu
Binding Energy
The loss in mass, or mass defect, is due to the conversion of mass to binding energy when the nucleus is formed. Binding energy is defined as the amount of energy that must be supplied to a nucleus to completely separate its nuclear particles (nucleons). It can also be understood as
Fixture checkpoint marker-0168-xray is recorded on fixture page 0168.
the amount of energy that would be released if the nucleus was formed from the separate particles. Binding energy is the energy equivalent of the mass defect. Since the mass defect was converted to binding energy (BE) when the nucleus was formed, it is possible to calculate the binding energy using a conversion factor derived by the mass-energy relationship from Einstein's Theory of Relativity.
Einstein's famous equation relating mass and energy is E = mc2 where c is the velocity of light (c = 2.998 x 108 m/sec). The energy equivalent of 1 amu can be determined by inserting this quantity of mass into Einstein's equation and applying conversion factors.

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1 amu
1.6606 x 10 27 kg
1 amu
2.998 x 108 m
sec
2
1 N
1
kg m
sec2
1 J
1 N m
1.4924 x 10 10 J 1 MeV
1.6022 x 10 13 J
931.5 MeV
Conversion Factors:
Fixture checkpoint marker-0169-xray is recorded on fixture page 0169.
1 amu = 1.6606 x 10 -27 kg
1 newton = 1 kg-m/sec2
1 joule = 1 newton-meter
1 MeV = 1.6022 x 10-13 joules
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Atomic and Nuclear Physics DOE-HDBK-1019/1-93 MASS DEFECT AND BINDING ENERGY
Since 1 amu is equivalent to 931.5 MeV of energy, the binding energy can be calculated using Equation (1-2).
(1-2)
B.E. ∆m
931.5 MeV
1 amu
Example:
Calculate the mass defect and binding energy for uranium-235. One uranium-235 atom has a mass of 235.043924 amu.
Solution:
Step 1: Calculate the mass defect using Equation (1-1).
∆m Z m
p
me A Z mn matom
∆m 92 1.007826 amu 235 92 1.008665 amu 235.043924 amu
∆m 1.91517 amu
Step 2: Use the mass defect and Equation (1-2) to calculate the binding energy.
B.E. ∆m
931.5 MeV
1 amu
1.91517 amu
931.5 MeV
1 amu
1784 MeV
Energy Levels of Atoms

are more tightly bound in the atom than others. For example, only 7.38 eV is required to remove the outermost electron from a lead atom, while 88,000 eV is required to remove the innermost electron. The process of removing an electron from an atom is called ionization, and the energy required to remove the electron is called the ionization energy.

In a neutral atom (number of electrons = Z) it is possible for the electrons to be in a variety of different orbits, each with a different energy level. The state of lowest energy is the one in which the atom is normally found and is called the ground state. When the atom possesses more energy than its ground state energy, it is said to be in an excited state.

An atom cannot stay in the excited state for an indefinite period of time. An excited atom will eventually transition to either a lower-energy excited state, or directly to its ground state, by emitting a discrete bundle of electromagnetic energy called an x-ray. The energy of the x-ray will be equal to the difference between the energy levels of the atom and will typically range from several eV to 100,000 eV in magnitude.

Energy Levels of the Nucleus

The nucleons in the nucleus of an atom, like the electrons that circle the nucleus, exist in shells that correspond to energy states. The energy shells of the nucleus are less defined and less understood than those of the electrons. There is a state of lowest energy (the ground state) and discrete possible excited states for a nucleus. Where the discrete energy states for the electrons of an atom are measured in eV or keV, the energy levels of the nucleus are considerably greater and typically measured in MeV.

A nucleus that is in the excited state will not remain at that energy level for an indefinite period.

Fixture checkpoint marker-0170-xray is recorded on fixture page 0170.

Like the electrons in an excited atom, the nucleons in an excited nucleus will transition towards their lowest energy configuration and in doing so emit a discrete bundle of electromagnetic radiation called a gamma ray (γ-ray). The only differences between x-rays and γ-rays are their energy levels and whether they are emitted from the electron shell or from the nucleus.

The ground state and the excited states of a nucleus can be depicted in a nuclear energy-level diagram. The nuclear energy-level diagram consists of a stack of horizontal bars, one bar for each of the excited states of the nucleus. The vertical distance between the bar representing an excited state and the bar representing the ground state is proportional to the energy level of the excited state with respect to the ground state. This difference in energy between the ground state and the excited state is called the excitation energy of the excited state. The ground state of a nuclide has zero excitation energy. The bars for the excited states are labeled with their respective energy levels. Figure 7 is the energy level diagram for nickel-60.

Figure 7 Energy Level Diagram - Nickel-60

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Summary
The important information in this chapter is summarized below.
Mass Defect and Binding Energy Summary
Mass defect is the difference between the mass of the atom and the sum of the masses of its constituent parts.
Binding energy is the amount of energy that must be supplied to a nucleus to completely separate its nuclear particles. Binding energy is the energy equivalent of the mass defect.
Mass defect can be calculated by using the equation below.
$\Delta m = [Z(mp + me) + (A-Z)mn] - matom$
Binding energy can be calculated by multiplying the mass defect by the factor of 931.5 MeV per amu.

MODES OF RADIOACTIVE DECAY
Most atoms found in nature are stable and do not emit particles or energy that change form over time. Some atoms, however, do not have stable nuclei. These atoms emit radiation in order to achieve a more stable configuration.
EO 2.1 DESCRIBE the following processes:
a. Alpha decay d. Electron capture
b. Beta-minus decay e. Internal conversions
c. Beta-plus decay f. Isomeric transitions
EO 2.2 Given a Chart of the Nuclides, WRITE the radioactive decay chain for a nuclide.
EO 2.3 EXPLAIN why one or more gamma rays typically accompany particle emission.
EO 2.4 Given the stability curve on the Chart of the Nuclides, DETERMINE the type of radioactive decay that the nuclides in each region of the chart will typically undergo.
Stability of Nuclei
Fixture checkpoint marker-0171-xray is recorded on fixture page 0171.
As mass numbers become larger, the ratio of neutrons to protons in the nucleus becomes larger for the stable nuclei. Non-stable nuclei may have an excess or deficiency of neutrons and undergo a transformation process known as beta (β) decay. Non-stable nuclei can also undergo a variety of other processes such as alpha (α) or neutron (n) decay. As a result of these decay processes, the final nucleus is in a more stable or more tightly bound configuration.
Natural Radioactivity
In 1896, the French physicist Becquerel discovered that crystals of a uranium salt emitted rays that were similar to x-rays in that they were highly penetrating, could affect a photographic plate, and induced electrical conductivity in gases. Becquerel's discovery was followed in 1898 by the identification of two other radioactive elements, polonium and radium, by Pierre and Marie Curie.

Heavy elements, such as uranium or thorium, and their unstable decay chain elements emit radiation in their naturally occurring state. Uranium and thorium, present since their creation at the beginning of geological time, have an extremely slow rate of decay. All naturally occurring nuclides with atomic numbers greater than 82 are radioactive.

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Whenever a nucleus can attain a more stable (i.e., more tightly bound) configuration by emitting radiation, a spontaneous disintegration process known as radioactive decay or nuclear decay may occur. In practice, this "radiation" may be electromagnetic radiation, particles, or both.
Detailed studies of radioactive decay and nuclear reaction processes have led to the formulation of useful conservation principles. The four principles of most interest in this module are discussed below.
Conservation of electric charge implies that charges are neither created nor destroyed. Single positive and negative charges may, however, neutralize each other. It is also possible for a neutral particle to produce one charge of each sign.
Conservation of mass number does not allow a net change in the number of nucleons. However, the conversion of a proton to a neutron and vice versa is allowed.
Conservation of mass and energy implies that the total of the kinetic energy and the energy equivalent of the mass in a system must be conserved in all decays and reactions. Mass can be converted to energy and energy can be converted to mass, but the sum of mass and energy must be constant.
Conservation of momentum is responsible for the distribution of the available kinetic energy among product nuclei, particles, and/or radiation. The total amount is the same before and after the reaction even though it may be distributed differently among entirely different nuclides and/or particles.
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■ ■ ■ ■ ■
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Alpha Decay (■)
Alpha decay is the emission of alpha particles (helium nuclei) which may be represented as either $^{4}_{2}He$ or ■. When an unstable nucleus ejects an alpha particle, the atomic number is reduced by 2 $^{4}_{2}$ 4
and the mass number decreased by 4. An example is uranium-234 which decays by the ejection of an alpha particle accompanied by the emission of a 0.068 MeV gamma.
The combined kinetic energy of the daughter nucleus (Thorium-230) and the ■ particle is designated as KE. The sum of the KE and the gamma energy is equal to the difference in mass between the original nucleus (Uranium-234) and the final particles (equivalent to the binding energy released, since ■m = BE). The alpha particle will carry off as much as 98% of the kinetic energy and, in most cases, can be considered to carry off all the kinetic energy.
Beta Decay (■)
Beta decay is the emission of electrons of nuclear rather than orbital origin. These particles are
Fixture checkpoint marker-0172-xray is recorded on fixture page 0172.
electrons that have been expelled by excited nuclei and may have a charge of either sign.
If both energy and momentum are to be conserved, a third type of particle, the neutrino, ■, must be involved. The neutrino is associated with positive electron emission, and its antiparticle, the antineutrino, , is emitted with a negative electron. These uncharged particles have only the weakest interaction with matter, no mass, and travel at the speed of light. For all practical purposes, they pass through all materials with so few interactions that the energy they possess cannot be recovered. The neutrinos and antineutrinos are included here only because they carry a portion of the kinetic energy that would otherwise belong to the beta particle, and therefore, must be considered for energy and momentum to be conserved. They are normally ignored since they are not significant in the context of nuclear reactor applications.

to a proton, thus increasing the atomic number by one and leaving the mass number unchanged. This is a common mode of decay for nuclei with an excess of neutrons, such as fission fragments below and to the right of the neutron-proton stability curve (refer to Figure 6). An example of a typical beta minus-decay reaction is shown below.

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Figure 9 Orbital Electron Capture

Positively charged electrons (beta-plus) are known as positrons. Except for sign, they are nearly identical to their negatively charged cousins. When a positron, represented as e, ■, or simply as +1 +1 0 0

Fixture checkpoint marker-0173-xray is recorded on fixture page 0173.

e or ■ , is ejected from the nucleus, the atomic number is decreased by one and the mass number + + remains unchanged. A proton has been converted to a neutron. An example of a typical positron (beta-plus) decay is shown below.

Electron Capture (EC, K-capture)

Nuclei having an excess of protons may capture an electron from one of the inner orbits which immediately combines with a proton in the nucleus to form a neutron. This process is called electron capture (EC). The electron is normally captured from the innermost orbit (the K-shell), and, consequently, this process is sometimes called K-capture. The following example depicts electron capture.

A neutrino is formed at the same time that the neutron is formed, and energy carried off by it serves to conserve momentum. Any energy that is available due to the atomic mass of the product being appreciably less than that of the parent will appear as gamma radiation. Also, there will always be characteristic x-rays given off when an electron from one of the higher energy shells moves in to fill the vacancy in the K-shell.

Electron capture is shown graphically in Figure 8.

Electron capture and positron emission result in the production of the same daughter product, and they exist as competing processes.

For positron emission to occur, however, the mass of the daughter product must be less than the mass of the parent by an amount equal to at least twice the mass of an electron. This mass difference between the parent and daughter is necessary to account for two items present in the parent but not in the daughter. One item is the positron ejected from the nucleus of the parent. The other item is that the daughter product has one less orbital electron than the parent. If this requirement is not met,

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Gamma Emission (■)
Gamma radiation is a high-energy electromagnetic radiation that originates in the nucleus. It is emitted in the form of photons, discrete bundles of energy that have both wave and particle properties. Often a daughter nuclide is left in an excited state after a radioactive parent nucleus undergoes a transformation by alpha decay, beta decay, or electron capture. The nucleus will drop to the ground state by the emission of gamma radiation.
Internal Conversion
The usual method for an excited nucleus to go from the excited state to the ground state is by emission of gamma radiation. However, in some cases the gamma ray (photon) emerges from the nucleus only to interact with one of the innermost orbital electrons and, as a result, the energy of the photon is transferred to the electron. The gamma ray is then said to have undergone internal conversion. The conversion electron is ejected from the atom with kinetic energy equal to the gamma energy minus the binding energy of the orbital electron. An orbital electron then drops to a lower energy state to fill the vacancy, and this is accompanied by the emission of characteristic x-rays.
Isomers and Isomeric Transition
Fixture checkpoint marker-0174-xray is recorded on fixture page 0174.
Isomeric transition commonly occurs immediately after particle emission; however, the nucleus may remain in an excited state for a measurable period of time before dropping to the ground state at its own characteristic rate. A nucleus that remains in such an excited state is known as a nuclear isomer because it differs in energy and behavior from other nuclei with the same atomic number and mass number. The decay of an excited nuclear isomer to a lower energy level is called an isomeric transition. It is also possible for the excited isomer to decay by some alternate means, for example, by beta emission.
An example of gamma emission accompanying particle emission is illustrated by the decay of nitrogen-16 below.
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Decay Chains
When an unstable nucleus decays, the resulting daughter nucleus is not necessarily stable. The nucleus resulting from the decay of a parent is often itself unstable, and will undergo an additional

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It is possible to trace the steps of an unstable atom as it goes through multiple decays trying to achieve stability. The list of the original unstable nuclide, the nuclides that are involved as intermediate steps in the decay, and the final stable nuclide is known as the decay chain. One common method for stating the decay chain is to state each of the nuclides involved in the standard X format. Arrows are used between nuclides to indicate where decays occur, with the type of decay A Z
indicated above the arrow and the half-life below the arrow. The half-life for decay will be discussed in the next chapter.
Example:
Write the decay chains for rubidium-91 and actinium-215. Continue the chains until a stable nuclide or a nuclide with a half-life greater than 1 x 10 years is reached. 6
Solution:
Predicting Type of Decay
Radioactive nuclides tend to decay in a way that results in a daughter nuclide that lies closer to the line of stability. Due to this, it is possible to predict the type of decay that a nuclide will undergo based on its location relative to the line of stability on the Chart of the Nuclides.
Figure 9 illustrates the type of decay nuclides in different regions of the chart will typically undergo. Nuclides that are below and to the right of the line of stability will usually undergo ■ decay. - Nuclides that are above and to the left of the line of stability will usually undergo either ■ decay or + electron capture. Most nuclides that will undergo ■ decay are found in the upper right hand region of the chart. These are general rules that have many exceptions, especially in the region of the heavy nuclides.
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Figure 9 Types of Radioactive Decay Relative to the Line of Stability
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Atomic and Nuclear Physics DOE-HDBK-1019/1-93 MODES OF RADIOACTIVE DECAY
Summary
The important information in this chapter is summarized below.
Fixture checkpoint marker-0175-xray is recorded on fixture page 0175.
Modes of Radioactive Decay Summary
Alpha decay is the emission of an alpha particle (2 protons and 2 neutrons) from an unstable nucleus. The daughter nuclide has an atomic number 2 less than the parent nuclide and a mass number 4 less than the parent nuclide. The daughter nucleus commonly releases its excitation energy by gamma emission.
Beta-minus decay effectively converts a neutron to a proton and an electron, which is immediately ejected from the nucleus. The daughter nuclide has its atomic number increased by 1 and the same mass number compared to the parent.
Beta-plus decay effectively converts a proton to a neutron and a positron, which is immediately ejected from the nucleus. The daughter nuclide has its atomic number decreased by 1 and the same mass number compared to the parent.
In electron capture, the nucleus absorbs an electron from the innermost orbit. This electron combines with a proton to form a neutron.
Internal conversion occurs when a gamma ray, emitted by the nucleus as it goes from the excited state to the ground state, interacts with one of the innermost electrons of the same atom. The electron is ejected from the atom.
An isomeric transition is the decay of an excited nucleus to a lower-energy level by the emission of a gamma ray.

as it tries to achieve stability.
Many modes of radioactive decay result in a daughter nuclide that has an energy level above the ground state. This excitation energy is usually released immediately in the form of a gamma ray.
The type of decay that a nuclide will typically undergo can be determined by its relationship to the line of stability on the Chart of the Nuclides. Nuclides that lie below and to the right of the line of stability will typically beta minus decay. Nuclides above and to the left of the line will typically either beta plus decay or electron capture. Most alpha emitters are found in the upper, right-hand corner of the chart.

RADIOACTIVITY
The rate at which a sample of radioactive material decays is not constant. As individual atoms of the material decay, there are fewer of those types of atoms remaining. Since the rate of decay is directly proportional to the number of atoms, the rate of decay will decrease as the number of atoms decreases.
EO 2.5 DEFINE the following terms:
a. Radioactivity d. Radioactive decay constant
b. Curie e. Radioactive half-life
c. Becquerel
EO 2.6 Given the number of atoms and either the half-life or decay constant of a nuclide, CALCULATE the activity.
EO 2.7 Given the initial activity and the decay constant of a nuclide, CALCULATE the activity at any later time.
EO 2.8 CONVERT between the half-life and decay constant for a nuclide.
EO 2.9 Given the Chart of the Nuclides and the original activity, PLOT the radioactive decay curve for a nuclide on either linear or semi-log coordinates.
EO 2.10 DEFINE the following terms:
a. Radioactive equilibrium
b. Transient radioactive equilibrium
Radioactive Decay Rates
Radioactivity is the property of certain nuclides of spontaneously emitting particles or gamma
Fixture checkpoint marker-0176-xray is recorded on fixture page 0176.
radiation. The decay of radioactive nuclides occurs in a random manner, and the precise time at which a single nucleus will decay cannot be determined. However, the average behavior of a very large sample can be predicted accurately by using statistical methods. These studies have revealed that there is a certain probability that in a given time interval a certain fraction of the nuclei within a sample of a particular nuclide will decay. This probability per unit time that an atom of a nuclide will decay is known as the radioactive decay constant, λ. The units for the decay constant are inverse time such as 1/second, 1/minute, 1/hour, or 1/year. These decay constant units can also be expressed as second-1, minute-1, hour-1, and year-1.

■■

measured in the number of disintegrations that occur per second. For a sample containing millions of atoms, the activity is the product of the decay constant and the number of atoms present in the sample.

The relationship between the activity, number of atoms, and decay constant is shown in Equation (1-3).

A = ■ N (1-3)

where:

A = Activity of the nuclide (disintegrations/second)

■ = decay constant of the nuclide (second)-1

N = Number of atoms of the nuclide in the sample

Since ■ is a constant, the activity and the number of atoms are always proportional.

Units of Measurement for Radioactivity

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Two common units to measure the activity of a substance are the curie (Ci) and becquerel (Bq). A curie is a unit of measure of the rate of radioactive decay equal to 3.7 x 10 disintegrations 10 per second. This is approximately equivalent to the number of disintegrations that one gram of radium-226 will undergo in one second. A becquerel is a more fundamental unit of measure of radioactive decay that is equal to 1 disintegration per second. Currently, the curie is more widely used in the United States, but usage of the becquerel can be expected to broaden as the metric system slowly comes into wider use. The conversion between curies and becquerels is shown below.

1 curie = 3.7 x 10 becquerels 10

Variation of Radioactivity Over Time

The rate at which a given radionuclide sample decays is stated in Equation (1-3) as being equal to the product of the number of atoms and the decay constant. From this basic relationship it is possible to use calculus to derive an expression which can be used to calculate how the number of atoms present will change over time. The derivation is beyond the scope of this text, but Equation (1-4) is the useful result.

(1-4)

where:

N = number of atoms present at time t

N = number of atoms initially present o

■ = decay constant (time)-1

t = time

A Ao

e

t

A Aoe

t

A

Ao

e

t

ln A

Ao

t

t

ln

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Ao
t
1/2
ln
1
2
t
1/2
ln 2 0.693
RADIOACTIVITY DOE-HDBK-1019/1-93 Atomic and Nuclear Physics
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Since the activity and the number of atoms are always proportional, they may be used interchangeably to describe any given radionuclide population. Therefore, the following is true.
(1-5)
where:
A = activity present at time t
A = activity initially present o
= decay constant (time)
-1
Fixture checkpoint marker-0178-xray is recorded on fixture page 0178.
t = time
Radioactive Half-Life
One of the most useful terms for estimating how quickly a nuclide will decay is the radioactive half-life. The radioactive half-life is defined as the amount of time required for the activity to decrease to one-half of its original value. A relationship between the half-life and decay constant can be developed from Equation (1-5). The half-life can be calculated by solving Equation (1-5) for the time, t, when the current activity, A, equals one-half the initial activity A .
o
First, solve Equation (1-5) for t.
If A is equal to one-half of Ao, then A/Ao is equal to one-half. Substituting this in the equation above yields an expression for t .
1/2
(1-6)
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The basic features of decay of a radionuclide sample are shown by the graph in Figure 10.
Figure 10 Radioactive Decay as a Function of Time in Units of Half-Life
Assuming an initial number of atoms No, the population, and consequently, the activity may be noted to decrease by one-half of this value in a time of one half-life. Additional decreases occur so that whenever one half-life elapses, the number of atoms drops to one-half of what its value was at the beginning of that time interval. After five half-lives have elapsed, only 1/32, or 3.1%, of the original number of atoms remains. After seven half-lives, only 1/128, or 0.78%, of the atoms remains. The number of atoms existing after 5 to 7 half-lives can usually be assumed to be negligible. The Chemistry Fundamentals Handbook contains additional information on calculating the number of atoms contained within a sample.
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Example:
A sample of material contains 20 micrograms of californium-252.
Californium-252 has a half-life of 2.638 years.
Calculate:
(a) The number of californium-252 atoms initially present
(b) The activity of the californium-252 in curies
(c) The number of californium-252 atoms that will remain in 12 years
(d) The time it will take for the activity to reach 0.001 curies
Solution:
(a) The number of atoms of californium-252 can be determined as below.
(b) First, use Equation (1-6) to calculate the decay constant.
Use this value for the decay constant in Equation (1-3) to determine the activity.
N No
e
t
4.78 x 1016 e
(0.263/yr) (12 yr)
2.04 x 1015
A Aoe
t
A
Ao
e
t
ln A
Ao
Fixture checkpoint marker-0179-xray is recorded on fixture page 0179.
t
t
ln
A
Ao
t
ln
0.001 Ci
0.0108 Ci
0.263 year 1
t 9.05 years
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(c) The number of californium atoms that will remain in 12 years can be calculated from Equation (1-4).
(d) The time that it will take for the activity to reach 0.001 Ci can be determined from Equation (1-5). First, solve Equation (1-5) for time.
Inserting the appropriate values in the right side of this equation will result in the required time.
Plotting Radioactive Decay
It is useful to plot the activity of a nuclide as it changes over time. Plots of this type can be

showing activity on either a linear or a logarithmic scale. The decay of the activity of a single nuclide on a logarithmic scale will plot as a straight line because the decay is exponential.
t
1/2
0.693
0.693
t
1/2
0.693
7.13 seconds
0.0972 second 1
A Aoe
t
RADIOACTIVITY DOE-HDBK-1019/1-93 Atomic and Nuclear Physics
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Example:
Plot the radioactive decay curve for nitrogen-16 over a period of 100 seconds. The initial activity is 142 curies and the half-life of nitrogen-16 is 7.13 seconds. Plot the curve on both linear rectangular coordinates and on a semi-log scale.
Solution:
First, use Equation (1-6) to calculate the decay constant corresponding to a half-life of 7.13 seconds.
Use the decay constant determined above to calculate the activity at various times using Equation (1-5).
Time Activity
0 seconds 142 Ci
20 seconds 20.3 Ci
40 seconds 2.91 Ci
60 seconds 0.416 Ci
80 seconds 0.0596 Ci
100 seconds 0.00853 Ci
■ ■
■ ■
Fixture checkpoint marker-0180-xray is recorded on fixture page 0180.
■ ■
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Figure 11 Linear and Semi-log Plots of Nitrogen-16 Decay
Plotting the data points calculated above on both linear and semilog scales results in the graphs shown in Figure 11.
If a substance contains more than one radioactive nuclide, the total activity is the sum of the individual activities of each nuclide. As an example, consider a sample of material that contained 1 x 10 atoms of iron-59 that has a half-life of 44.51 days (■ = 1.80 x 10 sec), 6 -7 -1
1 x 10 atoms of manganese-54 that has a half-life of 312.2 days (■ = 2.57 x 10 sec), and 1 6 -8 -1
x 10 atoms of cobalt-60 that has a half-life of 1925 days (■ = 4.17 x 10 sec). 6 -9 -1
The initial activity of each of the nuclides would be the product of the number of atoms and the decay constant.

Figure 12 Combined Decay of Iron-56, Manganese-54, and Cobalt-60

Plotting the manner in which the activities of each of the three nuclides decay over time demonstrates that initially the activity of the shortest-lived nuclide (iron-59) dominates the total activity, then manganese-54 dominates. After almost all of the iron and manganese have decayed away, the only contributor to activity will be the cobalt-60. A plot of this combined decay is shown in Figure 12.

Radioactive Equilibrium

Radioactive equilibrium exists when a radioactive nuclide is decaying at the same rate at which it is being produced. Since the production rate and decay rate are equal, the number of atoms present remains constant over time.

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An example of radioactive equilibrium is the concentration of sodium-24 in the coolant circulating through a sodium-cooled nuclear reactor. Assume that the sodium-24 is being produced at a rate of 1 x 10 atoms per second. If the sodium-24 were stable and did not decay, 6 the amount of sodium-24 present after some period of time could be calculated by multiplying the production rate by the amount of time. Plotting the amount of material present would result in the graph in Figure 13.

However, sodium-24 is not stable, and it decays with a half-life of 14.96 hours. If no sodium-24 is present initially and production starts at a rate of 1 x 10 atoms per second, the rate 6 of decay will initially be zero because there is no sodium-24 present to decay. The rate of decay of sodium-24 will increase as the amount of sodium-24 increases.

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Figure 13 Cumulative Production of Sodium-24 Over Time

The amount of sodium-24 present will initially increase rapidly, then it will increase at a continually decreasing rate until the rate of decay is equal to the rate of production. It is possible to calculate how much sodium-24 will be present at equilibrium by setting the production rate (R) equal to the decay rate (■ N).

where:

R = production rate (atoms/second)
■ = decay constant (second)-1
N = number of atoms

It is possible to calculate the equilibrium value for sodium-24 being produced at a rate of 1 x 10 atoms/second. 6

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Figure 14 Approach of Sodium-24 to Equilibrium

The development of the equation to calculate how the amount of sodium-24 changes over time as it approaches the equilibrium value is beyond the scope of this handbook. However, the

This equation can be used to calculate the values of the amount of sodium-24 present at different times. As the time increases, the exponential term approaches zero, and the number of atoms present will approach R/■. A plot of the approach of sodium-24 to equilibrium is shown in Figure 14.

Transient Radioactive Equilibrium

Transient radioactive equilibrium occurs when the parent nuclide and the daughter nuclide decay at essentially the same rate.

For transient equilibrium to occur, the parent must have a long half-life when compared to the daughter. An example of this type of compound decay process is barium-140, which decays by beta emission to lanthanum-140, which in turn decays by beta emission to stable cerium-140.

Figure 15 Transient Equilibrium in the Decay of Barium-140

The decay constant for barium-140 is considerably smaller than the decay constant for lanthanum-140. Remember that the rate of decay of both the parent and daughter can be represented as ■N. Although the decay constant for barium-140 is smaller, the actual rate of decay (■N) is initially larger than that of lanthanum-140 because of the great difference in their

Fixture checkpoint marker-0182-xray is recorded on fixture page 0182.

initial concentrations. As the concentration of the daughter increases, the rate of decay of the daughter will approach and eventually match the decay rate of the parent. When this occurs, they are said to be in transient equilibrium. A plot of the barium-lanthanum-cerium decay chain reaching transient equilibrium is shown in Figure 15.

Secular equilibrium occurs when the parent has an extremely long half-life. In the long decay chain for a naturally radioactive element, such as thorium-232, where all of the elements in the chain are in secular equilibrium, each of the descendants has built up to an equilibrium amount and all decay at the rate set by the original parent. The only exception is the final stable element on the end of the chain. Its number of atoms is constantly increasing.

Summary

The important information in this chapter is summarized on the following page.

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■■
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Radioactivity Summary

& Radioactivity is the decay of unstable atoms by the emission of particles and electromagnetic radiation.

& A curie (Ci) is a unit of radioactivity equal to 3.7 x 10 disintegrations per 10 second.

& A becquerel (Bq) is a unit of radioactivity equal to 1 disintegration per second.

& The radioactive decay constant (■) is the probability per unit time that an atom will decay.

& The radioactive half-life is the amount of time required for the activity to decrease to one-half its original value.

& The activity of a substance can be calculated from the number of atoms and the decay constant based on the equation below.

& The amount of activity remaining after a particular time can be calculated from the equation below.

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& Plots of radioactive decay can be useful to describe the variation of activity over time. If decay is plotted using semi-log scale the plot results in a straight line.
& Radioactive equilibrium exists when the production rate of a material is equal to the removal rate.
& Transient radioactive equilibrium exists when the parent nuclide and the daughter nuclide decay at essentially the same rate. This occurs only when the parent has a long half-life compared to the daughter.
Atomic and Nuclear Physics DOE-HDBK-1019/1-93 NEUTRON INTERACTIONS
NEUTRON INTERACTIONS
Neutrons can cause many different types of interactions. The neutron may simply scatter off the nucleus in two different ways, or it may actually be absorbed into the nucleus. If a neutron is absorbed into the nucleus, it may result in the emission of a gamma ray or a subatomic particle, or it may cause the nucleus to fission.
EO 3.1 DESCRIBE the following scattering interactions between a neutron and a nucleus:
a. Elastic scattering
b. Inelastic scattering
EO 3.2 STATE the conservation laws that apply to an elastic collision between a neutron and a nucleus.
EO 3.3 DESCRIBE the following reactions where a neutron is absorbed in a nucleus:
a. Radiative capture
b. Particle ejection
Fixture checkpoint marker-0183-xray is recorded on fixture page 0183.
Scattering
A neutron scattering reaction occurs when a nucleus, after having been struck by a neutron, emits a single neutron. Despite the fact that the initial and final neutrons do not need to be (and often are not) the same, the net effect of the reaction is as if the projectile neutron had merely "bounced off," or scattered from, the nucleus. The two categories of scattering reactions, elastic and inelastic scattering, are described in the following paragraphs.
Elastic Scattering
In an elastic scattering reaction between a neutron and a target nucleus, there is no energy transferred into nuclear excitation. Momentum and kinetic energy of the "system" are conserved although there is usually some transfer of kinetic energy from the neutron to the target nucleus. The target nucleus gains the amount of kinetic energy that the neutron loses.
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mn
vn,i mTvT,i mnvn,f mTvT,f
1
2
m v
2
1
2
mn
v
2

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1
2
mT
v
2
T,i
1
2
mn
v
2
n,f
1
2
mT
v
2
T,f
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Figure 16 Elastic Scattering
Figure 16 illustrates the process of elastic scattering of a neutron off a target nucleus. In the elastic scattering reaction, the conservation of momentum and kinetic energy is represented by the equations below.
Conservation of momentum (mv)
Conservation of kinetic energy
where:
mn = mass of the neutron
m = mass of the target nucleus T
vn,i = initial neutron velocity
vn,f = final neutron velocity
Fixture checkpoint marker-0184-xray is recorded on fixture page 0184.
vT,i = initial target velocity
vT,f = final target velocity
Atomic and Nuclear Physics DOE-HDBK-1019/1-93 NEUTRON INTERACTIONS
Elastic scattering of neutrons by nuclei can occur in two ways. The more unusual of the two interactions is the absorption of the neutron, forming a compound nucleus, followed by the re-emission of a neutron in such a way that the total kinetic energy is conserved and the nucleus returns to its ground state. This is known as resonance elastic scattering and is very dependent upon the initial kinetic energy possessed by the neutron. Due to formation of the compound nucleus, it is also referred to as compound elastic scattering. The second, more usual method, is termed potential elastic scattering and can be understood by visualizing the neutrons and nuclei to be much like billiard balls with impenetrable surfaces. Potential scattering takes place with incident neutrons that have an energy of up to about 1 MeV. In potential scattering, the neutron does not actually touch the nucleus and a compound nucleus is not formed. Instead, the neutron is acted on and scattered by the short range nuclear forces when it approaches close enough to the nucleus.
Inelastic Scattering

compound nucleus. The compound nucleus will then emit a neutron of lower kinetic energy which leaves the original nucleus in an excited state. The nucleus will usually, by one or more gamma emissions, emit this excess energy to reach its ground state. Figure 17 shows the process of inelastic scattering.

For the nucleus that has reached its ground state, the sum of the kinetic energy of the exit

Figure 17 Inelastic Scattering

neutron, the target nucleus, and the total gamma energy emitted is equal to the initial kinetic energy of the incident neutron.

1
0
n
238
92
U
239
92
U
239
92
U
0
0
1
0
n
10
5
B
11
5
B
7
3
Li
4
2

Fixture checkpoint marker-0185-xray is recorded on fixture page 0185.

Absorption Reactions

Most absorption reactions result in the loss of a neutron coupled with the production of a charged particle or gamma ray. When the product nucleus is radioactive, additional radiation is emitted at some later time. Radiative capture, particle ejection, and fission are all categorized as absorption reactions and are briefly described below.

Radiative Capture

In radiative capture the incident neutron enters the target nucleus forming a compound nucleus. The compound nucleus then decays to its ground state by gamma emission. An example of a

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Particle Ejection
In a particle ejection reaction the incident particle enters the target nucleus forming a compound nucleus. The newly formed compound nucleus has been excited to a high enough energy level to cause it to eject a new particle while the incident neutron remains in the nucleus. After the new particle is ejected, the remaining nucleus may or may not exist in an excited state depending upon the mass-energy balance of the reaction. An example of a particle ejection reaction is shown below.
Fission
One of the most important interactions that neutrons can cause is fission, in which the nucleus that absorbs the neutron actually splits into two similarly sized parts. Fission will be discussed in detail in the next chapter.
Atomic and Nuclear Physics DOE-HDBK-1019/1-93 NEUTRON INTERACTIONS
Summary
The important information in this chapter is summarized below.
Neutron Interactions Summary
Fixture checkpoint marker-0186-xray is recorded on fixture page 0186.
Interactions where a neutron scatters off a target nucleus are either elastic or inelastic. In elastic scattering, kinetic energy and momentum are conserved and no energy is transferred into excitation energy of the target nucleus. In inelastic scattering, some amount of kinetic energy is transferred into excitation energy of the target nucleus.
The conservation principles that apply to an elastic collision are conservation of kinetic energy and conservation of momentum.
Radiative capture is the absorption of a neutron by the target nucleus, resulting in an excited nucleus which subsequently (typically within a small fraction of a second) releases its excitation energy in the form of a gamma ray.
Particle ejection occurs when a neutron is absorbed by a target nucleus, resulting in the formation of a compound nucleus. The compound nucleus immediately ejects a particle (for example, alpha or proton).
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1
0
n
235
92
U
236
92
U
140
55
Cs
93
37
Rb 3
1
0
n

NUCLEAR FISSION
Nuclear fission is a process in which an atom splits and releases energy, fission products, and neutrons. The neutrons released by fission can, in turn, cause the fission of other atoms.
EO 4.1 EXPLAIN the fission process using the liquid drop model of a nucleus.
EO 4.2 DEFINE the following terms:
a. Excitation energy (E_{exc})
b. Critical energy (E_{crit})
EO 4.3 DEFINE the following terms:
a. Fissile material
b. Fissionable material
c. Fertile material
EO 4.4 DESCRIBE the processes of transmutation, conversion, and breeding.
EO 4.5 DESCRIBE the curve of Binding Energy per Nucleon versus mass number and give a qualitative description of the reasons for its shape.
EO 4.6 EXPLAIN why only the heaviest nuclei are easily fissioned.
EO 4.7 EXPLAIN why uranium-235 fissions with thermal neutrons and uranium-238 fissions only with fast neutrons.
Fixture checkpoint marker-0187-xray is recorded on fixture page 0187.
Fission
In the fission reaction the incident neutron enters the heavy target nucleus, forming a compound nucleus that is excited to such a high energy level ($E > E_{crit}$) that the nucleus "splits" (E_{exc}) into two large fragments plus some neutrons. An example of a typical fission reaction is shown below.
A large amount of energy is released in the form of radiation and fragment kinetic energy.

Liquid Drop Model of a Nucleus
The nucleus is held together by the attractive nuclear force between nucleons, which was discussed in a previous chapter. The characteristics of the nuclear force are listed below.
(a) very short range, with essentially no effect beyond nuclear dimensions (10^{-13} cm)
(b) stronger than the repulsive electrostatic forces within the nucleus
(c) independent of nucleon pairing, in that the attractive forces between pairs of neutrons are no different than those between pairs of protons or a neutron and a proton
(d) saturable, that is, a nucleon can attract only a few of its nearest neighbors
One theory of fission considers the fissioning of a nucleus similar in some respects to the splitting of a liquid drop. This analogy is justifiable to some extent by the fact that a liquid drop is held together by molecular forces that tend to make the drop spherical in shape and that try to resist any deformation in the same manner as nuclear forces are assumed to hold the nucleus together. By considering the nucleus as a liquid drop, the fission process can be described.
Referring to Figure 18(A), the nucleus in the ground state is undistorted, and its attractive nuclear forces are greater than the repulsive electrostatic forces between the protons within the nucleus. When an incident particle (in this instance a neutron) is absorbed by the target nucleus, a

mass involved in the reaction and exists in an excited state. The excitation energy added to the compound nucleus is equal to the binding energy contributed by the incident particle plus the kinetic energy possessed by that particle. Figure 18(B) illustrates the excitation energy thus imparted to the compound nucleus, which may cause it to oscillate and become distorted. If the excitation energy is greater than a certain critical energy, the oscillations may cause the compound nucleus to become dumbbell-shaped. When this happens, the attractive nuclear forces (short-range) in the neck area are small due to saturation, while the repulsive electrostatic forces (long-range) are only slightly less than before. When the repulsive electrostatic forces exceed the attractive nuclear forces, nuclear fission occurs, as illustrated in Figure 18(C).

Figure 18 Liquid Drop Model of Fission

Critical Energy

The measure of how far the energy level of a nucleus is above its ground state is called the excitation energy (Eexc). For fission to occur, the excitation energy must be above a particular value for that nuclide. The critical energy (Ecrit) is the minimum excitation energy required for fission to occur.

Fissile Material

A fissile material is composed of nuclides for which fission is possible with neutrons of any energy level. What is especially significant about these nuclides is their ability to be fissioned with zero kinetic energy neutrons (thermal neutrons). Thermal neutrons have very low kinetic energy levels (essentially zero) because they are roughly in equilibrium with the thermal motion of surrounding materials. Therefore, in order to be classified as fissile, a material must be capable of fissioning after absorbing a thermal neutron. Consequently, they impart essentially no kinetic energy to the reaction. Fission is possible in these materials with thermal neutrons, since the change in binding energy supplied by the neutron addition alone is high enough to exceed the critical energy. Some examples of fissile nuclides are uranium-235, uranium-233, and plutonium-239.

Fixture checkpoint marker-0188-xray is recorded on fixture page 0188.

Fissionable Material

A fissionable material is composed of nuclides for which fission with neutrons is possible. All fissile nuclides fall into this category. However, also included are those nuclides that can be fissioned only with high energy neutrons. The change in binding energy that occurs as the result of neutron absorption results in a nuclear excitation energy level that is less than the required critical energy. Therefore, the additional excitation energy must be supplied by the kinetic energy of the incident neutron. The reason for this difference between fissile and fissionable materials is the so-called odd-even effect for nuclei. It has been observed that nuclei with even numbers of neutrons and/or protons are more stable than those with odd numbers. Therefore, adding a neutron to change a nucleus with an odd number of neutrons to a nucleus with an even number of neutrons produces an appreciably higher binding energy than adding a neutron to a nucleus already possessing an even number of neutrons. Some examples of nuclides requiring high energy neutrons to cause fission are thorium-232, uranium-238, and plutonium-240. Table 4 indicates the critical energy (Ecrit) and the binding energy change for an added neutron (BEn) to target nuclei of interest. For fission to be possible, the change in binding energy plus the kinetic energy must equal or exceed the critical energy ($\Delta BE + KE > Ecrit$).

TABLE 4

SYNTHETIC FIXTURE PAGE 0189 OF 1200

Target
Nucleus
Critical Energy
Ecrit
Binding Energy of
Last Neutron BEn
BEn - Ecrit
23
9
2
0Th 7.5 MeV 5.4 MeV -2.1 MeV
23
9
8
2U 7.0 MeV 5.5 MeV -1.5 MeV
23
9
5
2U 6.5 MeV 6.8 MeV +0.3 MeV
23
9
3
2U 6.0 MeV 7.0 MeV +1.0 MeV
23
9
9
4Pu 5.0 MeV 6.6 MeV +1.6 MeV
Uranium-235 fissions with thermal neutrons because the binding energy released by the absorption of a neutron is greater than the critical energy for fission; therefore uranium-235 is a fissile material. The binding energy released by uranium-238 absorbing a thermal neutron is less than the critical energy, so additional energy must be possessed by the neutron for fission to be possible. Consequently, uranium-238 is a fissionable material.

Fertile Material
All of the neutron absorption reactions that do not result in fission lead to the production of new
Fixture checkpoint marker-0189-xray is recorded on fixture page 0189.
nuclides through the process known as transmutation. These nuclides can, in turn, be transmuted again or may undergo radioactive decay to produce still different nuclides. The nuclides that are produced by this process are referred to as transmutation products. Because several of the fissile nuclides do not exist in nature, they can only be produced by nuclear reactions (transmutation). The target nuclei for such reactions are said to be fertile. Fertile materials are materials that can undergo transmutation to become fissile materials. Figure 19 traces the transmutation mechanism by which two fertile nuclides, thorium-232 and uranium-238, produce uranium-233 and plutonium-239, respectively.
Figure 19 Conversion of Fertile Nuclides to Fissile Nuclides
If a reactor contains fertile material in addition to its fissile fuel, some new fuel will be produced as the original fuel is burned up. This is called conversion. Reactors that are specifically

of fissionable fuel produced is greater than the amount of fuel burnup. If less fuel is produced than used, the process is called conversion, and the reactor is termed a "converter."

Binding Energy Per Nucleon (BE/A)

As the number of particles in a nucleus increases, the total binding energy also increases. The rate of increase, however, is not uniform. This lack of uniformity results in a variation in the amount of binding energy associated with each nucleon within the nucleus. This variation in the binding energy per nucleon (BE/A) is easily seen when the average BE/A is plotted versus atomic mass number (A), as shown in Figure 20.

Figure 20 Binding Energy per Nucleon vs. Mass Number

Figure 20 illustrates that as the atomic mass number increases, the binding energy per nucleon decreases for A > 60. The BE/A curve reaches a maximum value of 8.79 MeV at A = 56 and

Fixture checkpoint marker-0190-xray is recorded on fixture page 0190.

decreases to about 7.6 MeV for A = 238. The general shape of the BE/A curve can be explained using the general properties of nuclear forces. The nucleus is held together by very short-range attractive forces that exist between nucleons. On the other hand, the nucleus is being forced apart by long range repulsive electrostatic (coulomb) forces that exist between all the protons in the nucleus.

As the atomic number and the atomic mass number increase, the repulsive electrostatic forces within the nucleus increase due to the greater number of protons in the heavy elements. To overcome this increased repulsion, the proportion of neutrons in the nucleus must increase to maintain stability. This increase in the neutron-to-proton ratio only partially compensates for the growing proton-proton repulsive force in the heavier, naturally occurring elements. Because the repulsive forces are increasing, less energy must be supplied, on the average, to remove a nucleon from the nucleus. The BE/A has decreased. The BE/A of a nucleus is an indication of its degree of stability. Generally, the more stable nuclides have higher BE/A than the less stable ones. The increase in the BE/A as the atomic mass number decreases from 260 to 60 is the primary reason for the energy liberation in the fission process. In addition, the increase in the BE/A as the atomic mass number increases from 1 to 60 is the reason for the energy liberation in the fusion process, which is the opposite reaction of fission.

The heaviest nuclei require only a small distortion from a spherical shape (small energy addition) for the relatively large coulomb forces forcing the two halves of the nucleus apart to overcome the attractive nuclear forces holding the two halves together. Consequently, the heaviest nuclei are easily fissionable compared to lighter nuclei.

Summary

The important information in this chapter is summarized on the following page.

Nuclear Fission Summary

The fission process can be explained using the liquid drop model of a nucleus. In the ground state the nucleus is nearly spherical in shape. After the absorption of a neutron, the nucleus will be in an excited state and start to oscillate and become distorted. If the oscillations cause the nucleus to become shaped like a dumbbell, the repulsive electrostatic forces will overcome the short-range attractive nuclear forces, and the nucleus will split in two.

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Critical energy is the minimum excitation energy that a nucleus must have before it can fission.

Fissile material is material for which fission is possible with neutrons that have zero kinetic energy. Fissionable material is material for which fission caused by neutron absorption is possible provided the kinetic energy added with the binding energy is greater than the critical energy. Fertile material is material that can undergo transmutation to become fissile material.

Transmutation is the process of neutron absorption and subsequent decay, which changes one nuclide to another nuclide. Conversion is the process of transmuting fertile material into fissile material in a reactor, where the amount of fissile material produced is less than the amount of fissile material consumed. Breeding is the same as conversion, except the amount of fissile material produced is more than the amount of fissile material consumed.

The curve of binding energy per nucleon increases quickly through the light nuclides and reaches a maximum at a mass number of about 56. The curve decreases slowly for mass numbers greater than 60.

The heaviest nuclei are easily fissionable because they require only a small distortion from the spherical shape to allow the coulomb forces to overcoming the attractive nuclear force, forcing the two halves of the nucleus apart.

Uranium-235 fissions with thermal neutrons because the binding energy released

Fixture checkpoint marker-0191-xray is recorded on fixture page 0191.

by the absorption of a neutron is greater than the critical energy for fission. The binding energy released by uranium-238 absorbing a neutron is less than the critical energy, so additional kinetic energy must be possessed by the neutron for fission to be possible.

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1
0
n
235
92
U
236
92
U
140
55
Cs
93
37
Rb 3
1
0
n

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ENERGY RELEASE FROM FISSION

Fission of heavy nuclides converts a small amount of mass into an enormous

determined based on either the change in mass that occurs during the reaction or by the difference in binding energy per nucleon between the fissile nuclide and the fission products.

EO 4.8 CHARACTERIZE the fission products in terms of mass groupings and radioactivity.

EO 4.9 Given the nuclides involved and their masses, CALCULATE the energy released from fission.

EO 4.10 Given the curve of Binding Energy per nucleon versus mass number, CALCULATE the energy released from fission.

Calculation of Fission Energy

Nuclear fission results in the release of enormous quantities of energy. It is necessary to be able to calculate the amount of energy that will be produced. The logical manner in which to pursue this is to first investigate a typical fission reaction such as the one listed below.

It can be seen that when the compound nucleus splits, it breaks into two fission fragments, rubidium-93, cesium-140, and some neutrons. Both fission products then decay by multiple - emissions as a result of the high neutron-to-proton ratio possessed by these nuclides.

In most cases, the resultant fission fragments have masses that vary widely. Figure 21 gives the percent yield for atomic mass numbers. The most probable pair of fission fragments for the thermal fission of the fuel uranium-235 have masses of about 95 and 140. Note that the vertical axis of the fission yield curve is on a logarithmic scale. Therefore, the formation of fission fragments of mass numbers of about 95 and 140 is highly likely.

Figure 21 Uranium-235 Fission Yield vs.
Mass Number

Referring now to the binding energy per nucleon curve (Figure 20), we can estimate the amount of energy released by our "typical" fission by plotting this reaction on the curve and calculating the change in binding energy (ΔBE) between the reactants on the left-hand side of the fission

Fixture checkpoint marker-0192-xray is recorded on fixture page 0192.

equation and the products on the right-hand side. Plotting the reactant and product nuclides on the curve shows that the total binding energy of the system after fission is greater than the total binding energy of the system before fission. When there is an increase in the total binding energy of a system, the system has become more stable by releasing an amount of energy equal to the increase in total binding energy of the system. Therefore, in the fission process, the energy liberated is equal to the increase in the total binding energy of the system.

Figure 22 graphically depicts that the binding energy per nucleon for the products (C,

Figure 22 Change in Binding Energy for Typical Fission

rubidium-93 and B, cesium-140) is greater than that for the reactant (A, uranium-235). The total binding energy for a nucleus can be found by multiplying the binding energy per nucleon by the number of nucleons.

TABLE 5
Binding Energies Calculated from
Binding Energy per Nucleon Curve

Nuclide	B.E. per Nucleon (BE/A)	Mass Number (A)

SYNTHETIC FIXTURE PAGE 0193 OF 1200

(BE/A) x (A)
9
3
3
7Rb 8.7 MeV 93 809 MeV
14
5
0
5Cs 8.4 MeV 140 1176 MeV
23
9
5
2U 7.6 MeV 235 1786 MeV
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BE BEproducts BEreactants
BERb 93 BECs 140 BEU 235
809 MeV 1176 MeV 1786 MeV
199 MeV
1
0
n
235
92
U
236
92
U
140
55
Cs
93
37
Rb 3
1
Fixture checkpoint marker-0193-xray is recorded on fixture page 0193.
0
n
Mass difference Mass of Reactants Mass of Products
236.052589 amu 235.85208 amu
0.200509 amu
EInst 0.020059 amu
931.5 MeV
amu
186.8 MeV
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The energy released will be equivalent to the difference in binding energy (BE) between the reactants and the products.

conservation of mass-energy. During the fission process, there is a decrease in the mass of the system. There must, therefore, be energy liberated equal to the energy equivalent of the mass lost in the process. This method is more accurate than the previously illustrated method and is used when actually calculating the energy liberated during the fission process.
Again, referring to the "typical" fission reaction.
E , the instantaneous energy, is the energy released immediately after the fission process. It Inst is equal to the energy equivalent of the mass lost in the fission process. It can be calculated as shown below.
Mass of the Reactants Mass of the Products
U 235.043924 amu Rb 92.91699 amu 235
92
Fixture checkpoint marker-0194-xray is recorded on fixture page 0194.
93
37
n 1.008665 amu Cs 139.90910 amu 1
0
140
55
3 (n) 3.02599 amu 1
0
236.052589 amu 235.85208 amu
This mass difference can be converted to an energy equivalent.
93
37
Rb
93
38
Sr
93
39
Y
93
40
Zr
93
41
Nb
140
55
Cs
140
56
Ba
140
57
La
140
58

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EDecay mRb 93 mNb 93 4 melectron
931.5MeV
amu
92.91699 amu 92.90638 amu 4 0.0005486 amu
931.5 MeV
amu
0.008416 amu
931.5 MeV
amu
7.84 MeV
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The total energy released per fission will vary from the fission to the next depending on what fission products are formed, but the average total energy released per fission of uranium-235 with a thermal neutron is 200 MeV.
As illustrated in the preceding example, the majority of the energy liberated in the fission process is released immediately after the fission occurs and appears as the kinetic energy of the fission fragments, kinetic energy of the fission neutrons, and instantaneous gamma rays. The Fixture checkpoint marker-0195-xray is recorded on fixture page 0195.
remaining energy is released over a period of time after the fission occurs and appears as kinetic energy of the beta, neutrino, and decay gamma rays.
Estimation of Decay Energy
In addition to this instantaneous energy release during the actual fission reaction, there is additional energy released when the fission fragments decay by emission. This additional
-
energy is called decay energy, EDecay. The decay chains for rubidium-93 and cesium-140 are shown below.
The energy released during the decay for each chain will be equivalent to the mass difference between the original fission product and the sum of the final stable nuclide and the beta particles emitted.
The energy released in the decay chain of rubidium-93 is calculated below.
EDecay mRb 93 mNb 93 3 melectron
931.5MeV
amu
139.90910 amu 139.90543 amu 3 0.0005486 amu
931.5 MeV
amu
0.000202 amu
931.5 MeV
amu
1.89 MeV
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The energy released in the decay chain of cesium-140 is calculated below.
The total decay energy is the sum of the energies of the two chains, or 9.73 MeV.
Distribution of Fission Energy
The average energy distribution for the energy released per fission with a thermal neutron in uranium-235 is shown in Tables 6 and 7.

Instantaneous Energy from Fission
Kinetic Energy of Fission Products 167 Mev
Energy of Fission Neutrons 5 MeV
Instantaneous Gamma-ray Energy 5 MeV
Capture Gamma-ray Energy 10 MeV
Total Instantaneous Energy 187 MeV
TABLE 7
Delayed Energy from Fission
Beta Particles From Fission Products 7 Mev
Gamma-rays from Fission Products 6 MeV
Neutrinos 10 MeV
Total Delayed Energy 23 MeV
ENERGY RELEASE FROM FISSION DOE-HDBK-1019/1-93 Atomic and Nuclear Physics
Because the 10 MeV of neutrino energy shown in Table 7 is not absorbed in the reactor, the average value of 200 MeV per fission is still accurate. Note in Table 6 that some fission neutrons undergo radiative capture and the resultant gamma ray emission provides an additional 10 MeV of instantaneous energy, which contributes to the total of 187 MeV instantaneous energy.
All of the energy released, with the exception of the neutrino energy, is ultimately transformed into heat through a number of processes. The fission fragments, with their high positive charge and kinetic energy, cause ionization directly as they rip orbital electrons from the surrounding atoms. In this ionization process, kinetic energy is transferred to the surrounding atoms of the fuel material, resulting in an increase in temperature. The beta particles and gamma rays also give up their energy through ionization, and the fission neutrons interact and lose their energy through elastic scattering. Of the 200 MeV released per fission, about seven percent (13 MeV)
Fixture checkpoint marker-0196-xray is recorded on fixture page 0196.
is released at some time after the instant of fission. When a reactor is shut down, fissions essentially cease, but energy is still being released from the decay of fission products. The heat produced by this decay energy is referred to as "decay heat." Although decay energy represents about seven percent of reactor heat production during reactor operation, once the reactor is shut down the decay heat production drops off quickly to a small fraction of its value while operating. The decay heat produced is significant, however, and systems must be provided to keep the reactor cool even after shutdown.
Summary
The important information in this chapter is summarized below.
Energy Release From Fission Summary
Fission products have some general characteristics in common.
They generally decay by β- emission.
The most common mass numbers are grouped near 95 and 140.
The energy released by fission can be calculated based on the difference in mass between the masses of the reactants before fission and the fission fragments and fission neutrons after fission.
Another method to determine the energy released by fission is based on the change in binding energy per nucleon between the fissile nuclide and the fission products.

SYNTHETIC FIXTURE PAGE 0197 OF 1200

Different types of radiation interact with matter in widely different ways. A large, massive, charged alpha particle cannot penetrate a piece of paper and even has a limited range in dry air. A neutrino, at the other extreme, has a low probability of interacting with any matter, even if it passed through the diameter of the earth.

EO 5.1 DESCRIBE interactions of the following with matter:

a. Alpha particle c. Positron

b. Beta particle d. Neutron

EO 5.2 DESCRIBE the following ways that gamma radiation interacts with matter:

a. Photoelectric effect

b. Compton scattering

c. Pair production

Interaction of Radiation With Matter

Radiation can be classified into two general groups, charged and uncharged; therefore, it may be expected that interactions with matter fall into two general types. Charged particles directly ionize the media through which they pass, while uncharged particles and photons can cause ionization only indirectly or by secondary radiation.

A moving charged particle has an electrical field surrounding it, which interacts with the atomic structure of the medium through which it is passing. This interaction decelerates the particle and accelerates electrons in the atoms of the medium. The accelerated electrons may acquire enough energy to escape from the parent atom. This process, whereby radiation "strips" off orbital electrons, is called ionization. Uncharged moving particles have no electrical field, so they can only lose energy and cause ionization by such means as collisions or scattering. A photon can lose energy by the photoelectric effect, Compton effect, or pair production.

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I

m z

2

K.E.

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Because ionizing radiation creates ions in pairs, the intensity of ionization or the specific ionization is defined as the number of ion-pairs formed per centimeter of travel in a given material. The amount of ionization produced by a charged particle per unit path length, which is a measure of its ionizing power, is roughly proportional to the particle's mass and the square of its charge as illustrated in the equation below.

where:

I is the ionizing power

m is the mass of the particle

z is the number of unit charges it carries

K.E. is its kinetic energy

Since m for an alpha particle is about 7300 times as large as m for a beta particle, and z is twice as great, an alpha will produce much more ionization per unit path length than a beta particle of the same energy. This phenomenon occurs because the larger alpha particle moves slower for a given energy and thus acts on a given electron for a longer time.

Alpha radiation is normally produced from the radioactive decay of heavy nuclides and from certain nuclear reactions. The alpha particle consists of 2 neutrons and 2 protons, so it is essentially the same as the nucleus of a helium atom. Because it has no electrons, the alpha particle has a charge of +2. This positive charge causes the alpha particle to strip electrons from the orbits of atoms in its vicinity. As the alpha particle passes through material, it removes electrons from the orbits of atoms it passes near. Energy is required to remove electrons and the energy of the alpha particle is reduced by each reaction. Eventually the particle will expend its kinetic energy, gain 2 electrons in orbit, and become a helium atom. Because of its strong positive charge and large mass, the alpha particle deposits a large amount of energy in a short distance of travel. This rapid, large deposition of energy limits the penetration of alpha particles. The most energetic alpha particles are stopped by a few centimeters of air or a sheet of paper.

Beta-Minus Radiation

A beta-minus particle is an electron that has been ejected at a high velocity from an unstable nucleus. An electron has a small mass and an electrical charge of -1. Beta particles cause ionization by displacing electrons from atom orbits. The ionization occurs from collisions with orbiting electrons. Each collision removes kinetic energy from the beta particle, causing it to slow down. Eventually the beta particle will be slowed enough to allow it to be captured as an

2 electrons 0.000549 amu

electron

931.5 MeV

amu

1.02 MeV

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orbiting electron in an atom. Although more penetrating than the alpha, the beta is relatively easy to stop and has a low power of penetration. Even the most energetic beta radiation can be stopped by a few millimeters of metal.

Positron Radiation

Positively charged electrons are called positrons. Except for the positive charge, they are identical to beta-minus particles and interact with matter in a similar manner. Positrons are very short-lived, however, and quickly are annihilated by interaction with a negatively charged electron, producing two gammas with a combined energy (calculated below) equal to the rest mass of the positive and negative electrons.

Neutron Radiation

Neutrons have no electrical charge. They have nearly the same mass as a proton (a hydrogen atom nucleus). A neutron has hundreds of times more mass than an electron, but 1/4 the mass of an alpha particle. The source of neutrons is primarily nuclear reactions, such as fission, but

Fixture checkpoint marker-0198-xray is recorded on fixture page 0198.

they may also be produced from the decay of radioactive nuclides. Because of its lack of charge, the neutron is difficult to stop and has a high penetrating power.

Neutrons are attenuated (reduced in energy and numbers) by three major interactions, elastic scatter, inelastic scatter, and absorption. In elastic scatter, a neutron collides with a nucleus and bounces off. This reaction transmits some of the kinetic energy of the neutron to the nucleus of the atom, resulting in the neutron being slowed, and the atom receives some kinetic energy (motion). This process is sometimes referred to as "the billiard ball effect."

As the mass of the nucleus approaches the mass of the neutron, this reaction becomes more

In the inelastic scatter reaction, the same neutron/nucleus collision occurs as in elastic scatter. However, in this reaction, the nucleus receives some internal energy as well as kinetic energy. This slows the neutron, but leaves the nucleus in an excited state. When the nucleus decays to its original energy level, it normally emits a gamma ray.

In the absorption reaction, the neutron is actually absorbed into the nucleus of an atom. The neutron is captured, but the atom is left in an excited state. If the nucleus emits one or more gamma rays to reach a stable level, the process is called radiative capture. This reaction occurs at most neutron energy levels, but is more probable at lower energy levels.

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Gamma Radiation

Gamma radiation is electromagnetic radiation. It is commonly referred to as a gamma ray and is very similar to an x-ray. The difference is that gamma rays are emitted from the nucleus of an atom, and x-rays are produced by orbiting electrons. The x-ray is produced when orbiting electrons move to a lower energy orbit or when fast-moving electrons approaching an atom are deflected and decelerated as they react with the atom's electrical field (called Bremsstrahlung).

Fixture checkpoint marker-0199-xray is recorded on fixture page 0199.

The gamma ray is produced by the decay of excited nuclei and by nuclear reactions. Because the gamma ray has no mass and no charge, it is difficult to stop and has a very high penetrating power. A small fraction of the original gamma stream will pass through several feet of concrete or several meters of water.

There are three methods of attenuating gamma rays. The first method is referred to as the photo-electric effect. When a low energy gamma strikes an atom, the total energy of the gamma is expended in ejecting an electron from orbit. The result is ionization of the atom and expulsion of a high energy electron. This reaction is most predominant with low energy gammas interacting in materials with high atomic weight and rarely occurs with gammas having an energy above 1 MeV. Annihilation of the gamma results. Any gamma energy in excess of the binding energy of the electron is carried off by the electron in the form of kinetic energy.

The second method of attenuation of gammas is called Compton scattering. The gamma interacts with an orbital or free electron; however, in this case, the photon loses only a fraction of its energy. The actual energy loss depending on the scattering angle of the gamma. The gamma continues on at lower energy, and the energy difference is absorbed by the electron. This reaction becomes important for gamma energies of about 0.1 MeV and higher.

At higher energy levels, a third method of attenuation is predominant. This method is pair-production. When a high energy gamma passes close enough to a heavy nucleus, the gamma completely disappears, and an electron and a positron are formed. For this reaction to take place, the original gamma must have at least 1.02 MeV energy. Any energy greater than 1.02 MeV becomes kinetic energy shared between the electron and positron. The probability of pair■production incre

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Interaction of Radiation with Matter Summary

An alpha particle deposits a large amount of energy in a short distance of travel due to its large mass and charge.

Beta-minus particles interact with the electrons orbiting the nucleus of atoms, causing ionization by displacing the electrons. The beta particle loses energy with each interaction. After the beta particle loses enough energy, it is captured in the

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Positrons interact with matter much the same way as beta minus particles. After the positron has lost most of its energy by ionizing atoms, it is annihilated by interaction with an electron. The electron-positron pair disappear and are replaced by two gammas, each with the energy equivalent of the mass of an electron (0.51 MeV).
Neutrons interact with matter by elastic scattering, inelastic scattering, or absorption.
Photoelectric effect is where a gamma interacts with an electron orbiting an atom. The entire energy of the gamma is transferred to the electron, and the electron is ejected from its orbit.
In Compton scattering a gamma interacts with an orbital electron, but only part of the gamma energy is transferred to the electron. The electron is ejected from its orbit, and the gamma is scattered off at a lower energy.
In pair-production, a gamma interacts with the electric field of a nucleus and is converted into an electron-positron pair. The gamma must have an energy greater than 1.02 MeV for this to occur.
Summary
The important information in this chapter is summarized below.
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Department of Energy
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Fundamentals Handbook
NUCLEAR PHYSICS
AND REACTOR THEORY
Module 2
Reactor Theory (Neutron Characteristics)
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TERMINAL OBJECTIVE
1.0 Without references, EXPLAIN how neutron sources produce neutrons.
ENABLING OBJECTIVES
1.1 DEFINE the following terms:
a. Intrinsic neutron source
b. Installed neutron source
1.2 LIST three examples of reactions that produce neutrons in intrinsic neutron sources.
1.3 LIST three examples of reactions that produce neutrons in installed neutron sources.

TERMINAL OBJECTIVE
2.0 Given the necessary information for calculations, EXPLAIN basic concepts in reactor physics and perform calculations.
ENABLING OBJECTIVES
2.1 DEFINE the following terms:
a. Atom density d. Barn
b. Neutron flux e. Macroscopic cross section
c. Microscopic cross section f. Mean free path
2.2 EXPRESS macroscopic cross section in terms of microscopic cross section.
2.3 DESCRIBE how the absorption cross section of typical nuclides varies with neutron energy at energies below the resonance absorption region.
2.4 DESCRIBE the cause of resonance absorption in terms of nuclear energy levels.
2.5 DESCRIBE the energy dependence of resonance absorption peaks for typical light and heavy nuclei.
2.6 EXPRESS mean free path in terms of macroscopic cross section.
2.7 Given the number densities (or total density and component fractions) and microscopic
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cross sections of components, CALCULATE the macroscopic cross section for a mixture.
2.8 CALCULATE a macroscopic cross section given a material density, atomic mass, and microscopic cross section.
2.9 EXPLAIN neutron shadowing or self-shielding.
2.10 Given the neutron flux and macroscopic cross section, CALCULATE the reaction rate.
2.11 DESCRIBE the relationship between neutron flux and reactor power.

ENABLING OBJECTIVES (Cont.)
2.12 DEFINE the following concepts:

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b. Moderator e. Macroscopic slowing down power
c. Moderating ratio
2.13 LIST three desirable characteristics of a moderator.
2.14 Given an average fractional energy loss per collision, CALCULATE the energy loss after a specified number of collisions.
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Reactor Theory (Neutron Characteristics) DOE-HDBK-1019/1-93 OBJECTIVES
TERMINAL OBJECTIVE
3.0 Without references, EXPLAIN the production process and effects on fission of prompt and delayed neutrons.
ENABLING OBJECTIVES
3.1 STATE the origin of prompt neutrons and delayed neutrons.
3.2 STATE the approximate fraction of neutrons that are born as delayed neutrons from the fission of the following nuclear fuels:
Fixture checkpoint marker-0203-xray is recorded on fixture page 0203.
a. Uranium-235
b. Plutonium-239
3.3 EXPLAIN the mechanism for production of delayed neutrons.
3.4 EXPLAIN prompt and delayed neutron generation times.
3.5 Given prompt and delayed neutron generation times and delayed neutron fraction, CALCULATE the average generation time.
3.6 EXPLAIN the effect of delayed neutrons on reactor control.
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OBJECTIVES DOE-HDBK-1019/1-93 Reactor Theory (Neutron Characteristics)
TERMINAL OBJECTIVE
4.0 Without references, DESCRIBE the neutron energy spectrum for the type of reactor presented in this module.
ENABLING OBJECTIVES
4.1 STATE the average energy at which prompt neutrons are produced.
4.2 DESCRIBE the neutron energy spectrum in the following reactors:
a. Fast reactor
b. Thermal reactor
4.3 EXPLAIN the reason for the particular shape of the fast, intermediate, and slow energy regions of the neutron flux spectrum for a thermal reactor.
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Reactor Theory (Neutron Characteristics) DOE-HDBK-1019/1-93 NEUTRON SOURCES
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NEUTRON SOURCES
Neutrons from a variety of sources are always present in a reactor core. This is true even when the reactor is shut down. Some of these neutrons are produced by naturally occurring (intrinsic) neutron sources, while others may be the result of fabricated (installed) neutron sources that are incorporated into the design of the reactor. The neutrons produced by sources other than neutron-induced fission are often grouped together and classified as source neutrons.
EO 1.1 DEFINE the following terms:
a. Intrinsic neutron source
b. Installed neutron source
EO 1.2 LIST three examples of reactions that produce neutrons in

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EO 1.3 LIST three examples of reactions that produce neutrons in installed neutron sources.

Neutron Sources

In addition to neutron-induced fission, neutrons are produced by other reactions. The neutrons produced by reactions other than neutron-induced fission are called source neutrons. Source neutrons are important because they ensure that the neutron population remains high enough to allow a visible indication of neutron level on the most sensitive monitoring instruments while the reactor is shutdown and during the startup sequence. This verifies instrument operability and allows monitoring of neutron population changes. Source neutrons can be classified as either intrinsic or installed neutron sources.

Intrinsic Neutron Sources

Some neutrons will be produced in the materials present in the reactor due to a variety of unavoidable reactions that occur because of the nature of these materials. Intrinsic neutron sources are those neutron-producing reactions that always occur in reactor materials.

■ ■ ■ ■
■ ■ ■
■ ■

NEUTRON SOURCES DOE-HDBK-1019/1-93 Reactor Theory (Neutron Characteristics)
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A limited number of neutrons will always be present, even in a reactor core that has never been operated, due to spontaneous fission of some heavy nuclides that are present in the fuel. Uranium■238, ur
Fixture checkpoint marker-0204-xray is recorded on fixture page 0204.
of the rate at which different heavy nuclides produce neutrons by spontaneous fission. Californium-252 is not an intrinsic neutron source, but will be discussed in the section on installed neutron sources.

TABLE 1

Neutron Production by Spontaneous Fission

Nuclide T (Fission) T (1/2 1/2 ■-decay) neutrons/sec/gram
U 1.8 x 10 years 6.8 x 10 years 8.0 x 10 235
92
17 8 -4
U 8.0 x 10 years 4.5 x 10 years 1.6 x 10 238
92
15 9 -2
Pu 5.5 x 10 years 2.4 x 10 years 3.0 x 10 239
94
5 4 -2
Pu 1.2 x 10 years 6.6 x 10 years 1.0 x 10 240
94
11 3 3
Cf 66.0 years 2.65 years 2.3 x 10 252
98
12

Another intrinsic neutron source is a reaction involving natural boron and fuel. In some reactors, natural boron is loaded into the reactor core as a neutron absorber to improve reactor control or increase core life-time. Boron-11 (80.1% of natural boron) undergoes a reaction with the alpha particle emitted by the radioactive decay of heavy nuclides in the fuel to yield a neutron as shown below.

of the short path length of the alpha particle. For a reactor core with this configuration, this (■,n) reaction is an important source of neutrons for reactor startup.

In a reactor that has been operated, another source of neutrons becomes significant. Neutrons may be produced by the interaction of a gamma ray and a deuterium nucleus. This reaction is commonly referred to as a photoneutron reaction because it is initiated by electromagnetic radiation and results in the production of a neutron. The photoneutron reaction is shown below.

■ ■ ■
■ ■
■ ■ ■
■ ■

There is an abundant supply of high energy gammas in a reactor that has been operated because many of the fission products are gamma emitters. All water-cooled reactors have some deuterium present in the coolant in the reactor core because a small fraction of natural hydrogen is the isotope deuterium. The atom percentage of deuterium in the water ranges from close to the naturally occurring value (0.015%) for light water reactors to above 90% deuterium for heavy water reactors. Therefore, the required conditions for production of photoneutrons exist.

The supply of gamma rays decreases with time after shutdown as the gamma emitters decay; therefore, the photoneutron production rate also decreases. In a few particular reactors, additional D O (heavy water) may be added to the reactor to increase the production of 2 photoneutrons following a long shutdown period.

Installed Neutron Sources

Because intrinsic neutron sources can be relatively weak or dependent upon the recent power history of the reactor, many reactors have artificial sources of neutrons installed. These neutron sources ensure that shutdown neutron levels are high enough to be detected by the nuclear instruments at all times. This provides a true picture of reactor conditions and any change in these conditions. An installed neutron source is an assembly placed in or near the reactor for

Fixture checkpoint marker-0205-xray is recorded on fixture page 0205.

the sole purpose of producing source neutrons.

One strong source of neutrons is the artificial nuclide californium-252, which emits neutrons at the rate of about 2 x 10 neutrons per second per gram as the result of spontaneous fission. 12 Important drawbacks for some applications may be its high cost and its short half-life (2.65 years).

Many installed neutron sources use the (■,n) reaction with beryllium. These sources are composed of a mixture of metallic beryllium (100% beryllium-9) with a small quantity of an alpha particle emitter, such as a compound of radium, polonium, or plutonium. The reaction that occurs is shown below.

The beryllium is intimately (homogeneously) mixed with the alpha emitter and is usually enclosed in a stainless steel capsule.

Another type of installed neutron source that is widely used is a photoneutron source that employs the (■,n) reaction with beryllium. Beryllium is used for photoneutron sources because its stable isotope beryllium-9 has a weakly attached last neutron with a binding energy of only 1.66 MeV. Thus, a gamma ray with greater energy than 1.66 MeV can cause neutrons to be ejected by the (■,n) reaction as shown below.

Neutron Sources Summary

the reactor for other purposes such as fuel, burnable poison, or moderator.
& Installed neutron sources are materials or components placed in the reactor specifically for the purpose of producing source neutrons.
& Examples of intrinsic neutron sources are listed below.
Spontaneous fission of heavy nuclides in fuel, such as uranium-238, uranium-235, and plutonium-239, results in fission fragments and free neutrons.
Boron-11 mixed with the fuel undergoes an alpha-neutron reaction and becomes nitrogen-14.
Deuterium present in the reactor coolant undergoes a gamma■neutron reaction and becomes hydrogen-1.
& Examples of installed neutron sources are listed below.
Spontaneous fission of californium-252 results in fission fragments and free neutrons.
Beryllium-9 undergoes an alpha-neutron reaction (alpha from the decay of plutonium, polonium, or radium) and becomes carbon-12.
Beryllium-9 undergoes a gamma-neutron reaction (high energy gamma from decay of antimony-124) and becomes beryllium-8.
Many startup sources of this type use antimony and beryllium because after activation with neutrons the radioactive antimony becomes an emitter of high energy gammas. The photoneutron sources of this type are constructed somewhat differently from the (■,n) types. One design incorporates a capsule of irradiated antimony enclosed in a beryllium sleeve. The entire assembly is then encased in a stainless steel cladding. A large reactor may have several neutron sources of this type installed within the core.
Summary
The important information in this chapter is summarized below.
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Reactor Theory (Neutron Characteristics) NUCLEAR CROSS SECTIONS AND NEUTRON FLUX
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NUCLEAR CROSS SECTIONS AND NEUTRON FLUX
To determine the frequency of neutron interactions, it is necessary to describe the availability of neutrons to cause interaction and the probability of a neutron interacting with material. The availability of neutrons and the probability of interaction are quantified by the neutron flux and nuclear cross section.
EO 2.1 DEFINE the following terms:
a. Atom density d. Barn
Fixture checkpoint marker-0206-xray is recorded on fixture page 0206.
b. Neutron flux e. Macroscopic cross section
c. Microscopic cross section f. Mean free path
EO 2.2 EXPRESS macroscopic cross section in terms of microscopic cross section.
EO 2.3 DESCRIBE how the absorption cross section of typical nuclides varies with neutron energy at energies below the resonance absorption region.
EO 2.4 DESCRIBE the cause of resonance absorption in terms of nuclear energy levels.
EO 2.5 DESCRIBE the energy dependence of resonance absorption peaks for typical light and heavy nuclei.
EO 2.6 EXPRESS mean free path in terms of macroscopic cross section.

fractions) and microscopic cross sections of components,
CALCULATE the macroscopic cross section for a mixture.
EO 2.8 CALCULATE a macroscopic cross section given a material density,
atomic mass, and microscopic cross section.
EO 2.9 EXPLAIN neutron shadowing or self-shielding.
'

Introduction

Fission neutrons are born with an average energy of about 2 MeV. These fast neutrons interact with the reactor core materials in various absorption and scattering reactions. Collisions that

Fixture checkpoint marker-0207-xray is recorded on fixture page 0207.

result in scattering are useful in slowing neutrons to thermal energies. Thermal neutrons may be absorbed by fissile nuclei to produce more fissions or be absorbed in fertile material for conversion to fissionable fuel. Absorption of neutrons in structural components, coolant, and other non-fuel material results in the removal of neutrons without fulfilling any useful purpose.

To safely and efficiently operate a nuclear reactor it is necessary to predict the probability that a particular absorption or scattering reaction will occur. Once these probabilities are known, if the availability of neutrons can be determined, then the rate at which these nuclear reactions take place can be predicted.

Atom Density

One important property of a material is the atom density. The atom density is the number of atoms of a given type per unit volume of the material. To calculate the atom density of a substance use Equation (2-1).

(2-1)

where:

N = atom density (atoms/cm)3
' = density (g/cm)3
N = Avogadro's number (6.022 x 10 atoms/mole) A
23
M = gram atomic weight
N
NA
M
2.699 g
cm3
6.022 x 1023 atoms
mole
26.9815 g
mole
6.024 x 1022 atoms
cm3

Example:

A block of aluminum has a density of 2.699 g/cm . If the gram atomic weight of

aluminum is 26.9815 g, calculate the atom density of the aluminum.
Solution:
Cross Sections
The probability of a neutron interacting with a nucleus for a particular reaction is dependent upon not only the kind of nucleus involved, but also the energy of the neutron. Accordingly, the absorption of a thermal neutron in most materials is much more probable than the absorption of a fast neutron. Also, the probability of interaction will vary depending upon the type of reaction involved.
The probability of a particular reaction occurring between a neutron and a nucleus is called the microscopic cross section ( ) of the nucleus for the particular reaction. This cross section will vary with the energy of the neutron. The microscopic cross section may also be regarded as the effective area the nucleus presents to the neutron for the particular reaction. The larger the effective area, the greater the probability for reaction.
Because the microscopic cross section is an area, it is expressed in units of area, or square centimeters. A square centimeter is tremendously large in comparison to the effective area of a nucleus, and it has been suggested that a physicist once referred to the measure of a square centimeter as being "as big as a barn" when applied to nuclear processes. The name has persisted and microscopic cross sections are expressed in terms of barns. The relationship between barns and cm is shown below. 2
Fixture checkpoint marker-0208-xray is recorded on fixture page 0208.
1 barn = 10-24 cm2
N
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NUCLEAR CROSS SECTIONS AND NEUTRON FLUX Reactor Theory (Neutron Characteristics)
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Whether a neutron will interact with a certain volume of material depends not only on the microscopic cross section of the individual nuclei but also on the number of nuclei within that volume. Therefore, it is necessary to define another kind of cross section known as the macroscopic cross section ( ). The macroscopic cross section is the probability of a given reaction occurring per unit travel of the neutron. is related to the microscopic cross section ( ) by the relationship shown below.
(2-2)
where:
= macroscopic cross section (cm)
-1
N = atom density of material (atoms/cm)
3
= microscopic cross-section (cm)
2
The difference between the microscopic and macroscopic cross sections is extremely important and is restated for clarity. The microscopic cross section ( ) represents the effective target area that a single nucleus presents to a bombarding particle. The units are given in barns or cm .
2
The macroscopic cross section ( ) represents the effective target area that is presented by all of the nuclei contained in 1 cm of the material. The units are given as 1/cm or cm .
3 -1
A neutron interacts with an atom of the material it enters in two basic ways. It will either interact through a scattering interaction or through an absorption reaction. The probability of

. The probability of a neutron scattering off of a particular nucleus is the microscopic cross a section for scattering, s
. The sum of the microscopic cross section for absorption and the microscopic cross section for scattering is the total microscopic cross section, . T
= + T a s
Both the absorption and the scattering microscopic cross sections can be further divided. For instance, the scattering cross section is the sum of the elastic scattering cross section (se) and the inelastic scattering cross section ( ). si
= + s se si
The microscopic absorption cross section ( ) includes all reactions except scattering. However, a for most purposes it is sufficient to merely separate it into two categories, fission (f) and capture ( ). Radiative capture of neutrons was described in the Neutron Interactions chapter c of Module 1.
= + a f c
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Reactor Theory (Neutron Characteristics) NUCLEAR CROSS SECTIONS AND NEUTRON FLUX
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Figure 1 Typical Neutron Absorption Cross Section vs. Neutron Energy
The variation of absorption cross sections with neutron energy is often complicated. For many elements the absorption cross sections are small, ranging from a fraction of a barn to a few barns for slow (or thermal) neutrons.
For a considerable number of nuclides of moderately high (or high) mass numbers, an examination of the variation of the absorption cross section with the energy of the incident neutron reveals the existence of three regions on a curve of absorption cross section versus neutron energy. This cross section is illustrated in Figure 1. First, the cross section decreases steadily with increasing neutron energy in a low energy region, which includes the thermal range
Fixture checkpoint marker-0209-xray is recorded on fixture page 0209.
(E < 1 eV). In this region the absorption cross section, which is often high, is inversely proportional to the velocity (v). This region is frequently referred to as the "1/v region," because the absorption cross section is proportional to 1/v, which is the reciprocal of neutron velocity. Following the 1/v region, there occurs the "resonance region" in which the cross sections rise sharply to high values called "resonance peaks" for neutrons of certain energies, and then fall again. These energies are called resonance energies and are a result of the affinity of the nucleus for neutrons whose energies closely match its discrete, quantum energy levels. That is, when the binding energy of a neutron plus the kinetic energy of the neutron are exactly equal to the amount required to raise a compound nucleus from its ground state to a quantum level, resonance absorption occurs. The following example problem further illustrates this point.
■
(
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NUCLEAR CROSS SECTIONS AND NEUTRON FLUX Reactor Theory (Neutron Characteristics)
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Assuming that uranium-236 has a nuclear quantum energy level at 6.8 MeV above its ground state, calculate the kinetic energy a neutron must possess to undergo resonant absorption in uranium-235 at this resonance energy level.
BE = [Mass(U) + Mass(neutron) - Mass(U)] x 931 MeV/amu 235 236
BE = (235.043925 + 1.008665 - 236.045563) x 931 MeV/amu
BE = (0.007025 amu) x 931 MeV/amu = 6.54 MeV

The difference between the binding energy and the quantum energy level equals the amount of kinetic energy the neutron must possess. The typical heavy nucleus will have many closely■spaced resonanc complex and have more possible configurations and corresponding energy states. Light nuclei, being less complex, have fewer possible energy states and fewer resonances that are sparsely distributed at higher energy levels.

For higher neutron energies, the absorption cross section steadily decreases as the energy of the neutron increases. This is called the "fast neutron region." In this region the absorption cross sections are usually less than 10 barns.

With the exception of hydrogen, for which the value is fairly large, the elastic scattering cross sections are generally small, for example, 5 barns to 10 barns. This is close to the magnitude of the actual geometric cross sectional area expected for atomic nuclei. In potential scattering, the cross section is essentially constant and independent of neutron energy. Resonance elastic scattering and inelastic scattering exhibit resonance peaks similar to those associated with absorption cross sections. The resonances occur at lower energies for heavy nuclei than for light nuclei. In general, the variations in scattering cross sections are very small when compared to the variations that occur in absorption cross sections.

Mean Free Path

If a neutron has a certain probability of undergoing a particular interaction in one centimeter of travel, then the inverse of this value describes how far the neutron will travel (in the average case) before undergoing an interaction. This average distance traveled by a neutron before interaction is known as the mean free path for that interaction and is represented by the symbol ■. The relationship between the mean free path (■) and the macroscopic cross section (*) is shown below.

(2-3)

'

(

)

Calculation of Macroscopic Cross Section and Mean Free Path

Most materials are composed of several elements, and because most elements are composed of several isotopes, most materials involve many cross sections, one for each isotope involved.

Fixture checkpoint marker-0210-xray is recorded on fixture page 0210.

Therefore, to include all the isotopes within a given material, it is necessary to determine the macroscopic cross section for each isotope and then sum all the individual macroscopic cross sections. Equation (2-4) can be used to determine the macroscopic cross section for a composite material.

* = N) + N) + N) + ... N) (2-4) 1 1 2 2 3 3 nn

where:

N = the number of nuclei per cm of the n element n

3 th

) = the microscopic cross section of the n element n

th

The following example problems illustrate the calculation of the macroscopic cross section for a single element and for combinations of materials.

Example 1:

Find the macroscopic thermal neutron absorption cross section for iron, which has a

and the gram atomic weight is 55.847 g.
Solution:
Step 1: Using Equation (2-1), calculate the atom density of iron.
Step 2: Use this atom density in Equation (2-2) to calculate the macroscopic cross section.
'
'

Fixture checkpoint marker-0211-xray is recorded on fixture page 0211.
Example 2:
An alloy is composed of 95% aluminum and 5% silicon (by weight). The density of the alloy is 2.66 g/cm . Properties of aluminum and silicon are shown below. 3

Element	Gram Atomic Weight	)	) a (barns)	s (barns)
Aluminum	26.9815	0.23	1.49	
Silicon	28.0855	0.16	2.20	

1. Calculate the atom densities for the aluminum and silicon.
2. Determine the absorption and scattering macroscopic cross sections for thermal neutrons.
3. Calculate the mean free paths for absorption and scattering.
Solution:
Step 1: The density of the aluminum will be 95% of the total density. Using Equation (2-1) yields the atom densities.
a NAl a,Al NSi a,Si
5.64 x 1022 atoms
cm3
0.23 x 10 24 cm22.85 x 1021 atoms
cm3
0.16 x 10 24 cm2
0.0134 cm 1
s NAl s,Al NSi s,Si
5.64 x 1022 atoms
cm3
1.49 x 10 24 cm22.85 x 1021 atoms
cm3
2.20 x 10 24 cm2
0.0903 cm 1
a
1
a
1
0.01345 cm 1
74.3 cm
s
1
s
1

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11.1 cm
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Step 2: The macroscopic cross sections for absorption and scattering are calculated using Equation (2-4).
Step 3: The mean free paths are calculated by inserting the macroscopic cross sections calculated above into Equation (2-3).
Thus, a neutron must travel an average of 74.3 cm to interact by absorption in this alloy, but it must travel only 11.1 cm to interact by scattering.
)
)
)
)
(■
(■
DOE-HDBK-1019/1-93
Fixture checkpoint marker-0212-xray is recorded on fixture page 0212.
NUCLEAR CROSS SECTIONS AND NEUTRON FLUX Reactor Theory (Neutron Characteristics)
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Effects of Temperature on Cross Section
As discussed, the microscopic absorption cross section varies significantly as neutron energy varies. The microscopic cross sections provided on most charts and tables are measured for a standard neutron velocity of 2200 meters/second, which corresponds to an ambient temperature of 68(F. Therefore, if our material is at a higher temperature, the absorption cross section will be lower than the value for 68(F, and any cross sections which involve absorption (for example,) ,) ,)) must be corrected for the existing temperature. acf
The following formula is used to correct microscopic cross sections for temperature. Although the example illustrates absorption cross section, the same formula may be used to correct capture and fission cross sections.
where:
) = microscopic cross section corrected for temperature
) = microscopic cross section at reference temperature (68(F or 20(C) o
T = reference temperature (68(F) in degrees Rankine ((R) or Kelvin ((K) o
T = temperature for which corrected value is being calculated
NOTE: When using this formula, all temperatures must be converted to (R or (K.
(R = (F + 460
(K = (C + 273
Example:
What is the value of)f for uranium-235 for thermal neutrons at 500(F? Uranium-235 has a) of 583 barns at 68(F. f
Solution:
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Reactor Theory (Neutron Characteristics) NUCLEAR CROSS SECTIONS AND NEUTRON FLUX
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Neutron Flux
Macroscopic cross sections for neutron reactions with materials determine the probability of one neutron undergoing a specific reaction per centimeter of travel through that material. If one

neutrons are traveling through the material and how many centimeters they travel each second. It is convenient to consider the number of neutrons existing in one cubic centimeter at any one instant and the total distance they travel each second while in that cubic centimeter. The number of neutrons existing in a cm of material at any instant is called neutron density and is 3 represented by the symbol n with units of neutrons/cm . The total distance these neutrons can 3 travel each second will be determined by their velocity.

A good way of defining neutron flux (1) is to consider it to be the total path length covered by all neutrons in one cubic centimeter during one second. Mathematically, this is the equation below.

1 = n v (2-5)

where:

1 = neutron flux (neutrons/cm -sec) 2

n = neutron density (neutrons/cm)3

v = neutron velocity (cm/sec)

The term neutron flux in some applications (for example, cross section measurement) is used as parallel beams of neutrons traveling in a single direction. The intensity (I) of a neutron beam is the product of the neutron density times the average neutron velocity. The directional beam intensity is equal to the number of neutrons per unit area and time (neutrons/cm -sec) falling on 2 a surface perpendicular to the direction of the beam.

One can think of the neutron flux in a reactor as being comprised of many neutron beams traveling in various directions. Then, the neutron flux becomes the scalar sum of these directional flux intensities (added as numbers and not vectors), that is, 1 = I + I + I +...I . 123 n

Since the atoms in a reactor do not interact preferentially with neutrons from any particular direction, all of these directional beams contribute to the total rate of reaction. In reality, at a

Fixture checkpoint marker-0213-xray is recorded on fixture page 0213.

given point within a reactor, neutrons will be traveling in all directions.

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NUCLEAR CROSS SECTIONS AND NEUTRON FLUX Reactor Theory (Neutron Characteristics)

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Self-Shielding

In some locations within the reactor, the flux level may be significantly lower than in other areas due to a phenomenon referred to as neutron shadowing or self-shielding. For example, the interior of a fuel pin or pellet will "see" a lower average flux level than the outer surfaces since an appreciable fraction of the neutrons will have been absorbed and therefore cannot reach the interior of the fuel pin. This is especially important at resonance energies, where the absorption cross sections are large.

Summary

The important information in this chapter is summarized below.

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Reactor Theory (Neutron Characteristics) NUCLEAR CROSS SECTIONS AND NEUTRON FLUX

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Nuclear Cross Section and Neutron Flux Summary

& Atom density (N) is the number of atoms of a given type per unit volume of material.

& Microscopic cross section ()) is the probability of a given reaction occurring between a neutron and a nucleus.

& Microscopic cross sections are measured in units of barns, where 1 barn = 10-24 cm .2

unit length of travel of the neutron. The units for macroscopic cross section are cm . -1
& The mean free path (■) is the average distance that a neutron travels in a material between interactions.
& Neutron flux (1) is the total path length traveled by all neutrons in one cubic centimeter of material during one second.
& The macroscopic cross section for a material can be calculated using the equation below.
* = N)
& The absorption cross section for a material usually has three distinct regions. At low neutron energies (<1 eV) the cross section is inversely proportional to the neutron velocity.
& Resonance absorption occurs when the sum of the kinetic energy of the neutron and its binding energy is equal to an allowed nuclear energy level of the nucleus.
& Resonance peaks exist at intermediate energy levels. For higher neutron energies, the absorption cross section steadily decreases as the neutron energy increases.
& The mean free path equals 1/*.
& The macroscopic cross section for a mixture of materials can be calculated using the equation below.
* = N 1 1 2 2 3 3 nn) + N) + N) + ... N)
& Self-shielding is where the local neutron flux is depressed within a material due to neutron absorption near the surface of the material.
1)
REACTION RATES DOE-HDBK-1019/1-93 Reactor Theory (Neutron Characteristics) NP-02 Page 18 Rev. 0
REACTION RATES
It is possible to determine the rate at which a nuclear reaction will take place based on the neutron flux, cross section for the interaction, and atom density of the target. This relationship illustrates how a change in one of these items affects the reaction rate.
EO 2.10 Given the neutron flux and macroscopic cross section,
Fixture checkpoint marker-0214-xray is recorded on fixture page 0214.
CALCULATE the reaction rate.
EO 2.11 DESCRIBE the relationship between neutron flux and reactor power.
Reaction Rates
If the total path length of all the neutrons in a cubic centimeter in a second is known, (neutron flux (1)), and if the probability of having an interaction per centimeter path length is also known (macroscopic cross section (*)), multiply them together to get the number of interactions taking place in that cubic centimeter in one second. This value is known as the reaction rate and is denoted by the symbol R. The reaction rate can be calculated by the equation shown below.
R = 1 * (2-6)
where:
R = reaction rate (reactions/sec)
1 = neutron flux (neutrons/cm -sec) 2
* = macroscopic cross section (cm) -1
Substituting the fact that * = N) into Equation (2-6) yields the equation below.
where:

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N = atom density (atoms/cm)3
Rf f
1 x 1013 neutrons
cm2
sec
0.1 cm 1
1 x 1012 fissions
cm3
sec
Rf f
f
Rf
1.29 x 1012 fissions
cm3
sec
3 x 1013 neutrons
cm2
sec
0.043 cm 1

The reaction rate calculated will depend on which macroscopic cross section is used in the calculation. Normally, the reaction rate of greatest interest is the fission reaction rate.
Example:
If a one cubic centimeter section of a reactor has a macroscopic fission cross section of 0.1 cm , and if the thermal neutron flux is 10 neutrons/cm -sec, what is the fission rate -1 13 2 in that cubic centimeter?
Solution:
In addition to using Equation (2-6) to determine the reaction rate based on the physical properties of material, it is also possible to algebraically manipulate the equation to determine physical properties if the reaction rate is known.
Example:
A reactor operating at a flux level of 3 x 10 neutrons/cm -sec contains 10 atoms of 13 2 20 uranium-235 per cm . The reaction rate is 1.29 x 10 fission/cm . Calculate and . 3 12 3 f f
Solution:
Fixture checkpoint marker-0215-xray is recorded on fixture page 0215.
Step 1: The macroscopic cross section can be determined by solving Equation (2-6) for and substituting the appropriate values. f
Rf N f
f
Rf
N
1.29 x 1012 fissions
cm3
sec

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cm3
3 x 1013 neutrons
cm2
sec
4.3 x 10 22 cm2 1 barn
1 x 10 24 cm2
430 barns
1 watt 1 erg
1 x 10 7 watt sec
1 MeV
1.602 x 10 6
erg
1 fission
200 MeV
3.12 x 1010 fissions
Fixture checkpoint marker-0216-xray is recorded on fixture page 0216.
second
REACTION RATES DOE-HDBK-1019/1-93 Reactor Theory (Neutron Characteristics)
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Step 2: To find the microscopic cross section, replace f with (N x f) and solve
for .
f
Reactor Power Calculation
Multiplying the reaction rate per unit volume by the total volume of the core results in the total number of reactions occurring in the core per unit time. If the amount of energy involved in each reaction were known, it would be possible to determine the rate of energy release (power) due to a certain reaction.
In a reactor where the average energy per fission is 200 MeV, it is possible to determine the number of fissions per second that are necessary to produce one watt of power using the following conversion factors.
1 fission = 200 MeV
1 MeV = 1.602 x 10 ergs -6
1 erg = 1 x 10 watt-sec -7
This is equivalent to stating that 3.12 x 10 fissions release 1 watt-second of energy. 10
P
th f V
3.12 x 1010 fissions
watt sec
Reactor Theory (Neutron Characteristics) DOE-HDBK-1019/1-93 REACTION RATES
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The power released in a reactor can be calculated based on Equation (2-6). Multiplying the reaction rate by the volume of the reactor results in the total fission rate for the entire reactor. Dividing by the number of fissions per watt-sec results in the power released by fission in the reactor in units of watts. This relationship is shown mathematically in Equation (2-7) below.
(2-7)
where:
P = power (watts)
= thermal neutron flux (neutrons/cm -sec) th

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= macroscopic cross section for fission (cm) f
-1
V = volume of core (cm)
3
Relationship Between Neutron Flux and Reactor Power
In an operating reactor the volume of the reactor is constant. Over a relatively short period of time (days or weeks), the number density of the fuel atoms is also relatively constant. Since the atom density and microscopic cross section are constant, the macroscopic cross section must also be constant. Examining Equation (2-7), it is apparent that if the reactor volume and macroscopic cross section are constant, then the reactor power and the neutron flux are directly proportional. This is true for day-to-day operation. The neutron flux for a given power level will increase very slowly over a period of months due to the burnup of the fuel and resulting decrease in atom density and macroscopic cross section.
REACTION RATES DOE-HDBK-1019/1-93 Reactor Theory (Neutron Characteristics)
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Reaction Rates Summary
& The reaction rate is the number of interactions of a particular type occurring in a cubic centimeter of material in a second.
& The reaction rate can be calculated by the equation below.
R = 1 *
& Over a period of several days, while the atom density of the fuel can be considered constant, the neutron flux is directly proportional to reactor power.
Fixture checkpoint marker-0217-xray is recorded on fixture page 0217.
Summary
The important information in this chapter is summarized below.
Reactor Theory (Neutron Characteristics) DOE-HDBK-1019/1-93 NEUTRON MODERATION
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NEUTRON MODERATION
In thermal reactors, the neutrons that cause fission are at a much lower energy than the energy level at which they were born from fission. In this type of reactor, specific materials must be included in the reactor design to reduce the energy level of the neutrons in an efficient manner.
EO 2.12 DEFINE the following concepts:
a. Thermalization d. Average logarithmic energy decrement
b. Moderator e. Macroscopic slowing down power
c. Moderating ratio
EO 2.13 LIST three desirable characteristics of a moderator.
EO 2.14 Given an average fractional energy loss per collision, CALCULATE the energy loss after a specified number of collisions.
Neutron Slowing Down and Thermalization
Fission neutrons are produced at an average energy level of 2 MeV and immediately begin to slow down as the result of numerous scattering reactions with a variety of target nuclei. After a number of collisions with nuclei, the speed of a neutron is reduced to such an extent that it has approximately the same average kinetic energy as the atoms (or molecules) of the medium in which the neutron is undergoing elastic scattering. This energy, which is only a small fraction of an electron volt at ordinary temperatures (0.025 eV at 20(C), is frequently referred to as the thermal energy, since it depends upon the temperature. Neutrons whose energies have been

reducing the energy of a neutron to the thermal region by elastic scattering is referred to as thermalization, slowing down, or moderation. The material used for the purpose of thermalizing neutrons is called a moderator. A good moderator reduces the speed of neutrons in a small number of collisions, but does not absorb them to any great extent. Slowing the neutrons in as few collisions as possible is desirable in order to reduce the amount of neutron leakage from the core and also to reduce the number of resonance absorptions in non-fuel materials. Neutron leakage and resonance absorption will be discussed in the next module.

ln Ei

ln Ef

ln

Ei

Ef

2

A

2

3

NEUTRON MODERATION DOE-HDBK-1019/1-93 Reactor Theory (Neutron Characteristics)

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The ideal moderating material (moderator) should have the following nuclear properties.

large scattering cross section

small absorption cross section

large energy loss per collision

A convenient measure of energy loss per collision is the logarithmic energy decrement. The average logarithmic energy decrement is the average decrease per collision in the logarithm of the neutron energy. This quantity is represented by the symbol (Greek letter xi).

(2-8)

where:

= average logarithmic energy decrement

Ei = average initial neutron energy

Fixture checkpoint marker-0218-xray is recorded on fixture page 0218.

Ef = average final neutron energy

The symbol is commonly called the average logarithmic energy decrement because of the fact that a neutron loses, on the average, a fixed fraction of its energy per scattering collision. Since the fraction of energy retained by a neutron in a single elastic collision is a constant for a given material, is also a constant. Because it is a constant for each type of material and does not depend upon the initial neutron energy, is a convenient quantity for assessing the moderating ability of a material.

The values for the lighter nuclei are tabulated in a variety of sources. The following commonly used approximation may be used when a tabulated value is not available.

This approximation is relatively accurate for mass numbers (A) greater than 10, but for some low values of A it may be in error by over three percent.

N

ln Ehigh ln Elow

ln

Ehigh

Elow

N

ln

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Elow
ln
2 x 106eV
0.025 eV
0.948
19.2 collisions
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Since represents the average logarithmic energy loss per collision, the total number of collisions necessary for a neutron to lose a given amount of energy may be determined by dividing into the difference of the natural logarithms of the energy range in question. The number of collisions (N) to travel from any energy, E , to any lower energy, E , can be calculated as shown below. high low
Example:
How many collisions are required to slow a neutron from an energy of 2 MeV to a thermal energy of 0.025 eV, using water as the moderator? Water has a value of 0.948 for .
Solution:
Sometimes it is convenient, based upon information known, to work with an average fractional energy loss per collision as opposed to an average logarithmic fraction. If the initial neutron energy level and the average fractional energy loss per collision are known, the final energy level for a given number of collisions may be computed using the following formula.
! (
NEUTRON MODERATION DOE-HDBK-1019/1-93 Reactor Theory (Neutron Characteristics)
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(2-9)
where:
E = initial neutron energy o
E = neutron energy after N collisions N
x = average fractional energy loss per collision
N = number of collisions
Example:
If the average fractional energy loss per collision in hydrogen is 0.63, what will be the energy of a 2 MeV neutron after (a) 5 collisions? (b) 10 collisions?
Solution:
a)
b)
Fixture checkpoint marker-0219-xray is recorded on fixture page 0219.
Macroscopic Slowing Down Power
Although the logarithmic energy decrement is a convenient measure of the ability of a material to slow neutrons, it does not measure all necessary properties of a moderator. A better measure of the capabilities of a material is the macroscopic slowing down power. The macroscopic slowing down power (MSDP) is the product of the logarithmic energy decrement and the macroscopic cross section for scattering in the material. Equation (2-10) illustrates how to calculate the macroscopic slowing down power.
(2-10)
MR
s
a

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Moderating Ratio
Macroscopic slowing down power indicates how rapidly a neutron will slow down in the material in question, but it still does not fully explain the effectiveness of the material as a moderator. An element such as boron has a high logarithmic energy decrement and a good slowing down power, but it is a poor moderator because of its high probability of absorbing neutrons.
The most complete measure of the effectiveness of a moderator is the moderating ratio. The moderating ratio is the ratio of the macroscopic slowing down power to the macroscopic cross section for absorption. The higher the moderating ratio, the more effectively the material performs as a moderator. Equation (2-11) shows how to calculate the moderating ratio of a material.
(2-11)
Moderating properties of different materials are compared in Table 2.
Fixture checkpoint marker-0220-xray is recorded on fixture page 0220.
TABLE 2
Moderating Properties of Materials
Material Collisions to Slowing Down
Number of Macroscopic
Thermalize Power
Moderating
Ratio
H2O 0.927 19 1.425 62
D2O 0.510 35 0.177 4830
Helium 0.427 42 9 x 10 51 -6
Beryllium 0.207 86 0.154 126
Boron 0.171 105 0.092 0.00086
Carbon 0.158 114 0.083 216
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Summary
The important information in this chapter is summarized below.
Neutron Moderation Summary
& Thermalization is the process of reducing the energy level of a neutron from the energy level at which it is produced to an energy level in the thermal range.
& The moderator is the reactor material that is present for the purpose of thermalizing neutrons.
& Moderating ratio is the ratio of the macroscopic slowing down power to the macroscopic cross section for absorption.
& The average logarithmic energy decrement (!) is the average change in the logarithm of neutron energy per collision.
& Macroscopic slowing down power is the product of the average logarithmic energy decrement and the macroscopic cross section for scattering.
& There are three desirable characteristics of a moderator.
1. large scattering cross section
2. small absorption cross section
3. large energy loss per collision
& The energy loss after a specified number of collisions can be calculated using the equation below.

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N
■
■ ■
Reactor Theory (Neutron Characteristics) DOE-HDBK-1019/1-93 PROMPT AND DELAYED NEUTRONS
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PROMPT AND DELAYED NEUTRONS
Not all neutrons are released at the same time following fission. Most neutrons are released virtually instantaneously and are called prompt neutrons. A very small fraction of neutrons are released after the decay of fission products and are called delayed neutrons. Although delayed neutrons are a very small fraction of the total number of neutrons, they play an extremely important role in the control of the reactor.
EO 3.1 STATE the origin of prompt neutrons and delayed neutrons.
EO 3.2 STATE the approximate fraction of neutrons that are born as delayed neutrons from the fission of the following nuclear fuels:
a. Uranium-235
b. Plutonium-239
EO 3.3 EXPLAIN the mechanism for production of delayed neutrons.
EO 3.4 EXPLAIN prompt and delayed neutron generation times.
EO 3.5 Given prompt and delayed neutron generation times and
Fixture checkpoint marker-0221-xray is recorded on fixture page 0221.
delayed neutron fraction, CALCULATE the average generation time.
EO 3.6 EXPLAIN the effect of delayed neutrons on reactor control.
Neutron Classification
The great majority (over 99%) of the neutrons produced in fission are released within about 10-13 seconds of the actual fission event. These are called prompt neutrons. A small portion of fission neutrons are delayed neutrons, which are produced for some time after the fission process has taken place. The delayed neutrons are emitted immediately following the first beta decay of a fission fragment known as a delayed neutron precursor. An example of a delayed neutron precursor is bromine-87, shown below.
PROMPT AND DELAYED NEUTRONS DOE-HDBK-1019/1-93 Reactor Theory (Neutron Characteristics)
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For most applications, it is convenient to combine the known precursors into groups with appropriately averaged properties. These groups vary somewhat depending on the fissile material in use. Table 3 lists the characteristics for the six precursor groups resulting from thermal fission of uranium-235. The fraction of all neutrons that are produced by each of these precursors is called the delayed neutron fraction for that precursor. The total fraction of all neutrons born as delayed neutrons is called the delayed neutron fraction (■). The fraction of delayed neutrons produced varies depending on the predominant fissile nuclide in use. The delayed neutron fractions (■) for the fissile nuclides of most interest are as follows: uranium-233 (0.0026), uranium-235 (0.0065), uranium-238 (0.0148), and plutonium-239 (0.0021).
TABLE 3
Delayed Neutron Precursor Groups
for Thermal Fission in Uranium-235
Group Half-Life Delayed Neutron Average Energy
(sec) Fraction (MeV)
1 55.7 0.00021 0.25

3 6.2 0.00127 0.41
4 2.3 0.00257 0.45
5 0.61 0.00075 0.41
6 0.23 0.00027 -
Total - 0.0065 -
Neutron Generation Time
The neutron generation time is the time required for neutrons from one generation to cause the fissions that produce the next generation of neutrons. The generation time for prompt neutrons (5* - pronounced "ell-star") is the total time from birth to rebirth. Three time intervals are involved: (a) the time it takes a fast neutron to slow down to thermal energy, (b) the time the now thermal neutron exists prior to absorption in fuel, and (c) the time required for a fissionable nucleus to emit a fast neutron after neutron absorption.
Fast neutrons slow to thermal energies or leak out of the reactor in 10 seconds to 10 seconds, -4 -6 depending on the moderator. In water moderated reactors, thermal neutrons tend to exist for about 10 seconds before they are absorbed. Fission and fast neutron production following -4 neutron absorption in a fissionable nucleus occurs in about 10 seconds. Thus, fast reactors -13 have an 5* of about 10 seconds, while thermal reactors have an 5* of about 10 seconds + 10 -6 -6 -4 seconds, which is about 10 seconds to 10 seconds. -4 -5
■ ■ ■ ■
■ ■ ■ ■
■ ■
Reactor Theory (Neutron Characteristics) DOE-HDBK-1019/1-93 PROMPT AND DELAYED NEUTRONS
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On the other hand, the average generation time for the six delayed neutron groups is the total time from the birth of the fast neutron to the emission of the delayed neutron. Again, three time intervals are involved: (a) the time it takes a fast neutron to slow down to thermal energy, (b) the time the thermal neutron exists prior to absorption, and (c) the average time from neutron
Fixture checkpoint marker-0222-xray is recorded on fixture page 0222.
absorption to neutron emission by the six precursor groups. The average time for decay of precursors from uranium-235 is 12.5 seconds. The other terms in the delayed neutron generation time are insignificant when compared to this value, and the average delayed neutron generation time becomes ■ 12.5 seconds.
A neutron generation time in the range of 10 seconds to 10 seconds or faster could result in -4 -5 very rapid power excursions, and control would not be possible without the dependence upon delayed neutrons to slow down the rate of the reaction. The average generation time, and hence the rate that power can rise, is determined largely by the delayed neutron generation time. The following equation shows this mathematically.
(2-12)
Example:
Assume a prompt neutron generation time for a particular reactor of 5 x 10 seconds and -5 a delayed neutron generation time of 12.5 seconds. If ■ is 0.0065, calculate the average generation time.
Solution:
This example demonstrates the effect delayed neutrons have on the neutron generation time and thus reactor control. If a reactor were to be operated in a sustained chain reaction using only prompt neutrons (■ = 0), the generation time from the previous example would be about 5 x 10 seconds. However, by operating the reactor such that a 0.0065 fraction of neutrons are -5 delayed, the generation life time is extended to 0.0813 seconds, providing time for adequate

delayed neutrons are extremely important to the control and maintenance of a sustained fission chain reaction.

Summary

The important information in this chapter is summarized on the following page.

■ ■ ■ ■

PROMPT AND DELAYED NEUTRONS DOE-HDBK-1019/1-93 Reactor Theory (Neutron Characteristics) NP-02 Page 32 Rev. 0

Prompt and Delayed Neutrons Summary

& Prompt neutrons are released directly from fission within 10 seconds of the -13 fission event.

& Delayed neutrons are released from the decay of fission products that are called delayed neutron precursors. Delayed neutron precursors are grouped according to half-life. Half-lives vary from fractions of a second to almost a minute.

& The fraction of neutrons born as delayed neutrons is different for different fuel materials. Following are values for some common fuel materials.

Uranium-235 0.0065

Plutonium-239 0.0021

& Delayed neutrons are produced by a classification of fission products known as delayed neutron precursors. When a delayed neutron precursor undergoes a ■ decay, it results in an excited daughter nucleus which immediately ejects a - neutron. Therefore, these delayed neutrons appear with a half-life of the delayed neutron precursor.

& The delayed neutron generation time is the total time from the birth of the fast neutron to the emission of the delayed neutron in the next generation. Delayed neutron generation times are dominated by the half-life of the delayed neutron precursor. The average delayed neutron generation time is about 12.5 seconds.

& A prompt neutron generation time is the sum of the amount of time it takes a fast neutron to thermalize, the amount of time the neutron exists as a thermal neutron before it is absorbed, and the amount of time between a fissionable nuclide absorbing a neutron and fission neutrons being released. Prompt neutron generation time is about 5 x 10 seconds. -5

& The average neutron generation time can be calculated from the prompt and delayed neutron generation times and the delayed neutron fraction using Equation (2-12).

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& Delayed neutrons are responsible for the ability to control the rate at which power can rise in a reactor. If only prompt neutrons existed, reactor control would not be possible due to the rapid power changes.

Reactor Theory (Neutron Characteristics) DOE-HDBK-1019/1-93 NEUTRON FLUX SPECTRUM Rev. 0 Page 33 NP-02

Figure 2 Prompt Fission Neutron Energy Spectrum for Thermal Fission of Uranium-235

NEUTRON FLUX SPECTRUM

The number of neutrons that exist at a given energy level varies. A plot of either the fraction of neutrons or the neutron flux at a given energy versus the energy level is called a neutron energy spectrum. The neutron energy spectrum varies widely for different types of reactors.

EO 4.1 STATE the average energy at which prompt neutrons are

EO 4.2 DESCRIBE the neutron energy spectrum in the following reactors:
a. Fast reactor
b. Thermal reactor
EO 4.3 EXPLAIN the reason for the particular shape of the fast, intermediate, and slow energy regions of the neutron flux spectrum for a thermal reactor.
Prompt Neutron Energies
The neutrons produced by fission are high energy neutrons, and almost all fission neutrons have
Fixture checkpoint marker-0224-xray is recorded on fixture page 0224.
energies between 0.1 MeV and 10 MeV. The neutron energy distribution, or spectrum, may best be described by plotting the fraction of neutrons per MeV as a function of neutron energy, as shown in Figure 2. From this figure it can be seen that the most probable neutron energy is about 0.7 MeV. Also, from this data it can be shown that the average energy of fission neutrons is about 2 MeV. Figure 2 is the neutron energy spectrum for thermal fission in uranium-235. The values will vary slightly for other nuclides.
NEUTRON FLUX SPECTRUM DOE-HDBK-1019/1-93 Reactor Theory (Neutron Characteristics) NP-02 Page 34 Rev. 0
Figure 3 Comparison of Neutron Flux Spectra for Thermal and Fast Breeder Reactor
Thermal and Fast Breeder Reactor Neutron Spectra
The spectrum of neutron energies produced by fission varies significantly from the energy spectrum, or flux, existing in a reactor at a given time. Figure 3 illustrates the difference in neutron flux spectra between a thermal reactor and a fast breeder reactor. The energy distribution of neutrons from fission is essentially the same for both reactors, so the differences in the curve shapes may be attributed to the neutron moderation or slowing down effects.
No attempt is made to thermalize or slow down neutrons in the fast breeder reactor (liquid metal cooled); therefore, an insignificant number of neutrons exist in the thermal range. For the thermal reactor (water moderated), the spectrum of neutrons in the fast region (> 0.1 MeV) has a shape similar to that for the spectrum of neutrons emitted by the fission process.
In the thermal reactor, the flux in the intermediate energy region (1 eV to 0.1 MeV) has approximately a 1/E dependence. That is, if the energy (E) is halved, the flux doubles. This 1/E dependence is caused by the slowing down process, where elastic collisions remove a constant fraction of the neutron energy per collision (on the average), independent of energy; thus, the neutron loses larger amounts of energy per collision at higher energies than at lower

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neutrons to tend to "pile up" at lower energies, that is, a greater number of neutrons exist at the lower energies as a result of this behavior.
vp
2 k T
m
vp
2 k T
m
2 1.38 x 10 16 erg
K
293 K
1.66 x 10 24 g
2.2 x 105 cm
sec
1 m
100 cm
2200 m
sec
Fixture checkpoint marker-0225-xray is recorded on fixture page 0225.
Reactor Theory (Neutron Characteristics) DOE-HDBK-1019/1-93 NEUTRON FLUX SPECTRUM
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In the thermal region the neutrons achieve a thermal equilibrium with the atoms of the moderator material. In any given collision they may gain or lose energy, and over successive collisions will gain as much energy as they lose. These thermal neutrons, even at a specific temperature, do not all have the same energy or velocity; there is a distribution of energies, usually referred to as the Maxwell distribution (e.g., Figure 2). The energies of most thermal neutrons lie close to the most probable energy, but there is a spread of neutrons above and below this value.
Most Probable Neutron Velocities
The most probable velocity (vp) of a thermal neutron is determined by the temperature of the medium and can be determined by Equation (2-13) .
(2-13)
where:
v = most probable velocity of neutron (cm/sec) p
k = Boltzman's constant (1.38 x 10 erg/ K) -16
T = absolute temperature in degrees Kelvin (K)
m = mass of neutron (1.66 x 10 grams) -24
Example:
Calculate the most probable velocities for neutrons in thermal equilibrium with their surroundings at the following temperatures. a) 20 C, b) 260 C.
Solution:
a) Calculate the most probable velocity for 20 C using Equation (2-13).
vp
2 k T
m
2 1.38 x 10 16 erg
K
533 K
1.66 x 10 24 g

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DOE-HDBK-1019/1-93
NUCLEAR PHYSICS AND REACTOR THEORY
ABSTRACT
The Nuclear Physics and Reactor Theory Handbook was developed to assist nuclear facility operating contractors in providing operators, maintenance personnel, and the technical staff with the necessary fundamentals training to ensure a basic understanding of nuclear physics and reactor theory. The handbook includes information on atomic and nuclear physics; neutron characteristics; reactor theory and nuclear parameters; and the theory of reactor operation. This information will provide personnel with a foundation for understanding the scientific principles that are associated with various DOE nuclear facility operations and maintenance.
Key Words: Training Material, Atomic Physics, The Chart of the Nuclides, Radioactivity, Radioactive Decay, Neutron Interaction, Fission, Reactor Theory, Neutron Characteristics, Neutron Life Cycle, Reactor Kinetics
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NUCLEAR PHYSICS AND REACTOR THEORY
FOREWORD
The Department of Energy (DOE) Fundamentals Handbooks consist of ten academic subjects, which include Mathematics; Classical Physics; Thermodynamics, Heat Transfer, and Fixture checkpoint marker-0226-xray is recorded on fixture page 0226.
Fluid Flow; Instrumentation and Control; Electrical Science; Material Science; Mechanical Science; Chemistry; Engineering Symbology, Prints, and Drawings; and Nuclear Physics and Reactor Theory. The handbooks are provided as an aid to DOE nuclear facility contractors.
These handbooks were first published as Reactor Operator Fundamentals Manuals in 1985 for use by DOE category A reactors. The subject areas, subject matter content, and level of detail of the Reactor Operator Fundamentals Manuals were determined from several sources. DOE Category A reactor training managers determined which materials should be included, and served as a primary reference in the initial development phase. Training guidelines from the commercial nuclear power industry, results of job and task analyses, and independent input from contractors and operations-oriented personnel were all considered and included to some degree in developing the text material and learning objectives.
The DOE Fundamentals Handbooks represent the needs of various DOE nuclear facilities' fundamental training requirements. To increase their applicability to nonreactor nuclear facilities, the Reactor Operator Fundamentals Manual learning objectives were distributed to the Nuclear Facility Training Coordination Program Steering Committee for review and comment. To update their reactor-specific content, DOE Category A reactor training managers also reviewed and commented on the content. On the basis of feedback from these sources, information that applied to two or more DOE nuclear facilities was considered generic and was included. The final draft of each of the handbooks was then reviewed by these two groups. This approach has resulted in revised modular handbooks that contain sufficient detail such that each facility may adjust the content to fit their specific needs.
Each handbook contains an abstract, a foreword, an overview, learning objectives, and text

DOE contractors to suit their specific training needs. Each handbook is supported by a separate examination bank with an answer key.

The DOE Fundamentals Handbooks have been prepared for the Assistant Secretary for Nuclear Energy, Office of Nuclear Safety Policy and Standards, by the DOE Training Coordination Program. This program is managed by EG&G Idaho, Inc.

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NUCLEAR PHYSICS AND REACTOR THEORY

OVERVIEW

The Department of Energy Fundamentals Handbook entitled Nuclear Physics and Reactor Theory was prepared as an information resource for personnel who are responsible for the operation of the Department's nuclear facilities. Almost all processes that take place in a nuclear facility involves the transfer of some type of energy. A basic understanding of nuclear physics and reactor theory is necessary for DOE nuclear facility operators, maintenance personnel, and the technical staff to safely operate and maintain the facility and facility support systems. The information in this handbook is presented to provide a foundation for applying engineering concepts to the job. This knowledge will help personnel understand the impact that their actions may have on the safe and reliable operation of facility components and systems.

The Nuclear Physics and Reactor Theory handbook consists of four modules that are contained in two volumes. The following is a brief description of the information presented in each module of the handbook.

Volume 1 of 2

Module 1 - Atomic and Nuclear Physics

Introduces concepts of atomic physics including the atomic nature of matter, the chart of the nuclides, radioactivity and radioactive decay, neutron interactions and fission, and the interaction of radiation with matter.

Module 2 - Reactor Theory (Nuclear Parameters)

Provides information on reactor theory and neutron characteristics. Includes topics such as neutron sources, neutron flux, neutron cross sections, reaction rates, neutron moderation, and prompt and delayed neutrons.

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Module 3 - Reactor Theory (Nuclear Parameters)

Explains the nuclear parameters associated with reactor theory. Topics include the neutron life cycle, reactivity and reactivity coefficients, neutron poisons, and control rods.

Module 4 - Reactor Theory (Reactor Operations)

Introduces the reactor operations aspect of reactor theory. Topics include subcritical multiplication, reactor kinetics, and reactor operation.

The information contained in this handbook is not all-encompassing. An attempt to present the entire subject of nuclear physics and reactor theory would be impractical. However, the Nuclear Physics and Reactor Theory handbook presents enough information to provide the reader with the fundamental knowledge necessary to understand the advanced theoretical concepts presented in other subject areas, and to understand basic system and equipment operation.

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Atomic and Nuclear Physics DOE-HDBK-1019/1-93 OBJECTIVES
TERMINAL OBJECTIVE
1.0 Given sufficient information, DESCRIBE atoms, including components, structure, and nomenclature.
ENABLING OBJECTIVES
1.1 STATE the characteristics of the following atomic particles, including mass, charge, and location within the atom:
a. Proton
b. Neutron
c. Electron
1.2 DESCRIBE the Bohr model of an atom.
1.3 DEFINE the following terms:
a. Nuclide c. Atomic number
b. Isotope d. Mass number
1.4 Given the standard A
ZX notation for a particular nuclide, DETERMINE the following:
a. Number of protons
b. Number of neutrons
c. Number of electrons
1.5 DESCRIBE the three forces that act on particles within the nucleus and affect the stability of the nucleus.
1.6 DEFINE the following terms:
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a. Enriched uranium
b. Depleted uranium
1.7 DEFINE the following terms:
a. Mass defect
b. Binding energy
1.8 Given the atomic mass for a nuclide and the atomic masses of a neutron, proton, and electron, CALCULATE the mass defect and binding energy of the nuclide.

OBJECTIVES DOE-HDBK-1019/1-93 Atomic and Nuclear Physics
TERMINAL OBJECTIVE
2.0 Given necessary references, DESCRIBE the various modes of radioactive decay.
ENABLING OBJECTIVES
2.1 DESCRIBE the following processes:
a. Alpha decay d. Electron capture
b. Beta-minus decay e. Internal conversions
c. Beta-plus decay f. Isomeric transitions
2.2 Given a Chart of the Nuclides, WRITE the radioactive decay chain for a nuclide.

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2.4 Given the stability curve on the Chart of the Nuclides, DETERMINE the type of radioactive decay that the nuclides in each region of the chart will typically undergo.
2.5 DEFINE the following terms:
a. Radioactivity d. Radioactive decay constant
b. Curie e. Radioactive half-life
c. Becquerel
2.6 Given the number of atoms and either the half-life or decay constant of a nuclide, CALCULATE the activity.
2.7 Given the initial activity and the decay constant of a nuclide, CALCULATE the activity at any later time.
2.8 CONVERT between the half-life and decay constant for a nuclide.
2.9 Given the Chart of the Nuclides and the original activity, PLOT the radioactive decay curve for a nuclide on either linear or semi-log coordinates.
2.10 DEFINE the following terms:
a. Radioactive equilibrium
b. Transient radioactive equilibrium
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Atomic and Nuclear Physics DOE-HDBK-1019/1-93 OBJECTIVES
TERMINAL OBJECTIVE
3.0 Without references, DESCRIBE the different nuclear interactions initiated by neutrons.
ENABLING OBJECTIVES
3.1 DESCRIBE the following scattering interactions between a neutron and a nucleus:
a. Elastic scattering
b. Inelastic scattering
3.2 STATE the conservation laws that apply to an elastic collision between a neutron and a nucleus.
3.3 DESCRIBE the following reactions where a neutron is absorbed in a nucleus:
a. Radiative capture
b. Particle ejection
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OBJECTIVES DOE-HDBK-1019/1-93 Atomic and Nuclear Physics
TERMINAL OBJECTIVE
4.0 Without references, DESCRIBE the fission process.
ENABLING OBJECTIVES
4.1 EXPLAIN the fission process using the liquid drop model of a nucleus.
4.2 DEFINE the following terms:
a. Excitation energy
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b. Critical energy
4.3 DEFINE the following terms:
a. Fissile material
b. Fissionable material
c. Fertile material
4.4 DESCRIBE the processes of transmutation, conversion, and breeding.
4.5 DESCRIBE the curve of Binding Energy per Nucleon versus mass number and give a qualitative description of the reasons for its shape.
4.6 EXPLAIN why only the heaviest nuclei are easily fissioned.
4.7 EXPLAIN why uranium-235 fissions with thermal neutrons and uranium-238 fissions only

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4.8 CHARACTERIZE the fission products in terms of mass groupings and radioactivity.
4.9 Given the nuclides involved and their masses, CALCULATE the energy released from fission.
4.10 Given the curve of Binding Energy per Nucleon versus mass number, CALCULATE the energy released from fission.
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Atomic and Nuclear Physics DOE-HDBK-1019/1-93 OBJECTIVES
TERMINAL OBJECTIVE
5.0 Without references, DESCRIBE how the various types of radiation interact with matter.
ENABLING OBJECTIVES
5.1 DESCRIBE interactions of the following with matter:
a. Alpha particle c. Positron
b. Beta particle d. Neutron
5.2 DESCRIBE the following ways that gamma radiation interacts with matter:
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a. Compton scattering
b. Photoelectric effect
c. Pair production
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OBJECTIVES DOE-HDBK-1019/1-93 Atomic and Nuclear Physics
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Atomic and Nuclear Physics DOE-HDBK-1019/1-93 ATOMIC NATURE OF MATTER
ATOMIC NATURE OF MATTER
All matter is composed of atoms. The atom is the smallest amount of matter that retains the properties of an element. Atoms themselves are composed of smaller particles, but these smaller particles no longer have the same properties as the overall element.
EO 1.1 STATE the characteristics of the following atomic particles, including mass, charge, and location within the atom:
a. Proton
b. Neutron
c. Electron
EO 1.2 DESCRIBE the Bohr model of an atom.
EO 1.3 DEFINE the following terms:
a. Nuclide c. Atomic number
b. Isotope d. Mass number
EO 1.4 Given the standard A ZX notation for a particular nuclide, DETERMINE the following:
a. Number of protons
b. Number of neutrons
c. Number of electrons
EO 1.5 DESCRIBE the three forces that act on particles within the nucleus and affect the stability of the nucleus.
Structure of Matter
Early Greek philosophers speculated that the earth was made up of different combinations of basic substances, or elements. They considered these basic elements to be earth, air, water, and

of a combination of basic elements, but they incorrectly identified the elements.
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ATOMIC NATURE OF MATTER DOE-HDBK-1019/1-93 Atomic and Nuclear Physics
In 1661 the English chemist Robert Boyle published the modern criterion for an element. He defined an element to be a basic substance that cannot be broken down into any simpler substance after it is isolated from a compound, but can be combined with other elements to form compounds. To date, 105 different elements have been confirmed to exist, and researchers claim to have discovered three additional elements. Of the 105 confirmed elements, 90 exist in nature and 15 are man-made.
Another basic concept of matter that the Greeks debated was whether matter was continuous or discrete. That is, whether matter could be continuously divided and subdivided into ever smaller particles or whether eventually an indivisible particle would be encountered. Democritus in about 450 B.C. argued that substances were ultimately composed of small, indivisible particles that he labeled atoms. He further suggested that different substances were composed of different atoms or combinations of atoms, and that one substance could be converted into another by rearranging the atoms. It was impossible to conclusively prove or disprove this proposal for more than 2000 years.
The modern proof for the atomic nature of matter was first proposed by the English chemist John Dalton in 1803. Dalton stated that each chemical element possesses a particular kind of atom, and any quantity of the element is made up of identical atoms of this kind. What distinguishes one element from another element is the kind of atom of which it consists, and the basic physical
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difference between kinds of atoms is their weight.
Subatomic Particles
For almost 100 years after Dalton established the atomic nature of atoms, it was considered impossible to divide the atom into even smaller parts. All of the results of chemical experiments during this time indicated that the atom was indivisible. Eventually, experimentation into electricity and radioactivity indicated that particles of matter smaller than the atom did indeed exist. In 1906, J. J. Thompson won the Nobel Prize in physics for establishing the existence of electrons. Electrons are negatively-charged particles that have 1/1835 the mass of the hydrogen atom. Soon after the discovery of electrons, protons were discovered. Protons are relatively large particles that have almost the same mass as a hydrogen atom and a positive charge equal in magnitude (but opposite in sign) to that of the electron. The third subatomic particle to be discovered, the neutron, was not found until 1932. The neutron has almost the same mass as the proton, but it is electrically neutral.
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Atomic and Nuclear Physics DOE-HDBK-1019/1-93 ATOMIC NATURE OF MATTER
Bohr Model of the Atom
The British physicist Ernest Rutherford postulated that the positive charge in an atom is concentrated in a small region called a nucleus at the center of the atom with electrons existing in orbits around it. Niels Bohr, coupling Rutherford's postulation with the quantum theory introduced by Max Planck, proposed that the atom consists of a dense nucleus of protons surrounded by electrons traveling in discrete orbits at fixed distances from the nucleus. An electron in one of these orbits or shells has a specific or discrete quantity of energy (quantum). When an electron moves from one allowed orbit to another allowed orbit, the energy difference between the two states is emitted or absorbed in the form of a single quantum of radiant energy called a photon. Figure 1 is Bohr's model of the hydrogen atom showing an electron as having just dropped from the third shell to the first shell with the emission of a photon that has an

theory was the first to successfully account for the discrete energy levels of this radiation as measured in the laboratory. Although Bohr's atomic model is designed specifically to explain the hydrogen atom, his theories apply generally to the structure of all atoms. Additional information on electron shell theory can be found in the Chemistry Fundamentals Handbook.
Figure 1 Bohr's Model of the Hydrogen Atom

Properties of the three subatomic particles are listed in Table 1.
TABLE 1
Properties of Subatomic Particles

Particle	Location	Charge	Mass
Neutron	Nucleus	none	1.008665 amu
Proton	Nucleus	+1	1.007277 amu
Electron	Shells around nucleus	-1	0.0005486 amu

Measuring Units on the Atomic Scale
The size and mass of atoms are so small that the use of normal measuring units, while possible, is often inconvenient. Units of measure have been defined for mass and energy on the atomic scale to make measurements more convenient to express. The unit of measure for mass is the atomic mass unit (amu). One atomic mass unit is equal to 1.66 x 10-24 grams. The reason for this particular value for the atomic mass unit will be discussed in a later chapter. Note from Table 1 that the mass of a neutron and a proton are both about 1 amu. The unit for energy is the electron volt (eV). The electron volt is the amount of energy acquired by a single electron when it falls through a potential difference of one volt. One electron volt is equivalent to 1.602 x 10-19 joules or 1.18 x 10-19 foot-pounds.
Nuclides
The total number of protons in the nucleus of an atom is called the atomic number of the atom and is given the symbol Z. The number of electrons in an electrically-neutral atom is the same as the number of protons in the nucleus. The number of neutrons in a nucleus is known as the
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neutron number and is given the symbol N. The mass number of the nucleus is the total number of nucleons, that is, protons and neutrons in the nucleus. The mass number is given the symbol A and can be found by the equation Z + N = A.
Each of the chemical elements has a unique atomic number because the atoms of different elements contain a different number of protons. The atomic number of an atom identifies the particular element.

Each type of atom that contains a unique combination of
Figure 2 Nomenclature for
Identifying Nuclides
protons and neutrons is called a nuclide. Not all combinations of numbers of protons and neutrons are possible, but about 2500 specific nuclides with unique combinations of neutrons and protons have been identified. Each nuclide is denoted by the chemical symbol of the element with the atomic number written as a subscript and the mass number written as a superscript, as shown in Figure 2. Because each element has a

one of the three is necessary to identify the element. For this reason nuclides can also be identified by either the chemical name or the chemical symbol followed by the mass number (for example, U-235 or uranium-235). Another common format is to use the abbreviation of the chemical element with the mass number superscripted (for example, 235U). In this handbook the format used in the text will usually be the element's name followed by the mass number. In equations and tables, the format in Figure 2 will usually be used.

Example:

State the name of the element and the number of protons, electrons, and neutrons in the nuclides listed below.

1
1H
10
5B
14
7N
11
4
4
8Cd
23
9
9
4Pu

Solution:

The name of the element can be found from the Periodic Table (refer to Chemistry Fundamentals Handbook) or the Chart of the Nuclides (to be discussed later). The number of protons and electrons are equal to Z. The number of neutrons is equal to Z - A.

Nuclide Element Protons Electrons Neutrons

1
1H hydrogen 1 1 0

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10
5B boron 5 5 5
14
7N nitrogen 7 7 7
11
4
4
8Cd cadmium 48 48 66
23
9
9
4Pu plutonium 94 94 145

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Isotopes are nuclides that have the same atomic number and are therefore the same element, but differ in the number of neutrons. Most elements have a few stable isotopes and several unstable, radioactive isotopes. For example, oxygen has three stable isotopes that can be found in nature (oxygen-16, oxygen-17, and oxygen-18) and eight radioactive isotopes. Another example is hydrogen, which has two stable isotopes (hydrogen-1 and hydrogen-2) and a single radioactive isotope (hydrogen-3).

The isotopes of hydrogen are unique in that they are each commonly referred to by a unique name instead of the common chemical element name. Hydrogen-1 is almost always referred to as hydrogen, but the term protium is infrequently used also. Hydrogen-2 is commonly called deuterium and symbolized 2

1D. Hydrogen-3 is commonly called tritium and symbolized 31T. This

text will normally use the symbology 2

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1H and 31H for deuterium and tritium, respectively.

Atomic and Nuclear Radii

The size of an atom is difficult to define exactly due to the fact that the electron cloud, formed by the electrons moving in their various orbitals, does not have a distinct outer edge. A reasonable measure of atomic size is given by the average distance of the outermost electron from the nucleus. Except for a few of the lightest atoms, the average atomic radii are approximately the same for all atoms, about 2 x 10 -8 cm.

Like the atom the nucleus does not have a sharp outer boundary. Experiments have shown that the nucleus is shaped like a sphere with a radius that depends on the atomic mass number of the atom. The relationship between the atomic mass number and the radius of the nucleus is shown in the following equation.

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r = (1.25 x 10 -13 cm) A1/3

where:

r = radius of the nucleus (cm)

A = atomic mass number (dimensionless)

The values of the nuclear radii for some light, intermediate, and heavy nuclides are shown in Table 2.

TABLE 2

Calculated Values for Nuclear Radii

Nuclide Radius of Nucleus

1

1H 1.25 x 10 -13 cm

10

5B 2.69 x 10 -13 cm

5

2

6

6Fe 4.78 x 10 -13 cm

17

7

8

2Hf 7.01 x 10 -13 cm

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9
8
2U 7.74 x 10 -13 cm
25
9
2
8Cf 7.89 x 10 -13 cm
From the table, it is clear that the radius of a typical atom (e.g. 2 x 10 -8 cm) is more than 25,000 times larger than the radius of the largest nucleus.
Nuclear Forces
In the Bohr model of the atom, the nucleus consists of positively-charged protons and electrically■neutra as nucleons. One problem that the Bohr model of the atom presented was accounting for an attractive force to overcome the repulsive force between protons.
Two forces present in the nucleus are (1) electrostatic forces between charged particles and (2) gravitational forces between any two objects that have mass. It is possible to calculate the magnitude of the gravitational force and electrostatic force based upon principles from classical physics.
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Fg
Fixture checkpoint marker-0238-xray is recorded on fixture page 0238.
G m1 m2
r
2
Fe
K Q1 Q2
r
2
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Newton stated that the gravitational force between two bodies is directly proportional to the masses of the two bodies and inversely proportional to the square of the distance between the bodies. This relationship is shown in the equation below.
where:
F = gravitational force (newtons) g
m = mass of first body (kilograms) 1
m = mass of second body (kilograms) 2
G = gravitational constant (6.67 x 10 N-m /kg)
-11 2 2
r = distance between particles (meters)
The equation illustrates that the larger the masses of the objects or the smaller the distance between the objects, the greater the gravitational force. So even though the masses of nucleons are very small, the fact that the distance between nucleons is extremely short may make the gravitational force significant. It is necessary to calculate the value for the gravitational force and compare it to the value for other forces to determine the significance of the gravitational force in the nucleus. The gravitational force between two protons that are separated by a distance of 10 meters is about 10 newtons. -20 -24
Coulomb's Law can be used to calculate the force between two protons. The electrostatic force is directly proportional to the electrical charges of the two particles and inversely proportional

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equation.
where:
F = electrostatic force (newtons) e
K = electrostatic constant (9.0 x 10 N-m /C)
9 2 2
Q = charge of first particle (coulombs) 1
Q = charge of second particle (coulombs) 2
r = distance between particles (meters)
Using this equation, the electrostatic force between two protons that are separated by a distance of 10 meters is about 10 newtons. Comparing this result with the calculation of the
-20 12
gravitational force (10 newtons) shows that the gravitational force is so small that it can be
-24
neglected.
Atomic and Nuclear Physics DOE-HDBK-1019/1-93 ATOMIC NATURE OF MATTER
If only the electrostatic and gravitational forces existed in the nucleus, then it would be impossible to have stable nuclei composed of protons and neutrons. The gravitational forces are much too small to hold the nucleons together compared to the electrostatic forces repelling the protons. Since stable atoms of neutrons and protons do exist, there must be another attractive force acting within the nucleus. This force is called the nuclear force.
The nuclear force is a strong attractive force that is independent of charge. It acts equally only between pairs of neutrons, pairs of protons, or a neutron and a proton. The nuclear force has a very short range; it acts only over distances approximately equal to the diameter of the nucleus (10 -13 cm). The attractive nuclear force between all nucleons drops off with distance much faster than the repulsive electrostatic force between protons.
TABLE 3
Fixture checkpoint marker-0239-xray is recorded on fixture page 0239.
Forces Acting in the Nucleus
Force Interaction Range
Gravitational Very weak attractive force between all nucleons Relatively long
Electrostatic Strong repulsive force between like charged particles (protons) Relatively long
Nuclear Force Strong attractive force between all nucleons Extremely short
In stable atoms, the attractive and repulsive forces in the nucleus balance. If the forces do not balance, the atom cannot be stable, and the nucleus will emit radiation in an attempt to achieve a more stable configuration.
Summary
The important information in this chapter is summarized on the following page.
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ATOMIC NATURE OF MATTER DOE-HDBK-1019/1-93 Atomic and Nuclear Physics
Atomic Nature of Matter Summary
Atoms consist of three basic subatomic particles. These particles are the proton, the neutron, and the electron.
Protons are particles that have a positive charge, have about the same mass as a hydrogen atom, and exist in the nucleus of an atom.
Neutrons are particles that have no electrical charge, have about the same mass as a

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Electrons are particles that have a negative charge, have a mass about eighteen hundred times smaller than the mass of a hydrogen atom, and exist in orbital shells around the nucleus of an atom.
The Bohr model of the atom consists of a dense nucleus of protons and neutrons (nucleons) surrounded by electrons traveling in discrete orbits at fixed distances from the nucleus.
Nuclides are atoms that contain a particular number of protons and neutrons.
Isotopes are nuclides that have the same atomic number and are therefore the same element, but differ in the number of neutrons.
The atomic number of an atom is the number of protons in the nucleus.
The mass number of an atom is the total number of nucleons (protons and neutrons) in the nucleus.
The notation A
ZX is used to identify a specific nuclide. "Z" represents the atomic number, which is equal to the number of protons. "A" represents the mass number, which is equal to the number of nucleons. "X" represents the chemical symbol of the element.
Number of protons = Z
Number of electrons = Z
Number of neutrons = A - Z
The stability of a nucleus is determined by the different forces interacting within it. The electrostatic force is a relatively long-range, strong, repulsive force that acts between the positively charged protons. The nuclear force is a relatively short-range attractive force between all nucleons. The gravitational force the long range, relatively weak attraction between masses, is negligible compared to the other forces.
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Atomic and Nuclear Physics DOE-HDBK-1019/1-93 CHART OF THE NUCLIDES
CHART OF THE NUCLIDES
The Chart of the Nuclides, like the Periodic Table, is a convenient format for presenting a large amount of scientific information in an organized manner.
EO 1.6 DEFINE the following terms:
Fixture checkpoint marker-0240-xray is recorded on fixture page 0240.
a. Enriched uranium
b. Depleted uranium
Chart of the Nuclides
A tabulated chart called the Chart of the Nuclides lists the stable and unstable nuclides in addition to pertinent information about each one. Figure 3 shows a small portion of a typical chart. This chart plots a box for each individual nuclide, with the number of protons (Z) on the vertical axis and the number of neutrons (N = A - Z) on the horizontal axis.
The completely gray squares indicate stable isotopes. Those in white squares are artificially radioactive, meaning that they are produced by artificial techniques and do not occur naturally. By consulting a complete chart, other types of isotopes can be found, such as naturally occurring radioactive types (but none are found in the region of the chart that is illustrated in Figure 3).
Located in the box on the far left of each horizontal row is general information about the element. The box contains the chemical symbol of the element in addition to the average atomic weight of the naturally occurring substance and the average thermal neutron absorption cross

atomic number Z but different mass number A) of each element are listed to the right.
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CHART OF THE NUCLIDES DOE-HDBK-1019/1-93 Atomic and Nuclear Physics
Figure 3 Nuclide Chart for Atomic Numbers 1 to 6
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Atomic and Nuclear Physics DOE-HDBK-1019/1-93 CHART OF THE NUCLIDES
Information for Stable Nuclides
For the stable isotopes, in addition to the symbol and the atomic mass number, the number percentage of each isotope in the naturally occurring element is listed, as well as the thermal neutron activation cross section and the mass in atomic mass units (amu). A typical block for Fixture checkpoint marker-0241-xray is recorded on fixture page 0241.
a stable nuclide from the Chart of the Nuclides is shown in Figure 4.
Figure 4 Stable Nuclides
Information for Unstable Nuclides
For unstable isotopes the additional information includes the half life, the mode of decay (for example, β^-, α), the total disintegration energy in MeV (million electron volts), and the mass in amu when available. A typical block for an unstable nuclide from the Chart of the Nuclides is shown in Figure 5.
Figure 5 Unstable Nuclides
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CHART OF THE NUCLIDES DOE-HDBK-1019/1-93 Atomic and Nuclear Physics
Neutron - Proton Ratios
Figure 6 shows the distribution of the stable nuclides plotted on the same axes as the Chart of the Nuclides. As the mass numbers become higher, the ratio of neutrons to protons in the nucleus becomes larger. For helium-4 (2 protons and 2 neutrons) and oxygen-16 (8 protons and 8 neutrons) this ratio is unity. For indium-115 (49 protons and 66 neutrons) the ratio of neutrons to protons has increased to 1.35, and for uranium-238 (92 protons and 146 neutrons) the neutron■to-proton
Figure 6 Neutron - Proton Plot of the Stable Nuclides
If a heavy nucleus were to split into two fragments, each fragment would form a nucleus that would have approximately the same neutron-to-proton ratio as the heavy nucleus. This high neutron-to-proton ratio places the fragments below and to the right of the stability curve displayed by Figure 6. The instability caused by this excess of neutrons is generally rectified by successive beta emissions, each of which converts a neutron to a proton and moves the nucleus toward a more stable neutron-to-proton ratio.
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■
Atomic and Nuclear Physics DOE-HDBK-1019/1-93 CHART OF THE NUCLIDES
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Natural Abundance of Isotopes
The relative abundance of an isotope in nature compared to other isotopes of the same element is relatively constant. The Chart of the Nuclides presents the relative abundance of the naturally occurring isotopes of an element in units of atom percent. Atom percent is the percentage of the atoms of an element that are of a particular isotope. Atom percent is abbreviated as a/o.
For example, if a cup of water contains 8.23 x 10 atoms of oxygen, and the isotopic abundance 24
of oxygen-18 is 0.20%, then there are 1.65 x 10 atoms of oxygen-18 in the cup. 22
The atomic weight for an element is defined as the average atomic weight of the isotopes of the element. The atomic weight for an element can be calculated by summing the products of the

Example:
Calculate the atomic weight for the element lithium. Lithium-6 has an atom percent abundance of 7.5% and an atomic mass of 6.015122 amu. Lithium-7 has an atomic abundance of 92.5% and an atomic mass of 7.016003 amu.
Solution:
The other common measurement of isotopic abundance is weight percent (w/o). Weight percent is the percent weight of an element that is a particular isotope. For example, if a sample of material contained 100 kg of uranium that was 28 w/o uranium-235, then 28 kg of uranium-235 was present in the sample.
Enriched and Depleted Uranium
Natural uranium mined from the earth contains the isotopes uranium-238, uranium-235 and uranium-234. The majority (99.2745%) of all the atoms in natural uranium are uranium-238. Most of the remaining atoms (0.72%) are uranium-235, and a slight trace (0.0055%) are uranium-234. Although all isotopes of uranium have similar chemical properties, each of the isotopes has significantly different nuclear properties. For reasons that will be discussed in later modules, the isotope uranium-235 is usually the desired material for use in reactors.
A vast amount of equipment and energy are expended in processes that separate the isotopes of
Fixture checkpoint marker-0242-xray is recorded on fixture page 0242.
uranium (and other elements). The details of these processes are beyond the scope of this module. These processes are called enrichment processes because they selectively increase the proportion of a particular isotope. The enrichment process typically starts with feed material that has the proportion of isotopes that occur naturally. The process results in two types of
CHART OF THE NUCLIDES DOE-HDBK-1019/1-93 Atomic and Nuclear Physics
In the case of uranium, the natural uranium ore is 0.72 a/o uranium-235. The desired outcome of the enrichment process is to produce enriched uranium. Enriched uranium is defined as uranium in which the isotope uranium-235 has a concentration greater than its natural value. The enrichment process will also result in the byproduct of depleted uranium. Depleted uranium is defined as uranium in which the isotope uranium-235 has a concentration less than its natural value. Although depleted uranium is referred to as a by-product of the enrichment process, it does have uses in the nuclear field and in commercial and defense industries.
Summary
The important information in this chapter is summarized below.
Chart of the Nuclides Summary
Enriched uranium is uranium in which the isotope uranium-235 has a concentration greater than its natural value of 0.7%.
Depleted uranium is uranium in which the isotope uranium-235 has a concentration less than its natural value of 0.7%.
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Atomic and Nuclear Physics DOE-HDBK-1019/1-93 MASS DEFECT AND BINDING ENERGY
MASS DEFECT AND BINDING ENERGY
The separate laws of Conservation of Mass and Conservation of Energy are not applied strictly on the nuclear level. It is possible to convert between mass and energy. Instead of two separate conservation laws, a single conservation law states that the sum of mass and energy is conserved. Mass does not magically appear and disappear at random. A decrease in mass will be accompanied by a corresponding increase in energy and vice versa.
EO 1.7 DEFINE the following terms:
a. Mass defect

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EO 1.8 Given the atomic mass for a nuclide and the atomic masses of a neutron, proton, and electron, CALCULATE the mass defect and binding energy of the nuclide.

Mass Defect

Careful measurements have shown that the mass of a particular atom is always slightly less than the sum of the masses of the individual neutrons, protons, and electrons of which the atom consists. The difference between the mass of the atom and the sum of the masses of its parts is called the mass defect (Δm). The mass defect can be calculated using Equation (1-1). In calculating the mass defect it is important to use the full accuracy of mass measurements because the difference in mass is small compared to the mass of the atom. Rounding off the masses of atoms and particles to three or four significant digits prior to the calculation will result in a calculated mass defect of zero.

Δm = [Z(mp + me) + (A-Z)mn] - matom (1-1)

where:

Δm = mass defect (amu)
mp = mass of a proton (1.007277 amu)
mn = mass of a neutron (1.008665 amu)
me = mass of an electron (0.000548597 amu)
matom = mass of nuclide A
ZX (amu)
Z = atomic number (number of protons)
A = mass number (number of nucleons)

Fixture checkpoint marker-0243-xray is recorded on fixture page 0243.

Example:

Calculate the mass defect for lithium-7. The mass of lithium-7 is 7.016003 amu.

Solution:

Δ m Z m
p
me A Z mn matom
Δ m 3 1.007826 amu 7 3 1.008665 amu 7.016003 amu
Δ m 0.0421335 amu

Binding Energy

The loss in mass, or mass defect, is due to the conversion of mass to binding energy when the nucleus is formed. Binding energy is defined as the amount of energy that must be supplied to a nucleus to completely separate its nuclear particles (nucleons). It can also be understood as the amount of energy that would be released if the nucleus was formed from the separate particles. Binding energy is the energy equivalent of the mass defect. Since the mass defect was converted to binding energy (BE) when the nucleus was formed, it is possible to calculate the binding energy using a conversion factor derived by the mass-energy relationship from Einstein's Theory of Relativity.

Einstein's famous equation relating mass and energy is E = mc2 where c is the velocity of light (c = 2.998 x 108 m/sec). The energy equivalent of 1 amu can be determined by inserting this quantity of mass into Einstein's equation and applying conversion factors.

E m c 2
1 amu
1.6606 x 10 27 kg

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2.998 x 108 m
sec
2
1 N
1
kg m
sec2
1 J
1 N m
1.4924 x 10 10 J 1 MeV
1.6022 x 10 13 J
931.5 MeV
Conversion Factors:
1 amu = 1.6606 x 10 -27 kg
1 newton = 1 kg-m/sec2
1 joule = 1 newton-meter
1 MeV = 1.6022 x 10-13 joules
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Atomic and Nuclear Physics DOE-HDBK-1019/1-93 MASS DEFECT AND BINDING ENERGY
Since 1 amu is equivalent to 931.5 MeV of energy, the binding energy can be calculated using Equation (1-2).
(1-2)
B.E. ∆m
931.5 MeV
1 amu
Example:
Calculate the mass defect and binding energy for uranium-235. One uranium-235 atom has a mass of 235.043924 amu.
Solution:
Step 1: Calculate the mass defect using Equation (1-1).
∆m Z m
Fixture checkpoint marker-0244-xray is recorded on fixture page 0244.
p
me A Z mn matom
∆m 92 1.007826 amu 235 92 1.008665 amu 235.043924 amu
∆m 1.91517 amu
Step 2: Use the mass defect and Equation (1-2) to calculate the binding energy.
B.E. ∆m
931.5 MeV
1 amu
1.91517 amu
931.5 MeV
1 amu
1784 MeV
Energy Levels of Atoms
The electrons that circle the nucleus move in fairly well-defined orbits. Some of these electrons are more tightly bound in the atom than others. For example, only 7.38 eV is required to remove the outermost electron from a lead atom, while 88,000 eV is required to remove the

the energy required to remove the electron is called the ionization energy.
In a neutral atom (number of electrons = Z) it is possible for the electrons to be in a variety of different orbits, each with a different energy level. The state of lowest energy is the one in which the atom is normally found and is called the ground state. When the atom possesses more energy than its ground state energy, it is said to be in an excited state.

An atom cannot stay in the excited state for an indefinite period of time. An excited atom will eventually transition to either a lower-energy excited state, or directly to its ground state, by emitting a discrete bundle of electromagnetic energy called an x-ray. The energy of the x-ray will be equal to the difference between the energy levels of the atom and will typically range from several eV to 100,000 eV in magnitude.

Energy Levels of the Nucleus

The nucleons in the nucleus of an atom, like the electrons that circle the nucleus, exist in shells that correspond to energy states. The energy shells of the nucleus are less defined and less understood than those of the electrons. There is a state of lowest energy (the ground state) and discrete possible excited states for a nucleus. Where the discrete energy states for the electrons of an atom are measured in eV or keV, the energy levels of the nucleus are considerably greater and typically measured in MeV.
A nucleus that is in the excited state will not remain at that energy level for an indefinite period. Like the electrons in an excited atom, the nucleons in an excited nucleus will transition towards their lowest energy configuration and in doing so emit a discrete bundle of electromagnetic radiation called a gamma ray (γ-ray). The only differences between x-rays and γ-rays are their energy levels and whether they are emitted from the electron shell or from the nucleus.
The ground state and the excited states of
Figure 7 Energy Level Diagram - Nickel-60
a nucleus can be depicted in a nuclear energy-level diagram. The nuclear energy-level diagram consists of a stack of horizontal bars, one bar for each of the excited states of the nucleus. The vertical distance between the bar representing an excited state and the bar representing the ground state is proportional to the energy level of the excited state with respect to the ground state. This difference in energy between the ground state and the excited state is called the excitation energy
Fixture checkpoint marker-0245-xray is recorded on fixture page 0245.
of the excited state. The ground state of a nuclide has zero excitation energy. The bars for the excited states are labeled with their respective energy levels. Figure 7 is the energy level diagram for nickel-60.

Summary

The important information in this chapter is summarized below.

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Mass defect is the difference between the mass of the atom and the sum of the masses of its constituent parts.
Binding energy is the amount of energy that must be supplied to a nucleus to completely separate its nuclear particles. Binding energy is the energy equivalent of the mass defect.
Mass defect can be calculated by using the equation below.
$\Delta m = [Z(mp + me) + (A-Z)mn] - matom$
Binding energy can be calculated by multiplying the mass defect by the factor of 931.5 MeV per amu.
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MODES OF RADIOACTIVE DECAY DOE-HDBK-1019/1-93 Atomic and Nuclear Physics
MODES OF RADIOACTIVE DECAY
Most atoms found in nature are stable and do not emit particles or energy that change form over time. Some atoms, however, do not have stable nuclei. These atoms emit radiation in order to achieve a more stable configuration.
Fixture checkpoint marker-0246-xray is recorded on fixture page 0246.
EO 2.1 DESCRIBE the following processes:
a. Alpha decay d. Electron capture
b. Beta-minus decay e. Internal conversions
c. Beta-plus decay f. Isomeric transitions
EO 2.2 Given a Chart of the Nuclides, WRITE the radioactive decay chain for a nuclide.
EO 2.3 EXPLAIN why one or more gamma rays typically accompany particle emission.
EO 2.4 Given the stability curve on the Chart of the Nuclides, DETERMINE the type of radioactive decay that the nuclides in each region of the chart will typically undergo.
Stability of Nuclei
As mass numbers become larger, the ratio of neutrons to protons in the nucleus becomes larger for the stable nuclei. Non-stable nuclei may have an excess or deficiency of neutrons and undergo a transformation process known as beta (β) decay. Non-stable nuclei can also undergo a variety of other processes such as alpha (α) or neutron (n) decay. As a result of these decay processes, the final nucleus is in a more stable or more tightly bound configuration.
Natural Radioactivity
In 1896, the French physicist Becquerel discovered that crystals of a uranium salt emitted rays that were similar to x-rays in that they were highly penetrating, could affect a photographic plate, and induced electrical conductivity in gases. Becquerel's discovery was followed in 1898 by the identification of two other radioactive elements, polonium and radium, by Pierre and Marie Curie.
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Atomic and Nuclear Physics DOE-HDBK-1019/1-93 MODES OF RADIOACTIVE DECAY
Heavy elements, such as uranium or thorium, and their unstable decay chain elements emit radiation in their naturally occurring state. Uranium and thorium, present since their creation at the beginning of geological time, have an extremely slow rate of decay. All naturally occurring nuclides with atomic numbers greater than 82 are radioactive.
Nuclear Decay
Whenever a nucleus can attain a more stable (i.e., more tightly bound) configuration by emitting radiation, a spontaneous disintegration process known as radioactive decay or nuclear decay may

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Detailed studies of radioactive decay and nuclear reaction processes have led to the formulation of useful conservation principles. The four principles of most interest in this module are discussed below.

Conservation of electric charge implies that charges are neither created nor destroyed. Single positive and negative charges may, however, neutralize each other. It is also possible for a neutral particle to produce one charge of each sign.

Conservation of mass number does not allow a net change in the number of nucleons. However, the conversion of a proton to a neutron and vice versa is allowed.

Conservation of mass and energy implies that the total of the kinetic energy and the energy equivalent of the mass in a system must be conserved in all decays and reactions. Mass can be converted to energy and energy can be converted to mass, but the sum of mass and energy must be constant.

Conservation of momentum is responsible for the distribution of the available kinetic energy among product nuclei, particles, and/or radiation. The total amount is the same before and after the reaction even though it may be distributed differently among entirely different nuclides and/or particles.

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Alpha Decay (■)

Fixture checkpoint marker-0247-xray is recorded on fixture page 0247.

Alpha decay is the emission of alpha particles (helium nuclei) which may be represented as either He or ■. When an unstable nucleus ejects an alpha particle, the atomic number is reduced by 2 4 4 2 2

and the mass number decreased by 4. An example is uranium-234 which decays by the ejection of an alpha particle accompanied by the emission of a 0.068 MeV gamma.

The combined kinetic energy of the daughter nucleus (Thorium-230) and the ■ particle is designated as KE. The sum of the KE and the gamma energy is equal to the difference in mass between the original nucleus (Uranium-234) and the final particles (equivalent to the binding energy released, since ■m = BE). The alpha particle will carry off as much as 98% of the kinetic energy and, in most cases, can be considered to carry off all the kinetic energy.

Beta Decay (■)

Beta decay is the emission of electrons of nuclear rather than orbital origin. These particles are electrons that have been expelled by excited nuclei and may have a charge of either sign.

If both energy and momentum are to be conserved, a third type of particle, the neutrino, ■, must be involved. The neutrino is associated with positive electron emission, and its antiparticle, the antineutrino, , is emitted with a negative electron. These uncharged particles have only the weakest interaction with matter, no mass, and travel at the speed of light. For all practical purposes, they pass through all materials with so few interactions that the energy they possess cannot be recovered. The neutrinos and antineutrinos are included here only because they carry a portion of the kinetic energy that would otherwise belong to the beta particle, and therefore, must be considered for energy and momentum to be conserved. They are normally ignored since they are not significant in the context of nuclear reactor applications.

Negative electron emission, represented as e, -1 -1■, or simply as e or ■ , effectively converts a neutro to a proton, thus increasing the atomic number by one and leaving the mass number unchanged. This is a common mode of decay for nuclei with an excess of neutrons, such as fission fragments

typical beta minus-decay reaction is shown below.
■ ■
■
■ ■
■ ■ ■ ■
Atomic and Nuclear Physics DOE-HDBK-1019/1-93 MODES OF RADIOACTIVE DECAY
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Figure 9 Orbital Electron Capture
Positively charged electrons (beta-plus) are known as positrons. Except for sign, they are nearly identical to their negatively charged cousins. When a positron, represented as e, ■, or simply as +1 +1 0 0
e or ■ , is ejected from the nucleus, the atomic number is decreased by one and the mass number + + remains unchanged. A proton has been converted to a neutron. An example of a typical positron (beta-plus) decay is shown below.
Electron Capture (EC, K-capture)
Nuclei having an excess of protons may capture an electron from one of the inner orbits which immediately combines with a proton in the nucleus to form a neutron. This process is called electron capture (EC). The electron is normally captured from the innermost orbit (the K-shell), and, consequently, this process is sometimes called K-capture. The following example depicts electron capture.
A neutrino is formed at the same time that the neutron is formed, and energy carried off by it serves to conserve momentum. Any energy that is available due to the atomic mass of the product being appreciably less than that of the parent will appear as gamma radiation. Also, there will always
Fixture checkpoint marker-0248-xray is recorded on fixture page 0248.
be characteristic x-rays given off when an electron from one of the higher energy shells moves in to fill the vacancy in the K-shell.
Electron capture is shown graphically in Figure 8.
Electron capture and positron emission result in the production of the same daughter product, and they exist as competing processes.
For positron emission to occur, however, the mass of the daughter product must be less than the mass of the parent by an amount equal to at least twice the mass of an electron. This mass difference between the parent and daughter is necessary to account for two items present in the parent but not in the daughter. One item is the positron ejected from the nucleus of the parent. The other item is that the daughter product has one less orbital electron than the parent. If this requirement is not met, then orbital electron capture takes place exclusively.
■
■ ■ ■ ■

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Gamma Emission (■)
Gamma radiation is a high-energy electromagnetic radiation that originates in the nucleus. It is emitted in the form of photons, discrete bundles of energy that have both wave and particle properties. Often a daughter nuclide is left in an excited state after a radioactive parent nucleus undergoes a transformation by alpha decay, beta decay, or electron capture. The nucleus will drop to the ground state by the emission of gamma radiation.
Internal Conversion
The usual method for an excited nucleus to go from the excited state to the ground state is by emission of gamma radiation. However, in some cases the gamma ray (photon) emerges from the nucleus only to interact with one of the innermost orbital electrons and, as a result, the energy of the photon is transferred to the electron. The gamma ray is then said to have undergone internal conversion. The conversion electron is ejected from the atom with kinetic energy equal to the gamma energy minus the binding energy of the orbital electron. An orbital electron then drops to a lower energy state to fill the vacancy, and this is accompanied by the emission of characteristic x-rays.
Isomers and Isomeric Transition
Isomeric transition commonly occurs immediately after particle emission; however, the nucleus may remain in an excited state for a measurable period of time before dropping to the ground state at its own characteristic rate. A nucleus that remains in such an excited state is known as a nuclear isomer because it differs in energy and behavior from other nuclei with the same atomic number and mass number. The decay of an excited nuclear isomer to a lower energy level is called an isomeric transition. It is also possible for the excited isomer to decay by some alternate means, for example, by beta emission.
An example of gamma emission accompanying particle emission is illustrated by the decay of nitrogen-16 below.
■
■
■
■
■
■
■
■
■
Fixture checkpoint marker-0249-xray is recorded on fixture page 0249.
■
■
■
Atomic and Nuclear Physics DOE-HDBK-1019/1-93 MODES OF RADIOACTIVE DECAY
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Decay Chains
When an unstable nucleus decays, the resulting daughter nucleus is not necessarily stable. The nucleus resulting from the decay of a parent is often itself unstable, and will undergo an additional decay. This is especially common among the larger nuclides.
It is possible to trace the steps of an unstable atom as it goes through multiple decays trying to achieve stability. The list of the original unstable nuclide, the nuclides that are involved as

common method for stating the decay chain is to state each of the nuclides involved in the standard X format. Arrows are used between nuclides to indicate where decays occur, with the type of decay A Z
indicated above the arrow and the half-life below the arrow. The half-life for decay will be discussed in the next chapter.
Example:
Write the decay chains for rubidium-91 and actinium-215. Continue the chains until a stable nuclide or a nuclide with a half-life greater than 1 x 10 years is reached. 6
Solution:
Predicting Type of Decay
Radioactive nuclides tend to decay in a way that results in a daughter nuclide that lies closer to the line of stability. Due to this, it is possible to predict the type of decay that a nuclide will undergo based on its location relative to the line of stability on the Chart of the Nuclides.
Fixture checkpoint marker-0250-xray is recorded on fixture page 0250.
Figure 9 illustrates the type of decay nuclides in different regions of the chart will typically undergo. Nuclides that are below and to the right of the line of stability will usually undergo ■ decay. - Nuclides that are above and to the left of the line of stability will usually undergo either ■ decay or + electron capture. Most nuclides that will undergo ■ decay are found in the upper right hand region of the chart. These are general rules that have many exceptions, especially in the region of the heavy nuclides.
MODES OF RADIOACTIVE DECAY DOE-HDBK-1019/1-93 Atomic and Nuclear Physics
Figure 9 Types of Radioactive Decay Relative to the Line of Stability
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Atomic and Nuclear Physics DOE-HDBK-1019/1-93 MODES OF RADIOACTIVE DECAY
Summary
The important information in this chapter is summarized below.
Modes of Radioactive Decay Summary
Alpha decay is the emission of an alpha particle (2 protons and 2 neutrons) from an unstable nucleus. The daughter nuclide has an atomic number 2 less than the parent nuclide and a mass number 4 less than the parent nuclide. The daughter nucleus commonly releases its excitation energy by gamma emission.
Beta-minus decay effectively converts a neutron to a proton and an electron, which is immediately ejected from the nucleus. The daughter nuclide has its atomic number increased by 1 and the same mass number compared to the parent.
Beta-plus decay effectively converts a proton to a neutron and a positron, which is immediately ejected from the nucleus. The daughter nuclide has its atomic number decreased by 1 and the same mass number compared to the parent.
In electron capture, the nucleus absorbs an electron from the innermost orbit. This electron combines with a proton to form a neutron.
Internal conversion occurs when a gamma ray, emitted by the nucleus as it goes from the excited state to the ground state, interacts with one of the innermost electrons of the same atom. The electron is ejected from the atom.
An isomeric transition is the decay of an excited nucleus to a lower-energy level by the emission of a gamma ray.
Decay chains can be found by tracing the steps an unstable atom goes through as it tries to achieve stability.
Many modes of radioactive decay result in a daughter nuclide that has an energy

immediately in the form of a gamma ray.
The type of decay that a nuclide will typically undergo can be determined by its relationship to the line of stability on the Chart of the Nuclides. Nuclides that lie below and to the right of the line of stability will typically beta minus decay. Nuclides above and to the left of the line will typically either beta plus decay or electron capture. Most alpha emitters are found in the upper, right-hand corner of the chart.
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RADIOACTIVITY DOE-HDBK-1019/1-93 Atomic and Nuclear Physics
RADIOACTIVITY
The rate at which a sample of radioactive material decays is not constant. As individual atoms of the material decay, there are fewer of those types of atoms remaining. Since the rate of decay is directly proportional to the number of atoms, the rate of decay will decrease as the number of atoms decreases.
EO 2.5 DEFINE the following terms:
a. Radioactivity d. Radioactive decay constant
b. Curie e. Radioactive half-life
c. Becquerel
EO 2.6 Given the number of atoms and either the half-life or decay constant of a nuclide, CALCULATE the activity.
Fixture checkpoint marker-0251-xray is recorded on fixture page 0251.
EO 2.7 Given the initial activity and the decay constant of a nuclide, CALCULATE the activity at any later time.
EO 2.8 CONVERT between the half-life and decay constant for a nuclide.
EO 2.9 Given the Chart of the Nuclides and the original activity, PLOT the radioactive decay curve for a nuclide on either linear or semi-log coordinates.
EO 2.10 DEFINE the following terms:
a. Radioactive equilibrium
b. Transient radioactive equilibrium
Radioactive Decay Rates
Radioactivity is the property of certain nuclides of spontaneously emitting particles or gamma radiation. The decay of radioactive nuclides occurs in a random manner, and the precise time at which a single nucleus will decay cannot be determined. However, the average behavior of a very large sample can be predicted accurately by using statistical methods. These studies have revealed that there is a certain probability that in a given time interval a certain fraction of the nuclei within a sample of a particular nuclide will decay. This probability per unit time that an atom of a nuclide will decay is known as the radioactive decay constant, λ. The units for the decay constant are inverse time such as 1/second, 1/minute, 1/hour, or 1/year. These decay constant units can also be expressed as second-1, minute-1, hour-1, and year-1.
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■■
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The activity (A) of a sample is the rate of decay of that sample. This rate of decay is usually measured in the number of disintegrations that occur per second. For a sample containing millions of atoms, the activity is the product of the decay constant and the number of atoms

The relationship between the activity, number of atoms, and decay constant is shown in Equation (1-3).

A = ■ N (1-3)

where:

A = Activity of the nuclide (disintegrations/second)

■ = decay constant of the nuclide (second)-1

N = Number of atoms of the nuclide in the sample

Since ■ is a constant, the activity and the number of atoms are always proportional.

Units of Measurement for Radioactivity

Two common units to measure the activity of a substance are the curie (Ci) and becquerel (Bq). A curie is a unit of measure of the rate of radioactive decay equal to 3.7 x 10 disintegrations 10 per second. This is approximately equivalent to the number of disintegrations that one gram of radium-226 will undergo in one second. A becquerel is a more fundamental unit of measure of radioactive decay that is equal to 1 disintegration per second. Currently, the curie is more widely used in the United States, but usage of the becquerel can be expected to broaden as the metric system slowly comes into wider use. The conversion between curies and becquerels is shown below.

1 curie = 3.7 x 10 becquerels 10

Variation of Radioactivity Over Time

The rate at which a given radionuclide sample decays is stated in Equation (1-3) as being equal to the product of the number of atoms and the decay constant. From this basic relationship it is possible to use calculus to derive an expression which can be used to calculate how the number of atoms present will change over time. The derivation is beyond the scope of this text, but Equation (1-4) is the useful result.

(1-4)

where:

N = number of atoms present at time t

Fixture checkpoint marker-0252-xray is recorded on fixture page 0252.

N = number of atoms initially present o

■ = decay constant (time)-1

t = time

A Ao

e

t

A Aoe

t

A

Ao

e

t

ln A

Ao

t

t

ln

A

Ao

t

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ln
1
2
t
1/2
ln 2 0.693
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Since the activity and the number of atoms are always proportional, they may be used interchangeably to describe any given radionuclide population. Therefore, the following is true.
(1-5)
where:
A = activity present at time t
A = activity initially present o
= decay constant (time)
-1
t = time
Radioactive Half-Life
One of the most useful terms for estimating how quickly a nuclide will decay is the radioactive half-life. The radioactive half-life is defined as the amount of time required for the activity to decrease to one-half of its original value. A relationship between the half-life and decay constant can be developed from Equation (1-5). The half-life can be calculated by solving Equation (1-5) for the time, t, when the current activity, A, equals one-half the initial activity A .
o
First, solve Equation (1-5) for t.
If A is equal to one-half of Ao, then A/Ao is equal to one-half. Substituting this in the equation above yields an expression for t .
1/2
(1-6)
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The basic features of decay of a radionuclide sample are shown by the graph in Figure 10.
Figure 10 Radioactive Decay as a Function of Time in Units of Half-Life
Assuming an initial number of atoms No, the population, and consequently, the activity may be noted to decrease by one-half of this value in a time of one half-life. Additional decreases occur
Fixture checkpoint marker-0253-xray is recorded on fixture page 0253.
so that whenever one half-life elapses, the number of atoms drops to one-half of what its value was at the beginning of that time interval. After five half-lives have elapsed, only 1/32, or 3.1%, of the original number of atoms remains. After seven half-lives, only 1/128, or 0.78%, of the atoms remains. The number of atoms existing after 5 to 7 half-lives can usually be assumed to be negligible. The Chemistry Fundamentals Handbook contains additional information on calculating the number of atoms contained within a sample.
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■
■
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Example:
A sample of material contains 20 micrograms of californium-252.

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Calculate:
(a) The number of californium-252 atoms initially present
(b) The activity of the californium-252 in curies
(c) The number of californium-252 atoms that will remain in 12 years
(d) The time it will take for the activity to reach 0.001 curies
Solution:
(a) The number of atoms of californium-252 can be determined as below.
(b) First, use Equation (1-6) to calculate the decay constant.
Use this value for the decay constant in Equation (1-3) to determine the activity.
N No
e
Fixture checkpoint marker-0254-xray is recorded on fixture page 0254.
t
4.78 x 1016 e
(0.263/yr) (12 yr)
2.04 x 1015
A Aoe
t
A
Ao
e
t
ln A
Ao
t
t
ln
A
Ao
t
ln
0.001 Ci
0.0108 Ci
0.263 year 1
t 9.05 years
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(c) The number of californium atoms that will remain in 12 years can be calculated from Equation (1-4).
(d) The time that it will take for the activity to reach 0.001 Ci can be determined from Equation (1-5). First, solve Equation (1-5) for time.
Inserting the appropriate values in the right side of this equation will result in the required time.
Plotting Radioactive Decay
It is useful to plot the activity of a nuclide as it changes over time. Plots of this type can be used to determine when the activity will fall below a certain level. This plot is usually done showing activity on either a linear or a logarithmic scale. The decay of the activity of a single nuclide on a logarithmic scale will plot as a straight line because the decay is exponential.

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1/2
0.693
0.693
t
1/2
0.693
7.13 seconds
0.0972 second 1
A Aoe
t
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Example:
Plot the radioactive decay curve for nitrogen-16 over a period of 100 seconds. The initial activity is 142 curies and the half-life of nitrogen-16 is 7.13 seconds. Plot the curve on both linear rectangular coordinates and on a semi-log scale.
Solution:
First, use Equation (1-6) to calculate the decay constant corresponding to a half-life of
Fixture checkpoint marker-0255-xray is recorded on fixture page 0255.
7.13 seconds.
Use the decay constant determined above to calculate the activity at various times using Equation (1-5).
Time Activity
0 seconds 142 Ci
20 seconds 20.3 Ci
40 seconds 2.91 Ci
60 seconds 0.416 Ci
80 seconds 0.0596 Ci
100 seconds 0.00853 Ci
■ ■
■ ■
■ ■
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Figure 11 Linear and Semi-log Plots of Nitrogen-16 Decay
Plotting the data points calculated above on both linear and semilog scales results in the graphs shown in Figure 11.
If a substance contains more than one radioactive nuclide, the total activity is the sum of the individual activities of each nuclide. As an example, consider a sample of material that contained 1 x 10 atoms of iron-59 that has a half-life of 44.51 days (■ = 1.80 x 10 sec), 6 -7 -1
1 x 10 atoms of manganese-54 that has a half-life of 312.2 days (■ = 2.57 x 10 sec), and 1 6 -8 -1
x 10 atoms of cobalt-60 that has a half-life of 1925 days (■ = 4.17 x 10 sec). 6 -9 -1
The initial activity of each of the nuclides would be the product of the number of atoms and the decay constant.
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Figure 12 Combined Decay of Iron-56, Manganese-54, and Cobalt-60
Plotting the manner in which the activities of each of the three nuclides decay over time

activity, then manganese-54 dominates. After almost all of the iron and manganese have decayed away, the only contributor to activity will be the cobalt-60. A plot of this combined decay is shown in Figure 12.

Radioactive Equilibrium

Radioactive equilibrium exists when a radioactive nuclide is decaying at the same rate at which it is being produced. Since the production rate and decay rate are equal, the number of atoms present remains constant over time.

An example of radioactive equilibrium is the concentration of sodium-24 in the coolant circulating through a sodium-cooled nuclear reactor. Assume that the sodium-24 is being produced at a rate of 1 x 10 atoms per second. If the sodium-24 were stable and did not decay, 6 the amount of sodium-24 present after some period of time could be calculated by multiplying the production rate by the amount of time. Plotting the amount of material present would result in the graph in Figure 13.

However, sodium-24 is not stable, and it decays with a half-life of 14.96 hours. If no sodium-24 is present initially and production starts at a rate of 1 x 10 atoms per second, the rate 6 of decay will initially be zero because there is no sodium-24 present to decay. The rate of decay of sodium-24 will increase as the amount of sodium-24 increases.

■
■
■
■

Figure 13 Cumulative Production of Sodium-24 Over Time

The amount of sodium-24 present will initially increase rapidly, then it will increase at a

Fixture checkpoint marker-0256-xray is recorded on fixture page 0256.

continually decreasing rate until the rate of decay is equal to the rate of production. It is possible to calculate how much sodium-24 will be present at equilibrium by setting the production rate (R) equal to the decay rate (■ N).

where:

R = production rate (atoms/second)
■ = decay constant (second)-1
N = number of atoms

It is possible to calculate the equilibrium value for sodium-24 being produced at a rate of 1 x 10 atoms/second. 6

■ ■ ■■
■
■
■
■

Figure 14 Approach of Sodium-24 to Equilibrium

The development of the equation to calculate how the amount of sodium-24 changes over time as it approaches the equilibrium value is beyond the scope of this handbook. However, the equation is presented below.

This equation can be used to calculate the values of the amount of sodium-24 present at different times. As the time increases, the exponential term approaches zero, and the number of atoms

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Figure 14.
Transient Radioactive Equilibrium
Transient radioactive equilibrium occurs when the parent nuclide and the daughter nuclide decay at essentially the same rate.
For transient equilibrium to occur, the parent must have a long half-life when compared to the daughter. An example of this type of compound decay process is barium-140, which decays by beta emission to lanthanum-140, which in turn decays by beta emission to stable cerium-140.
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Figure 15 Transient Equilibrium in the Decay of Barium-140
The decay constant for barium-140 is considerably smaller than the decay constant for lanthanum-140. Remember that the rate of decay of both the parent and daughter can be represented as ■N. Although the decay constant for barium-140 is smaller, the actual rate of decay (■N) is initially larger than that of lanthanum-140 because of the great difference in their initial concentrations. As the concentration of the daughter increases, the rate of decay of the daughter will approach and eventually match the decay rate of the parent. When this occurs, they are said to be in transient equilibrium. A plot of the barium-lanthanum-cerium decay chain reaching transient equilibrium is shown in Figure 15.
Secular equilibrium occurs when the parent has an extremely long half-life. In the long decay chain for a naturally radioactive element, such as thorium-232, where all of the elements in the chain are in secular equilibrium, each of the descendants has built up to an equilibrium amount and all decay at the rate set by the original parent. The only exception is the final stable element on the end of the chain. Its number of atoms is constantly increasing.
Summary
The important information in this chapter is summarized on the following page.
■
■■
■
RADIOACTIVITY DOE-HDBK-1019/1-93 Atomic and Nuclear Physics
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Radioactivity Summary
& Radioactivity is the decay of unstable atoms by the emission of particles and
Fixture checkpoint marker-0257-xray is recorded on fixture page 0257.
electromagnetic radiation.
& A curie (Ci) is a unit of radioactivity equal to 3.7 x 10 disintegrations per 10
second.
& A becquerel (Bq) is a unit of radioactivity equal to 1 disintegration per second.
& The radioactive decay constant (■) is the probability per unit time that an atom
will decay.
& The radioactive half-life is the amount of time required for the activity to
decrease to one-half its original value.
& The activity of a substance can be calculated from the number of atoms and the
decay constant based on the equation below.
& The amount of activity remaining after a particular time can be calculated from
the equation below.
& The relationship between the decay constant and the half-life is shown below.
& Plots of radioactive decay can be useful to describe the variation of activity over
time. If decay is plotted using semi-log scale the plot results in a straight line.

the removal rate.
& Transient radioactive equilibrium exists when the parent nuclide and the daughter nuclide decay at essentially the same rate. This occurs only when the parent has a long half-life compared to the daughter.
Atomic and Nuclear Physics DOE-HDBK-1019/1-93 NEUTRON INTERACTIONS
NEUTRON INTERACTIONS
Neutrons can cause many different types of interactions. The neutron may simply scatter off the nucleus in two different ways, or it may actually be absorbed into the nucleus. If a neutron is absorbed into the nucleus, it may result in the emission of a gamma ray or a subatomic particle, or it may cause the nucleus to fission.
EO 3.1 DESCRIBE the following scattering interactions between a neutron and a nucleus:
a. Elastic scattering
b. Inelastic scattering
EO 3.2 STATE the conservation laws that apply to an elastic collision between a neutron and a nucleus.
EO 3.3 DESCRIBE the following reactions where a neutron is absorbed in a nucleus:
a. Radiative capture
b. Particle ejection
Scattering
A neutron scattering reaction occurs when a nucleus, after having been struck by a neutron, emits a single neutron. Despite the fact that the initial and final neutrons do not need to be (and often are not) the same, the net effect of the reaction is as if the projectile neutron had merely "bounced off," or scattered from, the nucleus. The two categories of scattering reactions, elastic and inelastic scattering, are described in the following paragraphs.
Elastic Scattering
In an elastic scattering reaction between a neutron and a target nucleus, there is no energy transferred into nuclear excitation. Momentum and kinetic energy of the "system" are conserved although there is usually some transfer of kinetic energy from the neutron to the target nucleus. The target nucleus gains the amount of kinetic energy that the neutron loses.
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mn
vn,i mTvT,i mnvn,f mTvT,f
1
2
m v
2
Fixture checkpoint marker-0258-xray is recorded on fixture page 0258.
1
2
mn
v
2
n,i
1
2

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v
2
T,i
1
2
mn
v
2
n,f
1
2
mT
v
2
T,f
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Fixture checkpoint marker-0259-xray is recorded on fixture page 0259.
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Figure 16 Elastic Scattering
Figure 16 illustrates the process of elastic scattering of a neutron off a target nucleus. In the elastic scattering reaction, the conservation of momentum and kinetic energy is represented by the equations below.
Conservation of momentum (mv)
Conservation of kinetic energy
where:
mn = mass of the neutron
m = mass of the target nucleus T
vn,i = initial neutron velocity
vn,f = final neutron velocity
vT,i = initial target velocity
vT,f = final target velocity
Atomic and Nuclear Physics DOE-HDBK-1019/1-93 NEUTRON INTERACTIONS
Elastic scattering of neutrons by nuclei can occur in two ways. The more unusual of the two interactions is the absorption of the neutron, forming a compound nucleus, followed by the re-emission of a neutron in such a way that the total kinetic energy is conserved and the nucleus returns to its ground state. This is known as resonance elastic scattering and is very dependent upon the initial kinetic energy possessed by the neutron. Due to formation of the compound nucleus, it is also referred to as compound elastic scattering. The second, more usual method, is termed potential elastic scattering and can be understood by visualizing the neutrons and nuclei to be much like billiard balls with impenetrable surfaces. Potential scattering takes place with incident neutrons that have an energy of up to about 1 MeV. In potential scattering, the neutron does not actually touch the nucleus and a compound nucleus is not formed. Instead, the neutron is acted on and scattered by the short range nuclear forces when it approaches close enough to the nucleus.
Inelastic Scattering
In inelastic scattering, the incident neutron is absorbed by the target nucleus, forming a compound nucleus. The compound nucleus will then emit a neutron of lower kinetic energy which leaves the original nucleus in an excited state. The nucleus will usually, by one or more

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process of inelastic scattering.
For the nucleus that has reached its ground state, the sum of the kinetic energy of the exit
Figure 17 Inelastic Scattering
neutron, the target nucleus, and the total gamma energy emitted is equal to the initial kinetic energy of the incident neutron.
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1
0
n
238
92
U
239
92
U
239
92
U
0
0
1
0
n
Fixture checkpoint marker-0260-xray is recorded on fixture page 0260.
10
5
B
11
5
B
7
3
Li
4
2
NEUTRON INTERACTIONS DOE-HDBK-1019/1-93 Atomic and Nuclear Physics
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Absorption Reactions
Most absorption reactions result in the loss of a neutron coupled with the production of a charged particle or gamma ray. When the product nucleus is radioactive, additional radiation is emitted at some later time. Radiative capture, particle ejection, and fission are all categorized as absorption reactions and are briefly described below.
Radiative Capture
In radiative capture the incident neutron enters the target nucleus forming a compound nucleus. The compound nucleus then decays to its ground state by gamma emission. An example of a radiative capture reaction is shown below.
Particle Ejection
In a particle ejection reaction the incident particle enters the target nucleus forming a compound

to cause it to eject a new particle while the incident neutron remains in the nucleus. After the new particle is ejected, the remaining nucleus may or may not exist in an excited state depending upon the mass-energy balance of the reaction. An example of a particle ejection reaction is shown below.

Fission

One of the most important interactions that neutrons can cause is fission, in which the nucleus that absorbs the neutron actually splits into two similarly sized parts. Fission will be discussed in detail in the next chapter.

Atomic and Nuclear Physics DOE-HDBK-1019/1-93 NEUTRON INTERACTIONS

Summary

The important information in this chapter is summarized below.

Neutron Interactions Summary

Interactions where a neutron scatters off a target nucleus are either elastic or inelastic. In elastic scattering, kinetic energy and momentum are conserved and no energy is transferred into excitation energy of the target nucleus. In inelastic scattering, some amount of kinetic energy is transferred into excitation energy of the target nucleus.

The conservation principles that apply to an elastic collision are conservation of kinetic energy and conservation of momentum.

Radiative capture is the absorption of a neutron by the target nucleus, resulting in an excited nucleus which subsequently (typically within a small fraction of a second) releases its excitation energy in the form of a gamma ray.

Particle ejection occurs when a neutron is absorbed by a target nucleus, resulting in the formation of a compound nucleus. The compound nucleus immediately ejects a particle (for example, alpha or proton).

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1
0
n
235

Fixture checkpoint marker-0261-xray is recorded on fixture page 0261.

92
U
236
92
U
140
55
Cs
93
37
Rb 3
1
0
n

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NUCLEAR FISSION

products, and neutrons. The neutrons released by fission can, in turn, cause the fission of other atoms.

EO 4.1 EXPLAIN the fission process using the liquid drop model of a nucleus.

EO 4.2 DEFINE the following terms:

a. Excitation energy (E_{exc})

b. Critical energy (E_{crit})

EO 4.3 DEFINE the following terms:

a. Fissile material

b. Fissionable material

c. Fertile material

EO 4.4 DESCRIBE the processes of transmutation, conversion, and breeding.

EO 4.5 DESCRIBE the curve of Binding Energy per Nucleon versus mass number and give a qualitative description of the reasons for its shape.

EO 4.6 EXPLAIN why only the heaviest nuclei are easily fissioned.

EO 4.7 EXPLAIN why uranium-235 fissions with thermal neutrons and uranium-238 fissions only with fast neutrons.

Fission

In the fission reaction the incident neutron enters the heavy target nucleus, forming a compound nucleus that is excited to such a high energy level ($E > E_{exc} > E_{crit}$) that the nucleus "splits" (fissions) into two large fragments plus some neutrons. An example of a typical fission reaction is shown below.

A large amount of energy is released in the form of radiation and fragment kinetic energy.

Liquid Drop Model of a Nucleus

The nucleus is held together by the attractive nuclear force between nucleons, which was discussed in a previous chapter. The characteristics of the nuclear force are listed below.

(a) very short range, with essentially no effect beyond nuclear dimensions (10^{-13} cm)

(b) stronger than the repulsive electrostatic forces within the nucleus

(c) independent of nucleon pairing, in that the attractive forces between pairs of neutrons are no different than those between pairs of protons or a neutron and a proton

(d) saturable, that is, a nucleon can attract only a few of its nearest neighbors

One theory of fission considers the fissioning of a nucleus similar in some respects to the splitting

Fixture checkpoint marker-0262-xray is recorded on fixture page 0262.

of a liquid drop. This analogy is justifiable to some extent by the fact that a liquid drop is held together by molecular forces that tend to make the drop spherical in shape and that try to resist any deformation in the same manner as nuclear forces are assumed to hold the nucleus together. By considering the nucleus as a liquid drop, the fission process can be described.

Referring to Figure 18(A), the nucleus in the ground state is undistorted, and its attractive nuclear forces are greater than the repulsive electrostatic forces between the protons within the nucleus. When an incident particle (in this instance a neutron) is absorbed by the target nucleus, a compound nucleus is formed. The compound nucleus temporarily contains all the charge and mass involved in the reaction and exists in an excited state. The excitation energy added to the compound nucleus is equal to the binding energy contributed by the incident particle plus the

imparted to the compound nucleus, which may cause it to oscillate and become distorted. If the excitation energy is greater than a certain critical energy, the oscillations may cause the compound nucleus to become dumbbell-shaped. When this happens, the attractive nuclear forces (short-range) in the neck area are small due to saturation, while the repulsive electrostatic forces (long-range) are only slightly less than before. When the repulsive electrostatic forces exceed the attractive nuclear forces, nuclear fission occurs, as illustrated in Figure 18(C).
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Figure 18 Liquid Drop Model of Fission
Critical Energy
The measure of how far the energy level of a nucleus is above its ground state is called the excitation energy (Eexc). For fission to occur, the excitation energy must be above a particular value for that nuclide. The critical energy (Ecrit) is the minimum excitation energy required for fission to occur.
Fixture checkpoint marker-0263-xray is recorded on fixture page 0263.
Fissile Material
A fissile material is composed of nuclides for which fission is possible with neutrons of any energy level. What is especially significant about these nuclides is their ability to be fissioned with zero kinetic energy neutrons (thermal neutrons). Thermal neutrons have very low kinetic energy levels (essentially zero) because they are roughly in equilibrium with the thermal motion of surrounding materials. Therefore, in order to be classified as fissile, a material must be capable of fissioning after absorbing a thermal neutron. Consequently, they impart essentially no kinetic energy to the reaction. Fission is possible in these materials with thermal neutrons, since the change in binding energy supplied by the neutron addition alone is high enough to exceed the critical energy. Some examples of fissile nuclides are uranium-235, uranium-233, and plutonium-239.
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Atomic and Nuclear Physics DOE-HDBK-1019/1-93 NUCLEAR FISSION
Fissionable Material
A fissionable material is composed of nuclides for which fission with neutrons is possible. All fissile nuclides fall into this category. However, also included are those nuclides that can be fissioned only with high energy neutrons. The change in binding energy that occurs as the result of neutron absorption results in a nuclear excitation energy level that is less than the required critical energy. Therefore, the additional excitation energy must be supplied by the kinetic energy of the incident neutron. The reason for this difference between fissile and fissionable materials is the so-called odd-even effect for nuclei. It has been observed that nuclei with even numbers of neutrons and/or protons are more stable than those with odd numbers. Therefore, adding a neutron to change a nucleus with an odd number of neutrons to a nucleus with an even number of neutrons produces an appreciably higher binding energy than adding a neutron to a nucleus already possessing an even number of neutrons. Some examples of nuclides requiring high energy neutrons to cause fission are thorium-232, uranium-238, and plutonium-240. Table 4 indicates the critical energy (Ecrit) and the binding energy change for an added neutron (BEn) to target nuclei of interest. For fission to be possible, the change in binding energy plus the kinetic energy must equal or exceed the critical energy ($\Delta BE + KE > Ecrit$).
TABLE 4
Critical Energies Compared to Binding Energy of Last Neutron
Target
Nucleus

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Ecrit
Binding Energy of
Last Neutron BEn
BEn - Ecrit
23
9
2
0Th 7.5 MeV 5.4 MeV -2.1 MeV
23
9
8
2U 7.0 MeV 5.5 MeV -1.5 MeV
23
9
5
2U 6.5 MeV 6.8 MeV +0.3 MeV
23
9
3
2U 6.0 MeV 7.0 MeV +1.0 MeV
23
Fixture checkpoint marker-0264-xray is recorded on fixture page 0264.
9
9
4Pu 5.0 MeV 6.6 MeV +1.6 MeV
Uranium-235 fissions with thermal neutrons because the binding energy released by the absorption of a neutron is greater than the critical energy for fission; therefore uranium-235 is a fissile material. The binding energy released by uranium-238 absorbing a thermal neutron is less than the critical energy, so additional energy must be possessed by the neutron for fission to be possible. Consequently, uranium-238 is a fissionable material.
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Fertile Material
All of the neutron absorption reactions that do not result in fission lead to the production of new nuclides through the process known as transmutation. These nuclides can, in turn, be transmuted again or may undergo radioactive decay to produce still different nuclides. The nuclides that are produced by this process are referred to as transmutation products. Because several of the fissile nuclides do not exist in nature, they can only be produced by nuclear reactions (transmutation). The target nuclei for such reactions are said to be fertile. Fertile materials are materials that can undergo transmutation to become fissile materials. Figure 19 traces the transmutation mechanism by which two fertile nuclides, thorium-232 and uranium-238, produce uranium-233 and plutonium-239, respectively.
Figure 19 Conversion of Fertile Nuclides to Fissile Nuclides
If a reactor contains fertile material in addition to its fissile fuel, some new fuel will be produced as the original fuel is burned up. This is called conversion. Reactors that are specifically designed to produce fissionable fuel are called "breeder" reactors. In such reactors, the amount of fissionable fuel produced is greater than the amount of fuel burnup. If less fuel is produced than used, the process is called conversion, and the reactor is termed a "converter."

Atomic and Nuclear Physics DOE-HDBK-1019/1-93 NUCLEAR FISSION
Binding Energy Per Nucleon (BE/A)
As the number of particles in a nucleus increases, the total binding energy also increases. The rate of increase, however, is not uniform. This lack of uniformity results in a variation in the amount of binding energy associated with each nucleon within the nucleus. This variation in the binding energy per nucleon (BE/A) is easily seen when the average BE/A is plotted versus atomic mass number (A), as shown in Figure 20.
Figure 20 Binding Energy per Nucleon vs. Mass Number
Figure 20 illustrates that as the atomic mass number increases, the binding energy per nucleon decreases for A > 60. The BE/A curve reaches a maximum value of 8.79 MeV at A = 56 and decreases to about 7.6 MeV for A = 238. The general shape of the BE/A curve can be explained using the general properties of nuclear forces. The nucleus is held together by very short-range attractive forces that exist between nucleons. On the other hand, the nucleus is being forced apart by long range repulsive electrostatic (coulomb) forces that exist between all the protons in the nucleus.
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As the atomic number and the atomic mass number increase, the repulsive electrostatic forces within the nucleus increase due to the greater number of protons in the heavy elements. To overcome this increased repulsion, the proportion of neutrons in the nucleus must increase to maintain stability. This increase in the neutron-to-proton ratio only partially compensates for the growing proton-proton repulsive force in the heavier, naturally occurring elements. Because the repulsive forces are increasing, less energy must be supplied, on the average, to remove a nucleon from the nucleus. The BE/A has decreased. The BE/A of a nucleus is an indication of its degree of stability. Generally, the more stable nuclides have higher BE/A than the less stable ones. The increase in the BE/A as the atomic mass number decreases from 260 to 60 is the primary reason for the energy liberation in the fission process. In addition, the increase in the BE/A as the atomic mass number increases from 1 to 60 is the reason for the energy liberation in the fusion process, Fixture checkpoint marker-0265-xray is recorded on fixture page 0265.
which is the opposite reaction of fission.
The heaviest nuclei require only a small distortion from a spherical shape (small energy addition) for the relatively large coulomb forces forcing the two halves of the nucleus apart to overcome the attractive nuclear forces holding the two halves together. Consequently, the heaviest nuclei are easily fissionable compared to lighter nuclei.
Summary
The important information in this chapter is summarized on the following page.
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Atomic and Nuclear Physics DOE-HDBK-1019/1-93 NUCLEAR FISSION
Nuclear Fission Summary
The fission process can be explained using the liquid drop model of a nucleus.
In the ground state the nucleus is nearly spherical in shape. After the absorption of a neutron, the nucleus will be in an excited state and start to oscillate and become distorted. If the oscillations cause the nucleus to become shaped like a dumbbell, the repulsive electrostatic forces will overcome the short-range attractive nuclear forces, and the nucleus will split in two.
Excitation energy is the amount of energy a nucleus has above its ground state.
Critical energy is the minimum excitation energy that a nucleus must have before it can fission.

zero kinetic energy. Fissionable material is material for which fission caused by neutron absorption is possible provided the kinetic energy added with the binding energy is greater than the critical energy. Fertile material is material that can undergo transmutation to become fissile material.

Transmutation is the process of neutron absorption and subsequent decay, which changes one nuclide to another nuclide. Conversion is the process of transmuting fertile material into fissile material in a reactor, where the amount of fissile material produced is less than the amount of fissile material consumed. Breeding is the same as conversion, except the amount of fissile material produced is more than the amount of fissile material consumed.

The curve of binding energy per nucleon increases quickly through the light nuclides and reaches a maximum at a mass number of about 56. The curve decreases slowly for mass numbers greater than 60.

The heaviest nuclei are easily fissionable because they require only a small distortion from the spherical shape to allow the coulomb forces to overcoming the attractive nuclear force, forcing the two halves of the nucleus apart.

Uranium-235 fissions with thermal neutrons because the binding energy released by the absorption of a neutron is greater than the critical energy for fission. The binding energy released by uranium-238 absorbing a neutron is less than the critical energy, so additional kinetic energy must be possessed by the neutron for fission to be possible.

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1
0
n
235
92
U
236
92
U
140
55
Cs
93

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37
Rb 3
1
0
n

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ENERGY RELEASE FROM FISSION

Fission of heavy nuclides converts a small amount of mass into an enormous amount of energy. The amount of energy released by fission can be determined based on either the change in mass that occurs during the reaction or by the difference in binding energy per nucleon between the fissile nuclide

EO 4.8 CHARACTERIZE the fission products in terms of mass groupings and radioactivity.
EO 4.9 Given the nuclides involved and their masses, CALCULATE the energy released from fission.
EO 4.10 Given the curve of Binding Energy per nucleon versus mass number, CALCULATE the energy released from fission.
Calculation of Fission Energy
Nuclear fission results in the release of enormous quantities of energy. It is necessary to be able to calculate the amount of energy that will be produced. The logical manner in which to pursue this is to first investigate a typical fission reaction such as the one listed below.
It can be seen that when the compound nucleus splits, it breaks into two fission fragments, rubidium-93, cesium-140, and some neutrons. Both fission products then decay by multiple -
Fixture checkpoint marker-0267-xray is recorded on fixture page 0267.
emissions as a result of the high neutron-to-proton ratio possessed by these nuclides.
In most cases, the resultant fission fragments have masses that vary widely. Figure 21 gives the percent yield for atomic mass numbers. The most probable pair of fission fragments for the thermal fission of the fuel uranium-235 have masses of about 95 and 140. Note that the vertical axis of the fission yield curve is on a logarithmic scale. Therefore, the formation of fission fragments of mass numbers of about 95 and 140 is highly likely.
Atomic and Nuclear Physics DOE-HDBK-1019/1-93 ENERGY RELEASE FROM FISSION
Figure 21 Uranium-235 Fission Yield vs.
Mass Number
Referring now to the binding energy per nucleon curve (Figure 20), we can estimate the amount of energy released by our "typical" fission by plotting this reaction on the curve and calculating the change in binding energy (ΔBE) between the reactants on the left-hand side of the fission equation and the products on the right-hand side. Plotting the reactant and product nuclides on the curve shows that the total binding energy of the system after fission is greater than the total binding energy of the system before fission. When there is an increase in the total binding energy of a system, the system has become more stable by releasing an amount of energy equal to the increase in total binding energy of the system. Therefore, in the fission process, the energy liberated is equal to the increase in the total binding energy of the system.
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Figure 22 graphically depicts that the binding energy per nucleon for the products (C,
Figure 22 Change in Binding Energy for Typical Fission
rubidium-93 and B, cesium-140) is greater than that for the reactant (A, uranium-235). The total binding energy for a nucleus can be found by multiplying the binding energy per nucleon by the number of nucleons.
TABLE 5
Binding Energies Calculated from
Binding Energy per Nucleon Curve
Nuclide B.E. per Nucleon
(BE/A)
Mass Number
(A)
Binding Energy
(BE/A) x (A)
9

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3
7Rb 8.7 MeV 93 809 MeV
14
5
0
5Cs 8.4 MeV 140 1176 MeV
23
9
5
2U 7.6 MeV 235 1786 MeV
NP-01 Page 58 Rev. 0
BE BEproducts BEreactants
BERb 93 BECs 140 BEU 235
809 MeV 1176 MeV 1786 MeV
199 MeV
1
0
n
235
Fixture checkpoint marker-0268-xray is recorded on fixture page 0268.
92
U
236
92
U
140
55
Cs
93
37
Rb 3
1
0
n
Mass difference Mass of Reactants Mass of Products
236.052589 amu 235.85208 amu
0.200509 amu
EInst 0.020059 amu
931.5 MeV
amu
186.8 MeV
Atomic and Nuclear Physics DOE-HDBK-1019/1-93 ENERGY RELEASE FROM FISSION
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The energy released will be equivalent to the difference in binding energy (BE) between the reactants and the products.
The energy liberation during the fission process can also be explained from the standpoint of the conservation of mass-energy. During the fission process, there is a decrease in the mass of the system. There must, therefore, be energy liberated equal to the energy equivalent of the mass

used when actually calculating the energy liberated during the fission process.
Again, referring to the "typical" fission reaction.
E , the instantaneous energy, is the energy released immediately after the fission process. It Inst
is equal to the energy equivalent of the mass lost in the fission process. It can be calculated as
shown below.
Mass of the Reactants Mass of the Products
U 235.043924 amu Rb 92.91699 amu 235
92
93
37
n 1.008665 amu Cs 139.90910 amu 1
0
140
55
3 (n) 3.02599 amu 1
0
236.052589 amu 235.85208 amu
This mass difference can be converted to an energy equivalent.
93
37
Rb
93
38
Sr
93
39
Fixture checkpoint marker-0269-xray is recorded on fixture page 0269.
Y
93
40
Zr
93
41
Nb
140
55
Cs
140
56
Ba
140
57
La
140
58
Ce
EDecay mRb 93 mNb 93 4 melectron
931.5MeV

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92.91699 amu 92.90638 amu 4 0.0005486 amu
931.5 MeV
amu
0.008416 amu
931.5 MeV
amu
7.84 MeV
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The total energy released per fission will vary from the fission to the next depending on what fission products are formed, but the average total energy released per fission of uranium-235 with a thermal neutron is 200 MeV.
As illustrated in the preceding example, the majority of the energy liberated in the fission process is released immediately after the fission occurs and appears as the kinetic energy of the fission fragments, kinetic energy of the fission neutrons, and instantaneous gamma rays. The remaining energy is released over a period of time after the fission occurs and appears as kinetic energy of the beta, neutrino, and decay gamma rays.
Estimation of Decay Energy
In addition to this instantaneous energy release during the actual fission reaction, there is additional energy released when the fission fragments decay by emission. This additional
-
energy is called decay energy, EDecay. The decay chains for rubidium-93 and cesium-140 are shown below.
The energy released during the decay for each chain will be equivalent to the mass difference between the original fission product and the sum of the final stable nuclide and the beta particles emitted.
The energy released in the decay chain of rubidium-93 is calculated below.
EDecay mRb 93 mNb 93 3 melectron
931.5MeV
amu
139.90910 amu 139.90543 amu 3 0.0005486 amu
931.5 MeV
amu
Fixture checkpoint marker-0270-xray is recorded on fixture page 0270.
0.000202 amu
931.5 MeV
amu
1.89 MeV
Atomic and Nuclear Physics DOE-HDBK-1019/1-93 ENERGY RELEASE FROM FISSION
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The energy released in the decay chain of cesium-140 is calculated below.
The total decay energy is the sum of the energies of the two chains, or 9.73 MeV.
Distribution of Fission Energy
The average energy distribution for the energy released per fission with a thermal neutron in uranium-235 is shown in Tables 6 and 7.
TABLE 6
Instantaneous Energy from Fission
Kinetic Energy of Fission Products 167 Mev

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Instantaneous Gamma-ray Energy 5 MeV
Capture Gamma-ray Energy 10 MeV
Total Instantaneous Energy 187 MeV
TABLE 7
Delayed Energy from Fission
Beta Particles From Fission Products 7 Mev
Gamma-rays from Fission Products 6 MeV
Neutrinos 10 MeV
Total Delayed Energy 23 MeV
ENERGY RELEASE FROM FISSION DOE-HDBK-1019/1-93 Atomic and Nuclear Physics
Fixture checkpoint marker-0271-xray is recorded on fixture page 0271.
Because the 10 MeV of neutrino energy shown in Table 7 is not absorbed in the reactor, the average value of 200 MeV per fission is still accurate. Note in Table 6 that some fission neutrons undergo radiative capture and the resultant gamma ray emission provides an additional 10 MeV of instantaneous energy, which contributes to the total of 187 MeV instantaneous energy.
All of the energy released, with the exception of the neutrino energy, is ultimately transformed into heat through a number of processes. The fission fragments, with their high positive charge and kinetic energy, cause ionization directly as they rip orbital electrons from the surrounding atoms. In this ionization process, kinetic energy is transferred to the surrounding atoms of the fuel material, resulting in an increase in temperature. The beta particles and gamma rays also give up their energy through ionization, and the fission neutrons interact and lose their energy through elastic scattering. Of the 200 MeV released per fission, about seven percent (13 MeV) is released at some time after the instant of fission. When a reactor is shut down, fissions essentially cease, but energy is still being released from the decay of fission products. The heat produced by this decay energy is referred to as "decay heat." Although decay energy represents about seven percent of reactor heat production during reactor operation, once the reactor is shut down the decay heat production drops off quickly to a small fraction of its value while operating. The decay heat produced is significant, however, and systems must be provided to keep the reactor cool even after shutdown.
Summary
The important information in this chapter is summarized below.
Energy Release From Fission Summary
Fission products have some general characteristics in common.
They generally decay by β- emission.
The most common mass numbers are grouped near 95 and 140.
The energy released by fission can be calculated based on the difference in mass between the masses of the reactants before fission and the fission fragments and fission neutrons after fission.
Another method to determine the energy released by fission is based on the change in binding energy per nucleon between the fissile nuclide and the fission products.
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Atomic and Nuclear Physics INTERACTION OF RADIATION WITH MATTER
INTERACTION OF RADIATION WITH MATTER
Different types of radiation interact with matter in widely different ways. A large, massive, charged alpha particle cannot penetrate a piece of paper and even has

probability of interacting with any matter, even if it passed through the diameter of the earth.

EO 5.1 DESCRIBE interactions of the following with matter:

a. Alpha particle c. Positron

b. Beta particle d. Neutron

EO 5.2 DESCRIBE the following ways that gamma radiation interacts with matter:

a. Photoelectric effect

b. Compton scattering

c. Pair production

Interaction of Radiation With Matter

Radiation can be classified into two general groups, charged and uncharged; therefore, it may be expected that interactions with matter fall into two general types. Charged particles directly ionize the media through which they pass, while uncharged particles and photons can cause ionization only indirectly or by secondary radiation.

A moving charged particle has an electrical field surrounding it, which interacts with the atomic structure of the medium through which it is passing. This interaction decelerates the particle and

Fixture checkpoint marker-0272-xray is recorded on fixture page 0272.

accelerates electrons in the atoms of the medium. The accelerated electrons may acquire enough energy to escape from the parent atom. This process, whereby radiation "strips" off orbital electrons, is called ionization. Uncharged moving particles have no electrical field, so they can only lose energy and cause ionization by such means as collisions or scattering. A photon can lose energy by the photoelectric effect, Compton effect, or pair production.

I

m z

2

K.E.

Because ionizing radiation creates ions in pairs, the intensity of ionization or the specific ionization is defined as the number of ion-pairs formed per centimeter of travel in a given material. The amount of ionization produced by a charged particle per unit path length, which is a measure of its ionizing power, is roughly proportional to the particle's mass and the square of its charge as illustrated in the equation below.

where:

I is the ionizing power

m is the mass of the particle

z is the number of unit charges it carries

K.E. is its kinetic energy

Since m for an alpha particle is about 7300 times as large as m for a beta particle, and z is twice as great, an alpha will produce much more ionization per unit path length than a beta particle of the same energy. This phenomenon occurs because the larger alpha particle moves slower for a given energy and thus acts on a given electron for a longer time.

Alpha Radiation

Alpha radiation is normally produced from the radioactive decay of heavy nuclides and from certain nuclear reactions. The alpha particle consists of 2 neutrons and 2 protons, so it is

particle has a charge of +2. This positive charge causes the alpha particle to strip electrons from the orbits of atoms in its vicinity. As the alpha particle passes through material, it removes electrons from the orbits of atoms it passes near. Energy is required to remove electrons and the energy of the alpha particle is reduced by each reaction. Eventually the particle will expend its kinetic energy, gain 2 electrons in orbit, and become a helium atom. Because of its strong positive charge and large mass, the alpha particle deposits a large amount of energy in a short distance of travel. This rapid, large deposition of energy limits the penetration of alpha particles. The most energetic alpha particles are stopped by a few centimeters of air or a sheet of paper.

Beta-Minus Radiation

A beta-minus particle is an electron that has been ejected at a high velocity from an unstable nucleus. An electron has a small mass and an electrical charge of -1. Beta particles cause ionization by displacing electrons from atom orbits. The ionization occurs from collisions with orbiting electrons. Each collision removes kinetic energy from the beta particle, causing it to slow down. Eventually the beta particle will be slowed enough to allow it to be captured as an

2 electrons 0.000549 amu

electron

931.5 MeV

amu

1.02 MeV

orbiting electron in an atom. Although more penetrating than the alpha, the beta is relatively

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easy to stop and has a low power of penetration. Even the most energetic beta radiation can be stopped by a few millimeters of metal.

Positron Radiation

Positively charged electrons are called positrons. Except for the positive charge, they are identical to beta-minus particles and interact with matter in a similar manner. Positrons are very short-lived, however, and quickly are annihilated by interaction with a negatively charged electron, producing two gammas with a combined energy (calculated below) equal to the rest mass of the positive and negative electrons.

Neutron Radiation

Neutrons have no electrical charge. They have nearly the same mass as a proton (a hydrogen atom nucleus). A neutron has hundreds of times more mass than an electron, but 1/4 the mass of an alpha particle. The source of neutrons is primarily nuclear reactions, such as fission, but they may also be produced from the decay of radioactive nuclides. Because of its lack of charge, the neutron is difficult to stop and has a high penetrating power.

Neutrons are attenuated (reduced in energy and numbers) by three major interactions, elastic scatter, inelastic scatter, and absorption. In elastic scatter, a neutron collides with a nucleus and bounces off. This reaction transmits some of the kinetic energy of the neutron to the nucleus of the atom, resulting in the neutron being slowed, and the atom receives some kinetic energy (motion). This process is sometimes referred to as "the billiard ball effect."

As the mass of the nucleus approaches the mass of the neutron, this reaction becomes more effective in slowing the neutron. Hydrogenous material attenuates neutrons most effectively.

In the inelastic scatter reaction, the same neutron/nucleus collision occurs as in elastic scatter. However, in this reaction, the nucleus receives some internal energy as well as kinetic energy.

its original energy level, it normally emits a gamma ray.
In the absorption reaction, the neutron is actually absorbed into the nucleus of an atom. The neutron is captured, but the atom is left in an excited state. If the nucleus emits one or more gamma rays to reach a stable level, the process is called radiative capture. This reaction occurs at most neutron energy levels, but is more probable at lower energy levels.
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INTERACTION OF RADIATION WITH MATTER Atomic and Nuclear Physics
Gamma Radiation
Gamma radiation is electromagnetic radiation. It is commonly referred to as a gamma ray and is very similar to an x-ray. The difference is that gamma rays are emitted from the nucleus of an atom, and x-rays are produced by orbiting electrons. The x-ray is produced when orbiting electrons move to a lower energy orbit or when fast-moving electrons approaching an atom are deflected and decelerated as they react with the atom's electrical field (called Bremsstrahlung). The gamma ray is produced by the decay of excited nuclei and by nuclear reactions. Because the gamma ray has no mass and no charge, it is difficult to stop and has a very high penetrating power. A small fraction of the original gamma stream will pass through several feet of concrete or several meters of water.
There are three methods of attenuating gamma rays. The first method is referred to as the photo-electric effect. When a low energy gamma strikes an atom, the total energy of the gamma is expended in ejecting an electron from orbit. The result is ionization of the atom and expulsion of a high energy electron. This reaction is most predominant with low energy gammas interacting in materials with high atomic weight and rarely occurs with gammas having an energy above 1 MeV. Annihilation of the gamma results. Any gamma energy in excess of the binding energy of the electron is carried off by the electron in the form of kinetic energy.
The second method of attenuation of gammas is called Compton scattering. The gamma interacts with an orbital or free electron; however, in this case, the photon loses only a fraction of its energy. The actual energy loss depending on the scattering angle of the gamma. The gamma continues on at lower energy, and the energy difference is absorbed by the electron. This reaction becomes important for gamma energies of about 0.1 MeV and higher.
At higher energy levels, a third method of attenuation is predominant. This method is pair-production. When a high energy gamma passes close enough to a heavy nucleus, the gamma
Fixture checkpoint marker-0274-xray is recorded on fixture page 0274.
completely disappears, and an electron and a positron are formed. For this reaction to take place, the original gamma must have at least 1.02 MeV energy. Any energy greater than 1.02 MeV becomes kinetic energy shared between the electron and positron. The probability of pair■production incre
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DOE-HDBK-1019/1-93
Atomic and Nuclear Physics INTERACTION OF RADIATION WITH MATTER
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Interaction of Radiation with Matter Summary
An alpha particle deposits a large amount of energy in a short distance of travel due to its large mass and charge.
Beta-minus particles interact with the electrons orbiting the nucleus of atoms, causing ionization by displacing the electrons. The beta particle loses energy with each interaction. After the beta particle loses enough energy, it is captured in the orbital shells of an atom.
Positrons interact with matter much the same way as beta minus particles. After the positron has lost most of its energy by ionizing atoms, it is annihilated by

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by two gammas, each with the energy equivalent of the mass of an electron (0.51 MeV).
Neutrons interact with matter by elastic scattering, inelastic scattering, or absorption.
Photoelectric effect is where a gamma interacts with an electron orbiting an atom. The entire energy of the gamma is transferred to the electron, and the electron is ejected from its orbit.
In Compton scattering a gamma interacts with an orbital electron, but only part of the gamma energy is transferred to the electron. The electron is ejected from its orbit, and the gamma is scattered off at a lower energy.
In pair-production, a gamma interacts with the electric field of a nucleus and is converted into an electron-positron pair. The gamma must have an energy greater than 1.02 MeV for this to occur.
Summary
The important information in this chapter is summarized below.
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Department of Energy
Fundamentals Handbook
NUCLEAR PHYSICS
AND REACTOR THEORY
Module 2
Reactor Theory (Neutron Characteristics)
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TERMINAL OBJECTIVE
1.0 Without references, EXPLAIN how neutron sources produce neutrons.
ENABLING OBJECTIVES
1.1 DEFINE the following terms:
a. Intrinsic neutron source
b. Installed neutron source
1.2 LIST three examples of reactions that produce neutrons in intrinsic neutron sources.
1.3 LIST three examples of reactions that produce neutrons in installed neutron sources.

TERMINAL OBJECTIVE
2.0 Given the necessary information for calculations, EXPLAIN basic concepts in reactor physics and perform calculations.
ENABLING OBJECTIVES
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2.1 DEFINE the following terms:
a. Atom density d. Barn
b. Neutron flux e. Macroscopic cross section
c. Microscopic cross section f. Mean free path
2.2 EXPRESS macroscopic cross section in terms of microscopic cross section.
2.3 DESCRIBE how the absorption cross section of typical nuclides varies with neutron energy at energies below the resonance absorption region.
2.4 DESCRIBE the cause of resonance absorption in terms of nuclear energy levels.
2.5 DESCRIBE the energy dependence of resonance absorption peaks for typical light and heavy nuclei.
2.6 EXPRESS mean free path in terms of macroscopic cross section.
2.7 Given the number densities (or total density and component fractions) and microscopic cross sections of components, CALCULATE the macroscopic cross section for a mixture.
2.8 CALCULATE a macroscopic cross section given a material density, atomic mass, and microscopic cross section.
2.9 EXPLAIN neutron shadowing or self-shielding.
2.10 Given the neutron flux and macroscopic cross section, CALCULATE the reaction rate.
2.11 DESCRIBE the relationship between neutron flux and reactor power.

ENABLING OBJECTIVES (Cont.)
2.12 DEFINE the following concepts:
a. Thermalization d. Average logarithmic energy decrement
b. Moderator e. Macroscopic slowing down power
c. Moderating ratio

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2.14 Given an average fractional energy loss per collision, CALCULATE the energy loss after a specified number of collisions.
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Reactor Theory (Neutron Characteristics) DOE-HDBK-1019/1-93 OBJECTIVES
TERMINAL OBJECTIVE
3.0 Without references, EXPLAIN the production process and effects on fission of prompt and delayed neutrons.
ENABLING OBJECTIVES
3.1 STATE the origin of prompt neutrons and delayed neutrons.
3.2 STATE the approximate fraction of neutrons that are born as delayed neutrons from the fission of the following nuclear fuels:
a. Uranium-235
b. Plutonium-239
3.3 EXPLAIN the mechanism for production of delayed neutrons.
3.4 EXPLAIN prompt and delayed neutron generation times.
3.5 Given prompt and delayed neutron generation times and delayed neutron fraction, CALCULATE the average generation time.
3.6 EXPLAIN the effect of delayed neutrons on reactor control.
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OBJECTIVES DOE-HDBK-1019/1-93 Reactor Theory (Neutron Characteristics)
TERMINAL OBJECTIVE
4.0 Without references, DESCRIBE the neutron energy spectrum for the type of reactor presented in this module.
ENABLING OBJECTIVES
4.1 STATE the average energy at which prompt neutrons are produced.
4.2 DESCRIBE the neutron energy spectrum in the following reactors:
a. Fast reactor
b. Thermal reactor
4.3 EXPLAIN the reason for the particular shape of the fast, intermediate, and slow energy
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regions of the neutron flux spectrum for a thermal reactor.
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Reactor Theory (Neutron Characteristics) DOE-HDBK-1019/1-93 NEUTRON SOURCES
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NEUTRON SOURCES
Neutrons from a variety of sources are always present in a reactor core. This is true even when the reactor is shut down. Some of these neutrons are produced by naturally occurring (intrinsic) neutron sources, while others may be the result of fabricated (installed) neutron sources that are incorporated into the design of the reactor. The neutrons produced by sources other than neutron-induced fission are often grouped together and classified as source neutrons.
EO 1.1 DEFINE the following terms:
a. Intrinsic neutron source
b. Installed neutron source
EO 1.2 LIST three examples of reactions that produce neutrons in intrinsic neutron sources.
EO 1.3 LIST three examples of reactions that produce neutrons in installed neutron sources.

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In addition to neutron-induced fission, neutrons are produced by other reactions. The neutrons produced by reactions other than neutron-induced fission are called source neutrons. Source neutrons are important because they ensure that the neutron population remains high enough to allow a visible indication of neutron level on the most sensitive monitoring instruments while the reactor is shutdown and during the startup sequence. This verifies instrument operability and allows monitoring of neutron population changes. Source neutrons can be classified as either intrinsic or installed neutron sources.

Intrinsic Neutron Sources

Some neutrons will be produced in the materials present in the reactor due to a variety of unavoidable reactions that occur because of the nature of these materials. Intrinsic neutron sources are those neutron-producing reactions that always occur in reactor materials.

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■ ■

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A limited number of neutrons will always be present, even in a reactor core that has never been operated, due to spontaneous fission of some heavy nuclides that are present in the fuel. Uranium■238, ur of the rate at which different heavy nuclides produce neutrons by spontaneous fission.

Californium-252 is not an intrinsic neutron source, but will be discussed in the section on installed neutron sources.

TABLE 1

Neutron Production by Spontaneous Fission

Nuclide T (Fission) T (1/2 1/2 ■-decay) neutrons/sec/gram

U 1.8 x 10 years 6.8 x 10 years 8.0 x 10 235
92
17 8 -4

U 8.0 x 10 years 4.5 x 10 years 1.6 x 10 238
92
15 9 -2

Pu 5.5 x 10 years 2.4 x 10 years 3.0 x 10 239
94
5 4 -2

Pu 1.2 x 10 years 6.6 x 10 years 1.0 x 10 240
94
11 3 3

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Cf 66.0 years 2.65 years 2.3 x 10 252
98
12

Another intrinsic neutron source is a reaction involving natural boron and fuel. In some reactors, natural boron is loaded into the reactor core as a neutron absorber to improve reactor control or increase core life-time. Boron-11 (80.1% of natural boron) undergoes a reaction with the alpha particle emitted by the radioactive decay of heavy nuclides in the fuel to yield a neutron as shown below.

The boron-11 must be mixed with, or in very close proximity to, the fuel for this reaction because of the short path length of the alpha particle. For a reactor core with this configuration, this (■,n) reaction is an important source of neutrons for reactor startup.

be produced by the interaction of a gamma ray and a deuterium nucleus. This reaction is commonly referred to as a photoneutron reaction because it is initiated by electromagnetic radiation and results in the production of a neutron. The photoneutron reaction is shown below.

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■ ■

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There is an abundant supply of high energy gammas in a reactor that has been operated because many of the fission products are gamma emitters. All water-cooled reactors have some deuterium present in the coolant in the reactor core because a small fraction of natural hydrogen is the isotope deuterium. The atom percentage of deuterium in the water ranges from close to
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the naturally occurring value (0.015%) for light water reactors to above 90% deuterium for heavy water reactors. Therefore, the required conditions for production of photoneutrons exist.

The supply of gamma rays decreases with time after shutdown as the gamma emitters decay; therefore, the photoneutron production rate also decreases. In a few particular reactors, additional D O (heavy water) may be added to the reactor to increase the production of 2 photoneutrons following a long shutdown period.

Installed Neutron Sources

Because intrinsic neutron sources can be relatively weak or dependent upon the recent power history of the reactor, many reactors have artificial sources of neutrons installed. These neutron sources ensure that shutdown neutron levels are high enough to be detected by the nuclear instruments at all times. This provides a true picture of reactor conditions and any change in these conditions. An installed neutron source is an assembly placed in or near the reactor for the sole purpose of producing source neutrons.

One strong source of neutrons is the artificial nuclide californium-252, which emits neutrons at the rate of about 2 x 10 neutrons per second per gram as the result of spontaneous fission. 12 Important drawbacks for some applications may be its high cost and its short half-life (2.65 years).

Many installed neutron sources use the (■,n) reaction with beryllium. These sources are composed of a mixture of metallic beryllium (100% beryllium-9) with a small quantity of an alpha particle emitter, such as a compound of radium, polonium, or plutonium. The reaction that occurs is shown below.

The beryllium is intimately (homogeneously) mixed with the alpha emitter and is usually enclosed in a stainless steel capsule.

Another type of installed neutron source that is widely used is a photoneutron source that employs the (■,n) reaction with beryllium. Beryllium is used for photoneutron sources because its stable isotope beryllium-9 has a weakly attached last neutron with a binding energy of only 1.66 MeV. Thus, a gamma ray with greater energy than 1.66 MeV can cause neutrons to be ejected by the (■,n) reaction as shown below.

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Neutron Sources Summary

& Intrinsic neutron sources are sources of neutrons from materials that are in the reactor for other purposes such as fuel, burnable poison, or moderator.
& Installed neutron sources are materials or components placed in the reactor

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& Examples of intrinsic neutron sources are listed below.
Spontaneous fission of heavy nuclides in fuel, such as uranium-238, uranium-235, and plutonium-239, results in fission fragments and free neutrons.
Boron-11 mixed with the fuel undergoes an alpha-neutron reaction and becomes nitrogen-14.
Deuterium present in the reactor coolant undergoes a gamma■neutron reaction and becomes hydrogen-1.
& Examples of installed neutron sources are listed below.
Spontaneous fission of californium-252 results in fission fragments and free neutrons.
Beryllium-9 undergoes an alpha-neutron reaction (alpha from the decay of plutonium, polonium, or radium) and becomes carbon-12.
Beryllium-9 undergoes a gamma-neutron reaction (high energy gamma from decay of antimony-124) and becomes beryllium-8.
Many startup sources of this type use antimony and beryllium because after activation with neutrons the radioactive antimony becomes an emitter of high energy gammas. The photoneutron sources of this type are constructed somewhat differently from the (■,n) types. One design incorporates a capsule of irradiated antimony enclosed in a beryllium sleeve. The entire assembly is then encased in a stainless steel cladding. A large reactor may have several neutron sources of this type installed within the core.
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Summary
The important information in this chapter is summarized below.
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Reactor Theory (Neutron Characteristics) NUCLEAR CROSS SECTIONS AND NEUTRON FLUX
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NUCLEAR CROSS SECTIONS AND NEUTRON FLUX
To determine the frequency of neutron interactions, it is necessary to describe the availability of neutrons to cause interaction and the probability of a neutron interacting with material. The availability of neutrons and the probability of interaction are quantified by the neutron flux and nuclear cross section.
EO 2.1 DEFINE the following terms:
a. Atom density d. Barn
b. Neutron flux e. Macroscopic cross section
c. Microscopic cross section f. Mean free path
EO 2.2 EXPRESS macroscopic cross section in terms of microscopic cross section.
EO 2.3 DESCRIBE how the absorption cross section of typical nuclides varies with neutron energy at energies below the resonance absorption region.
EO 2.4 DESCRIBE the cause of resonance absorption in terms of nuclear energy levels.
EO 2.5 DESCRIBE the energy dependence of resonance absorption peaks for typical light and heavy nuclei.
EO 2.6 EXPRESS mean free path in terms of macroscopic cross section.
EO 2.7 Given the number densities (or total density and component fractions) and microscopic cross sections of components, CALCULATE the macroscopic cross section for a mixture.

atomic mass, and microscopic cross section.
EO 2.9 EXPLAIN neutron shadowing or self-shielding.
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NUCLEAR CROSS SECTIONS AND NEUTRON FLUX Reactor Theory (Neutron Characteristics)
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Introduction
Fission neutrons are born with an average energy of about 2 MeV. These fast neutrons interact with the reactor core materials in various absorption and scattering reactions. Collisions that result in scattering are useful in slowing neutrons to thermal energies. Thermal neutrons may be absorbed by fissile nuclei to produce more fissions or be absorbed in fertile material for conversion to fissionable fuel. Absorption of neutrons in structural components, coolant, and other non-fuel material results in the removal of neutrons without fulfilling any useful purpose. To safely and efficiently operate a nuclear reactor it is necessary to predict the probability that a particular absorption or scattering reaction will occur. Once these probabilities are known, if the availability of neutrons can be determined, then the rate at which these nuclear reactions take place can be predicted.
Atom Density
One important property of a material is the atom density. The atom density is the number of atoms of a given type per unit volume of the material. To calculate the atom density of a substance use Equation (2-1).
(2-1)
where:
N = atom density (atoms/cm)3
' = density (g/cm)3
N = Avogadro's number (6.022 x 10 atoms/mole) A
23
Fixture checkpoint marker-0282-xray is recorded on fixture page 0282.
M = gram atomic weight
N
NA
M
2.699 g
cm3
6.022 x 1023 atoms
mole
26.9815 g
mole
6.024 x 1022 atoms
cm3
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Example:
A block of aluminum has a density of 2.699 g/cm . If the gram atomic weight of
3
aluminum is 26.9815 g, calculate the atom density of the aluminum.
Solution:

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The probability of a neutron interacting with a nucleus for a particular reaction is dependent upon not only the kind of nucleus involved, but also the energy of the neutron. Accordingly, the absorption of a thermal neutron in most materials is much more probable than the absorption of a fast neutron. Also, the probability of interaction will vary depending upon the type of reaction involved.

The probability of a particular reaction occurring between a neutron and a nucleus is called the microscopic cross section ( ) of the nucleus for the particular reaction. This cross section will vary with the energy of the neutron. The microscopic cross section may also be regarded as the effective area the nucleus presents to the neutron for the particular reaction. The larger the effective area, the greater the probability for reaction.

Because the microscopic cross section is an area, it is expressed in units of area, or square centimeters. A square centimeter is tremendously large in comparison to the effective area of a nucleus, and it has been suggested that a physicist once referred to the measure of a square centimeter as being "as big as a barn" when applied to nuclear processes. The name has persisted and microscopic cross sections are expressed in terms of barns. The relationship between barns and cm is shown below. 2

1 barn = 10-24 cm2

N

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NUCLEAR CROSS SECTIONS AND NEUTRON FLUX Reactor Theory (Neutron Characteristics)

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Whether a neutron will interact with a certain volume of material depends not only on the microscopic cross section of the individual nuclei but also on the number of nuclei within that volume. Therefore, it is necessary to define another kind of cross section known as the macroscopic cross section ( ). The macroscopic cross section is the probability of a given reaction occurring per unit travel of the neutron. is related to the microscopic cross section ( ) by the relationship shown below.

(2-2)

where:

= macroscopic cross section (cm)

-1

N = atom density of material (atoms/cm)

3

= microscopic cross-section (cm)

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2

The difference between the microscopic and macroscopic cross sections is extremely important and is restated for clarity. The microscopic cross section ( ) represents the effective target area that a single nucleus presents to a bombarding particle. The units are given in barns or cm .

2

The macroscopic cross section ( ) represents the effective target area that is presented by all of the nuclei contained in 1 cm of the material. The units are given as 1/cm or cm .

3 -1

A neutron interacts with an atom of the material it enters in two basic ways. It will either interact through a scattering interaction or through an absorption reaction. The probability of a neutron being absorbed by a particular atom is the microscopic cross section for absorption, . The probability of a neutron scattering off of a particular nucleus is the microscopic cross a section for scattering, s

microscopic cross section for scattering is the total microscopic cross section, . T
= + T a s
Both the absorption and the scattering microscopic cross sections can be further divided. For instance, the scattering cross section is the sum of the elastic scattering cross section (se) and the inelastic scattering cross section ( ). si
= + s se si
The microscopic absorption cross section ( ) includes all reactions except scattering. However, a for most purposes it is sufficient to merely separate it into two categories, fission (f) and capture ( ). Radiative capture of neutrons was described in the Neutron Interactions chapter c of Module 1.
= + a f c
Fixture checkpoint marker-0284-xray is recorded on fixture page 0284.

Figure 1 Typical Neutron Absorption Cross Section vs. Neutron Energy
The variation of absorption cross sections with neutron energy is often complicated. For many elements the absorption cross sections are small, ranging from a fraction of a barn to a few barns for slow (or thermal) neutrons.
For a considerable number of nuclides of moderately high (or high) mass numbers, an examination of the variation of the absorption cross section with the energy of the incident neutron reveals the existence of three regions on a curve of absorption cross section versus neutron energy. This cross section is illustrated in Figure 1. First, the cross section decreases steadily with increasing neutron energy in a low energy region, which includes the thermal range (E < 1 eV). In this region the absorption cross section, which is often high, is inversely proportional to the velocity (v). This region is frequently referred to as the "1/v region," because the absorption cross section is proportional to 1/v, which is the reciprocal of neutron velocity. Following the 1/v region, there occurs the "resonance region" in which the cross sections rise sharply to high values called "resonance peaks" for neutrons of certain energies, and then fall again. These energies are called resonance energies and are a result of the affinity of the nucleus for neutrons whose energies closely match its discrete, quantum energy levels. That is, when the binding energy of a neutron plus the kinetic energy of the neutron are exactly equal to the amount required to raise a compound nucleus from its ground state to a quantum level, resonance absorption occurs. The following example problem further illustrates this point.
■
(

Assuming that uranium-236 has a nuclear quantum energy level at 6.8 MeV above its ground state, calculate the kinetic energy a neutron must possess to undergo resonant absorption in uranium-235 at this resonance energy level.
BE = [Mass(U) + Mass(neutron) - Mass(U)] x 931 MeV/amu 235 236
BE = (235.043925 + 1.008665 - 236.045563) x 931 MeV/amu
BE = (0.007025 amu) x 931 MeV/amu = 6.54 MeV
6.8 MeV - 6.54 MeV = 0.26 MeV
The difference between the binding energy and the quantum energy level equals the amount of kinetic energy the neutron must possess. The typical heavy nucleus will have many closely■spaced resonanc

being less complex, have fewer possible energy states and fewer resonances that are sparsely distributed at higher energy levels.

For higher neutron energies, the absorption cross section steadily decreases as the energy of the neutron increases. This is called the "fast neutron region." In this region the absorption cross sections are usually less than 10 barns.

With the exception of hydrogen, for which the value is fairly large, the elastic scattering cross sections are generally small, for example, 5 barns to 10 barns. This is close to the magnitude of the actual geometric cross sectional area expected for atomic nuclei. In potential scattering, the cross section is essentially constant and independent of neutron energy. Resonance elastic scattering and inelastic scattering exhibit resonance peaks similar to those associated with absorption cross sections. The resonances occur at lower energies for heavy nuclei than for light nuclei. In general, the variations in scattering cross sections are very small when compared to the variations that occur in absorption cross sections.

Mean Free Path

If a neutron has a certain probability of undergoing a particular interaction in one centimeter of travel, then the inverse of this value describes how far the neutron will travel (in the average case) before undergoing an interaction. This average distance traveled by a neutron before interaction is known as the mean free path for that interaction and is represented by the symbol

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■. The relationship between the mean free path (■) and the macroscopic cross section (*) is shown below.

(2-3)

'

(

)

Calculation of Macroscopic Cross Section and Mean Free Path

Most materials are composed of several elements, and because most elements are composed of several isotopes, most materials involve many cross sections, one for each isotope involved. Therefore, to include all the isotopes within a given material, it is necessary to determine the macroscopic cross section for each isotope and then sum all the individual macroscopic cross sections. Equation (2-4) can be used to determine the macroscopic cross section for a composite material.

* = N) + N) + N) + ... N) (2-4) 1 1 2 2 3 3 nn

where:

N = the number of nuclei per cm of the n element n

3 th

) = the microscopic cross section of the n element n

th

The following example problems illustrate the calculation of the macroscopic cross section for a single element and for combinations of materials.

Example 1:

Find the macroscopic thermal neutron absorption cross section for iron, which has a density of 7.86 g/cm . The microscopic cross section for absorption of iron is 2.56 barns 3 and the gram atomic weight is 55.847 g.

Solution:

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Step 2: Use this atom density in Equation (2-2) to calculate the macroscopic cross section.

'

'

Example 2:

An alloy is composed of 95% aluminum and 5% silicon (by weight). The density of the alloy is 2.66 g/cm . Properties of aluminum and silicon are shown below. 3

Element Gram Atomic Weight)) a (barns) s (barns)
Aluminum 26.9815 0.23 1.49
Silicon 28.0855 0.16 2.20

1. Calculate the atom densities for the aluminum and silicon.
2. Determine the absorption and scattering macroscopic cross sections for thermal neutrons.
3. Calculate the mean free paths for absorption and scattering.

Solution:

Step 1: The density of the aluminum will be 95% of the total density. Using Equation (2-1) yields the atom densities.

a NAl a,Al NSi a,Si
5.64 x 1022 atoms
cm3
0.23 x 10 24 cm22.85 x 1021 atoms
cm3

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0.16 x 10 24 cm2
0.0134 cm 1
s NAl s,Al NSi s,Si
5.64 x 1022 atoms
cm3
1.49 x 10 24 cm22.85 x 1021 atoms
cm3
2.20 x 10 24 cm2
0.0903 cm 1
a
1
a
1
0.01345 cm 1
74.3 cm
s
1
s
1
0.0903 cm 1
11.1 cm

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Step 2: The macroscopic cross sections for absorption and scattering are calculated using Equation (2-4).
Step 3: The mean free paths are calculated by inserting the macroscopic cross sections calculated above into Equation (2-3).
Thus, a neutron must travel an average of 74.3 cm to interact by absorption in this alloy, but it must travel only 11.1 cm to interact by scattering.
)
)
)
)
(■
(■
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Effects of Temperature on Cross Section
As discussed, the microscopic absorption cross section varies significantly as neutron energy varies. The microscopic cross sections provided on most charts and tables are measured for a standard neutron velocity of 2200 meters/second, which corresponds to an ambient temperature of 68(F. Therefore, if our material is at a higher temperature, the absorption cross section will be lower than the value for 68(F, and any cross sections which involve absorption (for example,) ,) ,)) must be corrected for the existing temperature. acf
The following formula is used to correct microscopic cross sections for temperature. Although the example illustrates absorption cross section, the same formula may be used to correct capture and fission cross sections.
where:
) = microscopic cross section corrected for temperature
) = microscopic cross section at reference temperature (68(F or 20(C) o
T = reference temperature (68(F) in degrees Rankine ((R) or Kelvin ((K) o
T = temperature for which corrected value is being calculated
NOTE: When using this formula, all temperatures must be converted to (R or (K.
Fixture checkpoint marker-0287-xray is recorded on fixture page 0287.
(R = (F + 460
(K = (C + 273
Example:
What is the value of)f for uranium-235 for thermal neutrons at 500(F? Uranium-235 has a) of 583 barns at 68(F. f
Solution:
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Reactor Theory (Neutron Characteristics) NUCLEAR CROSS SECTIONS AND NEUTRON FLUX
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Neutron Flux
Macroscopic cross sections for neutron reactions with materials determine the probability of one neutron undergoing a specific reaction per centimeter of travel through that material. If one wants to determine how many reactions will actually occur, it is necessary to know how many neutrons are traveling through the material and how many centimeters they travel each second. It is convenient to consider the number of neutrons existing in one cubic centimeter at any one

of neutrons existing in a cm of material at any instant is called neutron density and is 3
represented by the symbol n with units of neutrons/cm . The total distance these neutrons can 3
travel each second will be determined by their velocity.
A good way of defining neutron flux (1) is to consider it to be the total path length covered by all neutrons in one cubic centimeter during one second. Mathematically, this is the equation below.
1 = n v (2-5)
where:
1 = neutron flux (neutrons/cm -sec) 2
n = neutron density (neutrons/cm)3
v = neutron velocity (cm/sec)
The term neutron flux in some applications (for example, cross section measurement) is used as parallel beams of neutrons traveling in a single direction. The intensity (I) of a neutron beam is the product of the neutron density times the average neutron velocity. The directional beam intensity is equal to the number of neutrons per unit area and time (neutrons/cm -sec) falling on 2
a surface perpendicular to the direction of the beam.
One can think of the neutron flux in a reactor as being comprised of many neutron beams traveling in various directions. Then, the neutron flux becomes the scalar sum of these directional flux intensities (added as numbers and not vectors), that is, 1 = I + I + I +...I . 123 n
Since the atoms in a reactor do not interact preferentially with neutrons from any particular direction, all of these directional beams contribute to the total rate of reaction. In reality, at a given point within a reactor, neutrons will be traveling in all directions.
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Self-Shielding
In some locations within the reactor, the flux level may be significantly lower than in other areas due to a phenomenon referred to as neutron shadowing or self-shielding. For example, the interior of a fuel pin or pellet will "see" a lower average flux level than the outer surfaces since an appreciable fraction of the neutrons will have been absorbed and therefore cannot reach the interior of the fuel pin. This is especially important at resonance energies, where the absorption cross sections are large.
Summary
The important information in this chapter is summarized below.
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Reactor Theory (Neutron Characteristics) NUCLEAR CROSS SECTIONS AND NEUTRON FLUX
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Nuclear Cross Section and Neutron Flux Summary
& Atom density (N) is the number of atoms of a given type per unit volume of
Fixture checkpoint marker-0288-xray is recorded on fixture page 0288.
material.
& Microscopic cross section ()) is the probability of a given reaction occurring between a neutron and a nucleus.
& Microscopic cross sections are measured in units of barns, where 1 barn = 10-24
cm .2
& Macroscopic cross section (*) is the probability of a given reaction occurring per unit length of travel of the neutron. The units for macroscopic cross section are
cm . -1

between interactions.
& Neutron flux (1) is the total path length traveled by all neutrons in one cubic centimeter of material during one second.
& The macroscopic cross section for a material can be calculated using the equation below.
* = N)
& The absorption cross section for a material usually has three distinct regions. At low neutron energies (<1 eV) the cross section is inversely proportional to the neutron velocity.
& Resonance absorption occurs when the sum of the kinetic energy of the neutron and its binding energy is equal to an allowed nuclear energy level of the nucleus.
& Resonance peaks exist at intermediate energy levels. For higher neutron energies, the absorption cross section steadily decreases as the neutron energy increases.
& The mean free path equals 1/*.
& The macroscopic cross section for a mixture of materials can be calculated using the equation below.
Fixture checkpoint marker-0289-xray is recorded on fixture page 0289.
* = N 1 1 2 2 3 3 nn) + N) + N) + ... N)
& Self-shielding is where the local neutron flux is depressed within a material due to neutron absorption near the surface of the material.
1)
REACTION RATES DOE-HDBK-1019/1-93 Reactor Theory (Neutron Characteristics) NP-02 Page 18 Rev. 0
REACTION RATES
It is possible to determine the rate at which a nuclear reaction will take place based on the neutron flux, cross section for the interaction, and atom density of the target. This relationship illustrates how a change in one of these items affects the reaction rate.
EO 2.10 Given the neutron flux and macroscopic cross section, CALCULATE the reaction rate.
EO 2.11 DESCRIBE the relationship between neutron flux and reactor power.
Reaction Rates
If the total path length of all the neutrons in a cubic centimeter in a second is known, (neutron flux (1)), and if the probability of having an interaction per centimeter path length is also known (macroscopic cross section (*)), multiply them together to get the number of interactions taking place in that cubic centimeter in one second. This value is known as the reaction rate and is denoted by the symbol R. The reaction rate can be calculated by the equation shown below.
R = 1 * (2-6)
where:
R = reaction rate (reactions/sec)
1 = neutron flux (neutrons/cm -sec) 2
* = macroscopic cross section (cm) -1
Substituting the fact that * = N) into Equation (2-6) yields the equation below.
where:
) = microscopic cross section (cm)2
N = atom density (atoms/cm)3
Rf f

cm2
sec
0.1 cm 1
1 x 1012 fissions
cm3
sec
Rf f
f
Rf
1.29 x 1012 fissions
cm3
sec
3 x 1013 neutrons
cm2
sec
0.043 cm 1
Reactor Theory (Neutron Characteristics) DOE-HDBK-1019/1-93 REACTION RATES
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The reaction rate calculated will depend on which macroscopic cross section is used in the calculation. Normally, the reaction rate of greatest interest is the fission reaction rate.
Example:
If a one cubic centimeter section of a reactor has a macroscopic fission cross section of
0.1 cm , and if the thermal neutron flux is 10 neutrons/cm -sec, what is the fission rate -1 13 2
Fixture checkpoint marker-0290-xray is recorded on fixture page 0290.
in that cubic centimeter?
Solution:
In addition to using Equation (2-6) to determine the reaction rate based on the physical properties of material, it is also possible to algebraically manipulate the equation to determine physical properties if the reaction rate is known.
Example:
A reactor operating at a flux level of 3 x 10 neutrons/cm -sec contains 10 atoms of
13 2 20
uranium-235 per cm . The reaction rate is 1.29 x 10 fission/cm . Calculate and .
3 12 3
f f
Solution:
Step 1: The macroscopic cross section can be determined by solving
Equation (2-6) for and substituting the appropriate values. f
Rf N f
f
Rf
N
1.29 x 1012 fissions
cm3
sec
1 x 1020 atoms
cm3
3 x 1013 neutrons

sec
4.3 x 10 22 cm2 1 barn
1 x 10 24 cm2
430 barns
1 watt 1 erg
1 x 10 7 watt sec
1 MeV
1.602 x 10 6
erg
1 fission
200 MeV
3.12 x 1010 fissions
second
REACTION RATES DOE-HDBK-1019/1-93 Reactor Theory (Neutron Characteristics)
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Step 2: To find the microscopic cross section, replace f with (N x f) and solve
for .
f
Reactor Power Calculation
Multiplying the reaction rate per unit volume by the total volume of the core results in the total number of reactions occurring in the core per unit time. If the amount of energy involved in each reaction were known, it would be possible to determine the rate of energy release (power) due to a certain reaction.
In a reactor where the average energy per fission is 200 MeV, it is possible to determine the number of fissions per second that are necessary to produce one watt of power using the following conversion factors.
1 fission = 200 MeV
1 MeV = 1.602 x 10 ergs -6
1 erg = 1 x 10 watt-sec -7
This is equivalent to stating that 3.12 x 10 fissions release 1 watt-second of energy. 10
Fixture checkpoint marker-0291-xray is recorded on fixture page 0291.
P
th f V
3.12 x 1010 fissions
watt sec
Reactor Theory (Neutron Characteristics) DOE-HDBK-1019/1-93 REACTION RATES
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The power released in a reactor can be calculated based on Equation (2-6). Multiplying the reaction rate by the volume of the reactor results in the total fission rate for the entire reactor. Dividing by the number of fissions per watt-sec results in the power released by fission in the reactor in units of watts. This relationship is shown mathematically in Equation (2-7) below.
(2-7)
where:
P = power (watts)
= thermal neutron flux (neutrons/cm -sec) th
2
= macroscopic cross section for fission (cm) f
-1

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3
Relationship Between Neutron Flux and Reactor Power
In an operating reactor the volume of the reactor is constant. Over a relatively short period of time (days or weeks), the number density of the fuel atoms is also relatively constant. Since the atom density and microscopic cross section are constant, the macroscopic cross section must also be constant. Examining Equation (2-7), it is apparent that if the reactor volume and macroscopic cross section are constant, then the reactor power and the neutron flux are directly proportional. This is true for day-to-day operation. The neutron flux for a given power level will increase very slowly over a period of months due to the burnup of the fuel and resulting decrease in atom density and macroscopic cross section.
REACTION RATES DOE-HDBK-1019/1-93 Reactor Theory (Neutron Characteristics)
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Reaction Rates Summary
& The reaction rate is the number of interactions of a particular type occurring in a cubic centimeter of material in a second.
& The reaction rate can be calculated by the equation below.
R = 1 *
& Over a period of several days, while the atom density of the fuel can be considered constant, the neutron flux is directly proportional to reactor power.
Summary
The important information in this chapter is summarized below.
Reactor Theory (Neutron Characteristics) DOE-HDBK-1019/1-93 NEUTRON MODERATION
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NEUTRON MODERATION
In thermal reactors, the neutrons that cause fission are at a much lower energy than the energy level at which they were born from fission. In this type of reactor, specific materials must be included in the reactor design to reduce the energy level of the neutrons in an efficient manner.
EO 2.12 DEFINE the following concepts:
a. Thermalization d. Average logarithmic energy decrement
b. Moderator e. Macroscopic slowing down power
c. Moderating ratio
EO 2.13 LIST three desirable characteristics of a moderator.
EO 2.14 Given an average fractional energy loss per collision, CALCULATE the energy loss after a specified number of collisions.
Neutron Slowing Down and Thermalization
Fixture checkpoint marker-0292-xray is recorded on fixture page 0292.
Fission neutrons are produced at an average energy level of 2 MeV and immediately begin to slow down as the result of numerous scattering reactions with a variety of target nuclei. After a number of collisions with nuclei, the speed of a neutron is reduced to such an extent that it has approximately the same average kinetic energy as the atoms (or molecules) of the medium in which the neutron is undergoing elastic scattering. This energy, which is only a small fraction of an electron volt at ordinary temperatures (0.025 eV at 20(C), is frequently referred to as the thermal energy, since it depends upon the temperature. Neutrons whose energies have been reduced to values in this region (< 1 eV) are designated thermal neutrons. The process of reducing the energy of a neutron to the thermal region by elastic scattering is referred to as thermalization, slowing down, or moderation. The material used for the purpose of thermalizing

number of collisions, but does not absorb them to any great extent. Slowing the neutrons in as few collisions as possible is desirable in order to reduce the amount of neutron leakage from the core and also to reduce the number of resonance absorptions in non-fuel materials. Neutron leakage and resonance absorption will be discussed in the next module.
ln Ei
ln Ef
ln
Ei
Ef
2
A
2
3
NEUTRON MODERATION DOE-HDBK-1019/1-93 Reactor Theory (Neutron Characteristics)
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NP-02 Page 24 Rev. 0
The ideal moderating material (moderator) should have the following nuclear properties.
large scattering cross section
small absorption cross section
large energy loss per collision
A convenient measure of energy loss per collision is the logarithmic energy decrement. The average logarithmic energy decrement is the average decrease per collision in the logarithm of the neutron energy. This quantity is represented by the symbol (Greek letter xi).
(2-8)
where:
= average logarithmic energy decrement
Ei = average initial neutron energy
Ef = average final neutron energy
The symbol is commonly called the average logarithmic energy decrement because of the fact that a neutron loses, on the average, a fixed fraction of its energy per scattering collision. Since the fraction of energy retained by a neutron in a single elastic collision is a constant for a given material, is also a constant. Because it is a constant for each type of material and does not depend upon the initial neutron energy, is a convenient quantity for assessing the moderating ability of a material.
The values for the lighter nuclei are tabulated in a variety of sources. The following commonly used approximation may be used when a tabulated value is not available.
This approximation is relatively accurate for mass numbers (A) greater than 10, but for some low values of A it may be in error by over three percent.
N
ln Ehigh ln Elow
ln
Ehigh
Elow
N
ln
Ehigh
Elow
ln

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0.025 eV
0.948
19.2 collisions
Reactor Theory (Neutron Characteristics) DOE-HDBK-1019/1-93 NEUTRON MODERATION
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Since represents the average logarithmic energy loss per collision, the total number of collisions necessary for a neutron to lose a given amount of energy may be determined by dividing into the difference of the natural logarithms of the energy range in question. The number of collisions (N) to travel from any energy, E , to any lower energy, E , can be calculated as shown below. high low
Example:
How many collisions are required to slow a neutron from an energy of 2 MeV to a thermal energy of 0.025 eV, using water as the moderator? Water has a value of 0.948 for .
Solution:
Sometimes it is convenient, based upon information known, to work with an average fractional energy loss per collision as opposed to an average logarithmic fraction. If the initial neutron energy level and the average fractional energy loss per collision are known, the final energy level for a given number of collisions may be computed using the following formula.
! (
NEUTRON MODERATION DOE-HDBK-1019/1-93 Reactor Theory (Neutron Characteristics)
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Fixture checkpoint marker-0294-xray is recorded on fixture page 0294.
(2-9)
where:
E = initial neutron energy o
E = neutron energy after N collisions N
x = average fractional energy loss per collision
N = number of collisions
Example:
If the average fractional energy loss per collision in hydrogen is 0.63, what will be the energy of a 2 MeV neutron after (a) 5 collisions? (b) 10 collisions?
Solution:
a)
b)
Macroscopic Slowing Down Power
Although the logarithmic energy decrement is a convenient measure of the ability of a material to slow neutrons, it does not measure all necessary properties of a moderator. A better measure of the capabilities of a material is the macroscopic slowing down power. The macroscopic slowing down power (MSDP) is the product of the logarithmic energy decrement and the macroscopic cross section for scattering in the material. Equation (2-10) illustrates how to calculate the macroscopic slowing down power.
(2-10)
MR
s
a
Reactor Theory (Neutron Characteristics) DOE-HDBK-1019/1-93 NEUTRON MODERATION
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Moderating Ratio

in question, but it still does not fully explain the effectiveness of the material as a moderator. An element such as boron has a high logarithmic energy decrement and a good slowing down power, but it is a poor moderator because of its high probability of absorbing neutrons.
The most complete measure of the effectiveness of a moderator is the moderating ratio. The moderating ratio is the ratio of the macroscopic slowing down power to the macroscopic cross section for absorption. The higher the moderating ratio, the more effectively the material performs as a moderator. Equation (2-11) shows how to calculate the moderating ratio of a material.
(2-11)
Moderating properties of different materials are compared in Table 2.
TABLE 2
Moderating Properties of Materials
Material Collisions to Slowing Down
Number of Macroscopic
Thermalize Power
Moderating
Ratio
H2O 0.927 19 1.425 62
D2O 0.510 35 0.177 4830
Helium 0.427 42 9 x 10 51 -6
Beryllium 0.207 86 0.154 126
Boron 0.171 105 0.092 0.00086
Carbon 0.158 114 0.083 216
NEUTRON MODERATION DOE-HDBK-1019/1-93 Reactor Theory (Neutron Characteristics)
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Summary
The important information in this chapter is summarized below.
Neutron Moderation Summary
Fixture checkpoint marker-0295-xray is recorded on fixture page 0295.
& Thermalization is the process of reducing the energy level of a neutron from the energy level at which it is produced to an energy level in the thermal range.
& The moderator is the reactor material that is present for the purpose of thermalizing neutrons.
& Moderating ratio is the ratio of the macroscopic slowing down power to the macroscopic cross section for absorption.
& The average logarithmic energy decrement (!) is the average change in the logarithm of neutron energy per collision.
& Macroscopic slowing down power is the product of the average logarithmic energy decrement and the macroscopic cross section for scattering.
& There are three desirable characteristics of a moderator.
1. large scattering cross section
2. small absorption cross section
3. large energy loss per collision
& The energy loss after a specified number of collisions can be calculated using the equation below.
E = E (1 - x) N o
N
■

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Reactor Theory (Neutron Characteristics) DOE-HDBK-1019/1-93 PROMPT AND DELAYED NEUTRONS
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PROMPT AND DELAYED NEUTRONS
Not all neutrons are released at the same time following fission. Most neutrons are released virtually instantaneously and are called prompt neutrons. A very small fraction of neutrons are released after the decay of fission products and are called delayed neutrons. Although delayed neutrons are a very small fraction of the total number of neutrons, they play an extremely important role in the control of the reactor.
EO 3.1 STATE the origin of prompt neutrons and delayed neutrons.
EO 3.2 STATE the approximate fraction of neutrons that are born as delayed neutrons from the fission of the following nuclear fuels:
a. Uranium-235
b. Plutonium-239
EO 3.3 EXPLAIN the mechanism for production of delayed neutrons.
EO 3.4 EXPLAIN prompt and delayed neutron generation times.
EO 3.5 Given prompt and delayed neutron generation times and delayed neutron fraction, CALCULATE the average generation time.
EO 3.6 EXPLAIN the effect of delayed neutrons on reactor control.
Neutron Classification
The great majority (over 99%) of the neutrons produced in fission are released within about 10-13 seconds of the actual fission event. These are called prompt neutrons. A small portion of fission neutrons are delayed neutrons, which are produced for some time after the fission process has taken place. The delayed neutrons are emitted immediately following the first beta decay of a fission fragment known as a delayed neutron precursor. An example of a delayed neutron precursor is bromine-87, shown below.
PROMPT AND DELAYED NEUTRONS DOE-HDBK-1019/1-93 Reactor Theory (Neutron Characteristics)
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For most applications, it is convenient to combine the known precursors into groups with appropriately averaged properties. These groups vary somewhat depending on the fissile material in use. Table 3 lists the characteristics for the six precursor groups resulting from thermal fission of uranium-235. The fraction of all neutrons that are produced by each of these precursors is called the delayed neutron fraction for that precursor. The total fraction of all neutrons born as delayed neutrons is called the delayed neutron fraction (■). The fraction of
Fixture checkpoint marker-0296-xray is recorded on fixture page 0296.
delayed neutrons produced varies depending on the predominant fissile nuclide in use. The delayed neutron fractions (■) for the fissile nuclides of most interest are as follows: uranium-233 (0.0026), uranium-235 (0.0065), uranium-238 (0.0148), and plutonium-239 (0.0021).
TABLE 3
Delayed Neutron Precursor Groups
for Thermal Fission in Uranium-235

Group	Half-Life (sec)	Delayed Neutron Fraction	Average Energy (MeV)
1	55.7	0.00021	0.25
2	22.7	0.00142	0.46
3	6.2	0.00127	0.41
4	2.3	0.00257	0.45

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6 0.23 0.00027 -
Total - 0.0065 -
Neutron Generation Time
The neutron generation time is the time required for neutrons from one generation to cause the fissions that produce the next generation of neutrons. The generation time for prompt neutrons (5* - pronounced "ell-star") is the total time from birth to rebirth. Three time intervals are involved: (a) the time it takes a fast neutron to slow down to thermal energy, (b) the time the now thermal neutron exists prior to absorption in fuel, and (c) the time required for a fissionable nucleus to emit a fast neutron after neutron absorption.
Fast neutrons slow to thermal energies or leak out of the reactor in 10 seconds to 10 seconds, -4 -6 depending on the moderator. In water moderated reactors, thermal neutrons tend to exist for about 10 seconds before they are absorbed. Fission and fast neutron production following -4
Fixture checkpoint marker-0297-xray is recorded on fixture page 0297.
neutron absorption in a fissionable nucleus occurs in about 10 seconds. Thus, fast reactors -13 have an 5* of about 10 seconds, while thermal reactors have an 5* of about 10 seconds + 10 -6 -6 -4 seconds, which is about 10 seconds to 10 seconds. -4 -5
■ ■ ■ ■
■ ■ ■ ■
■ ■
Reactor Theory (Neutron Characteristics) DOE-HDBK-1019/1-93 PROMPT AND DELAYED NEUTRONS Rev. 0 Page 31 NP-02
On the other hand, the average generation time for the six delayed neutron groups is the total time from the birth of the fast neutron to the emission of the delayed neutron. Again, three time intervals are involved: (a) the time it takes a fast neutron to slow down to thermal energy, (b) the time the thermal neutron exists prior to absorption, and (c) the average time from neutron absorption to neutron emission by the six precursor groups. The average time for decay of precursors from uranium-235 is 12.5 seconds. The other terms in the delayed neutron generation time are insignificant when compared to this value, and the average delayed neutron generation time becomes ■ 12.5 seconds.
A neutron generation time in the range of 10 seconds to 10 seconds or faster could result in -4 -5 very rapid power excursions, and control would not be possible without the dependence upon delayed neutrons to slow down the rate of the reaction. The average generation time, and hence the rate that power can rise, is determined largely by the delayed neutron generation time. The following equation shows this mathematically.
(2-12)
Example:
Assume a prompt neutron generation time for a particular reactor of 5 x 10 seconds and -5 a delayed neutron generation time of 12.5 seconds. If ■ is 0.0065, calculate the average generation time.
Solution:
This example demonstrates the effect delayed neutrons have on the neutron generation time and thus reactor control. If a reactor were to be operated in a sustained chain reaction using only prompt neutrons (■ = 0), the generation time from the previous example would be about 5 x 10 seconds. However, by operating the reactor such that a 0.0065 fraction of neutrons are -5 delayed, the generation life time is extended to 0.0813 seconds, providing time for adequate operator control. Therefore, although only a small fraction of the total neutron population, delayed neutrons are extremely important to the control and maintenance of a sustained fission chain reaction.

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The important information in this chapter is summarized on the following page.

■■■■

PROMPT AND DELAYED NEUTRONS DOE-HDBK-1019/1-93 Reactor Theory (Neutron Characteristics)
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Prompt and Delayed Neutrons Summary

& Prompt neutrons are released directly from fission within 10 seconds of the -13 fission event.

& Delayed neutrons are released from the decay of fission products that are called delayed neutron precursors. Delayed neutron precursors are grouped according to half-life. Half-lives vary from fractions of a second to almost a minute.

& The fraction of neutrons born as delayed neutrons is different for different fuel materials. Following are values for some common fuel materials.

Uranium-235 0.0065
Plutonium-239 0.0021

& Delayed neutrons are produced by a classification of fission products known as delayed neutron precursors. When a delayed neutron precursor undergoes a ■ decay, it results in an excited daughter nucleus which immediately ejects a - neutron. Therefore, these delayed neutrons appear with a half-life of the delayed neutron precursor.

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& The delayed neutron generation time is the total time from the birth of the fast neutron to the emission of the delayed neutron in the next generation. Delayed neutron generation times are dominated by the half-life of the delayed neutron precursor. The average delayed neutron generation time is about 12.5 seconds.

& A prompt neutron generation time is the sum of the amount of time it takes a fast neutron to thermalize, the amount of time the neutron exists as a thermal neutron before it is absorbed, and the amount of time between a fissionable nuclide absorbing a neutron and fission neutrons being released. Prompt neutron generation time is about 5 x 10 seconds. -5

& The average neutron generation time can be calculated from the prompt and delayed neutron generation times and the delayed neutron fraction using Equation (2-12).

& Delayed neutrons are responsible for the ability to control the rate at which power can rise in a reactor. If only prompt neutrons existed, reactor control would not be possible due to the rapid power changes.

Reactor Theory (Neutron Characteristics) DOE-HDBK-1019/1-93 NEUTRON FLUX SPECTRUM
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Figure 2 Prompt Fission Neutron Energy Spectrum for
Thermal Fission of Uranium-235

NEUTRON FLUX SPECTRUM

The number of neutrons that exist at a given energy level varies. A plot of either the fraction of neutrons or the neutron flux at a given energy versus the energy level is called a neutron energy spectrum. The neutron energy spectrum varies widely for different types of reactors.

EO 4.1 STATE the average energy at which prompt neutrons are produced.

EO 4.2 DESCRIBE the neutron energy spectrum in the following reactors:

b. Thermal reactor

EO 4.3 EXPLAIN the reason for the particular shape of the fast, intermediate, and slow energy regions of the neutron flux spectrum for a thermal reactor.

Prompt Neutron Energies

The neutrons produced by fission are high energy neutrons, and almost all fission neutrons have energies between 0.1 MeV and 10 MeV. The neutron energy distribution, or spectrum, may best be described by plotting the fraction of neutrons per MeV as a function of neutron energy, as shown in Figure 2. From this figure it can be seen that the most probable neutron energy is about 0.7 MeV. Also, from this data it can be shown that the average energy of fission neutrons is about 2 MeV. Figure 2 is the neutron energy spectrum for thermal fission in uranium-235. The values will vary slightly for other nuclides.

NEUTRON FLUX SPECTRUM DOE-HDBK-1019/1-93 Reactor Theory (Neutron Characteristics)

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Figure 3 Comparison of Neutron Flux Spectra for Thermal and Fast Breeder Reactor

Thermal and Fast Breeder Reactor Neutron Spectra

The spectrum of neutron energies produced by fission varies significantly from the energy spectrum, or flux, existing in a reactor at a given time. Figure 3 illustrates the difference in neutron flux spectra between a thermal reactor and a fast breeder reactor. The energy distribution of neutrons from fission is essentially the same for both reactors, so the differences in the curve shapes may be attributed to the neutron moderation or slowing down effects.

No attempt is made to thermalize or slow down neutrons in the fast breeder reactor (liquid metal cooled); therefore, an insignificant number of neutrons exist in the thermal range. For the thermal reactor (water moderated), the spectrum of neutrons in the fast region (> 0.1 MeV) has a shape similar to that for the spectrum of neutrons emitted by the fission process.

In the thermal reactor, the flux in the intermediate energy region (1 eV to 0.1 MeV) has approximately a 1/E dependence. That is, if the energy (E) is halved, the flux doubles. This 1/E dependence is caused by the slowing down process, where elastic collisions remove a constant fraction of the neutron energy per collision (on the average), independent of energy; thus, the neutron loses larger amounts of energy per collision at higher energies than at lower energies. The fact that the neutrons lose a constant fraction of energy per collision causes the neutrons to tend to "pile up" at lower energies, that is, a greater number of neutrons exist at the lower energies as a result of this behavior.

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2 k T
m
vp
2 k T
m
2 1.38 x 10 16 erg
K
293 K
1.66 x 10 24 g
2.2 x 105 cm
sec
1 m
100 cm
2200 m
sec
Reactor Theory (Neutron Characteristics) DOE-HDBK-1019/1-93 NEUTRON FLUX SPECTRUM
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In the thermal region the neutrons achieve a thermal equilibrium with the atoms of the moderator material. In any given collision they may gain or lose energy, and over successive collisions will gain as much energy as they lose. These thermal neutrons, even at a specific temperature, do not all have the same energy or velocity; there is a distribution of energies, usually referred to as the Maxwell distribution (e.g., Figure 2). The energies of most thermal neutrons lie close to the most probable energy, but there is a spread of neutrons above and below this value.
Most Probable Neutron Velocities
The most probable velocity (vp) of a thermal neutron is determined by the temperature of the medium and can be determined by Equation (2-13) .
(2-13)
where:
v = most probable velocity of neutron (cm/sec) p
k = Boltzman's constant (1.38 x 10 erg/ K) -16
T = absolute temperature in degrees Kelvin (K)
m = mass of neutron (1.66 x 10 grams) -24
Example:
Fixture checkpoint marker-0300-xray is recorded on fixture page 0300.
Calculate the most probable velocities for neutrons in thermal equilibrium with their surroundings at the following temperatures. a) 20 C, b) 260 C.
Solution:
a) Calculate the most probable velocity for 20 C using Equation (2-13).
vp
2 k T
m
2 1.38 x 10 16 erg
K
533 K
1.66 x 10 24 g
2.977 x 105 cm
sec
1 m

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DOE-HDBK-1019/1-93
NUCLEAR PHYSICS AND REACTOR THEORY
ABSTRACT
The Nuclear Physics and Reactor Theory Handbook was developed to assist nuclear facility operating contractors in providing operators, maintenance personnel, and the technical staff with the necessary fundamentals training to ensure a basic understanding of nuclear physics and reactor theory. The handbook includes information on atomic and nuclear physics; neutron
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characteristics; reactor theory and nuclear parameters; and the theory of reactor operation. This information will provide personnel with a foundation for understanding the scientific principles that are associated with various DOE nuclear facility operations and maintenance.
Key Words: Training Material, Atomic Physics, The Chart of the Nuclides, Radioactivity, Radioactive Decay, Neutron Interaction, Fission, Reactor Theory, Neutron Characteristics, Neutron Life Cycle, Reactor Kinetics
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NUCLEAR PHYSICS AND REACTOR THEORY
FOREWORD
The Department of Energy (DOE) Fundamentals Handbooks consist of ten academic subjects, which include Mathematics; Classical Physics; Thermodynamics, Heat Transfer, and Fluid Flow; Instrumentation and Control; Electrical Science; Material Science; Mechanical Science; Chemistry; Engineering Symbology, Prints, and Drawings; and Nuclear Physics and Reactor Theory. The handbooks are provided as an aid to DOE nuclear facility contractors.
These handbooks were first published as Reactor Operator Fundamentals Manuals in 1985 for use by DOE category A reactors. The subject areas, subject matter content, and level of detail of the Reactor Operator Fundamentals Manuals were determined from several sources. DOE Category A reactor training managers determined which materials should be included, and served as a primary reference in the initial development phase. Training guidelines from the commercial nuclear power industry, results of job and task analyses, and independent input from contractors and operations-oriented personnel were all considered and included to some degree in developing the text material and learning objectives.
The DOE Fundamentals Handbooks represent the needs of various DOE nuclear facilities' fundamental training requirements. To increase their applicability to nonreactor nuclear facilities, the Reactor Operator Fundamentals Manual learning objectives were distributed to the Nuclear Facility Training Coordination Program Steering Committee for review and comment. To update their reactor-specific content, DOE Category A reactor training managers also reviewed and commented on the content. On the basis of feedback from these sources, information that applied to two or more DOE nuclear facilities was considered generic and was included. The final draft of each of the handbooks was then reviewed by these two groups. This approach has resulted in revised modular handbooks that contain sufficient detail such that each facility may adjust the content to fit their specific needs.
Each handbook contains an abstract, a foreword, an overview, learning objectives, and text material, and is divided into modules so that content and order may be modified by individual DOE contractors to suit their specific training needs. Each handbook is supported by a separate examination bank with an answer key.

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Nuclear Energy, Office of Nuclear Safety Policy and Standards, by the DOE Training Coordination Program. This program is managed by EG&G Idaho, Inc.
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NUCLEAR PHYSICS AND REACTOR THEORY
OVERVIEW
The Department of Energy Fundamentals Handbook entitled Nuclear Physics and Reactor Theory was prepared as an information resource for personnel who are responsible for the operation of the Department's nuclear facilities. Almost all processes that take place in a nuclear facility involves the transfer of some type of energy. A basic understanding of nuclear physics and reactor theory is necessary for DOE nuclear facility operators, maintenance personnel, and the technical staff to safely operate and maintain the facility and facility support systems. The information in this handbook is presented to provide a foundation for applying engineering concepts to the job. This knowledge will help personnel understand the impact that their actions may have on the safe and reliable operation of facility components and systems.
The Nuclear Physics and Reactor Theory handbook consists of four modules that are contained in two volumes. The following is a brief description of the information presented in Fixture checkpoint marker-0302-xray is recorded on fixture page 0302.
each module of the handbook.
Volume 1 of 2
Module 1 - Atomic and Nuclear Physics
Introduces concepts of atomic physics including the atomic nature of matter, the chart of the nuclides, radioactivity and radioactive decay, neutron interactions and fission, and the interaction of radiation with matter.
Module 2 - Reactor Theory (Nuclear Parameters)
Provides information on reactor theory and neutron characteristics. Includes topics such as neutron sources, neutron flux, neutron cross sections, reaction rates, neutron moderation, and prompt and delayed neutrons.
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Volume 2 of 2
Module 3 - Reactor Theory (Nuclear Parameters)
Explains the nuclear parameters associated with reactor theory. Topics include the neutron life cycle, reactivity and reactivity coefficients, neutron poisons, and control rods.
Module 4 - Reactor Theory (Reactor Operations)
Introduces the reactor operations aspect of reactor theory. Topics include subcritical multiplication, reactor kinetics, and reactor operation.
The information contained in this handbook is not all-encompassing. An attempt to present the entire subject of nuclear physics and reactor theory would be impractical. However, the Nuclear Physics and Reactor Theory handbook presents enough information to provide the reader with the fundamental knowledge necessary to understand the advanced theoretical concepts presented in other subject areas, and to understand basic system and equipment operation.
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Atomic and Nuclear Physics DOE-HDBK-1019/1-93 OBJECTIVES
TERMINAL OBJECTIVE
1.0 Given sufficient information, DESCRIBE atoms, including components, structure, and nomenclature.
ENABLING OBJECTIVES
1.1 STATE the characteristics of the following atomic particles, including mass, charge, and location within the atom:
a. Proton
b. Neutron
c. Electron
Fixture checkpoint marker-0306-xray is recorded on fixture page 0306.
1.2 DESCRIBE the Bohr model of an atom.
1.3 DEFINE the following terms:
a. Nuclide c. Atomic number
b. Isotope d. Mass number
1.4 Given the standard A
ZX notation for a particular nuclide, DETERMINE the following:
a. Number of protons
b. Number of neutrons
c. Number of electrons
1.5 DESCRIBE the three forces that act on particles within the nucleus and affect the stability of the nucleus.
1.6 DEFINE the following terms:
a. Enriched uranium
b. Depleted uranium
1.7 DEFINE the following terms:
a. Mass defect
b. Binding energy
1.8 Given the atomic mass for a nuclide and the atomic masses of a neutron, proton, and electron, CALCULATE the mass defect and binding energy of the nuclide.
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OBJECTIVES DOE-HDBK-1019/1-93 Atomic and Nuclear Physics
TERMINAL OBJECTIVE
2.0 Given necessary references, DESCRIBE the various modes of radioactive decay.
ENABLING OBJECTIVES
2.1 DESCRIBE the following processes:
a. Alpha decay d. Electron capture
b. Beta-minus decay e. Internal conversions
c. Beta-plus decay f. Isomeric transitions
2.2 Given a Chart of the Nuclides, WRITE the radioactive decay chain for a nuclide.
2.3 EXPLAIN why one or more gamma rays typically accompany particle emission.
2.4 Given the stability curve on the Chart of the Nuclides, DETERMINE the type of radioactive decay that the nuclides in each region of the chart will typically undergo.

a. Radioactivity d. Radioactive decay constant
b. Curie e. Radioactive half-life
c. Becquerel
2.6 Given the number of atoms and either the half-life or decay constant of a nuclide, CALCULATE the activity.
2.7 Given the initial activity and the decay constant of a nuclide, CALCULATE the activity at any later time.
2.8 CONVERT between the half-life and decay constant for a nuclide.
2.9 Given the Chart of the Nuclides and the original activity, PLOT the radioactive decay curve for a nuclide on either linear or semi-log coordinates.
2.10 DEFINE the following terms:
a. Radioactive equilibrium
b. Transient radioactive equilibrium
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Atomic and Nuclear Physics DOE-HDBK-1019/1-93 OBJECTIVES
TERMINAL OBJECTIVE
3.0 Without references, DESCRIBE the different nuclear interactions initiated by neutrons.
ENABLING OBJECTIVES
3.1 DESCRIBE the following scattering interactions between a neutron and a nucleus:
a. Elastic scattering
b. Inelastic scattering
3.2 STATE the conservation laws that apply to an elastic collision between a neutron and a
Fixture checkpoint marker-0307-xray is recorded on fixture page 0307.
nucleus.
3.3 DESCRIBE the following reactions where a neutron is absorbed in a nucleus:
a. Radiative capture
b. Particle ejection
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OBJECTIVES DOE-HDBK-1019/1-93 Atomic and Nuclear Physics
TERMINAL OBJECTIVE
4.0 Without references, DESCRIBE the fission process.
ENABLING OBJECTIVES
4.1 EXPLAIN the fission process using the liquid drop model of a nucleus.
4.2 DEFINE the following terms:
a. Excitation energy
b. Critical energy
4.3 DEFINE the following terms:
a. Fissile material
b. Fissionable material
c. Fertile material
4.4 DESCRIBE the processes of transmutation, conversion, and breeding.
4.5 DESCRIBE the curve of Binding Energy per Nucleon versus mass number and give a qualitative description of the reasons for its shape.
4.6 EXPLAIN why only the heaviest nuclei are easily fissioned.
4.7 EXPLAIN why uranium-235 fissions with thermal neutrons and uranium-238 fissions only with fast neutrons.
4.8 CHARACTERIZE the fission products in terms of mass groupings and radioactivity.
4.9 Given the nuclides involved and their masses, CALCULATE the energy released from

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4.10 Given the curve of Binding Energy per Nucleon versus mass number, CALCULATE the energy released from fission.
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Atomic and Nuclear Physics DOE-HDBK-1019/1-93 OBJECTIVES
TERMINAL OBJECTIVE
5.0 Without references, DESCRIBE how the various types of radiation interact with matter.
ENABLING OBJECTIVES
5.1 DESCRIBE interactions of the following with matter:
a. Alpha particle c. Positron
b. Beta particle d. Neutron
5.2 DESCRIBE the following ways that gamma radiation interacts with matter:
a. Compton scattering
b. Photoelectric effect
c. Pair production
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OBJECTIVES DOE-HDBK-1019/1-93 Atomic and Nuclear Physics
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NP-01 Page xii Rev. 0
Atomic and Nuclear Physics DOE-HDBK-1019/1-93 ATOMIC NATURE OF MATTER
ATOMIC NATURE OF MATTER
All matter is composed of atoms. The atom is the smallest amount of matter that retains the properties of an element. Atoms themselves are composed of smaller particles, but these smaller particles no longer have the same properties as the overall element.
EO 1.1 STATE the characteristics of the following atomic particles, including mass, charge, and location within the atom:
a. Proton
b. Neutron
c. Electron
Fixture checkpoint marker-0308-xray is recorded on fixture page 0308.
EO 1.2 DESCRIBE the Bohr model of an atom.
EO 1.3 DEFINE the following terms:
a. Nuclide c. Atomic number
b. Isotope d. Mass number
EO 1.4 Given the standard A ZX notation for a particular nuclide, DETERMINE the following:
a. Number of protons
b. Number of neutrons
c. Number of electrons
EO 1.5 DESCRIBE the three forces that act on particles within the nucleus and affect the stability of the nucleus.
Structure of Matter
Early Greek philosophers speculated that the earth was made up of different combinations of basic substances, or elements. They considered these basic elements to be earth, air, water, and fire. Modern science shows that the early Greeks held the correct concept that matter consists of a combination of basic elements, but they incorrectly identified the elements.
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In 1661 the English chemist Robert Boyle published the modern criterion for an element. He defined an element to be a basic substance that cannot be broken down into any simpler substance after it is isolated from a compound, but can be combined with other elements to form compounds. To date, 105 different elements have been confirmed to exist, and researchers claim to have discovered three additional elements. Of the 105 confirmed elements, 90 exist in nature and 15 are man-made.

Another basic concept of matter that the Greeks debated was whether matter was continuous or discrete. That is, whether matter could be continuously divided and subdivided into ever smaller particles or whether eventually an indivisible particle would be encountered. Democritus in about 450 B.C. argued that substances were ultimately composed of small, indivisible particles that he labeled atoms. He further suggested that different substances were composed of different atoms or combinations of atoms, and that one substance could be converted into another by rearranging the atoms. It was impossible to conclusively prove or disprove this proposal for more than 2000 years.

The modern proof for the atomic nature of matter was first proposed by the English chemist John Dalton in 1803. Dalton stated that each chemical element possesses a particular kind of atom, and any quantity of the element is made up of identical atoms of this kind. What distinguishes one element from another element is the kind of atom of which it consists, and the basic physical difference between kinds of atoms is their weight.

Subatomic Particles

For almost 100 years after Dalton established the atomic nature of atoms, it was considered impossible to divide the atom into even smaller parts. All of the results of chemical experiments during this time indicated that the atom was indivisible. Eventually, experimentation into electricity and radioactivity indicated that particles of matter smaller than the atom did indeed exist. In 1906, J. J. Thompson won the Nobel Prize in physics for establishing the existence of electrons. Electrons are negatively-charged particles that have 1/1835 the mass of the hydrogen atom. Soon after the discovery of electrons, protons were discovered. Protons are relatively large particles that have almost the same mass as a hydrogen atom and a positive charge equal in magnitude (but opposite in sign) to that of the electron. The third subatomic particle to be discovered, the neutron, was not found until 1932. The neutron has almost the same mass as the proton, but it is electrically neutral.

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Atomic and Nuclear Physics DOE-HDBK-1019/1-93 ATOMIC NATURE OF MATTER

Bohr Model of the Atom

The British physicist Ernest Rutherford postulated that the positive charge in an atom is concentrated in a small region called a nucleus at the center of the atom with electrons existing

Fixture checkpoint marker-0309-xray is recorded on fixture page 0309.

in orbits around it. Niels Bohr, coupling Rutherford's postulation with the quantum theory introduced by Max Planck, proposed that the atom consists of a dense nucleus of protons surrounded by electrons traveling in discrete orbits at fixed distances from the nucleus. An electron in one of these orbits or shells has a specific or discrete quantity of energy (quantum). When an electron moves from one allowed orbit to another allowed orbit, the energy difference between the two states is emitted or absorbed in the form of a single quantum of radiant energy called a photon. Figure 1 is Bohr's model of the hydrogen atom showing an electron as having just dropped from the third shell to the first shell with the emission of a photon that has an energy = hv. (h = Planck's constant = 6.63 x 10-34 J-s and v = frequency of the photon.) Bohr's theory was the first to successfully account for the discrete energy levels of this radiation as measured in the laboratory. Although Bohr's atomic model is designed specifically to explain

information on electron shell theory can be found in the Chemistry Fundamentals Handbook.
Figure 1 Bohr's Model of the Hydrogen Atom

Properties of the three subatomic particles are listed in Table 1.
TABLE 1
Properties of Subatomic Particles
Particle Location Charge Mass
Neutron Nucleus none 1.008665 amu
Proton Nucleus +1 1.007277 amu
Electron Shells around nucleus -1 0.0005486 amu
Measuring Units on the Atomic Scale
The size and mass of atoms are so small that the use of normal measuring units, while possible,
Fixture checkpoint marker-0310-xray is recorded on fixture page 0310.
is often inconvenient. Units of measure have been defined for mass and energy on the atomic scale to make measurements more convenient to express. The unit of measure for mass is the atomic mass unit (amu). One atomic mass unit is equal to 1.66 x 10-24 grams. The reason for this particular value for the atomic mass unit will be discussed in a later chapter. Note from Table 1 that the mass of a neutron and a proton are both about 1 amu. The unit for energy is the electron volt (eV). The electron volt is the amount of energy acquired by a single electron when it falls through a potential difference of one volt. One electron volt is equivalent to 1.602 x 10-19 joules or 1.18 x 10-19 foot-pounds.
Nuclides
The total number of protons in the nucleus of an atom is called the atomic number of the atom and is given the symbol Z. The number of electrons in an electrically-neutral atom is the same as the number of protons in the nucleus. The number of neutrons in a nucleus is known as the neutron number and is given the symbol N. The mass number of the nucleus is the total number of nucleons, that is, protons and neutrons in the nucleus. The mass number is given the symbol A and can be found by the equation Z + N = A.
Each of the chemical elements has a unique atomic number because the atoms of different elements contain a different number of protons. The atomic number of an atom identifies the particular element.

Each type of atom that contains a unique combination of
Figure 2 Nomenclature for
Identifying Nuclides
protons and neutrons is called a nuclide. Not all combinations of numbers of protons and neutrons are possible, but about 2500 specific nuclides with unique combinations of neutrons and protons have been identified. Each nuclide is denoted by the chemical symbol of the element with the atomic number written as a subscript and the mass number written as a superscript, as shown in Figure 2. Because each element has a unique name, chemical symbol, and atomic number, only one of the three is necessary to identify the element. For this reason nuclides can also be identified by either the

mass number (for example, U-235 or uranium-235).
Another common format is to use the abbreviation of the
chemical element with the mass number superscripted (for example, 235U). In this handbook the format used in the text will usually be the element's name followed by the mass number. In equations and tables, the format in Figure 2 will usually be used.
Example:
State the name of the element and the number of protons, electrons, and neutrons in the nuclides listed below.
1
1H
10
5B
14
7N
11
4
4
8Cd
23
9
Fixture checkpoint marker-0311-xray is recorded on fixture page 0311.
9
4Pu
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ATOMIC NATURE OF MATTER DOE-HDBK-1019/1-93 Atomic and Nuclear Physics
Solution:
The name of the element can be found from the Periodic Table (refer to Chemistry Fundamentals Handbook) or the Chart of the Nuclides (to be discussed later). The number of protons and electrons are equal to Z. The number of neutrons is equal to Z - A.
Nuclide Element Protons Electrons Neutrons
1
1H hydrogen 1 1 0
10
5B boron 5 5 5
14
7N nitrogen 7 7 7
11
4
4
8Cd cadmium 48 48 66
23
9
9
4Pu plutonium 94 94 145
Isotopes
Isotopes are nuclides that have the same atomic number and are therefore the same element, but differ in the number of neutrons. Most elements have a few stable isotopes and several unstable,

(oxygen-16, oxygen-17, and oxygen-18) and eight radioactive isotopes. Another example is hydrogen, which has two stable isotopes (hydrogen-1 and hydrogen-2) and a single radioactive isotope (hydrogen-3).

The isotopes of hydrogen are unique in that they are each commonly referred to by a unique name instead of the common chemical element name. Hydrogen-1 is almost always referred to as hydrogen, but the term protium is infrequently used also. Hydrogen-2 is commonly called deuterium and symbolized 2

1D. Hydrogen-3 is commonly called tritium and symbolized 31T. This

text will normally use the symbology 2

1H and 31H for deuterium and tritium, respectively.

Atomic and Nuclear Radii

The size of an atom is difficult to define exactly due to the fact that the electron cloud, formed by the electrons moving in their various orbitals, does not have a distinct outer edge. A reasonable measure of atomic size is given by the average distance of the outermost electron from the nucleus. Except for a few of the lightest atoms, the average atomic radii are approximately the same for all atoms, about 2 x 10 -8 cm.

Like the atom the nucleus does not have a sharp outer boundary. Experiments have shown that the nucleus is shaped like a sphere with a radius that depends on the atomic mass number of the atom. The relationship between the atomic mass number and the radius of the nucleus is shown in the following equation.

r = (1.25 x 10 -13 cm) A1/3

where:

r = radius of the nucleus (cm)

A = atomic mass number (dimensionless)

The values of the nuclear radii for some light, intermediate, and heavy nuclides are shown in

Fixture checkpoint marker-0312-xray is recorded on fixture page 0312.

Table 2.

TABLE 2

Calculated Values for Nuclear

Radii

Nuclide Radius of Nucleus

1

1H 1.25 x 10 -13 cm

10

5B 2.69 x 10 -13 cm

5

2

6

6Fe 4.78 x 10 -13 cm

17

7

8

2Hf 7.01 x 10 -13 cm

23

9

8

SYNTHETIC FIXTURE PAGE 0313 OF 1200

25
9
2
8Cf 7.89 x 10 -13 cm
From the table, it is clear that the radius of a typical atom (e.g. 2 x 10 -8 cm) is more than 25,000 times larger than the radius of the largest nucleus.
Nuclear Forces
In the Bohr model of the atom, the nucleus consists of positively-charged protons and electrically■neutra as nucleons. One problem that the Bohr model of the atom presented was accounting for an attractive force to overcome the repulsive force between protons.
Two forces present in the nucleus are (1) electrostatic forces between charged particles and (2) gravitational forces between any two objects that have mass. It is possible to calculate the magnitude of the gravitational force and electrostatic force based upon principles from classical physics.
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Fg
G m1 m2
r
2
Fe
K Q1 Q2
r
2
ATOMIC NATURE OF MATTER DOE-HDBK-1019/1-93 Atomic and Nuclear Physics
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Newton stated that the gravitational force between two bodies is directly proportional to the masses of the two bodies and inversely proportional to the square of the distance between the bodies. This relationship is shown in the equation below.
where:
F = gravitational force (newtons) g
m = mass of first body (kilograms) 1
m = mass of second body (kilograms) 2
G = gravitational constant (6.67 x 10 N-m /kg)
-11 2 2
Fixture checkpoint marker-0313-xray is recorded on fixture page 0313.
r = distance between particles (meters)
The equation illustrates that the larger the masses of the objects or the smaller the distance between the objects, the greater the gravitational force. So even though the masses of nucleons are very small, the fact that the distance between nucleons is extremely short may make the gravitational force significant. It is necessary to calculate the value for the gravitational force and compare it to the value for other forces to determine the significance of the gravitational force in the nucleus. The gravitational force between two protons that are separated by a distance of 10 meters is about 10 newtons. -20 -24
Coulomb's Law can be used to calculate the force between two protons. The electrostatic force is directly proportional to the electrical charges of the two particles and inversely proportional to the square of the distance between the particles. Coulomb's Law is stated as the following equation.
where:

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K = electrostatic constant (9.0 x 10 N-m /C)
9 2 2
Q = charge of first particle (coulombs) 1
Q = charge of second particle (coulombs) 2
r = distance between particles (meters)
Using this equation, the electrostatic force between two protons that are separated by a distance of 10 meters is about 10 newtons. Comparing this result with the calculation of the
-20 12
gravitational force (10 newtons) shows that the gravitational force is so small that it can be
-24
neglected.
Fixture checkpoint marker-0314-xray is recorded on fixture page 0314.
Atomic and Nuclear Physics DOE-HDBK-1019/1-93 ATOMIC NATURE OF MATTER
If only the electrostatic and gravitational forces existed in the nucleus, then it would be impossible to have stable nuclei composed of protons and neutrons. The gravitational forces are much too small to hold the nucleons together compared to the electrostatic forces repelling the protons. Since stable atoms of neutrons and protons do exist, there must be another attractive force acting within the nucleus. This force is called the nuclear force.
The nuclear force is a strong attractive force that is independent of charge. It acts equally only between pairs of neutrons, pairs of protons, or a neutron and a proton. The nuclear force has a very short range; it acts only over distances approximately equal to the diameter of the nucleus (10 -13 cm). The attractive nuclear force between all nucleons drops off with distance much faster than the repulsive electrostatic force between protons.
TABLE 3
Forces Acting in the Nucleus
Force Interaction Range
Gravitational Very weak attractive force between all nucleons Relatively long
Electrostatic Strong repulsive force between like charged particles (protons) Relatively long
Nuclear Force Strong attractive force between all nucleons Extremely short
In stable atoms, the attractive and repulsive forces in the nucleus balance. If the forces do not balance, the atom cannot be stable, and the nucleus will emit radiation in an attempt to achieve a more stable configuration.
Summary
The important information in this chapter is summarized on the following page.
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ATOMIC NATURE OF MATTER DOE-HDBK-1019/1-93 Atomic and Nuclear Physics
Atomic Nature of Matter Summary
Atoms consist of three basic subatomic particles. These particles are the proton, the neutron, and the electron.
Protons are particles that have a positive charge, have about the same mass as a hydrogen atom, and exist in the nucleus of an atom.
Neutrons are particles that have no electrical charge, have about the same mass as a hydrogen atom, and exist in the nucleus of an atom.
Electrons are particles that have a negative charge, have a mass about eighteen hundred times smaller than the mass of a hydrogen atom, and exist in orbital shells

SYNTHETIC FIXTURE PAGE 0315 OF 1200

The Bohr model of the atom consists of a dense nucleus of protons and neutrons (nucleons) surrounded by electrons traveling in discrete orbits at fixed distances from the nucleus.
Nuclides are atoms that contain a particular number of protons and neutrons.
Isotopes are nuclides that have the same atomic number and are therefore the same element, but differ in the number of neutrons.
The atomic number of an atom is the number of protons in the nucleus.
The mass number of an atom is the total number of nucleons (protons and neutrons) in the nucleus.
The notation A
ZX is used to identify a specific nuclide. "Z" represents the atomic number, which is equal to the number of protons. "A" represents the mass number, which is equal to the number of nucleons. "X" represents the chemical symbol of the element.
Number of protons = Z
Number of electrons = Z
Number of neutrons = A - Z
The stability of a nucleus is determined by the different forces interacting within
Fixture checkpoint marker-0315-xray is recorded on fixture page 0315.
it. The electrostatic force is a relatively long-range, strong, repulsive force that acts between the positively charged protons. The nuclear force is a relatively short-range attractive force between all nucleons. The gravitational force the long range, relatively weak attraction between masses, is negligible compared to the other forces.
NP-01 Page 10 Rev. 0
Atomic and Nuclear Physics DOE-HDBK-1019/1-93 CHART OF THE NUCLIDES
CHART OF THE NUCLIDES
The Chart of the Nuclides, like the Periodic Table, is a convenient format for presenting a large amount of scientific information in an organized manner.
EO 1.6 DEFINE the following terms:
a. Enriched uranium
b. Depleted uranium
Chart of the Nuclides
A tabulated chart called the Chart of the Nuclides lists the stable and unstable nuclides in addition to pertinent information about each one. Figure 3 shows a small portion of a typical chart. This chart plots a box for each individual nuclide, with the number of protons (Z) on the vertical axis and the number of neutrons (N = A - Z) on the horizontal axis.
The completely gray squares indicate stable isotopes. Those in white squares are artificially radioactive, meaning that they are produced by artificial techniques and do not occur naturally. By consulting a complete chart, other types of isotopes can be found, such as naturally occurring radioactive types (but none are found in the region of the chart that is illustrated in Figure 3).
Located in the box on the far left of each horizontal row is general information about the element. The box contains the chemical symbol of the element in addition to the average atomic weight of the naturally occurring substance and the average thermal neutron absorption cross section, which will be discussed in a later module. The known isotopes (elements with the same atomic number Z but different mass number A) of each element are listed to the right.
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Figure 3 Nuclide Chart for Atomic Numbers 1 to 6
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Atomic and Nuclear Physics DOE-HDBK-1019/1-93 CHART OF THE NUCLIDES
Information for Stable Nuclides
For the stable isotopes, in addition to the symbol and the atomic mass number, the number percentage of each isotope in the naturally occurring element is listed, as well as the thermal neutron activation cross section and the mass in atomic mass units (amu). A typical block for a stable nuclide from the Chart of the Nuclides is shown in Figure 4.
Figure 4 Stable Nuclides
Information for Unstable Nuclides
For unstable isotopes the additional information includes the half life, the mode of decay (for example, β^-, α), the total disintegration energy in MeV (million electron volts), and the mass in amu when available. A typical block for an unstable nuclide from the Chart of the Nuclides is shown in Figure 5.
Figure 5 Unstable Nuclides
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CHART OF THE NUCLIDES DOE-HDBK-1019/1-93 Atomic and Nuclear Physics
Neutron - Proton Ratios
Figure 6 shows the distribution of the stable nuclides plotted on the same axes as the Chart of the Nuclides. As the mass numbers become higher, the ratio of neutrons to protons in the nucleus becomes larger. For helium-4 (2 protons and 2 neutrons) and oxygen-16 (8 protons and 8 neutrons) this ratio is unity. For indium-115 (49 protons and 66 neutrons) the ratio of neutrons to protons has increased to 1.35, and for uranium-238 (92 protons and 146 neutrons) the neutron■to-proton
Figure 6 Neutron - Proton Plot of the Stable Nuclides
Fixture checkpoint marker-0316-xray is recorded on fixture page 0316.
If a heavy nucleus were to split into two fragments, each fragment would form a nucleus that would have approximately the same neutron-to-proton ratio as the heavy nucleus. This high neutron-to-proton ratio places the fragments below and to the right of the stability curve displayed by Figure 6. The instability caused by this excess of neutrons is generally rectified by successive beta emissions, each of which converts a neutron to a proton and moves the nucleus toward a more stable neutron-to-proton ratio.
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■
Atomic and Nuclear Physics DOE-HDBK-1019/1-93 CHART OF THE NUCLIDES
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Natural Abundance of Isotopes
The relative abundance of an isotope in nature compared to other isotopes of the same element is relatively constant. The Chart of the Nuclides presents the relative abundance of the naturally occurring isotopes of an element in units of atom percent. Atom percent is the percentage of the atoms of an element that are of a particular isotope. Atom percent is abbreviated as a/o.
For example, if a cup of water contains 8.23 x 10 atoms of oxygen, and the isotopic abundance 24
of oxygen-18 is 0.20%, then there are 1.65 x 10 atoms of oxygen-18 in the cup. 22
The atomic weight for an element is defined as the average atomic weight of the isotopes of the element. The atomic weight for an element can be calculated by summing the products of the isotopic abundance of the isotope with the atomic mass of the isotope.
Example:
Calculate the atomic weight for the element lithium. Lithium-6 has an atom percent

abundance of 92.5% and an atomic mass of 7.016003 amu.
Solution:
The other common measurement of isotopic abundance is weight percent (w/o). Weight percent is the percent weight of an element that is a particular isotope. For example, if a sample of material contained 100 kg of uranium that was 28 w/o uranium-235, then 28 kg of uranium-235 was present in the sample.
Enriched and Depleted Uranium
Natural uranium mined from the earth contains the isotopes uranium-238, uranium-235 and uranium-234. The majority (99.2745%) of all the atoms in natural uranium are uranium-238. Most of the remaining atoms (0.72%) are uranium-235, and a slight trace (0.0055%) are uranium-234. Although all isotopes of uranium have similar chemical properties, each of the isotopes has significantly different nuclear properties. For reasons that will be discussed in later modules, the isotope uranium-235 is usually the desired material for use in reactors.
A vast amount of equipment and energy are expended in processes that separate the isotopes of uranium (and other elements). The details of these processes are beyond the scope of this module. These processes are called enrichment processes because they selectively increase the proportion of a particular isotope. The enrichment process typically starts with feed material that has the proportion of isotopes that occur naturally. The process results in two types of
CHART OF THE NUCLIDES DOE-HDBK-1019/1-93 Atomic and Nuclear Physics
In the case of uranium, the natural uranium ore is 0.72 a/o uranium-235. The desired outcome of the enrichment process is to produce enriched uranium. Enriched uranium is defined as uranium in which the isotope uranium-235 has a concentration greater than its natural value. The enrichment process will also result in the byproduct of depleted uranium. Depleted uranium is defined as uranium in which the isotope uranium-235 has a concentration less than its natural value. Although depleted uranium is referred to as a by-product of the enrichment process, it does have uses in the nuclear field and in commercial and defense industries.
Summary
The important information in this chapter is summarized below.
Chart of the Nuclides Summary
Enriched uranium is uranium in which the isotope uranium-235 has a concentration greater than its natural value of 0.7%.
Depleted uranium is uranium in which the isotope uranium-235 has a
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concentration less than its natural value of 0.7%.
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Atomic and Nuclear Physics DOE-HDBK-1019/1-93 MASS DEFECT AND BINDING ENERGY
MASS DEFECT AND BINDING ENERGY
The separate laws of Conservation of Mass and Conservation of Energy are not applied strictly on the nuclear level. It is possible to convert between mass and energy. Instead of two separate conservation laws, a single conservation law states that the sum of mass and energy is conserved. Mass does not magically appear and disappear at random. A decrease in mass will be accompanied by a corresponding increase in energy and vice versa.
EO 1.7 DEFINE the following terms:
a. Mass defect
b. Binding energy
EO 1.8 Given the atomic mass for a nuclide and the atomic masses of a neutron, proton, and electron, CALCULATE the mass defect

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Mass Defect
Careful measurements have shown that the mass of a particular atom is always slightly less than the sum of the masses of the individual neutrons, protons, and electrons of which the atom consists. The difference between the mass of the atom and the sum of the masses of its parts is called the mass defect (Δm). The mass defect can be calculated using Equation (1-1). In calculating the mass defect it is important to use the full accuracy of mass measurements because the difference in mass is small compared to the mass of the atom. Rounding off the masses of atoms and particles to three or four significant digits prior to the calculation will result in a calculated mass defect of zero.
Δm = [Z(mp + me) + (A-Z)mn] - matom (1-1)
where:
Δm = mass defect (amu)
mp = mass of a proton (1.007277 amu)
mn = mass of a neutron (1.008665 amu)
me = mass of an electron (0.000548597 amu)
matom = mass of nuclide A
ZX (amu)
Z = atomic number (number of protons)
A = mass number (number of nucleons)
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MASS DEFECT AND BINDING ENERGY DOE-HDBK-1019/1-93 Atomic and Nuclear Physics
Example:
Calculate the mass defect for lithium-7. The mass of lithium-7 is 7.016003 amu.
Solution:
Δ m Z m
p
me A Z mn matom
Δ m 3 1.007826 amu 7 3 1.008665 amu 7.016003 amu
Δ m 0.0421335 amu
Binding Energy
The loss in mass, or mass defect, is due to the conversion of mass to binding energy when the nucleus is formed. Binding energy is defined as the amount of energy that must be supplied to a nucleus to completely separate its nuclear particles (nucleons). It can also be understood as the amount of energy that would be released if the nucleus was formed from the separate particles. Binding energy is the energy equivalent of the mass defect. Since the mass defect was converted to binding energy (BE) when the nucleus was formed, it is possible to calculate the binding energy using a conversion factor derived by the mass-energy relationship from Einstein's Theory of Relativity.
Einstein's famous equation relating mass and energy is E = mc2 where c is the velocity of light
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(c = 2.998 x 108 m/sec). The energy equivalent of 1 amu can be determined by inserting this quantity of mass into Einstein's equation and applying conversion factors.
E m c 2
1 amu
1.6606 x 10 27 kg
1 amu
2.998 x 108 m
sec

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1 N
1
kg m
sec2
1 J
1 N m
1.4924 x 10 10 J 1 MeV
1.6022 x 10 13 J
931.5 MeV
Conversion Factors:
1 amu = 1.6606 x 10 -27 kg
1 newton = 1 kg-m/sec2
1 joule = 1 newton-meter
1 MeV = 1.6022 x 10-13 joules
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Atomic and Nuclear Physics DOE-HDBK-1019/1-93 MASS DEFECT AND BINDING ENERGY
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Since 1 amu is equivalent to 931.5 MeV of energy, the binding energy can be calculated using Equation (1-2).
(1-2)
B.E. ∆m
931.5 MeV
1 amu
Example:
Calculate the mass defect and binding energy for uranium-235. One uranium-235 atom has a mass of 235.043924 amu.
Solution:
Step 1: Calculate the mass defect using Equation (1-1).
∆m Z m
p
me A Z mn matom
∆m 92 1.007826 amu 235 92 1.008665 amu 235.043924 amu
∆m 1.91517 amu
Step 2: Use the mass defect and Equation (1-2) to calculate the binding energy.
B.E. ∆m
931.5 MeV
1 amu
1.91517 amu
931.5 MeV
1 amu
1784 MeV
Energy Levels of Atoms
The electrons that circle the nucleus move in fairly well-defined orbits. Some of these electrons are more tightly bound in the atom than others. For example, only 7.38 eV is required to remove the outermost electron from a lead atom, while 88,000 eV is required to remove the innermost electron. The process of removing an electron from an atom is called ionization, and the energy required to remove the electron is called the ionization energy.
In a neutral atom (number of electrons = Z) it is possible for the electrons to be in a variety of

the atom is normally found and is called the ground state. When the atom possesses more energy than its ground state energy, it is said to be in an excited state.

An atom cannot stay in the excited state for an indefinite period of time. An excited atom will eventually transition to either a lower-energy excited state, or directly to its ground state, by emitting a discrete bundle of electromagnetic energy called an x-ray. The energy of the x-ray will be equal to the difference between the energy levels of the atom and will typically range from several eV to 100,000 eV in magnitude.

Energy Levels of the Nucleus

The nucleons in the nucleus of an atom, like the electrons that circle the nucleus, exist in shells that correspond to energy states. The energy shells of the nucleus are less defined and less understood than those of the electrons. There is a state of lowest energy (the ground state) and discrete possible excited states for a nucleus. Where the discrete energy states for the electrons of an atom are measured in eV or keV, the energy levels of the nucleus are considerably greater and typically measured in MeV.

A nucleus that is in the excited state will not remain at that energy level for an indefinite period. Like the electrons in an excited atom, the nucleons in an excited nucleus will transition towards their lowest energy configuration and in doing so emit a discrete bundle of electromagnetic radiation called a gamma ray (γ-ray). The only differences between x-rays and γ-rays are their energy levels and whether they are emitted from the electron shell or from the nucleus.

The ground state and the excited states of

Figure 7 Energy Level Diagram - Nickel-60

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a nucleus can be depicted in a nuclear energy-level diagram. The nuclear energy-level diagram consists of a stack of horizontal bars, one bar for each of the excited states of the nucleus. The vertical distance between the bar representing an excited state and the bar representing the ground state is proportional to the energy level of the excited state with respect to the ground state. This difference in energy between the ground state and the excited state is called the excitation energy of the excited state. The ground state of a nuclide has zero excitation energy. The bars for the excited states are labeled with their respective energy levels. Figure 7 is the energy level diagram for nickel-60.

Summary

The important information in this chapter is summarized below.

Mass Defect and Binding Energy Summary

Mass defect is the difference between the mass of the atom and the sum of the masses of its constituent parts.

completely separate its nuclear particles. Binding energy is the energy equivalent of the mass defect.

Mass defect can be calculated by using the equation below.

$\Delta m = [Z(mp + me) + (A-Z)mn] - matom$

Binding energy can be calculated by multiplying the mass defect by the factor of 931.5 MeV per amu.

MODES OF RADIOACTIVE DECAY

Most atoms found in nature are stable and do not emit particles or energy that change form over time. Some atoms, however, do not have stable nuclei. These atoms emit radiation in order to achieve a more stable configuration.

EO 2.1 DESCRIBE the following processes:

a. Alpha decay
b. Beta-minus decay
c. Beta-plus decay
d. Electron capture
e. Internal conversions
f. Isomeric transitions

EO 2.2 Given a Chart of the Nuclides, WRITE the radioactive decay chain for a nuclide.

EO 2.3 EXPLAIN why one or more gamma rays typically accompany particle emission.

EO 2.4 Given the stability curve on the Chart of the Nuclides, DETERMINE the type of radioactive decay that the nuclides in each region of the chart will typically undergo.

Stability of Nuclei

As mass numbers become larger, the ratio of neutrons to protons in the nucleus becomes larger for the stable nuclei. Non-stable nuclei may have an excess or deficiency of neutrons and undergo a transformation process known as beta (β) decay. Non-stable nuclei can also undergo a variety of other processes such as alpha (α) or neutron (n) decay. As a result of these decay processes, the final nucleus is in a more stable or more tightly bound configuration.

Natural Radioactivity

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In 1896, the French physicist Becquerel discovered that crystals of a uranium salt emitted rays that were similar to x-rays in that they were highly penetrating, could affect a photographic plate, and induced electrical conductivity in gases. Becquerel's discovery was followed in 1898 by the identification of two other radioactive elements, polonium and radium, by Pierre and Marie Curie.

Heavy elements, such as uranium or thorium, and their unstable decay chain elements emit radiation in their naturally occurring state. Uranium and thorium, present since their creation at the beginning of geological time, have an extremely slow rate of decay. All naturally occurring nuclides with atomic numbers greater than 82 are radioactive.

Nuclear Decay

Whenever a nucleus can attain a more stable (i.e., more tightly bound) configuration by emitting radiation, a spontaneous disintegration process known as radioactive decay or nuclear decay may occur. In practice, this "radiation" may be electromagnetic radiation, particles, or both.

Detailed studies of radioactive decay and nuclear reaction processes have led to the formulation of useful conservation principles. The four principles of most interest in this module are

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Conservation of electric charge implies that charges are neither created nor destroyed. Single positive and negative charges may, however, neutralize each other. It is also possible for a neutral particle to produce one charge of each sign.
Conservation of mass number does not allow a net change in the number of nucleons. However, the conversion of a proton to a neutron and vice versa is allowed.
Conservation of mass and energy implies that the total of the kinetic energy and the energy equivalent of the mass in a system must be conserved in all decays and reactions. Mass can be converted to energy and energy can be converted to mass, but the sum of mass and energy must be constant.
Conservation of momentum is responsible for the distribution of the available kinetic energy among product nuclei, particles, and/or radiation. The total amount is the same before and after the reaction even though it may be distributed differently among entirely different nuclides and/or particles.
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■ ■ ■ ■ ■ ■
■ ■ ■ ■ ■
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Alpha Decay (■)
Alpha decay is the emission of alpha particles (helium nuclei) which may be represented as either He or ■. When an unstable nucleus ejects an alpha particle, the atomic number is reduced by 2 4 4 2 2
and the mass number decreased by 4. An example is uranium-234 which decays by the ejection of an alpha particle accompanied by the emission of a 0.068 MeV gamma.
The combined kinetic energy of the daughter nucleus (Thorium-230) and the ■ particle is designated as KE. The sum of the KE and the gamma energy is equal to the difference in mass between the original nucleus (Uranium-234) and the final particles (equivalent to the binding energy released, since ■m = BE). The alpha particle will carry off as much as 98% of the kinetic energy and, in most cases, can be considered to carry off all the kinetic energy.
Beta Decay (■)
Beta decay is the emission of electrons of nuclear rather than orbital origin. These particles are electrons that have been expelled by excited nuclei and may have a charge of either sign.
If both energy and momentum are to be conserved, a third type of particle, the neutrino, ■, must be involved. The neutrino is associated with positive electron emission, and its antiparticle, the antineutrino, , is emitted with a negative electron. These uncharged particles have only the weakest interaction with matter, no mass, and travel at the speed of light. For all practical purposes, they pass through all materials with so few interactions that the energy they possess cannot be
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recovered. The neutrinos and antineutrinos are included here only because they carry a portion of the kinetic energy that would otherwise belong to the beta particle, and therefore, must be considered for energy and momentum to be conserved. They are normally ignored since they are not significant in the context of nuclear reactor applications.
Negative electron emission, represented as e, -1 -1■, or simply as e or ■ , effectively converts a neutro to a proton, thus increasing the atomic number by one and leaving the mass number unchanged. This is a common mode of decay for nuclei with an excess of neutrons, such as fission fragments below and to the right of the neutron-proton stability curve (refer to Figure 6). An example of a typical beta minus-decay reaction is shown below.
■ ■

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■ ■
■ ■ ■ ■

Figure 9 Orbital Electron Capture
Positively charged electrons (beta-plus) are known as positrons. Except for sign, they are nearly identical to their negatively charged cousins. When a positron, represented as e, ■, or simply as +1 +1
0 0
e or ■ , is ejected from the nucleus, the atomic number is decreased by one and the mass number + +
remains unchanged. A proton has been converted to a neutron. An example of a typical positron (beta-plus) decay is shown below.
Electron Capture (EC, K-capture)
Nuclei having an excess of protons may capture an electron from one of the inner orbits which immediately combines with a proton in the nucleus to form a neutron. This process is called electron
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capture (EC). The electron is normally captured from the innermost orbit (the K-shell), and, consequently, this process is sometimes called K-capture. The following example depicts electron capture.
A neutrino is formed at the same time that the neutron is formed, and energy carried off by it serves to conserve momentum. Any energy that is available due to the atomic mass of the product being appreciably less than that of the parent will appear as gamma radiation. Also, there will always be characteristic x-rays given off when an electron from one of the higher energy shells moves in to fill the vacancy in the K-shell.
Electron capture is shown graphically in Figure 8.
Electron capture and positron emission result in the production of the same daughter product, and they exist as competing processes.
For positron emission to occur, however, the mass of the daughter product must be less than the mass of the parent by an amount equal to at least twice the mass of an electron. This mass difference between the parent and daughter is necessary to account for two items present in the parent but not in the daughter. One item is the positron ejected from the nucleus of the parent. The other item is that the daughter product has one less orbital electron than the parent. If this requirement is not met, then orbital electron capture takes place exclusively.
■
■ ■ ■ ■
■ ■

Gamma radiation is a high-energy electromagnetic radiation that originates in the nucleus. It is emitted in the form of photons, discrete bundles of energy that have both wave and particle properties. Often a daughter nuclide is left in an excited state after a radioactive parent nucleus undergoes a transformation by alpha decay, beta decay, or electron capture. The nucleus will drop to the ground state by the emission of gamma radiation.

Internal Conversion

The usual method for an excited nucleus to go from the excited state to the ground state is by emission of gamma radiation. However, in some cases the gamma ray (photon) emerges from the nucleus only to interact with one of the innermost orbital electrons and, as a result, the energy of the photon is transferred to the electron. The gamma ray is then said to have undergone internal conversion. The conversion electron is ejected from the atom with kinetic energy equal to the gamma energy minus the binding energy of the orbital electron. An orbital electron then drops to a lower energy state to fill the vacancy, and this is accompanied by the emission of characteristic x-rays.

Isomers and Isomeric Transition

Isomeric transition commonly occurs immediately after particle emission; however, the nucleus may remain in an excited state for a measurable period of time before dropping to the ground state at its own characteristic rate. A nucleus that remains in such an excited state is known as a nuclear isomer because it differs in energy and behavior from other nuclei with the same atomic number and mass number. The decay of an excited nuclear isomer to a lower energy level is called an isomeric transition. It is also possible for the excited isomer to decay by some alternate means, for example,

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by beta emission.

An example of gamma emission accompanying particle emission is illustrated by the decay of nitrogen-16 below.

■
■
■
■
■
■
■
■
■
■
■
■

Decay Chains

When an unstable nucleus decays, the resulting daughter nucleus is not necessarily stable. The nucleus resulting from the decay of a parent is often itself unstable, and will undergo an additional decay. This is especially common among the larger nuclides.

It is possible to trace the steps of an unstable atom as it goes through multiple decays trying to achieve stability. The list of the original unstable nuclide, the nuclides that are involved as intermediate steps in the decay, and the final stable nuclide is known as the decay chain. One common method for stating the decay chain is to state each of the nuclides involved in the standard X format. Arrows are used between nuclides to indicate where decays occur, with the type of decay A

indicated above the arrow and the half-life below the arrow. The half-life for decay will be discussed in the next chapter.

Example:

Write the decay chains for rubidium-91 and actinium-215. Continue the chains until a stable nuclide or a nuclide with a half-life greater than 1 x 10 years is reached. 6

Solution:

Predicting Type of Decay

Radioactive nuclides tend to decay in a way that results in a daughter nuclide that lies closer to the line of stability. Due to this, it is possible to predict the type of decay that a nuclide will undergo based on its location relative to the line of stability on the Chart of the Nuclides.

Figure 9 illustrates the type of decay nuclides in different regions of the chart will typically undergo. Nuclides that are below and to the right of the line of stability will usually undergo ■ decay. - Nuclides that are above and to the left of the line of stability will usually undergo either ■ decay or + electron capture. Most nuclides that will undergo ■ decay are found in the upper right hand region of the chart. These are general rules that have many exceptions, especially in the region of the heavy nuclides.

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Figure 9 Types of Radioactive Decay Relative to the Line of Stability

Summary

The important information in this chapter is summarized below.

Modes of Radioactive Decay Summary

Alpha decay is the emission of an alpha particle (2 protons and 2 neutrons) from an unstable nucleus. The daughter nuclide has an atomic number 2 less than the parent nuclide and a mass number 4 less than the parent nuclide. The daughter nucleus commonly releases its excitation energy by gamma emission.

Beta-minus decay effectively converts a neutron to a proton and an electron, Fixture checkpoint marker-0325-xray is recorded on fixture page 0325. which is immediately ejected from the nucleus. The daughter nuclide has its atomic number increased by 1 and the same mass number compared to the parent.

Beta-plus decay effectively converts a proton to a neutron and a positron, which is immediately ejected from the nucleus. The daughter nuclide has its atomic number decreased by 1 and the same mass number compared to the parent.

In electron capture, the nucleus absorbs an electron from the innermost orbit. This electron combines with a proton to form a neutron.

Internal conversion occurs when a gamma ray, emitted by the nucleus as it goes from the excited state to the ground state, interacts with one of the innermost electrons of the same atom. The electron is ejected from the atom.

An isomeric transition is the decay of an excited nucleus to a lower-energy level by the emission of a gamma ray.

Decay chains can be found by tracing the steps an unstable atom goes through as it tries to achieve stability.

Many modes of radioactive decay result in a daughter nuclide that has an energy level above the ground state. This excitation energy is usually released immediately in the form of a gamma ray.

The type of decay that a nuclide will typically undergo can be determined by its

lie below and to the right of the line of stability will typically beta minus decay. Nuclides above and to the left of the line will typically either beta plus decay or electron capture. Most alpha emitters are found in the upper, right-hand corner of the chart.

RADIOACTIVITY

The rate at which a sample of radioactive material decays is not constant. As individual atoms of the material decay, there are fewer of those types of atoms remaining. Since the rate of decay is directly proportional to the number of atoms, the rate of decay will decrease as the number of atoms decreases.

EO 2.5 DEFINE the following terms:
a. Radioactivity d. Radioactive decay constant
b. Curie e. Radioactive half-life
c. Becquerel

EO 2.6 Given the number of atoms and either the half-life or decay constant of a nuclide, CALCULATE the activity.

EO 2.7 Given the initial activity and the decay constant of a nuclide, CALCULATE the activity at any later time.

EO 2.8 CONVERT between the half-life and decay constant for a nuclide.

EO 2.9 Given the Chart of the Nuclides and the original activity, PLOT the radioactive decay curve for a nuclide on either linear or semi-log coordinates.

EO 2.10 DEFINE the following terms:
a. Radioactive equilibrium
b. Transient radioactive equilibrium

Radioactive Decay Rates

Radioactivity is the property of certain nuclides of spontaneously emitting particles or gamma radiation. The decay of radioactive nuclides occurs in a random manner, and the precise time at which a single nucleus will decay cannot be determined. However, the average behavior of a very large sample can be predicted accurately by using statistical methods. These studies have revealed that there is a certain probability that in a given time interval a certain fraction of the nuclei within a sample of a particular nuclide will decay. This probability per unit time that an atom of a nuclide will decay is known as the radioactive decay constant, λ. The units for the

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decay constant are inverse time such as 1/second, 1/minute, 1/hour, or 1/year. These decay constant units can also be expressed as second-1, minute-1, hour-1, and year-1.

■■

The activity (A) of a sample is the rate of decay of that sample. This rate of decay is usually measured in the number of disintegrations that occur per second. For a sample containing millions of atoms, the activity is the product of the decay constant and the number of atoms present in the sample.

The relationship between the activity, number of atoms, and decay constant is shown in Equation (1-3).

where:
A = Activity of the nuclide (disintegrations/second)
■ = decay constant of the nuclide (second)-1
N = Number of atoms of the nuclide in the sample
Since ■ is a constant, the activity and the number of atoms are always proportional.
Units of Measurement for Radioactivity
Two common units to measure the activity of a substance are the curie (Ci) and becquerel (Bq). A curie is a unit of measure of the rate of radioactive decay equal to 3.7 x 10 disintegrations 10 per second. This is approximately equivalent to the number of disintegrations that one gram of radium-226 will undergo in one second. A becquerel is a more fundamental unit of measure of radioactive decay that is equal to 1 disintegration per second. Currently, the curie is more widely used in the United States, but usage of the becquerel can be expected to broaden as the Fixture checkpoint marker-0327-xray is recorded on fixture page 0327.
metric system slowly comes into wider use. The conversion between curies and becquerels is shown below.
1 curie = 3.7 x 10 becquerels 10
Variation of Radioactivity Over Time
The rate at which a given radionuclide sample decays is stated in Equation (1-3) as being equal to the product of the number of atoms and the decay constant. From this basic relationship it is possible to use calculus to derive an expression which can be used to calculate how the number of atoms present will change over time. The derivation is beyond the scope of this text, but Equation (1-4) is the useful result.
(1-4)
where:
N = number of atoms present at time t
N = number of atoms initially present o
■ = decay constant (time)-1
t = time
A Ao
e
t
A Aoe
t
A
Ao
e
t
ln A
Ao
t
t
ln
A
Ao
t
1/2
ln
1

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t
1/2
ln 2 0.693
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Since the activity and the number of atoms are always proportional, they may be used interchangeably to describe any given radionuclide population. Therefore, the following is true.
(1-5)
where:
A = activity present at time t
A = activity initially present o
= decay constant (time)
-1
t = time
Radioactive Half-Life
One of the most useful terms for estimating how quickly a nuclide will decay is the radioactive half-life. The radioactive half-life is defined as the amount of time required for the activity to decrease to one-half of its original value. A relationship between the half-life and decay constant can be developed from Equation (1-5). The half-life can be calculated by solving Equation (1-5)
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for the time, t, when the current activity, A, equals one-half the initial activity A .
o
First, solve Equation (1-5) for t.
If A is equal to one-half of Ao, then A/Ao is equal to one-half. Substituting this in the equation above yields an expression for t .
1/2
(1-6)
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The basic features of decay of a radionuclide sample are shown by the graph in Figure 10.
Figure 10 Radioactive Decay as a Function of Time in Units of Half-Life
Assuming an initial number of atoms No, the population, and consequently, the activity may be noted to decrease by one-half of this value in a time of one half-life. Additional decreases occur so that whenever one half-life elapses, the number of atoms drops to one-half of what its value was at the beginning of that time interval. After five half-lives have elapsed, only 1/32, or 3.1%, of the original number of atoms remains. After seven half-lives, only 1/128, or 0.78%, of the atoms remains. The number of atoms existing after 5 to 7 half-lives can usually be assumed to be negligible. The Chemistry Fundamentals Handbook contains additional information on calculating the number of atoms contained within a sample.
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■
■
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Example:
A sample of material contains 20 micrograms of californium-252.
Californium-252 has a half-life of 2.638 years.
Calculate:
(a) The number of californium-252 atoms initially present

(c) The number of californium-252 atoms that will remain in 12 years
(d) The time it will take for the activity to reach 0.001 curies
Solution:
(a) The number of atoms of californium-252 can be determined as below.
(b) First, use Equation (1-6) to calculate the decay constant.
Use this value for the decay constant in Equation (1-3) to determine the activity.
N No
e
t
4.78 x 1016 e
(0.263/yr) (12 yr)
2.04 x 1015
A Aoe
t
A
Ao
e
t
ln A
Ao
t
t
ln
A
Ao
t
Fixture checkpoint marker-0329-xray is recorded on fixture page 0329.
ln
0.001 Ci
0.0108 Ci
0.263 year 1
t 9.05 years
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(c) The number of californium atoms that will remain in 12 years can be calculated from Equation (1-4).
(d) The time that it will take for the activity to reach 0.001 Ci can be determined from Equation (1-5). First, solve Equation (1-5) for time.
Inserting the appropriate values in the right side of this equation will result in the required time.
Plotting Radioactive Decay
It is useful to plot the activity of a nuclide as it changes over time. Plots of this type can be used to determine when the activity will fall below a certain level. This plot is usually done showing activity on either a linear or a logarithmic scale. The decay of the activity of a single nuclide on a logarithmic scale will plot as a straight line because the decay is exponential.
t
1/2
0.693

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t
1/2
0.693
7.13 seconds
0.0972 second 1
A Aoe
t
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Example:
Plot the radioactive decay curve for nitrogen-16 over a period of 100 seconds. The initial activity is 142 curies and the half-life of nitrogen-16 is 7.13 seconds. Plot the curve on both linear rectangular coordinates and on a semi-log scale.
Solution:
First, use Equation (1-6) to calculate the decay constant corresponding to a half-life of 7.13 seconds.
Use the decay constant determined above to calculate the activity at various times using Equation (1-5).
Time Activity
0 seconds 142 Ci
20 seconds 20.3 Ci
40 seconds 2.91 Ci
60 seconds 0.416 Ci
80 seconds 0.0596 Ci
100 seconds 0.00853 Ci
■ ■
■ ■
■ ■
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Figure 11 Linear and Semi-log Plots of Nitrogen-16 Decay
Plotting the data points calculated above on both linear and semilog scales results in the graphs shown in Figure 11.
Fixture checkpoint marker-0330-xray is recorded on fixture page 0330.
If a substance contains more than one radioactive nuclide, the total activity is the sum of the individual activities of each nuclide. As an example, consider a sample of material that contained 1 x 10 atoms of iron-59 that has a half-life of 44.51 days (■ = 1.80 x 10 sec), 6 -7 -1
1 x 10 atoms of manganese-54 that has a half-life of 312.2 days (■ = 2.57 x 10 sec), and 1 6 -8 -1
x 10 atoms of cobalt-60 that has a half-life of 1925 days (■ = 4.17 x 10 sec). 6 -9 -1
The initial activity of each of the nuclides would be the product of the number of atoms and the decay constant.
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Figure 12 Combined Decay of Iron-56, Manganese-54, and Cobalt-60
Plotting the manner in which the activities of each of the three nuclides decay over time demonstrates that initially the activity of the shortest-lived nuclide (iron-59) dominates the total activity, then manganese-54 dominates. After almost all of the iron and manganese have decayed away, the only contributor to activity will be the cobalt-60. A plot of this combined

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Radioactive Equilibrium
Radioactive equilibrium exists when a radioactive nuclide is decaying at the same rate at which it is being produced. Since the production rate and decay rate are equal, the number of atoms present remains constant over time.
An example of radioactive equilibrium is the concentration of sodium-24 in the coolant circulating through a sodium-cooled nuclear reactor. Assume that the sodium-24 is being produced at a rate of 1 x 10 atoms per second. If the sodium-24 were stable and did not decay, 6 the amount of sodium-24 present after some period of time could be calculated by multiplying the production rate by the amount of time. Plotting the amount of material present would result in the graph in Figure 13.
Fixture checkpoint marker-0331-xray is recorded on fixture page 0331.
However, sodium-24 is not stable, and it decays with a half-life of 14.96 hours. If no sodium-24 is present initially and production starts at a rate of 1 x 10 atoms per second, the rate 6 of decay will initially be zero because there is no sodium-24 present to decay. The rate of decay of sodium-24 will increase as the amount of sodium-24 increases.
■
■
■
■
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Figure 13 Cumulative Production of Sodium-24 Over Time
The amount of sodium-24 present will initially increase rapidly, then it will increase at a continually decreasing rate until the rate of decay is equal to the rate of production. It is possible to calculate how much sodium-24 will be present at equilibrium by setting the production rate (R) equal to the decay rate (■ N).
where:
R = production rate (atoms/second)
■ = decay constant (second)-1
N = number of atoms
It is possible to calculate the equilibrium value for sodium-24 being produced at a rate of 1 x 10 atoms/second. 6
■ ■ ■■
■
■
■
■
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Figure 14 Approach of Sodium-24 to Equilibrium
The development of the equation to calculate how the amount of sodium-24 changes over time as it approaches the equilibrium value is beyond the scope of this handbook. However, the equation is presented below.
This equation can be used to calculate the values of the amount of sodium-24 present at different times. As the time increases, the exponential term approaches zero, and the number of atoms present will approach R/■. A plot of the approach of sodium-24 to equilibrium is shown in Figure 14.
Transient Radioactive Equilibrium

decay at essentially the same rate.
For transient equilibrium to occur, the parent must have a long half-life when compared to the daughter. An example of this type of compound decay process is barium-140, which decays by beta emission to lanthanum-140, which in turn decays by beta emission to stable cerium-140.
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Figure 15 Transient Equilibrium in the Decay of Barium-140
The decay constant for barium-140 is considerably smaller than the decay constant for lanthanum-140. Remember that the rate of decay of both the parent and daughter can be represented as ■N. Although the decay constant for barium-140 is smaller, the actual rate of decay (■N) is initially larger than that of lanthanum-140 because of the great difference in their initial concentrations. As the concentration of the daughter increases, the rate of decay of the daughter will approach and eventually match the decay rate of the parent. When this occurs, they are said to be in transient equilibrium. A plot of the barium-lanthanum-cerium decay chain reaching transient equilibrium is shown in Figure 15.
Secular equilibrium occurs when the parent has an extremely long half-life. In the long decay chain for a naturally radioactive element, such as thorium-232, where all of the elements in the
Fixture checkpoint marker-0332-xray is recorded on fixture page 0332.
chain are in secular equilibrium, each of the descendants has built up to an equilibrium amount and all decay at the rate set by the original parent. The only exception is the final stable element on the end of the chain. Its number of atoms is constantly increasing.
Summary
The important information in this chapter is summarized on the following page.
■
■■
■
RADIOACTIVITY DOE-HDBK-1019/1-93 Atomic and Nuclear Physics
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Radioactivity Summary
& Radioactivity is the decay of unstable atoms by the emission of particles and electromagnetic radiation.
& A curie (Ci) is a unit of radioactivity equal to 3.7 x 10 disintegrations per 10 second.
& A becquerel (Bq) is a unit of radioactivity equal to 1 disintegration per second.
& The radioactive decay constant (■) is the probability per unit time that an atom will decay.
& The radioactive half-life is the amount of time required for the activity to decrease to one-half its original value.
& The activity of a substance can be calculated from the number of atoms and the decay constant based on the equation below.
& The amount of activity remaining after a particular time can be calculated from the equation below.
& The relationship between the decay constant and the half-life is shown below.
& Plots of radioactive decay can be useful to describe the variation of activity over time. If decay is plotted using semi-log scale the plot results in a straight line.
& Radioactive equilibrium exists when the production rate of a material is equal to the removal rate.
& Transient radioactive equilibrium exists when the parent nuclide and the daughter

long half-life compared to the daughter.
Atomic and Nuclear Physics DOE-HDBK-1019/1-93 NEUTRON INTERACTIONS
NEUTRON INTERACTIONS
Neutrons can cause many different types of interactions. The neutron may simply scatter off the nucleus in two different ways, or it may actually be absorbed into the nucleus. If a neutron is absorbed into the nucleus, it may result in the emission of a gamma ray or a subatomic particle, or it may cause the nucleus to fission.
EO 3.1 DESCRIBE the following scattering interactions between a neutron and a nucleus:
a. Elastic scattering
b. Inelastic scattering
EO 3.2 STATE the conservation laws that apply to an elastic collision between a neutron and a nucleus.
EO 3.3 DESCRIBE the following reactions where a neutron is absorbed in a nucleus:
a. Radiative capture
b. Particle ejection
Scattering
A neutron scattering reaction occurs when a nucleus, after having been struck by a neutron, emits a single neutron. Despite the fact that the initial and final neutrons do not need to be (and often are not) the same, the net effect of the reaction is as if the projectile neutron had merely "bounced off," or scattered from, the nucleus. The two categories of scattering reactions, elastic and inelastic scattering, are described in the following paragraphs.
Fixture checkpoint marker-0333-xray is recorded on fixture page 0333.
Elastic Scattering
In an elastic scattering reaction between a neutron and a target nucleus, there is no energy transferred into nuclear excitation. Momentum and kinetic energy of the "system" are conserved although there is usually some transfer of kinetic energy from the neutron to the target nucleus. The target nucleus gains the amount of kinetic energy that the neutron loses.
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mn
vn,i mTvT,i mnvn,f mTvT,f
1
2
m v
2
1
2
mn
v
2
n,i
1
2
mT
v
2

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1
2
mn
v
2
n,f
1
2
mT
v
2
T,f
NEUTRON INTERACTIONS DOE-HDBK-1019/1-93 Atomic and Nuclear Physics
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Figure 16 Elastic Scattering
Figure 16 illustrates the process of elastic scattering of a neutron off a target nucleus. In the elastic scattering reaction, the conservation of momentum and kinetic energy is represented by the equations below.
Conservation of momentum (mv)
Conservation of kinetic energy
where:
mn = mass of the neutron
m = mass of the target nucleus T
vn,i = initial neutron velocity
vn,f = final neutron velocity
vT,i = initial target velocity
vT,f = final target velocity
Atomic and Nuclear Physics DOE-HDBK-1019/1-93 NEUTRON INTERACTIONS
Elastic scattering of neutrons by nuclei can occur in two ways. The more unusual of the two interactions is the absorption of the neutron, forming a compound nucleus, followed by the re-emission of a neutron in such a way that the total kinetic energy is conserved and the nucleus
Fixture checkpoint marker-0334-xray is recorded on fixture page 0334.
returns to its ground state. This is known as resonance elastic scattering and is very dependent upon the initial kinetic energy possessed by the neutron. Due to formation of the compound nucleus, it is also referred to as compound elastic scattering. The second, more usual method, is termed potential elastic scattering and can be understood by visualizing the neutrons and nuclei to be much like billiard balls with impenetrable surfaces. Potential scattering takes place with incident neutrons that have an energy of up to about 1 MeV. In potential scattering, the neutron does not actually touch the nucleus and a compound nucleus is not formed. Instead, the neutron is acted on and scattered by the short range nuclear forces when it approaches close enough to the nucleus.
Inelastic Scattering
In inelastic scattering, the incident neutron is absorbed by the target nucleus, forming a compound nucleus. The compound nucleus will then emit a neutron of lower kinetic energy which leaves the original nucleus in an excited state. The nucleus will usually, by one or more gamma emissions, emit this excess energy to reach its ground state. Figure 17 shows the process of inelastic scattering.
For the nucleus that has reached its ground state, the sum of the kinetic energy of the exit

neutron, the target nucleus, and the total gamma energy emitted is equal to the initial kinetic energy of the incident neutron.

1
0
n
238
92
U
239
92
U
239
92
U
0
0
1
0
n
10
5
B
11
5
B
7
3
Li
4
2

Absorption Reactions

Most absorption reactions result in the loss of a neutron coupled with the production of a charged particle or gamma ray. When the product nucleus is radioactive, additional radiation is emitted at some later time. Radiative capture, particle ejection, and fission are all categorized as absorption reactions and are briefly described below.

Fixture checkpoint marker-0335-xray is recorded on fixture page 0335.

Radiative Capture

In radiative capture the incident neutron enters the target nucleus forming a compound nucleus. The compound nucleus then decays to its ground state by gamma emission. An example of a radiative capture reaction is shown below.

Particle Ejection

In a particle ejection reaction the incident particle enters the target nucleus forming a compound nucleus. The newly formed compound nucleus has been excited to a high enough energy level to cause it to eject a new particle while the incident neutron remains in the nucleus. After the new particle is ejected, the remaining nucleus may or may not exist in an excited state depending

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shown below.
Fission
One of the most important interactions that neutrons can cause is fission, in which the nucleus that absorbs the neutron actually splits into two similarly sized parts. Fission will be discussed in detail in the next chapter.
Atomic and Nuclear Physics DOE-HDBK-1019/1-93 NEUTRON INTERACTIONS
Summary
The important information in this chapter is summarized below.
Neutron Interactions Summary
Interactions where a neutron scatters off a target nucleus are either elastic or inelastic. In elastic scattering, kinetic energy and momentum are conserved and no energy is transferred into excitation energy of the target nucleus. In inelastic scattering, some amount of kinetic energy is transferred into excitation energy of the target nucleus.
The conservation principles that apply to an elastic collision are conservation of
Fixture checkpoint marker-0336-xray is recorded on fixture page 0336.
kinetic energy and conservation of momentum.
Radiative capture is the absorption of a neutron by the target nucleus, resulting in an excited nucleus which subsequently (typically within a small fraction of a second) releases its excitation energy in the form of a gamma ray.
Particle ejection occurs when a neutron is absorbed by a target nucleus, resulting in the formation of a compound nucleus. The compound nucleus immediately ejects a particle (for example, alpha or proton).
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1
0
n
235
92
U
236
92
U
140
55
Cs
93
37
Rb 3
1
0
n
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NUCLEAR FISSION
Nuclear fission is a process in which an atom splits and releases energy, fission products, and neutrons. The neutrons released by fission can, in turn, cause the fission of other atoms.

of a nucleus.
EO 4.2 DEFINE the following terms:
a. Excitation energy (E_{exc})
b. Critical energy (E_{crit})
EO 4.3 DEFINE the following terms:
a. Fissile material
b. Fissionable material
c. Fertile material
EO 4.4 DESCRIBE the processes of transmutation, conversion, and breeding.
EO 4.5 DESCRIBE the curve of Binding Energy per Nucleon versus mass number and give a qualitative description of the reasons for its shape.
EO 4.6 EXPLAIN why only the heaviest nuclei are easily fissioned.
EO 4.7 EXPLAIN why uranium-235 fissions with thermal neutrons and uranium-238 fissions only with fast neutrons.
Fission
In the fission reaction the incident neutron enters the heavy target nucleus, forming a compound nucleus that is excited to such a high energy level ($E > E_{crit}$) that the nucleus "splits" (E_{exc}) (fissions) into two large fragments plus some neutrons. An example of a typical fission reaction is shown below.
A large amount of energy is released in the form of radiation and fragment kinetic energy.
Fixture checkpoint marker-0337-xray is recorded on fixture page 0337.
Atomic and Nuclear Physics DOE-HDBK-1019/1-93 NUCLEAR FISSION
Liquid Drop Model of a Nucleus
The nucleus is held together by the attractive nuclear force between nucleons, which was discussed in a previous chapter. The characteristics of the nuclear force are listed below.
(a) very short range, with essentially no effect beyond nuclear dimensions (10^{-13} cm)
(b) stronger than the repulsive electrostatic forces within the nucleus
(c) independent of nucleon pairing, in that the attractive forces between pairs of neutrons are no different than those between pairs of protons or a neutron and a proton
(d) saturable, that is, a nucleon can attract only a few of its nearest neighbors
One theory of fission considers the fissioning of a nucleus similar in some respects to the splitting of a liquid drop. This analogy is justifiable to some extent by the fact that a liquid drop is held together by molecular forces that tend to make the drop spherical in shape and that try to resist any deformation in the same manner as nuclear forces are assumed to hold the nucleus together. By considering the nucleus as a liquid drop, the fission process can be described.
Referring to Figure 18(A), the nucleus in the ground state is undistorted, and its attractive nuclear forces are greater than the repulsive electrostatic forces between the protons within the nucleus. When an incident particle (in this instance a neutron) is absorbed by the target nucleus, a compound nucleus is formed. The compound nucleus temporarily contains all the charge and mass involved in the reaction and exists in an excited state. The excitation energy added to the compound nucleus is equal to the binding energy contributed by the incident particle plus the kinetic energy possessed by that particle. Figure 18(B) illustrates the excitation energy thus imparted to the compound nucleus, which may cause it to oscillate and become distorted. If the excitation energy is greater than a certain critical energy, the oscillations may cause the

(short-range) in the neck area are small due to saturation, while the repulsive electrostatic forces (long-range) are only slightly less than before. When the repulsive electrostatic forces exceed the attractive nuclear forces, nuclear fission occurs, as illustrated in Figure 18(C).

Figure 18 Liquid Drop Model of Fission

Critical Energy

The measure of how far the energy level of a nucleus is above its ground state is called the excitation energy (Eexc). For fission to occur, the excitation energy must be above a particular value for that nuclide. The critical energy (Ecrit) is the minimum excitation energy required for fission to occur.

Fissile Material

A fissile material is composed of nuclides for which fission is possible with neutrons of any energy level. What is especially significant about these nuclides is their ability to be fissioned with zero kinetic energy neutrons (thermal neutrons). Thermal neutrons have very low kinetic energy levels (essentially zero) because they are roughly in equilibrium with the thermal motion of surrounding materials. Therefore, in order to be classified as fissile, a material must be capable of fissioning after absorbing a thermal neutron. Consequently, they impart essentially no kinetic energy to the reaction. Fission is possible in these materials with thermal neutrons, since the change in binding energy supplied by the neutron addition alone is high enough to exceed the critical energy. Some examples of fissile nuclides are uranium-235, uranium-233, and plutonium-239.

Fissionable Material

A fissionable material is composed of nuclides for which fission with neutrons is possible. All fissile nuclides fall into this category. However, also included are those nuclides that can be fissioned only with high energy neutrons. The change in binding energy that occurs as the result of neutron absorption results in a nuclear excitation energy level that is less than the required

Fixture checkpoint marker-0338-xray is recorded on fixture page 0338.

critical energy. Therefore, the additional excitation energy must be supplied by the kinetic energy of the incident neutron. The reason for this difference between fissile and fissionable materials is the so-called odd-even effect for nuclei. It has been observed that nuclei with even numbers of neutrons and/or protons are more stable than those with odd numbers. Therefore, adding a neutron to change a nucleus with an odd number of neutrons to a nucleus with an even number of neutrons produces an appreciably higher binding energy than adding a neutron to a nucleus already possessing an even number of neutrons. Some examples of nuclides requiring high energy neutrons to cause fission are thorium-232, uranium-238, and plutonium-240. Table 4 indicates the critical energy (Ecrit) and the binding energy change for an added neutron (BEn) to target nuclei of interest. For fission to be possible, the change in binding energy plus the kinetic energy must equal or exceed the critical energy ($\Delta BE + KE > Ecrit$).

TABLE 4
Critical Energies Compared to Binding Energy of Last Neutron

Target Nucleus	Critical Energy Ecrit	Binding Energy of

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BEn - Ecrit
23
9
2
0Th 7.5 MeV 5.4 MeV -2.1 MeV
23
9
8
2U 7.0 MeV 5.5 MeV -1.5 MeV
23
9
5
2U 6.5 MeV 6.8 MeV +0.3 MeV
23
9
3
2U 6.0 MeV 7.0 MeV +1.0 MeV
23
9
9
4Pu 5.0 MeV 6.6 MeV +1.6 MeV
Uranium-235 fissions with thermal neutrons because the binding energy released by the absorption of a neutron is greater than the critical energy for fission; therefore uranium-235 is a fissile material. The binding energy released by uranium-238 absorbing a thermal neutron is less than the critical energy, so additional energy must be possessed by the neutron for fission to be possible. Consequently, uranium-238 is a fissionable material.
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NUCLEAR FISSION DOE-HDBK-1019/1-93 Atomic and Nuclear Physics
Fertile Material
All of the neutron absorption reactions that do not result in fission lead to the production of new nuclides through the process known as transmutation. These nuclides can, in turn, be transmuted again or may undergo radioactive decay to produce still different nuclides. The nuclides that are produced by this process are referred to as transmutation products. Because several of the fissile nuclides do not exist in nature, they can only be produced by nuclear reactions (transmutation). The target nuclei for such reactions are said to be fertile. Fertile materials are materials that can undergo transmutation to become fissile materials. Figure 19 traces the transmutation mechanism
Fixture checkpoint marker-0339-xray is recorded on fixture page 0339.
by which two fertile nuclides, thorium-232 and uranium-238, produce uranium-233 and plutonium-239, respectively.
Figure 19 Conversion of Fertile Nuclides to Fissile Nuclides
If a reactor contains fertile material in addition to its fissile fuel, some new fuel will be produced as the original fuel is burned up. This is called conversion. Reactors that are specifically designed to produce fissionable fuel are called "breeder" reactors. In such reactors, the amount of fissionable fuel produced is greater than the amount of fuel burnup. If less fuel is produced than used, the process is called conversion, and the reactor is termed a "converter."
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Binding Energy Per Nucleon (BE/A)

rate of increase, however, is not uniform. This lack of uniformity results in a variation in the amount of binding energy associated with each nucleon within the nucleus. This variation in the binding energy per nucleon (BE/A) is easily seen when the average BE/A is plotted versus atomic mass number (A), as shown in Figure 20.

Figure 20 Binding Energy per Nucleon vs. Mass Number

Figure 20 illustrates that as the atomic mass number increases, the binding energy per nucleon decreases for A > 60. The BE/A curve reaches a maximum value of 8.79 MeV at A = 56 and decreases to about 7.6 MeV for A = 238. The general shape of the BE/A curve can be explained using the general properties of nuclear forces. The nucleus is held together by very short-range attractive forces that exist between nucleons. On the other hand, the nucleus is being forced apart by long range repulsive electrostatic (coulomb) forces that exist between all the protons in the nucleus.

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As the atomic number and the atomic mass number increase, the repulsive electrostatic forces within the nucleus increase due to the greater number of protons in the heavy elements. To overcome this increased repulsion, the proportion of neutrons in the nucleus must increase to maintain stability. This increase in the neutron-to-proton ratio only partially compensates for the growing proton-proton repulsive force in the heavier, naturally occurring elements. Because the repulsive forces are increasing, less energy must be supplied, on the average, to remove a nucleon from the nucleus. The BE/A has decreased. The BE/A of a nucleus is an indication of its degree of stability. Generally, the more stable nuclides have higher BE/A than the less stable ones. The increase in the BE/A as the atomic mass number decreases from 260 to 60 is the primary reason for the energy liberation in the fission process. In addition, the increase in the BE/A as the atomic mass number increases from 1 to 60 is the reason for the energy liberation in the fusion process, which is the opposite reaction of fission.

The heaviest nuclei require only a small distortion from a spherical shape (small energy addition) for the relatively large coulomb forces forcing the two halves of the nucleus apart to overcome the attractive nuclear forces holding the two halves together. Consequently, the heaviest nuclei are easily fissionable compared to lighter nuclei.

Summary

The important information in this chapter is summarized on the following page.

Nuclear Fission Summary

The fission process can be explained using the liquid drop model of a nucleus. In the ground state the nucleus is nearly spherical in shape. After the absorption of a neutron, the nucleus will be in an excited state and start to oscillate and become distorted. If the oscillations cause the nucleus to become shaped like a dumbbell, the repulsive electrostatic forces will overcome the short-range attractive nuclear forces, and the nucleus will split in two.

Excitation energy is the amount of energy a nucleus has above its ground state. Critical energy is the minimum excitation energy that a nucleus must have before it can fission.

Fissile material is material for which fission is possible with neutrons that have zero kinetic energy. Fissionable material is material for which fission caused by neutron absorption is possible provided the kinetic energy added with the binding

undergo transmutation to become fissile material.
Transmutation is the process of neutron absorption and subsequent decay, which changes one nuclide to another nuclide. Conversion is the process of transmuting fertile material into fissile material in a reactor, where the amount of fissile material produced is less than the amount of fissile material consumed. Breeding is the same as conversion, except the amount of fissile material produced is more than the amount of fissile material consumed.
The curve of binding energy per nucleon increases quickly through the light nuclides and reaches a maximum at a mass number of about 56. The curve decreases slowly for mass numbers greater than 60.
The heaviest nuclei are easily fissionable because they require only a small distortion from the spherical shape to allow the coulomb forces to overcoming the attractive nuclear force, forcing the two halves of the nucleus apart.
Uranium-235 fissions with thermal neutrons because the binding energy released by the absorption of a neutron is greater than the critical energy for fission. The binding energy released by uranium-238 absorbing a neutron is less than the critical energy, so additional kinetic energy must be possessed by the neutron for fission to be possible.
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1
Fixture checkpoint marker-0341-xray is recorded on fixture page 0341.
0
n
235
92
U
236
92
U
140
55
Cs
93
37
Rb 3
1
0
n
ENERGY RELEASE FROM FISSION DOE-HDBK-1019/1-93 Atomic and Nuclear Physics
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ENERGY RELEASE FROM FISSION
Fission of heavy nuclides converts a small amount of mass into an enormous amount of energy. The amount of energy released by fission can be determined based on either the change in mass that occurs during the reaction or by the difference in binding energy per nucleon between the fissile nuclide and the fission products.
EO 4.8 CHARACTERIZE the fission products in terms of mass groupings and radioactivity.

the energy released from fission.
EO 4.10 Given the curve of Binding Energy per nucleon versus mass number, CALCULATE the energy released from fission.
Calculation of Fission Energy
Nuclear fission results in the release of enormous quantities of energy. It is necessary to be able to calculate the amount of energy that will be produced. The logical manner in which to pursue this is to first investigate a typical fission reaction such as the one listed below.
It can be seen that when the compound nucleus splits, it breaks into two fission fragments, rubidium-93, cesium-140, and some neutrons. Both fission products then decay by multiple - emissions as a result of the high neutron-to-proton ratio possessed by these nuclides.
In most cases, the resultant fission fragments have masses that vary widely. Figure 21 gives the percent yield for atomic mass numbers. The most probable pair of fission fragments for the thermal fission of the fuel uranium-235 have masses of about 95 and 140. Note that the vertical axis of the fission yield curve is on a logarithmic scale. Therefore, the formation of fission fragments of mass numbers of about 95 and 140 is highly likely.
Atomic and Nuclear Physics DOE-HDBK-1019/1-93 ENERGY RELEASE FROM FISSION
Figure 21 Uranium-235 Fission Yield vs.
Mass Number
Referring now to the binding energy per nucleon curve (Figure 20), we can estimate the amount of energy released by our "typical" fission by plotting this reaction on the curve and calculating the change in binding energy (ΔBE) between the reactants on the left-hand side of the fission equation and the products on the right-hand side. Plotting the reactant and product nuclides on the curve shows that the total binding energy of the system after fission is greater than the total binding energy of the system before fission. When there is an increase in the total binding energy of a system, the system has become more stable by releasing an amount of energy equal to the increase in total binding energy of the system. Therefore, in the fission process, the energy liberated is equal to the increase in the total binding energy of the system.
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Figure 22 graphically depicts that the binding energy per nucleon for the products (C,
Figure 22 Change in Binding Energy for Typical Fission
rubidium-93 and B, cesium-140) is greater than that for the reactant (A, uranium-235). The total binding energy for a nucleus can be found by multiplying the binding energy per nucleon by the number of nucleons.
TABLE 5
Binding Energies Calculated from
Binding Energy per Nucleon Curve
Nuclide B.E. per Nucleon
(BE/A)
Mass Number
(A)
Binding Energy
(BE/A) x (A)
9
3
3
7Rb 8.7 MeV 93 809 MeV

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5
0
5Cs 8.4 MeV 140 1176 MeV
23
9
5
2U 7.6 MeV 235 1786 MeV
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BE BEproducts BEreactants
BERb 93 BECs 140 BEU 235
809 MeV 1176 MeV 1786 MeV
199 MeV
1
0
n
235
92
U
236
92
U
140
55
Cs
93
37
Rb 3
1
0
n
Mass difference Mass of Reactants Mass of Products
236.052589 amu 235.85208 amu
0.200509 amu
EInst 0.020059 amu
Fixture checkpoint marker-0343-xray is recorded on fixture page 0343.
931.5 MeV
amu
186.8 MeV
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The energy released will be equivalent to the difference in binding energy (BE) between the reactants and the products.
The energy liberation during the fission process can also be explained from the standpoint of the conservation of mass-energy. During the fission process, there is a decrease in the mass of the system. There must, therefore, be energy liberated equal to the energy equivalent of the mass lost in the process. This method is more accurate than the previously illustrated method and is used when actually calculating the energy liberated during the fission process.
Again, referring to the "typical" fission reaction.

is equal to the energy equivalent of the mass lost in the fission process. It can be calculated as shown below.
Mass of the Reactants Mass of the Products
U 235.043924 amu Rb 92.91699 amu 235
92
93
37
n 1.008665 amu Cs 139.90910 amu 1
0
140
55
Fixture checkpoint marker-0344-xray is recorded on fixture page 0344.
3 (n) 3.02599 amu 1
0
236.052589 amu 235.85208 amu
This mass difference can be converted to an energy equivalent.
93
37
Rb
93
38
Sr
93
39
Y
93
40
Zr
93
41
Nb
140
55
Cs
140
56
Ba
140
57
La
140
58
Ce
EDecay mRb 93 mNb 93 4 melectron
931.5MeV
amu
92.91699 amu 92.90638 amu 4 0.0005486 amu
931.5 MeV

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0.008416 amu
931.5 MeV
amu
7.84 MeV
ENERGY RELEASE FROM FISSION DOE-HDBK-1019/1-93 Atomic and Nuclear Physics
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The total energy released per fission will vary from the fission to the next depending on what fission products are formed, but the average total energy released per fission of uranium-235 with a thermal neutron is 200 MeV.
As illustrated in the preceding example, the majority of the energy liberated in the fission process is released immediately after the fission occurs and appears as the kinetic energy of the fission fragments, kinetic energy of the fission neutrons, and instantaneous gamma rays. The remaining energy is released over a period of time after the fission occurs and appears as kinetic energy of the beta, neutrino, and decay gamma rays.
Estimation of Decay Energy
In addition to this instantaneous energy release during the actual fission reaction, there is additional energy released when the fission fragments decay by emission. This additional
-
Fixture checkpoint marker-0345-xray is recorded on fixture page 0345.
energy is called decay energy, EDecay. The decay chains for rubidium-93 and cesium-140 are shown below.
The energy released during the decay for each chain will be equivalent to the mass difference between the original fission product and the sum of the final stable nuclide and the beta particles emitted.
The energy released in the decay chain of rubidium-93 is calculated below.
EDecay mRb 93 mNb 93 3 melectron
931.5MeV
amu
139.90910 amu 139.90543 amu 3 0.0005486 amu
931.5 MeV
amu
0.000202 amu
931.5 MeV
amu
1.89 MeV
Atomic and Nuclear Physics DOE-HDBK-1019/1-93 ENERGY RELEASE FROM FISSION
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The energy released in the decay chain of cesium-140 is calculated below.
The total decay energy is the sum of the energies of the two chains, or 9.73 MeV.
Distribution of Fission Energy
The average energy distribution for the energy released per fission with a thermal neutron in uranium-235 is shown in Tables 6 and 7.
TABLE 6
Instantaneous Energy from Fission
Kinetic Energy of Fission Products 167 Mev
Energy of Fission Neutrons 5 MeV
Instantaneous Gamma-ray Energy 5 MeV
Capture Gamma-ray Energy 10 MeV

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TABLE 7
Delayed Energy from Fission
Beta Particles From Fission Products 7 Mev
Gamma-rays from Fission Products 6 MeV
Neutrinos 10 MeV
Total Delayed Energy 23 MeV
ENERGY RELEASE FROM FISSION DOE-HDBK-1019/1-93 Atomic and Nuclear Physics
Because the 10 MeV of neutrino energy shown in Table 7 is not absorbed in the reactor, the average value of 200 MeV per fission is still accurate. Note in Table 6 that some fission neutrons undergo radiative capture and the resultant gamma ray emission provides an additional 10 MeV of instantaneous energy, which contributes to the total of 187 MeV instantaneous energy.
All of the energy released, with the exception of the neutrino energy, is ultimately transformed into heat through a number of processes. The fission fragments, with their high positive charge and kinetic energy, cause ionization directly as they rip orbital electrons from the surrounding atoms. In this ionization process, kinetic energy is transferred to the surrounding atoms of the fuel material, resulting in an increase in temperature. The beta particles and gamma rays also give up their energy through ionization, and the fission neutrons interact and lose their energy through elastic scattering. Of the 200 MeV released per fission, about seven percent (13 MeV) is released at some time after the instant of fission. When a reactor is shut down, fissions essentially cease, but energy is still being released from the decay of fission products. The heat produced by this decay energy is referred to as "decay heat." Although decay energy represents about seven percent of reactor heat production during reactor operation, once the reactor is shut down the decay heat production drops off quickly to a small fraction of its value while operating. The decay heat produced is significant, however, and systems must be provided to keep the
Fixture checkpoint marker-0346-xray is recorded on fixture page 0346.
reactor cool even after shutdown.
Summary
The important information in this chapter is summarized below.
Energy Release From Fission Summary
Fission products have some general characteristics in common.
They generally decay by β- emission.
The most common mass numbers are grouped near 95 and 140.
The energy released by fission can be calculated based on the difference in mass between the masses of the reactants before fission and the fission fragments and fission neutrons after fission.
Another method to determine the energy released by fission is based on the change in binding energy per nucleon between the fissile nuclide and the fission products.
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Atomic and Nuclear Physics INTERACTION OF RADIATION WITH MATTER
INTERACTION OF RADIATION WITH MATTER
Different types of radiation interact with matter in widely different ways. A large, massive, charged alpha particle cannot penetrate a piece of paper and even has a limited range in dry air. A neutrino, at the other extreme, has a low probability of interacting with any matter, even if it passed through the diameter of the earth.

a. Alpha particle c. Positron
b. Beta particle d. Neutron
EO 5.2 DESCRIBE the following ways that gamma radiation interacts with matter:
a. Photoelectric effect
b. Compton scattering
c. Pair production
Interaction of Radiation With Matter
Radiation can be classified into two general groups, charged and uncharged; therefore, it may be expected that interactions with matter fall into two general types. Charged particles directly ionize the media through which they pass, while uncharged particles and photons can cause ionization only indirectly or by secondary radiation.
A moving charged particle has an electrical field surrounding it, which interacts with the atomic structure of the medium through which it is passing. This interaction decelerates the particle and accelerates electrons in the atoms of the medium. The accelerated electrons may acquire enough energy to escape from the parent atom. This process, whereby radiation "strips" off orbital electrons, is called ionization. Uncharged moving particles have no electrical field, so they can only lose energy and cause ionization by such means as collisions or scattering. A photon can lose energy by the photoelectric effect, Compton effect, or pair production.
Rev. 0 Page 63 NP-01
I
m z
2
K.E.
DOE-HDBK-1019/1-93
INTERACTION OF RADIATION WITH MATTER Atomic and Nuclear Physics
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Because ionizing radiation creates ions in pairs, the intensity of ionization or the specific ionization is defined as the number of ion-pairs formed per centimeter of travel in a given material. The amount of ionization produced by a charged particle per unit path length, which is a measure of its ionizing power, is roughly proportional to the particle's mass and the square of its charge as illustrated in the equation below.
Fixture checkpoint marker-0347-xray is recorded on fixture page 0347.
where:
I is the ionizing power
m is the mass of the particle
z is the number of unit charges it carries
K.E. is its kinetic energy
Since m for an alpha particle is about 7300 times as large as m for a beta particle, and z is twice as great, an alpha will produce much more ionization per unit path length than a beta particle of the same energy. This phenomenon occurs because the larger alpha particle moves slower for a given energy and thus acts on a given electron for a longer time.
Alpha Radiation
Alpha radiation is normally produced from the radioactive decay of heavy nuclides and from certain nuclear reactions. The alpha particle consists of 2 neutrons and 2 protons, so it is essentially the same as the nucleus of a helium atom. Because it has no electrons, the alpha particle has a charge of +2. This positive charge causes the alpha particle to strip electrons from the orbits of atoms in its vicinity. As the alpha particle passes through material, it removes

the energy of the alpha particle is reduced by each reaction. Eventually the particle will expend its kinetic energy, gain 2 electrons in orbit, and become a helium atom. Because of its strong positive charge and large mass, the alpha particle deposits a large amount of energy in a short distance of travel. This rapid, large deposition of energy limits the penetration of alpha particles. The most energetic alpha particles are stopped by a few centimeters of air or a sheet of paper.

Beta-Minus Radiation

A beta-minus particle is an electron that has been ejected at a high velocity from an unstable nucleus. An electron has a small mass and an electrical charge of -1. Beta particles cause ionization by displacing electrons from atom orbits. The ionization occurs from collisions with orbiting electrons. Each collision removes kinetic energy from the beta particle, causing it to slow down. Eventually the beta particle will be slowed enough to allow it to be captured as an

2 electrons 0.000549 amu

electron

931.5 MeV

amu

1.02 MeV

orbiting electron in an atom. Although more penetrating than the alpha, the beta is relatively easy to stop and has a low power of penetration. Even the most energetic beta radiation can be stopped by a few millimeters of metal.

Positron Radiation

Positively charged electrons are called positrons. Except for the positive charge, they are identical to beta-minus particles and interact with matter in a similar manner. Positrons are very short-lived, however, and quickly are annihilated by interaction with a negatively charged electron, producing two gammas with a combined energy (calculated below) equal to the rest mass of the positive and negative electrons.

Neutron Radiation

Neutrons have no electrical charge. They have nearly the same mass as a proton (a hydrogen atom nucleus). A neutron has hundreds of times more mass than an electron, but 1/4 the mass of an alpha particle. The source of neutrons is primarily nuclear reactions, such as fission, but they may also be produced from the decay of radioactive nuclides. Because of its lack of charge, the neutron is difficult to stop and has a high penetrating power.

Neutrons are attenuated (reduced in energy and numbers) by three major interactions, elastic scatter, inelastic scatter, and absorption. In elastic scatter, a neutron collides with a nucleus and bounces off. This reaction transmits some of the kinetic energy of the neutron to the nucleus of the atom, resulting in the neutron being slowed, and the atom receives some kinetic energy

Fixture checkpoint marker-0348-xray is recorded on fixture page 0348.

(motion). This process is sometimes referred to as "the billiard ball effect."

As the mass of the nucleus approaches the mass of the neutron, this reaction becomes more effective in slowing the neutron. Hydrogenous material attenuates neutrons most effectively.

In the inelastic scatter reaction, the same neutron/nucleus collision occurs as in elastic scatter. However, in this reaction, the nucleus receives some internal energy as well as kinetic energy. This slows the neutron, but leaves the nucleus in an excited state. When the nucleus decays to its original energy level, it normally emits a gamma ray.

In the absorption reaction, the neutron is actually absorbed into the nucleus of an atom. The

gamma rays to reach a stable level, the process is called radiative capture. This reaction occurs at most neutron energy levels, but is more probable at lower energy levels.
DOE-HDBK-1019/1-93
INTERACTION OF RADIATION WITH MATTER Atomic and Nuclear Physics
Gamma Radiation
Gamma radiation is electromagnetic radiation. It is commonly referred to as a gamma ray and is very similar to an x-ray. The difference is that gamma rays are emitted from the nucleus of an atom, and x-rays are produced by orbiting electrons. The x-ray is produced when orbiting electrons move to a lower energy orbit or when fast-moving electrons approaching an atom are deflected and decelerated as they react with the atom's electrical field (called Bremsstrahlung). The gamma ray is produced by the decay of excited nuclei and by nuclear reactions. Because the gamma ray has no mass and no charge, it is difficult to stop and has a very high penetrating power. A small fraction of the original gamma stream will pass through several feet of concrete or several meters of water.
There are three methods of attenuating gamma rays. The first method is referred to as the photo-electric effect. When a low energy gamma strikes an atom, the total energy of the gamma
Fixture checkpoint marker-0349-xray is recorded on fixture page 0349.
is expended in ejecting an electron from orbit. The result is ionization of the atom and expulsion of a high energy electron. This reaction is most predominant with low energy gammas interacting in materials with high atomic weight and rarely occurs with gammas having an energy above 1 MeV. Annihilation of the gamma results. Any gamma energy in excess of the binding energy of the electron is carried off by the electron in the form of kinetic energy.
The second method of attenuation of gammas is called Compton scattering. The gamma interacts with an orbital or free electron; however, in this case, the photon loses only a fraction of its energy. The actual energy loss depending on the scattering angle of the gamma. The gamma continues on at lower energy, and the energy difference is absorbed by the electron. This reaction becomes important for gamma energies of about 0.1 MeV and higher.
At higher energy levels, a third method of attenuation is predominant. This method is pair-production. When a high energy gamma passes close enough to a heavy nucleus, the gamma completely disappears, and an electron and a positron are formed. For this reaction to take place, the original gamma must have at least 1.02 MeV energy. Any energy greater than 1.02 MeV becomes kinetic energy shared between the electron and positron. The probability of pair■production incre
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Atomic and Nuclear Physics INTERACTION OF RADIATION WITH MATTER
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Interaction of Radiation with Matter Summary
An alpha particle deposits a large amount of energy in a short distance of travel due to its large mass and charge.
Beta-minus particles interact with the electrons orbiting the nucleus of atoms, causing ionization by displacing the electrons. The beta particle loses energy with each interaction. After the beta particle loses enough energy, it is captured in the orbital shells of an atom.
Positrons interact with matter much the same way as beta minus particles. After the positron has lost most of its energy by ionizing atoms, it is annihilated by interaction with an electron. The electron-positron pair disappear and are replaced by two gammas, each with the energy equivalent of the mass of an electron (0.51 MeV).

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absorption.
Photoelectric effect is where a gamma interacts with an electron orbiting an atom. The entire energy of the gamma is transferred to the electron, and the electron is ejected from its orbit.
In Compton scattering a gamma interacts with an orbital electron, but only part of the gamma energy is transferred to the electron. The electron is ejected from its orbit, and the gamma is scattered off at a lower energy.
In pair-production, a gamma interacts with the electric field of a nucleus and is converted into an electron-positron pair. The gamma must have an energy greater than 1.02 MeV for this to occur.
Summary
The important information in this chapter is summarized below.
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INTERACTION OF RADIATION WITH MATTER Atomic and Nuclear Physics
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Department of Energy
Fundamentals Handbook
NUCLEAR PHYSICS
AND REACTOR THEORY
Module 2
Reactor Theory (Neutron Characteristics)
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OBJECTIVES DOE-HDBK-1019/1-93 Reactor Theory (Neutron Characteristics)
TERMINAL OBJECTIVE
1.0 Without references, EXPLAIN how neutron sources produce neutrons.
ENABLING OBJECTIVES
1.1 DEFINE the following terms:
a. Intrinsic neutron source
b. Installed neutron source
1.2 LIST three examples of reactions that produce neutrons in intrinsic neutron sources.
1.3 LIST three examples of reactions that produce neutrons in installed neutron sources.
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Reactor Theory (Neutron Characteristics) DOE-HDBK-1019/1-93 OBJECTIVES
TERMINAL OBJECTIVE
2.0 Given the necessary information for calculations, EXPLAIN basic concepts in reactor physics and perform calculations.
ENABLING OBJECTIVES
2.1 DEFINE the following terms:
a. Atom density d. Barn
b. Neutron flux e. Macroscopic cross section
c. Microscopic cross section f. Mean free path
2.2 EXPRESS macroscopic cross section in terms of microscopic cross section.
2.3 DESCRIBE how the absorption cross section of typical nuclides varies with neutron energy at energies below the resonance absorption region.
2.4 DESCRIBE the cause of resonance absorption in terms of nuclear energy levels.
2.5 DESCRIBE the energy dependence of resonance absorption peaks for typical light and heavy nuclei.
2.6 EXPRESS mean free path in terms of macroscopic cross section.
2.7 Given the number densities (or total density and component fractions) and microscopic cross sections of components, CALCULATE the macroscopic cross section for a mixture.
2.8 CALCULATE a macroscopic cross section given a material density, atomic mass, and microscopic cross section.
2.9 EXPLAIN neutron shadowing or self-shielding.
2.10 Given the neutron flux and macroscopic cross section, CALCULATE the reaction rate.
2.11 DESCRIBE the relationship between neutron flux and reactor power.
Fixture checkpoint marker-0352-xray is recorded on fixture page 0352.
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OBJECTIVES DOE-HDBK-1019/1-93 Reactor Theory (Neutron Characteristics)
ENABLING OBJECTIVES (Cont.)
2.12 DEFINE the following concepts:
a. Thermalization d. Average logarithmic energy decrement
b. Moderator e. Macroscopic slowing down power
c. Moderating ratio
2.13 LIST three desirable characteristics of a moderator.
2.14 Given an average fractional energy loss per collision, CALCULATE the energy loss after a specified number of collisions.

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Reactor Theory (Neutron Characteristics) DOE-HDBK-1019/1-93 OBJECTIVES
TERMINAL OBJECTIVE
3.0 Without references, EXPLAIN the production process and effects on fission of prompt and delayed neutrons.
ENABLING OBJECTIVES
3.1 STATE the origin of prompt neutrons and delayed neutrons.
3.2 STATE the approximate fraction of neutrons that are born as delayed neutrons from the fission of the following nuclear fuels:
a. Uranium-235
b. Plutonium-239
3.3 EXPLAIN the mechanism for production of delayed neutrons.
3.4 EXPLAIN prompt and delayed neutron generation times.
3.5 Given prompt and delayed neutron generation times and delayed neutron fraction, CALCULATE the average generation time.
Fixture checkpoint marker-0353-xray is recorded on fixture page 0353.
3.6 EXPLAIN the effect of delayed neutrons on reactor control.
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OBJECTIVES DOE-HDBK-1019/1-93 Reactor Theory (Neutron Characteristics)
TERMINAL OBJECTIVE
4.0 Without references, DESCRIBE the neutron energy spectrum for the type of reactor presented in this module.
ENABLING OBJECTIVES
4.1 STATE the average energy at which prompt neutrons are produced.
4.2 DESCRIBE the neutron energy spectrum in the following reactors:
a. Fast reactor
b. Thermal reactor
4.3 EXPLAIN the reason for the particular shape of the fast, intermediate, and slow energy regions of the neutron flux spectrum for a thermal reactor.
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Reactor Theory (Neutron Characteristics) DOE-HDBK-1019/1-93 NEUTRON SOURCES
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NEUTRON SOURCES
Neutrons from a variety of sources are always present in a reactor core. This is true even when the reactor is shut down. Some of these neutrons are produced by naturally occurring (intrinsic) neutron sources, while others may be the result of fabricated (installed) neutron sources that are incorporated into the design of the reactor. The neutrons produced by sources other than neutron-induced fission are often grouped together and classified as source neutrons.
EO 1.1 DEFINE the following terms:
a. Intrinsic neutron source
b. Installed neutron source
EO 1.2 LIST three examples of reactions that produce neutrons in intrinsic neutron sources.
EO 1.3 LIST three examples of reactions that produce neutrons in installed neutron sources.
Neutron Sources
In addition to neutron-induced fission, neutrons are produced by other reactions. The neutrons produced by reactions other than neutron-induced fission are called source neutrons. Source

allow a visible indication of neutron level on the most sensitive monitoring instruments while the reactor is shutdown and during the startup sequence. This verifies instrument operability and allows monitoring of neutron population changes. Source neutrons can be classified as either intrinsic or installed neutron sources.

Intrinsic Neutron Sources

Some neutrons will be produced in the materials present in the reactor due to a variety of unavoidable reactions that occur because of the nature of these materials. Intrinsic neutron sources are those neutron-producing reactions that always occur in reactor materials.

■ ■ ■ ■
■ ■ ■
■ ■

NEUTRON SOURCES DOE-HDBK-1019/1-93 Reactor Theory (Neutron Characteristics)
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A limited number of neutrons will always be present, even in a reactor core that has never been operated, due to spontaneous fission of some heavy nuclides that are present in the fuel. Uranium■238, ur of the rate at which different heavy nuclides produce neutrons by spontaneous fission.

Californium-252 is not an intrinsic neutron source, but will be discussed in the section on installed neutron sources.

TABLE 1

Neutron Production by Spontaneous Fission

Nuclide T (Fission) T (1/2 1/2 ■-decay) neutrons/sec/gram

Fixture checkpoint marker-0354-xray is recorded on fixture page 0354.

U 1.8 x 10 years 6.8 x 10 years 8.0 x 10 235
92
17 8 -4

U 8.0 x 10 years 4.5 x 10 years 1.6 x 10 238
92
15 9 -2

Pu 5.5 x 10 years 2.4 x 10 years 3.0 x 10 239
94
5 4 -2

Pu 1.2 x 10 years 6.6 x 10 years 1.0 x 10 240
94
11 3 3

Cf 66.0 years 2.65 years 2.3 x 10 252
98
12

Another intrinsic neutron source is a reaction involving natural boron and fuel. In some reactors, natural boron is loaded into the reactor core as a neutron absorber to improve reactor control or increase core life-time. Boron-11 (80.1% of natural boron) undergoes a reaction with the alpha particle emitted by the radioactive decay of heavy nuclides in the fuel to yield a neutron as shown below.

The boron-11 must be mixed with, or in very close proximity to, the fuel for this reaction because of the short path length of the alpha particle. For a reactor core with this configuration, this (■,n) reaction is an important source of neutrons for reactor startup.

In a reactor that has been operated, another source of neutrons becomes significant. Neutrons may be produced by the interaction of a gamma ray and a deuterium nucleus. This reaction is commonly referred to as a photoneutron reaction because it is initiated by electromagnetic radiation and results

■ ■ ■
■ ■
■ ■ ■
■ ■

There is an abundant supply of high energy gammas in a reactor that has been operated because many of the fission products are gamma emitters. All water-cooled reactors have some deuterium present in the coolant in the reactor core because a small fraction of natural hydrogen is the isotope deuterium. The atom percentage of deuterium in the water ranges from close to the naturally occurring value (0.015%) for light water reactors to above 90% deuterium for heavy water reactors. Therefore, the required conditions for production of photoneutrons exist.

The supply of gamma rays decreases with time after shutdown as the gamma emitters decay; therefore, the photoneutron production rate also decreases. In a few particular reactors, additional D O (heavy water) may be added to the reactor to increase the production of 2 photoneutrons following a long shutdown period.

Installed Neutron Sources

Because intrinsic neutron sources can be relatively weak or dependent upon the recent power history of the reactor, many reactors have artificial sources of neutrons installed. These neutron sources ensure that shutdown neutron levels are high enough to be detected by the nuclear instruments at all times. This provides a true picture of reactor conditions and any change in these conditions. An installed neutron source is an assembly placed in or near the reactor for the sole purpose of producing source neutrons.

One strong source of neutrons is the artificial nuclide californium-252, which emits neutrons at the rate of about 2 x 10 neutrons per second per gram as the result of spontaneous fission. 12 Important drawbacks for some applications may be its high cost and its short half-life (2.65 years).

Many installed neutron sources use the (■,n) reaction with beryllium. These sources are
Fixture checkpoint marker-0355-xray is recorded on fixture page 0355.
composed of a mixture of metallic beryllium (100% beryllium-9) with a small quantity of an alpha particle emitter, such as a compound of radium, polonium, or plutonium. The reaction that occurs is shown below.

The beryllium is intimately (homogeneously) mixed with the alpha emitter and is usually enclosed in a stainless steel capsule.

Another type of installed neutron source that is widely used is a photoneutron source that employs the (■,n) reaction with beryllium. Beryllium is used for photoneutron sources because its stable isotope beryllium-9 has a weakly attached last neutron with a binding energy of only 1.66 MeV. Thus, a gamma ray with greater energy than 1.66 MeV can cause neutrons to be ejected by the (■,n) reaction as shown below.

Neutron Sources Summary

& Intrinsic neutron sources are sources of neutrons from materials that are in the reactor for other purposes such as fuel, burnable poison, or moderator.

& Installed neutron sources are materials or components placed in the reactor specifically for the purpose of producing source neutrons.

& Examples of intrinsic neutron sources are listed below.

Spontaneous fission of heavy nuclides in fuel, such as uranium-238,

free neutrons.
Boron-11 mixed with the fuel undergoes an alpha-neutron reaction
and becomes nitrogen-14.
Deuterium present in the reactor coolant undergoes a gamma■neutron reaction and becomes hydrogen-1.
& Examples of installed neutron sources are listed below.
Spontaneous fission of californium-252 results in fission fragments
and free neutrons.
Beryllium-9 undergoes an alpha-neutron reaction (alpha from the
decay of plutonium, polonium, or radium) and becomes carbon-12.
Beryllium-9 undergoes a gamma-neutron reaction (high energy
gamma from decay of antimony-124) and becomes beryllium-8.
Many startup sources of this type use antimony and beryllium because after activation with neutrons the radioactive antimony becomes an emitter of high energy gammas. The photoneutron sources of this type are constructed somewhat differently from the (■,n) types. One design incorporates a capsule of irradiated antimony enclosed in a beryllium sleeve. The entire assembly is then encased in a stainless steel cladding. A large reactor may have several neutron sources of this type installed within the core.
Summary
The important information in this chapter is summarized below.
DOE-HDBK-1019/1-93
Reactor Theory (Neutron Characteristics) NUCLEAR CROSS SECTIONS AND NEUTRON FLUX
Rev. 0 Page 5 NP-02
NUCLEAR CROSS SECTIONS AND NEUTRON FLUX
To determine the frequency of neutron interactions, it is necessary to describe the availability of neutrons to cause interaction and the probability of a neutron interacting with material. The availability of neutrons and the probability of interaction are quantified by the neutron flux and nuclear cross section.
EO 2.1 DEFINE the following terms:
a. Atom density d. Barn
b. Neutron flux e. Macroscopic cross section
c. Microscopic cross section f. Mean free path
EO 2.2 EXPRESS macroscopic cross section in terms of microscopic cross section.
EO 2.3 DESCRIBE how the absorption cross section of typical nuclides varies with neutron energy at energies below the resonance
Fixture checkpoint marker-0356-xray is recorded on fixture page 0356.
absorption region.
EO 2.4 DESCRIBE the cause of resonance absorption in terms of nuclear energy levels.
EO 2.5 DESCRIBE the energy dependence of resonance absorption peaks for typical light and heavy nuclei.
EO 2.6 EXPRESS mean free path in terms of macroscopic cross section.
EO 2.7 Given the number densities (or total density and component fractions) and microscopic cross sections of components, CALCULATE the macroscopic cross section for a mixture.
EO 2.8 CALCULATE a macroscopic cross section given a material density, atomic mass, and microscopic cross section.
EO 2.9 EXPLAIN neutron shadowing or self-shielding.

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Introduction

Fission neutrons are born with an average energy of about 2 MeV. These fast neutrons interact with the reactor core materials in various absorption and scattering reactions. Collisions that result in scattering are useful in slowing neutrons to thermal energies. Thermal neutrons may be absorbed by fissile nuclei to produce more fissions or be absorbed in fertile material for conversion to fissionable fuel. Absorption of neutrons in structural components, coolant, and other non-fuel material results in the removal of neutrons without fulfilling any useful purpose. To safely and efficiently operate a nuclear reactor it is necessary to predict the probability that a particular absorption or scattering reaction will occur. Once these probabilities are known, if

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the availability of neutrons can be determined, then the rate at which these nuclear reactions take place can be predicted.

Atom Density

One important property of a material is the atom density. The atom density is the number of atoms of a given type per unit volume of the material. To calculate the atom density of a substance use Equation (2-1).

(2-1)

where:

N = atom density (atoms/cm)3

' = density (g/cm)3

N = Avogadro's number (6.022 x 10 atoms/mole) A

23

M = gram atomic weight

N

NA

M

2.699 g

cm3

6.022 x 1023 atoms

mole

26.9815 g

mole

6.024 x 1022 atoms

cm3

Example:

A block of aluminum has a density of 2.699 g/cm . If the gram atomic weight of

3

aluminum is 26.9815 g, calculate the atom density of the aluminum.

Solution:

Cross Sections

The probability of a neutron interacting with a nucleus for a particular reaction is dependent upon not only the kind of nucleus involved, but also the energy of the neutron. Accordingly,

of a fast neutron. Also, the probability of interaction will vary depending upon the type of reaction involved.

The probability of a particular reaction occurring between a neutron and a nucleus is called the microscopic cross section ( ) of the nucleus for the particular reaction. This cross section will vary with the energy of the neutron. The microscopic cross section may also be regarded as the effective area the nucleus presents to the neutron for the particular reaction. The larger the effective area, the greater the probability for reaction.

Because the microscopic cross section is an area, it is expressed in units of area, or square centimeters. A square centimeter is tremendously large in comparison to the effective area of a nucleus, and it has been suggested that a physicist once referred to the measure of a square centimeter as being "as big as a barn" when applied to nuclear processes. The name has persisted and microscopic cross sections are expressed in terms of barns. The relationship between barns and cm is shown below. 2

1 barn = 10-24 cm2

N

Whether a neutron will interact with a certain volume of material depends not only on the

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microscopic cross section of the individual nuclei but also on the number of nuclei within that volume. Therefore, it is necessary to define another kind of cross section known as the macroscopic cross section ( ). The macroscopic cross section is the probability of a given reaction occurring per unit travel of the neutron. is related to the microscopic cross section ( ) by the relationship shown below.

(2-2)

where:

= macroscopic cross section (cm)

-1

N = atom density of material (atoms/cm)

3

= microscopic cross-section (cm)

2

The difference between the microscopic and macroscopic cross sections is extremely important and is restated for clarity. The microscopic cross section ( ) represents the effective target area that a single nucleus presents to a bombarding particle. The units are given in barns or cm .

2

The macroscopic cross section ( ) represents the effective target area that is presented by all of the nuclei contained in 1 cm of the material. The units are given as 1/cm or cm .

3 -1

A neutron interacts with an atom of the material it enters in two basic ways. It will either interact through a scattering interaction or through an absorption reaction. The probability of a neutron being absorbed by a particular atom is the microscopic cross section for absorption, . The probability of a neutron scattering off of a particular nucleus is the microscopic cross a section for scattering, s

. The sum of the microscopic cross section for absorption and the microscopic cross section for scattering is the total microscopic cross section, . T

= + T a s

instance, the scattering cross section is the sum of the elastic scattering cross section (se) and the inelastic scattering cross section ( ). si

= + s se si

The microscopic absorption cross section ( ) includes all reactions except scattering. However, a for most purposes it is sufficient to merely separate it into two categories, fission (f) and capture ( ). Radiative capture of neutrons was described in the Neutron Interactions chapter c of Module 1.

= + a f c

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Reactor Theory (Neutron Characteristics) NUCLEAR CROSS SECTIONS AND NEUTRON FLUX

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Figure 1 Typical Neutron Absorption Cross Section vs. Neutron Energy

The variation of absorption cross sections with neutron energy is often complicated. For many elements the absorption cross sections are small, ranging from a fraction of a barn to a few barns for slow (or thermal) neutrons.

For a considerable number of nuclides of moderately high (or high) mass numbers, an examination of the variation of the absorption cross section with the energy of the incident neutron reveals the existence of three regions on a curve of absorption cross section versus neutron energy. This cross section is illustrated in Figure 1. First, the cross section decreases steadily with increasing neutron energy in a low energy region, which includes the thermal range (E < 1 eV). In this region the absorption cross section, which is often high, is inversely proportional to the velocity (v). This region is frequently referred to as the "1/v region," because the absorption cross section is proportional to 1/v, which is the reciprocal of neutron velocity. Following the 1/v region, there occurs the "resonance region" in which the cross sections rise sharply to high values called "resonance peaks" for neutrons of certain energies, and then fall again. These energies are called resonance energies and are a result of the affinity

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of the nucleus for neutrons whose energies closely match its discrete, quantum energy levels. That is, when the binding energy of a neutron plus the kinetic energy of the neutron are exactly equal to the amount required to raise a compound nucleus from its ground state to a quantum level, resonance absorption occurs. The following example problem further illustrates this point.

■

(

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NUCLEAR CROSS SECTIONS AND NEUTRON FLUX Reactor Theory (Neutron Characteristics)

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Assuming that uranium-236 has a nuclear quantum energy level at 6.8 MeV above its ground state, calculate the kinetic energy a neutron must possess to undergo resonant absorption in uranium-235 at this resonance energy level.

BE = [Mass(U) + Mass(neutron) - Mass(U)] x 931 MeV/amu 235 236

BE = (235.043925 + 1.008665 - 236.045563) x 931 MeV/amu

BE = (0.007025 amu) x 931 MeV/amu = 6.54 MeV

6.8 MeV - 6.54 MeV = 0.26 MeV

The difference between the binding energy and the quantum energy level equals the amount of kinetic energy the neutron must possess. The typical heavy nucleus will have many closely■spaced resonanc complex and have more possible configurations and corresponding energy states. Light nuclei, being less complex, have fewer possible energy states and fewer resonances that are sparsely distributed at higher energy levels.

neutron increases. This is called the "fast neutron region." In this region the absorption cross sections are usually less than 10 barns.

With the exception of hydrogen, for which the value is fairly large, the elastic scattering cross sections are generally small, for example, 5 barns to 10 barns. This is close to the magnitude of the actual geometric cross sectional area expected for atomic nuclei. In potential scattering, the cross section is essentially constant and independent of neutron energy. Resonance elastic scattering and inelastic scattering exhibit resonance peaks similar to those associated with absorption cross sections. The resonances occur at lower energies for heavy nuclei than for light nuclei. In general, the variations in scattering cross sections are very small when compared to the variations that occur in absorption cross sections.

Mean Free Path

If a neutron has a certain probability of undergoing a particular interaction in one centimeter of travel, then the inverse of this value describes how far the neutron will travel (in the average case) before undergoing an interaction. This average distance traveled by a neutron before interaction is known as the mean free path for that interaction and is represented by the symbol ■. The relationship between the mean free path (■) and the macroscopic cross section (*) is shown below.

(2-3)

'

(

)

Calculation of Macroscopic Cross Section and Mean Free Path

Most materials are composed of several elements, and because most elements are composed of several isotopes, most materials involve many cross sections, one for each isotope involved. Therefore, to include all the isotopes within a given material, it is necessary to determine the macroscopic cross section for each isotope and then sum all the individual macroscopic cross sections. Equation (2-4) can be used to determine the macroscopic cross section for a composite material.

* = N) + N) + N) + ... N) (2-4) 1 1 2 2 3 3 nn

where:

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N = the number of nuclei per cm of the n element n

3 th

) = the microscopic cross section of the n element n

th

The following example problems illustrate the calculation of the macroscopic cross section for a single element and for combinations of materials.

Example 1:

Find the macroscopic thermal neutron absorption cross section for iron, which has a density of 7.86 g/cm . The microscopic cross section for absorption of iron is 2.56 barns 3 and the gram atomic weight is 55.847 g.

Solution:

Step 1: Using Equation (2-1), calculate the atom density of iron.

Step 2: Use this atom density in Equation (2-2) to calculate the macroscopic cross section.

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'

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Example 2:
An alloy is composed of 95% aluminum and 5% silicon (by weight). The density of the alloy is 2.66 g/cm . Properties of aluminum and silicon are shown below. 3

Element Gram Atomic Weight)) a (barns) s (barns)
Aluminum 26.9815 0.23 1.49
Silicon 28.0855 0.16 2.20

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1. Calculate the atom densities for the aluminum and silicon.
2. Determine the absorption and scattering macroscopic cross sections for thermal neutrons.
3. Calculate the mean free paths for absorption and scattering.

Solution:
Step 1: The density of the aluminum will be 95% of the total density. Using Equation (2-1) yields the atom densities.

a NAl a,Al NSi a,Si
5.64 x 1022 atoms
cm3
0.23 x 10 24 cm22.85 x 1021 atoms
cm3
0.16 x 10 24 cm2
0.0134 cm 1
s NAl s,Al NSi s,Si
5.64 x 1022 atoms
cm3
1.49 x 10 24 cm22.85 x 1021 atoms
cm3
2.20 x 10 24 cm2
0.0903 cm 1
a
1
a
1
0.01345 cm 1
74.3 cm
s
1
s
1
0.0903 cm 1
11.1 cm

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Step 2: The macroscopic cross sections for absorption and scattering are

Step 3: The mean free paths are calculated by inserting the macroscopic cross sections calculated above into Equation (2-3).
Thus, a neutron must travel an average of 74.3 cm to interact by absorption in this alloy, but it must travel only 11.1 cm to interact by scattering.
)
)
)
)
(■
(■
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Effects of Temperature on Cross Section
As discussed, the microscopic absorption cross section varies significantly as neutron energy varies. The microscopic cross sections provided on most charts and tables are measured for a standard neutron velocity of 2200 meters/second, which corresponds to an ambient temperature
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of 68(F. Therefore, if our material is at a higher temperature, the absorption cross section will be lower than the value for 68(F, and any cross sections which involve absorption (for example,) ,) ,)) must be corrected for the existing temperature. acf
The following formula is used to correct microscopic cross sections for temperature. Although the example illustrates absorption cross section, the same formula may be used to correct capture and fission cross sections.
where:
) = microscopic cross section corrected for temperature
) = microscopic cross section at reference temperature (68(F or 20(C) o
T = reference temperature (68(F) in degrees Rankine ((R) or Kelvin ((K) o
T = temperature for which corrected value is being calculated
NOTE: When using this formula, all temperatures must be converted to (R or (K.
(R = (F + 460
(K = (C + 273
Example:
What is the value of)f for uranium-235 for thermal neutrons at 500(F? Uranium-235 has a) of 583 barns at 68(F. f
Solution:
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Neutron Flux
Macroscopic cross sections for neutron reactions with materials determine the probability of one neutron undergoing a specific reaction per centimeter of travel through that material. If one wants to determine how many reactions will actually occur, it is necessary to know how many neutrons are traveling through the material and how many centimeters they travel each second. It is convenient to consider the number of neutrons existing in one cubic centimeter at any one instant and the total distance they travel each second while in that cubic centimeter. The number of neutrons existing in a cm of material at any instant is called neutron density and is 3 represented by the symbol n with units of neutrons/cm . The total distance these neutrons can 3

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A good way of defining neutron flux (1) is to consider it to be the total path length covered by all neutrons in one cubic centimeter during one second. Mathematically, this is the equation below.
1 = n v (2-5)
where:
1 = neutron flux (neutrons/cm -sec) 2
n = neutron density (neutrons/cm)3
v = neutron velocity (cm/sec)
The term neutron flux in some applications (for example, cross section measurement) is used as parallel beams of neutrons traveling in a single direction. The intensity (I) of a neutron beam is the product of the neutron density times the average neutron velocity. The directional beam intensity is equal to the number of neutrons per unit area and time (neutrons/cm -sec) falling on 2 a surface perpendicular to the direction of the beam.
One can think of the neutron flux in a reactor as being comprised of many neutron beams traveling in various directions. Then, the neutron flux becomes the scalar sum of these directional flux intensities (added as numbers and not vectors), that is, 1 = I + I + I +...I . 123 n
Since the atoms in a reactor do not interact preferentially with neutrons from any particular direction, all of these directional beams contribute to the total rate of reaction. In reality, at a given point within a reactor, neutrons will be traveling in all directions.
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Self-Shielding
In some locations within the reactor, the flux level may be significantly lower than in other areas
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due to a phenomenon referred to as neutron shadowing or self-shielding. For example, the interior of a fuel pin or pellet will "see" a lower average flux level than the outer surfaces since an appreciable fraction of the neutrons will have been absorbed and therefore cannot reach the interior of the fuel pin. This is especially important at resonance energies, where the absorption cross sections are large.
Summary
The important information in this chapter is summarized below.
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Reactor Theory (Neutron Characteristics) NUCLEAR CROSS SECTIONS AND NEUTRON FLUX
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Nuclear Cross Section and Neutron Flux Summary
& Atom density (N) is the number of atoms of a given type per unit volume of material.
& Microscopic cross section ()) is the probability of a given reaction occurring between a neutron and a nucleus.
& Microscopic cross sections are measured in units of barns, where 1 barn = 10-24 cm .2
& Macroscopic cross section (*) is the probability of a given reaction occurring per unit length of travel of the neutron. The units for macroscopic cross section are cm . -1
& The mean free path (■) is the average distance that a neutron travels in a material between interactions.
& Neutron flux (1) is the total path length traveled by all neutrons in one cubic

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& The macroscopic cross section for a material can be calculated using the equation below.
* = N)
& The absorption cross section for a material usually has three distinct regions. At low neutron energies (<1 eV) the cross section is inversely proportional to the neutron velocity.
& Resonance absorption occurs when the sum of the kinetic energy of the neutron and its binding energy is equal to an allowed nuclear energy level of the nucleus.
& Resonance peaks exist at intermediate energy levels. For higher neutron energies, the absorption cross section steadily decreases as the neutron energy increases.
& The mean free path equals 1/*.
& The macroscopic cross section for a mixture of materials can be calculated using the equation below.
* = N 1 1 2 2 3 3 nn) + N) + N) + ... N)
& Self-shielding is where the local neutron flux is depressed within a material due to neutron absorption near the surface of the material.
1)
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REACTION RATES
It is possible to determine the rate at which a nuclear reaction will take place based on the neutron flux, cross section for the interaction, and atom density of the target. This relationship illustrates how a change in one of these items affects the reaction rate.
EO 2.10 Given the neutron flux and macroscopic cross section, CALCULATE the reaction rate.
EO 2.11 DESCRIBE the relationship between neutron flux and reactor power.
Reaction Rates
If the total path length of all the neutrons in a cubic centimeter in a second is known, (neutron flux (1)), and if the probability of having an interaction per centimeter path length is also known
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(macroscopic cross section (*)), multiply them together to get the number of interactions taking place in that cubic centimeter in one second. This value is known as the reaction rate and is denoted by the symbol R. The reaction rate can be calculated by the equation shown below.
R = 1 * (2-6)
where:
R = reaction rate (reactions/sec)
1 = neutron flux (neutrons/cm -sec) 2
* = macroscopic cross section (cm) -1
Substituting the fact that * = N) into Equation (2-6) yields the equation below.
where:
) = microscopic cross section (cm)2
N = atom density (atoms/cm)3
Rf f
1 x 1013 neutrons
cm2
sec

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1 x 1012 fissions
cm3
sec
Rf f
f
Rf
1.29 x 1012 fissions
cm3
sec
3 x 1013 neutrons
cm2
sec
0.043 cm 1
Reactor Theory (Neutron Characteristics) DOE-HDBK-1019/1-93 REACTION RATES
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The reaction rate calculated will depend on which macroscopic cross section is used in the calculation. Normally, the reaction rate of greatest interest is the fission reaction rate.
Example:
If a one cubic centimeter section of a reactor has a macroscopic fission cross section of 0.1 cm , and if the thermal neutron flux is 10 neutrons/cm -sec, what is the fission rate -1 13 2 in that cubic centimeter?
Solution:
In addition to using Equation (2-6) to determine the reaction rate based on the physical properties of material, it is also possible to algebraically manipulate the equation to determine physical properties if the reaction rate is known.
Example:
A reactor operating at a flux level of 3 x 10 neutrons/cm -sec contains 10 atoms of 13 2 20
uranium-235 per cm . The reaction rate is 1.29 x 10 fission/cm . Calculate and .
3 12 3
f f
Solution:
Step 1: The macroscopic cross section can be determined by solving
Equation (2-6) for and substituting the appropriate values. f
Rf N f
f
Rf
N
Fixture checkpoint marker-0365-xray is recorded on fixture page 0365.
1.29 x 1012 fissions
cm3
sec
1 x 1020 atoms
cm3
3 x 1013 neutrons
cm2
sec
4.3 x 10 22 cm2 1 barn

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430 barns
1 watt 1 erg
1 x 10 7 watt sec
1 MeV
1.602 x 10 6
erg
1 fission
200 MeV
3.12 x 1010 fissions
second
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Step 2: To find the microscopic cross section, replace f with (N x f) and solve
for .
f
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Reactor Power Calculation
Multiplying the reaction rate per unit volume by the total volume of the core results in the total number of reactions occurring in the core per unit time. If the amount of energy involved in each reaction were known, it would be possible to determine the rate of energy release (power) due to a certain reaction.
In a reactor where the average energy per fission is 200 MeV, it is possible to determine the number of fissions per second that are necessary to produce one watt of power using the following conversion factors.
1 fission = 200 MeV
1 MeV = 1.602 x 10 ergs -6
1 erg = 1 x 10 watt-sec -7
This is equivalent to stating that 3.12 x 10 fissions release 1 watt-second of energy. 10
P
th f V
3.12 x 1010 fissions
watt sec
Reactor Theory (Neutron Characteristics) DOE-HDBK-1019/1-93 REACTION RATES
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The power released in a reactor can be calculated based on Equation (2-6). Multiplying the reaction rate by the volume of the reactor results in the total fission rate for the entire reactor. Dividing by the number of fissions per watt-sec results in the power released by fission in the reactor in units of watts. This relationship is shown mathematically in Equation (2-7) below.
(2-7)
where:
P = power (watts)
= thermal neutron flux (neutrons/cm -sec) th
2
= macroscopic cross section for fission (cm) f
-1
V = volume of core (cm)
3
Relationship Between Neutron Flux and Reactor Power

time (days or weeks), the number density of the fuel atoms is also relatively constant. Since the atom density and microscopic cross section are constant, the macroscopic cross section must also be constant. Examining Equation (2-7), it is apparent that if the reactor volume and macroscopic cross section are constant, then the reactor power and the neutron flux are directly proportional. This is true for day-to-day operation. The neutron flux for a given power level will increase very slowly over a period of months due to the burnup of the fuel and resulting decrease in atom density and macroscopic cross section.

Reaction Rates Summary

& The reaction rate is the number of interactions of a particular type occurring in a cubic centimeter of material in a second.

& The reaction rate can be calculated by the equation below.

R = 1 *

& Over a period of several days, while the atom density of the fuel can be considered constant, the neutron flux is directly proportional to reactor power.

Summary

The important information in this chapter is summarized below.

NEUTRON MODERATION

In thermal reactors, the neutrons that cause fission are at a much lower energy

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than the energy level at which they were born from fission. In this type of reactor, specific materials must be included in the reactor design to reduce the energy level of the neutrons in an efficient manner.

EO 2.12 DEFINE the following concepts:

a. Thermalization
b. Moderator
c. Moderating ratio
d. Average logarithmic energy decrement
e. Macroscopic slowing down power

EO 2.13 LIST three desirable characteristics of a moderator.

EO 2.14 Given an average fractional energy loss per collision, CALCULATE the energy loss after a specified number of collisions.

Neutron Slowing Down and Thermalization

Fission neutrons are produced at an average energy level of 2 MeV and immediately begin to slow down as the result of numerous scattering reactions with a variety of target nuclei. After a number of collisions with nuclei, the speed of a neutron is reduced to such an extent that it has approximately the same average kinetic energy as the atoms (or molecules) of the medium in which the neutron is undergoing elastic scattering. This energy, which is only a small fraction of an electron volt at ordinary temperatures (0.025 eV at 20(C), is frequently referred to as the thermal energy, since it depends upon the temperature. Neutrons whose energies have been reduced to values in this region (< 1 eV) are designated thermal neutrons. The process of reducing the energy of a neutron to the thermal region by elastic scattering is referred to as thermalization, slowing down, or moderation. The material used for the purpose of thermalizing neutrons is called a moderator. A good moderator reduces the speed of neutrons in a small number of collisions, but does not absorb them to any great extent. Slowing the neutrons in as few collisions as possible is desirable in order to reduce the amount of neutron leakage from the

leakage and resonance absorption will be discussed in the next module.
ln Ei
ln Ef
ln
Ei
Ef
2
A
2
3
NEUTRON MODERATION DOE-HDBK-1019/1-93 Reactor Theory (Neutron Characteristics)
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The ideal moderating material (moderator) should have the following nuclear properties.
large scattering cross section
small absorption cross section
large energy loss per collision
A convenient measure of energy loss per collision is the logarithmic energy decrement. The average logarithmic energy decrement is the average decrease per collision in the logarithm of the neutron energy. This quantity is represented by the symbol (Greek letter xi).
(2-8)
where:
= average logarithmic energy decrement
Ei = average initial neutron energy
Ef = average final neutron energy
The symbol is commonly called the average logarithmic energy decrement because of the fact that a neutron loses, on the average, a fixed fraction of its energy per scattering collision. Since the fraction of energy retained by a neutron in a single elastic collision is a constant for a given material, is also a constant. Because it is a constant for each type of material and does not depend upon the initial neutron energy, is a convenient quantity for assessing the moderating
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ability of a material.
The values for the lighter nuclei are tabulated in a variety of sources. The following commonly used approximation may be used when a tabulated value is not available.
This approximation is relatively accurate for mass numbers (A) greater than 10, but for some low values of A it may be in error by over three percent.
N
ln Ehigh ln Elow
ln
Ehigh
Elow
N
ln
Ehigh
Elow
ln
2 x 106eV
0.025 eV
0.948

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Since represents the average logarithmic energy loss per collision, the total number of collisions necessary for a neutron to lose a given amount of energy may be determined by dividing into the difference of the natural logarithms of the energy range in question. The number of collisions (N) to travel from any energy, E , to any lower energy, E , can be calculated as shown below. high low
Example:
How many collisions are required to slow a neutron from an energy of 2 MeV to a thermal energy of 0.025 eV, using water as the moderator? Water has a value of 0.948 for .
Solution:
Sometimes it is convenient, based upon information known, to work with an average fractional energy loss per collision as opposed to an average logarithmic fraction. If the initial neutron energy level and the average fractional energy loss per collision are known, the final energy level for a given number of collisions may be computed using the following formula.
! (
NEUTRON MODERATION DOE-HDBK-1019/1-93 Reactor Theory (Neutron Characteristics)
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(2-9)
where:
E = initial neutron energy o
E = neutron energy after N collisions N
x = average fractional energy loss per collision
N = number of collisions
Example:
If the average fractional energy loss per collision in hydrogen is 0.63, what will be the energy of a 2 MeV neutron after (a) 5 collisions? (b) 10 collisions?
Solution:
a)
b)
Macroscopic Slowing Down Power
Although the logarithmic energy decrement is a convenient measure of the ability of a material to slow neutrons, it does not measure all necessary properties of a moderator. A better measure of the capabilities of a material is the macroscopic slowing down power. The macroscopic slowing down power (MSDP) is the product of the logarithmic energy decrement and the macroscopic cross section for scattering in the material. Equation (2-10) illustrates how to
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calculate the macroscopic slowing down power.
(2-10)
MR
s
a
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Moderating Ratio
Macroscopic slowing down power indicates how rapidly a neutron will slow down in the material in question, but it still does not fully explain the effectiveness of the material as a moderator. An element such as boron has a high logarithmic energy decrement and a good slowing down power,

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The most complete measure of the effectiveness of a moderator is the moderating ratio. The moderating ratio is the ratio of the macroscopic slowing down power to the macroscopic cross section for absorption. The higher the moderating ratio, the more effectively the material performs as a moderator. Equation (2-11) shows how to calculate the moderating ratio of a material.
(2-11)
Moderating properties of different materials are compared in Table 2.
TABLE 2
Moderating Properties of Materials
Material Collisions to Slowing Down
Number of Macroscopic
Thermalize Power
Moderating
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Ratio
H2O 0.927 19 1.425 62
D2O 0.510 35 0.177 4830
Helium 0.427 42 9 x 10 51 -6
Beryllium 0.207 86 0.154 126
Boron 0.171 105 0.092 0.00086
Carbon 0.158 114 0.083 216
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Summary
The important information in this chapter is summarized below.
Neutron Moderation Summary
& Thermalization is the process of reducing the energy level of a neutron from the energy level at which it is produced to an energy level in the thermal range.
& The moderator is the reactor material that is present for the purpose of thermalizing neutrons.
& Moderating ratio is the ratio of the macroscopic slowing down power to the macroscopic cross section for absorption.
& The average logarithmic energy decrement (!) is the average change in the logarithm of neutron energy per collision.
& Macroscopic slowing down power is the product of the average logarithmic energy decrement and the macroscopic cross section for scattering.
& There are three desirable characteristics of a moderator.
1. large scattering cross section
2. small absorption cross section
3. large energy loss per collision
& The energy loss after a specified number of collisions can be calculated using the equation below.
E = E (1 - x) N o
N
■
■ ■
Reactor Theory (Neutron Characteristics) DOE-HDBK-1019/1-93 PROMPT AND DELAYED NEUTRONS
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Not all neutrons are released at the same time following fission. Most neutrons are released virtually instantaneously and are called prompt neutrons. A very small fraction of neutrons are released after the decay of fission products and are called delayed neutrons. Although delayed neutrons are a very small fraction of the total number of neutrons, they play an extremely important role in the control of the reactor.

EO 3.1 STATE the origin of prompt neutrons and delayed neutrons.

EO 3.2 STATE the approximate fraction of neutrons that are born as delayed neutrons from the fission of the following nuclear fuels:

a. Uranium-235

b. Plutonium-239

EO 3.3 EXPLAIN the mechanism for production of delayed neutrons.

EO 3.4 EXPLAIN prompt and delayed neutron generation times.

EO 3.5 Given prompt and delayed neutron generation times and delayed neutron fraction, CALCULATE the average generation time.

EO 3.6 EXPLAIN the effect of delayed neutrons on reactor control.

Neutron Classification

The great majority (over 99%) of the neutrons produced in fission are released within about 10-13 seconds of the actual fission event. These are called prompt neutrons. A small portion of fission

Fixture checkpoint marker-0371-xray is recorded on fixture page 0371.

neutrons are delayed neutrons, which are produced for some time after the fission process has taken place. The delayed neutrons are emitted immediately following the first beta decay of a fission fragment known as a delayed neutron precursor. An example of a delayed neutron precursor is bromine-87, shown below.

For most applications, it is convenient to combine the known precursors into groups with appropriately averaged properties. These groups vary somewhat depending on the fissile material in use. Table 3 lists the characteristics for the six precursor groups resulting from thermal fission of uranium-235. The fraction of all neutrons that are produced by each of these precursors is called the delayed neutron fraction for that precursor. The total fraction of all neutrons born as delayed neutrons is called the delayed neutron fraction (■). The fraction of delayed neutrons produced varies depending on the predominant fissile nuclide in use. The delayed neutron fractions (■) for the fissile nuclides of most interest are as follows: uranium-233 (0.0026), uranium-235 (0.0065), uranium-238 (0.0148), and plutonium-239 (0.0021).

TABLE 3

Delayed Neutron Precursor Groups
for Thermal Fission in Uranium-235

Group	Half-Life (sec)	Delayed Neutron Fraction	Average Energy (MeV)
1	55.7	0.00021	0.25
2	22.7	0.00142	0.46
3	6.2	0.00127	0.41
4	2.3	0.00257	0.45
5	0.61	0.00075	0.41
6	0.23	0.00027	-
Total	-	0.0065	-

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The neutron generation time is the time required for neutrons from one generation to cause the fissions that produce the next generation of neutrons. The generation time for prompt neutrons (5* - pronounced "ell-star") is the total time from birth to rebirth. Three time intervals are involved: (a) the time it takes a fast neutron to slow down to thermal energy, (b) the time the now thermal neutron exists prior to absorption in fuel, and (c) the time required for a fissionable nucleus to emit a fast neutron after neutron absorption.

Fast neutrons slow to thermal energies or leak out of the reactor in 10 seconds to 10 seconds, -4 -6 depending on the moderator. In water moderated reactors, thermal neutrons tend to exist for about 10 seconds before they are absorbed. Fission and fast neutron production following -4 neutron absorption in a fissionable nucleus occurs in about 10 seconds. Thus, fast reactors -13 have an 5* of about 10 seconds, while thermal reactors have an 5* of about 10 seconds + 10 -6 -6 -4 seconds, which is about 10 seconds to 10 seconds. -4 -5

■ ■ ■ ■

■ ■ ■ ■

■ ■

Reactor Theory (Neutron Characteristics) DOE-HDBK-1019/1-93 PROMPT AND DELAYED NEUTRONS Rev. 0 Page 31 NP-02

On the other hand, the average generation time for the six delayed neutron groups is the total time from the birth of the fast neutron to the emission of the delayed neutron. Again, three time intervals are involved: (a) the time it takes a fast neutron to slow down to thermal energy, (b) the time the thermal neutron exists prior to absorption, and (c) the average time from neutron absorption to neutron emission by the six precursor groups. The average time for decay of precursors from uranium-235 is 12.5 seconds. The other terms in the delayed neutron generation time are insignificant when compared to this value, and the average delayed neutron generation time becomes ■ 12.5 seconds.

A neutron generation time in the range of 10 seconds to 10 seconds or faster could result in -4 -5 very rapid power excursions, and control would not be possible without the dependence upon

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delayed neutrons to slow down the rate of the reaction. The average generation time, and hence the rate that power can rise, is determined largely by the delayed neutron generation time. The following equation shows this mathematically.

(2-12)

Example:

Assume a prompt neutron generation time for a particular reactor of 5 x 10 seconds and -5 a delayed neutron generation time of 12.5 seconds. If ■ is 0.0065, calculate the average generation time.

Solution:

This example demonstrates the effect delayed neutrons have on the neutron generation time and thus reactor control. If a reactor were to be operated in a sustained chain reaction using only prompt neutrons (■ = 0), the generation time from the previous example would be about 5 x 10 seconds. However, by operating the reactor such that a 0.0065 fraction of neutrons are -5 delayed, the generation life time is extended to 0.0813 seconds, providing time for adequate operator control. Therefore, although only a small fraction of the total neutron population, delayed neutrons are extremely important to the control and maintenance of a sustained fission chain reaction.

Summary

The important information in this chapter is summarized on the following page.

■ ■ ■ ■

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Prompt and Delayed Neutrons Summary
& Prompt neutrons are released directly from fission within 10 seconds of the -13 fission event.
& Delayed neutrons are released from the decay of fission products that are called delayed neutron precursors. Delayed neutron precursors are grouped according to half-life. Half-lives vary from fractions of a second to almost a minute.
& The fraction of neutrons born as delayed neutrons is different for different fuel materials. Following are values for some common fuel materials.
Uranium-235 0.0065
Plutonium-239 0.0021
& Delayed neutrons are produced by a classification of fission products known as delayed neutron precursors. When a delayed neutron precursor undergoes a ■ decay, it results in an excited daughter nucleus which immediately ejects a - neutron. Therefore, these delayed neutrons appear with a half-life of the delayed neutron precursor.
& The delayed neutron generation time is the total time from the birth of the fast neutron to the emission of the delayed neutron in the next generation. Delayed neutron generation times are dominated by the half-life of the delayed neutron precursor. The average delayed neutron generation time is about 12.5 seconds.
& A prompt neutron generation time is the sum of the amount of time it takes a fast neutron to thermalize, the amount of time the neutron exists as a thermal neutron before it is absorbed, and the amount of time between a fissionable nuclide absorbing a neutron and fission neutrons being released. Prompt neutron generation time is about 5 x 10 seconds. -5
& The average neutron generation time can be calculated from the prompt and delayed neutron generation times and the delayed neutron fraction using Equation (2-12).
& Delayed neutrons are responsible for the ability to control the rate at which power can rise in a reactor. If only prompt neutrons existed, reactor control would not be possible due to the rapid power changes.
Reactor Theory (Neutron Characteristics) DOE-HDBK-1019/1-93 NEUTRON FLUX SPECTRUM
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Figure 2 Prompt Fission Neutron Energy Spectrum for
Fixture checkpoint marker-0373-xray is recorded on fixture page 0373.
Thermal Fission of Uranium-235
NEUTRON FLUX SPECTRUM
The number of neutrons that exist at a given energy level varies. A plot of either the fraction of neutrons or the neutron flux at a given energy versus the energy level is called a neutron energy spectrum. The neutron energy spectrum varies widely for different types of reactors.
EO 4.1 STATE the average energy at which prompt neutrons are produced.
EO 4.2 DESCRIBE the neutron energy spectrum in the following reactors:
a. Fast reactor
b. Thermal reactor
EO 4.3 EXPLAIN the reason for the particular shape of the fast,

spectrum for a thermal reactor.
Prompt Neutron Energies
The neutrons produced by fission are high energy neutrons, and almost all fission neutrons have energies between 0.1 MeV and 10 MeV. The neutron energy distribution, or spectrum, may best be described by plotting the fraction of neutrons per MeV as a function of neutron energy, as
Fixture checkpoint marker-0374-xray is recorded on fixture page 0374.
shown in Figure 2. From this figure it can be seen that the most probable neutron energy is about 0.7 MeV. Also, from this data it can be shown that the average energy of fission neutrons is about 2 MeV. Figure 2 is the neutron energy spectrum for thermal fission in uranium-235. The values will vary slightly for other nuclides.
NEUTRON FLUX SPECTRUM DOE-HDBK-1019/1-93 Reactor Theory (Neutron Characteristics)
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Figure 3 Comparison of Neutron Flux Spectra for Thermal and Fast Breeder Reactor
Thermal and Fast Breeder Reactor Neutron Spectra
The spectrum of neutron energies produced by fission varies significantly from the energy spectrum, or flux, existing in a reactor at a given time. Figure 3 illustrates the difference in neutron flux spectra between a thermal reactor and a fast breeder reactor. The energy distribution of neutrons from fission is essentially the same for both reactors, so the differences in the curve shapes may be attributed to the neutron moderation or slowing down effects.
No attempt is made to thermalize or slow down neutrons in the fast breeder reactor (liquid metal cooled); therefore, an insignificant number of neutrons exist in the thermal range. For the thermal reactor (water moderated), the spectrum of neutrons in the fast region (> 0.1 MeV) has a shape similar to that for the spectrum of neutrons emitted by the fission process.
In the thermal reactor, the flux in the intermediate energy region (1 eV to 0.1 MeV) has approximately a 1/E dependence. That is, if the energy (E) is halved, the flux doubles. This 1/E dependence is caused by the slowing down process, where elastic collisions remove a constant fraction of the neutron energy per collision (on the average), independent of energy; thus, the neutron loses larger amounts of energy per collision at higher energies than at lower energies. The fact that the neutrons lose a constant fraction of energy per collision causes the neutrons to tend to "pile up" at lower energies, that is, a greater number of neutrons exist at the lower energies as a result of this behavior.
vp
2 k T
m

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2 k T
m
2 1.38 x 10 16 erg
K
293 K
1.66 x 10 24 g
2.2 x 105 cm
sec
1 m
100 cm
2200 m
sec
Reactor Theory (Neutron Characteristics) DOE-HDBK-1019/1-93 NEUTRON FLUX SPECTRUM
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In the thermal region the neutrons achieve a thermal equilibrium with the atoms of the moderator material. In any given collision they may gain or lose energy, and over successive collisions will gain as much energy as they lose. These thermal neutrons, even at a specific temperature, do not all have the same energy or velocity; there is a distribution of energies, usually referred
Fixture checkpoint marker-0375-xray is recorded on fixture page 0375.
to as the Maxwell distribution (e.g., Figure 2). The energies of most thermal neutrons lie close to the most probable energy, but there is a spread of neutrons above and below this value.
Most Probable Neutron Velocities
The most probable velocity (vp) of a thermal neutron is determined by the temperature of the medium and can be determined by Equation (2-13) .
(2-13)
where:
v = most probable velocity of neutron (cm/sec) p
k = Boltzman's constant (1.38 x 10 erg/ K) -16
T = absolute temperature in degrees Kelvin (K)
m = mass of neutron (1.66 x 10 grams) -24
Example:
Calculate the most probable velocities for neutrons in thermal equilibrium with their surroundings at the following temperatures. a) 20 C, b) 260 C.
Solution:
a) Calculate the most probable velocity for 20 C using Equation (2-13).
vp
2 k T
m
2 1.38 x 10 16 erg
K
533 K
1.66 x 10 24 g
2.977 x 105 cm
sec
1 m
100 cm
2977 m
sec

DOE-HDBK-1019/1-93
NUCLEAR PHYSICS AND REACTOR THEORY
ABSTRACT

The Nuclear Physics and Reactor Theory Handbook was developed to assist nuclear facility operating contractors in providing operators, maintenance personnel, and the technical staff with the necessary fundamentals training to ensure a basic understanding of nuclear physics and reactor theory. The handbook includes information on atomic and nuclear physics; neutron characteristics; reactor theory and nuclear parameters; and the theory of reactor operation. This information will provide personnel with a foundation for understanding the scientific principles that are associated with various DOE nuclear facility operations and maintenance.

Key Words: Training Material, Atomic Physics, The Chart of the Nuclides, Radioactivity, Radioactive Decay, Neutron Interaction, Fission, Reactor Theory, Neutron Characteristics, Neutron Life Cycle, Reactor Kinetics
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NUCLEAR PHYSICS AND REACTOR THEORY

FOREWORD

The Department of Energy (DOE) Fundamentals Handbooks consist of ten academic subjects, which include Mathematics; Classical Physics; Thermodynamics, Heat Transfer, and Fluid Flow; Instrumentation and Control; Electrical Science; Material Science; Mechanical Science; Chemistry; Engineering Symbology, Prints, and Drawings; and Nuclear Physics and Reactor Theory. The handbooks are provided as an aid to DOE nuclear facility contractors.

These handbooks were first published as Reactor Operator Fundamentals Manuals in 1985 for use by DOE category A reactors. The subject areas, subject matter content, and level of detail of the Reactor Operator Fundamentals Manuals were determined from several sources.

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DOE Category A reactor training managers determined which materials should be included, and served as a primary reference in the initial development phase. Training guidelines from the commercial nuclear power industry, results of job and task analyses, and independent input from contractors and operations-oriented personnel were all considered and included to some degree in developing the text material and learning objectives.

The DOE Fundamentals Handbooks represent the needs of various DOE nuclear facilities' fundamental training requirements. To increase their applicability to nonreactor nuclear facilities, the Reactor Operator Fundamentals Manual learning objectives were distributed to the Nuclear Facility Training Coordination Program Steering Committee for review and comment. To update their reactor-specific content, DOE Category A reactor training managers also reviewed and commented on the content. On the basis of feedback from these sources, information that applied to two or more DOE nuclear facilities was considered generic and was included. The final draft of each of the handbooks was then reviewed by these two groups. This approach has resulted in revised modular handbooks that contain sufficient detail such that each facility may adjust the content to fit their specific needs.

Each handbook contains an abstract, a foreword, an overview, learning objectives, and text material, and is divided into modules so that content and order may be modified by individual DOE contractors to suit their specific training needs. Each handbook is supported by a separate examination bank with an answer key.

The DOE Fundamentals Handbooks have been prepared for the Assistant Secretary for Nuclear Energy, Office of Nuclear Safety Policy and Standards, by the DOE Training Coordination Program. This program is managed by EG&G Idaho, Inc.

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NUCLEAR PHYSICS AND REACTOR THEORY
OVERVIEW
The Department of Energy Fundamentals Handbook entitled Nuclear Physics and Reactor Theory was prepared as an information resource for personnel who are responsible for the operation of the Department's nuclear facilities. Almost all processes that take place in a nuclear facility involves the transfer of some type of energy. A basic understanding of nuclear physics and reactor theory is necessary for DOE nuclear facility operators, maintenance personnel, and the technical staff to safely operate and maintain the facility and facility support systems. The information in this handbook is presented to provide a foundation for applying engineering concepts to the job. This knowledge will help personnel understand the impact that their actions may have on the safe and reliable operation of facility components and systems.
The Nuclear Physics and Reactor Theory handbook consists of four modules that are contained in two volumes. The following is a brief description of the information presented in each module of the handbook.
Volume 1 of 2
Module 1 - Atomic and Nuclear Physics
Introduces concepts of atomic physics including the atomic nature of matter, the chart of the nuclides, radioactivity and radioactive decay, neutron interactions and fission, and the interaction of radiation with matter.
Module 2 - Reactor Theory (Nuclear Parameters)
Provides information on reactor theory and neutron characteristics. Includes topics such as neutron sources, neutron flux, neutron cross sections, reaction rates, neutron moderation, and prompt and delayed neutrons.
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NUCLEAR PHYSICS AND REACTOR THEORY
OVERVIEW (Cont.)
Volume 2 of 2
Module 3 - Reactor Theory (Nuclear Parameters)
Explains the nuclear parameters associated with reactor theory. Topics include the neutron life cycle, reactivity and reactivity coefficients, neutron poisons, and
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control rods.
Module 4 - Reactor Theory (Reactor Operations)
Introduces the reactor operations aspect of reactor theory. Topics include subcritical multiplication, reactor kinetics, and reactor operation.
The information contained in this handbook is not all-encompassing. An attempt to present the entire subject of nuclear physics and reactor theory would be impractical. However, the Nuclear Physics and Reactor Theory handbook presents enough information to provide the reader with the fundamental knowledge necessary to understand the advanced theoretical concepts presented in other subject areas, and to understand basic system and equipment operation.
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Department of Energy
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NUCLEAR PHYSICS
AND REACTOR THEORY
Module 1

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Van Nostrand Reinhold Company, 1981.
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Atomic and Nuclear Physics DOE-HDBK-1019/1-93 OBJECTIVES
TERMINAL OBJECTIVE
1.0 Given sufficient information, DESCRIBE atoms, including components, structure, and nomenclature.
ENABLING OBJECTIVES
1.1 STATE the characteristics of the following atomic particles, including mass, charge, and location within the atom:
a. Proton
b. Neutron
c. Electron
1.2 DESCRIBE the Bohr model of an atom.
1.3 DEFINE the following terms:
a. Nuclide c. Atomic number
b. Isotope d. Mass number
1.4 Given the standard A
ZX notation for a particular nuclide, DETERMINE the following:
a. Number of protons
b. Number of neutrons
c. Number of electrons
1.5 DESCRIBE the three forces that act on particles within the nucleus and affect the stability of the nucleus.
1.6 DEFINE the following terms:
a. Enriched uranium
b. Depleted uranium
1.7 DEFINE the following terms:
a. Mass defect
b. Binding energy
1.8 Given the atomic mass for a nuclide and the atomic masses of a neutron, proton, and
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electron, CALCULATE the mass defect and binding energy of the nuclide.
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OBJECTIVES DOE-HDBK-1019/1-93 Atomic and Nuclear Physics
TERMINAL OBJECTIVE
2.0 Given necessary references, DESCRIBE the various modes of radioactive decay.
ENABLING OBJECTIVES
2.1 DESCRIBE the following processes:
a. Alpha decay d. Electron capture
b. Beta-minus decay e. Internal conversions
c. Beta-plus decay f. Isomeric transitions
2.2 Given a Chart of the Nuclides, WRITE the radioactive decay chain for a nuclide.
2.3 EXPLAIN why one or more gamma rays typically accompany particle emission.
2.4 Given the stability curve on the Chart of the Nuclides, DETERMINE the type of radioactive decay that the nuclides in each region of the chart will typically undergo.
2.5 DEFINE the following terms:
a. Radioactivity d. Radioactive decay constant
b. Curie e. Radioactive half-life

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2.6 Given the number of atoms and either the half-life or decay constant of a nuclide, CALCULATE the activity.
2.7 Given the initial activity and the decay constant of a nuclide, CALCULATE the activity at any later time.
2.8 CONVERT between the half-life and decay constant for a nuclide.
2.9 Given the Chart of the Nuclides and the original activity, PLOT the radioactive decay curve for a nuclide on either linear or semi-log coordinates.
2.10 DEFINE the following terms:
a. Radioactive equilibrium
b. Transient radioactive equilibrium
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Atomic and Nuclear Physics DOE-HDBK-1019/1-93 OBJECTIVES
TERMINAL OBJECTIVE
3.0 Without references, DESCRIBE the different nuclear interactions initiated by neutrons.
ENABLING OBJECTIVES
3.1 DESCRIBE the following scattering interactions between a neutron and a nucleus:
a. Elastic scattering
b. Inelastic scattering
3.2 STATE the conservation laws that apply to an elastic collision between a neutron and a nucleus.
3.3 DESCRIBE the following reactions where a neutron is absorbed in a nucleus:
a. Radiative capture
b. Particle ejection
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OBJECTIVES DOE-HDBK-1019/1-93 Atomic and Nuclear Physics
TERMINAL OBJECTIVE
4.0 Without references, DESCRIBE the fission process.
ENABLING OBJECTIVES
4.1 EXPLAIN the fission process using the liquid drop model of a nucleus.
4.2 DEFINE the following terms:
a. Excitation energy
b. Critical energy
4.3 DEFINE the following terms:
a. Fissile material
b. Fissionable material
c. Fertile material
4.4 DESCRIBE the processes of transmutation, conversion, and breeding.
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4.5 DESCRIBE the curve of Binding Energy per Nucleon versus mass number and give a qualitative description of the reasons for its shape.
4.6 EXPLAIN why only the heaviest nuclei are easily fissioned.
4.7 EXPLAIN why uranium-235 fissions with thermal neutrons and uranium-238 fissions only with fast neutrons.
4.8 CHARACTERIZE the fission products in terms of mass groupings and radioactivity.
4.9 Given the nuclides involved and their masses, CALCULATE the energy released from fission.
4.10 Given the curve of Binding Energy per Nucleon versus mass number, CALCULATE the energy released from fission.

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Atomic and Nuclear Physics DOE-HDBK-1019/1-93 OBJECTIVES
TERMINAL OBJECTIVE
5.0 Without references, DESCRIBE how the various types of radiation interact with matter.
ENABLING OBJECTIVES
5.1 DESCRIBE interactions of the following with matter:
a. Alpha particle c. Positron
b. Beta particle d. Neutron
5.2 DESCRIBE the following ways that gamma radiation interacts with matter:
a. Compton scattering
b. Photoelectric effect
c. Pair production
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OBJECTIVES DOE-HDBK-1019/1-93 Atomic and Nuclear Physics
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Atomic and Nuclear Physics DOE-HDBK-1019/1-93 ATOMIC NATURE OF MATTER
ATOMIC NATURE OF MATTER
All matter is composed of atoms. The atom is the smallest amount of matter that retains the properties of an element. Atoms themselves are composed of smaller particles, but these smaller particles no longer have the same properties as the overall element.
EO 1.1 STATE the characteristics of the following atomic particles, including mass, charge, and location within the atom:
a. Proton
b. Neutron
c. Electron
EO 1.2 DESCRIBE the Bohr model of an atom.
EO 1.3 DEFINE the following terms:
a. Nuclide c. Atomic number
b. Isotope d. Mass number
EO 1.4 Given the standard A ZX notation for a particular nuclide, DETERMINE the following:
a. Number of protons
b. Number of neutrons
c. Number of electrons
EO 1.5 DESCRIBE the three forces that act on particles within the nucleus and affect the stability of the nucleus.
Structure of Matter
Early Greek philosophers speculated that the earth was made up of different combinations of basic substances, or elements. They considered these basic elements to be earth, air, water, and fire. Modern science shows that the early Greeks held the correct concept that matter consists of a combination of basic elements, but they incorrectly identified the elements.
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ATOMIC NATURE OF MATTER DOE-HDBK-1019/1-93 Atomic and Nuclear Physics
In 1661 the English chemist Robert Boyle published the modern criterion for an element. He defined an element to be a basic substance that cannot be broken down into any simpler

compounds. To date, 105 different elements have been confirmed to exist, and researchers claim to have discovered three additional elements. Of the 105 confirmed elements, 90 exist in nature and 15 are man-made.

Another basic concept of matter that the Greeks debated was whether matter was continuous or discrete. That is, whether matter could be continuously divided and subdivided into ever smaller particles or whether eventually an indivisible particle would be encountered. Democritus in about 450 B.C. argued that substances were ultimately composed of small, indivisible particles that he labeled atoms. He further suggested that different substances were composed of different atoms or combinations of atoms, and that one substance could be converted into another by rearranging the atoms. It was impossible to conclusively prove or disprove this proposal for more than 2000 years.

The modern proof for the atomic nature of matter was first proposed by the English chemist John Dalton in 1803. Dalton stated that each chemical element possesses a particular kind of atom, and any quantity of the element is made up of identical atoms of this kind. What distinguishes one element from another element is the kind of atom of which it consists, and the basic physical difference between kinds of atoms is their weight.

Subatomic Particles

For almost 100 years after Dalton established the atomic nature of atoms, it was considered impossible to divide the atom into even smaller parts. All of the results of chemical experiments during this time indicated that the atom was indivisible. Eventually, experimentation into electricity and radioactivity indicated that particles of matter smaller than the atom did indeed

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exist. In 1906, J. J. Thompson won the Nobel Prize in physics for establishing the existence of electrons. Electrons are negatively-charged particles that have 1/1835 the mass of the hydrogen atom. Soon after the discovery of electrons, protons were discovered. Protons are relatively large particles that have almost the same mass as a hydrogen atom and a positive charge equal in magnitude (but opposite in sign) to that of the electron. The third subatomic particle to be discovered, the neutron, was not found until 1932. The neutron has almost the same mass as the proton, but it is electrically neutral.

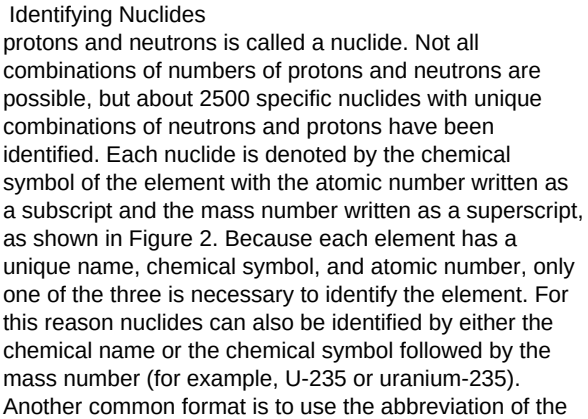
Bohr Model of the Atom

The British physicist Ernest Rutherford postulated that the positive charge in an atom is concentrated in a small region called a nucleus at the center of the atom with electrons existing in orbits around it. Niels Bohr, coupling Rutherford's postulation with the quantum theory introduced by Max Planck, proposed that the atom consists of a dense nucleus of protons surrounded by electrons traveling in discrete orbits at fixed distances from the nucleus. An electron in one of these orbits or shells has a specific or discrete quantity of energy (quantum). When an electron moves from one allowed orbit to another allowed orbit, the energy difference between the two states is emitted or absorbed in the form of a single quantum of radiant energy called a photon. Figure 1 is Bohr's model of the hydrogen atom showing an electron as having just dropped from the third shell to the first shell with the emission of a photon that has an energy = hv. (h = Planck's constant = 6.63 x 10-34 J-s and v = frequency of the photon.) Bohr's theory was the first to successfully account for the discrete energy levels of this radiation as measured in the laboratory. Although Bohr's atomic model is designed specifically to explain the hydrogen atom, his theories apply generally to the structure of all atoms. Additional information on electron shell theory can be found in the Chemistry Fundamentals Handbook.

Figure 1 Bohr's Model of the Hydrogen Atom

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Properties of the three subatomic particles are listed in Table 1.
TABLE 1
Properties of Subatomic Particles
Particle Location Charge Mass
Neutron Nucleus none 1.008665 amu
Proton Nucleus +1 1.007277 amu
Electron Shells around nucleus -1 0.0005486 amu
Measuring Units on the Atomic Scale
The size and mass of atoms are so small that the use of normal measuring units, while possible, is often inconvenient. Units of measure have been defined for mass and energy on the atomic scale to make measurements more convenient to express. The unit of measure for mass is the atomic mass unit (amu). One atomic mass unit is equal to 1.66 x 10-24 grams. The reason for this particular value for the atomic mass unit will be discussed in a later chapter. Note from Table 1 that the mass of a neutron and a proton are both about 1 amu. The unit for energy is the electron volt (eV). The electron volt is the amount of energy acquired by a single electron when it falls through a potential difference of one volt. One electron volt is equivalent to 1.602 x 10-19 joules or 1.18 x 10-19 foot-pounds.
Nuclides
The total number of protons in the nucleus of an atom is called the atomic number of the atom and is given the symbol Z. The number of electrons in an electrically-neutral atom is the same as the number of protons in the nucleus. The number of neutrons in a nucleus is known as the neutron number and is given the symbol N. The mass number of the nucleus is the total number of nucleons, that is, protons and neutrons in the nucleus. The mass number is given the symbol A and can be found by the equation Z + N = A.
Each of the chemical elements has a unique atomic number because the atoms of different elements contain a different number of protons. The atomic number of an atom identifies the particular element.
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Each type of atom that contains a unique combination of protons and neutrons is called a nuclide. Not all combinations of numbers of protons and neutrons are possible, but about 2500 specific nuclides with unique combinations of neutrons and protons have been identified. Each nuclide is denoted by the chemical symbol of the element with the atomic number written as a subscript and the mass number written as a superscript, as shown in Figure 2. Because each element has a unique name, chemical symbol, and atomic number, only one of the three is necessary to identify the element. For this reason nuclides can also be identified by either the chemical name or the chemical symbol followed by the mass number (for example, U-235 or uranium-235).
Another common format is to use the abbreviation of the


Figure 2 Nomenclature for Identifying Nuclides

format used in the text will usually be the element's name followed by the mass number. In equations and tables, the format in Figure 2 will usually be used.
Example:
State the name of the element and the number of protons, electrons, and neutrons in the nuclides listed below.
1
1H
10
5B
14
7N
11
4
4
8Cd
23
9
9
4Pu
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Solution:
The name of the element can be found from the Periodic Table (refer to Chemistry Fundamentals Handbook) or the Chart of the Nuclides (to be discussed later). The number of protons and electrons are equal to Z. The number of neutrons is equal to Z - A.
Nuclide Element Protons Electrons Neutrons
1
1H hydrogen 1 1 0
10
5B boron 5 5 5
14
7N nitrogen 7 7 7
11
4
Fixture checkpoint marker-0386-xray is recorded on fixture page 0386.
4
8Cd cadmium 48 48 66
23
9
9
4Pu plutonium 94 94 145
Isotopes
Isotopes are nuclides that have the same atomic number and are therefore the same element, but differ in the number of neutrons. Most elements have a few stable isotopes and several unstable, radioactive isotopes. For example, oxygen has three stable isotopes that can be found in nature (oxygen-16, oxygen-17, and oxygen-18) and eight radioactive isotopes. Another example is hydrogen, which has two stable isotopes (hydrogen-1 and hydrogen-2) and a single radioactive

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The isotopes of hydrogen are unique in that they are each commonly referred to by a unique name instead of the common chemical element name. Hydrogen-1 is almost always referred to as hydrogen, but the term protium is infrequently used also. Hydrogen-2 is commonly called deuterium and symbolized 2
1D. Hydrogen-3 is commonly called tritium and symbolized 31T. This
text will normally use the symbology 2
1H and 31H for deuterium and tritium, respectively.
Atomic and Nuclear Radii
The size of an atom is difficult to define exactly due to the fact that the electron cloud, formed by the electrons moving in their various orbitals, does not have a distinct outer edge. A reasonable measure of atomic size is given by the average distance of the outermost electron from the nucleus. Except for a few of the lightest atoms, the average atomic radii are
Fixture checkpoint marker-0387-xray is recorded on fixture page 0387.
approximately the same for all atoms, about 2 x 10 -8 cm.
Like the atom the nucleus does not have a sharp outer boundary. Experiments have shown that the nucleus is shaped like a sphere with a radius that depends on the atomic mass number of the atom. The relationship between the atomic mass number and the radius of the nucleus is shown in the following equation.

r = (1.25 x 10 -13 cm) A1/3
where:
r = radius of the nucleus (cm)
A = atomic mass number (dimensionless)
The values of the nuclear radii for some light, intermediate, and heavy nuclides are shown in Table 2.
TABLE 2
Calculated Values for Nuclear
Radii
Nuclide Radius of Nucleus
1
1H 1.25 x 10 -13 cm
10
5B 2.69 x 10 -13 cm
5
2
6
6Fe 4.78 x 10 -13 cm
17
7
8
2Hf 7.01 x 10 -13 cm
23
9
8
2U 7.74 x 10 -13 cm
25
9

8Cf 7.89 x 10 -13 cm
From the table, it is clear that the radius of a typical atom (e.g. 2 x 10 -8 cm) is more than 25,000 times larger than the radius of the largest nucleus.
Nuclear Forces
In the Bohr model of the atom, the nucleus consists of positively-charged protons and electrically■neutra as nucleons. One problem that the Bohr model of the atom presented was accounting for an attractive force to overcome the repulsive force between protons.
Two forces present in the nucleus are (1) electrostatic forces between charged particles and (2) gravitational forces between any two objects that have mass. It is possible to calculate the magnitude of the gravitational force and electrostatic force based upon principles from classical physics.

Fg
G m1 m2
r
2
Fe
K Q1 Q2
r
Fixture checkpoint marker-0388-xray is recorded on fixture page 0388.
2

Newton stated that the gravitational force between two bodies is directly proportional to the masses of the two bodies and inversely proportional to the square of the distance between the bodies. This relationship is shown in the equation below.
where:
F = gravitational force (newtons) g
m = mass of first body (kilograms) 1
m = mass of second body (kilograms) 2
G = gravitational constant (6.67 x 10 N-m /kg)
-11 2 2
r = distance between particles (meters)
The equation illustrates that the larger the masses of the objects or the smaller the distance between the objects, the greater the gravitational force. So even though the masses of nucleons are very small, the fact that the distance between nucleons is extremely short may make the gravitational force significant. It is necessary to calculate the value for the gravitational force and compare it to the value for other forces to determine the significance of the gravitational force in the nucleus. The gravitational force between two protons that are separated by a distance of 10 meters is about 10 newtons. -20 -24
Coulomb's Law can be used to calculate the force between two protons. The electrostatic force is directly proportional to the electrical charges of the two particles and inversely proportional to the square of the distance between the particles. Coulomb's Law is stated as the following equation.
where:
F = electrostatic force (newtons) e
K = electrostatic constant (9.0 x 10 N-m /C)
9 2 2

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Q = charge of second particle (coulombs) 2
r = distance between particles (meters)
Using this equation, the electrostatic force between two protons that are separated by a distance of 10 meters is about 10 newtons. Comparing this result with the calculation of the
-20 12
gravitational force (10 newtons) shows that the gravitational force is so small that it can be
-24
neglected.
Atomic and Nuclear Physics DOE-HDBK-1019/1-93 ATOMIC NATURE OF MATTER
If only the electrostatic and gravitational forces existed in the nucleus, then it would be impossible to have stable nuclei composed of protons and neutrons. The gravitational forces are much too small to hold the nucleons together compared to the electrostatic forces repelling the protons. Since stable atoms of neutrons and protons do exist, there must be another attractive force acting within the nucleus. This force is called the nuclear force.
The nuclear force is a strong attractive force that is independent of charge. It acts equally only between pairs of neutrons, pairs of protons, or a neutron and a proton. The nuclear force has a very short range; it acts only over distances approximately equal to the diameter of the nucleus (10 -13 cm). The attractive nuclear force between all nucleons drops off with distance much faster than the repulsive electrostatic force between protons.
TABLE 3
Forces Acting in the Nucleus
Force Interaction Range
Gravitational Very weak attractive force
between all nucleons Relatively long
Electrostatic Strong repulsive force between
like charged particles (protons) Relatively long
Fixture checkpoint marker-0389-xray is recorded on fixture page 0389.
Nuclear Force Strong attractive force between
all nucleons Extremely short
In stable atoms, the attractive and repulsive forces in the nucleus balance. If the forces do not balance, the atom cannot be stable, and the nucleus will emit radiation in an attempt to achieve a more stable configuration.
Summary
The important information in this chapter is summarized on the following page.
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ATOMIC NATURE OF MATTER DOE-HDBK-1019/1-93 Atomic and Nuclear Physics
Atomic Nature of Matter Summary
Atoms consist of three basic subatomic particles. These particles are the proton, the neutron, and the electron.
Protons are particles that have a positive charge, have about the same mass as a hydrogen atom, and exist in the nucleus of an atom.
Neutrons are particles that have no electrical charge, have about the same mass as a hydrogen atom, and exist in the nucleus of an atom.
Electrons are particles that have a negative charge, have a mass about eighteen hundred times smaller than the mass of a hydrogen atom, and exist in orbital shells around the nucleus of an atom.
The Bohr model of the atom consists of a dense nucleus of protons and neutrons (nucleons) surrounded by electrons traveling in discrete orbits at fixed distances

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Nuclides are atoms that contain a particular number of protons and neutrons.
Isotopes are nuclides that have the same atomic number and are therefore the same element, but differ in the number of neutrons.
The atomic number of an atom is the number of protons in the nucleus.
The mass number of an atom is the total number of nucleons (protons and neutrons) in the nucleus.
The notation A
ZX is used to identify a specific nuclide. "Z" represents the atomic number, which is equal to the number of protons. "A" represents the mass number, which is equal to the number of nucleons. "X" represents the chemical symbol of the element.
Number of protons = Z
Number of electrons = Z
Number of neutrons = A - Z
The stability of a nucleus is determined by the different forces interacting within it. The electrostatic force is a relatively long-range, strong, repulsive force that acts between the positively charged protons. The nuclear force is a relatively short-range attractive force between all nucleons. The gravitational force the long range, relatively weak attraction between masses, is negligible compared to the other forces.
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Atomic and Nuclear Physics DOE-HDBK-1019/1-93 CHART OF THE NUCLIDES
CHART OF THE NUCLIDES
The Chart of the Nuclides, like the Periodic Table, is a convenient format for presenting a large amount of scientific information in an organized manner.
EO 1.6 DEFINE the following terms:
a. Enriched uranium
b. Depleted uranium
Chart of the Nuclides
A tabulated chart called the Chart of the Nuclides lists the stable and unstable nuclides in addition to pertinent information about each one. Figure 3 shows a small portion of a typical chart. This chart plots a box for each individual nuclide, with the number of protons (Z) on the vertical axis
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and the number of neutrons (N = A - Z) on the horizontal axis.
The completely gray squares indicate stable isotopes. Those in white squares are artificially radioactive, meaning that they are produced by artificial techniques and do not occur naturally. By consulting a complete chart, other types of isotopes can be found, such as naturally occurring radioactive types (but none are found in the region of the chart that is illustrated in Figure 3). Located in the box on the far left of each horizontal row is general information about the element. The box contains the chemical symbol of the element in addition to the average atomic weight of the naturally occurring substance and the average thermal neutron absorption cross section, which will be discussed in a later module. The known isotopes (elements with the same atomic number Z but different mass number A) of each element are listed to the right.
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CHART OF THE NUCLIDES DOE-HDBK-1019/1-93 Atomic and Nuclear Physics
Figure 3 Nuclide Chart for Atomic Numbers 1 to 6
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Information for Stable Nuclides
For the stable isotopes, in addition to the symbol and the atomic mass number, the number percentage of each isotope in the naturally occurring element is listed, as well as the thermal neutron activation cross section and the mass in atomic mass units (amu). A typical block for a stable nuclide from the Chart of the Nuclides is shown in Figure 4.
Figure 4 Stable Nuclides
Information for Unstable Nuclides
For unstable isotopes the additional information includes the half life, the mode of decay (for example, β^-, α), the total disintegration energy in MeV (million electron volts), and the mass in
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amu when available. A typical block for an unstable nuclide from the Chart of the Nuclides is shown in Figure 5.
Figure 5 Unstable Nuclides
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CHART OF THE NUCLIDES DOE-HDBK-1019/1-93 Atomic and Nuclear Physics
Neutron - Proton Ratios
Figure 6 shows the distribution of the stable nuclides plotted on the same axes as the Chart of the Nuclides. As the mass numbers become higher, the ratio of neutrons to protons in the nucleus becomes larger. For helium-4 (2 protons and 2 neutrons) and oxygen-16 (8 protons and 8 neutrons) this ratio is unity. For indium-115 (49 protons and 66 neutrons) the ratio of neutrons to protons has increased to 1.35, and for uranium-238 (92 protons and 146 neutrons) the neutron■to-proton
Figure 6 Neutron - Proton Plot of the Stable Nuclides
If a heavy nucleus were to split into two fragments, each fragment would form a nucleus that would have approximately the same neutron-to-proton ratio as the heavy nucleus. This high neutron-to-proton ratio places the fragments below and to the right of the stability curve displayed by Figure 6. The instability caused by this excess of neutrons is generally rectified by successive beta emissions, each of which converts a neutron to a proton and moves the nucleus toward a more stable neutron-to-proton ratio.
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Atomic and Nuclear Physics DOE-HDBK-1019/1-93 CHART OF THE NUCLIDES
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Natural Abundance of Isotopes
The relative abundance of an isotope in nature compared to other isotopes of the same element is relatively constant. The Chart of the Nuclides presents the relative abundance of the naturally occurring isotopes of an element in units of atom percent. Atom percent is the percentage of the atoms of an element that are of a particular isotope. Atom percent is abbreviated as a/o.
For example, if a cup of water contains 8.23 x 10 atoms of oxygen, and the isotopic abundance 24
of oxygen-18 is 0.20%, then there are 1.65 x 10 atoms of oxygen-18 in the cup. 22
The atomic weight for an element is defined as the average atomic weight of the isotopes of the element. The atomic weight for an element can be calculated by summing the products of the isotopic abundance of the isotope with the atomic mass of the isotope.
Example:
Calculate the atomic weight for the element lithium. Lithium-6 has an atom percent abundance of 7.5% and an atomic mass of 6.015122 amu. Lithium-7 has an atomic abundance of 92.5% and an atomic mass of 7.016003 amu.
Solution:

is the percent weight of an element that is a particular isotope. For example, if a sample of material contained 100 kg of uranium that was 28 w/o uranium-235, then 28 kg of uranium-235 was present in the sample.

Enriched and Depleted Uranium

Natural uranium mined from the earth contains the isotopes uranium-238, uranium-235 and uranium-234. The majority (99.2745%) of all the atoms in natural uranium are uranium-238. Most of the remaining atoms (0.72%) are uranium-235, and a slight trace (0.0055%) are uranium-234. Although all isotopes of uranium have similar chemical properties, each of the isotopes has significantly different nuclear properties. For reasons that will be discussed in later modules, the isotope uranium-235 is usually the desired material for use in reactors.

A vast amount of equipment and energy are expended in processes that separate the isotopes of uranium (and other elements). The details of these processes are beyond the scope of this module. These processes are called enrichment processes because they selectively increase the proportion of a particular isotope. The enrichment process typically starts with feed material that has the proportion of isotopes that occur naturally. The process results in two types of

In the case of uranium, the natural uranium ore is 0.72 a/o uranium-235. The desired outcome

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of the enrichment process is to produce enriched uranium. Enriched uranium is defined as uranium in which the isotope uranium-235 has a concentration greater than its natural value. The enrichment process will also result in the byproduct of depleted uranium. Depleted uranium is defined as uranium in which the isotope uranium-235 has a concentration less than its natural value. Although depleted uranium is referred to as a by-product of the enrichment process, it does have uses in the nuclear field and in commercial and defense industries.

Summary

The important information in this chapter is summarized below.

Chart of the Nuclides Summary

Enriched uranium is uranium in which the isotope uranium-235 has a concentration greater than its natural value of 0.7%.

Depleted uranium is uranium in which the isotope uranium-235 has a concentration less than its natural value of 0.7%.

MASS DEFECT AND BINDING ENERGY

The separate laws of Conservation of Mass and Conservation of Energy are not applied strictly on the nuclear level. It is possible to convert between mass and energy. Instead of two separate conservation laws, a single conservation law states that the sum of mass and energy is conserved. Mass does not magically appear and disappear at random. A decrease in mass will be accompanied by a corresponding increase in energy and vice versa.

EO 1.7 DEFINE the following terms:

a. Mass defect

b. Binding energy

EO 1.8 Given the atomic mass for a nuclide and the atomic masses of a neutron, proton, and electron, CALCULATE the mass defect and binding energy of the nuclide.

Mass Defect

Careful measurements have shown that the mass of a particular atom is always slightly less than

consists. The difference between the mass of the atom and the sum of the masses of its parts is called the mass defect (∆m). The mass defect can be calculated using Equation (1-1). In calculating the mass defect it is important to use the full accuracy of mass measurements because the difference in mass is small compared to the mass of the atom. Rounding off the masses of atoms and particles to three or four significant digits prior to the calculation will result in a calculated mass defect of zero.

∆m = [Z(mp + me) + (A-Z)mn] - matom (1-1)

where:

∆m = mass defect (amu)

mp = mass of a proton (1.007277 amu)

mn = mass of a neutron (1.008665 amu)

me = mass of an electron (0.000548597 amu)

matom = mass of nuclide A

ZX (amu)

Z = atomic number (number of protons)

A = mass number (number of nucleons)

Example:

Calculate the mass defect for lithium-7. The mass of lithium-7 is 7.016003 amu.

Solution:

∆ m Z m

p

me A Z mn matom

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∆ m 3 1.007826 amu 7 3 1.008665 amu 7.016003 amu

∆ m 0.0421335 amu

Binding Energy

The loss in mass, or mass defect, is due to the conversion of mass to binding energy when the nucleus is formed. Binding energy is defined as the amount of energy that must be supplied to a nucleus to completely separate its nuclear particles (nucleons). It can also be understood as the amount of energy that would be released if the nucleus was formed from the separate particles. Binding energy is the energy equivalent of the mass defect. Since the mass defect was converted to binding energy (BE) when the nucleus was formed, it is possible to calculate the binding energy using a conversion factor derived by the mass-energy relationship from Einstein's Theory of Relativity.

Einstein's famous equation relating mass and energy is E = mc2 where c is the velocity of light (c = 2.998 x 108 m/sec). The energy equivalent of 1 amu can be determined by inserting this quantity of mass into Einstein's equation and applying conversion factors.

E m c 2

1 amu

1.6606 x 10 27 kg

1 amu

2.998 x 108 m

sec

2

1 N

1

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sec2
1 J
1 N m
1.4924 x 10 10 J 1 MeV
1.6022 x 10 13 J
931.5 MeV
Conversion Factors:
1 amu = 1.6606 x 10 -27 kg
1 newton = 1 kg-m/sec2
1 joule = 1 newton-meter
1 MeV = 1.6022 x 10-13 joules
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Atomic and Nuclear Physics DOE-HDBK-1019/1-93 MASS DEFECT AND BINDING ENERGY
Since 1 amu is equivalent to 931.5 MeV of energy, the binding energy can be calculated using Equation (1-2).
(1-2)
B.E. ∆m
931.5 MeV
1 amu
Example:
Calculate the mass defect and binding energy for uranium-235. One uranium-235 atom has a mass of 235.043924 amu.
Solution:
Step 1: Calculate the mass defect using Equation (1-1).
∆m Z m
p
me A Z mn matom
∆m 92 1.007826 amu 235 92 1.008665 amu 235.043924 amu
∆m 1.91517 amu
Step 2: Use the mass defect and Equation (1-2) to calculate the binding energy.
B.E. ∆m
Fixture checkpoint marker-0394-xray is recorded on fixture page 0394.
931.5 MeV
1 amu
1.91517 amu
931.5 MeV
1 amu
1784 MeV
Energy Levels of Atoms
The electrons that circle the nucleus move in fairly well-defined orbits. Some of these electrons are more tightly bound in the atom than others. For example, only 7.38 eV is required to remove the outermost electron from a lead atom, while 88,000 eV is required to remove the innermost electron. The process of removing an electron from an atom is called ionization, and the energy required to remove the electron is called the ionization energy.
In a neutral atom (number of electrons = Z) it is possible for the electrons to be in a variety of different orbits, each with a different energy level. The state of lowest energy is the one in which the atom is normally found and is called the ground state. When the atom possesses more energy than its ground state energy, it is said to be in an excited state.

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MASS DEFECT AND BINDING ENERGY DOE-HDBK-1019/1-93 Atomic and Nuclear Physics
An atom cannot stay in the excited state for an indefinite period of time. An excited atom will eventually transition to either a lower-energy excited state, or directly to its ground state, by emitting a discrete bundle of electromagnetic energy called an x-ray. The energy of the x-ray will be equal to the difference between the energy levels of the atom and will typically range from several eV to 100,000 eV in magnitude.
Energy Levels of the Nucleus
The nucleons in the nucleus of an atom, like the electrons that circle the nucleus, exist in shells that correspond to energy states. The energy shells of the nucleus are less defined and less understood than those of the electrons. There is a state of lowest energy (the ground state) and discrete possible excited states for a nucleus. Where the discrete energy states for the electrons of an atom are measured in eV or keV, the energy levels of the nucleus are considerably greater and typically measured in MeV.
A nucleus that is in the excited state will not remain at that energy level for an indefinite period. Like the electrons in an excited atom, the nucleons in an excited nucleus will transition towards their lowest energy configuration and in doing so emit a discrete bundle of electromagnetic radiation called a gamma ray (γ-ray). The only differences between x-rays and γ-rays are their energy levels and whether they are emitted from the electron shell or from the nucleus.
The ground state and the excited states of
Figure 7 Energy Level Diagram - Nickel-60
a nucleus can be depicted in a nuclear energy-level diagram. The nuclear energy-level diagram consists of a stack of horizontal bars, one bar for each of the excited states of the nucleus. The vertical distance between the bar representing an excited state and the bar representing the ground state is proportional to the energy level of the excited state with respect to the ground state. This difference in energy between the ground state and the excited state is called the excitation energy of the excited state. The ground state of a nuclide has zero excitation energy. The bars for the excited states are labeled with their respective energy levels. Figure 7 is the energy level diagram for nickel-60.
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Fixture checkpoint marker-0395-xray is recorded on fixture page 0395.
Atomic and Nuclear Physics DOE-HDBK-1019/1-93 MASS DEFECT AND BINDING ENERGY
Summary
The important information in this chapter is summarized below.
Mass Defect and Binding Energy Summary
Mass defect is the difference between the mass of the atom and the sum of the masses of its constituent parts.
Binding energy is the amount of energy that must be supplied to a nucleus to completely separate its nuclear particles. Binding energy is the energy equivalent of the mass defect.

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$\Delta m = [Z(mp + me) + (A-Z)mn] - matom$
Binding energy can be calculated by multiplying the mass defect by the factor of 931.5 MeV per amu.
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MODES OF RADIOACTIVE DECAY
Most atoms found in nature are stable and do not emit particles or energy that change form over time. Some atoms, however, do not have stable nuclei. These atoms emit radiation in order to achieve a more stable configuration.
EO 2.1 DESCRIBE the following processes:
a. Alpha decay d. Electron capture
b. Beta-minus decay e. Internal conversions
c. Beta-plus decay f. Isomeric transitions
EO 2.2 Given a Chart of the Nuclides, WRITE the radioactive decay chain for a nuclide.
Fixture checkpoint marker-0396-xray is recorded on fixture page 0396.
EO 2.3 EXPLAIN why one or more gamma rays typically accompany particle emission.
EO 2.4 Given the stability curve on the Chart of the Nuclides, DETERMINE the type of radioactive decay that the nuclides in each region of the chart will typically undergo.
Stability of Nuclei
As mass numbers become larger, the ratio of neutrons to protons in the nucleus becomes larger for the stable nuclei. Non-stable nuclei may have an excess or deficiency of neutrons and undergo a transformation process known as beta (β) decay. Non-stable nuclei can also undergo a variety of other processes such as alpha (α) or neutron (n) decay. As a result of these decay processes, the final nucleus is in a more stable or more tightly bound configuration.
Natural Radioactivity
In 1896, the French physicist Becquerel discovered that crystals of a uranium salt emitted rays that were similar to x-rays in that they were highly penetrating, could affect a photographic plate, and induced electrical conductivity in gases. Becquerel's discovery was followed in 1898 by the identification of two other radioactive elements, polonium and radium, by Pierre and Marie Curie.
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Atomic and Nuclear Physics DOE-HDBK-1019/1-93 MODES OF RADIOACTIVE DECAY
Heavy elements, such as uranium or thorium, and their unstable decay chain elements emit radiation in their naturally occurring state. Uranium and thorium, present since their creation at the beginning of geological time, have an extremely slow rate of decay. All naturally occurring nuclides with atomic numbers greater than 82 are radioactive.
Nuclear Decay
Whenever a nucleus can attain a more stable (i.e., more tightly bound) configuration by emitting radiation, a spontaneous disintegration process known as radioactive decay or nuclear decay may occur. In practice, this "radiation" may be electromagnetic radiation, particles, or both.
Detailed studies of radioactive decay and nuclear reaction processes have led to the formulation of useful conservation principles. The four principles of most interest in this module are discussed below.
Conservation of electric charge implies that charges are neither created nor destroyed. Single positive and negative charges may, however, neutralize each other. It is also possible for a

Conservation of mass number does not allow a net change in the number of nucleons. However, the conversion of a proton to a neutron and vice versa is allowed.
Conservation of mass and energy implies that the total of the kinetic energy and the energy equivalent of the mass in a system must be conserved in all decays and reactions. Mass can be converted to energy and energy can be converted to mass, but the sum of mass and energy must be constant.
Conservation of momentum is responsible for the distribution of the available kinetic energy among product nuclei, particles, and/or radiation. The total amount is the same before and after the reaction even though it may be distributed differently among entirely different nuclides and/or particles.
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■ ■ ■ ■ ■
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Alpha Decay (■)
Alpha decay is the emission of alpha particles (helium nuclei) which may be represented as either He or ■. When an unstable nucleus ejects an alpha particle, the atomic number is reduced by 2 4 4 2 2
and the mass number decreased by 4. An example is uranium-234 which decays by the ejection of an alpha particle accompanied by the emission of a 0.068 MeV gamma.
The combined kinetic energy of the daughter nucleus (Thorium-230) and the ■ particle is designated
Fixture checkpoint marker-0397-xray is recorded on fixture page 0397.
as KE. The sum of the KE and the gamma energy is equal to the difference in mass between the original nucleus (Uranium-234) and the final particles (equivalent to the binding energy released, since ■m = BE). The alpha particle will carry off as much as 98% of the kinetic energy and, in most cases, can be considered to carry off all the kinetic energy.
Beta Decay (■)
Beta decay is the emission of electrons of nuclear rather than orbital origin. These particles are electrons that have been expelled by excited nuclei and may have a charge of either sign.
If both energy and momentum are to be conserved, a third type of particle, the neutrino, ■, must be involved. The neutrino is associated with positive electron emission, and its antiparticle, the antineutrino, , is emitted with a negative electron. These uncharged particles have only the weakest interaction with matter, no mass, and travel at the speed of light. For all practical purposes, they pass through all materials with so few interactions that the energy they possess cannot be recovered. The neutrinos and antineutrinos are included here only because they carry a portion of the kinetic energy that would otherwise belong to the beta particle, and therefore, must be considered for energy and momentum to be conserved. They are normally ignored since they are not significant in the context of nuclear reactor applications.
Negative electron emission, represented as e, -1 -1■, or simply as e or ■ , effectively converts a neutro to a proton, thus increasing the atomic number by one and leaving the mass number unchanged. This is a common mode of decay for nuclei with an excess of neutrons, such as fission fragments below and to the right of the neutron-proton stability curve (refer to Figure 6). An example of a typical beta minus-decay reaction is shown below.
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Figure 9 Orbital Electron Capture
Positively charged electrons (beta-plus) are known as positrons. Except for sign, they are nearly identical to their negatively charged cousins. When a positron, represented as e, ■, or simply as +1 +1 0 0
e or ■ , is ejected from the nucleus, the atomic number is decreased by one and the mass number + + remains unchanged. A proton has been converted to a neutron. An example of a typical positron (beta-plus) decay is shown below.
Electron Capture (EC, K-capture)
Nuclei having an excess of protons may capture an electron from one of the inner orbits which immediately combines with a proton in the nucleus to form a neutron. This process is called electron capture (EC). The electron is normally captured from the innermost orbit (the K-shell), and, consequently, this process is sometimes called K-capture. The following example depicts electron capture.
A neutrino is formed at the same
time that the neutron is formed, and
energy carried off by it serves to
conserve momentum. Any energy
that is available due to the atomic
mass of the product being
appreciably less than that of the
parent will appear as gamma
radiation. Also, there will always
be characteristic x-rays given off
when an electron from one of the
higher energy shells moves in to fill
the vacancy in the K-shell.
Electron capture is shown
graphically in Figure 8.
Fixture checkpoint marker-0398-xray is recorded on fixture page 0398.
Electron capture and positron
emission result in the production of
the same daughter product, and
they exist as competing processes.
For positron emission to occur, however, the mass of the daughter product must be less than the mass of the parent by an amount equal to at least twice the mass of an electron. This mass difference between the parent and daughter is necessary to account for two items present in the parent but not in the daughter. One item is the positron ejected from the nucleus of the parent. The other item is that the daughter product has one less orbital electron than the parent. If this requirement is not met, then orbital electron capture takes place exclusively.
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■ ■ ■ ■
■ ■
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Gamma Emission (■)
Gamma radiation is a high-energy electromagnetic radiation that originates in the nucleus. It is emitted in the form of photons, discrete bundles of energy that have both wave and particle

undergoes a transformation by alpha decay, beta decay, or electron capture. The nucleus will drop to the ground state by the emission of gamma radiation.

Internal Conversion

The usual method for an excited nucleus to go from the excited state to the ground state is by emission of gamma radiation. However, in some cases the gamma ray (photon) emerges from the nucleus only to interact with one of the innermost orbital electrons and, as a result, the energy of the photon is transferred to the electron. The gamma ray is then said to have undergone internal conversion. The conversion electron is ejected from the atom with kinetic energy equal to the gamma energy minus the binding energy of the orbital electron. An orbital electron then drops to a lower energy state to fill the vacancy, and this is accompanied by the emission of characteristic x-rays.

Isomers and Isomeric Transition

Isomeric transition commonly occurs immediately after particle emission; however, the nucleus may remain in an excited state for a measurable period of time before dropping to the ground state at its own characteristic rate. A nucleus that remains in such an excited state is known as a nuclear isomer because it differs in energy and behavior from other nuclei with the same atomic number and mass number. The decay of an excited nuclear isomer to a lower energy level is called an isomeric transition. It is also possible for the excited isomer to decay by some alternate means, for example, by beta emission.

An example of gamma emission accompanying particle emission is illustrated by the decay of nitrogen-16 below.

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■
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Decay Chains

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When an unstable nucleus decays, the resulting daughter nucleus is not necessarily stable. The nucleus resulting from the decay of a parent is often itself unstable, and will undergo an additional decay. This is especially common among the larger nuclides.

It is possible to trace the steps of an unstable atom as it goes through multiple decays trying to achieve stability. The list of the original unstable nuclide, the nuclides that are involved as intermediate steps in the decay, and the final stable nuclide is known as the decay chain. One common method for stating the decay chain is to state each of the nuclides involved in the standard $^{A}_{Z}X$ format. Arrows are used between nuclides to indicate where decays occur, with the type of decay indicated above the arrow and the half-life below the arrow. The half-life for decay will be discussed in the next chapter.

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Write the decay chains for rubidium-91 and actinium-215. Continue the chains until a stable nuclide or a nuclide with a half-life greater than 1 x 10 years is reached. 6
Solution:
Predicting Type of Decay
Radioactive nuclides tend to decay in a way that results in a daughter nuclide that lies closer to the line of stability. Due to this, it is possible to predict the type of decay that a nuclide will undergo based on its location relative to the line of stability on the Chart of the Nuclides.
Figure 9 illustrates the type of decay nuclides in different regions of the chart will typically undergo. Nuclides that are below and to the right of the line of stability will usually undergo ■ decay. - Nuclides that are above and to the left of the line of stability will usually undergo either ■ decay or + electron capture. Most nuclides that will undergo ■ decay are found in the upper right hand region of the chart. These are general rules that have many exceptions, especially in the region of the heavy nuclides.
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Figure 9 Types of Radioactive Decay Relative to the Line of Stability
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Atomic and Nuclear Physics DOE-HDBK-1019/1-93 MODES OF RADIOACTIVE DECAY
Summary
The important information in this chapter is summarized below.
Modes of Radioactive Decay Summary
Alpha decay is the emission of an alpha particle (2 protons and 2 neutrons) from an unstable nucleus. The daughter nuclide has an atomic number 2 less than the parent nuclide and a mass number 4 less than the parent nuclide. The daughter nucleus commonly releases its excitation energy by gamma emission.
Beta-minus decay effectively converts a neutron to a proton and an electron, which is immediately ejected from the nucleus. The daughter nuclide has its atomic number increased by 1 and the same mass number compared to the parent.
Beta-plus decay effectively converts a proton to a neutron and a positron, which is immediately ejected from the nucleus. The daughter nuclide has its atomic number decreased by 1 and the same mass number compared to the parent.
In electron capture, the nucleus absorbs an electron from the innermost orbit. This electron combines with a proton to form a neutron.
Internal conversion occurs when a gamma ray, emitted by the nucleus as it goes from the excited state to the ground state, interacts with one of the innermost electrons of the same atom. The electron is ejected from the atom.
An isomeric transition is the decay of an excited nucleus to a lower-energy level by the emission of a gamma ray.
Decay chains can be found by tracing the steps an unstable atom goes through as it tries to achieve stability.
Many modes of radioactive decay result in a daughter nuclide that has an energy level above the ground state. This excitation energy is usually released immediately in the form of a gamma ray.
The type of decay that a nuclide will typically undergo can be determined by its relationship to the line of stability on the Chart of the Nuclides. Nuclides that lie below and to the right of the line of stability will typically beta minus decay. Nuclides above and to the left of the line will typically either beta plus decay or

of the chart.
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RADIOACTIVITY
The rate at which a sample of radioactive material decays is not constant. As individual atoms of the material decay, there are fewer of those types of atoms remaining. Since the rate of decay is directly proportional to the number of atoms, the rate of decay will decrease as the number of atoms decreases.
EO 2.5 DEFINE the following terms:
a. Radioactivity d. Radioactive decay constant
b. Curie e. Radioactive half-life
c. Becquerel
EO 2.6 Given the number of atoms and either the half-life or decay constant of a nuclide, CALCULATE the activity.
EO 2.7 Given the initial activity and the decay constant of a nuclide, CALCULATE the activity at any later time.
EO 2.8 CONVERT between the half-life and decay constant for a nuclide.
EO 2.9 Given the Chart of the Nuclides and the original activity, PLOT the radioactive decay curve for a nuclide on either linear or
Fixture checkpoint marker-0401-xray is recorded on fixture page 0401.
semi-log coordinates.
EO 2.10 DEFINE the following terms:
a. Radioactive equilibrium
b. Transient radioactive equilibrium
Radioactive Decay Rates
Radioactivity is the property of certain nuclides of spontaneously emitting particles or gamma radiation. The decay of radioactive nuclides occurs in a random manner, and the precise time at which a single nucleus will decay cannot be determined. However, the average behavior of a very large sample can be predicted accurately by using statistical methods. These studies have revealed that there is a certain probability that in a given time interval a certain fraction of the nuclei within a sample of a particular nuclide will decay. This probability per unit time that an atom of a nuclide will decay is known as the radioactive decay constant, λ. The units for the decay constant are inverse time such as 1/second, 1/minute, 1/hour, or 1/year. These decay constant units can also be expressed as second-1, minute-1, hour-1, and year-1.
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■■
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The activity (A) of a sample is the rate of decay of that sample. This rate of decay is usually measured in the number of disintegrations that occur per second. For a sample containing millions of atoms, the activity is the product of the decay constant and the number of atoms present in the sample.
The relationship between the activity, number of atoms, and decay constant is shown in Equation (1-3).
A = ■ N (1-3)
where:
A = Activity of the nuclide (disintegrations/second)

N = Number of atoms of the nuclide in the sample
Since ■ is a constant, the activity and the number of atoms are always proportional.
Units of Measurement for Radioactivity
Two common units to measure the activity of a substance are the curie (Ci) and becquerel (Bq). A curie is a unit of measure of the rate of radioactive decay equal to 3.7 x 10 disintegrations 10 per second. This is approximately equivalent to the number of disintegrations that one gram of radium-226 will undergo in one second. A becquerel is a more fundamental unit of measure of radioactive decay that is equal to 1 disintegration per second. Currently, the curie is more widely used in the United States, but usage of the becquerel can be expected to broaden as the metric system slowly comes into wider use. The conversion between curies and becquerels is shown below.
1 curie = 3.7 x 10 becquerels 10
Variation of Radioactivity Over Time
The rate at which a given radionuclide sample decays is stated in Equation (1-3) as being equal to the product of the number of atoms and the decay constant. From this basic relationship it is possible to use calculus to derive an expression which can be used to calculate how the number of atoms present will change over time. The derivation is beyond the scope of this text, but Equation (1-4) is the useful result.
(1-4)
where:
N = number of atoms present at time t
N = number of atoms initially present o
■ = decay constant (time)-1
t = time
A Ao
e
t
Fixture checkpoint marker-0402-xray is recorded on fixture page 0402.
A Aoe
t
A
Ao
e
t
ln A
Ao
t
t
ln
A
Ao
t
1/2
ln
1
2
t
1/2

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Since the activity and the number of atoms are always proportional, they may be used interchangeably to describe any given radionuclide population. Therefore, the following is true.
(1-5)
where:
A = activity present at time t
A = activity initially present o
= decay constant (time)
-1
t = time
Radioactive Half-Life
One of the most useful terms for estimating how quickly a nuclide will decay is the radioactive half-life. The radioactive half-life is defined as the amount of time required for the activity to decrease to one-half of its original value. A relationship between the half-life and decay constant can be developed from Equation (1-5). The half-life can be calculated by solving Equation (1-5) for the time, t, when the current activity, A, equals one-half the initial activity A .
o
First, solve Equation (1-5) for t.
If A is equal to one-half of Ao, then A/Ao is equal to one-half. Substituting this in the equation above yields an expression for t .
1/2
(1-6)

The basic features of decay of a radionuclide sample are shown by the graph in Figure 10.
Figure 10 Radioactive Decay as a Function of Time in Units of Half-Life
Assuming an initial number of atoms No, the population, and consequently, the activity may be noted to decrease by one-half of this value in a time of one half-life. Additional decreases occur so that whenever one half-life elapses, the number of atoms drops to one-half of what its value was at the beginning of that time interval. After five half-lives have elapsed, only 1/32, or 3.1%, of the original number of atoms remains. After seven half-lives, only 1/128, or 0.78%, of the atoms remains. The number of atoms existing after 5 to 7 half-lives can usually be assumed to be negligible. The Chemistry Fundamentals Handbook contains additional information on calculating the number of atoms contained within a sample.
Fixture checkpoint marker-0403-xray is recorded on fixture page 0403.

■
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Example:
A sample of material contains 20 micrograms of californium-252.
Californium-252 has a half-life of 2.638 years.
Calculate:
(a) The number of californium-252 atoms initially present
(b) The activity of the californium-252 in curies
(c) The number of californium-252 atoms that will remain in 12 years
(d) The time it will take for the activity to reach 0.001 curies

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(a) The number of atoms of californium-252 can be determined as below.
(b) First, use Equation (1-6) to calculate the decay constant.
Use this value for the decay constant in Equation (1-3) to determine the activity.
N No
e
t
4.78 x 1016 e
(0.263/yr) (12 yr)
2.04 x 1015
A Aoe
t
Fixture checkpoint marker-0404-xray is recorded on fixture page 0404.
A
Ao
e
t
ln A
Ao
t
t
ln
A
Ao
t
ln
0.001 Ci
0.0108 Ci
0.263 year 1
t 9.05 years
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(c) The number of californium atoms that will remain in 12 years can be calculated
from Equation (1-4).
(d) The time that it will take for the activity to reach 0.001 Ci can be determined
from Equation (1-5). First, solve Equation (1-5) for time.
Inserting the appropriate values in the right side of this equation will result in the
required time.
Plotting Radioactive Decay
It is useful to plot the activity of a nuclide as it changes over time. Plots of this type can be used to determine when the activity will fall below a certain level. This plot is usually done showing activity on either a linear or a logarithmic scale. The decay of the activity of a single nuclide on a logarithmic scale will plot as a straight line because the decay is exponential.
t
1/2
0.693
0.693
t
1/2

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7.13 seconds
0.0972 second 1
A Aoe
t
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Example:
Plot the radioactive decay curve for nitrogen-16 over a period of 100 seconds. The initial activity is 142 curies and the half-life of nitrogen-16 is 7.13 seconds. Plot the curve on both linear rectangular coordinates and on a semi-log scale.
Solution:
First, use Equation (1-6) to calculate the decay constant corresponding to a half-life of 7.13 seconds.
Use the decay constant determined above to calculate the activity at various times using Equation (1-5).
Time Activity
0 seconds 142 Ci
20 seconds 20.3 Ci
Fixture checkpoint marker-0405-xray is recorded on fixture page 0405.
40 seconds 2.91 Ci
60 seconds 0.416 Ci
80 seconds 0.0596 Ci
100 seconds 0.00853 Ci
■ ■
■ ■
■ ■
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Figure 11 Linear and Semi-log Plots of Nitrogen-16 Decay
Plotting the data points calculated above on both linear and semilog scales results in the graphs shown in Figure 11.
If a substance contains more than one radioactive nuclide, the total activity is the sum of the individual activities of each nuclide. As an example, consider a sample of material that contained 1 x 10 atoms of iron-59 that has a half-life of 44.51 days (■ = 1.80 x 10 sec), 6 -7 -1 1 x 10 atoms of manganese-54 that has a half-life of 312.2 days (■ = 2.57 x 10 sec), and 1 6 -8 -1 x 10 atoms of cobalt-60 that has a half-life of 1925 days (■ = 4.17 x 10 sec). 6 -9 -1
The initial activity of each of the nuclides would be the product of the number of atoms and the decay constant.
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Figure 12 Combined Decay of Iron-56, Manganese-54, and Cobalt-60
Plotting the manner in which the activities of each of the three nuclides decay over time demonstrates that initially the activity of the shortest-lived nuclide (iron-59) dominates the total activity, then manganese-54 dominates. After almost all of the iron and manganese have decayed away, the only contributor to activity will be the cobalt-60. A plot of this combined decay is shown in Figure 12.
Radioactive Equilibrium
Radioactive equilibrium exists when a radioactive nuclide is decaying at the same rate at which

present remains constant over time.
An example of radioactive equilibrium is the concentration of sodium-24 in the coolant circulating through a sodium-cooled nuclear reactor. Assume that the sodium-24 is being produced at a rate of 1 x 10 atoms per second. If the sodium-24 were stable and did not decay, 6 the amount of sodium-24 present after some period of time could be calculated by multiplying the production rate by the amount of time. Plotting the amount of material present would result in the graph in Figure 13.
However, sodium-24 is not stable, and it decays with a half-life of 14.96 hours. If no sodium-24 is present initially and production starts at a rate of 1 x 10 atoms per second, the rate 6 of decay will initially be zero because there is no sodium-24 present to decay. The rate of decay of sodium-24 will increase as the amount of sodium-24 increases.
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Figure 13 Cumulative Production of Sodium-24 Over Time
The amount of sodium-24 present will initially increase rapidly, then it will increase at a continually decreasing rate until the rate of decay is equal to the rate of production. It is possible to calculate how much sodium-24 will be present at equilibrium by setting the production rate (R) equal to the decay rate (■ N).
where:
R = production rate (atoms/second)
■ = decay constant (second)-1
Fixture checkpoint marker-0406-xray is recorded on fixture page 0406.
N = number of atoms
It is possible to calculate the equilibrium value for sodium-24 being produced at a rate of 1 x 10 atoms/second. 6
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Figure 14 Approach of Sodium-24 to Equilibrium
The development of the equation to calculate how the amount of sodium-24 changes over time as it approaches the equilibrium value is beyond the scope of this handbook. However, the equation is presented below.
This equation can be used to calculate the values of the amount of sodium-24 present at different times. As the time increases, the exponential term approaches zero, and the number of atoms present will approach R/■. A plot of the approach of sodium-24 to equilibrium is shown in Figure 14.
Transient Radioactive Equilibrium
Transient radioactive equilibrium occurs when the parent nuclide and the daughter nuclide decay at essentially the same rate.
For transient equilibrium to occur, the parent must have a long half-life when compared to the

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beta emission to lanthanum-140, which in turn decays by beta emission to stable cerium-140.
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Figure 15 Transient Equilibrium in the Decay of Barium-140
The decay constant for barium-140 is considerably smaller than the decay constant for lanthanum-140. Remember that the rate of decay of both the parent and daughter can be represented as ■N. Although the decay constant for barium-140 is smaller, the actual rate of decay (■N) is initially larger than that of lanthanum-140 because of the great difference in their initial concentrations. As the concentration of the daughter increases, the rate of decay of the daughter will approach and eventually match the decay rate of the parent. When this occurs, they are said to be in transient equilibrium. A plot of the barium-lanthanum-cerium decay chain reaching transient equilibrium is shown in Figure 15.
Secular equilibrium occurs when the parent has an extremely long half-life. In the long decay chain for a naturally radioactive element, such as thorium-232, where all of the elements in the chain are in secular equilibrium, each of the descendants has built up to an equilibrium amount and all decay at the rate set by the original parent. The only exception is the final stable element on the end of the chain. Its number of atoms is constantly increasing.
Summary
The important information in this chapter is summarized on the following page.
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■■
■
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Radioactivity Summary
& Radioactivity is the decay of unstable atoms by the emission of particles and electromagnetic radiation.
& A curie (Ci) is a unit of radioactivity equal to 3.7 x 10 disintegrations per 10 second.
& A becquerel (Bq) is a unit of radioactivity equal to 1 disintegration per second.
& The radioactive decay constant (■) is the probability per unit time that an atom will decay.
Fixture checkpoint marker-0407-xray is recorded on fixture page 0407.
& The radioactive half-life is the amount of time required for the activity to decrease to one-half its original value.
& The activity of a substance can be calculated from the number of atoms and the decay constant based on the equation below.
& The amount of activity remaining after a particular time can be calculated from the equation below.
& The relationship between the decay constant and the half-life is shown below.
& Plots of radioactive decay can be useful to describe the variation of activity over time. If decay is plotted using semi-log scale the plot results in a straight line.
& Radioactive equilibrium exists when the production rate of a material is equal to the removal rate.
& Transient radioactive equilibrium exists when the parent nuclide and the daughter nuclide decay at essentially the same rate. This occurs only when the parent has a long half-life compared to the daughter.
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Neutrons can cause many different types of interactions. The neutron may simply scatter off the nucleus in two different ways, or it may actually be absorbed into the nucleus. If a neutron is absorbed into the nucleus, it may result in the emission of a gamma ray or a subatomic particle, or it may cause the nucleus to fission.

EO 3.1 DESCRIBE the following scattering interactions between a neutron and a nucleus:

a. Elastic scattering

b. Inelastic scattering

EO 3.2 STATE the conservation laws that apply to an elastic collision between a neutron and a nucleus.

EO 3.3 DESCRIBE the following reactions where a neutron is absorbed in a nucleus:

a. Radiative capture

b. Particle ejection

Scattering

A neutron scattering reaction occurs when a nucleus, after having been struck by a neutron, emits a single neutron. Despite the fact that the initial and final neutrons do not need to be (and often are not) the same, the net effect of the reaction is as if the projectile neutron had merely "bounced off," or scattered from, the nucleus. The two categories of scattering reactions, elastic and inelastic scattering, are described in the following paragraphs.

Elastic Scattering

In an elastic scattering reaction between a neutron and a target nucleus, there is no energy transferred into nuclear excitation. Momentum and kinetic energy of the "system" are conserved although there is usually some transfer of kinetic energy from the neutron to the target nucleus. The target nucleus gains the amount of kinetic energy that the neutron loses.

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mn

vn,i mTvT,i mnvn,f mTvT,f

1

2

m v

2

1

2

mn

v

2

n,i

Fixture checkpoint marker-0408-xray is recorded on fixture page 0408.

1

2

mT

v

2

T,i

1

2

v
2
n,f
1
2
mT
v
2
T,f
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Figure 16 Elastic Scattering
Figure 16 illustrates the process of elastic scattering of a neutron off a target nucleus. In the elastic scattering reaction, the conservation of momentum and kinetic energy is represented by the equations below.
Conservation of momentum (mv)
Fixture checkpoint marker-0409-xray is recorded on fixture page 0409.
Conservation of kinetic energy
where:
mn = mass of the neutron
m = mass of the target nucleus T
vn,i = initial neutron velocity
vn,f = final neutron velocity
vT,i = initial target velocity
vT,f = final target velocity
Atomic and Nuclear Physics DOE-HDBK-1019/1-93 NEUTRON INTERACTIONS
Elastic scattering of neutrons by nuclei can occur in two ways. The more unusual of the two interactions is the absorption of the neutron, forming a compound nucleus, followed by the re-emission of a neutron in such a way that the total kinetic energy is conserved and the nucleus returns to its ground state. This is known as resonance elastic scattering and is very dependent upon the initial kinetic energy possessed by the neutron. Due to formation of the compound nucleus, it is also referred to as compound elastic scattering. The second, more usual method, is termed potential elastic scattering and can be understood by visualizing the neutrons and nuclei to be much like billiard balls with impenetrable surfaces. Potential scattering takes place with incident neutrons that have an energy of up to about 1 MeV. In potential scattering, the neutron does not actually touch the nucleus and a compound nucleus is not formed. Instead, the neutron is acted on and scattered by the short range nuclear forces when it approaches close enough to the nucleus.
Inelastic Scattering
In inelastic scattering, the incident neutron is absorbed by the target nucleus, forming a compound nucleus. The compound nucleus will then emit a neutron of lower kinetic energy which leaves the original nucleus in an excited state. The nucleus will usually, by one or more gamma emissions, emit this excess energy to reach its ground state. Figure 17 shows the process of inelastic scattering.
For the nucleus that has reached its ground state, the sum of the kinetic energy of the exit
Figure 17 Inelastic Scattering
neutron, the target nucleus, and the total gamma energy emitted is equal to the initial kinetic energy of the incident neutron.

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1
0
n
238
92
U
239
92
U
239
92
U
0
0
1
0
n
10
5
B
11
5
B
Fixture checkpoint marker-0410-xray is recorded on fixture page 0410.
7
3
Li
4
2
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Absorption Reactions
Most absorption reactions result in the loss of a neutron coupled with the production of a charged particle or gamma ray. When the product nucleus is radioactive, additional radiation is emitted at some later time. Radiative capture, particle ejection, and fission are all categorized as absorption reactions and are briefly described below.
Radiative Capture
In radiative capture the incident neutron enters the target nucleus forming a compound nucleus. The compound nucleus then decays to its ground state by gamma emission. An example of a radiative capture reaction is shown below.
Particle Ejection
In a particle ejection reaction the incident particle enters the target nucleus forming a compound nucleus. The newly formed compound nucleus has been excited to a high enough energy level to cause it to eject a new particle while the incident neutron remains in the nucleus. After the new particle is ejected, the remaining nucleus may or may not exist in an excited state depending upon the mass-energy balance of the reaction. An example of a particle ejection reaction is shown below.
Fission

that absorbs the neutron actually splits into two similarly sized parts. Fission will be discussed in detail in the next chapter.
Atomic and Nuclear Physics DOE-HDBK-1019/1-93 NEUTRON INTERACTIONS
Summary
The important information in this chapter is summarized below.
Neutron Interactions Summary
Interactions where a neutron scatters off a target nucleus are either elastic or inelastic. In elastic scattering, kinetic energy and momentum are conserved and no energy is transferred into excitation energy of the target nucleus. In inelastic scattering, some amount of kinetic energy is transferred into excitation energy of the target nucleus.
The conservation principles that apply to an elastic collision are conservation of kinetic energy and conservation of momentum.
Radiative capture is the absorption of a neutron by the target nucleus, resulting in an excited nucleus which subsequently (typically within a small fraction of a second) releases its excitation energy in the form of a gamma ray.
Particle ejection occurs when a neutron is absorbed by a target nucleus, resulting in the formation of a compound nucleus. The compound nucleus immediately ejects a particle (for example, alpha or proton).
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1
0
n
235
92
U
236
92
U
140
Fixture checkpoint marker-0411-xray is recorded on fixture page 0411.
55
Cs
93
37
Rb 3
1
0
n
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NUCLEAR FISSION
Nuclear fission is a process in which an atom splits and releases energy, fission products, and neutrons. The neutrons released by fission can, in turn, cause the fission of other atoms.
EO 4.1 EXPLAIN the fission process using the liquid drop model of a nucleus.
EO 4.2 DEFINE the following terms:

b. Critical energy (E_{crit})
EO 4.3 DEFINE the following terms:
a. Fissile material
b. Fissionable material
c. Fertile material
EO 4.4 DESCRIBE the processes of transmutation, conversion, and breeding.
EO 4.5 DESCRIBE the curve of Binding Energy per Nucleon versus mass number and give a qualitative description of the reasons for its shape.
EO 4.6 EXPLAIN why only the heaviest nuclei are easily fissioned.
EO 4.7 EXPLAIN why uranium-235 fissions with thermal neutrons and uranium-238 fissions only with fast neutrons.
Fission
In the fission reaction the incident neutron enters the heavy target nucleus, forming a compound nucleus that is excited to such a high energy level ($E > E_{crit}$) that the nucleus "splits" (fissions) into two large fragments plus some neutrons. An example of a typical fission reaction is shown below.
A large amount of energy is released in the form of radiation and fragment kinetic energy.
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Liquid Drop Model of a Nucleus
The nucleus is held together by the attractive nuclear force between nucleons, which was discussed in a previous chapter. The characteristics of the nuclear force are listed below.
(a) very short range, with essentially no effect beyond nuclear dimensions (10^{-13} cm)
(b) stronger than the repulsive electrostatic forces within the nucleus
(c) independent of nucleon pairing, in that the attractive forces between pairs of neutrons are no different than those between pairs of protons or a neutron and a proton
(d) saturable, that is, a nucleon can attract only a few of its nearest neighbors
One theory of fission considers the fissioning of a nucleus similar in some respects to the splitting of a liquid drop. This analogy is justifiable to some extent by the fact that a liquid drop is held together by molecular forces that tend to make the drop spherical in shape and that try to resist any deformation in the same manner as nuclear forces are assumed to hold the nucleus together. By considering the nucleus as a liquid drop, the fission process can be described.
Referring to Figure 18(A), the nucleus in the ground state is undistorted, and its attractive nuclear forces are greater than the repulsive electrostatic forces between the protons within the nucleus.
Fixture checkpoint marker-0412-xray is recorded on fixture page 0412.
When an incident particle (in this instance a neutron) is absorbed by the target nucleus, a compound nucleus is formed. The compound nucleus temporarily contains all the charge and mass involved in the reaction and exists in an excited state. The excitation energy added to the compound nucleus is equal to the binding energy contributed by the incident particle plus the kinetic energy possessed by that particle. Figure 18(B) illustrates the excitation energy thus imparted to the compound nucleus, which may cause it to oscillate and become distorted. If the excitation energy is greater than a certain critical energy, the oscillations may cause the compound nucleus to become dumbbell-shaped. When this happens, the attractive nuclear forces (short-range) in the neck area are small due to saturation, while the repulsive electrostatic forces (long-range) are only slightly less than before. When the repulsive electrostatic forces exceed

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Figure 18 Liquid Drop Model of Fission
Critical Energy
The measure of how far the energy level of a nucleus is above its ground state is called the excitation energy (Eexc). For fission to occur, the excitation energy must be above a particular value for that nuclide. The critical energy (Ecrit) is the minimum excitation energy required for fission to occur.
Fissile Material
A fissile material is composed of nuclides for which fission is possible with neutrons of any energy level. What is especially significant about these nuclides is their ability to be fissioned with zero kinetic energy neutrons (thermal neutrons). Thermal neutrons have very low kinetic energy levels (essentially zero) because they are roughly in equilibrium with the thermal motion of surrounding materials. Therefore, in order to be classified as fissile, a material must be
Fixture checkpoint marker-0413-xray is recorded on fixture page 0413.
capable of fissioning after absorbing a thermal neutron. Consequently, they impart essentially no kinetic energy to the reaction. Fission is possible in these materials with thermal neutrons, since the change in binding energy supplied by the neutron addition alone is high enough to exceed the critical energy. Some examples of fissile nuclides are uranium-235, uranium-233, and plutonium-239.

Fissionable Material
A fissionable material is composed of nuclides for which fission with neutrons is possible. All fissile nuclides fall into this category. However, also included are those nuclides that can be fissioned only with high energy neutrons. The change in binding energy that occurs as the result of neutron absorption results in a nuclear excitation energy level that is less than the required critical energy. Therefore, the additional excitation energy must be supplied by the kinetic energy of the incident neutron. The reason for this difference between fissile and fissionable materials is the so-called odd-even effect for nuclei. It has been observed that nuclei with even numbers of neutrons and/or protons are more stable than those with odd numbers. Therefore, adding a neutron to change a nucleus with an odd number of neutrons to a nucleus with an even number of neutrons produces an appreciably higher binding energy than adding a neutron to a nucleus already possessing an even number of neutrons. Some examples of nuclides requiring high energy neutrons to cause fission are thorium-232, uranium-238, and plutonium-240. Table 4 indicates the critical energy (Ecrit) and the binding energy change for an added neutron (BEn) to target nuclei of interest. For fission to be possible, the change in binding energy plus the kinetic energy must equal or exceed the critical energy ($\Delta BE + KE > Ecrit$).
TABLE 4
Critical Energies Compared to Binding Energy of Last Neutron
Target
Nucleus
Critical Energy
Ecrit
Binding Energy of
Last Neutron BEn
BEn - Ecrit
23

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2
0Th 7.5 MeV 5.4 MeV -2.1 MeV
23
9
8
2U 7.0 MeV 5.5 MeV -1.5 MeV
23
9
5
2U 6.5 MeV 6.8 MeV +0.3 MeV
23
9
3
2U 6.0 MeV 7.0 MeV +1.0 MeV
23
9
9
4Pu 5.0 MeV 6.6 MeV +1.6 MeV
Uranium-235 fissions with thermal neutrons because the binding energy released by the absorption of a neutron is greater than the critical energy for fission; therefore uranium-235 is a fissile material. The binding energy released by uranium-238 absorbing a thermal neutron is
Fixture checkpoint marker-0414-xray is recorded on fixture page 0414.
less than the critical energy, so additional energy must be possessed by the neutron for fission to be possible. Consequently, uranium-238 is a fissionable material.
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Fertile Material
All of the neutron absorption reactions that do not result in fission lead to the production of new nuclides through the process known as transmutation. These nuclides can, in turn, be transmuted again or may undergo radioactive decay to produce still different nuclides. The nuclides that are produced by this process are referred to as transmutation products. Because several of the fissile nuclides do not exist in nature, they can only be produced by nuclear reactions (transmutation). The target nuclei for such reactions are said to be fertile. Fertile materials are materials that can undergo transmutation to become fissile materials. Figure 19 traces the transmutation mechanism by which two fertile nuclides, thorium-232 and uranium-238, produce uranium-233 and plutonium-239, respectively.
Figure 19 Conversion of Fertile Nuclides to Fissile Nuclides
If a reactor contains fertile material in addition to its fissile fuel, some new fuel will be produced as the original fuel is burned up. This is called conversion. Reactors that are specifically designed to produce fissionable fuel are called "breeder" reactors. In such reactors, the amount of fissionable fuel produced is greater than the amount of fuel burnup. If less fuel is produced than used, the process is called conversion, and the reactor is termed a "converter."
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Atomic and Nuclear Physics DOE-HDBK-1019/1-93 NUCLEAR FISSION
Binding Energy Per Nucleon (BE/A)
As the number of particles in a nucleus increases, the total binding energy also increases. The rate of increase, however, is not uniform. This lack of uniformity results in a variation in the amount of binding energy associated with each nucleon within the nucleus. This variation in the

mass number (A), as shown in Figure 20.

Figure 20 Binding Energy per Nucleon vs. Mass Number

Figure 20 illustrates that as the atomic mass number increases, the binding energy per nucleon decreases for A > 60. The BE/A curve reaches a maximum value of 8.79 MeV at A = 56 and decreases to about 7.6 MeV for A = 238. The general shape of the BE/A curve can be explained using the general properties of nuclear forces. The nucleus is held together by very short-range attractive forces that exist between nucleons. On the other hand, the nucleus is being forced apart by long range repulsive electrostatic (coulomb) forces that exist between all the protons in the nucleus.

As the atomic number and the atomic mass number increase, the repulsive electrostatic forces within the nucleus increase due to the greater number of protons in the heavy elements. To overcome this increased repulsion, the proportion of neutrons in the nucleus must increase to maintain stability. This increase in the neutron-to-proton ratio only partially compensates for the growing proton-proton repulsive force in the heavier, naturally occurring elements. Because the repulsive forces are increasing, less energy must be supplied, on the average, to remove a nucleon from the nucleus. The BE/A has decreased. The BE/A of a nucleus is an indication of its degree of stability. Generally, the more stable nuclides have higher BE/A than the less stable ones. The increase in the BE/A as the atomic mass number decreases from 260 to 60 is the primary reason for the energy liberation in the fission process. In addition, the increase in the BE/A as the atomic mass number increases from 1 to 60 is the reason for the energy liberation in the fusion process, which is the opposite reaction of fission.

The heaviest nuclei require only a small distortion from a spherical shape (small energy addition) for the relatively large coulomb forces forcing the two halves of the nucleus apart to overcome the attractive nuclear forces holding the two halves together. Consequently, the heaviest nuclei are easily fissionable compared to lighter nuclei.

Summary

Fixture checkpoint marker-0415-xray is recorded on fixture page 0415.

The important information in this chapter is summarized on the following page.

Nuclear Fission Summary

The fission process can be explained using the liquid drop model of a nucleus. In the ground state the nucleus is nearly spherical in shape. After the absorption of a neutron, the nucleus will be in an excited state and start to oscillate and become distorted. If the oscillations cause the nucleus to become shaped like a dumbbell, the repulsive electrostatic forces will overcome the short-range attractive nuclear forces, and the nucleus will split in two.

Excitation energy is the amount of energy a nucleus has above its ground state. Critical energy is the minimum excitation energy that a nucleus must have before it can fission.

Fissile material is material for which fission is possible with neutrons that have zero kinetic energy. Fissionable material is material for which fission caused by neutron absorption is possible provided the kinetic energy added with the binding energy is greater than the critical energy. Fertile material is material that can undergo transmutation to become fissile material.

Transmutation is the process of neutron absorption and subsequent decay, which

fertile material into fissile material in a reactor, where the amount of fissile material produced is less than the amount of fissile material consumed. Breeding is the same as conversion, except the amount of fissile material produced is more than the amount of fissile material consumed.

The curve of binding energy per nucleon increases quickly through the light nuclides and reaches a maximum at a mass number of about 56. The curve decreases slowly for mass numbers greater than 60.

The heaviest nuclei are easily fissionable because they require only a small distortion from the spherical shape to allow the coulomb forces to overcoming the attractive nuclear force, forcing the two halves of the nucleus apart.

Uranium-235 fissions with thermal neutrons because the binding energy released by the absorption of a neutron is greater than the critical energy for fission. The binding energy released by uranium-238 absorbing a neutron is less than the critical energy, so additional kinetic energy must be possessed by the neutron for fission to be possible.

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1
0
n
235
92
U
236
92
U
140
55
Cs
93
37
Rb 3
1
0
n

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ENERGY RELEASE FROM FISSION

Fission of heavy nuclides converts a small amount of mass into an enormous amount of energy. The amount of energy released by fission can be determined based on either the change in mass that occurs during the reaction or by the difference in binding energy per nucleon between the fissile nuclide and the fission products.

EO 4.8 CHARACTERIZE the fission products in terms of mass groupings and radioactivity.

EO 4.9 Given the nuclides involved and their masses, CALCULATE the energy released from fission.

EO 4.10 Given the curve of Binding Energy per nucleon versus mass

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Calculation of Fission Energy

Nuclear fission results in the release of enormous quantities of energy. It is necessary to be able to calculate the amount of energy that will be produced. The logical manner in which to pursue this is to first investigate a typical fission reaction such as the one listed below.

It can be seen that when the compound nucleus splits, it breaks into two fission fragments, rubidium-93, cesium-140, and some neutrons. Both fission products then decay by multiple - emissions as a result of the high neutron-to-proton ratio possessed by these nuclides.

In most cases, the resultant fission fragments have masses that vary widely. Figure 21 gives the percent yield for atomic mass numbers. The most probable pair of fission fragments for the thermal fission of the fuel uranium-235 have masses of about 95 and 140. Note that the vertical axis of the fission yield curve is on a logarithmic scale. Therefore, the formation of fission fragments of mass numbers of about 95 and 140 is highly likely.

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Figure 21 Uranium-235 Fission Yield vs.

Mass Number

Referring now to the binding energy per nucleon curve (Figure 20), we can estimate the amount of energy released by our "typical" fission by plotting this reaction on the curve and calculating the change in binding energy (ΔBE) between the reactants on the left-hand side of the fission equation and the products on the right-hand side. Plotting the reactant and product nuclides on the curve shows that the total binding energy of the system after fission is greater than the total binding energy of the system before fission. When there is an increase in the total binding energy of a system, the system has become more stable by releasing an amount of energy equal to the increase in total binding energy of the system. Therefore, in the fission process, the energy liberated is equal to the increase in the total binding energy of the system.

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Figure 22 graphically depicts that the binding energy per nucleon for the products (C,

Figure 22 Change in Binding Energy for Typical Fission

rubidium-93 and B, cesium-140) is greater than that for the reactant (A, uranium-235). The total binding energy for a nucleus can be found by multiplying the binding energy per nucleon by the number of nucleons.

TABLE 5

Binding Energies Calculated from

Binding Energy per Nucleon Curve

Nuclide B.E. per Nucleon

(BE/A)

Mass Number

(A)

Binding Energy

(BE/A) x (A)

9

3

3

7Rb 8.7 MeV 93 809 MeV

14

5

0

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23
9
5
2U 7.6 MeV 235 1786 MeV
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BE BEproducts BEreactants
BERb 93 BECs 140 BEU 235
809 MeV 1176 MeV 1786 MeV
199 MeV
1
0
n
235
92
U
236
92
U
140
Fixture checkpoint marker-0418-xray is recorded on fixture page 0418.
55
Cs
93
37
Rb 3
1
0
n
Mass difference Mass of Reactants Mass of Products
236.052589 amu 235.85208 amu
0.200509 amu
EInst 0.020059 amu
931.5 MeV
amu
186.8 MeV
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The energy released will be equivalent to the difference in binding energy (BE) between the reactants and the products.
The energy liberation during the fission process can also be explained from the standpoint of the conservation of mass-energy. During the fission process, there is a decrease in the mass of the system. There must, therefore, be energy liberated equal to the energy equivalent of the mass lost in the process. This method is more accurate than the previously illustrated method and is used when actually calculating the energy liberated during the fission process.
Again, referring to the "typical" fission reaction.
E , the instantaneous energy, is the energy released immediately after the fission process. It Inst is equal to the energy equivalent of the mass lost in the fission process. It can be calculated as shown below.

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U 235.043924 amu Rb 92.91699 amu 235
92
93
37
n 1.008665 amu Cs 139.90910 amu 1
0
140
55
3 (n) 3.02599 amu 1
0
236.052589 amu 235.85208 amu
This mass difference can be converted to an energy equivalent.
93
37
Rb
93
38
Sr
93
39
Y
93
40
Zr
93
41
Fixture checkpoint marker-0419-xray is recorded on fixture page 0419.
Nb
140
55
Cs
140
56
Ba
140
57
La
140
58
Ce
EDecay mRb 93 mNb 93 4 melectron
931.5MeV
amu
92.91699 amu 92.90638 amu 4 0.0005486 amu
931.5 MeV
amu
0.008416 amu
931.5 MeV

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7.84 MeV
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The total energy released per fission will vary from the fission to the next depending on what fission products are formed, but the average total energy released per fission of uranium-235 with a thermal neutron is 200 MeV.
As illustrated in the preceding example, the majority of the energy liberated in the fission process is released immediately after the fission occurs and appears as the kinetic energy of the fission fragments, kinetic energy of the fission neutrons, and instantaneous gamma rays. The remaining energy is released over a period of time after the fission occurs and appears as kinetic energy of the beta, neutrino, and decay gamma rays.
Estimation of Decay Energy
In addition to this instantaneous energy release during the actual fission reaction, there is additional energy released when the fission fragments decay by emission. This additional
-
energy is called decay energy, EDecay. The decay chains for rubidium-93 and cesium-140 are shown below.
The energy released during the decay for each chain will be equivalent to the mass difference between the original fission product and the sum of the final stable nuclide and the beta particles emitted.
The energy released in the decay chain of rubidium-93 is calculated below.
EDecay mRb 93 mNb 93 3 melectron
931.5MeV
amu
139.90910 amu 139.90543 amu 3 0.0005486 amu
931.5 MeV
amu
0.000202 amu
931.5 MeV
amu
1.89 MeV
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Fixture checkpoint marker-0420-xray is recorded on fixture page 0420.
The energy released in the decay chain of cesium-140 is calculated below.
The total decay energy is the sum of the energies of the two chains, or 9.73 MeV.
Distribution of Fission Energy
The average energy distribution for the energy released per fission with a thermal neutron in uranium-235 is shown in Tables 6 and 7.
TABLE 6
Instantaneous Energy from Fission
Kinetic Energy of Fission Products 167 Mev
Energy of Fission Neutrons 5 MeV
Instantaneous Gamma-ray Energy 5 MeV
Capture Gamma-ray Energy 10 MeV
Total Instantaneous Energy 187 MeV
TABLE 7
Delayed Energy from Fission

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Gamma-rays from Fission Products 6 MeV
Neutrinos 10 MeV
Total Delayed Energy 23 MeV
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Because the 10 MeV of neutrino energy shown in Table 7 is not absorbed in the reactor, the average value of 200 MeV per fission is still accurate. Note in Table 6 that some fission neutrons undergo radiative capture and the resultant gamma ray emission provides an additional 10 MeV of instantaneous energy, which contributes to the total of 187 MeV instantaneous energy.
All of the energy released, with the exception of the neutrino energy, is ultimately transformed
Fixture checkpoint marker-0421-xray is recorded on fixture page 0421.
into heat through a number of processes. The fission fragments, with their high positive charge and kinetic energy, cause ionization directly as they rip orbital electrons from the surrounding atoms. In this ionization process, kinetic energy is transferred to the surrounding atoms of the fuel material, resulting in an increase in temperature. The beta particles and gamma rays also give up their energy through ionization, and the fission neutrons interact and lose their energy through elastic scattering. Of the 200 MeV released per fission, about seven percent (13 MeV) is released at some time after the instant of fission. When a reactor is shut down, fissions essentially cease, but energy is still being released from the decay of fission products. The heat produced by this decay energy is referred to as "decay heat." Although decay energy represents about seven percent of reactor heat production during reactor operation, once the reactor is shut down the decay heat production drops off quickly to a small fraction of its value while operating. The decay heat produced is significant, however, and systems must be provided to keep the reactor cool even after shutdown.
Summary
The important information in this chapter is summarized below.
Energy Release From Fission Summary
Fission products have some general characteristics in common.
They generally decay by β- emission.
The most common mass numbers are grouped near 95 and 140.
The energy released by fission can be calculated based on the difference in mass between the masses of the reactants before fission and the fission fragments and fission neutrons after fission.
Another method to determine the energy released by fission is based on the change in binding energy per nucleon between the fissile nuclide and the fission products.
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Atomic and Nuclear Physics INTERACTION OF RADIATION WITH MATTER
INTERACTION OF RADIATION WITH MATTER
Different types of radiation interact with matter in widely different ways. A large, massive, charged alpha particle cannot penetrate a piece of paper and even has a limited range in dry air. A neutrino, at the other extreme, has a low probability of interacting with any matter, even if it passed through the diameter of the earth.
EO 5.1 DESCRIBE interactions of the following with matter:
a. Alpha particle c. Positron
b. Beta particle d. Neutron

with matter:
a. Photoelectric effect
b. Compton scattering
c. Pair production
Interaction of Radiation With Matter
Radiation can be classified into two general groups, charged and uncharged; therefore, it may be expected that interactions with matter fall into two general types. Charged particles directly ionize the media through which they pass, while uncharged particles and photons can cause ionization only indirectly or by secondary radiation.
A moving charged particle has an electrical field surrounding it, which interacts with the atomic structure of the medium through which it is passing. This interaction decelerates the particle and accelerates electrons in the atoms of the medium. The accelerated electrons may acquire enough energy to escape from the parent atom. This process, whereby radiation "strips" off orbital electrons, is called ionization. Uncharged moving particles have no electrical field, so they can only lose energy and cause ionization by such means as collisions or scattering. A photon can lose energy by the photoelectric effect, Compton effect, or pair production.
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Fixture checkpoint marker-0422-xray is recorded on fixture page 0422.
I
m z
2
K.E.
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Because ionizing radiation creates ions in pairs, the intensity of ionization or the specific ionization is defined as the number of ion-pairs formed per centimeter of travel in a given material. The amount of ionization produced by a charged particle per unit path length, which is a measure of its ionizing power, is roughly proportional to the particle's mass and the square of its charge as illustrated in the equation below.
where:
I is the ionizing power
m is the mass of the particle
z is the number of unit charges it carries
K.E. is its kinetic energy
Since m for an alpha particle is about 7300 times as large as m for a beta particle, and z is twice as great, an alpha will produce much more ionization per unit path length than a beta particle of the same energy. This phenomenon occurs because the larger alpha particle moves slower for a given energy and thus acts on a given electron for a longer time.
Alpha Radiation
Alpha radiation is normally produced from the radioactive decay of heavy nuclides and from certain nuclear reactions. The alpha particle consists of 2 neutrons and 2 protons, so it is essentially the same as the nucleus of a helium atom. Because it has no electrons, the alpha particle has a charge of +2. This positive charge causes the alpha particle to strip electrons from the orbits of atoms in its vicinity. As the alpha particle passes through material, it removes electrons from the orbits of atoms it passes near. Energy is required to remove electrons and the energy of the alpha particle is reduced by each reaction. Eventually the particle will expend its kinetic energy, gain 2 electrons in orbit, and become a helium atom. Because of its strong

distance of travel. This rapid, large deposition of energy limits the penetration of alpha particles. The most energetic alpha particles are stopped by a few centimeters of air or a sheet of paper.

Beta-Minus Radiation

A beta-minus particle is an electron that has been ejected at a high velocity from an unstable nucleus. An electron has a small mass and an electrical charge of -1. Beta particles cause ionization by displacing electrons from atom orbits. The ionization occurs from collisions with orbiting electrons. Each collision removes kinetic energy from the beta particle, causing it to slow down. Eventually the beta particle will be slowed enough to allow it to be captured as an

2 electrons 0.000549 amu

electron

931.5 MeV

amu

1.02 MeV

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orbiting electron in an atom. Although more penetrating than the alpha, the beta is relatively easy to stop and has a low power of penetration. Even the most energetic beta radiation can be stopped by a few millimeters of metal.

Positron Radiation

Positively charged electrons are called positrons. Except for the positive charge, they are identical to beta-minus particles and interact with matter in a similar manner. Positrons are very short-lived, however, and quickly are annihilated by interaction with a negatively charged

Fixture checkpoint marker-0423-xray is recorded on fixture page 0423.

electron, producing two gammas with a combined energy (calculated below) equal to the rest mass of the positive and negative electrons.

Neutron Radiation

Neutrons have no electrical charge. They have nearly the same mass as a proton (a hydrogen atom nucleus). A neutron has hundreds of times more mass than an electron, but 1/4 the mass of an alpha particle. The source of neutrons is primarily nuclear reactions, such as fission, but they may also be produced from the decay of radioactive nuclides. Because of its lack of charge, the neutron is difficult to stop and has a high penetrating power.

Neutrons are attenuated (reduced in energy and numbers) by three major interactions, elastic scatter, inelastic scatter, and absorption. In elastic scatter, a neutron collides with a nucleus and bounces off. This reaction transmits some of the kinetic energy of the neutron to the nucleus of the atom, resulting in the neutron being slowed, and the atom receives some kinetic energy (motion). This process is sometimes referred to as "the billiard ball effect."

As the mass of the nucleus approaches the mass of the neutron, this reaction becomes more effective in slowing the neutron. Hydrogenous material attenuates neutrons most effectively.

In the inelastic scatter reaction, the same neutron/nucleus collision occurs as in elastic scatter. However, in this reaction, the nucleus receives some internal energy as well as kinetic energy. This slows the neutron, but leaves the nucleus in an excited state. When the nucleus decays to its original energy level, it normally emits a gamma ray.

In the absorption reaction, the neutron is actually absorbed into the nucleus of an atom. The neutron is captured, but the atom is left in an excited state. If the nucleus emits one or more gamma rays to reach a stable level, the process is called radiative capture. This reaction occurs at most neutron energy levels, but is more probable at lower energy levels.

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INTERACTION OF RADIATION WITH MATTER Atomic and Nuclear Physics
Gamma Radiation
Gamma radiation is electromagnetic radiation. It is commonly referred to as a gamma ray and is very similar to an x-ray. The difference is that gamma rays are emitted from the nucleus of an atom, and x-rays are produced by orbiting electrons. The x-ray is produced when orbiting electrons move to a lower energy orbit or when fast-moving electrons approaching an atom are deflected and decelerated as they react with the atom's electrical field (called Bremsstrahlung). The gamma ray is produced by the decay of excited nuclei and by nuclear reactions. Because the gamma ray has no mass and no charge, it is difficult to stop and has a very high penetrating power. A small fraction of the original gamma stream will pass through several feet of concrete or several meters of water.
There are three methods of attenuating gamma rays. The first method is referred to as the photo-electric effect. When a low energy gamma strikes an atom, the total energy of the gamma is expended in ejecting an electron from orbit. The result is ionization of the atom and expulsion of a high energy electron. This reaction is most predominant with low energy gammas interacting in materials with high atomic weight and rarely occurs with gammas having an energy above 1 MeV. Annihilation of the gamma results. Any gamma energy in excess of the binding energy of the electron is carried off by the electron in the form of kinetic energy.
The second method of attenuation of gammas is called Compton scattering. The gamma interacts with an orbital or free electron; however, in this case, the photon loses only a fraction of its energy. The actual energy loss depending on the scattering angle of the gamma. The gamma continues on at lower energy, and the energy difference is absorbed by the electron. This reaction becomes important for gamma energies of about 0.1 MeV and higher.
At higher energy levels, a third method of attenuation is predominant. This method is pair-production. When a high energy gamma passes close enough to a heavy nucleus, the gamma completely disappears, and an electron and a positron are formed. For this reaction to take place, the original gamma must have at least 1.02 MeV energy. Any energy greater than 1.02 MeV becomes kinetic energy shared between the electron and positron. The probability of pair■production incre
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Interaction of Radiation with Matter Summary
An alpha particle deposits a large amount of energy in a short distance of travel due to its large mass and charge.
Beta-minus particles interact with the electrons orbiting the nucleus of atoms, causing ionization by displacing the electrons. The beta particle loses energy with each interaction. After the beta particle loses enough energy, it is captured in the orbital shells of an atom.
Positrons interact with matter much the same way as beta minus particles. After the positron has lost most of its energy by ionizing atoms, it is annihilated by interaction with an electron. The electron-positron pair disappear and are replaced by two gammas, each with the energy equivalent of the mass of an electron (0.51 MeV).
Neutrons interact with matter by elastic scattering, inelastic scattering, or absorption.
Photoelectric effect is where a gamma interacts with an electron orbiting an atom.

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ejected from its orbit.
In Compton scattering a gamma interacts with an orbital electron, but only part of the gamma energy is transferred to the electron. The electron is ejected from its orbit, and the gamma is scattered off at a lower energy.
In pair-production, a gamma interacts with the electric field of a nucleus and is converted into an electron-positron pair. The gamma must have an energy greater than 1.02 MeV for this to occur.
Summary
The important information in this chapter is summarized below.
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NUCLEAR PHYSICS
AND REACTOR THEORY
Module 2
Reactor Theory (Neutron Characteristics)
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OBJECTIVES DOE-HDBK-1019/1-93 Reactor Theory (Neutron Characteristics)
TERMINAL OBJECTIVE
1.0 Without references, EXPLAIN how neutron sources produce neutrons.
ENABLING OBJECTIVES
1.1 DEFINE the following terms:
a. Intrinsic neutron source
b. Installed neutron source
1.2 LIST three examples of reactions that produce neutrons in intrinsic neutron sources.
1.3 LIST three examples of reactions that produce neutrons in installed neutron sources.
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Reactor Theory (Neutron Characteristics) DOE-HDBK-1019/1-93 OBJECTIVES
TERMINAL OBJECTIVE
2.0 Given the necessary information for calculations, EXPLAIN basic concepts in reactor physics and perform calculations.
ENABLING OBJECTIVES
2.1 DEFINE the following terms:
a. Atom density d. Barn
b. Neutron flux e. Macroscopic cross section
c. Microscopic cross section f. Mean free path
2.2 EXPRESS macroscopic cross section in terms of microscopic cross section.
2.3 DESCRIBE how the absorption cross section of typical nuclides varies with neutron
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energy at energies below the resonance absorption region.
2.4 DESCRIBE the cause of resonance absorption in terms of nuclear energy levels.
2.5 DESCRIBE the energy dependence of resonance absorption peaks for typical light and heavy nuclei.
2.6 EXPRESS mean free path in terms of macroscopic cross section.
2.7 Given the number densities (or total density and component fractions) and microscopic cross sections of components, CALCULATE the macroscopic cross section for a mixture.
2.8 CALCULATE a macroscopic cross section given a material density, atomic mass, and microscopic cross section.
2.9 EXPLAIN neutron shadowing or self-shielding.
2.10 Given the neutron flux and macroscopic cross section, CALCULATE the reaction rate.
2.11 DESCRIBE the relationship between neutron flux and reactor power.
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ENABLING OBJECTIVES (Cont.)
2.12 DEFINE the following concepts:
a. Thermalization d. Average logarithmic energy decrement
b. Moderator e. Macroscopic slowing down power
c. Moderating ratio
2.13 LIST three desirable characteristics of a moderator.
2.14 Given an average fractional energy loss per collision, CALCULATE the energy loss after a specified number of collisions.
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Reactor Theory (Neutron Characteristics) DOE-HDBK-1019/1-93 OBJECTIVES
TERMINAL OBJECTIVE

and delayed neutrons.
ENABLING OBJECTIVES
3.1 STATE the origin of prompt neutrons and delayed neutrons.
3.2 STATE the approximate fraction of neutrons that are born as delayed neutrons from the fission of the following nuclear fuels:
a. Uranium-235
b. Plutonium-239
3.3 EXPLAIN the mechanism for production of delayed neutrons.
3.4 EXPLAIN prompt and delayed neutron generation times.
3.5 Given prompt and delayed neutron generation times and delayed neutron fraction, CALCULATE the average generation time.
3.6 EXPLAIN the effect of delayed neutrons on reactor control.

TERMINAL OBJECTIVE
4.0 Without references, DESCRIBE the neutron energy spectrum for the type of reactor presented in this module.
ENABLING OBJECTIVES
4.1 STATE the average energy at which prompt neutrons are produced.
4.2 DESCRIBE the neutron energy spectrum in the following reactors:
a. Fast reactor
b. Thermal reactor
4.3 EXPLAIN the reason for the particular shape of the fast, intermediate, and slow energy regions of the neutron flux spectrum for a thermal reactor.

NEUTRON SOURCES
Neutrons from a variety of sources are always present in a reactor core. This is
Fixture checkpoint marker-0428-xray is recorded on fixture page 0428.
true even when the reactor is shut down. Some of these neutrons are produced by naturally occurring (intrinsic) neutron sources, while others may be the result of fabricated (installed) neutron sources that are incorporated into the design of the reactor. The neutrons produced by sources other than neutron-induced fission are often grouped together and classified as source neutrons.
EO 1.1 DEFINE the following terms:
a. Intrinsic neutron source
b. Installed neutron source
EO 1.2 LIST three examples of reactions that produce neutrons in intrinsic neutron sources.
EO 1.3 LIST three examples of reactions that produce neutrons in installed neutron sources.
Neutron Sources
In addition to neutron-induced fission, neutrons are produced by other reactions. The neutrons produced by reactions other than neutron-induced fission are called source neutrons. Source neutrons are important because they ensure that the neutron population remains high enough to allow a visible indication of neutron level on the most sensitive monitoring instruments while the reactor is shutdown and during the startup sequence. This verifies instrument operability and

intrinsic or installed neutron sources.
Intrinsic Neutron Sources
Some neutrons will be produced in the materials present in the reactor due to a variety of unavoidable reactions that occur because of the nature of these materials. Intrinsic neutron sources are those neutron-producing reactions that always occur in reactor materials.
■ ■ ■ ■
■ ■ ■
■ ■
NEUTRON SOURCES DOE-HDBK-1019/1-93 Reactor Theory (Neutron Characteristics)
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A limited number of neutrons will always be present, even in a reactor core that has never been operated, due to spontaneous fission of some heavy nuclides that are present in the fuel. Uranium■238, ur of the rate at which different heavy nuclides produce neutrons by spontaneous fission.
Californium-252 is not an intrinsic neutron source, but will be discussed in the section on installed neutron sources.
TABLE 1
Neutron Production by Spontaneous Fission
Nuclide T (Fission) T (1/2 1/2 ■-decay) neutrons/sec/gram
U 1.8 x 10 years 6.8 x 10 years 8.0 x 10 235
92
17 8 -4
U 8.0 x 10 years 4.5 x 10 years 1.6 x 10 238
92
15 9 -2
Pu 5.5 x 10 years 2.4 x 10 years 3.0 x 10 239
94
5 4 -2
Pu 1.2 x 10 years 6.6 x 10 years 1.0 x 10 240
94
11 3 3
Cf 66.0 years 2.65 years 2.3 x 10 252
98
12
Another intrinsic neutron source is a reaction involving natural boron and fuel. In some reactors, natural boron is loaded into the reactor core as a neutron absorber to improve reactor control or increase core life-time. Boron-11 (80.1% of natural boron) undergoes a reaction with the alpha
Fixture checkpoint marker-0429-xray is recorded on fixture page 0429.
particle emitted by the radioactive decay of heavy nuclides in the fuel to yield a neutron as shown below.
The boron-11 must be mixed with, or in very close proximity to, the fuel for this reaction because of the short path length of the alpha particle. For a reactor core with this configuration, this (■,n) reaction is an important source of neutrons for reactor startup.
In a reactor that has been operated, another source of neutrons becomes significant. Neutrons may be produced by the interaction of a gamma ray and a deuterium nucleus. This reaction is commonly referred to as a photoneutron reaction because it is initiated by electromagnetic radiation and results in the production of a neutron. The photoneutron reaction is shown below.
■ ■ ■
■ ■

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■ ■

Reactor Theory (Neutron Characteristics) DOE-HDBK-1019/1-93 NEUTRON SOURCES
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There is an abundant supply of high energy gammas in a reactor that has been operated because many of the fission products are gamma emitters. All water-cooled reactors have some deuterium present in the coolant in the reactor core because a small fraction of natural hydrogen is the isotope deuterium. The atom percentage of deuterium in the water ranges from close to the naturally occurring value (0.015%) for light water reactors to above 90% deuterium for heavy water reactors. Therefore, the required conditions for production of photoneutrons exist.

The supply of gamma rays decreases with time after shutdown as the gamma emitters decay; therefore, the photoneutron production rate also decreases. In a few particular reactors, additional D O (heavy water) may be added to the reactor to increase the production of 2 photoneutrons following a long shutdown period.

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Installed Neutron Sources

Because intrinsic neutron sources can be relatively weak or dependent upon the recent power history of the reactor, many reactors have artificial sources of neutrons installed. These neutron sources ensure that shutdown neutron levels are high enough to be detected by the nuclear instruments at all times. This provides a true picture of reactor conditions and any change in these conditions. An installed neutron source is an assembly placed in or near the reactor for the sole purpose of producing source neutrons.

One strong source of neutrons is the artificial nuclide californium-252, which emits neutrons at the rate of about 2 x 10 neutrons per second per gram as the result of spontaneous fission. 12 Important drawbacks for some applications may be its high cost and its short half-life (2.65 years).

Many installed neutron sources use the (■,n) reaction with beryllium. These sources are composed of a mixture of metallic beryllium (100% beryllium-9) with a small quantity of an alpha particle emitter, such as a compound of radium, polonium, or plutonium. The reaction that occurs is shown below.

The beryllium is intimately (homogeneously) mixed with the alpha emitter and is usually enclosed in a stainless steel capsule.

Another type of installed neutron source that is widely used is a photoneutron source that employs the (■,n) reaction with beryllium. Beryllium is used for photoneutron sources because its stable isotope beryllium-9 has a weakly attached last neutron with a binding energy of only 1.66 MeV. Thus, a gamma ray with greater energy than 1.66 MeV can cause neutrons to be ejected by the (■,n) reaction as shown below.

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Neutron Sources Summary

& Intrinsic neutron sources are sources of neutrons from materials that are in the reactor for other purposes such as fuel, burnable poison, or moderator.

& Installed neutron sources are materials or components placed in the reactor specifically for the purpose of producing source neutrons.

& Examples of intrinsic neutron sources are listed below.

Spontaneous fission of heavy nuclides in fuel, such as uranium-238, uranium-235, and plutonium-239, results in fission fragments and free neutrons.

Boron-11 mixed with the fuel undergoes an alpha-neutron reaction

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Deuterium present in the reactor coolant undergoes a gamma■neutron reaction and becomes hydrogen-1.
& Examples of installed neutron sources are listed below.
Spontaneous fission of californium-252 results in fission fragments
and free neutrons.
Beryllium-9 undergoes an alpha-neutron reaction (alpha from the
decay of plutonium, polonium, or radium) and becomes carbon-12.
Beryllium-9 undergoes a gamma-neutron reaction (high energy
gamma from decay of antimony-124) and becomes beryllium-8.
Many startup sources of this type use antimony and beryllium because after activation with neutrons the radioactive antimony becomes an emitter of high energy gammas. The photoneutron sources of this type are constructed somewhat differently from the (■,n) types. One design incorporates a capsule of irradiated antimony enclosed in a beryllium sleeve. The entire assembly is then encased in a stainless steel cladding. A large reactor may have several neutron sources of this type installed within the core.
Summary
The important information in this chapter is summarized below.
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Reactor Theory (Neutron Characteristics) NUCLEAR CROSS SECTIONS AND NEUTRON FLUX
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NUCLEAR CROSS SECTIONS AND NEUTRON FLUX
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To determine the frequency of neutron interactions, it is necessary to describe the availability of neutrons to cause interaction and the probability of a neutron interacting with material. The availability of neutrons and the probability of interaction are quantified by the neutron flux and nuclear cross section.
EO 2.1 DEFINE the following terms:
a. Atom density d. Barn
b. Neutron flux e. Macroscopic cross section
c. Microscopic cross section f. Mean free path
EO 2.2 EXPRESS macroscopic cross section in terms of microscopic cross section.
EO 2.3 DESCRIBE how the absorption cross section of typical nuclides varies with neutron energy at energies below the resonance absorption region.
EO 2.4 DESCRIBE the cause of resonance absorption in terms of nuclear energy levels.
EO 2.5 DESCRIBE the energy dependence of resonance absorption peaks for typical light and heavy nuclei.
EO 2.6 EXPRESS mean free path in terms of macroscopic cross section.
EO 2.7 Given the number densities (or total density and component fractions) and microscopic cross sections of components, CALCULATE the macroscopic cross section for a mixture.
EO 2.8 CALCULATE a macroscopic cross section given a material density, atomic mass, and microscopic cross section.
EO 2.9 EXPLAIN neutron shadowing or self-shielding.
'

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Introduction
Fission neutrons are born with an average energy of about 2 MeV. These fast neutrons interact with the reactor core materials in various absorption and scattering reactions. Collisions that result in scattering are useful in slowing neutrons to thermal energies. Thermal neutrons may be absorbed by fissile nuclei to produce more fissions or be absorbed in fertile material for conversion to fissionable fuel. Absorption of neutrons in structural components, coolant, and other non-fuel material results in the removal of neutrons without fulfilling any useful purpose. To safely and efficiently operate a nuclear reactor it is necessary to predict the probability that a particular absorption or scattering reaction will occur. Once these probabilities are known, if the availability of neutrons can be determined, then the rate at which these nuclear reactions take place can be predicted.
Atom Density
One important property of a material is the atom density. The atom density is the number of atoms of a given type per unit volume of the material. To calculate the atom density of a substance use Equation (2-1).
(2-1)
where:
N = atom density (atoms/cm)3
' = density (g/cm)3
N = Avogadro's number (6.022 x 10 atoms/mole) A
23
M = gram atomic weight
N
NA
M
2.699 g
cm3
Fixture checkpoint marker-0432-xray is recorded on fixture page 0432.
6.022 x 1023 atoms
mole
26.9815 g
mole
6.024 x 1022 atoms
cm3
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Reactor Theory (Neutron Characteristics) NUCLEAR CROSS SECTIONS AND NEUTRON FLUX
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Example:
A block of aluminum has a density of 2.699 g/cm . If the gram atomic weight of
3
aluminum is 26.9815 g, calculate the atom density of the aluminum.
Solution:
Cross Sections
The probability of a neutron interacting with a nucleus for a particular reaction is dependent upon not only the kind of nucleus involved, but also the energy of the neutron. Accordingly, the absorption of a thermal neutron in most materials is much more probable than the absorption of a fast neutron. Also, the probability of interaction will vary depending upon the type of reaction involved.

microscopic cross section ( ) of the nucleus for the particular reaction. This cross section will vary with the energy of the neutron. The microscopic cross section may also be regarded as the effective area the nucleus presents to the neutron for the particular reaction. The larger the effective area, the greater the probability for reaction.

Because the microscopic cross section is an area, it is expressed in units of area, or square centimeters. A square centimeter is tremendously large in comparison to the effective area of a nucleus, and it has been suggested that a physicist once referred to the measure of a square centimeter as being "as big as a barn" when applied to nuclear processes. The name has persisted and microscopic cross sections are expressed in terms of barns. The relationship between barns and cm is shown below. 2

1 barn = 10-24 cm2

N

Whether a neutron will interact with a certain volume of material depends not only on the microscopic cross section of the individual nuclei but also on the number of nuclei within that volume. Therefore, it is necessary to define another kind of cross section known as the macroscopic cross section ( ). The macroscopic cross section is the probability of a given reaction occurring per unit travel of the neutron. is related to the microscopic cross section ( ) by the relationship shown below.

(2-2)

where:

= macroscopic cross section (cm)

-1

N = atom density of material (atoms/cm)

3

= microscopic cross-section (cm)

2

The difference between the microscopic and macroscopic cross sections is extremely important and is restated for clarity. The microscopic cross section ( ) represents the effective target area that a single nucleus presents to a bombarding particle. The units are given in barns or cm .

2

The macroscopic cross section ( ) represents the effective target area that is presented by all of

Fixture checkpoint marker-0433-xray is recorded on fixture page 0433.

the nuclei contained in 1 cm of the material. The units are given as 1/cm or cm .

3 -1

A neutron interacts with an atom of the material it enters in two basic ways. It will either interact through a scattering interaction or through an absorption reaction. The probability of a neutron being absorbed by a particular atom is the microscopic cross section for absorption, . The probability of a neutron scattering off of a particular nucleus is the microscopic cross a section for scattering, s

. The sum of the microscopic cross section for absorption and the microscopic cross section for scattering is the total microscopic cross section, . T

= + T a s

Both the absorption and the scattering microscopic cross sections can be further divided. For instance, the scattering cross section is the sum of the elastic scattering cross section (se) and the inelastic scattering cross section ( ). si

The microscopic absorption cross section ( ) includes all reactions except scattering. However, a for most purposes it is sufficient to merely separate it into two categories, fission (f) and capture ( ). Radiative capture of neutrons was described in the Neutron Interactions chapter c of Module 1.

= + a f c

Figure 1 Typical Neutron Absorption Cross Section vs. Neutron Energy

The variation of absorption cross sections with neutron energy is often complicated. For many elements the absorption cross sections are small, ranging from a fraction of a barn to a few

Fixture checkpoint marker-0434-xray is recorded on fixture page 0434.

barns for slow (or thermal) neutrons.

For a considerable number of nuclides of moderately high (or high) mass numbers, an examination of the variation of the absorption cross section with the energy of the incident neutron reveals the existence of three regions on a curve of absorption cross section versus neutron energy. This cross section is illustrated in Figure 1. First, the cross section decreases steadily with increasing neutron energy in a low energy region, which includes the thermal range (E < 1 eV). In this region the absorption cross section, which is often high, is inversely proportional to the velocity (v). This region is frequently referred to as the "1/v region," because the absorption cross section is proportional to 1/v, which is the reciprocal of neutron velocity. Following the 1/v region, there occurs the "resonance region" in which the cross sections rise sharply to high values called "resonance peaks" for neutrons of certain energies, and then fall again. These energies are called resonance energies and are a result of the affinity of the nucleus for neutrons whose energies closely match its discrete, quantum energy levels. That is, when the binding energy of a neutron plus the kinetic energy of the neutron are exactly equal to the amount required to raise a compound nucleus from its ground state to a quantum level, resonance absorption occurs. The following example problem further illustrates this point.

■

(

Assuming that uranium-236 has a nuclear quantum energy level at 6.8 MeV above its ground state, calculate the kinetic energy a neutron must possess to undergo resonant absorption in uranium-235 at this resonance energy level.

BE = [Mass(U) + Mass(neutron) - Mass(U)] x 931 MeV/amu 235 236

BE = (235.043925 + 1.008665 - 236.045563) x 931 MeV/amu

BE = (0.007025 amu) x 931 MeV/amu = 6.54 MeV

6.8 MeV - 6.54 MeV = 0.26 MeV

The difference between the binding energy and the quantum energy level equals the amount of kinetic energy the neutron must possess. The typical heavy nucleus will have many closely■spaced resonanc complex and have more possible configurations and corresponding energy states. Light nuclei, being less complex, have fewer possible energy states and fewer resonances that are sparsely distributed at higher energy levels.

For higher neutron energies, the absorption cross section steadily decreases as the energy of the neutron increases. This is called the "fast neutron region." In this region the absorption cross sections are usually less than 10 barns.

sections are generally small, for example, 5 barns to 10 barns. This is close to the magnitude of the actual geometric cross sectional area expected for atomic nuclei. In potential scattering, the cross section is essentially constant and independent of neutron energy. Resonance elastic scattering and inelastic scattering exhibit resonance peaks similar to those associated with absorption cross sections. The resonances occur at lower energies for heavy nuclei than for light nuclei. In general, the variations in scattering cross sections are very small when compared to the variations that occur in absorption cross sections.

Mean Free Path

If a neutron has a certain probability of undergoing a particular interaction in one centimeter of travel, then the inverse of this value describes how far the neutron will travel (in the average case) before undergoing an interaction. This average distance traveled by a neutron before interaction is known as the mean free path for that interaction and is represented by the symbol ■. The relationship between the mean free path (■) and the macroscopic cross section (*) is shown below.

(2-3)

'

(

)

Fixture checkpoint marker-0435-xray is recorded on fixture page 0435.

Calculation of Macroscopic Cross Section and Mean Free Path

Most materials are composed of several elements, and because most elements are composed of several isotopes, most materials involve many cross sections, one for each isotope involved. Therefore, to include all the isotopes within a given material, it is necessary to determine the macroscopic cross section for each isotope and then sum all the individual macroscopic cross sections. Equation (2-4) can be used to determine the macroscopic cross section for a composite material.

* = N) + N) + N) + ... N) (2-4) 1 1 2 2 3 3 nn

where:

N = the number of nuclei per cm of the n element n

3 th

) = the microscopic cross section of the n element n

th

The following example problems illustrate the calculation of the macroscopic cross section for a single element and for combinations of materials.

Example 1:

Find the macroscopic thermal neutron absorption cross section for iron, which has a density of 7.86 g/cm . The microscopic cross section for absorption of iron is 2.56 barns 3 and the gram atomic weight is 55.847 g.

Solution:

Step 1: Using Equation (2-1), calculate the atom density of iron.

Step 2: Use this atom density in Equation (2-2) to calculate the macroscopic cross section.

'

'

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Example 2:
An alloy is composed of 95% aluminum and 5% silicon (by weight). The density of the alloy is 2.66 g/cm . Properties of aluminum and silicon are shown below. 3
Element Gram Atomic Weight)) a (barns) s (barns)
Aluminum 26.9815 0.23 1.49
Silicon 28.0855 0.16 2.20
1. Calculate the atom densities for the aluminum and silicon.
2. Determine the absorption and scattering macroscopic cross sections for thermal neutrons.
3. Calculate the mean free paths for absorption and scattering.
Solution:
Step 1: The density of the aluminum will be 95% of the total density. Using Equation (2-1) yields the atom densities.
a NAl a,Al NSi a,Si
5.64 x 1022 atoms
cm3
0.23 x 10 24 cm22.85 x 1021 atoms
cm3
0.16 x 10 24 cm2
0.0134 cm 1
s NAl s,Al NSi s,Si
5.64 x 1022 atoms
cm3
1.49 x 10 24 cm22.85 x 1021 atoms
Fixture checkpoint marker-0436-xray is recorded on fixture page 0436.
cm3
2.20 x 10 24 cm2
0.0903 cm 1
a
1
a
1
0.01345 cm 1
74.3 cm
s
1
s
1
0.0903 cm 1
11.1 cm
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Step 2: The macroscopic cross sections for absorption and scattering are calculated using Equation (2-4).
Step 3: The mean free paths are calculated by inserting the macroscopic cross sections calculated above into Equation (2-3).

must travel only 11.1 cm to interact by scattering.
)
)
)
)
(■
(■
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NUCLEAR CROSS SECTIONS AND NEUTRON FLUX Reactor Theory (Neutron Characteristics)
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Effects of Temperature on Cross Section
As discussed, the microscopic absorption cross section varies significantly as neutron energy varies. The microscopic cross sections provided on most charts and tables are measured for a standard neutron velocity of 2200 meters/second, which corresponds to an ambient temperature of 68(F. Therefore, if our material is at a higher temperature, the absorption cross section will be lower than the value for 68(F, and any cross sections which involve absorption (for example,) ,) ,)) must be corrected for the existing temperature. acf
The following formula is used to correct microscopic cross sections for temperature. Although the example illustrates absorption cross section, the same formula may be used to correct capture and fission cross sections.
where:
) = microscopic cross section corrected for temperature
) = microscopic cross section at reference temperature (68(F or 20(C) o
T = reference temperature (68(F) in degrees Rankine ((R) or Kelvin ((K) o
T = temperature for which corrected value is being calculated
NOTE: When using this formula, all temperatures must be converted to (R or (K.
(R = (F + 460
(K = (C + 273
Example:
What is the value of)f for uranium-235 for thermal neutrons at 500(F? Uranium-235 has a) of 583 barns at 68(F. f
Solution:
Fixture checkpoint marker-0437-xray is recorded on fixture page 0437.
DOE-HDBK-1019/1-93
Reactor Theory (Neutron Characteristics) NUCLEAR CROSS SECTIONS AND NEUTRON FLUX
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Neutron Flux
Macroscopic cross sections for neutron reactions with materials determine the probability of one neutron undergoing a specific reaction per centimeter of travel through that material. If one wants to determine how many reactions will actually occur, it is necessary to know how many neutrons are traveling through the material and how many centimeters they travel each second. It is convenient to consider the number of neutrons existing in one cubic centimeter at any one instant and the total distance they travel each second while in that cubic centimeter. The number of neutrons existing in a cm of material at any instant is called neutron density and is 3 represented by the symbol n with units of neutrons/cm . The total distance these neutrons can 3 travel each second will be determined by their velocity.
A good way of defining neutron flux (1) is to consider it to be the total path length covered by all neutrons in one cubic centimeter during one second. Mathematically, this is the equation

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1 = n v (2-5)
where:
1 = neutron flux (neutrons/cm -sec) 2
n = neutron density (neutrons/cm)3
v = neutron velocity (cm/sec)
The term neutron flux in some applications (for example, cross section measurement) is used as parallel beams of neutrons traveling in a single direction. The intensity (I) of a neutron beam is the product of the neutron density times the average neutron velocity. The directional beam intensity is equal to the number of neutrons per unit area and time (neutrons/cm -sec) falling on 2 a surface perpendicular to the direction of the beam.
One can think of the neutron flux in a reactor as being comprised of many neutron beams traveling in various directions. Then, the neutron flux becomes the scalar sum of these directional flux intensities (added as numbers and not vectors), that is, 1 = I + I + I +...I . 123 n
Since the atoms in a reactor do not interact preferentially with neutrons from any particular direction, all of these directional beams contribute to the total rate of reaction. In reality, at a given point within a reactor, neutrons will be traveling in all directions.
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NUCLEAR CROSS SECTIONS AND NEUTRON FLUX Reactor Theory (Neutron Characteristics)
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Self-Shielding
In some locations within the reactor, the flux level may be significantly lower than in other areas due to a phenomenon referred to as neutron shadowing or self-shielding. For example, the interior of a fuel pin or pellet will "see" a lower average flux level than the outer surfaces since an appreciable fraction of the neutrons will have been absorbed and therefore cannot reach the interior of the fuel pin. This is especially important at resonance energies, where the absorption cross sections are large.
Summary
The important information in this chapter is summarized below.
DOE-HDBK-1019/1-93
Reactor Theory (Neutron Characteristics) NUCLEAR CROSS SECTIONS AND NEUTRON FLUX
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Nuclear Cross Section and Neutron Flux Summary
& Atom density (N) is the number of atoms of a given type per unit volume of material.
& Microscopic cross section ()) is the probability of a given reaction occurring between a neutron and a nucleus.
& Microscopic cross sections are measured in units of barns, where 1 barn = 10-24 cm .2
& Macroscopic cross section (*) is the probability of a given reaction occurring per
Fixture checkpoint marker-0438-xray is recorded on fixture page 0438.
unit length of travel of the neutron. The units for macroscopic cross section are cm . -1
& The mean free path (■) is the average distance that a neutron travels in a material between interactions.
& Neutron flux (1) is the total path length traveled by all neutrons in one cubic centimeter of material during one second.
& The macroscopic cross section for a material can be calculated using the equation below.

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& The absorption cross section for a material usually has three distinct regions. At low neutron energies (<1 eV) the cross section is inversely proportional to the neutron velocity.
& Resonance absorption occurs when the sum of the kinetic energy of the neutron and its binding energy is equal to an allowed nuclear energy level of the nucleus.
& Resonance peaks exist at intermediate energy levels. For higher neutron energies, the absorption cross section steadily decreases as the neutron energy increases.
& The mean free path equals 1/*.
& The macroscopic cross section for a mixture of materials can be calculated using the equation below.
* = N 1 1 2 2 3 3 nn) + N) + N) + ... N)
& Self-shielding is where the local neutron flux is depressed within a material due to neutron absorption near the surface of the material.
1)
REACTION RATES DOE-HDBK-1019/1-93 Reactor Theory (Neutron Characteristics) NP-02 Page 18 Rev. 0
Fixture checkpoint marker-0439-xray is recorded on fixture page 0439.
REACTION RATES
It is possible to determine the rate at which a nuclear reaction will take place based on the neutron flux, cross section for the interaction, and atom density of the target. This relationship illustrates how a change in one of these items affects the reaction rate.
EO 2.10 Given the neutron flux and macroscopic cross section, CALCULATE the reaction rate.
EO 2.11 DESCRIBE the relationship between neutron flux and reactor power.
Reaction Rates
If the total path length of all the neutrons in a cubic centimeter in a second is known, (neutron flux (1)), and if the probability of having an interaction per centimeter path length is also known (macroscopic cross section (*)), multiply them together to get the number of interactions taking place in that cubic centimeter in one second. This value is known as the reaction rate and is denoted by the symbol R. The reaction rate can be calculated by the equation shown below.
R = 1 * (2-6)
where:
R = reaction rate (reactions/sec)
1 = neutron flux (neutrons/cm -sec) 2
* = macroscopic cross section (cm) -1
Substituting the fact that * = N) into Equation (2-6) yields the equation below.
where:
) = microscopic cross section (cm)2
N = atom density (atoms/cm)3
Rf f
1 x 1013 neutrons
cm2
sec
0.1 cm 1
1 x 1012 fissions
cm3

Rf f
f
Rf
1.29 x 1012 fissions
cm3
sec
3 x 1013 neutrons
cm2
sec
0.043 cm 1
Reactor Theory (Neutron Characteristics) DOE-HDBK-1019/1-93 REACTION RATES
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The reaction rate calculated will depend on which macroscopic cross section is used in the calculation. Normally, the reaction rate of greatest interest is the fission reaction rate.
Example:
If a one cubic centimeter section of a reactor has a macroscopic fission cross section of 0.1 cm , and if the thermal neutron flux is 10 neutrons/cm -sec, what is the fission rate -1 13 2 in that cubic centimeter?
Solution:
In addition to using Equation (2-6) to determine the reaction rate based on the physical properties of material, it is also possible to algebraically manipulate the equation to determine physical properties if the reaction rate is known.
Example:
Fixture checkpoint marker-0440-xray is recorded on fixture page 0440.
A reactor operating at a flux level of 3 x 10 neutrons/cm -sec contains 10 atoms of
13 2 20
uranium-235 per cm . The reaction rate is 1.29 x 10 fission/cm . Calculate and .
3 12 3
f f
Solution:
Step 1: The macroscopic cross section can be determined by solving
Equation (2-6) for and substituting the appropriate values. f
Rf N f
f
Rf
N
1.29 x 1012 fissions
cm3
sec
1 x 1020 atoms
cm3
3 x 1013 neutrons
cm2
sec
4.3 x 10 22 cm2 1 barn
1 x 10 24 cm2
430 barns
1 watt 1 erg

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1 MeV
1.602 x 10 6
erg
1 fission
200 MeV
3.12 x 1010 fissions
second
REACTION RATES DOE-HDBK-1019/1-93 Reactor Theory (Neutron Characteristics)
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Step 2: To find the microscopic cross section, replace f with (N x f) and solve
for .
f
Reactor Power Calculation
Multiplying the reaction rate per unit volume by the total volume of the core results in the total number of reactions occurring in the core per unit time. If the amount of energy involved in each reaction were known, it would be possible to determine the rate of energy release (power) due to a certain reaction.
In a reactor where the average energy per fission is 200 MeV, it is possible to determine the number of fissions per second that are necessary to produce one watt of power using the following conversion factors.
1 fission = 200 MeV
1 MeV = 1.602 x 10 ergs -6
1 erg = 1 x 10 watt-sec -7
This is equivalent to stating that 3.12 x 10 fissions release 1 watt-second of energy. 10
P
th f V
3.12 x 1010 fissions
watt sec
Reactor Theory (Neutron Characteristics) DOE-HDBK-1019/1-93 REACTION RATES
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Fixture checkpoint marker-0441-xray is recorded on fixture page 0441.
The power released in a reactor can be calculated based on Equation (2-6). Multiplying the reaction rate by the volume of the reactor results in the total fission rate for the entire reactor. Dividing by the number of fissions per watt-sec results in the power released by fission in the reactor in units of watts. This relationship is shown mathematically in Equation (2-7) below.
(2-7)
where:
P = power (watts)
= thermal neutron flux (neutrons/cm -sec) th
2
= macroscopic cross section for fission (cm) f
-1
V = volume of core (cm)
3
Relationship Between Neutron Flux and Reactor Power
In an operating reactor the volume of the reactor is constant. Over a relatively short period of time (days or weeks), the number density of the fuel atoms is also relatively constant. Since the atom density and microscopic cross section are constant, the macroscopic cross section must also

cross section are constant, then the reactor power and the neutron flux are directly proportional. This is true for day-to-day operation. The neutron flux for a given power level will increase very slowly over a period of months due to the burnup of the fuel and resulting decrease in atom density and macroscopic cross section.
REACTION RATES DOE-HDBK-1019/1-93 Reactor Theory (Neutron Characteristics)
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Reaction Rates Summary
& The reaction rate is the number of interactions of a particular type occurring in a cubic centimeter of material in a second.
& The reaction rate can be calculated by the equation below.
R = 1 *
& Over a period of several days, while the atom density of the fuel can be considered constant, the neutron flux is directly proportional to reactor power.
Summary
The important information in this chapter is summarized below.
Reactor Theory (Neutron Characteristics) DOE-HDBK-1019/1-93 NEUTRON MODERATION
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NEUTRON MODERATION
In thermal reactors, the neutrons that cause fission are at a much lower energy than the energy level at which they were born from fission. In this type of reactor, specific materials must be included in the reactor design to reduce the energy level of the neutrons in an efficient manner.
EO 2.12 DEFINE the following concepts:
a. Thermalization d. Average logarithmic energy decrement
b. Moderator e. Macroscopic slowing down power
c. Moderating ratio
EO 2.13 LIST three desirable characteristics of a moderator.
EO 2.14 Given an average fractional energy loss per collision, CALCULATE the energy loss after a specified number of collisions.
Neutron Slowing Down and Thermalization
Fission neutrons are produced at an average energy level of 2 MeV and immediately begin to slow down as the result of numerous scattering reactions with a variety of target nuclei. After a number of collisions with nuclei, the speed of a neutron is reduced to such an extent that it has approximately the same average kinetic energy as the atoms (or molecules) of the medium in which the neutron is undergoing elastic scattering. This energy, which is only a small fraction of an electron volt at ordinary temperatures (0.025 eV at 20(C), is frequently referred to as the
Fixture checkpoint marker-0442-xray is recorded on fixture page 0442.
thermal energy, since it depends upon the temperature. Neutrons whose energies have been reduced to values in this region (< 1 eV) are designated thermal neutrons. The process of reducing the energy of a neutron to the thermal region by elastic scattering is referred to as thermalization, slowing down, or moderation. The material used for the purpose of thermalizing neutrons is called a moderator. A good moderator reduces the speed of neutrons in a small number of collisions, but does not absorb them to any great extent. Slowing the neutrons in as few collisions as possible is desirable in order to reduce the amount of neutron leakage from the core and also to reduce the number of resonance absorptions in non-fuel materials. Neutron leakage and resonance absorption will be discussed in the next module.
ln Ei

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ln
Ei
Ef
2
A
2
3
NEUTRON MODERATION DOE-HDBK-1019/1-93 Reactor Theory (Neutron Characteristics)
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The ideal moderating material (moderator) should have the following nuclear properties.
large scattering cross section
small absorption cross section
large energy loss per collision
A convenient measure of energy loss per collision is the logarithmic energy decrement. The
Fixture checkpoint marker-0443-xray is recorded on fixture page 0443.
average logarithmic energy decrement is the average decrease per collision in the logarithm of the neutron energy. This quantity is represented by the symbol (Greek letter xi).
(2-8)
where:
= average logarithmic energy decrement
Ei = average initial neutron energy
Ef = average final neutron energy
The symbol is commonly called the average logarithmic energy decrement because of the fact that a neutron loses, on the average, a fixed fraction of its energy per scattering collision. Since the fraction of energy retained by a neutron in a single elastic collision is a constant for a given material, is also a constant. Because it is a constant for each type of material and does not depend upon the initial neutron energy, is a convenient quantity for assessing the moderating ability of a material.
The values for the lighter nuclei are tabulated in a variety of sources. The following commonly used approximation may be used when a tabulated value is not available.
This approximation is relatively accurate for mass numbers (A) greater than 10, but for some low values of A it may be in error by over three percent.
N
ln Ehigh ln Elow
ln
Ehigh
Elow
N
ln
Ehigh
Elow
ln
2 x 106eV
0.025 eV
0.948
19.2 collisions
Reactor Theory (Neutron Characteristics) DOE-HDBK-1019/1-93 NEUTRON MODERATION
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necessary for a neutron to lose a given amount of energy may be determined by dividing into the difference of the natural logarithms of the energy range in question. The number of collisions (N) to travel from any energy, E , to any lower energy, E , can be calculated as shown below. high low

Example:

How many collisions are required to slow a neutron from an energy of 2 MeV to a thermal energy of 0.025 eV, using water as the moderator? Water has a value of 0.948 for .

Solution:

Sometimes it is convenient, based upon information known, to work with an average fractional energy loss per collision as opposed to an average logarithmic fraction. If the initial neutron energy level and the average fractional energy loss per collision are known, the final energy level for a given number of collisions may be computed using the following formula.

! (

(2-9)

where:

E = initial neutron energy o

E = neutron energy after N collisions N

x = average fractional energy loss per collision

N = number of collisions

Fixture checkpoint marker-0444-xray is recorded on fixture page 0444.

Example:

If the average fractional energy loss per collision in hydrogen is 0.63, what will be the energy of a 2 MeV neutron after (a) 5 collisions? (b) 10 collisions?

Solution:

a)

b)

Macroscopic Slowing Down Power

Although the logarithmic energy decrement is a convenient measure of the ability of a material to slow neutrons, it does not measure all necessary properties of a moderator. A better measure of the capabilities of a material is the macroscopic slowing down power. The macroscopic slowing down power (MSDP) is the product of the logarithmic energy decrement and the macroscopic cross section for scattering in the material. Equation (2-10) illustrates how to calculate the macroscopic slowing down power.

(2-10)

MR

s

a

Moderating Ratio

Macroscopic slowing down power indicates how rapidly a neutron will slow down in the material in question, but it still does not fully explain the effectiveness of the material as a moderator. An element such as boron has a high logarithmic energy decrement and a good slowing down power, but it is a poor moderator because of its high probability of absorbing neutrons.

The most complete measure of the effectiveness of a moderator is the moderating ratio. The moderating ratio is the ratio of the macroscopic slowing down power to the macroscopic cross

performs as a moderator. Equation (2-11) shows how to calculate the moderating ratio of a material.

(2-11)

Moderating properties of different materials are compared in Table 2.

TABLE 2

Moderating Properties of Materials

Material	Collisions to Slowing Down	Number of Macroscopic Thermalize Power	Moderating Ratio
H2O	0.927	19	1.425 62
D2O	0.510	35	0.177 4830
Helium	0.427	42	9 x 10 51 -6
Beryllium	0.207	86	0.154 126
Boron	0.171	105	0.092 0.00086
Carbon	0.158	114	0.083 216

NEUTRON MODERATION DOE-HDBK-1019/1-93 Reactor Theory (Neutron Characteristics) NP-02 Page 28 Rev. 0

Summary

The important information in this chapter is summarized below.

Neutron Moderation Summary

& Thermalization is the process of reducing the energy level of a neutron from the energy level at which it is produced to an energy level in the thermal range.

& The moderator is the reactor material that is present for the purpose of thermalizing neutrons.

& Moderating ratio is the ratio of the macroscopic slowing down power to the macroscopic cross section for absorption.

Fixture checkpoint marker-0445-xray is recorded on fixture page 0445.

& The average logarithmic energy decrement (!) is the average change in the logarithm of neutron energy per collision.

& Macroscopic slowing down power is the product of the average logarithmic energy decrement and the macroscopic cross section for scattering.

& There are three desirable characteristics of a moderator.

1. large scattering cross section
2. small absorption cross section
3. large energy loss per collision

& The energy loss after a specified number of collisions can be calculated using the equation below.

E = E (1 - x) N o

N

■

■ ■

Reactor Theory (Neutron Characteristics) DOE-HDBK-1019/1-93 PROMPT AND DELAYED NEUTRONS Rev. 0 Page 29 NP-02

PROMPT AND DELAYED NEUTRONS

Not all neutrons are released at the same time following fission. Most neutrons are released virtually instantaneously and are called prompt neutrons. A very

called delayed neutrons. Although delayed neutrons are a very small fraction of the total number of neutrons, they play an extremely important role in the control of the reactor.

EO 3.1 STATE the origin of prompt neutrons and delayed neutrons.

EO 3.2 STATE the approximate fraction of neutrons that are born as delayed neutrons from the fission of the following nuclear fuels:

a. Uranium-235

b. Plutonium-239

EO 3.3 EXPLAIN the mechanism for production of delayed neutrons.

EO 3.4 EXPLAIN prompt and delayed neutron generation times.

EO 3.5 Given prompt and delayed neutron generation times and delayed neutron fraction, CALCULATE the average generation time.

EO 3.6 EXPLAIN the effect of delayed neutrons on reactor control.

Neutron Classification

The great majority (over 99%) of the neutrons produced in fission are released within about 10-13 seconds of the actual fission event. These are called prompt neutrons. A small portion of fission neutrons are delayed neutrons, which are produced for some time after the fission process has taken place. The delayed neutrons are emitted immediately following the first beta decay of a fission fragment known as a delayed neutron precursor. An example of a delayed neutron precursor is bromine-87, shown below.

PROMPT AND DELAYED NEUTRONS DOE-HDBK-1019/1-93 Reactor Theory (Neutron Characteristics) NP-02 Page 30 Rev. 0

For most applications, it is convenient to combine the known precursors into groups with appropriately averaged properties. These groups vary somewhat depending on the fissile material in use. Table 3 lists the characteristics for the six precursor groups resulting from thermal fission of uranium-235. The fraction of all neutrons that are produced by each of these precursors is called the delayed neutron fraction for that precursor. The total fraction of all neutrons born as delayed neutrons is called the delayed neutron fraction (■). The fraction of delayed neutrons produced varies depending on the predominant fissile nuclide in use. The delayed neutron fractions (■) for the fissile nuclides of most interest are as follows: uranium-233 (0.0026), uranium-235 (0.0065), uranium-238 (0.0148), and plutonium-239 (0.0021).

TABLE 3

Delayed Neutron Precursor Groups for Thermal Fission in Uranium-235

Fixture checkpoint marker-0446-xray is recorded on fixture page 0446.

Group	Half-Life (sec)	Delayed Neutron Fraction	Average Energy (MeV)
1	55.7	0.00021	0.25
2	22.7	0.00142	0.46
3	6.2	0.00127	0.41
4	2.3	0.00257	0.45
5	0.61	0.00075	0.41
6	0.23	0.00027	-
Total	-	0.0065	-

Neutron Generation Time

The neutron generation time is the time required for neutrons from one generation to cause the fissions that produce the next generation of neutrons. The generation time for prompt neutrons

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involved: (a) the time it takes a fast neutron to slow down to thermal energy, (b) the time the now thermal neutron exists prior to absorption in fuel, and (c) the time required for a fissionable nucleus to emit a fast neutron after neutron absorption.
Fast neutrons slow to thermal energies or leak out of the reactor in 10 seconds to 10 seconds, -4 -6 depending on the moderator. In water moderated reactors, thermal neutrons tend to exist for about 10 seconds before they are absorbed. Fission and fast neutron production following -4 neutron absorption in a fissionable nucleus occurs in about 10 seconds. Thus, fast reactors -13 have an 5* of about 10 seconds, while thermal reactors have an 5* of about 10 seconds + 10 -6 -6 -4 seconds, which is about 10 seconds to 10 seconds. -4 -5

■ ■ ■ ■
■ ■ ■ ■
■ ■

Fixture checkpoint marker-0447-xray is recorded on fixture page 0447.

On the other hand, the average generation time for the six delayed neutron groups is the total time from the birth of the fast neutron to the emission of the delayed neutron. Again, three time intervals are involved: (a) the time it takes a fast neutron to slow down to thermal energy, (b) the time the thermal neutron exists prior to absorption, and (c) the average time from neutron absorption to neutron emission by the six precursor groups. The average time for decay of precursors from uranium-235 is 12.5 seconds. The other terms in the delayed neutron generation time are insignificant when compared to this value, and the average delayed neutron generation time becomes ■ 12.5 seconds.
A neutron generation time in the range of 10 seconds to 10 seconds or faster could result in -4 -5 very rapid power excursions, and control would not be possible without the dependence upon delayed neutrons to slow down the rate of the reaction. The average generation time, and hence the rate that power can rise, is determined largely by the delayed neutron generation time. The following equation shows this mathematically.
(2-12)
Example:
Assume a prompt neutron generation time for a particular reactor of 5 x 10 seconds and -5 a delayed neutron generation time of 12.5 seconds. If ■ is 0.0065, calculate the average generation time.
Solution:
This example demonstrates the effect delayed neutrons have on the neutron generation time and thus reactor control. If a reactor were to be operated in a sustained chain reaction using only prompt neutrons (■ = 0), the generation time from the previous example would be about 5 x 10 seconds. However, by operating the reactor such that a 0.0065 fraction of neutrons are -5 delayed, the generation life time is extended to 0.0813 seconds, providing time for adequate operator control. Therefore, although only a small fraction of the total neutron population, delayed neutrons are extremely important to the control and maintenance of a sustained fission chain reaction.
Summary
The important information in this chapter is summarized on the following page.

■ ■ ■ ■

Prompt and Delayed Neutrons Summary

fission event.
& Delayed neutrons are released from the decay of fission products that are called delayed neutron precursors. Delayed neutron precursors are grouped according to half-life. Half-lives vary from fractions of a second to almost a minute.
& The fraction of neutrons born as delayed neutrons is different for different fuel materials. Following are values for some common fuel materials.
Uranium-235 0.0065
Plutonium-239 0.0021
& Delayed neutrons are produced by a classification of fission products known as delayed neutron precursors. When a delayed neutron precursor undergoes a ■ decay, it results in an excited daughter nucleus which immediately ejects a - neutron. Therefore, these delayed neutrons appear with a half-life of the delayed neutron precursor.
& The delayed neutron generation time is the total time from the birth of the fast neutron to the emission of the delayed neutron in the next generation. Delayed neutron generation times are dominated by the half-life of the delayed neutron precursor. The average delayed neutron generation time is about 12.5 seconds.
& A prompt neutron generation time is the sum of the amount of time it takes a fast neutron to thermalize, the amount of time the neutron exists as a thermal neutron
Fixture checkpoint marker-0448-xray is recorded on fixture page 0448.
before it is absorbed, and the amount of time between a fissionable nuclide absorbing a neutron and fission neutrons being released. Prompt neutron generation time is about 5 x 10 seconds. -5
& The average neutron generation time can be calculated from the prompt and delayed neutron generation times and the delayed neutron fraction using Equation (2-12).
& Delayed neutrons are responsible for the ability to control the rate at which power can rise in a reactor. If only prompt neutrons existed, reactor control would not be possible due to the rapid power changes.
Reactor Theory (Neutron Characteristics) DOE-HDBK-1019/1-93 NEUTRON FLUX SPECTRUM
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Figure 2 Prompt Fission Neutron Energy Spectrum for
Thermal Fission of Uranium-235
NEUTRON FLUX SPECTRUM
The number of neutrons that exist at a given energy level varies. A plot of either the fraction of neutrons or the neutron flux at a given energy versus the energy level is called a neutron energy spectrum. The neutron energy spectrum varies widely for different types of reactors.
EO 4.1 STATE the average energy at which prompt neutrons are produced.
EO 4.2 DESCRIBE the neutron energy spectrum in the following reactors:
a. Fast reactor
b. Thermal reactor
EO 4.3 EXPLAIN the reason for the particular shape of the fast, intermediate, and slow energy regions of the neutron flux spectrum for a thermal reactor.
Prompt Neutron Energies

are high energy neutrons, and almost all fission neutrons have energies between 0.1 MeV and 10 MeV. The neutron energy distribution, or spectrum, may best be described by plotting the fraction of neutrons per MeV as a function of neutron energy, as shown in Figure 2. From this figure it can be seen that the most probable neutron energy is about 0.7 MeV. Also, from this data it can be shown that the average energy of fission neutrons is about 2 MeV. Figure 2 is the neutron energy spectrum for thermal fission in uranium-235. The values will vary slightly for other nuclides.

NEUTRON FLUX SPECTRUM DOE-HDBK-1019/1-93 Reactor Theory (Neutron Characteristics) NP-02 Page 34 Rev. 0

Figure 3 Comparison of Neutron Flux Spectra for Thermal and Fast Breeder Reactor

Thermal and Fast Breeder Reactor Neutron Spectra

The spectrum of neutron energies produced by fission varies significantly from the energy spectrum, or flux, existing in a reactor at a given time. Figure 3 illustrates the difference in

Fixture checkpoint marker-0449-xray is recorded on fixture page 0449.

neutron flux spectra between a thermal reactor and a fast breeder reactor. The energy distribution of neutrons from fission is essentially the same for both reactors, so the differences in the curve shapes may be attributed to the neutron moderation or slowing down effects. No attempt is made to thermalize or slow down neutrons in the fast breeder reactor (liquid metal cooled); therefore, an insignificant number of neutrons exist in the thermal range. For the thermal reactor (water moderated), the spectrum of neutrons in the fast region (> 0.1 MeV) has a shape similar to that for the spectrum of neutrons emitted by the fission process.

In the thermal reactor, the flux in the intermediate energy region (1 eV to 0.1 MeV) has approximately a 1/E dependence. That is, if the energy (E) is halved, the flux doubles. This 1/E dependence is caused by the slowing down process, where elastic collisions remove a constant fraction of the neutron energy per collision (on the average), independent of energy; thus, the neutron loses larger amounts of energy per collision at higher energies than at lower energies. The fact that the neutrons lose a constant fraction of energy per collision causes the neutrons to tend to "pile up" at lower energies, that is, a greater number of neutrons exist at the lower energies as a result of this behavior.

vp

2 k T

m

vp

2 k T

m

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K
293 K
1.66 x 10 24 g
2.2 x 105 cm
sec
1 m
100 cm
2200 m
sec
Reactor Theory (Neutron Characteristics) DOE-HDBK-1019/1-93 NEUTRON FLUX SPECTRUM
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In the thermal region the neutrons achieve a thermal equilibrium with the atoms of the moderator material. In any given collision they may gain or lose energy, and over successive collisions will gain as much energy as they lose. These thermal neutrons, even at a specific temperature, do not all have the same energy or velocity; there is a distribution of energies, usually referred to as the Maxwell distribution (e.g., Figure 2). The energies of most thermal neutrons lie close to the most probable energy, but there is a spread of neutrons above and below this value.
Most Probable Neutron Velocities
The most probable velocity (vp) of a thermal neutron is determined by the temperature of the medium and can be determined by Equation (2-13) .
(2-13)
where:
v = most probable velocity of neutron (cm/sec) p
k = Boltzman's constant (1.38 x 10 erg/ K) -16
T = absolute temperature in degrees Kelvin (K)
m = mass of neutron (1.66 x 10 grams) -24
Example:
Calculate the most probable velocities for neutrons in thermal equilibrium with their surroundings at the following temperatures. a) 20 C, b) 260 C.
Solution:
a) Calculate the most probable velocity for 20 C using Equation (2-13).
vp
2 k T
Fixture checkpoint marker-0450-xray is recorded on fixture page 0450.
m
2 1.38 x 10 16 erg
K
533 K
1.66 x 10 24 g
2.977 x 105 cm
sec
1 m
100 cm
2977 m
sec
NEUTRON FLUX SPECTRUM DOE-HDBK-1019/1-93 Reactor Theory (Neutron Characteristics)
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b) Calculate the most probable velocity for 260 C using Equation (2-13).

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The Nuclear Physics and Reactor Theory Handbook was developed to assist nuclear facility operating contractors in providing operators, maintenance personnel, and the technical staff with the necessary fundamentals training to ensure a basic understanding of nuclear physics and reactor theory. The handbook includes information on atomic and nuclear physics; neutron characteristics; reactor theory and nuclear parameters; and the theory of reactor operation. This information will provide personnel with a foundation for understanding the scientific principles that are associated with various DOE nuclear facility operations and maintenance.

Key Words: Training Material, Atomic Physics, The Chart of the Nuclides, Radioactivity, Radioactive Decay, Neutron Interaction, Fission, Reactor Theory, Neutron Characteristics, Neutron Life Cycle, Reactor Kinetics

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FOREWORD

The Department of Energy (DOE) Fundamentals Handbooks consist of ten academic subjects, which include Mathematics; Classical Physics; Thermodynamics, Heat Transfer, and Fluid Flow; Instrumentation and Control; Electrical Science; Material Science; Mechanical Science; Chemistry; Engineering Symbology, Prints, and Drawings; and Nuclear Physics and Reactor Theory. The handbooks are provided as an aid to DOE nuclear facility contractors.

These handbooks were first published as Reactor Operator Fundamentals Manuals in 1985 for use by DOE category A reactors. The subject areas, subject matter content, and level of detail of the Reactor Operator Fundamentals Manuals were determined from several sources. DOE Category A reactor training managers determined which materials should be included, and served as a primary reference in the initial development phase. Training guidelines from the commercial nuclear power industry, results of job and task analyses, and independent input from contractors and operations-oriented personnel were all considered and included to some degree in developing the text material and learning objectives.

The DOE Fundamentals Handbooks represent the needs of various DOE nuclear facilities' fundamental training requirements. To increase their applicability to nonreactor nuclear facilities, the Reactor Operator Fundamentals Manual learning objectives were distributed to the Nuclear Facility Training Coordination Program Steering Committee for review and comment. To update their reactor-specific content, DOE Category A reactor training managers also reviewed and commented on the content. On the basis of feedback from these sources, information that applied to two or more DOE nuclear facilities was considered generic and was included. The final draft of each of the handbooks was then reviewed by these two groups. This approach has resulted in revised modular handbooks that contain sufficient detail such that each facility may adjust the content to fit their specific needs.

Each handbook contains an abstract, a foreword, an overview, learning objectives, and text material, and is divided into modules so that content and order may be modified by individual DOE contractors to suit their specific training needs. Each handbook is supported by a separate examination bank with an answer key.

The DOE Fundamentals Handbooks have been prepared for the Assistant Secretary for Nuclear Energy, Office of Nuclear Safety Policy and Standards, by the DOE Training Coordination Program. This program is managed by EG&G Idaho, Inc.

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The Department of Energy Fundamentals Handbook entitled Nuclear Physics and Reactor Theory was prepared as an information resource for personnel who are responsible for the operation of the Department's nuclear facilities. Almost all processes that take place in a nuclear facility involves the transfer of some type of energy. A basic understanding of nuclear physics and reactor theory is necessary for DOE nuclear facility operators, maintenance personnel, and the technical staff to safely operate and maintain the facility and facility support systems. The information in this handbook is presented to provide a foundation for applying engineering concepts to the job. This knowledge will help personnel understand the impact that their actions may have on the safe and reliable operation of facility components and systems.
The Nuclear Physics and Reactor Theory handbook consists of four modules that are contained in two volumes. The following is a brief description of the information presented in each module of the handbook.
Volume 1 of 2
Module 1 - Atomic and Nuclear Physics
Introduces concepts of atomic physics including the atomic nature of matter, the chart of the nuclides, radioactivity and radioactive decay, neutron interactions and fission, and the interaction of radiation with matter.
Fixture checkpoint marker-0452-xray is recorded on fixture page 0452.
Module 2 - Reactor Theory (Nuclear Parameters)
Provides information on reactor theory and neutron characteristics. Includes topics such as neutron sources, neutron flux, neutron cross sections, reaction rates, neutron moderation, and prompt and delayed neutrons.
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NUCLEAR PHYSICS AND REACTOR THEORY
OVERVIEW (Cont.)
Volume 2 of 2
Module 3 - Reactor Theory (Nuclear Parameters)
Explains the nuclear parameters associated with reactor theory. Topics include the neutron life cycle, reactivity and reactivity coefficients, neutron poisons, and control rods.
Module 4 - Reactor Theory (Reactor Operations)
Introduces the reactor operations aspect of reactor theory. Topics include subcritical multiplication, reactor kinetics, and reactor operation.
The information contained in this handbook is not all-encompassing. An attempt to present the entire subject of nuclear physics and reactor theory would be impractical. However, the Nuclear Physics and Reactor Theory handbook presents enough information to provide the reader with the fundamental knowledge necessary to understand the advanced theoretical concepts presented in other subject areas, and to understand basic system and equipment operation.
Rev. 0 NP
Department of Energy
Fundamentals Handbook
NUCLEAR PHYSICS
AND REACTOR THEORY
Module 1
Atomic and Nuclear Physics
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TERMINAL OBJECTIVE
1.0 Given sufficient information, DESCRIBE atoms, including components, structure, and nomenclature.
ENABLING OBJECTIVES
1.1 STATE the characteristics of the following atomic particles, including mass, charge, and location within the atom:
a. Proton
b. Neutron
c. Electron
1.2 DESCRIBE the Bohr model of an atom.
1.3 DEFINE the following terms:
a. Nuclide c. Atomic number
b. Isotope d. Mass number
1.4 Given the standard A
ZX notation for a particular nuclide, DETERMINE the following:
Fixture checkpoint marker-0456-xray is recorded on fixture page 0456.
a. Number of protons
b. Number of neutrons
c. Number of electrons
1.5 DESCRIBE the three forces that act on particles within the nucleus and affect the stability of the nucleus.
1.6 DEFINE the following terms:
a. Enriched uranium
b. Depleted uranium
1.7 DEFINE the following terms:
a. Mass defect
b. Binding energy
1.8 Given the atomic mass for a nuclide and the atomic masses of a neutron, proton, and electron, CALCULATE the mass defect and binding energy of the nuclide.
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OBJECTIVES DOE-HDBK-1019/1-93 Atomic and Nuclear Physics
TERMINAL OBJECTIVE
2.0 Given necessary references, DESCRIBE the various modes of radioactive decay.
ENABLING OBJECTIVES
2.1 DESCRIBE the following processes:
a. Alpha decay d. Electron capture
b. Beta-minus decay e. Internal conversions
c. Beta-plus decay f. Isomeric transitions
2.2 Given a Chart of the Nuclides, WRITE the radioactive decay chain for a nuclide.
2.3 EXPLAIN why one or more gamma rays typically accompany particle emission.
2.4 Given the stability curve on the Chart of the Nuclides, DETERMINE the type of radioactive decay that the nuclides in each region of the chart will typically undergo.
2.5 DEFINE the following terms:
a. Radioactivity d. Radioactive decay constant
b. Curie e. Radioactive half-life
c. Becquerel
2.6 Given the number of atoms and either the half-life or decay constant of a nuclide, CALCULATE the activity.

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at any later time.
2.8 CONVERT between the half-life and decay constant for a nuclide.
2.9 Given the Chart of the Nuclides and the original activity, PLOT the radioactive decay curve for a nuclide on either linear or semi-log coordinates.
2.10 DEFINE the following terms:
a. Radioactive equilibrium
b. Transient radioactive equilibrium
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Atomic and Nuclear Physics DOE-HDBK-1019/1-93 OBJECTIVES
TERMINAL OBJECTIVE
3.0 Without references, DESCRIBE the different nuclear interactions initiated by neutrons.
ENABLING OBJECTIVES
3.1 DESCRIBE the following scattering interactions between a neutron and a nucleus:
a. Elastic scattering
b. Inelastic scattering
3.2 STATE the conservation laws that apply to an elastic collision between a neutron and a nucleus.
3.3 DESCRIBE the following reactions where a neutron is absorbed in a nucleus:
a. Radiative capture
b. Particle ejection
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OBJECTIVES DOE-HDBK-1019/1-93 Atomic and Nuclear Physics
Fixture checkpoint marker-0457-xray is recorded on fixture page 0457.
TERMINAL OBJECTIVE
4.0 Without references, DESCRIBE the fission process.
ENABLING OBJECTIVES
4.1 EXPLAIN the fission process using the liquid drop model of a nucleus.
4.2 DEFINE the following terms:
a. Excitation energy
b. Critical energy
4.3 DEFINE the following terms:
a. Fissile material
b. Fissionable material
c. Fertile material
4.4 DESCRIBE the processes of transmutation, conversion, and breeding.
4.5 DESCRIBE the curve of Binding Energy per Nucleon versus mass number and give a qualitative description of the reasons for its shape.
4.6 EXPLAIN why only the heaviest nuclei are easily fissioned.
4.7 EXPLAIN why uranium-235 fissions with thermal neutrons and uranium-238 fissions only with fast neutrons.
4.8 CHARACTERIZE the fission products in terms of mass groupings and radioactivity.
4.9 Given the nuclides involved and their masses, CALCULATE the energy released from fission.
4.10 Given the curve of Binding Energy per Nucleon versus mass number, CALCULATE the energy released from fission.
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Atomic and Nuclear Physics DOE-HDBK-1019/1-93 OBJECTIVES
TERMINAL OBJECTIVE

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ENABLING OBJECTIVES
5.1 DESCRIBE interactions of the following with matter:
a. Alpha particle c. Positron
b. Beta particle d. Neutron
5.2 DESCRIBE the following ways that gamma radiation interacts with matter:
a. Compton scattering
b. Photoelectric effect
c. Pair production

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ATOMIC NATURE OF MATTER
All matter is composed of atoms. The atom is the smallest amount of matter that retains the properties of an element. Atoms themselves are composed of smaller particles, but these smaller particles no longer have the same properties as the overall element.
EO 1.1 STATE the characteristics of the following atomic particles, including mass, charge, and location within the atom:
a. Proton
b. Neutron
c. Electron
EO 1.2 DESCRIBE the Bohr model of an atom.
EO 1.3 DEFINE the following terms:
a. Nuclide c. Atomic number
b. Isotope d. Mass number
EO 1.4 Given the standard A
ZX notation for a particular nuclide,
Fixture checkpoint marker-0458-xray is recorded on fixture page 0458.
DETERMINE the following:
a. Number of protons
b. Number of neutrons
c. Number of electrons
EO 1.5 DESCRIBE the three forces that act on particles within the nucleus and affect the stability of the nucleus.
Structure of Matter
Early Greek philosophers speculated that the earth was made up of different combinations of basic substances, or elements. They considered these basic elements to be earth, air, water, and fire. Modern science shows that the early Greeks held the correct concept that matter consists of a combination of basic elements, but they incorrectly identified the elements.

In 1661 the English chemist Robert Boyle published the modern criterion for an element. He defined an element to be a basic substance that cannot be broken down into any simpler substance after it is isolated from a compound, but can be combined with other elements to form compounds. To date, 105 different elements have been confirmed to exist, and researchers claim to have discovered three additional elements. Of the 105 confirmed elements, 90 exist in nature

Another basic concept of matter that the Greeks debated was whether matter was continuous or discrete. That is, whether matter could be continuously divided and subdivided into ever smaller particles or whether eventually an indivisible particle would be encountered. Democritus in about 450 B.C. argued that substances were ultimately composed of small, indivisible particles that he labeled atoms. He further suggested that different substances were composed of different atoms or combinations of atoms, and that one substance could be converted into another by rearranging the atoms. It was impossible to conclusively prove or disprove this proposal for more than 2000 years.

The modern proof for the atomic nature of matter was first proposed by the English chemist John Dalton in 1803. Dalton stated that each chemical element possesses a particular kind of atom, and any quantity of the element is made up of identical atoms of this kind. What distinguishes one element from another element is the kind of atom of which it consists, and the basic physical difference between kinds of atoms is their weight.

Subatomic Particles

For almost 100 years after Dalton established the atomic nature of atoms, it was considered impossible to divide the atom into even smaller parts. All of the results of chemical experiments during this time indicated that the atom was indivisible. Eventually, experimentation into electricity and radioactivity indicated that particles of matter smaller than the atom did indeed exist. In 1906, J. J. Thompson won the Nobel Prize in physics for establishing the existence of electrons. Electrons are negatively-charged particles that have 1/1835 the mass of the hydrogen atom. Soon after the discovery of electrons, protons were discovered. Protons are relatively large particles that have almost the same mass as a hydrogen atom and a positive charge equal in magnitude (but opposite in sign) to that of the electron. The third subatomic particle to be discovered, the neutron, was not found until 1932. The neutron has almost the same mass as the proton, but it is electrically neutral.

Bohr Model of the Atom

The British physicist Ernest Rutherford postulated that the positive charge in an atom is concentrated in a small region called a nucleus at the center of the atom with electrons existing in orbits around it. Niels Bohr, coupling Rutherford's postulation with the quantum theory introduced by Max Planck, proposed that the atom consists of a dense nucleus of protons surrounded by electrons traveling in discrete orbits at fixed distances from the nucleus. An electron in one of these orbits or shells has a specific or discrete quantity of energy (quantum). When an electron moves from one allowed orbit to another allowed orbit, the energy difference between the two states is emitted or absorbed in the form of a single quantum of radiant energy

Fixture checkpoint marker-0459-xray is recorded on fixture page 0459.

called a photon. Figure 1 is Bohr's model of the hydrogen atom showing an electron as having just dropped from the third shell to the first shell with the emission of a photon that has an energy = hv. (h = Planck's constant = 6.63 x 10-34 J-s and v = frequency of the photon.) Bohr's theory was the first to successfully account for the discrete energy levels of this radiation as measured in the laboratory. Although Bohr's atomic model is designed specifically to explain the hydrogen atom, his theories apply generally to the structure of all atoms. Additional information on electron shell theory can be found in the Chemistry Fundamentals Handbook.

Figure 1 Bohr's Model of the Hydrogen Atom

Properties of the three subatomic particles are listed in Table 1.

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Properties of Subatomic Particles
Particle Location Charge Mass
Neutron Nucleus none 1.008665 amu
Proton Nucleus +1 1.007277 amu
Electron Shells around nucleus -1 0.0005486 amu
Measuring Units on the Atomic Scale
The size and mass of atoms are so small that the use of normal measuring units, while possible, is often inconvenient. Units of measure have been defined for mass and energy on the atomic scale to make measurements more convenient to express. The unit of measure for mass is the atomic mass unit (amu). One atomic mass unit is equal to 1.66 x 10-24 grams. The reason for this particular value for the atomic mass unit will be discussed in a later chapter. Note from Table 1 that the mass of a neutron and a proton are both about 1 amu. The unit for energy is the electron volt (eV). The electron volt is the amount of energy acquired by a single electron
Fixture checkpoint marker-0460-xray is recorded on fixture page 0460.
when it falls through a potential difference of one volt. One electron volt is equivalent to 1.602 x 10-19 joules or 1.18 x 10-19 foot-pounds.
Nuclides
The total number of protons in the nucleus of an atom is called the atomic number of the atom and is given the symbol Z. The number of electrons in an electrically-neutral atom is the same as the number of protons in the nucleus. The number of neutrons in a nucleus is known as the neutron number and is given the symbol N. The mass number of the nucleus is the total number of nucleons, that is, protons and neutrons in the nucleus. The mass number is given the symbol A and can be found by the equation Z + N = A.
Each of the chemical elements has a unique atomic number because the atoms of different elements contain a different number of protons. The atomic number of an atom identifies the particular element.
NP-01 Page 4 Rev. 0
Atomic and Nuclear Physics DOE-HDBK-1019/1-93 ATOMIC NATURE OF MATTER
Each type of atom that contains a unique combination of
Figure 2 Nomenclature for
Identifying Nuclides
protons and neutrons is called a nuclide. Not all combinations of numbers of protons and neutrons are possible, but about 2500 specific nuclides with unique combinations of neutrons and protons have been identified. Each nuclide is denoted by the chemical symbol of the element with the atomic number written as a subscript and the mass number written as a superscript, as shown in Figure 2. Because each element has a unique name, chemical symbol, and atomic number, only one of the three is necessary to identify the element. For this reason nuclides can also be identified by either the chemical name or the chemical symbol followed by the mass number (for example, U-235 or uranium-235).
Another common format is to use the abbreviation of the chemical element with the mass number superscripted (for example, 235U). In this handbook the format used in the text will usually be the element's name followed by the mass number. In equations and tables, the format in Figure 2 will usually be used.

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State the name of the element and the number of protons, electrons, and neutrons in the nuclides listed below.
1
1H
10
5B
14
7N
11
4
4
8Cd
23
9
9
4Pu
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ATOMIC NATURE OF MATTER DOE-HDBK-1019/1-93 Atomic and Nuclear Physics
Solution:
The name of the element can be found from the Periodic Table (refer to Chemistry
Fixture checkpoint marker-0461-xray is recorded on fixture page 0461.
Fundamentals Handbook) or the Chart of the Nuclides (to be discussed later). The number of protons and electrons are equal to Z. The number of neutrons is equal to Z - A.
Nuclide Element Protons Electrons Neutrons
1
1H hydrogen 1 1 0
10
5B boron 5 5 5
14
7N nitrogen 7 7 7
11
4
4
8Cd cadmium 48 48 66
23
9
9
4Pu plutonium 94 94 145
Isotopes
Isotopes are nuclides that have the same atomic number and are therefore the same element, but differ in the number of neutrons. Most elements have a few stable isotopes and several unstable, radioactive isotopes. For example, oxygen has three stable isotopes that can be found in nature (oxygen-16, oxygen-17, and oxygen-18) and eight radioactive isotopes. Another example is hydrogen, which has two stable isotopes (hydrogen-1 and hydrogen-2) and a single radioactive isotope (hydrogen-3).
The isotopes of hydrogen are unique in that they are each commonly referred to by a unique name instead of the common chemical element name. Hydrogen-1 is almost always referred to

deuterium and symbolized 2
1D. Hydrogen-3 is commonly called tritium and symbolized 31T. This
text will normally use the symbology 2
1H and 31H for deuterium and tritium, respectively.
Atomic and Nuclear Radii
The size of an atom is difficult to define exactly due to the fact that the electron cloud, formed by the electrons moving in their various orbitals, does not have a distinct outer edge. A reasonable measure of atomic size is given by the average distance of the outermost electron from the nucleus. Except for a few of the lightest atoms, the average atomic radii are approximately the same for all atoms, about 2 x 10 -8 cm.
Like the atom the nucleus does not have a sharp outer boundary. Experiments have shown that the nucleus is shaped like a sphere with a radius that depends on the atomic mass number of the atom. The relationship between the atomic mass number and the radius of the nucleus is shown in the following equation.

r = (1.25 x 10 -13 cm) A1/3
where:
r = radius of the nucleus (cm)
A = atomic mass number (dimensionless)
The values of the nuclear radii for some light, intermediate, and heavy nuclides are shown in Table 2.
TABLE 2
Calculated Values for Nuclear
Radii
Nuclide Radius of Nucleus
1
Fixture checkpoint marker-0462-xray is recorded on fixture page 0462.
1H 1.25 x 10 -13 cm
10
5B 2.69 x 10 -13 cm
5
2
6
6Fe 4.78 x 10 -13 cm
17
7
8
2Hf 7.01 x 10 -13 cm
23
9
8
2U 7.74 x 10 -13 cm
25
9
2
8Cf 7.89 x 10 -13 cm
From the table, it is clear that the radius of a typical atom (e.g. 2 x 10 -8 cm) is more than 25,000

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Nuclear Forces
In the Bohr model of the atom, the nucleus consists of positively-charged protons and electrically■neutra as nucleons. One problem that the Bohr model of the atom presented was accounting for an attractive force to overcome the repulsive force between protons.
Two forces present in the nucleus are (1) electrostatic forces between charged particles and (2) gravitational forces between any two objects that have mass. It is possible to calculate the magnitude of the gravitational force and electrostatic force based upon principles from classical physics.
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Fg
G m1 m2
r
2
Fe
K Q1 Q2
r
2
ATOMIC NATURE OF MATTER DOE-HDBK-1019/1-93 Atomic and Nuclear Physics
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Newton stated that the gravitational force between two bodies is directly proportional to the masses of the two bodies and inversely proportional to the square of the distance between the bodies. This relationship is shown in the equation below.
where:
F = gravitational force (newtons) g
m = mass of first body (kilograms) 1
m = mass of second body (kilograms) 2
G = gravitational constant (6.67 x 10 N-m /kg)
-11 2 2
r = distance between particles (meters)
The equation illustrates that the larger the masses of the objects or the smaller the distance between the objects, the greater the gravitational force. So even though the masses of nucleons are very small, the fact that the distance between nucleons is extremely short may make the gravitational force significant. It is necessary to calculate the value for the gravitational force and compare it to the value for other forces to determine the significance of the gravitational force
Fixture checkpoint marker-0463-xray is recorded on fixture page 0463.
in the nucleus. The gravitational force between two protons that are separated by a distance of 10 meters is about 10 newtons. -20 -24
Coulomb's Law can be used to calculate the force between two protons. The electrostatic force is directly proportional to the electrical charges of the two particles and inversely proportional to the square of the distance between the particles. Coulomb's Law is stated as the following equation.
where:
F = electrostatic force (newtons) e
K = electrostatic constant (9.0 x 10 N-m /C)
9 2 2
Q = charge of first particle (coulombs) 1
Q = charge of second particle (coulombs) 2
r = distance between particles (meters)

of 10 meters is about 10 newtons. Comparing this result with the calculation of the 10^{-20} 12 gravitational force (10^{-24} newtons) shows that the gravitational force is so small that it can be neglected.

If only the electrostatic and gravitational forces existed in the nucleus, then it would be impossible to have stable nuclei composed of protons and neutrons. The gravitational forces are much too small to hold the nucleons together compared to the electrostatic forces repelling the protons. Since stable atoms of neutrons and protons do exist, there must be another attractive force acting within the nucleus. This force is called the nuclear force.

Fixture checkpoint marker-0464-xray is recorded on fixture page 0464.

The nuclear force is a strong attractive force that is independent of charge. It acts equally only between pairs of neutrons, pairs of protons, or a neutron and a proton. The nuclear force has a very short range; it acts only over distances approximately equal to the diameter of the nucleus (10^{-13} cm). The attractive nuclear force between all nucleons drops off with distance much faster than the repulsive electrostatic force between protons.

TABLE 3

Forces Acting in the Nucleus

Force	Interaction	Range
Gravitational	Very weak attractive force between all nucleons	Relatively long
Electrostatic	Strong repulsive force between like charged particles (protons)	Relatively long
Nuclear Force	Strong attractive force between all nucleons	Extremely short

In stable atoms, the attractive and repulsive forces in the nucleus balance. If the forces do not balance, the atom cannot be stable, and the nucleus will emit radiation in an attempt to achieve a more stable configuration.

Summary

The important information in this chapter is summarized on the following page.

Atomic Nature of Matter Summary

Atoms consist of three basic subatomic particles. These particles are the proton, the neutron, and the electron.

Protons are particles that have a positive charge, have about the same mass as a hydrogen atom, and exist in the nucleus of an atom.

Neutrons are particles that have no electrical charge, have about the same mass as a hydrogen atom, and exist in the nucleus of an atom.

Electrons are particles that have a negative charge, have a mass about eighteen hundred times smaller than the mass of a hydrogen atom, and exist in orbital shells around the nucleus of an atom.

The Bohr model of the atom consists of a dense nucleus of protons and neutrons (nucleons) surrounded by electrons traveling in discrete orbits at fixed distances from the nucleus.

Nuclides are atoms that contain a particular number of protons and neutrons.

Isotopes are nuclides that have the same atomic number and are therefore the same

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The atomic number of an atom is the number of protons in the nucleus.
The mass number of an atom is the total number of nucleons (protons and neutrons) in the nucleus.
The notation A
ZX is used to identify a specific nuclide. "Z" represents the atomic number, which is equal to the number of protons. "A" represents the mass number, which is equal to the number of nucleons. "X" represents the chemical symbol of the element.
Number of protons = Z
Number of electrons = Z
Number of neutrons = A - Z
The stability of a nucleus is determined by the different forces interacting within it. The electrostatic force is a relatively long-range, strong, repulsive force that acts between the positively charged protons. The nuclear force is a relatively short-range attractive force between all nucleons. The gravitational force the long range, relatively weak attraction between masses, is negligible compared to the other forces.
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Fixture checkpoint marker-0465-xray is recorded on fixture page 0465.
Atomic and Nuclear Physics DOE-HDBK-1019/1-93 CHART OF THE NUCLIDES
CHART OF THE NUCLIDES
The Chart of the Nuclides, like the Periodic Table, is a convenient format for presenting a large amount of scientific information in an organized manner.
EO 1.6 DEFINE the following terms:
a. Enriched uranium
b. Depleted uranium
Chart of the Nuclides
A tabulated chart called the Chart of the Nuclides lists the stable and unstable nuclides in addition to pertinent information about each one. Figure 3 shows a small portion of a typical chart. This chart plots a box for each individual nuclide, with the number of protons (Z) on the vertical axis and the number of neutrons (N = A - Z) on the horizontal axis.
The completely gray squares indicate stable isotopes. Those in white squares are artificially radioactive, meaning that they are produced by artificial techniques and do not occur naturally. By consulting a complete chart, other types of isotopes can be found, such as naturally occurring radioactive types (but none are found in the region of the chart that is illustrated in Figure 3).
Located in the box on the far left of each horizontal row is general information about the element. The box contains the chemical symbol of the element in addition to the average atomic weight of the naturally occurring substance and the average thermal neutron absorption cross section, which will be discussed in a later module. The known isotopes (elements with the same atomic number Z but different mass number A) of each element are listed to the right.
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CHART OF THE NUCLIDES DOE-HDBK-1019/1-93 Atomic and Nuclear Physics
Figure 3 Nuclide Chart for Atomic Numbers 1 to 6
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Atomic and Nuclear Physics DOE-HDBK-1019/1-93 CHART OF THE NUCLIDES
Information for Stable Nuclides
For the stable isotopes, in addition to the symbol and the atomic mass number, the number

neutron activation cross section and the mass in atomic mass units (amu). A typical block for a stable nuclide from the Chart of the Nuclides is shown in Figure 4.

Figure 4 Stable Nuclides

Information for Unstable Nuclides

For unstable isotopes the additional information includes the half life, the mode of decay (for example, β^-, α), the total disintegration energy in MeV (million electron volts), and the mass in amu when available. A typical block for an unstable nuclide from the Chart of the Nuclides is shown in Figure 5.

Figure 5 Unstable Nuclides

Neutron - Proton Ratios

Figure 6 shows the distribution of the stable nuclides plotted on the same axes as the Chart of the Nuclides. As the mass numbers become higher, the ratio of neutrons to protons in the nucleus becomes larger. For helium-4 (2 protons and 2 neutrons) and oxygen-16 (8 protons and 8 neutrons) this ratio is unity. For indium-115 (49 protons and 66 neutrons) the ratio of neutrons to protons has increased to 1.35, and for uranium-238 (92 protons and 146 neutrons) the neutron■to-proton

Figure 6 Neutron - Proton Plot of the Stable Nuclides

If a heavy nucleus were to split into two fragments, each fragment would form a nucleus that would have approximately the same neutron-to-proton ratio as the heavy nucleus. This high neutron-to-proton ratio places the fragments below and to the right of the stability curve displayed by Figure 6. The instability caused by this excess of neutrons is generally rectified by successive beta emissions, each of which converts a neutron to a proton and moves the nucleus toward a more stable neutron-to-proton ratio.

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■

Natural Abundance of Isotopes

The relative abundance of an isotope in nature compared to other isotopes of the same element is relatively constant. The Chart of the Nuclides presents the relative abundance of the naturally occurring isotopes of an element in units of atom percent. Atom percent is the percentage of the atoms of an element that are of a particular isotope. Atom percent is abbreviated as a/o.

For example, if a cup of water contains 8.23×10^{24} atoms of oxygen, and the isotopic abundance of oxygen-18 is 0.20%, then there are 1.65×10^{22} atoms of oxygen-18 in the cup.

The atomic weight for an element is defined as the average atomic weight of the isotopes of the element. The atomic weight for an element can be calculated by summing the products of the isotopic abundance of the isotope with the atomic mass of the isotope.

Example:

Calculate the atomic weight for the element lithium. Lithium-6 has an atom percent abundance of 7.5% and an atomic mass of 6.015122 amu. Lithium-7 has an atomic abundance of 92.5% and an atomic mass of 7.016003 amu.

Solution:

The other common measurement of isotopic abundance is weight percent (w/o). Weight percent is the percent weight of an element that is a particular isotope. For example, if a sample of material contained 100 kg of uranium that was 28 w/o uranium-235, then 28 kg of uranium-235

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Enriched and Depleted Uranium
Natural uranium mined from the earth contains the isotopes uranium-238, uranium-235 and uranium-234. The majority (99.2745%) of all the atoms in natural uranium are uranium-238. Most of the remaining atoms (0.72%) are uranium-235, and a slight trace (0.0055%) are uranium-234. Although all isotopes of uranium have similar chemical properties, each of the isotopes has significantly different nuclear properties. For reasons that will be discussed in later modules, the isotope uranium-235 is usually the desired material for use in reactors.
A vast amount of equipment and energy are expended in processes that separate the isotopes of uranium (and other elements). The details of these processes are beyond the scope of this module. These processes are called enrichment processes because they selectively increase the proportion of a particular isotope. The enrichment process typically starts with feed material that has the proportion of isotopes that occur naturally. The process results in two types of
CHART OF THE NUCLIDES DOE-HDBK-1019/1-93 Atomic and Nuclear Physics
In the case of uranium, the natural uranium ore is 0.72 a/o uranium-235. The desired outcome of the enrichment process is to produce enriched uranium. Enriched uranium is defined as uranium in which the isotope uranium-235 has a concentration greater than its natural value. The enrichment process will also result in the byproduct of depleted uranium. Depleted uranium is defined as uranium in which the isotope uranium-235 has a concentration less than its natural value. Although depleted uranium is referred to as a by-product of the enrichment process, it does have uses in the nuclear field and in commercial and defense industries.
Summary
The important information in this chapter is summarized below.
Chart of the Nuclides Summary
Enriched uranium is uranium in which the isotope uranium-235 has a concentration greater than its natural value of 0.7%.
Depleted uranium is uranium in which the isotope uranium-235 has a concentration less than its natural value of 0.7%.
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Atomic and Nuclear Physics DOE-HDBK-1019/1-93 MASS DEFECT AND BINDING ENERGY
MASS DEFECT AND BINDING ENERGY
The separate laws of Conservation of Mass and Conservation of Energy are not applied strictly on the nuclear level. It is possible to convert between mass and
Fixture checkpoint marker-0467-xray is recorded on fixture page 0467.
energy. Instead of two separate conservation laws, a single conservation law states that the sum of mass and energy is conserved. Mass does not magically appear and disappear at random. A decrease in mass will be accompanied by a corresponding increase in energy and vice versa.
EO 1.7 DEFINE the following terms:
a. Mass defect
b. Binding energy
EO 1.8 Given the atomic mass for a nuclide and the atomic masses of a neutron, proton, and electron, CALCULATE the mass defect and binding energy of the nuclide.
Mass Defect
Careful measurements have shown that the mass of a particular atom is always slightly less than the sum of the masses of the individual neutrons, protons, and electrons of which the atom consists. The difference between the mass of the atom and the sum of the masses of its parts is called the mass defect (Δm). The mass defect can be calculated using Equation (1-1). In

the difference in mass is small compared to the mass of the atom. Rounding off the masses of atoms and particles to three or four significant digits prior to the calculation will result in a calculated mass defect of zero.

$\Delta m = [Z(mp + me) + (A-Z)mn] - matom$ (1-1)

where:

Δm = mass defect (amu)
mp = mass of a proton (1.007277 amu)
mn = mass of a neutron (1.008665 amu)
me = mass of an electron (0.000548597 amu)
matom = mass of nuclide A
ZX (amu)
Z = atomic number (number of protons)
A = mass number (number of nucleons)

Example:

Calculate the mass defect for lithium-7. The mass of lithium-7 is 7.016003 amu.

Solution:

Δ m Z m
p
me A Z mn matom
Δ m 3 1.007826 amu 7 3 1.008665 amu 7.016003 amu
Δ m 0.0421335 amu

Binding Energy

The loss in mass, or mass defect, is due to the conversion of mass to binding energy when the nucleus is formed. Binding energy is defined as the amount of energy that must be supplied to a nucleus to completely separate its nuclear particles (nucleons). It can also be understood as the amount of energy that would be released if the nucleus was formed from the separate particles. Binding energy is the energy equivalent of the mass defect. Since the mass defect was converted to binding energy (BE) when the nucleus was formed, it is possible to calculate the binding energy using a conversion factor derived by the mass-energy relationship from Einstein's Theory of Relativity.

Einstein's famous equation relating mass and energy is E = mc2 where c is the velocity of light (c = 2.998 x 108 m/sec). The energy equivalent of 1 amu can be determined by inserting this quantity of mass into Einstein's equation and applying conversion factors.

E m c 2
1 amu
1.6606 x 10 27 kg
1 amu

Fixture checkpoint marker-0468-xray is recorded on fixture page 0468.

2.998 x 108 m
sec
2
1 N
1
kg m
sec2
1 J

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1.4924 x 10 10 J 1 MeV
1.6022 x 10 13 J
931.5 MeV
Conversion Factors:
1 amu = 1.6606 x 10 -27 kg
1 newton = 1 kg-m/sec2
1 joule = 1 newton-meter
1 MeV = 1.6022 x 10-13 joules
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Atomic and Nuclear Physics DOE-HDBK-1019/1-93 MASS DEFECT AND BINDING ENERGY
Since 1 amu is equivalent to 931.5 MeV of energy, the binding energy can be calculated using Equation (1-2).
(1-2)
B.E. ∆m
931.5 MeV
1 amu
Fixture checkpoint marker-0469-xray is recorded on fixture page 0469.
Example:
Calculate the mass defect and binding energy for uranium-235. One uranium-235 atom has a mass of 235.043924 amu.
Solution:
Step 1: Calculate the mass defect using Equation (1-1).
∆m Z m
p
me A Z mn matom
∆m 92 1.007826 amu 235 92 1.008665 amu 235.043924 amu
∆m 1.91517 amu
Step 2: Use the mass defect and Equation (1-2) to calculate the binding energy.
B.E. ∆m
931.5 MeV
1 amu
1.91517 amu
931.5 MeV
1 amu
1784 MeV
Energy Levels of Atoms
The electrons that circle the nucleus move in fairly well-defined orbits. Some of these electrons are more tightly bound in the atom than others. For example, only 7.38 eV is required to remove the outermost electron from a lead atom, while 88,000 eV is required to remove the innermost electron. The process of removing an electron from an atom is called ionization, and the energy required to remove the electron is called the ionization energy.
In a neutral atom (number of electrons = Z) it is possible for the electrons to be in a variety of different orbits, each with a different energy level. The state of lowest energy is the one in which the atom is normally found and is called the ground state. When the atom possesses more energy than its ground state energy, it is said to be in an excited state.
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MASS DEFECT AND BINDING ENERGY DOE-HDBK-1019/1-93 Atomic and Nuclear Physics
An atom cannot stay in the excited state for an indefinite period of time. An excited atom will

emitting a discrete bundle of electromagnetic energy called an x-ray. The energy of the x-ray will be equal to the difference between the energy levels of the atom and will typically range from several eV to 100,000 eV in magnitude.

Energy Levels of the Nucleus

The nucleons in the nucleus of an atom, like the electrons that circle the nucleus, exist in shells that correspond to energy states. The energy shells of the nucleus are less defined and less understood than those of the electrons. There is a state of lowest energy (the ground state) and discrete possible excited states for a nucleus. Where the discrete energy states for the electrons of an atom are measured in eV or keV, the energy levels of the nucleus are considerably greater and typically measured in MeV.

A nucleus that is in the excited state will not remain at that energy level for an indefinite period. Like the electrons in an excited atom, the nucleons in an excited nucleus will transition towards their lowest energy configuration and in doing so emit a discrete bundle of electromagnetic radiation called a gamma ray (γ-ray). The only differences between x-rays and γ-rays are their energy levels and whether they are emitted from the electron shell or from the nucleus.

The ground state and the excited states of a nucleus can be depicted in a nuclear energy-level diagram. The nuclear energy-level diagram consists of a stack of horizontal bars, one bar for each of the excited states of the nucleus. The vertical distance between the bar representing an excited state and the bar representing the ground state is proportional to the energy level of the excited state with respect to the ground state. This difference in energy between the ground state and the excited state is called the excitation energy of the excited state. The ground state of a nuclide has zero excitation energy. The bars for the excited states are labeled with their respective energy levels. Figure 7 is the energy level diagram for nickel-60.

Figure 7 Energy Level Diagram - Nickel-60

Fixture checkpoint marker-0470-xray is recorded on fixture page 0470.

Summary

The important information in this chapter is summarized below.

Mass Defect and Binding Energy Summary

Mass defect is the difference between the mass of the atom and the sum of the masses of its constituent parts.

Binding energy is the amount of energy that must be supplied to a nucleus to completely separate its nuclear particles. Binding energy is the energy equivalent of the mass defect.

Mass defect can be calculated by using the equation below.

$\Delta m = [\, Z(mp + me) + (A\text{-}Z)mn \,] - matom$

Binding energy can be calculated by multiplying the mass defect by the factor

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MODES OF RADIOACTIVE DECAY
Most atoms found in nature are stable and do not emit particles or energy that change form over time. Some atoms, however, do not have stable nuclei. These atoms emit radiation in order to achieve a more stable configuration.
EO 2.1 DESCRIBE the following processes:
a. Alpha decay d. Electron capture
b. Beta-minus decay e. Internal conversions
c. Beta-plus decay f. Isomeric transitions
EO 2.2 Given a Chart of the Nuclides, WRITE the radioactive decay chain for a nuclide.
EO 2.3 EXPLAIN why one or more gamma rays typically accompany particle emission.
EO 2.4 Given the stability curve on the Chart of the Nuclides, DETERMINE the type of radioactive decay that the nuclides in each region of the chart will typically undergo.
Stability of Nuclei
As mass numbers become larger, the ratio of neutrons to protons in the nucleus becomes larger for the stable nuclei. Non-stable nuclei may have an excess or deficiency of neutrons and undergo a transformation process known as beta (β) decay. Non-stable nuclei can also undergo a variety of other processes such as alpha (α) or neutron (n) decay. As a result of these decay processes, the final nucleus is in a more stable or more tightly bound configuration.
Natural Radioactivity
In 1896, the French physicist Becquerel discovered that crystals of a uranium salt emitted rays that were similar to x-rays in that they were highly penetrating, could affect a photographic plate, and induced electrical conductivity in gases. Becquerel's discovery was followed in 1898 by the identification of two other radioactive elements, polonium and radium, by Pierre and Marie Curie.

Fixture checkpoint marker-0471-xray is recorded on fixture page 0471.

Heavy elements, such as uranium or thorium, and their unstable decay chain elements emit radiation in their naturally occurring state. Uranium and thorium, present since their creation at the beginning of geological time, have an extremely slow rate of decay. All naturally occurring nuclides with atomic numbers greater than 82 are radioactive.
Nuclear Decay
Whenever a nucleus can attain a more stable (i.e., more tightly bound) configuration by emitting radiation, a spontaneous disintegration process known as radioactive decay or nuclear decay may occur. In practice, this "radiation" may be electromagnetic radiation, particles, or both.
Detailed studies of radioactive decay and nuclear reaction processes have led to the formulation of useful conservation principles. The four principles of most interest in this module are discussed below.
Conservation of electric charge implies that charges are neither created nor destroyed. Single positive and negative charges may, however, neutralize each other. It is also possible for a neutral particle to produce one charge of each sign.
Conservation of mass number does not allow a net change in the number of nucleons. However, the conversion of a proton to a neutron and vice versa is allowed.

equivalent of the mass in a system must be conserved in all decays and reactions. Mass can be converted to energy and energy can be converted to mass, but the sum of mass and energy must be constant.

Conservation of momentum is responsible for the distribution of the available kinetic energy among product nuclei, particles, and/or radiation. The total amount is the same before and after the reaction even though it may be distributed differently among entirely different nuclides and/or particles.

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Alpha Decay (■)

Alpha decay is the emission of alpha particles (helium nuclei) which may be represented as either He or ■. When an unstable nucleus ejects an alpha particle, the atomic number is reduced by 2 4 4 2 2

and the mass number decreased by 4. An example is uranium-234 which decays by the ejection of an alpha particle accompanied by the emission of a 0.068 MeV gamma.

The combined kinetic energy of the daughter nucleus (Thorium-230) and the ■ particle is designated as KE. The sum of the KE and the gamma energy is equal to the difference in mass between the original nucleus (Uranium-234) and the final particles (equivalent to the binding energy released, since ■m = BE). The alpha particle will carry off as much as 98% of the kinetic energy and, in most cases, can be considered to carry off all the kinetic energy.

Beta Decay (■)

Beta decay is the emission of electrons of nuclear rather than orbital origin. These particles are electrons that have been expelled by excited nuclei and may have a charge of either sign.

If both energy and momentum are to be conserved, a third type of particle, the neutrino, ■, must be involved. The neutrino is associated with positive electron emission, and its antiparticle, the antineutrino, , is emitted with a negative electron. These uncharged particles have only the weakest interaction with matter, no mass, and travel at the speed of light. For all practical purposes, they pass through all materials with so few interactions that the energy they possess cannot be recovered. The neutrinos and antineutrinos are included here only because they carry a portion of the kinetic energy that would otherwise belong to the beta particle, and therefore, must be considered for energy and momentum to be conserved. They are normally ignored since they are not significant in the context of nuclear reactor applications.

Negative electron emission, represented as e, -1 -1■, or simply as e or ■ , effectively converts a neutro to a proton, thus increasing the atomic number by one and leaving the mass number unchanged.

Fixture checkpoint marker-0472-xray is recorded on fixture page 0472.

This is a common mode of decay for nuclei with an excess of neutrons, such as fission fragments below and to the right of the neutron-proton stability curve (refer to Figure 6). An example of a typical beta minus-decay reaction is shown below.

■ ■

■

■ ■

■ ■ ■ ■

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Figure 9 Orbital Electron Capture

identical to their negatively charged cousins. When a positron, represented as e, ■, or simply as +1 +1
0 0
e or ■ , is ejected from the nucleus, the atomic number is decreased by one and the mass number + +
remains unchanged. A proton has been converted to a neutron. An example of a typical positron (beta-plus) decay is shown below.

Electron Capture (EC, K-capture)

Nuclei having an excess of protons may capture an electron from one of the inner orbits which immediately combines with a proton in the nucleus to form a neutron. This process is called electron capture (EC). The electron is normally captured from the innermost orbit (the K-shell), and, consequently, this process is sometimes called K-capture. The following example depicts electron capture.

A neutrino is formed at the same
time that the neutron is formed, and
energy carried off by it serves to
Fixture checkpoint marker-0473-xray is recorded on fixture page 0473.
conserve momentum. Any energy
that is available due to the atomic
mass of the product being
appreciably less than that of the
parent will appear as gamma
radiation. Also, there will always
be characteristic x-rays given off
when an electron from one of the
higher energy shells moves in to fill
the vacancy in the K-shell.

Electron capture is shown
graphically in Figure 8.

Electron capture and positron
emission result in the production of
the same daughter product, and
they exist as competing processes.

For positron emission to occur, however, the mass of the daughter product must be less than the mass of the parent by an amount equal to at least twice the mass of an electron. This mass difference between the parent and daughter is necessary to account for two items present in the parent but not in the daughter. One item is the positron ejected from the nucleus of the parent. The other item is that the daughter product has one less orbital electron than the parent. If this requirement is not met, then orbital electron capture takes place exclusively.

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■ ■

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Gamma Emission (■)

Gamma radiation is a high-energy electromagnetic radiation that originates in the nucleus. It is emitted in the form of photons, discrete bundles of energy that have both wave and particle properties. Often a daughter nuclide is left in an excited state after a radioactive parent nucleus undergoes a transformation by alpha decay, beta decay, or electron capture. The nucleus will drop to the ground state by the emission of gamma radiation.

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The usual method for an excited nucleus to go from the excited state to the ground state is by emission of gamma radiation. However, in some cases the gamma ray (photon) emerges from the nucleus only to interact with one of the innermost orbital electrons and, as a result, the energy of the photon is transferred to the electron. The gamma ray is then said to have undergone internal conversion. The conversion electron is ejected from the atom with kinetic energy equal to the gamma energy minus the binding energy of the orbital electron. An orbital electron then drops to a lower energy state to fill the vacancy, and this is accompanied by the emission of characteristic x-rays.

Isomers and Isomeric Transition

Isomeric transition commonly occurs immediately after particle emission; however, the nucleus may remain in an excited state for a measurable period of time before dropping to the ground state at its own characteristic rate. A nucleus that remains in such an excited state is known as a nuclear isomer because it differs in energy and behavior from other nuclei with the same atomic number and mass number. The decay of an excited nuclear isomer to a lower energy level is called an isomeric transition. It is also possible for the excited isomer to decay by some alternate means, for example, by beta emission.

An example of gamma emission accompanying particle emission is illustrated by the decay of nitrogen-16 below.

■
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Decay Chains

When an unstable nucleus decays, the resulting daughter nucleus is not necessarily stable. The nucleus resulting from the decay of a parent is often itself unstable, and will undergo an additional decay. This is especially common among the larger nuclides.

It is possible to trace the steps of an unstable atom as it goes through multiple decays trying to achieve stability. The list of the original unstable nuclide, the nuclides that are involved as intermediate steps in the decay, and the final stable nuclide is known as the decay chain. One common method for stating the decay chain is to state each of the nuclides involved in the standard $^{A}_{Z}X$ format. Arrows are used between nuclides to indicate where decays occur, with the type of decay indicated above the arrow and the half-life below the arrow. The half-life for decay will be discussed in the next chapter.

Example:

Write the decay chains for rubidium-91 and actinium-215. Continue the chains until a stable nuclide or a nuclide with a half-life greater than 1 x 10 years is reached. 6

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Predicting Type of Decay
Radioactive nuclides tend to decay in a way that results in a daughter nuclide that lies closer to the line of stability. Due to this, it is possible to predict the type of decay that a nuclide will undergo based on its location relative to the line of stability on the Chart of the Nuclides.
Figure 9 illustrates the type of decay nuclides in different regions of the chart will typically undergo. Nuclides that are below and to the right of the line of stability will usually undergo ■ decay. - Nuclides that are above and to the left of the line of stability will usually undergo either ■ decay or + electron capture. Most nuclides that will undergo ■ decay are found in the upper right hand region of the chart. These are general rules that have many exceptions, especially in the region of the heavy nuclides.
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Figure 9 Types of Radioactive Decay Relative to the Line of Stability
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Atomic and Nuclear Physics DOE-HDBK-1019/1-93 MODES OF RADIOACTIVE DECAY
Summary
The important information in this chapter is summarized below.
Modes of Radioactive Decay Summary
Alpha decay is the emission of an alpha particle (2 protons and 2 neutrons) from an unstable nucleus. The daughter nuclide has an atomic number 2 less than the parent nuclide and a mass number 4 less than the parent nuclide. The daughter nucleus commonly releases its excitation energy by gamma emission.
Beta-minus decay effectively converts a neutron to a proton and an electron, which is immediately ejected from the nucleus. The daughter nuclide has its atomic number increased by 1 and the same mass number compared to the parent.
Beta-plus decay effectively converts a proton to a neutron and a positron, which is immediately ejected from the nucleus. The daughter nuclide has its atomic number decreased by 1 and the same mass number compared to the parent.
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In electron capture, the nucleus absorbs an electron from the innermost orbit. This electron combines with a proton to form a neutron.
Internal conversion occurs when a gamma ray, emitted by the nucleus as it goes from the excited state to the ground state, interacts with one of the innermost electrons of the same atom. The electron is ejected from the atom.
An isomeric transition is the decay of an excited nucleus to a lower-energy level by the emission of a gamma ray.
Decay chains can be found by tracing the steps an unstable atom goes through as it tries to achieve stability.
Many modes of radioactive decay result in a daughter nuclide that has an energy level above the ground state. This excitation energy is usually released immediately in the form of a gamma ray.
The type of decay that a nuclide will typically undergo can be determined by its relationship to the line of stability on the Chart of the Nuclides. Nuclides that lie below and to the right of the line of stability will typically beta minus decay. Nuclides above and to the left of the line will typically either beta plus decay or electron capture. Most alpha emitters are found in the upper, right-hand corner of the chart.
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RADIOACTIVITY
The rate at which a sample of radioactive material decays is not constant. As individual atoms of the material decay, there are fewer of those types of atoms remaining. Since the rate of decay is directly proportional to the number of atoms, the rate of decay will decrease as the number of atoms decreases.
EO 2.5 DEFINE the following terms:
a. Radioactivity d. Radioactive decay constant
b. Curie e. Radioactive half-life
c. Becquerel
EO 2.6 Given the number of atoms and either the half-life or decay constant of a nuclide, CALCULATE the activity.
EO 2.7 Given the initial activity and the decay constant of a nuclide, CALCULATE the activity at any later time.
EO 2.8 CONVERT between the half-life and decay constant for a nuclide.
EO 2.9 Given the Chart of the Nuclides and the original activity, PLOT the radioactive decay curve for a nuclide on either linear or semi-log coordinates.
EO 2.10 DEFINE the following terms:
a. Radioactive equilibrium
b. Transient radioactive equilibrium
Radioactive Decay Rates
Radioactivity is the property of certain nuclides of spontaneously emitting particles or gamma radiation. The decay of radioactive nuclides occurs in a random manner, and the precise time at which a single nucleus will decay cannot be determined. However, the average behavior of a very large sample can be predicted accurately by using statistical methods. These studies have revealed that there is a certain probability that in a given time interval a certain fraction of the nuclei within a sample of a particular nuclide will decay. This probability per unit time that an atom of a nuclide will decay is known as the radioactive decay constant, λ. The units for the decay constant are inverse time such as 1/second, 1/minute, 1/hour, or 1/year. These decay constant units can also be expressed as second-1, minute-1, hour-1, and year-1.

■■

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The activity (A) of a sample is the rate of decay of that sample. This rate of decay is usually measured in the number of disintegrations that occur per second. For a sample containing millions of atoms, the activity is the product of the decay constant and the number of atoms present in the sample.
The relationship between the activity, number of atoms, and decay constant is shown in Equation (1-3).
A = ■ N (1-3)
where:
A = Activity of the nuclide (disintegrations/second)
■ = decay constant of the nuclide (second)-1
N = Number of atoms of the nuclide in the sample
Since ■ is a constant, the activity and the number of atoms are always proportional.

Two common units to measure the activity of a substance are the curie (Ci) and becquerel (Bq). A curie is a unit of measure of the rate of radioactive decay equal to 3.7 x 10 disintegrations 10 per second. This is approximately equivalent to the number of disintegrations that one gram of radium-226 will undergo in one second. A becquerel is a more fundamental unit of measure of radioactive decay that is equal to 1 disintegration per second. Currently, the curie is more widely used in the United States, but usage of the becquerel can be expected to broaden as the metric system slowly comes into wider use. The conversion between curies and becquerels is shown below.

1 curie = 3.7 x 10 becquerels 10

Variation of Radioactivity Over Time

The rate at which a given radionuclide sample decays is stated in Equation (1-3) as being equal to the product of the number of atoms and the decay constant. From this basic relationship it

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is possible to use calculus to derive an expression which can be used to calculate how the number of atoms present will change over time. The derivation is beyond the scope of this text, but Equation (1-4) is the useful result.

(1-4)

where:

N = number of atoms present at time t

N = number of atoms initially present o

■ = decay constant (time)-1

t = time

A Ao

e

t

A Aoe

t

A

Ao

e

t

ln A

Ao

t

t

ln

A

Ao

t

1/2

ln

1

2

t

1/2

ln 2 0.693

interchangeably to describe any given radionuclide population. Therefore, the following is true.
(1-5)
where:
A = activity present at time t
A = activity initially present o
= decay constant (time)
-1
t = time
Radioactive Half-Life
One of the most useful terms for estimating how quickly a nuclide will decay is the radioactive half-life. The radioactive half-life is defined as the amount of time required for the activity to decrease to one-half of its original value. A relationship between the half-life and decay constant can be developed from Equation (1-5). The half-life can be calculated by solving Equation (1-5) for the time, t, when the current activity, A, equals one-half the initial activity A .
o
First, solve Equation (1-5) for t.
If A is equal to one-half of Ao, then A/Ao is equal to one-half. Substituting this in the equation above yields an expression for t .
1/2
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(1-6)
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The basic features of decay of a radionuclide sample are shown by the graph in Figure 10.
Figure 10 Radioactive Decay as a Function of Time in Units of Half-Life
Assuming an initial number of atoms No, the population, and consequently, the activity may be noted to decrease by one-half of this value in a time of one half-life. Additional decreases occur so that whenever one half-life elapses, the number of atoms drops to one-half of what its value was at the beginning of that time interval. After five half-lives have elapsed, only 1/32, or 3.1%, of the original number of atoms remains. After seven half-lives, only 1/128, or 0.78%, of the atoms remains. The number of atoms existing after 5 to 7 half-lives can usually be assumed to be negligible. The Chemistry Fundamentals Handbook contains additional information on calculating the number of atoms contained within a sample.
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Example:
A sample of material contains 20 micrograms of californium-252.
Californium-252 has a half-life of 2.638 years.
Calculate:
(a) The number of californium-252 atoms initially present
(b) The activity of the californium-252 in curies
(c) The number of californium-252 atoms that will remain in 12 years
(d) The time it will take for the activity to reach 0.001 curies
Solution:
(a) The number of atoms of californium-252 can be determined as below.
(b) First, use Equation (1-6) to calculate the decay constant.

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N No
e
t
4.78 x 1016 e
(0.263/yr) (12 yr)
2.04 x 1015
A Aoe
t
A
Ao
e
t
ln A
Ao
t
t
ln
A
Ao
t
ln
0.001 Ci
0.0108 Ci
0.263 year 1
t 9.05 years
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(c) The number of californium atoms that will remain in 12 years can be calculated from Equation (1-4).
(d) The time that it will take for the activity to reach 0.001 Ci can be determined from Equation (1-5). First, solve Equation (1-5) for time.
Inserting the appropriate values in the right side of this equation will result in the required time.
Plotting Radioactive Decay
It is useful to plot the activity of a nuclide as it changes over time. Plots of this type can be used to determine when the activity will fall below a certain level. This plot is usually done showing activity on either a linear or a logarithmic scale. The decay of the activity of a single nuclide on a logarithmic scale will plot as a straight line because the decay is exponential.
t
1/2
0.693
0.693
t
1/2
0.693
7.13 seconds
0.0972 second 1

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t
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Example:
Plot the radioactive decay curve for nitrogen-16 over a period of 100 seconds. The initial activity is 142 curies and the half-life of nitrogen-16 is 7.13 seconds. Plot the curve on both linear rectangular coordinates and on a semi-log scale.
Solution:
First, use Equation (1-6) to calculate the decay constant corresponding to a half-life of 7.13 seconds.
Use the decay constant determined above to calculate the activity at various times using Equation (1-5).
Time Activity
0 seconds 142 Ci
20 seconds 20.3 Ci
40 seconds 2.91 Ci
60 seconds 0.416 Ci
80 seconds 0.0596 Ci
100 seconds 0.00853 Ci
■ ■
■ ■
■ ■
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Figure 11 Linear and Semi-log Plots of Nitrogen-16 Decay
Plotting the data points calculated above on both linear and semilog scales results in the graphs shown in Figure 11.
If a substance contains more than one radioactive nuclide, the total activity is the sum of the individual activities of each nuclide. As an example, consider a sample of material that contained 1 x 10 atoms of iron-59 that has a half-life of 44.51 days (■ = 1.80 x 10 sec), 6 -7 -1
1 x 10 atoms of manganese-54 that has a half-life of 312.2 days (■ = 2.57 x 10 sec), and 1 6 -8 -1
x 10 atoms of cobalt-60 that has a half-life of 1925 days (■ = 4.17 x 10 sec). 6 -9 -1
The initial activity of each of the nuclides would be the product of the number of atoms and the
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decay constant.
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Figure 12 Combined Decay of Iron-56, Manganese-54, and Cobalt-60
Plotting the manner in which the activities of each of the three nuclides decay over time demonstrates that initially the activity of the shortest-lived nuclide (iron-59) dominates the total activity, then manganese-54 dominates. After almost all of the iron and manganese have decayed away, the only contributor to activity will be the cobalt-60. A plot of this combined decay is shown in Figure 12.
Radioactive Equilibrium
Radioactive equilibrium exists when a radioactive nuclide is decaying at the same rate at which it is being produced. Since the production rate and decay rate are equal, the number of atoms present remains constant over time.
An example of radioactive equilibrium is the concentration of sodium-24 in the coolant

produced at a rate of 1 x 10 atoms per second. If the sodium-24 were stable and did not decay, 6 the amount of sodium-24 present after some period of time could be calculated by multiplying the production rate by the amount of time. Plotting the amount of material present would result in the graph in Figure 13.

However, sodium-24 is not stable, and it decays with a half-life of 14.96 hours. If no sodium-24 is present initially and production starts at a rate of 1 x 10 atoms per second, the rate 6 of decay will initially be zero because there is no sodium-24 present to decay. The rate of decay of sodium-24 will increase as the amount of sodium-24 increases.

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Figure 13 Cumulative Production of Sodium-24 Over Time

The amount of sodium-24 present will initially increase rapidly, then it will increase at a continually decreasing rate until the rate of decay is equal to the rate of production. It is possible to calculate how much sodium-24 will be present at equilibrium by setting the production rate (R) equal to the decay rate (■ N).

where:

R = production rate (atoms/second)

■ = decay constant (second)-1

N = number of atoms

It is possible to calculate the equilibrium value for sodium-24 being produced at a rate of 1 x 10 atoms/second. 6

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Figure 14 Approach of Sodium-24 to Equilibrium

The development of the equation to calculate how the amount of sodium-24 changes over time as it approaches the equilibrium value is beyond the scope of this handbook. However, the equation is presented below.

This equation can be used to calculate the values of the amount of sodium-24 present at different times. As the time increases, the exponential term approaches zero, and the number of atoms present will approach R/■. A plot of the approach of sodium-24 to equilibrium is shown in Figure 14.

Transient Radioactive Equilibrium

Transient radioactive equilibrium occurs when the parent nuclide and the daughter nuclide decay at essentially the same rate.

For transient equilibrium to occur, the parent must have a long half-life when compared to the daughter. An example of this type of compound decay process is barium-140, which decays by beta emission to lanthanum-140, which in turn decays by beta emission to stable cerium-140.

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Figure 15 Transient Equilibrium in the Decay of Barium-140

The decay constant for barium-140 is considerably smaller than the decay constant for lanthanum-140. Remember that the rate of decay of both the parent and daughter can be represented as ■N. Although the decay constant for barium-140 is smaller, the actual rate of decay (■N) is initially larger than that of lanthanum-140 because of the great difference in their initial concentrations. As the concentration of the daughter increases, the rate of decay of the daughter will approach and eventually match the decay rate of the parent. When this occurs, they are said to be in transient equilibrium. A plot of the barium-lanthanum-cerium decay chain reaching transient equilibrium is shown in Figure 15.

Secular equilibrium occurs when the parent has an extremely long half-life. In the long decay chain for a naturally radioactive element, such as thorium-232, where all of the elements in the chain are in secular equilibrium, each of the descendants has built up to an equilibrium amount and all decay at the rate set by the original parent. The only exception is the final stable element on the end of the chain. Its number of atoms is constantly increasing.

Summary

The important information in this chapter is summarized on the following page.

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Radioactivity Summary

& Radioactivity is the decay of unstable atoms by the emission of particles and electromagnetic radiation.

& A curie (Ci) is a unit of radioactivity equal to 3.7 x 10 disintegrations per 10 second.

& A becquerel (Bq) is a unit of radioactivity equal to 1 disintegration per second.

& The radioactive decay constant (■) is the probability per unit time that an atom will decay.

& The radioactive half-life is the amount of time required for the activity to decrease to one-half its original value.

& The activity of a substance can be calculated from the number of atoms and the decay constant based on the equation below.

& The amount of activity remaining after a particular time can be calculated from the equation below.

& The relationship between the decay constant and the half-life is shown below.

& Plots of radioactive decay can be useful to describe the variation of activity over time. If decay is plotted using semi-log scale the plot results in a straight line.

& Radioactive equilibrium exists when the production rate of a material is equal to the removal rate.

& Transient radioactive equilibrium exists when the parent nuclide and the daughter nuclide decay at essentially the same rate. This occurs only when the parent has a long half-life compared to the daughter.

Atomic and Nuclear Physics DOE-HDBK-1019/1-93 NEUTRON INTERACTIONS

NEUTRON INTERACTIONS

Neutrons can cause many different types of interactions. The neutron may simply scatter off the nucleus in two different ways, or it may actually be absorbed into

emission of a gamma ray or a subatomic particle, or it may cause the nucleus to fission.

EO 3.1 DESCRIBE the following scattering interactions between a neutron and a nucleus:

a. Elastic scattering

b. Inelastic scattering

EO 3.2 STATE the conservation laws that apply to an elastic collision between a neutron and a nucleus.

EO 3.3 DESCRIBE the following reactions where a neutron is absorbed in a nucleus:

a. Radiative capture

b. Particle ejection

Scattering

A neutron scattering reaction occurs when a nucleus, after having been struck by a neutron, emits a single neutron. Despite the fact that the initial and final neutrons do not need to be (and often are not) the same, the net effect of the reaction is as if the projectile neutron had merely "bounced off," or scattered from, the nucleus. The two categories of scattering reactions, elastic and inelastic scattering, are described in the following paragraphs.

Elastic Scattering

In an elastic scattering reaction between a neutron and a target nucleus, there is no energy transferred into nuclear excitation. Momentum and kinetic energy of the "system" are conserved although there is usually some transfer of kinetic energy from the neutron to the target nucleus. The target nucleus gains the amount of kinetic energy that the neutron loses.

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mn
vn,i mTvT,i mnvn,f mTvT,f
1
2
m v
2
1
2
mn
v
2
n,i
1
2
mT
v
2
T,i
1
2
mn
v
2

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1
2
mT
v
2
T,f
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Figure 16 Elastic Scattering
Figure 16 illustrates the process of elastic scattering of a neutron off a target nucleus. In the elastic scattering reaction, the conservation of momentum and kinetic energy is represented by the equations below.
Conservation of momentum (mv)
Conservation of kinetic energy
where:
mn = mass of the neutron
m = mass of the target nucleus T
vn,i = initial neutron velocity
vn,f = final neutron velocity
vT,i = initial target velocity
vT,f = final target velocity
Atomic and Nuclear Physics DOE-HDBK-1019/1-93 NEUTRON INTERACTIONS
Elastic scattering of neutrons by nuclei can occur in two ways. The more unusual of the two interactions is the absorption of the neutron, forming a compound nucleus, followed by the re-emission of a neutron in such a way that the total kinetic energy is conserved and the nucleus returns to its ground state. This is known as resonance elastic scattering and is very dependent upon the initial kinetic energy possessed by the neutron. Due to formation of the compound nucleus, it is also referred to as compound elastic scattering. The second, more usual method, is termed potential elastic scattering and can be understood by visualizing the neutrons and nuclei to be much like billiard balls with impenetrable surfaces. Potential scattering takes place with incident neutrons that have an energy of up to about 1 MeV. In potential scattering, the
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neutron does not actually touch the nucleus and a compound nucleus is not formed. Instead, the neutron is acted on and scattered by the short range nuclear forces when it approaches close enough to the nucleus.
Inelastic Scattering
In inelastic scattering, the incident neutron is absorbed by the target nucleus, forming a compound nucleus. The compound nucleus will then emit a neutron of lower kinetic energy which leaves the original nucleus in an excited state. The nucleus will usually, by one or more gamma emissions, emit this excess energy to reach its ground state. Figure 17 shows the process of inelastic scattering.
For the nucleus that has reached its ground state, the sum of the kinetic energy of the exit
Figure 17 Inelastic Scattering
neutron, the target nucleus, and the total gamma energy emitted is equal to the initial kinetic energy of the incident neutron.
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1
0

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238
92
U
239
92
U
239
92
U
0
0
1
0
n
10
5
B
11
5
B
7
3
Li
4
2

Absorption Reactions

Most absorption reactions result in the loss of a neutron coupled with the production of a charged particle or gamma ray. When the product nucleus is radioactive, additional radiation is emitted at some later time. Radiative capture, particle ejection, and fission are all categorized as absorption reactions and are briefly described below.

Radiative Capture

In radiative capture the incident neutron enters the target nucleus forming a compound nucleus. The compound nucleus then decays to its ground state by gamma emission. An example of a radiative capture reaction is shown below.

Particle Ejection

In a particle ejection reaction the incident particle enters the target nucleus forming a compound Fixture checkpoint marker-0485-xray is recorded on fixture page 0485.
nucleus. The newly formed compound nucleus has been excited to a high enough energy level to cause it to eject a new particle while the incident neutron remains in the nucleus. After the new particle is ejected, the remaining nucleus may or may not exist in an excited state depending upon the mass-energy balance of the reaction. An example of a particle ejection reaction is shown below.

Fission

One of the most important interactions that neutrons can cause is fission, in which the nucleus that absorbs the neutron actually splits into two similarly sized parts. Fission will be discussed in detail in the next chapter.

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Summary
The important information in this chapter is summarized below.
Neutron Interactions Summary
Interactions where a neutron scatters off a target nucleus are either elastic or inelastic. In elastic scattering, kinetic energy and momentum are conserved and no energy is transferred into excitation energy of the target nucleus. In inelastic scattering, some amount of kinetic energy is transferred into excitation energy of the target nucleus.
The conservation principles that apply to an elastic collision are conservation of kinetic energy and conservation of momentum.
Radiative capture is the absorption of a neutron by the target nucleus, resulting in an excited nucleus which subsequently (typically within a small fraction of a second) releases its excitation energy in the form of a gamma ray.
Particle ejection occurs when a neutron is absorbed by a target nucleus, resulting in the formation of a compound nucleus. The compound nucleus immediately
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ejects a particle (for example, alpha or proton).
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1
0
n
235
92
U
236
92
U
140
55
Cs
93
37
Rb 3
1
0
n
NUCLEAR FISSION DOE-HDBK-1019/1-93 Atomic and Nuclear Physics
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NUCLEAR FISSION
Nuclear fission is a process in which an atom splits and releases energy, fission products, and neutrons. The neutrons released by fission can, in turn, cause the fission of other atoms.
EO 4.1 EXPLAIN the fission process using the liquid drop model of a nucleus.
EO 4.2 DEFINE the following terms:
a. Excitation energy (E) exc
b. Critical energy (E) crit
EO 4.3 DEFINE the following terms:

b. Fissionable material
c. Fertile material
EO 4.4 DESCRIBE the processes of transmutation, conversion, and breeding.
EO 4.5 DESCRIBE the curve of Binding Energy per Nucleon versus mass number and give a qualitative description of the reasons for its shape.
EO 4.6 EXPLAIN why only the heaviest nuclei are easily fissioned.
EO 4.7 EXPLAIN why uranium-235 fissions with thermal neutrons and uranium-238 fissions only with fast neutrons.
Fission
In the fission reaction the incident neutron enters the heavy target nucleus, forming a compound nucleus that is excited to such a high energy level (E > E) that the nucleus "splits" exc crit (fissions) into two large fragments plus some neutrons. An example of a typical fission reaction is shown below.
A large amount of energy is released in the form of radiation and fragment kinetic energy.
Atomic and Nuclear Physics DOE-HDBK-1019/1-93 NUCLEAR FISSION
Liquid Drop Model of a Nucleus
The nucleus is held together by the attractive nuclear force between nucleons, which was discussed in a previous chapter. The characteristics of the nuclear force are listed below.
(a) very short range, with essentially no effect beyond nuclear dimensions (10-13 cm)
Fixture checkpoint marker-0487-xray is recorded on fixture page 0487.
(b) stronger than the repulsive electrostatic forces within the nucleus
(c) independent of nucleon pairing, in that the attractive forces between pairs of neutrons are no different than those between pairs of protons or a neutron and a proton
(d) saturable, that is, a nucleon can attract only a few of its nearest neighbors
One theory of fission considers the fissioning of a nucleus similar in some respects to the splitting of a liquid drop. This analogy is justifiable to some extent by the fact that a liquid drop is held together by molecular forces that tend to make the drop spherical in shape and that try to resist any deformation in the same manner as nuclear forces are assumed to hold the nucleus together. By considering the nucleus as a liquid drop, the fission process can be described.
Referring to Figure 18(A), the nucleus in the ground state is undistorted, and its attractive nuclear forces are greater than the repulsive electrostatic forces between the protons within the nucleus. When an incident particle (in this instance a neutron) is absorbed by the target nucleus, a compound nucleus is formed. The compound nucleus temporarily contains all the charge and mass involved in the reaction and exists in an excited state. The excitation energy added to the compound nucleus is equal to the binding energy contributed by the incident particle plus the kinetic energy possessed by that particle. Figure 18(B) illustrates the excitation energy thus imparted to the compound nucleus, which may cause it to oscillate and become distorted. If the excitation energy is greater than a certain critical energy, the oscillations may cause the compound nucleus to become dumbbell-shaped. When this happens, the attractive nuclear forces (short-range) in the neck area are small due to saturation, while the repulsive electrostatic forces (long-range) are only slightly less than before. When the repulsive electrostatic forces exceed the attractive nuclear forces, nuclear fission occurs, as illustrated in Figure 18(C).

Critical Energy
The measure of how far the energy level of a nucleus is above its ground state is called the excitation energy (Eexc). For fission to occur, the excitation energy must be above a particular value for that nuclide. The critical energy (Ecrit) is the minimum excitation energy required for fission to occur.
Fissile Material
A fissile material is composed of nuclides for which fission is possible with neutrons of any energy level. What is especially significant about these nuclides is their ability to be fissioned with zero kinetic energy neutrons (thermal neutrons). Thermal neutrons have very low kinetic energy levels (essentially zero) because they are roughly in equilibrium with the thermal motion of surrounding materials. Therefore, in order to be classified as fissile, a material must be capable of fissioning after absorbing a thermal neutron. Consequently, they impart essentially no kinetic energy to the reaction. Fission is possible in these materials with thermal neutrons, since the change in binding energy supplied by the neutron addition alone is high enough to exceed the critical energy. Some examples of fissile nuclides are uranium-235, uranium-233, and plutonium-239.

Fissionable Material
A fissionable material is composed of nuclides for which fission with neutrons is possible. All fissile nuclides fall into this category. However, also included are those nuclides that can be fissioned only with high energy neutrons. The change in binding energy that occurs as the result of neutron absorption results in a nuclear excitation energy level that is less than the required critical energy. Therefore, the additional excitation energy must be supplied by the kinetic energy of the incident neutron. The reason for this difference between fissile and fissionable materials is the so-called odd-even effect for nuclei. It has been observed that nuclei with even numbers of neutrons and/or protons are more stable than those with odd numbers. Therefore, adding a neutron to change a nucleus with an odd number of neutrons to a nucleus with an even number of neutrons produces an appreciably higher binding energy than adding a neutron to a nucleus
Fixture checkpoint marker-0488-xray is recorded on fixture page 0488.
already possessing an even number of neutrons. Some examples of nuclides requiring high energy neutrons to cause fission are thorium-232, uranium-238, and plutonium-240. Table 4 indicates the critical energy (Ecrit) and the binding energy change for an added neutron (BEn) to target nuclei of interest. For fission to be possible, the change in binding energy plus the kinetic energy must equal or exceed the critical energy ($\Delta BE + KE > Ecrit$).
TABLE 4
Critical Energies Compared to Binding Energy of Last Neutron
Target
Nucleus
Critical Energy
Ecrit
Binding Energy of
Last Neutron BEn
BEn - Ecrit
23
9
2
0Th 7.5 MeV 5.4 MeV -2.1 MeV

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9
8
2U 7.0 MeV 5.5 MeV -1.5 MeV
23
9
5
2U 6.5 MeV 6.8 MeV +0.3 MeV
23
9
3
2U 6.0 MeV 7.0 MeV +1.0 MeV
23
9
9
4Pu 5.0 MeV 6.6 MeV +1.6 MeV
Uranium-235 fissions with thermal neutrons because the binding energy released by the absorption of a neutron is greater than the critical energy for fission; therefore uranium-235 is a fissile material. The binding energy released by uranium-238 absorbing a thermal neutron is less than the critical energy, so additional energy must be possessed by the neutron for fission to be possible. Consequently, uranium-238 is a fissionable material.
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Fertile Material
All of the neutron absorption reactions that do not result in fission lead to the production of new nuclides through the process known as transmutation. These nuclides can, in turn, be transmuted again or may undergo radioactive decay to produce still different nuclides. The nuclides that are produced by this process are referred to as transmutation products. Because several of the fissile nuclides do not exist in nature, they can only be produced by nuclear reactions (transmutation). The target nuclei for such reactions are said to be fertile. Fertile materials are materials that can undergo transmutation to become fissile materials. Figure 19 traces the transmutation mechanism by which two fertile nuclides, thorium-232 and uranium-238, produce uranium-233 and plutonium-239, respectively.
Figure 19 Conversion of Fertile Nuclides to Fissile Nuclides
If a reactor contains fertile material in addition to its fissile fuel, some new fuel will be produced as the original fuel is burned up. This is called conversion. Reactors that are specifically designed to produce fissionable fuel are called "breeder" reactors. In such reactors, the amount
Fixture checkpoint marker-0489-xray is recorded on fixture page 0489.
of fissionable fuel produced is greater than the amount of fuel burnup. If less fuel is produced than used, the process is called conversion, and the reactor is termed a "converter."
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Binding Energy Per Nucleon (BE/A)
As the number of particles in a nucleus increases, the total binding energy also increases. The rate of increase, however, is not uniform. This lack of uniformity results in a variation in the amount of binding energy associated with each nucleon within the nucleus. This variation in the binding energy per nucleon (BE/A) is easily seen when the average BE/A is plotted versus atomic mass number (A), as shown in Figure 20.
Figure 20 Binding Energy per Nucleon vs. Mass Number

decreases for A > 60. The BE/A curve reaches a maximum value of 8.79 MeV at A = 56 and decreases to about 7.6 MeV for A = 238. The general shape of the BE/A curve can be explained using the general properties of nuclear forces. The nucleus is held together by very short-range attractive forces that exist between nucleons. On the other hand, the nucleus is being forced apart by long range repulsive electrostatic (coulomb) forces that exist between all the protons in the nucleus.

As the atomic number and the atomic mass number increase, the repulsive electrostatic forces within the nucleus increase due to the greater number of protons in the heavy elements. To overcome this increased repulsion, the proportion of neutrons in the nucleus must increase to maintain stability. This increase in the neutron-to-proton ratio only partially compensates for the growing proton-proton repulsive force in the heavier, naturally occurring elements. Because the

Fixture checkpoint marker-0490-xray is recorded on fixture page 0490.

repulsive forces are increasing, less energy must be supplied, on the average, to remove a nucleon from the nucleus. The BE/A has decreased. The BE/A of a nucleus is an indication of its degree of stability. Generally, the more stable nuclides have higher BE/A than the less stable ones. The increase in the BE/A as the atomic mass number decreases from 260 to 60 is the primary reason for the energy liberation in the fission process. In addition, the increase in the BE/A as the atomic mass number increases from 1 to 60 is the reason for the energy liberation in the fusion process, which is the opposite reaction of fission.

The heaviest nuclei require only a small distortion from a spherical shape (small energy addition) for the relatively large coulomb forces forcing the two halves of the nucleus apart to overcome the attractive nuclear forces holding the two halves together. Consequently, the heaviest nuclei are easily fissionable compared to lighter nuclei.

Summary

The important information in this chapter is summarized on the following page.

Nuclear Fission Summary

The fission process can be explained using the liquid drop model of a nucleus. In the ground state the nucleus is nearly spherical in shape. After the absorption of a neutron, the nucleus will be in an excited state and start to oscillate and become distorted. If the oscillations cause the nucleus to become shaped like a dumbbell, the repulsive electrostatic forces will overcome the short-range attractive nuclear forces, and the nucleus will split in two.

Excitation energy is the amount of energy a nucleus has above its ground state. Critical energy is the minimum excitation energy that a nucleus must have before it can fission.

Fissile material is material for which fission is possible with neutrons that have zero kinetic energy. Fissionable material is material for which fission caused by neutron absorption is possible provided the kinetic energy added with the binding energy is greater than the critical energy. Fertile material is material that can undergo transmutation to become fissile material.

Transmutation is the process of neutron absorption and subsequent decay, which changes one nuclide to another nuclide. Conversion is the process of transmuting fertile material into fissile material in a reactor, where the amount of fissile material produced is less than the amount of fissile material consumed. Breeding

than the amount of fissile material consumed.
The curve of binding energy per nucleon increases quickly through the light nuclides and reaches a maximum at a mass number of about 56. The curve decreases slowly for mass numbers greater than 60.
The heaviest nuclei are easily fissionable because they require only a small distortion from the spherical shape to allow the coulomb forces to overcoming the attractive nuclear force, forcing the two halves of the nucleus apart.
Uranium-235 fissions with thermal neutrons because the binding energy released by the absorption of a neutron is greater than the critical energy for fission. The binding energy released by uranium-238 absorbing a neutron is less than the critical energy, so additional kinetic energy must be possessed by the neutron for fission to be possible.
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1
0
n
235
92
U
236
Fixture checkpoint marker-0491-xray is recorded on fixture page 0491.
92
U
140
55
Cs
93
37
Rb 3
1
0
n
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ENERGY RELEASE FROM FISSION
Fission of heavy nuclides converts a small amount of mass into an enormous amount of energy. The amount of energy released by fission can be determined based on either the change in mass that occurs during the reaction or by the difference in binding energy per nucleon between the fissile nuclide and the fission products.
EO 4.8 CHARACTERIZE the fission products in terms of mass groupings and radioactivity.
EO 4.9 Given the nuclides involved and their masses, CALCULATE the energy released from fission.
EO 4.10 Given the curve of Binding Energy per nucleon versus mass number, CALCULATE the energy released from fission.
Calculation of Fission Energy
Nuclear fission results in the release of enormous quantities of energy. It is necessary to be able

this is to first investigate a typical fission reaction such as the one listed below.
It can be seen that when the compound nucleus splits, it breaks into two fission fragments, rubidium-93, cesium-140, and some neutrons. Both fission products then decay by multiple - emissions as a result of the high neutron-to-proton ratio possessed by these nuclides.
In most cases, the resultant fission fragments have masses that vary widely. Figure 21 gives the percent yield for atomic mass numbers. The most probable pair of fission fragments for the thermal fission of the fuel uranium-235 have masses of about 95 and 140. Note that the vertical axis of the fission yield curve is on a logarithmic scale. Therefore, the formation of fission fragments of mass numbers of about 95 and 140 is highly likely.
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Figure 21 Uranium-235 Fission Yield vs.
Mass Number
Referring now to the binding energy per nucleon curve (Figure 20), we can estimate the amount of energy released by our "typical" fission by plotting this reaction on the curve and calculating the change in binding energy (ΔBE) between the reactants on the left-hand side of the fission equation and the products on the right-hand side. Plotting the reactant and product nuclides on the curve shows that the total binding energy of the system after fission is greater than the total binding energy of the system before fission. When there is an increase in the total binding energy of a system, the system has become more stable by releasing an amount of energy equal to the increase in total binding energy of the system. Therefore, in the fission process, the energy liberated is equal to the increase in the total binding energy of the system.
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Figure 22 graphically depicts that the binding energy per nucleon for the products (C,
Figure 22 Change in Binding Energy for Typical Fission
rubidium-93 and B, cesium-140) is greater than that for the reactant (A, uranium-235). The total binding energy for a nucleus can be found by multiplying the binding energy per nucleon by the
Fixture checkpoint marker-0492-xray is recorded on fixture page 0492.
number of nucleons.
TABLE 5
Binding Energies Calculated from
Binding Energy per Nucleon Curve
Nuclide B.E. per Nucleon
(BE/A)
Mass Number
(A)
Binding Energy
(BE/A) x (A)
9
3
3
7Rb 8.7 MeV 93 809 MeV
14
5
0
5Cs 8.4 MeV 140 1176 MeV
23
9

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2U 7.6 MeV 235 1786 MeV
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BE BEproducts BEreactants
BERb 93 BECs 140 BEU 235
809 MeV 1176 MeV 1786 MeV
199 MeV
1
0
n
235
92
U
236
92
U
140
55
Cs
93
37
Rb 3
1
0
n
Mass difference Mass of Reactants Mass of Products
236.052589 amu 235.85208 amu
0.200509 amu
EInst 0.020059 amu
931.5 MeV
amu
186.8 MeV
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The energy released will be equivalent to the difference in binding energy (BE) between the
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reactants and the products.
The energy liberation during the fission process can also be explained from the standpoint of the conservation of mass-energy. During the fission process, there is a decrease in the mass of the system. There must, therefore, be energy liberated equal to the energy equivalent of the mass lost in the process. This method is more accurate than the previously illustrated method and is used when actually calculating the energy liberated during the fission process.
Again, referring to the "typical" fission reaction.
E , the instantaneous energy, is the energy released immediately after the fission process. It Inst is equal to the energy equivalent of the mass lost in the fission process. It can be calculated as shown below.
Mass of the Reactants Mass of the Products
U 235.043924 amu Rb 92.91699 amu 235
92

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37
n 1.008665 amu Cs 139.90910 amu 1
0
140
55
3 (n) 3.02599 amu 1
0
236.052589 amu 235.85208 amu
This mass difference can be converted to an energy equivalent.
93
37
Fixture checkpoint marker-0494-xray is recorded on fixture page 0494.
Rb
93
38
Sr
93
39
Y
93
40
Zr
93
41
Nb
140
55
Cs
140
56
Ba
140
57
La
140
58
Ce
EDecay mRb 93 mNb 93 4 melectron
931.5MeV
amu
92.91699 amu 92.90638 amu 4 0.0005486 amu
931.5 MeV
amu
0.008416 amu
931.5 MeV
amu
7.84 MeV
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The total energy released per fission will vary from the fission to the next depending on what fission products are formed, but the average total energy released per fission of uranium-235 with a thermal neutron is 200 MeV.

As illustrated in the preceding example, the majority of the energy liberated in the fission process is released immediately after the fission occurs and appears as the kinetic energy of the fission fragments, kinetic energy of the fission neutrons, and instantaneous gamma rays. The remaining energy is released over a period of time after the fission occurs and appears as kinetic energy of the beta, neutrino, and decay gamma rays.

Estimation of Decay Energy

In addition to this instantaneous energy release during the actual fission reaction, there is additional energy released when the fission fragments decay by emission. This additional

-

energy is called decay energy, EDecay. The decay chains for rubidium-93 and cesium-140 are shown below.

The energy released during the decay for each chain will be equivalent to the mass difference between the original fission product and the sum of the final stable nuclide and the beta particles emitted.

The energy released in the decay chain of rubidium-93 is calculated below.

Fixture checkpoint marker-0495-xray is recorded on fixture page 0495.

EDecay mRb 93 mNb 93 3 melectron

931.5MeV

amu

139.90910 amu 139.90543 amu 3 0.0005486 amu

931.5 MeV

amu

0.000202 amu

931.5 MeV

amu

1.89 MeV

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The energy released in the decay chain of cesium-140 is calculated below.

The total decay energy is the sum of the energies of the two chains, or 9.73 MeV.

Distribution of Fission Energy

The average energy distribution for the energy released per fission with a thermal neutron in uranium-235 is shown in Tables 6 and 7.

TABLE 6

Instantaneous Energy from Fission

Kinetic Energy of Fission Products 167 Mev

Energy of Fission Neutrons 5 MeV

Instantaneous Gamma-ray Energy 5 MeV

Capture Gamma-ray Energy 10 MeV

Total Instantaneous Energy 187 MeV

TABLE 7

Delayed Energy from Fission

Beta Particles From Fission Products 7 Mev

Gamma-rays from Fission Products 6 MeV

Neutrinos 10 MeV

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Because the 10 MeV of neutrino energy shown in Table 7 is not absorbed in the reactor, the average value of 200 MeV per fission is still accurate. Note in Table 6 that some fission neutrons undergo radiative capture and the resultant gamma ray emission provides an additional 10 MeV of instantaneous energy, which contributes to the total of 187 MeV instantaneous energy.
All of the energy released, with the exception of the neutrino energy, is ultimately transformed into heat through a number of processes. The fission fragments, with their high positive charge and kinetic energy, cause ionization directly as they rip orbital electrons from the surrounding atoms. In this ionization process, kinetic energy is transferred to the surrounding atoms of the fuel material, resulting in an increase in temperature. The beta particles and gamma rays also give up their energy through ionization, and the fission neutrons interact and lose their energy through elastic scattering. Of the 200 MeV released per fission, about seven percent (13 MeV) is released at some time after the instant of fission. When a reactor is shut down, fissions essentially cease, but energy is still being released from the decay of fission products. The heat produced by this decay energy is referred to as "decay heat." Although decay energy represents about seven percent of reactor heat production during reactor operation, once the reactor is shut down the decay heat production drops off quickly to a small fraction of its value while operating. The decay heat produced is significant, however, and systems must be provided to keep the reactor cool even after shutdown.
Summary
The important information in this chapter is summarized below.
Energy Release From Fission Summary
Fission products have some general characteristics in common.
They generally decay by β- emission.
Fixture checkpoint marker-0496-xray is recorded on fixture page 0496.
The most common mass numbers are grouped near 95 and 140.
The energy released by fission can be calculated based on the difference in mass between the masses of the reactants before fission and the fission fragments and fission neutrons after fission.
Another method to determine the energy released by fission is based on the change in binding energy per nucleon between the fissile nuclide and the fission products.
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Atomic and Nuclear Physics INTERACTION OF RADIATION WITH MATTER
INTERACTION OF RADIATION WITH MATTER
Different types of radiation interact with matter in widely different ways. A large, massive, charged alpha particle cannot penetrate a piece of paper and even has a limited range in dry air. A neutrino, at the other extreme, has a low probability of interacting with any matter, even if it passed through the diameter of the earth.
EO 5.1 DESCRIBE interactions of the following with matter:
a. Alpha particle c. Positron
b. Beta particle d. Neutron
EO 5.2 DESCRIBE the following ways that gamma radiation interacts with matter:
a. Photoelectric effect

c. Pair production

Interaction of Radiation With Matter

Radiation can be classified into two general groups, charged and uncharged; therefore, it may be expected that interactions with matter fall into two general types. Charged particles directly ionize the media through which they pass, while uncharged particles and photons can cause ionization only indirectly or by secondary radiation.

A moving charged particle has an electrical field surrounding it, which interacts with the atomic structure of the medium through which it is passing. This interaction decelerates the particle and accelerates electrons in the atoms of the medium. The accelerated electrons may acquire enough energy to escape from the parent atom. This process, whereby radiation "strips" off orbital electrons, is called ionization. Uncharged moving particles have no electrical field, so they can only lose energy and cause ionization by such means as collisions or scattering. A photon can lose energy by the photoelectric effect, Compton effect, or pair production.

I

m z

2

K.E.

Because ionizing radiation creates ions in pairs, the intensity of ionization or the specific ionization is defined as the number of ion-pairs formed per centimeter of travel in a given material. The amount of ionization produced by a charged particle per unit path length, which is a measure of its ionizing power, is roughly proportional to the particle's mass and the square of its charge as illustrated in the equation below.

where:

I is the ionizing power

m is the mass of the particle

z is the number of unit charges it carries

K.E. is its kinetic energy

Since m for an alpha particle is about 7300 times as large as m for a beta particle, and z is twice

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as great, an alpha will produce much more ionization per unit path length than a beta particle of the same energy. This phenomenon occurs because the larger alpha particle moves slower for a given energy and thus acts on a given electron for a longer time.

Alpha Radiation

Alpha radiation is normally produced from the radioactive decay of heavy nuclides and from certain nuclear reactions. The alpha particle consists of 2 neutrons and 2 protons, so it is essentially the same as the nucleus of a helium atom. Because it has no electrons, the alpha particle has a charge of +2. This positive charge causes the alpha particle to strip electrons from the orbits of atoms in its vicinity. As the alpha particle passes through material, it removes electrons from the orbits of atoms it passes near. Energy is required to remove electrons and the energy of the alpha particle is reduced by each reaction. Eventually the particle will expend its kinetic energy, gain 2 electrons in orbit, and become a helium atom. Because of its strong positive charge and large mass, the alpha particle deposits a large amount of energy in a short distance of travel. This rapid, large deposition of energy limits the penetration of alpha particles. The most energetic alpha particles are stopped by a few centimeters of air or a sheet

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Beta-Minus Radiation
A beta-minus particle is an electron that has been ejected at a high velocity from an unstable nucleus. An electron has a small mass and an electrical charge of -1. Beta particles cause ionization by displacing electrons from atom orbits. The ionization occurs from collisions with orbiting electrons. Each collision removes kinetic energy from the beta particle, causing it to slow down. Eventually the beta particle will be slowed enough to allow it to be captured as an
2 electrons 0.000549 amu
electron
931.5 MeV
amu
1.02 MeV
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orbiting electron in an atom. Although more penetrating than the alpha, the beta is relatively easy to stop and has a low power of penetration. Even the most energetic beta radiation can be stopped by a few millimeters of metal.
Positron Radiation
Positively charged electrons are called positrons. Except for the positive charge, they are identical to beta-minus particles and interact with matter in a similar manner. Positrons are very short-lived, however, and quickly are annihilated by interaction with a negatively charged electron, producing two gammas with a combined energy (calculated below) equal to the rest mass of the positive and negative electrons.
Neutron Radiation
Neutrons have no electrical charge. They have nearly the same mass as a proton (a hydrogen atom nucleus). A neutron has hundreds of times more mass than an electron, but 1/4 the mass of an alpha particle. The source of neutrons is primarily nuclear reactions, such as fission, but they may also be produced from the decay of radioactive nuclides. Because of its lack of charge, the neutron is difficult to stop and has a high penetrating power.
Neutrons are attenuated (reduced in energy and numbers) by three major interactions, elastic scatter, inelastic scatter, and absorption. In elastic scatter, a neutron collides with a nucleus and bounces off. This reaction transmits some of the kinetic energy of the neutron to the nucleus of the atom, resulting in the neutron being slowed, and the atom receives some kinetic energy (motion). This process is sometimes referred to as "the billiard ball effect."
As the mass of the nucleus approaches the mass of the neutron, this reaction becomes more effective in slowing the neutron. Hydrogenous material attenuates neutrons most effectively.
In the inelastic scatter reaction, the same neutron/nucleus collision occurs as in elastic scatter. However, in this reaction, the nucleus receives some internal energy as well as kinetic energy. This slows the neutron, but leaves the nucleus in an excited state. When the nucleus decays to
Fixture checkpoint marker-0498-xray is recorded on fixture page 0498.
its original energy level, it normally emits a gamma ray.
In the absorption reaction, the neutron is actually absorbed into the nucleus of an atom. The neutron is captured, but the atom is left in an excited state. If the nucleus emits one or more gamma rays to reach a stable level, the process is called radiative capture. This reaction occurs at most neutron energy levels, but is more probable at lower energy levels.
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Gamma Radiation

is very similar to an x-ray. The difference is that gamma rays are emitted from the nucleus of an atom, and x-rays are produced by orbiting electrons. The x-ray is produced when orbiting electrons move to a lower energy orbit or when fast-moving electrons approaching an atom are deflected and decelerated as they react with the atom's electrical field (called Bremsstrahlung). The gamma ray is produced by the decay of excited nuclei and by nuclear reactions. Because the gamma ray has no mass and no charge, it is difficult to stop and has a very high penetrating power. A small fraction of the original gamma stream will pass through several feet of concrete or several meters of water.

There are three methods of attenuating gamma rays. The first method is referred to as the photo-electric effect. When a low energy gamma strikes an atom, the total energy of the gamma is expended in ejecting an electron from orbit. The result is ionization of the atom and expulsion of a high energy electron. This reaction is most predominant with low energy gammas interacting in materials with high atomic weight and rarely occurs with gammas having an energy above 1 MeV. Annihilation of the gamma results. Any gamma energy in excess of the binding energy of the electron is carried off by the electron in the form of kinetic energy.

The second method of attenuation of gammas is called Compton scattering. The gamma interacts

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with an orbital or free electron; however, in this case, the photon loses only a fraction of its energy. The actual energy loss depending on the scattering angle of the gamma. The gamma continues on at lower energy, and the energy difference is absorbed by the electron. This reaction becomes important for gamma energies of about 0.1 MeV and higher.

At higher energy levels, a third method of attenuation is predominant. This method is pair-production. When a high energy gamma passes close enough to a heavy nucleus, the gamma completely disappears, and an electron and a positron are formed. For this reaction to take place, the original gamma must have at least 1.02 MeV energy. Any energy greater than 1.02 MeV becomes kinetic energy shared between the electron and positron. The probability of pair■production incre

Interaction of Radiation with Matter Summary

An alpha particle deposits a large amount of energy in a short distance of travel due to its large mass and charge.

Beta-minus particles interact with the electrons orbiting the nucleus of atoms, causing ionization by displacing the electrons. The beta particle loses energy with each interaction. After the beta particle loses enough energy, it is captured in the orbital shells of an atom.

Positrons interact with matter much the same way as beta minus particles. After the positron has lost most of its energy by ionizing atoms, it is annihilated by interaction with an electron. The electron-positron pair disappear and are replaced by two gammas, each with the energy equivalent of the mass of an electron (0.51 MeV).

Neutrons interact with matter by elastic scattering, inelastic scattering, or absorption.

Photoelectric effect is where a gamma interacts with an electron orbiting an atom. The entire energy of the gamma is transferred to the electron, and the electron is ejected from its orbit.

In Compton scattering a gamma interacts with an orbital electron, but only part of

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orbit, and the gamma is scattered off at a lower energy.
In pair-production, a gamma interacts with the electric field of a nucleus and is converted into an electron-positron pair. The gamma must have an energy greater than 1.02 MeV for this to occur.
Summary
The important information in this chapter is summarized below.
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Table 3 Delayed Neutron Precursor Groups for Thermal Fission in Uranium-235 30
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1.0 Without references, EXPLAIN how neutron sources produce neutrons.
ENABLING OBJECTIVES
1.1 DEFINE the following terms:
a. Intrinsic neutron source
b. Installed neutron source
1.2 LIST three examples of reactions that produce neutrons in intrinsic neutron sources.
1.3 LIST three examples of reactions that produce neutrons in installed neutron sources.

TERMINAL OBJECTIVE
2.0 Given the necessary information for calculations, EXPLAIN basic concepts in reactor physics and perform calculations.
ENABLING OBJECTIVES
2.1 DEFINE the following terms:
a. Atom density d. Barn
b. Neutron flux e. Macroscopic cross section
c. Microscopic cross section f. Mean free path
2.2 EXPRESS macroscopic cross section in terms of microscopic cross section.
2.3 DESCRIBE how the absorption cross section of typical nuclides varies with neutron energy at energies below the resonance absorption region.
2.4 DESCRIBE the cause of resonance absorption in terms of nuclear energy levels.
2.5 DESCRIBE the energy dependence of resonance absorption peaks for typical light and heavy nuclei.
2.6 EXPRESS mean free path in terms of macroscopic cross section.
2.7 Given the number densities (or total density and component fractions) and microscopic cross sections of components, CALCULATE the macroscopic cross section for a mixture.
2.8 CALCULATE a macroscopic cross section given a material density, atomic mass, and microscopic cross section.
2.9 EXPLAIN neutron shadowing or self-shielding.
2.10 Given the neutron flux and macroscopic cross section, CALCULATE the reaction rate.
2.11 DESCRIBE the relationship between neutron flux and reactor power.

ENABLING OBJECTIVES (Cont.)
2.12 DEFINE the following concepts:
a. Thermalization d. Average logarithmic energy decrement
b. Moderator e. Macroscopic slowing down power
Fixture checkpoint marker-0502-xray is recorded on fixture page 0502.
c. Moderating ratio
2.13 LIST three desirable characteristics of a moderator.
2.14 Given an average fractional energy loss per collision, CALCULATE the energy loss after a specified number of collisions.

TERMINAL OBJECTIVE
3.0 Without references, EXPLAIN the production process and effects on fission of prompt and delayed neutrons.
ENABLING OBJECTIVES

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3.2 STATE the approximate fraction of neutrons that are born as delayed neutrons from the fission of the following nuclear fuels:
a. Uranium-235
b. Plutonium-239
3.3 EXPLAIN the mechanism for production of delayed neutrons.
3.4 EXPLAIN prompt and delayed neutron generation times.
3.5 Given prompt and delayed neutron generation times and delayed neutron fraction, CALCULATE the average generation time.
3.6 EXPLAIN the effect of delayed neutrons on reactor control.
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OBJECTIVES DOE-HDBK-1019/1-93 Reactor Theory (Neutron Characteristics)
TERMINAL OBJECTIVE
4.0 Without references, DESCRIBE the neutron energy spectrum for the type of reactor presented in this module.
Fixture checkpoint marker-0503-xray is recorded on fixture page 0503.
ENABLING OBJECTIVES
4.1 STATE the average energy at which prompt neutrons are produced.
4.2 DESCRIBE the neutron energy spectrum in the following reactors:
a. Fast reactor
b. Thermal reactor
4.3 EXPLAIN the reason for the particular shape of the fast, intermediate, and slow energy regions of the neutron flux spectrum for a thermal reactor.
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Reactor Theory (Neutron Characteristics) DOE-HDBK-1019/1-93 NEUTRON SOURCES
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NEUTRON SOURCES
Neutrons from a variety of sources are always present in a reactor core. This is true even when the reactor is shut down. Some of these neutrons are produced by naturally occurring (intrinsic) neutron sources, while others may be the result of fabricated (installed) neutron sources that are incorporated into the design of the reactor. The neutrons produced by sources other than neutron-induced fission are often grouped together and classified as source neutrons.
EO 1.1 DEFINE the following terms:
a. Intrinsic neutron source
b. Installed neutron source
EO 1.2 LIST three examples of reactions that produce neutrons in intrinsic neutron sources.
EO 1.3 LIST three examples of reactions that produce neutrons in installed neutron sources.
Neutron Sources
In addition to neutron-induced fission, neutrons are produced by other reactions. The neutrons produced by reactions other than neutron-induced fission are called source neutrons. Source neutrons are important because they ensure that the neutron population remains high enough to allow a visible indication of neutron level on the most sensitive monitoring instruments while the reactor is shutdown and during the startup sequence. This verifies instrument operability and allows monitoring of neutron population changes. Source neutrons can be classified as either intrinsic or installed neutron sources.
Intrinsic Neutron Sources

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unavoidable reactions that occur because of the nature of these materials. Intrinsic neutron sources are those neutron-producing reactions that always occur in reactor materials.

■ ■ ■ ■
■ ■ ■
■ ■

NEUTRON SOURCES DOE-HDBK-1019/1-93 Reactor Theory (Neutron Characteristics)
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A limited number of neutrons will always be present, even in a reactor core that has never been operated, due to spontaneous fission of some heavy nuclides that are present in the fuel. Uranium■238, ur of the rate at which different heavy nuclides produce neutrons by spontaneous fission.

Californium-252 is not an intrinsic neutron source, but will be discussed in the section on installed neutron sources.

TABLE 1

Neutron Production by Spontaneous Fission

Nuclide T (Fission) T (1/2 1/2 ■-decay) neutrons/sec/gram

U 1.8 x 10 years 6.8 x 10 years 8.0 x 10 235
92
17 8 -4

U 8.0 x 10 years 4.5 x 10 years 1.6 x 10 238
92
15 9 -2

Fixture checkpoint marker-0504-xray is recorded on fixture page 0504.

Pu 5.5 x 10 years 2.4 x 10 years 3.0 x 10 239
94
5 4 -2

Pu 1.2 x 10 years 6.6 x 10 years 1.0 x 10 240
94
11 3 3

Cf 66.0 years 2.65 years 2.3 x 10 252
98
12

Another intrinsic neutron source is a reaction involving natural boron and fuel. In some reactors, natural boron is loaded into the reactor core as a neutron absorber to improve reactor control or increase core life-time. Boron-11 (80.1% of natural boron) undergoes a reaction with the alpha particle emitted by the radioactive decay of heavy nuclides in the fuel to yield a neutron as shown below.

The boron-11 must be mixed with, or in very close proximity to, the fuel for this reaction because of the short path length of the alpha particle. For a reactor core with this configuration, this (■,n) reaction is an important source of neutrons for reactor startup.

In a reactor that has been operated, another source of neutrons becomes significant. Neutrons may be produced by the interaction of a gamma ray and a deuterium nucleus. This reaction is commonly referred to as a photoneutron reaction because it is initiated by electromagnetic radiation and results in the production of a neutron. The photoneutron reaction is shown below.

■ ■ ■
■ ■
■ ■ ■
■ ■

Reactor Theory (Neutron Characteristics) DOE-HDBK-1019/1-93 NEUTRON SOURCES

There is an abundant supply of high energy gammas in a reactor that has been operated because many of the fission products are gamma emitters. All water-cooled reactors have some deuterium present in the coolant in the reactor core because a small fraction of natural hydrogen is the isotope deuterium. The atom percentage of deuterium in the water ranges from close to the naturally occurring value (0.015%) for light water reactors to above 90% deuterium for heavy water reactors. Therefore, the required conditions for production of photoneutrons exist.
The supply of gamma rays decreases with time after shutdown as the gamma emitters decay; therefore, the photoneutron production rate also decreases. In a few particular reactors, additional D O (heavy water) may be added to the reactor to increase the production of 2 photoneutrons following a long shutdown period.
Installed Neutron Sources
Because intrinsic neutron sources can be relatively weak or dependent upon the recent power history of the reactor, many reactors have artificial sources of neutrons installed. These neutron sources ensure that shutdown neutron levels are high enough to be detected by the nuclear instruments at all times. This provides a true picture of reactor conditions and any change in these conditions. An installed neutron source is an assembly placed in or near the reactor for the sole purpose of producing source neutrons.
One strong source of neutrons is the artificial nuclide californium-252, which emits neutrons at the rate of about 2 x 10 neutrons per second per gram as the result of spontaneous fission. 12 Important drawbacks for some applications may be its high cost and its short half-life (2.65 years).
Many installed neutron sources use the (■,n) reaction with beryllium. These sources are composed of a mixture of metallic beryllium (100% beryllium-9) with a small quantity of an alpha particle emitter, such as a compound of radium, polonium, or plutonium. The reaction that occurs is shown below.
The beryllium is intimately (homogeneously) mixed with the alpha emitter and is usually enclosed in a stainless steel capsule.
Another type of installed neutron source that is widely used is a photoneutron source that
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employs the (■,n) reaction with beryllium. Beryllium is used for photoneutron sources because its stable isotope beryllium-9 has a weakly attached last neutron with a binding energy of only 1.66 MeV. Thus, a gamma ray with greater energy than 1.66 MeV can cause neutrons to be ejected by the (■,n) reaction as shown below.
NEUTRON SOURCES DOE-HDBK-1019/1-93 Reactor Theory (Neutron Characteristics)
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Neutron Sources Summary
& Intrinsic neutron sources are sources of neutrons from materials that are in the reactor for other purposes such as fuel, burnable poison, or moderator.
& Installed neutron sources are materials or components placed in the reactor specifically for the purpose of producing source neutrons.
& Examples of intrinsic neutron sources are listed below.
Spontaneous fission of heavy nuclides in fuel, such as uranium-238, uranium-235, and plutonium-239, results in fission fragments and free neutrons.
Boron-11 mixed with the fuel undergoes an alpha-neutron reaction and becomes nitrogen-14.
Deuterium present in the reactor coolant undergoes a gamma■neutron reaction and becomes hydrogen-1.
& Examples of installed neutron sources are listed below.

and free neutrons.
Beryllium-9 undergoes an alpha-neutron reaction (alpha from the decay of plutonium, polonium, or radium) and becomes carbon-12.
Beryllium-9 undergoes a gamma-neutron reaction (high energy gamma from decay of antimony-124) and becomes beryllium-8.
Many startup sources of this type use antimony and beryllium because after activation with neutrons the radioactive antimony becomes an emitter of high energy gammas. The photoneutron sources of this type are constructed somewhat differently from the (■,n) types. One design incorporates a capsule of irradiated antimony enclosed in a beryllium sleeve. The entire assembly is then encased in a stainless steel cladding. A large reactor may have several neutron sources of this type installed within the core.
Summary
The important information in this chapter is summarized below.
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Reactor Theory (Neutron Characteristics) NUCLEAR CROSS SECTIONS AND NEUTRON FLUX
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NUCLEAR CROSS SECTIONS AND NEUTRON FLUX
To determine the frequency of neutron interactions, it is necessary to describe the availability of neutrons to cause interaction and the probability of a neutron interacting with material. The availability of neutrons and the probability of interaction are quantified by the neutron flux and nuclear cross section.
EO 2.1 DEFINE the following terms:
a. Atom density d. Barn
b. Neutron flux e. Macroscopic cross section
c. Microscopic cross section f. Mean free path
EO 2.2 EXPRESS macroscopic cross section in terms of microscopic cross section.
EO 2.3 DESCRIBE how the absorption cross section of typical nuclides varies with neutron energy at energies below the resonance absorption region.
EO 2.4 DESCRIBE the cause of resonance absorption in terms of nuclear energy levels.
EO 2.5 DESCRIBE the energy dependence of resonance absorption peaks for typical light and heavy nuclei.
EO 2.6 EXPRESS mean free path in terms of macroscopic cross section.
Fixture checkpoint marker-0506-xray is recorded on fixture page 0506.
EO 2.7 Given the number densities (or total density and component fractions) and microscopic cross sections of components, CALCULATE the macroscopic cross section for a mixture.
EO 2.8 CALCULATE a macroscopic cross section given a material density, atomic mass, and microscopic cross section.
EO 2.9 EXPLAIN neutron shadowing or self-shielding.
'
DOE-HDBK-1019/1-93
NUCLEAR CROSS SECTIONS AND NEUTRON FLUX Reactor Theory (Neutron Characteristics)
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Introduction
Fission neutrons are born with an average energy of about 2 MeV. These fast neutrons interact

result in scattering are useful in slowing neutrons to thermal energies. Thermal neutrons may be absorbed by fissile nuclei to produce more fissions or be absorbed in fertile material for conversion to fissionable fuel. Absorption of neutrons in structural components, coolant, and other non-fuel material results in the removal of neutrons without fulfilling any useful purpose. To safely and efficiently operate a nuclear reactor it is necessary to predict the probability that a particular absorption or scattering reaction will occur. Once these probabilities are known, if the availability of neutrons can be determined, then the rate at which these nuclear reactions take place can be predicted.

Atom Density

One important property of a material is the atom density. The atom density is the number of atoms of a given type per unit volume of the material. To calculate the atom density of a substance use Equation (2-1).

Fixture checkpoint marker-0507-xray is recorded on fixture page 0507.

(2-1)

where:

N = atom density (atoms/cm)3

' = density (g/cm)3

N = Avogadro's number (6.022 x 10 atoms/mole) A

23

M = gram atomic weight

N

NA

M

2.699 g

cm3

6.022 x 1023 atoms

mole

26.9815 g

mole

6.024 x 1022 atoms

cm3

Example:

A block of aluminum has a density of 2.699 g/cm . If the gram atomic weight of

3

aluminum is 26.9815 g, calculate the atom density of the aluminum.

Solution:

Cross Sections

The probability of a neutron interacting with a nucleus for a particular reaction is dependent upon not only the kind of nucleus involved, but also the energy of the neutron. Accordingly, the absorption of a thermal neutron in most materials is much more probable than the absorption of a fast neutron. Also, the probability of interaction will vary depending upon the type of reaction involved.

The probability of a particular reaction occurring between a neutron and a nucleus is called the microscopic cross section ( ) of the nucleus for the particular reaction. This cross section will vary with the energy of the neutron. The microscopic cross section may also be regarded as the

effective area, the greater the probability for reaction.
Because the microscopic cross section is an area, it is expressed in units of area, or square centimeters. A square centimeter is tremendously large in comparison to the effective area of a nucleus, and it has been suggested that a physicist once referred to the measure of a square centimeter as being "as big as a barn" when applied to nuclear processes. The name has persisted and microscopic cross sections are expressed in terms of barns. The relationship between barns and cm is shown below. 2
1 barn = 10-24 cm2
N
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NUCLEAR CROSS SECTIONS AND NEUTRON FLUX Reactor Theory (Neutron Characteristics)
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Whether a neutron will interact with a certain volume of material depends not only on the microscopic cross section of the individual nuclei but also on the number of nuclei within that volume. Therefore, it is necessary to define another kind of cross section known as the macroscopic cross section ( ). The macroscopic cross section is the probability of a given reaction occurring per unit travel of the neutron. is related to the microscopic cross section ( ) by the relationship shown below.
(2-2)
Fixture checkpoint marker-0508-xray is recorded on fixture page 0508.
where:
= macroscopic cross section (cm)
-1
N = atom density of material (atoms/cm)
3
= microscopic cross-section (cm)
2
The difference between the microscopic and macroscopic cross sections is extremely important and is restated for clarity. The microscopic cross section ( ) represents the effective target area that a single nucleus presents to a bombarding particle. The units are given in barns or cm .
2
The macroscopic cross section ( ) represents the effective target area that is presented by all of the nuclei contained in 1 cm of the material. The units are given as 1/cm or cm .
3 -1
A neutron interacts with an atom of the material it enters in two basic ways. It will either interact through a scattering interaction or through an absorption reaction. The probability of a neutron being absorbed by a particular atom is the microscopic cross section for absorption, . The probability of a neutron scattering off of a particular nucleus is the microscopic cross a section for scattering, s
. The sum of the microscopic cross section for absorption and the microscopic cross section for scattering is the total microscopic cross section, . T
= + T a s
Both the absorption and the scattering microscopic cross sections can be further divided. For instance, the scattering cross section is the sum of the elastic scattering cross section (se) and the inelastic scattering cross section ( ). si
= + s se si
The microscopic absorption cross section ( ) includes all reactions except scattering. However, a for most purposes it is sufficient to merely separate it into two categories, fission (f) and

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= + a f c
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Reactor Theory (Neutron Characteristics) NUCLEAR CROSS SECTIONS AND NEUTRON FLUX
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Figure 1 Typical Neutron Absorption Cross Section vs. Neutron Energy
The variation of absorption cross sections with neutron energy is often complicated. For many elements the absorption cross sections are small, ranging from a fraction of a barn to a few barns for slow (or thermal) neutrons.
For a considerable number of nuclides of moderately high (or high) mass numbers, an examination of the variation of the absorption cross section with the energy of the incident neutron reveals the existence of three regions on a curve of absorption cross section versus neutron energy. This cross section is illustrated in Figure 1. First, the cross section decreases steadily with increasing neutron energy in a low energy region, which includes the thermal range (E < 1 eV). In this region the absorption cross section, which is often high, is inversely proportional to the velocity (v). This region is frequently referred to as the "1/v region," because the absorption cross section is proportional to 1/v, which is the reciprocal of neutron velocity. Following the 1/v region, there occurs the "resonance region" in which the cross sections rise sharply to high values called "resonance peaks" for neutrons of certain energies, and then fall again. These energies are called resonance energies and are a result of the affinity of the nucleus for neutrons whose energies closely match its discrete, quantum energy levels. That is, when the binding energy of a neutron plus the kinetic energy of the neutron are exactly equal to the amount required to raise a compound nucleus from its ground state to a quantum level, resonance absorption occurs. The following example problem further illustrates this point.
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Fixture checkpoint marker-0509-xray is recorded on fixture page 0509.
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NUCLEAR CROSS SECTIONS AND NEUTRON FLUX Reactor Theory (Neutron Characteristics)
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Assuming that uranium-236 has a nuclear quantum energy level at 6.8 MeV above its ground state, calculate the kinetic energy a neutron must possess to undergo resonant absorption in uranium-235 at this resonance energy level.
BE = [Mass(U) + Mass(neutron) - Mass(U)] x 931 MeV/amu 235 236
BE = (235.043925 + 1.008665 - 236.045563) x 931 MeV/amu
BE = (0.007025 amu) x 931 MeV/amu = 6.54 MeV
6.8 MeV - 6.54 MeV = 0.26 MeV
The difference between the binding energy and the quantum energy level equals the amount of kinetic energy the neutron must possess. The typical heavy nucleus will have many closely■spaced resonanc complex and have more possible configurations and corresponding energy states. Light nuclei, being less complex, have fewer possible energy states and fewer resonances that are sparsely distributed at higher energy levels.
For higher neutron energies, the absorption cross section steadily decreases as the energy of the neutron increases. This is called the "fast neutron region." In this region the absorption cross sections are usually less than 10 barns.
With the exception of hydrogen, for which the value is fairly large, the elastic scattering cross sections are generally small, for example, 5 barns to 10 barns. This is close to the magnitude of the actual geometric cross sectional area expected for atomic nuclei. In potential scattering,

scattering and inelastic scattering exhibit resonance peaks similar to those associated with absorption cross sections. The resonances occur at lower energies for heavy nuclei than for light nuclei. In general, the variations in scattering cross sections are very small when compared to the variations that occur in absorption cross sections.

Mean Free Path

If a neutron has a certain probability of undergoing a particular interaction in one centimeter of travel, then the inverse of this value describes how far the neutron will travel (in the average case) before undergoing an interaction. This average distance traveled by a neutron before interaction is known as the mean free path for that interaction and is represented by the symbol ■. The relationship between the mean free path (■) and the macroscopic cross section (*) is shown below.

(2-3)

'

(

)

Calculation of Macroscopic Cross Section and Mean Free Path

Most materials are composed of several elements, and because most elements are composed of several isotopes, most materials involve many cross sections, one for each isotope involved. Therefore, to include all the isotopes within a given material, it is necessary to determine the macroscopic cross section for each isotope and then sum all the individual macroscopic cross sections. Equation (2-4) can be used to determine the macroscopic cross section for a composite material.

* = N) + N) + N) + ... N) (2-4) 1 1 2 2 3 3 nn

where:

N = the number of nuclei per cm of the n element n

3 th

) = the microscopic cross section of the n element n

th

The following example problems illustrate the calculation of the macroscopic cross section for a single element and for combinations of materials.

Fixture checkpoint marker-0510-xray is recorded on fixture page 0510.

Example 1:

Find the macroscopic thermal neutron absorption cross section for iron, which has a density of 7.86 g/cm . The microscopic cross section for absorption of iron is 2.56 barns 3 and the gram atomic weight is 55.847 g.

Solution:

Step 1: Using Equation (2-1), calculate the atom density of iron.

Step 2: Use this atom density in Equation (2-2) to calculate the macroscopic cross section.

'

'

Example 2:

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alloy is 2.66 g/cm . Properties of aluminum and silicon are shown below. 3
Element Gram Atomic Weight)) a (barns) s (barns)
Aluminum 26.9815 0.23 1.49
Silicon 28.0855 0.16 2.20
1. Calculate the atom densities for the aluminum and silicon.
2. Determine the absorption and scattering macroscopic cross sections for thermal neutrons.
3. Calculate the mean free paths for absorption and scattering.
Solution:
Step 1: The density of the aluminum will be 95% of the total density. Using
Fixture checkpoint marker-0511-xray is recorded on fixture page 0511.
Equation (2-1) yields the atom densities.
a NAl a,Al NSi a,Si
5.64 x 1022 atoms
cm3
0.23 x 10 24 cm22.85 x 1021 atoms
cm3
0.16 x 10 24 cm2
0.0134 cm 1
s NAl s,Al NSi s,Si
5.64 x 1022 atoms
cm3
1.49 x 10 24 cm22.85 x 1021 atoms
cm3
2.20 x 10 24 cm2
0.0903 cm 1
a
1
a
1
0.01345 cm 1
74.3 cm
s
1
s
1
0.0903 cm 1
11.1 cm
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Step 2: The macroscopic cross sections for absorption and scattering are calculated using Equation (2-4).
Step 3: The mean free paths are calculated by inserting the macroscopic cross sections calculated above into Equation (2-3).
Thus, a neutron must travel an average of 74.3 cm to interact by absorption in this alloy, but it must travel only 11.1 cm to interact by scattering.
)

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)
)
(■
(■
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NUCLEAR CROSS SECTIONS AND NEUTRON FLUX Reactor Theory (Neutron Characteristics)
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Effects of Temperature on Cross Section
As discussed, the microscopic absorption cross section varies significantly as neutron energy varies. The microscopic cross sections provided on most charts and tables are measured for a standard neutron velocity of 2200 meters/second, which corresponds to an ambient temperature of 68(F. Therefore, if our material is at a higher temperature, the absorption cross section will be lower than the value for 68(F, and any cross sections which involve absorption (for example,) ,) ,)) must be corrected for the existing temperature. acf
The following formula is used to correct microscopic cross sections for temperature. Although the example illustrates absorption cross section, the same formula may be used to correct capture and fission cross sections.
Fixture checkpoint marker-0512-xray is recorded on fixture page 0512.
where:
) = microscopic cross section corrected for temperature
) = microscopic cross section at reference temperature (68(F or 20(C) o
T = reference temperature (68(F) in degrees Rankine ((R) or Kelvin ((K) o
T = temperature for which corrected value is being calculated
NOTE: When using this formula, all temperatures must be converted to (R or (K.
(R = (F + 460
(K = (C + 273
Example:
What is the value of)f for uranium-235 for thermal neutrons at 500(F? Uranium-235 has a) of 583 barns at 68(F. f
Solution:
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Reactor Theory (Neutron Characteristics) NUCLEAR CROSS SECTIONS AND NEUTRON FLUX
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Neutron Flux
Macroscopic cross sections for neutron reactions with materials determine the probability of one neutron undergoing a specific reaction per centimeter of travel through that material. If one wants to determine how many reactions will actually occur, it is necessary to know how many neutrons are traveling through the material and how many centimeters they travel each second. It is convenient to consider the number of neutrons existing in one cubic centimeter at any one instant and the total distance they travel each second while in that cubic centimeter. The number of neutrons existing in a cm of material at any instant is called neutron density and is 3 represented by the symbol n with units of neutrons/cm . The total distance these neutrons can 3 travel each second will be determined by their velocity.
A good way of defining neutron flux (1) is to consider it to be the total path length covered by all neutrons in one cubic centimeter during one second. Mathematically, this is the equation below.
1 = n v (2-5)
where:

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n = neutron density (neutrons/cm)3
v = neutron velocity (cm/sec)
The term neutron flux in some applications (for example, cross section measurement) is used as parallel beams of neutrons traveling in a single direction. The intensity (I) of a neutron beam is the product of the neutron density times the average neutron velocity. The directional beam intensity is equal to the number of neutrons per unit area and time (neutrons/cm -sec) falling on 2 a surface perpendicular to the direction of the beam.
One can think of the neutron flux in a reactor as being comprised of many neutron beams traveling in various directions. Then, the neutron flux becomes the scalar sum of these directional flux intensities (added as numbers and not vectors), that is, 1 = I + I + I +...I . 123 n
Since the atoms in a reactor do not interact preferentially with neutrons from any particular direction, all of these directional beams contribute to the total rate of reaction. In reality, at a given point within a reactor, neutrons will be traveling in all directions.
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Self-Shielding
In some locations within the reactor, the flux level may be significantly lower than in other areas due to a phenomenon referred to as neutron shadowing or self-shielding. For example, the interior of a fuel pin or pellet will "see" a lower average flux level than the outer surfaces since an appreciable fraction of the neutrons will have been absorbed and therefore cannot reach the interior of the fuel pin. This is especially important at resonance energies, where the absorption cross sections are large.
Summary
Fixture checkpoint marker-0513-xray is recorded on fixture page 0513.
The important information in this chapter is summarized below.
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Reactor Theory (Neutron Characteristics) NUCLEAR CROSS SECTIONS AND NEUTRON FLUX
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Nuclear Cross Section and Neutron Flux Summary
& Atom density (N) is the number of atoms of a given type per unit volume of material.
& Microscopic cross section ()) is the probability of a given reaction occurring between a neutron and a nucleus.
& Microscopic cross sections are measured in units of barns, where 1 barn = 10-24 cm .2
& Macroscopic cross section (*) is the probability of a given reaction occurring per unit length of travel of the neutron. The units for macroscopic cross section are cm . -1
& The mean free path (■) is the average distance that a neutron travels in a material between interactions.
& Neutron flux (1) is the total path length traveled by all neutrons in one cubic centimeter of material during one second.
& The macroscopic cross section for a material can be calculated using the equation below.
* = N)
& The absorption cross section for a material usually has three distinct regions. At low neutron energies (<1 eV) the cross section is inversely proportional to the

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& Resonance absorption occurs when the sum of the kinetic energy of the neutron and its binding energy is equal to an allowed nuclear energy level of the nucleus.
& Resonance peaks exist at intermediate energy levels. For higher neutron energies, the absorption cross section steadily decreases as the neutron energy increases.
& The mean free path equals 1/*.
& The macroscopic cross section for a mixture of materials can be calculated using the equation below.
* = N 1 1 2 2 3 3 nn) + N) + N) + ... N)
& Self-shielding is where the local neutron flux is depressed within a material due to neutron absorption near the surface of the material.
1)
REACTION RATES DOE-HDBK-1019/1-93 Reactor Theory (Neutron Characteristics)
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REACTION RATES
It is possible to determine the rate at which a nuclear reaction will take place based on the neutron flux, cross section for the interaction, and atom density of the target. This relationship illustrates how a change in one of these items affects the reaction rate.
EO 2.10 Given the neutron flux and macroscopic cross section, CALCULATE the reaction rate.
EO 2.11 DESCRIBE the relationship between neutron flux and reactor power.
Reaction Rates
If the total path length of all the neutrons in a cubic centimeter in a second is known, (neutron flux (1)), and if the probability of having an interaction per centimeter path length is also known (macroscopic cross section (*)), multiply them together to get the number of interactions taking place in that cubic centimeter in one second. This value is known as the reaction rate and is denoted by the symbol R. The reaction rate can be calculated by the equation shown below.
R = 1 * (2-6)
where:
R = reaction rate (reactions/sec)
Fixture checkpoint marker-0514-xray is recorded on fixture page 0514.
1 = neutron flux (neutrons/cm -sec) 2
* = macroscopic cross section (cm) -1
Substituting the fact that * = N) into Equation (2-6) yields the equation below.
where:
) = microscopic cross section (cm)2
N = atom density (atoms/cm)3
Rf f
1 x 1013 neutrons
cm2
sec
0.1 cm 1
1 x 1012 fissions
cm3
sec
Rf f
f

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1.29 x 1012 fissions
cm3
sec
3 x 1013 neutrons
cm2
sec
0.043 cm 1
Reactor Theory (Neutron Characteristics) DOE-HDBK-1019/1-93 REACTION RATES
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The reaction rate calculated will depend on which macroscopic cross section is used in the calculation. Normally, the reaction rate of greatest interest is the fission reaction rate.
Example:
If a one cubic centimeter section of a reactor has a macroscopic fission cross section of 0.1 cm , and if the thermal neutron flux is 10 neutrons/cm -sec, what is the fission rate -1 13 2 in that cubic centimeter?
Solution:
In addition to using Equation (2-6) to determine the reaction rate based on the physical properties of material, it is also possible to algebraically manipulate the equation to determine physical properties if the reaction rate is known.
Example:
A reactor operating at a flux level of 3 x 10 neutrons/cm -sec contains 10 atoms of
13 2 20
uranium-235 per cm . The reaction rate is 1.29 x 10 fission/cm . Calculate and .
3 12 3
f f
Solution:
Step 1: The macroscopic cross section can be determined by solving
Equation (2-6) for and substituting the appropriate values. f
Rf N f
f
Rf
N
1.29 x 1012 fissions
cm3
sec
1 x 1020 atoms
cm3
3 x 1013 neutrons
Fixture checkpoint marker-0515-xray is recorded on fixture page 0515.
cm2
sec
4.3 x 10 22 cm2 1 barn
1 x 10 24 cm2
430 barns
1 watt 1 erg
1 x 10 7 watt sec
1 MeV
1.602 x 10 6

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1 fission
200 MeV
3.12 x 1010 fissions
second
REACTION RATES DOE-HDBK-1019/1-93 Reactor Theory (Neutron Characteristics)
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Step 2: To find the microscopic cross section, replace f with (N x f) and solve
for .
f
Reactor Power Calculation
Multiplying the reaction rate per unit volume by the total volume of the core results in the total number of reactions occurring in the core per unit time. If the amount of energy involved in each reaction were known, it would be possible to determine the rate of energy release (power) due to a certain reaction.
In a reactor where the average energy per fission is 200 MeV, it is possible to determine the
Fixture checkpoint marker-0516-xray is recorded on fixture page 0516.
number of fissions per second that are necessary to produce one watt of power using the following conversion factors.
1 fission = 200 MeV
1 MeV = 1.602 x 10 ergs -6
1 erg = 1 x 10 watt-sec -7
This is equivalent to stating that 3.12 x 10 fissions release 1 watt-second of energy. 10
P
th f V
3.12 x 1010 fissions
watt sec
Reactor Theory (Neutron Characteristics) DOE-HDBK-1019/1-93 REACTION RATES
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The power released in a reactor can be calculated based on Equation (2-6). Multiplying the reaction rate by the volume of the reactor results in the total fission rate for the entire reactor. Dividing by the number of fissions per watt-sec results in the power released by fission in the reactor in units of watts. This relationship is shown mathematically in Equation (2-7) below.
(2-7)
where:
P = power (watts)
= thermal neutron flux (neutrons/cm -sec) th
2
= macroscopic cross section for fission (cm) f
-1
V = volume of core (cm)
3
Relationship Between Neutron Flux and Reactor Power
In an operating reactor the volume of the reactor is constant. Over a relatively short period of time (days or weeks), the number density of the fuel atoms is also relatively constant. Since the atom density and microscopic cross section are constant, the macroscopic cross section must also be constant. Examining Equation (2-7), it is apparent that if the reactor volume and macroscopic cross section are constant, then the reactor power and the neutron flux are directly proportional. This is true for day-to-day operation. The neutron flux for a given power level will increase very

density and macroscopic cross section.
REACTION RATES DOE-HDBK-1019/1-93 Reactor Theory (Neutron Characteristics)
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Reaction Rates Summary
& The reaction rate is the number of interactions of a particular type occurring in a cubic centimeter of material in a second.
& The reaction rate can be calculated by the equation below.
R = 1 *
& Over a period of several days, while the atom density of the fuel can be considered constant, the neutron flux is directly proportional to reactor power.
Summary
The important information in this chapter is summarized below.
Reactor Theory (Neutron Characteristics) DOE-HDBK-1019/1-93 NEUTRON MODERATION
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NEUTRON MODERATION
In thermal reactors, the neutrons that cause fission are at a much lower energy than the energy level at which they were born from fission. In this type of reactor, specific materials must be included in the reactor design to reduce the energy level of the neutrons in an efficient manner.
EO 2.12 DEFINE the following concepts:
a. Thermalization d. Average logarithmic energy decrement
b. Moderator e. Macroscopic slowing down power
Fixture checkpoint marker-0517-xray is recorded on fixture page 0517.
c. Moderating ratio
EO 2.13 LIST three desirable characteristics of a moderator.
EO 2.14 Given an average fractional energy loss per collision, CALCULATE the energy loss after a specified number of collisions.
Neutron Slowing Down and Thermalization
Fission neutrons are produced at an average energy level of 2 MeV and immediately begin to slow down as the result of numerous scattering reactions with a variety of target nuclei. After a number of collisions with nuclei, the speed of a neutron is reduced to such an extent that it has approximately the same average kinetic energy as the atoms (or molecules) of the medium in which the neutron is undergoing elastic scattering. This energy, which is only a small fraction of an electron volt at ordinary temperatures (0.025 eV at 20(C), is frequently referred to as the thermal energy, since it depends upon the temperature. Neutrons whose energies have been reduced to values in this region (< 1 eV) are designated thermal neutrons. The process of reducing the energy of a neutron to the thermal region by elastic scattering is referred to as thermalization, slowing down, or moderation. The material used for the purpose of thermalizing neutrons is called a moderator. A good moderator reduces the speed of neutrons in a small number of collisions, but does not absorb them to any great extent. Slowing the neutrons in as few collisions as possible is desirable in order to reduce the amount of neutron leakage from the core and also to reduce the number of resonance absorptions in non-fuel materials. Neutron leakage and resonance absorption will be discussed in the next module.
ln Ei
ln Ef
ln
Ei

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2
A
2
3
NEUTRON MODERATION DOE-HDBK-1019/1-93 Reactor Theory (Neutron Characteristics)
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The ideal moderating material (moderator) should have the following nuclear properties.
large scattering cross section
small absorption cross section
large energy loss per collision
A convenient measure of energy loss per collision is the logarithmic energy decrement. The average logarithmic energy decrement is the average decrease per collision in the logarithm of the neutron energy. This quantity is represented by the symbol (Greek letter xi).
(2-8)
where:
= average logarithmic energy decrement
Ei = average initial neutron energy
Ef = average final neutron energy
The symbol is commonly called the average logarithmic energy decrement because of the fact that a neutron loses, on the average, a fixed fraction of its energy per scattering collision. Since the fraction of energy retained by a neutron in a single elastic collision is a constant for a given material, is also a constant. Because it is a constant for each type of material and does not depend upon the initial neutron energy, is a convenient quantity for assessing the moderating ability of a material.
The values for the lighter nuclei are tabulated in a variety of sources. The following commonly used approximation may be used when a tabulated value is not available.
This approximation is relatively accurate for mass numbers (A) greater than 10, but for some low values of A it may be in error by over three percent.
N
Fixture checkpoint marker-0518-xray is recorded on fixture page 0518.
ln Ehigh ln Elow
ln
Ehigh
Elow
N
ln
Ehigh
Elow
ln
2 x 106eV
0.025 eV
0.948
19.2 collisions
Reactor Theory (Neutron Characteristics) DOE-HDBK-1019/1-93 NEUTRON MODERATION
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Since represents the average logarithmic energy loss per collision, the total number of collisions necessary for a neutron to lose a given amount of energy may be determined by dividing into the difference of the natural logarithms of the energy range in question. The number of collisions

Example:
How many collisions are required to slow a neutron from an energy of 2 MeV to a thermal energy of 0.025 eV, using water as the moderator? Water has a value of 0.948 for .
Solution:
Sometimes it is convenient, based upon information known, to work with an average fractional energy loss per collision as opposed to an average logarithmic fraction. If the initial neutron energy level and the average fractional energy loss per collision are known, the final energy level for a given number of collisions may be computed using the following formula.
! (
NEUTRON MODERATION DOE-HDBK-1019/1-93 Reactor Theory (Neutron Characteristics)
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(2-9)
where:
E = initial neutron energy o
E = neutron energy after N collisions N
x = average fractional energy loss per collision
N = number of collisions
Example:
If the average fractional energy loss per collision in hydrogen is 0.63, what will be the energy of a 2 MeV neutron after (a) 5 collisions? (b) 10 collisions?
Solution:
a)
b)
Macroscopic Slowing Down Power
Although the logarithmic energy decrement is a convenient measure of the ability of a material to slow neutrons, it does not measure all necessary properties of a moderator. A better measure of the capabilities of a material is the macroscopic slowing down power. The macroscopic slowing down power (MSDP) is the product of the logarithmic energy decrement and the macroscopic cross section for scattering in the material. Equation (2-10) illustrates how to calculate the macroscopic slowing down power.
(2-10)
MR
s
a
Reactor Theory (Neutron Characteristics) DOE-HDBK-1019/1-93 NEUTRON MODERATION
Fixture checkpoint marker-0519-xray is recorded on fixture page 0519.
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Moderating Ratio
Macroscopic slowing down power indicates how rapidly a neutron will slow down in the material in question, but it still does not fully explain the effectiveness of the material as a moderator. An element such as boron has a high logarithmic energy decrement and a good slowing down power, but it is a poor moderator because of its high probability of absorbing neutrons.
The most complete measure of the effectiveness of a moderator is the moderating ratio. The moderating ratio is the ratio of the macroscopic slowing down power to the macroscopic cross section for absorption. The higher the moderating ratio, the more effectively the material performs as a moderator. Equation (2-11) shows how to calculate the moderating ratio of a material.

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Moderating properties of different materials are compared in Table 2.
TABLE 2
Moderating Properties of Materials
Material Collisions to Slowing Down
Number of Macroscopic
Thermalize Power
Moderating
Ratio
H2O 0.927 19 1.425 62
D2O 0.510 35 0.177 4830
Helium 0.427 42 9 x 10 51 -6
Beryllium 0.207 86 0.154 126
Boron 0.171 105 0.092 0.00086
Fixture checkpoint marker-0520-xray is recorded on fixture page 0520.
Carbon 0.158 114 0.083 216
NEUTRON MODERATION DOE-HDBK-1019/1-93 Reactor Theory (Neutron Characteristics)
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Summary
The important information in this chapter is summarized below.
Neutron Moderation Summary
& Thermalization is the process of reducing the energy level of a neutron from the energy level at which it is produced to an energy level in the thermal range.
& The moderator is the reactor material that is present for the purpose of thermalizing neutrons.
& Moderating ratio is the ratio of the macroscopic slowing down power to the macroscopic cross section for absorption.
& The average logarithmic energy decrement (!) is the average change in the logarithm of neutron energy per collision.
& Macroscopic slowing down power is the product of the average logarithmic energy decrement and the macroscopic cross section for scattering.
& There are three desirable characteristics of a moderator.
1. large scattering cross section
2. small absorption cross section
3. large energy loss per collision
& The energy loss after a specified number of collisions can be calculated using the equation below.
E = E (1 - x) N o
N
■
■ ■
Reactor Theory (Neutron Characteristics) DOE-HDBK-1019/1-93 PROMPT AND DELAYED NEUTRONS
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PROMPT AND DELAYED NEUTRONS
Not all neutrons are released at the same time following fission. Most neutrons are released virtually instantaneously and are called prompt neutrons. A very small fraction of neutrons are released after the decay of fission products and are called delayed neutrons. Although delayed neutrons are a very small fraction of the total number of neutrons, they play an extremely important role in the control

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EO 3.1 STATE the origin of prompt neutrons and delayed neutrons.
EO 3.2 STATE the approximate fraction of neutrons that are born as delayed neutrons from the fission of the following nuclear fuels:
a. Uranium-235
b. Plutonium-239
EO 3.3 EXPLAIN the mechanism for production of delayed neutrons.
EO 3.4 EXPLAIN prompt and delayed neutron generation times.
EO 3.5 Given prompt and delayed neutron generation times and delayed neutron fraction, CALCULATE the average generation time.
EO 3.6 EXPLAIN the effect of delayed neutrons on reactor control.

Neutron Classification

The great majority (over 99%) of the neutrons produced in fission are released within about 10-13 seconds of the actual fission event. These are called prompt neutrons. A small portion of fission neutrons are delayed neutrons, which are produced for some time after the fission process has taken place. The delayed neutrons are emitted immediately following the first beta decay of a fission fragment known as a delayed neutron precursor. An example of a delayed neutron precursor is bromine-87, shown below.

PROMPT AND DELAYED NEUTRONS DOE-HDBK-1019/1-93 Reactor Theory (Neutron Characteristics) NP-02 Page 30 Rev. 0

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For most applications, it is convenient to combine the known precursors into groups with appropriately averaged properties. These groups vary somewhat depending on the fissile material in use. Table 3 lists the characteristics for the six precursor groups resulting from thermal fission of uranium-235. The fraction of all neutrons that are produced by each of these precursors is called the delayed neutron fraction for that precursor. The total fraction of all neutrons born as delayed neutrons is called the delayed neutron fraction (■). The fraction of delayed neutrons produced varies depending on the predominant fissile nuclide in use. The delayed neutron fractions (■) for the fissile nuclides of most interest are as follows: uranium-233 (0.0026), uranium-235 (0.0065), uranium-238 (0.0148), and plutonium-239 (0.0021).

TABLE 3
Delayed Neutron Precursor Groups
for Thermal Fission in Uranium-235

Group	Half-Life (sec)	Delayed Neutron Fraction	Average Energy (MeV)
1	55.7	0.00021	0.25
2	22.7	0.00142	0.46
3	6.2	0.00127	0.41
4	2.3	0.00257	0.45
5	0.61	0.00075	0.41
6	0.23	0.00027	-
Total	-	0.0065	-

Neutron Generation Time

The neutron generation time is the time required for neutrons from one generation to cause the fissions that produce the next generation of neutrons. The generation time for prompt neutrons (5* - pronounced "ell-star") is the total time from birth to rebirth. Three time intervals are involved: (a) the time it takes a fast neutron to slow down to thermal energy, (b) the time the now thermal neutron exists prior to absorption in fuel, and (c) the time required for a fissionable

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Fast neutrons slow to thermal energies or leak out of the reactor in 10 seconds to 10 seconds, -4 -6
depending on the moderator. In water moderated reactors, thermal neutrons tend to exist for
about 10 seconds before they are absorbed. Fission and fast neutron production following -4
neutron absorption in a fissionable nucleus occurs in about 10 seconds. Thus, fast reactors -13
have an 5* of about 10 seconds, while thermal reactors have an 5* of about 10 seconds + 10 -6 -6 -4
seconds, which is about 10 seconds to 10 seconds. -4 -5

■ ■ ■ ■
■ ■ ■ ■
■ ■

Reactor Theory (Neutron Characteristics) DOE-HDBK-1019/1-93 PROMPT AND DELAYED NEUTRONS
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On the other hand, the average generation time for the six delayed neutron groups is the total
time from the birth of the fast neutron to the emission of the delayed neutron. Again, three time
intervals are involved: (a) the time it takes a fast neutron to slow down to thermal energy,
(b) the time the thermal neutron exists prior to absorption, and (c) the average time from neutron
absorption to neutron emission by the six precursor groups. The average time for decay of
precursors from uranium-235 is 12.5 seconds. The other terms in the delayed neutron
generation time are insignificant when compared to this value, and the average delayed neutron
generation time becomes ■ 12.5 seconds.

A neutron generation time in the range of 10 seconds to 10 seconds or faster could result in -4 -5
very rapid power excursions, and control would not be possible without the dependence upon
delayed neutrons to slow down the rate of the reaction. The average generation time, and hence
the rate that power can rise, is determined largely by the delayed neutron generation time. The
following equation shows this mathematically.

(2-12)

Example:

Assume a prompt neutron generation time for a particular reactor of 5 x 10 seconds and -5
Fixture checkpoint marker-0522-xray is recorded on fixture page 0522.
a delayed neutron generation time of 12.5 seconds. If ■ is 0.0065, calculate the average
generation time.

Solution:

This example demonstrates the effect delayed neutrons have on the neutron generation time and
thus reactor control. If a reactor were to be operated in a sustained chain reaction using only
prompt neutrons (■ = 0), the generation time from the previous example would be about
5 x 10 seconds. However, by operating the reactor such that a 0.0065 fraction of neutrons are -5
delayed, the generation life time is extended to 0.0813 seconds, providing time for adequate
operator control. Therefore, although only a small fraction of the total neutron population,
delayed neutrons are extremely important to the control and maintenance of a sustained fission
chain reaction.

Summary

The important information in this chapter is summarized on the following page.

■ ■ ■ ■

PROMPT AND DELAYED NEUTRONS DOE-HDBK-1019/1-93 Reactor Theory (Neutron Characteristics)
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Prompt and Delayed Neutrons Summary

& Prompt neutrons are released directly from fission within 10 seconds of the -13
fission event.

& Delayed neutrons are released from the decay of fission products that are called

half-life. Half-lives vary from fractions of a second to almost a minute.
& The fraction of neutrons born as delayed neutrons is different for different fuel materials. Following are values for some common fuel materials.
Uranium-235 0.0065
Plutonium-239 0.0021
& Delayed neutrons are produced by a classification of fission products known as delayed neutron precursors. When a delayed neutron precursor undergoes a ■ decay, it results in an excited daughter nucleus which immediately ejects a - neutron. Therefore, these delayed neutrons appear with a half-life of the delayed neutron precursor.
& The delayed neutron generation time is the total time from the birth of the fast neutron to the emission of the delayed neutron in the next generation. Delayed neutron generation times are dominated by the half-life of the delayed neutron precursor. The average delayed neutron generation time is about 12.5 seconds.
& A prompt neutron generation time is the sum of the amount of time it takes a fast neutron to thermalize, the amount of time the neutron exists as a thermal neutron before it is absorbed, and the amount of time between a fissionable nuclide absorbing a neutron and fission neutrons being released. Prompt neutron generation time is about 5 x 10 seconds. -5
& The average neutron generation time can be calculated from the prompt and delayed neutron generation times and the delayed neutron fraction using Equation (2-12).
& Delayed neutrons are responsible for the ability to control the rate at which power can rise in a reactor. If only prompt neutrons existed, reactor control would not be possible due to the rapid power changes.
Reactor Theory (Neutron Characteristics) DOE-HDBK-1019/1-93 NEUTRON FLUX SPECTRUM
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Figure 2 Prompt Fission Neutron Energy Spectrum for
Thermal Fission of Uranium-235
NEUTRON FLUX SPECTRUM
The number of neutrons that exist at a given energy level varies. A plot of either the fraction of neutrons or the neutron flux at a given energy versus the energy level is called a neutron energy spectrum. The neutron energy spectrum varies widely for different types of reactors.
Fixture checkpoint marker-0523-xray is recorded on fixture page 0523.
EO 4.1 STATE the average energy at which prompt neutrons are produced.
EO 4.2 DESCRIBE the neutron energy spectrum in the following reactors:
a. Fast reactor
b. Thermal reactor
EO 4.3 EXPLAIN the reason for the particular shape of the fast, intermediate, and slow energy regions of the neutron flux spectrum for a thermal reactor.
Prompt Neutron Energies
The neutrons produced by fission are high energy neutrons, and almost all fission neutrons have

10 MeV. The neutron energy
distribution, or spectrum, may best
be described by plotting the
fraction of neutrons per MeV as a
function of neutron energy, as
shown in Figure 2. From this
figure it can be seen that the most
probable neutron energy is about
0.7 MeV. Also, from this data it
can be shown that the average
energy of fission neutrons is about
Fixture checkpoint marker-0524-xray is recorded on fixture page 0524.
2 MeV. Figure 2 is the neutron
energy spectrum for thermal
fission in uranium-235. The values
will vary slightly for other
nuclides.
NEUTRON FLUX SPECTRUM DOE-HDBK-1019/1-93 Reactor Theory (Neutron Characteristics)
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Figure 3 Comparison of Neutron Flux Spectra for
Thermal and Fast Breeder Reactor
Thermal and Fast Breeder Reactor Neutron Spectra
The spectrum of neutron energies produced by fission varies significantly from the energy spectrum, or flux, existing in a reactor at a given time. Figure 3 illustrates the difference in neutron flux spectra between a thermal reactor and a fast breeder reactor. The energy distribution of neutrons from fission is essentially the same for both reactors, so the differences in the curve shapes may be attributed to the neutron moderation or slowing down effects.
No attempt is made to thermalize or slow down neutrons in the fast breeder reactor (liquid metal cooled); therefore, an insignificant number of neutrons exist in the thermal range. For the thermal reactor (water moderated), the spectrum of neutrons in the fast region (> 0.1 MeV) has a shape similar to that for the spectrum of neutrons emitted by the fission process.
In the thermal reactor, the flux in the intermediate energy region (1 eV to 0.1 MeV) has approximately a 1/E dependence. That is, if the energy (E) is halved, the flux doubles. This 1/E dependence is caused by the slowing down process, where elastic collisions remove a constant fraction of the neutron energy per collision (on the average), independent of energy; thus, the neutron loses larger amounts of energy per collision at higher energies than at lower energies. The fact that the neutrons lose a constant fraction of energy per collision causes the neutrons to tend to "pile up" at lower energies, that is, a greater number of neutrons exist at the lower energies as a result of this behavior.
vp
2 k T
m
vp
2 k T
m
2 1.38 x 10 16 erg
K
293 K

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2.2 x 105 cm
sec
1 m
100 cm
2200 m
sec
Reactor Theory (Neutron Characteristics) DOE-HDBK-1019/1-93 NEUTRON FLUX SPECTRUM
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In the thermal region the neutrons achieve a thermal equilibrium with the atoms of the moderator material. In any given collision they may gain or lose energy, and over successive collisions will gain as much energy as they lose. These thermal neutrons, even at a specific temperature, do not all have the same energy or velocity; there is a distribution of energies, usually referred to as the Maxwell distribution (e.g., Figure 2). The energies of most thermal neutrons lie close to the most probable energy, but there is a spread of neutrons above and below this value.
Most Probable Neutron Velocities
The most probable velocity (vp) of a thermal neutron is determined by the temperature of the medium and can be determined by Equation (2-13) .
(2-13)
Fixture checkpoint marker-0525-xray is recorded on fixture page 0525.
where:
v = most probable velocity of neutron (cm/sec) p
k = Boltzman's constant (1.38 x 10 erg/ K) -16
T = absolute temperature in degrees Kelvin (K)
m = mass of neutron (1.66 x 10 grams) -24
Example:
Calculate the most probable velocities for neutrons in thermal equilibrium with their surroundings at the following temperatures. a) 20 C, b) 260 C.
Solution:
a) Calculate the most probable velocity for 20 C using Equation (2-13).
vp
2 k T
m
2 1.38 x 10 16 erg
K
533 K
1.66 x 10 24 g
2.977 x 105 cm
sec
1 m
100 cm
2977 m
sec
NEUTRON FLUX SPECTRUM DOE-HDBK-1019/1-93 Reactor Theory (Neutron Characteristics)
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b) Calculate the most probable velocity for 260 C using Equation (2-13).
From these calculations it is evident that the most probable velocity of a thermal neutron increases as temperature increases. The most probable velocity at 20 C is of particular importance since reference data, such as nuclear cross sections, are tabulated for a neutron

and reactor theory. The handbook includes information on atomic and nuclear physics; neutron characteristics; reactor theory and nuclear parameters; and the theory of reactor operation. This information will provide personnel with a foundation for understanding the scientific principles that are associated with various DOE nuclear facility operations and maintenance.

Key Words: Training Material, Atomic Physics, The Chart of the Nuclides, Radioactivity, Radioactive Decay, Neutron Interaction, Fission, Reactor Theory, Neutron Characteristics, Neutron Life Cycle, Reactor Kinetics

FOREWORD

The Department of Energy (DOE) Fundamentals Handbooks consist of ten academic subjects, which include Mathematics; Classical Physics; Thermodynamics, Heat Transfer, and Fluid Flow; Instrumentation and Control; Electrical Science; Material Science; Mechanical Science; Chemistry; Engineering Symbology, Prints, and Drawings; and Nuclear Physics and Reactor Theory. The handbooks are provided as an aid to DOE nuclear facility contractors.

These handbooks were first published as Reactor Operator Fundamentals Manuals in 1985 for use by DOE category A reactors. The subject areas, subject matter content, and level of detail of the Reactor Operator Fundamentals Manuals were determined from several sources. DOE Category A reactor training managers determined which materials should be included, and served as a primary reference in the initial development phase. Training guidelines from the commercial nuclear power industry, results of job and task analyses, and independent input from contractors and operations-oriented personnel were all considered and included to some degree in developing the text material and learning objectives.

The DOE Fundamentals Handbooks represent the needs of various DOE nuclear facilities' Fixture checkpoint marker-0526-xray is recorded on fixture page 0526.
fundamental training requirements. To increase their applicability to nonreactor nuclear facilities, the Reactor Operator Fundamentals Manual learning objectives were distributed to the Nuclear Facility Training Coordination Program Steering Committee for review and comment. To update their reactor-specific content, DOE Category A reactor training managers also reviewed and commented on the content. On the basis of feedback from these sources, information that applied to two or more DOE nuclear facilities was considered generic and was included. The final draft of each of the handbooks was then reviewed by these two groups. This approach has resulted in revised modular handbooks that contain sufficient detail such that each facility may adjust the content to fit their specific needs.

Each handbook contains an abstract, a foreword, an overview, learning objectives, and text material, and is divided into modules so that content and order may be modified by individual DOE contractors to suit their specific training needs. Each handbook is supported by a separate examination bank with an answer key.

The DOE Fundamentals Handbooks have been prepared for the Assistant Secretary for Nuclear Energy, Office of Nuclear Safety Policy and Standards, by the DOE Training Coordination Program. This program is managed by EG&G Idaho, Inc.

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DOE-HDBK-1019/1-93
NUCLEAR PHYSICS AND REACTOR THEORY

OVERVIEW

The Department of Energy Fundamentals Handbook entitled Nuclear Physics and Reactor Theory was prepared as an information resource for personnel who are responsible for the

facility involves the transfer of some type of energy. A basic understanding of nuclear physics and reactor theory is necessary for DOE nuclear facility operators, maintenance personnel, and the technical staff to safely operate and maintain the facility and facility support systems. The information in this handbook is presented to provide a foundation for applying engineering concepts to the job. This knowledge will help personnel understand the impact that their actions may have on the safe and reliable operation of facility components and systems.

The Nuclear Physics and Reactor Theory handbook consists of four modules that are contained in two volumes. The following is a brief description of the information presented in each module of the handbook.

Volume 1 of 2

Module 1 - Atomic and Nuclear Physics

Introduces concepts of atomic physics including the atomic nature of matter, the chart of the nuclides, radioactivity and radioactive decay, neutron interactions and fission, and the interaction of radiation with matter.

Module 2 - Reactor Theory (Nuclear Parameters)

Provides information on reactor theory and neutron characteristics. Includes topics such as neutron sources, neutron flux, neutron cross sections, reaction rates, neutron moderation, and prompt and delayed neutrons.

Volume 2 of 2

Module 3 - Reactor Theory (Nuclear Parameters)

Explains the nuclear parameters associated with reactor theory. Topics include the neutron life cycle, reactivity and reactivity coefficients, neutron poisons, and control rods.

Module 4 - Reactor Theory (Reactor Operations)

Introduces the reactor operations aspect of reactor theory. Topics include subcritical multiplication, reactor kinetics, and reactor operation.

The information contained in this handbook is not all-encompassing. An attempt to present the entire subject of nuclear physics and reactor theory would be impractical. However, Fixture checkpoint marker-0527-xray is recorded on fixture page 0527.
the Nuclear Physics and Reactor Theory handbook presents enough information to provide the reader with the fundamental knowledge necessary to understand the advanced theoretical concepts presented in other subject areas, and to understand basic system and equipment operation.

Department of Energy
Fundamentals Handbook
NUCLEAR PHYSICS
AND REACTOR THEORY
Module 1
Atomic and Nuclear Physics

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TERMINAL OBJECTIVE
1.0 Given sufficient information, DESCRIBE atoms, including components, structure, and

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ENABLING OBJECTIVES
1.1 STATE the characteristics of the following atomic particles, including mass, charge, and location within the atom:
a. Proton
b. Neutron
c. Electron
1.2 DESCRIBE the Bohr model of an atom.
1.3 DEFINE the following terms:
a. Nuclide c. Atomic number
b. Isotope d. Mass number
1.4 Given the standard A
ZX notation for a particular nuclide, DETERMINE the following:
a. Number of protons
b. Number of neutrons
c. Number of electrons
1.5 DESCRIBE the three forces that act on particles within the nucleus and affect the stability of the nucleus.
1.6 DEFINE the following terms:
a. Enriched uranium
b. Depleted uranium
1.7 DEFINE the following terms:
a. Mass defect
b. Binding energy
1.8 Given the atomic mass for a nuclide and the atomic masses of a neutron, proton, and electron, CALCULATE the mass defect and binding energy of the nuclide.

TERMINAL OBJECTIVE
2.0 Given necessary references, DESCRIBE the various modes of radioactive decay.
ENABLING OBJECTIVES
Fixture checkpoint marker-0531-xray is recorded on fixture page 0531.
2.1 DESCRIBE the following processes:
a. Alpha decay d. Electron capture
b. Beta-minus decay e. Internal conversions
c. Beta-plus decay f. Isomeric transitions
2.2 Given a Chart of the Nuclides, WRITE the radioactive decay chain for a nuclide.
2.3 EXPLAIN why one or more gamma rays typically accompany particle emission.
2.4 Given the stability curve on the Chart of the Nuclides, DETERMINE the type of radioactive decay that the nuclides in each region of the chart will typically undergo.
2.5 DEFINE the following terms:
a. Radioactivity d. Radioactive decay constant
b. Curie e. Radioactive half-life
c. Becquerel
2.6 Given the number of atoms and either the half-life or decay constant of a nuclide, CALCULATE the activity.
2.7 Given the initial activity and the decay constant of a nuclide, CALCULATE the activity at any later time.
2.8 CONVERT between the half-life and decay constant for a nuclide.

curve for a nuclide on either linear or semi-log coordinates.
2.10 DEFINE the following terms:
a. Radioactive equilibrium
b. Transient radioactive equilibrium

TERMINAL OBJECTIVE
3.0 Without references, DESCRIBE the different nuclear interactions initiated by neutrons.
ENABLING OBJECTIVES
3.1 DESCRIBE the following scattering interactions between a neutron and a nucleus:
a. Elastic scattering
b. Inelastic scattering
3.2 STATE the conservation laws that apply to an elastic collision between a neutron and a nucleus.
3.3 DESCRIBE the following reactions where a neutron is absorbed in a nucleus:
a. Radiative capture
b. Particle ejection

TERMINAL OBJECTIVE
4.0 Without references, DESCRIBE the fission process.
ENABLING OBJECTIVES
4.1 EXPLAIN the fission process using the liquid drop model of a nucleus.
4.2 DEFINE the following terms:
a. Excitation energy
b. Critical energy
4.3 DEFINE the following terms:
a. Fissile material
b. Fissionable material
c. Fertile material
4.4 DESCRIBE the processes of transmutation, conversion, and breeding.
4.5 DESCRIBE the curve of Binding Energy per Nucleon versus mass number and give a qualitative description of the reasons for its shape.
4.6 EXPLAIN why only the heaviest nuclei are easily fissioned.
4.7 EXPLAIN why uranium-235 fissions with thermal neutrons and uranium-238 fissions only with fast neutrons.
4.8 CHARACTERIZE the fission products in terms of mass groupings and radioactivity.
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4.9 Given the nuclides involved and their masses, CALCULATE the energy released from fission.
4.10 Given the curve of Binding Energy per Nucleon versus mass number, CALCULATE the energy released from fission.

TERMINAL OBJECTIVE
5.0 Without references, DESCRIBE how the various types of radiation interact with matter.
ENABLING OBJECTIVES
5.1 DESCRIBE interactions of the following with matter:

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b. Beta particle d. Neutron
5.2 DESCRIBE the following ways that gamma radiation interacts with matter:
a. Compton scattering
b. Photoelectric effect
c. Pair production

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ATOMIC NATURE OF MATTER
All matter is composed of atoms. The atom is the smallest amount of matter that retains the properties of an element. Atoms themselves are composed of smaller particles, but these smaller particles no longer have
Fixture checkpoint marker-0533-xray is recorded on fixture page 0533.
the same properties as the overall element.
EO 1.1 STATE the characteristics of the following atomic particles, including mass, charge, and location within the atom:
a. Proton
b. Neutron
c. Electron
EO 1.2 DESCRIBE the Bohr model of an atom.
EO 1.3 DEFINE the following terms:
a. Nuclide c. Atomic number
b. Isotope d. Mass number
EO 1.4 Given the standard A ZX notation for a particular nuclide, DETERMINE the following:
a. Number of protons
b. Number of neutrons
c. Number of electrons
EO 1.5 DESCRIBE the three forces that act on particles within the nucleus and affect the stability of the nucleus.
Structure of Matter
Early Greek philosophers speculated that the earth was made up of different combinations of basic substances, or elements. They considered these basic elements to be earth, air, water, and fire. Modern science shows that the early Greeks held the correct concept that matter consists of a combination of basic elements, but they incorrectly identified the elements.

In 1661 the English chemist Robert Boyle published the modern criterion for an element. He defined an element to be a basic substance that cannot be broken down into any simpler substance after it is isolated from a compound, but can be combined with other elements to form compounds. To date, 105 different elements have been confirmed to exist, and researchers claim to have discovered three additional elements. Of the 105 confirmed elements, 90 exist in nature and 15 are man-made.
Another basic concept of matter that the Greeks debated was whether matter was continuous or discrete. That is, whether matter could be continuously divided and subdivided into ever smaller

450 B.C. argued that substances were ultimately composed of small, indivisible particles that he labeled atoms. He further suggested that different substances were composed of different atoms or combinations of atoms, and that one substance could be converted into another by rearranging the atoms. It was impossible to conclusively prove or disprove this proposal for more than 2000 years.

The modern proof for the atomic nature of matter was first proposed by the English chemist John Dalton in 1803. Dalton stated that each chemical element possesses a particular kind of atom, and any quantity of the element is made up of identical atoms of this kind. What distinguishes one element from another element is the kind of atom of which it consists, and the basic physical difference between kinds of atoms is their weight.

Subatomic Particles

For almost 100 years after Dalton established the atomic nature of atoms, it was considered impossible to divide the atom into even smaller parts. All of the results of chemical experiments during this time indicated that the atom was indivisible. Eventually, experimentation into electricity and radioactivity indicated that particles of matter smaller than the atom did indeed exist. In 1906, J. J. Thompson won the Nobel Prize in physics for establishing the existence of electrons. Electrons are negatively-charged particles that have 1/1835 the mass of the hydrogen atom. Soon after the discovery of electrons, protons were discovered. Protons are relatively large particles that have almost the same mass as a hydrogen atom and a positive charge equal in magnitude (but opposite in sign) to that of the electron. The third subatomic particle to be discovered, the neutron, was not found until 1932. The neutron has almost the same mass as the Fixture checkpoint marker-0534-xray is recorded on fixture page 0534.

proton, but it is electrically neutral.

Bohr Model of the Atom

The British physicist Ernest Rutherford postulated that the positive charge in an atom is concentrated in a small region called a nucleus at the center of the atom with electrons existing in orbits around it. Niels Bohr, coupling Rutherford's postulation with the quantum theory introduced by Max Planck, proposed that the atom consists of a dense nucleus of protons surrounded by electrons traveling in discrete orbits at fixed distances from the nucleus. An electron in one of these orbits or shells has a specific or discrete quantity of energy (quantum). When an electron moves from one allowed orbit to another allowed orbit, the energy difference between the two states is emitted or absorbed in the form of a single quantum of radiant energy called a photon. Figure 1 is Bohr's model of the hydrogen atom showing an electron as having just dropped from the third shell to the first shell with the emission of a photon that has an energy = hv. (h = Planck's constant = 6.63 x 10-34 J-s and v = frequency of the photon.) Bohr's theory was the first to successfully account for the discrete energy levels of this radiation as measured in the laboratory. Although Bohr's atomic model is designed specifically to explain the hydrogen atom, his theories apply generally to the structure of all atoms. Additional information on electron shell theory can be found in the Chemistry Fundamentals Handbook.

Figure 1 Bohr's Model of the Hydrogen Atom

Properties of the three subatomic particles are listed in Table 1.

TABLE 1

Properties of Subatomic Particles

Particle	Location	Charge	Mass

SYNTHETIC FIXTURE PAGE 0535 OF 1200

Proton Nucleus +1 1.007277 amu
Electron Shells around nucleus -1 0.0005486 amu
Measuring Units on the Atomic Scale
The size and mass of atoms are so small that the use of normal measuring units, while possible, is often inconvenient. Units of measure have been defined for mass and energy on the atomic scale to make measurements more convenient to express. The unit of measure for mass is the atomic mass unit (amu). One atomic mass unit is equal to 1.66 x 10-24 grams. The reason for this particular value for the atomic mass unit will be discussed in a later chapter. Note from Table 1 that the mass of a neutron and a proton are both about 1 amu. The unit for energy is the electron volt (eV). The electron volt is the amount of energy acquired by a single electron when it falls through a potential difference of one volt. One electron volt is equivalent to 1.602 x 10-19 joules or 1.18 x 10-19 foot-pounds.
Nuclides
The total number of protons in the nucleus of an atom is called the atomic number of the atom and is given the symbol Z. The number of electrons in an electrically-neutral atom is the same as the number of protons in the nucleus. The number of neutrons in a nucleus is known as the neutron number and is given the symbol N. The mass number of the nucleus is the total number of nucleons, that is, protons and neutrons in the nucleus. The mass number is given the symbol A and can be found by the equation Z + N = A.
Each of the chemical elements has a unique atomic number because the atoms of different elements contain a different number of protons. The atomic number of an atom identifies the particular element.
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Atomic and Nuclear Physics DOE-HDBK-1019/1-93 ATOMIC NATURE OF MATTER
Each type of atom that contains a unique combination of
Figure 2 Nomenclature for
Identifying Nuclides
protons and neutrons is called a nuclide. Not all
Fixture checkpoint marker-0535-xray is recorded on fixture page 0535.
combinations of numbers of protons and neutrons are possible, but about 2500 specific nuclides with unique combinations of neutrons and protons have been identified. Each nuclide is denoted by the chemical symbol of the element with the atomic number written as a subscript and the mass number written as a superscript, as shown in Figure 2. Because each element has a unique name, chemical symbol, and atomic number, only one of the three is necessary to identify the element. For this reason nuclides can also be identified by either the chemical name or the chemical symbol followed by the mass number (for example, U-235 or uranium-235).
Another common format is to use the abbreviation of the chemical element with the mass number superscripted (for example, 235U). In this handbook the format used in the text will usually be the element's name followed by the mass number. In equations and tables, the format in Figure 2 will usually be used.
Example:
State the name of the element and the number of protons, electrons, and neutrons in the nuclides listed below.

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1H
10
5B
14
7N
11
4
4
8Cd
23
9
9
4Pu
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ATOMIC NATURE OF MATTER DOE-HDBK-1019/1-93 Atomic and Nuclear Physics
Solution:
The name of the element can be found from the Periodic Table (refer to Chemistry Fundamentals Handbook) or the Chart of the Nuclides (to be discussed later). The number of protons and electrons are equal to Z. The number of neutrons is equal to Z - A.
Nuclide Element Protons Electrons Neutrons
1
1H hydrogen 1 1 0
10
5B boron 5 5 5
14
7N nitrogen 7 7 7
11
4
4
8Cd cadmium 48 48 66
23
9
9
4Pu plutonium 94 94 145
Fixture checkpoint marker-0536-xray is recorded on fixture page 0536.
Isotopes
Isotopes are nuclides that have the same atomic number and are therefore the same element, but differ in the number of neutrons. Most elements have a few stable isotopes and several unstable, radioactive isotopes. For example, oxygen has three stable isotopes that can be found in nature (oxygen-16, oxygen-17, and oxygen-18) and eight radioactive isotopes. Another example is hydrogen, which has two stable isotopes (hydrogen-1 and hydrogen-2) and a single radioactive isotope (hydrogen-3).
The isotopes of hydrogen are unique in that they are each commonly referred to by a unique name instead of the common chemical element name. Hydrogen-1 is almost always referred to as hydrogen, but the term protium is infrequently used also. Hydrogen-2 is commonly called deuterium and symbolized 2
1D. Hydrogen-3 is commonly called tritium and symbolized 31T. This

SYNTHETIC FIXTURE PAGE 0537 OF 1200

1H and 31H for deuterium and tritium, respectively.
Atomic and Nuclear Radii
The size of an atom is difficult to define exactly due to the fact that the electron cloud, formed by the electrons moving in their various orbitals, does not have a distinct outer edge. A reasonable measure of atomic size is given by the average distance of the outermost electron from the nucleus. Except for a few of the lightest atoms, the average atomic radii are approximately the same for all atoms, about 2 x 10 -8 cm.
Like the atom the nucleus does not have a sharp outer boundary. Experiments have shown that the nucleus is shaped like a sphere with a radius that depends on the atomic mass number of the atom. The relationship between the atomic mass number and the radius of the nucleus is shown in the following equation.
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Fixture checkpoint marker-0537-xray is recorded on fixture page 0537.
Atomic and Nuclear Physics DOE-HDBK-1019/1-93 ATOMIC NATURE OF MATTER
r = (1.25 x 10 -13 cm) A1/3
where:
r = radius of the nucleus (cm)
A = atomic mass number (dimensionless)
The values of the nuclear radii for some light, intermediate, and heavy nuclides are shown in Table 2.
TABLE 2
Calculated Values for Nuclear
Radii
Nuclide Radius of Nucleus
1
1H 1.25 x 10 -13 cm
10
5B 2.69 x 10 -13 cm
5
2
6
6Fe 4.78 x 10 -13 cm
17
7
8
2Hf 7.01 x 10 -13 cm
23
9
8
2U 7.74 x 10 -13 cm
25
9
2
8Cf 7.89 x 10 -13 cm
From the table, it is clear that the radius of a typical atom (e.g. 2 x 10 -8 cm) is more than 25,000 times larger than the radius of the largest nucleus.
Nuclear Forces
In the Bohr model of the atom, the nucleus consists of positively-charged protons and electrically■neutra

attractive force to overcome the repulsive force between protons.
Two forces present in the nucleus are (1) electrostatic forces between charged particles and (2) gravitational forces between any two objects that have mass. It is possible to calculate the magnitude of the gravitational force and electrostatic force based upon principles from classical physics.
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Fg
G m1 m2
r
2
Fe
K Q1 Q2
r
2
ATOMIC NATURE OF MATTER DOE-HDBK-1019/1-93 Atomic and Nuclear Physics
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Newton stated that the gravitational force between two bodies is directly proportional to the masses of the two bodies and inversely proportional to the square of the distance between the bodies. This relationship is shown in the equation below.
Fixture checkpoint marker-0538-xray is recorded on fixture page 0538.
where:
F = gravitational force (newtons) g
m = mass of first body (kilograms) 1
m = mass of second body (kilograms) 2
G = gravitational constant (6.67 x 10 N-m /kg)
-11 2 2
r = distance between particles (meters)
The equation illustrates that the larger the masses of the objects or the smaller the distance between the objects, the greater the gravitational force. So even though the masses of nucleons are very small, the fact that the distance between nucleons is extremely short may make the gravitational force significant. It is necessary to calculate the value for the gravitational force and compare it to the value for other forces to determine the significance of the gravitational force in the nucleus. The gravitational force between two protons that are separated by a distance of 10 meters is about 10 newtons. -20 -24
Coulomb's Law can be used to calculate the force between two protons. The electrostatic force is directly proportional to the electrical charges of the two particles and inversely proportional to the square of the distance between the particles. Coulomb's Law is stated as the following equation.
where:
F = electrostatic force (newtons) e
K = electrostatic constant (9.0 x 10 N-m /C)
9 2 2
Q = charge of first particle (coulombs) 1
Q = charge of second particle (coulombs) 2
r = distance between particles (meters)
Using this equation, the electrostatic force between two protons that are separated by a distance of 10 meters is about 10 newtons. Comparing this result with the calculation of the
-20 12

-24
neglected.
Atomic and Nuclear Physics DOE-HDBK-1019/1-93 ATOMIC NATURE OF MATTER
If only the electrostatic and gravitational forces existed in the nucleus, then it would be impossible to have stable nuclei composed of protons and neutrons. The gravitational forces are much too small to hold the nucleons together compared to the electrostatic forces repelling the protons. Since stable atoms of neutrons and protons do exist, there must be another attractive force acting within the nucleus. This force is called the nuclear force.
The nuclear force is a strong attractive force that is independent of charge. It acts equally only between pairs of neutrons, pairs of protons, or a neutron and a proton. The nuclear force has a very short range; it acts only over distances approximately equal to the diameter of the nucleus (10 -13 cm). The attractive nuclear force between all nucleons drops off with distance much faster than the repulsive electrostatic force between protons.
TABLE 3
Forces Acting in the Nucleus
Force Interaction Range
Gravitational Very weak attractive force
between all nucleons Relatively long
Electrostatic Strong repulsive force between
like charged particles (protons) Relatively long
Nuclear Force Strong attractive force between
all nucleons Extremely short
In stable atoms, the attractive and repulsive forces in the nucleus balance. If the forces do not balance, the atom cannot be stable, and the nucleus will emit radiation in an attempt to achieve a more stable configuration.
Summary
Fixture checkpoint marker-0539-xray is recorded on fixture page 0539.
The important information in this chapter is summarized on the following page.
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ATOMIC NATURE OF MATTER DOE-HDBK-1019/1-93 Atomic and Nuclear Physics
Atomic Nature of Matter Summary
Atoms consist of three basic subatomic particles. These particles are the proton, the neutron, and the electron.
Protons are particles that have a positive charge, have about the same mass as a hydrogen atom, and exist in the nucleus of an atom.
Neutrons are particles that have no electrical charge, have about the same mass as a hydrogen atom, and exist in the nucleus of an atom.
Electrons are particles that have a negative charge, have a mass about eighteen hundred times smaller than the mass of a hydrogen atom, and exist in orbital shells around the nucleus of an atom.
The Bohr model of the atom consists of a dense nucleus of protons and neutrons (nucleons) surrounded by electrons traveling in discrete orbits at fixed distances from the nucleus.
Nuclides are atoms that contain a particular number of protons and neutrons.
Isotopes are nuclides that have the same atomic number and are therefore the same element, but differ in the number of neutrons.
The atomic number of an atom is the number of protons in the nucleus.
The mass number of an atom is the total number of nucleons (protons and neutrons) in

The notation A
ZX is used to identify a specific nuclide. "Z" represents the atomic number, which is equal to the number of protons. "A" represents the mass number, which is equal to the number of nucleons. "X" represents the chemical symbol of the element.
Number of protons = Z
Number of electrons = Z
Number of neutrons = A - Z
The stability of a nucleus is determined by the different forces interacting within it. The electrostatic force is a relatively long-range, strong, repulsive force that acts between the positively charged protons. The nuclear force is a relatively short-range attractive force between all nucleons. The gravitational force the long range, relatively weak attraction between masses, is negligible compared to the other forces.

CHART OF THE NUCLIDES
The Chart of the Nuclides, like the Periodic Table, is a convenient format for presenting a large amount of scientific information in an organized manner.
EO 1.6 DEFINE the following terms:
a. Enriched uranium
b. Depleted uranium
Chart of the Nuclides
A tabulated chart called the Chart of the Nuclides lists the stable and unstable nuclides in addition to pertinent information about each one. Figure 3 shows a small portion of a typical chart. This chart plots a box for each individual nuclide, with the number of protons (Z) on the vertical axis and the number of neutrons (N = A - Z) on the horizontal axis.
The completely gray squares indicate stable isotopes. Those in white squares are artificially radioactive, meaning that they are produced by artificial techniques and do not occur naturally. By consulting a complete chart, other types of isotopes can be found, such as naturally occurring radioactive types (but none are found in the region of the chart that is illustrated in Figure 3).
Located in the box on the far left of each horizontal row is general information about the
Fixture checkpoint marker-0540-xray is recorded on fixture page 0540.
element. The box contains the chemical symbol of the element in addition to the average atomic weight of the naturally occurring substance and the average thermal neutron absorption cross section, which will be discussed in a later module. The known isotopes (elements with the same atomic number Z but different mass number A) of each element are listed to the right.

Figure 3 Nuclide Chart for Atomic Numbers 1 to 6

Information for Stable Nuclides
For the stable isotopes, in addition to the symbol and the atomic mass number, the number percentage of each isotope in the naturally occurring element is listed, as well as the thermal neutron activation cross section and the mass in atomic mass units (amu). A typical block for a stable nuclide from the Chart of the Nuclides is shown in Figure 4.

Information for Unstable Nuclides
For unstable isotopes the additional information includes the half life, the mode of decay (for example, β-
, α), the total disintegration energy in MeV (million electron volts), and the mass in amu when available. A typical block for an unstable nuclide from the Chart of the Nuclides is shown in Figure 5.
Figure 5 Unstable Nuclides

Neutron - Proton Ratios
Fixture checkpoint marker-0541-xray is recorded on fixture page 0541.
Figure 6 shows the distribution of the stable nuclides plotted on the same axes as the Chart of the Nuclides. As the mass numbers become higher, the ratio of neutrons to protons in the nucleus becomes larger. For helium-4 (2 protons and 2 neutrons) and oxygen-16 (8 protons and 8 neutrons) this ratio is unity. For indium-115 (49 protons and 66 neutrons) the ratio of neutrons to protons has increased to 1.35, and for uranium-238 (92 protons and 146 neutrons) the neutron■to-proton
Figure 6 Neutron - Proton Plot of the Stable Nuclides
If a heavy nucleus were to split into two fragments, each fragment would form a nucleus that would have approximately the same neutron-to-proton ratio as the heavy nucleus. This high neutron-to-proton ratio places the fragments below and to the right of the stability curve displayed by Figure 6. The instability caused by this excess of neutrons is generally rectified by successive beta emissions, each of which converts a neutron to a proton and moves the nucleus toward a more stable neutron-to-proton ratio.

■

Natural Abundance of Isotopes
The relative abundance of an isotope in nature compared to other isotopes of the same element is relatively constant. The Chart of the Nuclides presents the relative abundance of the naturally occurring isotopes of an element in units of atom percent. Atom percent is the percentage of the atoms of an element that are of a particular isotope. Atom percent is abbreviated as a/o.
For example, if a cup of water contains 8.23 x 10 atoms of oxygen, and the isotopic abundance 24 of oxygen-18 is 0.20%, then there are 1.65 x 10 atoms of oxygen-18 in the cup. 22
The atomic weight for an element is defined as the average atomic weight of the isotopes of the element. The atomic weight for an element can be calculated by summing the products of the isotopic abundance of the isotope with the atomic mass of the isotope.
Example:
Calculate the atomic weight for the element lithium. Lithium-6 has an atom percent abundance of 7.5% and an atomic mass of 6.015122 amu. Lithium-7 has an atomic abundance of 92.5% and an atomic mass of 7.016003 amu.
Solution:
The other common measurement of isotopic abundance is weight percent (w/o). Weight percent is the percent weight of an element that is a particular isotope. For example, if a sample of material contained 100 kg of uranium that was 28 w/o uranium-235, then 28 kg of uranium-235 was present in the sample.
Enriched and Depleted Uranium
Natural uranium mined from the earth contains the isotopes uranium-238, uranium-235 and

Most of the remaining atoms (0.72%) are uranium-235, and a slight trace (0.0055%) are uranium-234. Although all isotopes of uranium have similar chemical properties, each of the isotopes has significantly different nuclear properties. For reasons that will be discussed in later modules, the isotope uranium-235 is usually the desired material for use in reactors.
A vast amount of equipment and energy are expended in processes that separate the isotopes of uranium (and other elements). The details of these processes are beyond the scope of this module. These processes are called enrichment processes because they selectively increase the proportion of a particular isotope. The enrichment process typically starts with feed material that has the proportion of isotopes that occur naturally. The process results in two types of
CHART OF THE NUCLIDES DOE-HDBK-1019/1-93 Atomic and Nuclear Physics
In the case of uranium, the natural uranium ore is 0.72 a/o uranium-235. The desired outcome of the enrichment process is to produce enriched uranium. Enriched uranium is defined as uranium in which the isotope uranium-235 has a concentration greater than its natural value. The enrichment process will also result in the byproduct of depleted uranium. Depleted uranium is defined as uranium in which the isotope uranium-235 has a concentration less than its natural value. Although depleted uranium is referred to as a by-product of the enrichment process, it does have uses in the nuclear field and in commercial and defense industries.
Fixture checkpoint marker-0542-xray is recorded on fixture page 0542.
Summary
The important information in this chapter is summarized below.
Chart of the Nuclides Summary
Enriched uranium is uranium in which the isotope uranium-235 has a concentration greater than its natural value of 0.7%.
Depleted uranium is uranium in which the isotope uranium-235 has a concentration less than its natural value of 0.7%.
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Atomic and Nuclear Physics DOE-HDBK-1019/1-93 MASS DEFECT AND BINDING ENERGY
MASS DEFECT AND BINDING ENERGY
The separate laws of Conservation of Mass and Conservation of Energy are not applied strictly on the nuclear level. It is possible to convert between mass and energy. Instead of two separate conservation laws, a single conservation law states that the sum of mass and energy is conserved. Mass does not magically appear and disappear at random. A decrease in mass will be accompanied by a corresponding increase in energy and vice versa.
EO 1.7 DEFINE the following terms:
a. Mass defect
b. Binding energy
EO 1.8 Given the atomic mass for a nuclide and the atomic masses of a neutron, proton, and electron, CALCULATE the mass defect and binding energy of the nuclide.
Mass Defect
Careful measurements have shown that the mass of a particular atom is always slightly less than the sum of the masses of the individual neutrons, protons, and electrons of which the atom consists. The difference between the mass of the atom and the sum of the masses of its parts is called the mass defect (Δm). The mass defect can be calculated using Equation (1-1). In calculating the mass defect it is important to use the full accuracy of mass measurements because the difference in mass is small compared to the mass of the atom. Rounding off the masses of atoms and particles to three or four significant digits prior to the calculation will result in a

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∆m = [Z(mp + me) + (A-Z)mn] - matom (1-1)
where:
∆m = mass defect (amu)
mp = mass of a proton (1.007277 amu)
mn = mass of a neutron (1.008665 amu)
me = mass of an electron (0.000548597 amu)
matom = mass of nuclide A
ZX (amu)
Z = atomic number (number of protons)
A = mass number (number of nucleons)
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MASS DEFECT AND BINDING ENERGY DOE-HDBK-1019/1-93 Atomic and Nuclear Physics
Example:
Calculate the mass defect for lithium-7. The mass of lithium-7 is 7.016003 amu.
Solution:
∆ m Z m
p
me A Z mn matom
∆ m 3 1.007826 amu 7 3 1.008665 amu 7.016003 amu
∆ m 0.0421335 amu
Binding Energy
The loss in mass, or mass defect, is due to the conversion of mass to binding energy when the nucleus is formed. Binding energy is defined as the amount of energy that must be supplied to a nucleus to completely separate its nuclear particles (nucleons). It can also be understood as Fixture checkpoint marker-0543-xray is recorded on fixture page 0543.
the amount of energy that would be released if the nucleus was formed from the separate particles. Binding energy is the energy equivalent of the mass defect. Since the mass defect was converted to binding energy (BE) when the nucleus was formed, it is possible to calculate the binding energy using a conversion factor derived by the mass-energy relationship from Einstein's Theory of Relativity.
Einstein's famous equation relating mass and energy is E = mc2 where c is the velocity of light (c = 2.998 x 108 m/sec). The energy equivalent of 1 amu can be determined by inserting this quantity of mass into Einstein's equation and applying conversion factors.
E m c 2
1 amu
1.6606 x 10 27 kg
1 amu
2.998 x 108 m
sec
2
1 N
1
kg m
sec2
1 J
1 N m
1.4924 x 10 10 J 1 MeV
1.6022 x 10 13 J

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Conversion Factors:
1 amu = 1.6606 x 10 -27 kg
1 newton = 1 kg-m/sec2
1 joule = 1 newton-meter
1 MeV = 1.6022 x 10-13 joules
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Atomic and Nuclear Physics DOE-HDBK-1019/1-93 MASS DEFECT AND BINDING ENERGY
Since 1 amu is equivalent to 931.5 MeV of energy, the binding energy can be calculated using Equation (1-2).
(1-2)
B.E. ∆m
931.5 MeV
1 amu
Example:
Calculate the mass defect and binding energy for uranium-235. One uranium-235 atom has a mass of 235.043924 amu.
Solution:
Step 1: Calculate the mass defect using Equation (1-1).
∆m Z m
p
me A Z mn matom
∆m 92 1.007826 amu 235 92 1.008665 amu 235.043924 amu
∆m 1.91517 amu
Step 2: Use the mass defect and Equation (1-2) to calculate the binding energy.
B.E. ∆m
931.5 MeV
1 amu
1.91517 amu
931.5 MeV
1 amu
1784 MeV
Fixture checkpoint marker-0544-xray is recorded on fixture page 0544.
Energy Levels of Atoms
The electrons that circle the nucleus move in fairly well-defined orbits. Some of these electrons are more tightly bound in the atom than others. For example, only 7.38 eV is required to remove the outermost electron from a lead atom, while 88,000 eV is required to remove the innermost electron. The process of removing an electron from an atom is called ionization, and the energy required to remove the electron is called the ionization energy.
In a neutral atom (number of electrons = Z) it is possible for the electrons to be in a variety of different orbits, each with a different energy level. The state of lowest energy is the one in which the atom is normally found and is called the ground state. When the atom possesses more energy than its ground state energy, it is said to be in an excited state.
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An atom cannot stay in the excited state for an indefinite period of time. An excited atom will eventually transition to either a lower-energy excited state, or directly to its ground state, by emitting a discrete bundle of electromagnetic energy called an x-ray. The energy of the x-ray will be equal to the difference between the energy levels of the atom and will typically range

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Energy Levels of the Nucleus

The nucleons in the nucleus of an atom, like the electrons that circle the nucleus, exist in shells that correspond to energy states. The energy shells of the nucleus are less defined and less understood than those of the electrons. There is a state of lowest energy (the ground state) and discrete possible excited states for a nucleus. Where the discrete energy states for the electrons of an atom are measured in eV or keV, the energy levels of the nucleus are considerably greater and typically measured in MeV.

A nucleus that is in the excited state will not remain at that energy level for an indefinite period. Like the electrons in an excited atom, the nucleons in an excited nucleus will transition towards their lowest energy configuration and in doing so emit a discrete bundle of electromagnetic radiation called a gamma ray (γ-ray). The only differences between x-rays and γ-rays are their energy levels and whether they are emitted from the electron shell or from the nucleus.

The ground state and the excited states of a nucleus can be depicted in a nuclear energy-level diagram. The nuclear energy-level diagram consists of a stack of horizontal bars, one bar for each of the excited states of the nucleus. The vertical distance between the bar representing an excited state and the bar representing the ground state is proportional to the energy level of the excited state with respect to the ground state. This difference in energy between the ground state and the excited state is called the excitation energy of the excited state. The ground state of a nuclide has zero excitation energy. The bars for the excited states are labeled with their respective energy levels. Figure 7 is the energy level diagram for nickel-60.

Figure 7 Energy Level Diagram - Nickel-60

Summary

The important information in this chapter is summarized below.

Mass Defect and Binding Energy Summary

Mass defect is the difference between the mass of the atom and the sum of the masses of its constituent parts.

Fixture checkpoint marker-0545-xray is recorded on fixture page 0545.

Binding energy is the amount of energy that must be supplied to a nucleus to completely separate its nuclear particles. Binding energy is the energy equivalent of the mass defect.

Mass defect can be calculated by using the equation below.

$\Delta m = [Z(mp + me) + (A-Z)mn] - matom$

Binding energy can be calculated by multiplying the mass defect by the factor of 931.5 MeV per amu.

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Most atoms found in nature are stable and do not emit particles or energy that change form over time. Some atoms, however, do not have stable nuclei. These atoms emit radiation in order to achieve a more stable configuration.
EO 2.1 DESCRIBE the following processes:
a. Alpha decay d. Electron capture
b. Beta-minus decay e. Internal conversions
c. Beta-plus decay f. Isomeric transitions
EO 2.2 Given a Chart of the Nuclides, WRITE the radioactive decay chain for a nuclide.
EO 2.3 EXPLAIN why one or more gamma rays typically accompany particle emission.
EO 2.4 Given the stability curve on the Chart of the Nuclides, DETERMINE the type of radioactive decay that the nuclides in each region of the chart will typically undergo.
Stability of Nuclei
Fixture checkpoint marker-0546-xray is recorded on fixture page 0546.
As mass numbers become larger, the ratio of neutrons to protons in the nucleus becomes larger for the stable nuclei. Non-stable nuclei may have an excess or deficiency of neutrons and undergo a transformation process known as beta (β) decay. Non-stable nuclei can also undergo a variety of other processes such as alpha (α) or neutron (n) decay. As a result of these decay processes, the final nucleus is in a more stable or more tightly bound configuration.
Natural Radioactivity
In 1896, the French physicist Becquerel discovered that crystals of a uranium salt emitted rays that were similar to x-rays in that they were highly penetrating, could affect a photographic plate, and induced electrical conductivity in gases. Becquerel's discovery was followed in 1898 by the identification of two other radioactive elements, polonium and radium, by Pierre and Marie Curie.
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Atomic and Nuclear Physics DOE-HDBK-1019/1-93 MODES OF RADIOACTIVE DECAY
Heavy elements, such as uranium or thorium, and their unstable decay chain elements emit radiation in their naturally occurring state. Uranium and thorium, present since their creation at the beginning of geological time, have an extremely slow rate of decay. All naturally occurring nuclides with atomic numbers greater than 82 are radioactive.
Nuclear Decay
Whenever a nucleus can attain a more stable (i.e., more tightly bound) configuration by emitting radiation, a spontaneous disintegration process known as radioactive decay or nuclear decay may occur. In practice, this "radiation" may be electromagnetic radiation, particles, or both.
Detailed studies of radioactive decay and nuclear reaction processes have led to the formulation of useful conservation principles. The four principles of most interest in this module are discussed below.
Conservation of electric charge implies that charges are neither created nor destroyed. Single positive and negative charges may, however, neutralize each other. It is also possible for a neutral particle to produce one charge of each sign.
Conservation of mass number does not allow a net change in the number of nucleons. However, the conversion of a proton to a neutron and vice versa is allowed.
Conservation of mass and energy implies that the total of the kinetic energy and the energy equivalent of the mass in a system must be conserved in all decays and reactions. Mass can be converted to energy and energy can be converted to mass, but the sum of mass and energy must

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Conservation of momentum is responsible for the distribution of the available kinetic energy among product nuclei, particles, and/or radiation. The total amount is the same before and after the reaction even though it may be distributed differently among entirely different nuclides and/or particles.
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■ ■ ■ ■ ■ ■
■ ■ ■ ■ ■
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Alpha Decay (■)
Alpha decay is the emission of alpha particles (helium nuclei) which may be represented as either He or ■. When an unstable nucleus ejects an alpha particle, the atomic number is reduced by 2 4 4 2 2
and the mass number decreased by 4. An example is uranium-234 which decays by the ejection of an alpha particle accompanied by the emission of a 0.068 MeV gamma.
The combined kinetic energy of the daughter nucleus (Thorium-230) and the ■ particle is designated as KE. The sum of the KE and the gamma energy is equal to the difference in mass between the original nucleus (Uranium-234) and the final particles (equivalent to the binding energy released, since ■m = BE). The alpha particle will carry off as much as 98% of the kinetic energy and, in most cases, can be considered to carry off all the kinetic energy.
Beta Decay (■)
Beta decay is the emission of electrons of nuclear rather than orbital origin. These particles are
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electrons that have been expelled by excited nuclei and may have a charge of either sign.
If both energy and momentum are to be conserved, a third type of particle, the neutrino, ■, must be involved. The neutrino is associated with positive electron emission, and its antiparticle, the antineutrino, , is emitted with a negative electron. These uncharged particles have only the weakest interaction with matter, no mass, and travel at the speed of light. For all practical purposes, they pass through all materials with so few interactions that the energy they possess cannot be recovered. The neutrinos and antineutrinos are included here only because they carry a portion of the kinetic energy that would otherwise belong to the beta particle, and therefore, must be considered for energy and momentum to be conserved. They are normally ignored since they are not significant in the context of nuclear reactor applications.
Negative electron emission, represented as e, -1 -1■, or simply as e or ■ , effectively converts a neutro to a proton, thus increasing the atomic number by one and leaving the mass number unchanged. This is a common mode of decay for nuclei with an excess of neutrons, such as fission fragments below and to the right of the neutron-proton stability curve (refer to Figure 6). An example of a typical beta minus-decay reaction is shown below.
■ ■
■
■ ■
■ ■ ■ ■
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Figure 9 Orbital Electron Capture
Positively charged electrons (beta-plus) are known as positrons. Except for sign, they are nearly identical to their negatively charged cousins. When a positron, represented as e, ■, or simply as +1 +1 0 0

remains unchanged. A proton has been converted to a neutron. An example of a typical positron (beta-plus) decay is shown below.

Electron Capture (EC, K-capture)

Nuclei having an excess of protons may capture an electron from one of the inner orbits which immediately combines with a proton in the nucleus to form a neutron. This process is called electron capture (EC). The electron is normally captured from the innermost orbit (the K-shell), and, consequently, this process is sometimes called K-capture. The following example depicts electron capture.

A neutrino is formed at the same time that the neutron is formed, and energy carried off by it serves to conserve momentum. Any energy that is available due to the atomic mass of the product being appreciably less than that of the parent will appear as gamma radiation. Also, there will always be characteristic x-rays given off when an electron from one of the higher energy shells moves in to fill the vacancy in the K-shell.

Electron capture is shown graphically in Figure 8.

Electron capture and positron emission result in the production of the same daughter product, and they exist as competing processes.

For positron emission to occur, however, the mass of the daughter product must be less than the mass of the parent by an amount equal to at least twice the mass of an electron. This mass difference

Fixture checkpoint marker-0548-xray is recorded on fixture page 0548.

between the parent and daughter is necessary to account for two items present in the parent but not in the daughter. One item is the positron ejected from the nucleus of the parent. The other item is that the daughter product has one less orbital electron than the parent. If this requirement is not met, then orbital electron capture takes place exclusively.

■

■ ■ ■ ■

■ ■

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Gamma Emission (■)

Gamma radiation is a high-energy electromagnetic radiation that originates in the nucleus. It is emitted in the form of photons, discrete bundles of energy that have both wave and particle properties. Often a daughter nuclide is left in an excited state after a radioactive parent nucleus undergoes a transformation by alpha decay, beta decay, or electron capture. The nucleus will drop to the ground state by the emission of gamma radiation.

Internal Conversion

The usual method for an excited nucleus to go from the excited state to the ground state is by emission of gamma radiation. However, in some cases the gamma ray (photon) emerges from the

photon is transferred to the electron. The gamma ray is then said to have undergone internal conversion. The conversion electron is ejected from the atom with kinetic energy equal to the gamma energy minus the binding energy of the orbital electron. An orbital electron then drops to a lower energy state to fill the vacancy, and this is accompanied by the emission of characteristic x-rays.

Isomers and Isomeric Transition

Isomeric transition commonly occurs immediately after particle emission; however, the nucleus may remain in an excited state for a measurable period of time before dropping to the ground state at its own characteristic rate. A nucleus that remains in such an excited state is known as a nuclear isomer because it differs in energy and behavior from other nuclei with the same atomic number and mass number. The decay of an excited nuclear isomer to a lower energy level is called an isomeric transition. It is also possible for the excited isomer to decay by some alternate means, for example, by beta emission.

An example of gamma emission accompanying particle emission is illustrated by the decay of nitrogen-16 below.

■
■
■
■
■
■
■
■
■
■
■

Decay Chains

When an unstable nucleus decays, the resulting daughter nucleus is not necessarily stable. The nucleus resulting from the decay of a parent is often itself unstable, and will undergo an additional decay. This is especially common among the larger nuclides.

It is possible to trace the steps of an unstable atom as it goes through multiple decays trying to achieve stability. The list of the original unstable nuclide, the nuclides that are involved as intermediate steps in the decay, and the final stable nuclide is known as the decay chain. One

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common method for stating the decay chain is to state each of the nuclides involved in the standard $^{A}_{Z}X$ format. Arrows are used between nuclides to indicate where decays occur, with the type of decay indicated above the arrow and the half-life below the arrow. The half-life for decay will be discussed in the next chapter.

Example:

Write the decay chains for rubidium-91 and actinium-215. Continue the chains until a stable nuclide or a nuclide with a half-life greater than 1×10^{6} years is reached.

Solution:

Predicting Type of Decay

Radioactive nuclides tend to decay in a way that results in a daughter nuclide that lies closer to the

based on its location relative to the line of stability on the Chart of the Nuclides.
Figure 9 illustrates the type of decay nuclides in different regions of the chart will typically undergo. Nuclides that are below and to the right of the line of stability will usually undergo ■ decay. - Nuclides that are above and to the left of the line of stability will usually undergo either ■ decay or + electron capture. Most nuclides that will undergo ■ decay are found in the upper right hand region of the chart. These are general rules that have many exceptions, especially in the region of the heavy nuclides.
MODES OF RADIOACTIVE DECAY DOE-HDBK-1019/1-93 Atomic and Nuclear Physics
Figure 9 Types of Radioactive Decay Relative to the Line of Stability
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Atomic and Nuclear Physics DOE-HDBK-1019/1-93 MODES OF RADIOACTIVE DECAY
Summary
The important information in this chapter is summarized below.
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Modes of Radioactive Decay Summary
Alpha decay is the emission of an alpha particle (2 protons and 2 neutrons) from an unstable nucleus. The daughter nuclide has an atomic number 2 less than the parent nuclide and a mass number 4 less than the parent nuclide. The daughter nucleus commonly releases its excitation energy by gamma emission.
Beta-minus decay effectively converts a neutron to a proton and an electron, which is immediately ejected from the nucleus. The daughter nuclide has its atomic number increased by 1 and the same mass number compared to the parent.
Beta-plus decay effectively converts a proton to a neutron and a positron, which is immediately ejected from the nucleus. The daughter nuclide has its atomic number decreased by 1 and the same mass number compared to the parent.
In electron capture, the nucleus absorbs an electron from the innermost orbit. This electron combines with a proton to form a neutron.
Internal conversion occurs when a gamma ray, emitted by the nucleus as it goes from the excited state to the ground state, interacts with one of the innermost electrons of the same atom. The electron is ejected from the atom.
An isomeric transition is the decay of an excited nucleus to a lower-energy level by the emission of a gamma ray.
Decay chains can be found by tracing the steps an unstable atom goes through as it tries to achieve stability.
Many modes of radioactive decay result in a daughter nuclide that has an energy level above the ground state. This excitation energy is usually released immediately in the form of a gamma ray.
The type of decay that a nuclide will typically undergo can be determined by its relationship to the line of stability on the Chart of the Nuclides. Nuclides that lie below and to the right of the line of stability will typically beta minus decay. Nuclides above and to the left of the line will typically either beta plus decay or electron capture. Most alpha emitters are found in the upper, right-hand corner of the chart.
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RADIOACTIVITY DOE-HDBK-1019/1-93 Atomic and Nuclear Physics
RADIOACTIVITY
The rate at which a sample of radioactive material decays is not constant. As

remaining. Since the rate of decay is directly proportional to the number of atoms, the rate of decay will decrease as the number of atoms decreases.
EO 2.5 DEFINE the following terms:
a. Radioactivity d. Radioactive decay constant
b. Curie e. Radioactive half-life
c. Becquerel
EO 2.6 Given the number of atoms and either the half-life or decay constant of a nuclide, CALCULATE the activity.
EO 2.7 Given the initial activity and the decay constant of a nuclide, CALCULATE the activity at any later time.
EO 2.8 CONVERT between the half-life and decay constant for a nuclide.
EO 2.9 Given the Chart of the Nuclides and the original activity, PLOT the radioactive decay curve for a nuclide on either linear or semi-log coordinates.
EO 2.10 DEFINE the following terms:
a. Radioactive equilibrium
b. Transient radioactive equilibrium
Radioactive Decay Rates
Radioactivity is the property of certain nuclides of spontaneously emitting particles or gamma
Fixture checkpoint marker-0551-xray is recorded on fixture page 0551.
radiation. The decay of radioactive nuclides occurs in a random manner, and the precise time at which a single nucleus will decay cannot be determined. However, the average behavior of a very large sample can be predicted accurately by using statistical methods. These studies have revealed that there is a certain probability that in a given time interval a certain fraction of the nuclei within a sample of a particular nuclide will decay. This probability per unit time that an atom of a nuclide will decay is known as the radioactive decay constant, λ. The units for the decay constant are inverse time such as 1/second, 1/minute, 1/hour, or 1/year. These decay constant units can also be expressed as second-1, minute-1, hour-1, and year-1.

■■

The activity (A) of a sample is the rate of decay of that sample. This rate of decay is usually measured in the number of disintegrations that occur per second. For a sample containing millions of atoms, the activity is the product of the decay constant and the number of atoms present in the sample.
The relationship between the activity, number of atoms, and decay constant is shown in Equation (1-3).
A = ■ N (1-3)
where:
A = Activity of the nuclide (disintegrations/second)
■ = decay constant of the nuclide (second)-1
N = Number of atoms of the nuclide in the sample
Since ■ is a constant, the activity and the number of atoms are always proportional.
Units of Measurement for Radioactivity
Two common units to measure the activity of a substance are the curie (Ci) and becquerel (Bq).
A curie is a unit of measure of the rate of radioactive decay equal to 3.7 x 10 disintegrations 10

radium-226 will undergo in one second. A becquerel is a more fundamental unit of measure of radioactive decay that is equal to 1 disintegration per second. Currently, the curie is more widely used in the United States, but usage of the becquerel can be expected to broaden as the metric system slowly comes into wider use. The conversion between curies and becquerels is shown below.

1 curie = 3.7 x 10 becquerels 10

Variation of Radioactivity Over Time

The rate at which a given radionuclide sample decays is stated in Equation (1-3) as being equal to the product of the number of atoms and the decay constant. From this basic relationship it is possible to use calculus to derive an expression which can be used to calculate how the number of atoms present will change over time. The derivation is beyond the scope of this text, but Equation (1-4) is the useful result.

(1-4)

where:

N = number of atoms present at time t

N = number of atoms initially present o

■ = decay constant (time)-1

t = time

A Ao

e

t

A Aoe

t

A

Ao

e

t

Fixture checkpoint marker-0552-xray is recorded on fixture page 0552.

ln A

Ao

t

t

ln

A

Ao

t

1/2

ln

1

2

t

1/2

ln 2 0.693

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Since the activity and the number of atoms are always proportional, they may be used interchangeably to describe any given radionuclide population. Therefore, the following is true.

(1-5)

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A = activity present at time t
A = activity initially present o
= decay constant (time)
-1
t = time
Radioactive Half-Life
One of the most useful terms for estimating how quickly a nuclide will decay is the radioactive half-life. The radioactive half-life is defined as the amount of time required for the activity to decrease to one-half of its original value. A relationship between the half-life and decay constant can be developed from Equation (1-5). The half-life can be calculated by solving Equation (1-5) for the time, t, when the current activity, A, equals one-half the initial activity A .
o
First, solve Equation (1-5) for t.
If A is equal to one-half of Ao, then A/Ao is equal to one-half. Substituting this in the equation above yields an expression for t .
1/2
(1-6)
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The basic features of decay of a radionuclide sample are shown by the graph in Figure 10.
Figure 10 Radioactive Decay as a Function of Time in Units of Half-Life
Assuming an initial number of atoms No, the population, and consequently, the activity may be noted to decrease by one-half of this value in a time of one half-life. Additional decreases occur so that whenever one half-life elapses, the number of atoms drops to one-half of what its value was at the beginning of that time interval. After five half-lives have elapsed, only 1/32, or 3.1%, of the original number of atoms remains. After seven half-lives, only 1/128, or 0.78%, of the atoms remains. The number of atoms existing after 5 to 7 half-lives can usually be assumed to be negligible. The Chemistry Fundamentals Handbook contains additional information on calculating the number of atoms contained within a sample.
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■
■
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Example:
Fixture checkpoint marker-0553-xray is recorded on fixture page 0553.
A sample of material contains 20 micrograms of californium-252.
Californium-252 has a half-life of 2.638 years.
Calculate:
(a) The number of californium-252 atoms initially present
(b) The activity of the californium-252 in curies
(c) The number of californium-252 atoms that will remain in 12 years
(d) The time it will take for the activity to reach 0.001 curies
Solution:
(a) The number of atoms of californium-252 can be determined as below.
(b) First, use Equation (1-6) to calculate the decay constant.
Use this value for the decay constant in Equation (1-3) to determine the activity.
N No
e

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4.78 x 1016 e
(0.263/yr) (12 yr)
2.04 x 1015
A Aoe
t
A
Ao
e
t
ln A
Ao
Fixture checkpoint marker-0554-xray is recorded on fixture page 0554.
t
t
ln
A
Ao
t
ln
0.001 Ci
0.0108 Ci
0.263 year 1
t 9.05 years
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(c) The number of californium atoms that will remain in 12 years can be calculated from Equation (1-4).
(d) The time that it will take for the activity to reach 0.001 Ci can be determined from Equation (1-5). First, solve Equation (1-5) for time.
Inserting the appropriate values in the right side of this equation will result in the required time.
Plotting Radioactive Decay
It is useful to plot the activity of a nuclide as it changes over time. Plots of this type can be used to determine when the activity will fall below a certain level. This plot is usually done showing activity on either a linear or a logarithmic scale. The decay of the activity of a single nuclide on a logarithmic scale will plot as a straight line because the decay is exponential.
t
1/2
0.693
0.693
t
1/2
0.693
7.13 seconds
0.0972 second 1
A Aoe
t
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Example:
Plot the radioactive decay curve for nitrogen-16 over a period of 100 seconds. The initial activity is 142 curies and the half-life of nitrogen-16 is 7.13 seconds. Plot the curve on both linear rectangular coordinates and on a semi-log scale.
Solution:
First, use Equation (1-6) to calculate the decay constant corresponding to a half-life of 7.13 seconds.
Use the decay constant determined above to calculate the activity at various times using Equation (1-5).
Time Activity
0 seconds 142 Ci
20 seconds 20.3 Ci
40 seconds 2.91 Ci
60 seconds 0.416 Ci
80 seconds 0.0596 Ci
100 seconds 0.00853 Ci
■ ■
■ ■
Fixture checkpoint marker-0555-xray is recorded on fixture page 0555.
■ ■

Figure 11 Linear and Semi-log Plots of Nitrogen-16 Decay
Plotting the data points calculated above on both linear and semilog scales results in the graphs shown in Figure 11.
If a substance contains more than one radioactive nuclide, the total activity is the sum of the individual activities of each nuclide. As an example, consider a sample of material that contained 1 x 10 atoms of iron-59 that has a half-life of 44.51 days (■ = 1.80 x 10 sec), 6 -7 -1 1 x 10 atoms of manganese-54 that has a half-life of 312.2 days (■ = 2.57 x 10 sec), and 1 6 -8 -1 x 10 atoms of cobalt-60 that has a half-life of 1925 days (■ = 4.17 x 10 sec). 6 -9 -1
The initial activity of each of the nuclides would be the product of the number of atoms and the decay constant.

Figure 12 Combined Decay of Iron-56, Manganese-54, and Cobalt-60
Plotting the manner in which the activities of each of the three nuclides decay over time demonstrates that initially the activity of the shortest-lived nuclide (iron-59) dominates the total activity, then manganese-54 dominates. After almost all of the iron and manganese have decayed away, the only contributor to activity will be the cobalt-60. A plot of this combined decay is shown in Figure 12.
Radioactive Equilibrium
Radioactive equilibrium exists when a radioactive nuclide is decaying at the same rate at which it is being produced. Since the production rate and decay rate are equal, the number of atoms present remains constant over time.
An example of radioactive equilibrium is the concentration of sodium-24 in the coolant circulating through a sodium-cooled nuclear reactor. Assume that the sodium-24 is being produced at a rate of 1 x 10 atoms per second. If the sodium-24 were stable and did not decay, 6 the amount of sodium-24 present after some period of time could be calculated by multiplying

in the graph in Figure 13.
However, sodium-24 is not stable, and it decays with a half-life of 14.96 hours. If no sodium-24 is present initially and production starts at a rate of 1 x 10 atoms per second, the rate 6 of decay will initially be zero because there is no sodium-24 present to decay. The rate of decay of sodium-24 will increase as the amount of sodium-24 increases.
■
■
■
■
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Figure 13 Cumulative Production of Sodium-24 Over Time
The amount of sodium-24 present will initially increase rapidly, then it will increase at a continually decreasing rate until the rate of decay is equal to the rate of production. It is possible to calculate how much sodium-24 will be present at equilibrium by setting the production rate (R) equal to the decay rate (■ N).
where:
R = production rate (atoms/second)
■ = decay constant (second)-1
N = number of atoms
It is possible to calculate the equilibrium value for sodium-24 being produced at a rate of 1 x 10 atoms/second. 6
■ ■ ■■
■
■
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■
■
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Figure 14 Approach of Sodium-24 to Equilibrium
The development of the equation to calculate how the amount of sodium-24 changes over time as it approaches the equilibrium value is beyond the scope of this handbook. However, the equation is presented below.
This equation can be used to calculate the values of the amount of sodium-24 present at different times. As the time increases, the exponential term approaches zero, and the number of atoms present will approach R/■. A plot of the approach of sodium-24 to equilibrium is shown in Figure 14.
Transient Radioactive Equilibrium
Transient radioactive equilibrium occurs when the parent nuclide and the daughter nuclide decay at essentially the same rate.
For transient equilibrium to occur, the parent must have a long half-life when compared to the daughter. An example of this type of compound decay process is barium-140, which decays by beta emission to lanthanum-140, which in turn decays by beta emission to stable cerium-140.
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Figure 15 Transient Equilibrium in the Decay of Barium-140
The decay constant for barium-140 is considerably smaller than the decay constant for

represented as ■N. Although the decay constant for barium-140 is smaller, the actual rate of decay (■N) is initially larger than that of lanthanum-140 because of the great difference in their initial concentrations. As the concentration of the daughter increases, the rate of decay of the daughter will approach and eventually match the decay rate of the parent. When this occurs, they are said to be in transient equilibrium. A plot of the barium-lanthanum-cerium decay chain reaching transient equilibrium is shown in Figure 15.

Secular equilibrium occurs when the parent has an extremely long half-life. In the long decay chain for a naturally radioactive element, such as thorium-232, where all of the elements in the chain are in secular equilibrium, each of the descendants has built up to an equilibrium amount and all decay at the rate set by the original parent. The only exception is the final stable element on the end of the chain. Its number of atoms is constantly increasing.

Summary

The important information in this chapter is summarized on the following page.

■

■■

■

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Radioactivity Summary

& Radioactivity is the decay of unstable atoms by the emission of particles and electromagnetic radiation.

& A curie (Ci) is a unit of radioactivity equal to 3.7 x 10 disintegrations per 10 second.

& A becquerel (Bq) is a unit of radioactivity equal to 1 disintegration per second.

& The radioactive decay constant (■) is the probability per unit time that an atom will decay.

& The radioactive half-life is the amount of time required for the activity to decrease to one-half its original value.

& The activity of a substance can be calculated from the number of atoms and the decay constant based on the equation below.

& The amount of activity remaining after a particular time can be calculated from the equation below.

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& The relationship between the decay constant and the half-life is shown below.

& Plots of radioactive decay can be useful to describe the variation of activity over time. If decay is plotted using semi-log scale the plot results in a straight line.

& Radioactive equilibrium exists when the production rate of a material is equal to the removal rate.

& Transient radioactive equilibrium exists when the parent nuclide and the daughter nuclide decay at essentially the same rate. This occurs only when the parent has a long half-life compared to the daughter.

Atomic and Nuclear Physics DOE-HDBK-1019/1-93 NEUTRON INTERACTIONS

NEUTRON INTERACTIONS

Neutrons can cause many different types of interactions. The neutron may simply scatter off the nucleus in two different ways, or it may actually be absorbed into the nucleus. If a neutron is absorbed into the nucleus, it may result in the emission of a gamma ray or a subatomic particle, or it may cause the nucleus to fission.

neutron and a nucleus:
a. Elastic scattering
b. Inelastic scattering
EO 3.2 STATE the conservation laws that apply to an elastic collision between a neutron and a nucleus.
EO 3.3 DESCRIBE the following reactions where a neutron is absorbed in a nucleus:
a. Radiative capture
b. Particle ejection
Scattering
A neutron scattering reaction occurs when a nucleus, after having been struck by a neutron, emits a single neutron. Despite the fact that the initial and final neutrons do not need to be (and often are not) the same, the net effect of the reaction is as if the projectile neutron had merely "bounced off," or scattered from, the nucleus. The two categories of scattering reactions, elastic and inelastic scattering, are described in the following paragraphs.
Elastic Scattering
In an elastic scattering reaction between a neutron and a target nucleus, there is no energy transferred into nuclear excitation. Momentum and kinetic energy of the "system" are conserved although there is usually some transfer of kinetic energy from the neutron to the target nucleus. The target nucleus gains the amount of kinetic energy that the neutron loses.
Rev. 0 Page 43 NP-01
mn
vn,i mTvT,i mnvn,f mTvT,f
1
2
m v
2
1
2
mn
v
2
n,i
1
2
mT
v
2
T,i
Fixture checkpoint marker-0558-xray is recorded on fixture page 0558.
1
2
mn
v
2
n,f
1
2

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v
2
T,f
NEUTRON INTERACTIONS DOE-HDBK-1019/1-93 Atomic and Nuclear Physics
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Figure 16 Elastic Scattering
Figure 16 illustrates the process of elastic scattering of a neutron off a target nucleus. In the elastic scattering reaction, the conservation of momentum and kinetic energy is represented by the equations below.
Conservation of momentum (mv)
Conservation of kinetic energy
where:
mn = mass of the neutron
m = mass of the target nucleus T
vn,i = initial neutron velocity
vn,f = final neutron velocity
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vT,i = initial target velocity
vT,f = final target velocity
Atomic and Nuclear Physics DOE-HDBK-1019/1-93 NEUTRON INTERACTIONS
Elastic scattering of neutrons by nuclei can occur in two ways. The more unusual of the two interactions is the absorption of the neutron, forming a compound nucleus, followed by the re-emission of a neutron in such a way that the total kinetic energy is conserved and the nucleus returns to its ground state. This is known as resonance elastic scattering and is very dependent upon the initial kinetic energy possessed by the neutron. Due to formation of the compound nucleus, it is also referred to as compound elastic scattering. The second, more usual method, is termed potential elastic scattering and can be understood by visualizing the neutrons and nuclei to be much like billiard balls with impenetrable surfaces. Potential scattering takes place with incident neutrons that have an energy of up to about 1 MeV. In potential scattering, the neutron does not actually touch the nucleus and a compound nucleus is not formed. Instead, the neutron is acted on and scattered by the short range nuclear forces when it approaches close enough to the nucleus.
Inelastic Scattering
In inelastic scattering, the incident neutron is absorbed by the target nucleus, forming a compound nucleus. The compound nucleus will then emit a neutron of lower kinetic energy which leaves the original nucleus in an excited state. The nucleus will usually, by one or more gamma emissions, emit this excess energy to reach its ground state. Figure 17 shows the process of inelastic scattering.
For the nucleus that has reached its ground state, the sum of the kinetic energy of the exit
Figure 17 Inelastic Scattering
neutron, the target nucleus, and the total gamma energy emitted is equal to the initial kinetic energy of the incident neutron.
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1
0
n
238
92

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239
92
U
239
92
U
0
0
1
0
n
10
5
B
11
5
B
7
3
Li
4
2
NEUTRON INTERACTIONS DOE-HDBK-1019/1-93 Atomic and Nuclear Physics
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Absorption Reactions
Most absorption reactions result in the loss of a neutron coupled with the production of a charged particle or gamma ray. When the product nucleus is radioactive, additional radiation is emitted at some later time. Radiative capture, particle ejection, and fission are all categorized as absorption reactions and are briefly described below.
Radiative Capture
In radiative capture the incident neutron enters the target nucleus forming a compound nucleus. The compound nucleus then decays to its ground state by gamma emission. An example of a radiative capture reaction is shown below.
Particle Ejection
In a particle ejection reaction the incident particle enters the target nucleus forming a compound nucleus. The newly formed compound nucleus has been excited to a high enough energy level to cause it to eject a new particle while the incident neutron remains in the nucleus. After the new particle is ejected, the remaining nucleus may or may not exist in an excited state depending upon the mass-energy balance of the reaction. An example of a particle ejection reaction is shown below.
Fission
One of the most important interactions that neutrons can cause is fission, in which the nucleus that absorbs the neutron actually splits into two similarly sized parts. Fission will be discussed in detail in the next chapter.
Atomic and Nuclear Physics DOE-HDBK-1019/1-93 NEUTRON INTERACTIONS
Summary
The important information in this chapter is summarized below.

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Interactions where a neutron scatters off a target nucleus are either elastic or inelastic. In elastic scattering, kinetic energy and momentum are conserved and no energy is transferred into excitation energy of the target nucleus. In inelastic scattering, some amount of kinetic energy is transferred into excitation energy of the target nucleus.
The conservation principles that apply to an elastic collision are conservation of kinetic energy and conservation of momentum.
Radiative capture is the absorption of a neutron by the target nucleus, resulting in an excited nucleus which subsequently (typically within a small fraction of a second) releases its excitation energy in the form of a gamma ray.
Particle ejection occurs when a neutron is absorbed by a target nucleus, resulting in the formation of a compound nucleus. The compound nucleus immediately ejects a particle (for example, alpha or proton).
Rev. 0 Page 47 NP-01
1
0
n
235
92
U
236
92
U
140
55
Cs
93
37
Rb 3
1
Fixture checkpoint marker-0561-xray is recorded on fixture page 0561.
0
n
NUCLEAR FISSION DOE-HDBK-1019/1-93 Atomic and Nuclear Physics
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NUCLEAR FISSION
Nuclear fission is a process in which an atom splits and releases energy, fission products, and neutrons. The neutrons released by fission can, in turn, cause the fission of other atoms.
EO 4.1 EXPLAIN the fission process using the liquid drop model of a nucleus.
EO 4.2 DEFINE the following terms:
a. Excitation energy (E_{exc})
b. Critical energy (E_{crit})
EO 4.3 DEFINE the following terms:
a. Fissile material
b. Fissionable material
c. Fertile material

and breeding.
EO 4.5 DESCRIBE the curve of Binding Energy per Nucleon
versus mass number and give a qualitative description of
the reasons for its shape.
EO 4.6 EXPLAIN why only the heaviest nuclei are easily fissioned.
EO 4.7 EXPLAIN why uranium-235 fissions with thermal
neutrons and uranium-238 fissions only with fast neutrons.
Fission
In the fission reaction the incident neutron enters the heavy target nucleus, forming a compound nucleus that is excited to such a high energy level (E > E) that the nucleus "splits" exc crit (fissions) into two large fragments plus some neutrons. An example of a typical fission reaction is shown below.
A large amount of energy is released in the form of radiation and fragment kinetic energy.
Atomic and Nuclear Physics DOE-HDBK-1019/1-93 NUCLEAR FISSION
Liquid Drop Model of a Nucleus
The nucleus is held together by the attractive nuclear force between nucleons, which was discussed in a previous chapter. The characteristics of the nuclear force are listed below.
(a) very short range, with essentially no effect beyond nuclear dimensions
(10-13 cm)
(b) stronger than the repulsive electrostatic forces within the nucleus
(c) independent of nucleon pairing, in that the attractive forces between pairs of
neutrons are no different than those between pairs of protons or a neutron and a
proton
(d) saturable, that is, a nucleon can attract only a few of its nearest neighbors
One theory of fission considers the fissioning of a nucleus similar in some respects to the splitting of a liquid drop. This analogy is justifiable to some extent by the fact that a liquid drop is held together by molecular forces that tend to make the drop spherical in shape and that try to resist any deformation in the same manner as nuclear forces are assumed to hold the nucleus together. By considering the nucleus as a liquid drop, the fission process can be described.
Referring to Figure 18(A), the nucleus in the ground state is undistorted, and its attractive nuclear forces are greater than the repulsive electrostatic forces between the protons within the nucleus. When an incident particle (in this instance a neutron) is absorbed by the target nucleus, a compound nucleus is formed. The compound nucleus temporarily contains all the charge and mass involved in the reaction and exists in an excited state. The excitation energy added to the compound nucleus is equal to the binding energy contributed by the incident particle plus the kinetic energy possessed by that particle. Figure 18(B) illustrates the excitation energy thus imparted to the compound nucleus, which may cause it to oscillate and become distorted. If the
Fixture checkpoint marker-0562-xray is recorded on fixture page 0562.
excitation energy is greater than a certain critical energy, the oscillations may cause the compound nucleus to become dumbbell-shaped. When this happens, the attractive nuclear forces (short-range) in the neck area are small due to saturation, while the repulsive electrostatic forces (long-range) are only slightly less than before. When the repulsive electrostatic forces exceed the attractive nuclear forces, nuclear fission occurs, as illustrated in Figure 18(C).
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NUCLEAR FISSION DOE-HDBK-1019/1-93 Atomic and Nuclear Physics
Figure 18 Liquid Drop Model of Fission
Critical Energy
The measure of how far the energy level of a nucleus is above its ground state is called the

value for that nuclide. The critical energy (Ecrit) is the minimum excitation energy required for fission to occur.

Fissile Material

A fissile material is composed of nuclides for which fission is possible with neutrons of any energy level. What is especially significant about these nuclides is their ability to be fissioned with zero kinetic energy neutrons (thermal neutrons). Thermal neutrons have very low kinetic energy levels (essentially zero) because they are roughly in equilibrium with the thermal motion of surrounding materials. Therefore, in order to be classified as fissile, a material must be capable of fissioning after absorbing a thermal neutron. Consequently, they impart essentially no kinetic energy to the reaction. Fission is possible in these materials with thermal neutrons, since the change in binding energy supplied by the neutron addition alone is high enough to exceed the critical energy. Some examples of fissile nuclides are uranium-235, uranium-233, and plutonium-239.

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Atomic and Nuclear Physics DOE-HDBK-1019/1-93 NUCLEAR FISSION

Fissionable Material

A fissionable material is composed of nuclides for which fission with neutrons is possible. All fissile nuclides fall into this category. However, also included are those nuclides that can be fissioned only with high energy neutrons. The change in binding energy that occurs as the result of neutron absorption results in a nuclear excitation energy level that is less than the required critical energy. Therefore, the additional excitation energy must be supplied by the kinetic energy of the incident neutron. The reason for this difference between fissile and fissionable materials is the so-called odd-even effect for nuclei. It has been observed that nuclei with even numbers of neutrons and/or protons are more stable than those with odd numbers. Therefore, adding a neutron to change a nucleus with an odd number of neutrons to a nucleus with an even number of neutrons produces an appreciably higher binding energy than adding a neutron to a nucleus already possessing an even number of neutrons. Some examples of nuclides requiring high energy neutrons to cause fission are thorium-232, uranium-238, and plutonium-240. Table 4 indicates the critical energy (Ecrit) and the binding energy change for an added neutron (BEn) to target nuclei of interest. For fission to be possible, the change in binding energy plus the kinetic energy must equal or exceed the critical energy ($\Delta BE + KE > Ecrit$).

TABLE 4

Critical Energies Compared to Binding Energy of Last Neutron

Target
Nucleus
Critical Energy
Ecrit
Binding Energy of
Last Neutron BEn
BEn - Ecrit
23
9
2
0Th 7.5 MeV 5.4 MeV -2.1 MeV
23
9
8

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23
9
5
2U 6.5 MeV 6.8 MeV +0.3 MeV
23
9
3
2U 6.0 MeV 7.0 MeV +1.0 MeV
23
9
9
4Pu 5.0 MeV 6.6 MeV +1.6 MeV
Uranium-235 fissions with thermal neutrons because the binding energy released by the absorption of a neutron is greater than the critical energy for fission; therefore uranium-235 is a fissile material. The binding energy released by uranium-238 absorbing a thermal neutron is less than the critical energy, so additional energy must be possessed by the neutron for fission to be possible. Consequently, uranium-238 is a fissionable material.

Fertile Material
All of the neutron absorption reactions that do not result in fission lead to the production of new
Fixture checkpoint marker-0564-xray is recorded on fixture page 0564.
nuclides through the process known as transmutation. These nuclides can, in turn, be transmuted again or may undergo radioactive decay to produce still different nuclides. The nuclides that are produced by this process are referred to as transmutation products. Because several of the fissile nuclides do not exist in nature, they can only be produced by nuclear reactions (transmutation). The target nuclei for such reactions are said to be fertile. Fertile materials are materials that can undergo transmutation to become fissile materials. Figure 19 traces the transmutation mechanism by which two fertile nuclides, thorium-232 and uranium-238, produce uranium-233 and plutonium-239, respectively.
Figure 19 Conversion of Fertile Nuclides to Fissile Nuclides
If a reactor contains fertile material in addition to its fissile fuel, some new fuel will be produced as the original fuel is burned up. This is called conversion. Reactors that are specifically designed to produce fissionable fuel are called "breeder" reactors. In such reactors, the amount of fissionable fuel produced is greater than the amount of fuel burnup. If less fuel is produced than used, the process is called conversion, and the reactor is termed a "converter."

Binding Energy Per Nucleon (BE/A)
As the number of particles in a nucleus increases, the total binding energy also increases. The rate of increase, however, is not uniform. This lack of uniformity results in a variation in the amount of binding energy associated with each nucleon within the nucleus. This variation in the binding energy per nucleon (BE/A) is easily seen when the average BE/A is plotted versus atomic mass number (A), as shown in Figure 20.
Figure 20 Binding Energy per Nucleon vs. Mass Number
Figure 20 illustrates that as the atomic mass number increases, the binding energy per nucleon decreases for A > 60. The BE/A curve reaches a maximum value of 8.79 MeV at A = 56 and decreases to about 7.6 MeV for A = 238. The general shape of the BE/A curve can be explained

attractive forces that exist between nucleons. On the other hand, the nucleus is being forced apart by long range repulsive electrostatic (coulomb) forces that exist between all the protons in the nucleus.

As the atomic number and the atomic mass number increase, the repulsive electrostatic forces within the nucleus increase due to the greater number of protons in the heavy elements. To overcome this increased repulsion, the proportion of neutrons in the nucleus must increase to maintain stability. This increase in the neutron-to-proton ratio only partially compensates for the growing proton-proton repulsive force in the heavier, naturally occurring elements. Because the repulsive forces are increasing, less energy must be supplied, on the average, to remove a nucleon from the nucleus. The BE/A has decreased. The BE/A of a nucleus is an indication of its degree of stability. Generally, the more stable nuclides have higher BE/A than the less stable ones. The increase in the BE/A as the atomic mass number decreases from 260 to 60 is the primary reason for the energy liberation in the fission process. In addition, the increase in the BE/A as the atomic mass number increases from 1 to 60 is the reason for the energy liberation in the fusion process, which is the opposite reaction of fission.

The heaviest nuclei require only a small distortion from a spherical shape (small energy addition) for the relatively large coulomb forces forcing the two halves of the nucleus apart to overcome the attractive nuclear forces holding the two halves together. Consequently, the heaviest nuclei are easily fissionable compared to lighter nuclei.

Summary

The important information in this chapter is summarized on the following page.

Nuclear Fission Summary

The fission process can be explained using the liquid drop model of a nucleus. In the ground state the nucleus is nearly spherical in shape. After the absorption

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of a neutron, the nucleus will be in an excited state and start to oscillate and become distorted. If the oscillations cause the nucleus to become shaped like a dumbbell, the repulsive electrostatic forces will overcome the short-range attractive nuclear forces, and the nucleus will split in two.

Excitation energy is the amount of energy a nucleus has above its ground state. Critical energy is the minimum excitation energy that a nucleus must have before it can fission.

Fissile material is material for which fission is possible with neutrons that have zero kinetic energy. Fissionable material is material for which fission caused by neutron absorption is possible provided the kinetic energy added with the binding energy is greater than the critical energy. Fertile material is material that can undergo transmutation to become fissile material.

Transmutation is the process of neutron absorption and subsequent decay, which changes one nuclide to another nuclide. Conversion is the process of transmuting fertile material into fissile material in a reactor, where the amount of fissile material produced is less than the amount of fissile material consumed. Breeding is the same as conversion, except the amount of fissile material produced is more than the amount of fissile material consumed.

The curve of binding energy per nucleon increases quickly through the light

decreases slowly for mass numbers greater than 60.
The heaviest nuclei are easily fissionable because they require only a small distortion from the spherical shape to allow the coulomb forces to overcoming the attractive nuclear force, forcing the two halves of the nucleus apart.
Uranium-235 fissions with thermal neutrons because the binding energy released by the absorption of a neutron is greater than the critical energy for fission. The binding energy released by uranium-238 absorbing a neutron is less than the critical energy, so additional kinetic energy must be possessed by the neutron for fission to be possible.
Rev. 0 Page 55 NP-01
1
0
n
235
92
U
236
92
U
140
55
Cs
93
37
Rb 3
1
0
n
ENERGY RELEASE FROM FISSION DOE-HDBK-1019/1-93 Atomic and Nuclear Physics
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ENERGY RELEASE FROM FISSION
Fission of heavy nuclides converts a small amount of mass into an enormous amount of energy. The amount of energy released by fission can be determined based on either the change in mass that occurs during the reaction or by the difference in binding energy per nucleon between the fissile nuclide
Fixture checkpoint marker-0566-xray is recorded on fixture page 0566.
and the fission products.
EO 4.8 CHARACTERIZE the fission products in terms of mass groupings and radioactivity.
EO 4.9 Given the nuclides involved and their masses, CALCULATE the energy released from fission.
EO 4.10 Given the curve of Binding Energy per nucleon versus mass number, CALCULATE the energy released from fission.
Calculation of Fission Energy
Nuclear fission results in the release of enormous quantities of energy. It is necessary to be able to calculate the amount of energy that will be produced. The logical manner in which to pursue this is to first investigate a typical fission reaction such as the one listed below.
It can be seen that when the compound nucleus splits, it breaks into two fission fragments,

emissions as a result of the high neutron-to-proton ratio possessed by these nuclides.
In most cases, the resultant fission fragments have masses that vary widely. Figure 21 gives the percent yield for atomic mass numbers. The most probable pair of fission fragments for the thermal fission of the fuel uranium-235 have masses of about 95 and 140. Note that the vertical axis of the fission yield curve is on a logarithmic scale. Therefore, the formation of fission fragments of mass numbers of about 95 and 140 is highly likely.
Atomic and Nuclear Physics DOE-HDBK-1019/1-93 ENERGY RELEASE FROM FISSION
Figure 21 Uranium-235 Fission Yield vs.
Mass Number
Referring now to the binding energy per nucleon curve (Figure 20), we can estimate the amount of energy released by our "typical" fission by plotting this reaction on the curve and calculating the change in binding energy (ΔBE) between the reactants on the left-hand side of the fission
Fixture checkpoint marker-0567-xray is recorded on fixture page 0567.
equation and the products on the right-hand side. Plotting the reactant and product nuclides on the curve shows that the total binding energy of the system after fission is greater than the total binding energy of the system before fission. When there is an increase in the total binding energy of a system, the system has become more stable by releasing an amount of energy equal to the increase in total binding energy of the system. Therefore, in the fission process, the energy liberated is equal to the increase in the total binding energy of the system.
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Figure 22 graphically depicts that the binding energy per nucleon for the products (C,
Figure 22 Change in Binding Energy for Typical Fission
rubidium-93 and B, cesium-140) is greater than that for the reactant (A, uranium-235). The total binding energy for a nucleus can be found by multiplying the binding energy per nucleon by the number of nucleons.
TABLE 5
Binding Energies Calculated from
Binding Energy per Nucleon Curve
Nuclide B.E. per Nucleon
(BE/A)
Mass Number
(A)
Binding Energy
(BE/A) x (A)
9
3
3
7Rb 8.7 MeV 93 809 MeV
14
5
0
5Cs 8.4 MeV 140 1176 MeV
23
9
5
2U 7.6 MeV 235 1786 MeV
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BERb 93 BECs 140 BEU 235
809 MeV 1176 MeV 1786 MeV
199 MeV
1
0
n
235
92
U
236
92
U
140
55
Cs
93
37
Rb 3
1
Fixture checkpoint marker-0568-xray is recorded on fixture page 0568.
0
n
Mass difference Mass of Reactants Mass of Products
236.052589 amu 235.85208 amu
0.200509 amu
EInst 0.020059 amu
931.5 MeV
amu
186.8 MeV
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The energy released will be equivalent to the difference in binding energy (BE) between the reactants and the products.
The energy liberation during the fission process can also be explained from the standpoint of the conservation of mass-energy. During the fission process, there is a decrease in the mass of the system. There must, therefore, be energy liberated equal to the energy equivalent of the mass lost in the process. This method is more accurate than the previously illustrated method and is used when actually calculating the energy liberated during the fission process.
Again, referring to the "typical" fission reaction.
E , the instantaneous energy, is the energy released immediately after the fission process. It Inst is equal to the energy equivalent of the mass lost in the fission process. It can be calculated as shown below.
Mass of the Reactants Mass of the Products
U 235.043924 amu Rb 92.91699 amu 235
92
93
37
n 1.008665 amu Cs 139.90910 amu 1

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140
55
3 (n) 3.02599 amu 1
0
236.052589 amu 235.85208 amu
This mass difference can be converted to an energy equivalent.
93
37
Rb
93
38
Sr
93
39
Y
93
40
Zr
93
41
Nb
140
55
Cs
140
56
Fixture checkpoint marker-0569-xray is recorded on fixture page 0569.
Ba
140
57
La
140
58
Ce
EDecay mRb 93 mNb 93 4 melectron
931.5MeV
amu
92.91699 amu 92.90638 amu 4 0.0005486 amu
931.5 MeV
amu
0.008416 amu
931.5 MeV
amu
7.84 MeV
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The total energy released per fission will vary from the fission to the next depending on what fission products are formed, but the average total energy released per fission of uranium-235

As illustrated in the preceding example, the majority of the energy liberated in the fission process is released immediately after the fission occurs and appears as the kinetic energy of the fission fragments, kinetic energy of the fission neutrons, and instantaneous gamma rays. The remaining energy is released over a period of time after the fission occurs and appears as kinetic energy of the beta, neutrino, and decay gamma rays.

Estimation of Decay Energy

In addition to this instantaneous energy release during the actual fission reaction, there is additional energy released when the fission fragments decay by emission. This additional

-

energy is called decay energy, EDecay. The decay chains for rubidium-93 and cesium-140 are shown below.

The energy released during the decay for each chain will be equivalent to the mass difference between the original fission product and the sum of the final stable nuclide and the beta particles emitted.

The energy released in the decay chain of rubidium-93 is calculated below.

EDecay mRb 93 mNb 93 3 melectron

931.5MeV

amu

139.90910 amu 139.90543 amu 3 0.0005486 amu

931.5 MeV

amu

0.000202 amu

931.5 MeV

amu

1.89 MeV

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The energy released in the decay chain of cesium-140 is calculated below.

The total decay energy is the sum of the energies of the two chains, or 9.73 MeV.

Distribution of Fission Energy

The average energy distribution for the energy released per fission with a thermal neutron in uranium-235 is shown in Tables 6 and 7.

TABLE 6

Fixture checkpoint marker-0570-xray is recorded on fixture page 0570.

Instantaneous Energy from Fission

Kinetic Energy of Fission Products 167 Mev

Energy of Fission Neutrons 5 MeV

Instantaneous Gamma-ray Energy 5 MeV

Capture Gamma-ray Energy 10 MeV

Total Instantaneous Energy 187 MeV

TABLE 7

Delayed Energy from Fission

Beta Particles From Fission Products 7 Mev

Gamma-rays from Fission Products 6 MeV

Neutrinos 10 MeV

Total Delayed Energy 23 MeV

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Because the 10 MeV of neutrino energy shown in Table 7 is not absorbed in the reactor, the

neutrons undergo radiative capture and the resultant gamma ray emission provides an additional 10 MeV of instantaneous energy, which contributes to the total of 187 MeV instantaneous energy.

All of the energy released, with the exception of the neutrino energy, is ultimately transformed into heat through a number of processes. The fission fragments, with their high positive charge and kinetic energy, cause ionization directly as they rip orbital electrons from the surrounding atoms. In this ionization process, kinetic energy is transferred to the surrounding atoms of the fuel material, resulting in an increase in temperature. The beta particles and gamma rays also give up their energy through ionization, and the fission neutrons interact and lose their energy through elastic scattering. Of the 200 MeV released per fission, about seven percent (13 MeV)

Fixture checkpoint marker-0571-xray is recorded on fixture page 0571.

is released at some time after the instant of fission. When a reactor is shut down, fissions essentially cease, but energy is still being released from the decay of fission products. The heat produced by this decay energy is referred to as "decay heat." Although decay energy represents about seven percent of reactor heat production during reactor operation, once the reactor is shut down the decay heat production drops off quickly to a small fraction of its value while operating. The decay heat produced is significant, however, and systems must be provided to keep the reactor cool even after shutdown.

Summary

The important information in this chapter is summarized below.

Energy Release From Fission Summary

Fission products have some general characteristics in common.

They generally decay by β- emission.

The most common mass numbers are grouped near 95 and 140.

The energy released by fission can be calculated based on the difference in mass between the masses of the reactants before fission and the fission fragments and fission neutrons after fission.

Another method to determine the energy released by fission is based on the change in binding energy per nucleon between the fissile nuclide and the fission products.

INTERACTION OF RADIATION WITH MATTER

Different types of radiation interact with matter in widely different ways. A large, massive, charged alpha particle cannot penetrate a piece of paper and even has a limited range in dry air. A neutrino, at the other extreme, has a low probability of interacting with any matter, even if it passed through the diameter of the earth.

EO 5.1 DESCRIBE interactions of the following with matter:

a. Alpha particle c. Positron

b. Beta particle d. Neutron

EO 5.2 DESCRIBE the following ways that gamma radiation interacts with matter:

a. Photoelectric effect

b. Compton scattering

c. Pair production

Interaction of Radiation With Matter

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expected that interactions with matter fall into two general types. Charged particles directly ionize the media through which they pass, while uncharged particles and photons can cause ionization only indirectly or by secondary radiation.

A moving charged particle has an electrical field surrounding it, which interacts with the atomic structure of the medium through which it is passing. This interaction decelerates the particle and accelerates electrons in the atoms of the medium. The accelerated electrons may acquire enough energy to escape from the parent atom. This process, whereby radiation "strips" off orbital electrons, is called ionization. Uncharged moving particles have no electrical field, so they can only lose energy and cause ionization by such means as collisions or scattering. A photon can lose energy by the photoelectric effect, Compton effect, or pair production.

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I

m z

2

K.E.

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Because ionizing radiation creates ions in pairs, the intensity of ionization or the specific ionization is defined as the number of ion-pairs formed per centimeter of travel in a given material. The amount of ionization produced by a charged particle per unit path length, which is a measure of its ionizing power, is roughly proportional to the particle's mass and the square of its charge as illustrated in the equation below.

where:

I is the ionizing power

m is the mass of the particle

z is the number of unit charges it carries

K.E. is its kinetic energy

Since m for an alpha particle is about 7300 times as large as m for a beta particle, and z is twice as great, an alpha will produce much more ionization per unit path length than a beta particle of the same energy. This phenomenon occurs because the larger alpha particle moves slower for a given energy and thus acts on a given electron for a longer time.

Alpha Radiation

Alpha radiation is normally produced from the radioactive decay of heavy nuclides and from certain nuclear reactions. The alpha particle consists of 2 neutrons and 2 protons, so it is essentially the same as the nucleus of a helium atom. Because it has no electrons, the alpha particle has a charge of +2. This positive charge causes the alpha particle to strip electrons from the orbits of atoms in its vicinity. As the alpha particle passes through material, it removes electrons from the orbits of atoms it passes near. Energy is required to remove electrons and the energy of the alpha particle is reduced by each reaction. Eventually the particle will expend its kinetic energy, gain 2 electrons in orbit, and become a helium atom. Because of its strong positive charge and large mass, the alpha particle deposits a large amount of energy in a short distance of travel. This rapid, large deposition of energy limits the penetration of alpha particles. The most energetic alpha particles are stopped by a few centimeters of air or a sheet of paper.

Beta-Minus Radiation

A beta-minus particle is an electron that has been ejected at a high velocity from an unstable

ionization by displacing electrons from atom orbits. The ionization occurs from collisions with orbiting electrons. Each collision removes kinetic energy from the beta particle, causing it to slow down. Eventually the beta particle will be slowed enough to allow it to be captured as an
2 electrons 0.000549 amu
electron
931.5 MeV
amu
1.02 MeV
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Atomic and Nuclear Physics INTERACTION OF RADIATION WITH MATTER
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orbiting electron in an atom. Although more penetrating than the alpha, the beta is relatively easy to stop and has a low power of penetration. Even the most energetic beta radiation can be stopped by a few millimeters of metal.

Positron Radiation

Positively charged electrons are called positrons. Except for the positive charge, they are identical to beta-minus particles and interact with matter in a similar manner. Positrons are very short-lived, however, and quickly are annihilated by interaction with a negatively charged electron, producing two gammas with a combined energy (calculated below) equal to the rest mass of the positive and negative electrons.

Neutron Radiation

Neutrons have no electrical charge. They have nearly the same mass as a proton (a hydrogen atom nucleus). A neutron has hundreds of times more mass than an electron, but 1/4 the mass of an alpha particle. The source of neutrons is primarily nuclear reactions, such as fission, but
Fixture checkpoint marker-0573-xray is recorded on fixture page 0573.
they may also be produced from the decay of radioactive nuclides. Because of its lack of charge, the neutron is difficult to stop and has a high penetrating power.

Neutrons are attenuated (reduced in energy and numbers) by three major interactions, elastic scatter, inelastic scatter, and absorption. In elastic scatter, a neutron collides with a nucleus and bounces off. This reaction transmits some of the kinetic energy of the neutron to the nucleus of the atom, resulting in the neutron being slowed, and the atom receives some kinetic energy (motion). This process is sometimes referred to as "the billiard ball effect."

As the mass of the nucleus approaches the mass of the neutron, this reaction becomes more effective in slowing the neutron. Hydrogenous material attenuates neutrons most effectively.

In the inelastic scatter reaction, the same neutron/nucleus collision occurs as in elastic scatter. However, in this reaction, the nucleus receives some internal energy as well as kinetic energy. This slows the neutron, but leaves the nucleus in an excited state. When the nucleus decays to its original energy level, it normally emits a gamma ray.

In the absorption reaction, the neutron is actually absorbed into the nucleus of an atom. The neutron is captured, but the atom is left in an excited state. If the nucleus emits one or more gamma rays to reach a stable level, the process is called radiative capture. This reaction occurs at most neutron energy levels, but is more probable at lower energy levels.

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Gamma Radiation

Gamma radiation is electromagnetic radiation. It is commonly referred to as a gamma ray and is very similar to an x-ray. The difference is that gamma rays are emitted from the nucleus of an atom, and x-rays are produced by orbiting electrons. The x-ray is produced when orbiting

deflected and decelerated as they react with the atom's electrical field (called Bremsstrahlung). The gamma ray is produced by the decay of excited nuclei and by nuclear reactions. Because the gamma ray has no mass and no charge, it is difficult to stop and has a very high penetrating power. A small fraction of the original gamma stream will pass through several feet of concrete or several meters of water.

There are three methods of attenuating gamma rays. The first method is referred to as the photo-electric effect. When a low energy gamma strikes an atom, the total energy of the gamma is expended in ejecting an electron from orbit. The result is ionization of the atom and expulsion of a high energy electron. This reaction is most predominant with low energy gammas interacting in materials with high atomic weight and rarely occurs with gammas having an energy above 1 MeV. Annihilation of the gamma results. Any gamma energy in excess of the binding energy of the electron is carried off by the electron in the form of kinetic energy.

The second method of attenuation of gammas is called Compton scattering. The gamma interacts with an orbital or free electron; however, in this case, the photon loses only a fraction of its energy. The actual energy loss depending on the scattering angle of the gamma. The gamma continues on at lower energy, and the energy difference is absorbed by the electron. This reaction becomes important for gamma energies of about 0.1 MeV and higher.

At higher energy levels, a third method of attenuation is predominant. This method is pair-production. When a high energy gamma passes close enough to a heavy nucleus, the gamma completely disappears, and an electron and a positron are formed. For this reaction to take place, the original gamma must have at least 1.02 MeV energy. Any energy greater than 1.02 MeV becomes kinetic energy shared between the electron and positron. The probability of pair■production incre

Interaction of Radiation with Matter Summary

An alpha particle deposits a large amount of energy in a short distance of travel due to its large mass and charge.

Beta-minus particles interact with the electrons orbiting the nucleus of atoms, causing ionization by displacing the electrons. The beta particle loses energy with

Fixture checkpoint marker-0574-xray is recorded on fixture page 0574.

each interaction. After the beta particle loses enough energy, it is captured in the orbital shells of an atom.

Positrons interact with matter much the same way as beta minus particles. After the positron has lost most of its energy by ionizing atoms, it is annihilated by interaction with an electron. The electron-positron pair disappear and are replaced by two gammas, each with the energy equivalent of the mass of an electron (0.51 MeV).

Neutrons interact with matter by elastic scattering, inelastic scattering, or absorption.

Photoelectric effect is where a gamma interacts with an electron orbiting an atom. The entire energy of the gamma is transferred to the electron, and the electron is ejected from its orbit.

In Compton scattering a gamma interacts with an orbital electron, but only part of the gamma energy is transferred to the electron. The electron is ejected from its orbit, and the gamma is scattered off at a lower energy.

In pair-production, a gamma interacts with the electric field of a nucleus and is

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than 1.02 MeV for this to occur.
Summary
The important information in this chapter is summarized below.
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Department of Energy
Fundamentals Handbook
NUCLEAR PHYSICS
AND REACTOR THEORY
Module 2
Reactor Theory (Neutron Characteristics)
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TERMINAL OBJECTIVE
1.0 Without references, EXPLAIN how neutron sources produce neutrons.
ENABLING OBJECTIVES

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a. Intrinsic neutron source
b. Installed neutron source
1.2 LIST three examples of reactions that produce neutrons in intrinsic neutron sources.
1.3 LIST three examples of reactions that produce neutrons in installed neutron sources.
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Reactor Theory (Neutron Characteristics) DOE-HDBK-1019/1-93 OBJECTIVES
TERMINAL OBJECTIVE
2.0 Given the necessary information for calculations, EXPLAIN basic concepts in reactor physics and perform calculations.
ENABLING OBJECTIVES
2.1 DEFINE the following terms:
a. Atom density d. Barn
b. Neutron flux e. Macroscopic cross section
c. Microscopic cross section f. Mean free path
2.2 EXPRESS macroscopic cross section in terms of microscopic cross section.
2.3 DESCRIBE how the absorption cross section of typical nuclides varies with neutron energy at energies below the resonance absorption region.
2.4 DESCRIBE the cause of resonance absorption in terms of nuclear energy levels.
2.5 DESCRIBE the energy dependence of resonance absorption peaks for typical light and heavy nuclei.
2.6 EXPRESS mean free path in terms of macroscopic cross section.
2.7 Given the number densities (or total density and component fractions) and microscopic
Fixture checkpoint marker-0577-xray is recorded on fixture page 0577.
cross sections of components, CALCULATE the macroscopic cross section for a mixture.
2.8 CALCULATE a macroscopic cross section given a material density, atomic mass, and microscopic cross section.
2.9 EXPLAIN neutron shadowing or self-shielding.
2.10 Given the neutron flux and macroscopic cross section, CALCULATE the reaction rate.
2.11 DESCRIBE the relationship between neutron flux and reactor power.
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OBJECTIVES DOE-HDBK-1019/1-93 Reactor Theory (Neutron Characteristics)
ENABLING OBJECTIVES (Cont.)
2.12 DEFINE the following concepts:
a. Thermalization d. Average logarithmic energy decrement
b. Moderator e. Macroscopic slowing down power
c. Moderating ratio
2.13 LIST three desirable characteristics of a moderator.
2.14 Given an average fractional energy loss per collision, CALCULATE the energy loss after a specified number of collisions.
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Reactor Theory (Neutron Characteristics) DOE-HDBK-1019/1-93 OBJECTIVES
TERMINAL OBJECTIVE
3.0 Without references, EXPLAIN the production process and effects on fission of prompt and delayed neutrons.
ENABLING OBJECTIVES
3.1 STATE the origin of prompt neutrons and delayed neutrons.
3.2 STATE the approximate fraction of neutrons that are born as delayed neutrons from the fission of the following nuclear fuels:

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b. Plutonium-239
3.3 EXPLAIN the mechanism for production of delayed neutrons.
3.4 EXPLAIN prompt and delayed neutron generation times.
3.5 Given prompt and delayed neutron generation times and delayed neutron fraction, CALCULATE the average generation time.
3.6 EXPLAIN the effect of delayed neutrons on reactor control.
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OBJECTIVES DOE-HDBK-1019/1-93 Reactor Theory (Neutron Characteristics)
TERMINAL OBJECTIVE
4.0 Without references, DESCRIBE the neutron energy spectrum for the type of reactor presented in this module.
ENABLING OBJECTIVES
4.1 STATE the average energy at which prompt neutrons are produced.
4.2 DESCRIBE the neutron energy spectrum in the following reactors:
a. Fast reactor
b. Thermal reactor
4.3 EXPLAIN the reason for the particular shape of the fast, intermediate, and slow energy regions of the neutron flux spectrum for a thermal reactor.
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Reactor Theory (Neutron Characteristics) DOE-HDBK-1019/1-93 NEUTRON SOURCES
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NEUTRON SOURCES
Neutrons from a variety of sources are always present in a reactor core. This is true even when the reactor is shut down. Some of these neutrons are produced by naturally occurring (intrinsic) neutron sources, while others may be the result of fabricated (installed) neutron sources that are incorporated into the design of the reactor. The neutrons produced by sources other than neutron-induced fission are often grouped together and classified as source neutrons.
EO 1.1 DEFINE the following terms:
Fixture checkpoint marker-0578-xray is recorded on fixture page 0578.
a. Intrinsic neutron source
b. Installed neutron source
EO 1.2 LIST three examples of reactions that produce neutrons in intrinsic neutron sources.
EO 1.3 LIST three examples of reactions that produce neutrons in installed neutron sources.
Neutron Sources
In addition to neutron-induced fission, neutrons are produced by other reactions. The neutrons produced by reactions other than neutron-induced fission are called source neutrons. Source neutrons are important because they ensure that the neutron population remains high enough to allow a visible indication of neutron level on the most sensitive monitoring instruments while the reactor is shutdown and during the startup sequence. This verifies instrument operability and allows monitoring of neutron population changes. Source neutrons can be classified as either intrinsic or installed neutron sources.
Intrinsic Neutron Sources
Some neutrons will be produced in the materials present in the reactor due to a variety of unavoidable reactions that occur because of the nature of these materials. Intrinsic neutron sources are those neutron-producing reactions that always occur in reactor materials.

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■ ■ ■
■ ■
NEUTRON SOURCES DOE-HDBK-1019/1-93 Reactor Theory (Neutron Characteristics)
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A limited number of neutrons will always be present, even in a reactor core that has never been operated, due to spontaneous fission of some heavy nuclides that are present in the fuel. Uranium■238, ur of the rate at which different heavy nuclides produce neutrons by spontaneous fission.
Californium-252 is not an intrinsic neutron source, but will be discussed in the section on installed neutron sources.
TABLE 1
Neutron Production by Spontaneous Fission
Nuclide T (Fission) T (1/2 1/2 ■-decay) neutrons/sec/gram
U 1.8 x 10 years 6.8 x 10 years 8.0 x 10 235
92
17 8 -4
U 8.0 x 10 years 4.5 x 10 years 1.6 x 10 238
92
15 9 -2
Pu 5.5 x 10 years 2.4 x 10 years 3.0 x 10 239
94
5 4 -2
Pu 1.2 x 10 years 6.6 x 10 years 1.0 x 10 240
94
11 3 3
Cf 66.0 years 2.65 years 2.3 x 10 252
98
12
Another intrinsic neutron source is a reaction involving natural boron and fuel. In some reactors, natural boron is loaded into the reactor core as a neutron absorber to improve reactor control or increase core life-time. Boron-11 (80.1% of natural boron) undergoes a reaction with the alpha particle emitted by the radioactive decay of heavy nuclides in the fuel to yield a neutron as shown below.
The boron-11 must be mixed with, or in very close proximity to, the fuel for this reaction because of the short path length of the alpha particle. For a reactor core with this configuration, this (■,n) reaction is an important source of neutrons for reactor startup.
In a reactor that has been operated, another source of neutrons becomes significant. Neutrons may
Fixture checkpoint marker-0579-xray is recorded on fixture page 0579.
be produced by the interaction of a gamma ray and a deuterium nucleus. This reaction is commonly referred to as a photoneutron reaction because it is initiated by electromagnetic radiation and results in the production of a neutron. The photoneutron reaction is shown below.
■ ■ ■
■ ■
■ ■ ■
■ ■
Reactor Theory (Neutron Characteristics) DOE-HDBK-1019/1-93 NEUTRON SOURCES
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There is an abundant supply of high energy gammas in a reactor that has been operated because many of the fission products are gamma emitters. All water-cooled reactors have some

is the isotope deuterium. The atom percentage of deuterium in the water ranges from close to the naturally occurring value (0.015%) for light water reactors to above 90% deuterium for heavy water reactors. Therefore, the required conditions for production of photoneutrons exist.
The supply of gamma rays decreases with time after shutdown as the gamma emitters decay; therefore, the photoneutron production rate also decreases. In a few particular reactors, additional D O (heavy water) may be added to the reactor to increase the production of 2 photoneutrons following a long shutdown period.
Installed Neutron Sources
Because intrinsic neutron sources can be relatively weak or dependent upon the recent power history of the reactor, many reactors have artificial sources of neutrons installed. These neutron sources ensure that shutdown neutron levels are high enough to be detected by the nuclear instruments at all times. This provides a true picture of reactor conditions and any change in these conditions. An installed neutron source is an assembly placed in or near the reactor for
Fixture checkpoint marker-0580-xray is recorded on fixture page 0580.
the sole purpose of producing source neutrons.
One strong source of neutrons is the artificial nuclide californium-252, which emits neutrons at the rate of about 2 x 10 neutrons per second per gram as the result of spontaneous fission. 12
Important drawbacks for some applications may be its high cost and its short half-life (2.65 years).
Many installed neutron sources use the (■,n) reaction with beryllium. These sources are composed of a mixture of metallic beryllium (100% beryllium-9) with a small quantity of an alpha particle emitter, such as a compound of radium, polonium, or plutonium. The reaction that occurs is shown below.
The beryllium is intimately (homogeneously) mixed with the alpha emitter and is usually enclosed in a stainless steel capsule.
Another type of installed neutron source that is widely used is a photoneutron source that employs the (■,n) reaction with beryllium. Beryllium is used for photoneutron sources because its stable isotope beryllium-9 has a weakly attached last neutron with a binding energy of only 1.66 MeV. Thus, a gamma ray with greater energy than 1.66 MeV can cause neutrons to be ejected by the (■,n) reaction as shown below.
NEUTRON SOURCES DOE-HDBK-1019/1-93 Reactor Theory (Neutron Characteristics)
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Neutron Sources Summary
& Intrinsic neutron sources are sources of neutrons from materials that are in the reactor for other purposes such as fuel, burnable poison, or moderator.
& Installed neutron sources are materials or components placed in the reactor specifically for the purpose of producing source neutrons.
& Examples of intrinsic neutron sources are listed below.
Spontaneous fission of heavy nuclides in fuel, such as uranium-238, uranium-235, and plutonium-239, results in fission fragments and free neutrons.
Boron-11 mixed with the fuel undergoes an alpha-neutron reaction and becomes nitrogen-14.
Deuterium present in the reactor coolant undergoes a gamma■neutron reaction and becomes hydrogen-1.
& Examples of installed neutron sources are listed below.
Spontaneous fission of californium-252 results in fission fragments and free neutrons.
Beryllium-9 undergoes an alpha-neutron reaction (alpha from the

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Beryllium-9 undergoes a gamma-neutron reaction (high energy
gamma from decay of antimony-124) and becomes beryllium-8.
Many startup sources of this type use antimony and beryllium because after activation with neutrons the radioactive antimony becomes an emitter of high energy gammas. The photoneutron sources of this type are constructed somewhat differently from the (■,n) types. One design incorporates a capsule of irradiated antimony enclosed in a beryllium sleeve. The entire assembly is then encased in a stainless steel cladding. A large reactor may have several neutron sources of this type installed within the core.
Summary
The important information in this chapter is summarized below.
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Reactor Theory (Neutron Characteristics) NUCLEAR CROSS SECTIONS AND NEUTRON FLUX
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NUCLEAR CROSS SECTIONS AND NEUTRON FLUX
To determine the frequency of neutron interactions, it is necessary to describe the availability of neutrons to cause interaction and the probability of a neutron interacting with material. The availability of neutrons and the probability of interaction are quantified by the neutron flux and nuclear cross section.
EO 2.1 DEFINE the following terms:
a. Atom density d. Barn
Fixture checkpoint marker-0581-xray is recorded on fixture page 0581.
b. Neutron flux e. Macroscopic cross section
c. Microscopic cross section f. Mean free path
EO 2.2 EXPRESS macroscopic cross section in terms of microscopic cross section.
EO 2.3 DESCRIBE how the absorption cross section of typical nuclides varies with neutron energy at energies below the resonance absorption region.
EO 2.4 DESCRIBE the cause of resonance absorption in terms of nuclear energy levels.
EO 2.5 DESCRIBE the energy dependence of resonance absorption peaks for typical light and heavy nuclei.
EO 2.6 EXPRESS mean free path in terms of macroscopic cross section.
EO 2.7 Given the number densities (or total density and component fractions) and microscopic cross sections of components, CALCULATE the macroscopic cross section for a mixture.
EO 2.8 CALCULATE a macroscopic cross section given a material density, atomic mass, and microscopic cross section.
EO 2.9 EXPLAIN neutron shadowing or self-shielding.
'
DOE-HDBK-1019/1-93
NUCLEAR CROSS SECTIONS AND NEUTRON FLUX Reactor Theory (Neutron Characteristics)
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Introduction
Fission neutrons are born with an average energy of about 2 MeV. These fast neutrons interact with the reactor core materials in various absorption and scattering reactions. Collisions that result in scattering are useful in slowing neutrons to thermal energies. Thermal neutrons may be absorbed by fissile nuclei to produce more fissions or be absorbed in fertile material for

other non-fuel material results in the removal of neutrons without fulfilling any useful purpose. To safely and efficiently operate a nuclear reactor it is necessary to predict the probability that a particular absorption or scattering reaction will occur. Once these probabilities are known, if the availability of neutrons can be determined, then the rate at which these nuclear reactions take place can be predicted.

Atom Density

One important property of a material is the atom density. The atom density is the number of atoms of a given type per unit volume of the material. To calculate the atom density of a substance use Equation (2-1).

(2-1)

where:

N = atom density (atoms/cm)3

' = density (g/cm)3

N = Avogadro's number (6.022 x 10 atoms/mole) A

23

M = gram atomic weight

N

NA

M

2.699 g

cm3

6.022 x 1023 atoms

mole

26.9815 g

mole

6.024 x 1022 atoms

cm3

Fixture checkpoint marker-0582-xray is recorded on fixture page 0582.

Example:

A block of aluminum has a density of 2.699 g/cm . If the gram atomic weight of

3

aluminum is 26.9815 g, calculate the atom density of the aluminum.

Solution:

Cross Sections

The probability of a neutron interacting with a nucleus for a particular reaction is dependent upon not only the kind of nucleus involved, but also the energy of the neutron. Accordingly, the absorption of a thermal neutron in most materials is much more probable than the absorption of a fast neutron. Also, the probability of interaction will vary depending upon the type of reaction involved.

The probability of a particular reaction occurring between a neutron and a nucleus is called the microscopic cross section ( ) of the nucleus for the particular reaction. This cross section will vary with the energy of the neutron. The microscopic cross section may also be regarded as the effective area the nucleus presents to the neutron for the particular reaction. The larger the effective area, the greater the probability for reaction.

Because the microscopic cross section is an area, it is expressed in units of area, or square

a nucleus, and it has been suggested that a physicist once referred to the measure of a square centimeter as being "as big as a barn" when applied to nuclear processes. The name has persisted and microscopic cross sections are expressed in terms of barns. The relationship between barns and cm is shown below. 2

1 barn = 10-24 cm2

N

Whether a neutron will interact with a certain volume of material depends not only on the microscopic cross section of the individual nuclei but also on the number of nuclei within that volume. Therefore, it is necessary to define another kind of cross section known as the macroscopic cross section ( ). The macroscopic cross section is the probability of a given reaction occurring per unit travel of the neutron. is related to the microscopic cross section ( ) by the relationship shown below.

(2-2)

where:

= macroscopic cross section (cm)

-1

N = atom density of material (atoms/cm)

3

= microscopic cross-section (cm)

2

The difference between the microscopic and macroscopic cross sections is extremely important and is restated for clarity. The microscopic cross section ( ) represents the effective target area that a single nucleus presents to a bombarding particle. The units are given in barns or cm .

2

The macroscopic cross section ( ) represents the effective target area that is presented by all of the nuclei contained in 1 cm of the material. The units are given as 1/cm or cm .

3 -1

A neutron interacts with an atom of the material it enters in two basic ways. It will either interact through a scattering interaction or through an absorption reaction. The probability of a neutron being absorbed by a particular atom is the microscopic cross section for absorption, . The probability of a neutron scattering off of a particular nucleus is the microscopic cross a

Fixture checkpoint marker-0583-xray is recorded on fixture page 0583.

section for scattering, s

. The sum of the microscopic cross section for absorption and the microscopic cross section for scattering is the total microscopic cross section, . T

= + T a s

Both the absorption and the scattering microscopic cross sections can be further divided. For instance, the scattering cross section is the sum of the elastic scattering cross section (se) and the inelastic scattering cross section ( ). si

= + s se si

The microscopic absorption cross section ( ) includes all reactions except scattering. However, a for most purposes it is sufficient to merely separate it into two categories, fission (f) and capture ( ). Radiative capture of neutrons was described in the Neutron Interactions chapter c of Module 1.

= + a f c

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Reactor Theory (Neutron Characteristics) NUCLEAR CROSS SECTIONS AND NEUTRON FLUX
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Figure 1 Typical Neutron Absorption Cross Section vs. Neutron Energy
The variation of absorption cross sections with neutron energy is often complicated. For many elements the absorption cross sections are small, ranging from a fraction of a barn to a few barns for slow (or thermal) neutrons.
For a considerable number of nuclides of moderately high (or high) mass numbers, an examination of the variation of the absorption cross section with the energy of the incident neutron reveals the existence of three regions on a curve of absorption cross section versus neutron energy. This cross section is illustrated in Figure 1. First, the cross section decreases steadily with increasing neutron energy in a low energy region, which includes the thermal range
Fixture checkpoint marker-0584-xray is recorded on fixture page 0584.
(E < 1 eV). In this region the absorption cross section, which is often high, is inversely proportional to the velocity (v). This region is frequently referred to as the "1/v region," because the absorption cross section is proportional to 1/v, which is the reciprocal of neutron velocity. Following the 1/v region, there occurs the "resonance region" in which the cross sections rise sharply to high values called "resonance peaks" for neutrons of certain energies, and then fall again. These energies are called resonance energies and are a result of the affinity of the nucleus for neutrons whose energies closely match its discrete, quantum energy levels. That is, when the binding energy of a neutron plus the kinetic energy of the neutron are exactly equal to the amount required to raise a compound nucleus from its ground state to a quantum level, resonance absorption occurs. The following example problem further illustrates this point.
■
(
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Assuming that uranium-236 has a nuclear quantum energy level at 6.8 MeV above its ground state, calculate the kinetic energy a neutron must possess to undergo resonant absorption in uranium-235 at this resonance energy level.
BE = [Mass(U) + Mass(neutron) - Mass(U)] x 931 MeV/amu 235 236
BE = (235.043925 + 1.008665 - 236.045563) x 931 MeV/amu
BE = (0.007025 amu) x 931 MeV/amu = 6.54 MeV
6.8 MeV - 6.54 MeV = 0.26 MeV
The difference between the binding energy and the quantum energy level equals the amount of kinetic energy the neutron must possess. The typical heavy nucleus will have many closely■spaced resonanc complex and have more possible configurations and corresponding energy states. Light nuclei, being less complex, have fewer possible energy states and fewer resonances that are sparsely distributed at higher energy levels.
For higher neutron energies, the absorption cross section steadily decreases as the energy of the neutron increases. This is called the "fast neutron region." In this region the absorption cross sections are usually less than 10 barns.
With the exception of hydrogen, for which the value is fairly large, the elastic scattering cross sections are generally small, for example, 5 barns to 10 barns. This is close to the magnitude of the actual geometric cross sectional area expected for atomic nuclei. In potential scattering, the cross section is essentially constant and independent of neutron energy. Resonance elastic scattering and inelastic scattering exhibit resonance peaks similar to those associated with absorption cross sections. The resonances occur at lower energies for heavy nuclei than for light

the variations that occur in absorption cross sections.

Mean Free Path

If a neutron has a certain probability of undergoing a particular interaction in one centimeter of travel, then the inverse of this value describes how far the neutron will travel (in the average case) before undergoing an interaction. This average distance traveled by a neutron before interaction is known as the mean free path for that interaction and is represented by the symbol ■. The relationship between the mean free path (■) and the macroscopic cross section (*) is shown below.

(2-3)

'

(

)

Calculation of Macroscopic Cross Section and Mean Free Path

Most materials are composed of several elements, and because most elements are composed of several isotopes, most materials involve many cross sections, one for each isotope involved.

Fixture checkpoint marker-0585-xray is recorded on fixture page 0585.

Therefore, to include all the isotopes within a given material, it is necessary to determine the macroscopic cross section for each isotope and then sum all the individual macroscopic cross sections. Equation (2-4) can be used to determine the macroscopic cross section for a composite material.

* = N) + N) + N) + ... N) (2-4) 1 1 2 2 3 3 nn

where:

N = the number of nuclei per cm of the n element n

3 th

) = the microscopic cross section of the n element n

th

The following example problems illustrate the calculation of the macroscopic cross section for a single element and for combinations of materials.

Example 1:

Find the macroscopic thermal neutron absorption cross section for iron, which has a density of 7.86 g/cm . The microscopic cross section for absorption of iron is 2.56 barns 3 and the gram atomic weight is 55.847 g.

Solution:

Step 1: Using Equation (2-1), calculate the atom density of iron.

Step 2: Use this atom density in Equation (2-2) to calculate the macroscopic cross section.

'

'

Example 2:

An alloy is composed of 95% aluminum and 5% silicon (by weight). The density of the alloy is 2.66 g/cm . Properties of aluminum and silicon are shown below. 3

Element Gram Atomic Weight)) a (barns) s (barns)

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Silicon 28.0855 0.16 2.20
1. Calculate the atom densities for the aluminum and silicon.
2. Determine the absorption and scattering macroscopic cross sections for thermal neutrons.
3. Calculate the mean free paths for absorption and scattering.
Solution:
Step 1: The density of the aluminum will be 95% of the total density. Using Equation (2-1) yields the atom densities.
a NAl a,Al NSi a,Si
5.64 x 1022 atoms
cm3
0.23 x 10 24 cm22.85 x 1021 atoms
cm3
0.16 x 10 24 cm2
0.0134 cm 1
s NAl s,Al NSi s,Si
5.64 x 1022 atoms
cm3
1.49 x 10 24 cm22.85 x 1021 atoms
cm3
2.20 x 10 24 cm2
0.0903 cm 1
a
1
a
Fixture checkpoint marker-0586-xray is recorded on fixture page 0586.
1
0.01345 cm 1
74.3 cm
s
1
s
1
0.0903 cm 1
11.1 cm
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Step 2: The macroscopic cross sections for absorption and scattering are calculated using Equation (2-4).
Step 3: The mean free paths are calculated by inserting the macroscopic cross sections calculated above into Equation (2-3).
Thus, a neutron must travel an average of 74.3 cm to interact by absorption in this alloy, but it must travel only 11.1 cm to interact by scattering.
)
)
)
)

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(■
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Effects of Temperature on Cross Section
As discussed, the microscopic absorption cross section varies significantly as neutron energy varies. The microscopic cross sections provided on most charts and tables are measured for a standard neutron velocity of 2200 meters/second, which corresponds to an ambient temperature of 68(F. Therefore, if our material is at a higher temperature, the absorption cross section will be lower than the value for 68(F, and any cross sections which involve absorption (for example,) ,) ,)) must be corrected for the existing temperature. acf
The following formula is used to correct microscopic cross sections for temperature. Although the example illustrates absorption cross section, the same formula may be used to correct capture and fission cross sections.
where:
) = microscopic cross section corrected for temperature
) = microscopic cross section at reference temperature (68(F or 20(C) o
T = reference temperature (68(F) in degrees Rankine ((R) or Kelvin ((K) o
T = temperature for which corrected value is being calculated
NOTE: When using this formula, all temperatures must be converted to (R or (K.
(R = (F + 460
(K = (C + 273
Example:
What is the value of)f for uranium-235 for thermal neutrons at 500(F? Uranium-235 has a) of 583 barns at 68(F. f
Solution:
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Neutron Flux
Macroscopic cross sections for neutron reactions with materials determine the probability of one neutron undergoing a specific reaction per centimeter of travel through that material. If one
Fixture checkpoint marker-0587-xray is recorded on fixture page 0587.
wants to determine how many reactions will actually occur, it is necessary to know how many neutrons are traveling through the material and how many centimeters they travel each second. It is convenient to consider the number of neutrons existing in one cubic centimeter at any one instant and the total distance they travel each second while in that cubic centimeter. The number of neutrons existing in a cm of material at any instant is called neutron density and is 3 represented by the symbol n with units of neutrons/cm . The total distance these neutrons can 3 travel each second will be determined by their velocity.
A good way of defining neutron flux (1) is to consider it to be the total path length covered by all neutrons in one cubic centimeter during one second. Mathematically, this is the equation below.
1 = n v (2-5)
where:
1 = neutron flux (neutrons/cm -sec) 2
n = neutron density (neutrons/cm)3
v = neutron velocity (cm/sec)

as parallel beams of neutrons traveling in a single direction. The intensity (I) of a neutron beam is the product of the neutron density times the average neutron velocity. The directional beam intensity is equal to the number of neutrons per unit area and time (neutrons/cm -sec) falling on 2 a surface perpendicular to the direction of the beam.

One can think of the neutron flux in a reactor as being comprised of many neutron beams traveling in various directions. Then, the neutron flux becomes the scalar sum of these directional flux intensities (added as numbers and not vectors), that is, 1 = I + I + I +...I . 123 n

Since the atoms in a reactor do not interact preferentially with neutrons from any particular direction, all of these directional beams contribute to the total rate of reaction. In reality, at a given point within a reactor, neutrons will be traveling in all directions.

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Self-Shielding

In some locations within the reactor, the flux level may be significantly lower than in other areas due to a phenomenon referred to as neutron shadowing or self-shielding. For example, the interior of a fuel pin or pellet will "see" a lower average flux level than the outer surfaces since an appreciable fraction of the neutrons will have been absorbed and therefore cannot reach the interior of the fuel pin. This is especially important at resonance energies, where the absorption cross sections are large.

Summary

The important information in this chapter is summarized below.

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Nuclear Cross Section and Neutron Flux Summary

& Atom density (N) is the number of atoms of a given type per unit volume of material.

& Microscopic cross section ()) is the probability of a given reaction occurring between a neutron and a nucleus.

& Microscopic cross sections are measured in units of barns, where 1 barn = 10-24 cm .2

& Macroscopic cross section (*) is the probability of a given reaction occurring per unit length of travel of the neutron. The units for macroscopic cross section are cm . -1

& The mean free path (■) is the average distance that a neutron travels in a material between interactions.

& Neutron flux (1) is the total path length traveled by all neutrons in one cubic centimeter of material during one second.

Fixture checkpoint marker-0588-xray is recorded on fixture page 0588.

& The macroscopic cross section for a material can be calculated using the equation below.

* = N)

& The absorption cross section for a material usually has three distinct regions. At low neutron energies (<1 eV) the cross section is inversely proportional to the neutron velocity.

& Resonance absorption occurs when the sum of the kinetic energy of the neutron and its binding energy is equal to an allowed nuclear energy level of the nucleus.

the absorption cross section steadily decreases as the neutron energy increases.
& The mean free path equals 1/*.
& The macroscopic cross section for a mixture of materials can be calculated using the equation below.
* = N 1 1 2 2 3 3 nn) + N) + N) + ... N)
& Self-shielding is where the local neutron flux is depressed within a material due to neutron absorption near the surface of the material.
1)
REACTION RATES DOE-HDBK-1019/1-93 Reactor Theory (Neutron Characteristics)
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REACTION RATES
It is possible to determine the rate at which a nuclear reaction will take place based on the neutron flux, cross section for the interaction, and atom density of the target. This relationship illustrates how a change in one of these items affects the reaction rate.
EO 2.10 Given the neutron flux and macroscopic cross section,
Fixture checkpoint marker-0589-xray is recorded on fixture page 0589.
CALCULATE the reaction rate.
EO 2.11 DESCRIBE the relationship between neutron flux and reactor power.
Reaction Rates
If the total path length of all the neutrons in a cubic centimeter in a second is known, (neutron flux (1)), and if the probability of having an interaction per centimeter path length is also known (macroscopic cross section (*)), multiply them together to get the number of interactions taking place in that cubic centimeter in one second. This value is known as the reaction rate and is denoted by the symbol R. The reaction rate can be calculated by the equation shown below.
R = 1 * (2-6)
where:
R = reaction rate (reactions/sec)
1 = neutron flux (neutrons/cm -sec) 2
* = macroscopic cross section (cm) -1
Substituting the fact that * = N) into Equation (2-6) yields the equation below.
where:
) = microscopic cross section (cm)2
N = atom density (atoms/cm)3
Rf f
1 x 1013 neutrons
cm2
sec
0.1 cm 1
1 x 1012 fissions
cm3
sec
Rf f
f
Rf
1.29 x 1012 fissions
cm3

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3 x 1013 neutrons
cm2
sec
0.043 cm 1
Reactor Theory (Neutron Characteristics) DOE-HDBK-1019/1-93 REACTION RATES
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The reaction rate calculated will depend on which macroscopic cross section is used in the calculation. Normally, the reaction rate of greatest interest is the fission reaction rate.
Example:
If a one cubic centimeter section of a reactor has a macroscopic fission cross section of 0.1 cm , and if the thermal neutron flux is 10 neutrons/cm -sec, what is the fission rate -1 13 2 in that cubic centimeter?
Solution:
In addition to using Equation (2-6) to determine the reaction rate based on the physical properties of material, it is also possible to algebraically manipulate the equation to determine physical properties if the reaction rate is known.
Example:
A reactor operating at a flux level of 3 x 10 neutrons/cm -sec contains 10 atoms of 13 2 20
uranium-235 per cm . The reaction rate is 1.29 x 10 fission/cm . Calculate and .
3 12 3
f f
Solution:
Fixture checkpoint marker-0590-xray is recorded on fixture page 0590.
Step 1: The macroscopic cross section can be determined by solving Equation (2-6) for and substituting the appropriate values. f
Rf N f
f
Rf
N
1.29 x 1012 fissions
cm3
sec
1 x 1020 atoms
cm3
3 x 1013 neutrons
cm2
sec
4.3 x 10 22 cm2 1 barn
1 x 10 24 cm2
430 barns
1 watt 1 erg
1 x 10 7 watt sec
1 MeV
1.602 x 10 6
erg
1 fission
200 MeV

second
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Step 2: To find the microscopic cross section, replace f with (N x f) and solve
for .
f
Reactor Power Calculation
Multiplying the reaction rate per unit volume by the total volume of the core results in the total number of reactions occurring in the core per unit time. If the amount of energy involved in each reaction were known, it would be possible to determine the rate of energy release (power) due to a certain reaction.
In a reactor where the average energy per fission is 200 MeV, it is possible to determine the number of fissions per second that are necessary to produce one watt of power using the following conversion factors.
1 fission = 200 MeV
1 MeV = 1.602 x 10 ergs -6
1 erg = 1 x 10 watt-sec -7
This is equivalent to stating that 3.12 x 10 fissions release 1 watt-second of energy. 10
P
th f V
3.12 x 1010 fissions
watt sec
Reactor Theory (Neutron Characteristics) DOE-HDBK-1019/1-93 REACTION RATES
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The power released in a reactor can be calculated based on Equation (2-6). Multiplying the reaction rate by the volume of the reactor results in the total fission rate for the entire reactor. Dividing by the number of fissions per watt-sec results in the power released by fission in the reactor in units of watts. This relationship is shown mathematically in Equation (2-7) below.
(2-7)
where:
Fixture checkpoint marker-0591-xray is recorded on fixture page 0591.
P = power (watts)
= thermal neutron flux (neutrons/cm -sec) th
2
= macroscopic cross section for fission (cm) f
-1
V = volume of core (cm)
3
Relationship Between Neutron Flux and Reactor Power
In an operating reactor the volume of the reactor is constant. Over a relatively short period of time (days or weeks), the number density of the fuel atoms is also relatively constant. Since the atom density and microscopic cross section are constant, the macroscopic cross section must also be constant. Examining Equation (2-7), it is apparent that if the reactor volume and macroscopic cross section are constant, then the reactor power and the neutron flux are directly proportional. This is true for day-to-day operation. The neutron flux for a given power level will increase very slowly over a period of months due to the burnup of the fuel and resulting decrease in atom density and macroscopic cross section.
REACTION RATES DOE-HDBK-1019/1-93 Reactor Theory (Neutron Characteristics)

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Reaction Rates Summary
& The reaction rate is the number of interactions of a particular type occurring in a cubic centimeter of material in a second.
& The reaction rate can be calculated by the equation below.
R = 1 *
& Over a period of several days, while the atom density of the fuel can be considered constant, the neutron flux is directly proportional to reactor power.
Summary
The important information in this chapter is summarized below.
Reactor Theory (Neutron Characteristics) DOE-HDBK-1019/1-93 NEUTRON MODERATION
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NEUTRON MODERATION
In thermal reactors, the neutrons that cause fission are at a much lower energy than the energy level at which they were born from fission. In this type of reactor, specific materials must be included in the reactor design to reduce the energy level of the neutrons in an efficient manner.
EO 2.12 DEFINE the following concepts:
a. Thermalization d. Average logarithmic energy decrement
b. Moderator e. Macroscopic slowing down power
c. Moderating ratio
EO 2.13 LIST three desirable characteristics of a moderator.
EO 2.14 Given an average fractional energy loss per collision, CALCULATE the energy loss after a specified number of collisions.
Neutron Slowing Down and Thermalization
Fission neutrons are produced at an average energy level of 2 MeV and immediately begin to slow down as the result of numerous scattering reactions with a variety of target nuclei. After a number of collisions with nuclei, the speed of a neutron is reduced to such an extent that it has approximately the same average kinetic energy as the atoms (or molecules) of the medium in which the neutron is undergoing elastic scattering. This energy, which is only a small fraction of an electron volt at ordinary temperatures (0.025 eV at 20(C), is frequently referred to as the thermal energy, since it depends upon the temperature. Neutrons whose energies have been reduced to values in this region (< 1 eV) are designated thermal neutrons. The process of reducing the energy of a neutron to the thermal region by elastic scattering is referred to as thermalization, slowing down, or moderation. The material used for the purpose of thermalizing neutrons is called a moderator. A good moderator reduces the speed of neutrons in a small number of collisions, but does not absorb them to any great extent. Slowing the neutrons in as
Fixture checkpoint marker-0592-xray is recorded on fixture page 0592.
few collisions as possible is desirable in order to reduce the amount of neutron leakage from the core and also to reduce the number of resonance absorptions in non-fuel materials. Neutron leakage and resonance absorption will be discussed in the next module.
ln Ei
ln Ef
ln
Ei
Ef
2
A

3
NEUTRON MODERATION DOE-HDBK-1019/1-93 Reactor Theory (Neutron Characteristics)
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The ideal moderating material (moderator) should have the following nuclear properties.
large scattering cross section
small absorption cross section
large energy loss per collision
A convenient measure of energy loss per collision is the logarithmic energy decrement. The average logarithmic energy decrement is the average decrease per collision in the logarithm of the neutron energy. This quantity is represented by the symbol (Greek letter xi).
(2-8)
where:
= average logarithmic energy decrement
Ei = average initial neutron energy
Fixture checkpoint marker-0593-xray is recorded on fixture page 0593.
Ef = average final neutron energy
The symbol is commonly called the average logarithmic energy decrement because of the fact that a neutron loses, on the average, a fixed fraction of its energy per scattering collision. Since the fraction of energy retained by a neutron in a single elastic collision is a constant for a given material, is also a constant. Because it is a constant for each type of material and does not depend upon the initial neutron energy, is a convenient quantity for assessing the moderating ability of a material.
The values for the lighter nuclei are tabulated in a variety of sources. The following commonly used approximation may be used when a tabulated value is not available.
This approximation is relatively accurate for mass numbers (A) greater than 10, but for some low values of A it may be in error by over three percent.
N
ln Ehigh ln Elow
ln
Ehigh
Elow
N
ln
Ehigh
Elow
ln
2 x 106eV
0.025 eV
0.948
19.2 collisions
Reactor Theory (Neutron Characteristics) DOE-HDBK-1019/1-93 NEUTRON MODERATION
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Since represents the average logarithmic energy loss per collision, the total number of collisions necessary for a neutron to lose a given amount of energy may be determined by dividing into the difference of the natural logarithms of the energy range in question. The number of collisions (N) to travel from any energy, E , to any lower energy, E , can be calculated as shown below. high low
Example:
How many collisions are required to slow a neutron from an energy of 2 MeV to a

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for .
Solution:
Sometimes it is convenient, based upon information known, to work with an average fractional energy loss per collision as opposed to an average logarithmic fraction. If the initial neutron energy level and the average fractional energy loss per collision are known, the final energy level for a given number of collisions may be computed using the following formula.
! (
NEUTRON MODERATION DOE-HDBK-1019/1-93 Reactor Theory (Neutron Characteristics)
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(2-9)
where:
E = initial neutron energy o
E = neutron energy after N collisions N
x = average fractional energy loss per collision
N = number of collisions
Example:
If the average fractional energy loss per collision in hydrogen is 0.63, what will be the energy of a 2 MeV neutron after (a) 5 collisions? (b) 10 collisions?
Solution:
a)
b)
Fixture checkpoint marker-0594-xray is recorded on fixture page 0594.
Macroscopic Slowing Down Power
Although the logarithmic energy decrement is a convenient measure of the ability of a material to slow neutrons, it does not measure all necessary properties of a moderator. A better measure of the capabilities of a material is the macroscopic slowing down power. The macroscopic slowing down power (MSDP) is the product of the logarithmic energy decrement and the macroscopic cross section for scattering in the material. Equation (2-10) illustrates how to calculate the macroscopic slowing down power.
(2-10)
MR
s
a
Reactor Theory (Neutron Characteristics) DOE-HDBK-1019/1-93 NEUTRON MODERATION
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Moderating Ratio
Macroscopic slowing down power indicates how rapidly a neutron will slow down in the material in question, but it still does not fully explain the effectiveness of the material as a moderator. An element such as boron has a high logarithmic energy decrement and a good slowing down power, but it is a poor moderator because of its high probability of absorbing neutrons.
The most complete measure of the effectiveness of a moderator is the moderating ratio. The moderating ratio is the ratio of the macroscopic slowing down power to the macroscopic cross section for absorption. The higher the moderating ratio, the more effectively the material performs as a moderator. Equation (2-11) shows how to calculate the moderating ratio of a material.
(2-11)
Moderating properties of different materials are compared in Table 2.
TABLE 2

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Material Collisions to Slowing Down
Number of Macroscopic
Thermalize Power
Moderating
Ratio
H2O 0.927 19 1.425 62
D2O 0.510 35 0.177 4830
Helium 0.427 42 9 x 10 51 -6
Beryllium 0.207 86 0.154 126
Boron 0.171 105 0.092 0.00086
Carbon 0.158 114 0.083 216
NEUTRON MODERATION DOE-HDBK-1019/1-93 Reactor Theory (Neutron Characteristics)
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Summary
The important information in this chapter is summarized below.
Neutron Moderation Summary
& Thermalization is the process of reducing the energy level of a neutron from the energy level at which it is produced to an energy level in the thermal range.
& The moderator is the reactor material that is present for the purpose of thermalizing neutrons.
& Moderating ratio is the ratio of the macroscopic slowing down power to the macroscopic cross section for absorption.
& The average logarithmic energy decrement (!) is the average change in the logarithm of neutron energy per collision.
& Macroscopic slowing down power is the product of the average logarithmic energy decrement and the macroscopic cross section for scattering.
& There are three desirable characteristics of a moderator.
1. large scattering cross section
Fixture checkpoint marker-0595-xray is recorded on fixture page 0595.
2. small absorption cross section
3. large energy loss per collision
& The energy loss after a specified number of collisions can be calculated using the equation below.
E = E (1 - x) N o
N
■
■ ■
Reactor Theory (Neutron Characteristics) DOE-HDBK-1019/1-93 PROMPT AND DELAYED NEUTRONS
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PROMPT AND DELAYED NEUTRONS
Not all neutrons are released at the same time following fission. Most neutrons are released virtually instantaneously and are called prompt neutrons. A very small fraction of neutrons are released after the decay of fission products and are called delayed neutrons. Although delayed neutrons are a very small fraction of the total number of neutrons, they play an extremely important role in the control of the reactor.
EO 3.1 STATE the origin of prompt neutrons and delayed neutrons.
EO 3.2 STATE the approximate fraction of neutrons that are born as

a. Uranium-235
b. Plutonium-239
EO 3.3 EXPLAIN the mechanism for production of delayed neutrons.
EO 3.4 EXPLAIN prompt and delayed neutron generation times.
EO 3.5 Given prompt and delayed neutron generation times and delayed neutron fraction, CALCULATE the average generation time.
EO 3.6 EXPLAIN the effect of delayed neutrons on reactor control.
Neutron Classification
The great majority (over 99%) of the neutrons produced in fission are released within about 10-13 seconds of the actual fission event. These are called prompt neutrons. A small portion of fission neutrons are delayed neutrons, which are produced for some time after the fission process has taken place. The delayed neutrons are emitted immediately following the first beta decay of a fission fragment known as a delayed neutron precursor. An example of a delayed neutron precursor is bromine-87, shown below.
PROMPT AND DELAYED NEUTRONS DOE-HDBK-1019/1-93 Reactor Theory (Neutron Characteristics)
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For most applications, it is convenient to combine the known precursors into groups with appropriately averaged properties. These groups vary somewhat depending on the fissile material in use. Table 3 lists the characteristics for the six precursor groups resulting from thermal fission of uranium-235. The fraction of all neutrons that are produced by each of these precursors is called the delayed neutron fraction for that precursor. The total fraction of all neutrons born as delayed neutrons is called the delayed neutron fraction (■). The fraction of delayed neutrons produced varies depending on the predominant fissile nuclide in use. The delayed neutron fractions (■) for the fissile nuclides of most interest are as follows: uranium-233 (0.0026), uranium-235 (0.0065), uranium-238 (0.0148), and plutonium-239 (0.0021).
TABLE 3
Delayed Neutron Precursor Groups
for Thermal Fission in Uranium-235
Group Half-Life Delayed Neutron Average Energy
(sec) Fraction (MeV)
1 55.7 0.00021 0.25
2 22.7 0.00142 0.46
3 6.2 0.00127 0.41
4 2.3 0.00257 0.45
Fixture checkpoint marker-0596-xray is recorded on fixture page 0596.
5 0.61 0.00075 0.41
6 0.23 0.00027 -
Total - 0.0065 -
Neutron Generation Time
The neutron generation time is the time required for neutrons from one generation to cause the fissions that produce the next generation of neutrons. The generation time for prompt neutrons (5* - pronounced "ell-star") is the total time from birth to rebirth. Three time intervals are involved: (a) the time it takes a fast neutron to slow down to thermal energy, (b) the time the now thermal neutron exists prior to absorption in fuel, and (c) the time required for a fissionable nucleus to emit a fast neutron after neutron absorption.
Fast neutrons slow to thermal energies or leak out of the reactor in 10 seconds to 10 seconds, -4 -6 depending on the moderator. In water moderated reactors, thermal neutrons tend to exist for

neutron absorption in a fissionable nucleus occurs in about 10 seconds. Thus, fast reactors -13
have an 5* of about 10 seconds, while thermal reactors have an 5* of about 10 seconds + 10 -6 -6 -4
seconds, which is about 10 seconds to 10 seconds. -4 -5

■ ■ ■ ■
■ ■ ■ ■
■ ■

On the other hand, the average generation time for the six delayed neutron groups is the total time from the birth of the fast neutron to the emission of the delayed neutron. Again, three time intervals are involved: (a) the time it takes a fast neutron to slow down to thermal energy, (b) the time the thermal neutron exists prior to absorption, and (c) the average time from neutron

Fixture checkpoint marker-0597-xray is recorded on fixture page 0597.

absorption to neutron emission by the six precursor groups. The average time for decay of precursors from uranium-235 is 12.5 seconds. The other terms in the delayed neutron generation time are insignificant when compared to this value, and the average delayed neutron generation time becomes ■ 12.5 seconds.

A neutron generation time in the range of 10 seconds to 10 seconds or faster could result in -4 -5 very rapid power excursions, and control would not be possible without the dependence upon delayed neutrons to slow down the rate of the reaction. The average generation time, and hence the rate that power can rise, is determined largely by the delayed neutron generation time. The following equation shows this mathematically.

(2-12)

Example:

Assume a prompt neutron generation time for a particular reactor of 5 x 10 seconds and -5 a delayed neutron generation time of 12.5 seconds. If ■ is 0.0065, calculate the average generation time.

Solution:

This example demonstrates the effect delayed neutrons have on the neutron generation time and thus reactor control. If a reactor were to be operated in a sustained chain reaction using only prompt neutrons (■ = 0), the generation time from the previous example would be about 5 x 10 seconds. However, by operating the reactor such that a 0.0065 fraction of neutrons are -5 delayed, the generation life time is extended to 0.0813 seconds, providing time for adequate operator control. Therefore, although only a small fraction of the total neutron population, delayed neutrons are extremely important to the control and maintenance of a sustained fission chain reaction.

Summary

The important information in this chapter is summarized on the following page.

■ ■ ■ ■

Prompt and Delayed Neutrons Summary

& Prompt neutrons are released directly from fission within 10 seconds of the -13 fission event.

& Delayed neutrons are released from the decay of fission products that are called delayed neutron precursors. Delayed neutron precursors are grouped according to half-life. Half-lives vary from fractions of a second to almost a minute.

& The fraction of neutrons born as delayed neutrons is different for different fuel

Uranium-235 0.0065
Plutonium-239 0.0021
& Delayed neutrons are produced by a classification of fission products known as delayed neutron precursors. When a delayed neutron precursor undergoes a ■ decay, it results in an excited daughter nucleus which immediately ejects a - neutron. Therefore, these delayed neutrons appear with a half-life of the delayed neutron precursor.
& The delayed neutron generation time is the total time from the birth of the fast neutron to the emission of the delayed neutron in the next generation. Delayed neutron generation times are dominated by the half-life of the delayed neutron precursor. The average delayed neutron generation time is about 12.5 seconds.
& A prompt neutron generation time is the sum of the amount of time it takes a fast neutron to thermalize, the amount of time the neutron exists as a thermal neutron before it is absorbed, and the amount of time between a fissionable nuclide absorbing a neutron and fission neutrons being released. Prompt neutron generation time is about 5 x 10 seconds. -5
& The average neutron generation time can be calculated from the prompt and delayed neutron generation times and the delayed neutron fraction using Equation (2-12).
Fixture checkpoint marker-0598-xray is recorded on fixture page 0598.
& Delayed neutrons are responsible for the ability to control the rate at which power can rise in a reactor. If only prompt neutrons existed, reactor control would not be possible due to the rapid power changes.
Reactor Theory (Neutron Characteristics) DOE-HDBK-1019/1-93 NEUTRON FLUX SPECTRUM
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Figure 2 Prompt Fission Neutron Energy Spectrum for
Thermal Fission of Uranium-235
NEUTRON FLUX SPECTRUM
The number of neutrons that exist at a given energy level varies. A plot of either the fraction of neutrons or the neutron flux at a given energy versus the energy level is called a neutron energy spectrum. The neutron energy spectrum varies widely for different types of reactors.
EO 4.1 STATE the average energy at which prompt neutrons are produced.
EO 4.2 DESCRIBE the neutron energy spectrum in the following reactors:
a. Fast reactor
b. Thermal reactor
EO 4.3 EXPLAIN the reason for the particular shape of the fast, intermediate, and slow energy regions of the neutron flux spectrum for a thermal reactor.
Prompt Neutron Energies
The neutrons produced by fission are high energy neutrons, and almost all fission neutrons have energies between 0.1 MeV and 10 MeV. The neutron energy distribution, or spectrum, may best

fraction of neutrons per MeV as a function of neutron energy, as shown in Figure 2. From this figure it can be seen that the most probable neutron energy is about 0.7 MeV. Also, from this data it can be shown that the average energy of fission neutrons is about 2 MeV. Figure 2 is the neutron energy spectrum for thermal fission in uranium-235. The values will vary slightly for other nuclides.

NEUTRON FLUX SPECTRUM DOE-HDBK-1019/1-93 Reactor Theory (Neutron Characteristics)
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Figure 3 Comparison of Neutron Flux Spectra for Thermal and Fast Breeder Reactor

Thermal and Fast Breeder Reactor Neutron Spectra

The spectrum of neutron energies produced by fission varies significantly from the energy spectrum, or flux, existing in a reactor at a given time. Figure 3 illustrates the difference in neutron flux spectra between a thermal reactor and a fast breeder reactor. The energy distribution of neutrons from fission is essentially the same for both reactors, so the differences in the curve shapes may be attributed to the neutron moderation or slowing down effects. No attempt is made to thermalize or slow down neutrons in the fast breeder reactor (liquid metal cooled); therefore, an insignificant number of neutrons exist in the thermal range. For the thermal reactor (water moderated), the spectrum of neutrons in the fast region (> 0.1 MeV) has

Fixture checkpoint marker-0599-xray is recorded on fixture page 0599.

a shape similar to that for the spectrum of neutrons emitted by the fission process. In the thermal reactor, the flux in the intermediate energy region (1 eV to 0.1 MeV) has approximately a 1/E dependence. That is, if the energy (E) is halved, the flux doubles. This 1/E dependence is caused by the slowing down process, where elastic collisions remove a constant fraction of the neutron energy per collision (on the average), independent of energy; thus, the neutron loses larger amounts of energy per collision at higher energies than at lower energies. The fact that the neutrons lose a constant fraction of energy per collision causes the neutrons to tend to "pile up" at lower energies, that is, a greater number of neutrons exist at the lower energies as a result of this behavior.

vp
2 k T
m
vp
2 k T
m
2 1.38 x 10 16 erg
K
293 K
1.66 x 10 24 g
2.2 x 105 cm
sec

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100 cm
2200 m
sec
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In the thermal region the neutrons achieve a thermal equilibrium with the atoms of the moderator material. In any given collision they may gain or lose energy, and over successive collisions will gain as much energy as they lose. These thermal neutrons, even at a specific temperature, do not all have the same energy or velocity; there is a distribution of energies, usually referred to as the Maxwell distribution (e.g., Figure 2). The energies of most thermal neutrons lie close to the most probable energy, but there is a spread of neutrons above and below this value.
Most Probable Neutron Velocities
The most probable velocity (vp) of a thermal neutron is determined by the temperature of the medium and can be determined by Equation (2-13) .
(2-13)
where:
v = most probable velocity of neutron (cm/sec) p
k = Boltzman's constant (1.38 x 10 erg/ K) -16
T = absolute temperature in degrees Kelvin (K)
m = mass of neutron (1.66 x 10 grams) -24
Example:
Calculate the most probable velocities for neutrons in thermal equilibrium with their surroundings at the following temperatures. a) 20 C, b) 260 C.
Solution:
a) Calculate the most probable velocity for 20 C using Equation (2-13).
vp
2 k T
m
2 1.38 x 10 16 erg
K
533 K
1.66 x 10 24 g
2.977 x 105 cm
Fixture checkpoint marker-0600-xray is recorded on fixture page 0600.
sec
1 m
100 cm
2977 m
sec
NEUTRON FLUX SPECTRUM DOE-HDBK-1019/1-93 Reactor Theory (Neutron Characteristics) NP-02 Page 36 Rev. 0
b) Calculate the most probable velocity for 260 C using Equation (2-13).
From these calculations it is evident that the most probable velocity of a thermal neutron increases as temperature increases. The most probable velocity at 20 C is of particular importance since reference data, such as nuclear cross sections, are tabulated for a neutron velocity of 2200 meters per second.
Reactor Theory (Neutron Characteristics) DOE-HDBK-1019/1-93 NEUTRON FLUX SPECTRUM Rev. 0 Page 37 NP-02

that are associated with various DOE nuclear facility operations and maintenance.
Key Words: Training Material, Atomic Physics, The Chart of the Nuclides, Radioactivity, Radioactive Decay, Neutron Interaction, Fission, Reactor Theory, Neutron Characteristics, Neutron Life Cycle, Reactor Kinetics
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DOE-HDBK-1019/1-93
NUCLEAR PHYSICS AND REACTOR THEORY
FOREWORD
The Department of Energy (DOE) Fundamentals Handbooks consist of ten academic subjects, which include Mathematics; Classical Physics; Thermodynamics, Heat Transfer, and Fixture checkpoint marker-0601-xray is recorded on fixture page 0601.
Fluid Flow; Instrumentation and Control; Electrical Science; Material Science; Mechanical Science; Chemistry; Engineering Symbology, Prints, and Drawings; and Nuclear Physics and Reactor Theory. The handbooks are provided as an aid to DOE nuclear facility contractors.
These handbooks were first published as Reactor Operator Fundamentals Manuals in 1985 for use by DOE category A reactors. The subject areas, subject matter content, and level of detail of the Reactor Operator Fundamentals Manuals were determined from several sources. DOE Category A reactor training managers determined which materials should be included, and served as a primary reference in the initial development phase. Training guidelines from the commercial nuclear power industry, results of job and task analyses, and independent input from contractors and operations-oriented personnel were all considered and included to some degree in developing the text material and learning objectives.
The DOE Fundamentals Handbooks represent the needs of various DOE nuclear facilities' fundamental training requirements. To increase their applicability to nonreactor nuclear facilities, the Reactor Operator Fundamentals Manual learning objectives were distributed to the Nuclear Facility Training Coordination Program Steering Committee for review and comment. To update their reactor-specific content, DOE Category A reactor training managers also reviewed and commented on the content. On the basis of feedback from these sources, information that applied to two or more DOE nuclear facilities was considered generic and was included. The final draft of each of the handbooks was then reviewed by these two groups. This approach has resulted in revised modular handbooks that contain sufficient detail such that each facility may adjust the content to fit their specific needs.
Each handbook contains an abstract, a foreword, an overview, learning objectives, and text material, and is divided into modules so that content and order may be modified by individual DOE contractors to suit their specific training needs. Each handbook is supported by a separate examination bank with an answer key.
The DOE Fundamentals Handbooks have been prepared for the Assistant Secretary for Nuclear Energy, Office of Nuclear Safety Policy and Standards, by the DOE Training Coordination Program. This program is managed by EG&G Idaho, Inc.
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DOE-HDBK-1019/1-93
NUCLEAR PHYSICS AND REACTOR THEORY
OVERVIEW
The Department of Energy Fundamentals Handbook entitled Nuclear Physics and Reactor Theory was prepared as an information resource for personnel who are responsible for the operation of the Department's nuclear facilities. Almost all processes that take place in a nuclear facility involves the transfer of some type of energy. A basic understanding of nuclear physics and reactor theory is necessary for DOE nuclear facility operators, maintenance personnel, and

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information in this handbook is presented to provide a foundation for applying engineering concepts to the job. This knowledge will help personnel understand the impact that their actions may have on the safe and reliable operation of facility components and systems.
The Nuclear Physics and Reactor Theory handbook consists of four modules that are contained in two volumes. The following is a brief description of the information presented in each module of the handbook.
Volume 1 of 2
Module 1 - Atomic and Nuclear Physics
Introduces concepts of atomic physics including the atomic nature of matter, the chart of the nuclides, radioactivity and radioactive decay, neutron interactions and fission, and the interaction of radiation with matter.
Module 2 - Reactor Theory (Nuclear Parameters)
Provides information on reactor theory and neutron characteristics. Includes topics such as neutron sources, neutron flux, neutron cross sections, reaction rates, neutron moderation, and prompt and delayed neutrons.
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NUCLEAR PHYSICS AND REACTOR THEORY
OVERVIEW (Cont.)
Volume 2 of 2
Module 3 - Reactor Theory (Nuclear Parameters)
Explains the nuclear parameters associated with reactor theory. Topics include the neutron life cycle, reactivity and reactivity coefficients, neutron poisons, and control rods.
Module 4 - Reactor Theory (Reactor Operations)
Introduces the reactor operations aspect of reactor theory. Topics include subcritical multiplication, reactor kinetics, and reactor operation.
The information contained in this handbook is not all-encompassing. An attempt to present the entire subject of nuclear physics and reactor theory would be impractical. However, the Nuclear Physics and Reactor Theory handbook presents enough information to provide the reader with the fundamental knowledge necessary to understand the advanced theoretical concepts presented in other subject areas, and to understand basic system and equipment operation.
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Department of Energy
Fundamentals Handbook
NUCLEAR PHYSICS
AND REACTOR THEORY
Module 1
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Fixture checkpoint marker-0605-xray is recorded on fixture page 0605.
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TERMINAL OBJECTIVE
1.0 Given sufficient information, DESCRIBE atoms, including components, structure, and nomenclature.
ENABLING OBJECTIVES
1.1 STATE the characteristics of the following atomic particles, including mass, charge, and

a. Proton
b. Neutron
c. Electron
1.2 DESCRIBE the Bohr model of an atom.
1.3 DEFINE the following terms:
a. Nuclide c. Atomic number
b. Isotope d. Mass number
1.4 Given the standard A
ZX notation for a particular nuclide, DETERMINE the following:
a. Number of protons
b. Number of neutrons
c. Number of electrons
1.5 DESCRIBE the three forces that act on particles within the nucleus and affect the stability of the nucleus.
1.6 DEFINE the following terms:
Fixture checkpoint marker-0606-xray is recorded on fixture page 0606.
a. Enriched uranium
b. Depleted uranium
1.7 DEFINE the following terms:
a. Mass defect
b. Binding energy
1.8 Given the atomic mass for a nuclide and the atomic masses of a neutron, proton, and electron, CALCULATE the mass defect and binding energy of the nuclide.
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OBJECTIVES DOE-HDBK-1019/1-93 Atomic and Nuclear Physics
TERMINAL OBJECTIVE
2.0 Given necessary references, DESCRIBE the various modes of radioactive decay.
ENABLING OBJECTIVES
2.1 DESCRIBE the following processes:
a. Alpha decay d. Electron capture
b. Beta-minus decay e. Internal conversions
c. Beta-plus decay f. Isomeric transitions
2.2 Given a Chart of the Nuclides, WRITE the radioactive decay chain for a nuclide.
2.3 EXPLAIN why one or more gamma rays typically accompany particle emission.
2.4 Given the stability curve on the Chart of the Nuclides, DETERMINE the type of radioactive decay that the nuclides in each region of the chart will typically undergo.
2.5 DEFINE the following terms:
a. Radioactivity d. Radioactive decay constant
b. Curie e. Radioactive half-life
c. Becquerel
2.6 Given the number of atoms and either the half-life or decay constant of a nuclide, CALCULATE the activity.
2.7 Given the initial activity and the decay constant of a nuclide, CALCULATE the activity at any later time.
2.8 CONVERT between the half-life and decay constant for a nuclide.
2.9 Given the Chart of the Nuclides and the original activity, PLOT the radioactive decay curve for a nuclide on either linear or semi-log coordinates.
2.10 DEFINE the following terms:

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b. Transient radioactive equilibrium
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Atomic and Nuclear Physics DOE-HDBK-1019/1-93 OBJECTIVES
TERMINAL OBJECTIVE
3.0 Without references, DESCRIBE the different nuclear interactions initiated by neutrons.
ENABLING OBJECTIVES
3.1 DESCRIBE the following scattering interactions between a neutron and a nucleus:
a. Elastic scattering
b. Inelastic scattering
3.2 STATE the conservation laws that apply to an elastic collision between a neutron and a nucleus.
3.3 DESCRIBE the following reactions where a neutron is absorbed in a nucleus:
a. Radiative capture
b. Particle ejection
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OBJECTIVES DOE-HDBK-1019/1-93 Atomic and Nuclear Physics
TERMINAL OBJECTIVE
4.0 Without references, DESCRIBE the fission process.
ENABLING OBJECTIVES
4.1 EXPLAIN the fission process using the liquid drop model of a nucleus.
4.2 DEFINE the following terms:
a. Excitation energy
Fixture checkpoint marker-0607-xray is recorded on fixture page 0607.
b. Critical energy
4.3 DEFINE the following terms:
a. Fissile material
b. Fissionable material
c. Fertile material
4.4 DESCRIBE the processes of transmutation, conversion, and breeding.
4.5 DESCRIBE the curve of Binding Energy per Nucleon versus mass number and give a qualitative description of the reasons for its shape.
4.6 EXPLAIN why only the heaviest nuclei are easily fissioned.
4.7 EXPLAIN why uranium-235 fissions with thermal neutrons and uranium-238 fissions only with fast neutrons.
4.8 CHARACTERIZE the fission products in terms of mass groupings and radioactivity.
4.9 Given the nuclides involved and their masses, CALCULATE the energy released from fission.
4.10 Given the curve of Binding Energy per Nucleon versus mass number, CALCULATE the energy released from fission.
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Atomic and Nuclear Physics DOE-HDBK-1019/1-93 OBJECTIVES
TERMINAL OBJECTIVE
5.0 Without references, DESCRIBE how the various types of radiation interact with matter.
ENABLING OBJECTIVES
5.1 DESCRIBE interactions of the following with matter:
a. Alpha particle c. Positron
b. Beta particle d. Neutron
5.2 DESCRIBE the following ways that gamma radiation interacts with matter:

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b. Photoelectric effect
c. Pair production
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OBJECTIVES DOE-HDBK-1019/1-93 Atomic and Nuclear Physics
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Atomic and Nuclear Physics DOE-HDBK-1019/1-93 ATOMIC NATURE OF MATTER
ATOMIC NATURE OF MATTER
All matter is composed of atoms. The atom is the smallest amount of matter that retains the properties of an element. Atoms themselves are composed of smaller particles, but these smaller particles no longer have the same properties as the overall element.
EO 1.1 STATE the characteristics of the following atomic particles, including mass, charge, and location within the atom:
a. Proton
b. Neutron
c. Electron
EO 1.2 DESCRIBE the Bohr model of an atom.
EO 1.3 DEFINE the following terms:
a. Nuclide c. Atomic number
b. Isotope d. Mass number
EO 1.4 Given the standard A ZX notation for a particular nuclide, DETERMINE the following:
a. Number of protons
b. Number of neutrons
c. Number of electrons
EO 1.5 DESCRIBE the three forces that act on particles within the nucleus and affect the stability of the nucleus.
Fixture checkpoint marker-0608-xray is recorded on fixture page 0608.
Structure of Matter
Early Greek philosophers speculated that the earth was made up of different combinations of basic substances, or elements. They considered these basic elements to be earth, air, water, and fire. Modern science shows that the early Greeks held the correct concept that matter consists of a combination of basic elements, but they incorrectly identified the elements.
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ATOMIC NATURE OF MATTER DOE-HDBK-1019/1-93 Atomic and Nuclear Physics
In 1661 the English chemist Robert Boyle published the modern criterion for an element. He defined an element to be a basic substance that cannot be broken down into any simpler substance after it is isolated from a compound, but can be combined with other elements to form compounds. To date, 105 different elements have been confirmed to exist, and researchers claim to have discovered three additional elements. Of the 105 confirmed elements, 90 exist in nature and 15 are man-made.
Another basic concept of matter that the Greeks debated was whether matter was continuous or discrete. That is, whether matter could be continuously divided and subdivided into ever smaller particles or whether eventually an indivisible particle would be encountered. Democritus in about 450 B.C. argued that substances were ultimately composed of small, indivisible particles that he labeled atoms. He further suggested that different substances were composed of different atoms

the atoms. It was impossible to conclusively prove or disprove this proposal for more than 2000 years.

The modern proof for the atomic nature of matter was first proposed by the English chemist John Dalton in 1803. Dalton stated that each chemical element possesses a particular kind of atom, and any quantity of the element is made up of identical atoms of this kind. What distinguishes one element from another element is the kind of atom of which it consists, and the basic physical difference between kinds of atoms is their weight.

Subatomic Particles

For almost 100 years after Dalton established the atomic nature of atoms, it was considered impossible to divide the atom into even smaller parts. All of the results of chemical experiments during this time indicated that the atom was indivisible. Eventually, experimentation into electricity and radioactivity indicated that particles of matter smaller than the atom did indeed exist. In 1906, J. J. Thompson won the Nobel Prize in physics for establishing the existence of electrons. Electrons are negatively-charged particles that have 1/1835 the mass of the hydrogen atom. Soon after the discovery of electrons, protons were discovered. Protons are relatively large particles that have almost the same mass as a hydrogen atom and a positive charge equal in magnitude (but opposite in sign) to that of the electron. The third subatomic particle to be discovered, the neutron, was not found until 1932. The neutron has almost the same mass as the proton, but it is electrically neutral.

Bohr Model of the Atom

The British physicist Ernest Rutherford postulated that the positive charge in an atom is concentrated in a small region called a nucleus at the center of the atom with electrons existing in orbits around it. Niels Bohr, coupling Rutherford's postulation with the quantum theory introduced by Max Planck, proposed that the atom consists of a dense nucleus of protons surrounded by electrons traveling in discrete orbits at fixed distances from the nucleus. An electron in one of these orbits or shells has a specific or discrete quantity of energy (quantum). When an electron moves from one allowed orbit to another allowed orbit, the energy difference between the two states is emitted or absorbed in the form of a single quantum of radiant energy called a photon. Figure 1 is Bohr's model of the hydrogen atom showing an electron as having just dropped from the third shell to the first shell with the emission of a photon that has an energy = hv. (h = Planck's constant = 6.63 x 10-34 J-s and v = frequency of the photon.) Bohr's theory was the first to successfully account for the discrete energy levels of this radiation as measured in the laboratory. Although Bohr's atomic model is designed specifically to explain the hydrogen atom, his theories apply generally to the structure of all atoms. Additional

Fixture checkpoint marker-0609-xray is recorded on fixture page 0609.

information on electron shell theory can be found in the Chemistry Fundamentals Handbook.

Figure 1 Bohr's Model of the Hydrogen Atom

Properties of the three subatomic particles are listed in Table 1.

TABLE 1

Properties of Subatomic Particles

Particle	Location	Charge	Mass
Neutron	Nucleus	none	1.008665 amu
Proton	Nucleus	+1	1.007277 amu
Electron	Shells around nucleus	-1	0.0005486 amu

SYNTHETIC FIXTURE PAGE 0610 OF 1200

The size and mass of atoms are so small that the use of normal measuring units, while possible, is often inconvenient. Units of measure have been defined for mass and energy on the atomic scale to make measurements more convenient to express. The unit of measure for mass is the atomic mass unit (amu). One atomic mass unit is equal to 1.66 x 10-24 grams. The reason for this particular value for the atomic mass unit will be discussed in a later chapter. Note from Table 1 that the mass of a neutron and a proton are both about 1 amu. The unit for energy is the electron volt (eV). The electron volt is the amount of energy acquired by a single electron when it falls through a potential difference of one volt. One electron volt is equivalent to 1.602 x 10-19 joules or 1.18 x 10-19 foot-pounds.

Nuclides

The total number of protons in the nucleus of an atom is called the atomic number of the atom and is given the symbol Z. The number of electrons in an electrically-neutral atom is the same as the number of protons in the nucleus. The number of neutrons in a nucleus is known as the Fixture checkpoint marker-0610-xray is recorded on fixture page 0610.

neutron number and is given the symbol N. The mass number of the nucleus is the total number of nucleons, that is, protons and neutrons in the nucleus. The mass number is given the symbol A and can be found by the equation Z + N = A.

Each of the chemical elements has a unique atomic number because the atoms of different elements contain a different number of protons. The atomic number of an atom identifies the particular element.

Each type of atom that contains a unique combination of protons and neutrons is called a nuclide. Not all combinations of numbers of protons and neutrons are possible, but about 2500 specific nuclides with unique combinations of neutrons and protons have been identified. Each nuclide is denoted by the chemical symbol of the element with the atomic number written as a subscript and the mass number written as a superscript, as shown in Figure 2. Because each element has a unique name, chemical symbol, and atomic number, only one of the three is necessary to identify the element. For this reason nuclides can also be identified by either the chemical name or the chemical symbol followed by the mass number (for example, U-235 or uranium-235).

Figure 2 Nomenclature for Identifying Nuclides

Another common format is to use the abbreviation of the chemical element with the mass number superscripted (for example, 235U). In this handbook the format used in the text will usually be the element's name followed by the mass number. In equations and tables, the format in Figure 2 will usually be used.

Example:

State the name of the element and the number of protons, electrons, and neutrons in the nuclides listed below.

1

1H

10

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14
7N
11
4
4
8Cd
23
9
9
4Pu
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Solution:
The name of the element can be found from the Periodic Table (refer to Chemistry Fundamentals Handbook) or the Chart of the Nuclides (to be discussed later). The number of protons and electrons are equal to Z. The number of neutrons is equal to Z - A.
Nuclide Element Protons Electrons Neutrons
1
1H hydrogen 1 1 0
Fixture checkpoint marker-0611-xray is recorded on fixture page 0611.
10
5B boron 5 5 5
14
7N nitrogen 7 7 7
11
4
4
8Cd cadmium 48 48 66
23
9
9
4Pu plutonium 94 94 145
Isotopes
Isotopes are nuclides that have the same atomic number and are therefore the same element, but differ in the number of neutrons. Most elements have a few stable isotopes and several unstable, radioactive isotopes. For example, oxygen has three stable isotopes that can be found in nature (oxygen-16, oxygen-17, and oxygen-18) and eight radioactive isotopes. Another example is hydrogen, which has two stable isotopes (hydrogen-1 and hydrogen-2) and a single radioactive isotope (hydrogen-3).
The isotopes of hydrogen are unique in that they are each commonly referred to by a unique name instead of the common chemical element name. Hydrogen-1 is almost always referred to as hydrogen, but the term protium is infrequently used also. Hydrogen-2 is commonly called deuterium and symbolized 2
1D. Hydrogen-3 is commonly called tritium and symbolized 31T. This text will normally use the symbology 2
1H and 31H for deuterium and tritium, respectively.
Atomic and Nuclear Radii

by the electrons moving in their various orbitals, does not have a distinct outer edge. A reasonable measure of atomic size is given by the average distance of the outermost electron from the nucleus. Except for a few of the lightest atoms, the average atomic radii are approximately the same for all atoms, about 2 x 10 -8 cm.

Like the atom the nucleus does not have a sharp outer boundary. Experiments have shown that the nucleus is shaped like a sphere with a radius that depends on the atomic mass number of the atom. The relationship between the atomic mass number and the radius of the nucleus is shown in the following equation.

r = (1.25 x 10 -13 cm) A1/3

where:

r = radius of the nucleus (cm)

A = atomic mass number (dimensionless)

The values of the nuclear radii for some light, intermediate, and heavy nuclides are shown in Table 2.

TABLE 2

Calculated Values for Nuclear Radii

Nuclide	Radius of Nucleus
1 1H	1.25 x 10 -13 cm
10 5B	2.69 x 10 -13 cm
5 2 6	Fixture checkpoint marker-0612-xray is recorded on fixture page 0612.
6Fe	4.78 x 10 -13 cm
17 7 8 2Hf	7.01 x 10 -13 cm
23 9 8 2U	7.74 x 10 -13 cm
25 9 2 8Cf	7.89 x 10 -13 cm

From the table, it is clear that the radius of a typical atom (e.g. 2 x 10 -8 cm) is more than 25,000 times larger than the radius of the largest nucleus.

Nuclear Forces

In the Bohr model of the atom, the nucleus consists of positively-charged protons and electrically■neutra as nucleons. One problem that the Bohr model of the atom presented was accounting for an attractive force to overcome the repulsive force between protons.

Two forces present in the nucleus are (1) electrostatic forces between charged particles and (2)

magnitude of the gravitational force and electrostatic force based upon principles from classical physics.
Rev. 0 Page 7 NP-01
Fg
G m1 m2
r
2
Fe
K Q1 Q2
r
2
ATOMIC NATURE OF MATTER DOE-HDBK-1019/1-93 Atomic and Nuclear Physics
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Newton stated that the gravitational force between two bodies is directly proportional to the masses of the two bodies and inversely proportional to the square of the distance between the bodies. This relationship is shown in the equation below.
where:
F = gravitational force (newtons) g
m = mass of first body (kilograms) 1
m = mass of second body (kilograms) 2
G = gravitational constant (6.67 x 10 N-m /kg)
-11 2 2
r = distance between particles (meters)
The equation illustrates that the larger the masses of the objects or the smaller the distance between the objects, the greater the gravitational force. So even though the masses of nucleons are very small, the fact that the distance between nucleons is extremely short may make the gravitational force significant. It is necessary to calculate the value for the gravitational force and compare it to the value for other forces to determine the significance of the gravitational force in the nucleus. The gravitational force between two protons that are separated by a distance of 10 meters is about 10 newtons. -20 -24
Coulomb's Law can be used to calculate the force between two protons. The electrostatic force is directly proportional to the electrical charges of the two particles and inversely proportional to the square of the distance between the particles. Coulomb's Law is stated as the following equation.
Fixture checkpoint marker-0613-xray is recorded on fixture page 0613.
where:
F = electrostatic force (newtons) e
K = electrostatic constant (9.0 x 10 N-m /C)
9 2 2
Q = charge of first particle (coulombs) 1
Q = charge of second particle (coulombs) 2
r = distance between particles (meters)
Using this equation, the electrostatic force between two protons that are separated by a distance of 10 meters is about 10 newtons. Comparing this result with the calculation of the
-20 12
gravitational force (10 newtons) shows that the gravitational force is so small that it can be
-24
neglected.

If only the electrostatic and gravitational forces existed in the nucleus, then it would be impossible to have stable nuclei composed of protons and neutrons. The gravitational forces are much too small to hold the nucleons together compared to the electrostatic forces repelling the protons. Since stable atoms of neutrons and protons do exist, there must be another attractive force acting within the nucleus. This force is called the nuclear force.

The nuclear force is a strong attractive force that is independent of charge. It acts equally only between pairs of neutrons, pairs of protons, or a neutron and a proton. The nuclear force has a very short range; it acts only over distances approximately equal to the diameter of the nucleus (10 -13 cm). The attractive nuclear force between all nucleons drops off with distance much faster than the repulsive electrostatic force between protons.

TABLE 3

Fixture checkpoint marker-0614-xray is recorded on fixture page 0614.

Forces Acting in the Nucleus

Force	Interaction Range
Gravitational	Very weak attractive force between all nucleons
Relatively long	
Electrostatic	Strong repulsive force between like charged particles (protons)
Relatively long	
Nuclear Force	Strong attractive force between all nucleons
Extremely short	

In stable atoms, the attractive and repulsive forces in the nucleus balance. If the forces do not balance, the atom cannot be stable, and the nucleus will emit radiation in an attempt to achieve a more stable configuration.

Summary

The important information in this chapter is summarized on the following page.

Atomic Nature of Matter Summary

Atoms consist of three basic subatomic particles. These particles are the proton, the neutron, and the electron.

Protons are particles that have a positive charge, have about the same mass as a hydrogen atom, and exist in the nucleus of an atom.

Neutrons are particles that have no electrical charge, have about the same mass as a hydrogen atom, and exist in the nucleus of an atom.

Electrons are particles that have a negative charge, have a mass about eighteen hundred times smaller than the mass of a hydrogen atom, and exist in orbital shells around the nucleus of an atom.

The Bohr model of the atom consists of a dense nucleus of protons and neutrons (nucleons) surrounded by electrons traveling in discrete orbits at fixed distances from the nucleus.

Nuclides are atoms that contain a particular number of protons and neutrons.

Isotopes are nuclides that have the same atomic number and are therefore the same element, but differ in the number of neutrons.

The atomic number of an atom is the number of protons in the nucleus.

The mass number of an atom is the total number of nucleons (protons and neutrons) in the nucleus.

The notation $^{A}_{Z}X$ is used to identify a specific nuclide. "Z" represents the atomic

number, which is equal to the number of nucleons. "X" represents the chemical symbol of the element.
Number of protons = Z
Number of electrons = Z
Number of neutrons = A - Z
The stability of a nucleus is determined by the different forces interacting within it. The electrostatic force is a relatively long-range, strong, repulsive force that acts between the positively charged protons. The nuclear force is a relatively short-range attractive force between all nucleons. The gravitational force the long range, relatively weak attraction between masses, is negligible compared to the other forces.

CHART OF THE NUCLIDES
The Chart of the Nuclides, like the Periodic Table, is a convenient format for presenting a large amount of scientific information in an organized manner.
EO 1.6 DEFINE the following terms:
Fixture checkpoint marker-0615-xray is recorded on fixture page 0615.
a. Enriched uranium
b. Depleted uranium
Chart of the Nuclides
A tabulated chart called the Chart of the Nuclides lists the stable and unstable nuclides in addition to pertinent information about each one. Figure 3 shows a small portion of a typical chart. This chart plots a box for each individual nuclide, with the number of protons (Z) on the vertical axis and the number of neutrons (N = A - Z) on the horizontal axis.
The completely gray squares indicate stable isotopes. Those in white squares are artificially radioactive, meaning that they are produced by artificial techniques and do not occur naturally. By consulting a complete chart, other types of isotopes can be found, such as naturally occurring radioactive types (but none are found in the region of the chart that is illustrated in Figure 3).
Located in the box on the far left of each horizontal row is general information about the element. The box contains the chemical symbol of the element in addition to the average atomic weight of the naturally occurring substance and the average thermal neutron absorption cross section, which will be discussed in a later module. The known isotopes (elements with the same atomic number Z but different mass number A) of each element are listed to the right.

Figure 3 Nuclide Chart for Atomic Numbers 1 to 6

Information for Stable Nuclides
For the stable isotopes, in addition to the symbol and the atomic mass number, the number percentage of each isotope in the naturally occurring element is listed, as well as the thermal neutron activation cross section and the mass in atomic mass units (amu). A typical block for a stable nuclide from the Chart of the Nuclides is shown in Figure 4.
Figure 4 Stable Nuclides
Information for Unstable Nuclides
For unstable isotopes the additional information includes the half life, the mode of decay (for

, α), the total disintegration energy in MeV (million electron volts), and the mass in amu when available. A typical block for an unstable nuclide from the Chart of the Nuclides is shown in Figure 5.

Figure 5 Unstable Nuclides

Neutron - Proton Ratios

Figure 6 shows the distribution of the stable nuclides plotted on the same axes as the Chart of the Nuclides. As the mass numbers become higher, the ratio of neutrons to protons in the nucleus becomes larger. For helium-4 (2 protons and 2 neutrons) and oxygen-16 (8 protons and 8 neutrons) this ratio is unity. For indium-115 (49 protons and 66 neutrons) the ratio of neutrons to protons has increased to 1.35, and for uranium-238 (92 protons and 146 neutrons) the neutron■to-proton

Figure 6 Neutron - Proton Plot of the Stable Nuclides

If a heavy nucleus were to split into two fragments, each fragment would form a nucleus that would have approximately the same neutron-to-proton ratio as the heavy nucleus. This high neutron-to-proton ratio places the fragments below and to the right of the stability curve displayed by Figure 6. The instability caused by this excess of neutrons is generally rectified by successive beta emissions, each of which converts a neutron to a proton and moves the nucleus toward a more stable neutron-to-proton ratio.

■

Natural Abundance of Isotopes

The relative abundance of an isotope in nature compared to other isotopes of the same element

Fixture checkpoint marker-0616-xray is recorded on fixture page 0616.

is relatively constant. The Chart of the Nuclides presents the relative abundance of the naturally occurring isotopes of an element in units of atom percent. Atom percent is the percentage of the atoms of an element that are of a particular isotope. Atom percent is abbreviated as a/o. For example, if a cup of water contains 8.23 x 10 atoms of oxygen, and the isotopic abundance 24 of oxygen-18 is 0.20%, then there are 1.65 x 10 atoms of oxygen-18 in the cup. 22

The atomic weight for an element is defined as the average atomic weight of the isotopes of the element. The atomic weight for an element can be calculated by summing the products of the isotopic abundance of the isotope with the atomic mass of the isotope.

Example:

Calculate the atomic weight for the element lithium. Lithium-6 has an atom percent abundance of 7.5% and an atomic mass of 6.015122 amu. Lithium-7 has an atomic abundance of 92.5% and an atomic mass of 7.016003 amu.

Solution:

The other common measurement of isotopic abundance is weight percent (w/o). Weight percent is the percent weight of an element that is a particular isotope. For example, if a sample of material contained 100 kg of uranium that was 28 w/o uranium-235, then 28 kg of uranium-235 was present in the sample.

Enriched and Depleted Uranium

Natural uranium mined from the earth contains the isotopes uranium-238, uranium-235 and uranium-234. The majority (99.2745%) of all the atoms in natural uranium are uranium-238. Most of the remaining atoms (0.72%) are uranium-235, and a slight trace (0.0055%) are uranium-234. Although all isotopes of uranium have similar chemical properties, each of the

modules, the isotope uranium-235 is usually the desired material for use in reactors.
A vast amount of equipment and energy are expended in processes that separate the isotopes of uranium (and other elements). The details of these processes are beyond the scope of this module. These processes are called enrichment processes because they selectively increase the proportion of a particular isotope. The enrichment process typically starts with feed material that has the proportion of isotopes that occur naturally. The process results in two types of
CHART OF THE NUCLIDES DOE-HDBK-1019/1-93 Atomic and Nuclear Physics
In the case of uranium, the natural uranium ore is 0.72 a/o uranium-235. The desired outcome of the enrichment process is to produce enriched uranium. Enriched uranium is defined as uranium in which the isotope uranium-235 has a concentration greater than its natural value. The enrichment process will also result in the byproduct of depleted uranium. Depleted uranium is defined as uranium in which the isotope uranium-235 has a concentration less than its natural value. Although depleted uranium is referred to as a by-product of the enrichment process, it does have uses in the nuclear field and in commercial and defense industries.
Summary
The important information in this chapter is summarized below.
Chart of the Nuclides Summary
Enriched uranium is uranium in which the isotope uranium-235 has a concentration greater than its natural value of 0.7%.
Depleted uranium is uranium in which the isotope uranium-235 has a concentration less than its natural value of 0.7%.
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Atomic and Nuclear Physics DOE-HDBK-1019/1-93 MASS DEFECT AND BINDING ENERGY
MASS DEFECT AND BINDING ENERGY
The separate laws of Conservation of Mass and Conservation of Energy are not applied strictly on the nuclear level. It is possible to convert between mass and energy. Instead of two separate conservation laws, a single conservation law states that the sum of mass and energy is conserved. Mass does not magically appear and disappear at random. A decrease in mass will be accompanied by a corresponding increase in energy and vice versa.
EO 1.7 DEFINE the following terms:
a. Mass defect
Fixture checkpoint marker-0617-xray is recorded on fixture page 0617.
b. Binding energy
EO 1.8 Given the atomic mass for a nuclide and the atomic masses of a neutron, proton, and electron, CALCULATE the mass defect and binding energy of the nuclide.
Mass Defect
Careful measurements have shown that the mass of a particular atom is always slightly less than the sum of the masses of the individual neutrons, protons, and electrons of which the atom consists. The difference between the mass of the atom and the sum of the masses of its parts is called the mass defect (Δm). The mass defect can be calculated using Equation (1-1). In calculating the mass defect it is important to use the full accuracy of mass measurements because the difference in mass is small compared to the mass of the atom. Rounding off the masses of atoms and particles to three or four significant digits prior to the calculation will result in a calculated mass defect of zero.

$$\Delta m = [\, Z(m_p + m_e) + (A-Z)m_n \,] - m_{atom} \quad (1\text{-}1)$$

where:

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mp = mass of a proton (1.007277 amu)
mn = mass of a neutron (1.008665 amu)
me = mass of an electron (0.000548597 amu)
matom = mass of nuclide A
ZX (amu)
Z = atomic number (number of protons)
A = mass number (number of nucleons)
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MASS DEFECT AND BINDING ENERGY DOE-HDBK-1019/1-93 Atomic and Nuclear Physics
Example:
Calculate the mass defect for lithium-7. The mass of lithium-7 is 7.016003 amu.
Solution:
∆ m Z m
p
me A Z mn matom
∆ m 3 1.007826 amu 7 3 1.008665 amu 7.016003 amu
∆ m 0.0421335 amu
Binding Energy
The loss in mass, or mass defect, is due to the conversion of mass to binding energy when the nucleus is formed. Binding energy is defined as the amount of energy that must be supplied to a nucleus to completely separate its nuclear particles (nucleons). It can also be understood as the amount of energy that would be released if the nucleus was formed from the separate particles. Binding energy is the energy equivalent of the mass defect. Since the mass defect was converted to binding energy (BE) when the nucleus was formed, it is possible to calculate the binding energy using a conversion factor derived by the mass-energy relationship from Einstein's Theory of Relativity.
Einstein's famous equation relating mass and energy is E = mc2 where c is the velocity of light (c = 2.998 x 108 m/sec). The energy equivalent of 1 amu can be determined by inserting this quantity of mass into Einstein's equation and applying conversion factors.
E m c 2
1 amu
1.6606 x 10 27 kg
1 amu
2.998 x 108 m
sec
2
1 N
1
kg m
Fixture checkpoint marker-0618-xray is recorded on fixture page 0618.
sec2
1 J
1 N m
1.4924 x 10 10 J 1 MeV
1.6022 x 10 13 J
931.5 MeV
Conversion Factors:
1 amu = 1.6606 x 10 -27 kg

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1 joule = 1 newton-meter
1 MeV = 1.6022 x 10-13 joules
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Atomic and Nuclear Physics DOE-HDBK-1019/1-93 MASS DEFECT AND BINDING ENERGY
Since 1 amu is equivalent to 931.5 MeV of energy, the binding energy can be calculated using Equation (1-2).
(1-2)
B.E. ∆m
931.5 MeV
1 amu
Example:
Calculate the mass defect and binding energy for uranium-235. One uranium-235 atom has a mass of 235.043924 amu.
Solution:
Step 1: Calculate the mass defect using Equation (1-1).
∆m Z m
Fixture checkpoint marker-0619-xray is recorded on fixture page 0619.
p
me A Z mn matom
∆m 92 1.007826 amu 235 92 1.008665 amu 235.043924 amu
∆m 1.91517 amu
Step 2: Use the mass defect and Equation (1-2) to calculate the binding energy.
B.E. ∆m
931.5 MeV
1 amu
1.91517 amu
931.5 MeV
1 amu
1784 MeV
Energy Levels of Atoms
The electrons that circle the nucleus move in fairly well-defined orbits. Some of these electrons are more tightly bound in the atom than others. For example, only 7.38 eV is required to remove the outermost electron from a lead atom, while 88,000 eV is required to remove the innermost electron. The process of removing an electron from an atom is called ionization, and the energy required to remove the electron is called the ionization energy.
In a neutral atom (number of electrons = Z) it is possible for the electrons to be in a variety of different orbits, each with a different energy level. The state of lowest energy is the one in which the atom is normally found and is called the ground state. When the atom possesses more energy than its ground state energy, it is said to be in an excited state.
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MASS DEFECT AND BINDING ENERGY DOE-HDBK-1019/1-93 Atomic and Nuclear Physics
An atom cannot stay in the excited state for an indefinite period of time. An excited atom will eventually transition to either a lower-energy excited state, or directly to its ground state, by emitting a discrete bundle of electromagnetic energy called an x-ray. The energy of the x-ray will be equal to the difference between the energy levels of the atom and will typically range from several eV to 100,000 eV in magnitude.
Energy Levels of the Nucleus
The nucleons in the nucleus of an atom, like the electrons that circle the nucleus, exist in shells

understood than those of the electrons. There is a state of lowest energy (the ground state) and discrete possible excited states for a nucleus. Where the discrete energy states for the electrons of an atom are measured in eV or keV, the energy levels of the nucleus are considerably greater and typically measured in MeV.

A nucleus that is in the excited state will not remain at that energy level for an indefinite period. Like the electrons in an excited atom, the nucleons in an excited nucleus will transition towards their lowest energy configuration and in doing so emit a discrete bundle of electromagnetic radiation called a gamma ray (γ-ray). The only differences between x-rays and γ-rays are their energy levels and whether they are emitted from the electron shell or from the nucleus.

The ground state and the excited states of a nucleus can be depicted in a nuclear energy-level diagram. The nuclear energy-level diagram consists of a stack of horizontal bars, one bar for each of the excited states of the nucleus. The vertical distance between the bar representing an excited state and the bar representing the ground state is proportional to the energy level of the excited state with respect to the ground state. This difference in energy between the ground state and the excited state is called the excitation energy of the excited state. The ground state of a nuclide has zero excitation energy. The bars for the excited states are labeled with their respective energy levels. Figure 7 is the energy level diagram for nickel-60.

Figure 7 Energy Level Diagram - Nickel-60

Fixture checkpoint marker-0620-xray is recorded on fixture page 0620.

Summary

The important information in this chapter is summarized below.

Mass Defect and Binding Energy Summary

Mass defect is the difference between the mass of the atom and the sum of the masses of its constituent parts.

Binding energy is the amount of energy that must be supplied to a nucleus to completely separate its nuclear particles. Binding energy is the energy equivalent of the mass defect.

Mass defect can be calculated by using the equation below.

$\Delta m = [Z(mp + me) + (A-Z)mn] - matom$

Binding energy can be calculated by multiplying the mass defect by the factor of 931.5 MeV per amu.

MODES OF RADIOACTIVE DECAY

Most atoms found in nature are stable and do not emit particles or energy that change form over time. Some atoms, however, do not have stable nuclei. These

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EO 2.1 DESCRIBE the following processes:
a. Alpha decay d. Electron capture
b. Beta-minus decay e. Internal conversions
c. Beta-plus decay f. Isomeric transitions
EO 2.2 Given a Chart of the Nuclides, WRITE the radioactive decay chain for a nuclide.
EO 2.3 EXPLAIN why one or more gamma rays typically accompany particle emission.
EO 2.4 Given the stability curve on the Chart of the Nuclides, DETERMINE the type of radioactive decay that the nuclides in each region of the chart will typically undergo.
Stability of Nuclei
As mass numbers become larger, the ratio of neutrons to protons in the nucleus becomes larger for the stable nuclei. Non-stable nuclei may have an excess or deficiency of neutrons and undergo a transformation process known as beta (β) decay. Non-stable nuclei can also undergo a variety of other processes such as alpha (α) or neutron (n) decay. As a result of these decay processes, the final nucleus is in a more stable or more tightly bound configuration.
Natural Radioactivity
In 1896, the French physicist Becquerel discovered that crystals of a uranium salt emitted rays that were similar to x-rays in that they were highly penetrating, could affect a photographic plate, and induced electrical conductivity in gases. Becquerel's discovery was followed in 1898 by the identification of two other radioactive elements, polonium and radium, by Pierre and Marie Curie.
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Heavy elements, such as uranium or thorium, and their unstable decay chain elements emit radiation in their naturally occurring state. Uranium and thorium, present since their creation at the beginning of geological time, have an extremely slow rate of decay. All naturally occurring nuclides with atomic numbers greater than 82 are radioactive.
Nuclear Decay
Fixture checkpoint marker-0621-xray is recorded on fixture page 0621.
Whenever a nucleus can attain a more stable (i.e., more tightly bound) configuration by emitting radiation, a spontaneous disintegration process known as radioactive decay or nuclear decay may occur. In practice, this "radiation" may be electromagnetic radiation, particles, or both.
Detailed studies of radioactive decay and nuclear reaction processes have led to the formulation of useful conservation principles. The four principles of most interest in this module are discussed below.
Conservation of electric charge implies that charges are neither created nor destroyed. Single positive and negative charges may, however, neutralize each other. It is also possible for a neutral particle to produce one charge of each sign.
Conservation of mass number does not allow a net change in the number of nucleons. However, the conversion of a proton to a neutron and vice versa is allowed.
Conservation of mass and energy implies that the total of the kinetic energy and the energy equivalent of the mass in a system must be conserved in all decays and reactions. Mass can be converted to energy and energy can be converted to mass, but the sum of mass and energy must be constant.
Conservation of momentum is responsible for the distribution of the available kinetic energy among product nuclei, particles, and/or radiation. The total amount is the same before and after

and/or particles.
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■ ■ ■ ■ ■ ■
■ ■ ■ ■ ■
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Alpha Decay (■)
Alpha decay is the emission of alpha particles (helium nuclei) which may be represented as either
He or ■. When an unstable nucleus ejects an alpha particle, the atomic number is reduced by 2 4 4
2 2
and the mass number decreased by 4. An example is uranium-234 which decays by the ejection of an alpha particle accompanied by the emission of a 0.068 MeV gamma.
The combined kinetic energy of the daughter nucleus (Thorium-230) and the ■ particle is designated as KE. The sum of the KE and the gamma energy is equal to the difference in mass between the original nucleus (Uranium-234) and the final particles (equivalent to the binding energy released, since ■m = BE). The alpha particle will carry off as much as 98% of the kinetic energy and, in most cases, can be considered to carry off all the kinetic energy.
Beta Decay (■)
Beta decay is the emission of electrons of nuclear rather than orbital origin. These particles are electrons that have been expelled by excited nuclei and may have a charge of either sign.
If both energy and momentum are to be conserved, a third type of particle, the neutrino, ■, must be involved. The neutrino is associated with positive electron emission, and its antiparticle, the antineutrino, , is emitted with a negative electron. These uncharged particles have only the weakest interaction with matter, no mass, and travel at the speed of light. For all practical purposes, they pass through all materials with so few interactions that the energy they possess cannot be recovered. The neutrinos and antineutrinos are included here only because they carry a portion of the kinetic energy that would otherwise belong to the beta particle, and therefore, must be considered for energy and momentum to be conserved. They are normally ignored since they are not significant in the context of nuclear reactor applications.
Negative electron emission, represented as e, -1 -1■, or simply as e or ■ , effectively converts a neutro to a proton, thus increasing the atomic number by one and leaving the mass number unchanged. This is a common mode of decay for nuclei with an excess of neutrons, such as fission fragments below and to the right of the neutron-proton stability curve (refer to Figure 6). An example of a typical beta minus-decay reaction is shown below.
■ ■
■
■ ■
Fixture checkpoint marker-0622-xray is recorded on fixture page 0622.
■ ■ ■ ■
Atomic and Nuclear Physics DOE-HDBK-1019/1-93 MODES OF RADIOACTIVE DECAY
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Figure 9 Orbital Electron Capture
Positively charged electrons (beta-plus) are known as positrons. Except for sign, they are nearly identical to their negatively charged cousins. When a positron, represented as e, ■, or simply as +1 +1
0 0
e or ■ , is ejected from the nucleus, the atomic number is decreased by one and the mass number + + remains unchanged. A proton has been converted to a neutron. An example of a typical positron (beta-plus) decay is shown below.

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Nuclei having an excess of protons may capture an electron from one of the inner orbits which immediately combines with a proton in the nucleus to form a neutron. This process is called electron capture (EC). The electron is normally captured from the innermost orbit (the K-shell), and, consequently, this process is sometimes called K-capture. The following example depicts electron capture.

A neutrino is formed at the same time that the neutron is formed, and energy carried off by it serves to conserve momentum. Any energy that is available due to the atomic mass of the product being appreciably less than that of the parent will appear as gamma radiation. Also, there will always

Fixture checkpoint marker-0623-xray is recorded on fixture page 0623.

be characteristic x-rays given off when an electron from one of the higher energy shells moves in to fill the vacancy in the K-shell.

Electron capture is shown graphically in Figure 8.

Electron capture and positron emission result in the production of the same daughter product, and they exist as competing processes.

For positron emission to occur, however, the mass of the daughter product must be less than the mass of the parent by an amount equal to at least twice the mass of an electron. This mass difference between the parent and daughter is necessary to account for two items present in the parent but not in the daughter. One item is the positron ejected from the nucleus of the parent. The other item is that the daughter product has one less orbital electron than the parent. If this requirement is not met, then orbital electron capture takes place exclusively.

■

■ ■ ■ ■

■ ■

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Gamma Emission (■)

Gamma radiation is a high-energy electromagnetic radiation that originates in the nucleus. It is emitted in the form of photons, discrete bundles of energy that have both wave and particle properties. Often a daughter nuclide is left in an excited state after a radioactive parent nucleus undergoes a transformation by alpha decay, beta decay, or electron capture. The nucleus will drop to the ground state by the emission of gamma radiation.

Internal Conversion

The usual method for an excited nucleus to go from the excited state to the ground state is by emission of gamma radiation. However, in some cases the gamma ray (photon) emerges from the nucleus only to interact with one of the innermost orbital electrons and, as a result, the energy of the photon is transferred to the electron. The gamma ray is then said to have undergone internal conversion. The conversion electron is ejected from the atom with kinetic energy equal to the

a lower energy state to fill the vacancy, and this is accompanied by the emission of characteristic x-rays.

Isomers and Isomeric Transition

Isomeric transition commonly occurs immediately after particle emission; however, the nucleus may remain in an excited state for a measurable period of time before dropping to the ground state at its own characteristic rate. A nucleus that remains in such an excited state is known as a nuclear isomer because it differs in energy and behavior from other nuclei with the same atomic number and mass number. The decay of an excited nuclear isomer to a lower energy level is called an isomeric transition. It is also possible for the excited isomer to decay by some alternate means, for example, by beta emission.

An example of gamma emission accompanying particle emission is illustrated by the decay of nitrogen-16 below.

■
■
■
■
■
■
■
■
■

Fixture checkpoint marker-0624-xray is recorded on fixture page 0624.

■
■
■

Decay Chains

When an unstable nucleus decays, the resulting daughter nucleus is not necessarily stable. The nucleus resulting from the decay of a parent is often itself unstable, and will undergo an additional decay. This is especially common among the larger nuclides.

It is possible to trace the steps of an unstable atom as it goes through multiple decays trying to achieve stability. The list of the original unstable nuclide, the nuclides that are involved as intermediate steps in the decay, and the final stable nuclide is known as the decay chain. One common method for stating the decay chain is to state each of the nuclides involved in the standard $^{A}_{Z}X$ format. Arrows are used between nuclides to indicate where decays occur, with the type of decay indicated above the arrow and the half-life below the arrow. The half-life for decay will be discussed in the next chapter.

Example:

Write the decay chains for rubidium-91 and actinium-215. Continue the chains until a stable nuclide or a nuclide with a half-life greater than 1×10^{6} years is reached.

Solution:

Predicting Type of Decay

Radioactive nuclides tend to decay in a way that results in a daughter nuclide that lies closer to the line of stability. Due to this, it is possible to predict the type of decay that a nuclide will undergo based on its location relative to the line of stability on the Chart of the Nuclides.

Figure 9 illustrates the type of decay nuclides in different regions of the chart will typically undergo.

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Nuclides that are above and to the left of the line of stability will usually undergo either ■ decay or + electron capture. Most nuclides that will undergo ■ decay are found in the upper right hand region of the chart. These are general rules that have many exceptions, especially in the region of the heavy nuclides.
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Figure 9 Types of Radioactive Decay Relative to the Line of Stability
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Summary
The important information in this chapter is summarized below.
Modes of Radioactive Decay Summary
Alpha decay is the emission of an alpha particle (2 protons and 2 neutrons) from an unstable nucleus. The daughter nuclide has an atomic number 2 less than the parent nuclide and a mass number 4 less than the parent nuclide. The daughter nucleus commonly releases its excitation energy by gamma emission.
Beta-minus decay effectively converts a neutron to a proton and an electron, which is immediately ejected from the nucleus. The daughter nuclide has its atomic number increased by 1 and the same mass number compared to the parent.
Beta-plus decay effectively converts a proton to a neutron and a positron, which is immediately ejected from the nucleus. The daughter nuclide has its atomic number decreased by 1 and the same mass number compared to the parent.
In electron capture, the nucleus absorbs an electron from the innermost orbit. This electron combines with a proton to form a neutron.
Internal conversion occurs when a gamma ray, emitted by the nucleus as it goes from the excited state to the ground state, interacts with one of the innermost electrons of the same atom. The electron is ejected from the atom.
An isomeric transition is the decay of an excited nucleus to a lower-energy level
Fixture checkpoint marker-0625-xray is recorded on fixture page 0625.
by the emission of a gamma ray.
Decay chains can be found by tracing the steps an unstable atom goes through as it tries to achieve stability.
Many modes of radioactive decay result in a daughter nuclide that has an energy level above the ground state. This excitation energy is usually released immediately in the form of a gamma ray.
The type of decay that a nuclide will typically undergo can be determined by its relationship to the line of stability on the Chart of the Nuclides. Nuclides that lie below and to the right of the line of stability will typically beta minus decay. Nuclides above and to the left of the line will typically either beta plus decay or electron capture. Most alpha emitters are found in the upper, right-hand corner of the chart.
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RADIOACTIVITY
The rate at which a sample of radioactive material decays is not constant. As individual atoms of the material decay, there are fewer of those types of atoms remaining. Since the rate of decay is directly proportional to the number of atoms, the rate of decay will decrease as the number of atoms decreases.

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a. Radioactivity d. Radioactive decay constant
b. Curie e. Radioactive half-life
c. Becquerel
EO 2.6 Given the number of atoms and either the half-life or decay constant of a nuclide, CALCULATE the activity.
EO 2.7 Given the initial activity and the decay constant of a nuclide, CALCULATE the activity at any later time.
EO 2.8 CONVERT between the half-life and decay constant for a nuclide.
EO 2.9 Given the Chart of the Nuclides and the original activity, PLOT the radioactive decay curve for a nuclide on either linear or semi-log coordinates.
EO 2.10 DEFINE the following terms:
a. Radioactive equilibrium
b. Transient radioactive equilibrium
Radioactive Decay Rates
Radioactivity is the property of certain nuclides of spontaneously emitting particles or gamma radiation. The decay of radioactive nuclides occurs in a random manner, and the precise time at which a single nucleus will decay cannot be determined. However, the average behavior of a very large sample can be predicted accurately by using statistical methods. These studies have revealed that there is a certain probability that in a given time interval a certain fraction of the nuclei within a sample of a particular nuclide will decay. This probability per unit time that an atom of a nuclide will decay is known as the radioactive decay constant, λ. The units for the decay constant are inverse time such as 1/second, 1/minute, 1/hour, or 1/year. These decay constant units can also be expressed as second-1, minute-1, hour-1, and year-1.

■■

The activity (A) of a sample is the rate of decay of that sample. This rate of decay is usually measured in the number of disintegrations that occur per second. For a sample containing millions of atoms, the activity is the product of the decay constant and the number of atoms present in the sample.
The relationship between the activity, number of atoms, and decay constant is shown in Equation (1-3).
Fixture checkpoint marker-0626-xray is recorded on fixture page 0626.
A = ■ N (1-3)
where:
A = Activity of the nuclide (disintegrations/second)
■ = decay constant of the nuclide (second)-1
N = Number of atoms of the nuclide in the sample
Since ■ is a constant, the activity and the number of atoms are always proportional.
Units of Measurement for Radioactivity
Two common units to measure the activity of a substance are the curie (Ci) and becquerel (Bq). A curie is a unit of measure of the rate of radioactive decay equal to 3.7 x 10 disintegrations 10 per second. This is approximately equivalent to the number of disintegrations that one gram of radium-226 will undergo in one second. A becquerel is a more fundamental unit of measure of radioactive decay that is equal to 1 disintegration per second. Currently, the curie is more

metric system slowly comes into wider use. The conversion between curies and becquerels is shown below.

1 curie = 3.7 x 10 becquerels 10

Variation of Radioactivity Over Time

The rate at which a given radionuclide sample decays is stated in Equation (1-3) as being equal to the product of the number of atoms and the decay constant. From this basic relationship it is possible to use calculus to derive an expression which can be used to calculate how the number of atoms present will change over time. The derivation is beyond the scope of this text, but Equation (1-4) is the useful result.

(1-4)

where:

N = number of atoms present at time t

Fixture checkpoint marker-0627-xray is recorded on fixture page 0627.

N = number of atoms initially present o

■ = decay constant (time)-1

t = time

A Ao

e

t

A Aoe

t

A

Ao

e

t

ln A

Ao

t

t

ln

A

Ao

t

1/2

ln

1

2

t

1/2

ln 2 0.693

Since the activity and the number of atoms are always proportional, they may be used interchangeably to describe any given radionuclide population. Therefore, the following is true.

(1-5)

where:

A = activity present at time t

A = activity initially present o

-1
t = time
Radioactive Half-Life
One of the most useful terms for estimating how quickly a nuclide will decay is the radioactive half-life. The radioactive half-life is defined as the amount of time required for the activity to decrease to one-half of its original value. A relationship between the half-life and decay constant can be developed from Equation (1-5). The half-life can be calculated by solving Equation (1-5) for the time, t, when the current activity, A, equals one-half the initial activity A .
o
First, solve Equation (1-5) for t.
If A is equal to one-half of Ao, then A/Ao is equal to one-half. Substituting this in the equation above yields an expression for t .
1/2
(1-6)
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The basic features of decay of a radionuclide sample are shown by the graph in Figure 10.
Figure 10 Radioactive Decay as a Function of Time in Units of Half-Life
Assuming an initial number of atoms No, the population, and consequently, the activity may be noted to decrease by one-half of this value in a time of one half-life. Additional decreases occur
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so that whenever one half-life elapses, the number of atoms drops to one-half of what its value was at the beginning of that time interval. After five half-lives have elapsed, only 1/32, or 3.1%, of the original number of atoms remains. After seven half-lives, only 1/128, or 0.78%, of the atoms remains. The number of atoms existing after 5 to 7 half-lives can usually be assumed to be negligible. The Chemistry Fundamentals Handbook contains additional information on calculating the number of atoms contained within a sample.
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■
■
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Example:
A sample of material contains 20 micrograms of californium-252.
Californium-252 has a half-life of 2.638 years.
Calculate:
(a) The number of californium-252 atoms initially present
(b) The activity of the californium-252 in curies
(c) The number of californium-252 atoms that will remain in 12 years
(d) The time it will take for the activity to reach 0.001 curies
Solution:
(a) The number of atoms of californium-252 can be determined as below.
(b) First, use Equation (1-6) to calculate the decay constant.
Use this value for the decay constant in Equation (1-3) to determine the activity.
N No
e
t
4.78 x 1016 e
(0.263/yr) (12 yr)

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A Aoe
t
A
Ao
e
t
ln A
Ao
t
t
ln
A
Ao
t
ln
0.001 Ci
0.0108 Ci
0.263 year 1
t 9.05 years
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(c) The number of californium atoms that will remain in 12 years can be calculated from Equation (1-4).
(d) The time that it will take for the activity to reach 0.001 Ci can be determined from Equation (1-5). First, solve Equation (1-5) for time.
Inserting the appropriate values in the right side of this equation will result in the
Fixture checkpoint marker-0629-xray is recorded on fixture page 0629.
required time.
Plotting Radioactive Decay
It is useful to plot the activity of a nuclide as it changes over time. Plots of this type can be used to determine when the activity will fall below a certain level. This plot is usually done showing activity on either a linear or a logarithmic scale. The decay of the activity of a single nuclide on a logarithmic scale will plot as a straight line because the decay is exponential.
t
1/2
0.693
0.693
t
1/2
0.693
7.13 seconds
0.0972 second 1
A Aoe
t
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Example:
Plot the radioactive decay curve for nitrogen-16 over a period of 100 seconds. The

curve on both linear rectangular coordinates and on a semi-log scale.
Solution:
First, use Equation (1-6) to calculate the decay constant corresponding to a half-life of 7.13 seconds.
Use the decay constant determined above to calculate the activity at various times using Equation (1-5).
Time Activity
0 seconds 142 Ci
20 seconds 20.3 Ci
40 seconds 2.91 Ci
60 seconds 0.416 Ci
80 seconds 0.0596 Ci
100 seconds 0.00853 Ci
■ ■
■ ■
■ ■
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Figure 11 Linear and Semi-log Plots of Nitrogen-16 Decay
Plotting the data points calculated above on both linear and semilog scales results in the graphs shown in Figure 11.
If a substance contains more than one radioactive nuclide, the total activity is the sum of the individual activities of each nuclide. As an example, consider a sample of material that contained 1 x 10 atoms of iron-59 that has a half-life of 44.51 days (■ = 1.80 x 10 sec), 6 -7 -1 1 x 10 atoms of manganese-54 that has a half-life of 312.2 days (■ = 2.57 x 10 sec), and 1 6 -8 -1 x 10 atoms of cobalt-60 that has a half-life of 1925 days (■ = 4.17 x 10 sec). 6 -9 -1
The initial activity of each of the nuclides would be the product of the number of atoms and the decay constant.
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Figure 12 Combined Decay of Iron-56, Manganese-54, and Cobalt-60
Plotting the manner in which the activities of each of the three nuclides decay over time demonstrates that initially the activity of the shortest-lived nuclide (iron-59) dominates the total
Fixture checkpoint marker-0630-xray is recorded on fixture page 0630.
activity, then manganese-54 dominates. After almost all of the iron and manganese have decayed away, the only contributor to activity will be the cobalt-60. A plot of this combined decay is shown in Figure 12.
Radioactive Equilibrium
Radioactive equilibrium exists when a radioactive nuclide is decaying at the same rate at which it is being produced. Since the production rate and decay rate are equal, the number of atoms present remains constant over time.
An example of radioactive equilibrium is the concentration of sodium-24 in the coolant circulating through a sodium-cooled nuclear reactor. Assume that the sodium-24 is being produced at a rate of 1 x 10 atoms per second. If the sodium-24 were stable and did not decay, 6 the amount of sodium-24 present after some period of time could be calculated by multiplying the production rate by the amount of time. Plotting the amount of material present would result in the graph in Figure 13.
However, sodium-24 is not stable, and it decays with a half-life of 14.96 hours. If no

of decay will initially be zero because there is no sodium-24 present to decay. The rate of decay of sodium-24 will increase as the amount of sodium-24 increases.

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■
■
■

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Figure 13 Cumulative Production of Sodium-24 Over Time

The amount of sodium-24 present will initially increase rapidly, then it will increase at a
Fixture checkpoint marker-0631-xray is recorded on fixture page 0631.
continually decreasing rate until the rate of decay is equal to the rate of production. It is possible to calculate how much sodium-24 will be present at equilibrium by setting the production rate (R) equal to the decay rate (■ N).

where:

R = production rate (atoms/second)
■ = decay constant (second)-1
N = number of atoms

It is possible to calculate the equilibrium value for sodium-24 being produced at a rate of 1 x 10 atoms/second. 6

■ ■ ■■
■
■
■
■

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Figure 14 Approach of Sodium-24 to Equilibrium

The development of the equation to calculate how the amount of sodium-24 changes over time as it approaches the equilibrium value is beyond the scope of this handbook. However, the equation is presented below.

This equation can be used to calculate the values of the amount of sodium-24 present at different times. As the time increases, the exponential term approaches zero, and the number of atoms present will approach R/■. A plot of the approach of sodium-24 to equilibrium is shown in Figure 14.

Transient Radioactive Equilibrium

Transient radioactive equilibrium occurs when the parent nuclide and the daughter nuclide decay at essentially the same rate.

For transient equilibrium to occur, the parent must have a long half-life when compared to the daughter. An example of this type of compound decay process is barium-140, which decays by beta emission to lanthanum-140, which in turn decays by beta emission to stable cerium-140.

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Figure 15 Transient Equilibrium in the Decay of Barium-140

The decay constant for barium-140 is considerably smaller than the decay constant for lanthanum-140. Remember that the rate of decay of both the parent and daughter can be represented as ■N. Although the decay constant for barium-140 is smaller, the actual rate of decay (■N) is initially larger than that of lanthanum-140 because of the great difference in their

daughter will approach and eventually match the decay rate of the parent. When this occurs, they are said to be in transient equilibrium. A plot of the barium-lanthanum-cerium decay chain reaching transient equilibrium is shown in Figure 15.

Secular equilibrium occurs when the parent has an extremely long half-life. In the long decay chain for a naturally radioactive element, such as thorium-232, where all of the elements in the chain are in secular equilibrium, each of the descendants has built up to an equilibrium amount and all decay at the rate set by the original parent. The only exception is the final stable element on the end of the chain. Its number of atoms is constantly increasing.

Summary

The important information in this chapter is summarized on the following page.

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■■

■

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Radioactivity Summary

& Radioactivity is the decay of unstable atoms by the emission of particles and Fixture checkpoint marker-0632-xray is recorded on fixture page 0632. electromagnetic radiation.

& A curie (Ci) is a unit of radioactivity equal to 3.7 x 10 disintegrations per 10 second.

& A becquerel (Bq) is a unit of radioactivity equal to 1 disintegration per second.

& The radioactive decay constant (■) is the probability per unit time that an atom will decay.

& The radioactive half-life is the amount of time required for the activity to decrease to one-half its original value.

& The activity of a substance can be calculated from the number of atoms and the decay constant based on the equation below.

& The amount of activity remaining after a particular time can be calculated from the equation below.

& The relationship between the decay constant and the half-life is shown below.

& Plots of radioactive decay can be useful to describe the variation of activity over time. If decay is plotted using semi-log scale the plot results in a straight line.

& Radioactive equilibrium exists when the production rate of a material is equal to the removal rate.

& Transient radioactive equilibrium exists when the parent nuclide and the daughter nuclide decay at essentially the same rate. This occurs only when the parent has a long half-life compared to the daughter.

Atomic and Nuclear Physics DOE-HDBK-1019/1-93 NEUTRON INTERACTIONS

NEUTRON INTERACTIONS

Neutrons can cause many different types of interactions. The neutron may simply scatter off the nucleus in two different ways, or it may actually be absorbed into the nucleus. If a neutron is absorbed into the nucleus, it may result in the emission of a gamma ray or a subatomic particle, or it may cause the nucleus to fission.

EO 3.1 DESCRIBE the following scattering interactions between a neutron and a nucleus:

a. Elastic scattering

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EO 3.2 STATE the conservation laws that apply to an elastic collision between a neutron and a nucleus.
EO 3.3 DESCRIBE the following reactions where a neutron is absorbed in a nucleus:
a. Radiative capture
b. Particle ejection
Scattering
A neutron scattering reaction occurs when a nucleus, after having been struck by a neutron, emits a single neutron. Despite the fact that the initial and final neutrons do not need to be (and often are not) the same, the net effect of the reaction is as if the projectile neutron had merely "bounced off," or scattered from, the nucleus. The two categories of scattering reactions, elastic and inelastic scattering, are described in the following paragraphs.
Elastic Scattering
In an elastic scattering reaction between a neutron and a target nucleus, there is no energy transferred into nuclear excitation. Momentum and kinetic energy of the "system" are conserved although there is usually some transfer of kinetic energy from the neutron to the target nucleus. The target nucleus gains the amount of kinetic energy that the neutron loses.
Rev. 0 Page 43 NP-01
mn
vn,i mTvT,i mnvn,f mTvT,f
1
2
m v
2
Fixture checkpoint marker-0633-xray is recorded on fixture page 0633.
1
2
mn
v
2
n,i
1
2
mT
v
2
T,i
1
2
mn
v
2
n,f
1
2
mT
v
2

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Figure 16 Elastic Scattering
Figure 16 illustrates the process of elastic scattering of a neutron off a target nucleus. In the elastic scattering reaction, the conservation of momentum and kinetic energy is represented by the equations below.
Conservation of momentum (mv)
Conservation of kinetic energy
where:
mn = mass of the neutron
m = mass of the target nucleus T
vn,i = initial neutron velocity
vn,f = final neutron velocity
vT,i = initial target velocity
vT,f = final target velocity
Atomic and Nuclear Physics DOE-HDBK-1019/1-93 NEUTRON INTERACTIONS
Elastic scattering of neutrons by nuclei can occur in two ways. The more unusual of the two interactions is the absorption of the neutron, forming a compound nucleus, followed by the re-emission of a neutron in such a way that the total kinetic energy is conserved and the nucleus returns to its ground state. This is known as resonance elastic scattering and is very dependent upon the initial kinetic energy possessed by the neutron. Due to formation of the compound nucleus, it is also referred to as compound elastic scattering. The second, more usual method, is termed potential elastic scattering and can be understood by visualizing the neutrons and nuclei to be much like billiard balls with impenetrable surfaces. Potential scattering takes place with incident neutrons that have an energy of up to about 1 MeV. In potential scattering, the neutron does not actually touch the nucleus and a compound nucleus is not formed. Instead, the neutron is acted on and scattered by the short range nuclear forces when it approaches close enough to the nucleus.
Inelastic Scattering
In inelastic scattering, the incident neutron is absorbed by the target nucleus, forming a compound nucleus. The compound nucleus will then emit a neutron of lower kinetic energy
Fixture checkpoint marker-0634-xray is recorded on fixture page 0634.
which leaves the original nucleus in an excited state. The nucleus will usually, by one or more gamma emissions, emit this excess energy to reach its ground state. Figure 17 shows the process of inelastic scattering.
For the nucleus that has reached its ground state, the sum of the kinetic energy of the exit
Figure 17 Inelastic Scattering
neutron, the target nucleus, and the total gamma energy emitted is equal to the initial kinetic energy of the incident neutron.
Rev. 0 Page 45 NP-01
1
0
n
238
92
U
239
92

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239
92
U
0
0
1
0
n
10
5
B
11
5
B
7
3
Li
4
2

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Absorption Reactions

Most absorption reactions result in the loss of a neutron coupled with the production of a charged particle or gamma ray. When the product nucleus is radioactive, additional radiation is emitted at some later time. Radiative capture, particle ejection, and fission are all categorized as absorption reactions and are briefly described below.

Radiative Capture

In radiative capture the incident neutron enters the target nucleus forming a compound nucleus. The compound nucleus then decays to its ground state by gamma emission. An example of a radiative capture reaction is shown below.

Particle Ejection

In a particle ejection reaction the incident particle enters the target nucleus forming a compound nucleus. The newly formed compound nucleus has been excited to a high enough energy level to cause it to eject a new particle while the incident neutron remains in the nucleus. After the new particle is ejected, the remaining nucleus may or may not exist in an excited state depending upon the mass-energy balance of the reaction. An example of a particle ejection reaction is shown below.

Fission

Fixture checkpoint marker-0635-xray is recorded on fixture page 0635.

One of the most important interactions that neutrons can cause is fission, in which the nucleus that absorbs the neutron actually splits into two similarly sized parts. Fission will be discussed in detail in the next chapter.

Atomic and Nuclear Physics DOE-HDBK-1019/1-93 NEUTRON INTERACTIONS

Summary

The important information in this chapter is summarized below.

Neutron Interactions Summary

Interactions where a neutron scatters off a target nucleus are either elastic or inelastic. In elastic scattering, kinetic energy and momentum are conserved and

scattering, some amount of kinetic energy is transferred into excitation energy of the target nucleus.
The conservation principles that apply to an elastic collision are conservation of kinetic energy and conservation of momentum.
Radiative capture is the absorption of a neutron by the target nucleus, resulting in an excited nucleus which subsequently (typically within a small fraction of a second) releases its excitation energy in the form of a gamma ray.
Particle ejection occurs when a neutron is absorbed by a target nucleus, resulting in the formation of a compound nucleus. The compound nucleus immediately ejects a particle (for example, alpha or proton).
Rev. 0 Page 47 NP-01
1
0
n
235
Fixture checkpoint marker-0636-xray is recorded on fixture page 0636.
92
U
236
92
U
140
55
Cs
93
37
Rb 3
1
0
n
NUCLEAR FISSION DOE-HDBK-1019/1-93 Atomic and Nuclear Physics
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NUCLEAR FISSION
Nuclear fission is a process in which an atom splits and releases energy, fission products, and neutrons. The neutrons released by fission can, in turn, cause the fission of other atoms.
EO 4.1 EXPLAIN the fission process using the liquid drop model of a nucleus.
EO 4.2 DEFINE the following terms:
a. Excitation energy (E) exc
b. Critical energy (E) crit
EO 4.3 DEFINE the following terms:
a. Fissile material
b. Fissionable material
c. Fertile material
EO 4.4 DESCRIBE the processes of transmutation, conversion, and breeding.
EO 4.5 DESCRIBE the curve of Binding Energy per Nucleon

the reasons for its shape.
EO 4.6 EXPLAIN why only the heaviest nuclei are easily fissioned.
EO 4.7 EXPLAIN why uranium-235 fissions with thermal
neutrons and uranium-238 fissions only with fast neutrons.
Fission
In the fission reaction the incident neutron enters the heavy target nucleus, forming a compound nucleus that is excited to such a high energy level (E > E) that the nucleus "splits" exc crit (fissions) into two large fragments plus some neutrons. An example of a typical fission reaction is shown below.
A large amount of energy is released in the form of radiation and fragment kinetic energy.
Atomic and Nuclear Physics DOE-HDBK-1019/1-93 NUCLEAR FISSION
Liquid Drop Model of a Nucleus
The nucleus is held together by the attractive nuclear force between nucleons, which was discussed in a previous chapter. The characteristics of the nuclear force are listed below.
(a) very short range, with essentially no effect beyond nuclear dimensions
(10-13 cm)
(b) stronger than the repulsive electrostatic forces within the nucleus
(c) independent of nucleon pairing, in that the attractive forces between pairs of neutrons are no different than those between pairs of protons or a neutron and a proton
(d) saturable, that is, a nucleon can attract only a few of its nearest neighbors
One theory of fission considers the fissioning of a nucleus similar in some respects to the splitting
Fixture checkpoint marker-0637-xray is recorded on fixture page 0637.
of a liquid drop. This analogy is justifiable to some extent by the fact that a liquid drop is held together by molecular forces that tend to make the drop spherical in shape and that try to resist any deformation in the same manner as nuclear forces are assumed to hold the nucleus together. By considering the nucleus as a liquid drop, the fission process can be described.
Referring to Figure 18(A), the nucleus in the ground state is undistorted, and its attractive nuclear forces are greater than the repulsive electrostatic forces between the protons within the nucleus. When an incident particle (in this instance a neutron) is absorbed by the target nucleus, a compound nucleus is formed. The compound nucleus temporarily contains all the charge and mass involved in the reaction and exists in an excited state. The excitation energy added to the compound nucleus is equal to the binding energy contributed by the incident particle plus the kinetic energy possessed by that particle. Figure 18(B) illustrates the excitation energy thus imparted to the compound nucleus, which may cause it to oscillate and become distorted. If the excitation energy is greater than a certain critical energy, the oscillations may cause the compound nucleus to become dumbbell-shaped. When this happens, the attractive nuclear forces (short-range) in the neck area are small due to saturation, while the repulsive electrostatic forces (long-range) are only slightly less than before. When the repulsive electrostatic forces exceed the attractive nuclear forces, nuclear fission occurs, as illustrated in Figure 18(C).

Figure 18 Liquid Drop Model of Fission
Critical Energy
The measure of how far the energy level of a nucleus is above its ground state is called the excitation energy (Eexc). For fission to occur, the excitation energy must be above a particular value for that nuclide. The critical energy (Ecrit) is the minimum excitation energy required for fission to occur.

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A fissile material is composed of nuclides for which fission is possible with neutrons of any energy level. What is especially significant about these nuclides is their ability to be fissioned with zero kinetic energy neutrons (thermal neutrons). Thermal neutrons have very low kinetic energy levels (essentially zero) because they are roughly in equilibrium with the thermal motion of surrounding materials. Therefore, in order to be classified as fissile, a material must be capable of fissioning after absorbing a thermal neutron. Consequently, they impart essentially no kinetic energy to the reaction. Fission is possible in these materials with thermal neutrons, since the change in binding energy supplied by the neutron addition alone is high enough to exceed the critical energy. Some examples of fissile nuclides are uranium-235, uranium-233, and plutonium-239.

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Fissionable Material

A fissionable material is composed of nuclides for which fission with neutrons is possible. All fissile nuclides fall into this category. However, also included are those nuclides that can be fissioned only with high energy neutrons. The change in binding energy that occurs as the result of neutron absorption results in a nuclear excitation energy level that is less than the required critical energy. Therefore, the additional excitation energy must be supplied by the kinetic energy of the incident neutron. The reason for this difference between fissile and fissionable materials is the so-called odd-even effect for nuclei. It has been observed that nuclei with even numbers of neutrons and/or protons are more stable than those with odd numbers. Therefore, adding a neutron to change a nucleus with an odd number of neutrons to a nucleus with an even number of neutrons produces an appreciably higher binding energy than adding a neutron to a nucleus already possessing an even number of neutrons. Some examples of nuclides requiring high energy neutrons to cause fission are thorium-232, uranium-238, and plutonium-240. Table 4 indicates the critical energy (Ecrit) and the binding energy change for an added neutron (BEn) to target nuclei of interest. For fission to be possible, the change in binding energy plus the kinetic energy must equal or exceed the critical energy ($\Delta BE + KE > Ecrit$).

TABLE 4

Fixture checkpoint marker-0638-xray is recorded on fixture page 0638.

Critical Energies Compared to Binding Energy of Last Neutron

Target
Nucleus
Critical Energy
Ecrit
Binding Energy of
Last Neutron BEn
BEn - Ecrit
23
9
2
0Th 7.5 MeV 5.4 MeV -2.1 MeV
23
9
8
2U 7.0 MeV 5.5 MeV -1.5 MeV
23
9

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2U 6.5 MeV 6.8 MeV +0.3 MeV
23
9
3
2U 6.0 MeV 7.0 MeV +1.0 MeV
23
9
9
4Pu 5.0 MeV 6.6 MeV +1.6 MeV
Uranium-235 fissions with thermal neutrons because the binding energy released by the absorption of a neutron is greater than the critical energy for fission; therefore uranium-235 is a fissile material. The binding energy released by uranium-238 absorbing a thermal neutron is less than the critical energy, so additional energy must be possessed by the neutron for fission to be possible. Consequently, uranium-238 is a fissionable material.
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Fertile Material
All of the neutron absorption reactions that do not result in fission lead to the production of new nuclides through the process known as transmutation. These nuclides can, in turn, be transmuted again or may undergo radioactive decay to produce still different nuclides. The nuclides that are produced by this process are referred to as transmutation products. Because several of the fissile nuclides do not exist in nature, they can only be produced by nuclear reactions (transmutation). The target nuclei for such reactions are said to be fertile. Fertile materials are materials that can undergo transmutation to become fissile materials. Figure 19 traces the transmutation mechanism by which two fertile nuclides, thorium-232 and uranium-238, produce uranium-233 and plutonium-239, respectively.
Figure 19 Conversion of Fertile Nuclides to Fissile Nuclides
If a reactor contains fertile material in addition to its fissile fuel, some new fuel will be produced as the original fuel is burned up. This is called conversion. Reactors that are specifically designed to produce fissionable fuel are called "breeder" reactors. In such reactors, the amount of fissionable fuel produced is greater than the amount of fuel burnup. If less fuel is produced than used, the process is called conversion, and the reactor is termed a "converter."
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Atomic and Nuclear Physics DOE-HDBK-1019/1-93 NUCLEAR FISSION
Binding Energy Per Nucleon (BE/A)
As the number of particles in a nucleus increases, the total binding energy also increases. The
Fixture checkpoint marker-0639-xray is recorded on fixture page 0639.
rate of increase, however, is not uniform. This lack of uniformity results in a variation in the amount of binding energy associated with each nucleon within the nucleus. This variation in the binding energy per nucleon (BE/A) is easily seen when the average BE/A is plotted versus atomic mass number (A), as shown in Figure 20.
Figure 20 Binding Energy per Nucleon vs. Mass Number
Figure 20 illustrates that as the atomic mass number increases, the binding energy per nucleon decreases for A > 60. The BE/A curve reaches a maximum value of 8.79 MeV at A = 56 and decreases to about 7.6 MeV for A = 238. The general shape of the BE/A curve can be explained using the general properties of nuclear forces. The nucleus is held together by very short-range attractive forces that exist between nucleons. On the other hand, the nucleus is being forced apart by long range repulsive electrostatic (coulomb) forces that exist between all the protons in the

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As the atomic number and the atomic mass number increase, the repulsive electrostatic forces within the nucleus increase due to the greater number of protons in the heavy elements. To overcome this increased repulsion, the proportion of neutrons in the nucleus must increase to maintain stability. This increase in the neutron-to-proton ratio only partially compensates for the growing proton-proton repulsive force in the heavier, naturally occurring elements. Because the repulsive forces are increasing, less energy must be supplied, on the average, to remove a nucleon from the nucleus. The BE/A has decreased. The BE/A of a nucleus is an indication of its degree of stability. Generally, the more stable nuclides have higher BE/A than the less stable ones. The increase in the BE/A as the atomic mass number decreases from 260 to 60 is the primary reason for the energy liberation in the fission process. In addition, the increase in the BE/A as the atomic mass number increases from 1 to 60 is the reason for the energy liberation in the fusion process, Fixture checkpoint marker-0640-xray is recorded on fixture page 0640.
which is the opposite reaction of fission.
The heaviest nuclei require only a small distortion from a spherical shape (small energy addition) for the relatively large coulomb forces forcing the two halves of the nucleus apart to overcome the attractive nuclear forces holding the two halves together. Consequently, the heaviest nuclei are easily fissionable compared to lighter nuclei.
Summary
The important information in this chapter is summarized on the following page.

Nuclear Fission Summary
The fission process can be explained using the liquid drop model of a nucleus. In the ground state the nucleus is nearly spherical in shape. After the absorption of a neutron, the nucleus will be in an excited state and start to oscillate and become distorted. If the oscillations cause the nucleus to become shaped like a dumbbell, the repulsive electrostatic forces will overcome the short-range attractive nuclear forces, and the nucleus will split in two.
Excitation energy is the amount of energy a nucleus has above its ground state. Critical energy is the minimum excitation energy that a nucleus must have before it can fission.
Fissile material is material for which fission is possible with neutrons that have zero kinetic energy. Fissionable material is material for which fission caused by neutron absorption is possible provided the kinetic energy added with the binding energy is greater than the critical energy. Fertile material is material that can undergo transmutation to become fissile material.
Transmutation is the process of neutron absorption and subsequent decay, which changes one nuclide to another nuclide. Conversion is the process of transmuting fertile material into fissile material in a reactor, where the amount of fissile material produced is less than the amount of fissile material consumed. Breeding is the same as conversion, except the amount of fissile material produced is more than the amount of fissile material consumed.
The curve of binding energy per nucleon increases quickly through the light nuclides and reaches a maximum at a mass number of about 56. The curve decreases slowly for mass numbers greater than 60.
The heaviest nuclei are easily fissionable because they require only a small

the attractive nuclear force, forcing the two halves of the nucleus apart.
Uranium-235 fissions with thermal neutrons because the binding energy released by the absorption of a neutron is greater than the critical energy for fission. The binding energy released by uranium-238 absorbing a neutron is less than the critical energy, so additional kinetic energy must be possessed by the neutron for fission to be possible.
Rev. 0 Page 55 NP-01
1
0
n
235
92
U
236
92
U
140
55
Cs
93
Fixture checkpoint marker-0641-xray is recorded on fixture page 0641.
37
Rb 3
1
0
n
ENERGY RELEASE FROM FISSION DOE-HDBK-1019/1-93 Atomic and Nuclear Physics
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ENERGY RELEASE FROM FISSION
Fission of heavy nuclides converts a small amount of mass into an enormous amount of energy. The amount of energy released by fission can be determined based on either the change in mass that occurs during the reaction or by the difference in binding energy per nucleon between the fissile nuclide and the fission products.
EO 4.8 CHARACTERIZE the fission products in terms of mass groupings and radioactivity.
EO 4.9 Given the nuclides involved and their masses, CALCULATE the energy released from fission.
EO 4.10 Given the curve of Binding Energy per nucleon versus mass number, CALCULATE the energy released from fission.
Calculation of Fission Energy
Nuclear fission results in the release of enormous quantities of energy. It is necessary to be able to calculate the amount of energy that will be produced. The logical manner in which to pursue this is to first investigate a typical fission reaction such as the one listed below.
It can be seen that when the compound nucleus splits, it breaks into two fission fragments, rubidium-93, cesium-140, and some neutrons. Both fission products then decay by multiple - emissions as a result of the high neutron-to-proton ratio possessed by these nuclides.
In most cases, the resultant fission fragments have masses that vary widely. Figure 21 gives the

thermal fission of the fuel uranium-235 have masses of about 95 and 140. Note that the vertical axis of the fission yield curve is on a logarithmic scale. Therefore, the formation of fission fragments of mass numbers of about 95 and 140 is highly likely.

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Figure 21 Uranium-235 Fission Yield vs.
Mass Number

Referring now to the binding energy per nucleon curve (Figure 20), we can estimate the amount of energy released by our "typical" fission by plotting this reaction on the curve and calculating the change in binding energy (ΔBE) between the reactants on the left-hand side of the fission equation and the products on the right-hand side. Plotting the reactant and product nuclides on the curve shows that the total binding energy of the system after fission is greater than the total binding energy of the system before fission. When there is an increase in the total binding energy of a system, the system has become more stable by releasing an amount of energy equal to the increase in total binding energy of the system. Therefore, in the fission process, the energy liberated is equal to the increase in the total binding energy of the system.

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Figure 22 graphically depicts that the binding energy per nucleon for the products (C,

Figure 22 Change in Binding Energy for Typical Fission

rubidium-93 and B, cesium-140) is greater than that for the reactant (A, uranium-235). The total binding energy for a nucleus can be found by multiplying the binding energy per nucleon by the number of nucleons.

TABLE 5
Binding Energies Calculated from
Binding Energy per Nucleon Curve
Nuclide B.E. per Nucleon
(BE/A)

Fixture checkpoint marker-0642-xray is recorded on fixture page 0642.

Mass Number
(A)
Binding Energy
(BE/A) x (A)
9
3
3
7Rb 8.7 MeV 93 809 MeV
14
5
0
5Cs 8.4 MeV 140 1176 MeV
23
9
5
2U 7.6 MeV 235 1786 MeV

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BE BEproducts BEreactants
BERb 93 BECs 140 BEU 235
809 MeV 1176 MeV 1786 MeV

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1
0
n
235
92
U
236
92
U
140
55
Cs
93
37
Rb 3
1
0
n
Mass difference Mass of Reactants Mass of Products
236.052589 amu 235.85208 amu
0.200509 amu
EInst 0.020059 amu
931.5 MeV
amu
186.8 MeV
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The energy released will be equivalent to the difference in binding energy (BE) between the reactants and the products.
The energy liberation during the fission process can also be explained from the standpoint of the conservation of mass-energy. During the fission process, there is a decrease in the mass of the system. There must, therefore, be energy liberated equal to the energy equivalent of the mass lost in the process. This method is more accurate than the previously illustrated method and is used when actually calculating the energy liberated during the fission process.
Fixture checkpoint marker-0643-xray is recorded on fixture page 0643.
Again, referring to the "typical" fission reaction.
E , the instantaneous energy, is the energy released immediately after the fission process. It Inst is equal to the energy equivalent of the mass lost in the fission process. It can be calculated as shown below.
Mass of the Reactants Mass of the Products
U 235.043924 amu Rb 92.91699 amu 235
92
93
37
n 1.008665 amu Cs 139.90910 amu 1
0
140
55

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0
236.052589 amu 235.85208 amu
This mass difference can be converted to an energy equivalent.
93
37
Rb
93
38
Sr
93
39
Fixture checkpoint marker-0644-xray is recorded on fixture page 0644.
Y
93
40
Zr
93
41
Nb
140
55
Cs
140
56
Ba
140
57
La
140
58
Ce
EDecay mRb 93 mNb 93 4 melectron
931.5MeV
amu
92.91699 amu 92.90638 amu 4 0.0005486 amu
931.5 MeV
amu
0.008416 amu
931.5 MeV
amu
7.84 MeV
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The total energy released per fission will vary from the fission to the next depending on what fission products are formed, but the average total energy released per fission of uranium-235 with a thermal neutron is 200 MeV.
As illustrated in the preceding example, the majority of the energy liberated in the fission process is released immediately after the fission occurs and appears as the kinetic energy of the

remaining energy is released over a period of time after the fission occurs and appears as kinetic energy of the beta, neutrino, and decay gamma rays.

Estimation of Decay Energy

In addition to this instantaneous energy release during the actual fission reaction, there is additional energy released when the fission fragments decay by emission. This additional

-

energy is called decay energy, EDecay. The decay chains for rubidium-93 and cesium-140 are shown below.

The energy released during the decay for each chain will be equivalent to the mass difference between the original fission product and the sum of the final stable nuclide and the beta particles emitted.

The energy released in the decay chain of rubidium-93 is calculated below.

EDecay mRb 93 mNb 93 3 melectron

931.5MeV

amu

139.90910 amu 139.90543 amu 3 0.0005486 amu

931.5 MeV

amu

Fixture checkpoint marker-0645-xray is recorded on fixture page 0645.

0.000202 amu

931.5 MeV

amu

1.89 MeV

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The energy released in the decay chain of cesium-140 is calculated below.

The total decay energy is the sum of the energies of the two chains, or 9.73 MeV.

Distribution of Fission Energy

The average energy distribution for the energy released per fission with a thermal neutron in uranium-235 is shown in Tables 6 and 7.

TABLE 6

Instantaneous Energy from Fission

Kinetic Energy of Fission Products 167 Mev

Energy of Fission Neutrons 5 MeV

Instantaneous Gamma-ray Energy 5 MeV

Capture Gamma-ray Energy 10 MeV

Total Instantaneous Energy 187 MeV

TABLE 7

Delayed Energy from Fission

Beta Particles From Fission Products 7 Mev

Gamma-rays from Fission Products 6 MeV

Neutrinos 10 MeV

Total Delayed Energy 23 MeV

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Because the 10 MeV of neutrino energy shown in Table 7 is not absorbed in the reactor, the average value of 200 MeV per fission is still accurate. Note in Table 6 that some fission neutrons undergo radiative capture and the resultant gamma ray emission provides an additional 10 MeV of instantaneous energy, which contributes to the total of 187 MeV instantaneous

All of the energy released, with the exception of the neutrino energy, is ultimately transformed into heat through a number of processes. The fission fragments, with their high positive charge and kinetic energy, cause ionization directly as they rip orbital electrons from the surrounding atoms. In this ionization process, kinetic energy is transferred to the surrounding atoms of the fuel material, resulting in an increase in temperature. The beta particles and gamma rays also give up their energy through ionization, and the fission neutrons interact and lose their energy through elastic scattering. Of the 200 MeV released per fission, about seven percent (13 MeV) is released at some time after the instant of fission. When a reactor is shut down, fissions essentially cease, but energy is still being released from the decay of fission products. The heat produced by this decay energy is referred to as "decay heat." Although decay energy represents about seven percent of reactor heat production during reactor operation, once the reactor is shut down the decay heat production drops off quickly to a small fraction of its value while operating. The decay heat produced is significant, however, and systems must be provided to keep the reactor cool even after shutdown.

Summary

The important information in this chapter is summarized below.

Energy Release From Fission Summary

Fission products have some general characteristics in common.

They generally decay by β- emission.

The most common mass numbers are grouped near 95 and 140.

The energy released by fission can be calculated based on the difference in mass between the masses of the reactants before fission and the fission fragments and fission neutrons after fission.

Another method to determine the energy released by fission is based on the change in binding energy per nucleon between the fissile nuclide and the fission

Fixture checkpoint marker-0646-xray is recorded on fixture page 0646.

products.

INTERACTION OF RADIATION WITH MATTER

Different types of radiation interact with matter in widely different ways. A large, massive, charged alpha particle cannot penetrate a piece of paper and even has a limited range in dry air. A neutrino, at the other extreme, has a low probability of interacting with any matter, even if it passed through the diameter of the earth.

EO 5.1 DESCRIBE interactions of the following with matter:

a. Alpha particle c. Positron

b. Beta particle d. Neutron

EO 5.2 DESCRIBE the following ways that gamma radiation interacts with matter:

a. Photoelectric effect

b. Compton scattering

c. Pair production

Interaction of Radiation With Matter

Radiation can be classified into two general groups, charged and uncharged; therefore, it may be expected that interactions with matter fall into two general types. Charged particles directly ionize the media through which they pass, while uncharged particles and photons can cause

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A moving charged particle has an electrical field surrounding it, which interacts with the atomic structure of the medium through which it is passing. This interaction decelerates the particle and accelerates electrons in the atoms of the medium. The accelerated electrons may acquire enough energy to escape from the parent atom. This process, whereby radiation "strips" off orbital electrons, is called ionization. Uncharged moving particles have no electrical field, so they can only lose energy and cause ionization by such means as collisions or scattering. A photon can lose energy by the photoelectric effect, Compton effect, or pair production.
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I
m z
2
K.E.
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Because ionizing radiation creates ions in pairs, the intensity of ionization or the specific ionization is defined as the number of ion-pairs formed per centimeter of travel in a given material. The amount of ionization produced by a charged particle per unit path length, which is a measure of its ionizing power, is roughly proportional to the particle's mass and the square of its charge as illustrated in the equation below.
where:
I is the ionizing power
m is the mass of the particle
z is the number of unit charges it carries
K.E. is its kinetic energy
Since m for an alpha particle is about 7300 times as large as m for a beta particle, and z is twice as great, an alpha will produce much more ionization per unit path length than a beta particle of the same energy. This phenomenon occurs because the larger alpha particle moves slower for a given energy and thus acts on a given electron for a longer time.
Alpha Radiation
Alpha radiation is normally produced from the radioactive decay of heavy nuclides and from certain nuclear reactions. The alpha particle consists of 2 neutrons and 2 protons, so it is
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essentially the same as the nucleus of a helium atom. Because it has no electrons, the alpha particle has a charge of +2. This positive charge causes the alpha particle to strip electrons from the orbits of atoms in its vicinity. As the alpha particle passes through material, it removes electrons from the orbits of atoms it passes near. Energy is required to remove electrons and the energy of the alpha particle is reduced by each reaction. Eventually the particle will expend its kinetic energy, gain 2 electrons in orbit, and become a helium atom. Because of its strong positive charge and large mass, the alpha particle deposits a large amount of energy in a short distance of travel. This rapid, large deposition of energy limits the penetration of alpha particles. The most energetic alpha particles are stopped by a few centimeters of air or a sheet of paper.
Beta-Minus Radiation
A beta-minus particle is an electron that has been ejected at a high velocity from an unstable nucleus. An electron has a small mass and an electrical charge of -1. Beta particles cause ionization by displacing electrons from atom orbits. The ionization occurs from collisions with orbiting electrons. Each collision removes kinetic energy from the beta particle, causing it to

2 electrons 0.000549 amu
electron
931.5 MeV
amu
1.02 MeV
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orbiting electron in an atom. Although more penetrating than the alpha, the beta is relatively easy to stop and has a low power of penetration. Even the most energetic beta radiation can be stopped by a few millimeters of metal.
Positron Radiation
Positively charged electrons are called positrons. Except for the positive charge, they are identical to beta-minus particles and interact with matter in a similar manner. Positrons are very short-lived, however, and quickly are annihilated by interaction with a negatively charged electron, producing two gammas with a combined energy (calculated below) equal to the rest mass of the positive and negative electrons.
Neutron Radiation
Neutrons have no electrical charge. They have nearly the same mass as a proton (a hydrogen atom nucleus). A neutron has hundreds of times more mass than an electron, but 1/4 the mass of an alpha particle. The source of neutrons is primarily nuclear reactions, such as fission, but they may also be produced from the decay of radioactive nuclides. Because of its lack of charge, the neutron is difficult to stop and has a high penetrating power.
Neutrons are attenuated (reduced in energy and numbers) by three major interactions, elastic scatter, inelastic scatter, and absorption. In elastic scatter, a neutron collides with a nucleus and bounces off. This reaction transmits some of the kinetic energy of the neutron to the nucleus of the atom, resulting in the neutron being slowed, and the atom receives some kinetic energy (motion). This process is sometimes referred to as "the billiard ball effect."
As the mass of the nucleus approaches the mass of the neutron, this reaction becomes more effective in slowing the neutron. Hydrogenous material attenuates neutrons most effectively.
In the inelastic scatter reaction, the same neutron/nucleus collision occurs as in elastic scatter. However, in this reaction, the nucleus receives some internal energy as well as kinetic energy. This slows the neutron, but leaves the nucleus in an excited state. When the nucleus decays to its original energy level, it normally emits a gamma ray.
In the absorption reaction, the neutron is actually absorbed into the nucleus of an atom. The neutron is captured, but the atom is left in an excited state. If the nucleus emits one or more gamma rays to reach a stable level, the process is called radiative capture. This reaction occurs at most neutron energy levels, but is more probable at lower energy levels.
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Gamma Radiation
Gamma radiation is electromagnetic radiation. It is commonly referred to as a gamma ray and is very similar to an x-ray. The difference is that gamma rays are emitted from the nucleus of an atom, and x-rays are produced by orbiting electrons. The x-ray is produced when orbiting electrons move to a lower energy orbit or when fast-moving electrons approaching an atom are deflected and decelerated as they react with the atom's electrical field (called Bremsstrahlung). The gamma ray is produced by the decay of excited nuclei and by nuclear reactions. Because

power. A small fraction of the original gamma stream will pass through several feet of concrete or several meters of water.

There are three methods of attenuating gamma rays. The first method is referred to as the photo-electric effect. When a low energy gamma strikes an atom, the total energy of the gamma is expended in ejecting an electron from orbit. The result is ionization of the atom and expulsion of a high energy electron. This reaction is most predominant with low energy gammas interacting in materials with high atomic weight and rarely occurs with gammas having an energy above 1 MeV. Annihilation of the gamma results. Any gamma energy in excess of the binding energy of the electron is carried off by the electron in the form of kinetic energy.

The second method of attenuation of gammas is called Compton scattering. The gamma interacts with an orbital or free electron; however, in this case, the photon loses only a fraction of its energy. The actual energy loss depending on the scattering angle of the gamma. The gamma continues on at lower energy, and the energy difference is absorbed by the electron. This reaction becomes important for gamma energies of about 0.1 MeV and higher.

At higher energy levels, a third method of attenuation is predominant. This method is pair-production. When a high energy gamma passes close enough to a heavy nucleus, the gamma

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completely disappears, and an electron and a positron are formed. For this reaction to take place, the original gamma must have at least 1.02 MeV energy. Any energy greater than 1.02 MeV becomes kinetic energy shared between the electron and positron. The probability of pair■production incre

Interaction of Radiation with Matter Summary

An alpha particle deposits a large amount of energy in a short distance of travel due to its large mass and charge.

Beta-minus particles interact with the electrons orbiting the nucleus of atoms, causing ionization by displacing the electrons. The beta particle loses energy with each interaction. After the beta particle loses enough energy, it is captured in the orbital shells of an atom.

Positrons interact with matter much the same way as beta minus particles. After the positron has lost most of its energy by ionizing atoms, it is annihilated by interaction with an electron. The electron-positron pair disappear and are replaced by two gammas, each with the energy equivalent of the mass of an electron (0.51 MeV).

Neutrons interact with matter by elastic scattering, inelastic scattering, or absorption.

Photoelectric effect is where a gamma interacts with an electron orbiting an atom. The entire energy of the gamma is transferred to the electron, and the electron is ejected from its orbit.

In Compton scattering a gamma interacts with an orbital electron, but only part of the gamma energy is transferred to the electron. The electron is ejected from its orbit, and the gamma is scattered off at a lower energy.

In pair-production, a gamma interacts with the electric field of a nucleus and is converted into an electron-positron pair. The gamma must have an energy greater than 1.02 MeV for this to occur.

Summary

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TERMINAL OBJECTIVE
1.0 Without references, EXPLAIN how neutron sources produce neutrons.
ENABLING OBJECTIVES
1.1 DEFINE the following terms:
a. Intrinsic neutron source
b. Installed neutron source

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1.3 LIST three examples of reactions that produce neutrons in installed neutron sources.
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Reactor Theory (Neutron Characteristics) DOE-HDBK-1019/1-93 OBJECTIVES
TERMINAL OBJECTIVE
2.0 Given the necessary information for calculations, EXPLAIN basic concepts in reactor physics and perform calculations.
ENABLING OBJECTIVES
2.1 DEFINE the following terms:
a. Atom density d. Barn
b. Neutron flux e. Macroscopic cross section
c. Microscopic cross section f. Mean free path
2.2 EXPRESS macroscopic cross section in terms of microscopic cross section.
2.3 DESCRIBE how the absorption cross section of typical nuclides varies with neutron energy at energies below the resonance absorption region.
2.4 DESCRIBE the cause of resonance absorption in terms of nuclear energy levels.
2.5 DESCRIBE the energy dependence of resonance absorption peaks for typical light and heavy nuclei.
2.6 EXPRESS mean free path in terms of macroscopic cross section.
2.7 Given the number densities (or total density and component fractions) and microscopic cross sections of components, CALCULATE the macroscopic cross section for a mixture.
2.8 CALCULATE a macroscopic cross section given a material density, atomic mass, and microscopic cross section.
2.9 EXPLAIN neutron shadowing or self-shielding.
2.10 Given the neutron flux and macroscopic cross section, CALCULATE the reaction rate.
2.11 DESCRIBE the relationship between neutron flux and reactor power.
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ENABLING OBJECTIVES (Cont.)
2.12 DEFINE the following concepts:
a. Thermalization d. Average logarithmic energy decrement
b. Moderator e. Macroscopic slowing down power
c. Moderating ratio
2.13 LIST three desirable characteristics of a moderator.
2.14 Given an average fractional energy loss per collision, CALCULATE the energy loss after a specified number of collisions.
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TERMINAL OBJECTIVE
3.0 Without references, EXPLAIN the production process and effects on fission of prompt and delayed neutrons.
ENABLING OBJECTIVES
3.1 STATE the origin of prompt neutrons and delayed neutrons.
3.2 STATE the approximate fraction of neutrons that are born as delayed neutrons from the fission of the following nuclear fuels:
a. Uranium-235
b. Plutonium-239
3.3 EXPLAIN the mechanism for production of delayed neutrons.

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3.5 Given prompt and delayed neutron generation times and delayed neutron fraction, CALCULATE the average generation time.
3.6 EXPLAIN the effect of delayed neutrons on reactor control.
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TERMINAL OBJECTIVE
4.0 Without references, DESCRIBE the neutron energy spectrum for the type of reactor presented in this module.
ENABLING OBJECTIVES
4.1 STATE the average energy at which prompt neutrons are produced.
4.2 DESCRIBE the neutron energy spectrum in the following reactors:
a. Fast reactor
b. Thermal reactor
4.3 EXPLAIN the reason for the particular shape of the fast, intermediate, and slow energy
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regions of the neutron flux spectrum for a thermal reactor.
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Reactor Theory (Neutron Characteristics) DOE-HDBK-1019/1-93 NEUTRON SOURCES
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NEUTRON SOURCES
Neutrons from a variety of sources are always present in a reactor core. This is true even when the reactor is shut down. Some of these neutrons are produced by naturally occurring (intrinsic) neutron sources, while others may be the result of fabricated (installed) neutron sources that are incorporated into the design of the reactor. The neutrons produced by sources other than neutron-induced fission are often grouped together and classified as source neutrons.
EO 1.1 DEFINE the following terms:
a. Intrinsic neutron source
b. Installed neutron source
EO 1.2 LIST three examples of reactions that produce neutrons in intrinsic neutron sources.
EO 1.3 LIST three examples of reactions that produce neutrons in installed neutron sources.
Neutron Sources
In addition to neutron-induced fission, neutrons are produced by other reactions. The neutrons produced by reactions other than neutron-induced fission are called source neutrons. Source neutrons are important because they ensure that the neutron population remains high enough to allow a visible indication of neutron level on the most sensitive monitoring instruments while the reactor is shutdown and during the startup sequence. This verifies instrument operability and allows monitoring of neutron population changes. Source neutrons can be classified as either intrinsic or installed neutron sources.
Intrinsic Neutron Sources
Some neutrons will be produced in the materials present in the reactor due to a variety of unavoidable reactions that occur because of the nature of these materials. Intrinsic neutron sources are those neutron-producing reactions that always occur in reactor materials.
■ ■ ■ ■
■ ■ ■
■ ■

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A limited number of neutrons will always be present, even in a reactor core that has never been operated, due to spontaneous fission of some heavy nuclides that are present in the fuel. Uranium■238, ur of the rate at which different heavy nuclides produce neutrons by spontaneous fission.
Californium-252 is not an intrinsic neutron source, but will be discussed in the section on installed neutron sources.
TABLE 1
Neutron Production by Spontaneous Fission
Nuclide T (Fission) T (1/2 1/2 ■-decay) neutrons/sec/gram
U 1.8 x 10 years 6.8 x 10 years 8.0 x 10 235
92
17 8 -4
U 8.0 x 10 years 4.5 x 10 years 1.6 x 10 238
92
15 9 -2
Pu 5.5 x 10 years 2.4 x 10 years 3.0 x 10 239
94
5 4 -2
Pu 1.2 x 10 years 6.6 x 10 years 1.0 x 10 240
94
11 3 3
Fixture checkpoint marker-0654-xray is recorded on fixture page 0654.
Cf 66.0 years 2.65 years 2.3 x 10 252
98
12
Another intrinsic neutron source is a reaction involving natural boron and fuel. In some reactors, natural boron is loaded into the reactor core as a neutron absorber to improve reactor control or increase core life-time. Boron-11 (80.1% of natural boron) undergoes a reaction with the alpha particle emitted by the radioactive decay of heavy nuclides in the fuel to yield a neutron as shown below.
The boron-11 must be mixed with, or in very close proximity to, the fuel for this reaction because of the short path length of the alpha particle. For a reactor core with this configuration, this (■,n) reaction is an important source of neutrons for reactor startup.
In a reactor that has been operated, another source of neutrons becomes significant. Neutrons may be produced by the interaction of a gamma ray and a deuterium nucleus. This reaction is commonly referred to as a photoneutron reaction because it is initiated by electromagnetic radiation and results in the production of a neutron. The photoneutron reaction is shown below.
■ ■ ■
■ ■
■ ■ ■
■ ■

There is an abundant supply of high energy gammas in a reactor that has been operated because many of the fission products are gamma emitters. All water-cooled reactors have some deuterium present in the coolant in the reactor core because a small fraction of natural hydrogen is the isotope deuterium. The atom percentage of deuterium in the water ranges from close to the naturally occurring value (0.015%) for light water reactors to above 90% deuterium for heavy

The supply of gamma rays decreases with time after shutdown as the gamma emitters decay; therefore, the photoneutron production rate also decreases. In a few particular reactors, additional D O (heavy water) may be added to the reactor to increase the production of 2 photoneutrons following a long shutdown period.

Installed Neutron Sources

Because intrinsic neutron sources can be relatively weak or dependent upon the recent power history of the reactor, many reactors have artificial sources of neutrons installed. These neutron sources ensure that shutdown neutron levels are high enough to be detected by the nuclear instruments at all times. This provides a true picture of reactor conditions and any change in these conditions. An installed neutron source is an assembly placed in or near the reactor for the sole purpose of producing source neutrons.

One strong source of neutrons is the artificial nuclide californium-252, which emits neutrons at the rate of about 2 x 10 neutrons per second per gram as the result of spontaneous fission. 12 Important drawbacks for some applications may be its high cost and its short half-life (2.65 years).

Many installed neutron sources use the (■,n) reaction with beryllium. These sources are composed of a mixture of metallic beryllium (100% beryllium-9) with a small quantity of an alpha particle emitter, such as a compound of radium, polonium, or plutonium. The reaction that occurs is shown below.

The beryllium is intimately (homogeneously) mixed with the alpha emitter and is usually enclosed in a stainless steel capsule.

Another type of installed neutron source that is widely used is a photoneutron source that employs the (■,n) reaction with beryllium. Beryllium is used for photoneutron sources because its stable isotope beryllium-9 has a weakly attached last neutron with a binding energy of only 1.66 MeV. Thus, a gamma ray with greater energy than 1.66 MeV can cause neutrons to be ejected by the (■,n) reaction as shown below.

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Neutron Sources Summary

& Intrinsic neutron sources are sources of neutrons from materials that are in the reactor for other purposes such as fuel, burnable poison, or moderator.

& Installed neutron sources are materials or components placed in the reactor specifically for the purpose of producing source neutrons.

& Examples of intrinsic neutron sources are listed below.

Spontaneous fission of heavy nuclides in fuel, such as uranium-238, uranium-235, and plutonium-239, results in fission fragments and free neutrons.

Boron-11 mixed with the fuel undergoes an alpha-neutron reaction and becomes nitrogen-14.

Deuterium present in the reactor coolant undergoes a gamma■neutron reaction and becomes hydrogen-1.

& Examples of installed neutron sources are listed below.

Spontaneous fission of californium-252 results in fission fragments and free neutrons.

Beryllium-9 undergoes an alpha-neutron reaction (alpha from the decay of plutonium, polonium, or radium) and becomes carbon-12.

Beryllium-9 undergoes a gamma-neutron reaction (high energy gamma from decay of antimony-124) and becomes beryllium-8.

neutrons the radioactive antimony becomes an emitter of high energy gammas. The photoneutron sources of this type are constructed somewhat differently from the (■,n) types. One design incorporates a capsule of irradiated antimony enclosed in a beryllium sleeve. The entire assembly is then encased in a stainless steel cladding. A large reactor may have several neutron sources of this type installed within the core.

Summary

The important information in this chapter is summarized below.

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Reactor Theory (Neutron Characteristics) NUCLEAR CROSS SECTIONS AND NEUTRON FLUX
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NUCLEAR CROSS SECTIONS AND NEUTRON FLUX

To determine the frequency of neutron interactions, it is necessary to describe the availability of neutrons to cause interaction and the probability of a neutron interacting with material. The availability of neutrons and the probability of interaction are quantified by the neutron flux and nuclear cross section.

EO 2.1 DEFINE the following terms:
a. Atom density d. Barn
b. Neutron flux e. Macroscopic cross section
c. Microscopic cross section f. Mean free path

EO 2.2 EXPRESS macroscopic cross section in terms of microscopic cross section.

EO 2.3 DESCRIBE how the absorption cross section of typical nuclides varies with neutron energy at energies below the resonance absorption region.

EO 2.4 DESCRIBE the cause of resonance absorption in terms of nuclear energy levels.

EO 2.5 DESCRIBE the energy dependence of resonance absorption peaks for typical light and heavy nuclei.

EO 2.6 EXPRESS mean free path in terms of macroscopic cross section.

EO 2.7 Given the number densities (or total density and component fractions) and microscopic cross sections of components, CALCULATE the macroscopic cross section for a mixture.

EO 2.8 CALCULATE a macroscopic cross section given a material density, atomic mass, and microscopic cross section.

EO 2.9 EXPLAIN neutron shadowing or self-shielding.

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'

Introduction

Fission neutrons are born with an average energy of about 2 MeV. These fast neutrons interact with the reactor core materials in various absorption and scattering reactions. Collisions that result in scattering are useful in slowing neutrons to thermal energies. Thermal neutrons may be absorbed by fissile nuclei to produce more fissions or be absorbed in fertile material for conversion to fissionable fuel. Absorption of neutrons in structural components, coolant, and other non-fuel material results in the removal of neutrons without fulfilling any useful purpose. To safely and efficiently operate a nuclear reactor it is necessary to predict the probability that

the availability of neutrons can be determined, then the rate at which these nuclear reactions take place can be predicted.

Atom Density

One important property of a material is the atom density. The atom density is the number of atoms of a given type per unit volume of the material. To calculate the atom density of a substance use Equation (2-1).

(2-1)

where:

N = atom density (atoms/cm)3

' = density (g/cm)3

N = Avogadro's number (6.022 x 10 atoms/mole) A

23

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M = gram atomic weight

N

NA

M

2.699 g

cm3

6.022 x 1023 atoms

mole

26.9815 g

mole

6.024 x 1022 atoms

cm3

Example:

A block of aluminum has a density of 2.699 g/cm . If the gram atomic weight of

3

aluminum is 26.9815 g, calculate the atom density of the aluminum.

Solution:

Cross Sections

The probability of a neutron interacting with a nucleus for a particular reaction is dependent upon not only the kind of nucleus involved, but also the energy of the neutron. Accordingly, the absorption of a thermal neutron in most materials is much more probable than the absorption of a fast neutron. Also, the probability of interaction will vary depending upon the type of reaction involved.

The probability of a particular reaction occurring between a neutron and a nucleus is called the microscopic cross section ( ) of the nucleus for the particular reaction. This cross section will vary with the energy of the neutron. The microscopic cross section may also be regarded as the effective area the nucleus presents to the neutron for the particular reaction. The larger the effective area, the greater the probability for reaction.

Because the microscopic cross section is an area, it is expressed in units of area, or square centimeters. A square centimeter is tremendously large in comparison to the effective area of a nucleus, and it has been suggested that a physicist once referred to the measure of a square centimeter as being "as big as a barn" when applied to nuclear processes. The name has

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between barns and cm is shown below. 2
1 barn = 10-24 cm2
N
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Whether a neutron will interact with a certain volume of material depends not only on the microscopic cross section of the individual nuclei but also on the number of nuclei within that volume. Therefore, it is necessary to define another kind of cross section known as the macroscopic cross section ( ). The macroscopic cross section is the probability of a given reaction occurring per unit travel of the neutron. is related to the microscopic cross section ( ) by the relationship shown below.
(2-2)
where:
= macroscopic cross section (cm)
-1
N = atom density of material (atoms/cm)
3
= microscopic cross-section (cm)
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2
The difference between the microscopic and macroscopic cross sections is extremely important and is restated for clarity. The microscopic cross section ( ) represents the effective target area that a single nucleus presents to a bombarding particle. The units are given in barns or cm .
2
The macroscopic cross section ( ) represents the effective target area that is presented by all of the nuclei contained in 1 cm of the material. The units are given as 1/cm or cm .
3 -1
A neutron interacts with an atom of the material it enters in two basic ways. It will either interact through a scattering interaction or through an absorption reaction. The probability of a neutron being absorbed by a particular atom is the microscopic cross section for absorption, . The probability of a neutron scattering off of a particular nucleus is the microscopic cross a section for scattering, s
. The sum of the microscopic cross section for absorption and the microscopic cross section for scattering is the total microscopic cross section, . T
= + T a s
Both the absorption and the scattering microscopic cross sections can be further divided. For instance, the scattering cross section is the sum of the elastic scattering cross section (se) and the inelastic scattering cross section ( ). si
= + s se si
The microscopic absorption cross section ( ) includes all reactions except scattering. However, a for most purposes it is sufficient to merely separate it into two categories, fission (f) and capture ( ). Radiative capture of neutrons was described in the Neutron Interactions chapter c of Module 1.
= + a f c
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Reactor Theory (Neutron Characteristics) NUCLEAR CROSS SECTIONS AND NEUTRON FLUX
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The variation of absorption cross sections with neutron energy is often complicated. For many elements the absorption cross sections are small, ranging from a fraction of a barn to a few barns for slow (or thermal) neutrons.

For a considerable number of nuclides of moderately high (or high) mass numbers, an examination of the variation of the absorption cross section with the energy of the incident neutron reveals the existence of three regions on a curve of absorption cross section versus neutron energy. This cross section is illustrated in Figure 1. First, the cross section decreases steadily with increasing neutron energy in a low energy region, which includes the thermal range (E < 1 eV). In this region the absorption cross section, which is often high, is inversely proportional to the velocity (v). This region is frequently referred to as the "1/v region," because the absorption cross section is proportional to 1/v, which is the reciprocal of neutron velocity. Following the 1/v region, there occurs the "resonance region" in which the cross sections rise sharply to high values called "resonance peaks" for neutrons of certain energies, and then fall again. These energies are called resonance energies and are a result of the affinity of the nucleus for neutrons whose energies closely match its discrete, quantum energy levels. That is, when the binding energy of a neutron plus the kinetic energy of the neutron are exactly equal to the amount required to raise a compound nucleus from its ground state to a quantum level, resonance absorption occurs. The following example problem further illustrates this point.

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Assuming that uranium-236 has a nuclear quantum energy level at 6.8 MeV above its ground state, calculate the kinetic energy a neutron must possess to undergo resonant absorption in uranium-235 at this resonance energy level.

Fixture checkpoint marker-0659-xray is recorded on fixture page 0659.

BE = [Mass(U) + Mass(neutron) - Mass(U)] x 931 MeV/amu 235 236

BE = (235.043925 + 1.008665 - 236.045563) x 931 MeV/amu

BE = (0.007025 amu) x 931 MeV/amu = 6.54 MeV

6.8 MeV - 6.54 MeV = 0.26 MeV

The difference between the binding energy and the quantum energy level equals the amount of kinetic energy the neutron must possess. The typical heavy nucleus will have many closely■spaced resonanc complex and have more possible configurations and corresponding energy states. Light nuclei, being less complex, have fewer possible energy states and fewer resonances that are sparsely distributed at higher energy levels.

For higher neutron energies, the absorption cross section steadily decreases as the energy of the neutron increases. This is called the "fast neutron region." In this region the absorption cross sections are usually less than 10 barns.

With the exception of hydrogen, for which the value is fairly large, the elastic scattering cross sections are generally small, for example, 5 barns to 10 barns. This is close to the magnitude of the actual geometric cross sectional area expected for atomic nuclei. In potential scattering, the cross section is essentially constant and independent of neutron energy. Resonance elastic scattering and inelastic scattering exhibit resonance peaks similar to those associated with absorption cross sections. The resonances occur at lower energies for heavy nuclei than for light nuclei. In general, the variations in scattering cross sections are very small when compared to the variations that occur in absorption cross sections.

Mean Free Path

travel, then the inverse of this value describes how far the neutron will travel (in the average case) before undergoing an interaction. This average distance traveled by a neutron before interaction is known as the mean free path for that interaction and is represented by the symbol ■. The relationship between the mean free path (■) and the macroscopic cross section (*) is shown below.

(2-3)

'

(

)

Calculation of Macroscopic Cross Section and Mean Free Path

Most materials are composed of several elements, and because most elements are composed of several isotopes, most materials involve many cross sections, one for each isotope involved. Therefore, to include all the isotopes within a given material, it is necessary to determine the macroscopic cross section for each isotope and then sum all the individual macroscopic cross sections. Equation (2-4) can be used to determine the macroscopic cross section for a composite material.

* = N) + N) + N) + ... N) (2-4) 1 1 2 2 3 3 nn

where:

N = the number of nuclei per cm of the n element n

3 th

) = the microscopic cross section of the n element n

th

The following example problems illustrate the calculation of the macroscopic cross section for a single element and for combinations of materials.

Example 1:

Find the macroscopic thermal neutron absorption cross section for iron, which has a density of 7.86 g/cm . The microscopic cross section for absorption of iron is 2.56 barns 3 and the gram atomic weight is 55.847 g.

Solution:

Step 1: Using Equation (2-1), calculate the atom density of iron.

Fixture checkpoint marker-0660-xray is recorded on fixture page 0660.

Step 2: Use this atom density in Equation (2-2) to calculate the macroscopic cross section.

'

'

Example 2:

An alloy is composed of 95% aluminum and 5% silicon (by weight). The density of the alloy is 2.66 g/cm . Properties of aluminum and silicon are shown below. 3

Element	Gram Atomic Weight	) a (barns)	) s (barns)
Aluminum	26.9815	0.23	1.49
Silicon	28.0855	0.16	2.20

1. Calculate the atom densities for the aluminum and silicon.

neutrons.
3. Calculate the mean free paths for absorption and scattering.
Solution:
Step 1: The density of the aluminum will be 95% of the total density. Using
Equation (2-1) yields the atom densities.
a NAl a,Al NSi a,Si
5.64 x 1022 atoms
cm3
0.23 x 10 24 cm22.85 x 1021 atoms
cm3
Fixture checkpoint marker-0661-xray is recorded on fixture page 0661.
0.16 x 10 24 cm2
0.0134 cm 1
s NAl s,Al NSi s,Si
5.64 x 1022 atoms
cm3
1.49 x 10 24 cm22.85 x 1021 atoms
cm3
2.20 x 10 24 cm2
0.0903 cm 1
a
1
a
1
0.01345 cm 1
74.3 cm
s
1
s
1
0.0903 cm 1
11.1 cm
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Step 2: The macroscopic cross sections for absorption and scattering are
calculated using Equation (2-4).
Step 3: The mean free paths are calculated by inserting the macroscopic cross
sections calculated above into Equation (2-3).
Thus, a neutron must travel an average of 74.3 cm to interact by absorption in this alloy, but it
must travel only 11.1 cm to interact by scattering.
)
)
)
)
(■
(■
DOE-HDBK-1019/1-93

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Effects of Temperature on Cross Section
As discussed, the microscopic absorption cross section varies significantly as neutron energy varies. The microscopic cross sections provided on most charts and tables are measured for a standard neutron velocity of 2200 meters/second, which corresponds to an ambient temperature of 68(F. Therefore, if our material is at a higher temperature, the absorption cross section will be lower than the value for 68(F, and any cross sections which involve absorption (for example,) ,) ,)) must be corrected for the existing temperature. acf
The following formula is used to correct microscopic cross sections for temperature. Although the example illustrates absorption cross section, the same formula may be used to correct capture and fission cross sections.
where:
) = microscopic cross section corrected for temperature
) = microscopic cross section at reference temperature (68(F or 20(C) o
T = reference temperature (68(F) in degrees Rankine ((R) or Kelvin ((K) o
T = temperature for which corrected value is being calculated
NOTE: When using this formula, all temperatures must be converted to (R or (K.
Fixture checkpoint marker-0662-xray is recorded on fixture page 0662.
(R = (F + 460
(K = (C + 273
Example:
What is the value of)f for uranium-235 for thermal neutrons at 500(F? Uranium-235 has a) of 583 barns at 68(F. f
Solution:
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Reactor Theory (Neutron Characteristics) NUCLEAR CROSS SECTIONS AND NEUTRON FLUX
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Neutron Flux
Macroscopic cross sections for neutron reactions with materials determine the probability of one neutron undergoing a specific reaction per centimeter of travel through that material. If one wants to determine how many reactions will actually occur, it is necessary to know how many neutrons are traveling through the material and how many centimeters they travel each second. It is convenient to consider the number of neutrons existing in one cubic centimeter at any one instant and the total distance they travel each second while in that cubic centimeter. The number of neutrons existing in a cm of material at any instant is called neutron density and is 3 represented by the symbol n with units of neutrons/cm . The total distance these neutrons can 3 travel each second will be determined by their velocity.
A good way of defining neutron flux (1) is to consider it to be the total path length covered by all neutrons in one cubic centimeter during one second. Mathematically, this is the equation below.
1 = n v (2-5)
where:
1 = neutron flux (neutrons/cm -sec) 2
n = neutron density (neutrons/cm)3
v = neutron velocity (cm/sec)
The term neutron flux in some applications (for example, cross section measurement) is used as parallel beams of neutrons traveling in a single direction. The intensity (I) of a neutron beam is the product of the neutron density times the average neutron velocity. The directional beam

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a surface perpendicular to the direction of the beam.
One can think of the neutron flux in a reactor as being comprised of many neutron beams traveling in various directions. Then, the neutron flux becomes the scalar sum of these directional flux intensities (added as numbers and not vectors), that is, 1 = I + I + I +...I . 123 n
Since the atoms in a reactor do not interact preferentially with neutrons from any particular direction, all of these directional beams contribute to the total rate of reaction. In reality, at a given point within a reactor, neutrons will be traveling in all directions.
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NUCLEAR CROSS SECTIONS AND NEUTRON FLUX Reactor Theory (Neutron Characteristics)
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Self-Shielding
In some locations within the reactor, the flux level may be significantly lower than in other areas due to a phenomenon referred to as neutron shadowing or self-shielding. For example, the interior of a fuel pin or pellet will "see" a lower average flux level than the outer surfaces since an appreciable fraction of the neutrons will have been absorbed and therefore cannot reach the interior of the fuel pin. This is especially important at resonance energies, where the absorption cross sections are large.
Summary
The important information in this chapter is summarized below.
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Reactor Theory (Neutron Characteristics) NUCLEAR CROSS SECTIONS AND NEUTRON FLUX
Rev. 0 Page 17 NP-02
Nuclear Cross Section and Neutron Flux Summary
& Atom density (N) is the number of atoms of a given type per unit volume of
Fixture checkpoint marker-0663-xray is recorded on fixture page 0663.
material.
& Microscopic cross section ()) is the probability of a given reaction occurring between a neutron and a nucleus.
& Microscopic cross sections are measured in units of barns, where 1 barn = 10-24 cm .2
& Macroscopic cross section (*) is the probability of a given reaction occurring per unit length of travel of the neutron. The units for macroscopic cross section are cm . -1
& The mean free path (■) is the average distance that a neutron travels in a material between interactions.
& Neutron flux (1) is the total path length traveled by all neutrons in one cubic centimeter of material during one second.
& The macroscopic cross section for a material can be calculated using the equation below.
* = N)
& The absorption cross section for a material usually has three distinct regions. At low neutron energies (<1 eV) the cross section is inversely proportional to the neutron velocity.
& Resonance absorption occurs when the sum of the kinetic energy of the neutron and its binding energy is equal to an allowed nuclear energy level of the nucleus.
& Resonance peaks exist at intermediate energy levels. For higher neutron energies, the absorption cross section steadily decreases as the neutron energy increases.
& The mean free path equals 1/*.

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the equation below.
* = N 1 1 2 2 3 3 nn) + N) + N) + ... N)
& Self-shielding is where the local neutron flux is depressed within a material due to neutron absorption near the surface of the material.
1)
REACTION RATES DOE-HDBK-1019/1-93 Reactor Theory (Neutron Characteristics)
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REACTION RATES
It is possible to determine the rate at which a nuclear reaction will take place based on the neutron flux, cross section for the interaction, and atom density of the target. This relationship illustrates how a change in one of these items affects the reaction rate.
EO 2.10 Given the neutron flux and macroscopic cross section, CALCULATE the reaction rate.
EO 2.11 DESCRIBE the relationship between neutron flux and reactor power.
Reaction Rates
If the total path length of all the neutrons in a cubic centimeter in a second is known, (neutron flux (1)), and if the probability of having an interaction per centimeter path length is also known (macroscopic cross section (*)), multiply them together to get the number of interactions taking place in that cubic centimeter in one second. This value is known as the reaction rate and is denoted by the symbol R. The reaction rate can be calculated by the equation shown below.
R = 1 * (2-6)
where:
R = reaction rate (reactions/sec)
1 = neutron flux (neutrons/cm -sec) 2
* = macroscopic cross section (cm) -1
Substituting the fact that * = N) into Equation (2-6) yields the equation below.
where:
) = microscopic cross section (cm)2
N = atom density (atoms/cm)3
Fixture checkpoint marker-0664-xray is recorded on fixture page 0664.
Rf f
1 x 1013 neutrons
cm2
sec
0.1 cm 1
1 x 1012 fissions
cm3
sec
Rf f
f
Rf
1.29 x 1012 fissions
cm3
sec
3 x 1013 neutrons
cm2

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0.043 cm 1
Reactor Theory (Neutron Characteristics) DOE-HDBK-1019/1-93 REACTION RATES
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The reaction rate calculated will depend on which macroscopic cross section is used in the calculation. Normally, the reaction rate of greatest interest is the fission reaction rate.
Example:
If a one cubic centimeter section of a reactor has a macroscopic fission cross section of 0.1 cm , and if the thermal neutron flux is 10 neutrons/cm -sec, what is the fission rate -1 13 2 in that cubic centimeter?
Solution:
In addition to using Equation (2-6) to determine the reaction rate based on the physical properties of material, it is also possible to algebraically manipulate the equation to determine physical properties if the reaction rate is known.
Example:
A reactor operating at a flux level of 3 x 10 neutrons/cm -sec contains 10 atoms of 13 2 20
uranium-235 per cm . The reaction rate is 1.29 x 10 fission/cm . Calculate and .
3 12 3
f f
Solution:
Step 1: The macroscopic cross section can be determined by solving
Equation (2-6) for and substituting the appropriate values. f
Rf N f
f
Rf
N
1.29 x 1012 fissions
cm3
sec
1 x 1020 atoms
cm3
3 x 1013 neutrons
cm2
sec
4.3 x 10 22 cm2 1 barn
1 x 10 24 cm2
430 barns
1 watt 1 erg
Fixture checkpoint marker-0665-xray is recorded on fixture page 0665.
1 x 10 7 watt sec
1 MeV
1.602 x 10 6
erg
1 fission
200 MeV
3.12 x 1010 fissions
second
REACTION RATES DOE-HDBK-1019/1-93 Reactor Theory (Neutron Characteristics)

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Step 2: To find the microscopic cross section, replace f with (N x f) and solve for .
f
Reactor Power Calculation
Multiplying the reaction rate per unit volume by the total volume of the core results in the total number of reactions occurring in the core per unit time. If the amount of energy involved in each reaction were known, it would be possible to determine the rate of energy release (power) due to a certain reaction.
In a reactor where the average energy per fission is 200 MeV, it is possible to determine the number of fissions per second that are necessary to produce one watt of power using the following conversion factors.
1 fission = 200 MeV
1 MeV = 1.602 x 10 ergs -6
1 erg = 1 x 10 watt-sec -7
This is equivalent to stating that 3.12 x 10 fissions release 1 watt-second of energy. 10
Fixture checkpoint marker-0666-xray is recorded on fixture page 0666.
P
th f V
3.12 x 1010 fissions
watt sec
Reactor Theory (Neutron Characteristics) DOE-HDBK-1019/1-93 REACTION RATES
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The power released in a reactor can be calculated based on Equation (2-6). Multiplying the reaction rate by the volume of the reactor results in the total fission rate for the entire reactor. Dividing by the number of fissions per watt-sec results in the power released by fission in the reactor in units of watts. This relationship is shown mathematically in Equation (2-7) below.
(2-7)
where:
P = power (watts)
= thermal neutron flux (neutrons/cm -sec) th
2
= macroscopic cross section for fission (cm) f
-1
V = volume of core (cm)
3
Relationship Between Neutron Flux and Reactor Power
In an operating reactor the volume of the reactor is constant. Over a relatively short period of time (days or weeks), the number density of the fuel atoms is also relatively constant. Since the atom density and microscopic cross section are constant, the macroscopic cross section must also be constant. Examining Equation (2-7), it is apparent that if the reactor volume and macroscopic cross section are constant, then the reactor power and the neutron flux are directly proportional. This is true for day-to-day operation. The neutron flux for a given power level will increase very slowly over a period of months due to the burnup of the fuel and resulting decrease in atom density and macroscopic cross section.
REACTION RATES DOE-HDBK-1019/1-93 Reactor Theory (Neutron Characteristics)
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Reaction Rates Summary
& The reaction rate is the number of interactions of a particular type occurring in a

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& The reaction rate can be calculated by the equation below.
R = 1 *
& Over a period of several days, while the atom density of the fuel can be considered constant, the neutron flux is directly proportional to reactor power.
Summary
The important information in this chapter is summarized below.
Reactor Theory (Neutron Characteristics) DOE-HDBK-1019/1-93 NEUTRON MODERATION
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NEUTRON MODERATION
In thermal reactors, the neutrons that cause fission are at a much lower energy than the energy level at which they were born from fission. In this type of reactor, specific materials must be included in the reactor design to reduce the energy level of the neutrons in an efficient manner.
EO 2.12 DEFINE the following concepts:
a. Thermalization d. Average logarithmic energy decrement
b. Moderator e. Macroscopic slowing down power
c. Moderating ratio
EO 2.13 LIST three desirable characteristics of a moderator.
EO 2.14 Given an average fractional energy loss per collision, CALCULATE the energy loss after a specified number of collisions.
Neutron Slowing Down and Thermalization
Fixture checkpoint marker-0667-xray is recorded on fixture page 0667.
Fission neutrons are produced at an average energy level of 2 MeV and immediately begin to slow down as the result of numerous scattering reactions with a variety of target nuclei. After a number of collisions with nuclei, the speed of a neutron is reduced to such an extent that it has approximately the same average kinetic energy as the atoms (or molecules) of the medium in which the neutron is undergoing elastic scattering. This energy, which is only a small fraction of an electron volt at ordinary temperatures (0.025 eV at 20(C), is frequently referred to as the thermal energy, since it depends upon the temperature. Neutrons whose energies have been reduced to values in this region (< 1 eV) are designated thermal neutrons. The process of reducing the energy of a neutron to the thermal region by elastic scattering is referred to as thermalization, slowing down, or moderation. The material used for the purpose of thermalizing neutrons is called a moderator. A good moderator reduces the speed of neutrons in a small number of collisions, but does not absorb them to any great extent. Slowing the neutrons in as few collisions as possible is desirable in order to reduce the amount of neutron leakage from the core and also to reduce the number of resonance absorptions in non-fuel materials. Neutron leakage and resonance absorption will be discussed in the next module.
ln Ei
ln Ef
ln
Ei
Ef
2
A
2
3
NEUTRON MODERATION DOE-HDBK-1019/1-93 Reactor Theory (Neutron Characteristics)

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The ideal moderating material (moderator) should have the following nuclear properties.
large scattering cross section
small absorption cross section
large energy loss per collision
A convenient measure of energy loss per collision is the logarithmic energy decrement. The average logarithmic energy decrement is the average decrease per collision in the logarithm of the neutron energy. This quantity is represented by the symbol (Greek letter xi).
(2-8)
where:
= average logarithmic energy decrement
Ei = average initial neutron energy
Ef = average final neutron energy
The symbol is commonly called the average logarithmic energy decrement because of the fact that a neutron loses, on the average, a fixed fraction of its energy per scattering collision. Since the fraction of energy retained by a neutron in a single elastic collision is a constant for a given material, is also a constant. Because it is a constant for each type of material and does not depend upon the initial neutron energy, is a convenient quantity for assessing the moderating ability of a material.
The values for the lighter nuclei are tabulated in a variety of sources. The following commonly used approximation may be used when a tabulated value is not available.
This approximation is relatively accurate for mass numbers (A) greater than 10, but for some low values of A it may be in error by over three percent.
N
ln Ehigh ln Elow
ln
Ehigh
Elow
N
ln
Fixture checkpoint marker-0668-xray is recorded on fixture page 0668.
Ehigh
Elow
ln
2 x 106eV
0.025 eV
0.948
19.2 collisions
Reactor Theory (Neutron Characteristics) DOE-HDBK-1019/1-93 NEUTRON MODERATION
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Since represents the average logarithmic energy loss per collision, the total number of collisions necessary for a neutron to lose a given amount of energy may be determined by dividing into the difference of the natural logarithms of the energy range in question. The number of collisions (N) to travel from any energy, E , to any lower energy, E , can be calculated as shown below. high low
Example:
How many collisions are required to slow a neutron from an energy of 2 MeV to a thermal energy of 0.025 eV, using water as the moderator? Water has a value of 0.948 for .
Solution:

energy loss per collision as opposed to an average logarithmic fraction. If the initial neutron energy level and the average fractional energy loss per collision are known, the final energy level for a given number of collisions may be computed using the following formula.

! (

NEUTRON MODERATION DOE-HDBK-1019/1-93 Reactor Theory (Neutron Characteristics)
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(2-9)

where:

E = initial neutron energy o

E = neutron energy after N collisions N

x = average fractional energy loss per collision

N = number of collisions

Example:

If the average fractional energy loss per collision in hydrogen is 0.63, what will be the energy of a 2 MeV neutron after (a) 5 collisions? (b) 10 collisions?

Solution:

a)

b)

Macroscopic Slowing Down Power

Although the logarithmic energy decrement is a convenient measure of the ability of a material to slow neutrons, it does not measure all necessary properties of a moderator. A better measure of the capabilities of a material is the macroscopic slowing down power. The macroscopic slowing down power (MSDP) is the product of the logarithmic energy decrement and the macroscopic cross section for scattering in the material. Equation (2-10) illustrates how to calculate the macroscopic slowing down power.

(2-10)

MR

s

a

Reactor Theory (Neutron Characteristics) DOE-HDBK-1019/1-93 NEUTRON MODERATION
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Moderating Ratio

Macroscopic slowing down power indicates how rapidly a neutron will slow down in the material in question, but it still does not fully explain the effectiveness of the material as a moderator. An element such as boron has a high logarithmic energy decrement and a good slowing down power, but it is a poor moderator because of its high probability of absorbing neutrons.

Fixture checkpoint marker-0669-xray is recorded on fixture page 0669.

The most complete measure of the effectiveness of a moderator is the moderating ratio. The moderating ratio is the ratio of the macroscopic slowing down power to the macroscopic cross section for absorption. The higher the moderating ratio, the more effectively the material performs as a moderator. Equation (2-11) shows how to calculate the moderating ratio of a material.

(2-11)

Moderating properties of different materials are compared in Table 2.

TABLE 2

Moderating Properties of Materials

Material Collisions to Slowing Down

Number of Macroscopic

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Moderating
Ratio
H2O 0.927 19 1.425 62
D2O 0.510 35 0.177 4830
Helium 0.427 42 9 x 10 51 -6
Beryllium 0.207 86 0.154 126
Boron 0.171 105 0.092 0.00086
Carbon 0.158 114 0.083 216
NEUTRON MODERATION DOE-HDBK-1019/1-93 Reactor Theory (Neutron Characteristics)
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Summary
The important information in this chapter is summarized below.
Neutron Moderation Summary
Fixture checkpoint marker-0670-xray is recorded on fixture page 0670.
& Thermalization is the process of reducing the energy level of a neutron from the energy level at which it is produced to an energy level in the thermal range.
& The moderator is the reactor material that is present for the purpose of thermalizing neutrons.
& Moderating ratio is the ratio of the macroscopic slowing down power to the macroscopic cross section for absorption.
& The average logarithmic energy decrement (!) is the average change in the logarithm of neutron energy per collision.
& Macroscopic slowing down power is the product of the average logarithmic energy decrement and the macroscopic cross section for scattering.
& There are three desirable characteristics of a moderator.
1. large scattering cross section
2. small absorption cross section
3. large energy loss per collision
& The energy loss after a specified number of collisions can be calculated using the equation below.
E = E (1 - x) N o
N
■
■ ■
Reactor Theory (Neutron Characteristics) DOE-HDBK-1019/1-93 PROMPT AND DELAYED NEUTRONS
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PROMPT AND DELAYED NEUTRONS
Not all neutrons are released at the same time following fission. Most neutrons are released virtually instantaneously and are called prompt neutrons. A very small fraction of neutrons are released after the decay of fission products and are called delayed neutrons. Although delayed neutrons are a very small fraction of the total number of neutrons, they play an extremely important role in the control of the reactor.
EO 3.1 STATE the origin of prompt neutrons and delayed neutrons.
EO 3.2 STATE the approximate fraction of neutrons that are born as delayed neutrons from the fission of the following nuclear fuels:
a. Uranium-235
b. Plutonium-239

EO 3.4 EXPLAIN prompt and delayed neutron generation times.
EO 3.5 Given prompt and delayed neutron generation times and delayed neutron fraction, CALCULATE the average generation time.
EO 3.6 EXPLAIN the effect of delayed neutrons on reactor control.

Neutron Classification

The great majority (over 99%) of the neutrons produced in fission are released within about 10-13 seconds of the actual fission event. These are called prompt neutrons. A small portion of fission neutrons are delayed neutrons, which are produced for some time after the fission process has taken place. The delayed neutrons are emitted immediately following the first beta decay of a fission fragment known as a delayed neutron precursor. An example of a delayed neutron precursor is bromine-87, shown below.

PROMPT AND DELAYED NEUTRONS DOE-HDBK-1019/1-93 Reactor Theory (Neutron Characteristics) NP-02 Page 30 Rev. 0

For most applications, it is convenient to combine the known precursors into groups with appropriately averaged properties. These groups vary somewhat depending on the fissile material in use. Table 3 lists the characteristics for the six precursor groups resulting from thermal fission of uranium-235. The fraction of all neutrons that are produced by each of these precursors is called the delayed neutron fraction for that precursor. The total fraction of all neutrons born as delayed neutrons is called the delayed neutron fraction (■). The fraction of

Fixture checkpoint marker-0671-xray is recorded on fixture page 0671.

delayed neutrons produced varies depending on the predominant fissile nuclide in use. The delayed neutron fractions (■) for the fissile nuclides of most interest are as follows: uranium-233 (0.0026), uranium-235 (0.0065), uranium-238 (0.0148), and plutonium-239 (0.0021).

TABLE 3
Delayed Neutron Precursor Groups
for Thermal Fission in Uranium-235

Group	Half-Life (sec)	Delayed Neutron Fraction	Average Energy (MeV)
1	55.7	0.00021	0.25
2	22.7	0.00142	0.46
3	6.2	0.00127	0.41
4	2.3	0.00257	0.45
5	0.61	0.00075	0.41
6	0.23	0.00027	-
Total	-	0.0065	-

Neutron Generation Time

The neutron generation time is the time required for neutrons from one generation to cause the fissions that produce the next generation of neutrons. The generation time for prompt neutrons (5* - pronounced "ell-star") is the total time from birth to rebirth. Three time intervals are involved: (a) the time it takes a fast neutron to slow down to thermal energy, (b) the time the now thermal neutron exists prior to absorption in fuel, and (c) the time required for a fissionable nucleus to emit a fast neutron after neutron absorption.

Fast neutrons slow to thermal energies or leak out of the reactor in 10 seconds to 10 seconds, -4 -6 depending on the moderator. In water moderated reactors, thermal neutrons tend to exist for about 10 seconds before they are absorbed. Fission and fast neutron production following -4 neutron absorption in a fissionable nucleus occurs in about 10 seconds. Thus, fast reactors -13 have an 5* of about 10 seconds, while thermal reactors have an 5* of about 10 seconds + 10 -6 -6 -4

■■■■
■■■■
■■

On the other hand, the average generation time for the six delayed neutron groups is the total time from the birth of the fast neutron to the emission of the delayed neutron. Again, three time intervals are involved: (a) the time it takes a fast neutron to slow down to thermal energy, (b) the time the thermal neutron exists prior to absorption, and (c) the average time from neutron absorption to neutron emission by the six precursor groups. The average time for decay of precursors from uranium-235 is 12.5 seconds. The other terms in the delayed neutron generation time are insignificant when compared to this value, and the average delayed neutron generation time becomes ■ 12.5 seconds.

A neutron generation time in the range of 10 seconds to 10 seconds or faster could result in -4 -5 very rapid power excursions, and control would not be possible without the dependence upon delayed neutrons to slow down the rate of the reaction. The average generation time, and hence the rate that power can rise, is determined largely by the delayed neutron generation time. The following equation shows this mathematically.

(2-12)

Example:

Assume a prompt neutron generation time for a particular reactor of 5 x 10 seconds and -5 a delayed neutron generation time of 12.5 seconds. If ■ is 0.0065, calculate the average generation time.

Solution:

This example demonstrates the effect delayed neutrons have on the neutron generation time and thus reactor control. If a reactor were to be operated in a sustained chain reaction using only prompt neutrons (■ = 0), the generation time from the previous example would be about

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5 x 10 seconds. However, by operating the reactor such that a 0.0065 fraction of neutrons are -5 delayed, the generation life time is extended to 0.0813 seconds, providing time for adequate operator control. Therefore, although only a small fraction of the total neutron population, delayed neutrons are extremely important to the control and maintenance of a sustained fission chain reaction.

Summary

The important information in this chapter is summarized on the following page.

■■■■

Prompt and Delayed Neutrons Summary

& Prompt neutrons are released directly from fission within 10 seconds of the -13 fission event.

& Delayed neutrons are released from the decay of fission products that are called delayed neutron precursors. Delayed neutron precursors are grouped according to half-life. Half-lives vary from fractions of a second to almost a minute.

& The fraction of neutrons born as delayed neutrons is different for different fuel materials. Following are values for some common fuel materials.

Uranium-235 0.0065
Plutonium-239 0.0021

delayed neutron precursors. When a delayed neutron precursor undergoes a ■ decay, it results in an excited daughter nucleus which immediately ejects a - neutron. Therefore, these delayed neutrons appear with a half-life of the delayed neutron precursor.

& The delayed neutron generation time is the total time from the birth of the fast neutron to the emission of the delayed neutron in the next generation. Delayed neutron generation times are dominated by the half-life of the delayed neutron precursor. The average delayed neutron generation time is about 12.5 seconds.

& A prompt neutron generation time is the sum of the amount of time it takes a fast neutron to thermalize, the amount of time the neutron exists as a thermal neutron before it is absorbed, and the amount of time between a fissionable nuclide absorbing a neutron and fission neutrons being released. Prompt neutron generation time is about 5 x 10 seconds. -5

& The average neutron generation time can be calculated from the prompt and delayed neutron generation times and the delayed neutron fraction using Equation (2-12).

& Delayed neutrons are responsible for the ability to control the rate at which power can rise in a reactor. If only prompt neutrons existed, reactor control would not be possible due to the rapid power changes.

Figure 2 Prompt Fission Neutron Energy Spectrum for Thermal Fission of Uranium-235

NEUTRON FLUX SPECTRUM

The number of neutrons that exist at a given energy level varies. A plot of either the fraction of neutrons or the neutron flux at a given energy versus the energy level is called a neutron energy spectrum. The neutron energy spectrum varies widely for different types of reactors.

EO 4.1 STATE the average energy at which prompt neutrons are produced.

EO 4.2 DESCRIBE the neutron energy spectrum in the following reactors:

a. Fast reactor

b. Thermal reactor

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EO 4.3 EXPLAIN the reason for the particular shape of the fast, intermediate, and slow energy regions of the neutron flux spectrum for a thermal reactor.

Prompt Neutron Energies

The neutrons produced by fission are high energy neutrons, and almost all fission neutrons have energies between 0.1 MeV and 10 MeV. The neutron energy distribution, or spectrum, may best be described by plotting the fraction of neutrons per MeV as a function of neutron energy, as

figure it can be seen that the most probable neutron energy is about 0.7 MeV. Also, from this data it can be shown that the average energy of fission neutrons is about 2 MeV. Figure 2 is the neutron energy spectrum for thermal fission in uranium-235. The values will vary slightly for other nuclides.

NEUTRON FLUX SPECTRUM DOE-HDBK-1019/1-93 Reactor Theory (Neutron Characteristics)

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Figure 3 Comparison of Neutron Flux Spectra for Thermal and Fast Breeder Reactor

Thermal and Fast Breeder Reactor Neutron Spectra

The spectrum of neutron energies produced by fission varies significantly from the energy spectrum, or flux, existing in a reactor at a given time. Figure 3 illustrates the difference in neutron flux spectra between a thermal reactor and a fast breeder reactor. The energy distribution of neutrons from fission is essentially the same for both reactors, so the differences in the curve shapes may be attributed to the neutron moderation or slowing down effects.

No attempt is made to thermalize or slow down neutrons in the fast breeder reactor (liquid metal cooled); therefore, an insignificant number of neutrons exist in the thermal range. For the thermal reactor (water moderated), the spectrum of neutrons in the fast region (> 0.1 MeV) has a shape similar to that for the spectrum of neutrons emitted by the fission process.

In the thermal reactor, the flux in the intermediate energy region (1 eV to 0.1 MeV) has approximately a 1/E dependence. That is, if the energy (E) is halved, the flux doubles. This 1/E dependence is caused by the slowing down process, where elastic collisions remove a constant fraction of the neutron energy per collision (on the average), independent of energy; thus, the neutron loses larger amounts of energy per collision at higher energies than at lower energies. The fact that the neutrons lose a constant fraction of energy per collision causes the neutrons to tend to "pile up" at lower energies, that is, a greater number of neutrons exist at the lower energies as a result of this behavior.

vp
2 k T
m
vp
2 k T
m
2 1.38 x 10 16 erg
K
293 K
1.66 x 10 24 g
2.2 x 105 cm
sec
1 m
100 cm
2200 m

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Reactor Theory (Neutron Characteristics) DOE-HDBK-1019/1-93 NEUTRON FLUX SPECTRUM
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In the thermal region the neutrons achieve a thermal equilibrium with the atoms of the moderator material. In any given collision they may gain or lose energy, and over successive collisions will gain as much energy as they lose. These thermal neutrons, even at a specific temperature, do not all have the same energy or velocity; there is a distribution of energies, usually referred to as the Maxwell distribution (e.g., Figure 2). The energies of most thermal neutrons lie close to the most probable energy, but there is a spread of neutrons above and below this value.
Most Probable Neutron Velocities
The most probable velocity (vp) of a thermal neutron is determined by the temperature of the medium and can be determined by Equation (2-13) .
(2-13)
where:
v = most probable velocity of neutron (cm/sec) p
k = Boltzman's constant (1.38 x 10 erg/ K) -16
T = absolute temperature in degrees Kelvin (K)
m = mass of neutron (1.66 x 10 grams) -24
Example:
Fixture checkpoint marker-0675-xray is recorded on fixture page 0675.
Calculate the most probable velocities for neutrons in thermal equilibrium with their surroundings at the following temperatures. a) 20 C, b) 260 C.
Solution:
a) Calculate the most probable velocity for 20 C using Equation (2-13).
vp
2 k T
m
2 1.38 x 10 16 erg
K
533 K
1.66 x 10 24 g
2.977 x 105 cm
sec
1 m
100 cm
2977 m
sec
NEUTRON FLUX SPECTRUM DOE-HDBK-1019/1-93 Reactor Theory (Neutron Characteristics)
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b) Calculate the most probable velocity for 260 C using Equation (2-13).
From these calculations it is evident that the most probable velocity of a thermal neutron increases as temperature increases. The most probable velocity at 20 C is of particular importance since reference data, such as nuclear cross sections, are tabulated for a neutron velocity of 2200 meters per second.
Reactor Theory (Neutron Characteristics) DOE-HDBK-1019/1-93 NEUTRON FLUX SPECTRUM
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Neutron Flux Spectrum Summary
& Prompt neutrons are born at energies between 0.1 MeV and 10 MeV. The average prompt neutron energy is about 2 MeV.

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FOREWORD
The Department of Energy (DOE) Fundamentals Handbooks consist of ten academic subjects, which include Mathematics; Classical Physics; Thermodynamics, Heat Transfer, and Fluid Flow; Instrumentation and Control; Electrical Science; Material Science; Mechanical Science; Chemistry; Engineering Symbology, Prints, and Drawings; and Nuclear Physics and Reactor Theory. The handbooks are provided as an aid to DOE nuclear facility contractors.
These handbooks were first published as Reactor Operator Fundamentals Manuals in 1985 for use by DOE category A reactors. The subject areas, subject matter content, and level of detail of the Reactor Operator Fundamentals Manuals were determined from several sources. DOE Category A reactor training managers determined which materials should be included, and served as a primary reference in the initial development phase. Training guidelines from the commercial nuclear power industry, results of job and task analyses, and independent input from contractors and operations-oriented personnel were all considered and included to some degree in developing the text material and learning objectives.
The DOE Fundamentals Handbooks represent the needs of various DOE nuclear facilities' fundamental training requirements. To increase their applicability to nonreactor nuclear facilities, the Reactor Operator Fundamentals Manual learning objectives were distributed to the Nuclear Facility Training Coordination Program Steering Committee for review and comment. To update their reactor-specific content, DOE Category A reactor training managers also reviewed and commented on the content. On the basis of feedback from these sources, information that applied to two or more DOE nuclear facilities was considered generic and was included. The final draft Fixture checkpoint marker-0676-xray is recorded on fixture page 0676.
of each of the handbooks was then reviewed by these two groups. This approach has resulted in revised modular handbooks that contain sufficient detail such that each facility may adjust the content to fit their specific needs.
Each handbook contains an abstract, a foreword, an overview, learning objectives, and text material, and is divided into modules so that content and order may be modified by individual DOE contractors to suit their specific training needs. Each handbook is supported by a separate examination bank with an answer key.
The DOE Fundamentals Handbooks have been prepared for the Assistant Secretary for Nuclear Energy, Office of Nuclear Safety Policy and Standards, by the DOE Training Coordination Program. This program is managed by EG&G Idaho, Inc.
Rev. 0 NP
DOE-HDBK-1019/1-93
NUCLEAR PHYSICS AND REACTOR THEORY
OVERVIEW
The Department of Energy Fundamentals Handbook entitled Nuclear Physics and Reactor Theory was prepared as an information resource for personnel who are responsible for the operation of the Department's nuclear facilities. Almost all processes that take place in a nuclear facility involves the transfer of some type of energy. A basic understanding of nuclear physics and reactor theory is necessary for DOE nuclear facility operators, maintenance personnel, and the technical staff to safely operate and maintain the facility and facility support systems. The information in this handbook is presented to provide a foundation for applying engineering concepts to the job. This knowledge will help personnel understand the impact that their actions

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The Nuclear Physics and Reactor Theory handbook consists of four modules that are contained in two volumes. The following is a brief description of the information presented in each module of the handbook.
Volume 1 of 2
Module 1 - Atomic and Nuclear Physics
Introduces concepts of atomic physics including the atomic nature of matter, the chart of the nuclides, radioactivity and radioactive decay, neutron interactions and fission, and the interaction of radiation with matter.
Module 2 - Reactor Theory (Nuclear Parameters)
Provides information on reactor theory and neutron characteristics. Includes topics such as neutron sources, neutron flux, neutron cross sections, reaction rates, neutron moderation, and prompt and delayed neutrons.
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NUCLEAR PHYSICS AND REACTOR THEORY
OVERVIEW (Cont.)
Volume 2 of 2
Module 3 - Reactor Theory (Nuclear Parameters)
Explains the nuclear parameters associated with reactor theory. Topics include the neutron life cycle, reactivity and reactivity coefficients, neutron poisons, and control rods.
Module 4 - Reactor Theory (Reactor Operations)
Introduces the reactor operations aspect of reactor theory. Topics include subcritical multiplication, reactor kinetics, and reactor operation.
The information contained in this handbook is not all-encompassing. An attempt to present the entire subject of nuclear physics and reactor theory would be impractical. However, the Nuclear Physics and Reactor Theory handbook presents enough information to provide the reader with the fundamental knowledge necessary to understand the advanced theoretical concepts presented in other subject areas, and to understand basic system and equipment operation.
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Department of Energy
Fundamentals Handbook
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NUCLEAR PHYSICS
AND REACTOR THEORY
Module 1
Atomic and Nuclear Physics
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TERMINAL OBJECTIVE
1.0 Given sufficient information, DESCRIBE atoms, including components, structure, and nomenclature.
ENABLING OBJECTIVES
1.1 STATE the characteristics of the following atomic particles, including mass, charge, and location within the atom:
a. Proton
b. Neutron

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1.2 DESCRIBE the Bohr model of an atom.
1.3 DEFINE the following terms:
a. Nuclide c. Atomic number
b. Isotope d. Mass number
1.4 Given the standard A
ZX notation for a particular nuclide, DETERMINE the following:
a. Number of protons
b. Number of neutrons
c. Number of electrons
1.5 DESCRIBE the three forces that act on particles within the nucleus and affect the stability of the nucleus.
1.6 DEFINE the following terms:
a. Enriched uranium
b. Depleted uranium
1.7 DEFINE the following terms:
a. Mass defect
b. Binding energy
1.8 Given the atomic mass for a nuclide and the atomic masses of a neutron, proton, and electron, CALCULATE the mass defect and binding energy of the nuclide.

TERMINAL OBJECTIVE
2.0 Given necessary references, DESCRIBE the various modes of radioactive decay.
ENABLING OBJECTIVES
2.1 DESCRIBE the following processes:
a. Alpha decay d. Electron capture
b. Beta-minus decay e. Internal conversions
c. Beta-plus decay f. Isomeric transitions
2.2 Given a Chart of the Nuclides, WRITE the radioactive decay chain for a nuclide.
2.3 EXPLAIN why one or more gamma rays typically accompany particle emission.
Fixture checkpoint marker-0681-xray is recorded on fixture page 0681.
2.4 Given the stability curve on the Chart of the Nuclides, DETERMINE the type of radioactive decay that the nuclides in each region of the chart will typically undergo.
2.5 DEFINE the following terms:
a. Radioactivity d. Radioactive decay constant
b. Curie e. Radioactive half-life
c. Becquerel
2.6 Given the number of atoms and either the half-life or decay constant of a nuclide, CALCULATE the activity.
2.7 Given the initial activity and the decay constant of a nuclide, CALCULATE the activity at any later time.
2.8 CONVERT between the half-life and decay constant for a nuclide.
2.9 Given the Chart of the Nuclides and the original activity, PLOT the radioactive decay curve for a nuclide on either linear or semi-log coordinates.
2.10 DEFINE the following terms:
a. Radioactive equilibrium
b. Transient radioactive equilibrium

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TERMINAL OBJECTIVE
3.0 Without references, DESCRIBE the different nuclear interactions initiated by neutrons.
ENABLING OBJECTIVES
3.1 DESCRIBE the following scattering interactions between a neutron and a nucleus:
a. Elastic scattering
b. Inelastic scattering
3.2 STATE the conservation laws that apply to an elastic collision between a neutron and a nucleus.
3.3 DESCRIBE the following reactions where a neutron is absorbed in a nucleus:
a. Radiative capture
b. Particle ejection
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OBJECTIVES DOE-HDBK-1019/1-93 Atomic and Nuclear Physics
TERMINAL OBJECTIVE
4.0 Without references, DESCRIBE the fission process.
ENABLING OBJECTIVES
4.1 EXPLAIN the fission process using the liquid drop model of a nucleus.
4.2 DEFINE the following terms:
a. Excitation energy
b. Critical energy
4.3 DEFINE the following terms:
a. Fissile material
b. Fissionable material
c. Fertile material
4.4 DESCRIBE the processes of transmutation, conversion, and breeding.
4.5 DESCRIBE the curve of Binding Energy per Nucleon versus mass number and give a qualitative description of the reasons for its shape.
4.6 EXPLAIN why only the heaviest nuclei are easily fissioned.
4.7 EXPLAIN why uranium-235 fissions with thermal neutrons and uranium-238 fissions only with fast neutrons.
4.8 CHARACTERIZE the fission products in terms of mass groupings and radioactivity.
4.9 Given the nuclides involved and their masses, CALCULATE the energy released from fission.
4.10 Given the curve of Binding Energy per Nucleon versus mass number, CALCULATE the energy released from fission.
NP-01 Page x Rev. 0
Atomic and Nuclear Physics DOE-HDBK-1019/1-93 OBJECTIVES
Fixture checkpoint marker-0682-xray is recorded on fixture page 0682.
TERMINAL OBJECTIVE
5.0 Without references, DESCRIBE how the various types of radiation interact with matter.
ENABLING OBJECTIVES
5.1 DESCRIBE interactions of the following with matter:
a. Alpha particle c. Positron
b. Beta particle d. Neutron
5.2 DESCRIBE the following ways that gamma radiation interacts with matter:
a. Compton scattering
b. Photoelectric effect
c. Pair production

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OBJECTIVES DOE-HDBK-1019/1-93 Atomic and Nuclear Physics
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NP-01 Page xii Rev. 0
Atomic and Nuclear Physics DOE-HDBK-1019/1-93 ATOMIC NATURE OF MATTER
ATOMIC NATURE OF MATTER
All matter is composed of atoms. The atom is the smallest amount of matter that retains the properties of an element. Atoms themselves are composed of smaller particles, but these smaller particles no longer have the same properties as the overall element.
EO 1.1 STATE the characteristics of the following atomic particles, including mass, charge, and location within the atom:
a. Proton
b. Neutron
c. Electron
Fixture checkpoint marker-0683-xray is recorded on fixture page 0683.
EO 1.2 DESCRIBE the Bohr model of an atom.
EO 1.3 DEFINE the following terms:
a. Nuclide c. Atomic number
b. Isotope d. Mass number
EO 1.4 Given the standard A ZX notation for a particular nuclide, DETERMINE the following:
a. Number of protons
b. Number of neutrons
c. Number of electrons
EO 1.5 DESCRIBE the three forces that act on particles within the nucleus and affect the stability of the nucleus.
Structure of Matter
Early Greek philosophers speculated that the earth was made up of different combinations of basic substances, or elements. They considered these basic elements to be earth, air, water, and fire. Modern science shows that the early Greeks held the correct concept that matter consists of a combination of basic elements, but they incorrectly identified the elements.
Rev. 0 Page 1 NP-01
ATOMIC NATURE OF MATTER DOE-HDBK-1019/1-93 Atomic and Nuclear Physics
In 1661 the English chemist Robert Boyle published the modern criterion for an element. He defined an element to be a basic substance that cannot be broken down into any simpler substance after it is isolated from a compound, but can be combined with other elements to form compounds. To date, 105 different elements have been confirmed to exist, and researchers claim to have discovered three additional elements. Of the 105 confirmed elements, 90 exist in nature and 15 are man-made.
Another basic concept of matter that the Greeks debated was whether matter was continuous or discrete. That is, whether matter could be continuously divided and subdivided into ever smaller particles or whether eventually an indivisible particle would be encountered. Democritus in about 450 B.C. argued that substances were ultimately composed of small, indivisible particles that he labeled atoms. He further suggested that different substances were composed of different atoms or combinations of atoms, and that one substance could be converted into another by rearranging the atoms. It was impossible to conclusively prove or disprove this proposal for more than 2000 years.

Dalton in 1803. Dalton stated that each chemical element possesses a particular kind of atom, and any quantity of the element is made up of identical atoms of this kind. What distinguishes one element from another element is the kind of atom of which it consists, and the basic physical difference between kinds of atoms is their weight.

Subatomic Particles

For almost 100 years after Dalton established the atomic nature of atoms, it was considered impossible to divide the atom into even smaller parts. All of the results of chemical experiments during this time indicated that the atom was indivisible. Eventually, experimentation into electricity and radioactivity indicated that particles of matter smaller than the atom did indeed exist. In 1906, J. J. Thompson won the Nobel Prize in physics for establishing the existence of electrons. Electrons are negatively-charged particles that have 1/1835 the mass of the hydrogen atom. Soon after the discovery of electrons, protons were discovered. Protons are relatively large particles that have almost the same mass as a hydrogen atom and a positive charge equal in magnitude (but opposite in sign) to that of the electron. The third subatomic particle to be discovered, the neutron, was not found until 1932. The neutron has almost the same mass as the proton, but it is electrically neutral.

Bohr Model of the Atom

The British physicist Ernest Rutherford postulated that the positive charge in an atom is concentrated in a small region called a nucleus at the center of the atom with electrons existing

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in orbits around it. Niels Bohr, coupling Rutherford's postulation with the quantum theory introduced by Max Planck, proposed that the atom consists of a dense nucleus of protons surrounded by electrons traveling in discrete orbits at fixed distances from the nucleus. An electron in one of these orbits or shells has a specific or discrete quantity of energy (quantum). When an electron moves from one allowed orbit to another allowed orbit, the energy difference between the two states is emitted or absorbed in the form of a single quantum of radiant energy called a photon. Figure 1 is Bohr's model of the hydrogen atom showing an electron as having just dropped from the third shell to the first shell with the emission of a photon that has an energy = hv. (h = Planck's constant = 6.63 x 10-34 J-s and v = frequency of the photon.) Bohr's theory was the first to successfully account for the discrete energy levels of this radiation as measured in the laboratory. Although Bohr's atomic model is designed specifically to explain the hydrogen atom, his theories apply generally to the structure of all atoms. Additional information on electron shell theory can be found in the Chemistry Fundamentals Handbook.

Figure 1 Bohr's Model of the Hydrogen Atom

Properties of the three subatomic particles are listed in Table 1.

TABLE 1

Properties of Subatomic Particles

Particle	Location	Charge	Mass
Neutron	Nucleus	none	1.008665 amu
Proton	Nucleus	+1	1.007277 amu
Electron	Shells around nucleus	-1	0.0005486 amu

Measuring Units on the Atomic Scale

The size and mass of atoms are so small that the use of normal measuring units, while possible, is often inconvenient. Units of measure have been defined for mass and energy on the atomic

atomic mass unit (amu). One atomic mass unit is equal to 1.66 x 10-24 grams. The reason for this particular value for the atomic mass unit will be discussed in a later chapter. Note from Table 1 that the mass of a neutron and a proton are both about 1 amu. The unit for energy is the electron volt (eV). The electron volt is the amount of energy acquired by a single electron when it falls through a potential difference of one volt. One electron volt is equivalent to 1.602 x 10-19 joules or 1.18 x 10-19 foot-pounds.

Nuclides

The total number of protons in the nucleus of an atom is called the atomic number of the atom and is given the symbol Z. The number of electrons in an electrically-neutral atom is the same as the number of protons in the nucleus. The number of neutrons in a nucleus is known as the neutron number and is given the symbol N. The mass number of the nucleus is the total number of nucleons, that is, protons and neutrons in the nucleus. The mass number is given the symbol A and can be found by the equation Z + N = A.

Each of the chemical elements has a unique atomic number because the atoms of different elements contain a different number of protons. The atomic number of an atom identifies the particular element.

Each type of atom that contains a unique combination of protons and neutrons is called a nuclide. Not all combinations of numbers of protons and neutrons are possible, but about 2500 specific nuclides with unique combinations of neutrons and protons have been identified. Each nuclide is denoted by the chemical symbol of the element with the atomic number written as a subscript and the mass number written as a superscript, as shown in Figure 2. Because each element has a unique name, chemical symbol, and atomic number, only one of the three is necessary to identify the element. For this reason nuclides can also be identified by either the chemical name or the chemical symbol followed by the mass number (for example, U-235 or uranium-235).

Figure 2 Nomenclature for Identifying Nuclides

Fixture checkpoint marker-0685-xray is recorded on fixture page 0685.

Another common format is to use the abbreviation of the chemical element with the mass number superscripted (for example, 235U). In this handbook the format used in the text will usually be the element's name followed by the mass number. In equations and tables, the format in Figure 2 will usually be used.

Example:

State the name of the element and the number of protons, electrons, and neutrons in the nuclides listed below.

1
1H
10
5B
14
7N

4
4
8Cd
23
9
9
4Pu
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Solution:
The name of the element can be found from the Periodic Table (refer to Chemistry Fundamentals Handbook) or the Chart of the Nuclides (to be discussed later). The number of protons and electrons are equal to Z. The number of neutrons is equal to Z - A.
Nuclide Element Protons Electrons Neutrons
1
1H hydrogen 1 1 0
10
5B boron 5 5 5
14
7N nitrogen 7 7 7
11
4
4
8Cd cadmium 48 48 66
23
9
9
4Pu plutonium 94 94 145
Isotopes
Isotopes are nuclides that have the same atomic number and are therefore the same element, but differ in the number of neutrons. Most elements have a few stable isotopes and several unstable, radioactive isotopes. For example, oxygen has three stable isotopes that can be found in nature (oxygen-16, oxygen-17, and oxygen-18) and eight radioactive isotopes. Another example is hydrogen, which has two stable isotopes (hydrogen-1 and hydrogen-2) and a single radioactive
Fixture checkpoint marker-0686-xray is recorded on fixture page 0686.
isotope (hydrogen-3).
The isotopes of hydrogen are unique in that they are each commonly referred to by a unique name instead of the common chemical element name. Hydrogen-1 is almost always referred to as hydrogen, but the term protium is infrequently used also. Hydrogen-2 is commonly called deuterium and symbolized 2
1D. Hydrogen-3 is commonly called tritium and symbolized 31T. This
text will normally use the symbology 2
1H and 31H for deuterium and tritium, respectively.
Atomic and Nuclear Radii
The size of an atom is difficult to define exactly due to the fact that the electron cloud, formed by the electrons moving in their various orbitals, does not have a distinct outer edge. A reasonable measure of atomic size is given by the average distance of the outermost electron

approximately the same for all atoms, about 2 x 10 -8 cm.
Like the atom the nucleus does not have a sharp outer boundary. Experiments have shown that the nucleus is shaped like a sphere with a radius that depends on the atomic mass number of the atom. The relationship between the atomic mass number and the radius of the nucleus is shown in the following equation.
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r = (1.25 x 10 -13 cm) A1/3
where:
r = radius of the nucleus (cm)
A = atomic mass number (dimensionless)
The values of the nuclear radii for some light, intermediate, and heavy nuclides are shown in
Fixture checkpoint marker-0687-xray is recorded on fixture page 0687.
Table 2.
TABLE 2
Calculated Values for Nuclear
Radii
Nuclide Radius of Nucleus
1
1H 1.25 x 10 -13 cm
10
5B 2.69 x 10 -13 cm
5
2
6
6Fe 4.78 x 10 -13 cm
17
7
8
2Hf 7.01 x 10 -13 cm
23
9
8
2U 7.74 x 10 -13 cm
25
9
2
8Cf 7.89 x 10 -13 cm
From the table, it is clear that the radius of a typical atom (e.g. 2 x 10 -8 cm) is more than 25,000 times larger than the radius of the largest nucleus.
Nuclear Forces
In the Bohr model of the atom, the nucleus consists of positively-charged protons and electrically■neutra as nucleons. One problem that the Bohr model of the atom presented was accounting for an attractive force to overcome the repulsive force between protons.
Two forces present in the nucleus are (1) electrostatic forces between charged particles and (2) gravitational forces between any two objects that have mass. It is possible to calculate the magnitude of the gravitational force and electrostatic force based upon principles from classical physics.

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Fg
G m1 m2
r
2
Fe
K Q1 Q2
r
2
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Newton stated that the gravitational force between two bodies is directly proportional to the masses of the two bodies and inversely proportional to the square of the distance between the bodies. This relationship is shown in the equation below.
where:
F = gravitational force (newtons) g
m = mass of first body (kilograms) 1
m = mass of second body (kilograms) 2
G = gravitational constant (6.67 x 10 N-m /kg)
-11 2 2
Fixture checkpoint marker-0688-xray is recorded on fixture page 0688.
r = distance between particles (meters)
The equation illustrates that the larger the masses of the objects or the smaller the distance between the objects, the greater the gravitational force. So even though the masses of nucleons are very small, the fact that the distance between nucleons is extremely short may make the gravitational force significant. It is necessary to calculate the value for the gravitational force and compare it to the value for other forces to determine the significance of the gravitational force in the nucleus. The gravitational force between two protons that are separated by a distance of 10 meters is about 10 newtons. -20 -24
Coulomb's Law can be used to calculate the force between two protons. The electrostatic force is directly proportional to the electrical charges of the two particles and inversely proportional to the square of the distance between the particles. Coulomb's Law is stated as the following equation.
where:
F = electrostatic force (newtons) e
K = electrostatic constant (9.0 x 10 N-m /C)
9 2 2
Q = charge of first particle (coulombs) 1
Q = charge of second particle (coulombs) 2
r = distance between particles (meters)
Using this equation, the electrostatic force between two protons that are separated by a distance of 10 meters is about 10 newtons. Comparing this result with the calculation of the
-20 12
gravitational force (10 newtons) shows that the gravitational force is so small that it can be
-24
neglected.
Atomic and Nuclear Physics DOE-HDBK-1019/1-93 ATOMIC NATURE OF MATTER
If only the electrostatic and gravitational forces existed in the nucleus, then it would be impossible to have stable nuclei composed of protons and neutrons. The gravitational forces are

protons. Since stable atoms of neutrons and protons do exist, there must be another attractive force acting within the nucleus. This force is called the nuclear force.

The nuclear force is a strong attractive force that is independent of charge. It acts equally only between pairs of neutrons, pairs of protons, or a neutron and a proton. The nuclear force has a very short range; it acts only over distances approximately equal to the diameter of the nucleus (10 -13 cm). The attractive nuclear force between all nucleons drops off with distance much faster than the repulsive electrostatic force between protons.

TABLE 3

Forces Acting in the Nucleus

Force	Interaction	Range
Gravitational	Very weak attractive force between all nucleons	Relatively long
Electrostatic	Strong repulsive force between like charged particles (protons)	Relatively long
Nuclear Force	Strong attractive force between all nucleons	Extremely short

In stable atoms, the attractive and repulsive forces in the nucleus balance. If the forces do not balance, the atom cannot be stable, and the nucleus will emit radiation in an attempt to achieve a more stable configuration.

Summary

The important information in this chapter is summarized on the following page.

Atomic Nature of Matter Summary

Atoms consist of three basic subatomic particles. These particles are the proton, the neutron, and the electron.

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Protons are particles that have a positive charge, have about the same mass as a hydrogen atom, and exist in the nucleus of an atom.

Neutrons are particles that have no electrical charge, have about the same mass as a hydrogen atom, and exist in the nucleus of an atom.

Electrons are particles that have a negative charge, have a mass about eighteen hundred times smaller than the mass of a hydrogen atom, and exist in orbital shells around the nucleus of an atom.

The Bohr model of the atom consists of a dense nucleus of protons and neutrons (nucleons) surrounded by electrons traveling in discrete orbits at fixed distances from the nucleus.

Nuclides are atoms that contain a particular number of protons and neutrons.

Isotopes are nuclides that have the same atomic number and are therefore the same element, but differ in the number of neutrons.

The atomic number of an atom is the number of protons in the nucleus.

The mass number of an atom is the total number of nucleons (protons and neutrons) in the nucleus.

The notation $^{A}_{Z}X$ is used to identify a specific nuclide. "Z" represents the atomic number, which is equal to the number of protons. "A" represents the mass number, which is equal to the number of nucleons. "X" represents the chemical symbol of the element.

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Number of electrons = Z
Number of neutrons = A - Z
The stability of a nucleus is determined by the different forces interacting within it. The electrostatic force is a relatively long-range, strong, repulsive force that acts between the positively charged protons. The nuclear force is a relatively short-range attractive force between all nucleons. The gravitational force the long range, relatively weak attraction between masses, is negligible compared to the other forces.
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Atomic and Nuclear Physics DOE-HDBK-1019/1-93 CHART OF THE NUCLIDES
CHART OF THE NUCLIDES
The Chart of the Nuclides, like the Periodic Table, is a convenient format for presenting a large amount of scientific information in an organized manner.
EO 1.6 DEFINE the following terms:
a. Enriched uranium
b. Depleted uranium
Chart of the Nuclides
A tabulated chart called the Chart of the Nuclides lists the stable and unstable nuclides in addition to pertinent information about each one. Figure 3 shows a small portion of a typical chart. This chart plots a box for each individual nuclide, with the number of protons (Z) on the vertical axis and the number of neutrons (N = A - Z) on the horizontal axis.
The completely gray squares indicate stable isotopes. Those in white squares are artificially radioactive, meaning that they are produced by artificial techniques and do not occur naturally. By consulting a complete chart, other types of isotopes can be found, such as naturally occurring radioactive types (but none are found in the region of the chart that is illustrated in Figure 3). Located in the box on the far left of each horizontal row is general information about the element. The box contains the chemical symbol of the element in addition to the average atomic weight of the naturally occurring substance and the average thermal neutron absorption cross section, which will be discussed in a later module. The known isotopes (elements with the same atomic number Z but different mass number A) of each element are listed to the right.
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CHART OF THE NUCLIDES DOE-HDBK-1019/1-93 Atomic and Nuclear Physics
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Figure 3 Nuclide Chart for Atomic Numbers 1 to 6
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Atomic and Nuclear Physics DOE-HDBK-1019/1-93 CHART OF THE NUCLIDES
Information for Stable Nuclides
For the stable isotopes, in addition to the symbol and the atomic mass number, the number percentage of each isotope in the naturally occurring element is listed, as well as the thermal neutron activation cross section and the mass in atomic mass units (amu). A typical block for a stable nuclide from the Chart of the Nuclides is shown in Figure 4.
Figure 4 Stable Nuclides
Information for Unstable Nuclides
For unstable isotopes the additional information includes the half life, the mode of decay (for example, β^-, α), the total disintegration energy in MeV (million electron volts), and the mass in amu when available. A typical block for an unstable nuclide from the Chart of the Nuclides is

Figure 5 Unstable Nuclides
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CHART OF THE NUCLIDES DOE-HDBK-1019/1-93 Atomic and Nuclear Physics
Neutron - Proton Ratios
Figure 6 shows the distribution of the stable nuclides plotted on the same axes as the Chart of the Nuclides. As the mass numbers become higher, the ratio of neutrons to protons in the nucleus becomes larger. For helium-4 (2 protons and 2 neutrons) and oxygen-16 (8 protons and 8 neutrons) this ratio is unity. For indium-115 (49 protons and 66 neutrons) the ratio of neutrons to protons has increased to 1.35, and for uranium-238 (92 protons and 146 neutrons) the neutron■to-proton
Figure 6 Neutron - Proton Plot of the Stable Nuclides
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If a heavy nucleus were to split into two fragments, each fragment would form a nucleus that would have approximately the same neutron-to-proton ratio as the heavy nucleus. This high neutron-to-proton ratio places the fragments below and to the right of the stability curve displayed by Figure 6. The instability caused by this excess of neutrons is generally rectified by successive beta emissions, each of which converts a neutron to a proton and moves the nucleus toward a more stable neutron-to-proton ratio.
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■
Atomic and Nuclear Physics DOE-HDBK-1019/1-93 CHART OF THE NUCLIDES
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Natural Abundance of Isotopes
The relative abundance of an isotope in nature compared to other isotopes of the same element is relatively constant. The Chart of the Nuclides presents the relative abundance of the naturally occurring isotopes of an element in units of atom percent. Atom percent is the percentage of the atoms of an element that are of a particular isotope. Atom percent is abbreviated as a/o.
For example, if a cup of water contains 8.23 x 10 atoms of oxygen, and the isotopic abundance 24 of oxygen-18 is 0.20%, then there are 1.65 x 10 atoms of oxygen-18 in the cup. 22
The atomic weight for an element is defined as the average atomic weight of the isotopes of the element. The atomic weight for an element can be calculated by summing the products of the isotopic abundance of the isotope with the atomic mass of the isotope.
Example:
Calculate the atomic weight for the element lithium. Lithium-6 has an atom percent abundance of 7.5% and an atomic mass of 6.015122 amu. Lithium-7 has an atomic abundance of 92.5% and an atomic mass of 7.016003 amu.
Solution:
The other common measurement of isotopic abundance is weight percent (w/o). Weight percent is the percent weight of an element that is a particular isotope. For example, if a sample of material contained 100 kg of uranium that was 28 w/o uranium-235, then 28 kg of uranium-235 was present in the sample.
Enriched and Depleted Uranium
Natural uranium mined from the earth contains the isotopes uranium-238, uranium-235 and uranium-234. The majority (99.2745%) of all the atoms in natural uranium are uranium-238. Most of the remaining atoms (0.72%) are uranium-235, and a slight trace (0.0055%) are uranium-234. Although all isotopes of uranium have similar chemical properties, each of the isotopes has significantly different nuclear properties. For reasons that will be discussed in later modules, the isotope uranium-235 is usually the desired material for use in reactors.
A vast amount of equipment and energy are expended in processes that separate the isotopes of

module. These processes are called enrichment processes because they selectively increase the proportion of a particular isotope. The enrichment process typically starts with feed material that has the proportion of isotopes that occur naturally. The process results in two types of
CHART OF THE NUCLIDES DOE-HDBK-1019/1-93 Atomic and Nuclear Physics
In the case of uranium, the natural uranium ore is 0.72 a/o uranium-235. The desired outcome of the enrichment process is to produce enriched uranium. Enriched uranium is defined as uranium in which the isotope uranium-235 has a concentration greater than its natural value. The enrichment process will also result in the byproduct of depleted uranium. Depleted uranium is defined as uranium in which the isotope uranium-235 has a concentration less than its natural value. Although depleted uranium is referred to as a by-product of the enrichment process, it does have uses in the nuclear field and in commercial and defense industries.
Summary
The important information in this chapter is summarized below.
Chart of the Nuclides Summary
Enriched uranium is uranium in which the isotope uranium-235 has a concentration greater than its natural value of 0.7%.
Depleted uranium is uranium in which the isotope uranium-235 has a
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concentration less than its natural value of 0.7%.
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Atomic and Nuclear Physics DOE-HDBK-1019/1-93 MASS DEFECT AND BINDING ENERGY
MASS DEFECT AND BINDING ENERGY
The separate laws of Conservation of Mass and Conservation of Energy are not applied strictly on the nuclear level. It is possible to convert between mass and energy. Instead of two separate conservation laws, a single conservation law states that the sum of mass and energy is conserved. Mass does not magically appear and disappear at random. A decrease in mass will be accompanied by a corresponding increase in energy and vice versa.
EO 1.7 DEFINE the following terms:
a. Mass defect
b. Binding energy
EO 1.8 Given the atomic mass for a nuclide and the atomic masses of a neutron, proton, and electron, CALCULATE the mass defect and binding energy of the nuclide.
Mass Defect
Careful measurements have shown that the mass of a particular atom is always slightly less than the sum of the masses of the individual neutrons, protons, and electrons of which the atom consists. The difference between the mass of the atom and the sum of the masses of its parts is called the mass defect (Δm). The mass defect can be calculated using Equation (1-1). In calculating the mass defect it is important to use the full accuracy of mass measurements because the difference in mass is small compared to the mass of the atom. Rounding off the masses of atoms and particles to three or four significant digits prior to the calculation will result in a calculated mass defect of zero.

$$\Delta m = [\, Z(mp + me) + (A-Z)mn \,] - matom \quad (1\text{-}1)$$

where:
Δm = mass defect (amu)
mp = mass of a proton (1.007277 amu)
mn = mass of a neutron (1.008665 amu)

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matom = mass of nuclide A
ZX (amu)
Z = atomic number (number of protons)
A = mass number (number of nucleons)
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MASS DEFECT AND BINDING ENERGY DOE-HDBK-1019/1-93 Atomic and Nuclear Physics
Example:
Calculate the mass defect for lithium-7. The mass of lithium-7 is 7.016003 amu.
Solution:
∆ m Z m
p
me A Z mn matom
∆ m 3 1.007826 amu 7 3 1.008665 amu 7.016003 amu
∆ m 0.0421335 amu
Binding Energy
The loss in mass, or mass defect, is due to the conversion of mass to binding energy when the nucleus is formed. Binding energy is defined as the amount of energy that must be supplied to a nucleus to completely separate its nuclear particles (nucleons). It can also be understood as the amount of energy that would be released if the nucleus was formed from the separate particles. Binding energy is the energy equivalent of the mass defect. Since the mass defect was converted to binding energy (BE) when the nucleus was formed, it is possible to calculate the binding energy using a conversion factor derived by the mass-energy relationship from Einstein's Theory of Relativity.
Einstein's famous equation relating mass and energy is E = mc2 where c is the velocity of light
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(c = 2.998 x 108 m/sec). The energy equivalent of 1 amu can be determined by inserting this quantity of mass into Einstein's equation and applying conversion factors.
E m c 2
1 amu
1.6606 x 10 27 kg
1 amu
2.998 x 108 m
sec
2
1 N
1
kg m
sec2
1 J
1 N m
1.4924 x 10 10 J 1 MeV
1.6022 x 10 13 J
931.5 MeV
Conversion Factors:
1 amu = 1.6606 x 10 -27 kg
1 newton = 1 kg-m/sec2
1 joule = 1 newton-meter
1 MeV = 1.6022 x 10-13 joules

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Atomic and Nuclear Physics DOE-HDBK-1019/1-93 MASS DEFECT AND BINDING ENERGY
Since 1 amu is equivalent to 931.5 MeV of energy, the binding energy can be calculated using Equation (1-2).
(1-2)
B.E. ∆m
931.5 MeV
1 amu
Example:
Calculate the mass defect and binding energy for uranium-235. One uranium-235 atom has a mass of 235.043924 amu.
Solution:
Step 1: Calculate the mass defect using Equation (1-1).
∆m Z m
p
me A Z mn matom
∆m 92 1.007826 amu 235 92 1.008665 amu 235.043924 amu
∆m 1.91517 amu
Step 2: Use the mass defect and Equation (1-2) to calculate the binding energy.
B.E. ∆m
931.5 MeV
1 amu
1.91517 amu
931.5 MeV
1 amu
1784 MeV
Energy Levels of Atoms
The electrons that circle the nucleus move in fairly well-defined orbits. Some of these electrons are more tightly bound in the atom than others. For example, only 7.38 eV is required to remove the outermost electron from a lead atom, while 88,000 eV is required to remove the innermost electron. The process of removing an electron from an atom is called ionization, and the energy required to remove the electron is called the ionization energy.
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In a neutral atom (number of electrons = Z) it is possible for the electrons to be in a variety of different orbits, each with a different energy level. The state of lowest energy is the one in which the atom is normally found and is called the ground state. When the atom possesses more energy than its ground state energy, it is said to be in an excited state.
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An atom cannot stay in the excited state for an indefinite period of time. An excited atom will eventually transition to either a lower-energy excited state, or directly to its ground state, by emitting a discrete bundle of electromagnetic energy called an x-ray. The energy of the x-ray will be equal to the difference between the energy levels of the atom and will typically range from several eV to 100,000 eV in magnitude.
Energy Levels of the Nucleus
The nucleons in the nucleus of an atom, like the electrons that circle the nucleus, exist in shells that correspond to energy states. The energy shells of the nucleus are less defined and less understood than those of the electrons. There is a state of lowest energy (the ground state) and discrete possible excited states for a nucleus. Where the discrete energy states for the electrons

and typically measured in MeV.
A nucleus that is in the excited state will not remain at that energy level for an indefinite period. Like the electrons in an excited atom, the nucleons in an excited nucleus will transition towards their lowest energy configuration and in doing so emit a discrete bundle of electromagnetic radiation called a gamma ray (γ-ray). The only differences between x-rays and γ-rays are their energy levels and whether they are emitted from the electron shell or from the nucleus.
The ground state and the excited states of
Figure 7 Energy Level Diagram - Nickel-60
a nucleus can be depicted in a nuclear energy-level diagram. The nuclear energy-level diagram consists of a stack of horizontal bars, one bar for each of the excited states of the nucleus. The vertical distance between the bar representing an excited state and the bar representing the ground state is proportional to the energy level of the excited state with respect to the ground state. This difference in energy between the ground state and the excited state is called the excitation energy of the excited state. The ground state of a nuclide has zero excitation energy. The bars for the excited states are labeled with their respective energy levels. Figure 7 is the energy level diagram for nickel-60.
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Atomic and Nuclear Physics DOE-HDBK-1019/1-93 MASS DEFECT AND BINDING ENERGY
Summary
The important information in this chapter is summarized below.
Mass Defect and Binding Energy Summary
Mass defect is the difference between the mass of the atom and the sum of the masses of its constituent parts.
Binding energy is the amount of energy that must be supplied to a nucleus to completely separate its nuclear particles. Binding energy is the energy equivalent of the mass defect.
Mass defect can be calculated by using the equation below.
$\Delta m = [Z(mp + me) + (A-Z)mn] - matom$
Binding energy can be calculated by multiplying the mass defect by the factor
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of 931.5 MeV per amu.
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MODES OF RADIOACTIVE DECAY DOE-HDBK-1019/1-93 Atomic and Nuclear Physics
MODES OF RADIOACTIVE DECAY
Most atoms found in nature are stable and do not emit particles or energy that change form over time. Some atoms, however, do not have stable nuclei. These atoms emit radiation in order to achieve a more stable configuration.
EO 2.1 DESCRIBE the following processes:
a. Alpha decay d. Electron capture

c. Beta-plus decay f. Isomeric transitions
EO 2.2 Given a Chart of the Nuclides, WRITE the radioactive decay chain for a nuclide.
EO 2.3 EXPLAIN why one or more gamma rays typically accompany particle emission.
EO 2.4 Given the stability curve on the Chart of the Nuclides, DETERMINE the type of radioactive decay that the nuclides in each region of the chart will typically undergo.
Stability of Nuclei
As mass numbers become larger, the ratio of neutrons to protons in the nucleus becomes larger for the stable nuclei. Non-stable nuclei may have an excess or deficiency of neutrons and undergo a transformation process known as beta (β) decay. Non-stable nuclei can also undergo a variety of other processes such as alpha (α) or neutron (n) decay. As a result of these decay processes, the final nucleus is in a more stable or more tightly bound configuration.
Natural Radioactivity
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In 1896, the French physicist Becquerel discovered that crystals of a uranium salt emitted rays that were similar to x-rays in that they were highly penetrating, could affect a photographic plate, and induced electrical conductivity in gases. Becquerel's discovery was followed in 1898 by the identification of two other radioactive elements, polonium and radium, by Pierre and Marie Curie.
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Atomic and Nuclear Physics DOE-HDBK-1019/1-93 MODES OF RADIOACTIVE DECAY
Heavy elements, such as uranium or thorium, and their unstable decay chain elements emit radiation in their naturally occurring state. Uranium and thorium, present since their creation at the beginning of geological time, have an extremely slow rate of decay. All naturally occurring nuclides with atomic numbers greater than 82 are radioactive.
Nuclear Decay
Whenever a nucleus can attain a more stable (i.e., more tightly bound) configuration by emitting radiation, a spontaneous disintegration process known as radioactive decay or nuclear decay may occur. In practice, this "radiation" may be electromagnetic radiation, particles, or both.
Detailed studies of radioactive decay and nuclear reaction processes have led to the formulation of useful conservation principles. The four principles of most interest in this module are discussed below.
Conservation of electric charge implies that charges are neither created nor destroyed. Single positive and negative charges may, however, neutralize each other. It is also possible for a neutral particle to produce one charge of each sign.
Conservation of mass number does not allow a net change in the number of nucleons. However, the conversion of a proton to a neutron and vice versa is allowed.
Conservation of mass and energy implies that the total of the kinetic energy and the energy equivalent of the mass in a system must be conserved in all decays and reactions. Mass can be converted to energy and energy can be converted to mass, but the sum of mass and energy must be constant.
Conservation of momentum is responsible for the distribution of the available kinetic energy among product nuclei, particles, and/or radiation. The total amount is the same before and after the reaction even though it may be distributed differently among entirely different nuclides and/or particles.
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Alpha Decay (■)
Alpha decay is the emission of alpha particles (helium nuclei) which may be represented as either
He or ■. When an unstable nucleus ejects an alpha particle, the atomic number is reduced by 2 4 4
2 2
and the mass number decreased by 4. An example is uranium-234 which decays by the ejection of an alpha particle accompanied by the emission of a 0.068 MeV gamma.
The combined kinetic energy of the daughter nucleus (Thorium-230) and the ■ particle is designated as KE. The sum of the KE and the gamma energy is equal to the difference in mass between the original nucleus (Uranium-234) and the final particles (equivalent to the binding energy released, since ■m = BE). The alpha particle will carry off as much as 98% of the kinetic energy and, in most cases, can be considered to carry off all the kinetic energy.
Beta Decay (■)
Beta decay is the emission of electrons of nuclear rather than orbital origin. These particles are electrons that have been expelled by excited nuclei and may have a charge of either sign.
If both energy and momentum are to be conserved, a third type of particle, the neutrino, ■, must be involved. The neutrino is associated with positive electron emission, and its antiparticle, the antineutrino, , is emitted with a negative electron. These uncharged particles have only the weakest interaction with matter, no mass, and travel at the speed of light. For all practical purposes, they pass through all materials with so few interactions that the energy they possess cannot be
Fixture checkpoint marker-0697-xray is recorded on fixture page 0697.
recovered. The neutrinos and antineutrinos are included here only because they carry a portion of the kinetic energy that would otherwise belong to the beta particle, and therefore, must be considered for energy and momentum to be conserved. They are normally ignored since they are not significant in the context of nuclear reactor applications.
Negative electron emission, represented as e, -1 -1■, or simply as e or ■ , effectively converts a neutro to a proton, thus increasing the atomic number by one and leaving the mass number unchanged. This is a common mode of decay for nuclei with an excess of neutrons, such as fission fragments below and to the right of the neutron-proton stability curve (refer to Figure 6). An example of a typical beta minus-decay reaction is shown below.
■ ■
■
■ ■
■ ■ ■ ■
Atomic and Nuclear Physics DOE-HDBK-1019/1-93 MODES OF RADIOACTIVE DECAY
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Figure 9 Orbital Electron Capture
Positively charged electrons (beta-plus) are known as positrons. Except for sign, they are nearly identical to their negatively charged cousins. When a positron, represented as e, ■, or simply as +1 +1
0 0
e or ■ , is ejected from the nucleus, the atomic number is decreased by one and the mass number + + remains unchanged. A proton has been converted to a neutron. An example of a typical positron (beta-plus) decay is shown below.
Electron Capture (EC, K-capture)
Nuclei having an excess of protons may capture an electron from one of the inner orbits which immediately combines with a proton in the nucleus to form a neutron. This process is called electron

consequently, this process is sometimes called K-capture. The following example depicts electron capture.

A neutrino is formed at the same time that the neutron is formed, and energy carried off by it serves to conserve momentum. Any energy that is available due to the atomic mass of the product being appreciably less than that of the parent will appear as gamma radiation. Also, there will always be characteristic x-rays given off when an electron from one of the higher energy shells moves in to fill the vacancy in the K-shell.

Electron capture is shown graphically in Figure 8.

Electron capture and positron emission result in the production of the same daughter product, and they exist as competing processes.

For positron emission to occur, however, the mass of the daughter product must be less than the mass of the parent by an amount equal to at least twice the mass of an electron. This mass difference between the parent and daughter is necessary to account for two items present in the parent but not in the daughter. One item is the positron ejected from the nucleus of the parent. The other item is that the daughter product has one less orbital electron than the parent. If this requirement is not met, then orbital electron capture takes place exclusively.

■

■ ■ ■ ■

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■ ■

Gamma Emission (■)

Gamma radiation is a high-energy electromagnetic radiation that originates in the nucleus. It is emitted in the form of photons, discrete bundles of energy that have both wave and particle properties. Often a daughter nuclide is left in an excited state after a radioactive parent nucleus undergoes a transformation by alpha decay, beta decay, or electron capture. The nucleus will drop to the ground state by the emission of gamma radiation.

Internal Conversion

The usual method for an excited nucleus to go from the excited state to the ground state is by emission of gamma radiation. However, in some cases the gamma ray (photon) emerges from the nucleus only to interact with one of the innermost orbital electrons and, as a result, the energy of the photon is transferred to the electron. The gamma ray is then said to have undergone internal conversion. The conversion electron is ejected from the atom with kinetic energy equal to the gamma energy minus the binding energy of the orbital electron. An orbital electron then drops to a lower energy state to fill the vacancy, and this is accompanied by the emission of characteristic x-rays.

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Isomeric transition commonly occurs immediately after particle emission; however, the nucleus may remain in an excited state for a measurable period of time before dropping to the ground state at its own characteristic rate. A nucleus that remains in such an excited state is known as a nuclear isomer because it differs in energy and behavior from other nuclei with the same atomic number and mass number. The decay of an excited nuclear isomer to a lower energy level is called an isomeric transition. It is also possible for the excited isomer to decay by some alternate means, for example, by beta emission.

An example of gamma emission accompanying particle emission is illustrated by the decay of nitrogen-16 below.

■
■
■
■
■
■
■
■
■
■
■

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Decay Chains

When an unstable nucleus decays, the resulting daughter nucleus is not necessarily stable. The nucleus resulting from the decay of a parent is often itself unstable, and will undergo an additional decay. This is especially common among the larger nuclides.

It is possible to trace the steps of an unstable atom as it goes through multiple decays trying to achieve stability. The list of the original unstable nuclide, the nuclides that are involved as intermediate steps in the decay, and the final stable nuclide is known as the decay chain. One common method for stating the decay chain is to state each of the nuclides involved in the standard X format. Arrows are used between nuclides to indicate where decays occur, with the type of decay A Z
indicated above the arrow and the half-life below the arrow. The half-life for decay will be discussed in the next chapter.

Example:

Fixture checkpoint marker-0699-xray is recorded on fixture page 0699.

Write the decay chains for rubidium-91 and actinium-215. Continue the chains until a stable nuclide or a nuclide with a half-life greater than 1 x 10 years is reached. 6

Solution:

Predicting Type of Decay

Radioactive nuclides tend to decay in a way that results in a daughter nuclide that lies closer to the line of stability. Due to this, it is possible to predict the type of decay that a nuclide will undergo based on its location relative to the line of stability on the Chart of the Nuclides.

Figure 9 illustrates the type of decay nuclides in different regions of the chart will typically undergo. Nuclides that are below and to the right of the line of stability will usually undergo ■ decay. - Nuclides that are above and to the left of the line of stability will usually undergo either ■ decay or + electron capture. Most nuclides that will undergo ■ decay are found in the upper right hand region

nuclides.
MODES OF RADIOACTIVE DECAY DOE-HDBK-1019/1-93 Atomic and Nuclear Physics
Figure 9 Types of Radioactive Decay Relative to the Line of Stability
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Atomic and Nuclear Physics DOE-HDBK-1019/1-93 MODES OF RADIOACTIVE DECAY
Summary
The important information in this chapter is summarized below.
Modes of Radioactive Decay Summary
Alpha decay is the emission of an alpha particle (2 protons and 2 neutrons) from an unstable nucleus. The daughter nuclide has an atomic number 2 less than the parent nuclide and a mass number 4 less than the parent nuclide. The daughter nucleus commonly releases its excitation energy by gamma emission.
Beta-minus decay effectively converts a neutron to a proton and an electron,
Fixture checkpoint marker-0700-xray is recorded on fixture page 0700.
which is immediately ejected from the nucleus. The daughter nuclide has its atomic number increased by 1 and the same mass number compared to the parent.
Beta-plus decay effectively converts a proton to a neutron and a positron, which is immediately ejected from the nucleus. The daughter nuclide has its atomic number decreased by 1 and the same mass number compared to the parent.
In electron capture, the nucleus absorbs an electron from the innermost orbit. This electron combines with a proton to form a neutron.
Internal conversion occurs when a gamma ray, emitted by the nucleus as it goes from the excited state to the ground state, interacts with one of the innermost electrons of the same atom. The electron is ejected from the atom.
An isomeric transition is the decay of an excited nucleus to a lower-energy level by the emission of a gamma ray.
Decay chains can be found by tracing the steps an unstable atom goes through as it tries to achieve stability.
Many modes of radioactive decay result in a daughter nuclide that has an energy level above the ground state. This excitation energy is usually released immediately in the form of a gamma ray.
The type of decay that a nuclide will typically undergo can be determined by its relationship to the line of stability on the Chart of the Nuclides. Nuclides that lie below and to the right of the line of stability will typically beta minus decay. Nuclides above and to the left of the line will typically either beta plus decay or electron capture. Most alpha emitters are found in the upper, right-hand corner of the chart.
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RADIOACTIVITY
The rate at which a sample of radioactive material decays is not constant. As individual atoms of the material decay, there are fewer of those types of atoms remaining. Since the rate of decay is directly proportional to the number of atoms, the rate of decay will decrease as the number of atoms decreases.
EO 2.5 DEFINE the following terms:
a. Radioactivity d. Radioactive decay constant
b. Curie e. Radioactive half-life

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EO 2.6 Given the number of atoms and either the half-life or decay constant of a nuclide, CALCULATE the activity.
EO 2.7 Given the initial activity and the decay constant of a nuclide, CALCULATE the activity at any later time.
EO 2.8 CONVERT between the half-life and decay constant for a nuclide.
EO 2.9 Given the Chart of the Nuclides and the original activity, PLOT the radioactive decay curve for a nuclide on either linear or semi-log coordinates.
EO 2.10 DEFINE the following terms:
a. Radioactive equilibrium
b. Transient radioactive equilibrium
Radioactive Decay Rates
Radioactivity is the property of certain nuclides of spontaneously emitting particles or gamma radiation. The decay of radioactive nuclides occurs in a random manner, and the precise time at which a single nucleus will decay cannot be determined. However, the average behavior of a very large sample can be predicted accurately by using statistical methods. These studies have revealed that there is a certain probability that in a given time interval a certain fraction of the nuclei within a sample of a particular nuclide will decay. This probability per unit time that an atom of a nuclide will decay is known as the radioactive decay constant, λ. The units for the
Fixture checkpoint marker-0701-xray is recorded on fixture page 0701.
decay constant are inverse time such as 1/second, 1/minute, 1/hour, or 1/year. These decay constant units can also be expressed as second-1, minute-1, hour-1, and year-1.

■■

The activity (A) of a sample is the rate of decay of that sample. This rate of decay is usually measured in the number of disintegrations that occur per second. For a sample containing millions of atoms, the activity is the product of the decay constant and the number of atoms present in the sample.
The relationship between the activity, number of atoms, and decay constant is shown in Equation (1-3).
A = ■ N (1-3)
where:
A = Activity of the nuclide (disintegrations/second)
■ = decay constant of the nuclide (second)-1
N = Number of atoms of the nuclide in the sample
Since ■ is a constant, the activity and the number of atoms are always proportional.
Units of Measurement for Radioactivity
Two common units to measure the activity of a substance are the curie (Ci) and becquerel (Bq). A curie is a unit of measure of the rate of radioactive decay equal to 3.7 x 10 disintegrations 10 per second. This is approximately equivalent to the number of disintegrations that one gram of radium-226 will undergo in one second. A becquerel is a more fundamental unit of measure of radioactive decay that is equal to 1 disintegration per second. Currently, the curie is more widely used in the United States, but usage of the becquerel can be expected to broaden as the metric system slowly comes into wider use. The conversion between curies and becquerels is shown below.

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Variation of Radioactivity Over Time

The rate at which a given radionuclide sample decays is stated in Equation (1-3) as being equal to the product of the number of atoms and the decay constant. From this basic relationship it is possible to use calculus to derive an expression which can be used to calculate how the number of atoms present will change over time. The derivation is beyond the scope of this text, but Equation (1-4) is the useful result.

(1-4)

where:

N = number of atoms present at time t

N = number of atoms initially present o

■ = decay constant (time)-1

t = time

A Ao

e

t

A Aoe

t

A

Ao

e

t

ln A

Ao

t

t

ln

A

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Ao

t

1/2

ln

1

2

t

1/2

ln 2 0.693

Since the activity and the number of atoms are always proportional, they may be used interchangeably to describe any given radionuclide population. Therefore, the following is true.

(1-5)

where:

A = activity present at time t

A = activity initially present o

= decay constant (time)

-1

t = time

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One of the most useful terms for estimating how quickly a nuclide will decay is the radioactive half-life. The radioactive half-life is defined as the amount of time required for the activity to decrease to one-half of its original value. A relationship between the half-life and decay constant can be developed from Equation (1-5). The half-life can be calculated by solving Equation (1-5) for the time, t, when the current activity, A, equals one-half the initial activity A .
o
First, solve Equation (1-5) for t.
If A is equal to one-half of Ao, then A/Ao is equal to one-half. Substituting this in the equation above yields an expression for t .
1/2
(1-6)
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The basic features of decay of a radionuclide sample are shown by the graph in Figure 10.
Figure 10 Radioactive Decay as a Function of Time in Units of Half-Life
Assuming an initial number of atoms No, the population, and consequently, the activity may be noted to decrease by one-half of this value in a time of one half-life. Additional decreases occur so that whenever one half-life elapses, the number of atoms drops to one-half of what its value was at the beginning of that time interval. After five half-lives have elapsed, only 1/32, or 3.1%, of the original number of atoms remains. After seven half-lives, only 1/128, or 0.78%, of the atoms remains. The number of atoms existing after 5 to 7 half-lives can usually be assumed to be negligible. The Chemistry Fundamentals Handbook contains additional information on calculating the number of atoms contained within a sample.
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■
■
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Example:
A sample of material contains 20 micrograms of californium-252.
Californium-252 has a half-life of 2.638 years.
Calculate:
(a) The number of californium-252 atoms initially present
(b) The activity of the californium-252 in curies
(c) The number of californium-252 atoms that will remain in 12 years
Fixture checkpoint marker-0703-xray is recorded on fixture page 0703.
(d) The time it will take for the activity to reach 0.001 curies
Solution:
(a) The number of atoms of californium-252 can be determined as below.
(b) First, use Equation (1-6) to calculate the decay constant.
Use this value for the decay constant in Equation (1-3) to determine the activity.
N No
e
t
4.78 x 1016 e
(0.263/yr) (12 yr)
2.04 x 1015
A Aoe
t

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Ao
e
t
ln A
Ao
t
t
ln
A
Ao
t
Fixture checkpoint marker-0704-xray is recorded on fixture page 0704.
ln
0.001 Ci
0.0108 Ci
0.263 year 1
t 9.05 years
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(c) The number of californium atoms that will remain in 12 years can be calculated from Equation (1-4).
(d) The time that it will take for the activity to reach 0.001 Ci can be determined from Equation (1-5). First, solve Equation (1-5) for time.
Inserting the appropriate values in the right side of this equation will result in the required time.
Plotting Radioactive Decay
It is useful to plot the activity of a nuclide as it changes over time. Plots of this type can be used to determine when the activity will fall below a certain level. This plot is usually done showing activity on either a linear or a logarithmic scale. The decay of the activity of a single nuclide on a logarithmic scale will plot as a straight line because the decay is exponential.
t
1/2
0.693
0.693
t
1/2
0.693
7.13 seconds
0.0972 second 1
A Aoe
t
RADIOACTIVITY DOE-HDBK-1019/1-93 Atomic and Nuclear Physics
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Example:
Plot the radioactive decay curve for nitrogen-16 over a period of 100 seconds. The initial activity is 142 curies and the half-life of nitrogen-16 is 7.13 seconds. Plot the curve on both linear rectangular coordinates and on a semi-log scale.
Solution:

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7.13 seconds.
Use the decay constant determined above to calculate the activity at various times using Equation (1-5).
Time Activity
0 seconds 142 Ci
20 seconds 20.3 Ci
40 seconds 2.91 Ci
60 seconds 0.416 Ci
80 seconds 0.0596 Ci
100 seconds 0.00853 Ci
■ ■
■ ■
■ ■
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Figure 11 Linear and Semi-log Plots of Nitrogen-16 Decay
Plotting the data points calculated above on both linear and semilog scales results in the graphs shown in Figure 11.
Fixture checkpoint marker-0705-xray is recorded on fixture page 0705.
If a substance contains more than one radioactive nuclide, the total activity is the sum of the individual activities of each nuclide. As an example, consider a sample of material that contained 1 x 10 atoms of iron-59 that has a half-life of 44.51 days (■ = 1.80 x 10 sec), 6 -7 -1 1 x 10 atoms of manganese-54 that has a half-life of 312.2 days (■ = 2.57 x 10 sec), and 1 6 -8 -1 x 10 atoms of cobalt-60 that has a half-life of 1925 days (■ = 4.17 x 10 sec). 6 -9 -1
The initial activity of each of the nuclides would be the product of the number of atoms and the decay constant.
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Figure 12 Combined Decay of Iron-56, Manganese-54, and Cobalt-60
Plotting the manner in which the activities of each of the three nuclides decay over time demonstrates that initially the activity of the shortest-lived nuclide (iron-59) dominates the total activity, then manganese-54 dominates. After almost all of the iron and manganese have decayed away, the only contributor to activity will be the cobalt-60. A plot of this combined decay is shown in Figure 12.
Radioactive Equilibrium
Radioactive equilibrium exists when a radioactive nuclide is decaying at the same rate at which it is being produced. Since the production rate and decay rate are equal, the number of atoms present remains constant over time.
An example of radioactive equilibrium is the concentration of sodium-24 in the coolant circulating through a sodium-cooled nuclear reactor. Assume that the sodium-24 is being produced at a rate of 1 x 10 atoms per second. If the sodium-24 were stable and did not decay, 6 the amount of sodium-24 present after some period of time could be calculated by multiplying the production rate by the amount of time. Plotting the amount of material present would result in the graph in Figure 13.
However, sodium-24 is not stable, and it decays with a half-life of 14.96 hours. If no sodium-24 is present initially and production starts at a rate of 1 x 10 atoms per second, the rate 6 of decay will initially be zero because there is no sodium-24 present to decay. The rate of decay of sodium-24 will increase as the amount of sodium-24 increases.

■
■
■
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Figure 13 Cumulative Production of Sodium-24 Over Time
The amount of sodium-24 present will initially increase rapidly, then it will increase at a continually decreasing rate until the rate of decay is equal to the rate of production. It is possible to calculate how much sodium-24 will be present at equilibrium by setting the production rate (R) equal to the decay rate (■ N).
where:
R = production rate (atoms/second)
■ = decay constant (second)-1
N = number of atoms
It is possible to calculate the equilibrium value for sodium-24 being produced at a rate of 1 x 10 atoms/second. 6
■ ■ ■■
■
■
■
■
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Figure 14 Approach of Sodium-24 to Equilibrium
The development of the equation to calculate how the amount of sodium-24 changes over time
Fixture checkpoint marker-0706-xray is recorded on fixture page 0706.
as it approaches the equilibrium value is beyond the scope of this handbook. However, the equation is presented below.
This equation can be used to calculate the values of the amount of sodium-24 present at different times. As the time increases, the exponential term approaches zero, and the number of atoms present will approach R/■. A plot of the approach of sodium-24 to equilibrium is shown in Figure 14.
Transient Radioactive Equilibrium
Transient radioactive equilibrium occurs when the parent nuclide and the daughter nuclide decay at essentially the same rate.
For transient equilibrium to occur, the parent must have a long half-life when compared to the daughter. An example of this type of compound decay process is barium-140, which decays by beta emission to lanthanum-140, which in turn decays by beta emission to stable cerium-140.
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Figure 15 Transient Equilibrium in the Decay of Barium-140
The decay constant for barium-140 is considerably smaller than the decay constant for lanthanum-140. Remember that the rate of decay of both the parent and daughter can be represented as ■N. Although the decay constant for barium-140 is smaller, the actual rate of decay (■N) is initially larger than that of lanthanum-140 because of the great difference in their initial concentrations. As the concentration of the daughter increases, the rate of decay of the daughter will approach and eventually match the decay rate of the parent. When this occurs, they are said to be in transient equilibrium. A plot of the barium-lanthanum-cerium decay chain

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Secular equilibrium occurs when the parent has an extremely long half-life. In the long decay chain for a naturally radioactive element, such as thorium-232, where all of the elements in the chain are in secular equilibrium, each of the descendants has built up to an equilibrium amount and all decay at the rate set by the original parent. The only exception is the final stable element on the end of the chain. Its number of atoms is constantly increasing.

Summary

The important information in this chapter is summarized on the following page.

■

■■

■

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Radioactivity Summary

& Radioactivity is the decay of unstable atoms by the emission of particles and electromagnetic radiation.

& A curie (Ci) is a unit of radioactivity equal to 3.7 x 10 disintegrations per 10 second.

& A becquerel (Bq) is a unit of radioactivity equal to 1 disintegration per second.

& The radioactive decay constant (■) is the probability per unit time that an atom will decay.

& The radioactive half-life is the amount of time required for the activity to decrease to one-half its original value.

& The activity of a substance can be calculated from the number of atoms and the decay constant based on the equation below.

& The amount of activity remaining after a particular time can be calculated from the equation below.

& The relationship between the decay constant and the half-life is shown below.

& Plots of radioactive decay can be useful to describe the variation of activity over time. If decay is plotted using semi-log scale the plot results in a straight line.

& Radioactive equilibrium exists when the production rate of a material is equal to the removal rate.

& Transient radioactive equilibrium exists when the parent nuclide and the daughter

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nuclide decay at essentially the same rate. This occurs only when the parent has a long half-life compared to the daughter.

Atomic and Nuclear Physics DOE-HDBK-1019/1-93 NEUTRON INTERACTIONS

NEUTRON INTERACTIONS

Neutrons can cause many different types of interactions. The neutron may simply scatter off the nucleus in two different ways, or it may actually be absorbed into the nucleus. If a neutron is absorbed into the nucleus, it may result in the emission of a gamma ray or a subatomic particle, or it may cause the nucleus to fission.

EO 3.1 DESCRIBE the following scattering interactions between a neutron and a nucleus:

a. Elastic scattering

b. Inelastic scattering

EO 3.2 STATE the conservation laws that apply to an elastic collision between a neutron and a nucleus.

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absorbed in a nucleus:
a. Radiative capture
b. Particle ejection
Scattering
A neutron scattering reaction occurs when a nucleus, after having been struck by a neutron, emits a single neutron. Despite the fact that the initial and final neutrons do not need to be (and often are not) the same, the net effect of the reaction is as if the projectile neutron had merely "bounced off," or scattered from, the nucleus. The two categories of scattering reactions, elastic and inelastic scattering, are described in the following paragraphs.
Elastic Scattering
In an elastic scattering reaction between a neutron and a target nucleus, there is no energy transferred into nuclear excitation. Momentum and kinetic energy of the "system" are conserved although there is usually some transfer of kinetic energy from the neutron to the target nucleus. The target nucleus gains the amount of kinetic energy that the neutron loses.
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mn
vn,i mTvT,i mnvn,f mTvT,f
1
2
m v
2
1
2
mn
v
2
n,i
1
2
mT
v
2
T,i
1
2
mn
v
2
n,f
Fixture checkpoint marker-0708-xray is recorded on fixture page 0708.
1
2
mT
v
2
T,f
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Figure 16 illustrates the process of elastic scattering of a neutron off a target nucleus. In the elastic scattering reaction, the conservation of momentum and kinetic energy is represented by the equations below.

Conservation of momentum (mv)

Conservation of kinetic energy

where:

mn = mass of the neutron

m = mass of the target nucleus T

vn,i = initial neutron velocity

vn,f = final neutron velocity

vT,i = initial target velocity

vT,f = final target velocity

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Elastic scattering of neutrons by nuclei can occur in two ways. The more unusual of the two interactions is the absorption of the neutron, forming a compound nucleus, followed by the re-emission of a neutron in such a way that the total kinetic energy is conserved and the nucleus

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returns to its ground state. This is known as resonance elastic scattering and is very dependent upon the initial kinetic energy possessed by the neutron. Due to formation of the compound nucleus, it is also referred to as compound elastic scattering. The second, more usual method, is termed potential elastic scattering and can be understood by visualizing the neutrons and nuclei to be much like billiard balls with impenetrable surfaces. Potential scattering takes place with incident neutrons that have an energy of up to about 1 MeV. In potential scattering, the neutron does not actually touch the nucleus and a compound nucleus is not formed. Instead, the neutron is acted on and scattered by the short range nuclear forces when it approaches close enough to the nucleus.

Inelastic Scattering

In inelastic scattering, the incident neutron is absorbed by the target nucleus, forming a compound nucleus. The compound nucleus will then emit a neutron of lower kinetic energy which leaves the original nucleus in an excited state. The nucleus will usually, by one or more gamma emissions, emit this excess energy to reach its ground state. Figure 17 shows the process of inelastic scattering.

For the nucleus that has reached its ground state, the sum of the kinetic energy of the exit

Figure 17 Inelastic Scattering

neutron, the target nucleus, and the total gamma energy emitted is equal to the initial kinetic energy of the incident neutron.

1
0
n
238
92
U
239
92
U
239
92

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0
0
1
0
n
10
5
B
11
5
B
7
3
Li
4
2

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Absorption Reactions

Most absorption reactions result in the loss of a neutron coupled with the production of a charged particle or gamma ray. When the product nucleus is radioactive, additional radiation is emitted at some later time. Radiative capture, particle ejection, and fission are all categorized as absorption reactions and are briefly described below.

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Radiative Capture

In radiative capture the incident neutron enters the target nucleus forming a compound nucleus. The compound nucleus then decays to its ground state by gamma emission. An example of a radiative capture reaction is shown below.

Particle Ejection

In a particle ejection reaction the incident particle enters the target nucleus forming a compound nucleus. The newly formed compound nucleus has been excited to a high enough energy level to cause it to eject a new particle while the incident neutron remains in the nucleus. After the new particle is ejected, the remaining nucleus may or may not exist in an excited state depending upon the mass-energy balance of the reaction. An example of a particle ejection reaction is shown below.

Fission

One of the most important interactions that neutrons can cause is fission, in which the nucleus that absorbs the neutron actually splits into two similarly sized parts. Fission will be discussed in detail in the next chapter.

Atomic and Nuclear Physics DOE-HDBK-1019/1-93 NEUTRON INTERACTIONS

Summary

The important information in this chapter is summarized below.

Neutron Interactions Summary

Interactions where a neutron scatters off a target nucleus are either elastic or inelastic. In elastic scattering, kinetic energy and momentum are conserved and no energy is transferred into excitation energy of the target nucleus. In inelastic scattering, some amount of kinetic energy is transferred into excitation energy of the target nucleus.

kinetic energy and conservation of momentum.
Radiative capture is the absorption of a neutron by the target nucleus, resulting in an excited nucleus which subsequently (typically within a small fraction of a second) releases its excitation energy in the form of a gamma ray.
Particle ejection occurs when a neutron is absorbed by a target nucleus, resulting in the formation of a compound nucleus. The compound nucleus immediately ejects a particle (for example, alpha or proton).

1
0
n
235
92
U
236
92
U
140
55
Cs
93
37
Rb 3
1
0
n

NUCLEAR FISSION
Nuclear fission is a process in which an atom splits and releases energy, fission
Fixture checkpoint marker-0711-xray is recorded on fixture page 0711.
products, and neutrons. The neutrons released by fission can, in turn, cause the fission of other atoms.
EO 4.1 EXPLAIN the fission process using the liquid drop model of a nucleus.
EO 4.2 DEFINE the following terms:
a. Excitation energy (E_{exc})
b. Critical energy (E_{crit})
EO 4.3 DEFINE the following terms:
a. Fissile material
b. Fissionable material
c. Fertile material
EO 4.4 DESCRIBE the processes of transmutation, conversion, and breeding.
EO 4.5 DESCRIBE the curve of Binding Energy per Nucleon versus mass number and give a qualitative description of the reasons for its shape.
EO 4.6 EXPLAIN why only the heaviest nuclei are easily fissioned.

neutrons and uranium-238 fissions only with fast neutrons.
Fission
In the fission reaction the incident neutron enters the heavy target nucleus, forming a compound nucleus that is excited to such a high energy level (E > E) that the nucleus "splits" exc crit (fissions) into two large fragments plus some neutrons. An example of a typical fission reaction is shown below.
A large amount of energy is released in the form of radiation and fragment kinetic energy.
Atomic and Nuclear Physics DOE-HDBK-1019/1-93 NUCLEAR FISSION
Liquid Drop Model of a Nucleus
The nucleus is held together by the attractive nuclear force between nucleons, which was discussed in a previous chapter. The characteristics of the nuclear force are listed below.
(a) very short range, with essentially no effect beyond nuclear dimensions (10-13 cm)
(b) stronger than the repulsive electrostatic forces within the nucleus
(c) independent of nucleon pairing, in that the attractive forces between pairs of neutrons are no different than those between pairs of protons or a neutron and a proton
(d) saturable, that is, a nucleon can attract only a few of its nearest neighbors
One theory of fission considers the fissioning of a nucleus similar in some respects to the splitting of a liquid drop. This analogy is justifiable to some extent by the fact that a liquid drop is held together by molecular forces that tend to make the drop spherical in shape and that try to resist any deformation in the same manner as nuclear forces are assumed to hold the nucleus together. By considering the nucleus as a liquid drop, the fission process can be described.
Referring to Figure 18(A), the nucleus in the ground state is undistorted, and its attractive nuclear forces are greater than the repulsive electrostatic forces between the protons within the nucleus. When an incident particle (in this instance a neutron) is absorbed by the target nucleus, a compound nucleus is formed. The compound nucleus temporarily contains all the charge and mass involved in the reaction and exists in an excited state. The excitation energy added to the compound nucleus is equal to the binding energy contributed by the incident particle plus the kinetic energy possessed by that particle. Figure 18(B) illustrates the excitation energy thus imparted to the compound nucleus, which may cause it to oscillate and become distorted. If the excitation energy is greater than a certain critical energy, the oscillations may cause the compound nucleus to become dumbbell-shaped. When this happens, the attractive nuclear forces (short-range) in the neck area are small due to saturation, while the repulsive electrostatic forces (long-range) are only slightly less than before. When the repulsive electrostatic forces exceed the attractive nuclear forces, nuclear fission occurs, as illustrated in Figure 18(C).
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Figure 18 Liquid Drop Model of Fission
Critical Energy
The measure of how far the energy level of a nucleus is above its ground state is called the excitation energy (Eexc). For fission to occur, the excitation energy must be above a particular value for that nuclide. The critical energy (Ecrit) is the minimum excitation energy required for fission to occur.
Fissile Material
A fissile material is composed of nuclides for which fission is possible with neutrons of any energy level. What is especially significant about these nuclides is their ability to be fissioned

energy levels (essentially zero) because they are roughly in equilibrium with the thermal motion of surrounding materials. Therefore, in order to be classified as fissile, a material must be capable of fissioning after absorbing a thermal neutron. Consequently, they impart essentially no kinetic energy to the reaction. Fission is possible in these materials with thermal neutrons, since the change in binding energy supplied by the neutron addition alone is high enough to exceed the critical energy. Some examples of fissile nuclides are uranium-235, uranium-233, and plutonium-239.

Fissionable Material

A fissionable material is composed of nuclides for which fission with neutrons is possible. All fissile nuclides fall into this category. However, also included are those nuclides that can be fissioned only with high energy neutrons. The change in binding energy that occurs as the result of neutron absorption results in a nuclear excitation energy level that is less than the required

Fixture checkpoint marker-0713-xray is recorded on fixture page 0713.

critical energy. Therefore, the additional excitation energy must be supplied by the kinetic energy of the incident neutron. The reason for this difference between fissile and fissionable materials is the so-called odd-even effect for nuclei. It has been observed that nuclei with even numbers of neutrons and/or protons are more stable than those with odd numbers. Therefore, adding a neutron to change a nucleus with an odd number of neutrons to a nucleus with an even number of neutrons produces an appreciably higher binding energy than adding a neutron to a nucleus already possessing an even number of neutrons. Some examples of nuclides requiring high energy neutrons to cause fission are thorium-232, uranium-238, and plutonium-240. Table 4 indicates the critical energy (Ecrit) and the binding energy change for an added neutron (BEn) to target nuclei of interest. For fission to be possible, the change in binding energy plus the kinetic energy must equal or exceed the critical energy (ΔBE + KE > Ecrit).

TABLE 4

Critical Energies Compared to Binding Energy of Last Neutron

Target
Nucleus
Critical Energy
Ecrit
Binding Energy of
Last Neutron BEn
BEn - Ecrit
23
9
2
0Th 7.5 MeV 5.4 MeV -2.1 MeV
23
9
8
2U 7.0 MeV 5.5 MeV -1.5 MeV
23
9
5
2U 6.5 MeV 6.8 MeV +0.3 MeV
23

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3
2U 6.0 MeV 7.0 MeV +1.0 MeV
23
9
9
4Pu 5.0 MeV 6.6 MeV +1.6 MeV
Uranium-235 fissions with thermal neutrons because the binding energy released by the absorption of a neutron is greater than the critical energy for fission; therefore uranium-235 is a fissile material. The binding energy released by uranium-238 absorbing a thermal neutron is less than the critical energy, so additional energy must be possessed by the neutron for fission to be possible. Consequently, uranium-238 is a fissionable material.

Fertile Material
All of the neutron absorption reactions that do not result in fission lead to the production of new nuclides through the process known as transmutation. These nuclides can, in turn, be transmuted again or may undergo radioactive decay to produce still different nuclides. The nuclides that are produced by this process are referred to as transmutation products. Because several of the fissile nuclides do not exist in nature, they can only be produced by nuclear reactions (transmutation). The target nuclei for such reactions are said to be fertile. Fertile materials are materials that can undergo transmutation to become fissile materials. Figure 19 traces the transmutation mechanism
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by which two fertile nuclides, thorium-232 and uranium-238, produce uranium-233 and plutonium-239, respectively.
Figure 19 Conversion of Fertile Nuclides to Fissile Nuclides
If a reactor contains fertile material in addition to its fissile fuel, some new fuel will be produced as the original fuel is burned up. This is called conversion. Reactors that are specifically designed to produce fissionable fuel are called "breeder" reactors. In such reactors, the amount of fissionable fuel produced is greater than the amount of fuel burnup. If less fuel is produced than used, the process is called conversion, and the reactor is termed a "converter."

Binding Energy Per Nucleon (BE/A)
As the number of particles in a nucleus increases, the total binding energy also increases. The rate of increase, however, is not uniform. This lack of uniformity results in a variation in the amount of binding energy associated with each nucleon within the nucleus. This variation in the binding energy per nucleon (BE/A) is easily seen when the average BE/A is plotted versus atomic mass number (A), as shown in Figure 20.
Figure 20 Binding Energy per Nucleon vs. Mass Number
Figure 20 illustrates that as the atomic mass number increases, the binding energy per nucleon decreases for A > 60. The BE/A curve reaches a maximum value of 8.79 MeV at A = 56 and decreases to about 7.6 MeV for A = 238. The general shape of the BE/A curve can be explained using the general properties of nuclear forces. The nucleus is held together by very short-range attractive forces that exist between nucleons. On the other hand, the nucleus is being forced apart by long range repulsive electrostatic (coulomb) forces that exist between all the protons in the nucleus.

within the nucleus increase due to the greater number of protons in the heavy elements. To overcome this increased repulsion, the proportion of neutrons in the nucleus must increase to maintain stability. This increase in the neutron-to-proton ratio only partially compensates for the growing proton-proton repulsive force in the heavier, naturally occurring elements. Because the repulsive forces are increasing, less energy must be supplied, on the average, to remove a nucleon from the nucleus. The BE/A has decreased. The BE/A of a nucleus is an indication of its degree of stability. Generally, the more stable nuclides have higher BE/A than the less stable ones. The increase in the BE/A as the atomic mass number decreases from 260 to 60 is the primary reason for the energy liberation in the fission process. In addition, the increase in the BE/A as the atomic mass number increases from 1 to 60 is the reason for the energy liberation in the fusion process, which is the opposite reaction of fission.

The heaviest nuclei require only a small distortion from a spherical shape (small energy addition) for the relatively large coulomb forces forcing the two halves of the nucleus apart to overcome the attractive nuclear forces holding the two halves together. Consequently, the heaviest nuclei are easily fissionable compared to lighter nuclei.

Summary

The important information in this chapter is summarized on the following page.

Nuclear Fission Summary

The fission process can be explained using the liquid drop model of a nucleus. In the ground state the nucleus is nearly spherical in shape. After the absorption of a neutron, the nucleus will be in an excited state and start to oscillate and become distorted. If the oscillations cause the nucleus to become shaped like a dumbbell, the repulsive electrostatic forces will overcome the short-range attractive nuclear forces, and the nucleus will split in two.

Excitation energy is the amount of energy a nucleus has above its ground state. Critical energy is the minimum excitation energy that a nucleus must have before

Fixture checkpoint marker-0715-xray is recorded on fixture page 0715.

it can fission.

Fissile material is material for which fission is possible with neutrons that have zero kinetic energy. Fissionable material is material for which fission caused by neutron absorption is possible provided the kinetic energy added with the binding energy is greater than the critical energy. Fertile material is material that can undergo transmutation to become fissile material.

Transmutation is the process of neutron absorption and subsequent decay, which changes one nuclide to another nuclide. Conversion is the process of transmuting fertile material into fissile material in a reactor, where the amount of fissile material produced is less than the amount of fissile material consumed. Breeding is the same as conversion, except the amount of fissile material produced is more than the amount of fissile material consumed.

The curve of binding energy per nucleon increases quickly through the light nuclides and reaches a maximum at a mass number of about 56. The curve decreases slowly for mass numbers greater than 60.

The heaviest nuclei are easily fissionable because they require only a small distortion from the spherical shape to allow the coulomb forces to overcoming the attractive nuclear force, forcing the two halves of the nucleus apart.

Uranium-235 fissions with thermal neutrons because the binding energy released

binding energy released by uranium-238 absorbing a neutron is less than the critical energy, so additional kinetic energy must be possessed by the neutron for fission to be possible.

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1
0
n
235
92
U
236
92
U
140
55
Cs
93
37
Rb 3
1
0
n

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ENERGY RELEASE FROM FISSION

Fission of heavy nuclides converts a small amount of mass into an enormous amount of energy. The amount of energy released by fission can be determined based on either the change in mass that occurs during the reaction or by the difference in binding energy per nucleon between the fissile nuclide and the fission products.

EO 4.8 CHARACTERIZE the fission products in terms of mass groupings and radioactivity.

EO 4.9 Given the nuclides involved and their masses, CALCULATE the energy released from fission.

EO 4.10 Given the curve of Binding Energy per nucleon versus mass

Fixture checkpoint marker-0716-xray is recorded on fixture page 0716.

number, CALCULATE the energy released from fission.

Calculation of Fission Energy

Nuclear fission results in the release of enormous quantities of energy. It is necessary to be able to calculate the amount of energy that will be produced. The logical manner in which to pursue this is to first investigate a typical fission reaction such as the one listed below.

It can be seen that when the compound nucleus splits, it breaks into two fission fragments, rubidium-93, cesium-140, and some neutrons. Both fission products then decay by multiple - emissions as a result of the high neutron-to-proton ratio possessed by these nuclides.

In most cases, the resultant fission fragments have masses that vary widely. Figure 21 gives the percent yield for atomic mass numbers. The most probable pair of fission fragments for the thermal fission of the fuel uranium-235 have masses of about 95 and 140. Note that the vertical axis of the fission yield curve is on a logarithmic scale. Therefore, the formation of fission

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Figure 21 Uranium-235 Fission Yield vs.
Mass Number
Referring now to the binding energy per nucleon curve (Figure 20), we can estimate the amount of energy released by our "typical" fission by plotting this reaction on the curve and calculating the change in binding energy (ΔBE) between the reactants on the left-hand side of the fission equation and the products on the right-hand side. Plotting the reactant and product nuclides on the curve shows that the total binding energy of the system after fission is greater than the total binding energy of the system before fission. When there is an increase in the total binding energy of a system, the system has become more stable by releasing an amount of energy equal to the increase in total binding energy of the system. Therefore, in the fission process, the energy liberated is equal to the increase in the total binding energy of the system.
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Figure 22 graphically depicts that the binding energy per nucleon for the products (C,
Figure 22 Change in Binding Energy for Typical Fission
rubidium-93 and B, cesium-140) is greater than that for the reactant (A, uranium-235). The total binding energy for a nucleus can be found by multiplying the binding energy per nucleon by the number of nucleons.
TABLE 5
Binding Energies Calculated from
Binding Energy per Nucleon Curve
Nuclide B.E. per Nucleon
(BE/A)
Mass Number
(A)
Binding Energy
(BE/A) x (A)
9
3
3
7Rb 8.7 MeV 93 809 MeV
14
5
0
5Cs 8.4 MeV 140 1176 MeV
23
9
5
2U 7.6 MeV 235 1786 MeV
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BE BEproducts BEreactants
BERb 93 BECs 140 BEU 235
809 MeV 1176 MeV 1786 MeV
199 MeV
1
0

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235
92
U
236
92
U
140
55
Cs
93
37
Rb 3
1
0
n
Mass difference Mass of Reactants Mass of Products
236.052589 amu 235.85208 amu
0.200509 amu
EInst 0.020059 amu
Fixture checkpoint marker-0718-xray is recorded on fixture page 0718.
931.5 MeV
amu
186.8 MeV
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The energy released will be equivalent to the difference in binding energy (BE) between the reactants and the products.
The energy liberation during the fission process can also be explained from the standpoint of the conservation of mass-energy. During the fission process, there is a decrease in the mass of the system. There must, therefore, be energy liberated equal to the energy equivalent of the mass lost in the process. This method is more accurate than the previously illustrated method and is used when actually calculating the energy liberated during the fission process.
Again, referring to the "typical" fission reaction.
E , the instantaneous energy, is the energy released immediately after the fission process. It Inst is equal to the energy equivalent of the mass lost in the fission process. It can be calculated as shown below.
Mass of the Reactants Mass of the Products
U 235.043924 amu Rb 92.91699 amu 235
92
93
37
n 1.008665 amu Cs 139.90910 amu 1
0
140
55
3 (n) 3.02599 amu 1
0
236.052589 amu 235.85208 amu

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93
37
Rb
93
38
Sr
93
39
Y
93
40
Zr
93
41
Nb
140
55
Cs
140
56
Ba
140
57
La
140
58
Fixture checkpoint marker-0719-xray is recorded on fixture page 0719.
Ce
EDecay mRb 93 mNb 93 4 melectron
931.5MeV
amu
92.91699 amu 92.90638 amu 4 0.0005486 amu
931.5 MeV
amu
0.008416 amu
931.5 MeV
amu
7.84 MeV
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The total energy released per fission will vary from the fission to the next depending on what fission products are formed, but the average total energy released per fission of uranium-235 with a thermal neutron is 200 MeV.
As illustrated in the preceding example, the majority of the energy liberated in the fission process is released immediately after the fission occurs and appears as the kinetic energy of the fission fragments, kinetic energy of the fission neutrons, and instantaneous gamma rays. The remaining energy is released over a period of time after the fission occurs and appears as kinetic energy of the beta, neutrino, and decay gamma rays.

In addition to this instantaneous energy release during the actual fission reaction, there is additional energy released when the fission fragments decay by emission. This additional
-
energy is called decay energy, EDecay. The decay chains for rubidium-93 and cesium-140 are shown below.
The energy released during the decay for each chain will be equivalent to the mass difference between the original fission product and the sum of the final stable nuclide and the beta particles emitted.
The energy released in the decay chain of rubidium-93 is calculated below.
EDecay mRb 93 mNb 93 3 melectron
931.5MeV
amu
139.90910 amu 139.90543 amu 3 0.0005486 amu
931.5 MeV
amu
0.000202 amu
931.5 MeV
amu
1.89 MeV
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The energy released in the decay chain of cesium-140 is calculated below.
The total decay energy is the sum of the energies of the two chains, or 9.73 MeV.
Distribution of Fission Energy
The average energy distribution for the energy released per fission with a thermal neutron in uranium-235 is shown in Tables 6 and 7.
TABLE 6
Instantaneous Energy from Fission
Kinetic Energy of Fission Products 167 Mev
Energy of Fission Neutrons 5 MeV
Instantaneous Gamma-ray Energy 5 MeV
Capture Gamma-ray Energy 10 MeV
Total Instantaneous Energy 187 MeV
Fixture checkpoint marker-0720-xray is recorded on fixture page 0720.
TABLE 7
Delayed Energy from Fission
Beta Particles From Fission Products 7 Mev
Gamma-rays from Fission Products 6 MeV
Neutrinos 10 MeV
Total Delayed Energy 23 MeV
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Because the 10 MeV of neutrino energy shown in Table 7 is not absorbed in the reactor, the average value of 200 MeV per fission is still accurate. Note in Table 6 that some fission neutrons undergo radiative capture and the resultant gamma ray emission provides an additional 10 MeV of instantaneous energy, which contributes to the total of 187 MeV instantaneous energy.
All of the energy released, with the exception of the neutrino energy, is ultimately transformed into heat through a number of processes. The fission fragments, with their high positive charge

atoms. In this ionization process, kinetic energy is transferred to the surrounding atoms of the fuel material, resulting in an increase in temperature. The beta particles and gamma rays also give up their energy through ionization, and the fission neutrons interact and lose their energy through elastic scattering. Of the 200 MeV released per fission, about seven percent (13 MeV) is released at some time after the instant of fission. When a reactor is shut down, fissions essentially cease, but energy is still being released from the decay of fission products. The heat produced by this decay energy is referred to as "decay heat." Although decay energy represents about seven percent of reactor heat production during reactor operation, once the reactor is shut down the decay heat production drops off quickly to a small fraction of its value while operating. The decay heat produced is significant, however, and systems must be provided to keep the
Fixture checkpoint marker-0721-xray is recorded on fixture page 0721.
reactor cool even after shutdown.

Summary

The important information in this chapter is summarized below.

Energy Release From Fission Summary

Fission products have some general characteristics in common.

They generally decay by β- emission.

The most common mass numbers are grouped near 95 and 140.

The energy released by fission can be calculated based on the difference in mass between the masses of the reactants before fission and the fission fragments and fission neutrons after fission.

Another method to determine the energy released by fission is based on the change in binding energy per nucleon between the fissile nuclide and the fission products.

INTERACTION OF RADIATION WITH MATTER

Different types of radiation interact with matter in widely different ways. A large, massive, charged alpha particle cannot penetrate a piece of paper and even has a limited range in dry air. A neutrino, at the other extreme, has a low probability of interacting with any matter, even if it passed through the diameter of the earth.

EO 5.1 DESCRIBE interactions of the following with matter:

a. Alpha particle c. Positron

b. Beta particle d. Neutron

EO 5.2 DESCRIBE the following ways that gamma radiation interacts with matter:

a. Photoelectric effect

b. Compton scattering

c. Pair production

Interaction of Radiation With Matter

Radiation can be classified into two general groups, charged and uncharged; therefore, it may be expected that interactions with matter fall into two general types. Charged particles directly ionize the media through which they pass, while uncharged particles and photons can cause ionization only indirectly or by secondary radiation.

A moving charged particle has an electrical field surrounding it, which interacts with the atomic structure of the medium through which it is passing. This interaction decelerates the particle and

energy to escape from the parent atom. This process, whereby radiation "strips" off orbital electrons, is called ionization. Uncharged moving particles have no electrical field, so they can only lose energy and cause ionization by such means as collisions or scattering. A photon can lose energy by the photoelectric effect, Compton effect, or pair production.
Rev. 0 Page 63 NP-01
I
m z
2
K.E.
DOE-HDBK-1019/1-93
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Because ionizing radiation creates ions in pairs, the intensity of ionization or the specific ionization is defined as the number of ion-pairs formed per centimeter of travel in a given material. The amount of ionization produced by a charged particle per unit path length, which is a measure of its ionizing power, is roughly proportional to the particle's mass and the square of its charge as illustrated in the equation below.
Fixture checkpoint marker-0722-xray is recorded on fixture page 0722.
where:
I is the ionizing power
m is the mass of the particle
z is the number of unit charges it carries
K.E. is its kinetic energy
Since m for an alpha particle is about 7300 times as large as m for a beta particle, and z is twice as great, an alpha will produce much more ionization per unit path length than a beta particle of the same energy. This phenomenon occurs because the larger alpha particle moves slower for a given energy and thus acts on a given electron for a longer time.
Alpha Radiation
Alpha radiation is normally produced from the radioactive decay of heavy nuclides and from certain nuclear reactions. The alpha particle consists of 2 neutrons and 2 protons, so it is essentially the same as the nucleus of a helium atom. Because it has no electrons, the alpha particle has a charge of +2. This positive charge causes the alpha particle to strip electrons from the orbits of atoms in its vicinity. As the alpha particle passes through material, it removes electrons from the orbits of atoms it passes near. Energy is required to remove electrons and the energy of the alpha particle is reduced by each reaction. Eventually the particle will expend its kinetic energy, gain 2 electrons in orbit, and become a helium atom. Because of its strong positive charge and large mass, the alpha particle deposits a large amount of energy in a short distance of travel. This rapid, large deposition of energy limits the penetration of alpha particles. The most energetic alpha particles are stopped by a few centimeters of air or a sheet of paper.
Beta-Minus Radiation
A beta-minus particle is an electron that has been ejected at a high velocity from an unstable nucleus. An electron has a small mass and an electrical charge of -1. Beta particles cause ionization by displacing electrons from atom orbits. The ionization occurs from collisions with orbiting electrons. Each collision removes kinetic energy from the beta particle, causing it to slow down. Eventually the beta particle will be slowed enough to allow it to be captured as an
2 electrons 0.000549 amu
electron

amu
1.02 MeV
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orbiting electron in an atom. Although more penetrating than the alpha, the beta is relatively easy to stop and has a low power of penetration. Even the most energetic beta radiation can be stopped by a few millimeters of metal.
Positron Radiation
Positively charged electrons are called positrons. Except for the positive charge, they are identical to beta-minus particles and interact with matter in a similar manner. Positrons are very short-lived, however, and quickly are annihilated by interaction with a negatively charged electron, producing two gammas with a combined energy (calculated below) equal to the rest mass of the positive and negative electrons.
Neutron Radiation
Neutrons have no electrical charge. They have nearly the same mass as a proton (a hydrogen atom nucleus). A neutron has hundreds of times more mass than an electron, but 1/4 the mass of an alpha particle. The source of neutrons is primarily nuclear reactions, such as fission, but they may also be produced from the decay of radioactive nuclides. Because of its lack of charge, the neutron is difficult to stop and has a high penetrating power.
Neutrons are attenuated (reduced in energy and numbers) by three major interactions, elastic scatter, inelastic scatter, and absorption. In elastic scatter, a neutron collides with a nucleus and bounces off. This reaction transmits some of the kinetic energy of the neutron to the nucleus of the atom, resulting in the neutron being slowed, and the atom receives some kinetic energy
Fixture checkpoint marker-0723-xray is recorded on fixture page 0723.
(motion). This process is sometimes referred to as "the billiard ball effect."
As the mass of the nucleus approaches the mass of the neutron, this reaction becomes more effective in slowing the neutron. Hydrogenous material attenuates neutrons most effectively.
In the inelastic scatter reaction, the same neutron/nucleus collision occurs as in elastic scatter. However, in this reaction, the nucleus receives some internal energy as well as kinetic energy. This slows the neutron, but leaves the nucleus in an excited state. When the nucleus decays to its original energy level, it normally emits a gamma ray.
In the absorption reaction, the neutron is actually absorbed into the nucleus of an atom. The neutron is captured, but the atom is left in an excited state. If the nucleus emits one or more gamma rays to reach a stable level, the process is called radiative capture. This reaction occurs at most neutron energy levels, but is more probable at lower energy levels.
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INTERACTION OF RADIATION WITH MATTER Atomic and Nuclear Physics
Gamma Radiation
Gamma radiation is electromagnetic radiation. It is commonly referred to as a gamma ray and is very similar to an x-ray. The difference is that gamma rays are emitted from the nucleus of an atom, and x-rays are produced by orbiting electrons. The x-ray is produced when orbiting electrons move to a lower energy orbit or when fast-moving electrons approaching an atom are deflected and decelerated as they react with the atom's electrical field (called Bremsstrahlung). The gamma ray is produced by the decay of excited nuclei and by nuclear reactions. Because the gamma ray has no mass and no charge, it is difficult to stop and has a very high penetrating power. A small fraction of the original gamma stream will pass through several feet of concrete or several meters of water.

photo-electric effect. When a low energy gamma strikes an atom, the total energy of the gamma is expended in ejecting an electron from orbit. The result is ionization of the atom and expulsion of a high energy electron. This reaction is most predominant with low energy gammas interacting in materials with high atomic weight and rarely occurs with gammas having an energy above 1 MeV. Annihilation of the gamma results. Any gamma energy in excess of the binding energy of the electron is carried off by the electron in the form of kinetic energy.

The second method of attenuation of gammas is called Compton scattering. The gamma interacts with an orbital or free electron; however, in this case, the photon loses only a fraction of its energy. The actual energy loss depending on the scattering angle of the gamma. The gamma continues on at lower energy, and the energy difference is absorbed by the electron. This reaction becomes important for gamma energies of about 0.1 MeV and higher.

At higher energy levels, a third method of attenuation is predominant. This method is pair-production. When a high energy gamma passes close enough to a heavy nucleus, the gamma completely disappears, and an electron and a positron are formed. For this reaction to take place, the original gamma must have at least 1.02 MeV energy. Any energy greater than 1.02 MeV becomes kinetic energy shared between the electron and positron. The probability of pair■production incre

Interaction of Radiation with Matter Summary

An alpha particle deposits a large amount of energy in a short distance of travel due to its large mass and charge.

Beta-minus particles interact with the electrons orbiting the nucleus of atoms, causing ionization by displacing the electrons. The beta particle loses energy with each interaction. After the beta particle loses enough energy, it is captured in the orbital shells of an atom.

Positrons interact with matter much the same way as beta minus particles. After the positron has lost most of its energy by ionizing atoms, it is annihilated by interaction with an electron. The electron-positron pair disappear and are replaced by two gammas, each with the energy equivalent of the mass of an electron (0.51

Fixture checkpoint marker-0724-xray is recorded on fixture page 0724.

MeV).

Neutrons interact with matter by elastic scattering, inelastic scattering, or absorption.

Photoelectric effect is where a gamma interacts with an electron orbiting an atom. The entire energy of the gamma is transferred to the electron, and the electron is ejected from its orbit.

In Compton scattering a gamma interacts with an orbital electron, but only part of the gamma energy is transferred to the electron. The electron is ejected from its orbit, and the gamma is scattered off at a lower energy.

In pair-production, a gamma interacts with the electric field of a nucleus and is converted into an electron-positron pair. The gamma must have an energy greater than 1.02 MeV for this to occur.

Summary

The important information in this chapter is summarized below.

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Department of Energy
Fundamentals Handbook
NUCLEAR PHYSICS
AND REACTOR THEORY
Module 2
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TERMINAL OBJECTIVE
1.0 Without references, EXPLAIN how neutron sources produce neutrons.
ENABLING OBJECTIVES
1.1 DEFINE the following terms:
a. Intrinsic neutron source
b. Installed neutron source
1.2 LIST three examples of reactions that produce neutrons in intrinsic neutron sources.
1.3 LIST three examples of reactions that produce neutrons in installed neutron sources.

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TERMINAL OBJECTIVE
2.0 Given the necessary information for calculations, EXPLAIN basic concepts in reactor physics and perform calculations.
ENABLING OBJECTIVES
2.1 DEFINE the following terms:
a. Atom density d. Barn
b. Neutron flux e. Macroscopic cross section
c. Microscopic cross section f. Mean free path
2.2 EXPRESS macroscopic cross section in terms of microscopic cross section.
2.3 DESCRIBE how the absorption cross section of typical nuclides varies with neutron energy at energies below the resonance absorption region.
2.4 DESCRIBE the cause of resonance absorption in terms of nuclear energy levels.
2.5 DESCRIBE the energy dependence of resonance absorption peaks for typical light and heavy nuclei.
2.6 EXPRESS mean free path in terms of macroscopic cross section.
2.7 Given the number densities (or total density and component fractions) and microscopic cross sections of components, CALCULATE the macroscopic cross section for a mixture.
2.8 CALCULATE a macroscopic cross section given a material density, atomic mass, and microscopic cross section.
2.9 EXPLAIN neutron shadowing or self-shielding.
2.10 Given the neutron flux and macroscopic cross section, CALCULATE the reaction rate.
2.11 DESCRIBE the relationship between neutron flux and reactor power.
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ENABLING OBJECTIVES (Cont.)
2.12 DEFINE the following concepts:
a. Thermalization d. Average logarithmic energy decrement
b. Moderator e. Macroscopic slowing down power
c. Moderating ratio
2.13 LIST three desirable characteristics of a moderator.
2.14 Given an average fractional energy loss per collision, CALCULATE the energy loss after a specified number of collisions.

TERMINAL OBJECTIVE
3.0 Without references, EXPLAIN the production process and effects on fission of prompt and delayed neutrons.
ENABLING OBJECTIVES
3.1 STATE the origin of prompt neutrons and delayed neutrons.
3.2 STATE the approximate fraction of neutrons that are born as delayed neutrons from the fission of the following nuclear fuels:
a. Uranium-235
b. Plutonium-239
3.3 EXPLAIN the mechanism for production of delayed neutrons.
3.4 EXPLAIN prompt and delayed neutron generation times.
3.5 Given prompt and delayed neutron generation times and delayed neutron fraction, CALCULATE the average generation time.

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TERMINAL OBJECTIVE
4.0 Without references, DESCRIBE the neutron energy spectrum for the type of reactor presented in this module.
ENABLING OBJECTIVES
4.1 STATE the average energy at which prompt neutrons are produced.
4.2 DESCRIBE the neutron energy spectrum in the following reactors:
a. Fast reactor
b. Thermal reactor
4.3 EXPLAIN the reason for the particular shape of the fast, intermediate, and slow energy regions of the neutron flux spectrum for a thermal reactor.
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Reactor Theory (Neutron Characteristics) DOE-HDBK-1019/1-93 NEUTRON SOURCES
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NEUTRON SOURCES
Neutrons from a variety of sources are always present in a reactor core. This is true even when the reactor is shut down. Some of these neutrons are produced by naturally occurring (intrinsic) neutron sources, while others may be the result of fabricated (installed) neutron sources that are incorporated into the design of the reactor. The neutrons produced by sources other than neutron-induced fission are often grouped together and classified as source neutrons.
EO 1.1 DEFINE the following terms:
a. Intrinsic neutron source
b. Installed neutron source
EO 1.2 LIST three examples of reactions that produce neutrons in intrinsic neutron sources.
EO 1.3 LIST three examples of reactions that produce neutrons in installed neutron sources.
Fixture checkpoint marker-0728-xray is recorded on fixture page 0728.
Neutron Sources
In addition to neutron-induced fission, neutrons are produced by other reactions. The neutrons produced by reactions other than neutron-induced fission are called source neutrons. Source neutrons are important because they ensure that the neutron population remains high enough to allow a visible indication of neutron level on the most sensitive monitoring instruments while the reactor is shutdown and during the startup sequence. This verifies instrument operability and allows monitoring of neutron population changes. Source neutrons can be classified as either intrinsic or installed neutron sources.
Intrinsic Neutron Sources
Some neutrons will be produced in the materials present in the reactor due to a variety of unavoidable reactions that occur because of the nature of these materials. Intrinsic neutron sources are those neutron-producing reactions that always occur in reactor materials.
■ ■ ■ ■
■ ■ ■
■ ■
NEUTRON SOURCES DOE-HDBK-1019/1-93 Reactor Theory (Neutron Characteristics)
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A limited number of neutrons will always be present, even in a reactor core that has never been

of the rate at which different heavy nuclides produce neutrons by spontaneous fission. Californium-252 is not an intrinsic neutron source, but will be discussed in the section on installed neutron sources.

TABLE 1

Neutron Production by Spontaneous Fission

Nuclide T (Fission) T (1/2 1/2 ■-decay) neutrons/sec/gram

U 1.8 x 10 years 6.8 x 10 years 8.0 x 10 235
92
17 8 -4

U 8.0 x 10 years 4.5 x 10 years 1.6 x 10 238
92
15 9 -2

Pu 5.5 x 10 years 2.4 x 10 years 3.0 x 10 239
94
5 4 -2

Pu 1.2 x 10 years 6.6 x 10 years 1.0 x 10 240
94
11 3 3

Cf 66.0 years 2.65 years 2.3 x 10 252
98
12

Another intrinsic neutron source is a reaction involving natural boron and fuel. In some reactors, natural boron is loaded into the reactor core as a neutron absorber to improve reactor control or increase core life-time. Boron-11 (80.1% of natural boron) undergoes a reaction with the alpha particle emitted by the radioactive decay of heavy nuclides in the fuel to yield a neutron as shown below.

The boron-11 must be mixed with, or in very close proximity to, the fuel for this reaction because of the short path length of the alpha particle. For a reactor core with this configuration, this (■,n) reaction is an important source of neutrons for reactor startup.

In a reactor that has been operated, another source of neutrons becomes significant. Neutrons may be produced by the interaction of a gamma ray and a deuterium nucleus. This reaction is commonly referred to as a photoneutron reaction because it is initiated by electromagnetic radiation and results in the production of a neutron. The photoneutron reaction is shown below.

■ ■ ■
■ ■
■ ■ ■

Fixture checkpoint marker-0729-xray is recorded on fixture page 0729.

■ ■

There is an abundant supply of high energy gammas in a reactor that has been operated because many of the fission products are gamma emitters. All water-cooled reactors have some deuterium present in the coolant in the reactor core because a small fraction of natural hydrogen is the isotope deuterium. The atom percentage of deuterium in the water ranges from close to the naturally occurring value (0.015%) for light water reactors to above 90% deuterium for heavy water reactors. Therefore, the required conditions for production of photoneutrons exist.

The supply of gamma rays decreases with time after shutdown as the gamma emitters decay; therefore, the photoneutron production rate also decreases. In a few particular reactors,

photoneutrons following a long shutdown period.
Installed Neutron Sources
Because intrinsic neutron sources can be relatively weak or dependent upon the recent power history of the reactor, many reactors have artificial sources of neutrons installed. These neutron sources ensure that shutdown neutron levels are high enough to be detected by the nuclear instruments at all times. This provides a true picture of reactor conditions and any change in these conditions. An installed neutron source is an assembly placed in or near the reactor for the sole purpose of producing source neutrons.
One strong source of neutrons is the artificial nuclide californium-252, which emits neutrons at the rate of about 2 x 10 neutrons per second per gram as the result of spontaneous fission. 12 Important drawbacks for some applications may be its high cost and its short half-life (2.65 years).
Many installed neutron sources use the (■,n) reaction with beryllium. These sources are
Fixture checkpoint marker-0730-xray is recorded on fixture page 0730.
composed of a mixture of metallic beryllium (100% beryllium-9) with a small quantity of an alpha particle emitter, such as a compound of radium, polonium, or plutonium. The reaction that occurs is shown below.
The beryllium is intimately (homogeneously) mixed with the alpha emitter and is usually enclosed in a stainless steel capsule.
Another type of installed neutron source that is widely used is a photoneutron source that employs the (■,n) reaction with beryllium. Beryllium is used for photoneutron sources because its stable isotope beryllium-9 has a weakly attached last neutron with a binding energy of only 1.66 MeV. Thus, a gamma ray with greater energy than 1.66 MeV can cause neutrons to be ejected by the (■,n) reaction as shown below.
NEUTRON SOURCES DOE-HDBK-1019/1-93 Reactor Theory (Neutron Characteristics)
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Neutron Sources Summary
& Intrinsic neutron sources are sources of neutrons from materials that are in the reactor for other purposes such as fuel, burnable poison, or moderator.
& Installed neutron sources are materials or components placed in the reactor specifically for the purpose of producing source neutrons.
& Examples of intrinsic neutron sources are listed below.
Spontaneous fission of heavy nuclides in fuel, such as uranium-238, uranium-235, and plutonium-239, results in fission fragments and free neutrons.
Boron-11 mixed with the fuel undergoes an alpha-neutron reaction and becomes nitrogen-14.
Deuterium present in the reactor coolant undergoes a gamma■neutron reaction and becomes hydrogen-1.
& Examples of installed neutron sources are listed below.
Spontaneous fission of californium-252 results in fission fragments and free neutrons.
Beryllium-9 undergoes an alpha-neutron reaction (alpha from the decay of plutonium, polonium, or radium) and becomes carbon-12.
Beryllium-9 undergoes a gamma-neutron reaction (high energy gamma from decay of antimony-124) and becomes beryllium-8.
Many startup sources of this type use antimony and beryllium because after activation with neutrons the radioactive antimony becomes an emitter of high energy gammas. The photoneutron sources of this type are constructed somewhat differently from the (■,n) types. One design

is then encased in a stainless steel cladding. A large reactor may have several neutron sources of this type installed within the core.

Summary

The important information in this chapter is summarized below.

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Reactor Theory (Neutron Characteristics) NUCLEAR CROSS SECTIONS AND NEUTRON FLUX
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NUCLEAR CROSS SECTIONS AND NEUTRON FLUX

To determine the frequency of neutron interactions, it is necessary to describe the availability of neutrons to cause interaction and the probability of a neutron interacting with material. The availability of neutrons and the probability of interaction are quantified by the neutron flux and nuclear cross section.

EO 2.1 DEFINE the following terms:

a. Atom density d. Barn
b. Neutron flux e. Macroscopic cross section
c. Microscopic cross section f. Mean free path

EO 2.2 EXPRESS macroscopic cross section in terms of microscopic cross section.

EO 2.3 DESCRIBE how the absorption cross section of typical nuclides varies with neutron energy at energies below the resonance

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absorption region.

EO 2.4 DESCRIBE the cause of resonance absorption in terms of nuclear energy levels.

EO 2.5 DESCRIBE the energy dependence of resonance absorption peaks for typical light and heavy nuclei.

EO 2.6 EXPRESS mean free path in terms of macroscopic cross section.

EO 2.7 Given the number densities (or total density and component fractions) and microscopic cross sections of components, CALCULATE the macroscopic cross section for a mixture.

EO 2.8 CALCULATE a macroscopic cross section given a material density, atomic mass, and microscopic cross section.

EO 2.9 EXPLAIN neutron shadowing or self-shielding.

'

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NUCLEAR CROSS SECTIONS AND NEUTRON FLUX Reactor Theory (Neutron Characteristics)
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Introduction

Fission neutrons are born with an average energy of about 2 MeV. These fast neutrons interact with the reactor core materials in various absorption and scattering reactions. Collisions that result in scattering are useful in slowing neutrons to thermal energies. Thermal neutrons may be absorbed by fissile nuclei to produce more fissions or be absorbed in fertile material for conversion to fissionable fuel. Absorption of neutrons in structural components, coolant, and other non-fuel material results in the removal of neutrons without fulfilling any useful purpose.

To safely and efficiently operate a nuclear reactor it is necessary to predict the probability that a particular absorption or scattering reaction will occur. Once these probabilities are known, if the availability of neutrons can be determined, then the rate at which these nuclear reactions take place can be predicted.

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One important property of a material is the atom density. The atom density is the number of atoms of a given type per unit volume of the material. To calculate the atom density of a substance use Equation (2-1).
(2-1)
where:
N = atom density (atoms/cm)3
' = density (g/cm)3
N = Avogadro's number (6.022 x 10 atoms/mole) A
23
M = gram atomic weight
N
NA
M
2.699 g
cm3
6.022 x 1023 atoms
mole
26.9815 g
mole
6.024 x 1022 atoms
cm3
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Reactor Theory (Neutron Characteristics) NUCLEAR CROSS SECTIONS AND NEUTRON FLUX
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Example:
A block of aluminum has a density of 2.699 g/cm . If the gram atomic weight of
3
Fixture checkpoint marker-0732-xray is recorded on fixture page 0732.
aluminum is 26.9815 g, calculate the atom density of the aluminum.
Solution:
Cross Sections
The probability of a neutron interacting with a nucleus for a particular reaction is dependent upon not only the kind of nucleus involved, but also the energy of the neutron. Accordingly, the absorption of a thermal neutron in most materials is much more probable than the absorption of a fast neutron. Also, the probability of interaction will vary depending upon the type of reaction involved.
The probability of a particular reaction occurring between a neutron and a nucleus is called the microscopic cross section ( ) of the nucleus for the particular reaction. This cross section will vary with the energy of the neutron. The microscopic cross section may also be regarded as the effective area the nucleus presents to the neutron for the particular reaction. The larger the effective area, the greater the probability for reaction.
Because the microscopic cross section is an area, it is expressed in units of area, or square centimeters. A square centimeter is tremendously large in comparison to the effective area of a nucleus, and it has been suggested that a physicist once referred to the measure of a square centimeter as being "as big as a barn" when applied to nuclear processes. The name has persisted and microscopic cross sections are expressed in terms of barns. The relationship between barns and cm is shown below. 2
1 barn = 10-24 cm2

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NUCLEAR CROSS SECTIONS AND NEUTRON FLUX Reactor Theory (Neutron Characteristics)
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Whether a neutron will interact with a certain volume of material depends not only on the microscopic cross section of the individual nuclei but also on the number of nuclei within that volume. Therefore, it is necessary to define another kind of cross section known as the macroscopic cross section ( ). The macroscopic cross section is the probability of a given reaction occurring per unit travel of the neutron. is related to the microscopic cross section ( ) by the relationship shown below.
(2-2)
where:
= macroscopic cross section (cm)
-1
N = atom density of material (atoms/cm)
3
= microscopic cross-section (cm)
2
The difference between the microscopic and macroscopic cross sections is extremely important and is restated for clarity. The microscopic cross section ( ) represents the effective target area that a single nucleus presents to a bombarding particle. The units are given in barns or cm .
2
The macroscopic cross section ( ) represents the effective target area that is presented by all of the nuclei contained in 1 cm of the material. The units are given as 1/cm or cm .
3 -1
A neutron interacts with an atom of the material it enters in two basic ways. It will either interact through a scattering interaction or through an absorption reaction. The probability of a neutron being absorbed by a particular atom is the microscopic cross section for absorption, . The probability of a neutron scattering off of a particular nucleus is the microscopic cross a section for scattering, s
. The sum of the microscopic cross section for absorption and the microscopic cross section for scattering is the total microscopic cross section, . T
= + T a s
Both the absorption and the scattering microscopic cross sections can be further divided. For instance, the scattering cross section is the sum of the elastic scattering cross section (se) and Fixture checkpoint marker-0733-xray is recorded on fixture page 0733.
the inelastic scattering cross section ( ). si
= + s se si
The microscopic absorption cross section ( ) includes all reactions except scattering. However, a for most purposes it is sufficient to merely separate it into two categories, fission (f) and capture ( ). Radiative capture of neutrons was described in the Neutron Interactions chapter c of Module 1.
= + a f c
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Reactor Theory (Neutron Characteristics) NUCLEAR CROSS SECTIONS AND NEUTRON FLUX
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Figure 1 Typical Neutron Absorption Cross Section vs. Neutron Energy
The variation of absorption cross sections with neutron energy is often complicated. For many elements the absorption cross sections are small, ranging from a fraction of a barn to a few

For a considerable number of nuclides of moderately high (or high) mass numbers, an examination of the variation of the absorption cross section with the energy of the incident neutron reveals the existence of three regions on a curve of absorption cross section versus neutron energy. This cross section is illustrated in Figure 1. First, the cross section decreases steadily with increasing neutron energy in a low energy region, which includes the thermal range (E < 1 eV). In this region the absorption cross section, which is often high, is inversely proportional to the velocity (v). This region is frequently referred to as the "1/v region," because the absorption cross section is proportional to 1/v, which is the reciprocal of neutron velocity. Following the 1/v region, there occurs the "resonance region" in which the cross sections rise sharply to high values called "resonance peaks" for neutrons of certain energies, and then fall again. These energies are called resonance energies and are a result of the affinity
Fixture checkpoint marker-0734-xray is recorded on fixture page 0734.
of the nucleus for neutrons whose energies closely match its discrete, quantum energy levels. That is, when the binding energy of a neutron plus the kinetic energy of the neutron are exactly equal to the amount required to raise a compound nucleus from its ground state to a quantum level, resonance absorption occurs. The following example problem further illustrates this point.
■
(

Assuming that uranium-236 has a nuclear quantum energy level at 6.8 MeV above its ground state, calculate the kinetic energy a neutron must possess to undergo resonant absorption in uranium-235 at this resonance energy level.
BE = [Mass(U) + Mass(neutron) - Mass(U)] x 931 MeV/amu 235 236
BE = (235.043925 + 1.008665 - 236.045563) x 931 MeV/amu
BE = (0.007025 amu) x 931 MeV/amu = 6.54 MeV
6.8 MeV - 6.54 MeV = 0.26 MeV
The difference between the binding energy and the quantum energy level equals the amount of kinetic energy the neutron must possess. The typical heavy nucleus will have many closely■spaced resonanc complex and have more possible configurations and corresponding energy states. Light nuclei, being less complex, have fewer possible energy states and fewer resonances that are sparsely distributed at higher energy levels.
For higher neutron energies, the absorption cross section steadily decreases as the energy of the neutron increases. This is called the "fast neutron region." In this region the absorption cross sections are usually less than 10 barns.
With the exception of hydrogen, for which the value is fairly large, the elastic scattering cross sections are generally small, for example, 5 barns to 10 barns. This is close to the magnitude of the actual geometric cross sectional area expected for atomic nuclei. In potential scattering, the cross section is essentially constant and independent of neutron energy. Resonance elastic scattering and inelastic scattering exhibit resonance peaks similar to those associated with absorption cross sections. The resonances occur at lower energies for heavy nuclei than for light nuclei. In general, the variations in scattering cross sections are very small when compared to the variations that occur in absorption cross sections.
Mean Free Path
If a neutron has a certain probability of undergoing a particular interaction in one centimeter of travel, then the inverse of this value describes how far the neutron will travel (in the average case) before undergoing an interaction. This average distance traveled by a neutron before

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■. The relationship between the mean free path (■) and the macroscopic cross section (*) is shown below.

(2-3)

'

(

)

Calculation of Macroscopic Cross Section and Mean Free Path

Most materials are composed of several elements, and because most elements are composed of several isotopes, most materials involve many cross sections, one for each isotope involved. Therefore, to include all the isotopes within a given material, it is necessary to determine the macroscopic cross section for each isotope and then sum all the individual macroscopic cross sections. Equation (2-4) can be used to determine the macroscopic cross section for a composite material.

* = N) + N) + N) + ... N) (2-4) 1 1 2 2 3 3 nn

where:

Fixture checkpoint marker-0735-xray is recorded on fixture page 0735.

N = the number of nuclei per cm of the n element n

3 th

) = the microscopic cross section of the n element n

th

The following example problems illustrate the calculation of the macroscopic cross section for a single element and for combinations of materials.

Example 1:

Find the macroscopic thermal neutron absorption cross section for iron, which has a density of 7.86 g/cm . The microscopic cross section for absorption of iron is 2.56 barns 3 and the gram atomic weight is 55.847 g.

Solution:

Step 1: Using Equation (2-1), calculate the atom density of iron.

Step 2: Use this atom density in Equation (2-2) to calculate the macroscopic cross section.

'

'

Example 2:

An alloy is composed of 95% aluminum and 5% silicon (by weight). The density of the alloy is 2.66 g/cm . Properties of aluminum and silicon are shown below. 3

Element	Gram Atomic Weight	) a (barns)	) s (barns)
Aluminum	26.9815	0.23	1.49
Silicon	28.0855	0.16	2.20

1. Calculate the atom densities for the aluminum and silicon.
2. Determine the absorption and scattering macroscopic cross sections for thermal neutrons.
3. Calculate the mean free paths for absorption and scattering.

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Step 1: The density of the aluminum will be 95% of the total density. Using
Equation (2-1) yields the atom densities.
a NAl a,Al NSi a,Si
5.64 x 1022 atoms
cm3
0.23 x 10 24 cm22.85 x 1021 atoms
cm3
0.16 x 10 24 cm2
0.0134 cm 1
s NAl s,Al NSi s,Si
5.64 x 1022 atoms
cm3
1.49 x 10 24 cm22.85 x 1021 atoms
cm3
2.20 x 10 24 cm2
0.0903 cm 1
a
1
a
1
0.01345 cm 1
74.3 cm
s
1
s
Fixture checkpoint marker-0736-xray is recorded on fixture page 0736.
1
0.0903 cm 1
11.1 cm
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Step 2: The macroscopic cross sections for absorption and scattering are
calculated using Equation (2-4).
Step 3: The mean free paths are calculated by inserting the macroscopic cross
sections calculated above into Equation (2-3).
Thus, a neutron must travel an average of 74.3 cm to interact by absorption in this alloy, but it
must travel only 11.1 cm to interact by scattering.
)
)
)
)
(■
(■
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NUCLEAR CROSS SECTIONS AND NEUTRON FLUX Reactor Theory (Neutron Characteristics)
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Effects of Temperature on Cross Section

varies. The microscopic cross sections provided on most charts and tables are measured for a standard neutron velocity of 2200 meters/second, which corresponds to an ambient temperature of 68(F. Therefore, if our material is at a higher temperature, the absorption cross section will be lower than the value for 68(F, and any cross sections which involve absorption (for example,) ,) ,)) must be corrected for the existing temperature. acf

The following formula is used to correct microscopic cross sections for temperature. Although the example illustrates absorption cross section, the same formula may be used to correct capture and fission cross sections.

where:

) = microscopic cross section corrected for temperature

) = microscopic cross section at reference temperature (68(F or 20(C) o

T = reference temperature (68(F) in degrees Rankine ((R) or Kelvin ((K) o

T = temperature for which corrected value is being calculated

NOTE: When using this formula, all temperatures must be converted to (R or (K.

(R = (F + 460

(K = (C + 273

Example:

What is the value of)f for uranium-235 for thermal neutrons at 500(F? Uranium-235 has a) of 583 barns at 68(F. f

Solution:

Neutron Flux

Macroscopic cross sections for neutron reactions with materials determine the probability of one neutron undergoing a specific reaction per centimeter of travel through that material. If one wants to determine how many reactions will actually occur, it is necessary to know how many neutrons are traveling through the material and how many centimeters they travel each second. It is convenient to consider the number of neutrons existing in one cubic centimeter at any one instant and the total distance they travel each second while in that cubic centimeter. The number of neutrons existing in a cm of material at any instant is called neutron density and is 3 represented by the symbol n with units of neutrons/cm . The total distance these neutrons can 3 Fixture checkpoint marker-0737-xray is recorded on fixture page 0737.
travel each second will be determined by their velocity.

A good way of defining neutron flux (1) is to consider it to be the total path length covered by all neutrons in one cubic centimeter during one second. Mathematically, this is the equation below.

1 = n v (2-5)

where:

1 = neutron flux (neutrons/cm -sec) 2

n = neutron density (neutrons/cm)3

v = neutron velocity (cm/sec)

The term neutron flux in some applications (for example, cross section measurement) is used as parallel beams of neutrons traveling in a single direction. The intensity (I) of a neutron beam is the product of the neutron density times the average neutron velocity. The directional beam intensity is equal to the number of neutrons per unit area and time (neutrons/cm -sec) falling on 2 a surface perpendicular to the direction of the beam.

One can think of the neutron flux in a reactor as being comprised of many neutron beams

directional flux intensities (added as numbers and not vectors), that is, 1 = I + I + I +...I . 123 n
Since the atoms in a reactor do not interact preferentially with neutrons from any particular direction, all of these directional beams contribute to the total rate of reaction. In reality, at a given point within a reactor, neutrons will be traveling in all directions.
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NUCLEAR CROSS SECTIONS AND NEUTRON FLUX Reactor Theory (Neutron Characteristics)
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Self-Shielding
In some locations within the reactor, the flux level may be significantly lower than in other areas due to a phenomenon referred to as neutron shadowing or self-shielding. For example, the interior of a fuel pin or pellet will "see" a lower average flux level than the outer surfaces since an appreciable fraction of the neutrons will have been absorbed and therefore cannot reach the interior of the fuel pin. This is especially important at resonance energies, where the absorption cross sections are large.
Summary
The important information in this chapter is summarized below.
DOE-HDBK-1019/1-93
Reactor Theory (Neutron Characteristics) NUCLEAR CROSS SECTIONS AND NEUTRON FLUX
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Nuclear Cross Section and Neutron Flux Summary
& Atom density (N) is the number of atoms of a given type per unit volume of material.
& Microscopic cross section ()) is the probability of a given reaction occurring between a neutron and a nucleus.
& Microscopic cross sections are measured in units of barns, where 1 barn = 10-24 cm .2
& Macroscopic cross section (*) is the probability of a given reaction occurring per unit length of travel of the neutron. The units for macroscopic cross section are cm . -1
& The mean free path (■) is the average distance that a neutron travels in a material between interactions.
& Neutron flux (1) is the total path length traveled by all neutrons in one cubic centimeter of material during one second.
& The macroscopic cross section for a material can be calculated using the equation below.
* = N)
& The absorption cross section for a material usually has three distinct regions. At low neutron energies (<1 eV) the cross section is inversely proportional to the neutron velocity.
Fixture checkpoint marker-0738-xray is recorded on fixture page 0738.
& Resonance absorption occurs when the sum of the kinetic energy of the neutron and its binding energy is equal to an allowed nuclear energy level of the nucleus.
& Resonance peaks exist at intermediate energy levels. For higher neutron energies, the absorption cross section steadily decreases as the neutron energy increases.
& The mean free path equals 1/*.
& The macroscopic cross section for a mixture of materials can be calculated using the equation below.
* = N 1 1 2 2 3 3 nn) + N) + N) + ... N)

neutron absorption near the surface of the material.
1)
REACTION RATES DOE-HDBK-1019/1-93 Reactor Theory (Neutron Characteristics)
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REACTION RATES
It is possible to determine the rate at which a nuclear reaction will take place based on the neutron flux, cross section for the interaction, and atom density of the target. This relationship illustrates how a change in one of these items affects the reaction rate.
EO 2.10 Given the neutron flux and macroscopic cross section, CALCULATE the reaction rate.
EO 2.11 DESCRIBE the relationship between neutron flux and reactor power.
Reaction Rates
If the total path length of all the neutrons in a cubic centimeter in a second is known, (neutron flux (1)), and if the probability of having an interaction per centimeter path length is also known
Fixture checkpoint marker-0739-xray is recorded on fixture page 0739.
(macroscopic cross section (*)), multiply them together to get the number of interactions taking place in that cubic centimeter in one second. This value is known as the reaction rate and is denoted by the symbol R. The reaction rate can be calculated by the equation shown below.
R = 1 * (2-6)
where:
R = reaction rate (reactions/sec)
1 = neutron flux (neutrons/cm -sec) 2
* = macroscopic cross section (cm) -1
Substituting the fact that * = N) into Equation (2-6) yields the equation below.
where:
) = microscopic cross section (cm)2
N = atom density (atoms/cm)3
Rf f
1 x 1013 neutrons
cm2
sec
0.1 cm 1
1 x 1012 fissions
cm3
sec
Rf f
f
Rf
1.29 x 1012 fissions
cm3
sec
3 x 1013 neutrons
cm2
sec
0.043 cm 1
Reactor Theory (Neutron Characteristics) DOE-HDBK-1019/1-93 REACTION RATES

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The reaction rate calculated will depend on which macroscopic cross section is used in the calculation. Normally, the reaction rate of greatest interest is the fission reaction rate.
Example:
If a one cubic centimeter section of a reactor has a macroscopic fission cross section of 0.1 cm , and if the thermal neutron flux is 10 neutrons/cm -sec, what is the fission rate -1 13 2 in that cubic centimeter?
Solution:
In addition to using Equation (2-6) to determine the reaction rate based on the physical properties of material, it is also possible to algebraically manipulate the equation to determine physical properties if the reaction rate is known.
Example:
A reactor operating at a flux level of 3 x 10 neutrons/cm -sec contains 10 atoms of 13 2 20 uranium-235 per cm . The reaction rate is 1.29 x 10 fission/cm . Calculate and . 3 12 3 f f
Solution:
Step 1: The macroscopic cross section can be determined by solving Equation (2-6) for and substituting the appropriate values. f
Rf N f
f
Rf
N
Fixture checkpoint marker-0740-xray is recorded on fixture page 0740.
1.29 x 1012 fissions
cm3
sec
1 x 1020 atoms
cm3
3 x 1013 neutrons
cm2
sec
4.3 x 10 22 cm2 1 barn
1 x 10 24 cm2
430 barns
1 watt 1 erg
1 x 10 7 watt sec
1 MeV
1.602 x 10 6
erg
1 fission
200 MeV
3.12 x 1010 fissions
second
REACTION RATES DOE-HDBK-1019/1-93 Reactor Theory (Neutron Characteristics)
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Step 2: To find the microscopic cross section, replace f with (N x f) and solve for .

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Reactor Power Calculation
Multiplying the reaction rate per unit volume by the total volume of the core results in the total number of reactions occurring in the core per unit time. If the amount of energy involved in each reaction were known, it would be possible to determine the rate of energy release (power) due to a certain reaction.
In a reactor where the average energy per fission is 200 MeV, it is possible to determine the number of fissions per second that are necessary to produce one watt of power using the following conversion factors.
1 fission = 200 MeV
1 MeV = 1.602 x 10 ergs -6
1 erg = 1 x 10 watt-sec -7
This is equivalent to stating that 3.12 x 10 fissions release 1 watt-second of energy. 10
P
th f V
3.12 x 1010 fissions
watt sec
Reactor Theory (Neutron Characteristics) DOE-HDBK-1019/1-93 REACTION RATES
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The power released in a reactor can be calculated based on Equation (2-6). Multiplying the reaction rate by the volume of the reactor results in the total fission rate for the entire reactor. Dividing by the number of fissions per watt-sec results in the power released by fission in the reactor in units of watts. This relationship is shown mathematically in Equation (2-7) below.
(2-7)
where:
P = power (watts)
= thermal neutron flux (neutrons/cm -sec) th
2
= macroscopic cross section for fission (cm) f
-1
V = volume of core (cm)
Fixture checkpoint marker-0741-xray is recorded on fixture page 0741.
3
Relationship Between Neutron Flux and Reactor Power
In an operating reactor the volume of the reactor is constant. Over a relatively short period of time (days or weeks), the number density of the fuel atoms is also relatively constant. Since the atom density and microscopic cross section are constant, the macroscopic cross section must also be constant. Examining Equation (2-7), it is apparent that if the reactor volume and macroscopic cross section are constant, then the reactor power and the neutron flux are directly proportional. This is true for day-to-day operation. The neutron flux for a given power level will increase very slowly over a period of months due to the burnup of the fuel and resulting decrease in atom density and macroscopic cross section.
REACTION RATES DOE-HDBK-1019/1-93 Reactor Theory (Neutron Characteristics)
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Reaction Rates Summary
& The reaction rate is the number of interactions of a particular type occurring in a cubic centimeter of material in a second.
& The reaction rate can be calculated by the equation below.
R = 1 *

constant, the neutron flux is directly proportional to reactor power.
Summary
The important information in this chapter is summarized below.
Reactor Theory (Neutron Characteristics) DOE-HDBK-1019/1-93 NEUTRON MODERATION
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NEUTRON MODERATION
In thermal reactors, the neutrons that cause fission are at a much lower energy than the energy level at which they were born from fission. In this type of reactor, specific materials must be included in the reactor design to reduce the energy level of the neutrons in an efficient manner.
EO 2.12 DEFINE the following concepts:
a. Thermalization d. Average logarithmic energy decrement
b. Moderator e. Macroscopic slowing down power
c. Moderating ratio
EO 2.13 LIST three desirable characteristics of a moderator.
EO 2.14 Given an average fractional energy loss per collision, CALCULATE the energy loss after a specified number of collisions.
Neutron Slowing Down and Thermalization
Fission neutrons are produced at an average energy level of 2 MeV and immediately begin to slow down as the result of numerous scattering reactions with a variety of target nuclei. After a number of collisions with nuclei, the speed of a neutron is reduced to such an extent that it has approximately the same average kinetic energy as the atoms (or molecules) of the medium in which the neutron is undergoing elastic scattering. This energy, which is only a small fraction of an electron volt at ordinary temperatures (0.025 eV at 20(C), is frequently referred to as the thermal energy, since it depends upon the temperature. Neutrons whose energies have been reduced to values in this region (< 1 eV) are designated thermal neutrons. The process of reducing the energy of a neutron to the thermal region by elastic scattering is referred to as thermalization, slowing down, or moderation. The material used for the purpose of thermalizing neutrons is called a moderator. A good moderator reduces the speed of neutrons in a small number of collisions, but does not absorb them to any great extent. Slowing the neutrons in as few collisions as possible is desirable in order to reduce the amount of neutron leakage from the core and also to reduce the number of resonance absorptions in non-fuel materials. Neutron leakage and resonance absorption will be discussed in the next module.
ln Ei
ln Ef
ln
Fixture checkpoint marker-0742-xray is recorded on fixture page 0742.
Ei
Ef
2
A
2
3
NEUTRON MODERATION DOE-HDBK-1019/1-93 Reactor Theory (Neutron Characteristics)
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The ideal moderating material (moderator) should have the following nuclear properties.
large scattering cross section

large energy loss per collision
A convenient measure of energy loss per collision is the logarithmic energy decrement. The average logarithmic energy decrement is the average decrease per collision in the logarithm of the neutron energy. This quantity is represented by the symbol (Greek letter xi).
(2-8)
where:
= average logarithmic energy decrement
Ei = average initial neutron energy
Ef = average final neutron energy
The symbol is commonly called the average logarithmic energy decrement because of the fact that a neutron loses, on the average, a fixed fraction of its energy per scattering collision. Since the fraction of energy retained by a neutron in a single elastic collision is a constant for a given material, is also a constant. Because it is a constant for each type of material and does not depend upon the initial neutron energy, is a convenient quantity for assessing the moderating
Fixture checkpoint marker-0743-xray is recorded on fixture page 0743.
ability of a material.
The values for the lighter nuclei are tabulated in a variety of sources. The following commonly used approximation may be used when a tabulated value is not available.
This approximation is relatively accurate for mass numbers (A) greater than 10, but for some low values of A it may be in error by over three percent.
N
ln Ehigh ln Elow
ln
Ehigh
Elow
N
ln
Ehigh
Elow
ln
2 x 106eV
0.025 eV
0.948
19.2 collisions
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Since represents the average logarithmic energy loss per collision, the total number of collisions necessary for a neutron to lose a given amount of energy may be determined by dividing into the difference of the natural logarithms of the energy range in question. The number of collisions (N) to travel from any energy, E , to any lower energy, E , can be calculated as shown below. high low
Example:
How many collisions are required to slow a neutron from an energy of 2 MeV to a thermal energy of 0.025 eV, using water as the moderator? Water has a value of 0.948 for .
Solution:
Sometimes it is convenient, based upon information known, to work with an average fractional energy loss per collision as opposed to an average logarithmic fraction. If the initial neutron energy level and the average fractional energy loss per collision are known, the final energy level

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! (
NEUTRON MODERATION DOE-HDBK-1019/1-93 Reactor Theory (Neutron Characteristics)
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(2-9)
where:
E = initial neutron energy o
E = neutron energy after N collisions N
x = average fractional energy loss per collision
N = number of collisions
Example:
If the average fractional energy loss per collision in hydrogen is 0.63, what will be the energy of a 2 MeV neutron after (a) 5 collisions? (b) 10 collisions?
Solution:
a)
b)
Macroscopic Slowing Down Power
Although the logarithmic energy decrement is a convenient measure of the ability of a material to slow neutrons, it does not measure all necessary properties of a moderator. A better measure of the capabilities of a material is the macroscopic slowing down power. The macroscopic slowing down power (MSDP) is the product of the logarithmic energy decrement and the macroscopic cross section for scattering in the material. Equation (2-10) illustrates how to
Fixture checkpoint marker-0744-xray is recorded on fixture page 0744.
calculate the macroscopic slowing down power.
(2-10)
MR
s
a
Reactor Theory (Neutron Characteristics) DOE-HDBK-1019/1-93 NEUTRON MODERATION
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Moderating Ratio
Macroscopic slowing down power indicates how rapidly a neutron will slow down in the material in question, but it still does not fully explain the effectiveness of the material as a moderator. An element such as boron has a high logarithmic energy decrement and a good slowing down power, but it is a poor moderator because of its high probability of absorbing neutrons.
The most complete measure of the effectiveness of a moderator is the moderating ratio. The moderating ratio is the ratio of the macroscopic slowing down power to the macroscopic cross section for absorption. The higher the moderating ratio, the more effectively the material performs as a moderator. Equation (2-11) shows how to calculate the moderating ratio of a material.
(2-11)
Moderating properties of different materials are compared in Table 2.
TABLE 2
Moderating Properties of Materials
Material Collisions to Slowing Down
Number of Macroscopic
Thermalize Power
Moderating
Ratio

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D2O 0.510 35 0.177 4830
Helium 0.427 42 9 x 10 51 -6
Beryllium 0.207 86 0.154 126
Boron 0.171 105 0.092 0.00086
Carbon 0.158 114 0.083 216
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Summary
The important information in this chapter is summarized below.
Neutron Moderation Summary
& Thermalization is the process of reducing the energy level of a neutron from the energy level at which it is produced to an energy level in the thermal range.
& The moderator is the reactor material that is present for the purpose of thermalizing neutrons.
& Moderating ratio is the ratio of the macroscopic slowing down power to the macroscopic cross section for absorption.
& The average logarithmic energy decrement (!) is the average change in the logarithm of neutron energy per collision.
& Macroscopic slowing down power is the product of the average logarithmic energy decrement and the macroscopic cross section for scattering.
& There are three desirable characteristics of a moderator.
1. large scattering cross section
2. small absorption cross section
3. large energy loss per collision
& The energy loss after a specified number of collisions can be calculated using the equation below.
E = E (1 - x) N o
N
Fixture checkpoint marker-0745-xray is recorded on fixture page 0745.
■
■ ■
Reactor Theory (Neutron Characteristics) DOE-HDBK-1019/1-93 PROMPT AND DELAYED NEUTRONS
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PROMPT AND DELAYED NEUTRONS
Not all neutrons are released at the same time following fission. Most neutrons are released virtually instantaneously and are called prompt neutrons. A very small fraction of neutrons are released after the decay of fission products and are called delayed neutrons. Although delayed neutrons are a very small fraction of the total number of neutrons, they play an extremely important role in the control of the reactor.
EO 3.1 STATE the origin of prompt neutrons and delayed neutrons.
EO 3.2 STATE the approximate fraction of neutrons that are born as delayed neutrons from the fission of the following nuclear fuels:
a. Uranium-235
b. Plutonium-239
EO 3.3 EXPLAIN the mechanism for production of delayed neutrons.
EO 3.4 EXPLAIN prompt and delayed neutron generation times.
EO 3.5 Given prompt and delayed neutron generation times and

time.
EO 3.6 EXPLAIN the effect of delayed neutrons on reactor control.
Neutron Classification
The great majority (over 99%) of the neutrons produced in fission are released within about 10-13 seconds of the actual fission event. These are called prompt neutrons. A small portion of fission neutrons are delayed neutrons, which are produced for some time after the fission process has taken place. The delayed neutrons are emitted immediately following the first beta decay of a fission fragment known as a delayed neutron precursor. An example of a delayed neutron precursor is bromine-87, shown below.
PROMPT AND DELAYED NEUTRONS DOE-HDBK-1019/1-93 Reactor Theory (Neutron Characteristics)
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For most applications, it is convenient to combine the known precursors into groups with appropriately averaged properties. These groups vary somewhat depending on the fissile material in use. Table 3 lists the characteristics for the six precursor groups resulting from thermal fission of uranium-235. The fraction of all neutrons that are produced by each of these precursors is called the delayed neutron fraction for that precursor. The total fraction of all neutrons born as delayed neutrons is called the delayed neutron fraction (■). The fraction of delayed neutrons produced varies depending on the predominant fissile nuclide in use. The delayed neutron fractions (■) for the fissile nuclides of most interest are as follows: uranium-233 (0.0026), uranium-235 (0.0065), uranium-238 (0.0148), and plutonium-239 (0.0021).
TABLE 3
Delayed Neutron Precursor Groups
for Thermal Fission in Uranium-235
Group Half-Life Delayed Neutron Average Energy
(sec) Fraction (MeV)
1 55.7 0.00021 0.25
2 22.7 0.00142 0.46
3 6.2 0.00127 0.41
4 2.3 0.00257 0.45
5 0.61 0.00075 0.41
6 0.23 0.00027 -
Total - 0.0065 -
Neutron Generation Time
The neutron generation time is the time required for neutrons from one generation to cause the fissions that produce the next generation of neutrons. The generation time for prompt neutrons
Fixture checkpoint marker-0746-xray is recorded on fixture page 0746.
(5* - pronounced "ell-star") is the total time from birth to rebirth. Three time intervals are involved: (a) the time it takes a fast neutron to slow down to thermal energy, (b) the time the now thermal neutron exists prior to absorption in fuel, and (c) the time required for a fissionable nucleus to emit a fast neutron after neutron absorption.
Fast neutrons slow to thermal energies or leak out of the reactor in 10 seconds to 10 seconds, -4 -6 depending on the moderator. In water moderated reactors, thermal neutrons tend to exist for about 10 seconds before they are absorbed. Fission and fast neutron production following -4 neutron absorption in a fissionable nucleus occurs in about 10 seconds. Thus, fast reactors -13 have an 5* of about 10 seconds, while thermal reactors have an 5* of about 10 seconds + 10 -6 -6 -4 seconds, which is about 10 seconds to 10 seconds. -4 -5
■■■■
■■■■

On the other hand, the average generation time for the six delayed neutron groups is the total time from the birth of the fast neutron to the emission of the delayed neutron. Again, three time intervals are involved: (a) the time it takes a fast neutron to slow down to thermal energy, (b) the time the thermal neutron exists prior to absorption, and (c) the average time from neutron absorption to neutron emission by the six precursor groups. The average time for decay of precursors from uranium-235 is 12.5 seconds. The other terms in the delayed neutron generation time are insignificant when compared to this value, and the average delayed neutron generation time becomes ■ 12.5 seconds.

A neutron generation time in the range of 10 seconds to 10 seconds or faster could result in -4 -5 very rapid power excursions, and control would not be possible without the dependence upon

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delayed neutrons to slow down the rate of the reaction. The average generation time, and hence the rate that power can rise, is determined largely by the delayed neutron generation time. The following equation shows this mathematically.

(2-12)

Example:

Assume a prompt neutron generation time for a particular reactor of 5 x 10 seconds and -5 a delayed neutron generation time of 12.5 seconds. If ■ is 0.0065, calculate the average generation time.

Solution:

This example demonstrates the effect delayed neutrons have on the neutron generation time and thus reactor control. If a reactor were to be operated in a sustained chain reaction using only prompt neutrons (■ = 0), the generation time from the previous example would be about 5 x 10 seconds. However, by operating the reactor such that a 0.0065 fraction of neutrons are -5 delayed, the generation life time is extended to 0.0813 seconds, providing time for adequate operator control. Therefore, although only a small fraction of the total neutron population, delayed neutrons are extremely important to the control and maintenance of a sustained fission chain reaction.

Summary

The important information in this chapter is summarized on the following page.

■ ■ ■ ■

Prompt and Delayed Neutrons Summary

& Prompt neutrons are released directly from fission within 10 seconds of the -13 fission event.

& Delayed neutrons are released from the decay of fission products that are called delayed neutron precursors. Delayed neutron precursors are grouped according to half-life. Half-lives vary from fractions of a second to almost a minute.

& The fraction of neutrons born as delayed neutrons is different for different fuel materials. Following are values for some common fuel materials.

Uranium-235 0.0065

Plutonium-239 0.0021

& Delayed neutrons are produced by a classification of fission products known as delayed neutron precursors. When a delayed neutron precursor undergoes a ■ decay, it results in an excited daughter nucleus which immediately ejects a -

neutron precursor.
& The delayed neutron generation time is the total time from the birth of the fast neutron to the emission of the delayed neutron in the next generation. Delayed neutron generation times are dominated by the half-life of the delayed neutron precursor. The average delayed neutron generation time is about 12.5 seconds.
& A prompt neutron generation time is the sum of the amount of time it takes a fast neutron to thermalize, the amount of time the neutron exists as a thermal neutron before it is absorbed, and the amount of time between a fissionable nuclide absorbing a neutron and fission neutrons being released. Prompt neutron generation time is about 5 x 10 seconds. -5
& The average neutron generation time can be calculated from the prompt and delayed neutron generation times and the delayed neutron fraction using Equation (2-12).
& Delayed neutrons are responsible for the ability to control the rate at which power can rise in a reactor. If only prompt neutrons existed, reactor control would not be possible due to the rapid power changes.
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Figure 2 Prompt Fission Neutron Energy Spectrum for
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Thermal Fission of Uranium-235
NEUTRON FLUX SPECTRUM
The number of neutrons that exist at a given energy level varies. A plot of either the fraction of neutrons or the neutron flux at a given energy versus the energy level is called a neutron energy spectrum. The neutron energy spectrum varies widely for different types of reactors.
EO 4.1 STATE the average energy at which prompt neutrons are produced.
EO 4.2 DESCRIBE the neutron energy spectrum in the following reactors:
a. Fast reactor
b. Thermal reactor
EO 4.3 EXPLAIN the reason for the particular shape of the fast, intermediate, and slow energy regions of the neutron flux spectrum for a thermal reactor.
Prompt Neutron Energies
The neutrons produced by fission are high energy neutrons, and almost all fission neutrons have energies between 0.1 MeV and 10 MeV. The neutron energy distribution, or spectrum, may best be described by plotting the fraction of neutrons per MeV as a function of neutron energy, as shown in Figure 2. From this figure it can be seen that the most probable neutron energy is about

can be shown that the average
energy of fission neutrons is about
2 MeV. Figure 2 is the neutron
energy spectrum for thermal
fission in uranium-235. The values
will vary slightly for other
nuclides.
NEUTRON FLUX SPECTRUM DOE-HDBK-1019/1-93 Reactor Theory (Neutron Characteristics)
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Figure 3 Comparison of Neutron Flux Spectra for
Thermal and Fast Breeder Reactor
Thermal and Fast Breeder Reactor Neutron Spectra
The spectrum of neutron energies produced by fission varies significantly from the energy spectrum, or flux, existing in a reactor at a given time. Figure 3 illustrates the difference in neutron flux spectra between a thermal reactor and a fast breeder reactor. The energy distribution of neutrons from fission is essentially the same for both reactors, so the differences in the curve shapes may be attributed to the neutron moderation or slowing down effects.
No attempt is made to thermalize or slow down neutrons in the fast breeder reactor (liquid metal cooled); therefore, an insignificant number of neutrons exist in the thermal range. For the thermal reactor (water moderated), the spectrum of neutrons in the fast region (> 0.1 MeV) has a shape similar to that for the spectrum of neutrons emitted by the fission process.
In the thermal reactor, the flux in the intermediate energy region (1 eV to 0.1 MeV) has approximately a 1/E dependence. That is, if the energy (E) is halved, the flux doubles. This 1/E dependence is caused by the slowing down process, where elastic collisions remove a constant fraction of the neutron energy per collision (on the average), independent of energy; thus, the neutron loses larger amounts of energy per collision at higher energies than at lower
Fixture checkpoint marker-0749-xray is recorded on fixture page 0749.
energies. The fact that the neutrons lose a constant fraction of energy per collision causes the neutrons to tend to "pile up" at lower energies, that is, a greater number of neutrons exist at the lower energies as a result of this behavior.
vp
2 k T
m
vp
2 k T
m
2 1.38 x 10 16 erg
K
293 K
1.66 x 10 24 g
2.2 x 105 cm
sec
1 m
100 cm
2200 m
sec
Reactor Theory (Neutron Characteristics) DOE-HDBK-1019/1-93 NEUTRON FLUX SPECTRUM
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material. In any given collision they may gain or lose energy, and over successive collisions will gain as much energy as they lose. These thermal neutrons, even at a specific temperature, do not all have the same energy or velocity; there is a distribution of energies, usually referred to as the Maxwell distribution (e.g., Figure 2). The energies of most thermal neutrons lie close to the most probable energy, but there is a spread of neutrons above and below this value.

Most Probable Neutron Velocities

The most probable velocity (vp) of a thermal neutron is determined by the temperature of the medium and can be determined by Equation (2-13) .

(2-13)

where:

v = most probable velocity of neutron (cm/sec) p

k = Boltzman's constant (1.38 x 10 erg/ K) -16

T = absolute temperature in degrees Kelvin (K)

m = mass of neutron (1.66 x 10 grams) -24

Example:

Calculate the most probable velocities for neutrons in thermal equilibrium with their surroundings at the following temperatures. a) 20 C, b) 260 C.

Solution:

a) Calculate the most probable velocity for 20 C using Equation (2-13).

vp

2 k T

m

2 1.38 x 10 16 erg

K

533 K

1.66 x 10 24 g

2.977 x 105 cm

sec

1 m

100 cm

2977 m

sec

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b) Calculate the most probable velocity for 260 C using Equation (2-13).

From these calculations it is evident that the most probable velocity of a thermal neutron increases as temperature increases. The most probable velocity at 20 C is of particular importance since reference data, such as nuclear cross sections, are tabulated for a neutron velocity of 2200 meters per second.

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Neutron Flux Spectrum Summary

& Prompt neutrons are born at energies between 0.1 MeV and 10 MeV. The average prompt neutron energy is about 2 MeV.

& Fast reactors have a neutron energy spectrum that has the same shape as the prompt neutron energy spectrum.

& Thermal reactors have a neutron energy spectrum that has two pronounced

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NUCLEAR PHYSICS AND REACTOR THEORY
FOREWORD
The Department of Energy (DOE) Fundamentals Handbooks consist of ten academic subjects, which include Mathematics; Classical Physics; Thermodynamics, Heat Transfer, and Fluid Flow; Instrumentation and Control; Electrical Science; Material Science; Mechanical Science; Chemistry; Engineering Symbology, Prints, and Drawings; and Nuclear Physics and Reactor Theory. The handbooks are provided as an aid to DOE nuclear facility contractors.
These handbooks were first published as Reactor Operator Fundamentals Manuals in 1985 for use by DOE category A reactors. The subject areas, subject matter content, and level of detail of the Reactor Operator Fundamentals Manuals were determined from several sources.
Fixture checkpoint marker-0751-xray is recorded on fixture page 0751.
DOE Category A reactor training managers determined which materials should be included, and served as a primary reference in the initial development phase. Training guidelines from the commercial nuclear power industry, results of job and task analyses, and independent input from contractors and operations-oriented personnel were all considered and included to some degree in developing the text material and learning objectives.
The DOE Fundamentals Handbooks represent the needs of various DOE nuclear facilities' fundamental training requirements. To increase their applicability to nonreactor nuclear facilities, the Reactor Operator Fundamentals Manual learning objectives were distributed to the Nuclear Facility Training Coordination Program Steering Committee for review and comment. To update their reactor-specific content, DOE Category A reactor training managers also reviewed and commented on the content. On the basis of feedback from these sources, information that applied to two or more DOE nuclear facilities was considered generic and was included. The final draft of each of the handbooks was then reviewed by these two groups. This approach has resulted in revised modular handbooks that contain sufficient detail such that each facility may adjust the content to fit their specific needs.
Each handbook contains an abstract, a foreword, an overview, learning objectives, and text material, and is divided into modules so that content and order may be modified by individual DOE contractors to suit their specific training needs. Each handbook is supported by a separate examination bank with an answer key.
The DOE Fundamentals Handbooks have been prepared for the Assistant Secretary for Nuclear Energy, Office of Nuclear Safety Policy and Standards, by the DOE Training Coordination Program. This program is managed by EG&G Idaho, Inc.
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NUCLEAR PHYSICS AND REACTOR THEORY
OVERVIEW
The Department of Energy Fundamentals Handbook entitled Nuclear Physics and Reactor Theory was prepared as an information resource for personnel who are responsible for the operation of the Department's nuclear facilities. Almost all processes that take place in a nuclear facility involves the transfer of some type of energy. A basic understanding of nuclear physics and reactor theory is necessary for DOE nuclear facility operators, maintenance personnel, and the technical staff to safely operate and maintain the facility and facility support systems. The information in this handbook is presented to provide a foundation for applying engineering concepts to the job. This knowledge will help personnel understand the impact that their actions may have on the safe and reliable operation of facility components and systems.
The Nuclear Physics and Reactor Theory handbook consists of four modules that are contained in two volumes. The following is a brief description of the information presented in

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Volume 1 of 2
Module 1 - Atomic and Nuclear Physics
Introduces concepts of atomic physics including the atomic nature of matter, the chart of the nuclides, radioactivity and radioactive decay, neutron interactions and fission, and the interaction of radiation with matter.
Module 2 - Reactor Theory (Nuclear Parameters)
Provides information on reactor theory and neutron characteristics. Includes topics such as neutron sources, neutron flux, neutron cross sections, reaction rates, neutron moderation, and prompt and delayed neutrons.
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NUCLEAR PHYSICS AND REACTOR THEORY
OVERVIEW (Cont.)
Volume 2 of 2
Module 3 - Reactor Theory (Nuclear Parameters)
Explains the nuclear parameters associated with reactor theory. Topics include the neutron life cycle, reactivity and reactivity coefficients, neutron poisons, and
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control rods.
Module 4 - Reactor Theory (Reactor Operations)
Introduces the reactor operations aspect of reactor theory. Topics include subcritical multiplication, reactor kinetics, and reactor operation.
The information contained in this handbook is not all-encompassing. An attempt to present the entire subject of nuclear physics and reactor theory would be impractical. However, the Nuclear Physics and Reactor Theory handbook presents enough information to provide the reader with the fundamental knowledge necessary to understand the advanced theoretical concepts presented in other subject areas, and to understand basic system and equipment operation.
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Fundamentals Handbook
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AND REACTOR THEORY
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TERMINAL OBJECTIVE
1.0 Given sufficient information, DESCRIBE atoms, including components, structure, and
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nomenclature.
ENABLING OBJECTIVES
1.1 STATE the characteristics of the following atomic particles, including mass, charge, and location within the atom:
a. Proton
b. Neutron
c. Electron
1.2 DESCRIBE the Bohr model of an atom.
1.3 DEFINE the following terms:

b. Isotope d. Mass number
1.4 Given the standard A
ZX notation for a particular nuclide, DETERMINE the following:
a. Number of protons
b. Number of neutrons
c. Number of electrons
1.5 DESCRIBE the three forces that act on particles within the nucleus and affect the stability of the nucleus.
1.6 DEFINE the following terms:
a. Enriched uranium
b. Depleted uranium
1.7 DEFINE the following terms:
a. Mass defect
b. Binding energy
1.8 Given the atomic mass for a nuclide and the atomic masses of a neutron, proton, and
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electron, CALCULATE the mass defect and binding energy of the nuclide.

TERMINAL OBJECTIVE
2.0 Given necessary references, DESCRIBE the various modes of radioactive decay.
ENABLING OBJECTIVES
2.1 DESCRIBE the following processes:
a. Alpha decay d. Electron capture
b. Beta-minus decay e. Internal conversions
c. Beta-plus decay f. Isomeric transitions
2.2 Given a Chart of the Nuclides, WRITE the radioactive decay chain for a nuclide.
2.3 EXPLAIN why one or more gamma rays typically accompany particle emission.
2.4 Given the stability curve on the Chart of the Nuclides, DETERMINE the type of radioactive decay that the nuclides in each region of the chart will typically undergo.
2.5 DEFINE the following terms:
a. Radioactivity d. Radioactive decay constant
b. Curie e. Radioactive half-life
c. Becquerel
2.6 Given the number of atoms and either the half-life or decay constant of a nuclide, CALCULATE the activity.
2.7 Given the initial activity and the decay constant of a nuclide, CALCULATE the activity at any later time.
2.8 CONVERT between the half-life and decay constant for a nuclide.
2.9 Given the Chart of the Nuclides and the original activity, PLOT the radioactive decay curve for a nuclide on either linear or semi-log coordinates.
2.10 DEFINE the following terms:
a. Radioactive equilibrium
b. Transient radioactive equilibrium

TERMINAL OBJECTIVE
3.0 Without references, DESCRIBE the different nuclear interactions initiated by neutrons.

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3.1 DESCRIBE the following scattering interactions between a neutron and a nucleus:
a. Elastic scattering
b. Inelastic scattering
3.2 STATE the conservation laws that apply to an elastic collision between a neutron and a nucleus.
3.3 DESCRIBE the following reactions where a neutron is absorbed in a nucleus:
a. Radiative capture
b. Particle ejection

TERMINAL OBJECTIVE
4.0 Without references, DESCRIBE the fission process.
ENABLING OBJECTIVES
4.1 EXPLAIN the fission process using the liquid drop model of a nucleus.
4.2 DEFINE the following terms:
a. Excitation energy
b. Critical energy
4.3 DEFINE the following terms:
a. Fissile material
b. Fissionable material
c. Fertile material
4.4 DESCRIBE the processes of transmutation, conversion, and breeding.
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4.5 DESCRIBE the curve of Binding Energy per Nucleon versus mass number and give a qualitative description of the reasons for its shape.
4.6 EXPLAIN why only the heaviest nuclei are easily fissioned.
4.7 EXPLAIN why uranium-235 fissions with thermal neutrons and uranium-238 fissions only with fast neutrons.
4.8 CHARACTERIZE the fission products in terms of mass groupings and radioactivity.
4.9 Given the nuclides involved and their masses, CALCULATE the energy released from fission.
4.10 Given the curve of Binding Energy per Nucleon versus mass number, CALCULATE the energy released from fission.

TERMINAL OBJECTIVE
5.0 Without references, DESCRIBE how the various types of radiation interact with matter.
ENABLING OBJECTIVES
5.1 DESCRIBE interactions of the following with matter:
a. Alpha particle c. Positron
b. Beta particle d. Neutron
5.2 DESCRIBE the following ways that gamma radiation interacts with matter:
a. Compton scattering
b. Photoelectric effect
c. Pair production

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Atomic and Nuclear Physics DOE-HDBK-1019/1-93 ATOMIC NATURE OF MATTER
ATOMIC NATURE OF MATTER
All matter is composed of atoms. The atom is the smallest amount of matter that retains the properties of an element. Atoms themselves are composed of smaller particles, but these smaller particles no longer have the same properties as the overall element.
EO 1.1 STATE the characteristics of the following atomic particles, including mass, charge, and location within the atom:
a. Proton
b. Neutron
c. Electron
EO 1.2 DESCRIBE the Bohr model of an atom.
EO 1.3 DEFINE the following terms:
a. Nuclide c. Atomic number
b. Isotope d. Mass number
EO 1.4 Given the standard A
ZX notation for a particular nuclide,
DETERMINE the following:
a. Number of protons
b. Number of neutrons
c. Number of electrons
EO 1.5 DESCRIBE the three forces that act on particles within the nucleus and affect the stability of the nucleus.
Structure of Matter
Early Greek philosophers speculated that the earth was made up of different combinations of basic substances, or elements. They considered these basic elements to be earth, air, water, and fire. Modern science shows that the early Greeks held the correct concept that matter consists of a combination of basic elements, but they incorrectly identified the elements.
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ATOMIC NATURE OF MATTER DOE-HDBK-1019/1-93 Atomic and Nuclear Physics
In 1661 the English chemist Robert Boyle published the modern criterion for an element. He defined an element to be a basic substance that cannot be broken down into any simpler substance after it is isolated from a compound, but can be combined with other elements to form compounds. To date, 105 different elements have been confirmed to exist, and researchers claim to have discovered three additional elements. Of the 105 confirmed elements, 90 exist in nature and 15 are man-made.
Another basic concept of matter that the Greeks debated was whether matter was continuous or discrete. That is, whether matter could be continuously divided and subdivided into ever smaller particles or whether eventually an indivisible particle would be encountered. Democritus in about 450 B.C. argued that substances were ultimately composed of small, indivisible particles that he labeled atoms. He further suggested that different substances were composed of different atoms or combinations of atoms, and that one substance could be converted into another by rearranging the atoms. It was impossible to conclusively prove or disprove this proposal for more than 2000 years.
The modern proof for the atomic nature of matter was first proposed by the English chemist John Dalton in 1803. Dalton stated that each chemical element possesses a particular kind of atom, and any quantity of the element is made up of identical atoms of this kind. What distinguishes

difference between kinds of atoms is their weight.

Subatomic Particles

For almost 100 years after Dalton established the atomic nature of atoms, it was considered impossible to divide the atom into even smaller parts. All of the results of chemical experiments during this time indicated that the atom was indivisible. Eventually, experimentation into electricity and radioactivity indicated that particles of matter smaller than the atom did indeed exist. In 1906, J. J. Thompson won the Nobel Prize in physics for establishing the existence of electrons. Electrons are negatively-charged particles that have 1/1835 the mass of the hydrogen atom. Soon after the discovery of electrons, protons were discovered. Protons are relatively large particles that have almost the same mass as a hydrogen atom and a positive charge equal in magnitude (but opposite in sign) to that of the electron. The third subatomic particle to be discovered, the neutron, was not found until 1932. The neutron has almost the same mass as the proton, but it is electrically neutral.

Bohr Model of the Atom

The British physicist Ernest Rutherford postulated that the positive charge in an atom is concentrated in a small region called a nucleus at the center of the atom with electrons existing in orbits around it. Niels Bohr, coupling Rutherford's postulation with the quantum theory introduced by Max Planck, proposed that the atom consists of a dense nucleus of protons surrounded by electrons traveling in discrete orbits at fixed distances from the nucleus. An electron in one of these orbits or shells has a specific or discrete quantity of energy (quantum). When an electron moves from one allowed orbit to another allowed orbit, the energy difference between the two states is emitted or absorbed in the form of a single quantum of radiant energy called a photon. Figure 1 is Bohr's model of the hydrogen atom showing an electron as having just dropped from the third shell to the first shell with the emission of a photon that has an energy = hv. (h = Planck's constant = 6.63 x 10-34 J-s and v = frequency of the photon.) Bohr's theory was the first to successfully account for the discrete energy levels of this radiation as measured in the laboratory. Although Bohr's atomic model is designed specifically to explain the hydrogen atom, his theories apply generally to the structure of all atoms. Additional information on electron shell theory can be found in the Chemistry Fundamentals Handbook.

Figure 1 Bohr's Model of the Hydrogen Atom

Properties of the three subatomic particles are listed in Table 1.

TABLE 1

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Properties of Subatomic Particles

Particle	Location	Charge	Mass
Neutron	Nucleus	none	1.008665 amu
Proton	Nucleus	+1	1.007277 amu
Electron	Shells around nucleus	-1	0.0005486 amu

Measuring Units on the Atomic Scale

The size and mass of atoms are so small that the use of normal measuring units, while possible, is often inconvenient. Units of measure have been defined for mass and energy on the atomic scale to make measurements more convenient to express. The unit of measure for mass is the atomic mass unit (amu). One atomic mass unit is equal to 1.66 x 10-24 grams. The reason for this particular value for the atomic mass unit will be discussed in a later chapter. Note from

the electron volt (eV). The electron volt is the amount of energy acquired by a single electron when it falls through a potential difference of one volt. One electron volt is equivalent to 1.602 x 10-19 joules or 1.18 x 10-19 foot-pounds.

Nuclides

The total number of protons in the nucleus of an atom is called the atomic number of the atom and is given the symbol Z. The number of electrons in an electrically-neutral atom is the same as the number of protons in the nucleus. The number of neutrons in a nucleus is known as the neutron number and is given the symbol N. The mass number of the nucleus is the total number of nucleons, that is, protons and neutrons in the nucleus. The mass number is given the symbol A and can be found by the equation Z + N = A.

Each of the chemical elements has a unique atomic number because the atoms of different elements contain a different number of protons. The atomic number of an atom identifies the particular element.

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Each type of atom that contains a unique combination of protons and neutrons is called a nuclide. Not all combinations of numbers of protons and neutrons are possible, but about 2500 specific nuclides with unique combinations of neutrons and protons have been identified. Each nuclide is denoted by the chemical symbol of the element with the atomic number written as a subscript and the mass number written as a superscript, as shown in Figure 2. Because each element has a unique name, chemical symbol, and atomic number, only one of the three is necessary to identify the element. For this reason nuclides can also be identified by either the chemical name or the chemical symbol followed by the mass number (for example, U-235 or uranium-235).

Figure 2 Nomenclature for Identifying Nuclides

Another common format is to use the abbreviation of the chemical element with the mass number superscripted (for example, 235U). In this handbook the format used in the text will usually be the element's name followed by the mass number. In equations and tables, the format in Figure 2 will usually be used.

Example:

State the name of the element and the number of protons, electrons, and neutrons in the nuclides listed below.

1
1H
10
5B
14
7N
11
4
4

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23
9
9
4Pu
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ATOMIC NATURE OF MATTER DOE-HDBK-1019/1-93 Atomic and Nuclear Physics
Solution:
The name of the element can be found from the Periodic Table (refer to Chemistry Fundamentals Handbook) or the Chart of the Nuclides (to be discussed later). The number of protons and electrons are equal to Z. The number of neutrons is equal to Z - A.
Nuclide Element Protons Electrons Neutrons
1
1H hydrogen 1 1 0
10
5B boron 5 5 5
14
7N nitrogen 7 7 7
11
4
Fixture checkpoint marker-0761-xray is recorded on fixture page 0761.
4
8Cd cadmium 48 48 66
23
9
9
4Pu plutonium 94 94 145
Isotopes
Isotopes are nuclides that have the same atomic number and are therefore the same element, but differ in the number of neutrons. Most elements have a few stable isotopes and several unstable, radioactive isotopes. For example, oxygen has three stable isotopes that can be found in nature (oxygen-16, oxygen-17, and oxygen-18) and eight radioactive isotopes. Another example is hydrogen, which has two stable isotopes (hydrogen-1 and hydrogen-2) and a single radioactive isotope (hydrogen-3).
The isotopes of hydrogen are unique in that they are each commonly referred to by a unique name instead of the common chemical element name. Hydrogen-1 is almost always referred to as hydrogen, but the term protium is infrequently used also. Hydrogen-2 is commonly called deuterium and symbolized 2
1D. Hydrogen-3 is commonly called tritium and symbolized 31T. This
text will normally use the symbology 2
1H and 31H for deuterium and tritium, respectively.
Atomic and Nuclear Radii
The size of an atom is difficult to define exactly due to the fact that the electron cloud, formed by the electrons moving in their various orbitals, does not have a distinct outer edge. A reasonable measure of atomic size is given by the average distance of the outermost electron from the nucleus. Except for a few of the lightest atoms, the average atomic radii are approximately the same for all atoms, about 2 x 10 -8 cm.
Like the atom the nucleus does not have a sharp outer boundary. Experiments have shown that

atom. The relationship between the atomic mass number and the radius of the nucleus is shown in the following equation.

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Atomic and Nuclear Physics DOE-HDBK-1019/1-93 ATOMIC NATURE OF MATTER

r = (1.25 x 10 -13 cm) A1/3

where:

r = radius of the nucleus (cm)

A = atomic mass number (dimensionless)

The values of the nuclear radii for some light, intermediate, and heavy nuclides are shown in Table 2.

TABLE 2

Calculated Values for Nuclear Radii

Nuclide Radius of Nucleus

1

1H 1.25 x 10 -13 cm

10

5B 2.69 x 10 -13 cm

5

2

6

6Fe 4.78 x 10 -13 cm

17

7

8

2Hf 7.01 x 10 -13 cm

23

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9

8

2U 7.74 x 10 -13 cm

25

9

2

8Cf 7.89 x 10 -13 cm

From the table, it is clear that the radius of a typical atom (e.g. 2 x 10 -8 cm) is more than 25,000 times larger than the radius of the largest nucleus.

Nuclear Forces

In the Bohr model of the atom, the nucleus consists of positively-charged protons and electrically■neutra as nucleons. One problem that the Bohr model of the atom presented was accounting for an attractive force to overcome the repulsive force between protons.

Two forces present in the nucleus are (1) electrostatic forces between charged particles and (2) gravitational forces between any two objects that have mass. It is possible to calculate the magnitude of the gravitational force and electrostatic force based upon principles from classical physics.

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Fg

G m1 m2

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2
Fe
K Q1 Q2
r
2
ATOMIC NATURE OF MATTER DOE-HDBK-1019/1-93 Atomic and Nuclear Physics
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Newton stated that the gravitational force between two bodies is directly proportional to the masses of the two bodies and inversely proportional to the square of the distance between the bodies. This relationship is shown in the equation below.
where:
F = gravitational force (newtons) g
m = mass of first body (kilograms) 1
m = mass of second body (kilograms) 2
G = gravitational constant (6.67 x 10 N-m /kg)
-11 2 2
r = distance between particles (meters)
The equation illustrates that the larger the masses of the objects or the smaller the distance between the objects, the greater the gravitational force. So even though the masses of nucleons are very small, the fact that the distance between nucleons is extremely short may make the gravitational force significant. It is necessary to calculate the value for the gravitational force and compare it to the value for other forces to determine the significance of the gravitational force in the nucleus. The gravitational force between two protons that are separated by a distance of 10 meters is about 10 newtons. -20 -24
Coulomb's Law can be used to calculate the force between two protons. The electrostatic force is directly proportional to the electrical charges of the two particles and inversely proportional to the square of the distance between the particles. Coulomb's Law is stated as the following equation.
where:
F = electrostatic force (newtons) e
K = electrostatic constant (9.0 x 10 N-m /C)
9 2 2
Q = charge of first particle (coulombs) 1
Q = charge of second particle (coulombs) 2
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r = distance between particles (meters)
Using this equation, the electrostatic force between two protons that are separated by a distance of 10 meters is about 10 newtons. Comparing this result with the calculation of the
-20 12
gravitational force (10 newtons) shows that the gravitational force is so small that it can be
-24
neglected.
Atomic and Nuclear Physics DOE-HDBK-1019/1-93 ATOMIC NATURE OF MATTER
If only the electrostatic and gravitational forces existed in the nucleus, then it would be impossible to have stable nuclei composed of protons and neutrons. The gravitational forces are much too small to hold the nucleons together compared to the electrostatic forces repelling the protons. Since stable atoms of neutrons and protons do exist, there must be another attractive force acting within the nucleus. This force is called the nuclear force.

between pairs of neutrons, pairs of protons, or a neutron and a proton. The nuclear force has a very short range; it acts only over distances approximately equal to the diameter of the nucleus (10 -13 cm). The attractive nuclear force between all nucleons drops off with distance much faster than the repulsive electrostatic force between protons.

TABLE 3

Forces Acting in the Nucleus

Force	Interaction Range
Gravitational	Very weak attractive force between all nucleons Relatively long
Electrostatic	Strong repulsive force between like charged particles (protons) Relatively long
Nuclear Force	Strong attractive force between all nucleons Extremely short

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In stable atoms, the attractive and repulsive forces in the nucleus balance. If the forces do not balance, the atom cannot be stable, and the nucleus will emit radiation in an attempt to achieve a more stable configuration.

Summary

The important information in this chapter is summarized on the following page.

Atomic Nature of Matter Summary

Atoms consist of three basic subatomic particles. These particles are the proton, the neutron, and the electron.

Protons are particles that have a positive charge, have about the same mass as a hydrogen atom, and exist in the nucleus of an atom.

Neutrons are particles that have no electrical charge, have about the same mass as a hydrogen atom, and exist in the nucleus of an atom.

Electrons are particles that have a negative charge, have a mass about eighteen hundred times smaller than the mass of a hydrogen atom, and exist in orbital shells around the nucleus of an atom.

The Bohr model of the atom consists of a dense nucleus of protons and neutrons (nucleons) surrounded by electrons traveling in discrete orbits at fixed distances from the nucleus.

Nuclides are atoms that contain a particular number of protons and neutrons.

Isotopes are nuclides that have the same atomic number and are therefore the same element, but differ in the number of neutrons.

The atomic number of an atom is the number of protons in the nucleus.

The mass number of an atom is the total number of nucleons (protons and neutrons) in the nucleus.

The notation $_{Z}^{A}X$ is used to identify a specific nuclide. "Z" represents the atomic number, which is equal to the number of protons. "A" represents the mass number, which is equal to the number of nucleons. "X" represents the chemical symbol of the element.

Number of protons = Z

Number of electrons = Z

Number of neutrons = A - Z

it. The electrostatic force is a relatively long-range, strong, repulsive force that acts between the positively charged protons. The nuclear force is a relatively short-range attractive force between all nucleons. The gravitational force the long range, relatively weak attraction between masses, is negligible compared to the other forces.

CHART OF THE NUCLIDES

The Chart of the Nuclides, like the Periodic Table, is a convenient format for presenting a large amount of scientific information in an organized manner.

EO 1.6 DEFINE the following terms:

a. Enriched uranium

b. Depleted uranium

Chart of the Nuclides

A tabulated chart called the Chart of the Nuclides lists the stable and unstable nuclides in addition to pertinent information about each one. Figure 3 shows a small portion of a typical chart. This chart plots a box for each individual nuclide, with the number of protons (Z) on the vertical axis

Fixture checkpoint marker-0765-xray is recorded on fixture page 0765.

and the number of neutrons (N = A - Z) on the horizontal axis.

The completely gray squares indicate stable isotopes. Those in white squares are artificially radioactive, meaning that they are produced by artificial techniques and do not occur naturally. By consulting a complete chart, other types of isotopes can be found, such as naturally occurring radioactive types (but none are found in the region of the chart that is illustrated in Figure 3). Located in the box on the far left of each horizontal row is general information about the element. The box contains the chemical symbol of the element in addition to the average atomic weight of the naturally occurring substance and the average thermal neutron absorption cross section, which will be discussed in a later module. The known isotopes (elements with the same atomic number Z but different mass number A) of each element are listed to the right.

Figure 3 Nuclide Chart for Atomic Numbers 1 to 6

Information for Stable Nuclides

For the stable isotopes, in addition to the symbol and the atomic mass number, the number percentage of each isotope in the naturally occurring element is listed, as well as the thermal neutron activation cross section and the mass in atomic mass units (amu). A typical block for a stable nuclide from the Chart of the Nuclides is shown in Figure 4.

Figure 4 Stable Nuclides

Information for Unstable Nuclides

For unstable isotopes the additional information includes the half life, the mode of decay (for example, β^-, α), the total disintegration energy in MeV (million electron volts), and the mass in amu when available. A typical block for an unstable nuclide from the Chart of the Nuclides is shown in Figure 5.

Figure 5 Unstable Nuclides

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Neutron - Proton Ratios

Figure 6 shows the distribution of the stable nuclides plotted on the same axes as the Chart of the Nuclides. As the mass numbers become higher, the ratio of neutrons to protons in the nucleus becomes larger. For helium-4 (2 protons and 2 neutrons) and oxygen-16 (8 protons and 8 neutrons) this ratio is unity. For indium-115 (49 protons and 66 neutrons) the ratio of neutrons to protons has increased to 1.35, and for uranium-238 (92 protons and 146 neutrons) the neutron■to-proton

Figure 6 Neutron - Proton Plot of the Stable Nuclides

If a heavy nucleus were to split into two fragments, each fragment would form a nucleus that would have approximately the same neutron-to-proton ratio as the heavy nucleus. This high neutron-to-proton ratio places the fragments below and to the right of the stability curve displayed by Figure 6. The instability caused by this excess of neutrons is generally rectified by successive beta emissions, each of which converts a neutron to a proton and moves the nucleus toward a more stable neutron-to-proton ratio.

■

Natural Abundance of Isotopes

The relative abundance of an isotope in nature compared to other isotopes of the same element is relatively constant. The Chart of the Nuclides presents the relative abundance of the naturally occurring isotopes of an element in units of atom percent. Atom percent is the percentage of the atoms of an element that are of a particular isotope. Atom percent is abbreviated as a/o.

For example, if a cup of water contains 8.23 x 10 atoms of oxygen, and the isotopic abundance 24 of oxygen-18 is 0.20%, then there are 1.65 x 10 atoms of oxygen-18 in the cup. 22

The atomic weight for an element is defined as the average atomic weight of the isotopes of the

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element. The atomic weight for an element can be calculated by summing the products of the isotopic abundance of the isotope with the atomic mass of the isotope.

Example:

Calculate the atomic weight for the element lithium. Lithium-6 has an atom percent abundance of 7.5% and an atomic mass of 6.015122 amu. Lithium-7 has an atomic abundance of 92.5% and an atomic mass of 7.016003 amu.

Solution:

The other common measurement of isotopic abundance is weight percent (w/o). Weight percent is the percent weight of an element that is a particular isotope. For example, if a sample of material contained 100 kg of uranium that was 28 w/o uranium-235, then 28 kg of uranium-235 was present in the sample.

Enriched and Depleted Uranium

Natural uranium mined from the earth contains the isotopes uranium-238, uranium-235 and uranium-234. The majority (99.2745%) of all the atoms in natural uranium are uranium-238. Most of the remaining atoms (0.72%) are uranium-235, and a slight trace (0.0055%) are uranium-234. Although all isotopes of uranium have similar chemical properties, each of the isotopes has significantly different nuclear properties. For reasons that will be discussed in later modules, the isotope uranium-235 is usually the desired material for use in reactors.

A vast amount of equipment and energy are expended in processes that separate the isotopes of uranium (and other elements). The details of these processes are beyond the scope of this module. These processes are called enrichment processes because they selectively increase the proportion of a particular isotope. The enrichment process typically starts with feed material

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CHART OF THE NUCLIDES DOE-HDBK-1019/1-93 Atomic and Nuclear Physics
In the case of uranium, the natural uranium ore is 0.72 a/o uranium-235. The desired outcome of the enrichment process is to produce enriched uranium. Enriched uranium is defined as uranium in which the isotope uranium-235 has a concentration greater than its natural value. The enrichment process will also result in the byproduct of depleted uranium. Depleted uranium is defined as uranium in which the isotope uranium-235 has a concentration less than its natural value. Although depleted uranium is referred to as a by-product of the enrichment process, it does have uses in the nuclear field and in commercial and defense industries.
Summary
The important information in this chapter is summarized below.
Chart of the Nuclides Summary
Enriched uranium is uranium in which the isotope uranium-235 has a concentration greater than its natural value of 0.7%.
Depleted uranium is uranium in which the isotope uranium-235 has a concentration less than its natural value of 0.7%.
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Atomic and Nuclear Physics DOE-HDBK-1019/1-93 MASS DEFECT AND BINDING ENERGY
MASS DEFECT AND BINDING ENERGY
The separate laws of Conservation of Mass and Conservation of Energy are not applied strictly on the nuclear level. It is possible to convert between mass and energy. Instead of two separate conservation laws, a single conservation law states that the sum of mass and energy is conserved. Mass does not magically appear and disappear at random. A decrease in mass will be accompanied by a corresponding increase in energy and vice versa.
EO 1.7 DEFINE the following terms:
a. Mass defect
b. Binding energy
EO 1.8 Given the atomic mass for a nuclide and the atomic masses of a neutron, proton, and electron, CALCULATE the mass defect and binding energy of the nuclide.
Mass Defect
Careful measurements have shown that the mass of a particular atom is always slightly less than
Fixture checkpoint marker-0767-xray is recorded on fixture page 0767.
the sum of the masses of the individual neutrons, protons, and electrons of which the atom consists. The difference between the mass of the atom and the sum of the masses of its parts is called the mass defect (Δm). The mass defect can be calculated using Equation (1-1). In calculating the mass defect it is important to use the full accuracy of mass measurements because the difference in mass is small compared to the mass of the atom. Rounding off the masses of atoms and particles to three or four significant digits prior to the calculation will result in a calculated mass defect of zero.
$\Delta m = [Z(mp + me) + (A-Z)mn] - matom$ (1-1)
where:
Δm = mass defect (amu)
mp = mass of a proton (1.007277 amu)
mn = mass of a neutron (1.008665 amu)
me = mass of an electron (0.000548597 amu)
matom = mass of nuclide A
ZX (amu)

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A = mass number (number of nucleons)
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MASS DEFECT AND BINDING ENERGY DOE-HDBK-1019/1-93 Atomic and Nuclear Physics
Example:
Calculate the mass defect for lithium-7. The mass of lithium-7 is 7.016003 amu.
Solution:
∆ m Z m
p
me A Z mn matom
∆ m 3 1.007826 amu 7 3 1.008665 amu 7.016003 amu
∆ m 0.0421335 amu
Binding Energy
The loss in mass, or mass defect, is due to the conversion of mass to binding energy when the nucleus is formed. Binding energy is defined as the amount of energy that must be supplied to a nucleus to completely separate its nuclear particles (nucleons). It can also be understood as the amount of energy that would be released if the nucleus was formed from the separate particles. Binding energy is the energy equivalent of the mass defect. Since the mass defect was converted to binding energy (BE) when the nucleus was formed, it is possible to calculate the binding energy using a conversion factor derived by the mass-energy relationship from Einstein's Theory of Relativity.
Einstein's famous equation relating mass and energy is E = mc2 where c is the velocity of light (c = 2.998 x 108 m/sec). The energy equivalent of 1 amu can be determined by inserting this quantity of mass into Einstein's equation and applying conversion factors.
E m c 2
1 amu
1.6606 x 10 27 kg
1 amu
2.998 x 108 m
sec
2
1 N
1
kg m
sec2
1 J
1 N m
1.4924 x 10 10 J 1 MeV
1.6022 x 10 13 J
931.5 MeV
Fixture checkpoint marker-0768-xray is recorded on fixture page 0768.
Conversion Factors:
1 amu = 1.6606 x 10 -27 kg
1 newton = 1 kg-m/sec2
1 joule = 1 newton-meter
1 MeV = 1.6022 x 10-13 joules
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Atomic and Nuclear Physics DOE-HDBK-1019/1-93 MASS DEFECT AND BINDING ENERGY
Since 1 amu is equivalent to 931.5 MeV of energy, the binding energy can be calculated using

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(1-2)
B.E. ∆m
931.5 MeV
1 amu
Example:
Calculate the mass defect and binding energy for uranium-235. One uranium-235 atom has a mass of 235.043924 amu.
Solution:
Step 1: Calculate the mass defect using Equation (1-1).
∆m Z m
p
me A Z mn matom
∆m 92 1.007826 amu 235 92 1.008665 amu 235.043924 amu
∆m 1.91517 amu
Step 2: Use the mass defect and Equation (1-2) to calculate the binding energy.
B.E. ∆m
Fixture checkpoint marker-0769-xray is recorded on fixture page 0769.
931.5 MeV
1 amu
1.91517 amu
931.5 MeV
1 amu
1784 MeV
Energy Levels of Atoms
The electrons that circle the nucleus move in fairly well-defined orbits. Some of these electrons are more tightly bound in the atom than others. For example, only 7.38 eV is required to remove the outermost electron from a lead atom, while 88,000 eV is required to remove the innermost electron. The process of removing an electron from an atom is called ionization, and the energy required to remove the electron is called the ionization energy.
In a neutral atom (number of electrons = Z) it is possible for the electrons to be in a variety of different orbits, each with a different energy level. The state of lowest energy is the one in which the atom is normally found and is called the ground state. When the atom possesses more energy than its ground state energy, it is said to be in an excited state.

An atom cannot stay in the excited state for an indefinite period of time. An excited atom will eventually transition to either a lower-energy excited state, or directly to its ground state, by emitting a discrete bundle of electromagnetic energy called an x-ray. The energy of the x-ray will be equal to the difference between the energy levels of the atom and will typically range from several eV to 100,000 eV in magnitude.
Energy Levels of the Nucleus
The nucleons in the nucleus of an atom, like the electrons that circle the nucleus, exist in shells that correspond to energy states. The energy shells of the nucleus are less defined and less understood than those of the electrons. There is a state of lowest energy (the ground state) and discrete possible excited states for a nucleus. Where the discrete energy states for the electrons of an atom are measured in eV or keV, the energy levels of the nucleus are considerably greater and typically measured in MeV.
A nucleus that is in the excited state will not remain at that energy level for an indefinite period.

their lowest energy configuration and in doing so emit a discrete bundle of electromagnetic radiation called a gamma ray (γ-ray). The only differences between x-rays and γ-rays are their energy levels and whether they are emitted from the electron shell or from the nucleus.

The ground state and the excited states of a nucleus can be depicted in a nuclear energy-level diagram. The nuclear energy-level diagram consists of a stack of horizontal bars, one bar for each of the excited states of the nucleus. The vertical distance between the bar representing an excited state and the bar representing the ground state is proportional to the energy level of the excited state with respect to the ground state. This difference in energy between the ground state and the excited state is called the excitation energy of the excited state. The ground state of a nuclide has zero excitation energy. The bars for the excited states are labeled with their respective energy levels. Figure 7 is the energy level diagram for nickel-60.

Figure 7 Energy Level Diagram - Nickel-60

Fixture checkpoint marker-0770-xray is recorded on fixture page 0770.

Summary

The important information in this chapter is summarized below.

Mass Defect and Binding Energy Summary

Mass defect is the difference between the mass of the atom and the sum of the masses of its constituent parts.

Binding energy is the amount of energy that must be supplied to a nucleus to completely separate its nuclear particles. Binding energy is the energy equivalent of the mass defect.

Mass defect can be calculated by using the equation below.

$\Delta m = [Z(mp + me) + (A-Z)mn] - matom$

Binding energy can be calculated by multiplying the mass defect by the factor of 931.5 MeV per amu.

MODES OF RADIOACTIVE DECAY

Most atoms found in nature are stable and do not emit particles or energy that change form over time. Some atoms, however, do not have stable nuclei. These atoms emit radiation in order to achieve a more stable configuration.

EO 2.1 DESCRIBE the following processes:

a. Alpha decay
b. Beta-minus decay
c. Beta-plus decay
d. Electron capture
e. Internal conversions
f. Isomeric transitions

EO 2.2 Given a Chart of the Nuclides, WRITE the radioactive decay

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EO 2.3 EXPLAIN why one or more gamma rays typically accompany particle emission.
EO 2.4 Given the stability curve on the Chart of the Nuclides, DETERMINE the type of radioactive decay that the nuclides in each region of the chart will typically undergo.
Stability of Nuclei
As mass numbers become larger, the ratio of neutrons to protons in the nucleus becomes larger for the stable nuclei. Non-stable nuclei may have an excess or deficiency of neutrons and undergo a transformation process known as beta (β) decay. Non-stable nuclei can also undergo a variety of other processes such as alpha (α) or neutron (n) decay. As a result of these decay processes, the final nucleus is in a more stable or more tightly bound configuration.
Natural Radioactivity
In 1896, the French physicist Becquerel discovered that crystals of a uranium salt emitted rays that were similar to x-rays in that they were highly penetrating, could affect a photographic plate, and induced electrical conductivity in gases. Becquerel's discovery was followed in 1898 by the identification of two other radioactive elements, polonium and radium, by Pierre and Marie Curie.
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Heavy elements, such as uranium or thorium, and their unstable decay chain elements emit radiation in their naturally occurring state. Uranium and thorium, present since their creation at the beginning of geological time, have an extremely slow rate of decay. All naturally occurring nuclides with atomic numbers greater than 82 are radioactive.
Nuclear Decay
Whenever a nucleus can attain a more stable (i.e., more tightly bound) configuration by emitting radiation, a spontaneous disintegration process known as radioactive decay or nuclear decay may occur. In practice, this "radiation" may be electromagnetic radiation, particles, or both.
Detailed studies of radioactive decay and nuclear reaction processes have led to the formulation of useful conservation principles. The four principles of most interest in this module are discussed below.
Fixture checkpoint marker-0771-xray is recorded on fixture page 0771.
Conservation of electric charge implies that charges are neither created nor destroyed. Single positive and negative charges may, however, neutralize each other. It is also possible for a neutral particle to produce one charge of each sign.
Conservation of mass number does not allow a net change in the number of nucleons. However, the conversion of a proton to a neutron and vice versa is allowed.
Conservation of mass and energy implies that the total of the kinetic energy and the energy equivalent of the mass in a system must be conserved in all decays and reactions. Mass can be converted to energy and energy can be converted to mass, but the sum of mass and energy must be constant.
Conservation of momentum is responsible for the distribution of the available kinetic energy among product nuclei, particles, and/or radiation. The total amount is the same before and after the reaction even though it may be distributed differently among entirely different nuclides and/or particles.
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Alpha Decay (■)

Alpha decay is the emission of alpha particles (helium nuclei) which may be represented as either He or ■. When an unstable nucleus ejects an alpha particle, the atomic number is reduced by 2 4 4 2 2

and the mass number decreased by 4. An example is uranium-234 which decays by the ejection of an alpha particle accompanied by the emission of a 0.068 MeV gamma.

The combined kinetic energy of the daughter nucleus (Thorium-230) and the ■ particle is designated as KE. The sum of the KE and the gamma energy is equal to the difference in mass between the original nucleus (Uranium-234) and the final particles (equivalent to the binding energy released, since ■m = BE). The alpha particle will carry off as much as 98% of the kinetic energy and, in most cases, can be considered to carry off all the kinetic energy.

Beta Decay (■)

Beta decay is the emission of electrons of nuclear rather than orbital origin. These particles are electrons that have been expelled by excited nuclei and may have a charge of either sign.

If both energy and momentum are to be conserved, a third type of particle, the neutrino, ■, must be involved. The neutrino is associated with positive electron emission, and its antiparticle, the antineutrino, , is emitted with a negative electron. These uncharged particles have only the weakest interaction with matter, no mass, and travel at the speed of light. For all practical purposes, they pass through all materials with so few interactions that the energy they possess cannot be recovered. The neutrinos and antineutrinos are included here only because they carry a portion of the kinetic energy that would otherwise belong to the beta particle, and therefore, must be considered for energy and momentum to be conserved. They are normally ignored since they are not significant in the context of nuclear reactor applications.

Negative electron emission, represented as e, -1 -1■, or simply as e or ■ , effectively converts a neutro to a proton, thus increasing the atomic number by one and leaving the mass number unchanged. This is a common mode of decay for nuclei with an excess of neutrons, such as fission fragments below and to the right of the neutron-proton stability curve (refer to Figure 6). An example of a typical beta minus-decay reaction is shown below.

■ ■

■

■ ■

■ ■ ■ ■

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Figure 9 Orbital Electron Capture

Positively charged electrons (beta-plus) are known as positrons. Except for sign, they are nearly identical to their negatively charged cousins. When a positron, represented as e, ■, or simply as +1 +1

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e or ■ , is ejected from the nucleus, the atomic number is decreased by one and the mass number + + remains unchanged. A proton has been converted to a neutron. An example of a typical positron (beta-plus) decay is shown below.

Electron Capture (EC, K-capture)

Nuclei having an excess of protons may capture an electron from one of the inner orbits which immediately combines with a proton in the nucleus to form a neutron. This process is called electron capture (EC). The electron is normally captured from the innermost orbit (the K-shell), and, consequently, this process is sometimes called K-capture. The following example depicts electron capture.

time that the neutron is formed, and
energy carried off by it serves to
conserve momentum. Any energy
that is available due to the atomic
mass of the product being
appreciably less than that of the
parent will appear as gamma
radiation. Also, there will always
be characteristic x-rays given off
when an electron from one of the
higher energy shells moves in to fill
the vacancy in the K-shell.
Electron capture is shown
graphically in Figure 8.
Fixture checkpoint marker-0773-xray is recorded on fixture page 0773.
Electron capture and positron
emission result in the production of
the same daughter product, and
they exist as competing processes.
For positron emission to occur, however, the mass of the daughter product must be less than the mass of the parent by an amount equal to at least twice the mass of an electron. This mass difference between the parent and daughter is necessary to account for two items present in the parent but not in the daughter. One item is the positron ejected from the nucleus of the parent. The other item is that the daughter product has one less orbital electron than the parent. If this requirement is not met, then orbital electron capture takes place exclusively.
■
■ ■ ■ ■
■ ■
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Gamma Emission (■)
Gamma radiation is a high-energy electromagnetic radiation that originates in the nucleus. It is emitted in the form of photons, discrete bundles of energy that have both wave and particle properties. Often a daughter nuclide is left in an excited state after a radioactive parent nucleus undergoes a transformation by alpha decay, beta decay, or electron capture. The nucleus will drop to the ground state by the emission of gamma radiation.
Internal Conversion
The usual method for an excited nucleus to go from the excited state to the ground state is by emission of gamma radiation. However, in some cases the gamma ray (photon) emerges from the nucleus only to interact with one of the innermost orbital electrons and, as a result, the energy of the photon is transferred to the electron. The gamma ray is then said to have undergone internal conversion. The conversion electron is ejected from the atom with kinetic energy equal to the gamma energy minus the binding energy of the orbital electron. An orbital electron then drops to a lower energy state to fill the vacancy, and this is accompanied by the emission of characteristic x-rays.
Isomers and Isomeric Transition
Isomeric transition commonly occurs immediately after particle emission; however, the nucleus may remain in an excited state for a measurable period of time before dropping to the ground state at its

because it differs in energy and behavior from other nuclei with the same atomic number and mass number. The decay of an excited nuclear isomer to a lower energy level is called an isomeric transition. It is also possible for the excited isomer to decay by some alternate means, for example, by beta emission.

An example of gamma emission accompanying particle emission is illustrated by the decay of nitrogen-16 below.

■
■
■
■
■
■
■
■
■
■
■

Decay Chains

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When an unstable nucleus decays, the resulting daughter nucleus is not necessarily stable. The nucleus resulting from the decay of a parent is often itself unstable, and will undergo an additional decay. This is especially common among the larger nuclides.

It is possible to trace the steps of an unstable atom as it goes through multiple decays trying to achieve stability. The list of the original unstable nuclide, the nuclides that are involved as intermediate steps in the decay, and the final stable nuclide is known as the decay chain. One common method for stating the decay chain is to state each of the nuclides involved in the standard X format. Arrows are used between nuclides to indicate where decays occur, with the type of decay A Z
indicated above the arrow and the half-life below the arrow. The half-life for decay will be discussed in the next chapter.

Example:

Write the decay chains for rubidium-91 and actinium-215. Continue the chains until a stable nuclide or a nuclide with a half-life greater than 1 x 10 years is reached. 6

Solution:

Predicting Type of Decay

Radioactive nuclides tend to decay in a way that results in a daughter nuclide that lies closer to the line of stability. Due to this, it is possible to predict the type of decay that a nuclide will undergo based on its location relative to the line of stability on the Chart of the Nuclides.

Figure 9 illustrates the type of decay nuclides in different regions of the chart will typically undergo. Nuclides that are below and to the right of the line of stability will usually undergo ■ decay. - Nuclides that are above and to the left of the line of stability will usually undergo either ■ decay or + electron capture. Most nuclides that will undergo ■ decay are found in the upper right hand region of the chart. These are general rules that have many exceptions, especially in the region of the heavy nuclides.

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Summary
The important information in this chapter is summarized below.
Modes of Radioactive Decay Summary
Alpha decay is the emission of an alpha particle (2 protons and 2 neutrons) from an unstable nucleus. The daughter nuclide has an atomic number 2 less than the parent nuclide and a mass number 4 less than the parent nuclide. The daughter nucleus commonly releases its excitation energy by gamma emission.
Beta-minus decay effectively converts a neutron to a proton and an electron, which is immediately ejected from the nucleus. The daughter nuclide has its atomic number increased by 1 and the same mass number compared to the parent.
Beta-plus decay effectively converts a proton to a neutron and a positron, which is immediately ejected from the nucleus. The daughter nuclide has its atomic number decreased by 1 and the same mass number compared to the parent.
In electron capture, the nucleus absorbs an electron from the innermost orbit. This electron combines with a proton to form a neutron.
Internal conversion occurs when a gamma ray, emitted by the nucleus as it goes from the excited state to the ground state, interacts with one of the innermost electrons of the same atom. The electron is ejected from the atom.
An isomeric transition is the decay of an excited nucleus to a lower-energy level by the emission of a gamma ray.
Decay chains can be found by tracing the steps an unstable atom goes through as it tries to achieve stability.
Many modes of radioactive decay result in a daughter nuclide that has an energy level above the ground state. This excitation energy is usually released immediately in the form of a gamma ray.
Fixture checkpoint marker-0775-xray is recorded on fixture page 0775.
The type of decay that a nuclide will typically undergo can be determined by its relationship to the line of stability on the Chart of the Nuclides. Nuclides that lie below and to the right of the line of stability will typically beta minus decay. Nuclides above and to the left of the line will typically either beta plus decay or electron capture. Most alpha emitters are found in the upper, right-hand corner of the chart.

RADIOACTIVITY
The rate at which a sample of radioactive material decays is not constant. As individual atoms of the material decay, there are fewer of those types of atoms remaining. Since the rate of decay is directly proportional to the number of atoms, the rate of decay will decrease as the number of atoms decreases.
EO 2.5 DEFINE the following terms:
a. Radioactivity d. Radioactive decay constant
b. Curie e. Radioactive half-life
c. Becquerel
EO 2.6 Given the number of atoms and either the half-life or decay constant of a nuclide, CALCULATE the activity.

CALCULATE the activity at any later time.
EO 2.8 CONVERT between the half-life and decay constant for a nuclide.
EO 2.9 Given the Chart of the Nuclides and the original activity, PLOT the radioactive decay curve for a nuclide on either linear or semi-log coordinates.
EO 2.10 DEFINE the following terms:
a. Radioactive equilibrium
b. Transient radioactive equilibrium
Radioactive Decay Rates
Radioactivity is the property of certain nuclides of spontaneously emitting particles or gamma radiation. The decay of radioactive nuclides occurs in a random manner, and the precise time at which a single nucleus will decay cannot be determined. However, the average behavior of a very large sample can be predicted accurately by using statistical methods. These studies have revealed that there is a certain probability that in a given time interval a certain fraction of the nuclei within a sample of a particular nuclide will decay. This probability per unit time that an atom of a nuclide will decay is known as the radioactive decay constant, λ. The units for the decay constant are inverse time such as 1/second, 1/minute, 1/hour, or 1/year. These decay constant units can also be expressed as second-1, minute-1, hour-1, and year-1.

■■

The activity (A) of a sample is the rate of decay of that sample. This rate of decay is usually measured in the number of disintegrations that occur per second. For a sample containing millions of atoms, the activity is the product of the decay constant and the number of atoms present in the sample.
The relationship between the activity, number of atoms, and decay constant is shown in Equation (1-3).
A = ■ N (1-3)
where:
A = Activity of the nuclide (disintegrations/second)
■ = decay constant of the nuclide (second)-1
N = Number of atoms of the nuclide in the sample
Since ■ is a constant, the activity and the number of atoms are always proportional.
Fixture checkpoint marker-0776-xray is recorded on fixture page 0776.
Units of Measurement for Radioactivity
Two common units to measure the activity of a substance are the curie (Ci) and becquerel (Bq). A curie is a unit of measure of the rate of radioactive decay equal to 3.7 x 10 disintegrations 10 per second. This is approximately equivalent to the number of disintegrations that one gram of radium-226 will undergo in one second. A becquerel is a more fundamental unit of measure of radioactive decay that is equal to 1 disintegration per second. Currently, the curie is more widely used in the United States, but usage of the becquerel can be expected to broaden as the metric system slowly comes into wider use. The conversion between curies and becquerels is shown below.
1 curie = 3.7 x 10 becquerels 10
Variation of Radioactivity Over Time
The rate at which a given radionuclide sample decays is stated in Equation (1-3) as being equal

is possible to use calculus to derive an expression which can be used to calculate how the number of atoms present will change over time. The derivation is beyond the scope of this text, but Equation (1-4) is the useful result.

(1-4)

where:

N = number of atoms present at time t

N = number of atoms initially present o

■ = decay constant (time)-1

t = time

A Ao

e

t

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A Aoe

t

A

Ao

e

t

ln A

Ao

t

t

ln

A

Ao

t

1/2

ln

1

2

t

1/2

ln 2 0.693

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Since the activity and the number of atoms are always proportional, they may be used interchangeably to describe any given radionuclide population. Therefore, the following is true.

(1-5)

where:

A = activity present at time t

A = activity initially present o

= decay constant (time)

-1

t = time

Radioactive Half-Life

One of the most useful terms for estimating how quickly a nuclide will decay is the radioactive half-life. The radioactive half-life is defined as the amount of time required for the activity to

can be developed from Equation (1-5). The half-life can be calculated by solving Equation (1-5) for the time, t, when the current activity, A, equals one-half the initial activity A .
o
First, solve Equation (1-5) for t.
If A is equal to one-half of Ao, then A/Ao is equal to one-half. Substituting this in the equation above yields an expression for t .
1/2
(1-6)
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The basic features of decay of a radionuclide sample are shown by the graph in Figure 10.
Figure 10 Radioactive Decay as a Function of Time in Units of Half-Life
Assuming an initial number of atoms No, the population, and consequently, the activity may be noted to decrease by one-half of this value in a time of one half-life. Additional decreases occur so that whenever one half-life elapses, the number of atoms drops to one-half of what its value was at the beginning of that time interval. After five half-lives have elapsed, only 1/32, or 3.1%, of the original number of atoms remains. After seven half-lives, only 1/128, or 0.78%, of the atoms remains. The number of atoms existing after 5 to 7 half-lives can usually be assumed to be negligible. The Chemistry Fundamentals Handbook contains additional information on calculating the number of atoms contained within a sample.
Fixture checkpoint marker-0778-xray is recorded on fixture page 0778.
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■
■
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Example:
A sample of material contains 20 micrograms of californium-252.
Californium-252 has a half-life of 2.638 years.
Calculate:
(a) The number of californium-252 atoms initially present
(b) The activity of the californium-252 in curies
(c) The number of californium-252 atoms that will remain in 12 years
(d) The time it will take for the activity to reach 0.001 curies
Solution:
(a) The number of atoms of californium-252 can be determined as below.
(b) First, use Equation (1-6) to calculate the decay constant.
Use this value for the decay constant in Equation (1-3) to determine the activity.
N No
e
t
4.78 x 1016 e
(0.263/yr) (12 yr)
2.04 x 1015
A Aoe
t
A
Ao
e

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ln A
Ao
t
t
ln
A
Ao
t
ln
0.001 Ci
0.0108 Ci
0.263 year 1
t 9.05 years
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(c) The number of californium atoms that will remain in 12 years can be calculated from Equation (1-4).
(d) The time that it will take for the activity to reach 0.001 Ci can be determined from Equation (1-5). First, solve Equation (1-5) for time.
Inserting the appropriate values in the right side of this equation will result in the required time.
Plotting Radioactive Decay
It is useful to plot the activity of a nuclide as it changes over time. Plots of this type can be used to determine when the activity will fall below a certain level. This plot is usually done showing activity on either a linear or a logarithmic scale. The decay of the activity of a single nuclide on a logarithmic scale will plot as a straight line because the decay is exponential.
Fixture checkpoint marker-0779-xray is recorded on fixture page 0779.
t
1/2
0.693
0.693
t
1/2
0.693
7.13 seconds
0.0972 second 1
A Aoe
t
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Example:
Plot the radioactive decay curve for nitrogen-16 over a period of 100 seconds. The initial activity is 142 curies and the half-life of nitrogen-16 is 7.13 seconds. Plot the curve on both linear rectangular coordinates and on a semi-log scale.
Solution:
First, use Equation (1-6) to calculate the decay constant corresponding to a half-life of 7.13 seconds.
Use the decay constant determined above to calculate the activity at various times using

Time Activity
0 seconds 142 Ci
20 seconds 20.3 Ci
40 seconds 2.91 Ci
60 seconds 0.416 Ci
80 seconds 0.0596 Ci
100 seconds 0.00853 Ci

■ ■
■ ■
■ ■

Figure 11 Linear and Semi-log Plots of Nitrogen-16 Decay

Plotting the data points calculated above on both linear and semilog scales results in the graphs shown in Figure 11.

If a substance contains more than one radioactive nuclide, the total activity is the sum of the individual activities of each nuclide. As an example, consider a sample of material that contained 1 x 10 atoms of iron-59 that has a half-life of 44.51 days (■ = 1.80 x 10 sec), 6 -7 -1 1 x 10 atoms of manganese-54 that has a half-life of 312.2 days (■ = 2.57 x 10 sec), and 1 6 -8 -1 x 10 atoms of cobalt-60 that has a half-life of 1925 days (■ = 4.17 x 10 sec). 6 -9 -1

The initial activity of each of the nuclides would be the product of the number of atoms and the decay constant.

Figure 12 Combined Decay of Iron-56, Manganese-54, and Cobalt-60

Plotting the manner in which the activities of each of the three nuclides decay over time demonstrates that initially the activity of the shortest-lived nuclide (iron-59) dominates the total activity, then manganese-54 dominates. After almost all of the iron and manganese have decayed away, the only contributor to activity will be the cobalt-60. A plot of this combined decay is shown in Figure 12.

Radioactive Equilibrium

Radioactive equilibrium exists when a radioactive nuclide is decaying at the same rate at which it is being produced. Since the production rate and decay rate are equal, the number of atoms

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present remains constant over time.

An example of radioactive equilibrium is the concentration of sodium-24 in the coolant circulating through a sodium-cooled nuclear reactor. Assume that the sodium-24 is being produced at a rate of 1 x 10 atoms per second. If the sodium-24 were stable and did not decay, 6 the amount of sodium-24 present after some period of time could be calculated by multiplying the production rate by the amount of time. Plotting the amount of material present would result in the graph in Figure 13.

However, sodium-24 is not stable, and it decays with a half-life of 14.96 hours. If no sodium-24 is present initially and production starts at a rate of 1 x 10 atoms per second, the rate 6 of decay will initially be zero because there is no sodium-24 present to decay. The rate of decay of sodium-24 will increase as the amount of sodium-24 increases.

■
■
■

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Figure 13 Cumulative Production of Sodium-24 Over Time
The amount of sodium-24 present will initially increase rapidly, then it will increase at a continually decreasing rate until the rate of decay is equal to the rate of production. It is possible to calculate how much sodium-24 will be present at equilibrium by setting the production rate (R) equal to the decay rate (■ N).
where:
R = production rate (atoms/second)
■ = decay constant (second)-1
Fixture checkpoint marker-0781-xray is recorded on fixture page 0781.
N = number of atoms
It is possible to calculate the equilibrium value for sodium-24 being produced at a rate of 1 x 10 atoms/second. 6
■ ■ ■■
■
■
■
■
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Figure 14 Approach of Sodium-24 to Equilibrium
The development of the equation to calculate how the amount of sodium-24 changes over time as it approaches the equilibrium value is beyond the scope of this handbook. However, the equation is presented below.
This equation can be used to calculate the values of the amount of sodium-24 present at different times. As the time increases, the exponential term approaches zero, and the number of atoms present will approach R/■. A plot of the approach of sodium-24 to equilibrium is shown in Figure 14.
Transient Radioactive Equilibrium
Transient radioactive equilibrium occurs when the parent nuclide and the daughter nuclide decay at essentially the same rate.
For transient equilibrium to occur, the parent must have a long half-life when compared to the daughter. An example of this type of compound decay process is barium-140, which decays by beta emission to lanthanum-140, which in turn decays by beta emission to stable cerium-140.
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Figure 15 Transient Equilibrium in the Decay of Barium-140
The decay constant for barium-140 is considerably smaller than the decay constant for lanthanum-140. Remember that the rate of decay of both the parent and daughter can be represented as ■N. Although the decay constant for barium-140 is smaller, the actual rate of decay (■N) is initially larger than that of lanthanum-140 because of the great difference in their initial concentrations. As the concentration of the daughter increases, the rate of decay of the daughter will approach and eventually match the decay rate of the parent. When this occurs, they are said to be in transient equilibrium. A plot of the barium-lanthanum-cerium decay chain reaching transient equilibrium is shown in Figure 15.
Secular equilibrium occurs when the parent has an extremely long half-life. In the long decay chain for a naturally radioactive element, such as thorium-232, where all of the elements in the

and all decay at the rate set by the original parent. The only exception is the final stable element on the end of the chain. Its number of atoms is constantly increasing.

Summary

The important information in this chapter is summarized on the following page.

■

■■

■

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Radioactivity Summary

& Radioactivity is the decay of unstable atoms by the emission of particles and electromagnetic radiation.

& A curie (Ci) is a unit of radioactivity equal to 3.7 x 10 disintegrations per 10 second.

& A becquerel (Bq) is a unit of radioactivity equal to 1 disintegration per second.

& The radioactive decay constant (■) is the probability per unit time that an atom will decay.

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& The radioactive half-life is the amount of time required for the activity to decrease to one-half its original value.

& The activity of a substance can be calculated from the number of atoms and the decay constant based on the equation below.

& The amount of activity remaining after a particular time can be calculated from the equation below.

& The relationship between the decay constant and the half-life is shown below.

& Plots of radioactive decay can be useful to describe the variation of activity over time. If decay is plotted using semi-log scale the plot results in a straight line.

& Radioactive equilibrium exists when the production rate of a material is equal to the removal rate.

& Transient radioactive equilibrium exists when the parent nuclide and the daughter nuclide decay at essentially the same rate. This occurs only when the parent has a long half-life compared to the daughter.

Atomic and Nuclear Physics DOE-HDBK-1019/1-93 NEUTRON INTERACTIONS

NEUTRON INTERACTIONS

Neutrons can cause many different types of interactions. The neutron may simply scatter off the nucleus in two different ways, or it may actually be absorbed into the nucleus. If a neutron is absorbed into the nucleus, it may result in the emission of a gamma ray or a subatomic particle, or it may cause the nucleus to fission.

EO 3.1 DESCRIBE the following scattering interactions between a neutron and a nucleus:

a. Elastic scattering

b. Inelastic scattering

EO 3.2 STATE the conservation laws that apply to an elastic collision between a neutron and a nucleus.

EO 3.3 DESCRIBE the following reactions where a neutron is absorbed in a nucleus:

a. Radiative capture

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Scattering
A neutron scattering reaction occurs when a nucleus, after having been struck by a neutron, emits a single neutron. Despite the fact that the initial and final neutrons do not need to be (and often are not) the same, the net effect of the reaction is as if the projectile neutron had merely "bounced off," or scattered from, the nucleus. The two categories of scattering reactions, elastic and inelastic scattering, are described in the following paragraphs.
Elastic Scattering
In an elastic scattering reaction between a neutron and a target nucleus, there is no energy transferred into nuclear excitation. Momentum and kinetic energy of the "system" are conserved although there is usually some transfer of kinetic energy from the neutron to the target nucleus. The target nucleus gains the amount of kinetic energy that the neutron loses.
Rev. 0 Page 43 NP-01
mn
vn,i mTvT,i mnvn,f mTvT,f
1
2
m v
2
1
2
mn
v
2
n,i
Fixture checkpoint marker-0783-xray is recorded on fixture page 0783.
1
2
mT
v
2
T,i
1
2
mn
v
2
n,f
1
2
mT
v
2
T,f
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Figure 16 Elastic Scattering
Figure 16 illustrates the process of elastic scattering of a neutron off a target nucleus. In the elastic scattering reaction, the conservation of momentum and kinetic energy is represented by

Conservation of momentum (mv)
Conservation of kinetic energy
where:
mn = mass of the neutron
m = mass of the target nucleus T
vn,i = initial neutron velocity
vn,f = final neutron velocity
vT,i = initial target velocity
vT,f = final target velocity
Atomic and Nuclear Physics DOE-HDBK-1019/1-93 NEUTRON INTERACTIONS
Elastic scattering of neutrons by nuclei can occur in two ways. The more unusual of the two interactions is the absorption of the neutron, forming a compound nucleus, followed by the re-emission of a neutron in such a way that the total kinetic energy is conserved and the nucleus returns to its ground state. This is known as resonance elastic scattering and is very dependent upon the initial kinetic energy possessed by the neutron. Due to formation of the compound nucleus, it is also referred to as compound elastic scattering. The second, more usual method, is termed potential elastic scattering and can be understood by visualizing the neutrons and nuclei to be much like billiard balls with impenetrable surfaces. Potential scattering takes place with incident neutrons that have an energy of up to about 1 MeV. In potential scattering, the neutron does not actually touch the nucleus and a compound nucleus is not formed. Instead, the neutron is acted on and scattered by the short range nuclear forces when it approaches close enough to the nucleus.
Inelastic Scattering
In inelastic scattering, the incident neutron is absorbed by the target nucleus, forming a compound nucleus. The compound nucleus will then emit a neutron of lower kinetic energy which leaves the original nucleus in an excited state. The nucleus will usually, by one or more gamma emissions, emit this excess energy to reach its ground state. Figure 17 shows the process of inelastic scattering.
For the nucleus that has reached its ground state, the sum of the kinetic energy of the exit
Figure 17 Inelastic Scattering
neutron, the target nucleus, and the total gamma energy emitted is equal to the initial kinetic
Fixture checkpoint marker-0784-xray is recorded on fixture page 0784.
energy of the incident neutron.

1
0
n
238
92
U
239
92
U
239
92
U
0
0

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0
n
10
5
B
11
5
B
7
3
Li
4
2
NEUTRON INTERACTIONS DOE-HDBK-1019/1-93 Atomic and Nuclear Physics
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Absorption Reactions
Most absorption reactions result in the loss of a neutron coupled with the production of a charged particle or gamma ray. When the product nucleus is radioactive, additional radiation is emitted at some later time. Radiative capture, particle ejection, and fission are all categorized as absorption reactions and are briefly described below.
Radiative Capture
In radiative capture the incident neutron enters the target nucleus forming a compound nucleus. The compound nucleus then decays to its ground state by gamma emission. An example of a radiative capture reaction is shown below.
Particle Ejection
In a particle ejection reaction the incident particle enters the target nucleus forming a compound nucleus. The newly formed compound nucleus has been excited to a high enough energy level to cause it to eject a new particle while the incident neutron remains in the nucleus. After the new particle is ejected, the remaining nucleus may or may not exist in an excited state depending upon the mass-energy balance of the reaction. An example of a particle ejection reaction is shown below.
Fission
One of the most important interactions that neutrons can cause is fission, in which the nucleus that absorbs the neutron actually splits into two similarly sized parts. Fission will be discussed in detail in the next chapter.
Atomic and Nuclear Physics DOE-HDBK-1019/1-93 NEUTRON INTERACTIONS
Summary
The important information in this chapter is summarized below.
Fixture checkpoint marker-0785-xray is recorded on fixture page 0785.
Neutron Interactions Summary
Interactions where a neutron scatters off a target nucleus are either elastic or inelastic. In elastic scattering, kinetic energy and momentum are conserved and no energy is transferred into excitation energy of the target nucleus. In inelastic scattering, some amount of kinetic energy is transferred into excitation energy of the target nucleus.
The conservation principles that apply to an elastic collision are conservation of kinetic energy and conservation of momentum.
Radiative capture is the absorption of a neutron by the target nucleus, resulting

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second) releases its excitation energy in the form of a gamma ray.
Particle ejection occurs when a neutron is absorbed by a target nucleus, resulting in the formation of a compound nucleus. The compound nucleus immediately ejects a particle (for example, alpha or proton).
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1
0
n
235
92
U
236
92
U
140
Fixture checkpoint marker-0786-xray is recorded on fixture page 0786.
55
Cs
93
37
Rb 3
1
0
n
NUCLEAR FISSION DOE-HDBK-1019/1-93 Atomic and Nuclear Physics
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NUCLEAR FISSION
Nuclear fission is a process in which an atom splits and releases energy, fission products, and neutrons. The neutrons released by fission can, in turn, cause the fission of other atoms.
EO 4.1 EXPLAIN the fission process using the liquid drop model of a nucleus.
EO 4.2 DEFINE the following terms:
a. Excitation energy (E) exc
b. Critical energy (E) crit
EO 4.3 DEFINE the following terms:
a. Fissile material
b. Fissionable material
c. Fertile material
EO 4.4 DESCRIBE the processes of transmutation, conversion, and breeding.
EO 4.5 DESCRIBE the curve of Binding Energy per Nucleon versus mass number and give a qualitative description of the reasons for its shape.
EO 4.6 EXPLAIN why only the heaviest nuclei are easily fissioned.
EO 4.7 EXPLAIN why uranium-235 fissions with thermal neutrons and uranium-238 fissions only with fast neutrons.
Fission

nucleus that is excited to such a high energy level (E > E) that the nucleus "splits" exc crit (fissions) into two large fragments plus some neutrons. An example of a typical fission reaction is shown below.

A large amount of energy is released in the form of radiation and fragment kinetic energy.

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Liquid Drop Model of a Nucleus

The nucleus is held together by the attractive nuclear force between nucleons, which was discussed in a previous chapter. The characteristics of the nuclear force are listed below.

(a) very short range, with essentially no effect beyond nuclear dimensions (10-13 cm)

(b) stronger than the repulsive electrostatic forces within the nucleus

(c) independent of nucleon pairing, in that the attractive forces between pairs of neutrons are no different than those between pairs of protons or a neutron and a proton

(d) saturable, that is, a nucleon can attract only a few of its nearest neighbors

One theory of fission considers the fissioning of a nucleus similar in some respects to the splitting of a liquid drop. This analogy is justifiable to some extent by the fact that a liquid drop is held together by molecular forces that tend to make the drop spherical in shape and that try to resist any deformation in the same manner as nuclear forces are assumed to hold the nucleus together. By considering the nucleus as a liquid drop, the fission process can be described.

Referring to Figure 18(A), the nucleus in the ground state is undistorted, and its attractive nuclear forces are greater than the repulsive electrostatic forces between the protons within the nucleus.

Fixture checkpoint marker-0787-xray is recorded on fixture page 0787.

When an incident particle (in this instance a neutron) is absorbed by the target nucleus, a compound nucleus is formed. The compound nucleus temporarily contains all the charge and mass involved in the reaction and exists in an excited state. The excitation energy added to the compound nucleus is equal to the binding energy contributed by the incident particle plus the kinetic energy possessed by that particle. Figure 18(B) illustrates the excitation energy thus imparted to the compound nucleus, which may cause it to oscillate and become distorted. If the excitation energy is greater than a certain critical energy, the oscillations may cause the compound nucleus to become dumbbell-shaped. When this happens, the attractive nuclear forces (short-range) in the neck area are small due to saturation, while the repulsive electrostatic forces (long-range) are only slightly less than before. When the repulsive electrostatic forces exceed the attractive nuclear forces, nuclear fission occurs, as illustrated in Figure 18(C).

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Figure 18 Liquid Drop Model of Fission

Critical Energy

The measure of how far the energy level of a nucleus is above its ground state is called the excitation energy (Eexc). For fission to occur, the excitation energy must be above a particular value for that nuclide. The critical energy (Ecrit) is the minimum excitation energy required for fission to occur.

Fissile Material

A fissile material is composed of nuclides for which fission is possible with neutrons of any energy level. What is especially significant about these nuclides is their ability to be fissioned with zero kinetic energy neutrons (thermal neutrons). Thermal neutrons have very low kinetic energy levels (essentially zero) because they are roughly in equilibrium with the thermal motion of surrounding materials. Therefore, in order to be classified as fissile, a material must be

no kinetic energy to the reaction. Fission is possible in these materials with thermal neutrons, since the change in binding energy supplied by the neutron addition alone is high enough to exceed the critical energy. Some examples of fissile nuclides are uranium-235, uranium-233, and plutonium-239.

Fissionable Material

A fissionable material is composed of nuclides for which fission with neutrons is possible. All fissile nuclides fall into this category. However, also included are those nuclides that can be fissioned only with high energy neutrons. The change in binding energy that occurs as the result of neutron absorption results in a nuclear excitation energy level that is less than the required critical energy. Therefore, the additional excitation energy must be supplied by the kinetic energy of the incident neutron. The reason for this difference between fissile and fissionable materials is the so-called odd-even effect for nuclei. It has been observed that nuclei with even numbers of neutrons and/or protons are more stable than those with odd numbers. Therefore, adding a neutron to change a nucleus with an odd number of neutrons to a nucleus with an even number of neutrons produces an appreciably higher binding energy than adding a neutron to a nucleus already possessing an even number of neutrons. Some examples of nuclides requiring high energy neutrons to cause fission are thorium-232, uranium-238, and plutonium-240. Table 4 indicates the critical energy (Ecrit) and the binding energy change for an added neutron (BEn) to target nuclei of interest. For fission to be possible, the change in binding energy plus the kinetic energy must equal or exceed the critical energy (ΔBE + KE > Ecrit).

TABLE 4

Critical Energies Compared to Binding Energy of Last Neutron

Target
Nucleus
Critical Energy
Ecrit
Binding Energy of
Fixture checkpoint marker-0788-xray is recorded on fixture page 0788.
Last Neutron BEn
BEn - Ecrit
23
9
2
0Th 7.5 MeV 5.4 MeV -2.1 MeV
23
9
8
2U 7.0 MeV 5.5 MeV -1.5 MeV
23
9
5
2U 6.5 MeV 6.8 MeV +0.3 MeV
23
9
3
2U 6.0 MeV 7.0 MeV +1.0 MeV

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9
9
4Pu 5.0 MeV 6.6 MeV +1.6 MeV
Uranium-235 fissions with thermal neutrons because the binding energy released by the absorption of a neutron is greater than the critical energy for fission; therefore uranium-235 is a fissile material. The binding energy released by uranium-238 absorbing a thermal neutron is less than the critical energy, so additional energy must be possessed by the neutron for fission to be possible. Consequently, uranium-238 is a fissionable material.
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Fertile Material
All of the neutron absorption reactions that do not result in fission lead to the production of new nuclides through the process known as transmutation. These nuclides can, in turn, be transmuted again or may undergo radioactive decay to produce still different nuclides. The nuclides that are produced by this process are referred to as transmutation products. Because several of the fissile nuclides do not exist in nature, they can only be produced by nuclear reactions (transmutation). The target nuclei for such reactions are said to be fertile. Fertile materials are materials that can undergo transmutation to become fissile materials. Figure 19 traces the transmutation mechanism by which two fertile nuclides, thorium-232 and uranium-238, produce uranium-233 and plutonium-239, respectively.
Figure 19 Conversion of Fertile Nuclides to Fissile Nuclides
If a reactor contains fertile material in addition to its fissile fuel, some new fuel will be produced as the original fuel is burned up. This is called conversion. Reactors that are specifically designed to produce fissionable fuel are called "breeder" reactors. In such reactors, the amount of fissionable fuel produced is greater than the amount of fuel burnup. If less fuel is produced than used, the process is called conversion, and the reactor is termed a "converter."
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Binding Energy Per Nucleon (BE/A)
As the number of particles in a nucleus increases, the total binding energy also increases. The rate of increase, however, is not uniform. This lack of uniformity results in a variation in the amount of binding energy associated with each nucleon within the nucleus. This variation in the binding energy per nucleon (BE/A) is easily seen when the average BE/A is plotted versus atomic mass number (A), as shown in Figure 20.
Figure 20 Binding Energy per Nucleon vs. Mass Number
Figure 20 illustrates that as the atomic mass number increases, the binding energy per nucleon
Fixture checkpoint marker-0789-xray is recorded on fixture page 0789.
decreases for A > 60. The BE/A curve reaches a maximum value of 8.79 MeV at A = 56 and decreases to about 7.6 MeV for A = 238. The general shape of the BE/A curve can be explained using the general properties of nuclear forces. The nucleus is held together by very short-range attractive forces that exist between nucleons. On the other hand, the nucleus is being forced apart by long range repulsive electrostatic (coulomb) forces that exist between all the protons in the nucleus.
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As the atomic number and the atomic mass number increase, the repulsive electrostatic forces within the nucleus increase due to the greater number of protons in the heavy elements. To overcome this increased repulsion, the proportion of neutrons in the nucleus must increase to

growing proton-proton repulsive force in the heavier, naturally occurring elements. Because the repulsive forces are increasing, less energy must be supplied, on the average, to remove a nucleon from the nucleus. The BE/A has decreased. The BE/A of a nucleus is an indication of its degree of stability. Generally, the more stable nuclides have higher BE/A than the less stable ones. The increase in the BE/A as the atomic mass number decreases from 260 to 60 is the primary reason for the energy liberation in the fission process. In addition, the increase in the BE/A as the atomic mass number increases from 1 to 60 is the reason for the energy liberation in the fusion process, which is the opposite reaction of fission.

The heaviest nuclei require only a small distortion from a spherical shape (small energy addition) for the relatively large coulomb forces forcing the two halves of the nucleus apart to overcome the attractive nuclear forces holding the two halves together. Consequently, the heaviest nuclei are easily fissionable compared to lighter nuclei.

Summary

Fixture checkpoint marker-0790-xray is recorded on fixture page 0790.

The important information in this chapter is summarized on the following page.

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Atomic and Nuclear Physics DOE-HDBK-1019/1-93 NUCLEAR FISSION

Nuclear Fission Summary

The fission process can be explained using the liquid drop model of a nucleus. In the ground state the nucleus is nearly spherical in shape. After the absorption of a neutron, the nucleus will be in an excited state and start to oscillate and become distorted. If the oscillations cause the nucleus to become shaped like a dumbbell, the repulsive electrostatic forces will overcome the short-range attractive nuclear forces, and the nucleus will split in two.

Excitation energy is the amount of energy a nucleus has above its ground state. Critical energy is the minimum excitation energy that a nucleus must have before it can fission.

Fissile material is material for which fission is possible with neutrons that have zero kinetic energy. Fissionable material is material for which fission caused by neutron absorption is possible provided the kinetic energy added with the binding energy is greater than the critical energy. Fertile material is material that can undergo transmutation to become fissile material.

Transmutation is the process of neutron absorption and subsequent decay, which changes one nuclide to another nuclide. Conversion is the process of transmuting fertile material into fissile material in a reactor, where the amount of fissile material produced is less than the amount of fissile material consumed. Breeding is the same as conversion, except the amount of fissile material produced is more than the amount of fissile material consumed.

The curve of binding energy per nucleon increases quickly through the light nuclides and reaches a maximum at a mass number of about 56. The curve decreases slowly for mass numbers greater than 60.

The heaviest nuclei are easily fissionable because they require only a small distortion from the spherical shape to allow the coulomb forces to overcoming the attractive nuclear force, forcing the two halves of the nucleus apart.

Uranium-235 fissions with thermal neutrons because the binding energy released by the absorption of a neutron is greater than the critical energy for fission. The binding energy released by uranium-238 absorbing a neutron is less than the critical energy, so additional kinetic energy must be possessed by the neutron for

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1
0
n
235
92
U
236
92
U
140
55
Cs
93
37
Rb 3
1
0
n
ENERGY RELEASE FROM FISSION DOE-HDBK-1019/1-93 Atomic and Nuclear Physics
Fixture checkpoint marker-0791-xray is recorded on fixture page 0791.
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ENERGY RELEASE FROM FISSION
Fission of heavy nuclides converts a small amount of mass into an enormous amount of energy. The amount of energy released by fission can be determined based on either the change in mass that occurs during the reaction or by the difference in binding energy per nucleon between the fissile nuclide and the fission products.
EO 4.8 CHARACTERIZE the fission products in terms of mass groupings and radioactivity.
EO 4.9 Given the nuclides involved and their masses, CALCULATE the energy released from fission.
EO 4.10 Given the curve of Binding Energy per nucleon versus mass number, CALCULATE the energy released from fission.
Calculation of Fission Energy
Nuclear fission results in the release of enormous quantities of energy. It is necessary to be able to calculate the amount of energy that will be produced. The logical manner in which to pursue this is to first investigate a typical fission reaction such as the one listed below.
It can be seen that when the compound nucleus splits, it breaks into two fission fragments, rubidium-93, cesium-140, and some neutrons. Both fission products then decay by multiple - emissions as a result of the high neutron-to-proton ratio possessed by these nuclides.
In most cases, the resultant fission fragments have masses that vary widely. Figure 21 gives the percent yield for atomic mass numbers. The most probable pair of fission fragments for the thermal fission of the fuel uranium-235 have masses of about 95 and 140. Note that the vertical axis of the fission yield curve is on a logarithmic scale. Therefore, the formation of fission fragments of mass numbers of about 95 and 140 is highly likely.
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Figure 21 Uranium-235 Fission Yield vs.

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Referring now to the binding energy per nucleon curve (Figure 20), we can estimate the amount of energy released by our "typical" fission by plotting this reaction on the curve and calculating the change in binding energy (ΔBE) between the reactants on the left-hand side of the fission equation and the products on the right-hand side. Plotting the reactant and product nuclides on the curve shows that the total binding energy of the system after fission is greater than the total binding energy of the system before fission. When there is an increase in the total binding energy of a system, the system has become more stable by releasing an amount of energy equal to the increase in total binding energy of the system. Therefore, in the fission process, the energy liberated is equal to the increase in the total binding energy of the system.

Figure 22 graphically depicts that the binding energy per nucleon for the products (C,

Figure 22 Change in Binding Energy for Typical Fission

rubidium-93 and B, cesium-140) is greater than that for the reactant (A, uranium-235). The total binding energy for a nucleus can be found by multiplying the binding energy per nucleon by the number of nucleons.

TABLE 5

Binding Energies Calculated from
Binding Energy per Nucleon Curve

Nuclide	B.E. per Nucleon (BE/A)	Mass Number (A)	Binding Energy (BE/A) x (A)
9			
3			

Fixture checkpoint marker-0792-xray is recorded on fixture page 0792.

3			
7Rb	8.7 MeV	93	809 MeV
14			
5			
0			
5Cs	8.4 MeV	140	1176 MeV
23			
9			
5			
2U	7.6 MeV	235	1786 MeV

BE BEproducts BEreactants
BERb 93 BECs 140 BEU 235
809 MeV 1176 MeV 1786 MeV
199 MeV
1
0
n
235
92

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236
92
U
140
55
Cs
93
37
Rb 3
1
0
n
Mass difference Mass of Reactants Mass of Products
236.052589 amu 235.85208 amu
0.200509 amu
EInst 0.020059 amu
931.5 MeV
amu
186.8 MeV
Atomic and Nuclear Physics DOE-HDBK-1019/1-93 ENERGY RELEASE FROM FISSION
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The energy released will be equivalent to the difference in binding energy (BE) between the reactants and the products.
The energy liberation during the fission process can also be explained from the standpoint of the conservation of mass-energy. During the fission process, there is a decrease in the mass of the system. There must, therefore, be energy liberated equal to the energy equivalent of the mass lost in the process. This method is more accurate than the previously illustrated method and is used when actually calculating the energy liberated during the fission process.
Again, referring to the "typical" fission reaction.
E , the instantaneous energy, is the energy released immediately after the fission process. It Inst is equal to the energy equivalent of the mass lost in the fission process. It can be calculated as shown below.
Mass of the Reactants Mass of the Products
U 235.043924 amu Rb 92.91699 amu 235
Fixture checkpoint marker-0793-xray is recorded on fixture page 0793.
92
93
37
n 1.008665 amu Cs 139.90910 amu 1
0
140
55
3 (n) 3.02599 amu 1
0
236.052589 amu 235.85208 amu
This mass difference can be converted to an energy equivalent.
93
37

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93
38
Sr
93
39
Y
93
40
Zr
93
41
Fixture checkpoint marker-0794-xray is recorded on fixture page 0794.
Nb
140
55
Cs
140
56
Ba
140
57
La
140
58
Ce
EDecay mRb 93 mNb 93 4 melectron
931.5MeV
amu
92.91699 amu 92.90638 amu 4 0.0005486 amu
931.5 MeV
amu
0.008416 amu
931.5 MeV
amu
7.84 MeV
ENERGY RELEASE FROM FISSION DOE-HDBK-1019/1-93 Atomic and Nuclear Physics
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The total energy released per fission will vary from the fission to the next depending on what fission products are formed, but the average total energy released per fission of uranium-235 with a thermal neutron is 200 MeV.
As illustrated in the preceding example, the majority of the energy liberated in the fission process is released immediately after the fission occurs and appears as the kinetic energy of the fission fragments, kinetic energy of the fission neutrons, and instantaneous gamma rays. The remaining energy is released over a period of time after the fission occurs and appears as kinetic energy of the beta, neutrino, and decay gamma rays.
Estimation of Decay Energy
In addition to this instantaneous energy release during the actual fission reaction, there is additional energy released when the fission fragments decay by emission. This additional

energy is called decay energy, EDecay. The decay chains for rubidium-93 and cesium-140 are shown below.

The energy released during the decay for each chain will be equivalent to the mass difference between the original fission product and the sum of the final stable nuclide and the beta particles emitted.

The energy released in the decay chain of rubidium-93 is calculated below.

EDecay mRb 93 mNb 93 3 melectron
931.5MeV
amu
139.90910 amu 139.90543 amu 3 0.0005486 amu
931.5 MeV
amu
0.000202 amu
931.5 MeV
amu
1.89 MeV

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Fixture checkpoint marker-0795-xray is recorded on fixture page 0795.

The energy released in the decay chain of cesium-140 is calculated below.

The total decay energy is the sum of the energies of the two chains, or 9.73 MeV.

Distribution of Fission Energy

The average energy distribution for the energy released per fission with a thermal neutron in uranium-235 is shown in Tables 6 and 7.

TABLE 6

Instantaneous Energy from Fission
Kinetic Energy of Fission Products 167 Mev
Energy of Fission Neutrons 5 MeV
Instantaneous Gamma-ray Energy 5 MeV
Capture Gamma-ray Energy 10 MeV
Total Instantaneous Energy 187 MeV

TABLE 7

Delayed Energy from Fission
Beta Particles From Fission Products 7 Mev
Gamma-rays from Fission Products 6 MeV
Neutrinos 10 MeV
Total Delayed Energy 23 MeV

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Because the 10 MeV of neutrino energy shown in Table 7 is not absorbed in the reactor, the average value of 200 MeV per fission is still accurate. Note in Table 6 that some fission neutrons undergo radiative capture and the resultant gamma ray emission provides an additional 10 MeV of instantaneous energy, which contributes to the total of 187 MeV instantaneous energy.

All of the energy released, with the exception of the neutrino energy, is ultimately transformed into heat through a number of processes. The fission fragments, with their high positive charge and kinetic energy, cause ionization directly as they rip orbital electrons from the surrounding atoms. In this ionization process, kinetic energy is transferred to the surrounding atoms of the fuel material, resulting in an increase in temperature. The beta particles and gamma rays also

through elastic scattering. Of the 200 MeV released per fission, about seven percent (13 MeV) is released at some time after the instant of fission. When a reactor is shut down, fissions essentially cease, but energy is still being released from the decay of fission products. The heat produced by this decay energy is referred to as "decay heat." Although decay energy represents about seven percent of reactor heat production during reactor operation, once the reactor is shut down the decay heat production drops off quickly to a small fraction of its value while operating. The decay heat produced is significant, however, and systems must be provided to keep the reactor cool even after shutdown.

Summary

The important information in this chapter is summarized below.

Energy Release From Fission Summary

Fission products have some general characteristics in common.

They generally decay by β- emission.

The most common mass numbers are grouped near 95 and 140.

The energy released by fission can be calculated based on the difference in mass between the masses of the reactants before fission and the fission fragments and fission neutrons after fission.

Another method to determine the energy released by fission is based on the change in binding energy per nucleon between the fissile nuclide and the fission products.

INTERACTION OF RADIATION WITH MATTER

Different types of radiation interact with matter in widely different ways. A large,

Fixture checkpoint marker-0796-xray is recorded on fixture page 0796.

massive, charged alpha particle cannot penetrate a piece of paper and even has a limited range in dry air. A neutrino, at the other extreme, has a low probability of interacting with any matter, even if it passed through the diameter of the earth.

EO 5.1 DESCRIBE interactions of the following with matter:

a. Alpha particle c. Positron

b. Beta particle d. Neutron

EO 5.2 DESCRIBE the following ways that gamma radiation interacts with matter:

a. Photoelectric effect

b. Compton scattering

c. Pair production

Interaction of Radiation With Matter

Radiation can be classified into two general groups, charged and uncharged; therefore, it may be expected that interactions with matter fall into two general types. Charged particles directly ionize the media through which they pass, while uncharged particles and photons can cause ionization only indirectly or by secondary radiation.

A moving charged particle has an electrical field surrounding it, which interacts with the atomic structure of the medium through which it is passing. This interaction decelerates the particle and accelerates electrons in the atoms of the medium. The accelerated electrons may acquire enough energy to escape from the parent atom. This process, whereby radiation "strips" off orbital electrons, is called ionization. Uncharged moving particles have no electrical field, so they can

lose energy by the photoelectric effect, Compton effect, or pair production.
Rev. 0 Page 63 NP-01
I
m z
2
K.E.
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Because ionizing radiation creates ions in pairs, the intensity of ionization or the specific ionization is defined as the number of ion-pairs formed per centimeter of travel in a given material. The amount of ionization produced by a charged particle per unit path length, which is a measure of its ionizing power, is roughly proportional to the particle's mass and the square of its charge as illustrated in the equation below.
where:
I is the ionizing power
m is the mass of the particle
z is the number of unit charges it carries
K.E. is its kinetic energy
Since m for an alpha particle is about 7300 times as large as m for a beta particle, and z is twice as great, an alpha will produce much more ionization per unit path length than a beta particle of the same energy. This phenomenon occurs because the larger alpha particle moves slower for a given energy and thus acts on a given electron for a longer time.
Alpha Radiation
Alpha radiation is normally produced from the radioactive decay of heavy nuclides and from certain nuclear reactions. The alpha particle consists of 2 neutrons and 2 protons, so it is essentially the same as the nucleus of a helium atom. Because it has no electrons, the alpha particle has a charge of +2. This positive charge causes the alpha particle to strip electrons from the orbits of atoms in its vicinity. As the alpha particle passes through material, it removes electrons from the orbits of atoms it passes near. Energy is required to remove electrons and the energy of the alpha particle is reduced by each reaction. Eventually the particle will expend its kinetic energy, gain 2 electrons in orbit, and become a helium atom. Because of its strong
Fixture checkpoint marker-0797-xray is recorded on fixture page 0797.
positive charge and large mass, the alpha particle deposits a large amount of energy in a short distance of travel. This rapid, large deposition of energy limits the penetration of alpha particles. The most energetic alpha particles are stopped by a few centimeters of air or a sheet of paper.
Beta-Minus Radiation
A beta-minus particle is an electron that has been ejected at a high velocity from an unstable nucleus. An electron has a small mass and an electrical charge of -1. Beta particles cause ionization by displacing electrons from atom orbits. The ionization occurs from collisions with orbiting electrons. Each collision removes kinetic energy from the beta particle, causing it to slow down. Eventually the beta particle will be slowed enough to allow it to be captured as an
2 electrons 0.000549 amu
electron
931.5 MeV
amu
1.02 MeV

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orbiting electron in an atom. Although more penetrating than the alpha, the beta is relatively easy to stop and has a low power of penetration. Even the most energetic beta radiation can be stopped by a few millimeters of metal.

Positron Radiation

Positively charged electrons are called positrons. Except for the positive charge, they are identical to beta-minus particles and interact with matter in a similar manner. Positrons are very short-lived, however, and quickly are annihilated by interaction with a negatively charged electron, producing two gammas with a combined energy (calculated below) equal to the rest mass of the positive and negative electrons.

Neutron Radiation

Neutrons have no electrical charge. They have nearly the same mass as a proton (a hydrogen atom nucleus). A neutron has hundreds of times more mass than an electron, but 1/4 the mass of an alpha particle. The source of neutrons is primarily nuclear reactions, such as fission, but they may also be produced from the decay of radioactive nuclides. Because of its lack of charge, the neutron is difficult to stop and has a high penetrating power.

Neutrons are attenuated (reduced in energy and numbers) by three major interactions, elastic scatter, inelastic scatter, and absorption. In elastic scatter, a neutron collides with a nucleus and bounces off. This reaction transmits some of the kinetic energy of the neutron to the nucleus of the atom, resulting in the neutron being slowed, and the atom receives some kinetic energy (motion). This process is sometimes referred to as "the billiard ball effect."

As the mass of the nucleus approaches the mass of the neutron, this reaction becomes more effective in slowing the neutron. Hydrogenous material attenuates neutrons most effectively.

In the inelastic scatter reaction, the same neutron/nucleus collision occurs as in elastic scatter. However, in this reaction, the nucleus receives some internal energy as well as kinetic energy. This slows the neutron, but leaves the nucleus in an excited state. When the nucleus decays to its original energy level, it normally emits a gamma ray.

In the absorption reaction, the neutron is actually absorbed into the nucleus of an atom. The neutron is captured, but the atom is left in an excited state. If the nucleus emits one or more gamma rays to reach a stable level, the process is called radiative capture. This reaction occurs at most neutron energy levels, but is more probable at lower energy levels.

Gamma Radiation

Gamma radiation is electromagnetic radiation. It is commonly referred to as a gamma ray and is very similar to an x-ray. The difference is that gamma rays are emitted from the nucleus of an atom, and x-rays are produced by orbiting electrons. The x-ray is produced when orbiting electrons move to a lower energy orbit or when fast-moving electrons approaching an atom are

Fixture checkpoint marker-0798-xray is recorded on fixture page 0798.

deflected and decelerated as they react with the atom's electrical field (called Bremsstrahlung). The gamma ray is produced by the decay of excited nuclei and by nuclear reactions. Because the gamma ray has no mass and no charge, it is difficult to stop and has a very high penetrating power. A small fraction of the original gamma stream will pass through several feet of concrete or several meters of water.

There are three methods of attenuating gamma rays. The first method is referred to as the photo-electric effect. When a low energy gamma strikes an atom, the total energy of the gamma is expended in ejecting an electron from orbit. The result is ionization of the atom and expulsion

interacting in materials with high atomic weight and rarely occurs with gammas having an energy above 1 MeV. Annihilation of the gamma results. Any gamma energy in excess of the binding energy of the electron is carried off by the electron in the form of kinetic energy.

The second method of attenuation of gammas is called Compton scattering. The gamma interacts with an orbital or free electron; however, in this case, the photon loses only a fraction of its energy. The actual energy loss depending on the scattering angle of the gamma. The gamma continues on at lower energy, and the energy difference is absorbed by the electron. This reaction becomes important for gamma energies of about 0.1 MeV and higher.

At higher energy levels, a third method of attenuation is predominant. This method is pair-production. When a high energy gamma passes close enough to a heavy nucleus, the gamma completely disappears, and an electron and a positron are formed. For this reaction to take place, the original gamma must have at least 1.02 MeV energy. Any energy greater than 1.02 MeV becomes kinetic energy shared between the electron and positron. The probability of pair■production incre

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Interaction of Radiation with Matter Summary

An alpha particle deposits a large amount of energy in a short distance of travel due to its large mass and charge.

Beta-minus particles interact with the electrons orbiting the nucleus of atoms, causing ionization by displacing the electrons. The beta particle loses energy with each interaction. After the beta particle loses enough energy, it is captured in the orbital shells of an atom.

Positrons interact with matter much the same way as beta minus particles. After the positron has lost most of its energy by ionizing atoms, it is annihilated by interaction with an electron. The electron-positron pair disappear and are replaced by two gammas, each with the energy equivalent of the mass of an electron (0.51 MeV).

Neutrons interact with matter by elastic scattering, inelastic scattering, or absorption.

Photoelectric effect is where a gamma interacts with an electron orbiting an atom. The entire energy of the gamma is transferred to the electron, and the electron is ejected from its orbit.

In Compton scattering a gamma interacts with an orbital electron, but only part of the gamma energy is transferred to the electron. The electron is ejected from its orbit, and the gamma is scattered off at a lower energy.

In pair-production, a gamma interacts with the electric field of a nucleus and is converted into an electron-positron pair. The gamma must have an energy greater than 1.02 MeV for this to occur.

Summary

The important information in this chapter is summarized below.

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NUCLEAR PHYSICS
AND REACTOR THEORY
Module 2
Reactor Theory (Neutron Characteristics)
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TERMINAL OBJECTIVE
1.0 Without references, EXPLAIN how neutron sources produce neutrons.
ENABLING OBJECTIVES
1.1 DEFINE the following terms:
a. Intrinsic neutron source
b. Installed neutron source
1.2 LIST three examples of reactions that produce neutrons in intrinsic neutron sources.
1.3 LIST three examples of reactions that produce neutrons in installed neutron sources.

TERMINAL OBJECTIVE
2.0 Given the necessary information for calculations, EXPLAIN basic concepts in reactor

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ENABLING OBJECTIVES
2.1 DEFINE the following terms:
a. Atom density d. Barn
b. Neutron flux e. Macroscopic cross section
c. Microscopic cross section f. Mean free path
2.2 EXPRESS macroscopic cross section in terms of microscopic cross section.
2.3 DESCRIBE how the absorption cross section of typical nuclides varies with neutron energy at energies below the resonance absorption region.
2.4 DESCRIBE the cause of resonance absorption in terms of nuclear energy levels.
2.5 DESCRIBE the energy dependence of resonance absorption peaks for typical light and heavy nuclei.
2.6 EXPRESS mean free path in terms of macroscopic cross section.
2.7 Given the number densities (or total density and component fractions) and microscopic cross sections of components, CALCULATE the macroscopic cross section for a mixture.
2.8 CALCULATE a macroscopic cross section given a material density, atomic mass, and microscopic cross section.
2.9 EXPLAIN neutron shadowing or self-shielding.
2.10 Given the neutron flux and macroscopic cross section, CALCULATE the reaction rate.
2.11 DESCRIBE the relationship between neutron flux and reactor power.

ENABLING OBJECTIVES (Cont.)
2.12 DEFINE the following concepts:
a. Thermalization d. Average logarithmic energy decrement
b. Moderator e. Macroscopic slowing down power
c. Moderating ratio
2.13 LIST three desirable characteristics of a moderator.
2.14 Given an average fractional energy loss per collision, CALCULATE the energy loss after a specified number of collisions.

TERMINAL OBJECTIVE
3.0 Without references, EXPLAIN the production process and effects on fission of prompt and delayed neutrons.
ENABLING OBJECTIVES
3.1 STATE the origin of prompt neutrons and delayed neutrons.
3.2 STATE the approximate fraction of neutrons that are born as delayed neutrons from the
Fixture checkpoint marker-0802-xray is recorded on fixture page 0802.
fission of the following nuclear fuels:
a. Uranium-235
b. Plutonium-239
3.3 EXPLAIN the mechanism for production of delayed neutrons.
3.4 EXPLAIN prompt and delayed neutron generation times.
3.5 Given prompt and delayed neutron generation times and delayed neutron fraction, CALCULATE the average generation time.
3.6 EXPLAIN the effect of delayed neutrons on reactor control.

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4.0 Without references, DESCRIBE the neutron energy spectrum for the type of reactor presented in this module.
ENABLING OBJECTIVES
4.1 STATE the average energy at which prompt neutrons are produced.
4.2 DESCRIBE the neutron energy spectrum in the following reactors:
a. Fast reactor
b. Thermal reactor
4.3 EXPLAIN the reason for the particular shape of the fast, intermediate, and slow energy regions of the neutron flux spectrum for a thermal reactor.
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Reactor Theory (Neutron Characteristics) DOE-HDBK-1019/1-93 NEUTRON SOURCES
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NEUTRON SOURCES
Neutrons from a variety of sources are always present in a reactor core. This is
Fixture checkpoint marker-0803-xray is recorded on fixture page 0803.
true even when the reactor is shut down. Some of these neutrons are produced by naturally occurring (intrinsic) neutron sources, while others may be the result of fabricated (installed) neutron sources that are incorporated into the design of the reactor. The neutrons produced by sources other than neutron-induced fission are often grouped together and classified as source neutrons.
EO 1.1 DEFINE the following terms:
a. Intrinsic neutron source
b. Installed neutron source
EO 1.2 LIST three examples of reactions that produce neutrons in intrinsic neutron sources.
EO 1.3 LIST three examples of reactions that produce neutrons in installed neutron sources.
Neutron Sources
In addition to neutron-induced fission, neutrons are produced by other reactions. The neutrons produced by reactions other than neutron-induced fission are called source neutrons. Source neutrons are important because they ensure that the neutron population remains high enough to allow a visible indication of neutron level on the most sensitive monitoring instruments while the reactor is shutdown and during the startup sequence. This verifies instrument operability and allows monitoring of neutron population changes. Source neutrons can be classified as either intrinsic or installed neutron sources.
Intrinsic Neutron Sources
Some neutrons will be produced in the materials present in the reactor due to a variety of unavoidable reactions that occur because of the nature of these materials. Intrinsic neutron sources are those neutron-producing reactions that always occur in reactor materials.
■ ■ ■ ■
■ ■ ■
■ ■
NEUTRON SOURCES DOE-HDBK-1019/1-93 Reactor Theory (Neutron Characteristics)
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A limited number of neutrons will always be present, even in a reactor core that has never been operated, due to spontaneous fission of some heavy nuclides that are present in the fuel. Uranium■238, ur of the rate at which different heavy nuclides produce neutrons by spontaneous fission.
Californium-252 is not an intrinsic neutron source, but will be discussed in the section on installed

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TABLE 1
Neutron Production by Spontaneous Fission
Nuclide T (Fission) T (1/2 1/2 ■-decay) neutrons/sec/gram
U 1.8 x 10 years 6.8 x 10 years 8.0 x 10 235
92
17 8 -4
U 8.0 x 10 years 4.5 x 10 years 1.6 x 10 238
92
15 9 -2
Pu 5.5 x 10 years 2.4 x 10 years 3.0 x 10 239
94
5 4 -2
Pu 1.2 x 10 years 6.6 x 10 years 1.0 x 10 240
94
11 3 3
Cf 66.0 years 2.65 years 2.3 x 10 252
98
12
Another intrinsic neutron source is a reaction involving natural boron and fuel. In some reactors, natural boron is loaded into the reactor core as a neutron absorber to improve reactor control or increase core life-time. Boron-11 (80.1% of natural boron) undergoes a reaction with the alpha
Fixture checkpoint marker-0804-xray is recorded on fixture page 0804.
particle emitted by the radioactive decay of heavy nuclides in the fuel to yield a neutron as shown below.
The boron-11 must be mixed with, or in very close proximity to, the fuel for this reaction because of the short path length of the alpha particle. For a reactor core with this configuration, this (■,n) reaction is an important source of neutrons for reactor startup.
In a reactor that has been operated, another source of neutrons becomes significant. Neutrons may be produced by the interaction of a gamma ray and a deuterium nucleus. This reaction is commonly referred to as a photoneutron reaction because it is initiated by electromagnetic radiation and results in the production of a neutron. The photoneutron reaction is shown below.
■ ■ ■
■ ■
■ ■ ■
■ ■
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There is an abundant supply of high energy gammas in a reactor that has been operated because many of the fission products are gamma emitters. All water-cooled reactors have some deuterium present in the coolant in the reactor core because a small fraction of natural hydrogen is the isotope deuterium. The atom percentage of deuterium in the water ranges from close to the naturally occurring value (0.015%) for light water reactors to above 90% deuterium for heavy water reactors. Therefore, the required conditions for production of photoneutrons exist.
The supply of gamma rays decreases with time after shutdown as the gamma emitters decay; therefore, the photoneutron production rate also decreases. In a few particular reactors, additional D O (heavy water) may be added to the reactor to increase the production of 2 photoneutrons following a long shutdown period.
Installed Neutron Sources

history of the reactor, many reactors have artificial sources of neutrons installed. These neutron sources ensure that shutdown neutron levels are high enough to be detected by the nuclear instruments at all times. This provides a true picture of reactor conditions and any change in these conditions. An installed neutron source is an assembly placed in or near the reactor for the sole purpose of producing source neutrons.

One strong source of neutrons is the artificial nuclide californium-252, which emits neutrons at the rate of about 2 x 10 neutrons per second per gram as the result of spontaneous fission. 12 Important drawbacks for some applications may be its high cost and its short half-life (2.65 years).

Many installed neutron sources use the (■,n) reaction with beryllium. These sources are composed of a mixture of metallic beryllium (100% beryllium-9) with a small quantity of an alpha particle emitter, such as a compound of radium, polonium, or plutonium. The reaction that occurs is shown below.

The beryllium is intimately (homogeneously) mixed with the alpha emitter and is usually enclosed in a stainless steel capsule.

Another type of installed neutron source that is widely used is a photoneutron source that employs the (■,n) reaction with beryllium. Beryllium is used for photoneutron sources because its stable isotope beryllium-9 has a weakly attached last neutron with a binding energy of only 1.66 MeV. Thus, a gamma ray with greater energy than 1.66 MeV can cause neutrons to be ejected by the (■,n) reaction as shown below.

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Neutron Sources Summary

& Intrinsic neutron sources are sources of neutrons from materials that are in the reactor for other purposes such as fuel, burnable poison, or moderator.

& Installed neutron sources are materials or components placed in the reactor specifically for the purpose of producing source neutrons.

& Examples of intrinsic neutron sources are listed below.

Fixture checkpoint marker-0805-xray is recorded on fixture page 0805.

Spontaneous fission of heavy nuclides in fuel, such as uranium-238, uranium-235, and plutonium-239, results in fission fragments and free neutrons.

Boron-11 mixed with the fuel undergoes an alpha-neutron reaction and becomes nitrogen-14.

Deuterium present in the reactor coolant undergoes a gamma■neutron reaction and becomes hydrogen-1.

& Examples of installed neutron sources are listed below.

Spontaneous fission of californium-252 results in fission fragments and free neutrons.

Beryllium-9 undergoes an alpha-neutron reaction (alpha from the decay of plutonium, polonium, or radium) and becomes carbon-12.

Beryllium-9 undergoes a gamma-neutron reaction (high energy gamma from decay of antimony-124) and becomes beryllium-8.

Many startup sources of this type use antimony and beryllium because after activation with neutrons the radioactive antimony becomes an emitter of high energy gammas. The photoneutron sources of this type are constructed somewhat differently from the (■,n) types. One design incorporates a capsule of irradiated antimony enclosed in a beryllium sleeve. The entire assembly is then encased in a stainless steel cladding. A large reactor may have several neutron sources of this type installed within the core.

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The important information in this chapter is summarized below.
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NUCLEAR CROSS SECTIONS AND NEUTRON FLUX
To determine the frequency of neutron interactions, it is necessary to describe the availability of neutrons to cause interaction and the probability of a neutron interacting with material. The availability of neutrons and the probability of interaction are quantified by the neutron flux and nuclear cross section.
EO 2.1 DEFINE the following terms:
a. Atom density d. Barn
b. Neutron flux e. Macroscopic cross section
c. Microscopic cross section f. Mean free path
EO 2.2 EXPRESS macroscopic cross section in terms of microscopic cross section.
EO 2.3 DESCRIBE how the absorption cross section of typical nuclides varies with neutron energy at energies below the resonance absorption region.
EO 2.4 DESCRIBE the cause of resonance absorption in terms of nuclear energy levels.
EO 2.5 DESCRIBE the energy dependence of resonance absorption peaks for typical light and heavy nuclei.
EO 2.6 EXPRESS mean free path in terms of macroscopic cross section.
EO 2.7 Given the number densities (or total density and component fractions) and microscopic cross sections of components, CALCULATE the macroscopic cross section for a mixture.
EO 2.8 CALCULATE a macroscopic cross section given a material density, atomic mass, and microscopic cross section.
EO 2.9 EXPLAIN neutron shadowing or self-shielding.
'
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NUCLEAR CROSS SECTIONS AND NEUTRON FLUX Reactor Theory (Neutron Characteristics)
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Introduction
Fission neutrons are born with an average energy of about 2 MeV. These fast neutrons interact
Fixture checkpoint marker-0806-xray is recorded on fixture page 0806.
with the reactor core materials in various absorption and scattering reactions. Collisions that result in scattering are useful in slowing neutrons to thermal energies. Thermal neutrons may be absorbed by fissile nuclei to produce more fissions or be absorbed in fertile material for conversion to fissionable fuel. Absorption of neutrons in structural components, coolant, and other non-fuel material results in the removal of neutrons without fulfilling any useful purpose.
To safely and efficiently operate a nuclear reactor it is necessary to predict the probability that a particular absorption or scattering reaction will occur. Once these probabilities are known, if the availability of neutrons can be determined, then the rate at which these nuclear reactions take place can be predicted.
Atom Density
One important property of a material is the atom density. The atom density is the number of atoms of a given type per unit volume of the material. To calculate the atom density of a

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(2-1)
where:
N = atom density (atoms/cm)3
' = density (g/cm)3
N = Avogadro's number (6.022 x 10 atoms/mole) A
23
M = gram atomic weight
N
NA
M
2.699 g
cm3
Fixture checkpoint marker-0807-xray is recorded on fixture page 0807.
6.022 x 1023 atoms
mole
26.9815 g
mole
6.024 x 1022 atoms
cm3
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Example:
A block of aluminum has a density of 2.699 g/cm . If the gram atomic weight of
3
aluminum is 26.9815 g, calculate the atom density of the aluminum.
Solution:
Cross Sections
The probability of a neutron interacting with a nucleus for a particular reaction is dependent upon not only the kind of nucleus involved, but also the energy of the neutron. Accordingly, the absorption of a thermal neutron in most materials is much more probable than the absorption of a fast neutron. Also, the probability of interaction will vary depending upon the type of reaction involved.
The probability of a particular reaction occurring between a neutron and a nucleus is called the microscopic cross section ( ) of the nucleus for the particular reaction. This cross section will vary with the energy of the neutron. The microscopic cross section may also be regarded as the effective area the nucleus presents to the neutron for the particular reaction. The larger the effective area, the greater the probability for reaction.
Because the microscopic cross section is an area, it is expressed in units of area, or square centimeters. A square centimeter is tremendously large in comparison to the effective area of a nucleus, and it has been suggested that a physicist once referred to the measure of a square centimeter as being "as big as a barn" when applied to nuclear processes. The name has persisted and microscopic cross sections are expressed in terms of barns. The relationship between barns and cm is shown below. 2
1 barn = 10-24 cm2
N
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NUCLEAR CROSS SECTIONS AND NEUTRON FLUX Reactor Theory (Neutron Characteristics)

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Whether a neutron will interact with a certain volume of material depends not only on the microscopic cross section of the individual nuclei but also on the number of nuclei within that volume. Therefore, it is necessary to define another kind of cross section known as the macroscopic cross section ( ). The macroscopic cross section is the probability of a given reaction occurring per unit travel of the neutron. is related to the microscopic cross section ( ) by the relationship shown below.

(2-2)

where:

= macroscopic cross section (cm)

-1

N = atom density of material (atoms/cm)

3

= microscopic cross-section (cm)

2

The difference between the microscopic and macroscopic cross sections is extremely important and is restated for clarity. The microscopic cross section ( ) represents the effective target area that a single nucleus presents to a bombarding particle. The units are given in barns or cm .

2

The macroscopic cross section ( ) represents the effective target area that is presented by all of

Fixture checkpoint marker-0808-xray is recorded on fixture page 0808.

the nuclei contained in 1 cm of the material. The units are given as 1/cm or cm .

3 -1

A neutron interacts with an atom of the material it enters in two basic ways. It will either interact through a scattering interaction or through an absorption reaction. The probability of a neutron being absorbed by a particular atom is the microscopic cross section for absorption, . The probability of a neutron scattering off of a particular nucleus is the microscopic cross a section for scattering, s

. The sum of the microscopic cross section for absorption and the microscopic cross section for scattering is the total microscopic cross section, . T

= + T a s

Both the absorption and the scattering microscopic cross sections can be further divided. For instance, the scattering cross section is the sum of the elastic scattering cross section (se) and the inelastic scattering cross section ( ). si

= + s se si

The microscopic absorption cross section ( ) includes all reactions except scattering. However, a for most purposes it is sufficient to merely separate it into two categories, fission (f) and capture ( ). Radiative capture of neutrons was described in the Neutron Interactions chapter c of Module 1.

= + a f c

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Figure 1 Typical Neutron Absorption Cross Section vs. Neutron Energy

The variation of absorption cross sections with neutron energy is often complicated. For many elements the absorption cross sections are small, ranging from a fraction of a barn to a few barns for slow (or thermal) neutrons.

For a considerable number of nuclides of moderately high (or high) mass numbers, an examination of the variation of the absorption cross section with the energy of the incident

neutron energy. This cross section is illustrated in Figure 1. First, the cross section decreases steadily with increasing neutron energy in a low energy region, which includes the thermal range (E < 1 eV). In this region the absorption cross section, which is often high, is inversely proportional to the velocity (v). This region is frequently referred to as the "1/v region," because the absorption cross section is proportional to 1/v, which is the reciprocal of neutron velocity. Following the 1/v region, there occurs the "resonance region" in which the cross sections rise sharply to high values called "resonance peaks" for neutrons of certain energies, and then fall again. These energies are called resonance energies and are a result of the affinity of the nucleus for neutrons whose energies closely match its discrete, quantum energy levels. That is, when the binding energy of a neutron plus the kinetic energy of the neutron are exactly equal to the amount required to raise a compound nucleus from its ground state to a quantum level, resonance absorption occurs. The following example problem further illustrates this point.

■

(

Assuming that uranium-236 has a nuclear quantum energy level at 6.8 MeV above its ground state, calculate the kinetic energy a neutron must possess to undergo resonant absorption in uranium-235 at this resonance energy level.

BE = [Mass(U) + Mass(neutron) - Mass(U)] x 931 MeV/amu 235 236

BE = (235.043925 + 1.008665 - 236.045563) x 931 MeV/amu

BE = (0.007025 amu) x 931 MeV/amu = 6.54 MeV

6.8 MeV - 6.54 MeV = 0.26 MeV

The difference between the binding energy and the quantum energy level equals the amount of kinetic energy the neutron must possess. The typical heavy nucleus will have many closely■spaced resonanc

Fixture checkpoint marker-0809-xray is recorded on fixture page 0809.

complex and have more possible configurations and corresponding energy states. Light nuclei, being less complex, have fewer possible energy states and fewer resonances that are sparsely distributed at higher energy levels.

For higher neutron energies, the absorption cross section steadily decreases as the energy of the neutron increases. This is called the "fast neutron region." In this region the absorption cross sections are usually less than 10 barns.

With the exception of hydrogen, for which the value is fairly large, the elastic scattering cross sections are generally small, for example, 5 barns to 10 barns. This is close to the magnitude of the actual geometric cross sectional area expected for atomic nuclei. In potential scattering, the cross section is essentially constant and independent of neutron energy. Resonance elastic scattering and inelastic scattering exhibit resonance peaks similar to those associated with absorption cross sections. The resonances occur at lower energies for heavy nuclei than for light nuclei. In general, the variations in scattering cross sections are very small when compared to the variations that occur in absorption cross sections.

Mean Free Path

If a neutron has a certain probability of undergoing a particular interaction in one centimeter of travel, then the inverse of this value describes how far the neutron will travel (in the average case) before undergoing an interaction. This average distance traveled by a neutron before interaction is known as the mean free path for that interaction and is represented by the symbol ■. The relationship between the mean free path (■) and the macroscopic cross section (*) is shown below.

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'
(
)
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Calculation of Macroscopic Cross Section and Mean Free Path
Most materials are composed of several elements, and because most elements are composed of several isotopes, most materials involve many cross sections, one for each isotope involved. Therefore, to include all the isotopes within a given material, it is necessary to determine the macroscopic cross section for each isotope and then sum all the individual macroscopic cross sections. Equation (2-4) can be used to determine the macroscopic cross section for a composite material.
* = N) + N) + N) + ... N) (2-4) 1 1 2 2 3 3 nn
where:
N = the number of nuclei per cm of the n element n
3 th
) = the microscopic cross section of the n element n
th
The following example problems illustrate the calculation of the macroscopic cross section for a single element and for combinations of materials.
Example 1:
Find the macroscopic thermal neutron absorption cross section for iron, which has a density of 7.86 g/cm . The microscopic cross section for absorption of iron is 2.56 barns 3 and the gram atomic weight is 55.847 g.
Solution:
Step 1: Using Equation (2-1), calculate the atom density of iron.
Step 2: Use this atom density in Equation (2-2) to calculate the macroscopic cross section.
'
'
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Example 2:
An alloy is composed of 95% aluminum and 5% silicon (by weight). The density of the alloy is 2.66 g/cm . Properties of aluminum and silicon are shown below. 3

Element	Gram Atomic Weight	) a (barns)	s (barns)
Aluminum	26.9815	0.23	1.49
Silicon	28.0855	0.16	2.20

1. Calculate the atom densities for the aluminum and silicon.
2. Determine the absorption and scattering macroscopic cross sections for thermal neutrons.
3. Calculate the mean free paths for absorption and scattering.
Solution:
Step 1: The density of the aluminum will be 95% of the total density. Using Equation (2-1) yields the atom densities.

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5.64 x 1022 atoms
cm3
0.23 x 10 24 cm22.85 x 1021 atoms
cm3
0.16 x 10 24 cm2
0.0134 cm 1
s NAl s,Al NSi s,Si
5.64 x 1022 atoms
cm3
1.49 x 10 24 cm22.85 x 1021 atoms
Fixture checkpoint marker-0811-xray is recorded on fixture page 0811.
cm3
2.20 x 10 24 cm2
0.0903 cm 1
a
1
a
1
0.01345 cm 1
74.3 cm
s
1
s
1
0.0903 cm 1
11.1 cm
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Step 2: The macroscopic cross sections for absorption and scattering are
calculated using Equation (2-4).
Step 3: The mean free paths are calculated by inserting the macroscopic cross
sections calculated above into Equation (2-3).
Thus, a neutron must travel an average of 74.3 cm to interact by absorption in this alloy, but it
must travel only 11.1 cm to interact by scattering.
)
)
)
)
(■
(■
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Effects of Temperature on Cross Section
As discussed, the microscopic absorption cross section varies significantly as neutron energy
varies. The microscopic cross sections provided on most charts and tables are measured for a
standard neutron velocity of 2200 meters/second, which corresponds to an ambient temperature

be lower than the value for 68(F, and any cross sections which involve absorption (for example,) ,) ,)) must be corrected for the existing temperature. acf

The following formula is used to correct microscopic cross sections for temperature. Although the example illustrates absorption cross section, the same formula may be used to correct capture and fission cross sections.

where:

) = microscopic cross section corrected for temperature

) = microscopic cross section at reference temperature (68(F or 20(C) o

T = reference temperature (68(F) in degrees Rankine ((R) or Kelvin ((K) o

T = temperature for which corrected value is being calculated

NOTE: When using this formula, all temperatures must be converted to (R or (K.

(R = (F + 460

(K = (C + 273

Example:

What is the value of)f for uranium-235 for thermal neutrons at 500(F? Uranium-235 has a) of 583 barns at 68(F. f

Solution:

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Reactor Theory (Neutron Characteristics) NUCLEAR CROSS SECTIONS AND NEUTRON FLUX

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Neutron Flux

Macroscopic cross sections for neutron reactions with materials determine the probability of one neutron undergoing a specific reaction per centimeter of travel through that material. If one wants to determine how many reactions will actually occur, it is necessary to know how many neutrons are traveling through the material and how many centimeters they travel each second. It is convenient to consider the number of neutrons existing in one cubic centimeter at any one instant and the total distance they travel each second while in that cubic centimeter. The number of neutrons existing in a cm of material at any instant is called neutron density and is 3 represented by the symbol n with units of neutrons/cm . The total distance these neutrons can 3 travel each second will be determined by their velocity.

A good way of defining neutron flux (1) is to consider it to be the total path length covered by all neutrons in one cubic centimeter during one second. Mathematically, this is the equation below.

1 = n v (2-5)

where:

1 = neutron flux (neutrons/cm -sec) 2

n = neutron density (neutrons/cm)3

v = neutron velocity (cm/sec)

The term neutron flux in some applications (for example, cross section measurement) is used as parallel beams of neutrons traveling in a single direction. The intensity (I) of a neutron beam is the product of the neutron density times the average neutron velocity. The directional beam intensity is equal to the number of neutrons per unit area and time (neutrons/cm -sec) falling on 2 a surface perpendicular to the direction of the beam.

One can think of the neutron flux in a reactor as being comprised of many neutron beams traveling in various directions. Then, the neutron flux becomes the scalar sum of these directional flux intensities (added as numbers and not vectors), that is, 1 = I + I + I +...I . 123 n

Since the atoms in a reactor do not interact preferentially with neutrons from any particular

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given point within a reactor, neutrons will be traveling in all directions.
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Self-Shielding
In some locations within the reactor, the flux level may be significantly lower than in other areas due to a phenomenon referred to as neutron shadowing or self-shielding. For example, the interior of a fuel pin or pellet will "see" a lower average flux level than the outer surfaces since an appreciable fraction of the neutrons will have been absorbed and therefore cannot reach the interior of the fuel pin. This is especially important at resonance energies, where the absorption cross sections are large.
Summary
The important information in this chapter is summarized below.
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Reactor Theory (Neutron Characteristics) NUCLEAR CROSS SECTIONS AND NEUTRON FLUX
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Nuclear Cross Section and Neutron Flux Summary
& Atom density (N) is the number of atoms of a given type per unit volume of material.
& Microscopic cross section ()) is the probability of a given reaction occurring between a neutron and a nucleus.
& Microscopic cross sections are measured in units of barns, where 1 barn = 10-24 cm .2
& Macroscopic cross section (*) is the probability of a given reaction occurring per
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unit length of travel of the neutron. The units for macroscopic cross section are cm . -1
& The mean free path (■) is the average distance that a neutron travels in a material between interactions.
& Neutron flux (1) is the total path length traveled by all neutrons in one cubic centimeter of material during one second.
& The macroscopic cross section for a material can be calculated using the equation below.
* = N)
& The absorption cross section for a material usually has three distinct regions. At low neutron energies (<1 eV) the cross section is inversely proportional to the neutron velocity.
& Resonance absorption occurs when the sum of the kinetic energy of the neutron and its binding energy is equal to an allowed nuclear energy level of the nucleus.
& Resonance peaks exist at intermediate energy levels. For higher neutron energies, the absorption cross section steadily decreases as the neutron energy increases.
& The mean free path equals 1/*.
& The macroscopic cross section for a mixture of materials can be calculated using the equation below.
* = N 1 1 2 2 3 3 nn) + N) + N) + ... N)
& Self-shielding is where the local neutron flux is depressed within a material due to neutron absorption near the surface of the material.
1)

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REACTION RATES
It is possible to determine the rate at which a nuclear reaction will take place based on the neutron flux, cross section for the interaction, and atom density of the target. This relationship illustrates how a change in one of these items affects the reaction rate.
EO 2.10 Given the neutron flux and macroscopic cross section, CALCULATE the reaction rate.
EO 2.11 DESCRIBE the relationship between neutron flux and reactor power.
Reaction Rates
If the total path length of all the neutrons in a cubic centimeter in a second is known, (neutron flux (1)), and if the probability of having an interaction per centimeter path length is also known (macroscopic cross section (*)), multiply them together to get the number of interactions taking place in that cubic centimeter in one second. This value is known as the reaction rate and is denoted by the symbol R. The reaction rate can be calculated by the equation shown below.
R = 1 * (2-6)
where:
R = reaction rate (reactions/sec)
1 = neutron flux (neutrons/cm -sec) 2
* = macroscopic cross section (cm) -1
Substituting the fact that * = N) into Equation (2-6) yields the equation below.
where:
) = microscopic cross section (cm)2
N = atom density (atoms/cm)3
Rf f
1 x 1013 neutrons
cm2
sec
0.1 cm 1
1 x 1012 fissions
Fixture checkpoint marker-0814-xray is recorded on fixture page 0814.
cm3
sec
Rf f
f
Rf
1.29 x 1012 fissions
cm3
sec
3 x 1013 neutrons
cm2
sec
0.043 cm 1

The reaction rate calculated will depend on which macroscopic cross section is used in the calculation. Normally, the reaction rate of greatest interest is the fission reaction rate.

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If a one cubic centimeter section of a reactor has a macroscopic fission cross section of
0.1 cm , and if the thermal neutron flux is 10 neutrons/cm -sec, what is the fission rate -1 13 2
in that cubic centimeter?
Solution:
In addition to using Equation (2-6) to determine the reaction rate based on the physical properties of material, it is also possible to algebraically manipulate the equation to determine physical properties if the reaction rate is known.
Example:
A reactor operating at a flux level of 3 x 10 neutrons/cm -sec contains 10 atoms of
13 2 20
uranium-235 per cm . The reaction rate is 1.29 x 10 fission/cm . Calculate and .
3 12 3
f f
Solution:
Step 1: The macroscopic cross section can be determined by solving
Equation (2-6) for and substituting the appropriate values. f
Rf N f
f
Rf
N
1.29 x 1012 fissions
cm3
sec
1 x 1020 atoms
cm3
3 x 1013 neutrons
cm2
sec
4.3 x 10 22 cm2 1 barn
1 x 10 24 cm2
430 barns
1 watt 1 erg
1 x 10 7 watt sec
1 MeV
1.602 x 10 6
erg
1 fission
200 MeV
Fixture checkpoint marker-0815-xray is recorded on fixture page 0815.
3.12 x 1010 fissions
second
REACTION RATES DOE-HDBK-1019/1-93 Reactor Theory (Neutron Characteristics)
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Step 2: To find the microscopic cross section, replace f with (N x f) and solve
for .
f
Reactor Power Calculation
Multiplying the reaction rate per unit volume by the total volume of the core results in the total

reaction were known, it would be possible to determine the rate of energy release (power) due to a certain reaction.

In a reactor where the average energy per fission is 200 MeV, it is possible to determine the number of fissions per second that are necessary to produce one watt of power using the following conversion factors.

1 fission = 200 MeV

1 MeV = 1.602 x 10 ergs -6

1 erg = 1 x 10 watt-sec -7

This is equivalent to stating that 3.12 x 10 fissions release 1 watt-second of energy. 10

P

th f V

3.12 x 1010 fissions

watt sec

Fixture checkpoint marker-0816-xray is recorded on fixture page 0816.

The power released in a reactor can be calculated based on Equation (2-6). Multiplying the reaction rate by the volume of the reactor results in the total fission rate for the entire reactor. Dividing by the number of fissions per watt-sec results in the power released by fission in the reactor in units of watts. This relationship is shown mathematically in Equation (2-7) below.

(2-7)

where:

P = power (watts)

= thermal neutron flux (neutrons/cm -sec) th

2

= macroscopic cross section for fission (cm) f

-1

V = volume of core (cm)

3

Relationship Between Neutron Flux and Reactor Power

In an operating reactor the volume of the reactor is constant. Over a relatively short period of time (days or weeks), the number density of the fuel atoms is also relatively constant. Since the atom density and microscopic cross section are constant, the macroscopic cross section must also be constant. Examining Equation (2-7), it is apparent that if the reactor volume and macroscopic cross section are constant, then the reactor power and the neutron flux are directly proportional. This is true for day-to-day operation. The neutron flux for a given power level will increase very slowly over a period of months due to the burnup of the fuel and resulting decrease in atom density and macroscopic cross section.

Reaction Rates Summary

& The reaction rate is the number of interactions of a particular type occurring in a cubic centimeter of material in a second.

& The reaction rate can be calculated by the equation below.

R = 1 *

& Over a period of several days, while the atom density of the fuel can be considered constant, the neutron flux is directly proportional to reactor power.

Summary

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NEUTRON MODERATION

In thermal reactors, the neutrons that cause fission are at a much lower energy than the energy level at which they were born from fission. In this type of reactor, specific materials must be included in the reactor design to reduce the energy level of the neutrons in an efficient manner.

EO 2.12 DEFINE the following concepts:

a. Thermalization d. Average logarithmic energy decrement
b. Moderator e. Macroscopic slowing down power
c. Moderating ratio

EO 2.13 LIST three desirable characteristics of a moderator.

EO 2.14 Given an average fractional energy loss per collision, CALCULATE the energy loss after a specified number of collisions.

Neutron Slowing Down and Thermalization

Fission neutrons are produced at an average energy level of 2 MeV and immediately begin to slow down as the result of numerous scattering reactions with a variety of target nuclei. After a number of collisions with nuclei, the speed of a neutron is reduced to such an extent that it has approximately the same average kinetic energy as the atoms (or molecules) of the medium in which the neutron is undergoing elastic scattering. This energy, which is only a small fraction of an electron volt at ordinary temperatures (0.025 eV at 20(C), is frequently referred to as the Fixture checkpoint marker-0817-xray is recorded on fixture page 0817.
thermal energy, since it depends upon the temperature. Neutrons whose energies have been reduced to values in this region (< 1 eV) are designated thermal neutrons. The process of reducing the energy of a neutron to the thermal region by elastic scattering is referred to as thermalization, slowing down, or moderation. The material used for the purpose of thermalizing neutrons is called a moderator. A good moderator reduces the speed of neutrons in a small number of collisions, but does not absorb them to any great extent. Slowing the neutrons in as few collisions as possible is desirable in order to reduce the amount of neutron leakage from the core and also to reduce the number of resonance absorptions in non-fuel materials. Neutron leakage and resonance absorption will be discussed in the next module.

ln Ei
ln Ef
ln
Ei
Ef
2
A
2
3

The ideal moderating material (moderator) should have the following nuclear properties.

large scattering cross section
small absorption cross section
large energy loss per collision

A convenient measure of energy loss per collision is the logarithmic energy decrement. The

the neutron energy. This quantity is represented by the symbol (Greek letter xi).
(2-8)
where:
= average logarithmic energy decrement
Ei = average initial neutron energy
Ef = average final neutron energy
The symbol is commonly called the average logarithmic energy decrement because of the fact that a neutron loses, on the average, a fixed fraction of its energy per scattering collision. Since the fraction of energy retained by a neutron in a single elastic collision is a constant for a given material, is also a constant. Because it is a constant for each type of material and does not depend upon the initial neutron energy, is a convenient quantity for assessing the moderating ability of a material.
The values for the lighter nuclei are tabulated in a variety of sources. The following commonly used approximation may be used when a tabulated value is not available.
This approximation is relatively accurate for mass numbers (A) greater than 10, but for some low values of A it may be in error by over three percent.
N
ln Ehigh ln Elow
ln
Ehigh
Elow
N
ln
Ehigh
Elow
ln
2 x 106eV
0.025 eV
0.948
Fixture checkpoint marker-0818-xray is recorded on fixture page 0818.
19.2 collisions
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Since represents the average logarithmic energy loss per collision, the total number of collisions necessary for a neutron to lose a given amount of energy may be determined by dividing into the difference of the natural logarithms of the energy range in question. The number of collisions (N) to travel from any energy, E , to any lower energy, E , can be calculated as shown below. high low
Example:
How many collisions are required to slow a neutron from an energy of 2 MeV to a thermal energy of 0.025 eV, using water as the moderator? Water has a value of 0.948 for .
Solution:
Sometimes it is convenient, based upon information known, to work with an average fractional energy loss per collision as opposed to an average logarithmic fraction. If the initial neutron energy level and the average fractional energy loss per collision are known, the final energy level for a given number of collisions may be computed using the following formula.
! (

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(2-9)
where:
E = initial neutron energy o
E = neutron energy after N collisions N
x = average fractional energy loss per collision
N = number of collisions
Example:
If the average fractional energy loss per collision in hydrogen is 0.63, what will be the energy of a 2 MeV neutron after (a) 5 collisions? (b) 10 collisions?
Solution:
a)
b)
Macroscopic Slowing Down Power
Although the logarithmic energy decrement is a convenient measure of the ability of a material to slow neutrons, it does not measure all necessary properties of a moderator. A better measure of the capabilities of a material is the macroscopic slowing down power. The macroscopic slowing down power (MSDP) is the product of the logarithmic energy decrement and the macroscopic cross section for scattering in the material. Equation (2-10) illustrates how to calculate the macroscopic slowing down power.
(2-10)
MR
s
a
Reactor Theory (Neutron Characteristics) DOE-HDBK-1019/1-93 NEUTRON MODERATION
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Moderating Ratio
Macroscopic slowing down power indicates how rapidly a neutron will slow down in the material in question, but it still does not fully explain the effectiveness of the material as a moderator. An element such as boron has a high logarithmic energy decrement and a good slowing down power, but it is a poor moderator because of its high probability of absorbing neutrons.
The most complete measure of the effectiveness of a moderator is the moderating ratio. The moderating ratio is the ratio of the macroscopic slowing down power to the macroscopic cross section for absorption. The higher the moderating ratio, the more effectively the material performs as a moderator. Equation (2-11) shows how to calculate the moderating ratio of a material.
(2-11)
Fixture checkpoint marker-0819-xray is recorded on fixture page 0819.
Moderating properties of different materials are compared in Table 2.
TABLE 2
Moderating Properties of Materials
Material Collisions to Slowing Down
Number of Macroscopic
Thermalize Power
Moderating
Ratio
H2O 0.927 19 1.425 62
D2O 0.510 35 0.177 4830
Helium 0.427 42 9 x 10 51 -6

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Boron 0.171 105 0.092 0.00086
Carbon 0.158 114 0.083 216
NEUTRON MODERATION DOE-HDBK-1019/1-93 Reactor Theory (Neutron Characteristics)
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Summary
The important information in this chapter is summarized below.
Neutron Moderation Summary
& Thermalization is the process of reducing the energy level of a neutron from the energy level at which it is produced to an energy level in the thermal range.
& The moderator is the reactor material that is present for the purpose of thermalizing neutrons.
& Moderating ratio is the ratio of the macroscopic slowing down power to the macroscopic cross section for absorption.
Fixture checkpoint marker-0820-xray is recorded on fixture page 0820.
& The average logarithmic energy decrement (!) is the average change in the logarithm of neutron energy per collision.
& Macroscopic slowing down power is the product of the average logarithmic energy decrement and the macroscopic cross section for scattering.
& There are three desirable characteristics of a moderator.
1. large scattering cross section
2. small absorption cross section
3. large energy loss per collision
& The energy loss after a specified number of collisions can be calculated using the equation below.
E = E (1 - x) N o
N
■
■ ■
Reactor Theory (Neutron Characteristics) DOE-HDBK-1019/1-93 PROMPT AND DELAYED NEUTRONS
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PROMPT AND DELAYED NEUTRONS
Not all neutrons are released at the same time following fission. Most neutrons are released virtually instantaneously and are called prompt neutrons. A very small fraction of neutrons are released after the decay of fission products and are called delayed neutrons. Although delayed neutrons are a very small fraction of the total number of neutrons, they play an extremely important role in the control of the reactor.
EO 3.1 STATE the origin of prompt neutrons and delayed neutrons.
EO 3.2 STATE the approximate fraction of neutrons that are born as delayed neutrons from the fission of the following nuclear fuels:
a. Uranium-235
b. Plutonium-239
EO 3.3 EXPLAIN the mechanism for production of delayed neutrons.
EO 3.4 EXPLAIN prompt and delayed neutron generation times.
EO 3.5 Given prompt and delayed neutron generation times and delayed neutron fraction, CALCULATE the average generation time.
EO 3.6 EXPLAIN the effect of delayed neutrons on reactor control.

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The great majority (over 99%) of the neutrons produced in fission are released within about 10-13 seconds of the actual fission event. These are called prompt neutrons. A small portion of fission neutrons are delayed neutrons, which are produced for some time after the fission process has taken place. The delayed neutrons are emitted immediately following the first beta decay of a fission fragment known as a delayed neutron precursor. An example of a delayed neutron precursor is bromine-87, shown below.

PROMPT AND DELAYED NEUTRONS DOE-HDBK-1019/1-93 Reactor Theory (Neutron Characteristics) NP-02 Page 30 Rev. 0

For most applications, it is convenient to combine the known precursors into groups with appropriately averaged properties. These groups vary somewhat depending on the fissile material in use. Table 3 lists the characteristics for the six precursor groups resulting from thermal fission of uranium-235. The fraction of all neutrons that are produced by each of these precursors is called the delayed neutron fraction for that precursor. The total fraction of all neutrons born as delayed neutrons is called the delayed neutron fraction (■). The fraction of delayed neutrons produced varies depending on the predominant fissile nuclide in use. The delayed neutron fractions (■) for the fissile nuclides of most interest are as follows: uranium-233 (0.0026), uranium-235 (0.0065), uranium-238 (0.0148), and plutonium-239 (0.0021).

TABLE 3

Delayed Neutron Precursor Groups
for Thermal Fission in Uranium-235

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Group	Half-Life (sec)	Delayed Neutron Fraction	Average Energy (MeV)
1	55.7	0.00021	0.25
2	22.7	0.00142	0.46
3	6.2	0.00127	0.41
4	2.3	0.00257	0.45
5	0.61	0.00075	0.41
6	0.23	0.00027	-
Total	-	0.0065	-

Neutron Generation Time

The neutron generation time is the time required for neutrons from one generation to cause the fissions that produce the next generation of neutrons. The generation time for prompt neutrons (5* - pronounced "ell-star") is the total time from birth to rebirth. Three time intervals are involved: (a) the time it takes a fast neutron to slow down to thermal energy, (b) the time the now thermal neutron exists prior to absorption in fuel, and (c) the time required for a fissionable nucleus to emit a fast neutron after neutron absorption.

Fast neutrons slow to thermal energies or leak out of the reactor in 10 seconds to 10 seconds, -4 -6 depending on the moderator. In water moderated reactors, thermal neutrons tend to exist for about 10 seconds before they are absorbed. Fission and fast neutron production following -4 neutron absorption in a fissionable nucleus occurs in about 10 seconds. Thus, fast reactors -13 have an 5* of about 10 seconds, while thermal reactors have an 5* of about 10 seconds + 10 -6 -6 -4 seconds, which is about 10 seconds to 10 seconds. -4 -5

■■■■
■■■■
■■

Reactor Theory (Neutron Characteristics) DOE-HDBK-1019/1-93 PROMPT AND DELAYED NEUTRONS Rev. 0 Page 31 NP-02

time from the birth of the fast neutron to the emission of the delayed neutron. Again, three time intervals are involved: (a) the time it takes a fast neutron to slow down to thermal energy, (b) the time the thermal neutron exists prior to absorption, and (c) the average time from neutron absorption to neutron emission by the six precursor groups. The average time for decay of precursors from uranium-235 is 12.5 seconds. The other terms in the delayed neutron generation time are insignificant when compared to this value, and the average delayed neutron generation time becomes ■ 12.5 seconds.

A neutron generation time in the range of 10 seconds to 10 seconds or faster could result in -4 -5 very rapid power excursions, and control would not be possible without the dependence upon delayed neutrons to slow down the rate of the reaction. The average generation time, and hence the rate that power can rise, is determined largely by the delayed neutron generation time. The following equation shows this mathematically.

(2-12)

Example:

Assume a prompt neutron generation time for a particular reactor of 5 x 10 seconds and -5 a delayed neutron generation time of 12.5 seconds. If ■ is 0.0065, calculate the average generation time.

Solution:

This example demonstrates the effect delayed neutrons have on the neutron generation time and thus reactor control. If a reactor were to be operated in a sustained chain reaction using only prompt neutrons (■ = 0), the generation time from the previous example would be about 5 x 10 seconds. However, by operating the reactor such that a 0.0065 fraction of neutrons are -5 delayed, the generation life time is extended to 0.0813 seconds, providing time for adequate operator control. Therefore, although only a small fraction of the total neutron population, delayed neutrons are extremely important to the control and maintenance of a sustained fission chain reaction.

Summary

Fixture checkpoint marker-0822-xray is recorded on fixture page 0822.

The important information in this chapter is summarized on the following page.

■ ■ ■ ■

PROMPT AND DELAYED NEUTRONS DOE-HDBK-1019/1-93 Reactor Theory (Neutron Characteristics) NP-02 Page 32 Rev. 0

Prompt and Delayed Neutrons Summary

& Prompt neutrons are released directly from fission within 10 seconds of the -13 fission event.

& Delayed neutrons are released from the decay of fission products that are called delayed neutron precursors. Delayed neutron precursors are grouped according to half-life. Half-lives vary from fractions of a second to almost a minute.

& The fraction of neutrons born as delayed neutrons is different for different fuel materials. Following are values for some common fuel materials.

Uranium-235 0.0065

Plutonium-239 0.0021

& Delayed neutrons are produced by a classification of fission products known as delayed neutron precursors. When a delayed neutron precursor undergoes a ■ decay, it results in an excited daughter nucleus which immediately ejects a - neutron. Therefore, these delayed neutrons appear with a half-life of the delayed neutron precursor.

& The delayed neutron generation time is the total time from the birth of the fast

neutron generation times are dominated by the half-life of the delayed neutron precursor. The average delayed neutron generation time is about 12.5 seconds.
& A prompt neutron generation time is the sum of the amount of time it takes a fast neutron to thermalize, the amount of time the neutron exists as a thermal neutron before it is absorbed, and the amount of time between a fissionable nuclide absorbing a neutron and fission neutrons being released. Prompt neutron generation time is about 5 x 10 seconds. -5
& The average neutron generation time can be calculated from the prompt and delayed neutron generation times and the delayed neutron fraction using Equation (2-12).
& Delayed neutrons are responsible for the ability to control the rate at which power can rise in a reactor. If only prompt neutrons existed, reactor control would not be possible due to the rapid power changes.
Reactor Theory (Neutron Characteristics) DOE-HDBK-1019/1-93 NEUTRON FLUX SPECTRUM
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Figure 2 Prompt Fission Neutron Energy Spectrum for
Thermal Fission of Uranium-235
NEUTRON FLUX SPECTRUM
The number of neutrons that exist at a given energy level varies. A plot of either the fraction of neutrons or the neutron flux at a given energy versus the energy level is called a neutron energy spectrum. The neutron energy spectrum varies widely for different types of reactors.
EO 4.1 STATE the average energy at which prompt neutrons are produced.
EO 4.2 DESCRIBE the neutron energy spectrum in the following reactors:
a. Fast reactor
b. Thermal reactor
EO 4.3 EXPLAIN the reason for the particular shape of the fast, intermediate, and slow energy regions of the neutron flux spectrum for a thermal reactor.
Prompt Neutron Energies
The neutrons produced by fission are high energy neutrons, and
Fixture checkpoint marker-0823-xray is recorded on fixture page 0823.
almost all fission neutrons have energies between 0.1 MeV and 10 MeV. The neutron energy distribution, or spectrum, may best be described by plotting the fraction of neutrons per MeV as a function of neutron energy, as shown in Figure 2. From this figure it can be seen that the most probable neutron energy is about 0.7 MeV. Also, from this data it can be shown that the average energy of fission neutrons is about

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energy spectrum for thermal
fission in uranium-235. The values
will vary slightly for other
nuclides.
NEUTRON FLUX SPECTRUM DOE-HDBK-1019/1-93 Reactor Theory (Neutron Characteristics)
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Figure 3 Comparison of Neutron Flux Spectra for
Thermal and Fast Breeder Reactor
Thermal and Fast Breeder Reactor Neutron Spectra
The spectrum of neutron energies produced by fission varies significantly from the energy spectrum, or flux, existing in a reactor at a given time. Figure 3 illustrates the difference in
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neutron flux spectra between a thermal reactor and a fast breeder reactor. The energy distribution of neutrons from fission is essentially the same for both reactors, so the differences in the curve shapes may be attributed to the neutron moderation or slowing down effects.
No attempt is made to thermalize or slow down neutrons in the fast breeder reactor (liquid metal cooled); therefore, an insignificant number of neutrons exist in the thermal range. For the thermal reactor (water moderated), the spectrum of neutrons in the fast region (> 0.1 MeV) has a shape similar to that for the spectrum of neutrons emitted by the fission process.
In the thermal reactor, the flux in the intermediate energy region (1 eV to 0.1 MeV) has approximately a 1/E dependence. That is, if the energy (E) is halved, the flux doubles. This 1/E dependence is caused by the slowing down process, where elastic collisions remove a constant fraction of the neutron energy per collision (on the average), independent of energy; thus, the neutron loses larger amounts of energy per collision at higher energies than at lower energies. The fact that the neutrons lose a constant fraction of energy per collision causes the neutrons to tend to "pile up" at lower energies, that is, a greater number of neutrons exist at the lower energies as a result of this behavior.
vp
2 k T
m
vp
2 k T
m
2 1.38 x 10 16 erg
K
293 K
1.66 x 10 24 g
2.2 x 105 cm
sec
1 m
100 cm
2200 m
sec
Reactor Theory (Neutron Characteristics) DOE-HDBK-1019/1-93 NEUTRON FLUX SPECTRUM
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In the thermal region the neutrons achieve a thermal equilibrium with the atoms of the moderator material. In any given collision they may gain or lose energy, and over successive collisions will gain as much energy as they lose. These thermal neutrons, even at a specific temperature,

to as the Maxwell distribution (e.g., Figure 2). The energies of most thermal neutrons lie close to the most probable energy, but there is a spread of neutrons above and below this value.
Most Probable Neutron Velocities
The most probable velocity (vp) of a thermal neutron is determined by the temperature of the medium and can be determined by Equation (2-13) .
(2-13)
where:
v = most probable velocity of neutron (cm/sec) p
k = Boltzman's constant (1.38 x 10 erg/ K) -16
T = absolute temperature in degrees Kelvin (K)
m = mass of neutron (1.66 x 10 grams) -24
Example:
Calculate the most probable velocities for neutrons in thermal equilibrium with their surroundings at the following temperatures. a) 20 C, b) 260 C.
Solution:
a) Calculate the most probable velocity for 20 C using Equation (2-13).
vp
2 k T
Fixture checkpoint marker-0825-xray is recorded on fixture page 0825.
m
2 1.38 x 10 16 erg
K
533 K
1.66 x 10 24 g
2.977 x 105 cm
sec
1 m
100 cm
2977 m
sec
NEUTRON FLUX SPECTRUM DOE-HDBK-1019/1-93 Reactor Theory (Neutron Characteristics) NP-02 Page 36 Rev. 0
b) Calculate the most probable velocity for 260 C using Equation (2-13).
From these calculations it is evident that the most probable velocity of a thermal neutron increases as temperature increases. The most probable velocity at 20 C is of particular importance since reference data, such as nuclear cross sections, are tabulated for a neutron velocity of 2200 meters per second.
Reactor Theory (Neutron Characteristics) DOE-HDBK-1019/1-93 NEUTRON FLUX SPECTRUM Rev. 0 Page 37 NP-02
Neutron Flux Spectrum Summary
& Prompt neutrons are born at energies between 0.1 MeV and 10 MeV. The average prompt neutron energy is about 2 MeV.
& Fast reactors have a neutron energy spectrum that has the same shape as the prompt neutron energy spectrum.
& Thermal reactors have a neutron energy spectrum that has two pronounced peaks, one in the thermal energy region where the neutrons are in thermal equilibrium with the core materials and another in the fast region at energies where neutrons are produced. The flux in the intermediate region (1 eV to

subjects, which include Mathematics; Classical Physics; Thermodynamics, Heat Transfer, and Fluid Flow; Instrumentation and Control; Electrical Science; Material Science; Mechanical Science; Chemistry; Engineering Symbology, Prints, and Drawings; and Nuclear Physics and Reactor Theory. The handbooks are provided as an aid to DOE nuclear facility contractors.
These handbooks were first published as Reactor Operator Fundamentals Manuals in 1985 for use by DOE category A reactors. The subject areas, subject matter content, and level of detail of the Reactor Operator Fundamentals Manuals were determined from several sources. DOE Category A reactor training managers determined which materials should be included, and served as a primary reference in the initial development phase. Training guidelines from the commercial nuclear power industry, results of job and task analyses, and independent input from contractors and operations-oriented personnel were all considered and included to some degree in developing the text material and learning objectives.
The DOE Fundamentals Handbooks represent the needs of various DOE nuclear facilities' fundamental training requirements. To increase their applicability to nonreactor nuclear facilities, the Reactor Operator Fundamentals Manual learning objectives were distributed to the Nuclear Facility Training Coordination Program Steering Committee for review and comment. To update their reactor-specific content, DOE Category A reactor training managers also reviewed and commented on the content. On the basis of feedback from these sources, information that applied to two or more DOE nuclear facilities was considered generic and was included. The final draft of each of the handbooks was then reviewed by these two groups. This approach has resulted in revised modular handbooks that contain sufficient detail such that each facility may adjust the content to fit their specific needs.
Each handbook contains an abstract, a foreword, an overview, learning objectives, and text material, and is divided into modules so that content and order may be modified by individual DOE contractors to suit their specific training needs. Each handbook is supported by a separate
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examination bank with an answer key.
The DOE Fundamentals Handbooks have been prepared for the Assistant Secretary for Nuclear Energy, Office of Nuclear Safety Policy and Standards, by the DOE Training Coordination Program. This program is managed by EG&G Idaho, Inc.
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NUCLEAR PHYSICS AND REACTOR THEORY
OVERVIEW
The Department of Energy Fundamentals Handbook entitled Nuclear Physics and Reactor Theory was prepared as an information resource for personnel who are responsible for the operation of the Department's nuclear facilities. Almost all processes that take place in a nuclear facility involves the transfer of some type of energy. A basic understanding of nuclear physics and reactor theory is necessary for DOE nuclear facility operators, maintenance personnel, and the technical staff to safely operate and maintain the facility and facility support systems. The information in this handbook is presented to provide a foundation for applying engineering concepts to the job. This knowledge will help personnel understand the impact that their actions may have on the safe and reliable operation of facility components and systems.
The Nuclear Physics and Reactor Theory handbook consists of four modules that are contained in two volumes. The following is a brief description of the information presented in each module of the handbook.
Volume 1 of 2
Module 1 - Atomic and Nuclear Physics

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chart of the nuclides, radioactivity and radioactive decay, neutron interactions and fission, and the interaction of radiation with matter.
Module 2 - Reactor Theory (Nuclear Parameters)
Provides information on reactor theory and neutron characteristics. Includes topics such as neutron sources, neutron flux, neutron cross sections, reaction rates, neutron moderation, and prompt and delayed neutrons.
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NUCLEAR PHYSICS AND REACTOR THEORY
OVERVIEW (Cont.)
Volume 2 of 2
Module 3 - Reactor Theory (Nuclear Parameters)
Explains the nuclear parameters associated with reactor theory. Topics include the neutron life cycle, reactivity and reactivity coefficients, neutron poisons, and control rods.
Module 4 - Reactor Theory (Reactor Operations)
Introduces the reactor operations aspect of reactor theory. Topics include subcritical multiplication, reactor kinetics, and reactor operation.
The information contained in this handbook is not all-encompassing. An attempt to present the entire subject of nuclear physics and reactor theory would be impractical. However, the Nuclear Physics and Reactor Theory handbook presents enough information to provide the reader with the fundamental knowledge necessary to understand the advanced theoretical concepts presented in other subject areas, and to understand basic system and equipment operation.
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Module 1
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TERMINAL OBJECTIVE
1.0 Given sufficient information, DESCRIBE atoms, including components, structure, and nomenclature.
ENABLING OBJECTIVES
1.1 STATE the characteristics of the following atomic particles, including mass, charge, and location within the atom:
a. Proton
b. Neutron
c. Electron
1.2 DESCRIBE the Bohr model of an atom.
1.3 DEFINE the following terms:
a. Nuclide c. Atomic number
b. Isotope d. Mass number
1.4 Given the standard A

a. Number of protons
b. Number of neutrons
c. Number of electrons
1.5 DESCRIBE the three forces that act on particles within the nucleus and affect the stability of the nucleus.
1.6 DEFINE the following terms:
a. Enriched uranium
b. Depleted uranium
1.7 DEFINE the following terms:
a. Mass defect
b. Binding energy
1.8 Given the atomic mass for a nuclide and the atomic masses of a neutron, proton, and electron, CALCULATE the mass defect and binding energy of the nuclide.
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OBJECTIVES DOE-HDBK-1019/1-93 Atomic and Nuclear Physics
TERMINAL OBJECTIVE
2.0 Given necessary references, DESCRIBE the various modes of radioactive decay.
ENABLING OBJECTIVES
2.1 DESCRIBE the following processes:
a. Alpha decay d. Electron capture
b. Beta-minus decay e. Internal conversions
c. Beta-plus decay f. Isomeric transitions
2.2 Given a Chart of the Nuclides, WRITE the radioactive decay chain for a nuclide.
2.3 EXPLAIN why one or more gamma rays typically accompany particle emission.
2.4 Given the stability curve on the Chart of the Nuclides, DETERMINE the type of radioactive decay that the nuclides in each region of the chart will typically undergo.
2.5 DEFINE the following terms:
a. Radioactivity d. Radioactive decay constant
b. Curie e. Radioactive half-life
c. Becquerel
Fixture checkpoint marker-0831-xray is recorded on fixture page 0831.
2.6 Given the number of atoms and either the half-life or decay constant of a nuclide, CALCULATE the activity.
2.7 Given the initial activity and the decay constant of a nuclide, CALCULATE the activity at any later time.
2.8 CONVERT between the half-life and decay constant for a nuclide.
2.9 Given the Chart of the Nuclides and the original activity, PLOT the radioactive decay curve for a nuclide on either linear or semi-log coordinates.
2.10 DEFINE the following terms:
a. Radioactive equilibrium
b. Transient radioactive equilibrium
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Atomic and Nuclear Physics DOE-HDBK-1019/1-93 OBJECTIVES
TERMINAL OBJECTIVE
3.0 Without references, DESCRIBE the different nuclear interactions initiated by neutrons.
ENABLING OBJECTIVES
3.1 DESCRIBE the following scattering interactions between a neutron and a nucleus:
a. Elastic scattering

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3.2 STATE the conservation laws that apply to an elastic collision between a neutron and a nucleus.
3.3 DESCRIBE the following reactions where a neutron is absorbed in a nucleus:
a. Radiative capture
b. Particle ejection

TERMINAL OBJECTIVE
4.0 Without references, DESCRIBE the fission process.
ENABLING OBJECTIVES
4.1 EXPLAIN the fission process using the liquid drop model of a nucleus.
4.2 DEFINE the following terms:
a. Excitation energy
b. Critical energy
4.3 DEFINE the following terms:
a. Fissile material
b. Fissionable material
c. Fertile material
4.4 DESCRIBE the processes of transmutation, conversion, and breeding.
4.5 DESCRIBE the curve of Binding Energy per Nucleon versus mass number and give a qualitative description of the reasons for its shape.
4.6 EXPLAIN why only the heaviest nuclei are easily fissioned.
4.7 EXPLAIN why uranium-235 fissions with thermal neutrons and uranium-238 fissions only with fast neutrons.
4.8 CHARACTERIZE the fission products in terms of mass groupings and radioactivity.
4.9 Given the nuclides involved and their masses, CALCULATE the energy released from fission.
4.10 Given the curve of Binding Energy per Nucleon versus mass number, CALCULATE the energy released from fission.

TERMINAL OBJECTIVE
5.0 Without references, DESCRIBE how the various types of radiation interact with matter.
ENABLING OBJECTIVES
5.1 DESCRIBE interactions of the following with matter:
a. Alpha particle c. Positron
b. Beta particle d. Neutron
Fixture checkpoint marker-0832-xray is recorded on fixture page 0832.
5.2 DESCRIBE the following ways that gamma radiation interacts with matter:
a. Compton scattering
b. Photoelectric effect
c. Pair production

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ATOMIC NATURE OF MATTER

matter that retains the properties of an element. Atoms themselves are composed of smaller particles, but these smaller particles no longer have the same properties as the overall element.

EO 1.1 STATE the characteristics of the following atomic particles, including mass, charge, and location within the atom:

a. Proton
b. Neutron
c. Electron

EO 1.2 DESCRIBE the Bohr model of an atom.

EO 1.3 DEFINE the following terms:

a. Nuclide c. Atomic number
b. Isotope d. Mass number

EO 1.4 Given the standard A
ZX notation for a particular nuclide,
Fixture checkpoint marker-0833-xray is recorded on fixture page 0833.
DETERMINE the following:

a. Number of protons
b. Number of neutrons
c. Number of electrons

EO 1.5 DESCRIBE the three forces that act on particles within the nucleus and affect the stability of the nucleus.

Structure of Matter

Early Greek philosophers speculated that the earth was made up of different combinations of basic substances, or elements. They considered these basic elements to be earth, air, water, and fire. Modern science shows that the early Greeks held the correct concept that matter consists of a combination of basic elements, but they incorrectly identified the elements.

In 1661 the English chemist Robert Boyle published the modern criterion for an element. He defined an element to be a basic substance that cannot be broken down into any simpler substance after it is isolated from a compound, but can be combined with other elements to form compounds. To date, 105 different elements have been confirmed to exist, and researchers claim to have discovered three additional elements. Of the 105 confirmed elements, 90 exist in nature and 15 are man-made.

Another basic concept of matter that the Greeks debated was whether matter was continuous or discrete. That is, whether matter could be continuously divided and subdivided into ever smaller particles or whether eventually an indivisible particle would be encountered. Democritus in about 450 B.C. argued that substances were ultimately composed of small, indivisible particles that he labeled atoms. He further suggested that different substances were composed of different atoms or combinations of atoms, and that one substance could be converted into another by rearranging the atoms. It was impossible to conclusively prove or disprove this proposal for more than 2000 years.

The modern proof for the atomic nature of matter was first proposed by the English chemist John Dalton in 1803. Dalton stated that each chemical element possesses a particular kind of atom, and any quantity of the element is made up of identical atoms of this kind. What distinguishes one element from another element is the kind of atom of which it consists, and the basic physical difference between kinds of atoms is their weight.

Subatomic Particles

impossible to divide the atom into even smaller parts. All of the results of chemical experiments during this time indicated that the atom was indivisible. Eventually, experimentation into electricity and radioactivity indicated that particles of matter smaller than the atom did indeed exist. In 1906, J. J. Thompson won the Nobel Prize in physics for establishing the existence of electrons. Electrons are negatively-charged particles that have 1/1835 the mass of the hydrogen atom. Soon after the discovery of electrons, protons were discovered. Protons are relatively large particles that have almost the same mass as a hydrogen atom and a positive charge equal in magnitude (but opposite in sign) to that of the electron. The third subatomic particle to be discovered, the neutron, was not found until 1932. The neutron has almost the same mass as the proton, but it is electrically neutral.

Bohr Model of the Atom

The British physicist Ernest Rutherford postulated that the positive charge in an atom is concentrated in a small region called a nucleus at the center of the atom with electrons existing in orbits around it. Niels Bohr, coupling Rutherford's postulation with the quantum theory introduced by Max Planck, proposed that the atom consists of a dense nucleus of protons surrounded by electrons traveling in discrete orbits at fixed distances from the nucleus. An electron in one of these orbits or shells has a specific or discrete quantity of energy (quantum). When an electron moves from one allowed orbit to another allowed orbit, the energy difference between the two states is emitted or absorbed in the form of a single quantum of radiant energy

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called a photon. Figure 1 is Bohr's model of the hydrogen atom showing an electron as having just dropped from the third shell to the first shell with the emission of a photon that has an energy = hv. (h = Planck's constant = 6.63 x 10-34 J-s and v = frequency of the photon.) Bohr's theory was the first to successfully account for the discrete energy levels of this radiation as measured in the laboratory. Although Bohr's atomic model is designed specifically to explain the hydrogen atom, his theories apply generally to the structure of all atoms. Additional information on electron shell theory can be found in the Chemistry Fundamentals Handbook.

Figure 1 Bohr's Model of the Hydrogen Atom

Properties of the three subatomic particles are listed in Table 1.

TABLE 1
Properties of Subatomic Particles

Particle	Location	Charge	Mass
Neutron	Nucleus	none	1.008665 amu
Proton	Nucleus	+1	1.007277 amu
Electron	Shells around nucleus	-1	0.0005486 amu

Measuring Units on the Atomic Scale

The size and mass of atoms are so small that the use of normal measuring units, while possible, is often inconvenient. Units of measure have been defined for mass and energy on the atomic scale to make measurements more convenient to express. The unit of measure for mass is the atomic mass unit (amu). One atomic mass unit is equal to 1.66 x 10-24 grams. The reason for this particular value for the atomic mass unit will be discussed in a later chapter. Note from Table 1 that the mass of a neutron and a proton are both about 1 amu. The unit for energy is the electron volt (eV). The electron volt is the amount of energy acquired by a single electron when it falls through a potential difference of one volt. One electron volt is equivalent to

SYNTHETIC FIXTURE PAGE 0835 OF 1200

Nuclides

The total number of protons in the nucleus of an atom is called the atomic number of the atom and is given the symbol Z. The number of electrons in an electrically-neutral atom is the same as the number of protons in the nucleus. The number of neutrons in a nucleus is known as the neutron number and is given the symbol N. The mass number of the nucleus is the total number of nucleons, that is, protons and neutrons in the nucleus. The mass number is given the symbol A and can be found by the equation Z + N = A.

Each of the chemical elements has a unique atomic number because the atoms of different elements contain a different number of protons. The atomic number of an atom identifies the particular element.

Each type of atom that contains a unique combination of protons and neutrons is called a nuclide. Not all combinations of numbers of protons and neutrons are possible, but about 2500 specific nuclides with unique combinations of neutrons and protons have been identified. Each nuclide is denoted by the chemical symbol of the element with the atomic number written as a subscript and the mass number written as a superscript, as shown in Figure 2. Because each element has a unique name, chemical symbol, and atomic number, only one of the three is necessary to identify the element. For this reason nuclides can also be identified by either the chemical name or the chemical symbol followed by the mass number (for example, U-235 or uranium-235).

Figure 2 Nomenclature for Identifying Nuclides

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Another common format is to use the abbreviation of the chemical element with the mass number superscripted (for example, 235U). In this handbook the format used in the text will usually be the element's name followed by the mass number. In equations and tables, the format in Figure 2 will usually be used.

Example:

State the name of the element and the number of protons, electrons, and neutrons in the nuclides listed below.

1
1H
10
5B
14
7N
11
4
4
8Cd
23
9

4Pu
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Solution:
The name of the element can be found from the Periodic Table (refer to Chemistry Fundamentals Handbook) or the Chart of the Nuclides (to be discussed later). The number of protons and electrons are equal to Z. The number of neutrons is equal to Z - A.
Nuclide Element Protons Electrons Neutrons
1
1H hydrogen 1 1 0
10
5B boron 5 5 5
14
7N nitrogen 7 7 7
11
4
4
8Cd cadmium 48 48 66
23
9
9
4Pu plutonium 94 94 145
Isotopes
Isotopes are nuclides that have the same atomic number and are therefore the same element, but differ in the number of neutrons. Most elements have a few stable isotopes and several unstable, radioactive isotopes. For example, oxygen has three stable isotopes that can be found in nature (oxygen-16, oxygen-17, and oxygen-18) and eight radioactive isotopes. Another example is hydrogen, which has two stable isotopes (hydrogen-1 and hydrogen-2) and a single radioactive isotope (hydrogen-3).
The isotopes of hydrogen are unique in that they are each commonly referred to by a unique name instead of the common chemical element name. Hydrogen-1 is almost always referred to as hydrogen, but the term protium is infrequently used also. Hydrogen-2 is commonly called deuterium and symbolized 2
1D. Hydrogen-3 is commonly called tritium and symbolized 31T. This
Fixture checkpoint marker-0836-xray is recorded on fixture page 0836.
text will normally use the symbology 2
1H and 31H for deuterium and tritium, respectively.
Atomic and Nuclear Radii
The size of an atom is difficult to define exactly due to the fact that the electron cloud, formed by the electrons moving in their various orbitals, does not have a distinct outer edge. A reasonable measure of atomic size is given by the average distance of the outermost electron from the nucleus. Except for a few of the lightest atoms, the average atomic radii are approximately the same for all atoms, about 2 x 10 -8 cm.
Like the atom the nucleus does not have a sharp outer boundary. Experiments have shown that the nucleus is shaped like a sphere with a radius that depends on the atomic mass number of the atom. The relationship between the atomic mass number and the radius of the nucleus is shown in the following equation.

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r = (1.25 x 10 -13 cm) A1/3
where:
r = radius of the nucleus (cm)
A = atomic mass number (dimensionless)
The values of the nuclear radii for some light, intermediate, and heavy nuclides are shown in Table 2.
TABLE 2
Calculated Values for Nuclear
Radii
Nuclide Radius of Nucleus
1
Fixture checkpoint marker-0837-xray is recorded on fixture page 0837.
1H 1.25 x 10 -13 cm
10
5B 2.69 x 10 -13 cm
5
2
6
6Fe 4.78 x 10 -13 cm
17
7
8
2Hf 7.01 x 10 -13 cm
23
9
8
2U 7.74 x 10 -13 cm
25
9
2
8Cf 7.89 x 10 -13 cm
From the table, it is clear that the radius of a typical atom (e.g. 2 x 10 -8 cm) is more than 25,000 times larger than the radius of the largest nucleus.
Nuclear Forces
In the Bohr model of the atom, the nucleus consists of positively-charged protons and electrically■neutra as nucleons. One problem that the Bohr model of the atom presented was accounting for an attractive force to overcome the repulsive force between protons.
Two forces present in the nucleus are (1) electrostatic forces between charged particles and (2) gravitational forces between any two objects that have mass. It is possible to calculate the magnitude of the gravitational force and electrostatic force based upon principles from classical physics.
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Fg
G m1 m2
r
2
Fe

r
2
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Newton stated that the gravitational force between two bodies is directly proportional to the masses of the two bodies and inversely proportional to the square of the distance between the bodies. This relationship is shown in the equation below.
where:
F = gravitational force (newtons) g
m = mass of first body (kilograms) 1
m = mass of second body (kilograms) 2
G = gravitational constant (6.67 x 10 N-m /kg)
-11 2 2
r = distance between particles (meters)
The equation illustrates that the larger the masses of the objects or the smaller the distance between the objects, the greater the gravitational force. So even though the masses of nucleons are very small, the fact that the distance between nucleons is extremely short may make the gravitational force significant. It is necessary to calculate the value for the gravitational force and compare it to the value for other forces to determine the significance of the gravitational force
Fixture checkpoint marker-0838-xray is recorded on fixture page 0838.
in the nucleus. The gravitational force between two protons that are separated by a distance of 10 meters is about 10 newtons. -20 -24
Coulomb's Law can be used to calculate the force between two protons. The electrostatic force is directly proportional to the electrical charges of the two particles and inversely proportional to the square of the distance between the particles. Coulomb's Law is stated as the following equation.
where:
F = electrostatic force (newtons) e
K = electrostatic constant (9.0 x 10 N-m /C)
9 2 2
Q = charge of first particle (coulombs) 1
Q = charge of second particle (coulombs) 2
r = distance between particles (meters)
Using this equation, the electrostatic force between two protons that are separated by a distance of 10 meters is about 10 newtons. Comparing this result with the calculation of the
-20 12
gravitational force (10 newtons) shows that the gravitational force is so small that it can be
-24
neglected.
Atomic and Nuclear Physics DOE-HDBK-1019/1-93 ATOMIC NATURE OF MATTER
If only the electrostatic and gravitational forces existed in the nucleus, then it would be impossible to have stable nuclei composed of protons and neutrons. The gravitational forces are much too small to hold the nucleons together compared to the electrostatic forces repelling the protons. Since stable atoms of neutrons and protons do exist, there must be another attractive force acting within the nucleus. This force is called the nuclear force.
The nuclear force is a strong attractive force that is independent of charge. It acts equally only between pairs of neutrons, pairs of protons, or a neutron and a proton. The nuclear force has a very short range; it acts only over distances approximately equal to the diameter of the nucleus

than the repulsive electrostatic force between protons.
TABLE 3
Forces Acting in the Nucleus
Force Interaction Range
Gravitational Very weak attractive force
between all nucleons Relatively long
Electrostatic Strong repulsive force between
like charged particles (protons) Relatively long
Nuclear Force Strong attractive force between
all nucleons Extremely short
In stable atoms, the attractive and repulsive forces in the nucleus balance. If the forces do not balance, the atom cannot be stable, and the nucleus will emit radiation in an attempt to achieve a more stable configuration.
Summary
The important information in this chapter is summarized on the following page.

Atomic Nature of Matter Summary
Atoms consist of three basic subatomic particles. These particles are the proton, the neutron, and the electron.
Protons are particles that have a positive charge, have about the same mass as a hydrogen atom, and exist in the nucleus of an atom.
Neutrons are particles that have no electrical charge, have about the same mass as a hydrogen atom, and exist in the nucleus of an atom.
Electrons are particles that have a negative charge, have a mass about eighteen hundred times smaller than the mass of a hydrogen atom, and exist in orbital shells
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around the nucleus of an atom.
The Bohr model of the atom consists of a dense nucleus of protons and neutrons (nucleons) surrounded by electrons traveling in discrete orbits at fixed distances from the nucleus.
Nuclides are atoms that contain a particular number of protons and neutrons.
Isotopes are nuclides that have the same atomic number and are therefore the same element, but differ in the number of neutrons.
The atomic number of an atom is the number of protons in the nucleus.
The mass number of an atom is the total number of nucleons (protons and neutrons) in the nucleus.
The notation A
ZX is used to identify a specific nuclide. "Z" represents the atomic number, which is equal to the number of protons. "A" represents the mass number, which is equal to the number of nucleons. "X" represents the chemical symbol of the element.
Number of protons = Z
Number of electrons = Z
Number of neutrons = A - Z
The stability of a nucleus is determined by the different forces interacting within it. The electrostatic force is a relatively long-range, strong, repulsive force that acts between the positively charged protons. The nuclear force is a relatively

long range, relatively weak attraction between masses, is negligible compared to the other forces.

CHART OF THE NUCLIDES

The Chart of the Nuclides, like the Periodic Table, is a convenient format for presenting a large amount of scientific information in an organized manner.

EO 1.6 DEFINE the following terms:

a. Enriched uranium

b. Depleted uranium

Chart of the Nuclides

A tabulated chart called the Chart of the Nuclides lists the stable and unstable nuclides in addition to pertinent information about each one. Figure 3 shows a small portion of a typical chart. This chart plots a box for each individual nuclide, with the number of protons (Z) on the vertical axis and the number of neutrons (N = A - Z) on the horizontal axis.

The completely gray squares indicate stable isotopes. Those in white squares are artificially radioactive, meaning that they are produced by artificial techniques and do not occur naturally. By consulting a complete chart, other types of isotopes can be found, such as naturally occurring radioactive types (but none are found in the region of the chart that is illustrated in Figure 3). Located in the box on the far left of each horizontal row is general information about the element. The box contains the chemical symbol of the element in addition to the average atomic weight of the naturally occurring substance and the average thermal neutron absorption cross section, which will be discussed in a later module. The known isotopes (elements with the same atomic number Z but different mass number A) of each element are listed to the right.

Figure 3 Nuclide Chart for Atomic Numbers 1 to 6

Information for Stable Nuclides

For the stable isotopes, in addition to the symbol and the atomic mass number, the number percentage of each isotope in the naturally occurring element is listed, as well as the thermal

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neutron activation cross section and the mass in atomic mass units (amu). A typical block for a stable nuclide from the Chart of the Nuclides is shown in Figure 4.

Figure 4 Stable Nuclides

Information for Unstable Nuclides

For unstable isotopes the additional information includes the half life, the mode of decay (for example, β^-, α), the total disintegration energy in MeV (million electron volts), and the mass in amu when available. A typical block for an unstable nuclide from the Chart of the Nuclides is shown in Figure 5.

Figure 5 Unstable Nuclides

Neutron - Proton Ratios

Figure 6 shows the distribution of the stable nuclides plotted on the same axes as the Chart of

nucleus becomes larger. For helium-4 (2 protons and 2 neutrons) and oxygen-16 (8 protons and 8 neutrons) this ratio is unity. For indium-115 (49 protons and 66 neutrons) the ratio of neutrons to protons has increased to 1.35, and for uranium-238 (92 protons and 146 neutrons) the neutron■to-proton
Figure 6 Neutron - Proton Plot of the Stable Nuclides
If a heavy nucleus were to split into two fragments, each fragment would form a nucleus that would have approximately the same neutron-to-proton ratio as the heavy nucleus. This high neutron-to-proton ratio places the fragments below and to the right of the stability curve displayed by Figure 6. The instability caused by this excess of neutrons is generally rectified by successive beta emissions, each of which converts a neutron to a proton and moves the nucleus toward a more stable neutron-to-proton ratio.
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■
Atomic and Nuclear Physics DOE-HDBK-1019/1-93 CHART OF THE NUCLIDES
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Natural Abundance of Isotopes
The relative abundance of an isotope in nature compared to other isotopes of the same element is relatively constant. The Chart of the Nuclides presents the relative abundance of the naturally occurring isotopes of an element in units of atom percent. Atom percent is the percentage of the atoms of an element that are of a particular isotope. Atom percent is abbreviated as a/o.
For example, if a cup of water contains 8.23 x 10 atoms of oxygen, and the isotopic abundance 24 of oxygen-18 is 0.20%, then there are 1.65 x 10 atoms of oxygen-18 in the cup. 22
The atomic weight for an element is defined as the average atomic weight of the isotopes of the element. The atomic weight for an element can be calculated by summing the products of the isotopic abundance of the isotope with the atomic mass of the isotope.
Example:
Calculate the atomic weight for the element lithium. Lithium-6 has an atom percent abundance of 7.5% and an atomic mass of 6.015122 amu. Lithium-7 has an atomic abundance of 92.5% and an atomic mass of 7.016003 amu.
Solution:
The other common measurement of isotopic abundance is weight percent (w/o). Weight percent is the percent weight of an element that is a particular isotope. For example, if a sample of material contained 100 kg of uranium that was 28 w/o uranium-235, then 28 kg of uranium-235 was present in the sample.
Enriched and Depleted Uranium
Natural uranium mined from the earth contains the isotopes uranium-238, uranium-235 and uranium-234. The majority (99.2745%) of all the atoms in natural uranium are uranium-238. Most of the remaining atoms (0.72%) are uranium-235, and a slight trace (0.0055%) are uranium-234. Although all isotopes of uranium have similar chemical properties, each of the isotopes has significantly different nuclear properties. For reasons that will be discussed in later modules, the isotope uranium-235 is usually the desired material for use in reactors.
A vast amount of equipment and energy are expended in processes that separate the isotopes of uranium (and other elements). The details of these processes are beyond the scope of this module. These processes are called enrichment processes because they selectively increase the proportion of a particular isotope. The enrichment process typically starts with feed material that has the proportion of isotopes that occur naturally. The process results in two types of
CHART OF THE NUCLIDES DOE-HDBK-1019/1-93 Atomic and Nuclear Physics
In the case of uranium, the natural uranium ore is 0.72 a/o uranium-235. The desired outcome

uranium in which the isotope uranium-235 has a concentration greater than its natural value. The enrichment process will also result in the byproduct of depleted uranium. Depleted uranium is defined as uranium in which the isotope uranium-235 has a concentration less than its natural value. Although depleted uranium is referred to as a by-product of the enrichment process, it does have uses in the nuclear field and in commercial and defense industries.

Summary

The important information in this chapter is summarized below.

Chart of the Nuclides Summary

Enriched uranium is uranium in which the isotope uranium-235 has a concentration greater than its natural value of 0.7%.

Depleted uranium is uranium in which the isotope uranium-235 has a concentration less than its natural value of 0.7%.

MASS DEFECT AND BINDING ENERGY

The separate laws of Conservation of Mass and Conservation of Energy are not applied strictly on the nuclear level. It is possible to convert between mass and

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energy. Instead of two separate conservation laws, a single conservation law states that the sum of mass and energy is conserved. Mass does not magically appear and disappear at random. A decrease in mass will be accompanied by a corresponding increase in energy and vice versa.

EO 1.7 DEFINE the following terms:

a. Mass defect

b. Binding energy

EO 1.8 Given the atomic mass for a nuclide and the atomic masses of a neutron, proton, and electron, CALCULATE the mass defect and binding energy of the nuclide.

Mass Defect

Careful measurements have shown that the mass of a particular atom is always slightly less than the sum of the masses of the individual neutrons, protons, and electrons of which the atom consists. The difference between the mass of the atom and the sum of the masses of its parts is called the mass defect (Δm). The mass defect can be calculated using Equation (1-1). In calculating the mass defect it is important to use the full accuracy of mass measurements because the difference in mass is small compared to the mass of the atom. Rounding off the masses of atoms and particles to three or four significant digits prior to the calculation will result in a calculated mass defect of zero.

$$\Delta m = [\, Z(mp + me) + (A-Z)mn \,] - matom \quad (1\text{-}1)$$

where:

Δm = mass defect (amu)

mp = mass of a proton (1.007277 amu)

mn = mass of a neutron (1.008665 amu)

me = mass of an electron (0.000548597 amu)

matom = mass of nuclide A

ZX (amu)

Z = atomic number (number of protons)

A = mass number (number of nucleons)

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Example:
Calculate the mass defect for lithium-7. The mass of lithium-7 is 7.016003 amu.
Solution:
∆ m Z m
p
me A Z mn matom
∆ m 3 1.007826 amu 7 3 1.008665 amu 7.016003 amu
∆ m 0.0421335 amu
Binding Energy
The loss in mass, or mass defect, is due to the conversion of mass to binding energy when the nucleus is formed. Binding energy is defined as the amount of energy that must be supplied to a nucleus to completely separate its nuclear particles (nucleons). It can also be understood as the amount of energy that would be released if the nucleus was formed from the separate particles. Binding energy is the energy equivalent of the mass defect. Since the mass defect was converted to binding energy (BE) when the nucleus was formed, it is possible to calculate the binding energy using a conversion factor derived by the mass-energy relationship from Einstein's Theory of Relativity.
Einstein's famous equation relating mass and energy is E = mc2 where c is the velocity of light (c = 2.998 x 108 m/sec). The energy equivalent of 1 amu can be determined by inserting this quantity of mass into Einstein's equation and applying conversion factors.
E m c 2
1 amu
1.6606 x 10 27 kg
1 amu
Fixture checkpoint marker-0843-xray is recorded on fixture page 0843.
2.998 x 108 m
sec
2
1 N
1
kg m
sec2
1 J
1 N m
1.4924 x 10 10 J 1 MeV
1.6022 x 10 13 J
931.5 MeV
Conversion Factors:
1 amu = 1.6606 x 10 -27 kg
1 newton = 1 kg-m/sec2
1 joule = 1 newton-meter
1 MeV = 1.6022 x 10-13 joules
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Atomic and Nuclear Physics DOE-HDBK-1019/1-93 MASS DEFECT AND BINDING ENERGY
Since 1 amu is equivalent to 931.5 MeV of energy, the binding energy can be calculated using Equation (1-2).
(1-2)
B.E. ∆m

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1 amu
Example:
Calculate the mass defect and binding energy for uranium-235. One uranium-235 atom has a mass of 235.043924 amu.
Solution:
Step 1: Calculate the mass defect using Equation (1-1).
Δm Z m
p
me A Z mn matom
Δm 92 1.007826 amu 235 92 1.008665 amu 235.043924 amu
Δm 1.91517 amu
Step 2: Use the mass defect and Equation (1-2) to calculate the binding energy.
B.E. Δm
931.5 MeV
1 amu
1.91517 amu
931.5 MeV
1 amu
1784 MeV
Energy Levels of Atoms
The electrons that circle the nucleus move in fairly well-defined orbits. Some of these electrons are more tightly bound in the atom than others. For example, only 7.38 eV is required to remove the outermost electron from a lead atom, while 88,000 eV is required to remove the innermost electron. The process of removing an electron from an atom is called ionization, and the energy required to remove the electron is called the ionization energy.
In a neutral atom (number of electrons = Z) it is possible for the electrons to be in a variety of different orbits, each with a different energy level. The state of lowest energy is the one in which the atom is normally found and is called the ground state. When the atom possesses more energy than its ground state energy, it is said to be in an excited state.

Fixture checkpoint marker-0844-xray is recorded on fixture page 0844.
An atom cannot stay in the excited state for an indefinite period of time. An excited atom will eventually transition to either a lower-energy excited state, or directly to its ground state, by emitting a discrete bundle of electromagnetic energy called an x-ray. The energy of the x-ray will be equal to the difference between the energy levels of the atom and will typically range from several eV to 100,000 eV in magnitude.
Energy Levels of the Nucleus
The nucleons in the nucleus of an atom, like the electrons that circle the nucleus, exist in shells that correspond to energy states. The energy shells of the nucleus are less defined and less understood than those of the electrons. There is a state of lowest energy (the ground state) and discrete possible excited states for a nucleus. Where the discrete energy states for the electrons of an atom are measured in eV or keV, the energy levels of the nucleus are considerably greater and typically measured in MeV.
A nucleus that is in the excited state will not remain at that energy level for an indefinite period. Like the electrons in an excited atom, the nucleons in an excited nucleus will transition towards their lowest energy configuration and in doing so emit a discrete bundle of electromagnetic radiation called a gamma ray (γ-ray). The only differences between x-rays and γ-rays are their

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The ground state and the excited states of
Figure 7 Energy Level Diagram - Nickel-60
a nucleus can be depicted in a nuclear
energy-level diagram. The nuclear
energy-level diagram consists of a stack of
horizontal bars, one bar for each of the
excited states of the nucleus. The vertical
distance between the bar representing an
excited state and the bar representing the
ground state is proportional to the energy
level of the excited state with respect to
the ground state. This difference in
energy between the ground state and the
excited state is called the excitation energy
of the excited state. The ground state of
a nuclide has zero excitation energy. The
bars for the excited states are labeled with
their respective energy levels. Figure 7 is
the energy level diagram for nickel-60.
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Atomic and Nuclear Physics DOE-HDBK-1019/1-93 MASS DEFECT AND BINDING ENERGY
Summary
The important information in this chapter is summarized below.
Mass Defect and Binding Energy Summary
Mass defect is the difference between the mass of the atom and the sum of the
masses of its constituent parts.
Binding energy is the amount of energy that must be supplied to a nucleus to
completely separate its nuclear particles. Binding energy is the energy equivalent
of the mass defect.
Mass defect can be calculated by using the equation below.
$\Delta m = [Z(mp + me) + (A-Z)mn] - matom$
Binding energy can be calculated by multiplying the mass defect by the factor
of 931.5 MeV per amu.
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MODES OF RADIOACTIVE DECAY DOE-HDBK-1019/1-93 Atomic and Nuclear Physics
MODES OF RADIOACTIVE DECAY
Most atoms found in nature are stable and do not emit particles or energy that
change form over time. Some atoms, however, do not have stable nuclei. These
Fixture checkpoint marker-0845-xray is recorded on fixture page 0845.
atoms emit radiation in order to achieve a more stable configuration.
EO 2.1 DESCRIBE the following processes:
a. Alpha decay d. Electron capture
b. Beta-minus decay e. Internal conversions
c. Beta-plus decay f. Isomeric transitions
EO 2.2 Given a Chart of the Nuclides, WRITE the radioactive decay
chain for a nuclide.
EO 2.3 EXPLAIN why one or more gamma rays typically accompany
particle emission.

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DETERMINE the type of radioactive decay that the nuclides in each region of the chart will typically undergo.

Stability of Nuclei

As mass numbers become larger, the ratio of neutrons to protons in the nucleus becomes larger for the stable nuclei. Non-stable nuclei may have an excess or deficiency of neutrons and undergo a transformation process known as beta (β) decay. Non-stable nuclei can also undergo a variety of other processes such as alpha (α) or neutron (n) decay. As a result of these decay processes, the final nucleus is in a more stable or more tightly bound configuration.

Natural Radioactivity

In 1896, the French physicist Becquerel discovered that crystals of a uranium salt emitted rays that were similar to x-rays in that they were highly penetrating, could affect a photographic plate, and induced electrical conductivity in gases. Becquerel's discovery was followed in 1898 by the identification of two other radioactive elements, polonium and radium, by Pierre and Marie Curie.

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Fixture checkpoint marker-0846-xray is recorded on fixture page 0846.

Atomic and Nuclear Physics DOE-HDBK-1019/1-93 MODES OF RADIOACTIVE DECAY

Heavy elements, such as uranium or thorium, and their unstable decay chain elements emit radiation in their naturally occurring state. Uranium and thorium, present since their creation at the beginning of geological time, have an extremely slow rate of decay. All naturally occurring nuclides with atomic numbers greater than 82 are radioactive.

Nuclear Decay

Whenever a nucleus can attain a more stable (i.e., more tightly bound) configuration by emitting radiation, a spontaneous disintegration process known as radioactive decay or nuclear decay may occur. In practice, this "radiation" may be electromagnetic radiation, particles, or both.

Detailed studies of radioactive decay and nuclear reaction processes have led to the formulation of useful conservation principles. The four principles of most interest in this module are discussed below.

Conservation of electric charge implies that charges are neither created nor destroyed. Single positive and negative charges may, however, neutralize each other. It is also possible for a neutral particle to produce one charge of each sign.

Conservation of mass number does not allow a net change in the number of nucleons. However, the conversion of a proton to a neutron and vice versa is allowed.

Conservation of mass and energy implies that the total of the kinetic energy and the energy equivalent of the mass in a system must be conserved in all decays and reactions. Mass can be converted to energy and energy can be converted to mass, but the sum of mass and energy must be constant.

Conservation of momentum is responsible for the distribution of the available kinetic energy among product nuclei, particles, and/or radiation. The total amount is the same before and after the reaction even though it may be distributed differently among entirely different nuclides and/or particles.

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■ ■ ■ ■ ■ ■

■ ■ ■ ■ ■

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Alpha Decay (■)

Alpha decay is the emission of alpha particles (helium nuclei) which may be represented as either

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2 2
and the mass number decreased by 4. An example is uranium-234 which decays by the ejection of an alpha particle accompanied by the emission of a 0.068 MeV gamma.
The combined kinetic energy of the daughter nucleus (Thorium-230) and the ■ particle is designated as KE. The sum of the KE and the gamma energy is equal to the difference in mass between the original nucleus (Uranium-234) and the final particles (equivalent to the binding energy released, since ■m = BE). The alpha particle will carry off as much as 98% of the kinetic energy and, in most cases, can be considered to carry off all the kinetic energy.
Beta Decay (■)
Beta decay is the emission of electrons of nuclear rather than orbital origin. These particles are electrons that have been expelled by excited nuclei and may have a charge of either sign.
If both energy and momentum are to be conserved, a third type of particle, the neutrino, ■, must be involved. The neutrino is associated with positive electron emission, and its antiparticle, the antineutrino, , is emitted with a negative electron. These uncharged particles have only the weakest interaction with matter, no mass, and travel at the speed of light. For all practical purposes, they pass through all materials with so few interactions that the energy they possess cannot be recovered. The neutrinos and antineutrinos are included here only because they carry a portion of the kinetic energy that would otherwise belong to the beta particle, and therefore, must be considered for energy and momentum to be conserved. They are normally ignored since they are not significant in the context of nuclear reactor applications.
Negative electron emission, represented as e, -1 -1■, or simply as e or ■ , effectively converts a neutro to a proton, thus increasing the atomic number by one and leaving the mass number unchanged.
Fixture checkpoint marker-0847-xray is recorded on fixture page 0847.
This is a common mode of decay for nuclei with an excess of neutrons, such as fission fragments below and to the right of the neutron-proton stability curve (refer to Figure 6). An example of a typical beta minus-decay reaction is shown below.
■ ■
■
■ ■
■ ■ ■ ■
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Figure 9 Orbital Electron Capture
Positively charged electrons (beta-plus) are known as positrons. Except for sign, they are nearly identical to their negatively charged cousins. When a positron, represented as e, ■, or simply as +1 +1 0 0
e or ■ , is ejected from the nucleus, the atomic number is decreased by one and the mass number + + remains unchanged. A proton has been converted to a neutron. An example of a typical positron (beta-plus) decay is shown below.
Electron Capture (EC, K-capture)
Nuclei having an excess of protons may capture an electron from one of the inner orbits which immediately combines with a proton in the nucleus to form a neutron. This process is called electron capture (EC). The electron is normally captured from the innermost orbit (the K-shell), and, consequently, this process is sometimes called K-capture. The following example depicts electron capture.
A neutrino is formed at the same
time that the neutron is formed, and
energy carried off by it serves to

that is available due to the atomic mass of the product being appreciably less than that of the parent will appear as gamma radiation. Also, there will always be characteristic x-rays given off when an electron from one of the higher energy shells moves in to fill the vacancy in the K-shell.
Electron capture is shown graphically in Figure 8.
Electron capture and positron emission result in the production of the same daughter product, and they exist as competing processes.
For positron emission to occur, however, the mass of the daughter product must be less than the mass of the parent by an amount equal to at least twice the mass of an electron. This mass difference between the parent and daughter is necessary to account for two items present in the parent but not in the daughter. One item is the positron ejected from the nucleus of the parent. The other item is that the daughter product has one less orbital electron than the parent. If this requirement is not met, then orbital electron capture takes place exclusively.
■
■ ■ ■ ■
■ ■

Gamma Emission (■)
Gamma radiation is a high-energy electromagnetic radiation that originates in the nucleus. It is emitted in the form of photons, discrete bundles of energy that have both wave and particle
Fixture checkpoint marker-0848-xray is recorded on fixture page 0848.
properties. Often a daughter nuclide is left in an excited state after a radioactive parent nucleus undergoes a transformation by alpha decay, beta decay, or electron capture. The nucleus will drop to the ground state by the emission of gamma radiation.
Internal Conversion
The usual method for an excited nucleus to go from the excited state to the ground state is by emission of gamma radiation. However, in some cases the gamma ray (photon) emerges from the nucleus only to interact with one of the innermost orbital electrons and, as a result, the energy of the photon is transferred to the electron. The gamma ray is then said to have undergone internal conversion. The conversion electron is ejected from the atom with kinetic energy equal to the gamma energy minus the binding energy of the orbital electron. An orbital electron then drops to a lower energy state to fill the vacancy, and this is accompanied by the emission of characteristic x-rays.
Isomers and Isomeric Transition
Isomeric transition commonly occurs immediately after particle emission; however, the nucleus may remain in an excited state for a measurable period of time before dropping to the ground state at its own characteristic rate. A nucleus that remains in such an excited state is known as a nuclear isomer because it differs in energy and behavior from other nuclei with the same atomic number and mass number. The decay of an excited nuclear isomer to a lower energy level is called an isomeric

by beta emission.
An example of gamma emission accompanying particle emission is illustrated by the decay of nitrogen-16 below.
■
■
■
■
■
■
■
■
■
■
■
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Decay Chains
When an unstable nucleus decays, the resulting daughter nucleus is not necessarily stable. The nucleus resulting from the decay of a parent is often itself unstable, and will undergo an additional decay. This is especially common among the larger nuclides.
It is possible to trace the steps of an unstable atom as it goes through multiple decays trying to achieve stability. The list of the original unstable nuclide, the nuclides that are involved as intermediate steps in the decay, and the final stable nuclide is known as the decay chain. One common method for stating the decay chain is to state each of the nuclides involved in the standard X format. Arrows are used between nuclides to indicate where decays occur, with the type of decay A Z
indicated above the arrow and the half-life below the arrow. The half-life for decay will be discussed in the next chapter.
Example:
Write the decay chains for rubidium-91 and actinium-215. Continue the chains until a stable nuclide or a nuclide with a half-life greater than 1 x 10 years is reached. 6
Solution:
Predicting Type of Decay
Radioactive nuclides tend to decay in a way that results in a daughter nuclide that lies closer to the line of stability. Due to this, it is possible to predict the type of decay that a nuclide will undergo
Fixture checkpoint marker-0849-xray is recorded on fixture page 0849.
based on its location relative to the line of stability on the Chart of the Nuclides.
Figure 9 illustrates the type of decay nuclides in different regions of the chart will typically undergo. Nuclides that are below and to the right of the line of stability will usually undergo ■ decay. - Nuclides that are above and to the left of the line of stability will usually undergo either ■ decay or + electron capture. Most nuclides that will undergo ■ decay are found in the upper right hand region of the chart. These are general rules that have many exceptions, especially in the region of the heavy nuclides.
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Figure 9 Types of Radioactive Decay Relative to the Line of Stability
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The important information in this chapter is summarized below.
Modes of Radioactive Decay Summary
Alpha decay is the emission of an alpha particle (2 protons and 2 neutrons) from an unstable nucleus. The daughter nuclide has an atomic number 2 less than the parent nuclide and a mass number 4 less than the parent nuclide. The daughter nucleus commonly releases its excitation energy by gamma emission.
Beta-minus decay effectively converts a neutron to a proton and an electron, which is immediately ejected from the nucleus. The daughter nuclide has its atomic number increased by 1 and the same mass number compared to the parent.
Beta-plus decay effectively converts a proton to a neutron and a positron, which is immediately ejected from the nucleus. The daughter nuclide has its atomic number decreased by 1 and the same mass number compared to the parent.
Fixture checkpoint marker-0850-xray is recorded on fixture page 0850.
In electron capture, the nucleus absorbs an electron from the innermost orbit. This electron combines with a proton to form a neutron.
Internal conversion occurs when a gamma ray, emitted by the nucleus as it goes from the excited state to the ground state, interacts with one of the innermost electrons of the same atom. The electron is ejected from the atom.
An isomeric transition is the decay of an excited nucleus to a lower-energy level by the emission of a gamma ray.
Decay chains can be found by tracing the steps an unstable atom goes through as it tries to achieve stability.
Many modes of radioactive decay result in a daughter nuclide that has an energy level above the ground state. This excitation energy is usually released immediately in the form of a gamma ray.
The type of decay that a nuclide will typically undergo can be determined by its relationship to the line of stability on the Chart of the Nuclides. Nuclides that lie below and to the right of the line of stability will typically beta minus decay. Nuclides above and to the left of the line will typically either beta plus decay or electron capture. Most alpha emitters are found in the upper, right-hand corner of the chart.
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RADIOACTIVITY
The rate at which a sample of radioactive material decays is not constant. As individual atoms of the material decay, there are fewer of those types of atoms remaining. Since the rate of decay is directly proportional to the number of atoms, the rate of decay will decrease as the number of atoms decreases.
EO 2.5 DEFINE the following terms:
a. Radioactivity d. Radioactive decay constant
b. Curie e. Radioactive half-life
c. Becquerel
EO 2.6 Given the number of atoms and either the half-life or decay constant of a nuclide, CALCULATE the activity.
EO 2.7 Given the initial activity and the decay constant of a nuclide, CALCULATE the activity at any later time.
EO 2.8 CONVERT between the half-life and decay constant for a

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EO 2.9 Given the Chart of the Nuclides and the original activity, PLOT the radioactive decay curve for a nuclide on either linear or semi-log coordinates.
EO 2.10 DEFINE the following terms:
a. Radioactive equilibrium
b. Transient radioactive equilibrium
Radioactive Decay Rates
Radioactivity is the property of certain nuclides of spontaneously emitting particles or gamma radiation. The decay of radioactive nuclides occurs in a random manner, and the precise time at which a single nucleus will decay cannot be determined. However, the average behavior of a very large sample can be predicted accurately by using statistical methods. These studies have revealed that there is a certain probability that in a given time interval a certain fraction of the nuclei within a sample of a particular nuclide will decay. This probability per unit time that an atom of a nuclide will decay is known as the radioactive decay constant, λ. The units for the decay constant are inverse time such as 1/second, 1/minute, 1/hour, or 1/year. These decay constant units can also be expressed as second-1, minute-1, hour-1, and year-1.

■■

Fixture checkpoint marker-0851-xray is recorded on fixture page 0851.
The activity (A) of a sample is the rate of decay of that sample. This rate of decay is usually measured in the number of disintegrations that occur per second. For a sample containing millions of atoms, the activity is the product of the decay constant and the number of atoms present in the sample.
The relationship between the activity, number of atoms, and decay constant is shown in Equation (1-3).
A = ■ N (1-3)
where:
A = Activity of the nuclide (disintegrations/second)
■ = decay constant of the nuclide (second)-1
N = Number of atoms of the nuclide in the sample
Since ■ is a constant, the activity and the number of atoms are always proportional.
Units of Measurement for Radioactivity
Two common units to measure the activity of a substance are the curie (Ci) and becquerel (Bq). A curie is a unit of measure of the rate of radioactive decay equal to 3.7 x 10 disintegrations 10 per second. This is approximately equivalent to the number of disintegrations that one gram of radium-226 will undergo in one second. A becquerel is a more fundamental unit of measure of radioactive decay that is equal to 1 disintegration per second. Currently, the curie is more widely used in the United States, but usage of the becquerel can be expected to broaden as the metric system slowly comes into wider use. The conversion between curies and becquerels is shown below.
1 curie = 3.7 x 10 becquerels 10
Variation of Radioactivity Over Time
The rate at which a given radionuclide sample decays is stated in Equation (1-3) as being equal to the product of the number of atoms and the decay constant. From this basic relationship it is possible to use calculus to derive an expression which can be used to calculate how the number of atoms present will change over time. The derivation is beyond the scope of this text,

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(1-4)
where:
N = number of atoms present at time t
N = number of atoms initially present o
■ = decay constant (time)-1
t = time
A Ao
e
t
A Aoe
t
A
Ao
e
t
ln A
Ao
t
t
ln
A
Ao
t
1/2
ln
1
2
Fixture checkpoint marker-0852-xray is recorded on fixture page 0852.
t
1/2
ln 2 0.693
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Since the activity and the number of atoms are always proportional, they may be used interchangeably to describe any given radionuclide population. Therefore, the following is true.
(1-5)
where:
A = activity present at time t
A = activity initially present o
= decay constant (time)
-1
t = time
Radioactive Half-Life
One of the most useful terms for estimating how quickly a nuclide will decay is the radioactive half-life. The radioactive half-life is defined as the amount of time required for the activity to decrease to one-half of its original value. A relationship between the half-life and decay constant can be developed from Equation (1-5). The half-life can be calculated by solving Equation (1-5) for the time, t, when the current activity, A, equals one-half the initial activity A .

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First, solve Equation (1-5) for t.
If A is equal to one-half of Ao, then A/Ao is equal to one-half. Substituting this in the equation above yields an expression for t .
1/2
(1-6)
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The basic features of decay of a radionuclide sample are shown by the graph in Figure 10.
Figure 10 Radioactive Decay as a Function of Time in Units of Half-Life
Assuming an initial number of atoms No, the population, and consequently, the activity may be noted to decrease by one-half of this value in a time of one half-life. Additional decreases occur so that whenever one half-life elapses, the number of atoms drops to one-half of what its value was at the beginning of that time interval. After five half-lives have elapsed, only 1/32, or 3.1%, of the original number of atoms remains. After seven half-lives, only 1/128, or 0.78%, of the atoms remains. The number of atoms existing after 5 to 7 half-lives can usually be assumed to be negligible. The Chemistry Fundamentals Handbook contains additional information on calculating the number of atoms contained within a sample.
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■
■
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Example:
A sample of material contains 20 micrograms of californium-252.
Californium-252 has a half-life of 2.638 years.
Calculate:
(a) The number of californium-252 atoms initially present
(b) The activity of the californium-252 in curies
(c) The number of californium-252 atoms that will remain in 12 years
(d) The time it will take for the activity to reach 0.001 curies
Solution:
(a) The number of atoms of californium-252 can be determined as below.
(b) First, use Equation (1-6) to calculate the decay constant.
Use this value for the decay constant in Equation (1-3) to determine the activity.
N No
Fixture checkpoint marker-0853-xray is recorded on fixture page 0853.
e
t
4.78 x 1016 e
(0.263/yr) (12 yr)
2.04 x 1015
A Aoe
t
A
Ao
e
t
ln A
Ao

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t
ln
A
Ao
t
ln
0.001 Ci
0.0108 Ci
0.263 year 1
t 9.05 years
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(c) The number of californium atoms that will remain in 12 years can be calculated from Equation (1-4).
(d) The time that it will take for the activity to reach 0.001 Ci can be determined from Equation (1-5). First, solve Equation (1-5) for time.
Inserting the appropriate values in the right side of this equation will result in the required time.
Plotting Radioactive Decay
It is useful to plot the activity of a nuclide as it changes over time. Plots of this type can be used to determine when the activity will fall below a certain level. This plot is usually done showing activity on either a linear or a logarithmic scale. The decay of the activity of a single nuclide on a logarithmic scale will plot as a straight line because the decay is exponential.
t
1/2
0.693
0.693
t
1/2
0.693
7.13 seconds
0.0972 second 1
A Aoe
t
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Example:
Plot the radioactive decay curve for nitrogen-16 over a period of 100 seconds. The initial activity is 142 curies and the half-life of nitrogen-16 is 7.13 seconds. Plot the curve on both linear rectangular coordinates and on a semi-log scale.
Solution:
First, use Equation (1-6) to calculate the decay constant corresponding to a half-life of 7.13 seconds.
Use the decay constant determined above to calculate the activity at various times using Equation (1-5).
Time Activity
0 seconds 142 Ci

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40 seconds 2.91 Ci
60 seconds 0.416 Ci
80 seconds 0.0596 Ci
100 seconds 0.00853 Ci

■ ■
■ ■
■ ■

Figure 11 Linear and Semi-log Plots of Nitrogen-16 Decay

Plotting the data points calculated above on both linear and semilog scales results in the graphs shown in Figure 11.

If a substance contains more than one radioactive nuclide, the total activity is the sum of the individual activities of each nuclide. As an example, consider a sample of material that contained 1 x 10 atoms of iron-59 that has a half-life of 44.51 days (■ = 1.80 x 10 sec), 6 -7 -1 1 x 10 atoms of manganese-54 that has a half-life of 312.2 days (■ = 2.57 x 10 sec), and 1 6 -8 -1 x 10 atoms of cobalt-60 that has a half-life of 1925 days (■ = 4.17 x 10 sec). 6 -9 -1

The initial activity of each of the nuclides would be the product of the number of atoms and the Fixture checkpoint marker-0855-xray is recorded on fixture page 0855.
decay constant.

Figure 12 Combined Decay of Iron-56, Manganese-54, and Cobalt-60

Plotting the manner in which the activities of each of the three nuclides decay over time demonstrates that initially the activity of the shortest-lived nuclide (iron-59) dominates the total activity, then manganese-54 dominates. After almost all of the iron and manganese have decayed away, the only contributor to activity will be the cobalt-60. A plot of this combined decay is shown in Figure 12.

Radioactive Equilibrium

Radioactive equilibrium exists when a radioactive nuclide is decaying at the same rate at which it is being produced. Since the production rate and decay rate are equal, the number of atoms present remains constant over time.

An example of radioactive equilibrium is the concentration of sodium-24 in the coolant circulating through a sodium-cooled nuclear reactor. Assume that the sodium-24 is being produced at a rate of 1 x 10 atoms per second. If the sodium-24 were stable and did not decay, 6 the amount of sodium-24 present after some period of time could be calculated by multiplying the production rate by the amount of time. Plotting the amount of material present would result in the graph in Figure 13.

However, sodium-24 is not stable, and it decays with a half-life of 14.96 hours. If no sodium-24 is present initially and production starts at a rate of 1 x 10 atoms per second, the rate 6 of decay will initially be zero because there is no sodium-24 present to decay. The rate of decay of sodium-24 will increase as the amount of sodium-24 increases.

■
■
■
■

The amount of sodium-24 present will initially increase rapidly, then it will increase at a continually decreasing rate until the rate of decay is equal to the rate of production. It is possible to calculate how much sodium-24 will be present at equilibrium by setting the production rate (R) equal to the decay rate (■ N).
where:
R = production rate (atoms/second)
■ = decay constant (second)-1
N = number of atoms
It is possible to calculate the equilibrium value for sodium-24 being produced at a rate of 1 x 10 atoms/second. 6
■ ■ ■■
■
■
■
■
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Figure 14 Approach of Sodium-24 to Equilibrium
The development of the equation to calculate how the amount of sodium-24 changes over time as it approaches the equilibrium value is beyond the scope of this handbook. However, the equation is presented below.
This equation can be used to calculate the values of the amount of sodium-24 present at different times. As the time increases, the exponential term approaches zero, and the number of atoms present will approach R/■. A plot of the approach of sodium-24 to equilibrium is shown in Figure 14.
Fixture checkpoint marker-0856-xray is recorded on fixture page 0856.
Transient Radioactive Equilibrium
Transient radioactive equilibrium occurs when the parent nuclide and the daughter nuclide decay at essentially the same rate.
For transient equilibrium to occur, the parent must have a long half-life when compared to the daughter. An example of this type of compound decay process is barium-140, which decays by beta emission to lanthanum-140, which in turn decays by beta emission to stable cerium-140.
Atomic and Nuclear Physics DOE-HDBK-1019/1-93 RADIOACTIVITY
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Figure 15 Transient Equilibrium in the Decay of Barium-140
The decay constant for barium-140 is considerably smaller than the decay constant for lanthanum-140. Remember that the rate of decay of both the parent and daughter can be represented as ■N. Although the decay constant for barium-140 is smaller, the actual rate of decay (■N) is initially larger than that of lanthanum-140 because of the great difference in their initial concentrations. As the concentration of the daughter increases, the rate of decay of the daughter will approach and eventually match the decay rate of the parent. When this occurs, they are said to be in transient equilibrium. A plot of the barium-lanthanum-cerium decay chain reaching transient equilibrium is shown in Figure 15.
Secular equilibrium occurs when the parent has an extremely long half-life. In the long decay chain for a naturally radioactive element, such as thorium-232, where all of the elements in the chain are in secular equilibrium, each of the descendants has built up to an equilibrium amount and all decay at the rate set by the original parent. The only exception is the final stable element on the end of the chain. Its number of atoms is constantly increasing.

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The important information in this chapter is summarized on the following page.

■

■■

■

RADIOACTIVITY DOE-HDBK-1019/1-93 Atomic and Nuclear Physics
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Radioactivity Summary
& Radioactivity is the decay of unstable atoms by the emission of particles and electromagnetic radiation.
& A curie (Ci) is a unit of radioactivity equal to 3.7 x 10 disintegrations per 10 second.
& A becquerel (Bq) is a unit of radioactivity equal to 1 disintegration per second.
& The radioactive decay constant (■) is the probability per unit time that an atom will decay.
& The radioactive half-life is the amount of time required for the activity to decrease to one-half its original value.
& The activity of a substance can be calculated from the number of atoms and the decay constant based on the equation below.
& The amount of activity remaining after a particular time can be calculated from the equation below.
& The relationship between the decay constant and the half-life is shown below.
& Plots of radioactive decay can be useful to describe the variation of activity over time. If decay is plotted using semi-log scale the plot results in a straight line.
& Radioactive equilibrium exists when the production rate of a material is equal to the removal rate.
& Transient radioactive equilibrium exists when the parent nuclide and the daughter nuclide decay at essentially the same rate. This occurs only when the parent has a long half-life compared to the daughter.
Atomic and Nuclear Physics DOE-HDBK-1019/1-93 NEUTRON INTERACTIONS
NEUTRON INTERACTIONS
Neutrons can cause many different types of interactions. The neutron may simply scatter off the nucleus in two different ways, or it may actually be absorbed into
Fixture checkpoint marker-0857-xray is recorded on fixture page 0857.
the nucleus. If a neutron is absorbed into the nucleus, it may result in the emission of a gamma ray or a subatomic particle, or it may cause the nucleus to fission.
EO 3.1 DESCRIBE the following scattering interactions between a neutron and a nucleus:
a. Elastic scattering
b. Inelastic scattering
EO 3.2 STATE the conservation laws that apply to an elastic collision between a neutron and a nucleus.
EO 3.3 DESCRIBE the following reactions where a neutron is absorbed in a nucleus:
a. Radiative capture
b. Particle ejection
Scattering
A neutron scattering reaction occurs when a nucleus, after having been struck by a neutron,

often are not) the same, the net effect of the reaction is as if the projectile neutron had merely "bounced off," or scattered from, the nucleus. The two categories of scattering reactions, elastic and inelastic scattering, are described in the following paragraphs.

Elastic Scattering

In an elastic scattering reaction between a neutron and a target nucleus, there is no energy transferred into nuclear excitation. Momentum and kinetic energy of the "system" are conserved although there is usually some transfer of kinetic energy from the neutron to the target nucleus. The target nucleus gains the amount of kinetic energy that the neutron loses.

mn
vn,i mTvT,i mnvn,f mTvT,f
1
2
m v
2
1
2
mn
v
2
n,i
1
2
mT
v
2
T,i
1
2
mn
v
2
n,f
1
2
mT
v
2
T,f

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Figure 16 Elastic Scattering

Figure 16 illustrates the process of elastic scattering of a neutron off a target nucleus. In the elastic scattering reaction, the conservation of momentum and kinetic energy is represented by the equations below.

Conservation of momentum (mv)

Conservation of kinetic energy

mn = mass of the neutron
m = mass of the target nucleus T
vn,i = initial neutron velocity
vn,f = final neutron velocity
vT,i = initial target velocity
vT,f = final target velocity
Atomic and Nuclear Physics DOE-HDBK-1019/1-93 NEUTRON INTERACTIONS
Elastic scattering of neutrons by nuclei can occur in two ways. The more unusual of the two interactions is the absorption of the neutron, forming a compound nucleus, followed by the re-emission of a neutron in such a way that the total kinetic energy is conserved and the nucleus returns to its ground state. This is known as resonance elastic scattering and is very dependent upon the initial kinetic energy possessed by the neutron. Due to formation of the compound nucleus, it is also referred to as compound elastic scattering. The second, more usual method, is termed potential elastic scattering and can be understood by visualizing the neutrons and nuclei to be much like billiard balls with impenetrable surfaces. Potential scattering takes place with incident neutrons that have an energy of up to about 1 MeV. In potential scattering, the
Fixture checkpoint marker-0859-xray is recorded on fixture page 0859.
neutron does not actually touch the nucleus and a compound nucleus is not formed. Instead, the neutron is acted on and scattered by the short range nuclear forces when it approaches close enough to the nucleus.
Inelastic Scattering
In inelastic scattering, the incident neutron is absorbed by the target nucleus, forming a compound nucleus. The compound nucleus will then emit a neutron of lower kinetic energy which leaves the original nucleus in an excited state. The nucleus will usually, by one or more gamma emissions, emit this excess energy to reach its ground state. Figure 17 shows the process of inelastic scattering.
For the nucleus that has reached its ground state, the sum of the kinetic energy of the exit
Figure 17 Inelastic Scattering
neutron, the target nucleus, and the total gamma energy emitted is equal to the initial kinetic energy of the incident neutron.

1
0
n
238
92
U
239
92
U
239
92
U
0
0
1
0
n

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5
B
11
5
B
7
3
Li
4
2
NEUTRON INTERACTIONS DOE-HDBK-1019/1-93 Atomic and Nuclear Physics
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Absorption Reactions
Most absorption reactions result in the loss of a neutron coupled with the production of a charged particle or gamma ray. When the product nucleus is radioactive, additional radiation is emitted at some later time. Radiative capture, particle ejection, and fission are all categorized as absorption reactions and are briefly described below.
Radiative Capture
In radiative capture the incident neutron enters the target nucleus forming a compound nucleus. The compound nucleus then decays to its ground state by gamma emission. An example of a radiative capture reaction is shown below.
Particle Ejection
In a particle ejection reaction the incident particle enters the target nucleus forming a compound
Fixture checkpoint marker-0860-xray is recorded on fixture page 0860.
nucleus. The newly formed compound nucleus has been excited to a high enough energy level to cause it to eject a new particle while the incident neutron remains in the nucleus. After the new particle is ejected, the remaining nucleus may or may not exist in an excited state depending upon the mass-energy balance of the reaction. An example of a particle ejection reaction is shown below.
Fission
One of the most important interactions that neutrons can cause is fission, in which the nucleus that absorbs the neutron actually splits into two similarly sized parts. Fission will be discussed in detail in the next chapter.
Atomic and Nuclear Physics DOE-HDBK-1019/1-93 NEUTRON INTERACTIONS
Summary
The important information in this chapter is summarized below.
Neutron Interactions Summary
Interactions where a neutron scatters off a target nucleus are either elastic or inelastic. In elastic scattering, kinetic energy and momentum are conserved and no energy is transferred into excitation energy of the target nucleus. In inelastic scattering, some amount of kinetic energy is transferred into excitation energy of the target nucleus.
The conservation principles that apply to an elastic collision are conservation of kinetic energy and conservation of momentum.
Radiative capture is the absorption of a neutron by the target nucleus, resulting in an excited nucleus which subsequently (typically within a small fraction of a second) releases its excitation energy in the form of a gamma ray.
Particle ejection occurs when a neutron is absorbed by a target nucleus, resulting

ejects a particle (for example, alpha or proton).
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1
0
n
235
92
U
236
92
U
140
55
Cs
93
37
Rb 3
1
0
n
NUCLEAR FISSION DOE-HDBK-1019/1-93 Atomic and Nuclear Physics
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NUCLEAR FISSION
Nuclear fission is a process in which an atom splits and releases energy, fission products, and neutrons. The neutrons released by fission can, in turn, cause the fission of other atoms.
EO 4.1 EXPLAIN the fission process using the liquid drop model of a nucleus.
EO 4.2 DEFINE the following terms:
a. Excitation energy (E_{exc})
Fixture checkpoint marker-0861-xray is recorded on fixture page 0861.
b. Critical energy (E_{crit})
EO 4.3 DEFINE the following terms:
a. Fissile material
b. Fissionable material
c. Fertile material
EO 4.4 DESCRIBE the processes of transmutation, conversion, and breeding.
EO 4.5 DESCRIBE the curve of Binding Energy per Nucleon versus mass number and give a qualitative description of the reasons for its shape.
EO 4.6 EXPLAIN why only the heaviest nuclei are easily fissioned.
EO 4.7 EXPLAIN why uranium-235 fissions with thermal neutrons and uranium-238 fissions only with fast neutrons.
Fission
In the fission reaction the incident neutron enters the heavy target nucleus, forming a compound nucleus that is excited to such a high energy level ($E > E_{crit}$) that the nucleus "splits" (fissions) into two large fragments plus some neutrons. An example of a typical fission

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A large amount of energy is released in the form of radiation and fragment kinetic energy.
Atomic and Nuclear Physics DOE-HDBK-1019/1-93 NUCLEAR FISSION
Liquid Drop Model of a Nucleus
The nucleus is held together by the attractive nuclear force between nucleons, which was discussed in a previous chapter. The characteristics of the nuclear force are listed below.
(a) very short range, with essentially no effect beyond nuclear dimensions
(10-13 cm)
(b) stronger than the repulsive electrostatic forces within the nucleus
(c) independent of nucleon pairing, in that the attractive forces between pairs of neutrons are no different than those between pairs of protons or a neutron and a proton
(d) saturable, that is, a nucleon can attract only a few of its nearest neighbors
One theory of fission considers the fissioning of a nucleus similar in some respects to the splitting of a liquid drop. This analogy is justifiable to some extent by the fact that a liquid drop is held together by molecular forces that tend to make the drop spherical in shape and that try to resist any deformation in the same manner as nuclear forces are assumed to hold the nucleus together. By considering the nucleus as a liquid drop, the fission process can be described.
Referring to Figure 18(A), the nucleus in the ground state is undistorted, and its attractive nuclear forces are greater than the repulsive electrostatic forces between the protons within the nucleus. When an incident particle (in this instance a neutron) is absorbed by the target nucleus, a compound nucleus is formed. The compound nucleus temporarily contains all the charge and mass involved in the reaction and exists in an excited state. The excitation energy added to the compound nucleus is equal to the binding energy contributed by the incident particle plus the kinetic energy possessed by that particle. Figure 18(B) illustrates the excitation energy thus imparted to the compound nucleus, which may cause it to oscillate and become distorted. If the excitation energy is greater than a certain critical energy, the oscillations may cause the compound nucleus to become dumbbell-shaped. When this happens, the attractive nuclear forces (short-range) in the neck area are small due to saturation, while the repulsive electrostatic forces (long-range) are only slightly less than before. When the repulsive electrostatic forces exceed the attractive nuclear forces, nuclear fission occurs, as illustrated in Figure 18(C).
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NUCLEAR FISSION DOE-HDBK-1019/1-93 Atomic and Nuclear Physics
Figure 18 Liquid Drop Model of Fission
Critical Energy
The measure of how far the energy level of a nucleus is above its ground state is called the excitation energy (Eexc). For fission to occur, the excitation energy must be above a particular value for that nuclide. The critical energy (Ecrit) is the minimum excitation energy required for
Fixture checkpoint marker-0862-xray is recorded on fixture page 0862.
fission to occur.
Fissile Material
A fissile material is composed of nuclides for which fission is possible with neutrons of any energy level. What is especially significant about these nuclides is their ability to be fissioned with zero kinetic energy neutrons (thermal neutrons). Thermal neutrons have very low kinetic energy levels (essentially zero) because they are roughly in equilibrium with the thermal motion of surrounding materials. Therefore, in order to be classified as fissile, a material must be capable of fissioning after absorbing a thermal neutron. Consequently, they impart essentially no kinetic energy to the reaction. Fission is possible in these materials with thermal neutrons, since the change in binding energy supplied by the neutron addition alone is high enough to

plutonium-239.
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Atomic and Nuclear Physics DOE-HDBK-1019/1-93 NUCLEAR FISSION
Fissionable Material
A fissionable material is composed of nuclides for which fission with neutrons is possible. All fissile nuclides fall into this category. However, also included are those nuclides that can be fissioned only with high energy neutrons. The change in binding energy that occurs as the result of neutron absorption results in a nuclear excitation energy level that is less than the required critical energy. Therefore, the additional excitation energy must be supplied by the kinetic energy of the incident neutron. The reason for this difference between fissile and fissionable materials is the so-called odd-even effect for nuclei. It has been observed that nuclei with even numbers of neutrons and/or protons are more stable than those with odd numbers. Therefore, adding a neutron to change a nucleus with an odd number of neutrons to a nucleus with an even number of neutrons produces an appreciably higher binding energy than adding a neutron to a nucleus
Fixture checkpoint marker-0863-xray is recorded on fixture page 0863.
already possessing an even number of neutrons. Some examples of nuclides requiring high energy neutrons to cause fission are thorium-232, uranium-238, and plutonium-240. Table 4 indicates the critical energy (Ecrit) and the binding energy change for an added neutron (BEn) to target nuclei of interest. For fission to be possible, the change in binding energy plus the kinetic energy must equal or exceed the critical energy ($\Delta BE + KE > Ecrit$).
TABLE 4
Critical Energies Compared to Binding Energy of Last Neutron
Target
Nucleus
Critical Energy
Ecrit
Binding Energy of
Last Neutron BEn
BEn - Ecrit
23
9
2
0Th 7.5 MeV 5.4 MeV -2.1 MeV
23
9
8
2U 7.0 MeV 5.5 MeV -1.5 MeV
23
9
5
2U 6.5 MeV 6.8 MeV +0.3 MeV
23
9
3
2U 6.0 MeV 7.0 MeV +1.0 MeV
23
9
9

Uranium-235 fissions with thermal neutrons because the binding energy released by the absorption of a neutron is greater than the critical energy for fission; therefore uranium-235 is a fissile material. The binding energy released by uranium-238 absorbing a thermal neutron is less than the critical energy, so additional energy must be possessed by the neutron for fission to be possible. Consequently, uranium-238 is a fissionable material.

Fertile Material

All of the neutron absorption reactions that do not result in fission lead to the production of new nuclides through the process known as transmutation. These nuclides can, in turn, be transmuted again or may undergo radioactive decay to produce still different nuclides. The nuclides that are produced by this process are referred to as transmutation products. Because several of the fissile nuclides do not exist in nature, they can only be produced by nuclear reactions (transmutation). The target nuclei for such reactions are said to be fertile. Fertile materials are materials that can undergo transmutation to become fissile materials. Figure 19 traces the transmutation mechanism by which two fertile nuclides, thorium-232 and uranium-238, produce uranium-233 and plutonium-239, respectively.

Figure 19 Conversion of Fertile Nuclides to Fissile Nuclides

If a reactor contains fertile material in addition to its fissile fuel, some new fuel will be produced as the original fuel is burned up. This is called conversion. Reactors that are specifically designed to produce fissionable fuel are called "breeder" reactors. In such reactors, the amount Fixture checkpoint marker-0864-xray is recorded on fixture page 0864.
of fissionable fuel produced is greater than the amount of fuel burnup. If less fuel is produced than used, the process is called conversion, and the reactor is termed a "converter."

Binding Energy Per Nucleon (BE/A)

As the number of particles in a nucleus increases, the total binding energy also increases. The rate of increase, however, is not uniform. This lack of uniformity results in a variation in the amount of binding energy associated with each nucleon within the nucleus. This variation in the binding energy per nucleon (BE/A) is easily seen when the average BE/A is plotted versus atomic mass number (A), as shown in Figure 20.

Figure 20 Binding Energy per Nucleon vs. Mass Number

Figure 20 illustrates that as the atomic mass number increases, the binding energy per nucleon decreases for A > 60. The BE/A curve reaches a maximum value of 8.79 MeV at A = 56 and decreases to about 7.6 MeV for A = 238. The general shape of the BE/A curve can be explained using the general properties of nuclear forces. The nucleus is held together by very short-range attractive forces that exist between nucleons. On the other hand, the nucleus is being forced apart by long range repulsive electrostatic (coulomb) forces that exist between all the protons in the nucleus.

As the atomic number and the atomic mass number increase, the repulsive electrostatic forces within the nucleus increase due to the greater number of protons in the heavy elements. To overcome this increased repulsion, the proportion of neutrons in the nucleus must increase to maintain stability. This increase in the neutron-to-proton ratio only partially compensates for the growing proton-proton repulsive force in the heavier, naturally occurring elements. Because the repulsive forces are increasing, less energy must be supplied, on the average, to remove a nucleon

of stability. Generally, the more stable nuclides have higher BE/A than the less stable ones. The increase in the BE/A as the atomic mass number decreases from 260 to 60 is the primary reason for the energy liberation in the fission process. In addition, the increase in the BE/A as the atomic mass number increases from 1 to 60 is the reason for the energy liberation in the fusion process, which is the opposite reaction of fission.

The heaviest nuclei require only a small distortion from a spherical shape (small energy addition) for the relatively large coulomb forces forcing the two halves of the nucleus apart to overcome the attractive nuclear forces holding the two halves together. Consequently, the heaviest nuclei are easily fissionable compared to lighter nuclei.

Summary

The important information in this chapter is summarized on the following page.

Nuclear Fission Summary

The fission process can be explained using the liquid drop model of a nucleus. In the ground state the nucleus is nearly spherical in shape. After the absorption of a neutron, the nucleus will be in an excited state and start to oscillate and become distorted. If the oscillations cause the nucleus to become shaped like a dumbbell, the repulsive electrostatic forces will overcome the short-range attractive nuclear forces, and the nucleus will split in two.

Excitation energy is the amount of energy a nucleus has above its ground state. Critical energy is the minimum excitation energy that a nucleus must have before it can fission.

Fissile material is material for which fission is possible with neutrons that have zero kinetic energy. Fissionable material is material for which fission caused by neutron absorption is possible provided the kinetic energy added with the binding energy is greater than the critical energy. Fertile material is material that can undergo transmutation to become fissile material.

Fixture checkpoint marker-0865-xray is recorded on fixture page 0865.

Transmutation is the process of neutron absorption and subsequent decay, which changes one nuclide to another nuclide. Conversion is the process of transmuting fertile material into fissile material in a reactor, where the amount of fissile material produced is less than the amount of fissile material consumed. Breeding is the same as conversion, except the amount of fissile material produced is more than the amount of fissile material consumed.

The curve of binding energy per nucleon increases quickly through the light nuclides and reaches a maximum at a mass number of about 56. The curve decreases slowly for mass numbers greater than 60.

The heaviest nuclei are easily fissionable because they require only a small distortion from the spherical shape to allow the coulomb forces to overcoming the attractive nuclear force, forcing the two halves of the nucleus apart.

Uranium-235 fissions with thermal neutrons because the binding energy released by the absorption of a neutron is greater than the critical energy for fission. The binding energy released by uranium-238 absorbing a neutron is less than the critical energy, so additional kinetic energy must be possessed by the neutron for fission to be possible.

1

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n
235
92
U
236
92
U
140
55
Cs
93
37
Rb 3
1
0
n
ENERGY RELEASE FROM FISSION DOE-HDBK-1019/1-93 Atomic and Nuclear Physics
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ENERGY RELEASE FROM FISSION
Fission of heavy nuclides converts a small amount of mass into an enormous amount of energy. The amount of energy released by fission can be determined based on either the change in mass that occurs during the reaction or by the difference in binding energy per nucleon between the fissile nuclide and the fission products.
EO 4.8 CHARACTERIZE the fission products in terms of mass groupings and radioactivity.
EO 4.9 Given the nuclides involved and their masses, CALCULATE the energy released from fission.
EO 4.10 Given the curve of Binding Energy per nucleon versus mass number, CALCULATE the energy released from fission.
Calculation of Fission Energy
Nuclear fission results in the release of enormous quantities of energy. It is necessary to be able to calculate the amount of energy that will be produced. The logical manner in which to pursue this is to first investigate a typical fission reaction such as the one listed below.
It can be seen that when the compound nucleus splits, it breaks into two fission fragments,
Fixture checkpoint marker-0866-xray is recorded on fixture page 0866.
rubidium-93, cesium-140, and some neutrons. Both fission products then decay by multiple - emissions as a result of the high neutron-to-proton ratio possessed by these nuclides.
In most cases, the resultant fission fragments have masses that vary widely. Figure 21 gives the percent yield for atomic mass numbers. The most probable pair of fission fragments for the thermal fission of the fuel uranium-235 have masses of about 95 and 140. Note that the vertical axis of the fission yield curve is on a logarithmic scale. Therefore, the formation of fission fragments of mass numbers of about 95 and 140 is highly likely.
Atomic and Nuclear Physics DOE-HDBK-1019/1-93 ENERGY RELEASE FROM FISSION
Figure 21 Uranium-235 Fission Yield vs.
Mass Number
Referring now to the binding energy per nucleon curve (Figure 20), we can estimate the amount of energy released by our "typical" fission by plotting this reaction on the curve and calculating

equation and the products on the right-hand side. Plotting the reactant and product nuclides on the curve shows that the total binding energy of the system after fission is greater than the total binding energy of the system before fission. When there is an increase in the total binding energy of a system, the system has become more stable by releasing an amount of energy equal to the increase in total binding energy of the system. Therefore, in the fission process, the energy liberated is equal to the increase in the total binding energy of the system.

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Figure 22 graphically depicts that the binding energy per nucleon for the products (C,

Figure 22 Change in Binding Energy for Typical Fission

rubidium-93 and B, cesium-140) is greater than that for the reactant (A, uranium-235). The total binding energy for a nucleus can be found by multiplying the binding energy per nucleon by the

Fixture checkpoint marker-0867-xray is recorded on fixture page 0867.

number of nucleons.

TABLE 5

Binding Energies Calculated from

Binding Energy per Nucleon Curve

Nuclide B.E. per Nucleon

(BE/A)

Mass Number

(A)

Binding Energy

(BE/A) x (A)

9

3

3

7Rb 8.7 MeV 93 809 MeV

14

5

0

5Cs 8.4 MeV 140 1176 MeV

23

9

5

2U 7.6 MeV 235 1786 MeV

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BE BEproducts BEreactants

BERb 93 BECs 140 BEU 235

809 MeV 1176 MeV 1786 MeV

199 MeV

1

0

n

235

92

U

236

92

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140
55
Cs
93
37
Rb 3
1
0
n
Mass difference Mass of Reactants Mass of Products
236.052589 amu 235.85208 amu
0.200509 amu
EInst 0.020059 amu
931.5 MeV
amu
186.8 MeV
Atomic and Nuclear Physics DOE-HDBK-1019/1-93 ENERGY RELEASE FROM FISSION
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The energy released will be equivalent to the difference in binding energy (BE) between the
Fixture checkpoint marker-0868-xray is recorded on fixture page 0868.
reactants and the products.
The energy liberation during the fission process can also be explained from the standpoint of the conservation of mass-energy. During the fission process, there is a decrease in the mass of the system. There must, therefore, be energy liberated equal to the energy equivalent of the mass lost in the process. This method is more accurate than the previously illustrated method and is used when actually calculating the energy liberated during the fission process.
Again, referring to the "typical" fission reaction.
E , the instantaneous energy, is the energy released immediately after the fission process. It Inst is equal to the energy equivalent of the mass lost in the fission process. It can be calculated as shown below.
Mass of the Reactants Mass of the Products
U 235.043924 amu Rb 92.91699 amu 235
92
93
37
n 1.008665 amu Cs 139.90910 amu 1
0
140
55
3 (n) 3.02599 amu 1
0
236.052589 amu 235.85208 amu
This mass difference can be converted to an energy equivalent.
93
37
Rb
93
38

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93
39
Y
93
40
Zr
93
41
Nb
140
55
Cs
140
56
Ba
140
57
La
140
58
Ce
EDecay mRb 93 mNb 93 4 melectron
931.5MeV
amu
92.91699 amu 92.90638 amu 4 0.0005486 amu
931.5 MeV
Fixture checkpoint marker-0869-xray is recorded on fixture page 0869.
amu
0.008416 amu
931.5 MeV
amu
7.84 MeV
ENERGY RELEASE FROM FISSION DOE-HDBK-1019/1-93 Atomic and Nuclear Physics
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The total energy released per fission will vary from the fission to the next depending on what fission products are formed, but the average total energy released per fission of uranium-235 with a thermal neutron is 200 MeV.
As illustrated in the preceding example, the majority of the energy liberated in the fission process is released immediately after the fission occurs and appears as the kinetic energy of the fission fragments, kinetic energy of the fission neutrons, and instantaneous gamma rays. The remaining energy is released over a period of time after the fission occurs and appears as kinetic energy of the beta, neutrino, and decay gamma rays.
Estimation of Decay Energy
In addition to this instantaneous energy release during the actual fission reaction, there is additional energy released when the fission fragments decay by emission. This additional
-
energy is called decay energy, EDecay. The decay chains for rubidium-93 and cesium-140 are shown below.

between the original fission product and the sum of the final stable nuclide and the beta particles emitted.
The energy released in the decay chain of rubidium-93 is calculated below.
EDecay mRb 93 mNb 93 3 melectron
931.5MeV
amu
139.90910 amu 139.90543 amu 3 0.0005486 amu
931.5 MeV
amu
0.000202 amu
931.5 MeV
amu
1.89 MeV
Atomic and Nuclear Physics DOE-HDBK-1019/1-93 ENERGY RELEASE FROM FISSION
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The energy released in the decay chain of cesium-140 is calculated below.
The total decay energy is the sum of the energies of the two chains, or 9.73 MeV.
Distribution of Fission Energy
The average energy distribution for the energy released per fission with a thermal neutron in uranium-235 is shown in Tables 6 and 7.
TABLE 6
Instantaneous Energy from Fission
Kinetic Energy of Fission Products 167 Mev
Energy of Fission Neutrons 5 MeV
Instantaneous Gamma-ray Energy 5 MeV
Capture Gamma-ray Energy 10 MeV
Total Instantaneous Energy 187 MeV
TABLE 7
Delayed Energy from Fission
Beta Particles From Fission Products 7 Mev
Gamma-rays from Fission Products 6 MeV
Neutrinos 10 MeV
Total Delayed Energy 23 MeV
Fixture checkpoint marker-0870-xray is recorded on fixture page 0870.
ENERGY RELEASE FROM FISSION DOE-HDBK-1019/1-93 Atomic and Nuclear Physics
Because the 10 MeV of neutrino energy shown in Table 7 is not absorbed in the reactor, the average value of 200 MeV per fission is still accurate. Note in Table 6 that some fission neutrons undergo radiative capture and the resultant gamma ray emission provides an additional 10 MeV of instantaneous energy, which contributes to the total of 187 MeV instantaneous energy.
All of the energy released, with the exception of the neutrino energy, is ultimately transformed into heat through a number of processes. The fission fragments, with their high positive charge and kinetic energy, cause ionization directly as they rip orbital electrons from the surrounding atoms. In this ionization process, kinetic energy is transferred to the surrounding atoms of the fuel material, resulting in an increase in temperature. The beta particles and gamma rays also give up their energy through ionization, and the fission neutrons interact and lose their energy through elastic scattering. Of the 200 MeV released per fission, about seven percent (13 MeV) is released at some time after the instant of fission. When a reactor is shut down, fissions

produced by this decay energy is referred to as "decay heat." Although decay energy represents about seven percent of reactor heat production during reactor operation, once the reactor is shut down the decay heat production drops off quickly to a small fraction of its value while operating. The decay heat produced is significant, however, and systems must be provided to keep the reactor cool even after shutdown.

Summary

The important information in this chapter is summarized below.

Energy Release From Fission Summary

Fission products have some general characteristics in common.

They generally decay by β- emission.

Fixture checkpoint marker-0871-xray is recorded on fixture page 0871.

The most common mass numbers are grouped near 95 and 140.

The energy released by fission can be calculated based on the difference in mass between the masses of the reactants before fission and the fission fragments and fission neutrons after fission.

Another method to determine the energy released by fission is based on the change in binding energy per nucleon between the fissile nuclide and the fission products.

INTERACTION OF RADIATION WITH MATTER

Different types of radiation interact with matter in widely different ways. A large, massive, charged alpha particle cannot penetrate a piece of paper and even has a limited range in dry air. A neutrino, at the other extreme, has a low probability of interacting with any matter, even if it passed through the diameter of the earth.

EO 5.1 DESCRIBE interactions of the following with matter:

a. Alpha particle c. Positron

b. Beta particle d. Neutron

EO 5.2 DESCRIBE the following ways that gamma radiation interacts with matter:

a. Photoelectric effect

b. Compton scattering

c. Pair production

Interaction of Radiation With Matter

Radiation can be classified into two general groups, charged and uncharged; therefore, it may be expected that interactions with matter fall into two general types. Charged particles directly ionize the media through which they pass, while uncharged particles and photons can cause ionization only indirectly or by secondary radiation.

A moving charged particle has an electrical field surrounding it, which interacts with the atomic structure of the medium through which it is passing. This interaction decelerates the particle and accelerates electrons in the atoms of the medium. The accelerated electrons may acquire enough energy to escape from the parent atom. This process, whereby radiation "strips" off orbital electrons, is called ionization. Uncharged moving particles have no electrical field, so they can only lose energy and cause ionization by such means as collisions or scattering. A photon can lose energy by the photoelectric effect, Compton effect, or pair production.

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m z
2
K.E.
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Because ionizing radiation creates ions in pairs, the intensity of ionization or the specific ionization is defined as the number of ion-pairs formed per centimeter of travel in a given material. The amount of ionization produced by a charged particle per unit path length, which is a measure of its ionizing power, is roughly proportional to the particle's mass and the square of its charge as illustrated in the equation below.
where:
I is the ionizing power
m is the mass of the particle
z is the number of unit charges it carries
K.E. is its kinetic energy
Since m for an alpha particle is about 7300 times as large as m for a beta particle, and z is twice
Fixture checkpoint marker-0872-xray is recorded on fixture page 0872.
as great, an alpha will produce much more ionization per unit path length than a beta particle of the same energy. This phenomenon occurs because the larger alpha particle moves slower for a given energy and thus acts on a given electron for a longer time.
Alpha Radiation
Alpha radiation is normally produced from the radioactive decay of heavy nuclides and from certain nuclear reactions. The alpha particle consists of 2 neutrons and 2 protons, so it is essentially the same as the nucleus of a helium atom. Because it has no electrons, the alpha particle has a charge of +2. This positive charge causes the alpha particle to strip electrons from the orbits of atoms in its vicinity. As the alpha particle passes through material, it removes electrons from the orbits of atoms it passes near. Energy is required to remove electrons and the energy of the alpha particle is reduced by each reaction. Eventually the particle will expend its kinetic energy, gain 2 electrons in orbit, and become a helium atom. Because of its strong positive charge and large mass, the alpha particle deposits a large amount of energy in a short distance of travel. This rapid, large deposition of energy limits the penetration of alpha particles. The most energetic alpha particles are stopped by a few centimeters of air or a sheet of paper.
Beta-Minus Radiation
A beta-minus particle is an electron that has been ejected at a high velocity from an unstable nucleus. An electron has a small mass and an electrical charge of -1. Beta particles cause ionization by displacing electrons from atom orbits. The ionization occurs from collisions with orbiting electrons. Each collision removes kinetic energy from the beta particle, causing it to slow down. Eventually the beta particle will be slowed enough to allow it to be captured as an
2 electrons 0.000549 amu
electron
931.5 MeV
amu
1.02 MeV
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easy to stop and has a low power of penetration. Even the most energetic beta radiation can be stopped by a few millimeters of metal.

Positron Radiation

Positively charged electrons are called positrons. Except for the positive charge, they are identical to beta-minus particles and interact with matter in a similar manner. Positrons are very short-lived, however, and quickly are annihilated by interaction with a negatively charged electron, producing two gammas with a combined energy (calculated below) equal to the rest mass of the positive and negative electrons.

Neutron Radiation

Neutrons have no electrical charge. They have nearly the same mass as a proton (a hydrogen atom nucleus). A neutron has hundreds of times more mass than an electron, but 1/4 the mass of an alpha particle. The source of neutrons is primarily nuclear reactions, such as fission, but they may also be produced from the decay of radioactive nuclides. Because of its lack of charge, the neutron is difficult to stop and has a high penetrating power.

Neutrons are attenuated (reduced in energy and numbers) by three major interactions, elastic scatter, inelastic scatter, and absorption. In elastic scatter, a neutron collides with a nucleus and bounces off. This reaction transmits some of the kinetic energy of the neutron to the nucleus of the atom, resulting in the neutron being slowed, and the atom receives some kinetic energy (motion). This process is sometimes referred to as "the billiard ball effect."

As the mass of the nucleus approaches the mass of the neutron, this reaction becomes more effective in slowing the neutron. Hydrogenous material attenuates neutrons most effectively.

In the inelastic scatter reaction, the same neutron/nucleus collision occurs as in elastic scatter. However, in this reaction, the nucleus receives some internal energy as well as kinetic energy. This slows the neutron, but leaves the nucleus in an excited state. When the nucleus decays to Fixture checkpoint marker-0873-xray is recorded on fixture page 0873.
its original energy level, it normally emits a gamma ray.

In the absorption reaction, the neutron is actually absorbed into the nucleus of an atom. The neutron is captured, but the atom is left in an excited state. If the nucleus emits one or more gamma rays to reach a stable level, the process is called radiative capture. This reaction occurs at most neutron energy levels, but is more probable at lower energy levels.

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Gamma Radiation

Gamma radiation is electromagnetic radiation. It is commonly referred to as a gamma ray and is very similar to an x-ray. The difference is that gamma rays are emitted from the nucleus of an atom, and x-rays are produced by orbiting electrons. The x-ray is produced when orbiting electrons move to a lower energy orbit or when fast-moving electrons approaching an atom are deflected and decelerated as they react with the atom's electrical field (called Bremsstrahlung). The gamma ray is produced by the decay of excited nuclei and by nuclear reactions. Because the gamma ray has no mass and no charge, it is difficult to stop and has a very high penetrating power. A small fraction of the original gamma stream will pass through several feet of concrete or several meters of water.

There are three methods of attenuating gamma rays. The first method is referred to as the photo-electric effect. When a low energy gamma strikes an atom, the total energy of the gamma is expended in ejecting an electron from orbit. The result is ionization of the atom and expulsion of a high energy electron. This reaction is most predominant with low energy gammas interacting in materials with high atomic weight and rarely occurs with gammas having an energy above 1 MeV. Annihilation of the gamma results. Any gamma energy in excess of the binding

The second method of attenuation of gammas is called Compton scattering. The gamma interacts with an orbital or free electron; however, in this case, the photon loses only a fraction of its energy. The actual energy loss depending on the scattering angle of the gamma. The gamma continues on at lower energy, and the energy difference is absorbed by the electron. This reaction becomes important for gamma energies of about 0.1 MeV and higher.
At higher energy levels, a third method of attenuation is predominant. This method is pair-production. When a high energy gamma passes close enough to a heavy nucleus, the gamma completely disappears, and an electron and a positron are formed. For this reaction to take place, the original gamma must have at least 1.02 MeV energy. Any energy greater than 1.02 MeV becomes kinetic energy shared between the electron and positron. The probability of pair■production incre
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Interaction of Radiation with Matter Summary
An alpha particle deposits a large amount of energy in a short distance of travel due to its large mass and charge.
Beta-minus particles interact with the electrons orbiting the nucleus of atoms, causing ionization by displacing the electrons. The beta particle loses energy with each interaction. After the beta particle loses enough energy, it is captured in the orbital shells of an atom.
Positrons interact with matter much the same way as beta minus particles. After the positron has lost most of its energy by ionizing atoms, it is annihilated by interaction with an electron. The electron-positron pair disappear and are replaced by two gammas, each with the energy equivalent of the mass of an electron (0.51 MeV).
Neutrons interact with matter by elastic scattering, inelastic scattering, or absorption.
Photoelectric effect is where a gamma interacts with an electron orbiting an atom. The entire energy of the gamma is transferred to the electron, and the electron is ejected from its orbit.
Fixture checkpoint marker-0874-xray is recorded on fixture page 0874.
In Compton scattering a gamma interacts with an orbital electron, but only part of the gamma energy is transferred to the electron. The electron is ejected from its orbit, and the gamma is scattered off at a lower energy.
In pair-production, a gamma interacts with the electric field of a nucleus and is converted into an electron-positron pair. The gamma must have an energy greater than 1.02 MeV for this to occur.
Summary
The important information in this chapter is summarized below.
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Department of Energy
Fundamentals Handbook
NUCLEAR PHYSICS
AND REACTOR THEORY

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Reactor Theory (Neutron Characteristics)
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TERMINAL OBJECTIVE
1.0 Without references, EXPLAIN how neutron sources produce neutrons.
ENABLING OBJECTIVES
1.1 DEFINE the following terms:
a. Intrinsic neutron source
b. Installed neutron source
1.2 LIST three examples of reactions that produce neutrons in intrinsic neutron sources.
1.3 LIST three examples of reactions that produce neutrons in installed neutron sources.

TERMINAL OBJECTIVE
2.0 Given the necessary information for calculations, EXPLAIN basic concepts in reactor physics and perform calculations.
ENABLING OBJECTIVES
2.1 DEFINE the following terms:

b. Neutron flux e. Macroscopic cross section
c. Microscopic cross section f. Mean free path
2.2 EXPRESS macroscopic cross section in terms of microscopic cross section.
2.3 DESCRIBE how the absorption cross section of typical nuclides varies with neutron energy at energies below the resonance absorption region.
2.4 DESCRIBE the cause of resonance absorption in terms of nuclear energy levels.
2.5 DESCRIBE the energy dependence of resonance absorption peaks for typical light and heavy nuclei.
2.6 EXPRESS mean free path in terms of macroscopic cross section.
2.7 Given the number densities (or total density and component fractions) and microscopic cross sections of components, CALCULATE the macroscopic cross section for a mixture.
2.8 CALCULATE a macroscopic cross section given a material density, atomic mass, and microscopic cross section.
2.9 EXPLAIN neutron shadowing or self-shielding.
2.10 Given the neutron flux and macroscopic cross section, CALCULATE the reaction rate.
2.11 DESCRIBE the relationship between neutron flux and reactor power.

ENABLING OBJECTIVES (Cont.)
2.12 DEFINE the following concepts:
a. Thermalization d. Average logarithmic energy decrement
b. Moderator e. Macroscopic slowing down power
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c. Moderating ratio
2.13 LIST three desirable characteristics of a moderator.
2.14 Given an average fractional energy loss per collision, CALCULATE the energy loss after a specified number of collisions.

TERMINAL OBJECTIVE
3.0 Without references, EXPLAIN the production process and effects on fission of prompt and delayed neutrons.
ENABLING OBJECTIVES
3.1 STATE the origin of prompt neutrons and delayed neutrons.
3.2 STATE the approximate fraction of neutrons that are born as delayed neutrons from the fission of the following nuclear fuels:
a. Uranium-235
b. Plutonium-239
3.3 EXPLAIN the mechanism for production of delayed neutrons.
3.4 EXPLAIN prompt and delayed neutron generation times.
3.5 Given prompt and delayed neutron generation times and delayed neutron fraction, CALCULATE the average generation time.
3.6 EXPLAIN the effect of delayed neutrons on reactor control.

TERMINAL OBJECTIVE
4.0 Without references, DESCRIBE the neutron energy spectrum for the type of reactor presented in this module.

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4.1 STATE the average energy at which prompt neutrons are produced.
4.2 DESCRIBE the neutron energy spectrum in the following reactors:
a. Fast reactor
b. Thermal reactor
4.3 EXPLAIN the reason for the particular shape of the fast, intermediate, and slow energy regions of the neutron flux spectrum for a thermal reactor.
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Reactor Theory (Neutron Characteristics) DOE-HDBK-1019/1-93 NEUTRON SOURCES
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NEUTRON SOURCES
Neutrons from a variety of sources are always present in a reactor core. This is true even when the reactor is shut down. Some of these neutrons are produced by naturally occurring (intrinsic) neutron sources, while others may be the result of fabricated (installed) neutron sources that are incorporated into the design of the reactor. The neutrons produced by sources other than neutron-induced fission are often grouped together and classified as source neutrons.
EO 1.1 DEFINE the following terms:
a. Intrinsic neutron source
b. Installed neutron source
EO 1.2 LIST three examples of reactions that produce neutrons in intrinsic neutron sources.
EO 1.3 LIST three examples of reactions that produce neutrons in installed neutron sources.
Neutron Sources
In addition to neutron-induced fission, neutrons are produced by other reactions. The neutrons produced by reactions other than neutron-induced fission are called source neutrons. Source neutrons are important because they ensure that the neutron population remains high enough to allow a visible indication of neutron level on the most sensitive monitoring instruments while the reactor is shutdown and during the startup sequence. This verifies instrument operability and
Fixture checkpoint marker-0878-xray is recorded on fixture page 0878.
allows monitoring of neutron population changes. Source neutrons can be classified as either intrinsic or installed neutron sources.
Intrinsic Neutron Sources
Some neutrons will be produced in the materials present in the reactor due to a variety of unavoidable reactions that occur because of the nature of these materials. Intrinsic neutron sources are those neutron-producing reactions that always occur in reactor materials.
■ ■ ■ ■
■ ■ ■
■ ■
NEUTRON SOURCES DOE-HDBK-1019/1-93 Reactor Theory (Neutron Characteristics)
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A limited number of neutrons will always be present, even in a reactor core that has never been operated, due to spontaneous fission of some heavy nuclides that are present in the fuel. Uranium■238, ur of the rate at which different heavy nuclides produce neutrons by spontaneous fission.
Californium-252 is not an intrinsic neutron source, but will be discussed in the section on installed neutron sources.
TABLE 1
Neutron Production by Spontaneous Fission

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U 1.8 x 10 years 6.8 x 10 years 8.0 x 10 235
92
17 8 -4
U 8.0 x 10 years 4.5 x 10 years 1.6 x 10 238
92
15 9 -2
Pu 5.5 x 10 years 2.4 x 10 years 3.0 x 10 239
94
5 4 -2
Pu 1.2 x 10 years 6.6 x 10 years 1.0 x 10 240
94
11 3 3
Cf 66.0 years 2.65 years 2.3 x 10 252
98
12

Another intrinsic neutron source is a reaction involving natural boron and fuel. In some reactors, natural boron is loaded into the reactor core as a neutron absorber to improve reactor control or increase core life-time. Boron-11 (80.1% of natural boron) undergoes a reaction with the alpha particle emitted by the radioactive decay of heavy nuclides in the fuel to yield a neutron as shown below.

The boron-11 must be mixed with, or in very close proximity to, the fuel for this reaction because of the short path length of the alpha particle. For a reactor core with this configuration, this (■,n) reaction is an important source of neutrons for reactor startup.

In a reactor that has been operated, another source of neutrons becomes significant. Neutrons may be produced by the interaction of a gamma ray and a deuterium nucleus. This reaction is commonly referred to as a photoneutron reaction because it is initiated by electromagnetic radiation and results in the production of a neutron. The photoneutron reaction is shown below.

■ ■ ■
■ ■
■ ■ ■
■ ■

Reactor Theory (Neutron Characteristics) DOE-HDBK-1019/1-93 NEUTRON SOURCES
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There is an abundant supply of high energy gammas in a reactor that has been operated because many of the fission products are gamma emitters. All water-cooled reactors have some deuterium present in the coolant in the reactor core because a small fraction of natural hydrogen
Fixture checkpoint marker-0879-xray is recorded on fixture page 0879.
is the isotope deuterium. The atom percentage of deuterium in the water ranges from close to the naturally occurring value (0.015%) for light water reactors to above 90% deuterium for heavy water reactors. Therefore, the required conditions for production of photoneutrons exist.

The supply of gamma rays decreases with time after shutdown as the gamma emitters decay; therefore, the photoneutron production rate also decreases. In a few particular reactors, additional D O (heavy water) may be added to the reactor to increase the production of 2 photoneutrons following a long shutdown period.

Installed Neutron Sources

Because intrinsic neutron sources can be relatively weak or dependent upon the recent power history of the reactor, many reactors have artificial sources of neutrons installed. These neutron sources ensure that shutdown neutron levels are high enough to be detected by the nuclear

these conditions. An installed neutron source is an assembly placed in or near the reactor for the sole purpose of producing source neutrons.
One strong source of neutrons is the artificial nuclide californium-252, which emits neutrons at the rate of about 2 x 10 neutrons per second per gram as the result of spontaneous fission. 12 Important drawbacks for some applications may be its high cost and its short half-life (2.65 years).
Many installed neutron sources use the (■,n) reaction with beryllium. These sources are composed of a mixture of metallic beryllium (100% beryllium-9) with a small quantity of an alpha particle emitter, such as a compound of radium, polonium, or plutonium. The reaction that occurs is shown below.
The beryllium is intimately (homogeneously) mixed with the alpha emitter and is usually enclosed in a stainless steel capsule.
Another type of installed neutron source that is widely used is a photoneutron source that
Fixture checkpoint marker-0880-xray is recorded on fixture page 0880.
employs the (■,n) reaction with beryllium. Beryllium is used for photoneutron sources because its stable isotope beryllium-9 has a weakly attached last neutron with a binding energy of only 1.66 MeV. Thus, a gamma ray with greater energy than 1.66 MeV can cause neutrons to be ejected by the (■,n) reaction as shown below.
NEUTRON SOURCES DOE-HDBK-1019/1-93 Reactor Theory (Neutron Characteristics)
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Neutron Sources Summary
& Intrinsic neutron sources are sources of neutrons from materials that are in the reactor for other purposes such as fuel, burnable poison, or moderator.
& Installed neutron sources are materials or components placed in the reactor specifically for the purpose of producing source neutrons.
& Examples of intrinsic neutron sources are listed below.
Spontaneous fission of heavy nuclides in fuel, such as uranium-238, uranium-235, and plutonium-239, results in fission fragments and free neutrons.
Boron-11 mixed with the fuel undergoes an alpha-neutron reaction and becomes nitrogen-14.
Deuterium present in the reactor coolant undergoes a gamma■neutron reaction and becomes hydrogen-1.
& Examples of installed neutron sources are listed below.
Spontaneous fission of californium-252 results in fission fragments and free neutrons.
Beryllium-9 undergoes an alpha-neutron reaction (alpha from the decay of plutonium, polonium, or radium) and becomes carbon-12.
Beryllium-9 undergoes a gamma-neutron reaction (high energy gamma from decay of antimony-124) and becomes beryllium-8.
Many startup sources of this type use antimony and beryllium because after activation with neutrons the radioactive antimony becomes an emitter of high energy gammas. The photoneutron sources of this type are constructed somewhat differently from the (■,n) types. One design incorporates a capsule of irradiated antimony enclosed in a beryllium sleeve. The entire assembly is then encased in a stainless steel cladding. A large reactor may have several neutron sources of this type installed within the core.
Summary
The important information in this chapter is summarized below.
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NUCLEAR CROSS SECTIONS AND NEUTRON FLUX
To determine the frequency of neutron interactions, it is necessary to describe the availability of neutrons to cause interaction and the probability of a neutron interacting with material. The availability of neutrons and the probability of interaction are quantified by the neutron flux and nuclear cross section.
EO 2.1 DEFINE the following terms:
a. Atom density d. Barn
b. Neutron flux e. Macroscopic cross section
c. Microscopic cross section f. Mean free path
EO 2.2 EXPRESS macroscopic cross section in terms of microscopic cross section.
EO 2.3 DESCRIBE how the absorption cross section of typical nuclides varies with neutron energy at energies below the resonance absorption region.
EO 2.4 DESCRIBE the cause of resonance absorption in terms of nuclear energy levels.
EO 2.5 DESCRIBE the energy dependence of resonance absorption peaks for typical light and heavy nuclei.
EO 2.6 EXPRESS mean free path in terms of macroscopic cross section.
Fixture checkpoint marker-0881-xray is recorded on fixture page 0881.
EO 2.7 Given the number densities (or total density and component fractions) and microscopic cross sections of components, CALCULATE the macroscopic cross section for a mixture.
EO 2.8 CALCULATE a macroscopic cross section given a material density, atomic mass, and microscopic cross section.
EO 2.9 EXPLAIN neutron shadowing or self-shielding.
'
DOE-HDBK-1019/1-93
NUCLEAR CROSS SECTIONS AND NEUTRON FLUX Reactor Theory (Neutron Characteristics)
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Introduction
Fission neutrons are born with an average energy of about 2 MeV. These fast neutrons interact with the reactor core materials in various absorption and scattering reactions. Collisions that result in scattering are useful in slowing neutrons to thermal energies. Thermal neutrons may be absorbed by fissile nuclei to produce more fissions or be absorbed in fertile material for conversion to fissionable fuel. Absorption of neutrons in structural components, coolant, and other non-fuel material results in the removal of neutrons without fulfilling any useful purpose. To safely and efficiently operate a nuclear reactor it is necessary to predict the probability that a particular absorption or scattering reaction will occur. Once these probabilities are known, if the availability of neutrons can be determined, then the rate at which these nuclear reactions take place can be predicted.
Atom Density
One important property of a material is the atom density. The atom density is the number of atoms of a given type per unit volume of the material. To calculate the atom density of a substance use Equation (2-1).
(2-1)
where:

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' = density (g/cm)3
N = Avogadro's number (6.022 x 10 atoms/mole) A
23
M = gram atomic weight
N
NA
M
2.699 g
cm3
6.022 x 1023 atoms
mole
26.9815 g
mole
6.024 x 1022 atoms
cm3
DOE-HDBK-1019/1-93
Reactor Theory (Neutron Characteristics) NUCLEAR CROSS SECTIONS AND NEUTRON FLUX
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Example:
A block of aluminum has a density of 2.699 g/cm . If the gram atomic weight of
3
aluminum is 26.9815 g, calculate the atom density of the aluminum.
Solution:
Cross Sections
The probability of a neutron interacting with a nucleus for a particular reaction is dependent upon not only the kind of nucleus involved, but also the energy of the neutron. Accordingly, the absorption of a thermal neutron in most materials is much more probable than the absorption
Fixture checkpoint marker-0882-xray is recorded on fixture page 0882.
of a fast neutron. Also, the probability of interaction will vary depending upon the type of reaction involved.
The probability of a particular reaction occurring between a neutron and a nucleus is called the microscopic cross section ( ) of the nucleus for the particular reaction. This cross section will vary with the energy of the neutron. The microscopic cross section may also be regarded as the effective area the nucleus presents to the neutron for the particular reaction. The larger the effective area, the greater the probability for reaction.
Because the microscopic cross section is an area, it is expressed in units of area, or square centimeters. A square centimeter is tremendously large in comparison to the effective area of a nucleus, and it has been suggested that a physicist once referred to the measure of a square centimeter as being "as big as a barn" when applied to nuclear processes. The name has persisted and microscopic cross sections are expressed in terms of barns. The relationship between barns and cm is shown below. 2
1 barn = 10-24 cm2
N
DOE-HDBK-1019/1-93
NUCLEAR CROSS SECTIONS AND NEUTRON FLUX Reactor Theory (Neutron Characteristics)
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Whether a neutron will interact with a certain volume of material depends not only on the microscopic cross section of the individual nuclei but also on the number of nuclei within that

macroscopic cross section ( ). The macroscopic cross section is the probability of a given reaction occurring per unit travel of the neutron. is related to the microscopic cross section ( ) by the relationship shown below.

(2-2)

where:

= macroscopic cross section (cm)

-1

N = atom density of material (atoms/cm)

3

= microscopic cross-section (cm)

2

The difference between the microscopic and macroscopic cross sections is extremely important and is restated for clarity. The microscopic cross section ( ) represents the effective target area that a single nucleus presents to a bombarding particle. The units are given in barns or cm .

2

The macroscopic cross section ( ) represents the effective target area that is presented by all of the nuclei contained in 1 cm of the material. The units are given as 1/cm or cm .

3 -1

A neutron interacts with an atom of the material it enters in two basic ways. It will either interact through a scattering interaction or through an absorption reaction. The probability of a neutron being absorbed by a particular atom is the microscopic cross section for absorption, . The probability of a neutron scattering off of a particular nucleus is the microscopic cross a section for scattering, s

. The sum of the microscopic cross section for absorption and the microscopic cross section for scattering is the total microscopic cross section, . T

= + T a s

Both the absorption and the scattering microscopic cross sections can be further divided. For instance, the scattering cross section is the sum of the elastic scattering cross section (se) and the inelastic scattering cross section ( ). si

= + s se si

The microscopic absorption cross section ( ) includes all reactions except scattering. However, a for most purposes it is sufficient to merely separate it into two categories, fission (f) and capture ( ). Radiative capture of neutrons was described in the Neutron Interactions chapter c of Module 1.

Fixture checkpoint marker-0883-xray is recorded on fixture page 0883.

= + a f c

Figure 1 Typical Neutron Absorption Cross Section vs. Neutron Energy

The variation of absorption cross sections with neutron energy is often complicated. For many elements the absorption cross sections are small, ranging from a fraction of a barn to a few barns for slow (or thermal) neutrons.

For a considerable number of nuclides of moderately high (or high) mass numbers, an examination of the variation of the absorption cross section with the energy of the incident neutron reveals the existence of three regions on a curve of absorption cross section versus neutron energy. This cross section is illustrated in Figure 1. First, the cross section decreases steadily with increasing neutron energy in a low energy region, which includes the thermal range

proportional to the velocity (v). This region is frequently referred to as the "1/v region," because the absorption cross section is proportional to 1/v, which is the reciprocal of neutron velocity. Following the 1/v region, there occurs the "resonance region" in which the cross sections rise sharply to high values called "resonance peaks" for neutrons of certain energies, and then fall again. These energies are called resonance energies and are a result of the affinity of the nucleus for neutrons whose energies closely match its discrete, quantum energy levels. That is, when the binding energy of a neutron plus the kinetic energy of the neutron are exactly equal to the amount required to raise a compound nucleus from its ground state to a quantum level, resonance absorption occurs. The following example problem further illustrates this point.

■

(

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Assuming that uranium-236 has a nuclear quantum energy level at 6.8 MeV above its ground state, calculate the kinetic energy a neutron must possess to undergo resonant absorption in uranium-235 at this resonance energy level.

BE = [Mass(U) + Mass(neutron) - Mass(U)] x 931 MeV/amu 235 236

BE = (235.043925 + 1.008665 - 236.045563) x 931 MeV/amu

BE = (0.007025 amu) x 931 MeV/amu = 6.54 MeV

6.8 MeV - 6.54 MeV = 0.26 MeV

The difference between the binding energy and the quantum energy level equals the amount of kinetic energy the neutron must possess. The typical heavy nucleus will have many closely■spaced resonanc complex and have more possible configurations and corresponding energy states. Light nuclei, being less complex, have fewer possible energy states and fewer resonances that are sparsely distributed at higher energy levels.

For higher neutron energies, the absorption cross section steadily decreases as the energy of the neutron increases. This is called the "fast neutron region." In this region the absorption cross sections are usually less than 10 barns.

With the exception of hydrogen, for which the value is fairly large, the elastic scattering cross sections are generally small, for example, 5 barns to 10 barns. This is close to the magnitude of the actual geometric cross sectional area expected for atomic nuclei. In potential scattering, the cross section is essentially constant and independent of neutron energy. Resonance elastic scattering and inelastic scattering exhibit resonance peaks similar to those associated with absorption cross sections. The resonances occur at lower energies for heavy nuclei than for light nuclei. In general, the variations in scattering cross sections are very small when compared to the variations that occur in absorption cross sections.

Mean Free Path

If a neutron has a certain probability of undergoing a particular interaction in one centimeter of travel, then the inverse of this value describes how far the neutron will travel (in the average case) before undergoing an interaction. This average distance traveled by a neutron before interaction is known as the mean free path for that interaction and is represented by the symbol ■. The relationship between the mean free path (■) and the macroscopic cross section (*) is shown below.

(2-3)

'

(

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DOE-HDBK-1019/1-93
Reactor Theory (Neutron Characteristics) NUCLEAR CROSS SECTIONS AND NEUTRON FLUX
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Calculation of Macroscopic Cross Section and Mean Free Path
Most materials are composed of several elements, and because most elements are composed of several isotopes, most materials involve many cross sections, one for each isotope involved. Therefore, to include all the isotopes within a given material, it is necessary to determine the macroscopic cross section for each isotope and then sum all the individual macroscopic cross sections. Equation (2-4) can be used to determine the macroscopic cross section for a composite material.
* = N) + N) + N) + ... N) (2-4) 1 1 2 2 3 3 nn
where:
N = the number of nuclei per cm of the n element n
3 th
) = the microscopic cross section of the n element n
th
The following example problems illustrate the calculation of the macroscopic cross section for a single element and for combinations of materials.
Fixture checkpoint marker-0885-xray is recorded on fixture page 0885.
Example 1:
Find the macroscopic thermal neutron absorption cross section for iron, which has a density of 7.86 g/cm . The microscopic cross section for absorption of iron is 2.56 barns 3 and the gram atomic weight is 55.847 g.
Solution:
Step 1: Using Equation (2-1), calculate the atom density of iron.
Step 2: Use this atom density in Equation (2-2) to calculate the macroscopic cross section.
'
'
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NUCLEAR CROSS SECTIONS AND NEUTRON FLUX Reactor Theory (Neutron Characteristics)
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Example 2:
An alloy is composed of 95% aluminum and 5% silicon (by weight). The density of the alloy is 2.66 g/cm . Properties of aluminum and silicon are shown below. 3
Element Gram Atomic Weight)) a (barns) s (barns)
Aluminum 26.9815 0.23 1.49
Silicon 28.0855 0.16 2.20
1. Calculate the atom densities for the aluminum and silicon.
2. Determine the absorption and scattering macroscopic cross sections for thermal neutrons.
3. Calculate the mean free paths for absorption and scattering.
Solution:
Step 1: The density of the aluminum will be 95% of the total density. Using Equation (2-1) yields the atom densities.
a NAl a,Al NSi a,Si
5.64 x 1022 atoms
cm3

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cm3
0.16 x 10 24 cm2
0.0134 cm 1
s NAl s,Al NSi s,Si
5.64 x 1022 atoms
cm3
1.49 x 10 24 cm22.85 x 1021 atoms
cm3
2.20 x 10 24 cm2
0.0903 cm 1
a
1
a
1
0.01345 cm 1
74.3 cm
s
1
s
1
0.0903 cm 1
11.1 cm
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Reactor Theory (Neutron Characteristics) NUCLEAR CROSS SECTIONS AND NEUTRON FLUX
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Fixture checkpoint marker-0886-xray is recorded on fixture page 0886.
Step 2: The macroscopic cross sections for absorption and scattering are calculated using Equation (2-4).
Step 3: The mean free paths are calculated by inserting the macroscopic cross sections calculated above into Equation (2-3).
Thus, a neutron must travel an average of 74.3 cm to interact by absorption in this alloy, but it must travel only 11.1 cm to interact by scattering.
)
)
)
)
(■
(■
DOE-HDBK-1019/1-93
NUCLEAR CROSS SECTIONS AND NEUTRON FLUX Reactor Theory (Neutron Characteristics)
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Effects of Temperature on Cross Section
As discussed, the microscopic absorption cross section varies significantly as neutron energy varies. The microscopic cross sections provided on most charts and tables are measured for a standard neutron velocity of 2200 meters/second, which corresponds to an ambient temperature of 68(F. Therefore, if our material is at a higher temperature, the absorption cross section will be lower than the value for 68(F, and any cross sections which involve absorption (for example,) ,) ,)) must be corrected for the existing temperature. acf

the example illustrates absorption cross section, the same formula may be used to correct capture and fission cross sections.
where:
) = microscopic cross section corrected for temperature
) = microscopic cross section at reference temperature (68(F or 20(C) o
T = reference temperature (68(F) in degrees Rankine ((R) or Kelvin ((K) o
T = temperature for which corrected value is being calculated
NOTE: When using this formula, all temperatures must be converted to (R or (K.
(R = (F + 460
(K = (C + 273
Example:
What is the value of)f for uranium-235 for thermal neutrons at 500(F? Uranium-235 has a) of 583 barns at 68(F. f
Solution:

Neutron Flux
Macroscopic cross sections for neutron reactions with materials determine the probability of one neutron undergoing a specific reaction per centimeter of travel through that material. If one wants to determine how many reactions will actually occur, it is necessary to know how many neutrons are traveling through the material and how many centimeters they travel each second. It is convenient to consider the number of neutrons existing in one cubic centimeter at any one instant and the total distance they travel each second while in that cubic centimeter. The number of neutrons existing in a cm of material at any instant is called neutron density and is 3 represented by the symbol n with units of neutrons/cm . The total distance these neutrons can 3 travel each second will be determined by their velocity.
A good way of defining neutron flux (1) is to consider it to be the total path length covered by all neutrons in one cubic centimeter during one second. Mathematically, this is the equation below.
1 = n v (2-5)
where:
Fixture checkpoint marker-0887-xray is recorded on fixture page 0887.
1 = neutron flux (neutrons/cm -sec) 2
n = neutron density (neutrons/cm)3
v = neutron velocity (cm/sec)
The term neutron flux in some applications (for example, cross section measurement) is used as parallel beams of neutrons traveling in a single direction. The intensity (I) of a neutron beam is the product of the neutron density times the average neutron velocity. The directional beam intensity is equal to the number of neutrons per unit area and time (neutrons/cm -sec) falling on 2 a surface perpendicular to the direction of the beam.
One can think of the neutron flux in a reactor as being comprised of many neutron beams traveling in various directions. Then, the neutron flux becomes the scalar sum of these directional flux intensities (added as numbers and not vectors), that is, 1 = I + I + I +...I . 123 n
Since the atoms in a reactor do not interact preferentially with neutrons from any particular direction, all of these directional beams contribute to the total rate of reaction. In reality, at a given point within a reactor, neutrons will be traveling in all directions.

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Self-Shielding
In some locations within the reactor, the flux level may be significantly lower than in other areas due to a phenomenon referred to as neutron shadowing or self-shielding. For example, the interior of a fuel pin or pellet will "see" a lower average flux level than the outer surfaces since an appreciable fraction of the neutrons will have been absorbed and therefore cannot reach the interior of the fuel pin. This is especially important at resonance energies, where the absorption cross sections are large.
Summary
The important information in this chapter is summarized below.
DOE-HDBK-1019/1-93
Reactor Theory (Neutron Characteristics) NUCLEAR CROSS SECTIONS AND NEUTRON FLUX
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Nuclear Cross Section and Neutron Flux Summary
& Atom density (N) is the number of atoms of a given type per unit volume of material.
& Microscopic cross section ()) is the probability of a given reaction occurring between a neutron and a nucleus.
& Microscopic cross sections are measured in units of barns, where 1 barn = 10-24 cm .2
& Macroscopic cross section (*) is the probability of a given reaction occurring per unit length of travel of the neutron. The units for macroscopic cross section are cm . -1
& The mean free path (■) is the average distance that a neutron travels in a material between interactions.
& Neutron flux (1) is the total path length traveled by all neutrons in one cubic centimeter of material during one second.
& The macroscopic cross section for a material can be calculated using the equation below.
* = N)
& The absorption cross section for a material usually has three distinct regions. At low neutron energies (<1 eV) the cross section is inversely proportional to the neutron velocity.
& Resonance absorption occurs when the sum of the kinetic energy of the neutron and its binding energy is equal to an allowed nuclear energy level of the nucleus.
& Resonance peaks exist at intermediate energy levels. For higher neutron energies, the absorption cross section steadily decreases as the neutron energy increases.
& The mean free path equals 1/*.
& The macroscopic cross section for a mixture of materials can be calculated using
Fixture checkpoint marker-0888-xray is recorded on fixture page 0888.
the equation below.
* = N 1 1 2 2 3 3 nn) + N) + N) + ... N)
& Self-shielding is where the local neutron flux is depressed within a material due to neutron absorption near the surface of the material.
1)
REACTION RATES DOE-HDBK-1019/1-93 Reactor Theory (Neutron Characteristics)
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REACTION RATES

based on the neutron flux, cross section for the interaction, and atom density of the target. This relationship illustrates how a change in one of these items affects the reaction rate.

EO 2.10 Given the neutron flux and macroscopic cross section, CALCULATE the reaction rate.

EO 2.11 DESCRIBE the relationship between neutron flux and reactor power.

Reaction Rates

If the total path length of all the neutrons in a cubic centimeter in a second is known, (neutron flux (1)), and if the probability of having an interaction per centimeter path length is also known (macroscopic cross section (*)), multiply them together to get the number of interactions taking place in that cubic centimeter in one second. This value is known as the reaction rate and is denoted by the symbol R. The reaction rate can be calculated by the equation shown below.

R = 1 * (2-6)

where:

R = reaction rate (reactions/sec)

Fixture checkpoint marker-0889-xray is recorded on fixture page 0889.

1 = neutron flux (neutrons/cm -sec) 2

* = macroscopic cross section (cm) -1

Substituting the fact that * = N) into Equation (2-6) yields the equation below.

where:

) = microscopic cross section (cm)2

N = atom density (atoms/cm)3

Rf f

1 x 1013 neutrons

cm2

sec

0.1 cm 1

1 x 1012 fissions

cm3

sec

Rf f

f

Rf

1.29 x 1012 fissions

cm3

sec

3 x 1013 neutrons

cm2

sec

0.043 cm 1

The reaction rate calculated will depend on which macroscopic cross section is used in the calculation. Normally, the reaction rate of greatest interest is the fission reaction rate.

Example:

If a one cubic centimeter section of a reactor has a macroscopic fission cross section of 0.1 cm , and if the thermal neutron flux is 10 neutrons/cm -sec, what is the fission rate -1 13 2

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Solution:
In addition to using Equation (2-6) to determine the reaction rate based on the physical properties of material, it is also possible to algebraically manipulate the equation to determine physical properties if the reaction rate is known.
Example:
A reactor operating at a flux level of 3 x 10 neutrons/cm -sec contains 10 atoms of
13 2 20
uranium-235 per cm . The reaction rate is 1.29 x 10 fission/cm . Calculate and .
3 12 3
f f
Solution:
Step 1: The macroscopic cross section can be determined by solving
Equation (2-6) for and substituting the appropriate values. f
Rf N f
f
Rf
N
1.29 x 1012 fissions
cm3
sec
1 x 1020 atoms
cm3
3 x 1013 neutrons
Fixture checkpoint marker-0890-xray is recorded on fixture page 0890.
cm2
sec
4.3 x 10 22 cm2 1 barn
1 x 10 24 cm2
430 barns
1 watt 1 erg
1 x 10 7 watt sec
1 MeV
1.602 x 10 6
erg
1 fission
200 MeV
3.12 x 1010 fissions
second
REACTION RATES DOE-HDBK-1019/1-93 Reactor Theory (Neutron Characteristics)
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Step 2: To find the microscopic cross section, replace f with (N x f) and solve
for .
f
Reactor Power Calculation
Multiplying the reaction rate per unit volume by the total volume of the core results in the total number of reactions occurring in the core per unit time. If the amount of energy involved in each reaction were known, it would be possible to determine the rate of energy release (power) due to a certain reaction.

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number of fissions per second that are necessary to produce one watt of power using the following conversion factors.
1 fission = 200 MeV
1 MeV = 1.602 x 10 ergs -6
1 erg = 1 x 10 watt-sec -7
This is equivalent to stating that 3.12 x 10 fissions release 1 watt-second of energy. 10
P
th f V
3.12 x 1010 fissions
watt sec
Reactor Theory (Neutron Characteristics) DOE-HDBK-1019/1-93 REACTION RATES
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The power released in a reactor can be calculated based on Equation (2-6). Multiplying the reaction rate by the volume of the reactor results in the total fission rate for the entire reactor. Dividing by the number of fissions per watt-sec results in the power released by fission in the reactor in units of watts. This relationship is shown mathematically in Equation (2-7) below.
(2-7)
where:
P = power (watts)
= thermal neutron flux (neutrons/cm -sec) th
2
= macroscopic cross section for fission (cm) f
-1
V = volume of core (cm)
3
Relationship Between Neutron Flux and Reactor Power
In an operating reactor the volume of the reactor is constant. Over a relatively short period of time (days or weeks), the number density of the fuel atoms is also relatively constant. Since the atom density and microscopic cross section are constant, the macroscopic cross section must also be constant. Examining Equation (2-7), it is apparent that if the reactor volume and macroscopic
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cross section are constant, then the reactor power and the neutron flux are directly proportional. This is true for day-to-day operation. The neutron flux for a given power level will increase very slowly over a period of months due to the burnup of the fuel and resulting decrease in atom density and macroscopic cross section.
REACTION RATES DOE-HDBK-1019/1-93 Reactor Theory (Neutron Characteristics)
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Reaction Rates Summary
& The reaction rate is the number of interactions of a particular type occurring in a cubic centimeter of material in a second.
& The reaction rate can be calculated by the equation below.
R = 1 *
& Over a period of several days, while the atom density of the fuel can be considered constant, the neutron flux is directly proportional to reactor power.
Summary
The important information in this chapter is summarized below.
Reactor Theory (Neutron Characteristics) DOE-HDBK-1019/1-93 NEUTRON MODERATION
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In thermal reactors, the neutrons that cause fission are at a much lower energy than the energy level at which they were born from fission. In this type of reactor, specific materials must be included in the reactor design to reduce the energy level of the neutrons in an efficient manner.

EO 2.12 DEFINE the following concepts:

a. Thermalization d. Average logarithmic energy decrement

b. Moderator e. Macroscopic slowing down power

c. Moderating ratio

EO 2.13 LIST three desirable characteristics of a moderator.

EO 2.14 Given an average fractional energy loss per collision, CALCULATE the energy loss after a specified number of collisions.

Neutron Slowing Down and Thermalization

Fission neutrons are produced at an average energy level of 2 MeV and immediately begin to slow down as the result of numerous scattering reactions with a variety of target nuclei. After a number of collisions with nuclei, the speed of a neutron is reduced to such an extent that it has approximately the same average kinetic energy as the atoms (or molecules) of the medium in which the neutron is undergoing elastic scattering. This energy, which is only a small fraction of an electron volt at ordinary temperatures (0.025 eV at 20(C), is frequently referred to as the thermal energy, since it depends upon the temperature. Neutrons whose energies have been reduced to values in this region (< 1 eV) are designated thermal neutrons. The process of reducing the energy of a neutron to the thermal region by elastic scattering is referred to as thermalization, slowing down, or moderation. The material used for the purpose of thermalizing neutrons is called a moderator. A good moderator reduces the speed of neutrons in a small number of collisions, but does not absorb them to any great extent. Slowing the neutrons in as few collisions as possible is desirable in order to reduce the amount of neutron leakage from the core and also to reduce the number of resonance absorptions in non-fuel materials. Neutron leakage and resonance absorption will be discussed in the next module.

ln Ei

ln Ef

ln

Ei

Ef

2

A

2

3

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NEUTRON MODERATION DOE-HDBK-1019/1-93 Reactor Theory (Neutron Characteristics)

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The ideal moderating material (moderator) should have the following nuclear properties.

large scattering cross section

small absorption cross section

large energy loss per collision

A convenient measure of energy loss per collision is the logarithmic energy decrement. The average logarithmic energy decrement is the average decrease per collision in the logarithm of the neutron energy. This quantity is represented by the symbol (Greek letter xi).

(2-8)

= average logarithmic energy decrement
Ei = average initial neutron energy
Ef = average final neutron energy
The symbol is commonly called the average logarithmic energy decrement because of the fact that a neutron loses, on the average, a fixed fraction of its energy per scattering collision. Since the fraction of energy retained by a neutron in a single elastic collision is a constant for a given material, is also a constant. Because it is a constant for each type of material and does not depend upon the initial neutron energy, is a convenient quantity for assessing the moderating ability of a material.
The values for the lighter nuclei are tabulated in a variety of sources. The following commonly used approximation may be used when a tabulated value is not available.
This approximation is relatively accurate for mass numbers (A) greater than 10, but for some low values of A it may be in error by over three percent.
N
Fixture checkpoint marker-0893-xray is recorded on fixture page 0893.
ln Ehigh ln Elow
ln
Ehigh
Elow
N
ln
Ehigh
Elow
ln
2 x 106eV
0.025 eV
0.948
19.2 collisions
Reactor Theory (Neutron Characteristics) DOE-HDBK-1019/1-93 NEUTRON MODERATION
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Since represents the average logarithmic energy loss per collision, the total number of collisions necessary for a neutron to lose a given amount of energy may be determined by dividing into the difference of the natural logarithms of the energy range in question. The number of collisions (N) to travel from any energy, E , to any lower energy, E , can be calculated as shown below. high low
Example:
How many collisions are required to slow a neutron from an energy of 2 MeV to a thermal energy of 0.025 eV, using water as the moderator? Water has a value of 0.948 for .
Solution:
Sometimes it is convenient, based upon information known, to work with an average fractional energy loss per collision as opposed to an average logarithmic fraction. If the initial neutron energy level and the average fractional energy loss per collision are known, the final energy level for a given number of collisions may be computed using the following formula.
! (
NEUTRON MODERATION DOE-HDBK-1019/1-93 Reactor Theory (Neutron Characteristics)
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(2-9)
where:

E = neutron energy after N collisions N
x = average fractional energy loss per collision
N = number of collisions
Example:
If the average fractional energy loss per collision in hydrogen is 0.63, what will be the energy of a 2 MeV neutron after (a) 5 collisions? (b) 10 collisions?
Solution:
a)
b)
Macroscopic Slowing Down Power
Although the logarithmic energy decrement is a convenient measure of the ability of a material to slow neutrons, it does not measure all necessary properties of a moderator. A better measure of the capabilities of a material is the macroscopic slowing down power. The macroscopic slowing down power (MSDP) is the product of the logarithmic energy decrement and the macroscopic cross section for scattering in the material. Equation (2-10) illustrates how to calculate the macroscopic slowing down power.
(2-10)
MR
s
a
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Moderating Ratio
Macroscopic slowing down power indicates how rapidly a neutron will slow down in the material in question, but it still does not fully explain the effectiveness of the material as a moderator. An element such as boron has a high logarithmic energy decrement and a good slowing down power, but it is a poor moderator because of its high probability of absorbing neutrons.
The most complete measure of the effectiveness of a moderator is the moderating ratio. The moderating ratio is the ratio of the macroscopic slowing down power to the macroscopic cross section for absorption. The higher the moderating ratio, the more effectively the material performs as a moderator. Equation (2-11) shows how to calculate the moderating ratio of a material.
(2-11)
Moderating properties of different materials are compared in Table 2.
TABLE 2
Moderating Properties of Materials
Material Collisions to Slowing Down
Number of Macroscopic
Thermalize Power
Moderating
Ratio
H2O 0.927 19 1.425 62
D2O 0.510 35 0.177 4830
Helium 0.427 42 9 x 10 51 -6
Beryllium 0.207 86 0.154 126
Boron 0.171 105 0.092 0.00086
Carbon 0.158 114 0.083 216

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Summary
The important information in this chapter is summarized below.
Neutron Moderation Summary
& Thermalization is the process of reducing the energy level of a neutron from the energy level at which it is produced to an energy level in the thermal range.
& The moderator is the reactor material that is present for the purpose of thermalizing neutrons.
& Moderating ratio is the ratio of the macroscopic slowing down power to the macroscopic cross section for absorption.
& The average logarithmic energy decrement (!) is the average change in the logarithm of neutron energy per collision.
& Macroscopic slowing down power is the product of the average logarithmic energy decrement and the macroscopic cross section for scattering.
& There are three desirable characteristics of a moderator.
1. large scattering cross section
2. small absorption cross section
3. large energy loss per collision
& The energy loss after a specified number of collisions can be calculated using the equation below.
E = E (1 - x) N o
N
■
■ ■

PROMPT AND DELAYED NEUTRONS
Not all neutrons are released at the same time following fission. Most neutrons
Fixture checkpoint marker-0895-xray is recorded on fixture page 0895.
are released virtually instantaneously and are called prompt neutrons. A very small fraction of neutrons are released after the decay of fission products and are called delayed neutrons. Although delayed neutrons are a very small fraction of the total number of neutrons, they play an extremely important role in the control of the reactor.
EO 3.1 STATE the origin of prompt neutrons and delayed neutrons.
EO 3.2 STATE the approximate fraction of neutrons that are born as delayed neutrons from the fission of the following nuclear fuels:
a. Uranium-235
b. Plutonium-239
EO 3.3 EXPLAIN the mechanism for production of delayed neutrons.
EO 3.4 EXPLAIN prompt and delayed neutron generation times.
EO 3.5 Given prompt and delayed neutron generation times and delayed neutron fraction, CALCULATE the average generation time.
EO 3.6 EXPLAIN the effect of delayed neutrons on reactor control.
Neutron Classification
The great majority (over 99%) of the neutrons produced in fission are released within about 10-13 seconds of the actual fission event. These are called prompt neutrons. A small portion of fission

taken place. The delayed neutrons are emitted immediately following the first beta decay of a fission fragment known as a delayed neutron precursor. An example of a delayed neutron precursor is bromine-87, shown below.

PROMPT AND DELAYED NEUTRONS DOE-HDBK-1019/1-93 Reactor Theory (Neutron Characteristics) NP-02 Page 30 Rev. 0

For most applications, it is convenient to combine the known precursors into groups with appropriately averaged properties. These groups vary somewhat depending on the fissile material in use. Table 3 lists the characteristics for the six precursor groups resulting from thermal fission of uranium-235. The fraction of all neutrons that are produced by each of these precursors is called the delayed neutron fraction for that precursor. The total fraction of all neutrons born as delayed neutrons is called the delayed neutron fraction (■). The fraction of delayed neutrons produced varies depending on the predominant fissile nuclide in use. The delayed neutron fractions (■) for the fissile nuclides of most interest are as follows: uranium-233 (0.0026), uranium-235 (0.0065), uranium-238 (0.0148), and plutonium-239 (0.0021).

TABLE 3

Delayed Neutron Precursor Groups
for Thermal Fission in Uranium-235

Group	Half-Life (sec)	Delayed Neutron Fraction	Average Energy (MeV)
1	55.7	0.00021	0.25
2	22.7	0.00142	0.46
3	6.2	0.00127	0.41
4	2.3	0.00257	0.45
5	0.61	0.00075	0.41
6	0.23	0.00027	-
Total	-	0.0065	-

Neutron Generation Time

The neutron generation time is the time required for neutrons from one generation to cause the fissions that produce the next generation of neutrons. The generation time for prompt neutrons (5* - pronounced "ell-star") is the total time from birth to rebirth. Three time intervals are involved: (a) the time it takes a fast neutron to slow down to thermal energy, (b) the time the now thermal neutron exists prior to absorption in fuel, and (c) the time required for a fissionable nucleus to emit a fast neutron after neutron absorption.

Fast neutrons slow to thermal energies or leak out of the reactor in 10 seconds to 10 seconds, -4 -6 depending on the moderator. In water moderated reactors, thermal neutrons tend to exist for

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about 10 seconds before they are absorbed. Fission and fast neutron production following -4 neutron absorption in a fissionable nucleus occurs in about 10 seconds. Thus, fast reactors -13 have an 5* of about 10 seconds, while thermal reactors have an 5* of about 10 seconds + 10 -6 -6 -4 seconds, which is about 10 seconds to 10 seconds. -4 -5

■■■■
■■■■
■■

Reactor Theory (Neutron Characteristics) DOE-HDBK-1019/1-93 PROMPT AND DELAYED NEUTRONS Rev. 0 Page 31 NP-02

On the other hand, the average generation time for the six delayed neutron groups is the total time from the birth of the fast neutron to the emission of the delayed neutron. Again, three time intervals are involved: (a) the time it takes a fast neutron to slow down to thermal energy,

absorption to neutron emission by the six precursor groups. The average time for decay of precursors from uranium-235 is 12.5 seconds. The other terms in the delayed neutron generation time are insignificant when compared to this value, and the average delayed neutron generation time becomes ■ 12.5 seconds.

A neutron generation time in the range of 10 seconds to 10 seconds or faster could result in -4 -5 very rapid power excursions, and control would not be possible without the dependence upon delayed neutrons to slow down the rate of the reaction. The average generation time, and hence the rate that power can rise, is determined largely by the delayed neutron generation time. The following equation shows this mathematically.

(2-12)

Example:

Assume a prompt neutron generation time for a particular reactor of 5 x 10 seconds and -5

Fixture checkpoint marker-0897-xray is recorded on fixture page 0897.

a delayed neutron generation time of 12.5 seconds. If ■ is 0.0065, calculate the average generation time.

Solution:

This example demonstrates the effect delayed neutrons have on the neutron generation time and thus reactor control. If a reactor were to be operated in a sustained chain reaction using only prompt neutrons (■ = 0), the generation time from the previous example would be about 5 x 10 seconds. However, by operating the reactor such that a 0.0065 fraction of neutrons are -5 delayed, the generation life time is extended to 0.0813 seconds, providing time for adequate operator control. Therefore, although only a small fraction of the total neutron population, delayed neutrons are extremely important to the control and maintenance of a sustained fission chain reaction.

Summary

The important information in this chapter is summarized on the following page.

■ ■ ■ ■

PROMPT AND DELAYED NEUTRONS DOE-HDBK-1019/1-93 Reactor Theory (Neutron Characteristics) NP-02 Page 32 Rev. 0

Prompt and Delayed Neutrons Summary

& Prompt neutrons are released directly from fission within 10 seconds of the -13 fission event.

& Delayed neutrons are released from the decay of fission products that are called delayed neutron precursors. Delayed neutron precursors are grouped according to half-life. Half-lives vary from fractions of a second to almost a minute.

& The fraction of neutrons born as delayed neutrons is different for different fuel materials. Following are values for some common fuel materials.

Uranium-235 0.0065

Plutonium-239 0.0021

& Delayed neutrons are produced by a classification of fission products known as delayed neutron precursors. When a delayed neutron precursor undergoes a ■ decay, it results in an excited daughter nucleus which immediately ejects a - neutron. Therefore, these delayed neutrons appear with a half-life of the delayed neutron precursor.

& The delayed neutron generation time is the total time from the birth of the fast neutron to the emission of the delayed neutron in the next generation. Delayed neutron generation times are dominated by the half-life of the delayed neutron precursor. The average delayed neutron generation time is about 12.5 seconds.

neutron to thermalize, the amount of time the neutron exists as a thermal neutron before it is absorbed, and the amount of time between a fissionable nuclide absorbing a neutron and fission neutrons being released. Prompt neutron generation time is about 5 x 10 seconds. -5

& The average neutron generation time can be calculated from the prompt and delayed neutron generation times and the delayed neutron fraction using Equation (2-12).

& Delayed neutrons are responsible for the ability to control the rate at which power can rise in a reactor. If only prompt neutrons existed, reactor control would not be possible due to the rapid power changes.

Reactor Theory (Neutron Characteristics) DOE-HDBK-1019/1-93 NEUTRON FLUX SPECTRUM
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Figure 2 Prompt Fission Neutron Energy Spectrum for
Thermal Fission of Uranium-235

NEUTRON FLUX SPECTRUM

The number of neutrons that exist at a given energy level varies. A plot of either the fraction of neutrons or the neutron flux at a given energy versus the energy level is called a neutron energy spectrum. The neutron energy spectrum varies widely for different types of reactors.

Fixture checkpoint marker-0898-xray is recorded on fixture page 0898.

EO 4.1 STATE the average energy at which prompt neutrons are produced.

EO 4.2 DESCRIBE the neutron energy spectrum in the following reactors:

a. Fast reactor

b. Thermal reactor

EO 4.3 EXPLAIN the reason for the particular shape of the fast, intermediate, and slow energy regions of the neutron flux spectrum for a thermal reactor.

Prompt Neutron Energies

The neutrons produced by fission are high energy neutrons, and almost all fission neutrons have energies between 0.1 MeV and 10 MeV. The neutron energy distribution, or spectrum, may best be described by plotting the fraction of neutrons per MeV as a function of neutron energy, as shown in Figure 2. From this figure it can be seen that the most probable neutron energy is about 0.7 MeV. Also, from this data it can be shown that the average energy of fission neutrons is about 2 MeV. Figure 2 is the neutron energy spectrum for thermal fission in uranium-235. The values

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nuclides.
NEUTRON FLUX SPECTRUM DOE-HDBK-1019/1-93 Reactor Theory (Neutron Characteristics)
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Figure 3 Comparison of Neutron Flux Spectra for
Thermal and Fast Breeder Reactor
Thermal and Fast Breeder Reactor Neutron Spectra
The spectrum of neutron energies produced by fission varies significantly from the energy spectrum, or flux, existing in a reactor at a given time. Figure 3 illustrates the difference in neutron flux spectra between a thermal reactor and a fast breeder reactor. The energy distribution of neutrons from fission is essentially the same for both reactors, so the differences in the curve shapes may be attributed to the neutron moderation or slowing down effects.
No attempt is made to thermalize or slow down neutrons in the fast breeder reactor (liquid metal cooled); therefore, an insignificant number of neutrons exist in the thermal range. For the thermal reactor (water moderated), the spectrum of neutrons in the fast region (> 0.1 MeV) has a shape similar to that for the spectrum of neutrons emitted by the fission process.
In the thermal reactor, the flux in the intermediate energy region (1 eV to 0.1 MeV) has approximately a 1/E dependence. That is, if the energy (E) is halved, the flux doubles. This 1/E dependence is caused by the slowing down process, where elastic collisions remove a constant fraction of the neutron energy per collision (on the average), independent of energy; thus, the neutron loses larger amounts of energy per collision at higher energies than at lower energies. The fact that the neutrons lose a constant fraction of energy per collision causes the neutrons to tend to "pile up" at lower energies, that is, a greater number of neutrons exist at the lower energies as a result of this behavior.
vp
2 k T
m
Fixture checkpoint marker-0899-xray is recorded on fixture page 0899.
vp
2 k T
m
2 1.38 x 10 16 erg
K
293 K
1.66 x 10 24 g
2.2 x 105 cm
sec
1 m
100 cm
2200 m
sec
Reactor Theory (Neutron Characteristics) DOE-HDBK-1019/1-93 NEUTRON FLUX SPECTRUM
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In the thermal region the neutrons achieve a thermal equilibrium with the atoms of the moderator material. In any given collision they may gain or lose energy, and over successive collisions will gain as much energy as they lose. These thermal neutrons, even at a specific temperature, do not all have the same energy or velocity; there is a distribution of energies, usually referred to as the Maxwell distribution (e.g., Figure 2). The energies of most thermal neutrons lie close to the most probable energy, but there is a spread of neutrons above and below this value.

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The most probable velocity (vp) of a thermal neutron is determined by the temperature of the medium and can be determined by Equation (2-13) .
(2-13)
where:
v = most probable velocity of neutron (cm/sec) p
k = Boltzman's constant (1.38 x 10 erg/ K) -16
T = absolute temperature in degrees Kelvin (K)
m = mass of neutron (1.66 x 10 grams) -24
Example:
Calculate the most probable velocities for neutrons in thermal equilibrium with their surroundings at the following temperatures. a) 20 C, b) 260 C.
Solution:
a) Calculate the most probable velocity for 20 C using Equation (2-13).
vp
2 k T
m
2 1.38 x 10 16 erg
K
533 K
1.66 x 10 24 g
2.977 x 105 cm
sec
1 m
100 cm
2977 m
sec
NEUTRON FLUX SPECTRUM DOE-HDBK-1019/1-93 Reactor Theory (Neutron Characteristics) NP-02 Page 36 Rev. 0
b) Calculate the most probable velocity for 260 C using Equation (2-13).
From these calculations it is evident that the most probable velocity of a thermal neutron increases as temperature increases. The most probable velocity at 20 C is of particular importance since reference data, such as nuclear cross sections, are tabulated for a neutron velocity of 2200 meters per second.
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Reactor Theory (Neutron Characteristics) DOE-HDBK-1019/1-93 NEUTRON FLUX SPECTRUM Rev. 0 Page 37 NP-02
Neutron Flux Spectrum Summary
& Prompt neutrons are born at energies between 0.1 MeV and 10 MeV. The average prompt neutron energy is about 2 MeV.
& Fast reactors have a neutron energy spectrum that has the same shape as the prompt neutron energy spectrum.
& Thermal reactors have a neutron energy spectrum that has two pronounced peaks, one in the thermal energy region where the neutrons are in thermal equilibrium with the core materials and another in the fast region at energies where neutrons are produced. The flux in the intermediate region (1 eV to 0.1 MeV) has a roughly 1/E dependence.
& The neutron flux spectrum for the fast energy region of a thermal reactor has a shape similar to that of the spectrum of neutrons emitted by the fission process.

Reactor Theory. The handbooks are provided as an aid to DOE nuclear facility contractors. These handbooks were first published as Reactor Operator Fundamentals Manuals in 1985 for use by DOE category A reactors. The subject areas, subject matter content, and level of detail of the Reactor Operator Fundamentals Manuals were determined from several sources. DOE Category A reactor training managers determined which materials should be included, and served as a primary reference in the initial development phase. Training guidelines from the commercial nuclear power industry, results of job and task analyses, and independent input from contractors and operations-oriented personnel were all considered and included to some degree in developing the text material and learning objectives.

The DOE Fundamentals Handbooks represent the needs of various DOE nuclear facilities'

Fixture checkpoint marker-0901-xray is recorded on fixture page 0901.

fundamental training requirements. To increase their applicability to nonreactor nuclear facilities, the Reactor Operator Fundamentals Manual learning objectives were distributed to the Nuclear Facility Training Coordination Program Steering Committee for review and comment. To update their reactor-specific content, DOE Category A reactor training managers also reviewed and commented on the content. On the basis of feedback from these sources, information that applied to two or more DOE nuclear facilities was considered generic and was included. The final draft of each of the handbooks was then reviewed by these two groups. This approach has resulted in revised modular handbooks that contain sufficient detail such that each facility may adjust the content to fit their specific needs.

Each handbook contains an abstract, a foreword, an overview, learning objectives, and text material, and is divided into modules so that content and order may be modified by individual DOE contractors to suit their specific training needs. Each handbook is supported by a separate examination bank with an answer key.

The DOE Fundamentals Handbooks have been prepared for the Assistant Secretary for Nuclear Energy, Office of Nuclear Safety Policy and Standards, by the DOE Training Coordination Program. This program is managed by EG&G Idaho, Inc.

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DOE-HDBK-1019/1-93

NUCLEAR PHYSICS AND REACTOR THEORY

OVERVIEW

The Department of Energy Fundamentals Handbook entitled Nuclear Physics and Reactor Theory was prepared as an information resource for personnel who are responsible for the operation of the Department's nuclear facilities. Almost all processes that take place in a nuclear facility involves the transfer of some type of energy. A basic understanding of nuclear physics and reactor theory is necessary for DOE nuclear facility operators, maintenance personnel, and the technical staff to safely operate and maintain the facility and facility support systems. The information in this handbook is presented to provide a foundation for applying engineering concepts to the job. This knowledge will help personnel understand the impact that their actions may have on the safe and reliable operation of facility components and systems.

The Nuclear Physics and Reactor Theory handbook consists of four modules that are contained in two volumes. The following is a brief description of the information presented in each module of the handbook.

Volume 1 of 2

Module 1 - Atomic and Nuclear Physics

Introduces concepts of atomic physics including the atomic nature of matter, the chart of the nuclides, radioactivity and radioactive decay, neutron interactions and fission, and the interaction of radiation with matter.

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Provides information on reactor theory and neutron characteristics. Includes topics such as neutron sources, neutron flux, neutron cross sections, reaction rates, neutron moderation, and prompt and delayed neutrons.
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NUCLEAR PHYSICS AND REACTOR THEORY
OVERVIEW (Cont.)
Volume 2 of 2
Module 3 - Reactor Theory (Nuclear Parameters)
Explains the nuclear parameters associated with reactor theory. Topics include the neutron life cycle, reactivity and reactivity coefficients, neutron poisons, and control rods.
Module 4 - Reactor Theory (Reactor Operations)
Introduces the reactor operations aspect of reactor theory. Topics include subcritical multiplication, reactor kinetics, and reactor operation.
The information contained in this handbook is not all-encompassing. An attempt to present the entire subject of nuclear physics and reactor theory would be impractical. However,
Fixture checkpoint marker-0902-xray is recorded on fixture page 0902.
the Nuclear Physics and Reactor Theory handbook presents enough information to provide the reader with the fundamental knowledge necessary to understand the advanced theoretical concepts presented in other subject areas, and to understand basic system and equipment operation.
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Department of Energy
Fundamentals Handbook
NUCLEAR PHYSICS
AND REACTOR THEORY
Module 1
Atomic and Nuclear Physics
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TERMINAL OBJECTIVE
1.0 Given sufficient information, DESCRIBE atoms, including components, structure, and nomenclature.
ENABLING OBJECTIVES
1.1 STATE the characteristics of the following atomic particles, including mass, charge, and location within the atom:
a. Proton
b. Neutron
Fixture checkpoint marker-0905-xray is recorded on fixture page 0905.
c. Electron
1.2 DESCRIBE the Bohr model of an atom.
1.3 DEFINE the following terms:
a. Nuclide c. Atomic number
b. Isotope d. Mass number
1.4 Given the standard A
ZX notation for a particular nuclide, DETERMINE the following:
a. Number of protons
b. Number of neutrons

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1.5 DESCRIBE the three forces that act on particles within the nucleus and affect the stability of the nucleus.
1.6 DEFINE the following terms:
a. Enriched uranium
b. Depleted uranium
1.7 DEFINE the following terms:
a. Mass defect
b. Binding energy
1.8 Given the atomic mass for a nuclide and the atomic masses of a neutron, proton, and electron, CALCULATE the mass defect and binding energy of the nuclide.
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OBJECTIVES DOE-HDBK-1019/1-93 Atomic and Nuclear Physics
TERMINAL OBJECTIVE
2.0 Given necessary references, DESCRIBE the various modes of radioactive decay.
ENABLING OBJECTIVES
Fixture checkpoint marker-0906-xray is recorded on fixture page 0906.
2.1 DESCRIBE the following processes:
a. Alpha decay d. Electron capture
b. Beta-minus decay e. Internal conversions
c. Beta-plus decay f. Isomeric transitions
2.2 Given a Chart of the Nuclides, WRITE the radioactive decay chain for a nuclide.
2.3 EXPLAIN why one or more gamma rays typically accompany particle emission.
2.4 Given the stability curve on the Chart of the Nuclides, DETERMINE the type of radioactive decay that the nuclides in each region of the chart will typically undergo.
2.5 DEFINE the following terms:
a. Radioactivity d. Radioactive decay constant
b. Curie e. Radioactive half-life
c. Becquerel
2.6 Given the number of atoms and either the half-life or decay constant of a nuclide, CALCULATE the activity.
2.7 Given the initial activity and the decay constant of a nuclide, CALCULATE the activity at any later time.
2.8 CONVERT between the half-life and decay constant for a nuclide.
2.9 Given the Chart of the Nuclides and the original activity, PLOT the radioactive decay curve for a nuclide on either linear or semi-log coordinates.
2.10 DEFINE the following terms:
a. Radioactive equilibrium
b. Transient radioactive equilibrium
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Atomic and Nuclear Physics DOE-HDBK-1019/1-93 OBJECTIVES
TERMINAL OBJECTIVE
3.0 Without references, DESCRIBE the different nuclear interactions initiated by neutrons.
ENABLING OBJECTIVES
3.1 DESCRIBE the following scattering interactions between a neutron and a nucleus:
a. Elastic scattering
b. Inelastic scattering
3.2 STATE the conservation laws that apply to an elastic collision between a neutron and a nucleus.

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a. Radiative capture
b. Particle ejection
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OBJECTIVES DOE-HDBK-1019/1-93 Atomic and Nuclear Physics
TERMINAL OBJECTIVE
4.0 Without references, DESCRIBE the fission process.
ENABLING OBJECTIVES
4.1 EXPLAIN the fission process using the liquid drop model of a nucleus.
4.2 DEFINE the following terms:
a. Excitation energy
b. Critical energy
4.3 DEFINE the following terms:
a. Fissile material
b. Fissionable material
c. Fertile material
4.4 DESCRIBE the processes of transmutation, conversion, and breeding.
4.5 DESCRIBE the curve of Binding Energy per Nucleon versus mass number and give a qualitative description of the reasons for its shape.
4.6 EXPLAIN why only the heaviest nuclei are easily fissioned.
4.7 EXPLAIN why uranium-235 fissions with thermal neutrons and uranium-238 fissions only with fast neutrons.
4.8 CHARACTERIZE the fission products in terms of mass groupings and radioactivity.
Fixture checkpoint marker-0907-xray is recorded on fixture page 0907.
4.9 Given the nuclides involved and their masses, CALCULATE the energy released from fission.
4.10 Given the curve of Binding Energy per Nucleon versus mass number, CALCULATE the energy released from fission.
NP-01 Page x Rev. 0
Atomic and Nuclear Physics DOE-HDBK-1019/1-93 OBJECTIVES
TERMINAL OBJECTIVE
5.0 Without references, DESCRIBE how the various types of radiation interact with matter.
ENABLING OBJECTIVES
5.1 DESCRIBE interactions of the following with matter:
a. Alpha particle c. Positron
b. Beta particle d. Neutron
5.2 DESCRIBE the following ways that gamma radiation interacts with matter:
a. Compton scattering
b. Photoelectric effect
c. Pair production
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OBJECTIVES DOE-HDBK-1019/1-93 Atomic and Nuclear Physics
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NP-01 Page xii Rev. 0
Atomic and Nuclear Physics DOE-HDBK-1019/1-93 ATOMIC NATURE OF MATTER
ATOMIC NATURE OF MATTER
All matter is composed of atoms. The atom is the smallest amount of matter that retains the properties of an element. Atoms themselves are composed of smaller particles, but these smaller particles no longer have

SYNTHETIC FIXTURE PAGE 0908 OF 1200

EO 1.1 STATE the characteristics of the following atomic particles, including mass, charge, and location within the atom:
a. Proton
b. Neutron
c. Electron
EO 1.2 DESCRIBE the Bohr model of an atom.
EO 1.3 DEFINE the following terms:
a. Nuclide c. Atomic number
b. Isotope d. Mass number
EO 1.4 Given the standard A ZX notation for a particular nuclide, DETERMINE the following:
a. Number of protons
b. Number of neutrons
c. Number of electrons
EO 1.5 DESCRIBE the three forces that act on particles within the nucleus and affect the stability of the nucleus.

Structure of Matter

Early Greek philosophers speculated that the earth was made up of different combinations of basic substances, or elements. They considered these basic elements to be earth, air, water, and fire. Modern science shows that the early Greeks held the correct concept that matter consists of a combination of basic elements, but they incorrectly identified the elements.

In 1661 the English chemist Robert Boyle published the modern criterion for an element. He defined an element to be a basic substance that cannot be broken down into any simpler substance after it is isolated from a compound, but can be combined with other elements to form compounds. To date, 105 different elements have been confirmed to exist, and researchers claim to have discovered three additional elements. Of the 105 confirmed elements, 90 exist in nature and 15 are man-made.

Fixture checkpoint marker-0908-xray is recorded on fixture page 0908.

Another basic concept of matter that the Greeks debated was whether matter was continuous or discrete. That is, whether matter could be continuously divided and subdivided into ever smaller particles or whether eventually an indivisible particle would be encountered. Democritus in about 450 B.C. argued that substances were ultimately composed of small, indivisible particles that he labeled atoms. He further suggested that different substances were composed of different atoms or combinations of atoms, and that one substance could be converted into another by rearranging the atoms. It was impossible to conclusively prove or disprove this proposal for more than 2000 years.

The modern proof for the atomic nature of matter was first proposed by the English chemist John Dalton in 1803. Dalton stated that each chemical element possesses a particular kind of atom, and any quantity of the element is made up of identical atoms of this kind. What distinguishes one element from another element is the kind of atom of which it consists, and the basic physical difference between kinds of atoms is their weight.

Subatomic Particles

For almost 100 years after Dalton established the atomic nature of atoms, it was considered impossible to divide the atom into even smaller parts. All of the results of chemical experiments during this time indicated that the atom was indivisible. Eventually, experimentation into

exist. In 1906, J. J. Thompson won the Nobel Prize in physics for establishing the existence of electrons. Electrons are negatively-charged particles that have 1/1835 the mass of the hydrogen atom. Soon after the discovery of electrons, protons were discovered. Protons are relatively large particles that have almost the same mass as a hydrogen atom and a positive charge equal in magnitude (but opposite in sign) to that of the electron. The third subatomic particle to be discovered, the neutron, was not found until 1932. The neutron has almost the same mass as the proton, but it is electrically neutral.

Bohr Model of the Atom

The British physicist Ernest Rutherford postulated that the positive charge in an atom is concentrated in a small region called a nucleus at the center of the atom with electrons existing in orbits around it. Niels Bohr, coupling Rutherford's postulation with the quantum theory introduced by Max Planck, proposed that the atom consists of a dense nucleus of protons surrounded by electrons traveling in discrete orbits at fixed distances from the nucleus. An electron in one of these orbits or shells has a specific or discrete quantity of energy (quantum). When an electron moves from one allowed orbit to another allowed orbit, the energy difference between the two states is emitted or absorbed in the form of a single quantum of radiant energy called a photon. Figure 1 is Bohr's model of the hydrogen atom showing an electron as having just dropped from the third shell to the first shell with the emission of a photon that has an energy = hv. (h = Planck's constant = 6.63 x 10-34 J-s and v = frequency of the photon.) Bohr's theory was the first to successfully account for the discrete energy levels of this radiation as measured in the laboratory. Although Bohr's atomic model is designed specifically to explain the hydrogen atom, his theories apply generally to the structure of all atoms. Additional information on electron shell theory can be found in the Chemistry Fundamentals Handbook.

Figure 1 Bohr's Model of the Hydrogen Atom

Properties of the three subatomic particles are listed in Table 1.

TABLE 1
Properties of Subatomic Particles

Particle	Location	Charge	Mass
Neutron	Nucleus	none	1.008665 amu
Proton	Nucleus	+1	1.007277 amu
Electron	Shells around nucleus	-1	0.0005486 amu

Measuring Units on the Atomic Scale

Fixture checkpoint marker-0909-xray is recorded on fixture page 0909.

The size and mass of atoms are so small that the use of normal measuring units, while possible, is often inconvenient. Units of measure have been defined for mass and energy on the atomic scale to make measurements more convenient to express. The unit of measure for mass is the atomic mass unit (amu). One atomic mass unit is equal to 1.66 x 10-24 grams. The reason for this particular value for the atomic mass unit will be discussed in a later chapter. Note from Table 1 that the mass of a neutron and a proton are both about 1 amu. The unit for energy is the electron volt (eV). The electron volt is the amount of energy acquired by a single electron when it falls through a potential difference of one volt. One electron volt is equivalent to 1.602 x 10-19 joules or 1.18 x 10-19 foot-pounds.

Nuclides

The total number of protons in the nucleus of an atom is called the atomic number of the atom

as the number of protons in the nucleus. The number of neutrons in a nucleus is known as the neutron number and is given the symbol N. The mass number of the nucleus is the total number of nucleons, that is, protons and neutrons in the nucleus. The mass number is given the symbol A and can be found by the equation Z + N = A.

Each of the chemical elements has a unique atomic number because the atoms of different elements contain a different number of protons. The atomic number of an atom identifies the particular element.

Each type of atom that contains a unique combination of protons and neutrons is called a nuclide. Not all combinations of numbers of protons and neutrons are possible, but about 2500 specific nuclides with unique combinations of neutrons and protons have been identified. Each nuclide is denoted by the chemical symbol of the element with the atomic number written as a subscript and the mass number written as a superscript, as shown in Figure 2. Because each element has a unique name, chemical symbol, and atomic number, only one of the three is necessary to identify the element. For this reason nuclides can also be identified by either the chemical name or the chemical symbol followed by the mass number (for example, U-235 or uranium-235).

Figure 2 Nomenclature for Identifying Nuclides

Fixture checkpoint marker-0910-xray is recorded on fixture page 0910.

Another common format is to use the abbreviation of the chemical element with the mass number superscripted (for example, 235U). In this handbook the format used in the text will usually be the element's name followed by the mass number. In equations and tables, the format in Figure 2 will usually be used.

Example:

State the name of the element and the number of protons, electrons, and neutrons in the nuclides listed below.

1
1H
10
5B
14
7N
11
4
4
8Cd
23
9
9
4Pu

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Solution:
The name of the element can be found from the Periodic Table (refer to Chemistry Fundamentals Handbook) or the Chart of the Nuclides (to be discussed later). The number of protons and electrons are equal to Z. The number of neutrons is equal to Z - A.
Nuclide Element Protons Electrons Neutrons
1
1H hydrogen 1 1 0
10
5B boron 5 5 5
14
7N nitrogen 7 7 7
11
4
4
8Cd cadmium 48 48 66
23
9
9
4Pu plutonium 94 94 145
Fixture checkpoint marker-0911-xray is recorded on fixture page 0911.
Isotopes
Isotopes are nuclides that have the same atomic number and are therefore the same element, but differ in the number of neutrons. Most elements have a few stable isotopes and several unstable, radioactive isotopes. For example, oxygen has three stable isotopes that can be found in nature (oxygen-16, oxygen-17, and oxygen-18) and eight radioactive isotopes. Another example is hydrogen, which has two stable isotopes (hydrogen-1 and hydrogen-2) and a single radioactive isotope (hydrogen-3).
The isotopes of hydrogen are unique in that they are each commonly referred to by a unique name instead of the common chemical element name. Hydrogen-1 is almost always referred to as hydrogen, but the term protium is infrequently used also. Hydrogen-2 is commonly called deuterium and symbolized 2
1D. Hydrogen-3 is commonly called tritium and symbolized 31T. This
text will normally use the symbology 2
1H and 31H for deuterium and tritium, respectively.
Atomic and Nuclear Radii
The size of an atom is difficult to define exactly due to the fact that the electron cloud, formed by the electrons moving in their various orbitals, does not have a distinct outer edge. A reasonable measure of atomic size is given by the average distance of the outermost electron from the nucleus. Except for a few of the lightest atoms, the average atomic radii are approximately the same for all atoms, about 2 x 10 -8 cm.
Like the atom the nucleus does not have a sharp outer boundary. Experiments have shown that the nucleus is shaped like a sphere with a radius that depends on the atomic mass number of the atom. The relationship between the atomic mass number and the radius of the nucleus is shown in the following equation.

r = (1.25 x 10 -13 cm) A1/3

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r = radius of the nucleus (cm)
A = atomic mass number (dimensionless)
The values of the nuclear radii for some light, intermediate, and heavy nuclides are shown in Table 2.
TABLE 2
Calculated Values for Nuclear Radii
Nuclide Radius of Nucleus
1
1H 1.25 x 10 -13 cm
10
5B 2.69 x 10 -13 cm
5
2
6
6Fe 4.78 x 10 -13 cm
17
7
8
2Hf 7.01 x 10 -13 cm
23
9
8
2U 7.74 x 10 -13 cm
25
9
2
Fixture checkpoint marker-0912-xray is recorded on fixture page 0912.
8Cf 7.89 x 10 -13 cm
From the table, it is clear that the radius of a typical atom (e.g. 2 x 10 -8 cm) is more than 25,000 times larger than the radius of the largest nucleus.
Nuclear Forces
In the Bohr model of the atom, the nucleus consists of positively-charged protons and electrically■neutra as nucleons. One problem that the Bohr model of the atom presented was accounting for an attractive force to overcome the repulsive force between protons.
Two forces present in the nucleus are (1) electrostatic forces between charged particles and (2) gravitational forces between any two objects that have mass. It is possible to calculate the magnitude of the gravitational force and electrostatic force based upon principles from classical physics.
Rev. 0 Page 7 NP-01
Fg
G m1 m2
r
2
Fe
K Q1 Q2
r
2

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Newton stated that the gravitational force between two bodies is directly proportional to the masses of the two bodies and inversely proportional to the square of the distance between the bodies. This relationship is shown in the equation below.
where:
F = gravitational force (newtons) g
m = mass of first body (kilograms) 1
m = mass of second body (kilograms) 2
G = gravitational constant (6.67 x 10 N-m /kg)
-11 2 2
r = distance between particles (meters)
The equation illustrates that the larger the masses of the objects or the smaller the distance between the objects, the greater the gravitational force. So even though the masses of nucleons are very small, the fact that the distance between nucleons is extremely short may make the gravitational force significant. It is necessary to calculate the value for the gravitational force and compare it to the value for other forces to determine the significance of the gravitational force in the nucleus. The gravitational force between two protons that are separated by a distance of 10 meters is about 10 newtons. -20 -24
Coulomb's Law can be used to calculate the force between two protons. The electrostatic force is directly proportional to the electrical charges of the two particles and inversely proportional to the square of the distance between the particles. Coulomb's Law is stated as the following equation.
where:
F = electrostatic force (newtons) e
K = electrostatic constant (9.0 x 10 N-m /C)
9 2 2
Q = charge of first particle (coulombs) 1
Q = charge of second particle (coulombs) 2
r = distance between particles (meters)
Using this equation, the electrostatic force between two protons that are separated by a distance of 10 meters is about 10 newtons. Comparing this result with the calculation of the
-20 12
gravitational force (10 newtons) shows that the gravitational force is so small that it can be
-24
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neglected.
Atomic and Nuclear Physics DOE-HDBK-1019/1-93 ATOMIC NATURE OF MATTER
If only the electrostatic and gravitational forces existed in the nucleus, then it would be impossible to have stable nuclei composed of protons and neutrons. The gravitational forces are much too small to hold the nucleons together compared to the electrostatic forces repelling the protons. Since stable atoms of neutrons and protons do exist, there must be another attractive force acting within the nucleus. This force is called the nuclear force.
The nuclear force is a strong attractive force that is independent of charge. It acts equally only between pairs of neutrons, pairs of protons, or a neutron and a proton. The nuclear force has a very short range; it acts only over distances approximately equal to the diameter of the nucleus (10 -13 cm). The attractive nuclear force between all nucleons drops off with distance much faster than the repulsive electrostatic force between protons.
TABLE 3

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Force Interaction Range
Gravitational Very weak attractive force
between all nucleons Relatively long
Electrostatic Strong repulsive force between
like charged particles (protons) Relatively long
Nuclear Force Strong attractive force between
all nucleons Extremely short
In stable atoms, the attractive and repulsive forces in the nucleus balance. If the forces do not balance, the atom cannot be stable, and the nucleus will emit radiation in an attempt to achieve a more stable configuration.
Summary
Fixture checkpoint marker-0914-xray is recorded on fixture page 0914.
The important information in this chapter is summarized on the following page.
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ATOMIC NATURE OF MATTER DOE-HDBK-1019/1-93 Atomic and Nuclear Physics
Atomic Nature of Matter Summary
Atoms consist of three basic subatomic particles. These particles are the proton, the neutron, and the electron.
Protons are particles that have a positive charge, have about the same mass as a hydrogen atom, and exist in the nucleus of an atom.
Neutrons are particles that have no electrical charge, have about the same mass as a hydrogen atom, and exist in the nucleus of an atom.
Electrons are particles that have a negative charge, have a mass about eighteen hundred times smaller than the mass of a hydrogen atom, and exist in orbital shells around the nucleus of an atom.
The Bohr model of the atom consists of a dense nucleus of protons and neutrons (nucleons) surrounded by electrons traveling in discrete orbits at fixed distances from the nucleus.
Nuclides are atoms that contain a particular number of protons and neutrons.
Isotopes are nuclides that have the same atomic number and are therefore the same element, but differ in the number of neutrons.
The atomic number of an atom is the number of protons in the nucleus.
The mass number of an atom is the total number of nucleons (protons and neutrons) in the nucleus.
The notation A
ZX is used to identify a specific nuclide. "Z" represents the atomic number, which is equal to the number of protons. "A" represents the mass number, which is equal to the number of nucleons. "X" represents the chemical symbol of the element.
Number of protons = Z
Number of electrons = Z
Number of neutrons = A - Z
The stability of a nucleus is determined by the different forces interacting within it. The electrostatic force is a relatively long-range, strong, repulsive force that acts between the positively charged protons. The nuclear force is a relatively short-range attractive force between all nucleons. The gravitational force the long range, relatively weak attraction between masses, is negligible compared to the other forces.

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Atomic and Nuclear Physics DOE-HDBK-1019/1-93 CHART OF THE NUCLIDES
CHART OF THE NUCLIDES
The Chart of the Nuclides, like the Periodic Table, is a convenient format for presenting a large amount of scientific information in an organized manner.
EO 1.6 DEFINE the following terms:
a. Enriched uranium
b. Depleted uranium
Chart of the Nuclides
A tabulated chart called the Chart of the Nuclides lists the stable and unstable nuclides in addition to pertinent information about each one. Figure 3 shows a small portion of a typical chart. This chart plots a box for each individual nuclide, with the number of protons (Z) on the vertical axis and the number of neutrons (N = A - Z) on the horizontal axis.
The completely gray squares indicate stable isotopes. Those in white squares are artificially radioactive, meaning that they are produced by artificial techniques and do not occur naturally. By consulting a complete chart, other types of isotopes can be found, such as naturally occurring radioactive types (but none are found in the region of the chart that is illustrated in Figure 3).
Located in the box on the far left of each horizontal row is general information about the
Fixture checkpoint marker-0915-xray is recorded on fixture page 0915.
element. The box contains the chemical symbol of the element in addition to the average atomic weight of the naturally occurring substance and the average thermal neutron absorption cross section, which will be discussed in a later module. The known isotopes (elements with the same atomic number Z but different mass number A) of each element are listed to the right.
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CHART OF THE NUCLIDES DOE-HDBK-1019/1-93 Atomic and Nuclear Physics
Figure 3 Nuclide Chart for Atomic Numbers 1 to 6
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Atomic and Nuclear Physics DOE-HDBK-1019/1-93 CHART OF THE NUCLIDES
Information for Stable Nuclides
For the stable isotopes, in addition to the symbol and the atomic mass number, the number percentage of each isotope in the naturally occurring element is listed, as well as the thermal neutron activation cross section and the mass in atomic mass units (amu). A typical block for a stable nuclide from the Chart of the Nuclides is shown in Figure 4.
Figure 4 Stable Nuclides
Information for Unstable Nuclides
For unstable isotopes the additional information includes the half life, the mode of decay (for example, β^-, α), the total disintegration energy in MeV (million electron volts), and the mass in amu when available. A typical block for an unstable nuclide from the Chart of the Nuclides is shown in Figure 5.
Figure 5 Unstable Nuclides
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CHART OF THE NUCLIDES DOE-HDBK-1019/1-93 Atomic and Nuclear Physics
Neutron - Proton Ratios
Figure 6 shows the distribution of the stable nuclides plotted on the same axes as the Chart of the Nuclides. As the mass numbers become higher, the ratio of neutrons to protons in the nucleus becomes larger. For helium-4 (2 protons and 2 neutrons) and oxygen-16 (8 protons and 8 neutrons) this ratio is unity. For indium-115 (49 protons and 66 neutrons) the ratio of neutrons

Figure 6 Neutron - Proton Plot of the Stable Nuclides

If a heavy nucleus were to split into two fragments, each fragment would form a nucleus that would have approximately the same neutron-to-proton ratio as the heavy nucleus. This high neutron-to-proton ratio places the fragments below and to the right of the stability curve displayed by Figure 6. The instability caused by this excess of neutrons is generally rectified by successive beta emissions, each of which converts a neutron to a proton and moves the nucleus toward a more stable neutron-to-proton ratio.

■

Natural Abundance of Isotopes

The relative abundance of an isotope in nature compared to other isotopes of the same element is relatively constant. The Chart of the Nuclides presents the relative abundance of the naturally occurring isotopes of an element in units of atom percent. Atom percent is the percentage of the atoms of an element that are of a particular isotope. Atom percent is abbreviated as a/o.

For example, if a cup of water contains 8.23 x 10 atoms of oxygen, and the isotopic abundance 24 of oxygen-18 is 0.20%, then there are 1.65 x 10 atoms of oxygen-18 in the cup. 22

The atomic weight for an element is defined as the average atomic weight of the isotopes of the element. The atomic weight for an element can be calculated by summing the products of the isotopic abundance of the isotope with the atomic mass of the isotope.

Example:

Calculate the atomic weight for the element lithium. Lithium-6 has an atom percent abundance of 7.5% and an atomic mass of 6.015122 amu. Lithium-7 has an atomic abundance of 92.5% and an atomic mass of 7.016003 amu.

Fixture checkpoint marker-0916-xray is recorded on fixture page 0916.

Solution:

The other common measurement of isotopic abundance is weight percent (w/o). Weight percent is the percent weight of an element that is a particular isotope. For example, if a sample of material contained 100 kg of uranium that was 28 w/o uranium-235, then 28 kg of uranium-235 was present in the sample.

Enriched and Depleted Uranium

Natural uranium mined from the earth contains the isotopes uranium-238, uranium-235 and uranium-234. The majority (99.2745%) of all the atoms in natural uranium are uranium-238. Most of the remaining atoms (0.72%) are uranium-235, and a slight trace (0.0055%) are uranium-234. Although all isotopes of uranium have similar chemical properties, each of the isotopes has significantly different nuclear properties. For reasons that will be discussed in later modules, the isotope uranium-235 is usually the desired material for use in reactors.

A vast amount of equipment and energy are expended in processes that separate the isotopes of uranium (and other elements). The details of these processes are beyond the scope of this module. These processes are called enrichment processes because they selectively increase the proportion of a particular isotope. The enrichment process typically starts with feed material that has the proportion of isotopes that occur naturally. The process results in two types of

In the case of uranium, the natural uranium ore is 0.72 a/o uranium-235. The desired outcome of the enrichment process is to produce enriched uranium. Enriched uranium is defined as uranium in which the isotope uranium-235 has a concentration greater than its natural value. The enrichment process will also result in the byproduct of depleted uranium. Depleted uranium is

value. Although depleted uranium is referred to as a by-product of the enrichment process, it does have uses in the nuclear field and in commercial and defense industries.

Summary

The important information in this chapter is summarized below.

Chart of the Nuclides Summary

Enriched uranium is uranium in which the isotope uranium-235 has a concentration greater than its natural value of 0.7%.

Depleted uranium is uranium in which the isotope uranium-235 has a concentration less than its natural value of 0.7%.

MASS DEFECT AND BINDING ENERGY

The separate laws of Conservation of Mass and Conservation of Energy are not applied strictly on the nuclear level. It is possible to convert between mass and energy. Instead of two separate conservation laws, a single conservation law states that the sum of mass and energy is conserved. Mass does not magically appear and disappear at random. A decrease in mass will be accompanied by a corresponding increase in energy and vice versa.

EO 1.7 DEFINE the following terms:

a. Mass defect

b. Binding energy

EO 1.8 Given the atomic mass for a nuclide and the atomic masses of a neutron, proton, and electron, CALCULATE the mass defect and binding energy of the nuclide.

Mass Defect

Careful measurements have shown that the mass of a particular atom is always slightly less than the sum of the masses of the individual neutrons, protons, and electrons of which the atom consists. The difference between the mass of the atom and the sum of the masses of its parts is called the mass defect (Δm). The mass defect can be calculated using Equation (1-1). In calculating the mass defect it is important to use the full accuracy of mass measurements because the difference in mass is small compared to the mass of the atom. Rounding off the masses of atoms and particles to three or four significant digits prior to the calculation will result in a

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calculated mass defect of zero.

$$\Delta m = [\, Z(mp + me) + (A-Z)mn \,] - matom \quad (1\text{-}1)$$

where:

Δm = mass defect (amu)

mp = mass of a proton (1.007277 amu)

mn = mass of a neutron (1.008665 amu)

me = mass of an electron (0.000548597 amu)

matom = mass of nuclide A

ZX (amu)

Z = atomic number (number of protons)

A = mass number (number of nucleons)

Example:

Calculate the mass defect for lithium-7. The mass of lithium-7 is 7.016003 amu.

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∆ m Z m
p
me A Z mn matom
∆ m 3 1.007826 amu 7 3 1.008665 amu 7.016003 amu
∆ m 0.0421335 amu
Binding Energy
The loss in mass, or mass defect, is due to the conversion of mass to binding energy when the nucleus is formed. Binding energy is defined as the amount of energy that must be supplied to a nucleus to completely separate its nuclear particles (nucleons). It can also be understood as the amount of energy that would be released if the nucleus was formed from the separate particles. Binding energy is the energy equivalent of the mass defect. Since the mass defect was converted to binding energy (BE) when the nucleus was formed, it is possible to calculate the binding energy using a conversion factor derived by the mass-energy relationship from Einstein's Theory of Relativity.
Einstein's famous equation relating mass and energy is E = mc2 where c is the velocity of light (c = 2.998 x 108 m/sec). The energy equivalent of 1 amu can be determined by inserting this quantity of mass into Einstein's equation and applying conversion factors.
E m c 2
1 amu
1.6606 x 10 27 kg
1 amu
2.998 x 108 m
sec
2
1 N
1
kg m
sec2
1 J
1 N m
1.4924 x 10 10 J 1 MeV
1.6022 x 10 13 J
931.5 MeV
Conversion Factors:
1 amu = 1.6606 x 10 -27 kg
1 newton = 1 kg-m/sec2
1 joule = 1 newton-meter
1 MeV = 1.6022 x 10-13 joules
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Fixture checkpoint marker-0918-xray is recorded on fixture page 0918.
Atomic and Nuclear Physics DOE-HDBK-1019/1-93 MASS DEFECT AND BINDING ENERGY
Since 1 amu is equivalent to 931.5 MeV of energy, the binding energy can be calculated using Equation (1-2).
(1-2)
B.E. ∆m
931.5 MeV
1 amu
Example:

has a mass of 235.043924 amu.
Solution:
Step 1: Calculate the mass defect using Equation (1-1).
Δm Z m
p
me A Z mn matom
Δm 92 1.007826 amu 235 92 1.008665 amu 235.043924 amu
Δm 1.91517 amu
Step 2: Use the mass defect and Equation (1-2) to calculate the binding energy.
B.E. Δm
931.5 MeV
1 amu
1.91517 amu
931.5 MeV
1 amu
1784 MeV
Fixture checkpoint marker-0919-xray is recorded on fixture page 0919.
Energy Levels of Atoms
The electrons that circle the nucleus move in fairly well-defined orbits. Some of these electrons are more tightly bound in the atom than others. For example, only 7.38 eV is required to remove the outermost electron from a lead atom, while 88,000 eV is required to remove the innermost electron. The process of removing an electron from an atom is called ionization, and the energy required to remove the electron is called the ionization energy.
In a neutral atom (number of electrons = Z) it is possible for the electrons to be in a variety of different orbits, each with a different energy level. The state of lowest energy is the one in which the atom is normally found and is called the ground state. When the atom possesses more energy than its ground state energy, it is said to be in an excited state.

An atom cannot stay in the excited state for an indefinite period of time. An excited atom will eventually transition to either a lower-energy excited state, or directly to its ground state, by emitting a discrete bundle of electromagnetic energy called an x-ray. The energy of the x-ray will be equal to the difference between the energy levels of the atom and will typically range from several eV to 100,000 eV in magnitude.
Energy Levels of the Nucleus
The nucleons in the nucleus of an atom, like the electrons that circle the nucleus, exist in shells that correspond to energy states. The energy shells of the nucleus are less defined and less understood than those of the electrons. There is a state of lowest energy (the ground state) and discrete possible excited states for a nucleus. Where the discrete energy states for the electrons of an atom are measured in eV or keV, the energy levels of the nucleus are considerably greater and typically measured in MeV.
A nucleus that is in the excited state will not remain at that energy level for an indefinite period. Like the electrons in an excited atom, the nucleons in an excited nucleus will transition towards their lowest energy configuration and in doing so emit a discrete bundle of electromagnetic radiation called a gamma ray (γ-ray). The only differences between x-rays and γ-rays are their energy levels and whether they are emitted from the electron shell or from the nucleus.
The ground state and the excited states of
Figure 7 Energy Level Diagram - Nickel-60

energy-level diagram. The nuclear energy-level diagram consists of a stack of horizontal bars, one bar for each of the excited states of the nucleus. The vertical distance between the bar representing an excited state and the bar representing the ground state is proportional to the energy level of the excited state with respect to the ground state. This difference in energy between the ground state and the excited state is called the excitation energy of the excited state. The ground state of a nuclide has zero excitation energy. The bars for the excited states are labeled with their respective energy levels. Figure 7 is the energy level diagram for nickel-60.

Summary

The important information in this chapter is summarized below.

Mass Defect and Binding Energy Summary

Mass defect is the difference between the mass of the atom and the sum of the masses of its constituent parts.

Fixture checkpoint marker-0920-xray is recorded on fixture page 0920.

Binding energy is the amount of energy that must be supplied to a nucleus to completely separate its nuclear particles. Binding energy is the energy equivalent of the mass defect.

Mass defect can be calculated by using the equation below.

$\Delta m = [Z(mp + me) + (A-Z)mn] - matom$

Binding energy can be calculated by multiplying the mass defect by the factor of 931.5 MeV per amu.

MODES OF RADIOACTIVE DECAY

Most atoms found in nature are stable and do not emit particles or energy that change form over time. Some atoms, however, do not have stable nuclei. These atoms emit radiation in order to achieve a more stable configuration.

EO 2.1 DESCRIBE the following processes:

a. Alpha decay
b. Beta-minus decay
c. Beta-plus decay
d. Electron capture
e. Internal conversions
f. Isomeric transitions

EO 2.2 Given a Chart of the Nuclides, WRITE the radioactive decay chain for a nuclide.

EO 2.3 EXPLAIN why one or more gamma rays typically accompany particle emission.

EO 2.4 Given the stability curve on the Chart of the Nuclides, DETERMINE the type of radioactive decay that the nuclides in each region of the chart will typically undergo.

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As mass numbers become larger, the ratio of neutrons to protons in the nucleus becomes larger for the stable nuclei. Non-stable nuclei may have an excess or deficiency of neutrons and undergo a transformation process known as beta (β) decay. Non-stable nuclei can also undergo a variety of other processes such as alpha (α) or neutron (n) decay. As a result of these decay processes, the final nucleus is in a more stable or more tightly bound configuration.

Natural Radioactivity

In 1896, the French physicist Becquerel discovered that crystals of a uranium salt emitted rays that were similar to x-rays in that they were highly penetrating, could affect a photographic plate, and induced electrical conductivity in gases. Becquerel's discovery was followed in 1898 by the identification of two other radioactive elements, polonium and radium, by Pierre and Marie Curie.

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Atomic and Nuclear Physics DOE-HDBK-1019/1-93 MODES OF RADIOACTIVE DECAY

Heavy elements, such as uranium or thorium, and their unstable decay chain elements emit radiation in their naturally occurring state. Uranium and thorium, present since their creation at the beginning of geological time, have an extremely slow rate of decay. All naturally occurring nuclides with atomic numbers greater than 82 are radioactive.

Nuclear Decay

Whenever a nucleus can attain a more stable (i.e., more tightly bound) configuration by emitting radiation, a spontaneous disintegration process known as radioactive decay or nuclear decay may occur. In practice, this "radiation" may be electromagnetic radiation, particles, or both.

Detailed studies of radioactive decay and nuclear reaction processes have led to the formulation of useful conservation principles. The four principles of most interest in this module are discussed below.

Conservation of electric charge implies that charges are neither created nor destroyed. Single positive and negative charges may, however, neutralize each other. It is also possible for a neutral particle to produce one charge of each sign.

Conservation of mass number does not allow a net change in the number of nucleons. However, the conversion of a proton to a neutron and vice versa is allowed.

Conservation of mass and energy implies that the total of the kinetic energy and the energy

Fixture checkpoint marker-0921-xray is recorded on fixture page 0921.

equivalent of the mass in a system must be conserved in all decays and reactions. Mass can be converted to energy and energy can be converted to mass, but the sum of mass and energy must be constant.

Conservation of momentum is responsible for the distribution of the available kinetic energy among product nuclei, particles, and/or radiation. The total amount is the same before and after the reaction even though it may be distributed differently among entirely different nuclides and/or particles.

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■ ■ ■ ■ ■

MODES OF RADIOACTIVE DECAY DOE-HDBK-1019/1-93 Atomic and Nuclear Physics

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Alpha Decay (■)

Alpha decay is the emission of alpha particles (helium nuclei) which may be represented as either $^{4}_{2}He$ or ■. When an unstable nucleus ejects an alpha particle, the atomic number is reduced by 2 and the mass number decreased by 4. An example is uranium-234 which decays by the ejection

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The combined kinetic energy of the daughter nucleus (Thorium-230) and the ■ particle is designated as KE. The sum of the KE and the gamma energy is equal to the difference in mass between the original nucleus (Uranium-234) and the final particles (equivalent to the binding energy released, since ■m = BE). The alpha particle will carry off as much as 98% of the kinetic energy and, in most cases, can be considered to carry off all the kinetic energy.

Beta Decay (■)

Beta decay is the emission of electrons of nuclear rather than orbital origin. These particles are electrons that have been expelled by excited nuclei and may have a charge of either sign.

If both energy and momentum are to be conserved, a third type of particle, the neutrino, ■, must be involved. The neutrino is associated with positive electron emission, and its antiparticle, the antineutrino, , is emitted with a negative electron. These uncharged particles have only the weakest interaction with matter, no mass, and travel at the speed of light. For all practical purposes, they pass through all materials with so few interactions that the energy they possess cannot be recovered. The neutrinos and antineutrinos are included here only because they carry a portion of the kinetic energy that would otherwise belong to the beta particle, and therefore, must be considered for energy and momentum to be conserved. They are normally ignored since they are not significant in the context of nuclear reactor applications.

Negative electron emission, represented as e, -1 -1■, or simply as e or ■ , effectively converts a neutro to a proton, thus increasing the atomic number by one and leaving the mass number unchanged. This is a common mode of decay for nuclei with an excess of neutrons, such as fission fragments below and to the right of the neutron-proton stability curve (refer to Figure 6). An example of a typical beta minus-decay reaction is shown below.

■ ■

■

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Figure 9 Orbital Electron Capture

Positively charged electrons (beta-plus) are known as positrons. Except for sign, they are nearly identical to their negatively charged cousins. When a positron, represented as e, ■, or simply as +1 +1 0 0

e or ■ , is ejected from the nucleus, the atomic number is decreased by one and the mass number + + remains unchanged. A proton has been converted to a neutron. An example of a typical positron (beta-plus) decay is shown below.

Electron Capture (EC, K-capture)

Nuclei having an excess of protons may capture an electron from one of the inner orbits which Fixture checkpoint marker-0922-xray is recorded on fixture page 0922.

immediately combines with a proton in the nucleus to form a neutron. This process is called electron capture (EC). The electron is normally captured from the innermost orbit (the K-shell), and, consequently, this process is sometimes called K-capture. The following example depicts electron capture.

A neutrino is formed at the same time that the neutron is formed, and energy carried off by it serves to conserve momentum. Any energy that is available due to the atomic mass of the product being

parent will appear as gamma
radiation. Also, there will always
be characteristic x-rays given off
when an electron from one of the
higher energy shells moves in to fill
the vacancy in the K-shell.
Electron capture is shown
graphically in Figure 8.
Electron capture and positron
emission result in the production of
the same daughter product, and
they exist as competing processes.
For positron emission to occur, however, the mass of the daughter product must be less than the mass of the parent by an amount equal to at least twice the mass of an electron. This mass difference
Fixture checkpoint marker-0923-xray is recorded on fixture page 0923.
between the parent and daughter is necessary to account for two items present in the parent but not in the daughter. One item is the positron ejected from the nucleus of the parent. The other item is that the daughter product has one less orbital electron than the parent. If this requirement is not met, then orbital electron capture takes place exclusively.
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Gamma Emission (■)
Gamma radiation is a high-energy electromagnetic radiation that originates in the nucleus. It is emitted in the form of photons, discrete bundles of energy that have both wave and particle properties. Often a daughter nuclide is left in an excited state after a radioactive parent nucleus undergoes a transformation by alpha decay, beta decay, or electron capture. The nucleus will drop to the ground state by the emission of gamma radiation.
Internal Conversion
The usual method for an excited nucleus to go from the excited state to the ground state is by emission of gamma radiation. However, in some cases the gamma ray (photon) emerges from the nucleus only to interact with one of the innermost orbital electrons and, as a result, the energy of the photon is transferred to the electron. The gamma ray is then said to have undergone internal conversion. The conversion electron is ejected from the atom with kinetic energy equal to the gamma energy minus the binding energy of the orbital electron. An orbital electron then drops to a lower energy state to fill the vacancy, and this is accompanied by the emission of characteristic x-rays.
Isomers and Isomeric Transition
Isomeric transition commonly occurs immediately after particle emission; however, the nucleus may remain in an excited state for a measurable period of time before dropping to the ground state at its own characteristic rate. A nucleus that remains in such an excited state is known as a nuclear isomer because it differs in energy and behavior from other nuclei with the same atomic number and mass number. The decay of an excited nuclear isomer to a lower energy level is called an isomeric transition. It is also possible for the excited isomer to decay by some alternate means, for example, by beta emission.
An example of gamma emission accompanying particle emission is illustrated by the decay of

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Decay Chains

When an unstable nucleus decays, the resulting daughter nucleus is not necessarily stable. The nucleus resulting from the decay of a parent is often itself unstable, and will undergo an additional decay. This is especially common among the larger nuclides.

It is possible to trace the steps of an unstable atom as it goes through multiple decays trying to achieve stability. The list of the original unstable nuclide, the nuclides that are involved as intermediate steps in the decay, and the final stable nuclide is known as the decay chain. One

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common method for stating the decay chain is to state each of the nuclides involved in the standard $^{A}_{Z}X$ format. Arrows are used between nuclides to indicate where decays occur, with the type of decay indicated above the arrow and the half-life below the arrow. The half-life for decay will be discussed in the next chapter.

Example:

Write the decay chains for rubidium-91 and actinium-215. Continue the chains until a stable nuclide or a nuclide with a half-life greater than 1 x 10 years is reached. 6

Solution:

Predicting Type of Decay

Radioactive nuclides tend to decay in a way that results in a daughter nuclide that lies closer to the line of stability. Due to this, it is possible to predict the type of decay that a nuclide will undergo based on its location relative to the line of stability on the Chart of the Nuclides.

Figure 9 illustrates the type of decay nuclides in different regions of the chart will typically undergo. Nuclides that are below and to the right of the line of stability will usually undergo ■ decay. - Nuclides that are above and to the left of the line of stability will usually undergo either ■ decay or + electron capture. Most nuclides that will undergo ■ decay are found in the upper right hand region of the chart. These are general rules that have many exceptions, especially in the region of the heavy nuclides.

Figure 9 Types of Radioactive Decay Relative to the Line of Stability

Summary

The important information in this chapter is summarized below.

Modes of Radioactive Decay Summary

an unstable nucleus. The daughter nuclide has an atomic number 2 less than the parent nuclide and a mass number 4 less than the parent nuclide. The daughter nucleus commonly releases its excitation energy by gamma emission.
Beta-minus decay effectively converts a neutron to a proton and an electron, which is immediately ejected from the nucleus. The daughter nuclide has its atomic number increased by 1 and the same mass number compared to the parent.
Beta-plus decay effectively converts a proton to a neutron and a positron, which is immediately ejected from the nucleus. The daughter nuclide has its atomic number decreased by 1 and the same mass number compared to the parent.
In electron capture, the nucleus absorbs an electron from the innermost orbit. This electron combines with a proton to form a neutron.
Internal conversion occurs when a gamma ray, emitted by the nucleus as it goes from the excited state to the ground state, interacts with one of the innermost electrons of the same atom. The electron is ejected from the atom.
An isomeric transition is the decay of an excited nucleus to a lower-energy level by the emission of a gamma ray.
Decay chains can be found by tracing the steps an unstable atom goes through as it tries to achieve stability.
Many modes of radioactive decay result in a daughter nuclide that has an energy level above the ground state. This excitation energy is usually released immediately in the form of a gamma ray.
The type of decay that a nuclide will typically undergo can be determined by its relationship to the line of stability on the Chart of the Nuclides. Nuclides that lie below and to the right of the line of stability will typically beta minus decay. Nuclides above and to the left of the line will typically either beta plus decay or electron capture. Most alpha emitters are found in the upper, right-hand corner of the chart.
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RADIOACTIVITY
The rate at which a sample of radioactive material decays is not constant. As individual atoms of the material decay, there are fewer of those types of atoms remaining. Since the rate of decay is directly proportional to the number of atoms, the rate of decay will decrease as the number of atoms decreases.
EO 2.5 DEFINE the following terms:
a. Radioactivity d. Radioactive decay constant
b. Curie e. Radioactive half-life
c. Becquerel
EO 2.6 Given the number of atoms and either the half-life or decay constant of a nuclide, CALCULATE the activity.
EO 2.7 Given the initial activity and the decay constant of a nuclide, CALCULATE the activity at any later time.
EO 2.8 CONVERT between the half-life and decay constant for a nuclide.
EO 2.9 Given the Chart of the Nuclides and the original activity, PLOT the radioactive decay curve for a nuclide on either linear or

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EO 2.10 DEFINE the following terms:
a. Radioactive equilibrium
b. Transient radioactive equilibrium
Radioactive Decay Rates
Radioactivity is the property of certain nuclides of spontaneously emitting particles or gamma radiation. The decay of radioactive nuclides occurs in a random manner, and the precise time at which a single nucleus will decay cannot be determined. However, the average behavior of a very large sample can be predicted accurately by using statistical methods. These studies have revealed that there is a certain probability that in a given time interval a certain fraction of the nuclei within a sample of a particular nuclide will decay. This probability per unit time that an atom of a nuclide will decay is known as the radioactive decay constant, λ. The units for the decay constant are inverse time such as 1/second, 1/minute, 1/hour, or 1/year. These decay constant units can also be expressed as second-1, minute-1, hour-1, and year-1.
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■■
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The activity (A) of a sample is the rate of decay of that sample. This rate of decay is usually measured in the number of disintegrations that occur per second. For a sample containing millions of atoms, the activity is the product of the decay constant and the number of atoms present in the sample.
The relationship between the activity, number of atoms, and decay constant is shown in Equation (1-3).
A = ■ N (1-3)
where:
A = Activity of the nuclide (disintegrations/second)
■ = decay constant of the nuclide (second)-1
N = Number of atoms of the nuclide in the sample
Since ■ is a constant, the activity and the number of atoms are always proportional.
Units of Measurement for Radioactivity
Two common units to measure the activity of a substance are the curie (Ci) and becquerel (Bq). A curie is a unit of measure of the rate of radioactive decay equal to 3.7 x 10 disintegrations 10 per second. This is approximately equivalent to the number of disintegrations that one gram of radium-226 will undergo in one second. A becquerel is a more fundamental unit of measure of radioactive decay that is equal to 1 disintegration per second. Currently, the curie is more
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widely used in the United States, but usage of the becquerel can be expected to broaden as the metric system slowly comes into wider use. The conversion between curies and becquerels is shown below.
1 curie = 3.7 x 10 becquerels 10
Variation of Radioactivity Over Time
The rate at which a given radionuclide sample decays is stated in Equation (1-3) as being equal to the product of the number of atoms and the decay constant. From this basic relationship it is possible to use calculus to derive an expression which can be used to calculate how the number of atoms present will change over time. The derivation is beyond the scope of this text, but Equation (1-4) is the useful result.
(1-4)
where:

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N = number of atoms initially present o
■ = decay constant (time)-1
t = time
A Ao
e
t
A Aoe
t
A
Ao
e
t
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ln A
Ao
t
t
ln
A
Ao
t
1/2
ln
1
2
t
1/2
ln 2 0.693
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Since the activity and the number of atoms are always proportional, they may be used interchangeably to describe any given radionuclide population. Therefore, the following is true.
(1-5)
where:
A = activity present at time t
A = activity initially present o
= decay constant (time)
-1
t = time
Radioactive Half-Life
One of the most useful terms for estimating how quickly a nuclide will decay is the radioactive half-life. The radioactive half-life is defined as the amount of time required for the activity to decrease to one-half of its original value. A relationship between the half-life and decay constant can be developed from Equation (1-5). The half-life can be calculated by solving Equation (1-5) for the time, t, when the current activity, A, equals one-half the initial activity A .
o
First, solve Equation (1-5) for t.
If A is equal to one-half of Ao, then A/Ao is equal to one-half. Substituting this in the equation

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1/2
(1-6)
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The basic features of decay of a radionuclide sample are shown by the graph in Figure 10.
Figure 10 Radioactive Decay as a Function of Time in Units of Half-Life
Assuming an initial number of atoms No, the population, and consequently, the activity may be noted to decrease by one-half of this value in a time of one half-life. Additional decreases occur so that whenever one half-life elapses, the number of atoms drops to one-half of what its value was at the beginning of that time interval. After five half-lives have elapsed, only 1/32, or 3.1%, of the original number of atoms remains. After seven half-lives, only 1/128, or 0.78%, of the atoms remains. The number of atoms existing after 5 to 7 half-lives can usually be assumed to be negligible. The Chemistry Fundamentals Handbook contains additional information on calculating the number of atoms contained within a sample.
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Example:
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A sample of material contains 20 micrograms of californium-252.
Californium-252 has a half-life of 2.638 years.
Calculate:
(a) The number of californium-252 atoms initially present
(b) The activity of the californium-252 in curies
(c) The number of californium-252 atoms that will remain in 12 years
(d) The time it will take for the activity to reach 0.001 curies
Solution:
(a) The number of atoms of californium-252 can be determined as below.
(b) First, use Equation (1-6) to calculate the decay constant.
Use this value for the decay constant in Equation (1-3) to determine the activity.
N No
e
t
4.78 x 1016 e
(0.263/yr) (12 yr)
2.04 x 1015
A Aoe
t
A
Ao
e
t
ln A
Ao
t
t
ln

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Ao
t
ln
0.001 Ci
0.0108 Ci
0.263 year 1
t 9.05 years
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(c) The number of californium atoms that will remain in 12 years can be calculated from Equation (1-4).
(d) The time that it will take for the activity to reach 0.001 Ci can be determined from Equation (1-5). First, solve Equation (1-5) for time.
Inserting the appropriate values in the right side of this equation will result in the required time.
Plotting Radioactive Decay
It is useful to plot the activity of a nuclide as it changes over time. Plots of this type can be used to determine when the activity will fall below a certain level. This plot is usually done showing activity on either a linear or a logarithmic scale. The decay of the activity of a single nuclide on a logarithmic scale will plot as a straight line because the decay is exponential.
t
1/2
0.693
0.693
t
1/2
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0.693
7.13 seconds
0.0972 second 1
A Aoe
t
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Example:
Plot the radioactive decay curve for nitrogen-16 over a period of 100 seconds. The initial activity is 142 curies and the half-life of nitrogen-16 is 7.13 seconds. Plot the curve on both linear rectangular coordinates and on a semi-log scale.
Solution:
First, use Equation (1-6) to calculate the decay constant corresponding to a half-life of 7.13 seconds.
Use the decay constant determined above to calculate the activity at various times using Equation (1-5).
Time Activity
0 seconds 142 Ci
20 seconds 20.3 Ci
40 seconds 2.91 Ci
60 seconds 0.416 Ci

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100 seconds 0.00853 Ci

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Figure 11 Linear and Semi-log Plots of Nitrogen-16 Decay

Plotting the data points calculated above on both linear and semilog scales results in the graphs shown in Figure 11.

If a substance contains more than one radioactive nuclide, the total activity is the sum of the individual activities of each nuclide. As an example, consider a sample of material that contained 1 x 10 atoms of iron-59 that has a half-life of 44.51 days (■ = 1.80 x 10 sec), 6 -7 -1 1 x 10 atoms of manganese-54 that has a half-life of 312.2 days (■ = 2.57 x 10 sec), and 1 6 -8 -1 x 10 atoms of cobalt-60 that has a half-life of 1925 days (■ = 4.17 x 10 sec). 6 -9 -1

The initial activity of each of the nuclides would be the product of the number of atoms and the decay constant.

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Figure 12 Combined Decay of Iron-56, Manganese-54, and Cobalt-60

Plotting the manner in which the activities of each of the three nuclides decay over time demonstrates that initially the activity of the shortest-lived nuclide (iron-59) dominates the total activity, then manganese-54 dominates. After almost all of the iron and manganese have decayed away, the only contributor to activity will be the cobalt-60. A plot of this combined decay is shown in Figure 12.

Radioactive Equilibrium

Radioactive equilibrium exists when a radioactive nuclide is decaying at the same rate at which it is being produced. Since the production rate and decay rate are equal, the number of atoms present remains constant over time.

An example of radioactive equilibrium is the concentration of sodium-24 in the coolant circulating through a sodium-cooled nuclear reactor. Assume that the sodium-24 is being produced at a rate of 1 x 10 atoms per second. If the sodium-24 were stable and did not decay, 6 the amount of sodium-24 present after some period of time could be calculated by multiplying the production rate by the amount of time. Plotting the amount of material present would result

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in the graph in Figure 13.

However, sodium-24 is not stable, and it decays with a half-life of 14.96 hours. If no sodium-24 is present initially and production starts at a rate of 1 x 10 atoms per second, the rate 6 of decay will initially be zero because there is no sodium-24 present to decay. The rate of decay of sodium-24 will increase as the amount of sodium-24 increases.

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Figure 13 Cumulative Production of Sodium-24 Over Time

The amount of sodium-24 present will initially increase rapidly, then it will increase at a continually decreasing rate until the rate of decay is equal to the rate of production. It is possible

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(R) equal to the decay rate (■ N).
where:
R = production rate (atoms/second)
■ = decay constant (second)-1
N = number of atoms
It is possible to calculate the equilibrium value for sodium-24 being produced at a rate of 1 x 10 atoms/second. 6
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Figure 14 Approach of Sodium-24 to Equilibrium
The development of the equation to calculate how the amount of sodium-24 changes over time as it approaches the equilibrium value is beyond the scope of this handbook. However, the equation is presented below.
This equation can be used to calculate the values of the amount of sodium-24 present at different times. As the time increases, the exponential term approaches zero, and the number of atoms present will approach R/■. A plot of the approach of sodium-24 to equilibrium is shown in Figure 14.
Transient Radioactive Equilibrium
Transient radioactive equilibrium occurs when the parent nuclide and the daughter nuclide decay at essentially the same rate.
For transient equilibrium to occur, the parent must have a long half-life when compared to the daughter. An example of this type of compound decay process is barium-140, which decays by beta emission to lanthanum-140, which in turn decays by beta emission to stable cerium-140.
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Figure 15 Transient Equilibrium in the Decay of Barium-140
The decay constant for barium-140 is considerably smaller than the decay constant for lanthanum-140. Remember that the rate of decay of both the parent and daughter can be represented as ■N. Although the decay constant for barium-140 is smaller, the actual rate of decay (■N) is initially larger than that of lanthanum-140 because of the great difference in their initial concentrations. As the concentration of the daughter increases, the rate of decay of the daughter will approach and eventually match the decay rate of the parent. When this occurs, they are said to be in transient equilibrium. A plot of the barium-lanthanum-cerium decay chain reaching transient equilibrium is shown in Figure 15.
Secular equilibrium occurs when the parent has an extremely long half-life. In the long decay chain for a naturally radioactive element, such as thorium-232, where all of the elements in the chain are in secular equilibrium, each of the descendants has built up to an equilibrium amount and all decay at the rate set by the original parent. The only exception is the final stable element on the end of the chain. Its number of atoms is constantly increasing.
Summary
The important information in this chapter is summarized on the following page.
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Radioactivity Summary
& Radioactivity is the decay of unstable atoms by the emission of particles and electromagnetic radiation.
& A curie (Ci) is a unit of radioactivity equal to 3.7 x 10 disintegrations per 10 second.
& A becquerel (Bq) is a unit of radioactivity equal to 1 disintegration per second.
& The radioactive decay constant (■) is the probability per unit time that an atom will decay.
& The radioactive half-life is the amount of time required for the activity to decrease to one-half its original value.
& The activity of a substance can be calculated from the number of atoms and the decay constant based on the equation below.
& The amount of activity remaining after a particular time can be calculated from the equation below.
Fixture checkpoint marker-0932-xray is recorded on fixture page 0932.
& The relationship between the decay constant and the half-life is shown below.
& Plots of radioactive decay can be useful to describe the variation of activity over time. If decay is plotted using semi-log scale the plot results in a straight line.
& Radioactive equilibrium exists when the production rate of a material is equal to the removal rate.
& Transient radioactive equilibrium exists when the parent nuclide and the daughter nuclide decay at essentially the same rate. This occurs only when the parent has a long half-life compared to the daughter.
Atomic and Nuclear Physics DOE-HDBK-1019/1-93 NEUTRON INTERACTIONS
NEUTRON INTERACTIONS
Neutrons can cause many different types of interactions. The neutron may simply scatter off the nucleus in two different ways, or it may actually be absorbed into the nucleus. If a neutron is absorbed into the nucleus, it may result in the emission of a gamma ray or a subatomic particle, or it may cause the nucleus to fission.
EO 3.1 DESCRIBE the following scattering interactions between a neutron and a nucleus:
a. Elastic scattering
b. Inelastic scattering
EO 3.2 STATE the conservation laws that apply to an elastic collision between a neutron and a nucleus.
EO 3.3 DESCRIBE the following reactions where a neutron is absorbed in a nucleus:
a. Radiative capture
b. Particle ejection
Scattering
A neutron scattering reaction occurs when a nucleus, after having been struck by a neutron, emits a single neutron. Despite the fact that the initial and final neutrons do not need to be (and often are not) the same, the net effect of the reaction is as if the projectile neutron had merely "bounced off," or scattered from, the nucleus. The two categories of scattering reactions, elastic

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Elastic Scattering
In an elastic scattering reaction between a neutron and a target nucleus, there is no energy transferred into nuclear excitation. Momentum and kinetic energy of the "system" are conserved although there is usually some transfer of kinetic energy from the neutron to the target nucleus. The target nucleus gains the amount of kinetic energy that the neutron loses.
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mn
vn,i mTvT,i mnvn,f mTvT,f
1
2
m v
2
1
2
mn
v
2
n,i
1
2
mT
v
2
T,i
Fixture checkpoint marker-0933-xray is recorded on fixture page 0933.
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mn
v
2
n,f
1
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mT
v
2
T,f
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Figure 16 Elastic Scattering
Figure 16 illustrates the process of elastic scattering of a neutron off a target nucleus. In the elastic scattering reaction, the conservation of momentum and kinetic energy is represented by the equations below.
Conservation of momentum (mv)
Conservation of kinetic energy
where:
mn = mass of the neutron
m = mass of the target nucleus T

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vn,f = final neutron velocity
vT,i = initial target velocity
vT,f = final target velocity
Atomic and Nuclear Physics DOE-HDBK-1019/1-93 NEUTRON INTERACTIONS
Elastic scattering of neutrons by nuclei can occur in two ways. The more unusual of the two interactions is the absorption of the neutron, forming a compound nucleus, followed by the re-emission of a neutron in such a way that the total kinetic energy is conserved and the nucleus returns to its ground state. This is known as resonance elastic scattering and is very dependent upon the initial kinetic energy possessed by the neutron. Due to formation of the compound nucleus, it is also referred to as compound elastic scattering. The second, more usual method, is termed potential elastic scattering and can be understood by visualizing the neutrons and nuclei to be much like billiard balls with impenetrable surfaces. Potential scattering takes place with incident neutrons that have an energy of up to about 1 MeV. In potential scattering, the neutron does not actually touch the nucleus and a compound nucleus is not formed. Instead, the neutron is acted on and scattered by the short range nuclear forces when it approaches close enough to the nucleus.
Inelastic Scattering
In inelastic scattering, the incident neutron is absorbed by the target nucleus, forming a compound nucleus. The compound nucleus will then emit a neutron of lower kinetic energy which leaves the original nucleus in an excited state. The nucleus will usually, by one or more gamma emissions, emit this excess energy to reach its ground state. Figure 17 shows the process of inelastic scattering.
For the nucleus that has reached its ground state, the sum of the kinetic energy of the exit
Figure 17 Inelastic Scattering
neutron, the target nucleus, and the total gamma energy emitted is equal to the initial kinetic energy of the incident neutron.
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1
0
n
238
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92
U
239
92
U
239
92
U
0
0
1
0
n
10
5
B

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5
B
7
3
Li
4
2
NEUTRON INTERACTIONS DOE-HDBK-1019/1-93 Atomic and Nuclear Physics
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Absorption Reactions
Most absorption reactions result in the loss of a neutron coupled with the production of a charged particle or gamma ray. When the product nucleus is radioactive, additional radiation is emitted at some later time. Radiative capture, particle ejection, and fission are all categorized as absorption reactions and are briefly described below.
Radiative Capture
In radiative capture the incident neutron enters the target nucleus forming a compound nucleus. The compound nucleus then decays to its ground state by gamma emission. An example of a radiative capture reaction is shown below.
Particle Ejection
In a particle ejection reaction the incident particle enters the target nucleus forming a compound nucleus. The newly formed compound nucleus has been excited to a high enough energy level to cause it to eject a new particle while the incident neutron remains in the nucleus. After the new particle is ejected, the remaining nucleus may or may not exist in an excited state depending upon the mass-energy balance of the reaction. An example of a particle ejection reaction is shown below.
Fission
One of the most important interactions that neutrons can cause is fission, in which the nucleus that absorbs the neutron actually splits into two similarly sized parts. Fission will be discussed in detail in the next chapter.
Atomic and Nuclear Physics DOE-HDBK-1019/1-93 NEUTRON INTERACTIONS
Summary
The important information in this chapter is summarized below.
Neutron Interactions Summary
Interactions where a neutron scatters off a target nucleus are either elastic or inelastic. In elastic scattering, kinetic energy and momentum are conserved and no energy is transferred into excitation energy of the target nucleus. In inelastic scattering, some amount of kinetic energy is transferred into excitation energy of the target nucleus.
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The conservation principles that apply to an elastic collision are conservation of kinetic energy and conservation of momentum.
Radiative capture is the absorption of a neutron by the target nucleus, resulting in an excited nucleus which subsequently (typically within a small fraction of a second) releases its excitation energy in the form of a gamma ray.
Particle ejection occurs when a neutron is absorbed by a target nucleus, resulting in the formation of a compound nucleus. The compound nucleus immediately ejects a particle (for example, alpha or proton).
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0
n
235
92
U
236
92
U
140
55
Cs
93
37
Rb 3
1
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0
n
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NUCLEAR FISSION
Nuclear fission is a process in which an atom splits and releases energy, fission products, and neutrons. The neutrons released by fission can, in turn, cause the fission of other atoms.
EO 4.1 EXPLAIN the fission process using the liquid drop model of a nucleus.
EO 4.2 DEFINE the following terms:
a. Excitation energy (E_{exc})
b. Critical energy (E_{crit})
EO 4.3 DEFINE the following terms:
a. Fissile material
b. Fissionable material
c. Fertile material
EO 4.4 DESCRIBE the processes of transmutation, conversion, and breeding.
EO 4.5 DESCRIBE the curve of Binding Energy per Nucleon versus mass number and give a qualitative description of the reasons for its shape.
EO 4.6 EXPLAIN why only the heaviest nuclei are easily fissioned.
EO 4.7 EXPLAIN why uranium-235 fissions with thermal neutrons and uranium-238 fissions only with fast neutrons.
Fission
In the fission reaction the incident neutron enters the heavy target nucleus, forming a compound nucleus that is excited to such a high energy level ($E > E_{crit}$) that the nucleus "splits" (E_{exc}) into two large fragments plus some neutrons. An example of a typical fission reaction is shown below.
A large amount of energy is released in the form of radiation and fragment kinetic energy.
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The nucleus is held together by the attractive nuclear force between nucleons, which was discussed in a previous chapter. The characteristics of the nuclear force are listed below.
(a) very short range, with essentially no effect beyond nuclear dimensions (10-13 cm)
(b) stronger than the repulsive electrostatic forces within the nucleus
(c) independent of nucleon pairing, in that the attractive forces between pairs of neutrons are no different than those between pairs of protons or a neutron and a proton
(d) saturable, that is, a nucleon can attract only a few of its nearest neighbors
One theory of fission considers the fissioning of a nucleus similar in some respects to the splitting of a liquid drop. This analogy is justifiable to some extent by the fact that a liquid drop is held together by molecular forces that tend to make the drop spherical in shape and that try to resist any deformation in the same manner as nuclear forces are assumed to hold the nucleus together. By considering the nucleus as a liquid drop, the fission process can be described.
Referring to Figure 18(A), the nucleus in the ground state is undistorted, and its attractive nuclear forces are greater than the repulsive electrostatic forces between the protons within the nucleus. When an incident particle (in this instance a neutron) is absorbed by the target nucleus, a compound nucleus is formed. The compound nucleus temporarily contains all the charge and mass involved in the reaction and exists in an excited state. The excitation energy added to the compound nucleus is equal to the binding energy contributed by the incident particle plus the kinetic energy possessed by that particle. Figure 18(B) illustrates the excitation energy thus imparted to the compound nucleus, which may cause it to oscillate and become distorted. If the
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excitation energy is greater than a certain critical energy, the oscillations may cause the compound nucleus to become dumbbell-shaped. When this happens, the attractive nuclear forces (short-range) in the neck area are small due to saturation, while the repulsive electrostatic forces (long-range) are only slightly less than before. When the repulsive electrostatic forces exceed the attractive nuclear forces, nuclear fission occurs, as illustrated in Figure 18(C).

Figure 18 Liquid Drop Model of Fission
Critical Energy
The measure of how far the energy level of a nucleus is above its ground state is called the excitation energy (Eexc). For fission to occur, the excitation energy must be above a particular value for that nuclide. The critical energy (Ecrit) is the minimum excitation energy required for fission to occur.
Fissile Material
A fissile material is composed of nuclides for which fission is possible with neutrons of any energy level. What is especially significant about these nuclides is their ability to be fissioned with zero kinetic energy neutrons (thermal neutrons). Thermal neutrons have very low kinetic energy levels (essentially zero) because they are roughly in equilibrium with the thermal motion of surrounding materials. Therefore, in order to be classified as fissile, a material must be capable of fissioning after absorbing a thermal neutron. Consequently, they impart essentially no kinetic energy to the reaction. Fission is possible in these materials with thermal neutrons, since the change in binding energy supplied by the neutron addition alone is high enough to exceed the critical energy. Some examples of fissile nuclides are uranium-235, uranium-233, and plutonium-239.

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Fissionable Material

A fissionable material is composed of nuclides for which fission with neutrons is possible. All fissile nuclides fall into this category. However, also included are those nuclides that can be fissioned only with high energy neutrons. The change in binding energy that occurs as the result of neutron absorption results in a nuclear excitation energy level that is less than the required critical energy. Therefore, the additional excitation energy must be supplied by the kinetic energy of the incident neutron. The reason for this difference between fissile and fissionable materials is the so-called odd-even effect for nuclei. It has been observed that nuclei with even numbers of neutrons and/or protons are more stable than those with odd numbers. Therefore, adding a neutron to change a nucleus with an odd number of neutrons to a nucleus with an even number of neutrons produces an appreciably higher binding energy than adding a neutron to a nucleus already possessing an even number of neutrons. Some examples of nuclides requiring high energy neutrons to cause fission are thorium-232, uranium-238, and plutonium-240. Table 4 indicates the critical energy (Ecrit) and the binding energy change for an added neutron (BEn) to target nuclei of interest. For fission to be possible, the change in binding energy plus the kinetic energy must equal or exceed the critical energy (ΔBE + KE > Ecrit).

TABLE 4

Critical Energies Compared to Binding Energy of Last Neutron

Target
Nucleus
Critical Energy
Ecrit
Binding Energy of
Last Neutron BEn
BEn - Ecrit

23
9
2
0Th 7.5 MeV 5.4 MeV -2.1 MeV

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23
9
8
2U 7.0 MeV 5.5 MeV -1.5 MeV

23
9
5
2U 6.5 MeV 6.8 MeV +0.3 MeV

23
9
3
2U 6.0 MeV 7.0 MeV +1.0 MeV

23
9
9
4Pu 5.0 MeV 6.6 MeV +1.6 MeV

Uranium-235 fissions with thermal neutrons because the binding energy released by the absorption of a neutron is greater than the critical energy for fission; therefore uranium-235 is

less than the critical energy, so additional energy must be possessed by the neutron for fission to be possible. Consequently, uranium-238 is a fissionable material.

Fertile Material

All of the neutron absorption reactions that do not result in fission lead to the production of new nuclides through the process known as transmutation. These nuclides can, in turn, be transmuted again or may undergo radioactive decay to produce still different nuclides. The nuclides that are produced by this process are referred to as transmutation products. Because several of the fissile nuclides do not exist in nature, they can only be produced by nuclear reactions (transmutation). The target nuclei for such reactions are said to be fertile. Fertile materials are materials that can undergo transmutation to become fissile materials. Figure 19 traces the transmutation mechanism by which two fertile nuclides, thorium-232 and uranium-238, produce uranium-233 and plutonium-239, respectively.

Figure 19 Conversion of Fertile Nuclides to Fissile Nuclides

If a reactor contains fertile material in addition to its fissile fuel, some new fuel will be produced as the original fuel is burned up. This is called conversion. Reactors that are specifically designed to produce fissionable fuel are called "breeder" reactors. In such reactors, the amount of fissionable fuel produced is greater than the amount of fuel burnup. If less fuel is produced than used, the process is called conversion, and the reactor is termed a "converter."

Binding Energy Per Nucleon (BE/A)

As the number of particles in a nucleus increases, the total binding energy also increases. The rate of increase, however, is not uniform. This lack of uniformity results in a variation in the amount of binding energy associated with each nucleon within the nucleus. This variation in the binding energy per nucleon (BE/A) is easily seen when the average BE/A is plotted versus atomic mass number (A), as shown in Figure 20.

Figure 20 Binding Energy per Nucleon vs. Mass Number

Figure 20 illustrates that as the atomic mass number increases, the binding energy per nucleon decreases for A > 60. The BE/A curve reaches a maximum value of 8.79 MeV at A = 56 and decreases to about 7.6 MeV for A = 238. The general shape of the BE/A curve can be explained using the general properties of nuclear forces. The nucleus is held together by very short-range attractive forces that exist between nucleons. On the other hand, the nucleus is being forced apart by long range repulsive electrostatic (coulomb) forces that exist between all the protons in the nucleus.

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As the atomic number and the atomic mass number increase, the repulsive electrostatic forces within the nucleus increase due to the greater number of protons in the heavy elements. To overcome this increased repulsion, the proportion of neutrons in the nucleus must increase to maintain stability. This increase in the neutron-to-proton ratio only partially compensates for the growing proton-proton repulsive force in the heavier, naturally occurring elements. Because the repulsive forces are increasing, less energy must be supplied, on the average, to remove a nucleon from the nucleus. The BE/A has decreased. The BE/A of a nucleus is an indication of its degree of stability. Generally, the more stable nuclides have higher BE/A than the less stable ones. The increase in the BE/A as the atomic mass number decreases from 260 to 60 is the primary reason

mass number increases from 1 to 60 is the reason for the energy liberation in the fusion process, which is the opposite reaction of fission.

The heaviest nuclei require only a small distortion from a spherical shape (small energy addition) for the relatively large coulomb forces forcing the two halves of the nucleus apart to overcome the attractive nuclear forces holding the two halves together. Consequently, the heaviest nuclei are easily fissionable compared to lighter nuclei.

Summary

The important information in this chapter is summarized on the following page.

Nuclear Fission Summary

The fission process can be explained using the liquid drop model of a nucleus. In the ground state the nucleus is nearly spherical in shape. After the absorption

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of a neutron, the nucleus will be in an excited state and start to oscillate and become distorted. If the oscillations cause the nucleus to become shaped like a dumbbell, the repulsive electrostatic forces will overcome the short-range attractive nuclear forces, and the nucleus will split in two.

Excitation energy is the amount of energy a nucleus has above its ground state. Critical energy is the minimum excitation energy that a nucleus must have before it can fission.

Fissile material is material for which fission is possible with neutrons that have zero kinetic energy. Fissionable material is material for which fission caused by neutron absorption is possible provided the kinetic energy added with the binding energy is greater than the critical energy. Fertile material is material that can undergo transmutation to become fissile material.

Transmutation is the process of neutron absorption and subsequent decay, which changes one nuclide to another nuclide. Conversion is the process of transmuting fertile material into fissile material in a reactor, where the amount of fissile material produced is less than the amount of fissile material consumed. Breeding is the same as conversion, except the amount of fissile material produced is more than the amount of fissile material consumed.

The curve of binding energy per nucleon increases quickly through the light nuclides and reaches a maximum at a mass number of about 56. The curve decreases slowly for mass numbers greater than 60.

The heaviest nuclei are easily fissionable because they require only a small distortion from the spherical shape to allow the coulomb forces to overcoming the attractive nuclear force, forcing the two halves of the nucleus apart.

Uranium-235 fissions with thermal neutrons because the binding energy released by the absorption of a neutron is greater than the critical energy for fission. The binding energy released by uranium-238 absorbing a neutron is less than the critical energy, so additional kinetic energy must be possessed by the neutron for fission to be possible.

1
0
n
235

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U
236
92
U
140
55
Cs
93
37
Rb 3
1
0
n
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ENERGY RELEASE FROM FISSION
Fission of heavy nuclides converts a small amount of mass into an enormous amount of energy. The amount of energy released by fission can be determined based on either the change in mass that occurs during the reaction or by the difference in binding energy per nucleon between the fissile nuclide
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and the fission products.
EO 4.8 CHARACTERIZE the fission products in terms of mass groupings and radioactivity.
EO 4.9 Given the nuclides involved and their masses, CALCULATE the energy released from fission.
EO 4.10 Given the curve of Binding Energy per nucleon versus mass number, CALCULATE the energy released from fission.
Calculation of Fission Energy
Nuclear fission results in the release of enormous quantities of energy. It is necessary to be able to calculate the amount of energy that will be produced. The logical manner in which to pursue this is to first investigate a typical fission reaction such as the one listed below.
It can be seen that when the compound nucleus splits, it breaks into two fission fragments, rubidium-93, cesium-140, and some neutrons. Both fission products then decay by multiple - emissions as a result of the high neutron-to-proton ratio possessed by these nuclides.
In most cases, the resultant fission fragments have masses that vary widely. Figure 21 gives the percent yield for atomic mass numbers. The most probable pair of fission fragments for the thermal fission of the fuel uranium-235 have masses of about 95 and 140. Note that the vertical axis of the fission yield curve is on a logarithmic scale. Therefore, the formation of fission fragments of mass numbers of about 95 and 140 is highly likely.
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Figure 21 Uranium-235 Fission Yield vs.
Mass Number
Referring now to the binding energy per nucleon curve (Figure 20), we can estimate the amount of energy released by our "typical" fission by plotting this reaction on the curve and calculating the change in binding energy (ΔBE) between the reactants on the left-hand side of the fission equation and the products on the right-hand side. Plotting the reactant and product nuclides on the curve shows that the total binding energy of the system after fission is greater than the total

energy of a system, the system has become more stable by releasing an amount of energy equal to the increase in total binding energy of the system. Therefore, in the fission process, the energy liberated is equal to the increase in the total binding energy of the system.

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Figure 22 graphically depicts that the binding energy per nucleon for the products (C,

Figure 22 Change in Binding Energy for Typical Fission

rubidium-93 and B, cesium-140) is greater than that for the reactant (A, uranium-235). The total binding energy for a nucleus can be found by multiplying the binding energy per nucleon by the number of nucleons.

TABLE 5

Binding Energies Calculated from
Binding Energy per Nucleon Curve

Nuclide	B.E. per Nucleon (BE/A)	Mass Number (A)	Binding Energy (BE/A) x (A)
9 3 3 7Rb	8.7 MeV	93	809 MeV
14 5 0 5Cs	8.4 MeV	140	1176 MeV

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23 9 5 2U	7.6 MeV	235	1786 MeV

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BE BEproducts BEreactants

BERb 93 BECs 140 BEU 235

809 MeV 1176 MeV 1786 MeV

199 MeV

1 0 n

235 92 U

236 92 U

140 55

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93
37
Rb 3
1
0
n
Mass difference Mass of Reactants Mass of Products
236.052589 amu 235.85208 amu
0.200509 amu
EInst 0.020059 amu
931.5 MeV
amu
186.8 MeV
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The energy released will be equivalent to the difference in binding energy (BE) between the reactants and the products.
The energy liberation during the fission process can also be explained from the standpoint of the conservation of mass-energy. During the fission process, there is a decrease in the mass of the system. There must, therefore, be energy liberated equal to the energy equivalent of the mass lost in the process. This method is more accurate than the previously illustrated method and is used when actually calculating the energy liberated during the fission process.
Again, referring to the "typical" fission reaction.
E , the instantaneous energy, is the energy released immediately after the fission process. It Inst is equal to the energy equivalent of the mass lost in the fission process. It can be calculated as shown below.
Mass of the Reactants Mass of the Products
U 235.043924 amu Rb 92.91699 amu 235
92
93
37
n 1.008665 amu Cs 139.90910 amu 1
0
140
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55
3 (n) 3.02599 amu 1
0
236.052589 amu 235.85208 amu
This mass difference can be converted to an energy equivalent.
93
37
Rb
93
38
Sr
93
39

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93
40
Zr
93
41
Nb
140
55
Cs
140
56
Fixture checkpoint marker-0944-xray is recorded on fixture page 0944.
Ba
140
57
La
140
58
Ce
EDecay mRb 93 mNb 93 4 melectron
931.5MeV
amu
92.91699 amu 92.90638 amu 4 0.0005486 amu
931.5 MeV
amu
0.008416 amu
931.5 MeV
amu
7.84 MeV
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The total energy released per fission will vary from the fission to the next depending on what fission products are formed, but the average total energy released per fission of uranium-235 with a thermal neutron is 200 MeV.
As illustrated in the preceding example, the majority of the energy liberated in the fission process is released immediately after the fission occurs and appears as the kinetic energy of the fission fragments, kinetic energy of the fission neutrons, and instantaneous gamma rays. The remaining energy is released over a period of time after the fission occurs and appears as kinetic energy of the beta, neutrino, and decay gamma rays.
Estimation of Decay Energy
In addition to this instantaneous energy release during the actual fission reaction, there is additional energy released when the fission fragments decay by emission. This additional
-
energy is called decay energy, EDecay. The decay chains for rubidium-93 and cesium-140 are shown below.
The energy released during the decay for each chain will be equivalent to the mass difference between the original fission product and the sum of the final stable nuclide and the beta particles emitted.

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EDecay mRb 93 mNb 93 3 melectron
931.5MeV
amu
139.90910 amu 139.90543 amu 3 0.0005486 amu
931.5 MeV
amu
0.000202 amu
931.5 MeV
amu
1.89 MeV
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The energy released in the decay chain of cesium-140 is calculated below.
The total decay energy is the sum of the energies of the two chains, or 9.73 MeV.
Distribution of Fission Energy
The average energy distribution for the energy released per fission with a thermal neutron in uranium-235 is shown in Tables 6 and 7.
TABLE 6
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Instantaneous Energy from Fission
Kinetic Energy of Fission Products 167 Mev
Energy of Fission Neutrons 5 MeV
Instantaneous Gamma-ray Energy 5 MeV
Capture Gamma-ray Energy 10 MeV
Total Instantaneous Energy 187 MeV
TABLE 7
Delayed Energy from Fission
Beta Particles From Fission Products 7 Mev
Gamma-rays from Fission Products 6 MeV
Neutrinos 10 MeV
Total Delayed Energy 23 MeV
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Because the 10 MeV of neutrino energy shown in Table 7 is not absorbed in the reactor, the average value of 200 MeV per fission is still accurate. Note in Table 6 that some fission neutrons undergo radiative capture and the resultant gamma ray emission provides an additional 10 MeV of instantaneous energy, which contributes to the total of 187 MeV instantaneous energy.
All of the energy released, with the exception of the neutrino energy, is ultimately transformed into heat through a number of processes. The fission fragments, with their high positive charge and kinetic energy, cause ionization directly as they rip orbital electrons from the surrounding atoms. In this ionization process, kinetic energy is transferred to the surrounding atoms of the fuel material, resulting in an increase in temperature. The beta particles and gamma rays also give up their energy through ionization, and the fission neutrons interact and lose their energy through elastic scattering. Of the 200 MeV released per fission, about seven percent (13 MeV) is released at some time after the instant of fission. When a reactor is shut down, fissions essentially cease, but energy is still being released from the decay of fission products. The heat produced by this decay energy is referred to as "decay heat." Although decay energy represents about seven percent of reactor heat production during reactor operation, once the reactor is shut

The decay heat produced is significant, however, and systems must be provided to keep the reactor cool even after shutdown.

Summary

The important information in this chapter is summarized below.

Energy Release From Fission Summary

Fission products have some general characteristics in common.

They generally decay by β- emission.

The most common mass numbers are grouped near 95 and 140.

The energy released by fission can be calculated based on the difference in mass between the masses of the reactants before fission and the fission fragments and fission neutrons after fission.

Another method to determine the energy released by fission is based on the change in binding energy per nucleon between the fissile nuclide and the fission products.

INTERACTION OF RADIATION WITH MATTER

Different types of radiation interact with matter in widely different ways. A large, massive, charged alpha particle cannot penetrate a piece of paper and even has a limited range in dry air. A neutrino, at the other extreme, has a low probability of interacting with any matter, even if it passed through the diameter of the earth.

EO 5.1 DESCRIBE interactions of the following with matter:

a. Alpha particle c. Positron

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b. Beta particle d. Neutron

EO 5.2 DESCRIBE the following ways that gamma radiation interacts with matter:

a. Photoelectric effect

b. Compton scattering

c. Pair production

Interaction of Radiation With Matter

Radiation can be classified into two general groups, charged and uncharged; therefore, it may be expected that interactions with matter fall into two general types. Charged particles directly ionize the media through which they pass, while uncharged particles and photons can cause ionization only indirectly or by secondary radiation.

A moving charged particle has an electrical field surrounding it, which interacts with the atomic structure of the medium through which it is passing. This interaction decelerates the particle and accelerates electrons in the atoms of the medium. The accelerated electrons may acquire enough energy to escape from the parent atom. This process, whereby radiation "strips" off orbital electrons, is called ionization. Uncharged moving particles have no electrical field, so they can only lose energy and cause ionization by such means as collisions or scattering. A photon can lose energy by the photoelectric effect, Compton effect, or pair production.

I

m z

2

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Because ionizing radiation creates ions in pairs, the intensity of ionization or the specific ionization is defined as the number of ion-pairs formed per centimeter of travel in a given material. The amount of ionization produced by a charged particle per unit path length, which is a measure of its ionizing power, is roughly proportional to the particle's mass and the square of its charge as illustrated in the equation below.
where:
I is the ionizing power
m is the mass of the particle
z is the number of unit charges it carries
K.E. is its kinetic energy
Since m for an alpha particle is about 7300 times as large as m for a beta particle, and z is twice as great, an alpha will produce much more ionization per unit path length than a beta particle of the same energy. This phenomenon occurs because the larger alpha particle moves slower for a given energy and thus acts on a given electron for a longer time.
Alpha Radiation
Alpha radiation is normally produced from the radioactive decay of heavy nuclides and from certain nuclear reactions. The alpha particle consists of 2 neutrons and 2 protons, so it is essentially the same as the nucleus of a helium atom. Because it has no electrons, the alpha particle has a charge of +2. This positive charge causes the alpha particle to strip electrons from the orbits of atoms in its vicinity. As the alpha particle passes through material, it removes electrons from the orbits of atoms it passes near. Energy is required to remove electrons and the energy of the alpha particle is reduced by each reaction. Eventually the particle will expend its kinetic energy, gain 2 electrons in orbit, and become a helium atom. Because of its strong positive charge and large mass, the alpha particle deposits a large amount of energy in a short distance of travel. This rapid, large deposition of energy limits the penetration of alpha particles. The most energetic alpha particles are stopped by a few centimeters of air or a sheet of paper.
Beta-Minus Radiation
A beta-minus particle is an electron that has been ejected at a high velocity from an unstable
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nucleus. An electron has a small mass and an electrical charge of -1. Beta particles cause ionization by displacing electrons from atom orbits. The ionization occurs from collisions with orbiting electrons. Each collision removes kinetic energy from the beta particle, causing it to slow down. Eventually the beta particle will be slowed enough to allow it to be captured as an
2 electrons 0.000549 amu
electron
931.5 MeV
amu
1.02 MeV
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orbiting electron in an atom. Although more penetrating than the alpha, the beta is relatively easy to stop and has a low power of penetration. Even the most energetic beta radiation can be stopped by a few millimeters of metal.

Positively charged electrons are called positrons. Except for the positive charge, they are identical to beta-minus particles and interact with matter in a similar manner. Positrons are very short-lived, however, and quickly are annihilated by interaction with a negatively charged electron, producing two gammas with a combined energy (calculated below) equal to the rest mass of the positive and negative electrons.

Neutron Radiation

Neutrons have no electrical charge. They have nearly the same mass as a proton (a hydrogen atom nucleus). A neutron has hundreds of times more mass than an electron, but 1/4 the mass of an alpha particle. The source of neutrons is primarily nuclear reactions, such as fission, but they may also be produced from the decay of radioactive nuclides. Because of its lack of charge, the neutron is difficult to stop and has a high penetrating power.

Neutrons are attenuated (reduced in energy and numbers) by three major interactions, elastic scatter, inelastic scatter, and absorption. In elastic scatter, a neutron collides with a nucleus and bounces off. This reaction transmits some of the kinetic energy of the neutron to the nucleus of the atom, resulting in the neutron being slowed, and the atom receives some kinetic energy (motion). This process is sometimes referred to as "the billiard ball effect."

As the mass of the nucleus approaches the mass of the neutron, this reaction becomes more effective in slowing the neutron. Hydrogenous material attenuates neutrons most effectively.

In the inelastic scatter reaction, the same neutron/nucleus collision occurs as in elastic scatter. However, in this reaction, the nucleus receives some internal energy as well as kinetic energy. This slows the neutron, but leaves the nucleus in an excited state. When the nucleus decays to its original energy level, it normally emits a gamma ray.

In the absorption reaction, the neutron is actually absorbed into the nucleus of an atom. The neutron is captured, but the atom is left in an excited state. If the nucleus emits one or more gamma rays to reach a stable level, the process is called radiative capture. This reaction occurs at most neutron energy levels, but is more probable at lower energy levels.

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Gamma Radiation

Gamma radiation is electromagnetic radiation. It is commonly referred to as a gamma ray and is very similar to an x-ray. The difference is that gamma rays are emitted from the nucleus of an atom, and x-rays are produced by orbiting electrons. The x-ray is produced when orbiting electrons move to a lower energy orbit or when fast-moving electrons approaching an atom are deflected and decelerated as they react with the atom's electrical field (called Bremsstrahlung). The gamma ray is produced by the decay of excited nuclei and by nuclear reactions. Because the gamma ray has no mass and no charge, it is difficult to stop and has a very high penetrating power. A small fraction of the original gamma stream will pass through several feet of concrete or several meters of water.

There are three methods of attenuating gamma rays. The first method is referred to as the

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photo-electric effect. When a low energy gamma strikes an atom, the total energy of the gamma is expended in ejecting an electron from orbit. The result is ionization of the atom and expulsion of a high energy electron. This reaction is most predominant with low energy gammas interacting in materials with high atomic weight and rarely occurs with gammas having an energy above 1 MeV. Annihilation of the gamma results. Any gamma energy in excess of the binding energy of the electron is carried off by the electron in the form of kinetic energy.

The second method of attenuation of gammas is called Compton scattering. The gamma interacts with an orbital or free electron; however, in this case, the photon loses only a fraction of its

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continues on at lower energy, and the energy difference is absorbed by the electron. This reaction becomes important for gamma energies of about 0.1 MeV and higher.
At higher energy levels, a third method of attenuation is predominant. This method is pair-production. When a high energy gamma passes close enough to a heavy nucleus, the gamma completely disappears, and an electron and a positron are formed. For this reaction to take place, the original gamma must have at least 1.02 MeV energy. Any energy greater than 1.02 MeV becomes kinetic energy shared between the electron and positron. The probability of pair■production incre
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Interaction of Radiation with Matter Summary
An alpha particle deposits a large amount of energy in a short distance of travel due to its large mass and charge.
Beta-minus particles interact with the electrons orbiting the nucleus of atoms, causing ionization by displacing the electrons. The beta particle loses energy with
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each interaction. After the beta particle loses enough energy, it is captured in the orbital shells of an atom.
Positrons interact with matter much the same way as beta minus particles. After the positron has lost most of its energy by ionizing atoms, it is annihilated by interaction with an electron. The electron-positron pair disappear and are replaced by two gammas, each with the energy equivalent of the mass of an electron (0.51 MeV).
Neutrons interact with matter by elastic scattering, inelastic scattering, or absorption.
Photoelectric effect is where a gamma interacts with an electron orbiting an atom. The entire energy of the gamma is transferred to the electron, and the electron is ejected from its orbit.
In Compton scattering a gamma interacts with an orbital electron, but only part of the gamma energy is transferred to the electron. The electron is ejected from its orbit, and the gamma is scattered off at a lower energy.
In pair-production, a gamma interacts with the electric field of a nucleus and is converted into an electron-positron pair. The gamma must have an energy greater than 1.02 MeV for this to occur.
Summary
The important information in this chapter is summarized below.
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Department of Energy
Fundamentals Handbook
NUCLEAR PHYSICS
AND REACTOR THEORY
Module 2
Reactor Theory (Neutron Characteristics)
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OBJECTIVES DOE-HDBK-1019/1-93 Reactor Theory (Neutron Characteristics)
TERMINAL OBJECTIVE
Fixture checkpoint marker-0951-xray is recorded on fixture page 0951.
1.0 Without references, EXPLAIN how neutron sources produce neutrons.
ENABLING OBJECTIVES
1.1 DEFINE the following terms:
a. Intrinsic neutron source
b. Installed neutron source
1.2 LIST three examples of reactions that produce neutrons in intrinsic neutron sources.
1.3 LIST three examples of reactions that produce neutrons in installed neutron sources.
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Reactor Theory (Neutron Characteristics) DOE-HDBK-1019/1-93 OBJECTIVES
TERMINAL OBJECTIVE
2.0 Given the necessary information for calculations, EXPLAIN basic concepts in reactor physics and perform calculations.
ENABLING OBJECTIVES
2.1 DEFINE the following terms:
a. Atom density d. Barn
b. Neutron flux e. Macroscopic cross section
c. Microscopic cross section f. Mean free path

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2.3 DESCRIBE how the absorption cross section of typical nuclides varies with neutron energy at energies below the resonance absorption region.
2.4 DESCRIBE the cause of resonance absorption in terms of nuclear energy levels.
2.5 DESCRIBE the energy dependence of resonance absorption peaks for typical light and heavy nuclei.
2.6 EXPRESS mean free path in terms of macroscopic cross section.
2.7 Given the number densities (or total density and component fractions) and microscopic cross sections of components, CALCULATE the macroscopic cross section for a mixture.
2.8 CALCULATE a macroscopic cross section given a material density, atomic mass, and microscopic cross section.
2.9 EXPLAIN neutron shadowing or self-shielding.
2.10 Given the neutron flux and macroscopic cross section, CALCULATE the reaction rate.
2.11 DESCRIBE the relationship between neutron flux and reactor power.

ENABLING OBJECTIVES (Cont.)
2.12 DEFINE the following concepts:
a. Thermalization d. Average logarithmic energy decrement
b. Moderator e. Macroscopic slowing down power
c. Moderating ratio
2.13 LIST three desirable characteristics of a moderator.
2.14 Given an average fractional energy loss per collision, CALCULATE the energy loss after a specified number of collisions.

TERMINAL OBJECTIVE
3.0 Without references, EXPLAIN the production process and effects on fission of prompt and delayed neutrons.
ENABLING OBJECTIVES
3.1 STATE the origin of prompt neutrons and delayed neutrons.
3.2 STATE the approximate fraction of neutrons that are born as delayed neutrons from the fission of the following nuclear fuels:
a. Uranium-235
b. Plutonium-239
3.3 EXPLAIN the mechanism for production of delayed neutrons.
3.4 EXPLAIN prompt and delayed neutron generation times.
3.5 Given prompt and delayed neutron generation times and delayed neutron fraction,
Fixture checkpoint marker-0952-xray is recorded on fixture page 0952.
CALCULATE the average generation time.
3.6 EXPLAIN the effect of delayed neutrons on reactor control.

TERMINAL OBJECTIVE
4.0 Without references, DESCRIBE the neutron energy spectrum for the type of reactor presented in this module.
ENABLING OBJECTIVES
4.1 STATE the average energy at which prompt neutrons are produced.
4.2 DESCRIBE the neutron energy spectrum in the following reactors:

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b. Thermal reactor
4.3 EXPLAIN the reason for the particular shape of the fast, intermediate, and slow energy regions of the neutron flux spectrum for a thermal reactor.
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Reactor Theory (Neutron Characteristics) DOE-HDBK-1019/1-93 NEUTRON SOURCES
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NEUTRON SOURCES
Neutrons from a variety of sources are always present in a reactor core. This is true even when the reactor is shut down. Some of these neutrons are produced by naturally occurring (intrinsic) neutron sources, while others may be the result of fabricated (installed) neutron sources that are incorporated into the design of the reactor. The neutrons produced by sources other than neutron-induced fission are often grouped together and classified as source neutrons.
EO 1.1 DEFINE the following terms:
Fixture checkpoint marker-0953-xray is recorded on fixture page 0953.
a. Intrinsic neutron source
b. Installed neutron source
EO 1.2 LIST three examples of reactions that produce neutrons in intrinsic neutron sources.
EO 1.3 LIST three examples of reactions that produce neutrons in installed neutron sources.
Neutron Sources
In addition to neutron-induced fission, neutrons are produced by other reactions. The neutrons produced by reactions other than neutron-induced fission are called source neutrons. Source neutrons are important because they ensure that the neutron population remains high enough to allow a visible indication of neutron level on the most sensitive monitoring instruments while the reactor is shutdown and during the startup sequence. This verifies instrument operability and allows monitoring of neutron population changes. Source neutrons can be classified as either intrinsic or installed neutron sources.
Intrinsic Neutron Sources
Some neutrons will be produced in the materials present in the reactor due to a variety of unavoidable reactions that occur because of the nature of these materials. Intrinsic neutron sources are those neutron-producing reactions that always occur in reactor materials.
■ ■ ■ ■
■ ■ ■
■ ■
NEUTRON SOURCES DOE-HDBK-1019/1-93 Reactor Theory (Neutron Characteristics)
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A limited number of neutrons will always be present, even in a reactor core that has never been operated, due to spontaneous fission of some heavy nuclides that are present in the fuel. Uranium■238, ur of the rate at which different heavy nuclides produce neutrons by spontaneous fission.
Californium-252 is not an intrinsic neutron source, but will be discussed in the section on installed neutron sources.
TABLE 1
Neutron Production by Spontaneous Fission
Nuclide T (Fission) T (1/2 1/2 ■-decay) neutrons/sec/gram
U 1.8 x 10 years 6.8 x 10 years 8.0 x 10 235
92

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U 8.0 x 10 years 4.5 x 10 years 1.6 x 10 238
92
15 9 -2
Pu 5.5 x 10 years 2.4 x 10 years 3.0 x 10 239
94
5 4 -2
Pu 1.2 x 10 years 6.6 x 10 years 1.0 x 10 240
94
11 3 3
Cf 66.0 years 2.65 years 2.3 x 10 252
98
12

Another intrinsic neutron source is a reaction involving natural boron and fuel. In some reactors, natural boron is loaded into the reactor core as a neutron absorber to improve reactor control or increase core life-time. Boron-11 (80.1% of natural boron) undergoes a reaction with the alpha particle emitted by the radioactive decay of heavy nuclides in the fuel to yield a neutron as shown below.

The boron-11 must be mixed with, or in very close proximity to, the fuel for this reaction because of the short path length of the alpha particle. For a reactor core with this configuration, this (■,n) reaction is an important source of neutrons for reactor startup.

In a reactor that has been operated, another source of neutrons becomes significant. Neutrons may

Fixture checkpoint marker-0954-xray is recorded on fixture page 0954.

be produced by the interaction of a gamma ray and a deuterium nucleus. This reaction is commonly referred to as a photoneutron reaction because it is initiated by electromagnetic radiation and results in the production of a neutron. The photoneutron reaction is shown below.

■ ■ ■
■ ■
■ ■ ■
■ ■

There is an abundant supply of high energy gammas in a reactor that has been operated because many of the fission products are gamma emitters. All water-cooled reactors have some deuterium present in the coolant in the reactor core because a small fraction of natural hydrogen is the isotope deuterium. The atom percentage of deuterium in the water ranges from close to the naturally occurring value (0.015%) for light water reactors to above 90% deuterium for heavy water reactors. Therefore, the required conditions for production of photoneutrons exist.

The supply of gamma rays decreases with time after shutdown as the gamma emitters decay; therefore, the photoneutron production rate also decreases. In a few particular reactors, additional D O (heavy water) may be added to the reactor to increase the production of 2 photoneutrons following a long shutdown period.

Installed Neutron Sources

Because intrinsic neutron sources can be relatively weak or dependent upon the recent power history of the reactor, many reactors have artificial sources of neutrons installed. These neutron sources ensure that shutdown neutron levels are high enough to be detected by the nuclear instruments at all times. This provides a true picture of reactor conditions and any change in these conditions. An installed neutron source is an assembly placed in or near the reactor for the sole purpose of producing source neutrons.

the rate of about 2 x 10 neutrons per second per gram as the result of spontaneous fission. 12
Important drawbacks for some applications may be its high cost and its short half-life (2.65 years).
Many installed neutron sources use the (■,n) reaction with beryllium. These sources are composed of a mixture of metallic beryllium (100% beryllium-9) with a small quantity of an alpha particle emitter, such as a compound of radium, polonium, or plutonium. The reaction that occurs is shown below.
The beryllium is intimately (homogeneously) mixed with the alpha emitter and is usually enclosed in a stainless steel capsule.
Another type of installed neutron source that is widely used is a photoneutron source that employs the (■,n) reaction with beryllium. Beryllium is used for photoneutron sources because its stable isotope beryllium-9 has a weakly attached last neutron with a binding energy of only 1.66 MeV. Thus, a gamma ray with greater energy than 1.66 MeV can cause neutrons to be ejected by the (■,n) reaction as shown below.
NEUTRON SOURCES DOE-HDBK-1019/1-93 Reactor Theory (Neutron Characteristics)
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Neutron Sources Summary
& Intrinsic neutron sources are sources of neutrons from materials that are in the reactor for other purposes such as fuel, burnable poison, or moderator.
& Installed neutron sources are materials or components placed in the reactor specifically for the purpose of producing source neutrons.
& Examples of intrinsic neutron sources are listed below.
Spontaneous fission of heavy nuclides in fuel, such as uranium-238, uranium-235, and plutonium-239, results in fission fragments and free neutrons.
Boron-11 mixed with the fuel undergoes an alpha-neutron reaction and becomes nitrogen-14.
Deuterium present in the reactor coolant undergoes a gamma■neutron reaction and becomes hydrogen-1.
Fixture checkpoint marker-0955-xray is recorded on fixture page 0955.
& Examples of installed neutron sources are listed below.
Spontaneous fission of californium-252 results in fission fragments and free neutrons.
Beryllium-9 undergoes an alpha-neutron reaction (alpha from the decay of plutonium, polonium, or radium) and becomes carbon-12.
Beryllium-9 undergoes a gamma-neutron reaction (high energy gamma from decay of antimony-124) and becomes beryllium-8.
Many startup sources of this type use antimony and beryllium because after activation with neutrons the radioactive antimony becomes an emitter of high energy gammas. The photoneutron sources of this type are constructed somewhat differently from the (■,n) types. One design incorporates a capsule of irradiated antimony enclosed in a beryllium sleeve. The entire assembly is then encased in a stainless steel cladding. A large reactor may have several neutron sources of this type installed within the core.
Summary
The important information in this chapter is summarized below.
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Reactor Theory (Neutron Characteristics) NUCLEAR CROSS SECTIONS AND NEUTRON FLUX
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NUCLEAR CROSS SECTIONS AND NEUTRON FLUX

availability of neutrons to cause interaction and the probability of a neutron interacting with material. The availability of neutrons and the probability of interaction are quantified by the neutron flux and nuclear cross section.

EO 2.1 DEFINE the following terms:
a. Atom density d. Barn
b. Neutron flux e. Macroscopic cross section
c. Microscopic cross section f. Mean free path

EO 2.2 EXPRESS macroscopic cross section in terms of microscopic cross section.

EO 2.3 DESCRIBE how the absorption cross section of typical nuclides varies with neutron energy at energies below the resonance absorption region.

EO 2.4 DESCRIBE the cause of resonance absorption in terms of nuclear energy levels.

EO 2.5 DESCRIBE the energy dependence of resonance absorption peaks for typical light and heavy nuclei.

EO 2.6 EXPRESS mean free path in terms of macroscopic cross section.

EO 2.7 Given the number densities (or total density and component fractions) and microscopic cross sections of components, CALCULATE the macroscopic cross section for a mixture.

EO 2.8 CALCULATE a macroscopic cross section given a material density, atomic mass, and microscopic cross section.

EO 2.9 EXPLAIN neutron shadowing or self-shielding.

'

Introduction

Fission neutrons are born with an average energy of about 2 MeV. These fast neutrons interact with the reactor core materials in various absorption and scattering reactions. Collisions that result in scattering are useful in slowing neutrons to thermal energies. Thermal neutrons may be absorbed by fissile nuclei to produce more fissions or be absorbed in fertile material for conversion to fissionable fuel. Absorption of neutrons in structural components, coolant, and other non-fuel material results in the removal of neutrons without fulfilling any useful purpose. To safely and efficiently operate a nuclear reactor it is necessary to predict the probability that

Fixture checkpoint marker-0956-xray is recorded on fixture page 0956.

a particular absorption or scattering reaction will occur. Once these probabilities are known, if the availability of neutrons can be determined, then the rate at which these nuclear reactions take place can be predicted.

Atom Density

One important property of a material is the atom density. The atom density is the number of atoms of a given type per unit volume of the material. To calculate the atom density of a substance use Equation (2-1).

(2-1)

where:

N = atom density (atoms/cm)3
' = density (g/cm)3
N = Avogadro's number (6.022 x 10 atoms/mole) A

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M = gram atomic weight
N
NA
M
2.699 g
cm3
6.022 x 1023 atoms
mole
26.9815 g
mole
6.024 x 1022 atoms
cm3
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Example:
A block of aluminum has a density of 2.699 g/cm . If the gram atomic weight of
3
aluminum is 26.9815 g, calculate the atom density of the aluminum.
Solution:
Cross Sections
The probability of a neutron interacting with a nucleus for a particular reaction is dependent upon not only the kind of nucleus involved, but also the energy of the neutron. Accordingly, the absorption of a thermal neutron in most materials is much more probable than the absorption of a fast neutron. Also, the probability of interaction will vary depending upon the type of reaction involved.
The probability of a particular reaction occurring between a neutron and a nucleus is called the microscopic cross section ( ) of the nucleus for the particular reaction. This cross section will vary with the energy of the neutron. The microscopic cross section may also be regarded as the effective area the nucleus presents to the neutron for the particular reaction. The larger the effective area, the greater the probability for reaction.
Because the microscopic cross section is an area, it is expressed in units of area, or square centimeters. A square centimeter is tremendously large in comparison to the effective area of a nucleus, and it has been suggested that a physicist once referred to the measure of a square centimeter as being "as big as a barn" when applied to nuclear processes. The name has persisted and microscopic cross sections are expressed in terms of barns. The relationship between barns and cm is shown below. 2
1 barn = 10-24 cm2
N
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Whether a neutron will interact with a certain volume of material depends not only on the microscopic cross section of the individual nuclei but also on the number of nuclei within that volume. Therefore, it is necessary to define another kind of cross section known as the macroscopic cross section ( ). The macroscopic cross section is the probability of a given reaction occurring per unit travel of the neutron. is related to the microscopic cross section

(2-2)
where:
= macroscopic cross section (cm)
-1
N = atom density of material (atoms/cm)
3
= microscopic cross-section (cm)
2
The difference between the microscopic and macroscopic cross sections is extremely important and is restated for clarity. The microscopic cross section ( ) represents the effective target area that a single nucleus presents to a bombarding particle. The units are given in barns or cm .
2
The macroscopic cross section ( ) represents the effective target area that is presented by all of the nuclei contained in 1 cm of the material. The units are given as 1/cm or cm .
3 -1
A neutron interacts with an atom of the material it enters in two basic ways. It will either interact through a scattering interaction or through an absorption reaction. The probability of a neutron being absorbed by a particular atom is the microscopic cross section for absorption, . The probability of a neutron scattering off of a particular nucleus is the microscopic cross a Fixture checkpoint marker-0958-xray is recorded on fixture page 0958.
section for scattering, s
. The sum of the microscopic cross section for absorption and the microscopic cross section for scattering is the total microscopic cross section, . T
= + T a s
Both the absorption and the scattering microscopic cross sections can be further divided. For instance, the scattering cross section is the sum of the elastic scattering cross section (se) and the inelastic scattering cross section ( ). si
= + s se si
The microscopic absorption cross section ( ) includes all reactions except scattering. However, a for most purposes it is sufficient to merely separate it into two categories, fission (f) and capture ( ). Radiative capture of neutrons was described in the Neutron Interactions chapter c of Module 1.
= + a f c
DOE-HDBK-1019/1-93
Reactor Theory (Neutron Characteristics) NUCLEAR CROSS SECTIONS AND NEUTRON FLUX
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Figure 1 Typical Neutron Absorption Cross Section vs. Neutron Energy
The variation of absorption cross sections with neutron energy is often complicated. For many elements the absorption cross sections are small, ranging from a fraction of a barn to a few barns for slow (or thermal) neutrons.
For a considerable number of nuclides of moderately high (or high) mass numbers, an examination of the variation of the absorption cross section with the energy of the incident neutron reveals the existence of three regions on a curve of absorption cross section versus neutron energy. This cross section is illustrated in Figure 1. First, the cross section decreases steadily with increasing neutron energy in a low energy region, which includes the thermal range (E < 1 eV). In this region the absorption cross section, which is often high, is inversely proportional to the velocity (v). This region is frequently referred to as the "1/v region," because the absorption cross section is proportional to 1/v, which is the reciprocal of neutron

sections rise sharply to high values called "resonance peaks" for neutrons of certain energies, and then fall again. These energies are called resonance energies and are a result of the affinity of the nucleus for neutrons whose energies closely match its discrete, quantum energy levels. That is, when the binding energy of a neutron plus the kinetic energy of the neutron are exactly equal to the amount required to raise a compound nucleus from its ground state to a quantum level, resonance absorption occurs. The following example problem further illustrates this point.

■

(

Assuming that uranium-236 has a nuclear quantum energy level at 6.8 MeV above its ground state, calculate the kinetic energy a neutron must possess to undergo resonant absorption in uranium-235 at this resonance energy level.

BE = [Mass(U) + Mass(neutron) - Mass(U)] x 931 MeV/amu 235 236

BE = (235.043925 + 1.008665 - 236.045563) x 931 MeV/amu

BE = (0.007025 amu) x 931 MeV/amu = 6.54 MeV

6.8 MeV - 6.54 MeV = 0.26 MeV

The difference between the binding energy and the quantum energy level equals the amount of kinetic energy the neutron must possess. The typical heavy nucleus will have many closely■spaced resonanc complex and have more possible configurations and corresponding energy states. Light nuclei, being less complex, have fewer possible energy states and fewer resonances that are sparsely distributed at higher energy levels.

For higher neutron energies, the absorption cross section steadily decreases as the energy of the neutron increases. This is called the "fast neutron region." In this region the absorption cross sections are usually less than 10 barns.

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With the exception of hydrogen, for which the value is fairly large, the elastic scattering cross sections are generally small, for example, 5 barns to 10 barns. This is close to the magnitude of the actual geometric cross sectional area expected for atomic nuclei. In potential scattering, the cross section is essentially constant and independent of neutron energy. Resonance elastic scattering and inelastic scattering exhibit resonance peaks similar to those associated with absorption cross sections. The resonances occur at lower energies for heavy nuclei than for light nuclei. In general, the variations in scattering cross sections are very small when compared to the variations that occur in absorption cross sections.

Mean Free Path

If a neutron has a certain probability of undergoing a particular interaction in one centimeter of travel, then the inverse of this value describes how far the neutron will travel (in the average case) before undergoing an interaction. This average distance traveled by a neutron before interaction is known as the mean free path for that interaction and is represented by the symbol ■. The relationship between the mean free path (■) and the macroscopic cross section (*) is shown below.

(2-3)

'

(

)

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Calculation of Macroscopic Cross Section and Mean Free Path
Most materials are composed of several elements, and because most elements are composed of several isotopes, most materials involve many cross sections, one for each isotope involved. Therefore, to include all the isotopes within a given material, it is necessary to determine the macroscopic cross section for each isotope and then sum all the individual macroscopic cross sections. Equation (2-4) can be used to determine the macroscopic cross section for a composite material.
* = N) + N) + N) + ... N) (2-4) 1 1 2 2 3 3 nn
where:
N = the number of nuclei per cm of the n element n
3 th
) = the microscopic cross section of the n element n
th
The following example problems illustrate the calculation of the macroscopic cross section for a single element and for combinations of materials.
Example 1:
Find the macroscopic thermal neutron absorption cross section for iron, which has a density of 7.86 g/cm . The microscopic cross section for absorption of iron is 2.56 barns 3 and the gram atomic weight is 55.847 g.
Solution:
Step 1: Using Equation (2-1), calculate the atom density of iron.
Step 2: Use this atom density in Equation (2-2) to calculate the macroscopic cross section.
'
'
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Example 2:
An alloy is composed of 95% aluminum and 5% silicon (by weight). The density of the alloy is 2.66 g/cm . Properties of aluminum and silicon are shown below. 3
Element Gram Atomic Weight)) a (barns) s (barns)
Aluminum 26.9815 0.23 1.49
Fixture checkpoint marker-0960-xray is recorded on fixture page 0960.
Silicon 28.0855 0.16 2.20
1. Calculate the atom densities for the aluminum and silicon.
2. Determine the absorption and scattering macroscopic cross sections for thermal neutrons.
3. Calculate the mean free paths for absorption and scattering.
Solution:
Step 1: The density of the aluminum will be 95% of the total density. Using Equation (2-1) yields the atom densities.
a NAl a,Al NSi a,Si
5.64 x 1022 atoms
cm3
0.23 x 10 24 cm22.85 x 1021 atoms
cm3
0.16 x 10 24 cm2

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s NAl s,Al NSi s,Si
5.64 x 1022 atoms
cm3
1.49 x 10 24 cm22.85 x 1021 atoms
cm3
2.20 x 10 24 cm2
0.0903 cm 1
a
1
a
Fixture checkpoint marker-0961-xray is recorded on fixture page 0961.
1
0.01345 cm 1
74.3 cm
s
1
s
1
0.0903 cm 1
11.1 cm
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Step 2: The macroscopic cross sections for absorption and scattering are calculated using Equation (2-4).
Step 3: The mean free paths are calculated by inserting the macroscopic cross sections calculated above into Equation (2-3).
Thus, a neutron must travel an average of 74.3 cm to interact by absorption in this alloy, but it must travel only 11.1 cm to interact by scattering.
)
)
)
)
(■
(■
DOE-HDBK-1019/1-93
NUCLEAR CROSS SECTIONS AND NEUTRON FLUX Reactor Theory (Neutron Characteristics)
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Effects of Temperature on Cross Section
As discussed, the microscopic absorption cross section varies significantly as neutron energy varies. The microscopic cross sections provided on most charts and tables are measured for a standard neutron velocity of 2200 meters/second, which corresponds to an ambient temperature of 68(F. Therefore, if our material is at a higher temperature, the absorption cross section will be lower than the value for 68(F, and any cross sections which involve absorption (for example,) ,) ,)) must be corrected for the existing temperature. acf
The following formula is used to correct microscopic cross sections for temperature. Although the example illustrates absorption cross section, the same formula may be used to correct capture and fission cross sections.

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) = microscopic cross section corrected for temperature
) = microscopic cross section at reference temperature (68(F or 20(C) o
T = reference temperature (68(F) in degrees Rankine ((R) or Kelvin ((K) o
T = temperature for which corrected value is being calculated
NOTE: When using this formula, all temperatures must be converted to (R or (K.
(R = (F + 460
(K = (C + 273
Example:
What is the value of)f for uranium-235 for thermal neutrons at 500(F? Uranium-235 has a) of 583 barns at 68(F. f
Solution:
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Neutron Flux
Macroscopic cross sections for neutron reactions with materials determine the probability of one neutron undergoing a specific reaction per centimeter of travel through that material. If one
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wants to determine how many reactions will actually occur, it is necessary to know how many neutrons are traveling through the material and how many centimeters they travel each second. It is convenient to consider the number of neutrons existing in one cubic centimeter at any one instant and the total distance they travel each second while in that cubic centimeter. The number of neutrons existing in a cm of material at any instant is called neutron density and is 3 represented by the symbol n with units of neutrons/cm . The total distance these neutrons can 3 travel each second will be determined by their velocity.
A good way of defining neutron flux (1) is to consider it to be the total path length covered by all neutrons in one cubic centimeter during one second. Mathematically, this is the equation below.
1 = n v (2-5)
where:
1 = neutron flux (neutrons/cm -sec) 2
n = neutron density (neutrons/cm)3
v = neutron velocity (cm/sec)
The term neutron flux in some applications (for example, cross section measurement) is used as parallel beams of neutrons traveling in a single direction. The intensity (I) of a neutron beam is the product of the neutron density times the average neutron velocity. The directional beam intensity is equal to the number of neutrons per unit area and time (neutrons/cm -sec) falling on 2 a surface perpendicular to the direction of the beam.
One can think of the neutron flux in a reactor as being comprised of many neutron beams traveling in various directions. Then, the neutron flux becomes the scalar sum of these directional flux intensities (added as numbers and not vectors), that is, 1 = I + I + I +...I . 123 n
Since the atoms in a reactor do not interact preferentially with neutrons from any particular direction, all of these directional beams contribute to the total rate of reaction. In reality, at a given point within a reactor, neutrons will be traveling in all directions.
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NUCLEAR CROSS SECTIONS AND NEUTRON FLUX Reactor Theory (Neutron Characteristics)
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Self-Shielding

due to a phenomenon referred to as neutron shadowing or self-shielding. For example, the interior of a fuel pin or pellet will "see" a lower average flux level than the outer surfaces since an appreciable fraction of the neutrons will have been absorbed and therefore cannot reach the interior of the fuel pin. This is especially important at resonance energies, where the absorption cross sections are large.

Summary

The important information in this chapter is summarized below.

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Reactor Theory (Neutron Characteristics) NUCLEAR CROSS SECTIONS AND NEUTRON FLUX

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Nuclear Cross Section and Neutron Flux Summary

& Atom density (N) is the number of atoms of a given type per unit volume of material.

& Microscopic cross section ()) is the probability of a given reaction occurring between a neutron and a nucleus.

& Microscopic cross sections are measured in units of barns, where 1 barn = 10-24 cm .2

& Macroscopic cross section (*) is the probability of a given reaction occurring per unit length of travel of the neutron. The units for macroscopic cross section are cm . -1

& The mean free path (■) is the average distance that a neutron travels in a material between interactions.

& Neutron flux (1) is the total path length traveled by all neutrons in one cubic centimeter of material during one second.

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& The macroscopic cross section for a material can be calculated using the equation below.

* = N)

& The absorption cross section for a material usually has three distinct regions. At low neutron energies (<1 eV) the cross section is inversely proportional to the neutron velocity.

& Resonance absorption occurs when the sum of the kinetic energy of the neutron and its binding energy is equal to an allowed nuclear energy level of the nucleus.

& Resonance peaks exist at intermediate energy levels. For higher neutron energies, the absorption cross section steadily decreases as the neutron energy increases.

& The mean free path equals 1/*.

& The macroscopic cross section for a mixture of materials can be calculated using the equation below.

* = N 1 1 2 2 3 3 nn) + N) + N) + ... N)

& Self-shielding is where the local neutron flux is depressed within a material due to neutron absorption near the surface of the material.

1)

REACTION RATES DOE-HDBK-1019/1-93 Reactor Theory (Neutron Characteristics)

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REACTION RATES

It is possible to determine the rate at which a nuclear reaction will take place based on the neutron flux, cross section for the interaction, and atom density of the target. This relationship illustrates how a change in one of these items affects the

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EO 2.10 Given the neutron flux and macroscopic cross section,
CALCULATE the reaction rate.
EO 2.11 DESCRIBE the relationship between neutron flux and reactor
power.
Reaction Rates
If the total path length of all the neutrons in a cubic centimeter in a second is known, (neutron flux (1)), and if the probability of having an interaction per centimeter path length is also known (macroscopic cross section (*)), multiply them together to get the number of interactions taking place in that cubic centimeter in one second. This value is known as the reaction rate and is denoted by the symbol R. The reaction rate can be calculated by the equation shown below.
R = 1 * (2-6)
where:
R = reaction rate (reactions/sec)
1 = neutron flux (neutrons/cm -sec) 2
* = macroscopic cross section (cm) -1
Substituting the fact that * = N) into Equation (2-6) yields the equation below.
where:
) = microscopic cross section (cm)2
N = atom density (atoms/cm)3
Rf f
1 x 1013 neutrons
cm2
sec
0.1 cm 1
1 x 1012 fissions
cm3
sec
Rf f
f
Rf
1.29 x 1012 fissions
Fixture checkpoint marker-0964-xray is recorded on fixture page 0964.
cm3
sec
3 x 1013 neutrons
cm2
sec
0.043 cm 1
Reactor Theory (Neutron Characteristics) DOE-HDBK-1019/1-93 REACTION RATES
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The reaction rate calculated will depend on which macroscopic cross section is used in the calculation. Normally, the reaction rate of greatest interest is the fission reaction rate.
Example:
If a one cubic centimeter section of a reactor has a macroscopic fission cross section of 0.1 cm , and if the thermal neutron flux is 10 neutrons/cm -sec, what is the fission rate -1 13 2 in that cubic centimeter?
Solution:
In addition to using Equation (2-6) to determine the reaction rate based on the physical properties

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properties if the reaction rate is known.
Example:
A reactor operating at a flux level of 3 x 10 neutrons/cm -sec contains 10 atoms of
13 2 20
uranium-235 per cm . The reaction rate is 1.29 x 10 fission/cm . Calculate and .
3 12 3
f f
Solution:
Step 1: The macroscopic cross section can be determined by solving
Equation (2-6) for and substituting the appropriate values. f
Rf N f
f
Rf
N
1.29 x 1012 fissions
cm3
sec
1 x 1020 atoms
cm3
3 x 1013 neutrons
cm2
sec
4.3 x 10 22 cm2 1 barn
1 x 10 24 cm2
430 barns
1 watt 1 erg
1 x 10 7 watt sec
1 MeV
1.602 x 10 6
erg
1 fission
200 MeV
3.12 x 1010 fissions
second
REACTION RATES DOE-HDBK-1019/1-93 Reactor Theory (Neutron Characteristics)
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Step 2: To find the microscopic cross section, replace f with (N x f) and solve
for .
Fixture checkpoint marker-0965-xray is recorded on fixture page 0965.
f
Reactor Power Calculation
Multiplying the reaction rate per unit volume by the total volume of the core results in the total number of reactions occurring in the core per unit time. If the amount of energy involved in each reaction were known, it would be possible to determine the rate of energy release (power) due to a certain reaction.
In a reactor where the average energy per fission is 200 MeV, it is possible to determine the number of fissions per second that are necessary to produce one watt of power using the following conversion factors.

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1 MeV = 1.602 x 10 ergs -6
1 erg = 1 x 10 watt-sec -7
This is equivalent to stating that 3.12 x 10 fissions release 1 watt-second of energy. 10
P
th f V
3.12 x 1010 fissions
watt sec
Reactor Theory (Neutron Characteristics) DOE-HDBK-1019/1-93 REACTION RATES
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The power released in a reactor can be calculated based on Equation (2-6). Multiplying the reaction rate by the volume of the reactor results in the total fission rate for the entire reactor. Dividing by the number of fissions per watt-sec results in the power released by fission in the reactor in units of watts. This relationship is shown mathematically in Equation (2-7) below.
(2-7)
where:
Fixture checkpoint marker-0966-xray is recorded on fixture page 0966.
P = power (watts)
= thermal neutron flux (neutrons/cm -sec) th
2
= macroscopic cross section for fission (cm) f
-1
V = volume of core (cm)
3
Relationship Between Neutron Flux and Reactor Power
In an operating reactor the volume of the reactor is constant. Over a relatively short period of time (days or weeks), the number density of the fuel atoms is also relatively constant. Since the atom density and microscopic cross section are constant, the macroscopic cross section must also be constant. Examining Equation (2-7), it is apparent that if the reactor volume and macroscopic cross section are constant, then the reactor power and the neutron flux are directly proportional. This is true for day-to-day operation. The neutron flux for a given power level will increase very slowly over a period of months due to the burnup of the fuel and resulting decrease in atom density and macroscopic cross section.
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Reaction Rates Summary
& The reaction rate is the number of interactions of a particular type occurring in a cubic centimeter of material in a second.
& The reaction rate can be calculated by the equation below.
R = 1 *
& Over a period of several days, while the atom density of the fuel can be considered constant, the neutron flux is directly proportional to reactor power.
Summary
The important information in this chapter is summarized below.
Reactor Theory (Neutron Characteristics) DOE-HDBK-1019/1-93 NEUTRON MODERATION
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NEUTRON MODERATION
In thermal reactors, the neutrons that cause fission are at a much lower energy than the energy level at which they were born from fission. In this type of

energy level of the neutrons in an efficient manner.
EO 2.12 DEFINE the following concepts:
a. Thermalization d. Average logarithmic energy decrement
b. Moderator e. Macroscopic slowing down power
c. Moderating ratio
EO 2.13 LIST three desirable characteristics of a moderator.
EO 2.14 Given an average fractional energy loss per collision, CALCULATE the energy loss after a specified number of collisions.
Neutron Slowing Down and Thermalization
Fission neutrons are produced at an average energy level of 2 MeV and immediately begin to slow down as the result of numerous scattering reactions with a variety of target nuclei. After a number of collisions with nuclei, the speed of a neutron is reduced to such an extent that it has approximately the same average kinetic energy as the atoms (or molecules) of the medium in which the neutron is undergoing elastic scattering. This energy, which is only a small fraction of an electron volt at ordinary temperatures (0.025 eV at 20(C), is frequently referred to as the thermal energy, since it depends upon the temperature. Neutrons whose energies have been reduced to values in this region (< 1 eV) are designated thermal neutrons. The process of reducing the energy of a neutron to the thermal region by elastic scattering is referred to as thermalization, slowing down, or moderation. The material used for the purpose of thermalizing neutrons is called a moderator. A good moderator reduces the speed of neutrons in a small number of collisions, but does not absorb them to any great extent. Slowing the neutrons in as
Fixture checkpoint marker-0967-xray is recorded on fixture page 0967.
few collisions as possible is desirable in order to reduce the amount of neutron leakage from the core and also to reduce the number of resonance absorptions in non-fuel materials. Neutron leakage and resonance absorption will be discussed in the next module.
ln Ei
ln Ef
ln
Ei
Ef
2
A
2
3
NEUTRON MODERATION DOE-HDBK-1019/1-93 Reactor Theory (Neutron Characteristics)
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The ideal moderating material (moderator) should have the following nuclear properties.
large scattering cross section
small absorption cross section
large energy loss per collision
A convenient measure of energy loss per collision is the logarithmic energy decrement. The average logarithmic energy decrement is the average decrease per collision in the logarithm of the neutron energy. This quantity is represented by the symbol (Greek letter xi).
(2-8)
where:
= average logarithmic energy decrement
Ei = average initial neutron energy

The symbol is commonly called the average logarithmic energy decrement because of the fact that a neutron loses, on the average, a fixed fraction of its energy per scattering collision. Since the fraction of energy retained by a neutron in a single elastic collision is a constant for a given material, is also a constant. Because it is a constant for each type of material and does not depend upon the initial neutron energy, is a convenient quantity for assessing the moderating ability of a material.

The values for the lighter nuclei are tabulated in a variety of sources. The following commonly used approximation may be used when a tabulated value is not available.

This approximation is relatively accurate for mass numbers (A) greater than 10, but for some low values of A it may be in error by over three percent.

N

ln Ehigh ln Elow

ln

Ehigh

Elow

N

ln

Ehigh

Elow

ln

2 x 106eV

0.025 eV

0.948

19.2 collisions

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Since represents the average logarithmic energy loss per collision, the total number of collisions necessary for a neutron to lose a given amount of energy may be determined by dividing into the difference of the natural logarithms of the energy range in question. The number of collisions

Fixture checkpoint marker-0968-xray is recorded on fixture page 0968.

(N) to travel from any energy, E , to any lower energy, E , can be calculated as shown below. high low

Example:

How many collisions are required to slow a neutron from an energy of 2 MeV to a thermal energy of 0.025 eV, using water as the moderator? Water has a value of 0.948 for .

Solution:

Sometimes it is convenient, based upon information known, to work with an average fractional energy loss per collision as opposed to an average logarithmic fraction. If the initial neutron energy level and the average fractional energy loss per collision are known, the final energy level for a given number of collisions may be computed using the following formula.

! (

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(2-9)

where:

E = initial neutron energy o

E = neutron energy after N collisions N

x = average fractional energy loss per collision

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Example:
If the average fractional energy loss per collision in hydrogen is 0.63, what will be the energy of a 2 MeV neutron after (a) 5 collisions? (b) 10 collisions?
Solution:
a)
b)
Macroscopic Slowing Down Power
Although the logarithmic energy decrement is a convenient measure of the ability of a material to slow neutrons, it does not measure all necessary properties of a moderator. A better measure of the capabilities of a material is the macroscopic slowing down power. The macroscopic slowing down power (MSDP) is the product of the logarithmic energy decrement and the macroscopic cross section for scattering in the material. Equation (2-10) illustrates how to calculate the macroscopic slowing down power.
(2-10)
MR
s
a
Reactor Theory (Neutron Characteristics) DOE-HDBK-1019/1-93 NEUTRON MODERATION
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Moderating Ratio
Macroscopic slowing down power indicates how rapidly a neutron will slow down in the material in question, but it still does not fully explain the effectiveness of the material as a moderator. An element such as boron has a high logarithmic energy decrement and a good slowing down power, but it is a poor moderator because of its high probability of absorbing neutrons.
The most complete measure of the effectiveness of a moderator is the moderating ratio. The moderating ratio is the ratio of the macroscopic slowing down power to the macroscopic cross section for absorption. The higher the moderating ratio, the more effectively the material performs as a moderator. Equation (2-11) shows how to calculate the moderating ratio of a material.
(2-11)
Moderating properties of different materials are compared in Table 2.
TABLE 2
Moderating Properties of Materials
Material Collisions to Slowing Down
Number of Macroscopic
Thermalize Power
Fixture checkpoint marker-0969-xray is recorded on fixture page 0969.
Moderating
Ratio
H2O 0.927 19 1.425 62
D2O 0.510 35 0.177 4830
Helium 0.427 42 9 x 10 51 -6
Beryllium 0.207 86 0.154 126
Boron 0.171 105 0.092 0.00086
Carbon 0.158 114 0.083 216
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Summary

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Neutron Moderation Summary
& Thermalization is the process of reducing the energy level of a neutron from the energy level at which it is produced to an energy level in the thermal range.
& The moderator is the reactor material that is present for the purpose of thermalizing neutrons.
& Moderating ratio is the ratio of the macroscopic slowing down power to the macroscopic cross section for absorption.
& The average logarithmic energy decrement (!) is the average change in the logarithm of neutron energy per collision.
& Macroscopic slowing down power is the product of the average logarithmic energy decrement and the macroscopic cross section for scattering.
& There are three desirable characteristics of a moderator.
1. large scattering cross section
Fixture checkpoint marker-0970-xray is recorded on fixture page 0970.
2. small absorption cross section
3. large energy loss per collision
& The energy loss after a specified number of collisions can be calculated using the equation below.
E = E (1 - x) N o
N
■
■ ■
Reactor Theory (Neutron Characteristics) DOE-HDBK-1019/1-93 PROMPT AND DELAYED NEUTRONS
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PROMPT AND DELAYED NEUTRONS
Not all neutrons are released at the same time following fission. Most neutrons are released virtually instantaneously and are called prompt neutrons. A very small fraction of neutrons are released after the decay of fission products and are called delayed neutrons. Although delayed neutrons are a very small fraction of the total number of neutrons, they play an extremely important role in the control of the reactor.
EO 3.1 STATE the origin of prompt neutrons and delayed neutrons.
EO 3.2 STATE the approximate fraction of neutrons that are born as delayed neutrons from the fission of the following nuclear fuels:
a. Uranium-235
b. Plutonium-239
EO 3.3 EXPLAIN the mechanism for production of delayed neutrons.
EO 3.4 EXPLAIN prompt and delayed neutron generation times.
EO 3.5 Given prompt and delayed neutron generation times and delayed neutron fraction, CALCULATE the average generation time.
EO 3.6 EXPLAIN the effect of delayed neutrons on reactor control.
Neutron Classification
The great majority (over 99%) of the neutrons produced in fission are released within about 10-13 seconds of the actual fission event. These are called prompt neutrons. A small portion of fission neutrons are delayed neutrons, which are produced for some time after the fission process has taken place. The delayed neutrons are emitted immediately following the first beta decay of a fission fragment known as a delayed neutron precursor. An example of a delayed neutron

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For most applications, it is convenient to combine the known precursors into groups with appropriately averaged properties. These groups vary somewhat depending on the fissile material in use. Table 3 lists the characteristics for the six precursor groups resulting from thermal fission of uranium-235. The fraction of all neutrons that are produced by each of these precursors is called the delayed neutron fraction for that precursor. The total fraction of all neutrons born as delayed neutrons is called the delayed neutron fraction (■). The fraction of delayed neutrons produced varies depending on the predominant fissile nuclide in use. The delayed neutron fractions (■) for the fissile nuclides of most interest are as follows: uranium-233 (0.0026), uranium-235 (0.0065), uranium-238 (0.0148), and plutonium-239 (0.0021).

TABLE 3

Delayed Neutron Precursor Groups
for Thermal Fission in Uranium-235

Group	Half-Life (sec)	Delayed Neutron Fraction	Average Energy (MeV)
1	55.7	0.00021	0.25
2	22.7	0.00142	0.46
3	6.2	0.00127	0.41
4	2.3	0.00257	0.45

Fixture checkpoint marker-0971-xray is recorded on fixture page 0971.

5	0.61	0.00075	0.41
6	0.23	0.00027	-
Total	-	0.0065	-

Neutron Generation Time

The neutron generation time is the time required for neutrons from one generation to cause the fissions that produce the next generation of neutrons. The generation time for prompt neutrons (5* - pronounced "ell-star") is the total time from birth to rebirth. Three time intervals are involved: (a) the time it takes a fast neutron to slow down to thermal energy, (b) the time the now thermal neutron exists prior to absorption in fuel, and (c) the time required for a fissionable nucleus to emit a fast neutron after neutron absorption.

Fast neutrons slow to thermal energies or leak out of the reactor in 10 seconds to 10 seconds, -4 -6 depending on the moderator. In water moderated reactors, thermal neutrons tend to exist for about 10 seconds before they are absorbed. Fission and fast neutron production following -4 neutron absorption in a fissionable nucleus occurs in about 10 seconds. Thus, fast reactors -13 have an 5* of about 10 seconds, while thermal reactors have an 5* of about 10 seconds + 10 -6 -6 -4 seconds, which is about 10 seconds to 10 seconds. -4 -5

■■■■
■■■■
■■

On the other hand, the average generation time for the six delayed neutron groups is the total time from the birth of the fast neutron to the emission of the delayed neutron. Again, three time intervals are involved: (a) the time it takes a fast neutron to slow down to thermal energy, (b) the time the thermal neutron exists prior to absorption, and (c) the average time from neutron absorption to neutron emission by the six precursor groups. The average time for decay of precursors from uranium-235 is 12.5 seconds. The other terms in the delayed neutron

generation time becomes ■ 12.5 seconds.
A neutron generation time in the range of 10 seconds to 10 seconds or faster could result in -4 -5
very rapid power excursions, and control would not be possible without the dependence upon
delayed neutrons to slow down the rate of the reaction. The average generation time, and hence
the rate that power can rise, is determined largely by the delayed neutron generation time. The
following equation shows this mathematically.
(2-12)
Example:
Assume a prompt neutron generation time for a particular reactor of 5 x 10 seconds and -5
a delayed neutron generation time of 12.5 seconds. If ■ is 0.0065, calculate the average
generation time.
Solution:
This example demonstrates the effect delayed neutrons have on the neutron generation time and
thus reactor control. If a reactor were to be operated in a sustained chain reaction using only
prompt neutrons (■ = 0), the generation time from the previous example would be about
5 x 10 seconds. However, by operating the reactor such that a 0.0065 fraction of neutrons are -5
delayed, the generation life time is extended to 0.0813 seconds, providing time for adequate
operator control. Therefore, although only a small fraction of the total neutron population,
delayed neutrons are extremely important to the control and maintenance of a sustained fission
chain reaction.
Summary
The important information in this chapter is summarized on the following page.
■ ■ ■ ■
PROMPT AND DELAYED NEUTRONS DOE-HDBK-1019/1-93 Reactor Theory (Neutron Characteristics)
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Prompt and Delayed Neutrons Summary
& Prompt neutrons are released directly from fission within 10 seconds of the -13
Fixture checkpoint marker-0972-xray is recorded on fixture page 0972.
fission event.
& Delayed neutrons are released from the decay of fission products that are called
delayed neutron precursors. Delayed neutron precursors are grouped according to
half-life. Half-lives vary from fractions of a second to almost a minute.
& The fraction of neutrons born as delayed neutrons is different for different fuel
materials. Following are values for some common fuel materials.
Uranium-235 0.0065
Plutonium-239 0.0021
& Delayed neutrons are produced by a classification of fission products known as
delayed neutron precursors. When a delayed neutron precursor undergoes a
■ decay, it results in an excited daughter nucleus which immediately ejects a -
neutron. Therefore, these delayed neutrons appear with a half-life of the delayed
neutron precursor.
& The delayed neutron generation time is the total time from the birth of the fast
neutron to the emission of the delayed neutron in the next generation. Delayed
neutron generation times are dominated by the half-life of the delayed neutron
precursor. The average delayed neutron generation time is about 12.5 seconds.
& A prompt neutron generation time is the sum of the amount of time it takes a fast
neutron to thermalize, the amount of time the neutron exists as a thermal neutron
before it is absorbed, and the amount of time between a fissionable nuclide

generation time is about 5 x 10 seconds. -5
& The average neutron generation time can be calculated from the prompt and delayed neutron generation times and the delayed neutron fraction using Equation (2-12).
& Delayed neutrons are responsible for the ability to control the rate at which power can rise in a reactor. If only prompt neutrons existed, reactor control would not be possible due to the rapid power changes.

Figure 2 Prompt Fission Neutron Energy Spectrum for
Thermal Fission of Uranium-235
NEUTRON FLUX SPECTRUM
The number of neutrons that exist at a given energy level varies. A plot of either the fraction of neutrons or the neutron flux at a given energy versus the energy level is called a neutron energy spectrum. The neutron energy spectrum varies widely for different types of reactors.
EO 4.1 STATE the average energy at which prompt neutrons are produced.
EO 4.2 DESCRIBE the neutron energy spectrum in the following reactors:
a. Fast reactor
b. Thermal reactor
EO 4.3 EXPLAIN the reason for the particular shape of the fast, intermediate, and slow energy regions of the neutron flux spectrum for a thermal reactor.
Prompt Neutron Energies
The neutrons produced by fission are high energy neutrons, and almost all fission neutrons have energies between 0.1 MeV and 10 MeV. The neutron energy distribution, or spectrum, may best be described by plotting the fraction of neutrons per MeV as a
Fixture checkpoint marker-0973-xray is recorded on fixture page 0973.
function of neutron energy, as shown in Figure 2. From this figure it can be seen that the most probable neutron energy is about 0.7 MeV. Also, from this data it can be shown that the average energy of fission neutrons is about 2 MeV. Figure 2 is the neutron energy spectrum for thermal fission in uranium-235. The values will vary slightly for other nuclides.

Figure 3 Comparison of Neutron Flux Spectra for
Thermal and Fast Breeder Reactor
Thermal and Fast Breeder Reactor Neutron Spectra
The spectrum of neutron energies produced by fission varies significantly from the energy spectrum, or flux, existing in a reactor at a given time. Figure 3 illustrates the difference in neutron flux spectra between a thermal reactor and a fast breeder reactor. The energy distribution of neutrons from fission is essentially the same for both reactors, so the differences in the curve shapes may be attributed to the neutron moderation or slowing down effects.
No attempt is made to thermalize or slow down neutrons in the fast breeder reactor (liquid metal cooled); therefore, an insignificant number of neutrons exist in the thermal range. For the thermal reactor (water moderated), the spectrum of neutrons in the fast region (> 0.1 MeV) has
Fixture checkpoint marker-0974-xray is recorded on fixture page 0974.
a shape similar to that for the spectrum of neutrons emitted by the fission process.
In the thermal reactor, the flux in the intermediate energy region (1 eV to 0.1 MeV) has approximately a 1/E dependence. That is, if the energy (E) is halved, the flux doubles. This 1/E dependence is caused by the slowing down process, where elastic collisions remove a constant fraction of the neutron energy per collision (on the average), independent of energy; thus, the neutron loses larger amounts of energy per collision at higher energies than at lower energies. The fact that the neutrons lose a constant fraction of energy per collision causes the neutrons to tend to "pile up" at lower energies, that is, a greater number of neutrons exist at the lower energies as a result of this behavior.
vp
2 k T
m
vp
2 k T
m
2 1.38 x 10 16 erg
K
293 K
1.66 x 10 24 g
2.2 x 105 cm
sec
1 m
100 cm
2200 m
sec
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In the thermal region the neutrons achieve a thermal equilibrium with the atoms of the moderator material. In any given collision they may gain or lose energy, and over successive collisions will gain as much energy as they lose. These thermal neutrons, even at a specific temperature, do not all have the same energy or velocity; there is a distribution of energies, usually referred to as the Maxwell distribution (e.g., Figure 2). The energies of most thermal neutrons lie close to the most probable energy, but there is a spread of neutrons above and below this value.
Most Probable Neutron Velocities
The most probable velocity (vp) of a thermal neutron is determined by the temperature of the medium and can be determined by Equation (2-13) .

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where:
v = most probable velocity of neutron (cm/sec) p
k = Boltzman's constant (1.38 x 10 erg/ K) -16
T = absolute temperature in degrees Kelvin (K)
m = mass of neutron (1.66 x 10 grams) -24
Example:
Calculate the most probable velocities for neutrons in thermal equilibrium with their surroundings at the following temperatures. a) 20 C, b) 260 C.
Solution:
a) Calculate the most probable velocity for 20 C using Equation (2-13).
vp
2 k T
m
2 1.38 x 10 16 erg
K
533 K
1.66 x 10 24 g
2.977 x 105 cm
Fixture checkpoint marker-0975-xray is recorded on fixture page 0975.
sec
1 m
100 cm
2977 m
sec
NEUTRON FLUX SPECTRUM DOE-HDBK-1019/1-93 Reactor Theory (Neutron Characteristics) NP-02 Page 36 Rev. 0
b) Calculate the most probable velocity for 260 C using Equation (2-13).
From these calculations it is evident that the most probable velocity of a thermal neutron increases as temperature increases. The most probable velocity at 20 C is of particular importance since reference data, such as nuclear cross sections, are tabulated for a neutron velocity of 2200 meters per second.
Reactor Theory (Neutron Characteristics) DOE-HDBK-1019/1-93 NEUTRON FLUX SPECTRUM Rev. 0 Page 37 NP-02
Neutron Flux Spectrum Summary
& Prompt neutrons are born at energies between 0.1 MeV and 10 MeV. The average prompt neutron energy is about 2 MeV.
& Fast reactors have a neutron energy spectrum that has the same shape as the prompt neutron energy spectrum.
& Thermal reactors have a neutron energy spectrum that has two pronounced peaks, one in the thermal energy region where the neutrons are in thermal equilibrium with the core materials and another in the fast region at energies where neutrons are produced. The flux in the intermediate region (1 eV to 0.1 MeV) has a roughly 1/E dependence.
& The neutron flux spectrum for the fast energy region of a thermal reactor has a shape similar to that of the spectrum of neutrons emitted by the fission process.
& The reason for the 1/E flux dependence at intermediate energy levels in a thermal reactor is due to the neutrons' tendency to lose a constant fraction of energy per collision. Since the neutrons lose a greater amount at the higher

detail of the Reactor Operator Fundamentals Manuals were determined from several sources. DOE Category A reactor training managers determined which materials should be included, and served as a primary reference in the initial development phase. Training guidelines from the commercial nuclear power industry, results of job and task analyses, and independent input from contractors and operations-oriented personnel were all considered and included to some degree in developing the text material and learning objectives.

The DOE Fundamentals Handbooks represent the needs of various DOE nuclear facilities' fundamental training requirements. To increase their applicability to nonreactor nuclear facilities, the Reactor Operator Fundamentals Manual learning objectives were distributed to the Nuclear Facility Training Coordination Program Steering Committee for review and comment. To update their reactor-specific content, DOE Category A reactor training managers also reviewed and commented on the content. On the basis of feedback from these sources, information that applied to two or more DOE nuclear facilities was considered generic and was included. The final draft of each of the handbooks was then reviewed by these two groups. This approach has resulted in revised modular handbooks that contain sufficient detail such that each facility may adjust the content to fit their specific needs.

Each handbook contains an abstract, a foreword, an overview, learning objectives, and text material, and is divided into modules so that content and order may be modified by individual DOE contractors to suit their specific training needs. Each handbook is supported by a separate examination bank with an answer key.

The DOE Fundamentals Handbooks have been prepared for the Assistant Secretary for Nuclear Energy, Office of Nuclear Safety Policy and Standards, by the DOE Training Coordination Program. This program is managed by EG&G Idaho, Inc.

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NUCLEAR PHYSICS AND REACTOR THEORY

OVERVIEW

The Department of Energy Fundamentals Handbook entitled Nuclear Physics and Reactor Theory was prepared as an information resource for personnel who are responsible for the operation of the Department's nuclear facilities. Almost all processes that take place in a nuclear facility involves the transfer of some type of energy. A basic understanding of nuclear physics and reactor theory is necessary for DOE nuclear facility operators, maintenance personnel, and the technical staff to safely operate and maintain the facility and facility support systems. The information in this handbook is presented to provide a foundation for applying engineering concepts to the job. This knowledge will help personnel understand the impact that their actions may have on the safe and reliable operation of facility components and systems.

The Nuclear Physics and Reactor Theory handbook consists of four modules that are contained in two volumes. The following is a brief description of the information presented in each module of the handbook.

Volume 1 of 2

Module 1 - Atomic and Nuclear Physics

Introduces concepts of atomic physics including the atomic nature of matter, the chart of the nuclides, radioactivity and radioactive decay, neutron interactions and fission, and the interaction of radiation with matter.

Module 2 - Reactor Theory (Nuclear Parameters)

Provides information on reactor theory and neutron characteristics. Includes topics such as neutron sources, neutron flux, neutron cross sections, reaction rates,

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NUCLEAR PHYSICS AND REACTOR THEORY
OVERVIEW (Cont.)
Volume 2 of 2
Module 3 - Reactor Theory (Nuclear Parameters)
Explains the nuclear parameters associated with reactor theory. Topics include the neutron life cycle, reactivity and reactivity coefficients, neutron poisons, and control rods.
Module 4 - Reactor Theory (Reactor Operations)
Introduces the reactor operations aspect of reactor theory. Topics include subcritical multiplication, reactor kinetics, and reactor operation.
The information contained in this handbook is not all-encompassing. An attempt to present the entire subject of nuclear physics and reactor theory would be impractical. However, the Nuclear Physics and Reactor Theory handbook presents enough information to provide the reader with the fundamental knowledge necessary to understand the advanced theoretical concepts presented in other subject areas, and to understand basic system and equipment operation.
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Department of Energy
Fundamentals Handbook
NUCLEAR PHYSICS
AND REACTOR THEORY
Module 1
Atomic and Nuclear Physics
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TERMINAL OBJECTIVE
1.0 Given sufficient information, DESCRIBE atoms, including components, structure, and nomenclature.
ENABLING OBJECTIVES
1.1 STATE the characteristics of the following atomic particles, including mass, charge, and location within the atom:
a. Proton
b. Neutron
c. Electron
1.2 DESCRIBE the Bohr model of an atom.
1.3 DEFINE the following terms:
a. Nuclide c. Atomic number
b. Isotope d. Mass number
1.4 Given the standard A
ZX notation for a particular nuclide, DETERMINE the following:
a. Number of protons
b. Number of neutrons
c. Number of electrons
1.5 DESCRIBE the three forces that act on particles within the nucleus and affect the stability of the nucleus.

a. Enriched uranium
b. Depleted uranium
1.7 DEFINE the following terms:
a. Mass defect
b. Binding energy
1.8 Given the atomic mass for a nuclide and the atomic masses of a neutron, proton, and electron, CALCULATE the mass defect and binding energy of the nuclide.

TERMINAL OBJECTIVE
2.0 Given necessary references, DESCRIBE the various modes of radioactive decay.
ENABLING OBJECTIVES
2.1 DESCRIBE the following processes:
a. Alpha decay d. Electron capture
b. Beta-minus decay e. Internal conversions
c. Beta-plus decay f. Isomeric transitions
2.2 Given a Chart of the Nuclides, WRITE the radioactive decay chain for a nuclide.
2.3 EXPLAIN why one or more gamma rays typically accompany particle emission.
2.4 Given the stability curve on the Chart of the Nuclides, DETERMINE the type of radioactive decay that the nuclides in each region of the chart will typically undergo.
2.5 DEFINE the following terms:
a. Radioactivity d. Radioactive decay constant
b. Curie e. Radioactive half-life
c. Becquerel
2.6 Given the number of atoms and either the half-life or decay constant of a nuclide, CALCULATE the activity.
2.7 Given the initial activity and the decay constant of a nuclide, CALCULATE the activity at any later time.
2.8 CONVERT between the half-life and decay constant for a nuclide.
2.9 Given the Chart of the Nuclides and the original activity, PLOT the radioactive decay
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curve for a nuclide on either linear or semi-log coordinates.
2.10 DEFINE the following terms:
a. Radioactive equilibrium
b. Transient radioactive equilibrium

TERMINAL OBJECTIVE
3.0 Without references, DESCRIBE the different nuclear interactions initiated by neutrons.
ENABLING OBJECTIVES
3.1 DESCRIBE the following scattering interactions between a neutron and a nucleus:
a. Elastic scattering
b. Inelastic scattering
3.2 STATE the conservation laws that apply to an elastic collision between a neutron and a nucleus.
3.3 DESCRIBE the following reactions where a neutron is absorbed in a nucleus:
a. Radiative capture
b. Particle ejection

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OBJECTIVES DOE-HDBK-1019/1-93 Atomic and Nuclear Physics
TERMINAL OBJECTIVE
4.0 Without references, DESCRIBE the fission process.
ENABLING OBJECTIVES
4.1 EXPLAIN the fission process using the liquid drop model of a nucleus.
4.2 DEFINE the following terms:
a. Excitation energy
b. Critical energy
4.3 DEFINE the following terms:
a. Fissile material
b. Fissionable material
c. Fertile material
4.4 DESCRIBE the processes of transmutation, conversion, and breeding.
4.5 DESCRIBE the curve of Binding Energy per Nucleon versus mass number and give a qualitative description of the reasons for its shape.
4.6 EXPLAIN why only the heaviest nuclei are easily fissioned.
4.7 EXPLAIN why uranium-235 fissions with thermal neutrons and uranium-238 fissions only with fast neutrons.
4.8 CHARACTERIZE the fission products in terms of mass groupings and radioactivity.
4.9 Given the nuclides involved and their masses, CALCULATE the energy released from fission.
4.10 Given the curve of Binding Energy per Nucleon versus mass number, CALCULATE the energy released from fission.
NP-01 Page x Rev. 0
Atomic and Nuclear Physics DOE-HDBK-1019/1-93 OBJECTIVES
TERMINAL OBJECTIVE
5.0 Without references, DESCRIBE how the various types of radiation interact with matter.
ENABLING OBJECTIVES
5.1 DESCRIBE interactions of the following with matter:
a. Alpha particle c. Positron
b. Beta particle d. Neutron
5.2 DESCRIBE the following ways that gamma radiation interacts with matter:
a. Compton scattering
b. Photoelectric effect
c. Pair production
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NP-01 Page xii Rev. 0
Atomic and Nuclear Physics DOE-HDBK-1019/1-93 ATOMIC NATURE OF MATTER
ATOMIC NATURE OF MATTER
All matter is composed of atoms. The atom is the smallest amount of matter that retains the properties of an element. Atoms themselves are composed of smaller particles, but these smaller particles no longer have the same properties as the overall element.
EO 1.1 STATE the characteristics of the following atomic particles, including mass, charge, and location within the atom:

b. Neutron
c. Electron
EO 1.2 DESCRIBE the Bohr model of an atom.
EO 1.3 DEFINE the following terms:
a. Nuclide c. Atomic number
b. Isotope d. Mass number
EO 1.4 Given the standard A
ZX notation for a particular nuclide,
DETERMINE the following:
a. Number of protons
b. Number of neutrons
c. Number of electrons
EO 1.5 DESCRIBE the three forces that act on particles within the nucleus
and affect the stability of the nucleus.
Fixture checkpoint marker-0983-xray is recorded on fixture page 0983.
Structure of Matter
Early Greek philosophers speculated that the earth was made up of different combinations of basic substances, or elements. They considered these basic elements to be earth, air, water, and fire. Modern science shows that the early Greeks held the correct concept that matter consists of a combination of basic elements, but they incorrectly identified the elements.

In 1661 the English chemist Robert Boyle published the modern criterion for an element. He defined an element to be a basic substance that cannot be broken down into any simpler substance after it is isolated from a compound, but can be combined with other elements to form compounds. To date, 105 different elements have been confirmed to exist, and researchers claim to have discovered three additional elements. Of the 105 confirmed elements, 90 exist in nature and 15 are man-made.
Another basic concept of matter that the Greeks debated was whether matter was continuous or discrete. That is, whether matter could be continuously divided and subdivided into ever smaller particles or whether eventually an indivisible particle would be encountered. Democritus in about 450 B.C. argued that substances were ultimately composed of small, indivisible particles that he labeled atoms. He further suggested that different substances were composed of different atoms or combinations of atoms, and that one substance could be converted into another by rearranging the atoms. It was impossible to conclusively prove or disprove this proposal for more than 2000 years.
The modern proof for the atomic nature of matter was first proposed by the English chemist John Dalton in 1803. Dalton stated that each chemical element possesses a particular kind of atom, and any quantity of the element is made up of identical atoms of this kind. What distinguishes one element from another element is the kind of atom of which it consists, and the basic physical difference between kinds of atoms is their weight.
Subatomic Particles
For almost 100 years after Dalton established the atomic nature of atoms, it was considered impossible to divide the atom into even smaller parts. All of the results of chemical experiments during this time indicated that the atom was indivisible. Eventually, experimentation into electricity and radioactivity indicated that particles of matter smaller than the atom did indeed exist. In 1906, J. J. Thompson won the Nobel Prize in physics for establishing the existence of electrons. Electrons are negatively-charged particles that have 1/1835 the mass of the hydrogen

large particles that have almost the same mass as a hydrogen atom and a positive charge equal in magnitude (but opposite in sign) to that of the electron. The third subatomic particle to be discovered, the neutron, was not found until 1932. The neutron has almost the same mass as the proton, but it is electrically neutral.

Bohr Model of the Atom

The British physicist Ernest Rutherford postulated that the positive charge in an atom is concentrated in a small region called a nucleus at the center of the atom with electrons existing in orbits around it. Niels Bohr, coupling Rutherford's postulation with the quantum theory introduced by Max Planck, proposed that the atom consists of a dense nucleus of protons surrounded by electrons traveling in discrete orbits at fixed distances from the nucleus. An electron in one of these orbits or shells has a specific or discrete quantity of energy (quantum). When an electron moves from one allowed orbit to another allowed orbit, the energy difference between the two states is emitted or absorbed in the form of a single quantum of radiant energy called a photon. Figure 1 is Bohr's model of the hydrogen atom showing an electron as having just dropped from the third shell to the first shell with the emission of a photon that has an energy = hv. (h = Planck's constant = 6.63 x 10-34 J-s and v = frequency of the photon.) Bohr's theory was the first to successfully account for the discrete energy levels of this radiation as measured in the laboratory. Although Bohr's atomic model is designed specifically to explain the hydrogen atom, his theories apply generally to the structure of all atoms. Additional

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information on electron shell theory can be found in the Chemistry Fundamentals Handbook.

Figure 1 Bohr's Model of the Hydrogen Atom

Properties of the three subatomic particles are listed in Table 1.

TABLE 1
Properties of Subatomic Particles

Particle	Location	Charge	Mass
Neutron	Nucleus	none	1.008665 amu
Proton	Nucleus	+1	1.007277 amu
Electron	Shells around nucleus	-1	0.0005486 amu

Measuring Units on the Atomic Scale

The size and mass of atoms are so small that the use of normal measuring units, while possible, is often inconvenient. Units of measure have been defined for mass and energy on the atomic scale to make measurements more convenient to express. The unit of measure for mass is the atomic mass unit (amu). One atomic mass unit is equal to 1.66 x 10-24 grams. The reason for this particular value for the atomic mass unit will be discussed in a later chapter. Note from Table 1 that the mass of a neutron and a proton are both about 1 amu. The unit for energy is the electron volt (eV). The electron volt is the amount of energy acquired by a single electron when it falls through a potential difference of one volt. One electron volt is equivalent to 1.602 x 10-19 joules or 1.18 x 10-19 foot-pounds.

Nuclides

The total number of protons in the nucleus of an atom is called the atomic number of the atom and is given the symbol Z. The number of electrons in an electrically-neutral atom is the same as the number of protons in the nucleus. The number of neutrons in a nucleus is known as the neutron number and is given the symbol N. The mass number of the nucleus is the total number

SYNTHETIC FIXTURE PAGE 0985 OF 1200

A and can be found by the equation Z + N = A.
Each of the chemical elements has a unique atomic number because the atoms of different elements contain a different number of protons. The atomic number of an atom identifies the particular element.
NP-01 Page 4 Rev. 0
Atomic and Nuclear Physics DOE-HDBK-1019/1-93 ATOMIC NATURE OF MATTER
Each type of atom that contains a unique combination of
Figure 2 Nomenclature for
Identifying Nuclides
protons and neutrons is called a nuclide. Not all combinations of numbers of protons and neutrons are possible, but about 2500 specific nuclides with unique combinations of neutrons and protons have been identified. Each nuclide is denoted by the chemical symbol of the element with the atomic number written as a subscript and the mass number written as a superscript, as shown in Figure 2. Because each element has a unique name, chemical symbol, and atomic number, only one of the three is necessary to identify the element. For this reason nuclides can also be identified by either the chemical name or the chemical symbol followed by the mass number (for example, U-235 or uranium-235).
Another common format is to use the abbreviation of the chemical element with the mass number superscripted (for example, 235U). In this handbook the format used in the text will usually be the element's name followed by the mass number. In equations and tables, the format in Figure 2 will usually be used.
Example:
State the name of the element and the number of protons, electrons, and neutrons in the nuclides listed below.
Fixture checkpoint marker-0985-xray is recorded on fixture page 0985.
1
1H
10
5B
14
7N
11
4
4
8Cd
23
9
9
4Pu
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ATOMIC NATURE OF MATTER DOE-HDBK-1019/1-93 Atomic and Nuclear Physics
Solution:
The name of the element can be found from the Periodic Table (refer to Chemistry

number of protons and electrons are equal to Z. The number of neutrons is equal to Z - A.

Nuclide Element Protons Electrons Neutrons
1
1H hydrogen 1 1 0
10
5B boron 5 5 5
14
7N nitrogen 7 7 7
11
4
4
8Cd cadmium 48 48 66
23
9
9
4Pu plutonium 94 94 145

Isotopes

Isotopes are nuclides that have the same atomic number and are therefore the same element, but differ in the number of neutrons. Most elements have a few stable isotopes and several unstable, radioactive isotopes. For example, oxygen has three stable isotopes that can be found in nature (oxygen-16, oxygen-17, and oxygen-18) and eight radioactive isotopes. Another example is hydrogen, which has two stable isotopes (hydrogen-1 and hydrogen-2) and a single radioactive isotope (hydrogen-3).

The isotopes of hydrogen are unique in that they are each commonly referred to by a unique name instead of the common chemical element name. Hydrogen-1 is almost always referred to as hydrogen, but the term protium is infrequently used also. Hydrogen-2 is commonly called deuterium and symbolized 2
1D. Hydrogen-3 is commonly called tritium and symbolized 31T. This text will normally use the symbology 2
1H and 31H for deuterium and tritium, respectively.

Atomic and Nuclear Radii

The size of an atom is difficult to define exactly due to the fact that the electron cloud, formed by the electrons moving in their various orbitals, does not have a distinct outer edge. A reasonable measure of atomic size is given by the average distance of the outermost electron

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from the nucleus. Except for a few of the lightest atoms, the average atomic radii are approximately the same for all atoms, about 2 x 10 -8 cm.

Like the atom the nucleus does not have a sharp outer boundary. Experiments have shown that the nucleus is shaped like a sphere with a radius that depends on the atomic mass number of the atom. The relationship between the atomic mass number and the radius of the nucleus is shown in the following equation.

r = (1.25 x 10 -13 cm) A1/3

where:

r = radius of the nucleus (cm)
A = atomic mass number (dimensionless)

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Table 2.
TABLE 2
Calculated Values for Nuclear
Radii
Nuclide Radius of Nucleus
1
1H 1.25 x 10 -13 cm
10
5B 2.69 x 10 -13 cm
5
2
6
Fixture checkpoint marker-0987-xray is recorded on fixture page 0987.
6Fe 4.78 x 10 -13 cm
17
7
8
2Hf 7.01 x 10 -13 cm
23
9
8
2U 7.74 x 10 -13 cm
25
9
2
8Cf 7.89 x 10 -13 cm
From the table, it is clear that the radius of a typical atom (e.g. 2 x 10 -8 cm) is more than 25,000 times larger than the radius of the largest nucleus.
Nuclear Forces
In the Bohr model of the atom, the nucleus consists of positively-charged protons and electrically■neutra as nucleons. One problem that the Bohr model of the atom presented was accounting for an attractive force to overcome the repulsive force between protons.
Two forces present in the nucleus are (1) electrostatic forces between charged particles and (2) gravitational forces between any two objects that have mass. It is possible to calculate the magnitude of the gravitational force and electrostatic force based upon principles from classical physics.
Rev. 0 Page 7 NP-01
Fg
G m1 m2
r
2
Fe
K Q1 Q2
r
2
ATOMIC NATURE OF MATTER DOE-HDBK-1019/1-93 Atomic and Nuclear Physics
NP-01 Page 8 Rev. 0
Newton stated that the gravitational force between two bodies is directly proportional to the

bodies. This relationship is shown in the equation below.
where:
F = gravitational force (newtons) g
m = mass of first body (kilograms) 1
m = mass of second body (kilograms) 2
G = gravitational constant (6.67 x 10 N-m /kg)
-11 2 2
r = distance between particles (meters)
The equation illustrates that the larger the masses of the objects or the smaller the distance between the objects, the greater the gravitational force. So even though the masses of nucleons are very small, the fact that the distance between nucleons is extremely short may make the gravitational force significant. It is necessary to calculate the value for the gravitational force and compare it to the value for other forces to determine the significance of the gravitational force in the nucleus. The gravitational force between two protons that are separated by a distance of 10 meters is about 10 newtons. -20 -24
Coulomb's Law can be used to calculate the force between two protons. The electrostatic force is directly proportional to the electrical charges of the two particles and inversely proportional to the square of the distance between the particles. Coulomb's Law is stated as the following equation.
Fixture checkpoint marker-0988-xray is recorded on fixture page 0988.
where:
F = electrostatic force (newtons) e
K = electrostatic constant (9.0 x 10 N-m /C)
9 2 2
Q = charge of first particle (coulombs) 1
Q = charge of second particle (coulombs) 2
r = distance between particles (meters)
Using this equation, the electrostatic force between two protons that are separated by a distance of 10 meters is about 10 newtons. Comparing this result with the calculation of the
-20 12
gravitational force (10 newtons) shows that the gravitational force is so small that it can be
-24
neglected.
Atomic and Nuclear Physics DOE-HDBK-1019/1-93 ATOMIC NATURE OF MATTER
If only the electrostatic and gravitational forces existed in the nucleus, then it would be impossible to have stable nuclei composed of protons and neutrons. The gravitational forces are much too small to hold the nucleons together compared to the electrostatic forces repelling the protons. Since stable atoms of neutrons and protons do exist, there must be another attractive force acting within the nucleus. This force is called the nuclear force.
The nuclear force is a strong attractive force that is independent of charge. It acts equally only between pairs of neutrons, pairs of protons, or a neutron and a proton. The nuclear force has a very short range; it acts only over distances approximately equal to the diameter of the nucleus (10 -13 cm). The attractive nuclear force between all nucleons drops off with distance much faster than the repulsive electrostatic force between protons.
TABLE 3
Forces Acting in the Nucleus
Force Interaction Range
Gravitational Very weak attractive force

Electrostatic Strong repulsive force between
like charged particles (protons) Relatively long
Nuclear Force Strong attractive force between
all nucleons Extremely short
In stable atoms, the attractive and repulsive forces in the nucleus balance. If the forces do not balance, the atom cannot be stable, and the nucleus will emit radiation in an attempt to achieve a more stable configuration.
Summary
The important information in this chapter is summarized on the following page.

Atomic Nature of Matter Summary
Atoms consist of three basic subatomic particles. These particles are the proton, the neutron, and the electron.
Protons are particles that have a positive charge, have about the same mass as a hydrogen atom, and exist in the nucleus of an atom.
Neutrons are particles that have no electrical charge, have about the same mass as a hydrogen atom, and exist in the nucleus of an atom.
Electrons are particles that have a negative charge, have a mass about eighteen hundred times smaller than the mass of a hydrogen atom, and exist in orbital shells around the nucleus of an atom.
The Bohr model of the atom consists of a dense nucleus of protons and neutrons (nucleons) surrounded by electrons traveling in discrete orbits at fixed distances from the nucleus.
Nuclides are atoms that contain a particular number of protons and neutrons.
Isotopes are nuclides that have the same atomic number and are therefore the same
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element, but differ in the number of neutrons.
The atomic number of an atom is the number of protons in the nucleus.
The mass number of an atom is the total number of nucleons (protons and neutrons) in the nucleus.
The notation A
ZX is used to identify a specific nuclide. "Z" represents the atomic number, which is equal to the number of protons. "A" represents the mass number, which is equal to the number of nucleons. "X" represents the chemical symbol of the element.
Number of protons = Z
Number of electrons = Z
Number of neutrons = A - Z
The stability of a nucleus is determined by the different forces interacting within it. The electrostatic force is a relatively long-range, strong, repulsive force that acts between the positively charged protons. The nuclear force is a relatively short-range attractive force between all nucleons. The gravitational force the long range, relatively weak attraction between masses, is negligible compared to the other forces.

CHART OF THE NUCLIDES

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for presenting a large amount of scientific information in an organized manner.

EO 1.6 DEFINE the following terms:

a. Enriched uranium

b. Depleted uranium

Chart of the Nuclides

A tabulated chart called the Chart of the Nuclides lists the stable and unstable nuclides in addition to pertinent information about each one. Figure 3 shows a small portion of a typical chart. This chart plots a box for each individual nuclide, with the number of protons (Z) on the vertical axis and the number of neutrons (N = A - Z) on the horizontal axis.

The completely gray squares indicate stable isotopes. Those in white squares are artificially radioactive, meaning that they are produced by artificial techniques and do not occur naturally. By consulting a complete chart, other types of isotopes can be found, such as naturally occurring radioactive types (but none are found in the region of the chart that is illustrated in Figure 3).

Located in the box on the far left of each horizontal row is general information about the element. The box contains the chemical symbol of the element in addition to the average atomic weight of the naturally occurring substance and the average thermal neutron absorption cross section, which will be discussed in a later module. The known isotopes (elements with the same atomic number Z but different mass number A) of each element are listed to the right.

Figure 3 Nuclide Chart for Atomic Numbers 1 to 6

Information for Stable Nuclides

For the stable isotopes, in addition to the symbol and the atomic mass number, the number percentage of each isotope in the naturally occurring element is listed, as well as the thermal neutron activation cross section and the mass in atomic mass units (amu). A typical block for a stable nuclide from the Chart of the Nuclides is shown in Figure 4.

Figure 4 Stable Nuclides

Information for Unstable Nuclides

For unstable isotopes the additional information includes the half life, the mode of decay (for example, β-

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, α), the total disintegration energy in MeV (million electron volts), and the mass in amu when available. A typical block for an unstable nuclide from the Chart of the Nuclides is shown in Figure 5.

Figure 5 Unstable Nuclides

Neutron - Proton Ratios

Figure 6 shows the distribution of the stable nuclides plotted on the same axes as the Chart of the Nuclides. As the mass numbers become higher, the ratio of neutrons to protons in the nucleus becomes larger. For helium-4 (2 protons and 2 neutrons) and oxygen-16 (8 protons and 8 neutrons) this ratio is unity. For indium-115 (49 protons and 66 neutrons) the ratio of neutrons to protons has increased to 1.35, and for uranium-238 (92 protons and 146 neutrons) the neutron■to-proton

Figure 6 Neutron - Proton Plot of the Stable Nuclides

If a heavy nucleus were to split into two fragments, each fragment would form a nucleus that

neutron-to-proton ratio places the fragments below and to the right of the stability curve displayed by Figure 6. The instability caused by this excess of neutrons is generally rectified by successive beta emissions, each of which converts a neutron to a proton and moves the nucleus toward a more stable neutron-to-proton ratio.

■

Natural Abundance of Isotopes

The relative abundance of an isotope in nature compared to other isotopes of the same element is relatively constant. The Chart of the Nuclides presents the relative abundance of the naturally occurring isotopes of an element in units of atom percent. Atom percent is the percentage of the atoms of an element that are of a particular isotope. Atom percent is abbreviated as a/o.

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For example, if a cup of water contains 8.23×10^{24} atoms of oxygen, and the isotopic abundance of oxygen-18 is 0.20%, then there are 1.65×10^{22} atoms of oxygen-18 in the cup.

The atomic weight for an element is defined as the average atomic weight of the isotopes of the element. The atomic weight for an element can be calculated by summing the products of the isotopic abundance of the isotope with the atomic mass of the isotope.

Example:

Calculate the atomic weight for the element lithium. Lithium-6 has an atom percent abundance of 7.5% and an atomic mass of 6.015122 amu. Lithium-7 has an atomic abundance of 92.5% and an atomic mass of 7.016003 amu.

Solution:

The other common measurement of isotopic abundance is weight percent (w/o). Weight percent is the percent weight of an element that is a particular isotope. For example, if a sample of material contained 100 kg of uranium that was 28 w/o uranium-235, then 28 kg of uranium-235 was present in the sample.

Enriched and Depleted Uranium

Natural uranium mined from the earth contains the isotopes uranium-238, uranium-235 and uranium-234. The majority (99.2745%) of all the atoms in natural uranium are uranium-238. Most of the remaining atoms (0.72%) are uranium-235, and a slight trace (0.0055%) are uranium-234. Although all isotopes of uranium have similar chemical properties, each of the isotopes has significantly different nuclear properties. For reasons that will be discussed in later modules, the isotope uranium-235 is usually the desired material for use in reactors.

A vast amount of equipment and energy are expended in processes that separate the isotopes of uranium (and other elements). The details of these processes are beyond the scope of this module. These processes are called enrichment processes because they selectively increase the proportion of a particular isotope. The enrichment process typically starts with feed material that has the proportion of isotopes that occur naturally. The process results in two types of

In the case of uranium, the natural uranium ore is 0.72 a/o uranium-235. The desired outcome of the enrichment process is to produce enriched uranium. Enriched uranium is defined as uranium in which the isotope uranium-235 has a concentration greater than its natural value. The enrichment process will also result in the byproduct of depleted uranium. Depleted uranium is defined as uranium in which the isotope uranium-235 has a concentration less than its natural value. Although depleted uranium is referred to as a by-product of the enrichment process, it does have uses in the nuclear field and in commercial and defense industries.

The important information in this chapter is summarized below.
Chart of the Nuclides Summary
Enriched uranium is uranium in which the isotope uranium-235 has a concentration greater than its natural value of 0.7%.
Depleted uranium is uranium in which the isotope uranium-235 has a concentration less than its natural value of 0.7%.
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Atomic and Nuclear Physics DOE-HDBK-1019/1-93 MASS DEFECT AND BINDING ENERGY
MASS DEFECT AND BINDING ENERGY
The separate laws of Conservation of Mass and Conservation of Energy are not applied strictly on the nuclear level. It is possible to convert between mass and energy. Instead of two separate conservation laws, a single conservation law states that the sum of mass and energy is conserved. Mass does not magically appear and disappear at random. A decrease in mass will be accompanied by a corresponding increase in energy and vice versa.
EO 1.7 DEFINE the following terms:
a. Mass defect
Fixture checkpoint marker-0992-xray is recorded on fixture page 0992.
b. Binding energy
EO 1.8 Given the atomic mass for a nuclide and the atomic masses of a neutron, proton, and electron, CALCULATE the mass defect and binding energy of the nuclide.
Mass Defect
Careful measurements have shown that the mass of a particular atom is always slightly less than the sum of the masses of the individual neutrons, protons, and electrons of which the atom consists. The difference between the mass of the atom and the sum of the masses of its parts is called the mass defect (Δm). The mass defect can be calculated using Equation (1-1). In calculating the mass defect it is important to use the full accuracy of mass measurements because the difference in mass is small compared to the mass of the atom. Rounding off the masses of atoms and particles to three or four significant digits prior to the calculation will result in a calculated mass defect of zero.
$\Delta m = [Z(mp + me) + (A-Z)mn] - matom$ (1-1)
where:
Δm = mass defect (amu)
mp = mass of a proton (1.007277 amu)
mn = mass of a neutron (1.008665 amu)
me = mass of an electron (0.000548597 amu)
matom = mass of nuclide A
ZX (amu)
Z = atomic number (number of protons)
A = mass number (number of nucleons)
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MASS DEFECT AND BINDING ENERGY DOE-HDBK-1019/1-93 Atomic and Nuclear Physics
Example:
Calculate the mass defect for lithium-7. The mass of lithium-7 is 7.016003 amu.
Solution:
Δ m Z m
p

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∆ m 3 1.007826 amu 7 3 1.008665 amu 7.016003 amu
∆ m 0.0421335 amu
Binding Energy
The loss in mass, or mass defect, is due to the conversion of mass to binding energy when the nucleus is formed. Binding energy is defined as the amount of energy that must be supplied to a nucleus to completely separate its nuclear particles (nucleons). It can also be understood as the amount of energy that would be released if the nucleus was formed from the separate particles. Binding energy is the energy equivalent of the mass defect. Since the mass defect was converted to binding energy (BE) when the nucleus was formed, it is possible to calculate the binding energy using a conversion factor derived by the mass-energy relationship from Einstein's Theory of Relativity.
Einstein's famous equation relating mass and energy is E = mc2 where c is the velocity of light (c = 2.998 x 108 m/sec). The energy equivalent of 1 amu can be determined by inserting this quantity of mass into Einstein's equation and applying conversion factors.
E m c 2
1 amu
1.6606 x 10 27 kg
1 amu
2.998 x 108 m
sec
2
1 N
1
kg m
Fixture checkpoint marker-0993-xray is recorded on fixture page 0993.
sec2
1 J
1 N m
1.4924 x 10 10 J 1 MeV
1.6022 x 10 13 J
931.5 MeV
Conversion Factors:
1 amu = 1.6606 x 10 -27 kg
1 newton = 1 kg-m/sec2
1 joule = 1 newton-meter
1 MeV = 1.6022 x 10-13 joules
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Atomic and Nuclear Physics DOE-HDBK-1019/1-93 MASS DEFECT AND BINDING ENERGY
Since 1 amu is equivalent to 931.5 MeV of energy, the binding energy can be calculated using Equation (1-2).
(1-2)
B.E. ∆m
931.5 MeV
1 amu
Example:
Calculate the mass defect and binding energy for uranium-235. One uranium-235 atom has a mass of 235.043924 amu.
Solution:

Δm Z m
p
me A Z mn matom
Δm 92 1.007826 amu 235 92 1.008665 amu 235.043924 amu
Δm 1.91517 amu
Step 2: Use the mass defect and Equation (1-2) to calculate the binding energy.
B.E. Δm
931.5 MeV
1 amu
1.91517 amu
931.5 MeV
1 amu
1784 MeV
Energy Levels of Atoms
The electrons that circle the nucleus move in fairly well-defined orbits. Some of these electrons are more tightly bound in the atom than others. For example, only 7.38 eV is required to remove the outermost electron from a lead atom, while 88,000 eV is required to remove the innermost electron. The process of removing an electron from an atom is called ionization, and the energy required to remove the electron is called the ionization energy.
In a neutral atom (number of electrons = Z) it is possible for the electrons to be in a variety of different orbits, each with a different energy level. The state of lowest energy is the one in which the atom is normally found and is called the ground state. When the atom possesses more energy than its ground state energy, it is said to be in an excited state.
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An atom cannot stay in the excited state for an indefinite period of time. An excited atom will eventually transition to either a lower-energy excited state, or directly to its ground state, by emitting a discrete bundle of electromagnetic energy called an x-ray. The energy of the x-ray will be equal to the difference between the energy levels of the atom and will typically range from several eV to 100,000 eV in magnitude.
Energy Levels of the Nucleus
Fixture checkpoint marker-0994-xray is recorded on fixture page 0994.
The nucleons in the nucleus of an atom, like the electrons that circle the nucleus, exist in shells that correspond to energy states. The energy shells of the nucleus are less defined and less understood than those of the electrons. There is a state of lowest energy (the ground state) and discrete possible excited states for a nucleus. Where the discrete energy states for the electrons of an atom are measured in eV or keV, the energy levels of the nucleus are considerably greater and typically measured in MeV.
A nucleus that is in the excited state will not remain at that energy level for an indefinite period. Like the electrons in an excited atom, the nucleons in an excited nucleus will transition towards their lowest energy configuration and in doing so emit a discrete bundle of electromagnetic radiation called a gamma ray (γ-ray). The only differences between x-rays and γ-rays are their energy levels and whether they are emitted from the electron shell or from the nucleus.
The ground state and the excited states of
Figure 7 Energy Level Diagram - Nickel-60
a nucleus can be depicted in a nuclear
energy-level diagram. The nuclear
energy-level diagram consists of a stack of

excited states of the nucleus. The vertical distance between the bar representing an excited state and the bar representing the ground state is proportional to the energy level of the excited state with respect to the ground state. This difference in energy between the ground state and the excited state is called the excitation energy of the excited state. The ground state of a nuclide has zero excitation energy. The bars for the excited states are labeled with their respective energy levels. Figure 7 is the energy level diagram for nickel-60.

Summary

The important information in this chapter is summarized below.

Mass Defect and Binding Energy Summary

Mass defect is the difference between the mass of the atom and the sum of the masses of its constituent parts.

Binding energy is the amount of energy that must be supplied to a nucleus to completely separate its nuclear particles. Binding energy is the energy equivalent of the mass defect.

Mass defect can be calculated by using the equation below.

$\Delta m = [Z(mp + me) + (A-Z)mn] - matom$

Binding energy can be calculated by multiplying the mass defect by the factor of 931.5 MeV per amu.

MODES OF RADIOACTIVE DECAY

Most atoms found in nature are stable and do not emit particles or energy that change form over time. Some atoms, however, do not have stable nuclei. These atoms emit radiation in order to achieve a more stable configuration.

EO 2.1 DESCRIBE the following processes:

a. Alpha decay
b. Beta-minus decay
c. Beta-plus decay
d. Electron capture
e. Internal conversions
f. Isomeric transitions

EO 2.2 Given a Chart of the Nuclides, WRITE the radioactive decay chain for a nuclide.

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EO 2.3 EXPLAIN why one or more gamma rays typically accompany particle emission.

EO 2.4 Given the stability curve on the Chart of the Nuclides, DETERMINE the type of radioactive decay that the nuclides in each region of the chart will typically undergo.

Stability of Nuclei

As mass numbers become larger, the ratio of neutrons to protons in the nucleus becomes larger for the stable nuclei. Non-stable nuclei may have an excess or deficiency of neutrons and

a variety of other processes such as alpha (α) or neutron (n) decay. As a result of these decay processes, the final nucleus is in a more stable or more tightly bound configuration.

Natural Radioactivity

In 1896, the French physicist Becquerel discovered that crystals of a uranium salt emitted rays that were similar to x-rays in that they were highly penetrating, could affect a photographic plate, and induced electrical conductivity in gases. Becquerel's discovery was followed in 1898 by the identification of two other radioactive elements, polonium and radium, by Pierre and Marie Curie.

Heavy elements, such as uranium or thorium, and their unstable decay chain elements emit radiation in their naturally occurring state. Uranium and thorium, present since their creation at the beginning of geological time, have an extremely slow rate of decay. All naturally occurring nuclides with atomic numbers greater than 82 are radioactive.

Nuclear Decay

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Whenever a nucleus can attain a more stable (i.e., more tightly bound) configuration by emitting radiation, a spontaneous disintegration process known as radioactive decay or nuclear decay may occur. In practice, this "radiation" may be electromagnetic radiation, particles, or both.

Detailed studies of radioactive decay and nuclear reaction processes have led to the formulation of useful conservation principles. The four principles of most interest in this module are discussed below.

Conservation of electric charge implies that charges are neither created nor destroyed. Single positive and negative charges may, however, neutralize each other. It is also possible for a neutral particle to produce one charge of each sign.

Conservation of mass number does not allow a net change in the number of nucleons. However, the conversion of a proton to a neutron and vice versa is allowed.

Conservation of mass and energy implies that the total of the kinetic energy and the energy equivalent of the mass in a system must be conserved in all decays and reactions. Mass can be converted to energy and energy can be converted to mass, but the sum of mass and energy must be constant.

Conservation of momentum is responsible for the distribution of the available kinetic energy among product nuclei, particles, and/or radiation. The total amount is the same before and after the reaction even though it may be distributed differently among entirely different nuclides and/or particles.

■ ■ ■ ■ ■ ■

■ ■ ■ ■ ■

Alpha Decay (■)

Alpha decay is the emission of alpha particles (helium nuclei) which may be represented as either $^{4}_{2}He$ or $^{4}_{2}■$. When an unstable nucleus ejects an alpha particle, the atomic number is reduced by 2 and the mass number decreased by 4. An example is uranium-234 which decays by the ejection of an alpha particle accompanied by the emission of a 0.068 MeV gamma.

The combined kinetic energy of the daughter nucleus (Thorium-230) and the ■ particle is designated as KE. The sum of the KE and the gamma energy is equal to the difference in mass between the

since ■m = BE). The alpha particle will carry off as much as 98% of the kinetic energy and, in most cases, can be considered to carry off all the kinetic energy.

Beta Decay (■)

Beta decay is the emission of electrons of nuclear rather than orbital origin. These particles are electrons that have been expelled by excited nuclei and may have a charge of either sign.

If both energy and momentum are to be conserved, a third type of particle, the neutrino, ■, must be involved. The neutrino is associated with positive electron emission, and its antiparticle, the antineutrino, , is emitted with a negative electron. These uncharged particles have only the weakest interaction with matter, no mass, and travel at the speed of light. For all practical purposes, they pass through all materials with so few interactions that the energy they possess cannot be recovered. The neutrinos and antineutrinos are included here only because they carry a portion of the kinetic energy that would otherwise belong to the beta particle, and therefore, must be considered for energy and momentum to be conserved. They are normally ignored since they are not significant in the context of nuclear reactor applications.

Negative electron emission, represented as e, -1 -1■, or simply as e or ■ , effectively converts a neutro to a proton, thus increasing the atomic number by one and leaving the mass number unchanged. This is a common mode of decay for nuclei with an excess of neutrons, such as fission fragments below and to the right of the neutron-proton stability curve (refer to Figure 6). An example of a typical beta minus-decay reaction is shown below.

■ ■

■

■ ■

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■ ■ ■ ■

Figure 9 Orbital Electron Capture

Positively charged electrons (beta-plus) are known as positrons. Except for sign, they are nearly identical to their negatively charged cousins. When a positron, represented as e, ■, or simply as +1 +1 0 0
e or ■ , is ejected from the nucleus, the atomic number is decreased by one and the mass number + + remains unchanged. A proton has been converted to a neutron. An example of a typical positron (beta-plus) decay is shown below.

Electron Capture (EC, K-capture)

Nuclei having an excess of protons may capture an electron from one of the inner orbits which immediately combines with a proton in the nucleus to form a neutron. This process is called electron capture (EC). The electron is normally captured from the innermost orbit (the K-shell), and, consequently, this process is sometimes called K-capture. The following example depicts electron capture.

A neutrino is formed at the same time that the neutron is formed, and energy carried off by it serves to conserve momentum. Any energy that is available due to the atomic mass of the product being appreciably less than that of the parent will appear as gamma radiation. Also, there will always

when an electron from one of the
higher energy shells moves in to fill
the vacancy in the K-shell.
Electron capture is shown
graphically in Figure 8.
Electron capture and positron
emission result in the production of
the same daughter product, and
they exist as competing processes.
For positron emission to occur, however, the mass of the daughter product must be less than the mass of the parent by an amount equal to at least twice the mass of an electron. This mass difference between the parent and daughter is necessary to account for two items present in the parent but not in the daughter. One item is the positron ejected from the nucleus of the parent. The other item is that the daughter product has one less orbital electron than the parent. If this requirement is not met, then orbital electron capture takes place exclusively.
■
■ ■ ■ ■
■ ■
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Gamma Emission (■)
Gamma radiation is a high-energy electromagnetic radiation that originates in the nucleus. It is emitted in the form of photons, discrete bundles of energy that have both wave and particle properties. Often a daughter nuclide is left in an excited state after a radioactive parent nucleus undergoes a transformation by alpha decay, beta decay, or electron capture. The nucleus will drop to the ground state by the emission of gamma radiation.
Internal Conversion
The usual method for an excited nucleus to go from the excited state to the ground state is by emission of gamma radiation. However, in some cases the gamma ray (photon) emerges from the
Fixture checkpoint marker-0998-xray is recorded on fixture page 0998.
nucleus only to interact with one of the innermost orbital electrons and, as a result, the energy of the photon is transferred to the electron. The gamma ray is then said to have undergone internal conversion. The conversion electron is ejected from the atom with kinetic energy equal to the gamma energy minus the binding energy of the orbital electron. An orbital electron then drops to a lower energy state to fill the vacancy, and this is accompanied by the emission of characteristic x-rays.
Isomers and Isomeric Transition
Isomeric transition commonly occurs immediately after particle emission; however, the nucleus may remain in an excited state for a measurable period of time before dropping to the ground state at its own characteristic rate. A nucleus that remains in such an excited state is known as a nuclear isomer because it differs in energy and behavior from other nuclei with the same atomic number and mass number. The decay of an excited nuclear isomer to a lower energy level is called an isomeric transition. It is also possible for the excited isomer to decay by some alternate means, for example, by beta emission.
An example of gamma emission accompanying particle emission is illustrated by the decay of nitrogen-16 below.
■
■

■
■
■
■
■
■
■
■
■

Decay Chains

When an unstable nucleus decays, the resulting daughter nucleus is not necessarily stable. The nucleus resulting from the decay of a parent is often itself unstable, and will undergo an additional decay. This is especially common among the larger nuclides.

It is possible to trace the steps of an unstable atom as it goes through multiple decays trying to achieve stability. The list of the original unstable nuclide, the nuclides that are involved as intermediate steps in the decay, and the final stable nuclide is known as the decay chain. One common method for stating the decay chain is to state each of the nuclides involved in the standard $^{A}_{Z}X$ format. Arrows are used between nuclides to indicate where decays occur, with the type of decay indicated above the arrow and the half-life below the arrow. The half-life for decay will be discussed in the next chapter.

Example:

Write the decay chains for rubidium-91 and actinium-215. Continue the chains until a stable nuclide or a nuclide with a half-life greater than 1 x 10 years is reached. 6

Solution:

Predicting Type of Decay

Radioactive nuclides tend to decay in a way that results in a daughter nuclide that lies closer to the line of stability. Due to this, it is possible to predict the type of decay that a nuclide will undergo based on its location relative to the line of stability on the Chart of the Nuclides.

Figure 9 illustrates the type of decay nuclides in different regions of the chart will typically undergo. Nuclides that are below and to the right of the line of stability will usually undergo ■ decay. - Nuclides that are above and to the left of the line of stability will usually undergo either ■ decay or + electron capture. Most nuclides that will undergo ■ decay are found in the upper right hand region of the chart. These are general rules that have many exceptions, especially in the region of the heavy

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nuclides.

Figure 9 Types of Radioactive Decay Relative to the Line of Stability

Summary

The important information in this chapter is summarized below.

Modes of Radioactive Decay Summary

Alpha decay is the emission of an alpha particle (2 protons and 2 neutrons) from an unstable nucleus. The daughter nuclide has an atomic number 2 less than the parent nuclide and a mass number 4 less than the parent nuclide. The daughter

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Beta-minus decay effectively converts a neutron to a proton and an electron, which is immediately ejected from the nucleus. The daughter nuclide has its atomic number increased by 1 and the same mass number compared to the parent.
Beta-plus decay effectively converts a proton to a neutron and a positron, which is immediately ejected from the nucleus. The daughter nuclide has its atomic number decreased by 1 and the same mass number compared to the parent.
In electron capture, the nucleus absorbs an electron from the innermost orbit. This electron combines with a proton to form a neutron.
Internal conversion occurs when a gamma ray, emitted by the nucleus as it goes from the excited state to the ground state, interacts with one of the innermost electrons of the same atom. The electron is ejected from the atom.
An isomeric transition is the decay of an excited nucleus to a lower-energy level
Fixture checkpoint marker-1000-xray is recorded on fixture page 1000.
by the emission of a gamma ray.
Decay chains can be found by tracing the steps an unstable atom goes through as it tries to achieve stability.
Many modes of radioactive decay result in a daughter nuclide that has an energy level above the ground state. This excitation energy is usually released immediately in the form of a gamma ray.
The type of decay that a nuclide will typically undergo can be determined by its relationship to the line of stability on the Chart of the Nuclides. Nuclides that lie below and to the right of the line of stability will typically beta minus decay. Nuclides above and to the left of the line will typically either beta plus decay or electron capture. Most alpha emitters are found in the upper, right-hand corner of the chart.
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RADIOACTIVITY
The rate at which a sample of radioactive material decays is not constant. As individual atoms of the material decay, there are fewer of those types of atoms remaining. Since the rate of decay is directly proportional to the number of atoms, the rate of decay will decrease as the number of atoms decreases.
EO 2.5 DEFINE the following terms:
a. Radioactivity d. Radioactive decay constant
b. Curie e. Radioactive half-life
c. Becquerel
EO 2.6 Given the number of atoms and either the half-life or decay constant of a nuclide, CALCULATE the activity.
EO 2.7 Given the initial activity and the decay constant of a nuclide, CALCULATE the activity at any later time.
EO 2.8 CONVERT between the half-life and decay constant for a nuclide.
EO 2.9 Given the Chart of the Nuclides and the original activity, PLOT the radioactive decay curve for a nuclide on either linear or semi-log coordinates.
EO 2.10 DEFINE the following terms:
a. Radioactive equilibrium

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Radioactive Decay Rates

Radioactivity is the property of certain nuclides of spontaneously emitting particles or gamma radiation. The decay of radioactive nuclides occurs in a random manner, and the precise time at which a single nucleus will decay cannot be determined. However, the average behavior of a very large sample can be predicted accurately by using statistical methods. These studies have revealed that there is a certain probability that in a given time interval a certain fraction of the nuclei within a sample of a particular nuclide will decay. This probability per unit time that an atom of a nuclide will decay is known as the radioactive decay constant, λ. The units for the decay constant are inverse time such as 1/second, 1/minute, 1/hour, or 1/year. These decay constant units can also be expressed as second-1, minute-1, hour-1, and year-1.

■■

The activity (A) of a sample is the rate of decay of that sample. This rate of decay is usually measured in the number of disintegrations that occur per second. For a sample containing millions of atoms, the activity is the product of the decay constant and the number of atoms present in the sample.

The relationship between the activity, number of atoms, and decay constant is shown in Equation (1-3).

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A = ■ N (1-3)

where:

A = Activity of the nuclide (disintegrations/second)

■ = decay constant of the nuclide (second)-1

N = Number of atoms of the nuclide in the sample

Since ■ is a constant, the activity and the number of atoms are always proportional.

Units of Measurement for Radioactivity

Two common units to measure the activity of a substance are the curie (Ci) and becquerel (Bq). A curie is a unit of measure of the rate of radioactive decay equal to 3.7 x 10 disintegrations 10 per second. This is approximately equivalent to the number of disintegrations that one gram of radium-226 will undergo in one second. A becquerel is a more fundamental unit of measure of radioactive decay that is equal to 1 disintegration per second. Currently, the curie is more widely used in the United States, but usage of the becquerel can be expected to broaden as the metric system slowly comes into wider use. The conversion between curies and becquerels is shown below.

1 curie = 3.7 x 10 becquerels 10

Variation of Radioactivity Over Time

The rate at which a given radionuclide sample decays is stated in Equation (1-3) as being equal to the product of the number of atoms and the decay constant. From this basic relationship it is possible to use calculus to derive an expression which can be used to calculate how the number of atoms present will change over time. The derivation is beyond the scope of this text, but Equation (1-4) is the useful result.

(1-4)

where:

N = number of atoms present at time t

N = number of atoms initially present o

■ = decay constant (time)-1

A Ao
e
t
A Aoe
t
A
Ao
e
t
ln A
Ao
t
t
ln
A
Ao
t
1/2
ln
1
2
t
1/2
ln 2 0.693
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Since the activity and the number of atoms are always proportional, they may be used
Fixture checkpoint marker-1002-xray is recorded on fixture page 1002.
interchangeably to describe any given radionuclide population. Therefore, the following is true.
(1-5)
where:
A = activity present at time t
A = activity initially present o
= decay constant (time)
-1
t = time
Radioactive Half-Life
One of the most useful terms for estimating how quickly a nuclide will decay is the radioactive half-life. The radioactive half-life is defined as the amount of time required for the activity to decrease to one-half of its original value. A relationship between the half-life and decay constant can be developed from Equation (1-5). The half-life can be calculated by solving Equation (1-5) for the time, t, when the current activity, A, equals one-half the initial activity A .
o
First, solve Equation (1-5) for t.
If A is equal to one-half of Ao, then A/Ao is equal to one-half. Substituting this in the equation above yields an expression for t .
1/2
(1-6)

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The basic features of decay of a radionuclide sample are shown by the graph in Figure 10.
Figure 10 Radioactive Decay as a Function of Time in Units of Half-Life
Assuming an initial number of atoms No, the population, and consequently, the activity may be noted to decrease by one-half of this value in a time of one half-life. Additional decreases occur so that whenever one half-life elapses, the number of atoms drops to one-half of what its value was at the beginning of that time interval. After five half-lives have elapsed, only 1/32, or 3.1%, of the original number of atoms remains. After seven half-lives, only 1/128, or 0.78%, of the atoms remains. The number of atoms existing after 5 to 7 half-lives can usually be assumed to be negligible. The Chemistry Fundamentals Handbook contains additional information on calculating the number of atoms contained within a sample.
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■
■
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Example:
A sample of material contains 20 micrograms of californium-252.
Californium-252 has a half-life of 2.638 years.
Calculate:
(a) The number of californium-252 atoms initially present
(b) The activity of the californium-252 in curies
(c) The number of californium-252 atoms that will remain in 12 years
(d) The time it will take for the activity to reach 0.001 curies
Solution:
(a) The number of atoms of californium-252 can be determined as below.
(b) First, use Equation (1-6) to calculate the decay constant.
Use this value for the decay constant in Equation (1-3) to determine the activity.
N No
e
t
4.78 x 1016 e
(0.263/yr) (12 yr)
2.04 x 1015
A Aoe
Fixture checkpoint marker-1003-xray is recorded on fixture page 1003.
t
A
Ao
e
t
ln A
Ao
t
t
ln
A
Ao
t

0.001 Ci
0.0108 Ci
0.263 year 1
t 9.05 years
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(c) The number of californium atoms that will remain in 12 years can be calculated from Equation (1-4).
(d) The time that it will take for the activity to reach 0.001 Ci can be determined from Equation (1-5). First, solve Equation (1-5) for time.
Inserting the appropriate values in the right side of this equation will result in the
Fixture checkpoint marker-1004-xray is recorded on fixture page 1004.
required time.
Plotting Radioactive Decay
It is useful to plot the activity of a nuclide as it changes over time. Plots of this type can be used to determine when the activity will fall below a certain level. This plot is usually done showing activity on either a linear or a logarithmic scale. The decay of the activity of a single nuclide on a logarithmic scale will plot as a straight line because the decay is exponential.
t
1/2
0.693
0.693
t
1/2
0.693
7.13 seconds
0.0972 second 1
A Aoe
t
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Example:
Plot the radioactive decay curve for nitrogen-16 over a period of 100 seconds. The initial activity is 142 curies and the half-life of nitrogen-16 is 7.13 seconds. Plot the curve on both linear rectangular coordinates and on a semi-log scale.
Solution:
First, use Equation (1-6) to calculate the decay constant corresponding to a half-life of 7.13 seconds.
Use the decay constant determined above to calculate the activity at various times using Equation (1-5).
Time Activity
0 seconds 142 Ci
20 seconds 20.3 Ci
40 seconds 2.91 Ci
60 seconds 0.416 Ci
80 seconds 0.0596 Ci
100 seconds 0.00853 Ci
■■

■ ■
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Figure 11 Linear and Semi-log Plots of Nitrogen-16 Decay
Plotting the data points calculated above on both linear and semilog scales results in the graphs shown in Figure 11.
If a substance contains more than one radioactive nuclide, the total activity is the sum of the individual activities of each nuclide. As an example, consider a sample of material that contained 1 x 10 atoms of iron-59 that has a half-life of 44.51 days (■ = 1.80 x 10 sec), 6 -7 -1 1 x 10 atoms of manganese-54 that has a half-life of 312.2 days (■ = 2.57 x 10 sec), and 1 6 -8 -1 x 10 atoms of cobalt-60 that has a half-life of 1925 days (■ = 4.17 x 10 sec). 6 -9 -1
The initial activity of each of the nuclides would be the product of the number of atoms and the decay constant.
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Figure 12 Combined Decay of Iron-56, Manganese-54, and Cobalt-60
Plotting the manner in which the activities of each of the three nuclides decay over time demonstrates that initially the activity of the shortest-lived nuclide (iron-59) dominates the total
Fixture checkpoint marker-1005-xray is recorded on fixture page 1005.
activity, then manganese-54 dominates. After almost all of the iron and manganese have decayed away, the only contributor to activity will be the cobalt-60. A plot of this combined decay is shown in Figure 12.
Radioactive Equilibrium
Radioactive equilibrium exists when a radioactive nuclide is decaying at the same rate at which it is being produced. Since the production rate and decay rate are equal, the number of atoms present remains constant over time.
An example of radioactive equilibrium is the concentration of sodium-24 in the coolant circulating through a sodium-cooled nuclear reactor. Assume that the sodium-24 is being produced at a rate of 1 x 10 atoms per second. If the sodium-24 were stable and did not decay, 6 the amount of sodium-24 present after some period of time could be calculated by multiplying the production rate by the amount of time. Plotting the amount of material present would result in the graph in Figure 13.
However, sodium-24 is not stable, and it decays with a half-life of 14.96 hours. If no sodium-24 is present initially and production starts at a rate of 1 x 10 atoms per second, the rate 6 of decay will initially be zero because there is no sodium-24 present to decay. The rate of decay of sodium-24 will increase as the amount of sodium-24 increases.
■
■
■
■
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Figure 13 Cumulative Production of Sodium-24 Over Time
The amount of sodium-24 present will initially increase rapidly, then it will increase at a continually decreasing rate until the rate of decay is equal to the rate of production. It is possible to calculate how much sodium-24 will be present at equilibrium by setting the production rate (R) equal to the decay rate (■ N).
where:

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■ = decay constant (second)-1
N = number of atoms
It is possible to calculate the equilibrium value for sodium-24 being produced at a rate of
1 x 10 atoms/second. 6
■ ■ ■■
■
■
■
■
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Figure 14 Approach of Sodium-24 to Equilibrium
The development of the equation to calculate how the amount of sodium-24 changes over time as it approaches the equilibrium value is beyond the scope of this handbook. However, the equation is presented below.
This equation can be used to calculate the values of the amount of sodium-24 present at different times. As the time increases, the exponential term approaches zero, and the number of atoms present will approach R/■. A plot of the approach of sodium-24 to equilibrium is shown in Figure 14.
Transient Radioactive Equilibrium
Transient radioactive equilibrium occurs when the parent nuclide and the daughter nuclide decay at essentially the same rate.
For transient equilibrium to occur, the parent must have a long half-life when compared to the daughter. An example of this type of compound decay process is barium-140, which decays by beta emission to lanthanum-140, which in turn decays by beta emission to stable cerium-140.
Fixture checkpoint marker-1006-xray is recorded on fixture page 1006.
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Figure 15 Transient Equilibrium in the Decay of Barium-140
The decay constant for barium-140 is considerably smaller than the decay constant for lanthanum-140. Remember that the rate of decay of both the parent and daughter can be represented as ■N. Although the decay constant for barium-140 is smaller, the actual rate of decay (■N) is initially larger than that of lanthanum-140 because of the great difference in their initial concentrations. As the concentration of the daughter increases, the rate of decay of the daughter will approach and eventually match the decay rate of the parent. When this occurs, they are said to be in transient equilibrium. A plot of the barium-lanthanum-cerium decay chain reaching transient equilibrium is shown in Figure 15.
Secular equilibrium occurs when the parent has an extremely long half-life. In the long decay chain for a naturally radioactive element, such as thorium-232, where all of the elements in the chain are in secular equilibrium, each of the descendants has built up to an equilibrium amount and all decay at the rate set by the original parent. The only exception is the final stable element on the end of the chain. Its number of atoms is constantly increasing.
Summary
The important information in this chapter is summarized on the following page.
■
■■
■

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Radioactivity Summary
& Radioactivity is the decay of unstable atoms by the emission of particles and electromagnetic radiation.
& A curie (Ci) is a unit of radioactivity equal to 3.7 x 10 disintegrations per 10 second.
& A becquerel (Bq) is a unit of radioactivity equal to 1 disintegration per second.
& The radioactive decay constant (■) is the probability per unit time that an atom will decay.
& The radioactive half-life is the amount of time required for the activity to decrease to one-half its original value.
& The activity of a substance can be calculated from the number of atoms and the decay constant based on the equation below.
& The amount of activity remaining after a particular time can be calculated from the equation below.
& The relationship between the decay constant and the half-life is shown below.
& Plots of radioactive decay can be useful to describe the variation of activity over time. If decay is plotted using semi-log scale the plot results in a straight line.
& Radioactive equilibrium exists when the production rate of a material is equal to the removal rate.
& Transient radioactive equilibrium exists when the parent nuclide and the daughter nuclide decay at essentially the same rate. This occurs only when the parent has a long half-life compared to the daughter.
Atomic and Nuclear Physics DOE-HDBK-1019/1-93 NEUTRON INTERACTIONS
NEUTRON INTERACTIONS
Neutrons can cause many different types of interactions. The neutron may simply scatter off the nucleus in two different ways, or it may actually be absorbed into the nucleus. If a neutron is absorbed into the nucleus, it may result in the emission of a gamma ray or a subatomic particle, or it may cause the nucleus to fission.
EO 3.1 DESCRIBE the following scattering interactions between a neutron and a nucleus:
a. Elastic scattering
Fixture checkpoint marker-1007-xray is recorded on fixture page 1007.
b. Inelastic scattering
EO 3.2 STATE the conservation laws that apply to an elastic collision between a neutron and a nucleus.
EO 3.3 DESCRIBE the following reactions where a neutron is absorbed in a nucleus:
a. Radiative capture
b. Particle ejection
Scattering
A neutron scattering reaction occurs when a nucleus, after having been struck by a neutron, emits a single neutron. Despite the fact that the initial and final neutrons do not need to be (and often are not) the same, the net effect of the reaction is as if the projectile neutron had merely "bounced off," or scattered from, the nucleus. The two categories of scattering reactions, elastic and inelastic scattering, are described in the following paragraphs.
Elastic Scattering
In an elastic scattering reaction between a neutron and a target nucleus, there is no energy

although there is usually some transfer of kinetic energy from the neutron to the target nucleus. The target nucleus gains the amount of kinetic energy that the neutron loses.
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mn
vn,i mTvT,i mnvn,f mTvT,f
1
2
m v
2
1
2
mn
v
2
n,i
1
2
mT
v
2
T,i
1
2
mn
v
2
n,f
1
2
mT
v
2
T,f
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Figure 16 Elastic Scattering
Figure 16 illustrates the process of elastic scattering of a neutron off a target nucleus. In the elastic scattering reaction, the conservation of momentum and kinetic energy is represented by the equations below.
Fixture checkpoint marker-1008-xray is recorded on fixture page 1008.
Conservation of momentum (mv)
Conservation of kinetic energy
where:
mn = mass of the neutron
m = mass of the target nucleus T
vn,i = initial neutron velocity
vn,f = final neutron velocity
vT,i = initial target velocity

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Elastic scattering of neutrons by nuclei can occur in two ways. The more unusual of the two interactions is the absorption of the neutron, forming a compound nucleus, followed by the re-emission of a neutron in such a way that the total kinetic energy is conserved and the nucleus returns to its ground state. This is known as resonance elastic scattering and is very dependent upon the initial kinetic energy possessed by the neutron. Due to formation of the compound nucleus, it is also referred to as compound elastic scattering. The second, more usual method, is termed potential elastic scattering and can be understood by visualizing the neutrons and nuclei to be much like billiard balls with impenetrable surfaces. Potential scattering takes place with incident neutrons that have an energy of up to about 1 MeV. In potential scattering, the neutron does not actually touch the nucleus and a compound nucleus is not formed. Instead, the neutron is acted on and scattered by the short range nuclear forces when it approaches close enough to the nucleus.

Inelastic Scattering

In inelastic scattering, the incident neutron is absorbed by the target nucleus, forming a compound nucleus. The compound nucleus will then emit a neutron of lower kinetic energy

Fixture checkpoint marker-1009-xray is recorded on fixture page 1009.

which leaves the original nucleus in an excited state. The nucleus will usually, by one or more gamma emissions, emit this excess energy to reach its ground state. Figure 17 shows the process of inelastic scattering.

For the nucleus that has reached its ground state, the sum of the kinetic energy of the exit

Figure 17 Inelastic Scattering

neutron, the target nucleus, and the total gamma energy emitted is equal to the initial kinetic energy of the incident neutron.

1
0
n
238
92
U
239
92
U
239
92
U
0
0
1
0
n
10
5
B
11
5
B

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3
Li
4
2
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Absorption Reactions
Most absorption reactions result in the loss of a neutron coupled with the production of a charged particle or gamma ray. When the product nucleus is radioactive, additional radiation is emitted at some later time. Radiative capture, particle ejection, and fission are all categorized as absorption reactions and are briefly described below.
Radiative Capture
In radiative capture the incident neutron enters the target nucleus forming a compound nucleus. The compound nucleus then decays to its ground state by gamma emission. An example of a radiative capture reaction is shown below.
Particle Ejection
In a particle ejection reaction the incident particle enters the target nucleus forming a compound nucleus. The newly formed compound nucleus has been excited to a high enough energy level to cause it to eject a new particle while the incident neutron remains in the nucleus. After the new particle is ejected, the remaining nucleus may or may not exist in an excited state depending upon the mass-energy balance of the reaction. An example of a particle ejection reaction is shown below.
Fission
Fixture checkpoint marker-1010-xray is recorded on fixture page 1010.
One of the most important interactions that neutrons can cause is fission, in which the nucleus that absorbs the neutron actually splits into two similarly sized parts. Fission will be discussed in detail in the next chapter.
Atomic and Nuclear Physics DOE-HDBK-1019/1-93 NEUTRON INTERACTIONS
Summary
The important information in this chapter is summarized below.
Neutron Interactions Summary
Interactions where a neutron scatters off a target nucleus are either elastic or inelastic. In elastic scattering, kinetic energy and momentum are conserved and no energy is transferred into excitation energy of the target nucleus. In inelastic scattering, some amount of kinetic energy is transferred into excitation energy of the target nucleus.
The conservation principles that apply to an elastic collision are conservation of kinetic energy and conservation of momentum.
Radiative capture is the absorption of a neutron by the target nucleus, resulting in an excited nucleus which subsequently (typically within a small fraction of a second) releases its excitation energy in the form of a gamma ray.
Particle ejection occurs when a neutron is absorbed by a target nucleus, resulting in the formation of a compound nucleus. The compound nucleus immediately ejects a particle (for example, alpha or proton).
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1
0
n

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92
U
236
92
U
140
55
Cs
93
37
Rb 3
1
0
n
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NUCLEAR FISSION
Nuclear fission is a process in which an atom splits and releases energy, fission products, and neutrons. The neutrons released by fission can, in turn, cause the fission of other atoms.
EO 4.1 EXPLAIN the fission process using the liquid drop model of a nucleus.
EO 4.2 DEFINE the following terms:
a. Excitation energy (E_{exc})
b. Critical energy (E_{crit})
EO 4.3 DEFINE the following terms:
a. Fissile material
b. Fissionable material
c. Fertile material
EO 4.4 DESCRIBE the processes of transmutation, conversion,
Fixture checkpoint marker-1011-xray is recorded on fixture page 1011.
and breeding.
EO 4.5 DESCRIBE the curve of Binding Energy per Nucleon versus mass number and give a qualitative description of the reasons for its shape.
EO 4.6 EXPLAIN why only the heaviest nuclei are easily fissioned.
EO 4.7 EXPLAIN why uranium-235 fissions with thermal neutrons and uranium-238 fissions only with fast neutrons.
Fission
In the fission reaction the incident neutron enters the heavy target nucleus, forming a compound nucleus that is excited to such a high energy level ($E > E_{crit}$) that the nucleus "splits" (E_{exc}) into two large fragments plus some neutrons. An example of a typical fission reaction is shown below.
A large amount of energy is released in the form of radiation and fragment kinetic energy.
Atomic and Nuclear Physics DOE-HDBK-1019/1-93 NUCLEAR FISSION
Liquid Drop Model of a Nucleus
The nucleus is held together by the attractive nuclear force between nucleons, which was discussed in a previous chapter. The characteristics of the nuclear force are listed below.

(10-13 cm)
(b) stronger than the repulsive electrostatic forces within the nucleus
(c) independent of nucleon pairing, in that the attractive forces between pairs of neutrons are no different than those between pairs of protons or a neutron and a proton
(d) saturable, that is, a nucleon can attract only a few of its nearest neighbors
One theory of fission considers the fissioning of a nucleus similar in some respects to the splitting of a liquid drop. This analogy is justifiable to some extent by the fact that a liquid drop is held together by molecular forces that tend to make the drop spherical in shape and that try to resist any deformation in the same manner as nuclear forces are assumed to hold the nucleus together. By considering the nucleus as a liquid drop, the fission process can be described.
Referring to Figure 18(A), the nucleus in the ground state is undistorted, and its attractive nuclear forces are greater than the repulsive electrostatic forces between the protons within the nucleus. When an incident particle (in this instance a neutron) is absorbed by the target nucleus, a compound nucleus is formed. The compound nucleus temporarily contains all the charge and mass involved in the reaction and exists in an excited state. The excitation energy added to the compound nucleus is equal to the binding energy contributed by the incident particle plus the kinetic energy possessed by that particle. Figure 18(B) illustrates the excitation energy thus imparted to the compound nucleus, which may cause it to oscillate and become distorted. If the excitation energy is greater than a certain critical energy, the oscillations may cause the compound nucleus to become dumbbell-shaped. When this happens, the attractive nuclear forces (short-range) in the neck area are small due to saturation, while the repulsive electrostatic forces (long-range) are only slightly less than before. When the repulsive electrostatic forces exceed the attractive nuclear forces, nuclear fission occurs, as illustrated in Figure 18(C).

Figure 18 Liquid Drop Model of Fission
Critical Energy
The measure of how far the energy level of a nucleus is above its ground state is called the excitation energy (Eexc). For fission to occur, the excitation energy must be above a particular value for that nuclide. The critical energy (Ecrit) is the minimum excitation energy required for fission to occur.
Fissile Material
A fissile material is composed of nuclides for which fission is possible with neutrons of any energy level. What is especially significant about these nuclides is their ability to be fissioned with zero kinetic energy neutrons (thermal neutrons). Thermal neutrons have very low kinetic energy levels (essentially zero) because they are roughly in equilibrium with the thermal motion
Fixture checkpoint marker-1012-xray is recorded on fixture page 1012.
of surrounding materials. Therefore, in order to be classified as fissile, a material must be capable of fissioning after absorbing a thermal neutron. Consequently, they impart essentially no kinetic energy to the reaction. Fission is possible in these materials with thermal neutrons, since the change in binding energy supplied by the neutron addition alone is high enough to exceed the critical energy. Some examples of fissile nuclides are uranium-235, uranium-233, and plutonium-239.

Fissionable Material
A fissionable material is composed of nuclides for which fission with neutrons is possible. All

fissioned only with high energy neutrons. The change in binding energy that occurs as the result of neutron absorption results in a nuclear excitation energy level that is less than the required critical energy. Therefore, the additional excitation energy must be supplied by the kinetic energy of the incident neutron. The reason for this difference between fissile and fissionable materials is the so-called odd-even effect for nuclei. It has been observed that nuclei with even numbers of neutrons and/or protons are more stable than those with odd numbers. Therefore, adding a neutron to change a nucleus with an odd number of neutrons to a nucleus with an even number of neutrons produces an appreciably higher binding energy than adding a neutron to a nucleus already possessing an even number of neutrons. Some examples of nuclides requiring high energy neutrons to cause fission are thorium-232, uranium-238, and plutonium-240. Table 4 indicates the critical energy (Ecrit) and the binding energy change for an added neutron (BEn) to target nuclei of interest. For fission to be possible, the change in binding energy plus the kinetic energy must equal or exceed the critical energy (ΔBE + KE > Ecrit).

TABLE 4

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Critical Energies Compared to Binding Energy of Last Neutron

Target
Nucleus
Critical Energy
Ecrit
Binding Energy of
Last Neutron BEn
BEn - Ecrit
23
9
2
0Th 7.5 MeV 5.4 MeV -2.1 MeV
23
9
8
2U 7.0 MeV 5.5 MeV -1.5 MeV
23
9
5
2U 6.5 MeV 6.8 MeV +0.3 MeV
23
9
3
2U 6.0 MeV 7.0 MeV +1.0 MeV
23
9
9
4Pu 5.0 MeV 6.6 MeV +1.6 MeV

Uranium-235 fissions with thermal neutrons because the binding energy released by the absorption of a neutron is greater than the critical energy for fission; therefore uranium-235 is a fissile material. The binding energy released by uranium-238 absorbing a thermal neutron is less than the critical energy, so additional energy must be possessed by the neutron for fission to be possible. Consequently, uranium-238 is a fissionable material.

NUCLEAR FISSION DOE-HDBK-1019/1-93 Atomic and Nuclear Physics
Fertile Material
All of the neutron absorption reactions that do not result in fission lead to the production of new nuclides through the process known as transmutation. These nuclides can, in turn, be transmuted again or may undergo radioactive decay to produce still different nuclides. The nuclides that are produced by this process are referred to as transmutation products. Because several of the fissile nuclides do not exist in nature, they can only be produced by nuclear reactions (transmutation). The target nuclei for such reactions are said to be fertile. Fertile materials are materials that can undergo transmutation to become fissile materials. Figure 19 traces the transmutation mechanism by which two fertile nuclides, thorium-232 and uranium-238, produce uranium-233 and plutonium-239, respectively.
Figure 19 Conversion of Fertile Nuclides to Fissile Nuclides
If a reactor contains fertile material in addition to its fissile fuel, some new fuel will be produced as the original fuel is burned up. This is called conversion. Reactors that are specifically designed to produce fissionable fuel are called "breeder" reactors. In such reactors, the amount of fissionable fuel produced is greater than the amount of fuel burnup. If less fuel is produced than used, the process is called conversion, and the reactor is termed a "converter."

Binding Energy Per Nucleon (BE/A)
As the number of particles in a nucleus increases, the total binding energy also increases. The
Fixture checkpoint marker-1014-xray is recorded on fixture page 1014.
rate of increase, however, is not uniform. This lack of uniformity results in a variation in the amount of binding energy associated with each nucleon within the nucleus. This variation in the binding energy per nucleon (BE/A) is easily seen when the average BE/A is plotted versus atomic mass number (A), as shown in Figure 20.
Figure 20 Binding Energy per Nucleon vs. Mass Number
Figure 20 illustrates that as the atomic mass number increases, the binding energy per nucleon decreases for A > 60. The BE/A curve reaches a maximum value of 8.79 MeV at A = 56 and decreases to about 7.6 MeV for A = 238. The general shape of the BE/A curve can be explained using the general properties of nuclear forces. The nucleus is held together by very short-range attractive forces that exist between nucleons. On the other hand, the nucleus is being forced apart by long range repulsive electrostatic (coulomb) forces that exist between all the protons in the nucleus.

As the atomic number and the atomic mass number increase, the repulsive electrostatic forces within the nucleus increase due to the greater number of protons in the heavy elements. To overcome this increased repulsion, the proportion of neutrons in the nucleus must increase to maintain stability. This increase in the neutron-to-proton ratio only partially compensates for the growing proton-proton repulsive force in the heavier, naturally occurring elements. Because the repulsive forces are increasing, less energy must be supplied, on the average, to remove a nucleon from the nucleus. The BE/A has decreased. The BE/A of a nucleus is an indication of its degree of stability. Generally, the more stable nuclides have higher BE/A than the less stable ones. The increase in the BE/A as the atomic mass number decreases from 260 to 60 is the primary reason for the energy liberation in the fission process. In addition, the increase in the BE/A as the atomic mass number increases from 1 to 60 is the reason for the energy liberation in the fusion process, which is the opposite reaction of fission.

for the relatively large coulomb forces forcing the two halves of the nucleus apart to overcome the attractive nuclear forces holding the two halves together. Consequently, the heaviest nuclei are easily fissionable compared to lighter nuclei.

Summary

The important information in this chapter is summarized on the following page.

Nuclear Fission Summary

The fission process can be explained using the liquid drop model of a nucleus. In the ground state the nucleus is nearly spherical in shape. After the absorption of a neutron, the nucleus will be in an excited state and start to oscillate and become distorted. If the oscillations cause the nucleus to become shaped like a dumbbell, the repulsive electrostatic forces will overcome the short-range attractive nuclear forces, and the nucleus will split in two.

Excitation energy is the amount of energy a nucleus has above its ground state. Critical energy is the minimum excitation energy that a nucleus must have before it can fission.

Fissile material is material for which fission is possible with neutrons that have zero kinetic energy. Fissionable material is material for which fission caused by neutron absorption is possible provided the kinetic energy added with the binding energy is greater than the critical energy. Fertile material is material that can undergo transmutation to become fissile material.

Transmutation is the process of neutron absorption and subsequent decay, which changes one nuclide to another nuclide. Conversion is the process of transmuting fertile material into fissile material in a reactor, where the amount of fissile material produced is less than the amount of fissile material consumed. Breeding is the same as conversion, except the amount of fissile material produced is more than the amount of fissile material consumed.

Fixture checkpoint marker-1015-xray is recorded on fixture page 1015.

The curve of binding energy per nucleon increases quickly through the light nuclides and reaches a maximum at a mass number of about 56. The curve decreases slowly for mass numbers greater than 60.

The heaviest nuclei are easily fissionable because they require only a small distortion from the spherical shape to allow the coulomb forces to overcoming the attractive nuclear force, forcing the two halves of the nucleus apart.

Uranium-235 fissions with thermal neutrons because the binding energy released by the absorption of a neutron is greater than the critical energy for fission. The binding energy released by uranium-238 absorbing a neutron is less than the critical energy, so additional kinetic energy must be possessed by the neutron for fission to be possible.

1
0
n
235
92
U
236

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U
140
55
Cs
93
37
Rb 3
1
0
n
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ENERGY RELEASE FROM FISSION
Fission of heavy nuclides converts a small amount of mass into an enormous amount of energy. The amount of energy released by fission can be determined based on either the change in mass that occurs during the reaction or by the difference in binding energy per nucleon between the fissile nuclide and the fission products.
EO 4.8 CHARACTERIZE the fission products in terms of mass groupings and radioactivity.
EO 4.9 Given the nuclides involved and their masses, CALCULATE the energy released from fission.
EO 4.10 Given the curve of Binding Energy per nucleon versus mass number, CALCULATE the energy released from fission.
Calculation of Fission Energy
Nuclear fission results in the release of enormous quantities of energy. It is necessary to be able to calculate the amount of energy that will be produced. The logical manner in which to pursue this is to first investigate a typical fission reaction such as the one listed below.
It can be seen that when the compound nucleus splits, it breaks into two fission fragments, rubidium-93, cesium-140, and some neutrons. Both fission products then decay by multiple - emissions as a result of the high neutron-to-proton ratio possessed by these nuclides.
In most cases, the resultant fission fragments have masses that vary widely. Figure 21 gives the percent yield for atomic mass numbers. The most probable pair of fission fragments for the thermal fission of the fuel uranium-235 have masses of about 95 and 140. Note that the vertical axis of the fission yield curve is on a logarithmic scale. Therefore, the formation of fission
Fixture checkpoint marker-1016-xray is recorded on fixture page 1016.
fragments of mass numbers of about 95 and 140 is highly likely.
Atomic and Nuclear Physics DOE-HDBK-1019/1-93 ENERGY RELEASE FROM FISSION
Figure 21 Uranium-235 Fission Yield vs.
Mass Number
Referring now to the binding energy per nucleon curve (Figure 20), we can estimate the amount of energy released by our "typical" fission by plotting this reaction on the curve and calculating the change in binding energy (ΔBE) between the reactants on the left-hand side of the fission equation and the products on the right-hand side. Plotting the reactant and product nuclides on the curve shows that the total binding energy of the system after fission is greater than the total binding energy of the system before fission. When there is an increase in the total binding energy of a system, the system has become more stable by releasing an amount of energy equal to the increase in total binding energy of the system. Therefore, in the fission process, the energy

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Figure 22 graphically depicts that the binding energy per nucleon for the products (C,
Figure 22 Change in Binding Energy for Typical Fission
rubidium-93 and B, cesium-140) is greater than that for the reactant (A, uranium-235). The total binding energy for a nucleus can be found by multiplying the binding energy per nucleon by the number of nucleons.
TABLE 5
Binding Energies Calculated from
Binding Energy per Nucleon Curve
Nuclide B.E. per Nucleon
(BE/A)
Fixture checkpoint marker-1017-xray is recorded on fixture page 1017.
Mass Number
(A)
Binding Energy
(BE/A) x (A)
9
3
3
7Rb 8.7 MeV 93 809 MeV
14
5
0
5Cs 8.4 MeV 140 1176 MeV
23
9
5
2U 7.6 MeV 235 1786 MeV
NP-01 Page 58 Rev. 0
BE BEproducts BEreactants
BERb 93 BECs 140 BEU 235
809 MeV 1176 MeV 1786 MeV
199 MeV
1
0
n
235
92
U
236
92
U
140
55
Cs
93
37

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1
0
n
Mass difference Mass of Reactants Mass of Products
236.052589 amu 235.85208 amu
0.200509 amu
EInst 0.020059 amu
931.5 MeV
amu
186.8 MeV
Atomic and Nuclear Physics DOE-HDBK-1019/1-93 ENERGY RELEASE FROM FISSION
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The energy released will be equivalent to the difference in binding energy (BE) between the reactants and the products.
The energy liberation during the fission process can also be explained from the standpoint of the conservation of mass-energy. During the fission process, there is a decrease in the mass of the system. There must, therefore, be energy liberated equal to the energy equivalent of the mass lost in the process. This method is more accurate than the previously illustrated method and is used when actually calculating the energy liberated during the fission process.
Fixture checkpoint marker-1018-xray is recorded on fixture page 1018.
Again, referring to the "typical" fission reaction.
E , the instantaneous energy, is the energy released immediately after the fission process. It Inst is equal to the energy equivalent of the mass lost in the fission process. It can be calculated as shown below.
Mass of the Reactants Mass of the Products
U 235.043924 amu Rb 92.91699 amu 235
92
93
37
n 1.008665 amu Cs 139.90910 amu 1
0
140
55
3 (n) 3.02599 amu 1
0
236.052589 amu 235.85208 amu
This mass difference can be converted to an energy equivalent.
93
37
Rb
93
38
Sr
93
39
Y
93
40

93
41
Nb
140
55
Cs
140
56
Ba
140
57
La
140
58
Ce
EDecay mRb 93 mNb 93 4 melectron
931.5MeV
amu
92.91699 amu 92.90638 amu 4 0.0005486 amu
931.5 MeV
amu
0.008416 amu
931.5 MeV
amu
7.84 MeV
ENERGY RELEASE FROM FISSION DOE-HDBK-1019/1-93 Atomic and Nuclear Physics
Fixture checkpoint marker-1019-xray is recorded on fixture page 1019.
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The total energy released per fission will vary from the fission to the next depending on what fission products are formed, but the average total energy released per fission of uranium-235 with a thermal neutron is 200 MeV.
As illustrated in the preceding example, the majority of the energy liberated in the fission process is released immediately after the fission occurs and appears as the kinetic energy of the fission fragments, kinetic energy of the fission neutrons, and instantaneous gamma rays. The remaining energy is released over a period of time after the fission occurs and appears as kinetic energy of the beta, neutrino, and decay gamma rays.
Estimation of Decay Energy
In addition to this instantaneous energy release during the actual fission reaction, there is additional energy released when the fission fragments decay by emission. This additional
-
energy is called decay energy, EDecay. The decay chains for rubidium-93 and cesium-140 are shown below.
The energy released during the decay for each chain will be equivalent to the mass difference between the original fission product and the sum of the final stable nuclide and the beta particles emitted.
The energy released in the decay chain of rubidium-93 is calculated below.
EDecay mRb 93 mNb 93 3 melectron
931.5MeV

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139.90910 amu 139.90543 amu 3 0.0005486 amu
931.5 MeV
amu
0.000202 amu
931.5 MeV
amu
1.89 MeV
Atomic and Nuclear Physics DOE-HDBK-1019/1-93 ENERGY RELEASE FROM FISSION
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The energy released in the decay chain of cesium-140 is calculated below.
The total decay energy is the sum of the energies of the two chains, or 9.73 MeV.
Distribution of Fission Energy
The average energy distribution for the energy released per fission with a thermal neutron in uranium-235 is shown in Tables 6 and 7.
TABLE 6
Instantaneous Energy from Fission
Kinetic Energy of Fission Products 167 Mev
Energy of Fission Neutrons 5 MeV
Instantaneous Gamma-ray Energy 5 MeV
Capture Gamma-ray Energy 10 MeV
Total Instantaneous Energy 187 MeV
TABLE 7
Delayed Energy from Fission
Beta Particles From Fission Products 7 Mev
Gamma-rays from Fission Products 6 MeV
Neutrinos 10 MeV
Total Delayed Energy 23 MeV
ENERGY RELEASE FROM FISSION DOE-HDBK-1019/1-93 Atomic and Nuclear Physics
Because the 10 MeV of neutrino energy shown in Table 7 is not absorbed in the reactor, the average value of 200 MeV per fission is still accurate. Note in Table 6 that some fission neutrons undergo radiative capture and the resultant gamma ray emission provides an additional 10 MeV of instantaneous energy, which contributes to the total of 187 MeV instantaneous energy.
Fixture checkpoint marker-1020-xray is recorded on fixture page 1020.
All of the energy released, with the exception of the neutrino energy, is ultimately transformed into heat through a number of processes. The fission fragments, with their high positive charge and kinetic energy, cause ionization directly as they rip orbital electrons from the surrounding atoms. In this ionization process, kinetic energy is transferred to the surrounding atoms of the fuel material, resulting in an increase in temperature. The beta particles and gamma rays also give up their energy through ionization, and the fission neutrons interact and lose their energy through elastic scattering. Of the 200 MeV released per fission, about seven percent (13 MeV) is released at some time after the instant of fission. When a reactor is shut down, fissions essentially cease, but energy is still being released from the decay of fission products. The heat produced by this decay energy is referred to as "decay heat." Although decay energy represents about seven percent of reactor heat production during reactor operation, once the reactor is shut down the decay heat production drops off quickly to a small fraction of its value while operating. The decay heat produced is significant, however, and systems must be provided to keep the reactor cool even after shutdown.

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The important information in this chapter is summarized below.
Energy Release From Fission Summary
Fission products have some general characteristics in common.
They generally decay by β- emission.
The most common mass numbers are grouped near 95 and 140.
The energy released by fission can be calculated based on the difference in mass between the masses of the reactants before fission and the fission fragments and fission neutrons after fission.
Another method to determine the energy released by fission is based on the change in binding energy per nucleon between the fissile nuclide and the fission
Fixture checkpoint marker-1021-xray is recorded on fixture page 1021.
products.
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DOE-HDBK-1019/1-93
Atomic and Nuclear Physics INTERACTION OF RADIATION WITH MATTER
INTERACTION OF RADIATION WITH MATTER
Different types of radiation interact with matter in widely different ways. A large, massive, charged alpha particle cannot penetrate a piece of paper and even has a limited range in dry air. A neutrino, at the other extreme, has a low probability of interacting with any matter, even if it passed through the diameter of the earth.
EO 5.1 DESCRIBE interactions of the following with matter:
a. Alpha particle c. Positron
b. Beta particle d. Neutron
EO 5.2 DESCRIBE the following ways that gamma radiation interacts with matter:
a. Photoelectric effect
b. Compton scattering
c. Pair production
Interaction of Radiation With Matter
Radiation can be classified into two general groups, charged and uncharged; therefore, it may be expected that interactions with matter fall into two general types. Charged particles directly ionize the media through which they pass, while uncharged particles and photons can cause ionization only indirectly or by secondary radiation.
A moving charged particle has an electrical field surrounding it, which interacts with the atomic structure of the medium through which it is passing. This interaction decelerates the particle and accelerates electrons in the atoms of the medium. The accelerated electrons may acquire enough energy to escape from the parent atom. This process, whereby radiation "strips" off orbital electrons, is called ionization. Uncharged moving particles have no electrical field, so they can only lose energy and cause ionization by such means as collisions or scattering. A photon can lose energy by the photoelectric effect, Compton effect, or pair production.
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I
m z
2
K.E.
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Because ionizing radiation creates ions in pairs, the intensity of ionization or the specific ionization is defined as the number of ion-pairs formed per centimeter of travel in a given material. The amount of ionization produced by a charged particle per unit path length, which is a measure of its ionizing power, is roughly proportional to the particle's mass and the square of its charge as illustrated in the equation below.

where:

I is the ionizing power

m is the mass of the particle

z is the number of unit charges it carries

K.E. is its kinetic energy

Since m for an alpha particle is about 7300 times as large as m for a beta particle, and z is twice as great, an alpha will produce much more ionization per unit path length than a beta particle of the same energy. This phenomenon occurs because the larger alpha particle moves slower for a given energy and thus acts on a given electron for a longer time.

Alpha Radiation

Alpha radiation is normally produced from the radioactive decay of heavy nuclides and from certain nuclear reactions. The alpha particle consists of 2 neutrons and 2 protons, so it is

Fixture checkpoint marker-1022-xray is recorded on fixture page 1022.

essentially the same as the nucleus of a helium atom. Because it has no electrons, the alpha particle has a charge of +2. This positive charge causes the alpha particle to strip electrons from the orbits of atoms in its vicinity. As the alpha particle passes through material, it removes electrons from the orbits of atoms it passes near. Energy is required to remove electrons and the energy of the alpha particle is reduced by each reaction. Eventually the particle will expend its kinetic energy, gain 2 electrons in orbit, and become a helium atom. Because of its strong positive charge and large mass, the alpha particle deposits a large amount of energy in a short distance of travel. This rapid, large deposition of energy limits the penetration of alpha particles. The most energetic alpha particles are stopped by a few centimeters of air or a sheet of paper.

Beta-Minus Radiation

A beta-minus particle is an electron that has been ejected at a high velocity from an unstable nucleus. An electron has a small mass and an electrical charge of -1. Beta particles cause ionization by displacing electrons from atom orbits. The ionization occurs from collisions with orbiting electrons. Each collision removes kinetic energy from the beta particle, causing it to slow down. Eventually the beta particle will be slowed enough to allow it to be captured as an

2 electrons 0.000549 amu

electron

931.5 MeV

amu

1.02 MeV

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Atomic and Nuclear Physics INTERACTION OF RADIATION WITH MATTER

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orbiting electron in an atom. Although more penetrating than the alpha, the beta is relatively easy to stop and has a low power of penetration. Even the most energetic beta radiation can be stopped by a few millimeters of metal.

Positron Radiation

Positively charged electrons are called positrons. Except for the positive charge, they are identical to beta-minus particles and interact with matter in a similar manner. Positrons are very

electron, producing two gammas with a combined energy (calculated below) equal to the rest mass of the positive and negative electrons.

Neutron Radiation

Neutrons have no electrical charge. They have nearly the same mass as a proton (a hydrogen atom nucleus). A neutron has hundreds of times more mass than an electron, but 1/4 the mass of an alpha particle. The source of neutrons is primarily nuclear reactions, such as fission, but they may also be produced from the decay of radioactive nuclides. Because of its lack of charge, the neutron is difficult to stop and has a high penetrating power.

Neutrons are attenuated (reduced in energy and numbers) by three major interactions, elastic scatter, inelastic scatter, and absorption. In elastic scatter, a neutron collides with a nucleus and bounces off. This reaction transmits some of the kinetic energy of the neutron to the nucleus of the atom, resulting in the neutron being slowed, and the atom receives some kinetic energy (motion). This process is sometimes referred to as "the billiard ball effect."

As the mass of the nucleus approaches the mass of the neutron, this reaction becomes more effective in slowing the neutron. Hydrogenous material attenuates neutrons most effectively.

In the inelastic scatter reaction, the same neutron/nucleus collision occurs as in elastic scatter. However, in this reaction, the nucleus receives some internal energy as well as kinetic energy. This slows the neutron, but leaves the nucleus in an excited state. When the nucleus decays to its original energy level, it normally emits a gamma ray.

In the absorption reaction, the neutron is actually absorbed into the nucleus of an atom. The neutron is captured, but the atom is left in an excited state. If the nucleus emits one or more gamma rays to reach a stable level, the process is called radiative capture. This reaction occurs at most neutron energy levels, but is more probable at lower energy levels.

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Gamma Radiation

Gamma radiation is electromagnetic radiation. It is commonly referred to as a gamma ray and is very similar to an x-ray. The difference is that gamma rays are emitted from the nucleus of an atom, and x-rays are produced by orbiting electrons. The x-ray is produced when orbiting electrons move to a lower energy orbit or when fast-moving electrons approaching an atom are deflected and decelerated as they react with the atom's electrical field (called Bremsstrahlung). The gamma ray is produced by the decay of excited nuclei and by nuclear reactions. Because the gamma ray has no mass and no charge, it is difficult to stop and has a very high penetrating power. A small fraction of the original gamma stream will pass through several feet of concrete or several meters of water.

There are three methods of attenuating gamma rays. The first method is referred to as the photo-electric effect. When a low energy gamma strikes an atom, the total energy of the gamma is expended in ejecting an electron from orbit. The result is ionization of the atom and expulsion of a high energy electron. This reaction is most predominant with low energy gammas interacting in materials with high atomic weight and rarely occurs with gammas having an energy above 1 MeV. Annihilation of the gamma results. Any gamma energy in excess of the binding energy of the electron is carried off by the electron in the form of kinetic energy.

The second method of attenuation of gammas is called Compton scattering. The gamma interacts with an orbital or free electron; however, in this case, the photon loses only a fraction of its energy. The actual energy loss depending on the scattering angle of the gamma. The gamma continues on at lower energy, and the energy difference is absorbed by the electron. This reaction becomes important for gamma energies of about 0.1 MeV and higher.

pair-production. When a high energy gamma passes close enough to a heavy nucleus, the gamma completely disappears, and an electron and a positron are formed. For this reaction to take place, the original gamma must have at least 1.02 MeV energy. Any energy greater than 1.02 MeV becomes kinetic energy shared between the electron and positron. The probability of pair■production incre
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Atomic and Nuclear Physics INTERACTION OF RADIATION WITH MATTER
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Interaction of Radiation with Matter Summary
An alpha particle deposits a large amount of energy in a short distance of travel due to its large mass and charge.
Beta-minus particles interact with the electrons orbiting the nucleus of atoms, causing ionization by displacing the electrons. The beta particle loses energy with each interaction. After the beta particle loses enough energy, it is captured in the orbital shells of an atom.
Positrons interact with matter much the same way as beta minus particles. After the positron has lost most of its energy by ionizing atoms, it is annihilated by interaction with an electron. The electron-positron pair disappear and are replaced by two gammas, each with the energy equivalent of the mass of an electron (0.51 MeV).
Neutrons interact with matter by elastic scattering, inelastic scattering, or absorption.
Photoelectric effect is where a gamma interacts with an electron orbiting an atom. The entire energy of the gamma is transferred to the electron, and the electron is ejected from its orbit.
In Compton scattering a gamma interacts with an orbital electron, but only part of the gamma energy is transferred to the electron. The electron is ejected from its orbit, and the gamma is scattered off at a lower energy.
In pair-production, a gamma interacts with the electric field of a nucleus and is converted into an electron-positron pair. The gamma must have an energy greater than 1.02 MeV for this to occur.
Fixture checkpoint marker-1024-xray is recorded on fixture page 1024.
Summary
The important information in this chapter is summarized below.
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Department of Energy
Fundamentals Handbook
NUCLEAR PHYSICS
AND REACTOR THEORY
Module 2
Reactor Theory (Neutron Characteristics)
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TERMINAL OBJECTIVE

1.0 Without references, EXPLAIN how neutron sources produce neutrons.

ENABLING OBJECTIVES

1.1 DEFINE the following terms:

a. Intrinsic neutron source

b. Installed neutron source

1.2 LIST three examples of reactions that produce neutrons in intrinsic neutron sources.

1.3 LIST three examples of reactions that produce neutrons in installed neutron sources.

TERMINAL OBJECTIVE

2.0 Given the necessary information for calculations, EXPLAIN basic concepts in reactor physics and perform calculations.

ENABLING OBJECTIVES

2.1 DEFINE the following terms:

a. Atom density d. Barn

b. Neutron flux e. Macroscopic cross section

c. Microscopic cross section f. Mean free path

2.2 EXPRESS macroscopic cross section in terms of microscopic cross section.

2.3 DESCRIBE how the absorption cross section of typical nuclides varies with neutron energy at energies below the resonance absorption region.

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2.5 DESCRIBE the energy dependence of resonance absorption peaks for typical light and heavy nuclei.
2.6 EXPRESS mean free path in terms of macroscopic cross section.
2.7 Given the number densities (or total density and component fractions) and microscopic cross sections of components, CALCULATE the macroscopic cross section for a mixture.
2.8 CALCULATE a macroscopic cross section given a material density, atomic mass, and microscopic cross section.
2.9 EXPLAIN neutron shadowing or self-shielding.
2.10 Given the neutron flux and macroscopic cross section, CALCULATE the reaction rate.
2.11 DESCRIBE the relationship between neutron flux and reactor power.
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OBJECTIVES DOE-HDBK-1019/1-93 Reactor Theory (Neutron Characteristics)
ENABLING OBJECTIVES (Cont.)
2.12 DEFINE the following concepts:
a. Thermalization d. Average logarithmic energy decrement
b. Moderator e. Macroscopic slowing down power
c. Moderating ratio
2.13 LIST three desirable characteristics of a moderator.
2.14 Given an average fractional energy loss per collision, CALCULATE the energy loss after a specified number of collisions.
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Reactor Theory (Neutron Characteristics) DOE-HDBK-1019/1-93 OBJECTIVES
Fixture checkpoint marker-1027-xray is recorded on fixture page 1027.
TERMINAL OBJECTIVE
3.0 Without references, EXPLAIN the production process and effects on fission of prompt and delayed neutrons.
ENABLING OBJECTIVES
3.1 STATE the origin of prompt neutrons and delayed neutrons.
3.2 STATE the approximate fraction of neutrons that are born as delayed neutrons from the fission of the following nuclear fuels:
a. Uranium-235
b. Plutonium-239
3.3 EXPLAIN the mechanism for production of delayed neutrons.
3.4 EXPLAIN prompt and delayed neutron generation times.
3.5 Given prompt and delayed neutron generation times and delayed neutron fraction, CALCULATE the average generation time.
3.6 EXPLAIN the effect of delayed neutrons on reactor control.
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OBJECTIVES DOE-HDBK-1019/1-93 Reactor Theory (Neutron Characteristics)
TERMINAL OBJECTIVE
4.0 Without references, DESCRIBE the neutron energy spectrum for the type of reactor presented in this module.
ENABLING OBJECTIVES
4.1 STATE the average energy at which prompt neutrons are produced.
4.2 DESCRIBE the neutron energy spectrum in the following reactors:
a. Fast reactor
b. Thermal reactor
4.3 EXPLAIN the reason for the particular shape of the fast, intermediate, and slow energy

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Reactor Theory (Neutron Characteristics) DOE-HDBK-1019/1-93 NEUTRON SOURCES
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NEUTRON SOURCES
Neutrons from a variety of sources are always present in a reactor core. This is true even when the reactor is shut down. Some of these neutrons are produced by naturally occurring (intrinsic) neutron sources, while others may be the result of fabricated (installed) neutron sources that are incorporated into the design of the reactor. The neutrons produced by sources other than neutron-induced fission are often grouped together and classified as source neutrons.
EO 1.1 DEFINE the following terms:
a. Intrinsic neutron source
b. Installed neutron source
EO 1.2 LIST three examples of reactions that produce neutrons in intrinsic neutron sources.
EO 1.3 LIST three examples of reactions that produce neutrons in installed neutron sources.
Neutron Sources
In addition to neutron-induced fission, neutrons are produced by other reactions. The neutrons produced by reactions other than neutron-induced fission are called source neutrons. Source neutrons are important because they ensure that the neutron population remains high enough to allow a visible indication of neutron level on the most sensitive monitoring instruments while the reactor is shutdown and during the startup sequence. This verifies instrument operability and allows monitoring of neutron population changes. Source neutrons can be classified as either intrinsic or installed neutron sources.
Intrinsic Neutron Sources
Some neutrons will be produced in the materials present in the reactor due to a variety of unavoidable reactions that occur because of the nature of these materials. Intrinsic neutron sources are those neutron-producing reactions that always occur in reactor materials.
Fixture checkpoint marker-1028-xray is recorded on fixture page 1028.
■ ■ ■ ■
■ ■ ■
■ ■
NEUTRON SOURCES DOE-HDBK-1019/1-93 Reactor Theory (Neutron Characteristics)
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A limited number of neutrons will always be present, even in a reactor core that has never been operated, due to spontaneous fission of some heavy nuclides that are present in the fuel. Uranium■238, ur of the rate at which different heavy nuclides produce neutrons by spontaneous fission.
Californium-252 is not an intrinsic neutron source, but will be discussed in the section on installed neutron sources.
TABLE 1
Neutron Production by Spontaneous Fission
Nuclide T (Fission) T (1/2 1/2 ■-decay) neutrons/sec/gram
U 1.8 x 10 years 6.8 x 10 years 8.0 x 10 235
92
17 8 -4
U 8.0 x 10 years 4.5 x 10 years 1.6 x 10 238
92

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Pu 5.5 x 10 years 2.4 x 10 years 3.0 x 10 239
94
5 4 -2
Pu 1.2 x 10 years 6.6 x 10 years 1.0 x 10 240
94
11 3 3
Cf 66.0 years 2.65 years 2.3 x 10 252
98
12
Another intrinsic neutron source is a reaction involving natural boron and fuel. In some reactors, natural boron is loaded into the reactor core as a neutron absorber to improve reactor control or increase core life-time. Boron-11 (80.1% of natural boron) undergoes a reaction with the alpha particle emitted by the radioactive decay of heavy nuclides in the fuel to yield a neutron as shown below.
The boron-11 must be mixed with, or in very close proximity to, the fuel for this reaction because of the short path length of the alpha particle. For a reactor core with this configuration, this (■,n) reaction is an important source of neutrons for reactor startup.
In a reactor that has been operated, another source of neutrons becomes significant. Neutrons may be produced by the interaction of a gamma ray and a deuterium nucleus. This reaction is commonly referred to as a photoneutron reaction because it is initiated by electromagnetic radiation and results in the production of a neutron. The photoneutron reaction is shown below.
■ ■ ■
■ ■
■ ■ ■
■ ■
Reactor Theory (Neutron Characteristics) DOE-HDBK-1019/1-93 NEUTRON SOURCES
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There is an abundant supply of high energy gammas in a reactor that has been operated because many of the fission products are gamma emitters. All water-cooled reactors have some deuterium present in the coolant in the reactor core because a small fraction of natural hydrogen is the isotope deuterium. The atom percentage of deuterium in the water ranges from close to the naturally occurring value (0.015%) for light water reactors to above 90% deuterium for heavy water reactors. Therefore, the required conditions for production of photoneutrons exist.
The supply of gamma rays decreases with time after shutdown as the gamma emitters decay; therefore, the photoneutron production rate also decreases. In a few particular reactors, additional D O (heavy water) may be added to the reactor to increase the production of 2
Fixture checkpoint marker-1029-xray is recorded on fixture page 1029.
photoneutrons following a long shutdown period.
Installed Neutron Sources
Because intrinsic neutron sources can be relatively weak or dependent upon the recent power history of the reactor, many reactors have artificial sources of neutrons installed. These neutron sources ensure that shutdown neutron levels are high enough to be detected by the nuclear instruments at all times. This provides a true picture of reactor conditions and any change in these conditions. An installed neutron source is an assembly placed in or near the reactor for the sole purpose of producing source neutrons.
One strong source of neutrons is the artificial nuclide californium-252, which emits neutrons at the rate of about 2 x 10 neutrons per second per gram as the result of spontaneous fission. 12
Important drawbacks for some applications may be its high cost and its short half-life (2.65

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Many installed neutron sources use the (■,n) reaction with beryllium. These sources are composed of a mixture of metallic beryllium (100% beryllium-9) with a small quantity of an alpha particle emitter, such as a compound of radium, polonium, or plutonium. The reaction that occurs is shown below.
The beryllium is intimately (homogeneously) mixed with the alpha emitter and is usually enclosed in a stainless steel capsule.
Another type of installed neutron source that is widely used is a photoneutron source that employs the (■,n) reaction with beryllium. Beryllium is used for photoneutron sources because its stable isotope beryllium-9 has a weakly attached last neutron with a binding energy of only 1.66 MeV. Thus, a gamma ray with greater energy than 1.66 MeV can cause neutrons to be ejected by the (■,n) reaction as shown below.
NEUTRON SOURCES DOE-HDBK-1019/1-93 Reactor Theory (Neutron Characteristics)
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Fixture checkpoint marker-1030-xray is recorded on fixture page 1030.
Neutron Sources Summary
& Intrinsic neutron sources are sources of neutrons from materials that are in the reactor for other purposes such as fuel, burnable poison, or moderator.
& Installed neutron sources are materials or components placed in the reactor specifically for the purpose of producing source neutrons.
& Examples of intrinsic neutron sources are listed below.
Spontaneous fission of heavy nuclides in fuel, such as uranium-238, uranium-235, and plutonium-239, results in fission fragments and free neutrons.
Boron-11 mixed with the fuel undergoes an alpha-neutron reaction and becomes nitrogen-14.
Deuterium present in the reactor coolant undergoes a gamma■neutron reaction and becomes hydrogen-1.
& Examples of installed neutron sources are listed below.
Spontaneous fission of californium-252 results in fission fragments and free neutrons.
Beryllium-9 undergoes an alpha-neutron reaction (alpha from the decay of plutonium, polonium, or radium) and becomes carbon-12.
Beryllium-9 undergoes a gamma-neutron reaction (high energy gamma from decay of antimony-124) and becomes beryllium-8.
Many startup sources of this type use antimony and beryllium because after activation with neutrons the radioactive antimony becomes an emitter of high energy gammas. The photoneutron sources of this type are constructed somewhat differently from the (■,n) types. One design incorporates a capsule of irradiated antimony enclosed in a beryllium sleeve. The entire assembly is then encased in a stainless steel cladding. A large reactor may have several neutron sources of this type installed within the core.
Summary
The important information in this chapter is summarized below.
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Reactor Theory (Neutron Characteristics) NUCLEAR CROSS SECTIONS AND NEUTRON FLUX
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NUCLEAR CROSS SECTIONS AND NEUTRON FLUX
To determine the frequency of neutron interactions, it is necessary to describe the availability of neutrons to cause interaction and the probability of a neutron interacting with material. The availability of neutrons and the probability of

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EO 2.1 DEFINE the following terms:
a. Atom density d. Barn
b. Neutron flux e. Macroscopic cross section
c. Microscopic cross section f. Mean free path
EO 2.2 EXPRESS macroscopic cross section in terms of microscopic cross section.
EO 2.3 DESCRIBE how the absorption cross section of typical nuclides varies with neutron energy at energies below the resonance absorption region.
EO 2.4 DESCRIBE the cause of resonance absorption in terms of nuclear energy levels.
EO 2.5 DESCRIBE the energy dependence of resonance absorption peaks for typical light and heavy nuclei.
EO 2.6 EXPRESS mean free path in terms of macroscopic cross section.
EO 2.7 Given the number densities (or total density and component fractions) and microscopic cross sections of components, CALCULATE the macroscopic cross section for a mixture.
EO 2.8 CALCULATE a macroscopic cross section given a material density, atomic mass, and microscopic cross section.
EO 2.9 EXPLAIN neutron shadowing or self-shielding.
Fixture checkpoint marker-1031-xray is recorded on fixture page 1031.
'

Introduction

Fission neutrons are born with an average energy of about 2 MeV. These fast neutrons interact with the reactor core materials in various absorption and scattering reactions. Collisions that result in scattering are useful in slowing neutrons to thermal energies. Thermal neutrons may be absorbed by fissile nuclei to produce more fissions or be absorbed in fertile material for conversion to fissionable fuel. Absorption of neutrons in structural components, coolant, and other non-fuel material results in the removal of neutrons without fulfilling any useful purpose. To safely and efficiently operate a nuclear reactor it is necessary to predict the probability that a particular absorption or scattering reaction will occur. Once these probabilities are known, if the availability of neutrons can be determined, then the rate at which these nuclear reactions take place can be predicted.

Atom Density

One important property of a material is the atom density. The atom density is the number of atoms of a given type per unit volume of the material. To calculate the atom density of a substance use Equation (2-1).

(2-1)

where:

N = atom density (atoms/cm)3
' = density (g/cm)3
N = Avogadro's number (6.022 x 10 atoms/mole) A
23
M = gram atomic weight
N

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M
2.699 g
cm3
6.022 x 1023 atoms
mole
26.9815 g
mole
6.024 x 1022 atoms
cm3
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Example:
A block of aluminum has a density of 2.699 g/cm . If the gram atomic weight of
3
aluminum is 26.9815 g, calculate the atom density of the aluminum.
Solution:
Cross Sections
The probability of a neutron interacting with a nucleus for a particular reaction is dependent upon not only the kind of nucleus involved, but also the energy of the neutron. Accordingly, the absorption of a thermal neutron in most materials is much more probable than the absorption of a fast neutron. Also, the probability of interaction will vary depending upon the type of reaction involved.
The probability of a particular reaction occurring between a neutron and a nucleus is called the microscopic cross section ( ) of the nucleus for the particular reaction. This cross section will vary with the energy of the neutron. The microscopic cross section may also be regarded as the effective area the nucleus presents to the neutron for the particular reaction. The larger the
Fixture checkpoint marker-1032-xray is recorded on fixture page 1032.
effective area, the greater the probability for reaction.
Because the microscopic cross section is an area, it is expressed in units of area, or square centimeters. A square centimeter is tremendously large in comparison to the effective area of a nucleus, and it has been suggested that a physicist once referred to the measure of a square centimeter as being "as big as a barn" when applied to nuclear processes. The name has persisted and microscopic cross sections are expressed in terms of barns. The relationship between barns and cm is shown below. 2
1 barn = 10-24 cm2
N
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NUCLEAR CROSS SECTIONS AND NEUTRON FLUX Reactor Theory (Neutron Characteristics)
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Whether a neutron will interact with a certain volume of material depends not only on the microscopic cross section of the individual nuclei but also on the number of nuclei within that volume. Therefore, it is necessary to define another kind of cross section known as the macroscopic cross section ( ). The macroscopic cross section is the probability of a given reaction occurring per unit travel of the neutron. is related to the microscopic cross section ( ) by the relationship shown below.
(2-2)
where:

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-1
N = atom density of material (atoms/cm)
3
= microscopic cross-section (cm)
2
The difference between the microscopic and macroscopic cross sections is extremely important and is restated for clarity. The microscopic cross section ( ) represents the effective target area that a single nucleus presents to a bombarding particle. The units are given in barns or cm .
2
The macroscopic cross section ( ) represents the effective target area that is presented by all of the nuclei contained in 1 cm of the material. The units are given as 1/cm or cm .
3 -1
A neutron interacts with an atom of the material it enters in two basic ways. It will either interact through a scattering interaction or through an absorption reaction. The probability of a neutron being absorbed by a particular atom is the microscopic cross section for absorption, . The probability of a neutron scattering off of a particular nucleus is the microscopic cross a section for scattering, s
. The sum of the microscopic cross section for absorption and the microscopic cross section for scattering is the total microscopic cross section, . T
= + T a s
Both the absorption and the scattering microscopic cross sections can be further divided. For instance, the scattering cross section is the sum of the elastic scattering cross section (se) and the inelastic scattering cross section ( ). si
= + s se si
The microscopic absorption cross section ( ) includes all reactions except scattering. However, a for most purposes it is sufficient to merely separate it into two categories, fission (f) and capture ( ). Radiative capture of neutrons was described in the Neutron Interactions chapter c of Module 1.
= + a f c
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Figure 1 Typical Neutron Absorption Cross Section vs. Neutron Energy
The variation of absorption cross sections with neutron energy is often complicated. For many
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elements the absorption cross sections are small, ranging from a fraction of a barn to a few barns for slow (or thermal) neutrons.
For a considerable number of nuclides of moderately high (or high) mass numbers, an examination of the variation of the absorption cross section with the energy of the incident neutron reveals the existence of three regions on a curve of absorption cross section versus neutron energy. This cross section is illustrated in Figure 1. First, the cross section decreases steadily with increasing neutron energy in a low energy region, which includes the thermal range (E < 1 eV). In this region the absorption cross section, which is often high, is inversely proportional to the velocity (v). This region is frequently referred to as the "1/v region," because the absorption cross section is proportional to 1/v, which is the reciprocal of neutron velocity. Following the 1/v region, there occurs the "resonance region" in which the cross sections rise sharply to high values called "resonance peaks" for neutrons of certain energies, and then fall again. These energies are called resonance energies and are a result of the affinity

That is, when the binding energy of a neutron plus the kinetic energy of the neutron are exactly equal to the amount required to raise a compound nucleus from its ground state to a quantum level, resonance absorption occurs. The following example problem further illustrates this point.
■
(
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NUCLEAR CROSS SECTIONS AND NEUTRON FLUX Reactor Theory (Neutron Characteristics)
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Assuming that uranium-236 has a nuclear quantum energy level at 6.8 MeV above its ground state, calculate the kinetic energy a neutron must possess to undergo resonant absorption in uranium-235 at this resonance energy level.
Fixture checkpoint marker-1034-xray is recorded on fixture page 1034.
BE = [Mass(U) + Mass(neutron) - Mass(U)] x 931 MeV/amu 235 236
BE = (235.043925 + 1.008665 - 236.045563) x 931 MeV/amu
BE = (0.007025 amu) x 931 MeV/amu = 6.54 MeV
6.8 MeV - 6.54 MeV = 0.26 MeV
The difference between the binding energy and the quantum energy level equals the amount of kinetic energy the neutron must possess. The typical heavy nucleus will have many closely■spaced resonanc complex and have more possible configurations and corresponding energy states. Light nuclei, being less complex, have fewer possible energy states and fewer resonances that are sparsely distributed at higher energy levels.
For higher neutron energies, the absorption cross section steadily decreases as the energy of the neutron increases. This is called the "fast neutron region." In this region the absorption cross sections are usually less than 10 barns.
With the exception of hydrogen, for which the value is fairly large, the elastic scattering cross sections are generally small, for example, 5 barns to 10 barns. This is close to the magnitude of the actual geometric cross sectional area expected for atomic nuclei. In potential scattering, the cross section is essentially constant and independent of neutron energy. Resonance elastic scattering and inelastic scattering exhibit resonance peaks similar to those associated with absorption cross sections. The resonances occur at lower energies for heavy nuclei than for light nuclei. In general, the variations in scattering cross sections are very small when compared to the variations that occur in absorption cross sections.
Mean Free Path
If a neutron has a certain probability of undergoing a particular interaction in one centimeter of travel, then the inverse of this value describes how far the neutron will travel (in the average case) before undergoing an interaction. This average distance traveled by a neutron before interaction is known as the mean free path for that interaction and is represented by the symbol ■. The relationship between the mean free path (■) and the macroscopic cross section (*) is shown below.
(2-3)
'
(
)
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Reactor Theory (Neutron Characteristics) NUCLEAR CROSS SECTIONS AND NEUTRON FLUX
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Calculation of Macroscopic Cross Section and Mean Free Path
Most materials are composed of several elements, and because most elements are composed of

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Therefore, to include all the isotopes within a given material, it is necessary to determine the macroscopic cross section for each isotope and then sum all the individual macroscopic cross sections. Equation (2-4) can be used to determine the macroscopic cross section for a composite material.

* = N) + N) + N) + ... N) (2-4) 1 1 2 2 3 3 nn

where:

N = the number of nuclei per cm of the n element n

3 th

) = the microscopic cross section of the n element n

th

The following example problems illustrate the calculation of the macroscopic cross section for a single element and for combinations of materials.

Example 1:

Find the macroscopic thermal neutron absorption cross section for iron, which has a density of 7.86 g/cm . The microscopic cross section for absorption of iron is 2.56 barns 3 and the gram atomic weight is 55.847 g.

Solution:

Step 1: Using Equation (2-1), calculate the atom density of iron.

Fixture checkpoint marker-1035-xray is recorded on fixture page 1035.

Step 2: Use this atom density in Equation (2-2) to calculate the macroscopic cross section.

'

'

Example 2:

An alloy is composed of 95% aluminum and 5% silicon (by weight). The density of the alloy is 2.66 g/cm . Properties of aluminum and silicon are shown below. 3

Element	Gram Atomic Weight	) a (barns)	s (barns)
Aluminum	26.9815	0.23	1.49
Silicon	28.0855	0.16	2.20

1. Calculate the atom densities for the aluminum and silicon.
2. Determine the absorption and scattering macroscopic cross sections for thermal neutrons.
3. Calculate the mean free paths for absorption and scattering.

Solution:

Step 1: The density of the aluminum will be 95% of the total density. Using Equation (2-1) yields the atom densities.

a NAl a,Al NSi a,Si

5.64 x 1022 atoms

cm3

0.23 x 10 24 cm22.85 x 1021 atoms

cm3

0.16 x 10 24 cm2

0.0134 cm 1

s NAl s,Al NSi s,Si

5.64 x 1022 atoms

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1.49 x 10 24 cm22.85 x 1021 atoms
cm3
2.20 x 10 24 cm2
0.0903 cm 1
a
1
a
1
0.01345 cm 1
74.3 cm
s
1
s
1
0.0903 cm 1
11.1 cm
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Step 2: The macroscopic cross sections for absorption and scattering are
calculated using Equation (2-4).
Step 3: The mean free paths are calculated by inserting the macroscopic cross
sections calculated above into Equation (2-3).
Thus, a neutron must travel an average of 74.3 cm to interact by absorption in this alloy, but it
must travel only 11.1 cm to interact by scattering.
Fixture checkpoint marker-1036-xray is recorded on fixture page 1036.
)
)
)
)
(■
(■
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NUCLEAR CROSS SECTIONS AND NEUTRON FLUX Reactor Theory (Neutron Characteristics)
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Effects of Temperature on Cross Section
As discussed, the microscopic absorption cross section varies significantly as neutron energy
varies. The microscopic cross sections provided on most charts and tables are measured for a
standard neutron velocity of 2200 meters/second, which corresponds to an ambient temperature
of 68(F. Therefore, if our material is at a higher temperature, the absorption cross section will
be lower than the value for 68(F, and any cross sections which involve absorption (for example,
) ,) ,)) must be corrected for the existing temperature. acf
The following formula is used to correct microscopic cross sections for temperature. Although
the example illustrates absorption cross section, the same formula may be used to correct capture
and fission cross sections.
where:
) = microscopic cross section corrected for temperature
) = microscopic cross section at reference temperature (68(F or 20(C) o

T = temperature for which corrected value is being calculated
NOTE: When using this formula, all temperatures must be converted to (R or (K.
(R = (F + 460
(K = (C + 273
Example:
What is the value of)f for uranium-235 for thermal neutrons at 500(F? Uranium-235 has a) of 583 barns at 68(F. f
Solution:
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Reactor Theory (Neutron Characteristics) NUCLEAR CROSS SECTIONS AND NEUTRON FLUX
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Neutron Flux
Macroscopic cross sections for neutron reactions with materials determine the probability of one neutron undergoing a specific reaction per centimeter of travel through that material. If one wants to determine how many reactions will actually occur, it is necessary to know how many neutrons are traveling through the material and how many centimeters they travel each second. It is convenient to consider the number of neutrons existing in one cubic centimeter at any one instant and the total distance they travel each second while in that cubic centimeter. The number of neutrons existing in a cm of material at any instant is called neutron density and is 3 represented by the symbol n with units of neutrons/cm . The total distance these neutrons can 3 travel each second will be determined by their velocity.
A good way of defining neutron flux (1) is to consider it to be the total path length covered by all neutrons in one cubic centimeter during one second. Mathematically, this is the equation below.
1 = n v (2-5)
where:
1 = neutron flux (neutrons/cm -sec) 2
n = neutron density (neutrons/cm)3
v = neutron velocity (cm/sec)
The term neutron flux in some applications (for example, cross section measurement) is used as parallel beams of neutrons traveling in a single direction. The intensity (I) of a neutron beam is the product of the neutron density times the average neutron velocity. The directional beam
Fixture checkpoint marker-1037-xray is recorded on fixture page 1037.
intensity is equal to the number of neutrons per unit area and time (neutrons/cm -sec) falling on 2 a surface perpendicular to the direction of the beam.
One can think of the neutron flux in a reactor as being comprised of many neutron beams traveling in various directions. Then, the neutron flux becomes the scalar sum of these directional flux intensities (added as numbers and not vectors), that is, 1 = I + I + I +...I . 123 n
Since the atoms in a reactor do not interact preferentially with neutrons from any particular direction, all of these directional beams contribute to the total rate of reaction. In reality, at a given point within a reactor, neutrons will be traveling in all directions.
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Self-Shielding
In some locations within the reactor, the flux level may be significantly lower than in other areas due to a phenomenon referred to as neutron shadowing or self-shielding. For example, the interior of a fuel pin or pellet will "see" a lower average flux level than the outer surfaces since

interior of the fuel pin. This is especially important at resonance energies, where the absorption cross sections are large.

Summary

The important information in this chapter is summarized below.

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Reactor Theory (Neutron Characteristics) NUCLEAR CROSS SECTIONS AND NEUTRON FLUX

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Nuclear Cross Section and Neutron Flux Summary

& Atom density (N) is the number of atoms of a given type per unit volume of material.

& Microscopic cross section ()) is the probability of a given reaction occurring between a neutron and a nucleus.

& Microscopic cross sections are measured in units of barns, where 1 barn = 10-24 cm .2

& Macroscopic cross section (*) is the probability of a given reaction occurring per unit length of travel of the neutron. The units for macroscopic cross section are cm . -1

& The mean free path (■) is the average distance that a neutron travels in a material between interactions.

& Neutron flux (1) is the total path length traveled by all neutrons in one cubic centimeter of material during one second.

& The macroscopic cross section for a material can be calculated using the equation below.

* = N)

& The absorption cross section for a material usually has three distinct regions. At low neutron energies (<1 eV) the cross section is inversely proportional to the neutron velocity.

& Resonance absorption occurs when the sum of the kinetic energy of the neutron and its binding energy is equal to an allowed nuclear energy level of the nucleus.

& Resonance peaks exist at intermediate energy levels. For higher neutron energies, the absorption cross section steadily decreases as the neutron energy increases.

& The mean free path equals 1/*.

& The macroscopic cross section for a mixture of materials can be calculated using the equation below.

* = N 1 1 2 2 3 3 nn) + N) + N) + ... N)

& Self-shielding is where the local neutron flux is depressed within a material due to neutron absorption near the surface of the material.

1)

REACTION RATES DOE-HDBK-1019/1-93 Reactor Theory (Neutron Characteristics)

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REACTION RATES

It is possible to determine the rate at which a nuclear reaction will take place based on the neutron flux, cross section for the interaction, and atom density of the target. This relationship illustrates how a change in one of these items affects the reaction rate.

EO 2.10 Given the neutron flux and macroscopic cross section, CALCULATE the reaction rate.

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power.
Reaction Rates
If the total path length of all the neutrons in a cubic centimeter in a second is known, (neutron flux (1)), and if the probability of having an interaction per centimeter path length is also known (macroscopic cross section (*)), multiply them together to get the number of interactions taking place in that cubic centimeter in one second. This value is known as the reaction rate and is denoted by the symbol R. The reaction rate can be calculated by the equation shown below.
R = 1 * (2-6)
where:
R = reaction rate (reactions/sec)
1 = neutron flux (neutrons/cm -sec) 2
* = macroscopic cross section (cm) -1
Substituting the fact that * = N) into Equation (2-6) yields the equation below.
where:
) = microscopic cross section (cm)2
N = atom density (atoms/cm)3
Fixture checkpoint marker-1039-xray is recorded on fixture page 1039.
Rf f
1 x 1013 neutrons
cm2
sec
0.1 cm 1
1 x 1012 fissions
cm3
sec
Rf f
f
Rf
1.29 x 1012 fissions
cm3
sec
3 x 1013 neutrons
cm2
sec
0.043 cm 1
Reactor Theory (Neutron Characteristics) DOE-HDBK-1019/1-93 REACTION RATES
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The reaction rate calculated will depend on which macroscopic cross section is used in the calculation. Normally, the reaction rate of greatest interest is the fission reaction rate.
Example:
If a one cubic centimeter section of a reactor has a macroscopic fission cross section of 0.1 cm , and if the thermal neutron flux is 10 neutrons/cm -sec, what is the fission rate -1 13 2 in that cubic centimeter?
Solution:
In addition to using Equation (2-6) to determine the reaction rate based on the physical properties of material, it is also possible to algebraically manipulate the equation to determine physical properties if the reaction rate is known.
Example:

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13 2 20
uranium-235 per cm . The reaction rate is 1.29 x 10 fission/cm . Calculate and .
3 12 3
f f
Solution:
Step 1: The macroscopic cross section can be determined by solving
Equation (2-6) for and substituting the appropriate values. f
Rf N f
f
Rf
N
1.29 x 1012 fissions
cm3
sec
1 x 1020 atoms
cm3
3 x 1013 neutrons
cm2
sec
4.3 x 10 22 cm2 1 barn
1 x 10 24 cm2
430 barns
1 watt 1 erg
Fixture checkpoint marker-1040-xray is recorded on fixture page 1040.
1 x 10 7 watt sec
1 MeV
1.602 x 10 6
erg
1 fission
200 MeV
3.12 x 1010 fissions
second
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Step 2: To find the microscopic cross section, replace f with (N x f) and solve
for .
f
Reactor Power Calculation
Multiplying the reaction rate per unit volume by the total volume of the core results in the total number of reactions occurring in the core per unit time. If the amount of energy involved in each reaction were known, it would be possible to determine the rate of energy release (power) due to a certain reaction.
In a reactor where the average energy per fission is 200 MeV, it is possible to determine the number of fissions per second that are necessary to produce one watt of power using the following conversion factors.
1 fission = 200 MeV
1 MeV = 1.602 x 10 ergs -6
1 erg = 1 x 10 watt-sec -7

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P
th f V
3.12 x 1010 fissions
watt sec
Reactor Theory (Neutron Characteristics) DOE-HDBK-1019/1-93 REACTION RATES
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The power released in a reactor can be calculated based on Equation (2-6). Multiplying the reaction rate by the volume of the reactor results in the total fission rate for the entire reactor. Dividing by the number of fissions per watt-sec results in the power released by fission in the reactor in units of watts. This relationship is shown mathematically in Equation (2-7) below.
(2-7)
where:
P = power (watts)
= thermal neutron flux (neutrons/cm -sec) th
2
= macroscopic cross section for fission (cm) f
-1
V = volume of core (cm)
3
Relationship Between Neutron Flux and Reactor Power
In an operating reactor the volume of the reactor is constant. Over a relatively short period of time (days or weeks), the number density of the fuel atoms is also relatively constant. Since the atom density and microscopic cross section are constant, the macroscopic cross section must also be constant. Examining Equation (2-7), it is apparent that if the reactor volume and macroscopic cross section are constant, then the reactor power and the neutron flux are directly proportional. This is true for day-to-day operation. The neutron flux for a given power level will increase very slowly over a period of months due to the burnup of the fuel and resulting decrease in atom density and macroscopic cross section.
REACTION RATES DOE-HDBK-1019/1-93 Reactor Theory (Neutron Characteristics)
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Fixture checkpoint marker-1041-xray is recorded on fixture page 1041.
Reaction Rates Summary
& The reaction rate is the number of interactions of a particular type occurring in a cubic centimeter of material in a second.
& The reaction rate can be calculated by the equation below.
R = 1 *
& Over a period of several days, while the atom density of the fuel can be considered constant, the neutron flux is directly proportional to reactor power.
Summary
The important information in this chapter is summarized below.
Reactor Theory (Neutron Characteristics) DOE-HDBK-1019/1-93 NEUTRON MODERATION
Rev. 0 Page 23 NP-02
NEUTRON MODERATION
In thermal reactors, the neutrons that cause fission are at a much lower energy than the energy level at which they were born from fission. In this type of reactor, specific materials must be included in the reactor design to reduce the energy level of the neutrons in an efficient manner.
EO 2.12 DEFINE the following concepts:

b. Moderator e. Macroscopic slowing down power
c. Moderating ratio
EO 2.13 LIST three desirable characteristics of a moderator.
EO 2.14 Given an average fractional energy loss per collision, CALCULATE the energy loss after a specified number of collisions.
Neutron Slowing Down and Thermalization
Fission neutrons are produced at an average energy level of 2 MeV and immediately begin to slow down as the result of numerous scattering reactions with a variety of target nuclei. After a number of collisions with nuclei, the speed of a neutron is reduced to such an extent that it has approximately the same average kinetic energy as the atoms (or molecules) of the medium in which the neutron is undergoing elastic scattering. This energy, which is only a small fraction of an electron volt at ordinary temperatures (0.025 eV at 20(C), is frequently referred to as the thermal energy, since it depends upon the temperature. Neutrons whose energies have been reduced to values in this region (< 1 eV) are designated thermal neutrons. The process of reducing the energy of a neutron to the thermal region by elastic scattering is referred to as thermalization, slowing down, or moderation. The material used for the purpose of thermalizing neutrons is called a moderator. A good moderator reduces the speed of neutrons in a small number of collisions, but does not absorb them to any great extent. Slowing the neutrons in as few collisions as possible is desirable in order to reduce the amount of neutron leakage from the core and also to reduce the number of resonance absorptions in non-fuel materials. Neutron leakage and resonance absorption will be discussed in the next module.
ln Ei
ln Ef
ln
Ei
Ef
2
A
2
3
NEUTRON MODERATION DOE-HDBK-1019/1-93 Reactor Theory (Neutron Characteristics)
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The ideal moderating material (moderator) should have the following nuclear properties.
large scattering cross section
small absorption cross section
large energy loss per collision
Fixture checkpoint marker-1042-xray is recorded on fixture page 1042.
A convenient measure of energy loss per collision is the logarithmic energy decrement. The average logarithmic energy decrement is the average decrease per collision in the logarithm of the neutron energy. This quantity is represented by the symbol (Greek letter xi).
(2-8)
where:
= average logarithmic energy decrement
Ei = average initial neutron energy
Ef = average final neutron energy
The symbol is commonly called the average logarithmic energy decrement because of the fact that a neutron loses, on the average, a fixed fraction of its energy per scattering collision. Since

material, is also a constant. Because it is a constant for each type of material and does not depend upon the initial neutron energy, is a convenient quantity for assessing the moderating ability of a material.

The values for the lighter nuclei are tabulated in a variety of sources. The following commonly used approximation may be used when a tabulated value is not available.

This approximation is relatively accurate for mass numbers (A) greater than 10, but for some low values of A it may be in error by over three percent.

N

ln Ehigh ln Elow

ln

Ehigh

Elow

N

ln

Fixture checkpoint marker-1043-xray is recorded on fixture page 1043.

Ehigh

Elow

ln

2 x 106eV

0.025 eV

0.948

19.2 collisions

Reactor Theory (Neutron Characteristics) DOE-HDBK-1019/1-93 NEUTRON MODERATION

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Since represents the average logarithmic energy loss per collision, the total number of collisions necessary for a neutron to lose a given amount of energy may be determined by dividing into the difference of the natural logarithms of the energy range in question. The number of collisions (N) to travel from any energy, E , to any lower energy, E , can be calculated as shown below. high low

Example:

How many collisions are required to slow a neutron from an energy of 2 MeV to a thermal energy of 0.025 eV, using water as the moderator? Water has a value of 0.948 for .

Solution:

Sometimes it is convenient, based upon information known, to work with an average fractional energy loss per collision as opposed to an average logarithmic fraction. If the initial neutron energy level and the average fractional energy loss per collision are known, the final energy level for a given number of collisions may be computed using the following formula.

! (

NEUTRON MODERATION DOE-HDBK-1019/1-93 Reactor Theory (Neutron Characteristics)

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(2-9)

where:

E = initial neutron energy o

E = neutron energy after N collisions N

x = average fractional energy loss per collision

N = number of collisions

Example:

If the average fractional energy loss per collision in hydrogen is 0.63, what will be the

SYNTHETIC FIXTURE PAGE 1044 OF 1200

Solution:
a)
b)
Macroscopic Slowing Down Power
Although the logarithmic energy decrement is a convenient measure of the ability of a material to slow neutrons, it does not measure all necessary properties of a moderator. A better measure of the capabilities of a material is the macroscopic slowing down power. The macroscopic slowing down power (MSDP) is the product of the logarithmic energy decrement and the macroscopic cross section for scattering in the material. Equation (2-10) illustrates how to calculate the macroscopic slowing down power.
(2-10)
MR
s
a
Reactor Theory (Neutron Characteristics) DOE-HDBK-1019/1-93 NEUTRON MODERATION
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Moderating Ratio
Macroscopic slowing down power indicates how rapidly a neutron will slow down in the material in question, but it still does not fully explain the effectiveness of the material as a moderator. An element such as boron has a high logarithmic energy decrement and a good slowing down power, but it is a poor moderator because of its high probability of absorbing neutrons.
Fixture checkpoint marker-1044-xray is recorded on fixture page 1044.
The most complete measure of the effectiveness of a moderator is the moderating ratio. The moderating ratio is the ratio of the macroscopic slowing down power to the macroscopic cross section for absorption. The higher the moderating ratio, the more effectively the material performs as a moderator. Equation (2-11) shows how to calculate the moderating ratio of a material.
(2-11)
Moderating properties of different materials are compared in Table 2.
TABLE 2
Moderating Properties of Materials
Material Collisions to Slowing Down
Number of Macroscopic
Thermalize Power
Moderating
Ratio
H2O 0.927 19 1.425 62
D2O 0.510 35 0.177 4830
Helium 0.427 42 9 x 10 51 -6
Beryllium 0.207 86 0.154 126
Boron 0.171 105 0.092 0.00086
Carbon 0.158 114 0.083 216
NEUTRON MODERATION DOE-HDBK-1019/1-93 Reactor Theory (Neutron Characteristics)
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Summary
The important information in this chapter is summarized below.
Neutron Moderation Summary
& Thermalization is the process of reducing the energy level of a neutron from the

& The moderator is the reactor material that is present for the purpose of thermalizing neutrons.
& Moderating ratio is the ratio of the macroscopic slowing down power to the macroscopic cross section for absorption.
& The average logarithmic energy decrement (!) is the average change in the logarithm of neutron energy per collision.
& Macroscopic slowing down power is the product of the average logarithmic energy decrement and the macroscopic cross section for scattering.
& There are three desirable characteristics of a moderator.
1. large scattering cross section
2. small absorption cross section
3. large energy loss per collision
& The energy loss after a specified number of collisions can be calculated using the equation below.
E = E (1 - x) N o
N
■
■ ■
Reactor Theory (Neutron Characteristics) DOE-HDBK-1019/1-93 PROMPT AND DELAYED NEUTRONS
Rev. 0 Page 29 NP-02
PROMPT AND DELAYED NEUTRONS
Not all neutrons are released at the same time following fission. Most neutrons are released virtually instantaneously and are called prompt neutrons. A very small fraction of neutrons are released after the decay of fission products and are called delayed neutrons. Although delayed neutrons are a very small fraction of the total number of neutrons, they play an extremely important role in the control of the reactor.
EO 3.1 STATE the origin of prompt neutrons and delayed neutrons.
Fixture checkpoint marker-1045-xray is recorded on fixture page 1045.
EO 3.2 STATE the approximate fraction of neutrons that are born as delayed neutrons from the fission of the following nuclear fuels:
a. Uranium-235
b. Plutonium-239
EO 3.3 EXPLAIN the mechanism for production of delayed neutrons.
EO 3.4 EXPLAIN prompt and delayed neutron generation times.
EO 3.5 Given prompt and delayed neutron generation times and delayed neutron fraction, CALCULATE the average generation time.
EO 3.6 EXPLAIN the effect of delayed neutrons on reactor control.
Neutron Classification
The great majority (over 99%) of the neutrons produced in fission are released within about 10-13 seconds of the actual fission event. These are called prompt neutrons. A small portion of fission neutrons are delayed neutrons, which are produced for some time after the fission process has taken place. The delayed neutrons are emitted immediately following the first beta decay of a fission fragment known as a delayed neutron precursor. An example of a delayed neutron precursor is bromine-87, shown below.
PROMPT AND DELAYED NEUTRONS DOE-HDBK-1019/1-93 Reactor Theory (Neutron Characteristics)
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appropriately averaged properties. These groups vary somewhat depending on the fissile material in use. Table 3 lists the characteristics for the six precursor groups resulting from thermal fission of uranium-235. The fraction of all neutrons that are produced by each of these precursors is called the delayed neutron fraction for that precursor. The total fraction of all neutrons born as delayed neutrons is called the delayed neutron fraction (■). The fraction of delayed neutrons produced varies depending on the predominant fissile nuclide in use. The delayed neutron fractions (■) for the fissile nuclides of most interest are as follows: uranium-233 (0.0026), uranium-235 (0.0065), uranium-238 (0.0148), and plutonium-239 (0.0021).

TABLE 3

Delayed Neutron Precursor Groups
for Thermal Fission in Uranium-235

Group	Half-Life (sec)	Delayed Neutron Fraction	Average Energy (MeV)
1	55.7	0.00021	0.25
2	22.7	0.00142	0.46
3	6.2	0.00127	0.41
4	2.3	0.00257	0.45
5	0.61	0.00075	0.41
6	0.23	0.00027	-
Total	-	0.0065	-

Neutron Generation Time

The neutron generation time is the time required for neutrons from one generation to cause the fissions that produce the next generation of neutrons. The generation time for prompt neutrons (5* - pronounced "ell-star") is the total time from birth to rebirth. Three time intervals are involved: (a) the time it takes a fast neutron to slow down to thermal energy, (b) the time the now thermal neutron exists prior to absorption in fuel, and (c) the time required for a fissionable nucleus to emit a fast neutron after neutron absorption.

Fast neutrons slow to thermal energies or leak out of the reactor in 10 seconds to 10 seconds, -4 -6 depending on the moderator. In water moderated reactors, thermal neutrons tend to exist for about 10 seconds before they are absorbed. Fission and fast neutron production following -4 neutron absorption in a fissionable nucleus occurs in about 10 seconds. Thus, fast reactors -13 have an 5* of about 10 seconds, while thermal reactors have an 5* of about 10 seconds + 10 -6 -6 -4 seconds, which is about 10 seconds to 10 seconds. -4 -5

■ ■ ■ ■

■ ■ ■ ■

Fixture checkpoint marker-1046-xray is recorded on fixture page 1046.

■ ■

On the other hand, the average generation time for the six delayed neutron groups is the total time from the birth of the fast neutron to the emission of the delayed neutron. Again, three time intervals are involved: (a) the time it takes a fast neutron to slow down to thermal energy, (b) the time the thermal neutron exists prior to absorption, and (c) the average time from neutron absorption to neutron emission by the six precursor groups. The average time for decay of precursors from uranium-235 is 12.5 seconds. The other terms in the delayed neutron generation time are insignificant when compared to this value, and the average delayed neutron generation time becomes ■ 12.5 seconds.

A neutron generation time in the range of 10 seconds to 10 seconds or faster could result in -4 -5

delayed neutrons to slow down the rate of the reaction. The average generation time, and hence the rate that power can rise, is determined largely by the delayed neutron generation time. The following equation shows this mathematically.

(2-12)

Example:

Assume a prompt neutron generation time for a particular reactor of 5 x 10 seconds and -5 a delayed neutron generation time of 12.5 seconds. If ■ is 0.0065, calculate the average generation time.

Solution:

This example demonstrates the effect delayed neutrons have on the neutron generation time and thus reactor control. If a reactor were to be operated in a sustained chain reaction using only prompt neutrons (■ = 0), the generation time from the previous example would be about

Fixture checkpoint marker-1047-xray is recorded on fixture page 1047.

5 x 10 seconds. However, by operating the reactor such that a 0.0065 fraction of neutrons are -5 delayed, the generation life time is extended to 0.0813 seconds, providing time for adequate operator control. Therefore, although only a small fraction of the total neutron population, delayed neutrons are extremely important to the control and maintenance of a sustained fission chain reaction.

Summary

The important information in this chapter is summarized on the following page.

■ ■ ■ ■

PROMPT AND DELAYED NEUTRONS DOE-HDBK-1019/1-93 Reactor Theory (Neutron Characteristics) NP-02 Page 32 Rev. 0

Prompt and Delayed Neutrons Summary

& Prompt neutrons are released directly from fission within 10 seconds of the -13 fission event.

& Delayed neutrons are released from the decay of fission products that are called delayed neutron precursors. Delayed neutron precursors are grouped according to half-life. Half-lives vary from fractions of a second to almost a minute.

& The fraction of neutrons born as delayed neutrons is different for different fuel materials. Following are values for some common fuel materials.

Uranium-235 0.0065

Plutonium-239 0.0021

& Delayed neutrons are produced by a classification of fission products known as delayed neutron precursors. When a delayed neutron precursor undergoes a ■ decay, it results in an excited daughter nucleus which immediately ejects a - neutron. Therefore, these delayed neutrons appear with a half-life of the delayed neutron precursor.

& The delayed neutron generation time is the total time from the birth of the fast neutron to the emission of the delayed neutron in the next generation. Delayed neutron generation times are dominated by the half-life of the delayed neutron precursor. The average delayed neutron generation time is about 12.5 seconds.

& A prompt neutron generation time is the sum of the amount of time it takes a fast neutron to thermalize, the amount of time the neutron exists as a thermal neutron before it is absorbed, and the amount of time between a fissionable nuclide absorbing a neutron and fission neutrons being released. Prompt neutron generation time is about 5 x 10 seconds. -5

& The average neutron generation time can be calculated from the prompt and

Equation (2-12).
& Delayed neutrons are responsible for the ability to control the rate at which power can rise in a reactor. If only prompt neutrons existed, reactor control would not be possible due to the rapid power changes.
Reactor Theory (Neutron Characteristics) DOE-HDBK-1019/1-93 NEUTRON FLUX SPECTRUM
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Figure 2 Prompt Fission Neutron Energy Spectrum for
Thermal Fission of Uranium-235
NEUTRON FLUX SPECTRUM
The number of neutrons that exist at a given energy level varies. A plot of either the fraction of neutrons or the neutron flux at a given energy versus the energy level is called a neutron energy spectrum. The neutron energy spectrum varies widely for different types of reactors.
EO 4.1 STATE the average energy at which prompt neutrons are produced.
EO 4.2 DESCRIBE the neutron energy spectrum in the following reactors:
a. Fast reactor
b. Thermal reactor
Fixture checkpoint marker-1048-xray is recorded on fixture page 1048.
EO 4.3 EXPLAIN the reason for the particular shape of the fast, intermediate, and slow energy regions of the neutron flux spectrum for a thermal reactor.
Prompt Neutron Energies
The neutrons produced by fission are high energy neutrons, and almost all fission neutrons have energies between 0.1 MeV and 10 MeV. The neutron energy distribution, or spectrum, may best be described by plotting the fraction of neutrons per MeV as a function of neutron energy, as shown in Figure 2. From this figure it can be seen that the most probable neutron energy is about 0.7 MeV. Also, from this data it can be shown that the average energy of fission neutrons is about 2 MeV. Figure 2 is the neutron energy spectrum for thermal fission in uranium-235. The values will vary slightly for other nuclides.
NEUTRON FLUX SPECTRUM DOE-HDBK-1019/1-93 Reactor Theory (Neutron Characteristics)
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Figure 3 Comparison of Neutron Flux Spectra for
Thermal and Fast Breeder Reactor

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The spectrum of neutron energies produced by fission varies significantly from the energy spectrum, or flux, existing in a reactor at a given time. Figure 3 illustrates the difference in neutron flux spectra between a thermal reactor and a fast breeder reactor. The energy distribution of neutrons from fission is essentially the same for both reactors, so the differences in the curve shapes may be attributed to the neutron moderation or slowing down effects.
No attempt is made to thermalize or slow down neutrons in the fast breeder reactor (liquid metal cooled); therefore, an insignificant number of neutrons exist in the thermal range. For the thermal reactor (water moderated), the spectrum of neutrons in the fast region (> 0.1 MeV) has a shape similar to that for the spectrum of neutrons emitted by the fission process.
In the thermal reactor, the flux in the intermediate energy region (1 eV to 0.1 MeV) has approximately a 1/E dependence. That is, if the energy (E) is halved, the flux doubles. This 1/E dependence is caused by the slowing down process, where elastic collisions remove a constant fraction of the neutron energy per collision (on the average), independent of energy; thus, the neutron loses larger amounts of energy per collision at higher energies than at lower energies. The fact that the neutrons lose a constant fraction of energy per collision causes the neutrons to tend to "pile up" at lower energies, that is, a greater number of neutrons exist at the lower energies as a result of this behavior.
vp
2 k T
m
vp
2 k T
m
2 1.38 x 10 16 erg
K
293 K
Fixture checkpoint marker-1049-xray is recorded on fixture page 1049.
1.66 x 10 24 g
2.2 x 105 cm
sec
1 m
100 cm
2200 m
sec
Reactor Theory (Neutron Characteristics) DOE-HDBK-1019/1-93 NEUTRON FLUX SPECTRUM
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In the thermal region the neutrons achieve a thermal equilibrium with the atoms of the moderator material. In any given collision they may gain or lose energy, and over successive collisions will gain as much energy as they lose. These thermal neutrons, even at a specific temperature, do not all have the same energy or velocity; there is a distribution of energies, usually referred to as the Maxwell distribution (e.g., Figure 2). The energies of most thermal neutrons lie close to the most probable energy, but there is a spread of neutrons above and below this value.
Most Probable Neutron Velocities
The most probable velocity (vp) of a thermal neutron is determined by the temperature of the medium and can be determined by Equation (2-13) .
(2-13)
where:
v = most probable velocity of neutron (cm/sec) p

T = absolute temperature in degrees Kelvin (K)
m = mass of neutron (1.66 x 10 grams) -24
Example:
Calculate the most probable velocities for neutrons in thermal equilibrium with their surroundings at the following temperatures. a) 20 C, b) 260 C.
Solution:
a) Calculate the most probable velocity for 20 C using Equation (2-13).
vp
2 k T
m
2 1.38 x 10 16 erg
K
533 K
1.66 x 10 24 g
2.977 x 105 cm
sec
1 m
100 cm
2977 m
sec
NEUTRON FLUX SPECTRUM DOE-HDBK-1019/1-93 Reactor Theory (Neutron Characteristics) NP-02 Page 36 Rev. 0
b) Calculate the most probable velocity for 260 C using Equation (2-13).
From these calculations it is evident that the most probable velocity of a thermal neutron increases as temperature increases. The most probable velocity at 20 C is of particular importance since reference data, such as nuclear cross sections, are tabulated for a neutron velocity of 2200 meters per second.
Reactor Theory (Neutron Characteristics) DOE-HDBK-1019/1-93 NEUTRON FLUX SPECTRUM Rev. 0 Page 37 NP-02
Neutron Flux Spectrum Summary
& Prompt neutrons are born at energies between 0.1 MeV and 10 MeV. The average prompt neutron energy is about 2 MeV.
& Fast reactors have a neutron energy spectrum that has the same shape as the Fixture checkpoint marker-1050-xray is recorded on fixture page 1050.
prompt neutron energy spectrum.
& Thermal reactors have a neutron energy spectrum that has two pronounced peaks, one in the thermal energy region where the neutrons are in thermal equilibrium with the core materials and another in the fast region at energies where neutrons are produced. The flux in the intermediate region (1 eV to 0.1 MeV) has a roughly 1/E dependence.
& The neutron flux spectrum for the fast energy region of a thermal reactor has a shape similar to that of the spectrum of neutrons emitted by the fission process.
& The reason for the 1/E flux dependence at intermediate energy levels in a thermal reactor is due to the neutrons' tendency to lose a constant fraction of energy per collision. Since the neutrons lose a greater amount at the higher energies, the neutrons tend to "pile up" at lower energies where they lose less energy per collision.
& The neutron flux spectrum for the slow region of a thermal reactor contains a

commercial nuclear power industry, results of job and task analyses, and independent input from contractors and operations-oriented personnel were all considered and included to some degree in developing the text material and learning objectives.

The DOE Fundamentals Handbooks represent the needs of various DOE nuclear facilities' fundamental training requirements. To increase their applicability to nonreactor nuclear facilities, the Reactor Operator Fundamentals Manual learning objectives were distributed to the Nuclear Facility Training Coordination Program Steering Committee for review and comment. To update their reactor-specific content, DOE Category A reactor training managers also reviewed and commented on the content. On the basis of feedback from these sources, information that applied to two or more DOE nuclear facilities was considered generic and was included. The final draft Fixture checkpoint marker-1051-xray is recorded on fixture page 1051.

of each of the handbooks was then reviewed by these two groups. This approach has resulted in revised modular handbooks that contain sufficient detail such that each facility may adjust the content to fit their specific needs.

Each handbook contains an abstract, a foreword, an overview, learning objectives, and text material, and is divided into modules so that content and order may be modified by individual DOE contractors to suit their specific training needs. Each handbook is supported by a separate examination bank with an answer key.

The DOE Fundamentals Handbooks have been prepared for the Assistant Secretary for Nuclear Energy, Office of Nuclear Safety Policy and Standards, by the DOE Training Coordination Program. This program is managed by EG&G Idaho, Inc.

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NUCLEAR PHYSICS AND REACTOR THEORY

OVERVIEW

The Department of Energy Fundamentals Handbook entitled Nuclear Physics and Reactor Theory was prepared as an information resource for personnel who are responsible for the operation of the Department's nuclear facilities. Almost all processes that take place in a nuclear facility involves the transfer of some type of energy. A basic understanding of nuclear physics and reactor theory is necessary for DOE nuclear facility operators, maintenance personnel, and the technical staff to safely operate and maintain the facility and facility support systems. The information in this handbook is presented to provide a foundation for applying engineering concepts to the job. This knowledge will help personnel understand the impact that their actions may have on the safe and reliable operation of facility components and systems.

The Nuclear Physics and Reactor Theory handbook consists of four modules that are contained in two volumes. The following is a brief description of the information presented in each module of the handbook.

Volume 1 of 2

Module 1 - Atomic and Nuclear Physics

Introduces concepts of atomic physics including the atomic nature of matter, the chart of the nuclides, radioactivity and radioactive decay, neutron interactions and fission, and the interaction of radiation with matter.

Module 2 - Reactor Theory (Nuclear Parameters)

Provides information on reactor theory and neutron characteristics. Includes topics such as neutron sources, neutron flux, neutron cross sections, reaction rates, neutron moderation, and prompt and delayed neutrons.

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OVERVIEW (Cont.)
Volume 2 of 2
Module 3 - Reactor Theory (Nuclear Parameters)
Explains the nuclear parameters associated with reactor theory. Topics include the neutron life cycle, reactivity and reactivity coefficients, neutron poisons, and control rods.
Module 4 - Reactor Theory (Reactor Operations)
Introduces the reactor operations aspect of reactor theory. Topics include subcritical multiplication, reactor kinetics, and reactor operation.
The information contained in this handbook is not all-encompassing. An attempt to present the entire subject of nuclear physics and reactor theory would be impractical. However, the Nuclear Physics and Reactor Theory handbook presents enough information to provide the reader with the fundamental knowledge necessary to understand the advanced theoretical concepts presented in other subject areas, and to understand basic system and equipment operation.
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Department of Energy
Fundamentals Handbook
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AND REACTOR THEORY
Module 1
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TERMINAL OBJECTIVE
1.0 Given sufficient information, DESCRIBE atoms, including components, structure, and nomenclature.
ENABLING OBJECTIVES
1.1 STATE the characteristics of the following atomic particles, including mass, charge, and location within the atom:
a. Proton
b. Neutron
c. Electron
1.2 DESCRIBE the Bohr model of an atom.
1.3 DEFINE the following terms:
a. Nuclide c. Atomic number
b. Isotope d. Mass number
1.4 Given the standard A
Fixture checkpoint marker-1055-xray is recorded on fixture page 1055.
ZX notation for a particular nuclide, DETERMINE the following:
a. Number of protons
b. Number of neutrons
c. Number of electrons
1.5 DESCRIBE the three forces that act on particles within the nucleus and affect the stability of the nucleus.
1.6 DEFINE the following terms:
a. Enriched uranium
b. Depleted uranium

a. Mass defect
b. Binding energy
1.8 Given the atomic mass for a nuclide and the atomic masses of a neutron, proton, and electron, CALCULATE the mass defect and binding energy of the nuclide.

TERMINAL OBJECTIVE
2.0 Given necessary references, DESCRIBE the various modes of radioactive decay.
ENABLING OBJECTIVES
2.1 DESCRIBE the following processes:
a. Alpha decay d. Electron capture
b. Beta-minus decay e. Internal conversions
c. Beta-plus decay f. Isomeric transitions
2.2 Given a Chart of the Nuclides, WRITE the radioactive decay chain for a nuclide.
2.3 EXPLAIN why one or more gamma rays typically accompany particle emission.
Fixture checkpoint marker-1056-xray is recorded on fixture page 1056.
2.4 Given the stability curve on the Chart of the Nuclides, DETERMINE the type of radioactive decay that the nuclides in each region of the chart will typically undergo.
2.5 DEFINE the following terms:
a. Radioactivity d. Radioactive decay constant
b. Curie e. Radioactive half-life
c. Becquerel
2.6 Given the number of atoms and either the half-life or decay constant of a nuclide, CALCULATE the activity.
2.7 Given the initial activity and the decay constant of a nuclide, CALCULATE the activity at any later time.
2.8 CONVERT between the half-life and decay constant for a nuclide.
2.9 Given the Chart of the Nuclides and the original activity, PLOT the radioactive decay curve for a nuclide on either linear or semi-log coordinates.
2.10 DEFINE the following terms:
a. Radioactive equilibrium
b. Transient radioactive equilibrium

TERMINAL OBJECTIVE
3.0 Without references, DESCRIBE the different nuclear interactions initiated by neutrons.
ENABLING OBJECTIVES
3.1 DESCRIBE the following scattering interactions between a neutron and a nucleus:
a. Elastic scattering
b. Inelastic scattering
3.2 STATE the conservation laws that apply to an elastic collision between a neutron and a nucleus.
3.3 DESCRIBE the following reactions where a neutron is absorbed in a nucleus:
a. Radiative capture
b. Particle ejection

TERMINAL OBJECTIVE

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ENABLING OBJECTIVES
4.1 EXPLAIN the fission process using the liquid drop model of a nucleus.
4.2 DEFINE the following terms:
a. Excitation energy
b. Critical energy
4.3 DEFINE the following terms:
a. Fissile material
b. Fissionable material
c. Fertile material
4.4 DESCRIBE the processes of transmutation, conversion, and breeding.
4.5 DESCRIBE the curve of Binding Energy per Nucleon versus mass number and give a qualitative description of the reasons for its shape.
4.6 EXPLAIN why only the heaviest nuclei are easily fissioned.
4.7 EXPLAIN why uranium-235 fissions with thermal neutrons and uranium-238 fissions only with fast neutrons.
4.8 CHARACTERIZE the fission products in terms of mass groupings and radioactivity.
4.9 Given the nuclides involved and their masses, CALCULATE the energy released from fission.
4.10 Given the curve of Binding Energy per Nucleon versus mass number, CALCULATE the energy released from fission.
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Atomic and Nuclear Physics DOE-HDBK-1019/1-93 OBJECTIVES
Fixture checkpoint marker-1057-xray is recorded on fixture page 1057.
TERMINAL OBJECTIVE
5.0 Without references, DESCRIBE how the various types of radiation interact with matter.
ENABLING OBJECTIVES
5.1 DESCRIBE interactions of the following with matter:
a. Alpha particle c. Positron
b. Beta particle d. Neutron
5.2 DESCRIBE the following ways that gamma radiation interacts with matter:
a. Compton scattering
b. Photoelectric effect
c. Pair production
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OBJECTIVES DOE-HDBK-1019/1-93 Atomic and Nuclear Physics
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NP-01 Page xii Rev. 0
Atomic and Nuclear Physics DOE-HDBK-1019/1-93 ATOMIC NATURE OF MATTER
ATOMIC NATURE OF MATTER
All matter is composed of atoms. The atom is the smallest amount of matter that retains the properties of an element. Atoms themselves are composed of smaller particles, but these smaller particles no longer have the same properties as the overall element.
EO 1.1 STATE the characteristics of the following atomic particles, including mass, charge, and location within the atom:
a. Proton
b. Neutron
c. Electron

EO 1.3 DEFINE the following terms:
a. Nuclide c. Atomic number
b. Isotope d. Mass number
EO 1.4 Given the standard A
ZX notation for a particular nuclide,
DETERMINE the following:
a. Number of protons
b. Number of neutrons
c. Number of electrons
EO 1.5 DESCRIBE the three forces that act on particles within the nucleus
and affect the stability of the nucleus.
Structure of Matter
Early Greek philosophers speculated that the earth was made up of different combinations of basic substances, or elements. They considered these basic elements to be earth, air, water, and fire. Modern science shows that the early Greeks held the correct concept that matter consists of a combination of basic elements, but they incorrectly identified the elements.

In 1661 the English chemist Robert Boyle published the modern criterion for an element. He defined an element to be a basic substance that cannot be broken down into any simpler substance after it is isolated from a compound, but can be combined with other elements to form compounds. To date, 105 different elements have been confirmed to exist, and researchers claim to have discovered three additional elements. Of the 105 confirmed elements, 90 exist in nature and 15 are man-made.
Another basic concept of matter that the Greeks debated was whether matter was continuous or discrete. That is, whether matter could be continuously divided and subdivided into ever smaller particles or whether eventually an indivisible particle would be encountered. Democritus in about 450 B.C. argued that substances were ultimately composed of small, indivisible particles that he labeled atoms. He further suggested that different substances were composed of different atoms
Fixture checkpoint marker-1058-xray is recorded on fixture page 1058.
or combinations of atoms, and that one substance could be converted into another by rearranging the atoms. It was impossible to conclusively prove or disprove this proposal for more than 2000 years.
The modern proof for the atomic nature of matter was first proposed by the English chemist John Dalton in 1803. Dalton stated that each chemical element possesses a particular kind of atom, and any quantity of the element is made up of identical atoms of this kind. What distinguishes one element from another element is the kind of atom of which it consists, and the basic physical difference between kinds of atoms is their weight.
Subatomic Particles
For almost 100 years after Dalton established the atomic nature of atoms, it was considered impossible to divide the atom into even smaller parts. All of the results of chemical experiments during this time indicated that the atom was indivisible. Eventually, experimentation into electricity and radioactivity indicated that particles of matter smaller than the atom did indeed exist. In 1906, J. J. Thompson won the Nobel Prize in physics for establishing the existence of electrons. Electrons are negatively-charged particles that have 1/1835 the mass of the hydrogen atom. Soon after the discovery of electrons, protons were discovered. Protons are relatively large particles that have almost the same mass as a hydrogen atom and a positive charge equal in magnitude (but opposite in sign) to that of the electron. The third subatomic particle to be

proton, but it is electrically neutral.
NP-01 Page 2 Rev. 0
Atomic and Nuclear Physics DOE-HDBK-1019/1-93 ATOMIC NATURE OF MATTER
Bohr Model of the Atom
The British physicist Ernest Rutherford postulated that the positive charge in an atom is concentrated in a small region called a nucleus at the center of the atom with electrons existing in orbits around it. Niels Bohr, coupling Rutherford's postulation with the quantum theory introduced by Max Planck, proposed that the atom consists of a dense nucleus of protons surrounded by electrons traveling in discrete orbits at fixed distances from the nucleus. An electron in one of these orbits or shells has a specific or discrete quantity of energy (quantum). When an electron moves from one allowed orbit to another allowed orbit, the energy difference between the two states is emitted or absorbed in the form of a single quantum of radiant energy called a photon. Figure 1 is Bohr's model of the hydrogen atom showing an electron as having just dropped from the third shell to the first shell with the emission of a photon that has an energy = hv. (h = Planck's constant = 6.63 x 10-34 J-s and v = frequency of the photon.) Bohr's theory was the first to successfully account for the discrete energy levels of this radiation as measured in the laboratory. Although Bohr's atomic model is designed specifically to explain the hydrogen atom, his theories apply generally to the structure of all atoms. Additional information on electron shell theory can be found in the Chemistry Fundamentals Handbook.
Figure 1 Bohr's Model of the Hydrogen Atom
Rev. 0 Page 3 NP-01
ATOMIC NATURE OF MATTER DOE-HDBK-1019/1-93 Atomic and Nuclear Physics
Properties of the three subatomic particles are listed in Table 1.
TABLE 1
Properties of Subatomic Particles
Particle Location Charge Mass
Neutron Nucleus none 1.008665 amu
Proton Nucleus +1 1.007277 amu
Electron Shells around nucleus -1 0.0005486 amu
Measuring Units on the Atomic Scale
The size and mass of atoms are so small that the use of normal measuring units, while possible, is often inconvenient. Units of measure have been defined for mass and energy on the atomic scale to make measurements more convenient to express. The unit of measure for mass is the atomic mass unit (amu). One atomic mass unit is equal to 1.66 x 10-24 grams. The reason for this particular value for the atomic mass unit will be discussed in a later chapter. Note from Table 1 that the mass of a neutron and a proton are both about 1 amu. The unit for energy is
Fixture checkpoint marker-1059-xray is recorded on fixture page 1059.
the electron volt (eV). The electron volt is the amount of energy acquired by a single electron when it falls through a potential difference of one volt. One electron volt is equivalent to 1.602 x 10-19 joules or 1.18 x 10-19 foot-pounds.
Nuclides
The total number of protons in the nucleus of an atom is called the atomic number of the atom and is given the symbol Z. The number of electrons in an electrically-neutral atom is the same as the number of protons in the nucleus. The number of neutrons in a nucleus is known as the neutron number and is given the symbol N. The mass number of the nucleus is the total number of nucleons, that is, protons and neutrons in the nucleus. The mass number is given the symbol A and can be found by the equation Z + N = A.
Each of the chemical elements has a unique atomic number because the atoms of different

particular element.

Each type of atom that contains a unique combination of protons and neutrons is called a nuclide. Not all combinations of numbers of protons and neutrons are possible, but about 2500 specific nuclides with unique combinations of neutrons and protons have been identified. Each nuclide is denoted by the chemical symbol of the element with the atomic number written as a subscript and the mass number written as a superscript, as shown in Figure 2. Because each element has a unique name, chemical symbol, and atomic number, only one of the three is necessary to identify the element. For this reason nuclides can also be identified by either the chemical name or the chemical symbol followed by the mass number (for example, U-235 or uranium-235).

Figure 2 Nomenclature for Identifying Nuclides

Fixture checkpoint marker-1060-xray is recorded on fixture page 1060.

Another common format is to use the abbreviation of the chemical element with the mass number superscripted (for example, 235U). In this handbook the format used in the text will usually be the element's name followed by the mass number. In equations and tables, the format in Figure 2 will usually be used.

Example:

State the name of the element and the number of protons, electrons, and neutrons in the nuclides listed below.

1
1H
10
5B
14
7N
11
4
4
8Cd
23
9
9
4Pu

Solution:

The name of the element can be found from the Periodic Table (refer to Chemistry Fundamentals Handbook) or the Chart of the Nuclides (to be discussed later). The number of protons and electrons are equal to Z. The number of neutrons is equal to Z - A.

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1
1H hydrogen 1 1 0
10
5B boron 5 5 5
14
7N nitrogen 7 7 7
11
4
4
8Cd cadmium 48 48 66
23
9
9
4Pu plutonium 94 94 145
Isotopes
Isotopes are nuclides that have the same atomic number and are therefore the same element, but differ in the number of neutrons. Most elements have a few stable isotopes and several unstable, radioactive isotopes. For example, oxygen has three stable isotopes that can be found in nature (oxygen-16, oxygen-17, and oxygen-18) and eight radioactive isotopes. Another example is hydrogen, which has two stable isotopes (hydrogen-1 and hydrogen-2) and a single radioactive
Fixture checkpoint marker-1061-xray is recorded on fixture page 1061.
isotope (hydrogen-3).
The isotopes of hydrogen are unique in that they are each commonly referred to by a unique name instead of the common chemical element name. Hydrogen-1 is almost always referred to as hydrogen, but the term protium is infrequently used also. Hydrogen-2 is commonly called deuterium and symbolized 2
1D. Hydrogen-3 is commonly called tritium and symbolized 31T. This
text will normally use the symbology 2
1H and 31H for deuterium and tritium, respectively.
Atomic and Nuclear Radii
The size of an atom is difficult to define exactly due to the fact that the electron cloud, formed by the electrons moving in their various orbitals, does not have a distinct outer edge. A reasonable measure of atomic size is given by the average distance of the outermost electron from the nucleus. Except for a few of the lightest atoms, the average atomic radii are approximately the same for all atoms, about 2 x 10 -8 cm.
Like the atom the nucleus does not have a sharp outer boundary. Experiments have shown that the nucleus is shaped like a sphere with a radius that depends on the atomic mass number of the atom. The relationship between the atomic mass number and the radius of the nucleus is shown in the following equation.
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Atomic and Nuclear Physics DOE-HDBK-1019/1-93 ATOMIC NATURE OF MATTER
r = (1.25 x 10 -13 cm) A1/3
where:
r = radius of the nucleus (cm)
A = atomic mass number (dimensionless)
The values of the nuclear radii for some light, intermediate, and heavy nuclides are shown in Table 2.
TABLE 2

SYNTHETIC FIXTURE PAGE 1062 OF 1200

Radii
Nuclide Radius of Nucleus
1
1H 1.25 x 10 -13 cm
10
5B 2.69 x 10 -13 cm
5
2
6
6Fe 4.78 x 10 -13 cm
17
7
8
2Hf 7.01 x 10 -13 cm
23
9
8
2U 7.74 x 10 -13 cm
25
9
2
8Cf 7.89 x 10 -13 cm
From the table, it is clear that the radius of a typical atom (e.g. 2 x 10 -8 cm) is more than 25,000 times larger than the radius of the largest nucleus.
Nuclear Forces
In the Bohr model of the atom, the nucleus consists of positively-charged protons and electrically■neutra as nucleons. One problem that the Bohr model of the atom presented was accounting for an
Fixture checkpoint marker-1062-xray is recorded on fixture page 1062.
attractive force to overcome the repulsive force between protons.
Two forces present in the nucleus are (1) electrostatic forces between charged particles and (2) gravitational forces between any two objects that have mass. It is possible to calculate the magnitude of the gravitational force and electrostatic force based upon principles from classical physics.
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Fg
G m1 m2
r
2
Fe
K Q1 Q2
r
2
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Newton stated that the gravitational force between two bodies is directly proportional to the masses of the two bodies and inversely proportional to the square of the distance between the bodies. This relationship is shown in the equation below.
where:

m = mass of first body (kilograms) 1
m = mass of second body (kilograms) 2
G = gravitational constant (6.67 x 10 N-m /kg)
-11 2 2
r = distance between particles (meters)
The equation illustrates that the larger the masses of the objects or the smaller the distance between the objects, the greater the gravitational force. So even though the masses of nucleons are very small, the fact that the distance between nucleons is extremely short may make the gravitational force significant. It is necessary to calculate the value for the gravitational force and compare it to the value for other forces to determine the significance of the gravitational force in the nucleus. The gravitational force between two protons that are separated by a distance of 10 meters is about 10 newtons. -20 -24
Coulomb's Law can be used to calculate the force between two protons. The electrostatic force is directly proportional to the electrical charges of the two particles and inversely proportional to the square of the distance between the particles. Coulomb's Law is stated as the following equation.
where:
F = electrostatic force (newtons) e
K = electrostatic constant (9.0 x 10 N-m /C)
9 2 2
Q = charge of first particle (coulombs) 1
Q = charge of second particle (coulombs) 2
r = distance between particles (meters)
Using this equation, the electrostatic force between two protons that are separated by a distance of 10 meters is about 10 newtons. Comparing this result with the calculation of the
-20 12
gravitational force (10 newtons) shows that the gravitational force is so small that it can be
-24
neglected.
Atomic and Nuclear Physics DOE-HDBK-1019/1-93 ATOMIC NATURE OF MATTER
If only the electrostatic and gravitational forces existed in the nucleus, then it would be impossible to have stable nuclei composed of protons and neutrons. The gravitational forces are much too small to hold the nucleons together compared to the electrostatic forces repelling the protons. Since stable atoms of neutrons and protons do exist, there must be another attractive
Fixture checkpoint marker-1063-xray is recorded on fixture page 1063.
force acting within the nucleus. This force is called the nuclear force.
The nuclear force is a strong attractive force that is independent of charge. It acts equally only between pairs of neutrons, pairs of protons, or a neutron and a proton. The nuclear force has a very short range; it acts only over distances approximately equal to the diameter of the nucleus (10 -13 cm). The attractive nuclear force between all nucleons drops off with distance much faster than the repulsive electrostatic force between protons.
TABLE 3
Forces Acting in the Nucleus
Force Interaction Range
Gravitational Very weak attractive force
between all nucleons Relatively long
Electrostatic Strong repulsive force between
like charged particles (protons) Relatively long

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all nucleons Extremely short
In stable atoms, the attractive and repulsive forces in the nucleus balance. If the forces do not balance, the atom cannot be stable, and the nucleus will emit radiation in an attempt to achieve a more stable configuration.
Summary
The important information in this chapter is summarized on the following page.
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ATOMIC NATURE OF MATTER DOE-HDBK-1019/1-93 Atomic and Nuclear Physics
Atomic Nature of Matter Summary
Atoms consist of three basic subatomic particles. These particles are the proton, the neutron, and the electron.
Fixture checkpoint marker-1064-xray is recorded on fixture page 1064.
Protons are particles that have a positive charge, have about the same mass as a hydrogen atom, and exist in the nucleus of an atom.
Neutrons are particles that have no electrical charge, have about the same mass as a hydrogen atom, and exist in the nucleus of an atom.
Electrons are particles that have a negative charge, have a mass about eighteen hundred times smaller than the mass of a hydrogen atom, and exist in orbital shells around the nucleus of an atom.
The Bohr model of the atom consists of a dense nucleus of protons and neutrons (nucleons) surrounded by electrons traveling in discrete orbits at fixed distances from the nucleus.
Nuclides are atoms that contain a particular number of protons and neutrons.
Isotopes are nuclides that have the same atomic number and are therefore the same element, but differ in the number of neutrons.
The atomic number of an atom is the number of protons in the nucleus.
The mass number of an atom is the total number of nucleons (protons and neutrons) in the nucleus.
The notation A
ZX is used to identify a specific nuclide. "Z" represents the atomic number, which is equal to the number of protons. "A" represents the mass number, which is equal to the number of nucleons. "X" represents the chemical symbol of the element.
Number of protons = Z
Number of electrons = Z
Number of neutrons = A - Z
The stability of a nucleus is determined by the different forces interacting within it. The electrostatic force is a relatively long-range, strong, repulsive force that acts between the positively charged protons. The nuclear force is a relatively short-range attractive force between all nucleons. The gravitational force the long range, relatively weak attraction between masses, is negligible compared to the other forces.
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Atomic and Nuclear Physics DOE-HDBK-1019/1-93 CHART OF THE NUCLIDES
CHART OF THE NUCLIDES
The Chart of the Nuclides, like the Periodic Table, is a convenient format for presenting a large amount of scientific information in an organized manner.

a. Enriched uranium
b. Depleted uranium
Chart of the Nuclides
A tabulated chart called the Chart of the Nuclides lists the stable and unstable nuclides in addition to pertinent information about each one. Figure 3 shows a small portion of a typical chart. This chart plots a box for each individual nuclide, with the number of protons (Z) on the vertical axis and the number of neutrons (N = A - Z) on the horizontal axis.
The completely gray squares indicate stable isotopes. Those in white squares are artificially radioactive, meaning that they are produced by artificial techniques and do not occur naturally. By consulting a complete chart, other types of isotopes can be found, such as naturally occurring radioactive types (but none are found in the region of the chart that is illustrated in Figure 3). Located in the box on the far left of each horizontal row is general information about the element. The box contains the chemical symbol of the element in addition to the average atomic weight of the naturally occurring substance and the average thermal neutron absorption cross section, which will be discussed in a later module. The known isotopes (elements with the same atomic number Z but different mass number A) of each element are listed to the right.

Fixture checkpoint marker-1065-xray is recorded on fixture page 1065.
Figure 3 Nuclide Chart for Atomic Numbers 1 to 6

Information for Stable Nuclides
For the stable isotopes, in addition to the symbol and the atomic mass number, the number percentage of each isotope in the naturally occurring element is listed, as well as the thermal neutron activation cross section and the mass in atomic mass units (amu). A typical block for a stable nuclide from the Chart of the Nuclides is shown in Figure 4.
Figure 4 Stable Nuclides
Information for Unstable Nuclides
For unstable isotopes the additional information includes the half life, the mode of decay (for example, β^-, α), the total disintegration energy in MeV (million electron volts), and the mass in amu when available. A typical block for an unstable nuclide from the Chart of the Nuclides is shown in Figure 5.
Figure 5 Unstable Nuclides

Neutron - Proton Ratios
Figure 6 shows the distribution of the stable nuclides plotted on the same axes as the Chart of the Nuclides. As the mass numbers become higher, the ratio of neutrons to protons in the nucleus becomes larger. For helium-4 (2 protons and 2 neutrons) and oxygen-16 (8 protons and 8 neutrons) this ratio is unity. For indium-115 (49 protons and 66 neutrons) the ratio of neutrons to protons has increased to 1.35, and for uranium-238 (92 protons and 146 neutrons) the neutron■to-proton
Figure 6 Neutron - Proton Plot of the Stable Nuclides
If a heavy nucleus were to split into two fragments, each fragment would form a nucleus that would have approximately the same neutron-to-proton ratio as the heavy nucleus. This high neutron-to-proton ratio places the fragments below and to the right of the stability curve displayed by Figure 6. The instability caused by this excess of neutrons is generally rectified

nucleus toward a more stable neutron-to-proton ratio.
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■
Atomic and Nuclear Physics DOE-HDBK-1019/1-93 CHART OF THE NUCLIDES
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Natural Abundance of Isotopes
The relative abundance of an isotope in nature compared to other isotopes of the same element is relatively constant. The Chart of the Nuclides presents the relative abundance of the naturally occurring isotopes of an element in units of atom percent. Atom percent is the percentage of the atoms of an element that are of a particular isotope. Atom percent is abbreviated as a/o.
For example, if a cup of water contains 8.23 x 10 atoms of oxygen, and the isotopic abundance 24 of oxygen-18 is 0.20%, then there are 1.65 x 10 atoms of oxygen-18 in the cup. 22
The atomic weight for an element is defined as the average atomic weight of the isotopes of the element. The atomic weight for an element can be calculated by summing the products of the isotopic abundance of the isotope with the atomic mass of the isotope.
Example:
Calculate the atomic weight for the element lithium. Lithium-6 has an atom percent abundance of 7.5% and an atomic mass of 6.015122 amu. Lithium-7 has an atomic abundance of 92.5% and an atomic mass of 7.016003 amu.
Solution:
The other common measurement of isotopic abundance is weight percent (w/o). Weight percent is the percent weight of an element that is a particular isotope. For example, if a sample of material contained 100 kg of uranium that was 28 w/o uranium-235, then 28 kg of uranium-235 was present in the sample.
Enriched and Depleted Uranium
Fixture checkpoint marker-1066-xray is recorded on fixture page 1066.
Natural uranium mined from the earth contains the isotopes uranium-238, uranium-235 and uranium-234. The majority (99.2745%) of all the atoms in natural uranium are uranium-238. Most of the remaining atoms (0.72%) are uranium-235, and a slight trace (0.0055%) are uranium-234. Although all isotopes of uranium have similar chemical properties, each of the isotopes has significantly different nuclear properties. For reasons that will be discussed in later modules, the isotope uranium-235 is usually the desired material for use in reactors.
A vast amount of equipment and energy are expended in processes that separate the isotopes of uranium (and other elements). The details of these processes are beyond the scope of this module. These processes are called enrichment processes because they selectively increase the proportion of a particular isotope. The enrichment process typically starts with feed material that has the proportion of isotopes that occur naturally. The process results in two types of
CHART OF THE NUCLIDES DOE-HDBK-1019/1-93 Atomic and Nuclear Physics
In the case of uranium, the natural uranium ore is 0.72 a/o uranium-235. The desired outcome of the enrichment process is to produce enriched uranium. Enriched uranium is defined as uranium in which the isotope uranium-235 has a concentration greater than its natural value. The enrichment process will also result in the byproduct of depleted uranium. Depleted uranium is defined as uranium in which the isotope uranium-235 has a concentration less than its natural value. Although depleted uranium is referred to as a by-product of the enrichment process, it does have uses in the nuclear field and in commercial and defense industries.
Summary
The important information in this chapter is summarized below.
Chart of the Nuclides Summary

concentration greater than its natural value of 0.7%.
Depleted uranium is uranium in which the isotope uranium-235 has a concentration less than its natural value of 0.7%.

MASS DEFECT AND BINDING ENERGY
The separate laws of Conservation of Mass and Conservation of Energy are not applied strictly on the nuclear level. It is possible to convert between mass and energy. Instead of two separate conservation laws, a single conservation law states that the sum of mass and energy is conserved. Mass does not magically appear and disappear at random. A decrease in mass will be accompanied by a corresponding increase in energy and vice versa.
EO 1.7 DEFINE the following terms:
a. Mass defect
b. Binding energy
EO 1.8 Given the atomic mass for a nuclide and the atomic masses of a neutron, proton, and electron, CALCULATE the mass defect and binding energy of the nuclide.
Mass Defect
Careful measurements have shown that the mass of a particular atom is always slightly less than the sum of the masses of the individual neutrons, protons, and electrons of which the atom consists. The difference between the mass of the atom and the sum of the masses of its parts is called the mass defect (Δm). The mass defect can be calculated using Equation (1-1). In calculating the mass defect it is important to use the full accuracy of mass measurements because the difference in mass is small compared to the mass of the atom. Rounding off the masses of atoms and particles to three or four significant digits prior to the calculation will result in a calculated mass defect of zero.
$\Delta m = [Z(mp + me) + (A-Z)mn] - matom$ (1-1)
where:
Δm = mass defect (amu)
mp = mass of a proton (1.007277 amu)
mn = mass of a neutron (1.008665 amu)
Fixture checkpoint marker-1067-xray is recorded on fixture page 1067.
me = mass of an electron (0.000548597 amu)
matom = mass of nuclide A
ZX (amu)
Z = atomic number (number of protons)
A = mass number (number of nucleons)

Example:
Calculate the mass defect for lithium-7. The mass of lithium-7 is 7.016003 amu.
Solution:
Δ m Z m
p
me A Z mn matom
Δ m 3 1.007826 amu 7 3 1.008665 amu 7.016003 amu
Δ m 0.0421335 amu

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The loss in mass, or mass defect, is due to the conversion of mass to binding energy when the nucleus is formed. Binding energy is defined as the amount of energy that must be supplied to a nucleus to completely separate its nuclear particles (nucleons). It can also be understood as the amount of energy that would be released if the nucleus was formed from the separate particles. Binding energy is the energy equivalent of the mass defect. Since the mass defect was converted to binding energy (BE) when the nucleus was formed, it is possible to calculate the binding energy using a conversion factor derived by the mass-energy relationship from Einstein's Theory of Relativity.
Einstein's famous equation relating mass and energy is E = mc2 where c is the velocity of light (c = 2.998 x 108 m/sec). The energy equivalent of 1 amu can be determined by inserting this quantity of mass into Einstein's equation and applying conversion factors.
E m c 2
1 amu
1.6606 x 10 27 kg
1 amu
2.998 x 108 m
sec
2
1 N
1
kg m
sec2
1 J
1 N m
1.4924 x 10 10 J 1 MeV
1.6022 x 10 13 J
931.5 MeV
Conversion Factors:
1 amu = 1.6606 x 10 -27 kg
1 newton = 1 kg-m/sec2
1 joule = 1 newton-meter
1 MeV = 1.6022 x 10-13 joules
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Atomic and Nuclear Physics DOE-HDBK-1019/1-93 MASS DEFECT AND BINDING ENERGY
Since 1 amu is equivalent to 931.5 MeV of energy, the binding energy can be calculated using Equation (1-2).
(1-2)
B.E. ∆m
931.5 MeV
Fixture checkpoint marker-1068-xray is recorded on fixture page 1068.
1 amu
Example:
Calculate the mass defect and binding energy for uranium-235. One uranium-235 atom has a mass of 235.043924 amu.
Solution:
Step 1: Calculate the mass defect using Equation (1-1).
∆m Z m
p

∆m 92 1.007826 amu 235 92 1.008665 amu 235.043924 amu
∆m 1.91517 amu
Step 2: Use the mass defect and Equation (1-2) to calculate the binding energy.
B.E. ∆m
931.5 MeV
1 amu
1.91517 amu
931.5 MeV
1 amu
1784 MeV
Energy Levels of Atoms
The electrons that circle the nucleus move in fairly well-defined orbits. Some of these electrons are more tightly bound in the atom than others. For example, only 7.38 eV is required to remove the outermost electron from a lead atom, while 88,000 eV is required to remove the innermost electron. The process of removing an electron from an atom is called ionization, and the energy required to remove the electron is called the ionization energy.
Fixture checkpoint marker-1069-xray is recorded on fixture page 1069.
In a neutral atom (number of electrons = Z) it is possible for the electrons to be in a variety of different orbits, each with a different energy level. The state of lowest energy is the one in which the atom is normally found and is called the ground state. When the atom possesses more energy than its ground state energy, it is said to be in an excited state.

An atom cannot stay in the excited state for an indefinite period of time. An excited atom will eventually transition to either a lower-energy excited state, or directly to its ground state, by emitting a discrete bundle of electromagnetic energy called an x-ray. The energy of the x-ray will be equal to the difference between the energy levels of the atom and will typically range from several eV to 100,000 eV in magnitude.
Energy Levels of the Nucleus
The nucleons in the nucleus of an atom, like the electrons that circle the nucleus, exist in shells that correspond to energy states. The energy shells of the nucleus are less defined and less understood than those of the electrons. There is a state of lowest energy (the ground state) and discrete possible excited states for a nucleus. Where the discrete energy states for the electrons of an atom are measured in eV or keV, the energy levels of the nucleus are considerably greater and typically measured in MeV.
A nucleus that is in the excited state will not remain at that energy level for an indefinite period. Like the electrons in an excited atom, the nucleons in an excited nucleus will transition towards their lowest energy configuration and in doing so emit a discrete bundle of electromagnetic radiation called a gamma ray (γ-ray). The only differences between x-rays and γ-rays are their energy levels and whether they are emitted from the electron shell or from the nucleus.
The ground state and the excited states of a nucleus can be depicted in a nuclear energy-level diagram. The nuclear energy-level diagram consists of a stack of horizontal bars, one bar for each of the excited states of the nucleus. The vertical distance between the bar representing an

Figure 7 Energy Level Diagram - Nickel-60

ground state is proportional to the energy
level of the excited state with respect to
the ground state. This difference in
energy between the ground state and the
excited state is called the excitation energy
of the excited state. The ground state of
a nuclide has zero excitation energy. The
bars for the excited states are labeled with
their respective energy levels. Figure 7 is
the energy level diagram for nickel-60.
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Atomic and Nuclear Physics DOE-HDBK-1019/1-93 MASS DEFECT AND BINDING ENERGY
Summary
The important information in this chapter is summarized below.
Mass Defect and Binding Energy Summary
Mass defect is the difference between the mass of the atom and the sum of the
masses of its constituent parts.
Binding energy is the amount of energy that must be supplied to a nucleus to
completely separate its nuclear particles. Binding energy is the energy equivalent
of the mass defect.
Mass defect can be calculated by using the equation below.
$\Delta m = [Z(mp + me) + (A-Z)mn] - matom$
Binding energy can be calculated by multiplying the mass defect by the factor
Fixture checkpoint marker-1070-xray is recorded on fixture page 1070.
of 931.5 MeV per amu.
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MODES OF RADIOACTIVE DECAY
Most atoms found in nature are stable and do not emit particles or energy that
change form over time. Some atoms, however, do not have stable nuclei. These
atoms emit radiation in order to achieve a more stable configuration.
EO 2.1 DESCRIBE the following processes:
a. Alpha decay d. Electron capture
b. Beta-minus decay e. Internal conversions
c. Beta-plus decay f. Isomeric transitions
EO 2.2 Given a Chart of the Nuclides, WRITE the radioactive decay
chain for a nuclide.
EO 2.3 EXPLAIN why one or more gamma rays typically accompany
particle emission.
EO 2.4 Given the stability curve on the Chart of the Nuclides,
DETERMINE the type of radioactive decay that the nuclides in
each region of the chart will typically undergo.
Stability of Nuclei
As mass numbers become larger, the ratio of neutrons to protons in the nucleus becomes larger for the stable nuclei. Non-stable nuclei may have an excess or deficiency of neutrons and undergo a transformation process known as beta (β) decay. Non-stable nuclei can also undergo a variety of other processes such as alpha (α) or neutron (n) decay. As a result of these decay processes, the final nucleus is in a more stable or more tightly bound configuration.

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In 1896, the French physicist Becquerel discovered that crystals of a uranium salt emitted rays that were similar to x-rays in that they were highly penetrating, could affect a photographic plate, and induced electrical conductivity in gases. Becquerel's discovery was followed in 1898 by the identification of two other radioactive elements, polonium and radium, by Pierre and Marie Curie.

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Atomic and Nuclear Physics DOE-HDBK-1019/1-93 MODES OF RADIOACTIVE DECAY

Heavy elements, such as uranium or thorium, and their unstable decay chain elements emit radiation in their naturally occurring state. Uranium and thorium, present since their creation at the beginning of geological time, have an extremely slow rate of decay. All naturally occurring nuclides with atomic numbers greater than 82 are radioactive.

Nuclear Decay

Whenever a nucleus can attain a more stable (i.e., more tightly bound) configuration by emitting radiation, a spontaneous disintegration process known as radioactive decay or nuclear decay may occur. In practice, this "radiation" may be electromagnetic radiation, particles, or both.

Detailed studies of radioactive decay and nuclear reaction processes have led to the formulation of useful conservation principles. The four principles of most interest in this module are discussed below.

Conservation of electric charge implies that charges are neither created nor destroyed. Single positive and negative charges may, however, neutralize each other. It is also possible for a neutral particle to produce one charge of each sign.

Conservation of mass number does not allow a net change in the number of nucleons. However, the conversion of a proton to a neutron and vice versa is allowed.

Conservation of mass and energy implies that the total of the kinetic energy and the energy equivalent of the mass in a system must be conserved in all decays and reactions. Mass can be converted to energy and energy can be converted to mass, but the sum of mass and energy must be constant.

Conservation of momentum is responsible for the distribution of the available kinetic energy among product nuclei, particles, and/or radiation. The total amount is the same before and after the reaction even though it may be distributed differently among entirely different nuclides

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and/or particles.

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■ ■ ■ ■ ■ ■

■ ■ ■ ■ ■

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Alpha Decay (■)

Alpha decay is the emission of alpha particles (helium nuclei) which may be represented as either $^{4}_{2}He$ or $^{4}_{2}■$. When an unstable nucleus ejects an alpha particle, the atomic number is reduced by 2 and the mass number decreased by 4. An example is uranium-234 which decays by the ejection of an alpha particle accompanied by the emission of a 0.068 MeV gamma.

The combined kinetic energy of the daughter nucleus (Thorium-230) and the ■ particle is designated as KE. The sum of the KE and the gamma energy is equal to the difference in mass between the original nucleus (Uranium-234) and the final particles (equivalent to the binding energy released, since ■m = BE). The alpha particle will carry off as much as 98% of the kinetic energy and, in most cases, can be considered to carry off all the kinetic energy.

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Beta decay is the emission of electrons of nuclear rather than orbital origin. These particles are electrons that have been expelled by excited nuclei and may have a charge of either sign.
If both energy and momentum are to be conserved, a third type of particle, the neutrino, ■, must be involved. The neutrino is associated with positive electron emission, and its antiparticle, the antineutrino, , is emitted with a negative electron. These uncharged particles have only the weakest interaction with matter, no mass, and travel at the speed of light. For all practical purposes, they pass through all materials with so few interactions that the energy they possess cannot be recovered. The neutrinos and antineutrinos are included here only because they carry a portion of the kinetic energy that would otherwise belong to the beta particle, and therefore, must be considered for energy and momentum to be conserved. They are normally ignored since they are not significant in the context of nuclear reactor applications.
Negative electron emission, represented as e, -1 -1■, or simply as e or ■ , effectively converts a neutro to a proton, thus increasing the atomic number by one and leaving the mass number unchanged. This is a common mode of decay for nuclei with an excess of neutrons, such as fission fragments below and to the right of the neutron-proton stability curve (refer to Figure 6). An example of a typical beta minus-decay reaction is shown below.
■ ■
■
■ ■
■ ■ ■ ■
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Figure 9 Orbital Electron Capture
Positively charged electrons (beta-plus) are known as positrons. Except for sign, they are nearly identical to their negatively charged cousins. When a positron, represented as e, ■, or simply as +1 +1 0 0
e or ■ , is ejected from the nucleus, the atomic number is decreased by one and the mass number + + remains unchanged. A proton has been converted to a neutron. An example of a typical positron (beta-plus) decay is shown below.
Electron Capture (EC, K-capture)
Nuclei having an excess of protons may capture an electron from one of the inner orbits which immediately combines with a proton in the nucleus to form a neutron. This process is called electron capture (EC). The electron is normally captured from the innermost orbit (the K-shell), and, consequently, this process is sometimes called K-capture. The following example depicts electron capture.
A neutrino is formed at the same time that the neutron is formed, and
Fixture checkpoint marker-1072-xray is recorded on fixture page 1072.
energy carried off by it serves to conserve momentum. Any energy that is available due to the atomic mass of the product being appreciably less than that of the parent will appear as gamma radiation. Also, there will always be characteristic x-rays given off when an electron from one of the higher energy shells moves in to fill

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Electron capture is shown
graphically in Figure 8.
Electron capture and positron
emission result in the production of
the same daughter product, and
they exist as competing processes.
For positron emission to occur, however, the mass of the daughter product must be less than the mass of the parent by an amount equal to at least twice the mass of an electron. This mass difference between the parent and daughter is necessary to account for two items present in the parent but not in the daughter. One item is the positron ejected from the nucleus of the parent. The other item is that the daughter product has one less orbital electron than the parent. If this requirement is not met, then orbital electron capture takes place exclusively.
■
■ ■ ■ ■
Fixture checkpoint marker-1073-xray is recorded on fixture page 1073.
■ ■
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Gamma Emission (■)
Gamma radiation is a high-energy electromagnetic radiation that originates in the nucleus. It is emitted in the form of photons, discrete bundles of energy that have both wave and particle properties. Often a daughter nuclide is left in an excited state after a radioactive parent nucleus undergoes a transformation by alpha decay, beta decay, or electron capture. The nucleus will drop to the ground state by the emission of gamma radiation.
Internal Conversion
The usual method for an excited nucleus to go from the excited state to the ground state is by emission of gamma radiation. However, in some cases the gamma ray (photon) emerges from the nucleus only to interact with one of the innermost orbital electrons and, as a result, the energy of the photon is transferred to the electron. The gamma ray is then said to have undergone internal conversion. The conversion electron is ejected from the atom with kinetic energy equal to the gamma energy minus the binding energy of the orbital electron. An orbital electron then drops to a lower energy state to fill the vacancy, and this is accompanied by the emission of characteristic x-rays.
Isomers and Isomeric Transition
Isomeric transition commonly occurs immediately after particle emission; however, the nucleus may remain in an excited state for a measurable period of time before dropping to the ground state at its own characteristic rate. A nucleus that remains in such an excited state is known as a nuclear isomer because it differs in energy and behavior from other nuclei with the same atomic number and mass number. The decay of an excited nuclear isomer to a lower energy level is called an isomeric transition. It is also possible for the excited isomer to decay by some alternate means, for example, by beta emission.
An example of gamma emission accompanying particle emission is illustrated by the decay of nitrogen-16 below.
■
■
■
■
■

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■
■
■
■
■
■

Decay Chains

When an unstable nucleus decays, the resulting daughter nucleus is not necessarily stable. The nucleus resulting from the decay of a parent is often itself unstable, and will undergo an additional decay. This is especially common among the larger nuclides.

It is possible to trace the steps of an unstable atom as it goes through multiple decays trying to achieve stability. The list of the original unstable nuclide, the nuclides that are involved as intermediate steps in the decay, and the final stable nuclide is known as the decay chain. One common method for stating the decay chain is to state each of the nuclides involved in the standard $^{A}_{Z}X$ format. Arrows are used between nuclides to indicate where decays occur, with the type of decay indicated above the arrow and the half-life below the arrow. The half-life for decay will be discussed in the next chapter.

Example:

Fixture checkpoint marker-1074-xray is recorded on fixture page 1074.

Write the decay chains for rubidium-91 and actinium-215. Continue the chains until a stable nuclide or a nuclide with a half-life greater than 1 x 10 years is reached. 6

Solution:

Predicting Type of Decay

Radioactive nuclides tend to decay in a way that results in a daughter nuclide that lies closer to the line of stability. Due to this, it is possible to predict the type of decay that a nuclide will undergo based on its location relative to the line of stability on the Chart of the Nuclides.

Figure 9 illustrates the type of decay nuclides in different regions of the chart will typically undergo. Nuclides that are below and to the right of the line of stability will usually undergo ■ decay. - Nuclides that are above and to the left of the line of stability will usually undergo either ■ decay or + electron capture. Most nuclides that will undergo ■ decay are found in the upper right hand region of the chart. These are general rules that have many exceptions, especially in the region of the heavy nuclides.

Figure 9 Types of Radioactive Decay Relative to the Line of Stability

Summary

The important information in this chapter is summarized below.

Modes of Radioactive Decay Summary

Alpha decay is the emission of an alpha particle (2 protons and 2 neutrons) from an unstable nucleus. The daughter nuclide has an atomic number 2 less than the parent nuclide and a mass number 4 less than the parent nuclide. The daughter nucleus commonly releases its excitation energy by gamma emission.

Beta-minus decay effectively converts a neutron to a proton and an electron, which is immediately ejected from the nucleus. The daughter nuclide has its

parent.
Beta-plus decay effectively converts a proton to a neutron and a positron, which is immediately ejected from the nucleus. The daughter nuclide has its atomic number decreased by 1 and the same mass number compared to the parent.
In electron capture, the nucleus absorbs an electron from the innermost orbit. This electron combines with a proton to form a neutron.
Internal conversion occurs when a gamma ray, emitted by the nucleus as it goes from the excited state to the ground state, interacts with one of the innermost electrons of the same atom. The electron is ejected from the atom.
An isomeric transition is the decay of an excited nucleus to a lower-energy level by the emission of a gamma ray.
Decay chains can be found by tracing the steps an unstable atom goes through as it tries to achieve stability.
Many modes of radioactive decay result in a daughter nuclide that has an energy level above the ground state. This excitation energy is usually released immediately in the form of a gamma ray.
The type of decay that a nuclide will typically undergo can be determined by its relationship to the line of stability on the Chart of the Nuclides. Nuclides that lie below and to the right of the line of stability will typically beta minus decay. Nuclides above and to the left of the line will typically either beta plus decay or electron capture. Most alpha emitters are found in the upper, right-hand corner of the chart.

RADIOACTIVITY
The rate at which a sample of radioactive material decays is not constant. As individual atoms of the material decay, there are fewer of those types of atoms remaining. Since the rate of decay is directly proportional to the number of
Fixture checkpoint marker-1075-xray is recorded on fixture page 1075.
atoms, the rate of decay will decrease as the number of atoms decreases.
EO 2.5 DEFINE the following terms:
a. Radioactivity d. Radioactive decay constant
b. Curie e. Radioactive half-life
c. Becquerel
EO 2.6 Given the number of atoms and either the half-life or decay constant of a nuclide, CALCULATE the activity.
EO 2.7 Given the initial activity and the decay constant of a nuclide, CALCULATE the activity at any later time.
EO 2.8 CONVERT between the half-life and decay constant for a nuclide.
EO 2.9 Given the Chart of the Nuclides and the original activity, PLOT the radioactive decay curve for a nuclide on either linear or semi-log coordinates.
EO 2.10 DEFINE the following terms:
a. Radioactive equilibrium
b. Transient radioactive equilibrium
Radioactive Decay Rates
Radioactivity is the property of certain nuclides of spontaneously emitting particles or gamma

at which a single nucleus will decay cannot be determined. However, the average behavior of a very large sample can be predicted accurately by using statistical methods. These studies have revealed that there is a certain probability that in a given time interval a certain fraction of the nuclei within a sample of a particular nuclide will decay. This probability per unit time that an atom of a nuclide will decay is known as the radioactive decay constant, λ. The units for the decay constant are inverse time such as 1/second, 1/minute, 1/hour, or 1/year. These decay constant units can also be expressed as second-1, minute-1, hour-1, and year-1.

■■

The activity (A) of a sample is the rate of decay of that sample. This rate of decay is usually measured in the number of disintegrations that occur per second. For a sample containing millions of atoms, the activity is the product of the decay constant and the number of atoms present in the sample.

The relationship between the activity, number of atoms, and decay constant is shown in Equation (1-3).

A = ■ N (1-3)

where:

A = Activity of the nuclide (disintegrations/second)
■ = decay constant of the nuclide (second)-1
N = Number of atoms of the nuclide in the sample

Since ■ is a constant, the activity and the number of atoms are always proportional.

Units of Measurement for Radioactivity

Two common units to measure the activity of a substance are the curie (Ci) and becquerel (Bq). A curie is a unit of measure of the rate of radioactive decay equal to 3.7 x 10 disintegrations 10 per second. This is approximately equivalent to the number of disintegrations that one gram of radium-226 will undergo in one second. A becquerel is a more fundamental unit of measure of radioactive decay that is equal to 1 disintegration per second. Currently, the curie is more widely used in the United States, but usage of the becquerel can be expected to broaden as the metric system slowly comes into wider use. The conversion between curies and becquerels is shown below.

1 curie = 3.7 x 10 becquerels 10

Variation of Radioactivity Over Time

The rate at which a given radionuclide sample decays is stated in Equation (1-3) as being equal Fixture checkpoint marker-1076-xray is recorded on fixture page 1076.
to the product of the number of atoms and the decay constant. From this basic relationship it is possible to use calculus to derive an expression which can be used to calculate how the number of atoms present will change over time. The derivation is beyond the scope of this text, but Equation (1-4) is the useful result.

(1-4)

where:

N = number of atoms present at time t
N = number of atoms initially present o
■ = decay constant (time)-1
t = time

A Ao
e

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A Aoe
t
A
Ao
e
t
ln A
Ao
t
t
ln
A
Fixture checkpoint marker-1077-xray is recorded on fixture page 1077.
Ao
t
1/2
ln
1
2
t
1/2
ln 2 0.693
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Since the activity and the number of atoms are always proportional, they may be used interchangeably to describe any given radionuclide population. Therefore, the following is true.
(1-5)
where:
A = activity present at time t
A = activity initially present o
= decay constant (time)
-1
t = time
Radioactive Half-Life
One of the most useful terms for estimating how quickly a nuclide will decay is the radioactive half-life. The radioactive half-life is defined as the amount of time required for the activity to decrease to one-half of its original value. A relationship between the half-life and decay constant can be developed from Equation (1-5). The half-life can be calculated by solving Equation (1-5) for the time, t, when the current activity, A, equals one-half the initial activity A .
o
First, solve Equation (1-5) for t.
If A is equal to one-half of Ao, then A/Ao is equal to one-half. Substituting this in the equation above yields an expression for t .
1/2
(1-6)
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The basic features of decay of a radionuclide sample are shown by the graph in Figure 10.
Figure 10 Radioactive Decay as a Function of Time in Units of Half-Life

noted to decrease by one-half of this value in a time of one half-life. Additional decreases occur so that whenever one half-life elapses, the number of atoms drops to one-half of what its value was at the beginning of that time interval. After five half-lives have elapsed, only 1/32, or 3.1%, of the original number of atoms remains. After seven half-lives, only 1/128, or 0.78%, of the atoms remains. The number of atoms existing after 5 to 7 half-lives can usually be assumed to be negligible. The Chemistry Fundamentals Handbook contains additional information on calculating the number of atoms contained within a sample.
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■
■
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Example:
A sample of material contains 20 micrograms of californium-252.
Californium-252 has a half-life of 2.638 years.
Calculate:
(a) The number of californium-252 atoms initially present
(b) The activity of the californium-252 in curies
(c) The number of californium-252 atoms that will remain in 12 years
Fixture checkpoint marker-1078-xray is recorded on fixture page 1078.
(d) The time it will take for the activity to reach 0.001 curies
Solution:
(a) The number of atoms of californium-252 can be determined as below.
(b) First, use Equation (1-6) to calculate the decay constant.
Use this value for the decay constant in Equation (1-3) to determine the activity.
N No
e
t
4.78 x 1016 e
(0.263/yr) (12 yr)
2.04 x 1015
A Aoe
t
A
Ao
e
t
ln A
Ao
t
t
ln
A
Ao
t
ln
0.001 Ci
0.0108 Ci

t 9.05 years
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(c) The number of californium atoms that will remain in 12 years can be calculated from Equation (1-4).
(d) The time that it will take for the activity to reach 0.001 Ci can be determined from Equation (1-5). First, solve Equation (1-5) for time.
Inserting the appropriate values in the right side of this equation will result in the required time.
Plotting Radioactive Decay
It is useful to plot the activity of a nuclide as it changes over time. Plots of this type can be used to determine when the activity will fall below a certain level. This plot is usually done showing activity on either a linear or a logarithmic scale. The decay of the activity of a single nuclide on a logarithmic scale will plot as a straight line because the decay is exponential.
t
1/2
0.693
0.693
t
1/2
0.693
7.13 seconds
0.0972 second 1
A Aoe
t
RADIOACTIVITY DOE-HDBK-1019/1-93 Atomic and Nuclear Physics
Fixture checkpoint marker-1079-xray is recorded on fixture page 1079.
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Example:
Plot the radioactive decay curve for nitrogen-16 over a period of 100 seconds. The initial activity is 142 curies and the half-life of nitrogen-16 is 7.13 seconds. Plot the curve on both linear rectangular coordinates and on a semi-log scale.
Solution:
First, use Equation (1-6) to calculate the decay constant corresponding to a half-life of 7.13 seconds.
Use the decay constant determined above to calculate the activity at various times using Equation (1-5).
Time Activity
0 seconds 142 Ci
20 seconds 20.3 Ci
40 seconds 2.91 Ci
60 seconds 0.416 Ci
80 seconds 0.0596 Ci
100 seconds 0.00853 Ci
■■
■■
■■
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Figure 11 Linear and Semi-log Plots of Nitrogen-16 Decay

Plotting the data points calculated above on both linear and semilog scales results in the graphs shown in Figure 11.

If a substance contains more than one radioactive nuclide, the total activity is the sum of the individual activities of each nuclide. As an example, consider a sample of material that contained 1 x 10 atoms of iron-59 that has a half-life of 44.51 days (■ = 1.80 x 10 sec), 6 -7 -1 1 x 10 atoms of manganese-54 that has a half-life of 312.2 days (■ = 2.57 x 10 sec), and 1 6 -8 -1 x 10 atoms of cobalt-60 that has a half-life of 1925 days (■ = 4.17 x 10 sec). 6 -9 -1

The initial activity of each of the nuclides would be the product of the number of atoms and the decay constant.

Figure 12 Combined Decay of Iron-56, Manganese-54, and Cobalt-60

Plotting the manner in which the activities of each of the three nuclides decay over time demonstrates that initially the activity of the shortest-lived nuclide (iron-59) dominates the total activity, then manganese-54 dominates. After almost all of the iron and manganese have decayed away, the only contributor to activity will be the cobalt-60. A plot of this combined decay is shown in Figure 12.

Radioactive Equilibrium

Radioactive equilibrium exists when a radioactive nuclide is decaying at the same rate at which it is being produced. Since the production rate and decay rate are equal, the number of atoms present remains constant over time.

An example of radioactive equilibrium is the concentration of sodium-24 in the coolant circulating through a sodium-cooled nuclear reactor. Assume that the sodium-24 is being produced at a rate of 1 x 10 atoms per second. If the sodium-24 were stable and did not decay, 6 the amount of sodium-24 present after some period of time could be calculated by multiplying the production rate by the amount of time. Plotting the amount of material present would result in the graph in Figure 13.

However, sodium-24 is not stable, and it decays with a half-life of 14.96 hours. If no sodium-24 is present initially and production starts at a rate of 1 x 10 atoms per second, the rate 6 of decay will initially be zero because there is no sodium-24 present to decay. The rate of decay of sodium-24 will increase as the amount of sodium-24 increases.

■

Fixture checkpoint marker-1080-xray is recorded on fixture page 1080.

■

■

■

Figure 13 Cumulative Production of Sodium-24 Over Time

The amount of sodium-24 present will initially increase rapidly, then it will increase at a continually decreasing rate until the rate of decay is equal to the rate of production. It is possible to calculate how much sodium-24 will be present at equilibrium by setting the production rate (R) equal to the decay rate (■ N).

where:

R = production rate (atoms/second)

■ = decay constant (second)-1

N = number of atoms

1 x 10 atoms/second. 6
■ ■ ■■
■
■
■
■
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Figure 14 Approach of Sodium-24 to Equilibrium
The development of the equation to calculate how the amount of sodium-24 changes over time
Fixture checkpoint marker-1081-xray is recorded on fixture page 1081.
as it approaches the equilibrium value is beyond the scope of this handbook. However, the equation is presented below.
This equation can be used to calculate the values of the amount of sodium-24 present at different times. As the time increases, the exponential term approaches zero, and the number of atoms present will approach R/■. A plot of the approach of sodium-24 to equilibrium is shown in Figure 14.
Transient Radioactive Equilibrium
Transient radioactive equilibrium occurs when the parent nuclide and the daughter nuclide decay at essentially the same rate.
For transient equilibrium to occur, the parent must have a long half-life when compared to the daughter. An example of this type of compound decay process is barium-140, which decays by beta emission to lanthanum-140, which in turn decays by beta emission to stable cerium-140.
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Figure 15 Transient Equilibrium in the Decay of Barium-140
The decay constant for barium-140 is considerably smaller than the decay constant for lanthanum-140. Remember that the rate of decay of both the parent and daughter can be represented as ■N. Although the decay constant for barium-140 is smaller, the actual rate of decay (■N) is initially larger than that of lanthanum-140 because of the great difference in their initial concentrations. As the concentration of the daughter increases, the rate of decay of the daughter will approach and eventually match the decay rate of the parent. When this occurs, they are said to be in transient equilibrium. A plot of the barium-lanthanum-cerium decay chain reaching transient equilibrium is shown in Figure 15.
Secular equilibrium occurs when the parent has an extremely long half-life. In the long decay chain for a naturally radioactive element, such as thorium-232, where all of the elements in the chain are in secular equilibrium, each of the descendants has built up to an equilibrium amount and all decay at the rate set by the original parent. The only exception is the final stable element on the end of the chain. Its number of atoms is constantly increasing.
Summary
The important information in this chapter is summarized on the following page.
■
■■
■
RADIOACTIVITY DOE-HDBK-1019/1-93 Atomic and Nuclear Physics
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Radioactivity Summary
& Radioactivity is the decay of unstable atoms by the emission of particles and

& A curie (Ci) is a unit of radioactivity equal to 3.7 x 10 disintegrations per 10 second.
& A becquerel (Bq) is a unit of radioactivity equal to 1 disintegration per second.
& The radioactive decay constant (■) is the probability per unit time that an atom will decay.
& The radioactive half-life is the amount of time required for the activity to decrease to one-half its original value.
& The activity of a substance can be calculated from the number of atoms and the decay constant based on the equation below.
& The amount of activity remaining after a particular time can be calculated from the equation below.
& The relationship between the decay constant and the half-life is shown below.
& Plots of radioactive decay can be useful to describe the variation of activity over time. If decay is plotted using semi-log scale the plot results in a straight line.
& Radioactive equilibrium exists when the production rate of a material is equal to the removal rate.
& Transient radioactive equilibrium exists when the parent nuclide and the daughter
Fixture checkpoint marker-1082-xray is recorded on fixture page 1082.
nuclide decay at essentially the same rate. This occurs only when the parent has a long half-life compared to the daughter.
Atomic and Nuclear Physics DOE-HDBK-1019/1-93 NEUTRON INTERACTIONS
NEUTRON INTERACTIONS
Neutrons can cause many different types of interactions. The neutron may simply scatter off the nucleus in two different ways, or it may actually be absorbed into the nucleus. If a neutron is absorbed into the nucleus, it may result in the emission of a gamma ray or a subatomic particle, or it may cause the nucleus to fission.
EO 3.1 DESCRIBE the following scattering interactions between a neutron and a nucleus:
a. Elastic scattering
b. Inelastic scattering
EO 3.2 STATE the conservation laws that apply to an elastic collision between a neutron and a nucleus.
EO 3.3 DESCRIBE the following reactions where a neutron is absorbed in a nucleus:
a. Radiative capture
b. Particle ejection
Scattering
A neutron scattering reaction occurs when a nucleus, after having been struck by a neutron, emits a single neutron. Despite the fact that the initial and final neutrons do not need to be (and often are not) the same, the net effect of the reaction is as if the projectile neutron had merely "bounced off," or scattered from, the nucleus. The two categories of scattering reactions, elastic and inelastic scattering, are described in the following paragraphs.
Elastic Scattering
In an elastic scattering reaction between a neutron and a target nucleus, there is no energy transferred into nuclear excitation. Momentum and kinetic energy of the "system" are conserved although there is usually some transfer of kinetic energy from the neutron to the target nucleus. The target nucleus gains the amount of kinetic energy that the neutron loses.

SYNTHETIC FIXTURE PAGE 1083 OF 1200

mn
vn,i mTvT,i mnvn,f mTvT,f
1
2
m v
2
1
2
mn
v
2
n,i
1
2
mT
v
2
T,i
1
2
mn
v
2
n,f
Fixture checkpoint marker-1083-xray is recorded on fixture page 1083.
1
2
mT
v
2
T,f
NEUTRON INTERACTIONS DOE-HDBK-1019/1-93 Atomic and Nuclear Physics
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Figure 16 Elastic Scattering
Figure 16 illustrates the process of elastic scattering of a neutron off a target nucleus. In the elastic scattering reaction, the conservation of momentum and kinetic energy is represented by the equations below.
Conservation of momentum (mv)
Conservation of kinetic energy
where:
mn = mass of the neutron
m = mass of the target nucleus T
vn,i = initial neutron velocity
vn,f = final neutron velocity
vT,i = initial target velocity
vT,f = final target velocity
Atomic and Nuclear Physics DOE-HDBK-1019/1-93 NEUTRON INTERACTIONS
Elastic scattering of neutrons by nuclei can occur in two ways. The more unusual of the two

re-emission of a neutron in such a way that the total kinetic energy is conserved and the nucleus returns to its ground state. This is known as resonance elastic scattering and is very dependent upon the initial kinetic energy possessed by the neutron. Due to formation of the compound nucleus, it is also referred to as compound elastic scattering. The second, more usual method, is termed potential elastic scattering and can be understood by visualizing the neutrons and nuclei to be much like billiard balls with impenetrable surfaces. Potential scattering takes place with incident neutrons that have an energy of up to about 1 MeV. In potential scattering, the neutron does not actually touch the nucleus and a compound nucleus is not formed. Instead, the neutron is acted on and scattered by the short range nuclear forces when it approaches close enough to the nucleus.

Inelastic Scattering

In inelastic scattering, the incident neutron is absorbed by the target nucleus, forming a compound nucleus. The compound nucleus will then emit a neutron of lower kinetic energy which leaves the original nucleus in an excited state. The nucleus will usually, by one or more gamma emissions, emit this excess energy to reach its ground state. Figure 17 shows the process of inelastic scattering.

For the nucleus that has reached its ground state, the sum of the kinetic energy of the exit

Figure 17 Inelastic Scattering

neutron, the target nucleus, and the total gamma energy emitted is equal to the initial kinetic energy of the incident neutron.

1
0
n
238
92
U
239
92
U
239

Fixture checkpoint marker-1084-xray is recorded on fixture page 1084.

92
U
0
0
1
0
n
10
5
B
11
5
B
7
3
Li

2
NEUTRON INTERACTIONS DOE-HDBK-1019/1-93 Atomic and Nuclear Physics
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Absorption Reactions
Most absorption reactions result in the loss of a neutron coupled with the production of a charged particle or gamma ray. When the product nucleus is radioactive, additional radiation is emitted at some later time. Radiative capture, particle ejection, and fission are all categorized as absorption reactions and are briefly described below.
Radiative Capture
In radiative capture the incident neutron enters the target nucleus forming a compound nucleus. The compound nucleus then decays to its ground state by gamma emission. An example of a radiative capture reaction is shown below.
Particle Ejection
In a particle ejection reaction the incident particle enters the target nucleus forming a compound nucleus. The newly formed compound nucleus has been excited to a high enough energy level to cause it to eject a new particle while the incident neutron remains in the nucleus. After the new particle is ejected, the remaining nucleus may or may not exist in an excited state depending upon the mass-energy balance of the reaction. An example of a particle ejection reaction is shown below.
Fission
One of the most important interactions that neutrons can cause is fission, in which the nucleus that absorbs the neutron actually splits into two similarly sized parts. Fission will be discussed in detail in the next chapter.
Atomic and Nuclear Physics DOE-HDBK-1019/1-93 NEUTRON INTERACTIONS
Summary
The important information in this chapter is summarized below.
Neutron Interactions Summary
Interactions where a neutron scatters off a target nucleus are either elastic or inelastic. In elastic scattering, kinetic energy and momentum are conserved and no energy is transferred into excitation energy of the target nucleus. In inelastic scattering, some amount of kinetic energy is transferred into excitation energy of the target nucleus.
The conservation principles that apply to an elastic collision are conservation of kinetic energy and conservation of momentum.
Radiative capture is the absorption of a neutron by the target nucleus, resulting in an excited nucleus which subsequently (typically within a small fraction of a second) releases its excitation energy in the form of a gamma ray.
Particle ejection occurs when a neutron is absorbed by a target nucleus, resulting
Fixture checkpoint marker-1085-xray is recorded on fixture page 1085.
in the formation of a compound nucleus. The compound nucleus immediately ejects a particle (for example, alpha or proton).
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1
0
n
235
92
U

92
U
140
55
Cs
93
37
Rb 3
1
0
n
NUCLEAR FISSION DOE-HDBK-1019/1-93 Atomic and Nuclear Physics
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NUCLEAR FISSION
Nuclear fission is a process in which an atom splits and releases energy, fission
Fixture checkpoint marker-1086-xray is recorded on fixture page 1086.
products, and neutrons. The neutrons released by fission can, in turn, cause
the fission of other atoms.
EO 4.1 EXPLAIN the fission process using the liquid drop model
of a nucleus.
EO 4.2 DEFINE the following terms:
a. Excitation energy (E) exc
b. Critical energy (E) crit
EO 4.3 DEFINE the following terms:
a. Fissile material
b. Fissionable material
c. Fertile material
EO 4.4 DESCRIBE the processes of transmutation, conversion,
and breeding.
EO 4.5 DESCRIBE the curve of Binding Energy per Nucleon
versus mass number and give a qualitative description of
the reasons for its shape.
EO 4.6 EXPLAIN why only the heaviest nuclei are easily fissioned.
EO 4.7 EXPLAIN why uranium-235 fissions with thermal
neutrons and uranium-238 fissions only with fast neutrons.
Fission
In the fission reaction the incident neutron enters the heavy target nucleus, forming a compound
nucleus that is excited to such a high energy level (E > E) that the nucleus "splits" exc crit
(fissions) into two large fragments plus some neutrons. An example of a typical fission
reaction is shown below.
A large amount of energy is released in the form of radiation and fragment kinetic energy.
Atomic and Nuclear Physics DOE-HDBK-1019/1-93 NUCLEAR FISSION
Liquid Drop Model of a Nucleus
The nucleus is held together by the attractive nuclear force between nucleons, which was
discussed in a previous chapter. The characteristics of the nuclear force are listed below.
(a) very short range, with essentially no effect beyond nuclear dimensions
(10-13 cm)
(b) stronger than the repulsive electrostatic forces within the nucleus

neutrons are no different than those between pairs of protons or a neutron and a proton

(d) saturable, that is, a nucleon can attract only a few of its nearest neighbors

One theory of fission considers the fissioning of a nucleus similar in some respects to the splitting of a liquid drop. This analogy is justifiable to some extent by the fact that a liquid drop is held together by molecular forces that tend to make the drop spherical in shape and that try to resist any deformation in the same manner as nuclear forces are assumed to hold the nucleus together. By considering the nucleus as a liquid drop, the fission process can be described.

Referring to Figure 18(A), the nucleus in the ground state is undistorted, and its attractive nuclear forces are greater than the repulsive electrostatic forces between the protons within the nucleus. When an incident particle (in this instance a neutron) is absorbed by the target nucleus, a compound nucleus is formed. The compound nucleus temporarily contains all the charge and mass involved in the reaction and exists in an excited state. The excitation energy added to the compound nucleus is equal to the binding energy contributed by the incident particle plus the kinetic energy possessed by that particle. Figure 18(B) illustrates the excitation energy thus imparted to the compound nucleus, which may cause it to oscillate and become distorted. If the excitation energy is greater than a certain critical energy, the oscillations may cause the compound nucleus to become dumbbell-shaped. When this happens, the attractive nuclear forces (short-range) in the neck area are small due to saturation, while the repulsive electrostatic forces (long-range) are only slightly less than before. When the repulsive electrostatic forces exceed the attractive nuclear forces, nuclear fission occurs, as illustrated in Figure 18(C).

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Figure 18 Liquid Drop Model of Fission

Critical Energy

The measure of how far the energy level of a nucleus is above its ground state is called the excitation energy (Eexc). For fission to occur, the excitation energy must be above a particular value for that nuclide. The critical energy (Ecrit) is the minimum excitation energy required for fission to occur.

Fissile Material

A fissile material is composed of nuclides for which fission is possible with neutrons of any energy level. What is especially significant about these nuclides is their ability to be fissioned with zero kinetic energy neutrons (thermal neutrons). Thermal neutrons have very low kinetic energy levels (essentially zero) because they are roughly in equilibrium with the thermal motion of surrounding materials. Therefore, in order to be classified as fissile, a material must be capable of fissioning after absorbing a thermal neutron. Consequently, they impart essentially no kinetic energy to the reaction. Fission is possible in these materials with thermal neutrons, since the change in binding energy supplied by the neutron addition alone is high enough to exceed the critical energy. Some examples of fissile nuclides are uranium-235, uranium-233, and plutonium-239.

Fissionable Material

A fissionable material is composed of nuclides for which fission with neutrons is possible. All fissile nuclides fall into this category. However, also included are those nuclides that can be fissioned only with high energy neutrons. The change in binding energy that occurs as the result of neutron absorption results in a nuclear excitation energy level that is less than the required

of the incident neutron. The reason for this difference between fissile and fissionable materials is the so-called odd-even effect for nuclei. It has been observed that nuclei with even numbers of neutrons and/or protons are more stable than those with odd numbers. Therefore, adding a neutron to change a nucleus with an odd number of neutrons to a nucleus with an even number of neutrons produces an appreciably higher binding energy than adding a neutron to a nucleus already possessing an even number of neutrons. Some examples of nuclides requiring high energy neutrons to cause fission are thorium-232, uranium-238, and plutonium-240. Table 4 indicates the critical energy (Ecrit) and the binding energy change for an added neutron (BEn) to target nuclei of interest. For fission to be possible, the change in binding energy plus the kinetic energy must equal or exceed the critical energy (ΔBE + KE > Ecrit).

TABLE 4

Critical Energies Compared to Binding Energy of Last Neutron

Target
Nucleus
Critical Energy
Ecrit
Binding Energy of
Last Neutron BEn
BEn - Ecrit
23
9
2
0Th 7.5 MeV 5.4 MeV -2.1 MeV
23
9
8
2U 7.0 MeV 5.5 MeV -1.5 MeV
23
9

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5
2U 6.5 MeV 6.8 MeV +0.3 MeV
23
9
3
2U 6.0 MeV 7.0 MeV +1.0 MeV
23
9
9
4Pu 5.0 MeV 6.6 MeV +1.6 MeV

Uranium-235 fissions with thermal neutrons because the binding energy released by the absorption of a neutron is greater than the critical energy for fission; therefore uranium-235 is a fissile material. The binding energy released by uranium-238 absorbing a thermal neutron is less than the critical energy, so additional energy must be possessed by the neutron for fission to be possible. Consequently, uranium-238 is a fissionable material.

Fertile Material

nuclides through the process known as transmutation. These nuclides can, in turn, be transmuted again or may undergo radioactive decay to produce still different nuclides. The nuclides that are produced by this process are referred to as transmutation products. Because several of the fissile nuclides do not exist in nature, they can only be produced by nuclear reactions (transmutation). The target nuclei for such reactions are said to be fertile. Fertile materials are materials that can undergo transmutation to become fissile materials. Figure 19 traces the transmutation mechanism by which two fertile nuclides, thorium-232 and uranium-238, produce uranium-233 and plutonium-239, respectively.

Figure 19 Conversion of Fertile Nuclides to Fissile Nuclides

If a reactor contains fertile material in addition to its fissile fuel, some new fuel will be produced as the original fuel is burned up. This is called conversion. Reactors that are specifically designed to produce fissionable fuel are called "breeder" reactors. In such reactors, the amount of fissionable fuel produced is greater than the amount of fuel burnup. If less fuel is produced than used, the process is called conversion, and the reactor is termed a "converter."

Binding Energy Per Nucleon (BE/A)

As the number of particles in a nucleus increases, the total binding energy also increases. The rate of increase, however, is not uniform. This lack of uniformity results in a variation in the amount of binding energy associated with each nucleon within the nucleus. This variation in the binding energy per nucleon (BE/A) is easily seen when the average BE/A is plotted versus atomic mass number (A), as shown in Figure 20.

Figure 20 Binding Energy per Nucleon vs. Mass Number

Figure 20 illustrates that as the atomic mass number increases, the binding energy per nucleon decreases for A > 60. The BE/A curve reaches a maximum value of 8.79 MeV at A = 56 and decreases to about 7.6 MeV for A = 238. The general shape of the BE/A curve can be explained using the general properties of nuclear forces. The nucleus is held together by very short-range attractive forces that exist between nucleons. On the other hand, the nucleus is being forced apart by long range repulsive electrostatic (coulomb) forces that exist between all the protons in the nucleus.

As the atomic number and the atomic mass number increase, the repulsive electrostatic forces within the nucleus increase due to the greater number of protons in the heavy elements. To overcome this increased repulsion, the proportion of neutrons in the nucleus must increase to maintain stability. This increase in the neutron-to-proton ratio only partially compensates for the

Fixture checkpoint marker-1089-xray is recorded on fixture page 1089.

growing proton-proton repulsive force in the heavier, naturally occurring elements. Because the repulsive forces are increasing, less energy must be supplied, on the average, to remove a nucleon from the nucleus. The BE/A has decreased. The BE/A of a nucleus is an indication of its degree of stability. Generally, the more stable nuclides have higher BE/A than the less stable ones. The increase in the BE/A as the atomic mass number decreases from 260 to 60 is the primary reason for the energy liberation in the fission process. In addition, the increase in the BE/A as the atomic mass number increases from 1 to 60 is the reason for the energy liberation in the fusion process, which is the opposite reaction of fission.

The heaviest nuclei require only a small distortion from a spherical shape (small energy addition) for the relatively large coulomb forces forcing the two halves of the nucleus apart to overcome the attractive nuclear forces holding the two halves together. Consequently, the heaviest nuclei

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Summary

The important information in this chapter is summarized on the following page.

Nuclear Fission Summary

The fission process can be explained using the liquid drop model of a nucleus. In the ground state the nucleus is nearly spherical in shape. After the absorption of a neutron, the nucleus will be in an excited state and start to oscillate and become distorted. If the oscillations cause the nucleus to become shaped like a dumbbell, the repulsive electrostatic forces will overcome the short-range attractive nuclear forces, and the nucleus will split in two.

Excitation energy is the amount of energy a nucleus has above its ground state. Critical energy is the minimum excitation energy that a nucleus must have before

Fixture checkpoint marker-1090-xray is recorded on fixture page 1090.

it can fission.

Fissile material is material for which fission is possible with neutrons that have zero kinetic energy. Fissionable material is material for which fission caused by neutron absorption is possible provided the kinetic energy added with the binding energy is greater than the critical energy. Fertile material is material that can undergo transmutation to become fissile material.

Transmutation is the process of neutron absorption and subsequent decay, which changes one nuclide to another nuclide. Conversion is the process of transmuting fertile material into fissile material in a reactor, where the amount of fissile material produced is less than the amount of fissile material consumed. Breeding is the same as conversion, except the amount of fissile material produced is more than the amount of fissile material consumed.

The curve of binding energy per nucleon increases quickly through the light nuclides and reaches a maximum at a mass number of about 56. The curve decreases slowly for mass numbers greater than 60.

The heaviest nuclei are easily fissionable because they require only a small distortion from the spherical shape to allow the coulomb forces to overcoming the attractive nuclear force, forcing the two halves of the nucleus apart.

Uranium-235 fissions with thermal neutrons because the binding energy released by the absorption of a neutron is greater than the critical energy for fission. The binding energy released by uranium-238 absorbing a neutron is less than the critical energy, so additional kinetic energy must be possessed by the neutron for fission to be possible.

1
0
n
235
92
U
236
92
U
140

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Cs
93
37
Rb 3
1
0
n
ENERGY RELEASE FROM FISSION DOE-HDBK-1019/1-93 Atomic and Nuclear Physics
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ENERGY RELEASE FROM FISSION
Fission of heavy nuclides converts a small amount of mass into an enormous amount of energy. The amount of energy released by fission can be determined based on either the change in mass that occurs during the reaction or by the difference in binding energy per nucleon between the fissile nuclide and the fission products.
EO 4.8 CHARACTERIZE the fission products in terms of mass groupings and radioactivity.
EO 4.9 Given the nuclides involved and their masses, CALCULATE the energy released from fission.
EO 4.10 Given the curve of Binding Energy per nucleon versus mass
Fixture checkpoint marker-1091-xray is recorded on fixture page 1091.
number, CALCULATE the energy released from fission.
Calculation of Fission Energy
Nuclear fission results in the release of enormous quantities of energy. It is necessary to be able to calculate the amount of energy that will be produced. The logical manner in which to pursue this is to first investigate a typical fission reaction such as the one listed below.
It can be seen that when the compound nucleus splits, it breaks into two fission fragments, rubidium-93, cesium-140, and some neutrons. Both fission products then decay by multiple - emissions as a result of the high neutron-to-proton ratio possessed by these nuclides.
In most cases, the resultant fission fragments have masses that vary widely. Figure 21 gives the percent yield for atomic mass numbers. The most probable pair of fission fragments for the thermal fission of the fuel uranium-235 have masses of about 95 and 140. Note that the vertical axis of the fission yield curve is on a logarithmic scale. Therefore, the formation of fission fragments of mass numbers of about 95 and 140 is highly likely.
Atomic and Nuclear Physics DOE-HDBK-1019/1-93 ENERGY RELEASE FROM FISSION
Figure 21 Uranium-235 Fission Yield vs.
Mass Number
Referring now to the binding energy per nucleon curve (Figure 20), we can estimate the amount of energy released by our "typical" fission by plotting this reaction on the curve and calculating the change in binding energy (ΔBE) between the reactants on the left-hand side of the fission equation and the products on the right-hand side. Plotting the reactant and product nuclides on the curve shows that the total binding energy of the system after fission is greater than the total binding energy of the system before fission. When there is an increase in the total binding energy of a system, the system has become more stable by releasing an amount of energy equal to the increase in total binding energy of the system. Therefore, in the fission process, the energy liberated is equal to the increase in the total binding energy of the system.
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Figure 22 Change in Binding Energy for Typical Fission

rubidium-93 and B, cesium-140) is greater than that for the reactant (A, uranium-235). The total binding energy for a nucleus can be found by multiplying the binding energy per nucleon by the number of nucleons.

TABLE 5

Binding Energies Calculated from Binding Energy per Nucleon Curve

Nuclide	B.E. per Nucleon (BE/A)	Mass Number (A)	Binding Energy (BE/A) x (A)
9 3 3 7Rb	8.7 MeV	93	809 MeV
14 5 0 5Cs	8.4 MeV	140	1176 MeV
23 9 5 2U	7.6 MeV	235	1786 MeV

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BE BEproducts BEreactants

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BERb 93 BECs 140 BEU 235

809 MeV 1176 MeV 1786 MeV

199 MeV

1 0 n

235 92 U

236 92 U

140 55 Cs

93 37 Rb 3 1 0

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Mass difference Mass of Reactants Mass of Products
236.052589 amu 235.85208 amu
0.200509 amu
EInst 0.020059 amu
931.5 MeV
amu
186.8 MeV
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The energy released will be equivalent to the difference in binding energy (BE) between the reactants and the products.
The energy liberation during the fission process can also be explained from the standpoint of the conservation of mass-energy. During the fission process, there is a decrease in the mass of the system. There must, therefore, be energy liberated equal to the energy equivalent of the mass lost in the process. This method is more accurate than the previously illustrated method and is used when actually calculating the energy liberated during the fission process.
Again, referring to the "typical" fission reaction.
E , the instantaneous energy, is the energy released immediately after the fission process. It Inst is equal to the energy equivalent of the mass lost in the fission process. It can be calculated as shown below.
Mass of the Reactants Mass of the Products
U 235.043924 amu Rb 92.91699 amu 235
92
93
37
n 1.008665 amu Cs 139.90910 amu 1
0
140
55
3 (n) 3.02599 amu 1
0
236.052589 amu 235.85208 amu
This mass difference can be converted to an energy equivalent.
93
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37
Rb
93
38
Sr
93
39
Y
93
40
Zr
93
41

140
55
Cs
140
56
Ba
140
57
La
140
58
Fixture checkpoint marker-1094-xray is recorded on fixture page 1094.
Ce
EDecay mRb 93 mNb 93 4 melectron
931.5MeV
amu
92.91699 amu 92.90638 amu 4 0.0005486 amu
931.5 MeV
amu
0.008416 amu
931.5 MeV
amu
7.84 MeV
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The total energy released per fission will vary from the fission to the next depending on what fission products are formed, but the average total energy released per fission of uranium-235 with a thermal neutron is 200 MeV.
As illustrated in the preceding example, the majority of the energy liberated in the fission process is released immediately after the fission occurs and appears as the kinetic energy of the fission fragments, kinetic energy of the fission neutrons, and instantaneous gamma rays. The remaining energy is released over a period of time after the fission occurs and appears as kinetic energy of the beta, neutrino, and decay gamma rays.
Estimation of Decay Energy
In addition to this instantaneous energy release during the actual fission reaction, there is additional energy released when the fission fragments decay by emission. This additional
-
energy is called decay energy, EDecay. The decay chains for rubidium-93 and cesium-140 are shown below.
The energy released during the decay for each chain will be equivalent to the mass difference between the original fission product and the sum of the final stable nuclide and the beta particles emitted.
The energy released in the decay chain of rubidium-93 is calculated below.
EDecay mRb 93 mNb 93 3 melectron
931.5MeV
amu
139.90910 amu 139.90543 amu 3 0.0005486 amu
931.5 MeV

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0.000202 amu
931.5 MeV
amu
1.89 MeV
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The energy released in the decay chain of cesium-140 is calculated below.
The total decay energy is the sum of the energies of the two chains, or 9.73 MeV.
Distribution of Fission Energy
The average energy distribution for the energy released per fission with a thermal neutron in uranium-235 is shown in Tables 6 and 7.
TABLE 6
Instantaneous Energy from Fission
Kinetic Energy of Fission Products 167 Mev
Energy of Fission Neutrons 5 MeV
Instantaneous Gamma-ray Energy 5 MeV
Capture Gamma-ray Energy 10 MeV
Total Instantaneous Energy 187 MeV
Fixture checkpoint marker-1095-xray is recorded on fixture page 1095.
TABLE 7
Delayed Energy from Fission
Beta Particles From Fission Products 7 Mev
Gamma-rays from Fission Products 6 MeV
Neutrinos 10 MeV
Total Delayed Energy 23 MeV
ENERGY RELEASE FROM FISSION DOE-HDBK-1019/1-93 Atomic and Nuclear Physics
Because the 10 MeV of neutrino energy shown in Table 7 is not absorbed in the reactor, the average value of 200 MeV per fission is still accurate. Note in Table 6 that some fission neutrons undergo radiative capture and the resultant gamma ray emission provides an additional 10 MeV of instantaneous energy, which contributes to the total of 187 MeV instantaneous energy.
All of the energy released, with the exception of the neutrino energy, is ultimately transformed into heat through a number of processes. The fission fragments, with their high positive charge and kinetic energy, cause ionization directly as they rip orbital electrons from the surrounding atoms. In this ionization process, kinetic energy is transferred to the surrounding atoms of the fuel material, resulting in an increase in temperature. The beta particles and gamma rays also give up their energy through ionization, and the fission neutrons interact and lose their energy through elastic scattering. Of the 200 MeV released per fission, about seven percent (13 MeV) is released at some time after the instant of fission. When a reactor is shut down, fissions essentially cease, but energy is still being released from the decay of fission products. The heat produced by this decay energy is referred to as "decay heat." Although decay energy represents about seven percent of reactor heat production during reactor operation, once the reactor is shut down the decay heat production drops off quickly to a small fraction of its value while operating. The decay heat produced is significant, however, and systems must be provided to keep the reactor cool even after shutdown.
Summary
The important information in this chapter is summarized below.
Energy Release From Fission Summary

They generally decay by β- emission.
The most common mass numbers are grouped near 95 and 140.
The energy released by fission can be calculated based on the difference in mass between the masses of the reactants before fission and the fission fragments and fission neutrons after fission.
Another method to determine the energy released by fission is based on the change in binding energy per nucleon between the fissile nuclide and the fission products.
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Atomic and Nuclear Physics INTERACTION OF RADIATION WITH MATTER
INTERACTION OF RADIATION WITH MATTER
Different types of radiation interact with matter in widely different ways. A large, massive, charged alpha particle cannot penetrate a piece of paper and even has a limited range in dry air. A neutrino, at the other extreme, has a low probability of interacting with any matter, even if it passed through the diameter of the earth.
EO 5.1 DESCRIBE interactions of the following with matter:
a. Alpha particle c. Positron
b. Beta particle d. Neutron
EO 5.2 DESCRIBE the following ways that gamma radiation interacts with matter:
a. Photoelectric effect
b. Compton scattering
c. Pair production
Fixture checkpoint marker-1096-xray is recorded on fixture page 1096.
Interaction of Radiation With Matter
Radiation can be classified into two general groups, charged and uncharged; therefore, it may be expected that interactions with matter fall into two general types. Charged particles directly ionize the media through which they pass, while uncharged particles and photons can cause ionization only indirectly or by secondary radiation.
A moving charged particle has an electrical field surrounding it, which interacts with the atomic structure of the medium through which it is passing. This interaction decelerates the particle and accelerates electrons in the atoms of the medium. The accelerated electrons may acquire enough energy to escape from the parent atom. This process, whereby radiation "strips" off orbital electrons, is called ionization. Uncharged moving particles have no electrical field, so they can only lose energy and cause ionization by such means as collisions or scattering. A photon can lose energy by the photoelectric effect, Compton effect, or pair production.
Rev. 0 Page 63 NP-01
I
m z
2
K.E.
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Because ionizing radiation creates ions in pairs, the intensity of ionization or the specific ionization is defined as the number of ion-pairs formed per centimeter of travel in a given

is a measure of its ionizing power, is roughly proportional to the particle's mass and the square of its charge as illustrated in the equation below.
where:
I is the ionizing power
m is the mass of the particle
z is the number of unit charges it carries
K.E. is its kinetic energy
Since m for an alpha particle is about 7300 times as large as m for a beta particle, and z is twice as great, an alpha will produce much more ionization per unit path length than a beta particle of the same energy. This phenomenon occurs because the larger alpha particle moves slower for a given energy and thus acts on a given electron for a longer time.
Alpha Radiation
Alpha radiation is normally produced from the radioactive decay of heavy nuclides and from certain nuclear reactions. The alpha particle consists of 2 neutrons and 2 protons, so it is essentially the same as the nucleus of a helium atom. Because it has no electrons, the alpha particle has a charge of +2. This positive charge causes the alpha particle to strip electrons from the orbits of atoms in its vicinity. As the alpha particle passes through material, it removes electrons from the orbits of atoms it passes near. Energy is required to remove electrons and the energy of the alpha particle is reduced by each reaction. Eventually the particle will expend its kinetic energy, gain 2 electrons in orbit, and become a helium atom. Because of its strong positive charge and large mass, the alpha particle deposits a large amount of energy in a short distance of travel. This rapid, large deposition of energy limits the penetration of alpha particles. The most energetic alpha particles are stopped by a few centimeters of air or a sheet of paper.
Beta-Minus Radiation
A beta-minus particle is an electron that has been ejected at a high velocity from an unstable nucleus. An electron has a small mass and an electrical charge of -1. Beta particles cause ionization by displacing electrons from atom orbits. The ionization occurs from collisions with orbiting electrons. Each collision removes kinetic energy from the beta particle, causing it to slow down. Eventually the beta particle will be slowed enough to allow it to be captured as an
2 electrons 0.000549 amu
electron
Fixture checkpoint marker-1097-xray is recorded on fixture page 1097.
931.5 MeV
amu
1.02 MeV

orbiting electron in an atom. Although more penetrating than the alpha, the beta is relatively easy to stop and has a low power of penetration. Even the most energetic beta radiation can be stopped by a few millimeters of metal.
Positron Radiation
Positively charged electrons are called positrons. Except for the positive charge, they are identical to beta-minus particles and interact with matter in a similar manner. Positrons are very short-lived, however, and quickly are annihilated by interaction with a negatively charged electron, producing two gammas with a combined energy (calculated below) equal to the rest mass of the positive and negative electrons.

Neutrons have no electrical charge. They have nearly the same mass as a proton (a hydrogen atom nucleus). A neutron has hundreds of times more mass than an electron, but 1/4 the mass of an alpha particle. The source of neutrons is primarily nuclear reactions, such as fission, but they may also be produced from the decay of radioactive nuclides. Because of its lack of charge, the neutron is difficult to stop and has a high penetrating power.

Neutrons are attenuated (reduced in energy and numbers) by three major interactions, elastic scatter, inelastic scatter, and absorption. In elastic scatter, a neutron collides with a nucleus and bounces off. This reaction transmits some of the kinetic energy of the neutron to the nucleus of the atom, resulting in the neutron being slowed, and the atom receives some kinetic energy (motion). This process is sometimes referred to as "the billiard ball effect."

As the mass of the nucleus approaches the mass of the neutron, this reaction becomes more effective in slowing the neutron. Hydrogenous material attenuates neutrons most effectively.

In the inelastic scatter reaction, the same neutron/nucleus collision occurs as in elastic scatter. However, in this reaction, the nucleus receives some internal energy as well as kinetic energy. This slows the neutron, but leaves the nucleus in an excited state. When the nucleus decays to its original energy level, it normally emits a gamma ray.

In the absorption reaction, the neutron is actually absorbed into the nucleus of an atom. The neutron is captured, but the atom is left in an excited state. If the nucleus emits one or more gamma rays to reach a stable level, the process is called radiative capture. This reaction occurs at most neutron energy levels, but is more probable at lower energy levels.

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INTERACTION OF RADIATION WITH MATTER Atomic and Nuclear Physics

Gamma Radiation

Gamma radiation is electromagnetic radiation. It is commonly referred to as a gamma ray and is very similar to an x-ray. The difference is that gamma rays are emitted from the nucleus of an atom, and x-rays are produced by orbiting electrons. The x-ray is produced when orbiting electrons move to a lower energy orbit or when fast-moving electrons approaching an atom are deflected and decelerated as they react with the atom's electrical field (called Bremsstrahlung). The gamma ray is produced by the decay of excited nuclei and by nuclear reactions. Because the gamma ray has no mass and no charge, it is difficult to stop and has a very high penetrating power. A small fraction of the original gamma stream will pass through several feet of concrete or several meters of water.

There are three methods of attenuating gamma rays. The first method is referred to as the photo-electric effect. When a low energy gamma strikes an atom, the total energy of the gamma is expended in ejecting an electron from orbit. The result is ionization of the atom and expulsion of a high energy electron. This reaction is most predominant with low energy gammas interacting in materials with high atomic weight and rarely occurs with gammas having an energy above 1 MeV. Annihilation of the gamma results. Any gamma energy in excess of the binding energy of the electron is carried off by the electron in the form of kinetic energy.

Fixture checkpoint marker-1098-xray is recorded on fixture page 1098.

The second method of attenuation of gammas is called Compton scattering. The gamma interacts with an orbital or free electron; however, in this case, the photon loses only a fraction of its energy. The actual energy loss depending on the scattering angle of the gamma. The gamma continues on at lower energy, and the energy difference is absorbed by the electron. This reaction becomes important for gamma energies of about 0.1 MeV and higher.

At higher energy levels, a third method of attenuation is predominant. This method is pair-production. When a high energy gamma passes close enough to a heavy nucleus, the gamma completely disappears, and an electron and a positron are formed. For this reaction to take place,

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becomes kinetic energy shared between the electron and positron. The probability of pair■production incre
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Atomic and Nuclear Physics INTERACTION OF RADIATION WITH MATTER
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Interaction of Radiation with Matter Summary
An alpha particle deposits a large amount of energy in a short distance of travel due to its large mass and charge.
Beta-minus particles interact with the electrons orbiting the nucleus of atoms, causing ionization by displacing the electrons. The beta particle loses energy with each interaction. After the beta particle loses enough energy, it is captured in the orbital shells of an atom.
Positrons interact with matter much the same way as beta minus particles. After the positron has lost most of its energy by ionizing atoms, it is annihilated by interaction with an electron. The electron-positron pair disappear and are replaced by two gammas, each with the energy equivalent of the mass of an electron (0.51
Fixture checkpoint marker-1099-xray is recorded on fixture page 1099.
MeV).
Neutrons interact with matter by elastic scattering, inelastic scattering, or absorption.
Photoelectric effect is where a gamma interacts with an electron orbiting an atom. The entire energy of the gamma is transferred to the electron, and the electron is ejected from its orbit.
In Compton scattering a gamma interacts with an orbital electron, but only part of the gamma energy is transferred to the electron. The electron is ejected from its orbit, and the gamma is scattered off at a lower energy.
In pair-production, a gamma interacts with the electric field of a nucleus and is converted into an electron-positron pair. The gamma must have an energy greater than 1.02 MeV for this to occur.
Summary
The important information in this chapter is summarized below.
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Department of Energy
Fundamentals Handbook
NUCLEAR PHYSICS
AND REACTOR THEORY
Module 2
Reactor Theory (Neutron Characteristics)
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OBJECTIVES DOE-HDBK-1019/1-93 Reactor Theory (Neutron Characteristics)
TERMINAL OBJECTIVE
1.0 Without references, EXPLAIN how neutron sources produce neutrons.
ENABLING OBJECTIVES
1.1 DEFINE the following terms:
a. Intrinsic neutron source
b. Installed neutron source
1.2 LIST three examples of reactions that produce neutrons in intrinsic neutron sources.
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1.3 LIST three examples of reactions that produce neutrons in installed neutron sources.
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Reactor Theory (Neutron Characteristics) DOE-HDBK-1019/1-93 OBJECTIVES
TERMINAL OBJECTIVE
2.0 Given the necessary information for calculations, EXPLAIN basic concepts in reactor physics and perform calculations.
ENABLING OBJECTIVES
2.1 DEFINE the following terms:
a. Atom density d. Barn
b. Neutron flux e. Macroscopic cross section
c. Microscopic cross section f. Mean free path
2.2 EXPRESS macroscopic cross section in terms of microscopic cross section.
2.3 DESCRIBE how the absorption cross section of typical nuclides varies with neutron energy at energies below the resonance absorption region.
2.4 DESCRIBE the cause of resonance absorption in terms of nuclear energy levels.
2.5 DESCRIBE the energy dependence of resonance absorption peaks for typical light and heavy nuclei.

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2.7 Given the number densities (or total density and component fractions) and microscopic cross sections of components, CALCULATE the macroscopic cross section for a mixture.
2.8 CALCULATE a macroscopic cross section given a material density, atomic mass, and microscopic cross section.
2.9 EXPLAIN neutron shadowing or self-shielding.
2.10 Given the neutron flux and macroscopic cross section, CALCULATE the reaction rate.
2.11 DESCRIBE the relationship between neutron flux and reactor power.
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OBJECTIVES DOE-HDBK-1019/1-93 Reactor Theory (Neutron Characteristics)
ENABLING OBJECTIVES (Cont.)
2.12 DEFINE the following concepts:
a. Thermalization d. Average logarithmic energy decrement
b. Moderator e. Macroscopic slowing down power
c. Moderating ratio
2.13 LIST three desirable characteristics of a moderator.
2.14 Given an average fractional energy loss per collision, CALCULATE the energy loss after a specified number of collisions.
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Reactor Theory (Neutron Characteristics) DOE-HDBK-1019/1-93 OBJECTIVES
TERMINAL OBJECTIVE
3.0 Without references, EXPLAIN the production process and effects on fission of prompt and delayed neutrons.
ENABLING OBJECTIVES
3.1 STATE the origin of prompt neutrons and delayed neutrons.
3.2 STATE the approximate fraction of neutrons that are born as delayed neutrons from the fission of the following nuclear fuels:
a. Uranium-235
b. Plutonium-239
3.3 EXPLAIN the mechanism for production of delayed neutrons.
3.4 EXPLAIN prompt and delayed neutron generation times.
3.5 Given prompt and delayed neutron generation times and delayed neutron fraction, CALCULATE the average generation time.
3.6 EXPLAIN the effect of delayed neutrons on reactor control.
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OBJECTIVES DOE-HDBK-1019/1-93 Reactor Theory (Neutron Characteristics)
TERMINAL OBJECTIVE
4.0 Without references, DESCRIBE the neutron energy spectrum for the type of reactor
Fixture checkpoint marker-1102-xray is recorded on fixture page 1102.
presented in this module.
ENABLING OBJECTIVES
4.1 STATE the average energy at which prompt neutrons are produced.
4.2 DESCRIBE the neutron energy spectrum in the following reactors:
a. Fast reactor
b. Thermal reactor
4.3 EXPLAIN the reason for the particular shape of the fast, intermediate, and slow energy regions of the neutron flux spectrum for a thermal reactor.
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Reactor Theory (Neutron Characteristics) DOE-HDBK-1019/1-93 NEUTRON SOURCES

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NEUTRON SOURCES
Neutrons from a variety of sources are always present in a reactor core. This is true even when the reactor is shut down. Some of these neutrons are produced by naturally occurring (intrinsic) neutron sources, while others may be the result of fabricated (installed) neutron sources that are incorporated into the design of the reactor. The neutrons produced by sources other than neutron-induced fission are often grouped together and classified as source neutrons.
EO 1.1 DEFINE the following terms:
a. Intrinsic neutron source
b. Installed neutron source
EO 1.2 LIST three examples of reactions that produce neutrons in intrinsic neutron sources.
EO 1.3 LIST three examples of reactions that produce neutrons in installed neutron sources.
Fixture checkpoint marker-1103-xray is recorded on fixture page 1103.
Neutron Sources
In addition to neutron-induced fission, neutrons are produced by other reactions. The neutrons produced by reactions other than neutron-induced fission are called source neutrons. Source neutrons are important because they ensure that the neutron population remains high enough to allow a visible indication of neutron level on the most sensitive monitoring instruments while the reactor is shutdown and during the startup sequence. This verifies instrument operability and allows monitoring of neutron population changes. Source neutrons can be classified as either intrinsic or installed neutron sources.
Intrinsic Neutron Sources
Some neutrons will be produced in the materials present in the reactor due to a variety of unavoidable reactions that occur because of the nature of these materials. Intrinsic neutron sources are those neutron-producing reactions that always occur in reactor materials.
■ ■ ■ ■
■ ■ ■
■ ■
NEUTRON SOURCES DOE-HDBK-1019/1-93 Reactor Theory (Neutron Characteristics)
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A limited number of neutrons will always be present, even in a reactor core that has never been operated, due to spontaneous fission of some heavy nuclides that are present in the fuel. Uranium■238, ur of the rate at which different heavy nuclides produce neutrons by spontaneous fission.
Californium-252 is not an intrinsic neutron source, but will be discussed in the section on installed neutron sources.
TABLE 1
Neutron Production by Spontaneous Fission
Nuclide T (Fission) T (1/2 1/2 ■-decay) neutrons/sec/gram
U 1.8 x 10 years 6.8 x 10 years 8.0 x 10 235
92
17 8 -4
U 8.0 x 10 years 4.5 x 10 years 1.6 x 10 238
92
15 9 -2
Pu 5.5 x 10 years 2.4 x 10 years 3.0 x 10 239
94

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Pu 1.2 x 10 years 6.6 x 10 years 1.0 x 10 240
94
11 3 3
Cf 66.0 years 2.65 years 2.3 x 10 252
98
12
Another intrinsic neutron source is a reaction involving natural boron and fuel. In some reactors, natural boron is loaded into the reactor core as a neutron absorber to improve reactor control or increase core life-time. Boron-11 (80.1% of natural boron) undergoes a reaction with the alpha particle emitted by the radioactive decay of heavy nuclides in the fuel to yield a neutron as shown below.
The boron-11 must be mixed with, or in very close proximity to, the fuel for this reaction because of the short path length of the alpha particle. For a reactor core with this configuration, this (■,n) reaction is an important source of neutrons for reactor startup.
In a reactor that has been operated, another source of neutrons becomes significant. Neutrons may be produced by the interaction of a gamma ray and a deuterium nucleus. This reaction is commonly referred to as a photoneutron reaction because it is initiated by electromagnetic radiation and results in the production of a neutron. The photoneutron reaction is shown below.
■ ■ ■
■ ■
■ ■ ■
Fixture checkpoint marker-1104-xray is recorded on fixture page 1104.
■ ■
Reactor Theory (Neutron Characteristics) DOE-HDBK-1019/1-93 NEUTRON SOURCES
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There is an abundant supply of high energy gammas in a reactor that has been operated because many of the fission products are gamma emitters. All water-cooled reactors have some deuterium present in the coolant in the reactor core because a small fraction of natural hydrogen is the isotope deuterium. The atom percentage of deuterium in the water ranges from close to the naturally occurring value (0.015%) for light water reactors to above 90% deuterium for heavy water reactors. Therefore, the required conditions for production of photoneutrons exist.
The supply of gamma rays decreases with time after shutdown as the gamma emitters decay; therefore, the photoneutron production rate also decreases. In a few particular reactors, additional D O (heavy water) may be added to the reactor to increase the production of 2 photoneutrons following a long shutdown period.
Installed Neutron Sources
Because intrinsic neutron sources can be relatively weak or dependent upon the recent power history of the reactor, many reactors have artificial sources of neutrons installed. These neutron sources ensure that shutdown neutron levels are high enough to be detected by the nuclear instruments at all times. This provides a true picture of reactor conditions and any change in these conditions. An installed neutron source is an assembly placed in or near the reactor for the sole purpose of producing source neutrons.
One strong source of neutrons is the artificial nuclide californium-252, which emits neutrons at the rate of about 2 x 10 neutrons per second per gram as the result of spontaneous fission. 12 Important drawbacks for some applications may be its high cost and its short half-life (2.65 years).
Many installed neutron sources use the (■,n) reaction with beryllium. These sources are composed of a mixture of metallic beryllium (100% beryllium-9) with a small quantity of an

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that occurs is shown below.
The beryllium is intimately (homogeneously) mixed with the alpha emitter and is usually enclosed in a stainless steel capsule.
Another type of installed neutron source that is widely used is a photoneutron source that employs the (■,n) reaction with beryllium. Beryllium is used for photoneutron sources because its stable isotope beryllium-9 has a weakly attached last neutron with a binding energy of only 1.66 MeV. Thus, a gamma ray with greater energy than 1.66 MeV can cause neutrons to be ejected by the (■,n) reaction as shown below.
NEUTRON SOURCES DOE-HDBK-1019/1-93 Reactor Theory (Neutron Characteristics)
NP-02 Page 4 Rev. 0
Neutron Sources Summary
& Intrinsic neutron sources are sources of neutrons from materials that are in the reactor for other purposes such as fuel, burnable poison, or moderator.
& Installed neutron sources are materials or components placed in the reactor specifically for the purpose of producing source neutrons.
& Examples of intrinsic neutron sources are listed below.
Spontaneous fission of heavy nuclides in fuel, such as uranium-238, uranium-235, and plutonium-239, results in fission fragments and free neutrons.
Boron-11 mixed with the fuel undergoes an alpha-neutron reaction and becomes nitrogen-14.
Deuterium present in the reactor coolant undergoes a gamma■neutron reaction and becomes hydrogen-1.
& Examples of installed neutron sources are listed below.
Spontaneous fission of californium-252 results in fission fragments and free neutrons.
Beryllium-9 undergoes an alpha-neutron reaction (alpha from the decay of plutonium, polonium, or radium) and becomes carbon-12.
Beryllium-9 undergoes a gamma-neutron reaction (high energy
Fixture checkpoint marker-1105-xray is recorded on fixture page 1105.
gamma from decay of antimony-124) and becomes beryllium-8.
Many startup sources of this type use antimony and beryllium because after activation with neutrons the radioactive antimony becomes an emitter of high energy gammas. The photoneutron sources of this type are constructed somewhat differently from the (■,n) types. One design incorporates a capsule of irradiated antimony enclosed in a beryllium sleeve. The entire assembly is then encased in a stainless steel cladding. A large reactor may have several neutron sources of this type installed within the core.
Summary
The important information in this chapter is summarized below.
DOE-HDBK-1019/1-93
Reactor Theory (Neutron Characteristics) NUCLEAR CROSS SECTIONS AND NEUTRON FLUX
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NUCLEAR CROSS SECTIONS AND NEUTRON FLUX
To determine the frequency of neutron interactions, it is necessary to describe the availability of neutrons to cause interaction and the probability of a neutron interacting with material. The availability of neutrons and the probability of interaction are quantified by the neutron flux and nuclear cross section.
EO 2.1 DEFINE the following terms:
a. Atom density d. Barn

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c. Microscopic cross section f. Mean free path
EO 2.2 EXPRESS macroscopic cross section in terms of microscopic cross section.
EO 2.3 DESCRIBE how the absorption cross section of typical nuclides varies with neutron energy at energies below the resonance absorption region.
EO 2.4 DESCRIBE the cause of resonance absorption in terms of nuclear energy levels.
EO 2.5 DESCRIBE the energy dependence of resonance absorption peaks for typical light and heavy nuclei.
EO 2.6 EXPRESS mean free path in terms of macroscopic cross section.
EO 2.7 Given the number densities (or total density and component fractions) and microscopic cross sections of components, CALCULATE the macroscopic cross section for a mixture.
EO 2.8 CALCULATE a macroscopic cross section given a material density, atomic mass, and microscopic cross section.
EO 2.9 EXPLAIN neutron shadowing or self-shielding.
'

Introduction

Fission neutrons are born with an average energy of about 2 MeV. These fast neutrons interact with the reactor core materials in various absorption and scattering reactions. Collisions that result in scattering are useful in slowing neutrons to thermal energies. Thermal neutrons may be absorbed by fissile nuclei to produce more fissions or be absorbed in fertile material for conversion to fissionable fuel. Absorption of neutrons in structural components, coolant, and other non-fuel material results in the removal of neutrons without fulfilling any useful purpose. To safely and efficiently operate a nuclear reactor it is necessary to predict the probability that a particular absorption or scattering reaction will occur. Once these probabilities are known, if the availability of neutrons can be determined, then the rate at which these nuclear reactions take place can be predicted.

Atom Density

One important property of a material is the atom density. The atom density is the number of atoms of a given type per unit volume of the material. To calculate the atom density of a
Fixture checkpoint marker-1106-xray is recorded on fixture page 1106.
substance use Equation (2-1).

(2-1)

where:

N = atom density (atoms/cm)3
' = density (g/cm)3
N = Avogadro's number (6.022 x 10 atoms/mole) A
23
M = gram atomic weight

N
NA
M
2.699 g

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6.022 x 1023 atoms
mole
26.9815 g
mole
6.024 x 1022 atoms
cm3
DOE-HDBK-1019/1-93
Reactor Theory (Neutron Characteristics) NUCLEAR CROSS SECTIONS AND NEUTRON FLUX
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Example:
A block of aluminum has a density of 2.699 g/cm . If the gram atomic weight of
3
Fixture checkpoint marker-1107-xray is recorded on fixture page 1107.
aluminum is 26.9815 g, calculate the atom density of the aluminum.
Solution:
Cross Sections
The probability of a neutron interacting with a nucleus for a particular reaction is dependent upon not only the kind of nucleus involved, but also the energy of the neutron. Accordingly, the absorption of a thermal neutron in most materials is much more probable than the absorption of a fast neutron. Also, the probability of interaction will vary depending upon the type of reaction involved.
The probability of a particular reaction occurring between a neutron and a nucleus is called the microscopic cross section ( ) of the nucleus for the particular reaction. This cross section will vary with the energy of the neutron. The microscopic cross section may also be regarded as the effective area the nucleus presents to the neutron for the particular reaction. The larger the effective area, the greater the probability for reaction.
Because the microscopic cross section is an area, it is expressed in units of area, or square centimeters. A square centimeter is tremendously large in comparison to the effective area of a nucleus, and it has been suggested that a physicist once referred to the measure of a square centimeter as being "as big as a barn" when applied to nuclear processes. The name has persisted and microscopic cross sections are expressed in terms of barns. The relationship between barns and cm is shown below. 2
1 barn = 10-24 cm2
N
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Whether a neutron will interact with a certain volume of material depends not only on the microscopic cross section of the individual nuclei but also on the number of nuclei within that volume. Therefore, it is necessary to define another kind of cross section known as the macroscopic cross section ( ). The macroscopic cross section is the probability of a given reaction occurring per unit travel of the neutron. is related to the microscopic cross section ( ) by the relationship shown below.
(2-2)
where:
= macroscopic cross section (cm)
-1
N = atom density of material (atoms/cm)

= microscopic cross-section (cm)
2
The difference between the microscopic and macroscopic cross sections is extremely important and is restated for clarity. The microscopic cross section ( ) represents the effective target area that a single nucleus presents to a bombarding particle. The units are given in barns or cm .
2
The macroscopic cross section ( ) represents the effective target area that is presented by all of the nuclei contained in 1 cm of the material. The units are given as 1/cm or cm .
3 -1
A neutron interacts with an atom of the material it enters in two basic ways. It will either interact through a scattering interaction or through an absorption reaction. The probability of a neutron being absorbed by a particular atom is the microscopic cross section for absorption, . The probability of a neutron scattering off of a particular nucleus is the microscopic cross a section for scattering, s
. The sum of the microscopic cross section for absorption and the microscopic cross section for scattering is the total microscopic cross section, . T
= + T a s
Both the absorption and the scattering microscopic cross sections can be further divided. For instance, the scattering cross section is the sum of the elastic scattering cross section (se) and Fixture checkpoint marker-1108-xray is recorded on fixture page 1108.
the inelastic scattering cross section ( ). si
= + s se si
The microscopic absorption cross section ( ) includes all reactions except scattering. However, a for most purposes it is sufficient to merely separate it into two categories, fission (f) and capture ( ). Radiative capture of neutrons was described in the Neutron Interactions chapter c of Module 1.
= + a f c
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Figure 1 Typical Neutron Absorption Cross Section vs. Neutron Energy
The variation of absorption cross sections with neutron energy is often complicated. For many elements the absorption cross sections are small, ranging from a fraction of a barn to a few barns for slow (or thermal) neutrons.
For a considerable number of nuclides of moderately high (or high) mass numbers, an examination of the variation of the absorption cross section with the energy of the incident neutron reveals the existence of three regions on a curve of absorption cross section versus neutron energy. This cross section is illustrated in Figure 1. First, the cross section decreases steadily with increasing neutron energy in a low energy region, which includes the thermal range (E < 1 eV). In this region the absorption cross section, which is often high, is inversely proportional to the velocity (v). This region is frequently referred to as the "1/v region," because the absorption cross section is proportional to 1/v, which is the reciprocal of neutron velocity. Following the 1/v region, there occurs the "resonance region" in which the cross sections rise sharply to high values called "resonance peaks" for neutrons of certain energies, and then fall again. These energies are called resonance energies and are a result of the affinity of the nucleus for neutrons whose energies closely match its discrete, quantum energy levels. That is, when the binding energy of a neutron plus the kinetic energy of the neutron are exactly equal to the amount required to raise a compound nucleus from its ground state to a quantum

■
(
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Assuming that uranium-236 has a nuclear quantum energy level at 6.8 MeV above its ground state, calculate the kinetic energy a neutron must possess to undergo resonant absorption in uranium-235 at this resonance energy level.
BE = [Mass(U) + Mass(neutron) - Mass(U)] x 931 MeV/amu 235 236
BE = (235.043925 + 1.008665 - 236.045563) x 931 MeV/amu
BE = (0.007025 amu) x 931 MeV/amu = 6.54 MeV
6.8 MeV - 6.54 MeV = 0.26 MeV
The difference between the binding energy and the quantum energy level equals the amount of kinetic energy the neutron must possess. The typical heavy nucleus will have many closely■spaced resonanc complex and have more possible configurations and corresponding energy states. Light nuclei, being less complex, have fewer possible energy states and fewer resonances that are sparsely distributed at higher energy levels.
For higher neutron energies, the absorption cross section steadily decreases as the energy of the neutron increases. This is called the "fast neutron region." In this region the absorption cross sections are usually less than 10 barns.
With the exception of hydrogen, for which the value is fairly large, the elastic scattering cross sections are generally small, for example, 5 barns to 10 barns. This is close to the magnitude of the actual geometric cross sectional area expected for atomic nuclei. In potential scattering, the cross section is essentially constant and independent of neutron energy. Resonance elastic scattering and inelastic scattering exhibit resonance peaks similar to those associated with absorption cross sections. The resonances occur at lower energies for heavy nuclei than for light
Fixture checkpoint marker-1109-xray is recorded on fixture page 1109.
nuclei. In general, the variations in scattering cross sections are very small when compared to the variations that occur in absorption cross sections.
Mean Free Path
If a neutron has a certain probability of undergoing a particular interaction in one centimeter of travel, then the inverse of this value describes how far the neutron will travel (in the average case) before undergoing an interaction. This average distance traveled by a neutron before interaction is known as the mean free path for that interaction and is represented by the symbol ■. The relationship between the mean free path (■) and the macroscopic cross section (*) is shown below.
(2-3)
'
(
)
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Reactor Theory (Neutron Characteristics) NUCLEAR CROSS SECTIONS AND NEUTRON FLUX
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Calculation of Macroscopic Cross Section and Mean Free Path
Most materials are composed of several elements, and because most elements are composed of several isotopes, most materials involve many cross sections, one for each isotope involved. Therefore, to include all the isotopes within a given material, it is necessary to determine the macroscopic cross section for each isotope and then sum all the individual macroscopic cross

material.
* = N) + N) + N) + ... N) (2-4) 1 1 2 2 3 3 nn
where:
N = the number of nuclei per cm of the n element n
3 th
) = the microscopic cross section of the n element n
th
The following example problems illustrate the calculation of the macroscopic cross section for a single element and for combinations of materials.
Example 1:
Find the macroscopic thermal neutron absorption cross section for iron, which has a density of 7.86 g/cm . The microscopic cross section for absorption of iron is 2.56 barns 3 and the gram atomic weight is 55.847 g.
Solution:
Step 1: Using Equation (2-1), calculate the atom density of iron.
Step 2: Use this atom density in Equation (2-2) to calculate the macroscopic cross section.
'
'

Example 2:
An alloy is composed of 95% aluminum and 5% silicon (by weight). The density of the alloy is 2.66 g/cm . Properties of aluminum and silicon are shown below. 3

Element	Gram Atomic Weight	) a (barns)	s (barns)
Aluminum	26.9815	0.23	1.49
Silicon	28.0855	0.16	2.20

1. Calculate the atom densities for the aluminum and silicon.
2. Determine the absorption and scattering macroscopic cross sections for thermal neutrons.
3. Calculate the mean free paths for absorption and scattering.
Solution:
Fixture checkpoint marker-1110-xray is recorded on fixture page 1110.
Step 1: The density of the aluminum will be 95% of the total density. Using Equation (2-1) yields the atom densities.
a NAl a,Al NSi a,Si
5.64 x 1022 atoms
cm3
0.23 x 10 24 cm22.85 x 1021 atoms
cm3
0.16 x 10 24 cm2
0.0134 cm 1
s NAl s,Al NSi s,Si
5.64 x 1022 atoms
cm3
1.49 x 10 24 cm22.85 x 1021 atoms
cm3

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0.0903 cm 1
a
1
a
1
0.01345 cm 1
74.3 cm
s
1
s
Fixture checkpoint marker-1111-xray is recorded on fixture page 1111.
1
0.0903 cm 1
11.1 cm
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Step 2: The macroscopic cross sections for absorption and scattering are
calculated using Equation (2-4).
Step 3: The mean free paths are calculated by inserting the macroscopic cross
sections calculated above into Equation (2-3).
Thus, a neutron must travel an average of 74.3 cm to interact by absorption in this alloy, but it must travel only 11.1 cm to interact by scattering.
)
)
)
)
(■
(■
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NUCLEAR CROSS SECTIONS AND NEUTRON FLUX Reactor Theory (Neutron Characteristics)
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Effects of Temperature on Cross Section
As discussed, the microscopic absorption cross section varies significantly as neutron energy varies. The microscopic cross sections provided on most charts and tables are measured for a standard neutron velocity of 2200 meters/second, which corresponds to an ambient temperature of 68(F. Therefore, if our material is at a higher temperature, the absorption cross section will be lower than the value for 68(F, and any cross sections which involve absorption (for example,) ,) ,)) must be corrected for the existing temperature. acf
The following formula is used to correct microscopic cross sections for temperature. Although the example illustrates absorption cross section, the same formula may be used to correct capture and fission cross sections.
where:
) = microscopic cross section corrected for temperature
) = microscopic cross section at reference temperature (68(F or 20(C) o
T = reference temperature (68(F) in degrees Rankine ((R) or Kelvin ((K) o
T = temperature for which corrected value is being calculated
NOTE: When using this formula, all temperatures must be converted to (R or (K.

(K = (C + 273
Example:
What is the value of)f for uranium-235 for thermal neutrons at 500(F? Uranium-235 has a) of 583 barns at 68(F. f
Solution:

Neutron Flux
Macroscopic cross sections for neutron reactions with materials determine the probability of one neutron undergoing a specific reaction per centimeter of travel through that material. If one wants to determine how many reactions will actually occur, it is necessary to know how many neutrons are traveling through the material and how many centimeters they travel each second. It is convenient to consider the number of neutrons existing in one cubic centimeter at any one instant and the total distance they travel each second while in that cubic centimeter. The number of neutrons existing in a cm of material at any instant is called neutron density and is 3 represented by the symbol n with units of neutrons/cm . The total distance these neutrons can 3
Fixture checkpoint marker-1112-xray is recorded on fixture page 1112.
travel each second will be determined by their velocity.
A good way of defining neutron flux (1) is to consider it to be the total path length covered by all neutrons in one cubic centimeter during one second. Mathematically, this is the equation below.
1 = n v (2-5)
where:
1 = neutron flux (neutrons/cm -sec) 2
n = neutron density (neutrons/cm)3
v = neutron velocity (cm/sec)
The term neutron flux in some applications (for example, cross section measurement) is used as parallel beams of neutrons traveling in a single direction. The intensity (I) of a neutron beam is the product of the neutron density times the average neutron velocity. The directional beam intensity is equal to the number of neutrons per unit area and time (neutrons/cm -sec) falling on 2 a surface perpendicular to the direction of the beam.
One can think of the neutron flux in a reactor as being comprised of many neutron beams traveling in various directions. Then, the neutron flux becomes the scalar sum of these directional flux intensities (added as numbers and not vectors), that is, 1 = I + I + I +...I . 123 n
Since the atoms in a reactor do not interact preferentially with neutrons from any particular direction, all of these directional beams contribute to the total rate of reaction. In reality, at a given point within a reactor, neutrons will be traveling in all directions.

Self-Shielding
In some locations within the reactor, the flux level may be significantly lower than in other areas due to a phenomenon referred to as neutron shadowing or self-shielding. For example, the interior of a fuel pin or pellet will "see" a lower average flux level than the outer surfaces since an appreciable fraction of the neutrons will have been absorbed and therefore cannot reach the interior of the fuel pin. This is especially important at resonance energies, where the absorption cross sections are large.

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The important information in this chapter is summarized below.
DOE-HDBK-1019/1-93
Reactor Theory (Neutron Characteristics) NUCLEAR CROSS SECTIONS AND NEUTRON FLUX
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Nuclear Cross Section and Neutron Flux Summary
& Atom density (N) is the number of atoms of a given type per unit volume of material.
& Microscopic cross section ()) is the probability of a given reaction occurring between a neutron and a nucleus.
& Microscopic cross sections are measured in units of barns, where 1 barn = 10-24 cm .2
& Macroscopic cross section (*) is the probability of a given reaction occurring per unit length of travel of the neutron. The units for macroscopic cross section are cm . -1
& The mean free path (■) is the average distance that a neutron travels in a material between interactions.
& Neutron flux (1) is the total path length traveled by all neutrons in one cubic centimeter of material during one second.
& The macroscopic cross section for a material can be calculated using the equation below.
* = N)
& The absorption cross section for a material usually has three distinct regions. At low neutron energies (<1 eV) the cross section is inversely proportional to the neutron velocity.
Fixture checkpoint marker-1113-xray is recorded on fixture page 1113.
& Resonance absorption occurs when the sum of the kinetic energy of the neutron and its binding energy is equal to an allowed nuclear energy level of the nucleus.
& Resonance peaks exist at intermediate energy levels. For higher neutron energies, the absorption cross section steadily decreases as the neutron energy increases.
& The mean free path equals 1/*.
& The macroscopic cross section for a mixture of materials can be calculated using the equation below.
* = N 1 1 2 2 3 3 nn) + N) + N) + ... N)
& Self-shielding is where the local neutron flux is depressed within a material due to neutron absorption near the surface of the material.
1)
REACTION RATES DOE-HDBK-1019/1-93 Reactor Theory (Neutron Characteristics) NP-02 Page 18 Rev. 0
REACTION RATES
It is possible to determine the rate at which a nuclear reaction will take place based on the neutron flux, cross section for the interaction, and atom density of the target. This relationship illustrates how a change in one of these items affects the reaction rate.
EO 2.10 Given the neutron flux and macroscopic cross section, CALCULATE the reaction rate.
EO 2.11 DESCRIBE the relationship between neutron flux and reactor power.
Reaction Rates

flux (1)), and if the probability of having an interaction per centimeter path length is also known (macroscopic cross section (*)), multiply them together to get the number of interactions taking place in that cubic centimeter in one second. This value is known as the reaction rate and is denoted by the symbol R. The reaction rate can be calculated by the equation shown below.
R = 1 * (2-6)
where:
R = reaction rate (reactions/sec)
1 = neutron flux (neutrons/cm -sec) 2
* = macroscopic cross section (cm) -1
Substituting the fact that * = N) into Equation (2-6) yields the equation below.
where:
) = microscopic cross section (cm)2
N = atom density (atoms/cm)3
Rf f
1 x 1013 neutrons
cm2
sec
0.1 cm 1
1 x 1012 fissions
cm3
sec
Rf f
f
Rf
1.29 x 1012 fissions
cm3
sec
3 x 1013 neutrons
cm2
sec
0.043 cm 1
Fixture checkpoint marker-1114-xray is recorded on fixture page 1114.
Reactor Theory (Neutron Characteristics) DOE-HDBK-1019/1-93 REACTION RATES
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The reaction rate calculated will depend on which macroscopic cross section is used in the calculation. Normally, the reaction rate of greatest interest is the fission reaction rate.
Example:
If a one cubic centimeter section of a reactor has a macroscopic fission cross section of
0.1 cm , and if the thermal neutron flux is 10 neutrons/cm -sec, what is the fission rate -1 13 2
in that cubic centimeter?
Solution:
In addition to using Equation (2-6) to determine the reaction rate based on the physical properties of material, it is also possible to algebraically manipulate the equation to determine physical properties if the reaction rate is known.
Example:
A reactor operating at a flux level of 3 x 10 neutrons/cm -sec contains 10 atoms of
13 2 20
uranium-235 per cm . The reaction rate is 1.29 x 10 fission/cm . Calculate and .

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f f
Solution:
Step 1: The macroscopic cross section can be determined by solving
Equation (2-6) for and substituting the appropriate values. f
Rf N f
f
Rf
N
1.29 x 1012 fissions
cm3
sec
1 x 1020 atoms
cm3
3 x 1013 neutrons
cm2
sec
4.3 x 10 22 cm2 1 barn
1 x 10 24 cm2
430 barns
1 watt 1 erg
1 x 10 7 watt sec
1 MeV
1.602 x 10 6
erg
1 fission
200 MeV
3.12 x 1010 fissions
second
REACTION RATES DOE-HDBK-1019/1-93 Reactor Theory (Neutron Characteristics)
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Step 2: To find the microscopic cross section, replace f with (N x f) and solve
for .
f
Reactor Power Calculation
Multiplying the reaction rate per unit volume by the total volume of the core results in the total number of reactions occurring in the core per unit time. If the amount of energy involved in each reaction were known, it would be possible to determine the rate of energy release (power) due to a certain reaction.
Fixture checkpoint marker-1115-xray is recorded on fixture page 1115.
In a reactor where the average energy per fission is 200 MeV, it is possible to determine the number of fissions per second that are necessary to produce one watt of power using the following conversion factors.
1 fission = 200 MeV
1 MeV = 1.602 x 10 ergs -6
1 erg = 1 x 10 watt-sec -7
This is equivalent to stating that 3.12 x 10 fissions release 1 watt-second of energy. 10
P
th f V

watt sec
Reactor Theory (Neutron Characteristics) DOE-HDBK-1019/1-93 REACTION RATES
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The power released in a reactor can be calculated based on Equation (2-6). Multiplying the reaction rate by the volume of the reactor results in the total fission rate for the entire reactor. Dividing by the number of fissions per watt-sec results in the power released by fission in the reactor in units of watts. This relationship is shown mathematically in Equation (2-7) below.
(2-7)
where:
P = power (watts)
= thermal neutron flux (neutrons/cm -sec) th
2
= macroscopic cross section for fission (cm) f
-1
V = volume of core (cm)
Fixture checkpoint marker-1116-xray is recorded on fixture page 1116.
3
Relationship Between Neutron Flux and Reactor Power
In an operating reactor the volume of the reactor is constant. Over a relatively short period of time (days or weeks), the number density of the fuel atoms is also relatively constant. Since the atom density and microscopic cross section are constant, the macroscopic cross section must also be constant. Examining Equation (2-7), it is apparent that if the reactor volume and macroscopic cross section are constant, then the reactor power and the neutron flux are directly proportional. This is true for day-to-day operation. The neutron flux for a given power level will increase very slowly over a period of months due to the burnup of the fuel and resulting decrease in atom density and macroscopic cross section.
REACTION RATES DOE-HDBK-1019/1-93 Reactor Theory (Neutron Characteristics)
NP-02 Page 22 Rev. 0
Reaction Rates Summary
& The reaction rate is the number of interactions of a particular type occurring in a cubic centimeter of material in a second.
& The reaction rate can be calculated by the equation below.
R = 1 *
& Over a period of several days, while the atom density of the fuel can be considered constant, the neutron flux is directly proportional to reactor power.
Summary
The important information in this chapter is summarized below.
Reactor Theory (Neutron Characteristics) DOE-HDBK-1019/1-93 NEUTRON MODERATION
Rev. 0 Page 23 NP-02
NEUTRON MODERATION
In thermal reactors, the neutrons that cause fission are at a much lower energy than the energy level at which they were born from fission. In this type of reactor, specific materials must be included in the reactor design to reduce the energy level of the neutrons in an efficient manner.
EO 2.12 DEFINE the following concepts:
a. Thermalization d. Average logarithmic energy decrement
b. Moderator e. Macroscopic slowing down power
c. Moderating ratio

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EO 2.14 Given an average fractional energy loss per collision, CALCULATE the energy loss after a specified number of collisions.

Neutron Slowing Down and Thermalization

Fission neutrons are produced at an average energy level of 2 MeV and immediately begin to slow down as the result of numerous scattering reactions with a variety of target nuclei. After a number of collisions with nuclei, the speed of a neutron is reduced to such an extent that it has approximately the same average kinetic energy as the atoms (or molecules) of the medium in which the neutron is undergoing elastic scattering. This energy, which is only a small fraction of an electron volt at ordinary temperatures (0.025 eV at 20(C), is frequently referred to as the thermal energy, since it depends upon the temperature. Neutrons whose energies have been reduced to values in this region (< 1 eV) are designated thermal neutrons. The process of reducing the energy of a neutron to the thermal region by elastic scattering is referred to as thermalization, slowing down, or moderation. The material used for the purpose of thermalizing neutrons is called a moderator. A good moderator reduces the speed of neutrons in a small number of collisions, but does not absorb them to any great extent. Slowing the neutrons in as few collisions as possible is desirable in order to reduce the amount of neutron leakage from the core and also to reduce the number of resonance absorptions in non-fuel materials. Neutron leakage and resonance absorption will be discussed in the next module.

ln Ei

ln Ef

ln

Fixture checkpoint marker-1117-xray is recorded on fixture page 1117.

Ei

Ef

2

A

2

3

NEUTRON MODERATION DOE-HDBK-1019/1-93 Reactor Theory (Neutron Characteristics) NP-02 Page 24 Rev. 0

The ideal moderating material (moderator) should have the following nuclear properties.

large scattering cross section

small absorption cross section

large energy loss per collision

A convenient measure of energy loss per collision is the logarithmic energy decrement. The average logarithmic energy decrement is the average decrease per collision in the logarithm of the neutron energy. This quantity is represented by the symbol (Greek letter xi).

(2-8)

where:

= average logarithmic energy decrement

Ei = average initial neutron energy

Ef = average final neutron energy

The symbol is commonly called the average logarithmic energy decrement because of the fact that a neutron loses, on the average, a fixed fraction of its energy per scattering collision. Since the fraction of energy retained by a neutron in a single elastic collision is a constant for a given material, is also a constant. Because it is a constant for each type of material and does not depend upon the initial neutron energy, is a convenient quantity for assessing the moderating

The values for the lighter nuclei are tabulated in a variety of sources. The following commonly used approximation may be used when a tabulated value is not available.
This approximation is relatively accurate for mass numbers (A) greater than 10, but for some low values of A it may be in error by over three percent.
N
ln Ehigh ln Elow
ln
Ehigh
Elow
N
ln
Ehigh
Elow
ln
2 x 106eV
0.025 eV
0.948
19.2 collisions
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Since represents the average logarithmic energy loss per collision, the total number of collisions necessary for a neutron to lose a given amount of energy may be determined by dividing into the difference of the natural logarithms of the energy range in question. The number of collisions (N) to travel from any energy, E , to any lower energy, E , can be calculated as shown below. high low
Example:
How many collisions are required to slow a neutron from an energy of 2 MeV to a thermal energy of 0.025 eV, using water as the moderator? Water has a value of 0.948 for .
Solution:
Fixture checkpoint marker-1118-xray is recorded on fixture page 1118.
Sometimes it is convenient, based upon information known, to work with an average fractional energy loss per collision as opposed to an average logarithmic fraction. If the initial neutron energy level and the average fractional energy loss per collision are known, the final energy level for a given number of collisions may be computed using the following formula.
! (
NEUTRON MODERATION DOE-HDBK-1019/1-93 Reactor Theory (Neutron Characteristics)
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(2-9)
where:
E = initial neutron energy o
E = neutron energy after N collisions N
x = average fractional energy loss per collision
N = number of collisions
Example:
If the average fractional energy loss per collision in hydrogen is 0.63, what will be the energy of a 2 MeV neutron after (a) 5 collisions? (b) 10 collisions?
Solution:
a)

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Macroscopic Slowing Down Power
Although the logarithmic energy decrement is a convenient measure of the ability of a material to slow neutrons, it does not measure all necessary properties of a moderator. A better measure of the capabilities of a material is the macroscopic slowing down power. The macroscopic slowing down power (MSDP) is the product of the logarithmic energy decrement and the macroscopic cross section for scattering in the material. Equation (2-10) illustrates how to calculate the macroscopic slowing down power.
(2-10)
MR
s
a
Reactor Theory (Neutron Characteristics) DOE-HDBK-1019/1-93 NEUTRON MODERATION
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Moderating Ratio
Macroscopic slowing down power indicates how rapidly a neutron will slow down in the material in question, but it still does not fully explain the effectiveness of the material as a moderator. An element such as boron has a high logarithmic energy decrement and a good slowing down power, but it is a poor moderator because of its high probability of absorbing neutrons.
The most complete measure of the effectiveness of a moderator is the moderating ratio. The moderating ratio is the ratio of the macroscopic slowing down power to the macroscopic cross section for absorption. The higher the moderating ratio, the more effectively the material performs as a moderator. Equation (2-11) shows how to calculate the moderating ratio of a material.
(2-11)
Moderating properties of different materials are compared in Table 2.
TABLE 2
Moderating Properties of Materials
Material Collisions to Slowing Down
Number of Macroscopic
Thermalize Power
Moderating
Ratio
H2O 0.927 19 1.425 62
D2O 0.510 35 0.177 4830
Helium 0.427 42 9 x 10 51 -6
Beryllium 0.207 86 0.154 126
Fixture checkpoint marker-1119-xray is recorded on fixture page 1119.
Boron 0.171 105 0.092 0.00086
Carbon 0.158 114 0.083 216
NEUTRON MODERATION DOE-HDBK-1019/1-93 Reactor Theory (Neutron Characteristics)
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Summary
The important information in this chapter is summarized below.
Neutron Moderation Summary
& Thermalization is the process of reducing the energy level of a neutron from the energy level at which it is produced to an energy level in the thermal range.
& The moderator is the reactor material that is present for the purpose of thermalizing neutrons.

macroscopic cross section for absorption.
& The average logarithmic energy decrement (!) is the average change in the logarithm of neutron energy per collision.
& Macroscopic slowing down power is the product of the average logarithmic energy decrement and the macroscopic cross section for scattering.
& There are three desirable characteristics of a moderator.
1. large scattering cross section
2. small absorption cross section
3. large energy loss per collision
& The energy loss after a specified number of collisions can be calculated using the equation below.
E = E (1 - x) N o
N
Fixture checkpoint marker-1120-xray is recorded on fixture page 1120.
■
■ ■
Reactor Theory (Neutron Characteristics) DOE-HDBK-1019/1-93 PROMPT AND DELAYED NEUTRONS Rev. 0 Page 29 NP-02
PROMPT AND DELAYED NEUTRONS
Not all neutrons are released at the same time following fission. Most neutrons are released virtually instantaneously and are called prompt neutrons. A very small fraction of neutrons are released after the decay of fission products and are called delayed neutrons. Although delayed neutrons are a very small fraction of the total number of neutrons, they play an extremely important role in the control of the reactor.
EO 3.1 STATE the origin of prompt neutrons and delayed neutrons.
EO 3.2 STATE the approximate fraction of neutrons that are born as delayed neutrons from the fission of the following nuclear fuels:
a. Uranium-235
b. Plutonium-239
EO 3.3 EXPLAIN the mechanism for production of delayed neutrons.
EO 3.4 EXPLAIN prompt and delayed neutron generation times.
EO 3.5 Given prompt and delayed neutron generation times and delayed neutron fraction, CALCULATE the average generation time.
EO 3.6 EXPLAIN the effect of delayed neutrons on reactor control.
Neutron Classification
The great majority (over 99%) of the neutrons produced in fission are released within about 10-13 seconds of the actual fission event. These are called prompt neutrons. A small portion of fission neutrons are delayed neutrons, which are produced for some time after the fission process has taken place. The delayed neutrons are emitted immediately following the first beta decay of a fission fragment known as a delayed neutron precursor. An example of a delayed neutron precursor is bromine-87, shown below.
PROMPT AND DELAYED NEUTRONS DOE-HDBK-1019/1-93 Reactor Theory (Neutron Characteristics) NP-02 Page 30 Rev. 0
For most applications, it is convenient to combine the known precursors into groups with appropriately averaged properties. These groups vary somewhat depending on the fissile material in use. Table 3 lists the characteristics for the six precursor groups resulting from

precursors is called the delayed neutron fraction for that precursor. The total fraction of all neutrons born as delayed neutrons is called the delayed neutron fraction (■). The fraction of delayed neutrons produced varies depending on the predominant fissile nuclide in use. The delayed neutron fractions (■) for the fissile nuclides of most interest are as follows: uranium-233 (0.0026), uranium-235 (0.0065), uranium-238 (0.0148), and plutonium-239 (0.0021).

TABLE 3

Delayed Neutron Precursor Groups
for Thermal Fission in Uranium-235

Group	Half-Life (sec)	Delayed Neutron Fraction	Average Energy (MeV)
1	55.7	0.00021	0.25
2	22.7	0.00142	0.46
3	6.2	0.00127	0.41
4	2.3	0.00257	0.45
5	0.61	0.00075	0.41
6	0.23	0.00027	-
Total	-	0.0065	-

Neutron Generation Time

The neutron generation time is the time required for neutrons from one generation to cause the fissions that produce the next generation of neutrons. The generation time for prompt neutrons Fixture checkpoint marker-1121-xray is recorded on fixture page 1121.
(5* - pronounced "ell-star") is the total time from birth to rebirth. Three time intervals are involved: (a) the time it takes a fast neutron to slow down to thermal energy, (b) the time the now thermal neutron exists prior to absorption in fuel, and (c) the time required for a fissionable nucleus to emit a fast neutron after neutron absorption.

Fast neutrons slow to thermal energies or leak out of the reactor in 10 seconds to 10 seconds, -4 -6 depending on the moderator. In water moderated reactors, thermal neutrons tend to exist for about 10 seconds before they are absorbed. Fission and fast neutron production following -4 neutron absorption in a fissionable nucleus occurs in about 10 seconds. Thus, fast reactors -13 have an 5* of about 10 seconds, while thermal reactors have an 5* of about 10 seconds + 10 -6 -6 -4 seconds, which is about 10 seconds to 10 seconds. -4 -5

■ ■ ■ ■
■ ■ ■ ■
■ ■

On the other hand, the average generation time for the six delayed neutron groups is the total time from the birth of the fast neutron to the emission of the delayed neutron. Again, three time intervals are involved: (a) the time it takes a fast neutron to slow down to thermal energy, (b) the time the thermal neutron exists prior to absorption, and (c) the average time from neutron absorption to neutron emission by the six precursor groups. The average time for decay of precursors from uranium-235 is 12.5 seconds. The other terms in the delayed neutron generation time are insignificant when compared to this value, and the average delayed neutron generation time becomes ■ 12.5 seconds.

A neutron generation time in the range of 10 seconds to 10 seconds or faster could result in -4 -5 very rapid power excursions, and control would not be possible without the dependence upon delayed neutrons to slow down the rate of the reaction. The average generation time, and hence the rate that power can rise, is determined largely by the delayed neutron generation time. The

(2-12)
Example:
Assume a prompt neutron generation time for a particular reactor of 5 x 10 seconds and -5
a delayed neutron generation time of 12.5 seconds. If ■ is 0.0065, calculate the average
generation time.
Solution:
This example demonstrates the effect delayed neutrons have on the neutron generation time and
thus reactor control. If a reactor were to be operated in a sustained chain reaction using only
prompt neutrons (■ = 0), the generation time from the previous example would be about
5 x 10 seconds. However, by operating the reactor such that a 0.0065 fraction of neutrons are -5
delayed, the generation life time is extended to 0.0813 seconds, providing time for adequate
operator control. Therefore, although only a small fraction of the total neutron population,
delayed neutrons are extremely important to the control and maintenance of a sustained fission
chain reaction.
Summary
The important information in this chapter is summarized on the following page.
■ ■ ■ ■
PROMPT AND DELAYED NEUTRONS DOE-HDBK-1019/1-93 Reactor Theory (Neutron Characteristics)
NP-02 Page 32 Rev. 0
Prompt and Delayed Neutrons Summary
& Prompt neutrons are released directly from fission within 10 seconds of the -13
fission event.
& Delayed neutrons are released from the decay of fission products that are called
delayed neutron precursors. Delayed neutron precursors are grouped according to
half-life. Half-lives vary from fractions of a second to almost a minute.
& The fraction of neutrons born as delayed neutrons is different for different fuel
materials. Following are values for some common fuel materials.
Fixture checkpoint marker-1122-xray is recorded on fixture page 1122.
Uranium-235 0.0065
Plutonium-239 0.0021
& Delayed neutrons are produced by a classification of fission products known as
delayed neutron precursors. When a delayed neutron precursor undergoes a
■ decay, it results in an excited daughter nucleus which immediately ejects a -
neutron. Therefore, these delayed neutrons appear with a half-life of the delayed
neutron precursor.
& The delayed neutron generation time is the total time from the birth of the fast
neutron to the emission of the delayed neutron in the next generation. Delayed
neutron generation times are dominated by the half-life of the delayed neutron
precursor. The average delayed neutron generation time is about 12.5 seconds.
& A prompt neutron generation time is the sum of the amount of time it takes a fast
neutron to thermalize, the amount of time the neutron exists as a thermal neutron
before it is absorbed, and the amount of time between a fissionable nuclide
absorbing a neutron and fission neutrons being released. Prompt neutron
generation time is about 5 x 10 seconds. -5
& The average neutron generation time can be calculated from the prompt and
delayed neutron generation times and the delayed neutron fraction using
Equation (2-12).
& Delayed neutrons are responsible for the ability to control the rate at which power

possible due to the rapid power changes.
Reactor Theory (Neutron Characteristics) DOE-HDBK-1019/1-93 NEUTRON FLUX SPECTRUM
Rev. 0 Page 33 NP-02
Figure 2 Prompt Fission Neutron Energy Spectrum for
Thermal Fission of Uranium-235
NEUTRON FLUX SPECTRUM
The number of neutrons that exist at a given energy level varies. A plot of either the fraction of neutrons or the neutron flux at a given energy versus the energy level is called a neutron energy spectrum. The neutron energy spectrum varies widely for different types of reactors.
EO 4.1 STATE the average energy at which prompt neutrons are produced.
EO 4.2 DESCRIBE the neutron energy spectrum in the following reactors:
a. Fast reactor
b. Thermal reactor
EO 4.3 EXPLAIN the reason for the particular shape of the fast, intermediate, and slow energy regions of the neutron flux spectrum for a thermal reactor.
Prompt Neutron Energies
The neutrons produced by fission are high energy neutrons, and almost all fission neutrons have energies between 0.1 MeV and 10 MeV. The neutron energy distribution, or spectrum, may best be described by plotting the fraction of neutrons per MeV as a function of neutron energy, as shown in Figure 2. From this figure it can be seen that the most probable neutron energy is about 0.7 MeV. Also, from this data it can be shown that the average
Fixture checkpoint marker-1123-xray is recorded on fixture page 1123.
energy of fission neutrons is about 2 MeV. Figure 2 is the neutron energy spectrum for thermal fission in uranium-235. The values will vary slightly for other nuclides.
NEUTRON FLUX SPECTRUM DOE-HDBK-1019/1-93 Reactor Theory (Neutron Characteristics)
NP-02 Page 34 Rev. 0
Figure 3 Comparison of Neutron Flux Spectra for
Thermal and Fast Breeder Reactor
Thermal and Fast Breeder Reactor Neutron Spectra
The spectrum of neutron energies produced by fission varies significantly from the energy spectrum, or flux, existing in a reactor at a given time. Figure 3 illustrates the difference in

distribution of neutrons from fission is essentially the same for both reactors, so the differences in the curve shapes may be attributed to the neutron moderation or slowing down effects.
No attempt is made to thermalize or slow down neutrons in the fast breeder reactor (liquid metal cooled); therefore, an insignificant number of neutrons exist in the thermal range. For the thermal reactor (water moderated), the spectrum of neutrons in the fast region (> 0.1 MeV) has a shape similar to that for the spectrum of neutrons emitted by the fission process.
In the thermal reactor, the flux in the intermediate energy region (1 eV to 0.1 MeV) has approximately a 1/E dependence. That is, if the energy (E) is halved, the flux doubles. This 1/E dependence is caused by the slowing down process, where elastic collisions remove a constant fraction of the neutron energy per collision (on the average), independent of energy; thus, the neutron loses larger amounts of energy per collision at higher energies than at lower
Fixture checkpoint marker-1124-xray is recorded on fixture page 1124.
energies. The fact that the neutrons lose a constant fraction of energy per collision causes the neutrons to tend to "pile up" at lower energies, that is, a greater number of neutrons exist at the lower energies as a result of this behavior.
vp
2 k T
m
vp
2 k T
m
2 1.38 x 10 16 erg
K
293 K
1.66 x 10 24 g
2.2 x 105 cm
sec
1 m
100 cm
2200 m
sec
Reactor Theory (Neutron Characteristics) DOE-HDBK-1019/1-93 NEUTRON FLUX SPECTRUM
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In the thermal region the neutrons achieve a thermal equilibrium with the atoms of the moderator material. In any given collision they may gain or lose energy, and over successive collisions will gain as much energy as they lose. These thermal neutrons, even at a specific temperature, do not all have the same energy or velocity; there is a distribution of energies, usually referred to as the Maxwell distribution (e.g., Figure 2). The energies of most thermal neutrons lie close to the most probable energy, but there is a spread of neutrons above and below this value.
Most Probable Neutron Velocities
The most probable velocity (vp) of a thermal neutron is determined by the temperature of the medium and can be determined by Equation (2-13) .
(2-13)
where:
v = most probable velocity of neutron (cm/sec) p
k = Boltzman's constant (1.38 x 10 erg/ K) -16
T = absolute temperature in degrees Kelvin (K)
m = mass of neutron (1.66 x 10 grams) -24

SYNTHETIC FIXTURE PAGE 1125 OF 1200

Calculate the most probable velocities for neutrons in thermal equilibrium with their surroundings at the following temperatures. a) 20 C, b) 260 C.
Solution:
a) Calculate the most probable velocity for 20 C using Equation (2-13).
vp
2 k T
m
2 1.38 x 10 16 erg
K
533 K
1.66 x 10 24 g
2.977 x 105 cm
sec
1 m
100 cm
2977 m
sec
NEUTRON FLUX SPECTRUM DOE-HDBK-1019/1-93 Reactor Theory (Neutron Characteristics)
Fixture checkpoint marker-1125-xray is recorded on fixture page 1125.
NP-02 Page 36 Rev. 0
b) Calculate the most probable velocity for 260 C using Equation (2-13).
From these calculations it is evident that the most probable velocity of a thermal neutron increases as temperature increases. The most probable velocity at 20 C is of particular importance since reference data, such as nuclear cross sections, are tabulated for a neutron velocity of 2200 meters per second.
Reactor Theory (Neutron Characteristics) DOE-HDBK-1019/1-93 NEUTRON FLUX SPECTRUM
Rev. 0 Page 37 NP-02
Neutron Flux Spectrum Summary
& Prompt neutrons are born at energies between 0.1 MeV and 10 MeV. The average prompt neutron energy is about 2 MeV.
& Fast reactors have a neutron energy spectrum that has the same shape as the prompt neutron energy spectrum.
& Thermal reactors have a neutron energy spectrum that has two pronounced peaks, one in the thermal energy region where the neutrons are in thermal equilibrium with the core materials and another in the fast region at energies where neutrons are produced. The flux in the intermediate region (1 eV to 0.1 MeV) has a roughly 1/E dependence.
& The neutron flux spectrum for the fast energy region of a thermal reactor has a shape similar to that of the spectrum of neutrons emitted by the fission process.
& The reason for the 1/E flux dependence at intermediate energy levels in a thermal reactor is due to the neutrons' tendency to lose a constant fraction of energy per collision. Since the neutrons lose a greater amount at the higher energies, the neutrons tend to "pile up" at lower energies where they lose less energy per collision.
& The neutron flux spectrum for the slow region of a thermal reactor contains a peak at the energy where the neutrons are in thermal equilibrium with the atoms of the surrounding material.
Summary

The DOE Fundamentals Handbooks represent the needs of various DOE nuclear facilities' fundamental training requirements. To increase their applicability to nonreactor nuclear facilities, the Reactor Operator Fundamentals Manual learning objectives were distributed to the Nuclear Facility Training Coordination Program Steering Committee for review and comment. To update their reactor-specific content, DOE Category A reactor training managers also reviewed and commented on the content. On the basis of feedback from these sources, information that applied to two or more DOE nuclear facilities was considered generic and was included. The final draft of each of the handbooks was then reviewed by these two groups. This approach has resulted in revised modular handbooks that contain sufficient detail such that each facility may adjust the content to fit their specific needs.

Each handbook contains an abstract, a foreword, an overview, learning objectives, and text material, and is divided into modules so that content and order may be modified by individual DOE contractors to suit their specific training needs. Each handbook is supported by a separate examination bank with an answer key.

The DOE Fundamentals Handbooks have been prepared for the Assistant Secretary for Nuclear Energy, Office of Nuclear Safety Policy and Standards, by the DOE Training Coordination Program. This program is managed by EG&G Idaho, Inc.

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DOE-HDBK-1019/1-93

NUCLEAR PHYSICS AND REACTOR THEORY

OVERVIEW

The Department of Energy Fundamentals Handbook entitled Nuclear Physics and Reactor Theory was prepared as an information resource for personnel who are responsible for the operation of the Department's nuclear facilities. Almost all processes that take place in a nuclear facility involves the transfer of some type of energy. A basic understanding of nuclear physics

Fixture checkpoint marker-1126-xray is recorded on fixture page 1126.

and reactor theory is necessary for DOE nuclear facility operators, maintenance personnel, and the technical staff to safely operate and maintain the facility and facility support systems. The information in this handbook is presented to provide a foundation for applying engineering concepts to the job. This knowledge will help personnel understand the impact that their actions may have on the safe and reliable operation of facility components and systems.

The Nuclear Physics and Reactor Theory handbook consists of four modules that are contained in two volumes. The following is a brief description of the information presented in each module of the handbook.

Volume 1 of 2

Module 1 - Atomic and Nuclear Physics

Introduces concepts of atomic physics including the atomic nature of matter, the chart of the nuclides, radioactivity and radioactive decay, neutron interactions and fission, and the interaction of radiation with matter.

Module 2 - Reactor Theory (Nuclear Parameters)

Provides information on reactor theory and neutron characteristics. Includes topics such as neutron sources, neutron flux, neutron cross sections, reaction rates, neutron moderation, and prompt and delayed neutrons.

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NUCLEAR PHYSICS AND REACTOR THEORY

OVERVIEW (Cont.)

Volume 2 of 2

SYNTHETIC FIXTURE PAGE 1127 OF 1200

Explains the nuclear parameters associated with reactor theory. Topics include the neutron life cycle, reactivity and reactivity coefficients, neutron poisons, and control rods.
Module 4 - Reactor Theory (Reactor Operations)
Introduces the reactor operations aspect of reactor theory. Topics include subcritical multiplication, reactor kinetics, and reactor operation.
The information contained in this handbook is not all-encompassing. An attempt to present the entire subject of nuclear physics and reactor theory would be impractical. However, the Nuclear Physics and Reactor Theory handbook presents enough information to provide the reader with the fundamental knowledge necessary to understand the advanced theoretical concepts presented in other subject areas, and to understand basic system and equipment operation.
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Department of Energy
Fundamentals Handbook
NUCLEAR PHYSICS
AND REACTOR THEORY
Module 1
Atomic and Nuclear Physics
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Atomic and Nuclear Physics DOE-HDBK-1019/1-93 OBJECTIVES

TERMINAL OBJECTIVE

1.0 Given sufficient information, DESCRIBE atoms, including components, structure, and Fixture checkpoint marker-1130-xray is recorded on fixture page 1130.
nomenclature.

ENABLING OBJECTIVES

1.1 STATE the characteristics of the following atomic particles, including mass, charge, and location within the atom:
a. Proton
b. Neutron
c. Electron

1.2 DESCRIBE the Bohr model of an atom.

1.3 DEFINE the following terms:
a. Nuclide c. Atomic number
b. Isotope d. Mass number

1.4 Given the standard A
ZX notation for a particular nuclide, DETERMINE the following:
a. Number of protons
b. Number of neutrons
c. Number of electrons

1.5 DESCRIBE the three forces that act on particles within the nucleus and affect the stability of the nucleus.

1.6 DEFINE the following terms:
a. Enriched uranium
b. Depleted uranium

1.7 DEFINE the following terms:
a. Mass defect
b. Binding energy

electron, CALCULATE the mass defect and binding energy of the nuclide.
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OBJECTIVES DOE-HDBK-1019/1-93 Atomic and Nuclear Physics
TERMINAL OBJECTIVE
2.0 Given necessary references, DESCRIBE the various modes of radioactive decay.
ENABLING OBJECTIVES
2.1 DESCRIBE the following processes:
a. Alpha decay d. Electron capture
b. Beta-minus decay e. Internal conversions
c. Beta-plus decay f. Isomeric transitions
2.2 Given a Chart of the Nuclides, WRITE the radioactive decay chain for a nuclide.
2.3 EXPLAIN why one or more gamma rays typically accompany particle emission.
2.4 Given the stability curve on the Chart of the Nuclides, DETERMINE the type of radioactive decay that the nuclides in each region of the chart will typically undergo.
2.5 DEFINE the following terms:
a. Radioactivity d. Radioactive decay constant
b. Curie e. Radioactive half-life
c. Becquerel
2.6 Given the number of atoms and either the half-life or decay constant of a nuclide, CALCULATE the activity.
2.7 Given the initial activity and the decay constant of a nuclide, CALCULATE the activity at any later time.
2.8 CONVERT between the half-life and decay constant for a nuclide.
2.9 Given the Chart of the Nuclides and the original activity, PLOT the radioactive decay curve for a nuclide on either linear or semi-log coordinates.
2.10 DEFINE the following terms:
a. Radioactive equilibrium
b. Transient radioactive equilibrium
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Atomic and Nuclear Physics DOE-HDBK-1019/1-93 OBJECTIVES
Fixture checkpoint marker-1131-xray is recorded on fixture page 1131.
TERMINAL OBJECTIVE
3.0 Without references, DESCRIBE the different nuclear interactions initiated by neutrons.
ENABLING OBJECTIVES
3.1 DESCRIBE the following scattering interactions between a neutron and a nucleus:
a. Elastic scattering
b. Inelastic scattering
3.2 STATE the conservation laws that apply to an elastic collision between a neutron and a nucleus.
3.3 DESCRIBE the following reactions where a neutron is absorbed in a nucleus:
a. Radiative capture
b. Particle ejection
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OBJECTIVES DOE-HDBK-1019/1-93 Atomic and Nuclear Physics
TERMINAL OBJECTIVE
4.0 Without references, DESCRIBE the fission process.
ENABLING OBJECTIVES
4.1 EXPLAIN the fission process using the liquid drop model of a nucleus.

SYNTHETIC FIXTURE PAGE 1132 OF 1200

a. Excitation energy
b. Critical energy
4.3 DEFINE the following terms:
a. Fissile material
b. Fissionable material
c. Fertile material
4.4 DESCRIBE the processes of transmutation, conversion, and breeding.
4.5 DESCRIBE the curve of Binding Energy per Nucleon versus mass number and give a qualitative description of the reasons for its shape.
4.6 EXPLAIN why only the heaviest nuclei are easily fissioned.
4.7 EXPLAIN why uranium-235 fissions with thermal neutrons and uranium-238 fissions only with fast neutrons.
4.8 CHARACTERIZE the fission products in terms of mass groupings and radioactivity.
4.9 Given the nuclides involved and their masses, CALCULATE the energy released from fission.
4.10 Given the curve of Binding Energy per Nucleon versus mass number, CALCULATE the energy released from fission.
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Atomic and Nuclear Physics DOE-HDBK-1019/1-93 OBJECTIVES
TERMINAL OBJECTIVE
5.0 Without references, DESCRIBE how the various types of radiation interact with matter.
ENABLING OBJECTIVES
5.1 DESCRIBE interactions of the following with matter:
a. Alpha particle c. Positron
b. Beta particle d. Neutron
5.2 DESCRIBE the following ways that gamma radiation interacts with matter:
a. Compton scattering
b. Photoelectric effect
c. Pair production
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OBJECTIVES DOE-HDBK-1019/1-93 Atomic and Nuclear Physics
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NP-01 Page xii Rev. 0
Atomic and Nuclear Physics DOE-HDBK-1019/1-93 ATOMIC NATURE OF MATTER
ATOMIC NATURE OF MATTER
All matter is composed of atoms. The atom is the smallest amount of matter that retains the properties of an element. Atoms themselves are
Fixture checkpoint marker-1132-xray is recorded on fixture page 1132.
composed of smaller particles, but these smaller particles no longer have the same properties as the overall element.
EO 1.1 STATE the characteristics of the following atomic particles, including mass, charge, and location within the atom:
a. Proton
b. Neutron
c. Electron
EO 1.2 DESCRIBE the Bohr model of an atom.
EO 1.3 DEFINE the following terms:
a. Nuclide c. Atomic number

SYNTHETIC FIXTURE PAGE 1133 OF 1200

EO 1.4 Given the standard A
ZX notation for a particular nuclide,
DETERMINE the following:
a. Number of protons
b. Number of neutrons
c. Number of electrons
EO 1.5 DESCRIBE the three forces that act on particles within the nucleus
and affect the stability of the nucleus.
Structure of Matter
Early Greek philosophers speculated that the earth was made up of different combinations of basic substances, or elements. They considered these basic elements to be earth, air, water, and fire. Modern science shows that the early Greeks held the correct concept that matter consists of a combination of basic elements, but they incorrectly identified the elements.
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Fixture checkpoint marker-1133-xray is recorded on fixture page 1133.
ATOMIC NATURE OF MATTER DOE-HDBK-1019/1-93 Atomic and Nuclear Physics
In 1661 the English chemist Robert Boyle published the modern criterion for an element. He defined an element to be a basic substance that cannot be broken down into any simpler substance after it is isolated from a compound, but can be combined with other elements to form compounds. To date, 105 different elements have been confirmed to exist, and researchers claim to have discovered three additional elements. Of the 105 confirmed elements, 90 exist in nature and 15 are man-made.
Another basic concept of matter that the Greeks debated was whether matter was continuous or discrete. That is, whether matter could be continuously divided and subdivided into ever smaller particles or whether eventually an indivisible particle would be encountered. Democritus in about 450 B.C. argued that substances were ultimately composed of small, indivisible particles that he labeled atoms. He further suggested that different substances were composed of different atoms or combinations of atoms, and that one substance could be converted into another by rearranging the atoms. It was impossible to conclusively prove or disprove this proposal for more than 2000 years.
The modern proof for the atomic nature of matter was first proposed by the English chemist John Dalton in 1803. Dalton stated that each chemical element possesses a particular kind of atom, and any quantity of the element is made up of identical atoms of this kind. What distinguishes one element from another element is the kind of atom of which it consists, and the basic physical difference between kinds of atoms is their weight.
Subatomic Particles
For almost 100 years after Dalton established the atomic nature of atoms, it was considered impossible to divide the atom into even smaller parts. All of the results of chemical experiments during this time indicated that the atom was indivisible. Eventually, experimentation into electricity and radioactivity indicated that particles of matter smaller than the atom did indeed exist. In 1906, J. J. Thompson won the Nobel Prize in physics for establishing the existence of electrons. Electrons are negatively-charged particles that have 1/1835 the mass of the hydrogen atom. Soon after the discovery of electrons, protons were discovered. Protons are relatively large particles that have almost the same mass as a hydrogen atom and a positive charge equal in magnitude (but opposite in sign) to that of the electron. The third subatomic particle to be discovered, the neutron, was not found until 1932. The neutron has almost the same mass as the proton, but it is electrically neutral.
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Bohr Model of the Atom

The British physicist Ernest Rutherford postulated that the positive charge in an atom is concentrated in a small region called a nucleus at the center of the atom with electrons existing in orbits around it. Niels Bohr, coupling Rutherford's postulation with the quantum theory introduced by Max Planck, proposed that the atom consists of a dense nucleus of protons surrounded by electrons traveling in discrete orbits at fixed distances from the nucleus. An electron in one of these orbits or shells has a specific or discrete quantity of energy (quantum). When an electron moves from one allowed orbit to another allowed orbit, the energy difference between the two states is emitted or absorbed in the form of a single quantum of radiant energy called a photon. Figure 1 is Bohr's model of the hydrogen atom showing an electron as having just dropped from the third shell to the first shell with the emission of a photon that has an energy = hv. (h = Planck's constant = 6.63 x 10-34 J-s and v = frequency of the photon.) Bohr's theory was the first to successfully account for the discrete energy levels of this radiation as measured in the laboratory. Although Bohr's atomic model is designed specifically to explain the hydrogen atom, his theories apply generally to the structure of all atoms. Additional information on electron shell theory can be found in the Chemistry Fundamentals Handbook.

Figure 1 Bohr's Model of the Hydrogen Atom

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Properties of the three subatomic particles are listed in Table 1.

TABLE 1

Fixture checkpoint marker-1134-xray is recorded on fixture page 1134.

Properties of Subatomic Particles

Particle	Location	Charge	Mass
Neutron	Nucleus	none	1.008665 amu
Proton	Nucleus	+1	1.007277 amu
Electron	Shells around nucleus	-1	0.0005486 amu

Measuring Units on the Atomic Scale

The size and mass of atoms are so small that the use of normal measuring units, while possible, is often inconvenient. Units of measure have been defined for mass and energy on the atomic scale to make measurements more convenient to express. The unit of measure for mass is the atomic mass unit (amu). One atomic mass unit is equal to 1.66 x 10-24 grams. The reason for this particular value for the atomic mass unit will be discussed in a later chapter. Note from Table 1 that the mass of a neutron and a proton are both about 1 amu. The unit for energy is the electron volt (eV). The electron volt is the amount of energy acquired by a single electron when it falls through a potential difference of one volt. One electron volt is equivalent to 1.602 x 10-19 joules or 1.18 x 10-19 foot-pounds.

Nuclides

The total number of protons in the nucleus of an atom is called the atomic number of the atom and is given the symbol Z. The number of electrons in an electrically-neutral atom is the same as the number of protons in the nucleus. The number of neutrons in a nucleus is known as the neutron number and is given the symbol N. The mass number of the nucleus is the total number of nucleons, that is, protons and neutrons in the nucleus. The mass number is given the symbol A and can be found by the equation Z + N = A.

Each of the chemical elements has a unique atomic number because the atoms of different elements contain a different number of protons. The atomic number of an atom identifies the particular element.

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Each type of atom that contains a unique combination of
Figure 2 Nomenclature for
Identifying Nuclides
protons and neutrons is called a nuclide. Not all combinations of numbers of protons and neutrons are possible, but about 2500 specific nuclides with unique combinations of neutrons and protons have been identified. Each nuclide is denoted by the chemical symbol of the element with the atomic number written as a subscript and the mass number written as a superscript, as shown in Figure 2. Because each element has a unique name, chemical symbol, and atomic number, only one of the three is necessary to identify the element. For this reason nuclides can also be identified by either the chemical name or the chemical symbol followed by the mass number (for example, U-235 or uranium-235). Another common format is to use the abbreviation of the chemical element with the mass number superscripted (for example, 235U). In this handbook the format used in the text will usually be the element's name followed by the mass number. In equations and tables, the format in Figure 2 will usually be used.

Example:

State the name of the element and the number of protons, electrons, and neutrons in the nuclides listed below.

1
1H
10
5B
14

Fixture checkpoint marker-1135-xray is recorded on fixture page 1135.

7N
11
4
4
8Cd
23
9
9
4Pu

Solution:

The name of the element can be found from the Periodic Table (refer to Chemistry Fundamentals Handbook) or the Chart of the Nuclides (to be discussed later). The number of protons and electrons are equal to Z. The number of neutrons is equal to Z - A.

Nuclide Element Protons Electrons Neutrons
1
1H hydrogen 1 1 0

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5B boron 5 5 5
14
7N nitrogen 7 7 7
11
4
4
8Cd cadmium 48 48 66
23
9
9
4Pu plutonium 94 94 145
Isotopes
Isotopes are nuclides that have the same atomic number and are therefore the same element, but differ in the number of neutrons. Most elements have a few stable isotopes and several unstable, radioactive isotopes. For example, oxygen has three stable isotopes that can be found in nature (oxygen-16, oxygen-17, and oxygen-18) and eight radioactive isotopes. Another example is hydrogen, which has two stable isotopes (hydrogen-1 and hydrogen-2) and a single radioactive isotope (hydrogen-3).
The isotopes of hydrogen are unique in that they are each commonly referred to by a unique name instead of the common chemical element name. Hydrogen-1 is almost always referred to as hydrogen, but the term protium is infrequently used also. Hydrogen-2 is commonly called deuterium and symbolized 2
1D. Hydrogen-3 is commonly called tritium and symbolized 31T. This
text will normally use the symbology 2
1H and 31H for deuterium and tritium, respectively.
Atomic and Nuclear Radii
The size of an atom is difficult to define exactly due to the fact that the electron cloud, formed by the electrons moving in their various orbitals, does not have a distinct outer edge. A reasonable measure of atomic size is given by the average distance of the outermost electron from the nucleus. Except for a few of the lightest atoms, the average atomic radii are approximately the same for all atoms, about 2 x 10 -8 cm.
Like the atom the nucleus does not have a sharp outer boundary. Experiments have shown that the nucleus is shaped like a sphere with a radius that depends on the atomic mass number of the atom. The relationship between the atomic mass number and the radius of the nucleus is shown in the following equation.
Fixture checkpoint marker-1136-xray is recorded on fixture page 1136.

r = (1.25 x 10 -13 cm) A1/3
where:
r = radius of the nucleus (cm)
A = atomic mass number (dimensionless)
The values of the nuclear radii for some light, intermediate, and heavy nuclides are shown in Table 2.
TABLE 2
Calculated Values for Nuclear
Radii
Nuclide Radius of Nucleus

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1H 1.25 x 10 -13 cm
10
5B 2.69 x 10 -13 cm
5
2
6
6Fe 4.78 x 10 -13 cm
17
7
8
2Hf 7.01 x 10 -13 cm
23
Fixture checkpoint marker-1137-xray is recorded on fixture page 1137.
9
8
2U 7.74 x 10 -13 cm
25
9
2
8Cf 7.89 x 10 -13 cm
From the table, it is clear that the radius of a typical atom (e.g. 2 x 10 -8 cm) is more than 25,000 times larger than the radius of the largest nucleus.
Nuclear Forces
In the Bohr model of the atom, the nucleus consists of positively-charged protons and electrically■neutra as nucleons. One problem that the Bohr model of the atom presented was accounting for an attractive force to overcome the repulsive force between protons.
Two forces present in the nucleus are (1) electrostatic forces between charged particles and (2) gravitational forces between any two objects that have mass. It is possible to calculate the magnitude of the gravitational force and electrostatic force based upon principles from classical physics.
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Fg
G m1 m2
r
2
Fe
K Q1 Q2
r
2
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Newton stated that the gravitational force between two bodies is directly proportional to the masses of the two bodies and inversely proportional to the square of the distance between the bodies. This relationship is shown in the equation below.
where:
F = gravitational force (newtons) g
m = mass of first body (kilograms) 1
m = mass of second body (kilograms) 2

-11 2 2
r = distance between particles (meters)
The equation illustrates that the larger the masses of the objects or the smaller the distance between the objects, the greater the gravitational force. So even though the masses of nucleons are very small, the fact that the distance between nucleons is extremely short may make the gravitational force significant. It is necessary to calculate the value for the gravitational force and compare it to the value for other forces to determine the significance of the gravitational force in the nucleus. The gravitational force between two protons that are separated by a distance of 10 meters is about 10 newtons. -20 -24
Coulomb's Law can be used to calculate the force between two protons. The electrostatic force is directly proportional to the electrical charges of the two particles and inversely proportional to the square of the distance between the particles. Coulomb's Law is stated as the following equation.
where:
F = electrostatic force (newtons) e
K = electrostatic constant (9.0 x 10 N-m /C)
9 2 2
Q = charge of first particle (coulombs) 1
Q = charge of second particle (coulombs) 2
Fixture checkpoint marker-1138-xray is recorded on fixture page 1138.
r = distance between particles (meters)
Using this equation, the electrostatic force between two protons that are separated by a distance of 10 meters is about 10 newtons. Comparing this result with the calculation of the
-20 12
gravitational force (10 newtons) shows that the gravitational force is so small that it can be
-24
neglected.
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If only the electrostatic and gravitational forces existed in the nucleus, then it would be impossible to have stable nuclei composed of protons and neutrons. The gravitational forces are much too small to hold the nucleons together compared to the electrostatic forces repelling the protons. Since stable atoms of neutrons and protons do exist, there must be another attractive force acting within the nucleus. This force is called the nuclear force.
The nuclear force is a strong attractive force that is independent of charge. It acts equally only between pairs of neutrons, pairs of protons, or a neutron and a proton. The nuclear force has a very short range; it acts only over distances approximately equal to the diameter of the nucleus (10 -13 cm). The attractive nuclear force between all nucleons drops off with distance much faster than the repulsive electrostatic force between protons.
TABLE 3
Forces Acting in the Nucleus
Force Interaction Range
Gravitational Very weak attractive force
between all nucleons Relatively long
Electrostatic Strong repulsive force between
like charged particles (protons) Relatively long
Nuclear Force Strong attractive force between
all nucleons Extremely short
In stable atoms, the attractive and repulsive forces in the nucleus balance. If the forces do not

a more stable configuration.
Summary
The important information in this chapter is summarized on the following page.
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ATOMIC NATURE OF MATTER DOE-HDBK-1019/1-93 Atomic and Nuclear Physics
Atomic Nature of Matter Summary
Atoms consist of three basic subatomic particles. These particles are the proton, the neutron, and the electron.
Protons are particles that have a positive charge, have about the same mass as a hydrogen atom, and exist in the nucleus of an atom.
Neutrons are particles that have no electrical charge, have about the same mass as a hydrogen atom, and exist in the nucleus of an atom.
Electrons are particles that have a negative charge, have a mass about eighteen hundred times smaller than the mass of a hydrogen atom, and exist in orbital shells around the nucleus of an atom.
The Bohr model of the atom consists of a dense nucleus of protons and neutrons (nucleons) surrounded by electrons traveling in discrete orbits at fixed distances from the nucleus.
Nuclides are atoms that contain a particular number of protons and neutrons.
Isotopes are nuclides that have the same atomic number and are therefore the same element, but differ in the number of neutrons.
The atomic number of an atom is the number of protons in the nucleus.
The mass number of an atom is the total number of nucleons (protons and neutrons) in the nucleus.
The notation A
ZX is used to identify a specific nuclide. "Z" represents the atomic
Fixture checkpoint marker-1139-xray is recorded on fixture page 1139.
number, which is equal to the number of protons. "A" represents the mass number, which is equal to the number of nucleons. "X" represents the chemical symbol of the element.
Number of protons = Z
Number of electrons = Z
Number of neutrons = A - Z
The stability of a nucleus is determined by the different forces interacting within it. The electrostatic force is a relatively long-range, strong, repulsive force that acts between the positively charged protons. The nuclear force is a relatively short-range attractive force between all nucleons. The gravitational force the long range, relatively weak attraction between masses, is negligible compared to the other forces.
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Atomic and Nuclear Physics DOE-HDBK-1019/1-93 CHART OF THE NUCLIDES
CHART OF THE NUCLIDES
The Chart of the Nuclides, like the Periodic Table, is a convenient format for presenting a large amount of scientific information in an organized manner.
EO 1.6 DEFINE the following terms:
a. Enriched uranium
b. Depleted uranium

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A tabulated chart called the Chart of the Nuclides lists the stable and unstable nuclides in addition to pertinent information about each one. Figure 3 shows a small portion of a typical chart. This chart plots a box for each individual nuclide, with the number of protons (Z) on the vertical axis and the number of neutrons (N = A - Z) on the horizontal axis.

The completely gray squares indicate stable isotopes. Those in white squares are artificially radioactive, meaning that they are produced by artificial techniques and do not occur naturally. By consulting a complete chart, other types of isotopes can be found, such as naturally occurring radioactive types (but none are found in the region of the chart that is illustrated in Figure 3).

Located in the box on the far left of each horizontal row is general information about the element. The box contains the chemical symbol of the element in addition to the average atomic weight of the naturally occurring substance and the average thermal neutron absorption cross section, which will be discussed in a later module. The known isotopes (elements with the same atomic number Z but different mass number A) of each element are listed to the right.

Figure 3 Nuclide Chart for Atomic Numbers 1 to 6

Information for Stable Nuclides

For the stable isotopes, in addition to the symbol and the atomic mass number, the number percentage of each isotope in the naturally occurring element is listed, as well as the thermal neutron activation cross section and the mass in atomic mass units (amu). A typical block for a stable nuclide from the Chart of the Nuclides is shown in Figure 4.

Figure 4 Stable Nuclides

Information for Unstable Nuclides

For unstable isotopes the additional information includes the half life, the mode of decay (for example, β^-, α), the total disintegration energy in MeV (million electron volts), and the mass in amu when available. A typical block for an unstable nuclide from the Chart of the Nuclides is shown in Figure 5.

Figure 5 Unstable Nuclides

Fixture checkpoint marker-1140-xray is recorded on fixture page 1140.

Neutron - Proton Ratios

Figure 6 shows the distribution of the stable nuclides plotted on the same axes as the Chart of the Nuclides. As the mass numbers become higher, the ratio of neutrons to protons in the nucleus becomes larger. For helium-4 (2 protons and 2 neutrons) and oxygen-16 (8 protons and 8 neutrons) this ratio is unity. For indium-115 (49 protons and 66 neutrons) the ratio of neutrons to protons has increased to 1.35, and for uranium-238 (92 protons and 146 neutrons) the neutron■to-proton

Figure 6 Neutron - Proton Plot of the Stable Nuclides

If a heavy nucleus were to split into two fragments, each fragment would form a nucleus that would have approximately the same neutron-to-proton ratio as the heavy nucleus. This high neutron-to-proton ratio places the fragments below and to the right of the stability curve displayed by Figure 6. The instability caused by this excess of neutrons is generally rectified by successive beta emissions, each of which converts a neutron to a proton and moves the nucleus toward a more stable neutron-to-proton ratio.

Natural Abundance of Isotopes

The relative abundance of an isotope in nature compared to other isotopes of the same element is relatively constant. The Chart of the Nuclides presents the relative abundance of the naturally occurring isotopes of an element in units of atom percent. Atom percent is the percentage of the atoms of an element that are of a particular isotope. Atom percent is abbreviated as a/o.

For example, if a cup of water contains 8.23 x 10 atoms of oxygen, and the isotopic abundance 24 of oxygen-18 is 0.20%, then there are 1.65 x 10 atoms of oxygen-18 in the cup. 22

The atomic weight for an element is defined as the average atomic weight of the isotopes of the

Fixture checkpoint marker-1141-xray is recorded on fixture page 1141.

element. The atomic weight for an element can be calculated by summing the products of the isotopic abundance of the isotope with the atomic mass of the isotope.

Example:

Calculate the atomic weight for the element lithium. Lithium-6 has an atom percent abundance of 7.5% and an atomic mass of 6.015122 amu. Lithium-7 has an atomic abundance of 92.5% and an atomic mass of 7.016003 amu.

Solution:

The other common measurement of isotopic abundance is weight percent (w/o). Weight percent is the percent weight of an element that is a particular isotope. For example, if a sample of material contained 100 kg of uranium that was 28 w/o uranium-235, then 28 kg of uranium-235 was present in the sample.

Enriched and Depleted Uranium

Natural uranium mined from the earth contains the isotopes uranium-238, uranium-235 and uranium-234. The majority (99.2745%) of all the atoms in natural uranium are uranium-238. Most of the remaining atoms (0.72%) are uranium-235, and a slight trace (0.0055%) are uranium-234. Although all isotopes of uranium have similar chemical properties, each of the isotopes has significantly different nuclear properties. For reasons that will be discussed in later modules, the isotope uranium-235 is usually the desired material for use in reactors.

A vast amount of equipment and energy are expended in processes that separate the isotopes of uranium (and other elements). The details of these processes are beyond the scope of this module. These processes are called enrichment processes because they selectively increase the proportion of a particular isotope. The enrichment process typically starts with feed material that has the proportion of isotopes that occur naturally. The process results in two types of

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In the case of uranium, the natural uranium ore is 0.72 a/o uranium-235. The desired outcome of the enrichment process is to produce enriched uranium. Enriched uranium is defined as uranium in which the isotope uranium-235 has a concentration greater than its natural value. The enrichment process will also result in the byproduct of depleted uranium. Depleted uranium is defined as uranium in which the isotope uranium-235 has a concentration less than its natural value. Although depleted uranium is referred to as a by-product of the enrichment process, it does have uses in the nuclear field and in commercial and defense industries.

Summary

The important information in this chapter is summarized below.

Chart of the Nuclides Summary

Enriched uranium is uranium in which the isotope uranium-235 has a concentration greater than its natural value of 0.7%.

Depleted uranium is uranium in which the isotope uranium-235 has a

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MASS DEFECT AND BINDING ENERGY
The separate laws of Conservation of Mass and Conservation of Energy are not applied strictly on the nuclear level. It is possible to convert between mass and energy. Instead of two separate conservation laws, a single conservation law states that the sum of mass and energy is conserved. Mass does not magically appear and disappear at random. A decrease in mass will be accompanied by a corresponding increase in energy and vice versa.
EO 1.7 DEFINE the following terms:
a. Mass defect
b. Binding energy
EO 1.8 Given the atomic mass for a nuclide and the atomic masses of a neutron, proton, and electron, CALCULATE the mass defect and binding energy of the nuclide.
Mass Defect
Careful measurements have shown that the mass of a particular atom is always slightly less than
Fixture checkpoint marker-1142-xray is recorded on fixture page 1142.
the sum of the masses of the individual neutrons, protons, and electrons of which the atom consists. The difference between the mass of the atom and the sum of the masses of its parts is called the mass defect (Δm). The mass defect can be calculated using Equation (1-1). In calculating the mass defect it is important to use the full accuracy of mass measurements because the difference in mass is small compared to the mass of the atom. Rounding off the masses of atoms and particles to three or four significant digits prior to the calculation will result in a calculated mass defect of zero.

$$\Delta m = [\, Z(mp + me) + (A-Z)mn \,] - matom \quad (1\text{-}1)$$

where:
Δm = mass defect (amu)
mp = mass of a proton (1.007277 amu)
mn = mass of a neutron (1.008665 amu)
me = mass of an electron (0.000548597 amu)
matom = mass of nuclide A
ZX (amu)
Z = atomic number (number of protons)
A = mass number (number of nucleons)

Example:
Calculate the mass defect for lithium-7. The mass of lithium-7 is 7.016003 amu.
Solution:
Δ m Z m
p
me A Z mn matom
Δ m 3 1.007826 amu 7 3 1.008665 amu 7.016003 amu
Δ m 0.0421335 amu
Binding Energy
The loss in mass, or mass defect, is due to the conversion of mass to binding energy when the nucleus is formed. Binding energy is defined as the amount of energy that must be supplied to

the amount of energy that would be released if the nucleus was formed from the separate particles. Binding energy is the energy equivalent of the mass defect. Since the mass defect was converted to binding energy (BE) when the nucleus was formed, it is possible to calculate the binding energy using a conversion factor derived by the mass-energy relationship from Einstein's Theory of Relativity.

Einstein's famous equation relating mass and energy is E = mc2 where c is the velocity of light (c = 2.998 x 108 m/sec). The energy equivalent of 1 amu can be determined by inserting this quantity of mass into Einstein's equation and applying conversion factors.

E m c 2

1 amu

1.6606 x 10 27 kg

1 amu

2.998 x 108 m

sec

2

1 N

1

kg m

sec2

1 J

1 N m

1.4924 x 10 10 J 1 MeV

1.6022 x 10 13 J

931.5 MeV

Fixture checkpoint marker-1143-xray is recorded on fixture page 1143.

Conversion Factors:

1 amu = 1.6606 x 10 -27 kg

1 newton = 1 kg-m/sec2

1 joule = 1 newton-meter

1 MeV = 1.6022 x 10-13 joules

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Atomic and Nuclear Physics DOE-HDBK-1019/1-93 MASS DEFECT AND BINDING ENERGY

Since 1 amu is equivalent to 931.5 MeV of energy, the binding energy can be calculated using Equation (1-2).

(1-2)

B.E. ∆m

931.5 MeV

1 amu

Example:

Calculate the mass defect and binding energy for uranium-235. One uranium-235 atom has a mass of 235.043924 amu.

Solution:

Step 1: Calculate the mass defect using Equation (1-1).

∆m Z m

p

me A Z mn matom

∆m 92 1.007826 amu 235 92 1.008665 amu 235.043924 amu

∆m 1.91517 amu

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B.E. Δm
931.5 MeV
1 amu
1.91517 amu
931.5 MeV
1 amu
1784 MeV
Energy Levels of Atoms
The electrons that circle the nucleus move in fairly well-defined orbits. Some of these electrons are more tightly bound in the atom than others. For example, only 7.38 eV is required to remove the outermost electron from a lead atom, while 88,000 eV is required to remove the innermost electron. The process of removing an electron from an atom is called ionization, and the energy required to remove the electron is called the ionization energy.
In a neutral atom (number of electrons = Z) it is possible for the electrons to be in a variety of different orbits, each with a different energy level. The state of lowest energy is the one in which the atom is normally found and is called the ground state. When the atom possesses more energy than its ground state energy, it is said to be in an excited state.
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MASS DEFECT AND BINDING ENERGY DOE-HDBK-1019/1-93 Atomic and Nuclear Physics
An atom cannot stay in the excited state for an indefinite period of time. An excited atom will eventually transition to either a lower-energy excited state, or directly to its ground state, by emitting a discrete bundle of electromagnetic energy called an x-ray. The energy of the x-ray will be equal to the difference between the energy levels of the atom and will typically range from several eV to 100,000 eV in magnitude.
Energy Levels of the Nucleus
The nucleons in the nucleus of an atom, like the electrons that circle the nucleus, exist in shells that correspond to energy states. The energy shells of the nucleus are less defined and less understood than those of the electrons. There is a state of lowest energy (the ground state) and discrete possible excited states for a nucleus. Where the discrete energy states for the electrons of an atom are measured in eV or keV, the energy levels of the nucleus are considerably greater and typically measured in MeV.
Fixture checkpoint marker-1144-xray is recorded on fixture page 1144.
A nucleus that is in the excited state will not remain at that energy level for an indefinite period. Like the electrons in an excited atom, the nucleons in an excited nucleus will transition towards their lowest energy configuration and in doing so emit a discrete bundle of electromagnetic radiation called a gamma ray (γ-ray). The only differences between x-rays and γ-rays are their energy levels and whether they are emitted from the electron shell or from the nucleus.
The ground state and the excited states of
Figure 7 Energy Level Diagram - Nickel-60
a nucleus can be depicted in a nuclear energy-level diagram. The nuclear energy-level diagram consists of a stack of horizontal bars, one bar for each of the excited states of the nucleus. The vertical distance between the bar representing an excited state and the bar representing the ground state is proportional to the energy level of the excited state with respect to

energy between the ground state and the
excited state is called the excitation energy
of the excited state. The ground state of
a nuclide has zero excitation energy. The
bars for the excited states are labeled with
their respective energy levels. Figure 7 is
the energy level diagram for nickel-60.
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Atomic and Nuclear Physics DOE-HDBK-1019/1-93 MASS DEFECT AND BINDING ENERGY
Summary
The important information in this chapter is summarized below.
Mass Defect and Binding Energy Summary
Mass defect is the difference between the mass of the atom and the sum of the
masses of its constituent parts.
Binding energy is the amount of energy that must be supplied to a nucleus to
completely separate its nuclear particles. Binding energy is the energy equivalent
of the mass defect.
Mass defect can be calculated by using the equation below.
$\Delta m = [Z(mp + me) + (A-Z)mn] - matom$
Binding energy can be calculated by multiplying the mass defect by the factor
of 931.5 MeV per amu.
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MODES OF RADIOACTIVE DECAY DOE-HDBK-1019/1-93 Atomic and Nuclear Physics
MODES OF RADIOACTIVE DECAY
Most atoms found in nature are stable and do not emit particles or energy that
change form over time. Some atoms, however, do not have stable nuclei. These
atoms emit radiation in order to achieve a more stable configuration.
EO 2.1 DESCRIBE the following processes:
a. Alpha decay d. Electron capture
b. Beta-minus decay e. Internal conversions
c. Beta-plus decay f. Isomeric transitions
EO 2.2 Given a Chart of the Nuclides, WRITE the radioactive decay
chain for a nuclide.
EO 2.3 EXPLAIN why one or more gamma rays typically accompany
particle emission.
EO 2.4 Given the stability curve on the Chart of the Nuclides,
DETERMINE the type of radioactive decay that the nuclides in
each region of the chart will typically undergo.
Fixture checkpoint marker-1145-xray is recorded on fixture page 1145.
Stability of Nuclei
As mass numbers become larger, the ratio of neutrons to protons in the nucleus becomes larger for the stable nuclei. Non-stable nuclei may have an excess or deficiency of neutrons and undergo a transformation process known as beta (β) decay. Non-stable nuclei can also undergo a variety of other processes such as alpha (α) or neutron (n) decay. As a result of these decay processes, the final nucleus is in a more stable or more tightly bound configuration.
Natural Radioactivity
In 1896, the French physicist Becquerel discovered that crystals of a uranium salt emitted rays that were similar to x-rays in that they were highly penetrating, could affect a photographic

by the identification of two other radioactive elements, polonium and radium, by Pierre and Marie Curie.

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Atomic and Nuclear Physics DOE-HDBK-1019/1-93 MODES OF RADIOACTIVE DECAY

Heavy elements, such as uranium or thorium, and their unstable decay chain elements emit radiation in their naturally occurring state. Uranium and thorium, present since their creation at the beginning of geological time, have an extremely slow rate of decay. All naturally occurring nuclides with atomic numbers greater than 82 are radioactive.

Nuclear Decay

Whenever a nucleus can attain a more stable (i.e., more tightly bound) configuration by emitting radiation, a spontaneous disintegration process known as radioactive decay or nuclear decay may occur. In practice, this "radiation" may be electromagnetic radiation, particles, or both.

Detailed studies of radioactive decay and nuclear reaction processes have led to the formulation of useful conservation principles. The four principles of most interest in this module are discussed below.

Fixture checkpoint marker-1146-xray is recorded on fixture page 1146.

Conservation of electric charge implies that charges are neither created nor destroyed. Single positive and negative charges may, however, neutralize each other. It is also possible for a neutral particle to produce one charge of each sign.

Conservation of mass number does not allow a net change in the number of nucleons. However, the conversion of a proton to a neutron and vice versa is allowed.

Conservation of mass and energy implies that the total of the kinetic energy and the energy equivalent of the mass in a system must be conserved in all decays and reactions. Mass can be converted to energy and energy can be converted to mass, but the sum of mass and energy must be constant.

Conservation of momentum is responsible for the distribution of the available kinetic energy among product nuclei, particles, and/or radiation. The total amount is the same before and after the reaction even though it may be distributed differently among entirely different nuclides and/or particles.

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Alpha Decay (■)

Alpha decay is the emission of alpha particles (helium nuclei) which may be represented as either $^{4}_{2}He$ or ■. When an unstable nucleus ejects an alpha particle, the atomic number is reduced by 2 and the mass number decreased by 4. An example is uranium-234 which decays by the ejection of an alpha particle accompanied by the emission of a 0.068 MeV gamma.

The combined kinetic energy of the daughter nucleus (Thorium-230) and the ■ particle is designated as KE. The sum of the KE and the gamma energy is equal to the difference in mass between the original nucleus (Uranium-234) and the final particles (equivalent to the binding energy released, since ■m = BE). The alpha particle will carry off as much as 98% of the kinetic energy and, in most cases, can be considered to carry off all the kinetic energy.

Beta Decay (■)

Beta decay is the emission of electrons of nuclear rather than orbital origin. These particles are electrons that have been expelled by excited nuclei and may have a charge of either sign.

involved. The neutrino is associated with positive electron emission, and its antiparticle, the antineutrino, , is emitted with a negative electron. These uncharged particles have only the weakest interaction with matter, no mass, and travel at the speed of light. For all practical purposes, they pass through all materials with so few interactions that the energy they possess cannot be recovered. The neutrinos and antineutrinos are included here only because they carry a portion of the kinetic energy that would otherwise belong to the beta particle, and therefore, must be considered for energy and momentum to be conserved. They are normally ignored since they are not significant in the context of nuclear reactor applications.

Negative electron emission, represented as e, -1 -1■, or simply as e or ■ , effectively converts a neutro to a proton, thus increasing the atomic number by one and leaving the mass number unchanged. This is a common mode of decay for nuclei with an excess of neutrons, such as fission fragments below and to the right of the neutron-proton stability curve (refer to Figure 6). An example of a typical beta minus-decay reaction is shown below.

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Figure 9 Orbital Electron Capture

Positively charged electrons (beta-plus) are known as positrons. Except for sign, they are nearly identical to their negatively charged cousins. When a positron, represented as e, ■, or simply as +1 +1

Fixture checkpoint marker-1147-xray is recorded on fixture page 1147.

0 0

e or ■ , is ejected from the nucleus, the atomic number is decreased by one and the mass number + + remains unchanged. A proton has been converted to a neutron. An example of a typical positron (beta-plus) decay is shown below.

Electron Capture (EC, K-capture)

Nuclei having an excess of protons may capture an electron from one of the inner orbits which immediately combines with a proton in the nucleus to form a neutron. This process is called electron capture (EC). The electron is normally captured from the innermost orbit (the K-shell), and, consequently, this process is sometimes called K-capture. The following example depicts electron capture.

A neutrino is formed at the same time that the neutron is formed, and energy carried off by it serves to conserve momentum. Any energy that is available due to the atomic mass of the product being appreciably less than that of the parent will appear as gamma radiation. Also, there will always be characteristic x-rays given off when an electron from one of the higher energy shells moves in to fill the vacancy in the K-shell.

Electron capture is shown graphically in Figure 8.

emission result in the production of
the same daughter product, and
they exist as competing processes.
For positron emission to occur, however, the mass of the daughter product must be less than the mass of the parent by an amount equal to at least twice the mass of an electron. This mass difference between the parent and daughter is necessary to account for two items present in the parent but not in the daughter. One item is the positron ejected from the nucleus of the parent. The other item is that the daughter product has one less orbital electron than the parent. If this requirement is not met, then orbital electron capture takes place exclusively.
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■ ■ ■ ■
■ ■
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Gamma Emission (■)
Gamma radiation is a high-energy electromagnetic radiation that originates in the nucleus. It is emitted in the form of photons, discrete bundles of energy that have both wave and particle properties. Often a daughter nuclide is left in an excited state after a radioactive parent nucleus undergoes a transformation by alpha decay, beta decay, or electron capture. The nucleus will drop to the ground state by the emission of gamma radiation.
Internal Conversion
The usual method for an excited nucleus to go from the excited state to the ground state is by emission of gamma radiation. However, in some cases the gamma ray (photon) emerges from the nucleus only to interact with one of the innermost orbital electrons and, as a result, the energy of the photon is transferred to the electron. The gamma ray is then said to have undergone internal conversion. The conversion electron is ejected from the atom with kinetic energy equal to the gamma energy minus the binding energy of the orbital electron. An orbital electron then drops to a lower energy state to fill the vacancy, and this is accompanied by the emission of characteristic x-rays.
Fixture checkpoint marker-1148-xray is recorded on fixture page 1148.
Isomers and Isomeric Transition
Isomeric transition commonly occurs immediately after particle emission; however, the nucleus may remain in an excited state for a measurable period of time before dropping to the ground state at its own characteristic rate. A nucleus that remains in such an excited state is known as a nuclear isomer because it differs in energy and behavior from other nuclei with the same atomic number and mass number. The decay of an excited nuclear isomer to a lower energy level is called an isomeric transition. It is also possible for the excited isomer to decay by some alternate means, for example, by beta emission.
An example of gamma emission accompanying particle emission is illustrated by the decay of nitrogen-16 below.
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Decay Chains

When an unstable nucleus decays, the resulting daughter nucleus is not necessarily stable. The nucleus resulting from the decay of a parent is often itself unstable, and will undergo an additional decay. This is especially common among the larger nuclides.

It is possible to trace the steps of an unstable atom as it goes through multiple decays trying to achieve stability. The list of the original unstable nuclide, the nuclides that are involved as intermediate steps in the decay, and the final stable nuclide is known as the decay chain. One common method for stating the decay chain is to state each of the nuclides involved in the standard X format. Arrows are used between nuclides to indicate where decays occur, with the type of decay A Z
indicated above the arrow and the half-life below the arrow. The half-life for decay will be discussed in the next chapter.

Example:

Write the decay chains for rubidium-91 and actinium-215. Continue the chains until a stable nuclide or a nuclide with a half-life greater than 1 x 10 years is reached. 6

Solution:

Predicting Type of Decay

Radioactive nuclides tend to decay in a way that results in a daughter nuclide that lies closer to the line of stability. Due to this, it is possible to predict the type of decay that a nuclide will undergo based on its location relative to the line of stability on the Chart of the Nuclides.

Figure 9 illustrates the type of decay nuclides in different regions of the chart will typically undergo. Nuclides that are below and to the right of the line of stability will usually undergo ■ decay. - Nuclides that are above and to the left of the line of stability will usually undergo either ■ decay or + electron capture. Most nuclides that will undergo ■ decay are found in the upper right hand region of the chart. These are general rules that have many exceptions, especially in the region of the heavy nuclides.

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Figure 9 Types of Radioactive Decay Relative to the Line of Stability

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Summary

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The important information in this chapter is summarized below.

Modes of Radioactive Decay Summary

Alpha decay is the emission of an alpha particle (2 protons and 2 neutrons) from an unstable nucleus. The daughter nuclide has an atomic number 2 less than the parent nuclide and a mass number 4 less than the parent nuclide. The daughter nucleus commonly releases its excitation energy by gamma emission.

Beta-minus decay effectively converts a neutron to a proton and an electron, which is immediately ejected from the nucleus. The daughter nuclide has its atomic number increased by 1 and the same mass number compared to the parent.

Beta-plus decay effectively converts a proton to a neutron and a positron, which

number decreased by 1 and the same mass number compared to the parent.
In electron capture, the nucleus absorbs an electron from the innermost orbit. This electron combines with a proton to form a neutron.
Internal conversion occurs when a gamma ray, emitted by the nucleus as it goes from the excited state to the ground state, interacts with one of the innermost electrons of the same atom. The electron is ejected from the atom.
An isomeric transition is the decay of an excited nucleus to a lower-energy level by the emission of a gamma ray.
Decay chains can be found by tracing the steps an unstable atom goes through as it tries to achieve stability.
Many modes of radioactive decay result in a daughter nuclide that has an energy level above the ground state. This excitation energy is usually released immediately in the form of a gamma ray.
Fixture checkpoint marker-1150-xray is recorded on fixture page 1150.
The type of decay that a nuclide will typically undergo can be determined by its relationship to the line of stability on the Chart of the Nuclides. Nuclides that lie below and to the right of the line of stability will typically beta minus decay. Nuclides above and to the left of the line will typically either beta plus decay or electron capture. Most alpha emitters are found in the upper, right-hand corner of the chart.
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RADIOACTIVITY
The rate at which a sample of radioactive material decays is not constant. As individual atoms of the material decay, there are fewer of those types of atoms remaining. Since the rate of decay is directly proportional to the number of atoms, the rate of decay will decrease as the number of atoms decreases.
EO 2.5 DEFINE the following terms:
a. Radioactivity d. Radioactive decay constant
b. Curie e. Radioactive half-life
c. Becquerel
EO 2.6 Given the number of atoms and either the half-life or decay constant of a nuclide, CALCULATE the activity.
EO 2.7 Given the initial activity and the decay constant of a nuclide, CALCULATE the activity at any later time.
EO 2.8 CONVERT between the half-life and decay constant for a nuclide.
EO 2.9 Given the Chart of the Nuclides and the original activity, PLOT the radioactive decay curve for a nuclide on either linear or semi-log coordinates.
EO 2.10 DEFINE the following terms:
a. Radioactive equilibrium
b. Transient radioactive equilibrium
Radioactive Decay Rates
Radioactivity is the property of certain nuclides of spontaneously emitting particles or gamma radiation. The decay of radioactive nuclides occurs in a random manner, and the precise time at which a single nucleus will decay cannot be determined. However, the average behavior of a very large sample can be predicted accurately by using statistical methods. These studies have

nuclei within a sample of a particular nuclide will decay. This probability per unit time that an atom of a nuclide will decay is known as the radioactive decay constant, λ. The units for the decay constant are inverse time such as 1/second, 1/minute, 1/hour, or 1/year. These decay constant units can also be expressed as second-1, minute-1, hour-1, and year-1.
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■■
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The activity (A) of a sample is the rate of decay of that sample. This rate of decay is usually measured in the number of disintegrations that occur per second. For a sample containing millions of atoms, the activity is the product of the decay constant and the number of atoms present in the sample.
The relationship between the activity, number of atoms, and decay constant is shown in Equation (1-3).
A = ■ N (1-3)
where:
A = Activity of the nuclide (disintegrations/second)
■ = decay constant of the nuclide (second)-1
N = Number of atoms of the nuclide in the sample
Since ■ is a constant, the activity and the number of atoms are always proportional.
Fixture checkpoint marker-1151-xray is recorded on fixture page 1151.
Units of Measurement for Radioactivity
Two common units to measure the activity of a substance are the curie (Ci) and becquerel (Bq). A curie is a unit of measure of the rate of radioactive decay equal to 3.7 x 10 disintegrations 10 per second. This is approximately equivalent to the number of disintegrations that one gram of radium-226 will undergo in one second. A becquerel is a more fundamental unit of measure of radioactive decay that is equal to 1 disintegration per second. Currently, the curie is more widely used in the United States, but usage of the becquerel can be expected to broaden as the metric system slowly comes into wider use. The conversion between curies and becquerels is shown below.
1 curie = 3.7 x 10 becquerels 10
Variation of Radioactivity Over Time
The rate at which a given radionuclide sample decays is stated in Equation (1-3) as being equal to the product of the number of atoms and the decay constant. From this basic relationship it is possible to use calculus to derive an expression which can be used to calculate how the number of atoms present will change over time. The derivation is beyond the scope of this text, but Equation (1-4) is the useful result.
(1-4)
where:
N = number of atoms present at time t
N = number of atoms initially present o
■ = decay constant (time)-1
t = time
A Ao
e
t
A Aoe
t

Ao
e
t
ln A
Ao
t
t
ln
A
Ao
t
1/2
ln
1
2
t
1/2
ln 2 0.693
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Since the activity and the number of atoms are always proportional, they may be used interchangeably to describe any given radionuclide population. Therefore, the following is true.
(1-5)
where:
A = activity present at time t
A = activity initially present o
= decay constant (time)
Fixture checkpoint marker-1152-xray is recorded on fixture page 1152.
-1
t = time
Radioactive Half-Life
One of the most useful terms for estimating how quickly a nuclide will decay is the radioactive half-life. The radioactive half-life is defined as the amount of time required for the activity to decrease to one-half of its original value. A relationship between the half-life and decay constant can be developed from Equation (1-5). The half-life can be calculated by solving Equation (1-5) for the time, t, when the current activity, A, equals one-half the initial activity A .
o
First, solve Equation (1-5) for t.
If A is equal to one-half of Ao, then A/Ao is equal to one-half. Substituting this in the equation above yields an expression for t .
1/2
(1-6)
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The basic features of decay of a radionuclide sample are shown by the graph in Figure 10.
Figure 10 Radioactive Decay as a Function of Time in Units of Half-Life
Assuming an initial number of atoms No, the population, and consequently, the activity may be noted to decrease by one-half of this value in a time of one half-life. Additional decreases occur so that whenever one half-life elapses, the number of atoms drops to one-half of what its value

3.1%, of the original number of atoms remains. After seven half-lives, only 1/128, or 0.78%, of the atoms remains. The number of atoms existing after 5 to 7 half-lives can usually be assumed to be negligible. The Chemistry Fundamentals Handbook contains additional information on calculating the number of atoms contained within a sample.
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Example:
A sample of material contains 20 micrograms of californium-252.
Californium-252 has a half-life of 2.638 years.
Calculate:
(a) The number of californium-252 atoms initially present
(b) The activity of the californium-252 in curies
(c) The number of californium-252 atoms that will remain in 12 years
(d) The time it will take for the activity to reach 0.001 curies
Solution:
(a) The number of atoms of californium-252 can be determined as below.
(b) First, use Equation (1-6) to calculate the decay constant.
Use this value for the decay constant in Equation (1-3) to determine the activity.
N No
e
t
4.78 x 1016 e
(0.263/yr) (12 yr)
2.04 x 1015
A Aoe
t
A
Ao
e
t
ln A
Fixture checkpoint marker-1153-xray is recorded on fixture page 1153.
Ao
t
t
ln
A
Ao
t
ln
0.001 Ci
0.0108 Ci
0.263 year 1
t 9.05 years
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(c) The number of californium atoms that will remain in 12 years can be calculated from Equation (1-4).
(d) The time that it will take for the activity to reach 0.001 Ci can be determined from Equation (1-5). First, solve Equation (1-5) for time.
Inserting the appropriate values in the right side of this equation will result in the required time.
Plotting Radioactive Decay
It is useful to plot the activity of a nuclide as it changes over time. Plots of this type can be used to determine when the activity will fall below a certain level. This plot is usually done showing activity on either a linear or a logarithmic scale. The decay of the activity of a single nuclide on a logarithmic scale will plot as a straight line because the decay is exponential.
Fixture checkpoint marker-1154-xray is recorded on fixture page 1154.
t
1/2
0.693
0.693
t
1/2
0.693
7.13 seconds
0.0972 second 1
A Aoe
t
RADIOACTIVITY DOE-HDBK-1019/1-93 Atomic and Nuclear Physics
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Example:
Plot the radioactive decay curve for nitrogen-16 over a period of 100 seconds. The initial activity is 142 curies and the half-life of nitrogen-16 is 7.13 seconds. Plot the curve on both linear rectangular coordinates and on a semi-log scale.
Solution:
First, use Equation (1-6) to calculate the decay constant corresponding to a half-life of 7.13 seconds.
Use the decay constant determined above to calculate the activity at various times using Equation (1-5).
Time Activity
0 seconds 142 Ci
20 seconds 20.3 Ci
40 seconds 2.91 Ci
60 seconds 0.416 Ci
80 seconds 0.0596 Ci
100 seconds 0.00853 Ci
■■
■■
■■
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Figure 11 Linear and Semi-log Plots of Nitrogen-16 Decay
Plotting the data points calculated above on both linear and semilog scales results in the graphs

If a substance contains more than one radioactive nuclide, the total activity is the sum of the individual activities of each nuclide. As an example, consider a sample of material that contained 1 x 10 atoms of iron-59 that has a half-life of 44.51 days (■ = 1.80 x 10 sec), 6 -7 -1 1 x 10 atoms of manganese-54 that has a half-life of 312.2 days (■ = 2.57 x 10 sec), and 1 6 -8 -1 x 10 atoms of cobalt-60 that has a half-life of 1925 days (■ = 4.17 x 10 sec). 6 -9 -1
The initial activity of each of the nuclides would be the product of the number of atoms and the decay constant.
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Figure 12 Combined Decay of Iron-56, Manganese-54, and Cobalt-60
Plotting the manner in which the activities of each of the three nuclides decay over time demonstrates that initially the activity of the shortest-lived nuclide (iron-59) dominates the total activity, then manganese-54 dominates. After almost all of the iron and manganese have decayed away, the only contributor to activity will be the cobalt-60. A plot of this combined decay is shown in Figure 12.
Radioactive Equilibrium
Radioactive equilibrium exists when a radioactive nuclide is decaying at the same rate at which it is being produced. Since the production rate and decay rate are equal, the number of atoms
Fixture checkpoint marker-1155-xray is recorded on fixture page 1155.
present remains constant over time.
An example of radioactive equilibrium is the concentration of sodium-24 in the coolant circulating through a sodium-cooled nuclear reactor. Assume that the sodium-24 is being produced at a rate of 1 x 10 atoms per second. If the sodium-24 were stable and did not decay, 6 the amount of sodium-24 present after some period of time could be calculated by multiplying the production rate by the amount of time. Plotting the amount of material present would result in the graph in Figure 13.
However, sodium-24 is not stable, and it decays with a half-life of 14.96 hours. If no sodium-24 is present initially and production starts at a rate of 1 x 10 atoms per second, the rate 6 of decay will initially be zero because there is no sodium-24 present to decay. The rate of decay of sodium-24 will increase as the amount of sodium-24 increases.
■
■
■
■
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Figure 13 Cumulative Production of Sodium-24 Over Time
The amount of sodium-24 present will initially increase rapidly, then it will increase at a continually decreasing rate until the rate of decay is equal to the rate of production. It is possible to calculate how much sodium-24 will be present at equilibrium by setting the production rate (R) equal to the decay rate (■ N).
where:
R = production rate (atoms/second)
■ = decay constant (second)-1
N = number of atoms
It is possible to calculate the equilibrium value for sodium-24 being produced at a rate of 1 x 10 atoms/second. 6
■ ■ ■■

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■

Figure 14 Approach of Sodium-24 to Equilibrium

The development of the equation to calculate how the amount of sodium-24 changes over time as it approaches the equilibrium value is beyond the scope of this handbook. However, the equation is presented below.

This equation can be used to calculate the values of the amount of sodium-24 present at different times. As the time increases, the exponential term approaches zero, and the number of atoms present will approach R/■. A plot of the approach of sodium-24 to equilibrium is shown in Figure 14.

Transient Radioactive Equilibrium

Transient radioactive equilibrium occurs when the parent nuclide and the daughter nuclide decay at essentially the same rate.

For transient equilibrium to occur, the parent must have a long half-life when compared to the daughter. An example of this type of compound decay process is barium-140, which decays by beta emission to lanthanum-140, which in turn decays by beta emission to stable cerium-140.

Figure 15 Transient Equilibrium in the Decay of Barium-140

The decay constant for barium-140 is considerably smaller than the decay constant for lanthanum-140. Remember that the rate of decay of both the parent and daughter can be represented as ■N. Although the decay constant for barium-140 is smaller, the actual rate of

Fixture checkpoint marker-1156-xray is recorded on fixture page 1156.

decay (■N) is initially larger than that of lanthanum-140 because of the great difference in their initial concentrations. As the concentration of the daughter increases, the rate of decay of the daughter will approach and eventually match the decay rate of the parent. When this occurs, they are said to be in transient equilibrium. A plot of the barium-lanthanum-cerium decay chain reaching transient equilibrium is shown in Figure 15.

Secular equilibrium occurs when the parent has an extremely long half-life. In the long decay chain for a naturally radioactive element, such as thorium-232, where all of the elements in the chain are in secular equilibrium, each of the descendants has built up to an equilibrium amount and all decay at the rate set by the original parent. The only exception is the final stable element on the end of the chain. Its number of atoms is constantly increasing.

Summary

The important information in this chapter is summarized on the following page.

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Radioactivity Summary

& Radioactivity is the decay of unstable atoms by the emission of particles and electromagnetic radiation.

& A curie (Ci) is a unit of radioactivity equal to 3.7 x 10 disintegrations per 10 second.

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& The radioactive decay constant (■) is the probability per unit time that an atom will decay.
& The radioactive half-life is the amount of time required for the activity to decrease to one-half its original value.
& The activity of a substance can be calculated from the number of atoms and the decay constant based on the equation below.
& The amount of activity remaining after a particular time can be calculated from the equation below.
& The relationship between the decay constant and the half-life is shown below.
& Plots of radioactive decay can be useful to describe the variation of activity over time. If decay is plotted using semi-log scale the plot results in a straight line.
& Radioactive equilibrium exists when the production rate of a material is equal to the removal rate.
& Transient radioactive equilibrium exists when the parent nuclide and the daughter nuclide decay at essentially the same rate. This occurs only when the parent has a long half-life compared to the daughter.
Atomic and Nuclear Physics DOE-HDBK-1019/1-93 NEUTRON INTERACTIONS
NEUTRON INTERACTIONS
Neutrons can cause many different types of interactions. The neutron may simply scatter off the nucleus in two different ways, or it may actually be absorbed into the nucleus. If a neutron is absorbed into the nucleus, it may result in the emission of a gamma ray or a subatomic particle, or it may cause the nucleus to fission.
EO 3.1 DESCRIBE the following scattering interactions between a neutron and a nucleus:
a. Elastic scattering
b. Inelastic scattering
EO 3.2 STATE the conservation laws that apply to an elastic collision between a neutron and a nucleus.
EO 3.3 DESCRIBE the following reactions where a neutron is absorbed in a nucleus:
a. Radiative capture
Fixture checkpoint marker-1157-xray is recorded on fixture page 1157.
b. Particle ejection
Scattering
A neutron scattering reaction occurs when a nucleus, after having been struck by a neutron, emits a single neutron. Despite the fact that the initial and final neutrons do not need to be (and often are not) the same, the net effect of the reaction is as if the projectile neutron had merely "bounced off," or scattered from, the nucleus. The two categories of scattering reactions, elastic and inelastic scattering, are described in the following paragraphs.
Elastic Scattering
In an elastic scattering reaction between a neutron and a target nucleus, there is no energy transferred into nuclear excitation. Momentum and kinetic energy of the "system" are conserved although there is usually some transfer of kinetic energy from the neutron to the target nucleus. The target nucleus gains the amount of kinetic energy that the neutron loses.
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mn
vn,i mTvT,i mnvn,f mTvT,f

2
m v
2
1
2
mn
v
2
n,i
1
2
mT
v
2
T,i
1
2
mn
v
2
n,f
1
2
mT
v
2
T,f
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Figure 16 Elastic Scattering
Figure 16 illustrates the process of elastic scattering of a neutron off a target nucleus. In the elastic scattering reaction, the conservation of momentum and kinetic energy is represented by the equations below.
Conservation of momentum (mv)
Conservation of kinetic energy
where:
mn = mass of the neutron
m = mass of the target nucleus T
vn,i = initial neutron velocity
Fixture checkpoint marker-1158-xray is recorded on fixture page 1158.
vn,f = final neutron velocity
vT,i = initial target velocity
vT,f = final target velocity
Atomic and Nuclear Physics DOE-HDBK-1019/1-93 NEUTRON INTERACTIONS
Elastic scattering of neutrons by nuclei can occur in two ways. The more unusual of the two interactions is the absorption of the neutron, forming a compound nucleus, followed by the re-emission of a neutron in such a way that the total kinetic energy is conserved and the nucleus returns to its ground state. This is known as resonance elastic scattering and is very dependent

nucleus, it is also referred to as compound elastic scattering. The second, more usual method, is termed potential elastic scattering and can be understood by visualizing the neutrons and nuclei to be much like billiard balls with impenetrable surfaces. Potential scattering takes place with incident neutrons that have an energy of up to about 1 MeV. In potential scattering, the neutron does not actually touch the nucleus and a compound nucleus is not formed. Instead, the neutron is acted on and scattered by the short range nuclear forces when it approaches close enough to the nucleus.

Inelastic Scattering

In inelastic scattering, the incident neutron is absorbed by the target nucleus, forming a compound nucleus. The compound nucleus will then emit a neutron of lower kinetic energy which leaves the original nucleus in an excited state. The nucleus will usually, by one or more gamma emissions, emit this excess energy to reach its ground state. Figure 17 shows the process of inelastic scattering.

For the nucleus that has reached its ground state, the sum of the kinetic energy of the exit

Figure 17 Inelastic Scattering

neutron, the target nucleus, and the total gamma energy emitted is equal to the initial kinetic

Fixture checkpoint marker-1159-xray is recorded on fixture page 1159.

energy of the incident neutron.

1
0
n
238
92
U
239
92
U
239
92
U
0
0
1
0
n
10
5
B
11
5
B
7
3
Li
4
2

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Absorption Reactions
Most absorption reactions result in the loss of a neutron coupled with the production of a charged particle or gamma ray. When the product nucleus is radioactive, additional radiation is emitted at some later time. Radiative capture, particle ejection, and fission are all categorized as absorption reactions and are briefly described below.
Radiative Capture
In radiative capture the incident neutron enters the target nucleus forming a compound nucleus. The compound nucleus then decays to its ground state by gamma emission. An example of a radiative capture reaction is shown below.
Particle Ejection
In a particle ejection reaction the incident particle enters the target nucleus forming a compound nucleus. The newly formed compound nucleus has been excited to a high enough energy level to cause it to eject a new particle while the incident neutron remains in the nucleus. After the new particle is ejected, the remaining nucleus may or may not exist in an excited state depending upon the mass-energy balance of the reaction. An example of a particle ejection reaction is shown below.
Fission
One of the most important interactions that neutrons can cause is fission, in which the nucleus that absorbs the neutron actually splits into two similarly sized parts. Fission will be discussed in detail in the next chapter.
Atomic and Nuclear Physics DOE-HDBK-1019/1-93 NEUTRON INTERACTIONS
Summary
The important information in this chapter is summarized below.
Fixture checkpoint marker-1160-xray is recorded on fixture page 1160.
Neutron Interactions Summary
Interactions where a neutron scatters off a target nucleus are either elastic or inelastic. In elastic scattering, kinetic energy and momentum are conserved and no energy is transferred into excitation energy of the target nucleus. In inelastic scattering, some amount of kinetic energy is transferred into excitation energy of the target nucleus.
The conservation principles that apply to an elastic collision are conservation of kinetic energy and conservation of momentum.
Radiative capture is the absorption of a neutron by the target nucleus, resulting in an excited nucleus which subsequently (typically within a small fraction of a second) releases its excitation energy in the form of a gamma ray.
Particle ejection occurs when a neutron is absorbed by a target nucleus, resulting in the formation of a compound nucleus. The compound nucleus immediately ejects a particle (for example, alpha or proton).
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1
0
n
235
92
U
236
92
U

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55
Cs
93
37
Rb 3
1
0
n
NUCLEAR FISSION DOE-HDBK-1019/1-93 Atomic and Nuclear Physics
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NUCLEAR FISSION
Nuclear fission is a process in which an atom splits and releases energy, fission products, and neutrons. The neutrons released by fission can, in turn, cause the fission of other atoms.
EO 4.1 EXPLAIN the fission process using the liquid drop model of a nucleus.
EO 4.2 DEFINE the following terms:
a. Excitation energy (E) exc
b. Critical energy (E) crit
EO 4.3 DEFINE the following terms:
a. Fissile material
b. Fissionable material
c. Fertile material
EO 4.4 DESCRIBE the processes of transmutation, conversion, and breeding.
EO 4.5 DESCRIBE the curve of Binding Energy per Nucleon versus mass number and give a qualitative description of the reasons for its shape.
EO 4.6 EXPLAIN why only the heaviest nuclei are easily fissioned.
EO 4.7 EXPLAIN why uranium-235 fissions with thermal
Fixture checkpoint marker-1161-xray is recorded on fixture page 1161.
neutrons and uranium-238 fissions only with fast neutrons.
Fission
In the fission reaction the incident neutron enters the heavy target nucleus, forming a compound nucleus that is excited to such a high energy level (E > E) that the nucleus "splits" exc crit (fissions) into two large fragments plus some neutrons. An example of a typical fission reaction is shown below.
A large amount of energy is released in the form of radiation and fragment kinetic energy.
Atomic and Nuclear Physics DOE-HDBK-1019/1-93 NUCLEAR FISSION
Liquid Drop Model of a Nucleus
The nucleus is held together by the attractive nuclear force between nucleons, which was discussed in a previous chapter. The characteristics of the nuclear force are listed below.
(a) very short range, with essentially no effect beyond nuclear dimensions (10-13 cm)
(b) stronger than the repulsive electrostatic forces within the nucleus
(c) independent of nucleon pairing, in that the attractive forces between pairs of neutrons are no different than those between pairs of protons or a neutron and a proton

One theory of fission considers the fissioning of a nucleus similar in some respects to the splitting of a liquid drop. This analogy is justifiable to some extent by the fact that a liquid drop is held together by molecular forces that tend to make the drop spherical in shape and that try to resist any deformation in the same manner as nuclear forces are assumed to hold the nucleus together. By considering the nucleus as a liquid drop, the fission process can be described.

Referring to Figure 18(A), the nucleus in the ground state is undistorted, and its attractive nuclear forces are greater than the repulsive electrostatic forces between the protons within the nucleus. When an incident particle (in this instance a neutron) is absorbed by the target nucleus, a compound nucleus is formed. The compound nucleus temporarily contains all the charge and mass involved in the reaction and exists in an excited state. The excitation energy added to the compound nucleus is equal to the binding energy contributed by the incident particle plus the kinetic energy possessed by that particle. Figure 18(B) illustrates the excitation energy thus imparted to the compound nucleus, which may cause it to oscillate and become distorted. If the excitation energy is greater than a certain critical energy, the oscillations may cause the compound nucleus to become dumbbell-shaped. When this happens, the attractive nuclear forces (short-range) in the neck area are small due to saturation, while the repulsive electrostatic forces (long-range) are only slightly less than before. When the repulsive electrostatic forces exceed the attractive nuclear forces, nuclear fission occurs, as illustrated in Figure 18(C).

Figure 18 Liquid Drop Model of Fission

Critical Energy

The measure of how far the energy level of a nucleus is above its ground state is called the excitation energy (Eexc). For fission to occur, the excitation energy must be above a particular value for that nuclide. The critical energy (Ecrit) is the minimum excitation energy required for fission to occur.

Fissile Material

A fissile material is composed of nuclides for which fission is possible with neutrons of any energy level. What is especially significant about these nuclides is their ability to be fissioned with zero kinetic energy neutrons (thermal neutrons). Thermal neutrons have very low kinetic energy levels (essentially zero) because they are roughly in equilibrium with the thermal motion of surrounding materials. Therefore, in order to be classified as fissile, a material must be capable of fissioning after absorbing a thermal neutron. Consequently, they impart essentially no kinetic energy to the reaction. Fission is possible in these materials with thermal neutrons, since the change in binding energy supplied by the neutron addition alone is high enough to exceed the critical energy. Some examples of fissile nuclides are uranium-235, uranium-233, and plutonium-239.

Fixture checkpoint marker-1162-xray is recorded on fixture page 1162.

Fissionable Material

A fissionable material is composed of nuclides for which fission with neutrons is possible. All fissile nuclides fall into this category. However, also included are those nuclides that can be fissioned only with high energy neutrons. The change in binding energy that occurs as the result of neutron absorption results in a nuclear excitation energy level that is less than the required critical energy. Therefore, the additional excitation energy must be supplied by the kinetic energy of the incident neutron. The reason for this difference between fissile and fissionable materials is the so-called odd-even effect for nuclei. It has been observed that nuclei with even numbers

neutron to change a nucleus with an odd number of neutrons to a nucleus with an even number of neutrons produces an appreciably higher binding energy than adding a neutron to a nucleus already possessing an even number of neutrons. Some examples of nuclides requiring high energy neutrons to cause fission are thorium-232, uranium-238, and plutonium-240. Table 4 indicates the critical energy (Ecrit) and the binding energy change for an added neutron (BEn) to target nuclei of interest. For fission to be possible, the change in binding energy plus the kinetic energy must equal or exceed the critical energy (ΔBE + KE > Ecrit).

TABLE 4

Critical Energies Compared to Binding Energy of Last Neutron

Target
Nucleus
Critical Energy
Ecrit
Binding Energy of
Fixture checkpoint marker-1163-xray is recorded on fixture page 1163.
Last Neutron BEn
BEn - Ecrit
23
9
2
0Th 7.5 MeV 5.4 MeV -2.1 MeV
23
9
8
2U 7.0 MeV 5.5 MeV -1.5 MeV
23
9
5
2U 6.5 MeV 6.8 MeV +0.3 MeV
23
9
3
2U 6.0 MeV 7.0 MeV +1.0 MeV
23
9
9
4Pu 5.0 MeV 6.6 MeV +1.6 MeV

Uranium-235 fissions with thermal neutrons because the binding energy released by the absorption of a neutron is greater than the critical energy for fission; therefore uranium-235 is a fissile material. The binding energy released by uranium-238 absorbing a thermal neutron is less than the critical energy, so additional energy must be possessed by the neutron for fission to be possible. Consequently, uranium-238 is a fissionable material.

Fertile Material

All of the neutron absorption reactions that do not result in fission lead to the production of new nuclides through the process known as transmutation. These nuclides can, in turn, be transmuted again or may undergo radioactive decay to produce still different nuclides. The nuclides that are

nuclides do not exist in nature, they can only be produced by nuclear reactions (transmutation). The target nuclei for such reactions are said to be fertile. Fertile materials are materials that can undergo transmutation to become fissile materials. Figure 19 traces the transmutation mechanism by which two fertile nuclides, thorium-232 and uranium-238, produce uranium-233 and plutonium-239, respectively.

Figure 19 Conversion of Fertile Nuclides to Fissile Nuclides

If a reactor contains fertile material in addition to its fissile fuel, some new fuel will be produced as the original fuel is burned up. This is called conversion. Reactors that are specifically designed to produce fissionable fuel are called "breeder" reactors. In such reactors, the amount of fissionable fuel produced is greater than the amount of fuel burnup. If less fuel is produced than used, the process is called conversion, and the reactor is termed a "converter."

Binding Energy Per Nucleon (BE/A)

As the number of particles in a nucleus increases, the total binding energy also increases. The rate of increase, however, is not uniform. This lack of uniformity results in a variation in the amount of binding energy associated with each nucleon within the nucleus. This variation in the binding energy per nucleon (BE/A) is easily seen when the average BE/A is plotted versus atomic mass number (A), as shown in Figure 20.

Figure 20 Binding Energy per Nucleon vs. Mass Number

Figure 20 illustrates that as the atomic mass number increases, the binding energy per nucleon

Fixture checkpoint marker-1164-xray is recorded on fixture page 1164.

decreases for A > 60. The BE/A curve reaches a maximum value of 8.79 MeV at A = 56 and decreases to about 7.6 MeV for A = 238. The general shape of the BE/A curve can be explained using the general properties of nuclear forces. The nucleus is held together by very short-range attractive forces that exist between nucleons. On the other hand, the nucleus is being forced apart by long range repulsive electrostatic (coulomb) forces that exist between all the protons in the nucleus.

As the atomic number and the atomic mass number increase, the repulsive electrostatic forces within the nucleus increase due to the greater number of protons in the heavy elements. To overcome this increased repulsion, the proportion of neutrons in the nucleus must increase to maintain stability. This increase in the neutron-to-proton ratio only partially compensates for the growing proton-proton repulsive force in the heavier, naturally occurring elements. Because the repulsive forces are increasing, less energy must be supplied, on the average, to remove a nucleon from the nucleus. The BE/A has decreased. The BE/A of a nucleus is an indication of its degree of stability. Generally, the more stable nuclides have higher BE/A than the less stable ones. The increase in the BE/A as the atomic mass number decreases from 260 to 60 is the primary reason for the energy liberation in the fission process. In addition, the increase in the BE/A as the atomic mass number increases from 1 to 60 is the reason for the energy liberation in the fusion process, which is the opposite reaction of fission.

The heaviest nuclei require only a small distortion from a spherical shape (small energy addition) for the relatively large coulomb forces forcing the two halves of the nucleus apart to overcome the attractive nuclear forces holding the two halves together. Consequently, the heaviest nuclei are easily fissionable compared to lighter nuclei.

Summary

The important information in this chapter is summarized on the following page.

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Atomic and Nuclear Physics DOE-HDBK-1019/1-93 NUCLEAR FISSION
Nuclear Fission Summary
The fission process can be explained using the liquid drop model of a nucleus. In the ground state the nucleus is nearly spherical in shape. After the absorption of a neutron, the nucleus will be in an excited state and start to oscillate and become distorted. If the oscillations cause the nucleus to become shaped like a dumbbell, the repulsive electrostatic forces will overcome the short-range attractive nuclear forces, and the nucleus will split in two.
Excitation energy is the amount of energy a nucleus has above its ground state. Critical energy is the minimum excitation energy that a nucleus must have before it can fission.
Fissile material is material for which fission is possible with neutrons that have zero kinetic energy. Fissionable material is material for which fission caused by neutron absorption is possible provided the kinetic energy added with the binding energy is greater than the critical energy. Fertile material is material that can undergo transmutation to become fissile material.
Transmutation is the process of neutron absorption and subsequent decay, which changes one nuclide to another nuclide. Conversion is the process of transmuting fertile material into fissile material in a reactor, where the amount of fissile material produced is less than the amount of fissile material consumed. Breeding is the same as conversion, except the amount of fissile material produced is more than the amount of fissile material consumed.
The curve of binding energy per nucleon increases quickly through the light nuclides and reaches a maximum at a mass number of about 56. The curve decreases slowly for mass numbers greater than 60.
The heaviest nuclei are easily fissionable because they require only a small distortion from the spherical shape to allow the coulomb forces to overcoming the attractive nuclear force, forcing the two halves of the nucleus apart.
Fixture checkpoint marker-1165-xray is recorded on fixture page 1165.
Uranium-235 fissions with thermal neutrons because the binding energy released by the absorption of a neutron is greater than the critical energy for fission. The binding energy released by uranium-238 absorbing a neutron is less than the critical energy, so additional kinetic energy must be possessed by the neutron for fission to be possible.
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1
0
n
235
92
U
236
92
U
140
55
Cs
93

Rb 3
1
0
n
ENERGY RELEASE FROM FISSION DOE-HDBK-1019/1-93 Atomic and Nuclear Physics
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ENERGY RELEASE FROM FISSION
Fission of heavy nuclides converts a small amount of mass into an enormous amount of energy. The amount of energy released by fission can be determined based on either the change in mass that occurs during the reaction or by the difference in binding energy per nucleon between the fissile nuclide and the fission products.
EO 4.8 CHARACTERIZE the fission products in terms of mass groupings and radioactivity.
EO 4.9 Given the nuclides involved and their masses, CALCULATE the energy released from fission.
EO 4.10 Given the curve of Binding Energy per nucleon versus mass number, CALCULATE the energy released from fission.
Calculation of Fission Energy
Nuclear fission results in the release of enormous quantities of energy. It is necessary to be able to calculate the amount of energy that will be produced. The logical manner in which to pursue this is to first investigate a typical fission reaction such as the one listed below.
It can be seen that when the compound nucleus splits, it breaks into two fission fragments, rubidium-93, cesium-140, and some neutrons. Both fission products then decay by multiple - emissions as a result of the high neutron-to-proton ratio possessed by these nuclides.
In most cases, the resultant fission fragments have masses that vary widely. Figure 21 gives the percent yield for atomic mass numbers. The most probable pair of fission fragments for the thermal fission of the fuel uranium-235 have masses of about 95 and 140. Note that the vertical axis of the fission yield curve is on a logarithmic scale. Therefore, the formation of fission fragments of mass numbers of about 95 and 140 is highly likely.
Atomic and Nuclear Physics DOE-HDBK-1019/1-93 ENERGY RELEASE FROM FISSION
Figure 21 Uranium-235 Fission Yield vs.
Mass Number
Referring now to the binding energy per nucleon curve (Figure 20), we can estimate the amount of energy released by our "typical" fission by plotting this reaction on the curve and calculating
Fixture checkpoint marker-1166-xray is recorded on fixture page 1166.
the change in binding energy (ΔBE) between the reactants on the left-hand side of the fission equation and the products on the right-hand side. Plotting the reactant and product nuclides on the curve shows that the total binding energy of the system after fission is greater than the total binding energy of the system before fission. When there is an increase in the total binding energy of a system, the system has become more stable by releasing an amount of energy equal to the increase in total binding energy of the system. Therefore, in the fission process, the energy liberated is equal to the increase in the total binding energy of the system.
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Figure 22 graphically depicts that the binding energy per nucleon for the products (C,
Figure 22 Change in Binding Energy for Typical Fission
rubidium-93 and B, cesium-140) is greater than that for the reactant (A, uranium-235). The total

number of nucleons.
TABLE 5
Binding Energies Calculated from
Binding Energy per Nucleon Curve
Nuclide B.E. per Nucleon
(BE/A)
Mass Number
(A)
Binding Energy
(BE/A) x (A)
9
3
Fixture checkpoint marker-1167-xray is recorded on fixture page 1167.
3
7Rb 8.7 MeV 93 809 MeV
14
5
0
5Cs 8.4 MeV 140 1176 MeV
23
9
5
2U 7.6 MeV 235 1786 MeV
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BE BEproducts BEreactants
BERb 93 BECs 140 BEU 235
809 MeV 1176 MeV 1786 MeV
199 MeV
1
0
n
235
92
U
236
92
U
140
55
Cs
93
37
Rb 3
1
0
n
Mass difference Mass of Reactants Mass of Products
236.052589 amu 235.85208 amu

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EInst 0.020059 amu
931.5 MeV
amu
186.8 MeV
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The energy released will be equivalent to the difference in binding energy (BE) between the reactants and the products.
The energy liberation during the fission process can also be explained from the standpoint of the conservation of mass-energy. During the fission process, there is a decrease in the mass of the system. There must, therefore, be energy liberated equal to the energy equivalent of the mass lost in the process. This method is more accurate than the previously illustrated method and is used when actually calculating the energy liberated during the fission process.
Again, referring to the "typical" fission reaction.
E , the instantaneous energy, is the energy released immediately after the fission process. It Inst is equal to the energy equivalent of the mass lost in the fission process. It can be calculated as shown below.
Mass of the Reactants Mass of the Products
U 235.043924 amu Rb 92.91699 amu 235
Fixture checkpoint marker-1168-xray is recorded on fixture page 1168.
92
93
37
n 1.008665 amu Cs 139.90910 amu 1
0
140
55
3 (n) 3.02599 amu 1
0
236.052589 amu 235.85208 amu
This mass difference can be converted to an energy equivalent.
93
37
Rb
93
38
Sr
93
39
Y
93
40
Zr
93
41
Nb
140
55

140
56
Ba
140
57
La
140
58
Ce
EDecay mRb 93 mNb 93 4 melectron
931.5MeV
amu
92.91699 amu 92.90638 amu 4 0.0005486 amu
931.5 MeV
amu
0.008416 amu
931.5 MeV
amu
7.84 MeV
ENERGY RELEASE FROM FISSION DOE-HDBK-1019/1-93 Atomic and Nuclear Physics
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The total energy released per fission will vary from the fission to the next depending on what fission products are formed, but the average total energy released per fission of uranium-235 with a thermal neutron is 200 MeV.
As illustrated in the preceding example, the majority of the energy liberated in the fission process is released immediately after the fission occurs and appears as the kinetic energy of the
Fixture checkpoint marker-1169-xray is recorded on fixture page 1169.
fission fragments, kinetic energy of the fission neutrons, and instantaneous gamma rays. The remaining energy is released over a period of time after the fission occurs and appears as kinetic energy of the beta, neutrino, and decay gamma rays.
Estimation of Decay Energy
In addition to this instantaneous energy release during the actual fission reaction, there is additional energy released when the fission fragments decay by emission. This additional
-
energy is called decay energy, EDecay. The decay chains for rubidium-93 and cesium-140 are shown below.
The energy released during the decay for each chain will be equivalent to the mass difference between the original fission product and the sum of the final stable nuclide and the beta particles emitted.
The energy released in the decay chain of rubidium-93 is calculated below.
EDecay mRb 93 mNb 93 3 melectron
931.5MeV
amu
139.90910 amu 139.90543 amu 3 0.0005486 amu
931.5 MeV
amu
0.000202 amu
931.5 MeV

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1.89 MeV
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The energy released in the decay chain of cesium-140 is calculated below.
The total decay energy is the sum of the energies of the two chains, or 9.73 MeV.
Distribution of Fission Energy
The average energy distribution for the energy released per fission with a thermal neutron in uranium-235 is shown in Tables 6 and 7.
TABLE 6
Instantaneous Energy from Fission
Kinetic Energy of Fission Products 167 Mev
Energy of Fission Neutrons 5 MeV
Instantaneous Gamma-ray Energy 5 MeV
Capture Gamma-ray Energy 10 MeV
Total Instantaneous Energy 187 MeV
TABLE 7
Delayed Energy from Fission
Beta Particles From Fission Products 7 Mev
Gamma-rays from Fission Products 6 MeV
Neutrinos 10 MeV
Total Delayed Energy 23 MeV
ENERGY RELEASE FROM FISSION DOE-HDBK-1019/1-93 Atomic and Nuclear Physics
Because the 10 MeV of neutrino energy shown in Table 7 is not absorbed in the reactor, the average value of 200 MeV per fission is still accurate. Note in Table 6 that some fission neutrons undergo radiative capture and the resultant gamma ray emission provides an additional 10 MeV of instantaneous energy, which contributes to the total of 187 MeV instantaneous energy.
All of the energy released, with the exception of the neutrino energy, is ultimately transformed into heat through a number of processes. The fission fragments, with their high positive charge and kinetic energy, cause ionization directly as they rip orbital electrons from the surrounding atoms. In this ionization process, kinetic energy is transferred to the surrounding atoms of the fuel material, resulting in an increase in temperature. The beta particles and gamma rays also give up their energy through ionization, and the fission neutrons interact and lose their energy
Fixture checkpoint marker-1170-xray is recorded on fixture page 1170.
through elastic scattering. Of the 200 MeV released per fission, about seven percent (13 MeV) is released at some time after the instant of fission. When a reactor is shut down, fissions essentially cease, but energy is still being released from the decay of fission products. The heat produced by this decay energy is referred to as "decay heat." Although decay energy represents about seven percent of reactor heat production during reactor operation, once the reactor is shut down the decay heat production drops off quickly to a small fraction of its value while operating. The decay heat produced is significant, however, and systems must be provided to keep the reactor cool even after shutdown.
Summary
The important information in this chapter is summarized below.
Energy Release From Fission Summary
Fission products have some general characteristics in common.
They generally decay by β- emission.
The most common mass numbers are grouped near 95 and 140.

between the masses of the reactants before fission and the fission fragments and fission neutrons after fission.

Another method to determine the energy released by fission is based on the change in binding energy per nucleon between the fissile nuclide and the fission products.

INTERACTION OF RADIATION WITH MATTER

Different types of radiation interact with matter in widely different ways. A large,

Fixture checkpoint marker-1171-xray is recorded on fixture page 1171.

massive, charged alpha particle cannot penetrate a piece of paper and even has a limited range in dry air. A neutrino, at the other extreme, has a low probability of interacting with any matter, even if it passed through the diameter of the earth.

EO 5.1 DESCRIBE interactions of the following with matter:

a. Alpha particle c. Positron
b. Beta particle d. Neutron

EO 5.2 DESCRIBE the following ways that gamma radiation interacts with matter:

a. Photoelectric effect
b. Compton scattering
c. Pair production

Interaction of Radiation With Matter

Radiation can be classified into two general groups, charged and uncharged; therefore, it may be expected that interactions with matter fall into two general types. Charged particles directly ionize the media through which they pass, while uncharged particles and photons can cause ionization only indirectly or by secondary radiation.

A moving charged particle has an electrical field surrounding it, which interacts with the atomic structure of the medium through which it is passing. This interaction decelerates the particle and accelerates electrons in the atoms of the medium. The accelerated electrons may acquire enough energy to escape from the parent atom. This process, whereby radiation "strips" off orbital electrons, is called ionization. Uncharged moving particles have no electrical field, so they can only lose energy and cause ionization by such means as collisions or scattering. A photon can lose energy by the photoelectric effect, Compton effect, or pair production.

I
m z
2
K.E.

Because ionizing radiation creates ions in pairs, the intensity of ionization or the specific ionization is defined as the number of ion-pairs formed per centimeter of travel in a given material. The amount of ionization produced by a charged particle per unit path length, which is a measure of its ionizing power, is roughly proportional to the particle's mass and the square of its charge as illustrated in the equation below.

I is the ionizing power
m is the mass of the particle
z is the number of unit charges it carries
K.E. is its kinetic energy
Since m for an alpha particle is about 7300 times as large as m for a beta particle, and z is twice as great, an alpha will produce much more ionization per unit path length than a beta particle of the same energy. This phenomenon occurs because the larger alpha particle moves slower for a given energy and thus acts on a given electron for a longer time.
Alpha Radiation
Alpha radiation is normally produced from the radioactive decay of heavy nuclides and from certain nuclear reactions. The alpha particle consists of 2 neutrons and 2 protons, so it is essentially the same as the nucleus of a helium atom. Because it has no electrons, the alpha particle has a charge of +2. This positive charge causes the alpha particle to strip electrons from the orbits of atoms in its vicinity. As the alpha particle passes through material, it removes electrons from the orbits of atoms it passes near. Energy is required to remove electrons and the energy of the alpha particle is reduced by each reaction. Eventually the particle will expend its kinetic energy, gain 2 electrons in orbit, and become a helium atom. Because of its strong
Fixture checkpoint marker-1172-xray is recorded on fixture page 1172.
positive charge and large mass, the alpha particle deposits a large amount of energy in a short distance of travel. This rapid, large deposition of energy limits the penetration of alpha particles. The most energetic alpha particles are stopped by a few centimeters of air or a sheet of paper.
Beta-Minus Radiation
A beta-minus particle is an electron that has been ejected at a high velocity from an unstable nucleus. An electron has a small mass and an electrical charge of -1. Beta particles cause ionization by displacing electrons from atom orbits. The ionization occurs from collisions with orbiting electrons. Each collision removes kinetic energy from the beta particle, causing it to slow down. Eventually the beta particle will be slowed enough to allow it to be captured as an
2 electrons 0.000549 amu
electron
931.5 MeV
amu
1.02 MeV
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orbiting electron in an atom. Although more penetrating than the alpha, the beta is relatively easy to stop and has a low power of penetration. Even the most energetic beta radiation can be stopped by a few millimeters of metal.
Positron Radiation
Positively charged electrons are called positrons. Except for the positive charge, they are identical to beta-minus particles and interact with matter in a similar manner. Positrons are very short-lived, however, and quickly are annihilated by interaction with a negatively charged electron, producing two gammas with a combined energy (calculated below) equal to the rest mass of the positive and negative electrons.
Neutron Radiation
Neutrons have no electrical charge. They have nearly the same mass as a proton (a hydrogen atom nucleus). A neutron has hundreds of times more mass than an electron, but 1/4 the mass

they may also be produced from the decay of radioactive nuclides. Because of its lack of charge, the neutron is difficult to stop and has a high penetrating power.

Neutrons are attenuated (reduced in energy and numbers) by three major interactions, elastic scatter, inelastic scatter, and absorption. In elastic scatter, a neutron collides with a nucleus and bounces off. This reaction transmits some of the kinetic energy of the neutron to the nucleus of the atom, resulting in the neutron being slowed, and the atom receives some kinetic energy (motion). This process is sometimes referred to as "the billiard ball effect."

As the mass of the nucleus approaches the mass of the neutron, this reaction becomes more effective in slowing the neutron. Hydrogenous material attenuates neutrons most effectively.

In the inelastic scatter reaction, the same neutron/nucleus collision occurs as in elastic scatter. However, in this reaction, the nucleus receives some internal energy as well as kinetic energy. This slows the neutron, but leaves the nucleus in an excited state. When the nucleus decays to its original energy level, it normally emits a gamma ray.

In the absorption reaction, the neutron is actually absorbed into the nucleus of an atom. The neutron is captured, but the atom is left in an excited state. If the nucleus emits one or more gamma rays to reach a stable level, the process is called radiative capture. This reaction occurs at most neutron energy levels, but is more probable at lower energy levels.

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Gamma Radiation

Gamma radiation is electromagnetic radiation. It is commonly referred to as a gamma ray and is very similar to an x-ray. The difference is that gamma rays are emitted from the nucleus of an atom, and x-rays are produced by orbiting electrons. The x-ray is produced when orbiting electrons move to a lower energy orbit or when fast-moving electrons approaching an atom are

Fixture checkpoint marker-1173-xray is recorded on fixture page 1173.

deflected and decelerated as they react with the atom's electrical field (called Bremsstrahlung). The gamma ray is produced by the decay of excited nuclei and by nuclear reactions. Because the gamma ray has no mass and no charge, it is difficult to stop and has a very high penetrating power. A small fraction of the original gamma stream will pass through several feet of concrete or several meters of water.

There are three methods of attenuating gamma rays. The first method is referred to as the photo-electric effect. When a low energy gamma strikes an atom, the total energy of the gamma is expended in ejecting an electron from orbit. The result is ionization of the atom and expulsion of a high energy electron. This reaction is most predominant with low energy gammas interacting in materials with high atomic weight and rarely occurs with gammas having an energy above 1 MeV. Annihilation of the gamma results. Any gamma energy in excess of the binding energy of the electron is carried off by the electron in the form of kinetic energy.

The second method of attenuation of gammas is called Compton scattering. The gamma interacts with an orbital or free electron; however, in this case, the photon loses only a fraction of its energy. The actual energy loss depending on the scattering angle of the gamma. The gamma continues on at lower energy, and the energy difference is absorbed by the electron. This reaction becomes important for gamma energies of about 0.1 MeV and higher.

At higher energy levels, a third method of attenuation is predominant. This method is pair-production. When a high energy gamma passes close enough to a heavy nucleus, the gamma completely disappears, and an electron and a positron are formed. For this reaction to take place, the original gamma must have at least 1.02 MeV energy. Any energy greater than 1.02 MeV becomes kinetic energy shared between the electron and positron. The probability of pair■production incre

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Atomic and Nuclear Physics INTERACTION OF RADIATION WITH MATTER
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Interaction of Radiation with Matter Summary
An alpha particle deposits a large amount of energy in a short distance of travel due to its large mass and charge.
Beta-minus particles interact with the electrons orbiting the nucleus of atoms, causing ionization by displacing the electrons. The beta particle loses energy with each interaction. After the beta particle loses enough energy, it is captured in the orbital shells of an atom.
Positrons interact with matter much the same way as beta minus particles. After the positron has lost most of its energy by ionizing atoms, it is annihilated by interaction with an electron. The electron-positron pair disappear and are replaced by two gammas, each with the energy equivalent of the mass of an electron (0.51 MeV).
Neutrons interact with matter by elastic scattering, inelastic scattering, or absorption.
Photoelectric effect is where a gamma interacts with an electron orbiting an atom. The entire energy of the gamma is transferred to the electron, and the electron is ejected from its orbit.
In Compton scattering a gamma interacts with an orbital electron, but only part of the gamma energy is transferred to the electron. The electron is ejected from its orbit, and the gamma is scattered off at a lower energy.
In pair-production, a gamma interacts with the electric field of a nucleus and is converted into an electron-positron pair. The gamma must have an energy greater than 1.02 MeV for this to occur.
Summary
The important information in this chapter is summarized below.
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INTERACTION OF RADIATION WITH MATTER Atomic and Nuclear Physics
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Department of Energy
Fundamentals Handbook
NUCLEAR PHYSICS
AND REACTOR THEORY
Module 2
Reactor Theory (Neutron Characteristics)
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OBJECTIVES DOE-HDBK-1019/1-93 Reactor Theory (Neutron Characteristics)
TERMINAL OBJECTIVE
1.0 Without references, EXPLAIN how neutron sources produce neutrons.
ENABLING OBJECTIVES
1.1 DEFINE the following terms:
a. Intrinsic neutron source
b. Installed neutron source
1.2 LIST three examples of reactions that produce neutrons in intrinsic neutron sources.
1.3 LIST three examples of reactions that produce neutrons in installed neutron sources.

Reactor Theory (Neutron Characteristics) DOE-HDBK-1019/1-93 OBJECTIVES
TERMINAL OBJECTIVE
2.0 Given the necessary information for calculations, EXPLAIN basic concepts in reactor physics and perform calculations.
ENABLING OBJECTIVES
2.1 DEFINE the following terms:
a. Atom density d. Barn
b. Neutron flux e. Macroscopic cross section
c. Microscopic cross section f. Mean free path
2.2 EXPRESS macroscopic cross section in terms of microscopic cross section.
2.3 DESCRIBE how the absorption cross section of typical nuclides varies with neutron energy at energies below the resonance absorption region.
2.4 DESCRIBE the cause of resonance absorption in terms of nuclear energy levels.
2.5 DESCRIBE the energy dependence of resonance absorption peaks for typical light and heavy nuclei.
2.6 EXPRESS mean free path in terms of macroscopic cross section.
2.7 Given the number densities (or total density and component fractions) and microscopic cross sections of components, CALCULATE the macroscopic cross section for a mixture.

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microscopic cross section.
2.9 EXPLAIN neutron shadowing or self-shielding.
2.10 Given the neutron flux and macroscopic cross section, CALCULATE the reaction rate.
2.11 DESCRIBE the relationship between neutron flux and reactor power.
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OBJECTIVES DOE-HDBK-1019/1-93 Reactor Theory (Neutron Characteristics)
ENABLING OBJECTIVES (Cont.)
2.12 DEFINE the following concepts:
a. Thermalization d. Average logarithmic energy decrement
b. Moderator e. Macroscopic slowing down power
c. Moderating ratio
2.13 LIST three desirable characteristics of a moderator.
2.14 Given an average fractional energy loss per collision, CALCULATE the energy loss after a specified number of collisions.
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Reactor Theory (Neutron Characteristics) DOE-HDBK-1019/1-93 OBJECTIVES
TERMINAL OBJECTIVE
3.0 Without references, EXPLAIN the production process and effects on fission of prompt and delayed neutrons.
ENABLING OBJECTIVES
3.1 STATE the origin of prompt neutrons and delayed neutrons.
3.2 STATE the approximate fraction of neutrons that are born as delayed neutrons from the
Fixture checkpoint marker-1177-xray is recorded on fixture page 1177.
fission of the following nuclear fuels:
a. Uranium-235
b. Plutonium-239
3.3 EXPLAIN the mechanism for production of delayed neutrons.
3.4 EXPLAIN prompt and delayed neutron generation times.
3.5 Given prompt and delayed neutron generation times and delayed neutron fraction, CALCULATE the average generation time.
3.6 EXPLAIN the effect of delayed neutrons on reactor control.
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OBJECTIVES DOE-HDBK-1019/1-93 Reactor Theory (Neutron Characteristics)
TERMINAL OBJECTIVE
4.0 Without references, DESCRIBE the neutron energy spectrum for the type of reactor presented in this module.
ENABLING OBJECTIVES
4.1 STATE the average energy at which prompt neutrons are produced.
4.2 DESCRIBE the neutron energy spectrum in the following reactors:
a. Fast reactor
b. Thermal reactor
4.3 EXPLAIN the reason for the particular shape of the fast, intermediate, and slow energy regions of the neutron flux spectrum for a thermal reactor.
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Reactor Theory (Neutron Characteristics) DOE-HDBK-1019/1-93 NEUTRON SOURCES
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NEUTRON SOURCES
Neutrons from a variety of sources are always present in a reactor core. This is

naturally occurring (intrinsic) neutron sources, while others may be the result of fabricated (installed) neutron sources that are incorporated into the design of the reactor. The neutrons produced by sources other than neutron-induced fission are often grouped together and classified as source neutrons.

EO 1.1 DEFINE the following terms:

a. Intrinsic neutron source

b. Installed neutron source

EO 1.2 LIST three examples of reactions that produce neutrons in intrinsic neutron sources.

EO 1.3 LIST three examples of reactions that produce neutrons in installed neutron sources.

Neutron Sources

In addition to neutron-induced fission, neutrons are produced by other reactions. The neutrons produced by reactions other than neutron-induced fission are called source neutrons. Source neutrons are important because they ensure that the neutron population remains high enough to allow a visible indication of neutron level on the most sensitive monitoring instruments while the reactor is shutdown and during the startup sequence. This verifies instrument operability and allows monitoring of neutron population changes. Source neutrons can be classified as either intrinsic or installed neutron sources.

Intrinsic Neutron Sources

Some neutrons will be produced in the materials present in the reactor due to a variety of unavoidable reactions that occur because of the nature of these materials. Intrinsic neutron sources are those neutron-producing reactions that always occur in reactor materials.

■ ■ ■ ■
■ ■ ■
■ ■

NEUTRON SOURCES DOE-HDBK-1019/1-93 Reactor Theory (Neutron Characteristics)

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A limited number of neutrons will always be present, even in a reactor core that has never been

Fixture checkpoint marker-1178-xray is recorded on fixture page 1178.

operated, due to spontaneous fission of some heavy nuclides that are present in the fuel. Uranium■238, ur of the rate at which different heavy nuclides produce neutrons by spontaneous fission.

Californium-252 is not an intrinsic neutron source, but will be discussed in the section on installed neutron sources.

TABLE 1

Neutron Production by Spontaneous Fission

Nuclide T (Fission) T (1/2 1/2 ■-decay) neutrons/sec/gram

U 1.8 x 10 years 6.8 x 10 years 8.0 x 10 235
92
17 8 -4

U 8.0 x 10 years 4.5 x 10 years 1.6 x 10 238
92
15 9 -2

Pu 5.5 x 10 years 2.4 x 10 years 3.0 x 10 239
94
5 4 -2

Pu 1.2 x 10 years 6.6 x 10 years 1.0 x 10 240
94

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Cf 66.0 years 2.65 years 2.3 x 10 252
98
12
Another intrinsic neutron source is a reaction involving natural boron and fuel. In some reactors, natural boron is loaded into the reactor core as a neutron absorber to improve reactor control or increase core life-time. Boron-11 (80.1% of natural boron) undergoes a reaction with the alpha particle emitted by the radioactive decay of heavy nuclides in the fuel to yield a neutron as shown below.
The boron-11 must be mixed with, or in very close proximity to, the fuel for this reaction because of the short path length of the alpha particle. For a reactor core with this configuration, this (■,n) reaction is an important source of neutrons for reactor startup.
In a reactor that has been operated, another source of neutrons becomes significant. Neutrons may be produced by the interaction of a gamma ray and a deuterium nucleus. This reaction is commonly referred to as a photoneutron reaction because it is initiated by electromagnetic radiation and results in the production of a neutron. The photoneutron reaction is shown below.
■ ■ ■
■ ■
■ ■ ■
■ ■
Reactor Theory (Neutron Characteristics) DOE-HDBK-1019/1-93 NEUTRON SOURCES
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There is an abundant supply of high energy gammas in a reactor that has been operated because many of the fission products are gamma emitters. All water-cooled reactors have some deuterium present in the coolant in the reactor core because a small fraction of natural hydrogen is the isotope deuterium. The atom percentage of deuterium in the water ranges from close to the naturally occurring value (0.015%) for light water reactors to above 90% deuterium for heavy water reactors. Therefore, the required conditions for production of photoneutrons exist.
The supply of gamma rays decreases with time after shutdown as the gamma emitters decay; therefore, the photoneutron production rate also decreases. In a few particular reactors, additional D O (heavy water) may be added to the reactor to increase the production of 2 photoneutrons following a long shutdown period.
Installed Neutron Sources
Because intrinsic neutron sources can be relatively weak or dependent upon the recent power history of the reactor, many reactors have artificial sources of neutrons installed. These neutron sources ensure that shutdown neutron levels are high enough to be detected by the nuclear instruments at all times. This provides a true picture of reactor conditions and any change in
Fixture checkpoint marker-1179-xray is recorded on fixture page 1179.
these conditions. An installed neutron source is an assembly placed in or near the reactor for the sole purpose of producing source neutrons.
One strong source of neutrons is the artificial nuclide californium-252, which emits neutrons at the rate of about 2 x 10 neutrons per second per gram as the result of spontaneous fission. 12 Important drawbacks for some applications may be its high cost and its short half-life (2.65 years).
Many installed neutron sources use the (■,n) reaction with beryllium. These sources are composed of a mixture of metallic beryllium (100% beryllium-9) with a small quantity of an alpha particle emitter, such as a compound of radium, polonium, or plutonium. The reaction that occurs is shown below.
The beryllium is intimately (homogeneously) mixed with the alpha emitter and is usually

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Another type of installed neutron source that is widely used is a photoneutron source that employs the (■,n) reaction with beryllium. Beryllium is used for photoneutron sources because its stable isotope beryllium-9 has a weakly attached last neutron with a binding energy of only 1.66 MeV. Thus, a gamma ray with greater energy than 1.66 MeV can cause neutrons to be ejected by the (■,n) reaction as shown below.
NEUTRON SOURCES DOE-HDBK-1019/1-93 Reactor Theory (Neutron Characteristics)
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Neutron Sources Summary
& Intrinsic neutron sources are sources of neutrons from materials that are in the reactor for other purposes such as fuel, burnable poison, or moderator.
& Installed neutron sources are materials or components placed in the reactor specifically for the purpose of producing source neutrons.
& Examples of intrinsic neutron sources are listed below.
Fixture checkpoint marker-1180-xray is recorded on fixture page 1180.
Spontaneous fission of heavy nuclides in fuel, such as uranium-238, uranium-235, and plutonium-239, results in fission fragments and free neutrons.
Boron-11 mixed with the fuel undergoes an alpha-neutron reaction and becomes nitrogen-14.
Deuterium present in the reactor coolant undergoes a gamma■neutron reaction and becomes hydrogen-1.
& Examples of installed neutron sources are listed below.
Spontaneous fission of californium-252 results in fission fragments and free neutrons.
Beryllium-9 undergoes an alpha-neutron reaction (alpha from the decay of plutonium, polonium, or radium) and becomes carbon-12.
Beryllium-9 undergoes a gamma-neutron reaction (high energy gamma from decay of antimony-124) and becomes beryllium-8.
Many startup sources of this type use antimony and beryllium because after activation with neutrons the radioactive antimony becomes an emitter of high energy gammas. The photoneutron sources of this type are constructed somewhat differently from the (■,n) types. One design incorporates a capsule of irradiated antimony enclosed in a beryllium sleeve. The entire assembly is then encased in a stainless steel cladding. A large reactor may have several neutron sources of this type installed within the core.
Summary
The important information in this chapter is summarized below.
DOE-HDBK-1019/1-93
Reactor Theory (Neutron Characteristics) NUCLEAR CROSS SECTIONS AND NEUTRON FLUX
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NUCLEAR CROSS SECTIONS AND NEUTRON FLUX
To determine the frequency of neutron interactions, it is necessary to describe the availability of neutrons to cause interaction and the probability of a neutron interacting with material. The availability of neutrons and the probability of interaction are quantified by the neutron flux and nuclear cross section.
EO 2.1 DEFINE the following terms:
a. Atom density d. Barn
b. Neutron flux e. Macroscopic cross section
c. Microscopic cross section f. Mean free path
EO 2.2 EXPRESS macroscopic cross section in terms of microscopic cross

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EO 2.3 DESCRIBE how the absorption cross section of typical nuclides varies with neutron energy at energies below the resonance absorption region.
EO 2.4 DESCRIBE the cause of resonance absorption in terms of nuclear energy levels.
EO 2.5 DESCRIBE the energy dependence of resonance absorption peaks for typical light and heavy nuclei.
EO 2.6 EXPRESS mean free path in terms of macroscopic cross section.
EO 2.7 Given the number densities (or total density and component fractions) and microscopic cross sections of components, CALCULATE the macroscopic cross section for a mixture.
EO 2.8 CALCULATE a macroscopic cross section given a material density, atomic mass, and microscopic cross section.
EO 2.9 EXPLAIN neutron shadowing or self-shielding.
'

Introduction

Fission neutrons are born with an average energy of about 2 MeV. These fast neutrons interact
Fixture checkpoint marker-1181-xray is recorded on fixture page 1181.
with the reactor core materials in various absorption and scattering reactions. Collisions that result in scattering are useful in slowing neutrons to thermal energies. Thermal neutrons may be absorbed by fissile nuclei to produce more fissions or be absorbed in fertile material for conversion to fissionable fuel. Absorption of neutrons in structural components, coolant, and other non-fuel material results in the removal of neutrons without fulfilling any useful purpose. To safely and efficiently operate a nuclear reactor it is necessary to predict the probability that a particular absorption or scattering reaction will occur. Once these probabilities are known, if the availability of neutrons can be determined, then the rate at which these nuclear reactions take place can be predicted.

Atom Density

One important property of a material is the atom density. The atom density is the number of atoms of a given type per unit volume of the material. To calculate the atom density of a substance use Equation (2-1).

(2-1)

where:

N = atom density (atoms/cm)3
' = density (g/cm)3
N = Avogadro's number (6.022 x 10 atoms/mole) A
23
M = gram atomic weight
N
NA
M
2.699 g
cm3
6.022 x 1023 atoms
mole

mole
6.024 x 1022 atoms
cm3
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Reactor Theory (Neutron Characteristics) NUCLEAR CROSS SECTIONS AND NEUTRON FLUX
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Example:
A block of aluminum has a density of 2.699 g/cm . If the gram atomic weight of
3
aluminum is 26.9815 g, calculate the atom density of the aluminum.
Solution:
Cross Sections
The probability of a neutron interacting with a nucleus for a particular reaction is dependent upon not only the kind of nucleus involved, but also the energy of the neutron. Accordingly, the absorption of a thermal neutron in most materials is much more probable than the absorption of a fast neutron. Also, the probability of interaction will vary depending upon the type of reaction involved.
The probability of a particular reaction occurring between a neutron and a nucleus is called the microscopic cross section ( ) of the nucleus for the particular reaction. This cross section will vary with the energy of the neutron. The microscopic cross section may also be regarded as the effective area the nucleus presents to the neutron for the particular reaction. The larger the effective area, the greater the probability for reaction.
Because the microscopic cross section is an area, it is expressed in units of area, or square centimeters. A square centimeter is tremendously large in comparison to the effective area of a nucleus, and it has been suggested that a physicist once referred to the measure of a square centimeter as being "as big as a barn" when applied to nuclear processes. The name has persisted and microscopic cross sections are expressed in terms of barns. The relationship
Fixture checkpoint marker-1182-xray is recorded on fixture page 1182.
between barns and cm is shown below. 2
1 barn = 10-24 cm2
N
DOE-HDBK-1019/1-93
NUCLEAR CROSS SECTIONS AND NEUTRON FLUX Reactor Theory (Neutron Characteristics)
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Whether a neutron will interact with a certain volume of material depends not only on the microscopic cross section of the individual nuclei but also on the number of nuclei within that volume. Therefore, it is necessary to define another kind of cross section known as the macroscopic cross section ( ). The macroscopic cross section is the probability of a given reaction occurring per unit travel of the neutron. is related to the microscopic cross section ( ) by the relationship shown below.
(2-2)
where:
= macroscopic cross section (cm)
-1
N = atom density of material (atoms/cm)
3
= microscopic cross-section (cm)
2

and is restated for clarity. The microscopic cross section ( ) represents the effective target area that a single nucleus presents to a bombarding particle. The units are given in barns or cm .
2
The macroscopic cross section ( ) represents the effective target area that is presented by all of the nuclei contained in 1 cm of the material. The units are given as 1/cm or cm .
3 -1
A neutron interacts with an atom of the material it enters in two basic ways. It will either interact through a scattering interaction or through an absorption reaction. The probability of a neutron being absorbed by a particular atom is the microscopic cross section for absorption, . The probability of a neutron scattering off of a particular nucleus is the microscopic cross a section for scattering, s
. The sum of the microscopic cross section for absorption and the microscopic cross section for scattering is the total microscopic cross section, . T
= + T a s
Both the absorption and the scattering microscopic cross sections can be further divided. For instance, the scattering cross section is the sum of the elastic scattering cross section (se) and the inelastic scattering cross section ( ). si
= + s se si
The microscopic absorption cross section ( ) includes all reactions except scattering. However, a for most purposes it is sufficient to merely separate it into two categories, fission (f) and capture ( ). Radiative capture of neutrons was described in the Neutron Interactions chapter c of Module 1.
= + a f c
DOE-HDBK-1019/1-93
Reactor Theory (Neutron Characteristics) NUCLEAR CROSS SECTIONS AND NEUTRON FLUX
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Figure 1 Typical Neutron Absorption Cross Section vs. Neutron Energy
The variation of absorption cross sections with neutron energy is often complicated. For many elements the absorption cross sections are small, ranging from a fraction of a barn to a few barns for slow (or thermal) neutrons.
For a considerable number of nuclides of moderately high (or high) mass numbers, an examination of the variation of the absorption cross section with the energy of the incident neutron reveals the existence of three regions on a curve of absorption cross section versus neutron energy. This cross section is illustrated in Figure 1. First, the cross section decreases
Fixture checkpoint marker-1183-xray is recorded on fixture page 1183.
steadily with increasing neutron energy in a low energy region, which includes the thermal range (E < 1 eV). In this region the absorption cross section, which is often high, is inversely proportional to the velocity (v). This region is frequently referred to as the "1/v region," because the absorption cross section is proportional to 1/v, which is the reciprocal of neutron velocity. Following the 1/v region, there occurs the "resonance region" in which the cross sections rise sharply to high values called "resonance peaks" for neutrons of certain energies, and then fall again. These energies are called resonance energies and are a result of the affinity of the nucleus for neutrons whose energies closely match its discrete, quantum energy levels. That is, when the binding energy of a neutron plus the kinetic energy of the neutron are exactly equal to the amount required to raise a compound nucleus from its ground state to a quantum level, resonance absorption occurs. The following example problem further illustrates this point.
■
(

NUCLEAR CROSS SECTIONS AND NEUTRON FLUX Reactor Theory (Neutron Characteristics)
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Assuming that uranium-236 has a nuclear quantum energy level at 6.8 MeV above its ground state, calculate the kinetic energy a neutron must possess to undergo resonant absorption in uranium-235 at this resonance energy level.
BE = [Mass(U) + Mass(neutron) - Mass(U)] x 931 MeV/amu 235 236
BE = (235.043925 + 1.008665 - 236.045563) x 931 MeV/amu
BE = (0.007025 amu) x 931 MeV/amu = 6.54 MeV
6.8 MeV - 6.54 MeV = 0.26 MeV
The difference between the binding energy and the quantum energy level equals the amount of kinetic energy the neutron must possess. The typical heavy nucleus will have many closely■spaced resonanc
Fixture checkpoint marker-1184-xray is recorded on fixture page 1184.
complex and have more possible configurations and corresponding energy states. Light nuclei, being less complex, have fewer possible energy states and fewer resonances that are sparsely distributed at higher energy levels.
For higher neutron energies, the absorption cross section steadily decreases as the energy of the neutron increases. This is called the "fast neutron region." In this region the absorption cross sections are usually less than 10 barns.
With the exception of hydrogen, for which the value is fairly large, the elastic scattering cross sections are generally small, for example, 5 barns to 10 barns. This is close to the magnitude of the actual geometric cross sectional area expected for atomic nuclei. In potential scattering, the cross section is essentially constant and independent of neutron energy. Resonance elastic scattering and inelastic scattering exhibit resonance peaks similar to those associated with absorption cross sections. The resonances occur at lower energies for heavy nuclei than for light nuclei. In general, the variations in scattering cross sections are very small when compared to the variations that occur in absorption cross sections.
Mean Free Path
If a neutron has a certain probability of undergoing a particular interaction in one centimeter of travel, then the inverse of this value describes how far the neutron will travel (in the average case) before undergoing an interaction. This average distance traveled by a neutron before interaction is known as the mean free path for that interaction and is represented by the symbol ■. The relationship between the mean free path (■) and the macroscopic cross section (*) is shown below.
(2-3)
'
(
)
DOE-HDBK-1019/1-93
Reactor Theory (Neutron Characteristics) NUCLEAR CROSS SECTIONS AND NEUTRON FLUX
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Calculation of Macroscopic Cross Section and Mean Free Path
Most materials are composed of several elements, and because most elements are composed of several isotopes, most materials involve many cross sections, one for each isotope involved. Therefore, to include all the isotopes within a given material, it is necessary to determine the macroscopic cross section for each isotope and then sum all the individual macroscopic cross sections. Equation (2-4) can be used to determine the macroscopic cross section for a composite material.
* = N) + N) + N) + ... N) (2-4) 1 1 2 2 3 3 nn

N = the number of nuclei per cm of the n element n
3 th
) = the microscopic cross section of the n element n
th
The following example problems illustrate the calculation of the macroscopic cross section for a single element and for combinations of materials.
Example 1:
Find the macroscopic thermal neutron absorption cross section for iron, which has a density of 7.86 g/cm . The microscopic cross section for absorption of iron is 2.56 barns 3 and the gram atomic weight is 55.847 g.
Solution:
Step 1: Using Equation (2-1), calculate the atom density of iron.
Step 2: Use this atom density in Equation (2-2) to calculate the macroscopic cross section.
'
'

Fixture checkpoint marker-1185-xray is recorded on fixture page 1185.

Example 2:
An alloy is composed of 95% aluminum and 5% silicon (by weight). The density of the alloy is 2.66 g/cm . Properties of aluminum and silicon are shown below. 3

Element	Gram Atomic Weight	)	) a (barns)	s (barns)
Aluminum	26.9815		0.23	1.49
Silicon	28.0855		0.16	2.20

1. Calculate the atom densities for the aluminum and silicon.
2. Determine the absorption and scattering macroscopic cross sections for thermal neutrons.
3. Calculate the mean free paths for absorption and scattering.
Solution:
Step 1: The density of the aluminum will be 95% of the total density. Using Equation (2-1) yields the atom densities.
a NAl a,Al NSi a,Si
5.64 x 1022 atoms
cm3
0.23 x 10 24 cm22.85 x 1021 atoms
cm3
0.16 x 10 24 cm2
0.0134 cm 1
s NAl s,Al NSi s,Si
5.64 x 1022 atoms
cm3
1.49 x 10 24 cm22.85 x 1021 atoms
cm3
2.20 x 10 24 cm2
0.0903 cm 1
a

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a
1
0.01345 cm 1
74.3 cm
s
1
s
1
0.0903 cm 1
11.1 cm
DOE-HDBK-1019/1-93
Reactor Theory (Neutron Characteristics) NUCLEAR CROSS SECTIONS AND NEUTRON FLUX
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Step 2: The macroscopic cross sections for absorption and scattering are calculated using Equation (2-4).
Step 3: The mean free paths are calculated by inserting the macroscopic cross sections calculated above into Equation (2-3).
Thus, a neutron must travel an average of 74.3 cm to interact by absorption in this alloy, but it must travel only 11.1 cm to interact by scattering.
)
)
)
)
(■
(■
Fixture checkpoint marker-1186-xray is recorded on fixture page 1186.
DOE-HDBK-1019/1-93
NUCLEAR CROSS SECTIONS AND NEUTRON FLUX Reactor Theory (Neutron Characteristics)
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Effects of Temperature on Cross Section
As discussed, the microscopic absorption cross section varies significantly as neutron energy varies. The microscopic cross sections provided on most charts and tables are measured for a standard neutron velocity of 2200 meters/second, which corresponds to an ambient temperature of 68(F. Therefore, if our material is at a higher temperature, the absorption cross section will be lower than the value for 68(F, and any cross sections which involve absorption (for example,) ,) ,)) must be corrected for the existing temperature. acf
The following formula is used to correct microscopic cross sections for temperature. Although the example illustrates absorption cross section, the same formula may be used to correct capture and fission cross sections.
where:
) = microscopic cross section corrected for temperature
) = microscopic cross section at reference temperature (68(F or 20(C) o
T = reference temperature (68(F) in degrees Rankine ((R) or Kelvin ((K) o
T = temperature for which corrected value is being calculated
NOTE: When using this formula, all temperatures must be converted to (R or (K.
(R = (F + 460
(K = (C + 273
Example:

has a) of 583 barns at 68(F. f
Solution:
DOE-HDBK-1019/1-93
Reactor Theory (Neutron Characteristics) NUCLEAR CROSS SECTIONS AND NEUTRON FLUX
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Neutron Flux
Macroscopic cross sections for neutron reactions with materials determine the probability of one neutron undergoing a specific reaction per centimeter of travel through that material. If one wants to determine how many reactions will actually occur, it is necessary to know how many neutrons are traveling through the material and how many centimeters they travel each second. It is convenient to consider the number of neutrons existing in one cubic centimeter at any one instant and the total distance they travel each second while in that cubic centimeter. The number of neutrons existing in a cm of material at any instant is called neutron density and is 3 represented by the symbol n with units of neutrons/cm . The total distance these neutrons can 3 travel each second will be determined by their velocity.
A good way of defining neutron flux (1) is to consider it to be the total path length covered by all neutrons in one cubic centimeter during one second. Mathematically, this is the equation below.
1 = n v (2-5)
where:
1 = neutron flux (neutrons/cm -sec) 2
n = neutron density (neutrons/cm)3
v = neutron velocity (cm/sec)
The term neutron flux in some applications (for example, cross section measurement) is used as parallel beams of neutrons traveling in a single direction. The intensity (I) of a neutron beam is the product of the neutron density times the average neutron velocity. The directional beam intensity is equal to the number of neutrons per unit area and time (neutrons/cm -sec) falling on 2 a surface perpendicular to the direction of the beam.
One can think of the neutron flux in a reactor as being comprised of many neutron beams traveling in various directions. Then, the neutron flux becomes the scalar sum of these directional flux intensities (added as numbers and not vectors), that is, 1 = I + I + I +...I . 123 n
Since the atoms in a reactor do not interact preferentially with neutrons from any particular
Fixture checkpoint marker-1187-xray is recorded on fixture page 1187.
direction, all of these directional beams contribute to the total rate of reaction. In reality, at a given point within a reactor, neutrons will be traveling in all directions.
DOE-HDBK-1019/1-93
NUCLEAR CROSS SECTIONS AND NEUTRON FLUX Reactor Theory (Neutron Characteristics)
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Self-Shielding
In some locations within the reactor, the flux level may be significantly lower than in other areas due to a phenomenon referred to as neutron shadowing or self-shielding. For example, the interior of a fuel pin or pellet will "see" a lower average flux level than the outer surfaces since an appreciable fraction of the neutrons will have been absorbed and therefore cannot reach the interior of the fuel pin. This is especially important at resonance energies, where the absorption cross sections are large.
Summary
The important information in this chapter is summarized below.
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Nuclear Cross Section and Neutron Flux Summary
& Atom density (N) is the number of atoms of a given type per unit volume of material.
& Microscopic cross section ()) is the probability of a given reaction occurring between a neutron and a nucleus.
& Microscopic cross sections are measured in units of barns, where 1 barn = 10-24 cm .2
& Macroscopic cross section (*) is the probability of a given reaction occurring per unit length of travel of the neutron. The units for macroscopic cross section are cm . -1
& The mean free path (■) is the average distance that a neutron travels in a material between interactions.
& Neutron flux (1) is the total path length traveled by all neutrons in one cubic centimeter of material during one second.
& The macroscopic cross section for a material can be calculated using the equation below.
* = N)
& The absorption cross section for a material usually has three distinct regions. At low neutron energies (<1 eV) the cross section is inversely proportional to the neutron velocity.
& Resonance absorption occurs when the sum of the kinetic energy of the neutron and its binding energy is equal to an allowed nuclear energy level of the nucleus.
& Resonance peaks exist at intermediate energy levels. For higher neutron energies, the absorption cross section steadily decreases as the neutron energy increases.
& The mean free path equals 1/*.
& The macroscopic cross section for a mixture of materials can be calculated using the equation below.
* = N 1 1 2 2 3 3 nn) + N) + N) + ... N)
& Self-shielding is where the local neutron flux is depressed within a material due to neutron absorption near the surface of the material.
1)
REACTION RATES DOE-HDBK-1019/1-93 Reactor Theory (Neutron Characteristics)
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REACTION RATES
It is possible to determine the rate at which a nuclear reaction will take place based on the neutron flux, cross section for the interaction, and atom density of the target. This relationship illustrates how a change in one of these items affects the reaction rate.
Fixture checkpoint marker-1188-xray is recorded on fixture page 1188.
EO 2.10 Given the neutron flux and macroscopic cross section,
CALCULATE the reaction rate.
EO 2.11 DESCRIBE the relationship between neutron flux and reactor power.
Reaction Rates
If the total path length of all the neutrons in a cubic centimeter in a second is known, (neutron flux (1)), and if the probability of having an interaction per centimeter path length is also known (macroscopic cross section (*)), multiply them together to get the number of interactions taking

denoted by the symbol R. The reaction rate can be calculated by the equation shown below.
R = 1 * (2-6)
where:
R = reaction rate (reactions/sec)
1 = neutron flux (neutrons/cm -sec) 2
* = macroscopic cross section (cm) -1
Substituting the fact that * = N) into Equation (2-6) yields the equation below.
where:
) = microscopic cross section (cm)2
N = atom density (atoms/cm)3
Rf f
1 x 1013 neutrons
cm2
sec
0.1 cm 1
1 x 1012 fissions
Fixture checkpoint marker-1189-xray is recorded on fixture page 1189.
cm3
sec
Rf f
f
Rf
1.29 x 1012 fissions
cm3
sec
3 x 1013 neutrons
cm2
sec
0.043 cm 1
Reactor Theory (Neutron Characteristics) DOE-HDBK-1019/1-93 REACTION RATES
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The reaction rate calculated will depend on which macroscopic cross section is used in the calculation. Normally, the reaction rate of greatest interest is the fission reaction rate.
Example:
If a one cubic centimeter section of a reactor has a macroscopic fission cross section of 0.1 cm , and if the thermal neutron flux is 10 neutrons/cm -sec, what is the fission rate -1 13 2 in that cubic centimeter?
Solution:
In addition to using Equation (2-6) to determine the reaction rate based on the physical properties of material, it is also possible to algebraically manipulate the equation to determine physical properties if the reaction rate is known.
Example:
A reactor operating at a flux level of 3 x 10 neutrons/cm -sec contains 10 atoms of 13 2 20 uranium-235 per cm . The reaction rate is 1.29 x 10 fission/cm . Calculate and .
3 12 3
f f
Solution:

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Equation (2-6) for and substituting the appropriate values. f
Rf N f
f
Rf
N
1.29 x 1012 fissions
cm3
sec
1 x 1020 atoms
cm3
3 x 1013 neutrons
cm2
sec
4.3 x 10 22 cm2 1 barn
1 x 10 24 cm2
430 barns
1 watt 1 erg
1 x 10 7 watt sec
1 MeV
1.602 x 10 6
erg
1 fission
200 MeV
Fixture checkpoint marker-1190-xray is recorded on fixture page 1190.
3.12 x 1010 fissions
second
REACTION RATES DOE-HDBK-1019/1-93 Reactor Theory (Neutron Characteristics)
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Step 2: To find the microscopic cross section, replace f with (N x f) and solve
for .
f
Reactor Power Calculation
Multiplying the reaction rate per unit volume by the total volume of the core results in the total number of reactions occurring in the core per unit time. If the amount of energy involved in each reaction were known, it would be possible to determine the rate of energy release (power) due to a certain reaction.
In a reactor where the average energy per fission is 200 MeV, it is possible to determine the number of fissions per second that are necessary to produce one watt of power using the following conversion factors.
1 fission = 200 MeV
1 MeV = 1.602 x 10 ergs -6
1 erg = 1 x 10 watt-sec -7
This is equivalent to stating that 3.12 x 10 fissions release 1 watt-second of energy. 10
P
th f V
3.12 x 1010 fissions
watt sec
Reactor Theory (Neutron Characteristics) DOE-HDBK-1019/1-93 REACTION RATES

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The power released in a reactor can be calculated based on Equation (2-6). Multiplying the reaction rate by the volume of the reactor results in the total fission rate for the entire reactor. Dividing by the number of fissions per watt-sec results in the power released by fission in the reactor in units of watts. This relationship is shown mathematically in Equation (2-7) below.
(2-7)
where:
P = power (watts)
= thermal neutron flux (neutrons/cm -sec) th
2
= macroscopic cross section for fission (cm) f
-1
V = volume of core (cm)
3
Relationship Between Neutron Flux and Reactor Power
In an operating reactor the volume of the reactor is constant. Over a relatively short period of time (days or weeks), the number density of the fuel atoms is also relatively constant. Since the atom density and microscopic cross section are constant, the macroscopic cross section must also be constant. Examining Equation (2-7), it is apparent that if the reactor volume and macroscopic cross section are constant, then the reactor power and the neutron flux are directly proportional. This is true for day-to-day operation. The neutron flux for a given power level will increase very slowly over a period of months due to the burnup of the fuel and resulting decrease in atom density and macroscopic cross section.
REACTION RATES DOE-HDBK-1019/1-93 Reactor Theory (Neutron Characteristics)
NP-02 Page 22 Rev. 0
Reaction Rates Summary
& The reaction rate is the number of interactions of a particular type occurring in a cubic centimeter of material in a second.
& The reaction rate can be calculated by the equation below.
R = 1 *
& Over a period of several days, while the atom density of the fuel can be considered
Fixture checkpoint marker-1191-xray is recorded on fixture page 1191.
constant, the neutron flux is directly proportional to reactor power.
Summary
The important information in this chapter is summarized below.
Reactor Theory (Neutron Characteristics) DOE-HDBK-1019/1-93 NEUTRON MODERATION
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NEUTRON MODERATION
In thermal reactors, the neutrons that cause fission are at a much lower energy than the energy level at which they were born from fission. In this type of reactor, specific materials must be included in the reactor design to reduce the energy level of the neutrons in an efficient manner.
EO 2.12 DEFINE the following concepts:
a. Thermalization d. Average logarithmic energy decrement
b. Moderator e. Macroscopic slowing down power
c. Moderating ratio
EO 2.13 LIST three desirable characteristics of a moderator.
EO 2.14 Given an average fractional energy loss per collision, CALCULATE the energy loss after a specified number of

Neutron Slowing Down and Thermalization

Fission neutrons are produced at an average energy level of 2 MeV and immediately begin to slow down as the result of numerous scattering reactions with a variety of target nuclei. After a number of collisions with nuclei, the speed of a neutron is reduced to such an extent that it has approximately the same average kinetic energy as the atoms (or molecules) of the medium in which the neutron is undergoing elastic scattering. This energy, which is only a small fraction of an electron volt at ordinary temperatures (0.025 eV at 20(C), is frequently referred to as the thermal energy, since it depends upon the temperature. Neutrons whose energies have been reduced to values in this region (< 1 eV) are designated thermal neutrons. The process of reducing the energy of a neutron to the thermal region by elastic scattering is referred to as thermalization, slowing down, or moderation. The material used for the purpose of thermalizing neutrons is called a moderator. A good moderator reduces the speed of neutrons in a small number of collisions, but does not absorb them to any great extent. Slowing the neutrons in as few collisions as possible is desirable in order to reduce the amount of neutron leakage from the core and also to reduce the number of resonance absorptions in non-fuel materials. Neutron leakage and resonance absorption will be discussed in the next module.

ln Ei

ln Ef

ln

Ei

Ef

2

A

2

3

NEUTRON MODERATION DOE-HDBK-1019/1-93 Reactor Theory (Neutron Characteristics)

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The ideal moderating material (moderator) should have the following nuclear properties.

large scattering cross section

small absorption cross section

large energy loss per collision

A convenient measure of energy loss per collision is the logarithmic energy decrement. The average logarithmic energy decrement is the average decrease per collision in the logarithm of the neutron energy. This quantity is represented by the symbol (Greek letter xi).

(2-8)

where:

= average logarithmic energy decrement

Fixture checkpoint marker-1192-xray is recorded on fixture page 1192.

Ei = average initial neutron energy

Ef = average final neutron energy

The symbol is commonly called the average logarithmic energy decrement because of the fact that a neutron loses, on the average, a fixed fraction of its energy per scattering collision. Since the fraction of energy retained by a neutron in a single elastic collision is a constant for a given material, is also a constant. Because it is a constant for each type of material and does not depend upon the initial neutron energy, is a convenient quantity for assessing the moderating ability of a material.

The values for the lighter nuclei are tabulated in a variety of sources. The following commonly used approximation may be used when a tabulated value is not available.

values of A it may be in error by over three percent.
N
ln Ehigh ln Elow
ln
Ehigh
Elow
N
ln
Ehigh
Elow
ln
2 x 106eV
0.025 eV
0.948
Fixture checkpoint marker-1193-xray is recorded on fixture page 1193.
19.2 collisions
Reactor Theory (Neutron Characteristics) DOE-HDBK-1019/1-93 NEUTRON MODERATION
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Since represents the average logarithmic energy loss per collision, the total number of collisions necessary for a neutron to lose a given amount of energy may be determined by dividing into the difference of the natural logarithms of the energy range in question. The number of collisions (N) to travel from any energy, E , to any lower energy, E , can be calculated as shown below. high low
Example:
How many collisions are required to slow a neutron from an energy of 2 MeV to a thermal energy of 0.025 eV, using water as the moderator? Water has a value of 0.948 for .
Solution:
Sometimes it is convenient, based upon information known, to work with an average fractional energy loss per collision as opposed to an average logarithmic fraction. If the initial neutron energy level and the average fractional energy loss per collision are known, the final energy level for a given number of collisions may be computed using the following formula.
! (
NEUTRON MODERATION DOE-HDBK-1019/1-93 Reactor Theory (Neutron Characteristics)
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(2-9)
where:
E = initial neutron energy o
E = neutron energy after N collisions N
x = average fractional energy loss per collision
N = number of collisions
Example:
If the average fractional energy loss per collision in hydrogen is 0.63, what will be the energy of a 2 MeV neutron after (a) 5 collisions? (b) 10 collisions?
Solution:
a)
b)
Macroscopic Slowing Down Power
Although the logarithmic energy decrement is a convenient measure of the ability of a material

of the capabilities of a material is the macroscopic slowing down power. The macroscopic slowing down power (MSDP) is the product of the logarithmic energy decrement and the macroscopic cross section for scattering in the material. Equation (2-10) illustrates how to calculate the macroscopic slowing down power.

(2-10)

MR

s

a

Reactor Theory (Neutron Characteristics) DOE-HDBK-1019/1-93 NEUTRON MODERATION Rev. 0 Page 27 NP-02

Moderating Ratio

Macroscopic slowing down power indicates how rapidly a neutron will slow down in the material in question, but it still does not fully explain the effectiveness of the material as a moderator. An element such as boron has a high logarithmic energy decrement and a good slowing down power, but it is a poor moderator because of its high probability of absorbing neutrons.

The most complete measure of the effectiveness of a moderator is the moderating ratio. The moderating ratio is the ratio of the macroscopic slowing down power to the macroscopic cross section for absorption. The higher the moderating ratio, the more effectively the material performs as a moderator. Equation (2-11) shows how to calculate the moderating ratio of a material.

(2-11)

Fixture checkpoint marker-1194-xray is recorded on fixture page 1194.

Moderating properties of different materials are compared in Table 2.

TABLE 2

Moderating Properties of Materials

Material Collisions to Slowing Down

Number of Macroscopic

Thermalize Power

Moderating

Ratio

H2O 0.927 19 1.425 62

D2O 0.510 35 0.177 4830

Helium 0.427 42 9 x 10 51 -6

Beryllium 0.207 86 0.154 126

Boron 0.171 105 0.092 0.00086

Carbon 0.158 114 0.083 216

NEUTRON MODERATION DOE-HDBK-1019/1-93 Reactor Theory (Neutron Characteristics) NP-02 Page 28 Rev. 0

Summary

The important information in this chapter is summarized below.

Neutron Moderation Summary

& Thermalization is the process of reducing the energy level of a neutron from the energy level at which it is produced to an energy level in the thermal range.

& The moderator is the reactor material that is present for the purpose of thermalizing neutrons.

& Moderating ratio is the ratio of the macroscopic slowing down power to the macroscopic cross section for absorption.

& The average logarithmic energy decrement (!) is the average change in the

& Macroscopic slowing down power is the product of the average logarithmic energy decrement and the macroscopic cross section for scattering.
& There are three desirable characteristics of a moderator.
1. large scattering cross section
2. small absorption cross section
3. large energy loss per collision
& The energy loss after a specified number of collisions can be calculated using the equation below.
E = E (1 - x) N o
N
■
■ ■
Reactor Theory (Neutron Characteristics) DOE-HDBK-1019/1-93 PROMPT AND DELAYED NEUTRONS Rev. 0 Page 29 NP-02
PROMPT AND DELAYED NEUTRONS
Not all neutrons are released at the same time following fission. Most neutrons are released virtually instantaneously and are called prompt neutrons. A very small fraction of neutrons are released after the decay of fission products and are called delayed neutrons. Although delayed neutrons are a very small fraction of the total number of neutrons, they play an extremely important role in the control of the reactor.
EO 3.1 STATE the origin of prompt neutrons and delayed neutrons.
EO 3.2 STATE the approximate fraction of neutrons that are born as delayed neutrons from the fission of the following nuclear fuels:
a. Uranium-235
b. Plutonium-239
EO 3.3 EXPLAIN the mechanism for production of delayed neutrons.
EO 3.4 EXPLAIN prompt and delayed neutron generation times.
Fixture checkpoint marker-1195-xray is recorded on fixture page 1195.
EO 3.5 Given prompt and delayed neutron generation times and delayed neutron fraction, CALCULATE the average generation time.
EO 3.6 EXPLAIN the effect of delayed neutrons on reactor control.
Neutron Classification
The great majority (over 99%) of the neutrons produced in fission are released within about 10-13 seconds of the actual fission event. These are called prompt neutrons. A small portion of fission neutrons are delayed neutrons, which are produced for some time after the fission process has taken place. The delayed neutrons are emitted immediately following the first beta decay of a fission fragment known as a delayed neutron precursor. An example of a delayed neutron precursor is bromine-87, shown below.
PROMPT AND DELAYED NEUTRONS DOE-HDBK-1019/1-93 Reactor Theory (Neutron Characteristics) NP-02 Page 30 Rev. 0
For most applications, it is convenient to combine the known precursors into groups with appropriately averaged properties. These groups vary somewhat depending on the fissile material in use. Table 3 lists the characteristics for the six precursor groups resulting from thermal fission of uranium-235. The fraction of all neutrons that are produced by each of these precursors is called the delayed neutron fraction for that precursor. The total fraction of all neutrons born as delayed neutrons is called the delayed neutron fraction (■). The fraction of

delayed neutron fractions (■) for the fissile nuclides of most interest are as follows: uranium-233 (0.0026), uranium-235 (0.0065), uranium-238 (0.0148), and plutonium-239 (0.0021).

TABLE 3
Delayed Neutron Precursor Groups
for Thermal Fission in Uranium-235

Group	Half-Life (sec)	Delayed Neutron Fraction	Average Energy (MeV)
1	55.7	0.00021	0.25
2	22.7	0.00142	0.46
3	6.2	0.00127	0.41
4	2.3	0.00257	0.45
5	0.61	0.00075	0.41
6	0.23	0.00027	-
Total	-	0.0065	-

Neutron Generation Time

The neutron generation time is the time required for neutrons from one generation to cause the fissions that produce the next generation of neutrons. The generation time for prompt neutrons (5* - pronounced "ell-star") is the total time from birth to rebirth. Three time intervals are involved: (a) the time it takes a fast neutron to slow down to thermal energy, (b) the time the now thermal neutron exists prior to absorption in fuel, and (c) the time required for a fissionable nucleus to emit a fast neutron after neutron absorption.

Fast neutrons slow to thermal energies or leak out of the reactor in 10 seconds to 10 seconds, -4 -6 depending on the moderator. In water moderated reactors, thermal neutrons tend to exist for about 10 seconds before they are absorbed. Fission and fast neutron production following -4 neutron absorption in a fissionable nucleus occurs in about 10 seconds. Thus, fast reactors -13 have an 5* of about 10 seconds, while thermal reactors have an 5* of about 10 seconds + 10 -6 -6 -4 seconds, which is about 10 seconds to 10 seconds. -4 -5

■■■■
■■■■
■■

On the other hand, the average generation time for the six delayed neutron groups is the total time from the birth of the fast neutron to the emission of the delayed neutron. Again, three time intervals are involved: (a) the time it takes a fast neutron to slow down to thermal energy,

Fixture checkpoint marker-1196-xray is recorded on fixture page 1196.

(b) the time the thermal neutron exists prior to absorption, and (c) the average time from neutron absorption to neutron emission by the six precursor groups. The average time for decay of precursors from uranium-235 is 12.5 seconds. The other terms in the delayed neutron generation time are insignificant when compared to this value, and the average delayed neutron generation time becomes ■ 12.5 seconds.

A neutron generation time in the range of 10 seconds to 10 seconds or faster could result in -4 -5 very rapid power excursions, and control would not be possible without the dependence upon delayed neutrons to slow down the rate of the reaction. The average generation time, and hence the rate that power can rise, is determined largely by the delayed neutron generation time. The following equation shows this mathematically.

(2-12)

Example:

a delayed neutron generation time of 12.5 seconds. If ■ is 0.0065, calculate the average generation time.

Solution:

This example demonstrates the effect delayed neutrons have on the neutron generation time and thus reactor control. If a reactor were to be operated in a sustained chain reaction using only prompt neutrons (■ = 0), the generation time from the previous example would be about 5 x 10 seconds. However, by operating the reactor such that a 0.0065 fraction of neutrons are -5 delayed, the generation life time is extended to 0.0813 seconds, providing time for adequate operator control. Therefore, although only a small fraction of the total neutron population, delayed neutrons are extremely important to the control and maintenance of a sustained fission chain reaction.

Summary

Fixture checkpoint marker-1197-xray is recorded on fixture page 1197.

The important information in this chapter is summarized on the following page.

■ ■ ■ ■

PROMPT AND DELAYED NEUTRONS DOE-HDBK-1019/1-93 Reactor Theory (Neutron Characteristics) NP-02 Page 32 Rev. 0

Prompt and Delayed Neutrons Summary

& Prompt neutrons are released directly from fission within 10 seconds of the -13 fission event.

& Delayed neutrons are released from the decay of fission products that are called delayed neutron precursors. Delayed neutron precursors are grouped according to half-life. Half-lives vary from fractions of a second to almost a minute.

& The fraction of neutrons born as delayed neutrons is different for different fuel materials. Following are values for some common fuel materials.

Uranium-235 0.0065

Plutonium-239 0.0021

& Delayed neutrons are produced by a classification of fission products known as delayed neutron precursors. When a delayed neutron precursor undergoes a ■ decay, it results in an excited daughter nucleus which immediately ejects a - neutron. Therefore, these delayed neutrons appear with a half-life of the delayed neutron precursor.

& The delayed neutron generation time is the total time from the birth of the fast neutron to the emission of the delayed neutron in the next generation. Delayed neutron generation times are dominated by the half-life of the delayed neutron precursor. The average delayed neutron generation time is about 12.5 seconds.

& A prompt neutron generation time is the sum of the amount of time it takes a fast neutron to thermalize, the amount of time the neutron exists as a thermal neutron before it is absorbed, and the amount of time between a fissionable nuclide absorbing a neutron and fission neutrons being released. Prompt neutron generation time is about 5 x 10 seconds. -5

& The average neutron generation time can be calculated from the prompt and delayed neutron generation times and the delayed neutron fraction using Equation (2-12).

& Delayed neutrons are responsible for the ability to control the rate at which power can rise in a reactor. If only prompt neutrons existed, reactor control would not be possible due to the rapid power changes.

Reactor Theory (Neutron Characteristics) DOE-HDBK-1019/1-93 NEUTRON FLUX SPECTRUM

Figure 2 Prompt Fission Neutron Energy Spectrum for
Thermal Fission of Uranium-235
NEUTRON FLUX SPECTRUM
The number of neutrons that exist at a given energy level varies. A plot of either the fraction of neutrons or the neutron flux at a given energy versus the energy level is called a neutron energy spectrum. The neutron energy spectrum varies widely for different types of reactors.
EO 4.1 STATE the average energy at which prompt neutrons are produced.
EO 4.2 DESCRIBE the neutron energy spectrum in the following reactors:
a. Fast reactor
b. Thermal reactor
EO 4.3 EXPLAIN the reason for the particular shape of the fast, intermediate, and slow energy regions of the neutron flux spectrum for a thermal reactor.
Prompt Neutron Energies
The neutrons produced by fission are high energy neutrons, and
Fixture checkpoint marker-1198-xray is recorded on fixture page 1198.
almost all fission neutrons have energies between 0.1 MeV and 10 MeV. The neutron energy distribution, or spectrum, may best be described by plotting the fraction of neutrons per MeV as a function of neutron energy, as shown in Figure 2. From this figure it can be seen that the most probable neutron energy is about 0.7 MeV. Also, from this data it can be shown that the average energy of fission neutrons is about 2 MeV. Figure 2 is the neutron energy spectrum for thermal fission in uranium-235. The values will vary slightly for other nuclides.
NEUTRON FLUX SPECTRUM DOE-HDBK-1019/1-93 Reactor Theory (Neutron Characteristics)
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Figure 3 Comparison of Neutron Flux Spectra for
Thermal and Fast Breeder Reactor
Thermal and Fast Breeder Reactor Neutron Spectra
The spectrum of neutron energies produced by fission varies significantly from the energy spectrum, or flux, existing in a reactor at a given time. Figure 3 illustrates the difference in neutron flux spectra between a thermal reactor and a fast breeder reactor. The energy distribution of neutrons from fission is essentially the same for both reactors, so the differences in the curve shapes may be attributed to the neutron moderation or slowing down effects.

cooled); therefore, an insignificant number of neutrons exist in the thermal range. For the thermal reactor (water moderated), the spectrum of neutrons in the fast region (> 0.1 MeV) has a shape similar to that for the spectrum of neutrons emitted by the fission process.
In the thermal reactor, the flux in the intermediate energy region (1 eV to 0.1 MeV) has approximately a 1/E dependence. That is, if the energy (E) is halved, the flux doubles. This 1/E dependence is caused by the slowing down process, where elastic collisions remove a constant fraction of the neutron energy per collision (on the average), independent of energy; thus, the neutron loses larger amounts of energy per collision at higher energies than at lower energies. The fact that the neutrons lose a constant fraction of energy per collision causes the neutrons to tend to "pile up" at lower energies, that is, a greater number of neutrons exist at the lower energies as a result of this behavior.
vp
2 k T
m
vp
2 k T
m
2 1.38 x 10 16 erg
K
293 K
1.66 x 10 24 g
2.2 x 105 cm
sec
1 m
100 cm
2200 m
Fixture checkpoint marker-1199-xray is recorded on fixture page 1199.
sec
Reactor Theory (Neutron Characteristics) DOE-HDBK-1019/1-93 NEUTRON FLUX SPECTRUM Rev. 0 Page 35 NP-02
In the thermal region the neutrons achieve a thermal equilibrium with the atoms of the moderator material. In any given collision they may gain or lose energy, and over successive collisions will gain as much energy as they lose. These thermal neutrons, even at a specific temperature, do not all have the same energy or velocity; there is a distribution of energies, usually referred to as the Maxwell distribution (e.g., Figure 2). The energies of most thermal neutrons lie close to the most probable energy, but there is a spread of neutrons above and below this value.
Most Probable Neutron Velocities
The most probable velocity (vp) of a thermal neutron is determined by the temperature of the medium and can be determined by Equation (2-13) .
(2-13)
where:
v = most probable velocity of neutron (cm/sec) p
k = Boltzman's constant (1.38 x 10 erg/ K) -16
T = absolute temperature in degrees Kelvin (K)
m = mass of neutron (1.66 x 10 grams) -24
Example:
Calculate the most probable velocities for neutrons in thermal equilibrium with their surroundings at the following temperatures. a) 20 C, b) 260 C.

SYNTHETIC FIXTURE PAGE 1200 OF 1200

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DOE FUNDAMENTALS HANDBOOK
NUCLEAR PHYSICS
AND REACTOR THEORY
Volume 1 of 2
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ABSTRACT
The Nuclear Physics and Reactor Theory Handbook was developed to assist nuclear facility operating contractors in providing operators, maintenance personnel, and the technical staff with the necessary fundamentals training to ensure a basic understanding of nuclear physics and reactor theory. The handbook includes information on atomic and nuclear physics; neutron characteristics; reactor theory and nuclear parameters; and the theory of reactor operation. This information will provide personnel with a foundation for understanding the scientific principles that are associated with various DOE nuclear facility operations and maintenance.
Key Words: Training Material, Atomic Physics, The Chart of the Nuclides, Radioactivity, Radioactive Decay, Neutron Interaction, Fission, Reactor Theory, Neutron Characteristics, Neutron Life Cycle, Reactor Kinetics
Rev. 0 NP

FOREWORD
The Department of Energy (DOE) Fundamentals Handbooks consist of ten academic
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subjects, which include Mathematics; Classical Physics; Thermodynamics, Heat Transfer, and Fluid Flow; Instrumentation and Control; Electrical Science; Material Science; Mechanical Science; Chemistry; Engineering Symbology, Prints, and Drawings; and Nuclear Physics and Reactor Theory. The handbooks are provided as an aid to DOE nuclear facility contractors.
These handbooks were first published as Reactor Operator Fundamentals Manuals in 1985 for use by DOE category A reactors. The subject areas, subject matter content, and level of detail of the Reactor Operator Fundamentals Manuals were determined from several sources. DOE Category A reactor training managers determined which materials should be included, and served as a primary reference in the initial development phase. Training guidelines from the commercial nuclear power industry, results of job and task analyses, and independent input from contractors and operations-oriented personnel were all considered and included to some degree in developing the text material and learning objectives.
The DOE Fundamentals Handbooks represent the needs of various DOE nuclear facilities' fundamental training requirements. To increase their applicability to nonreactor nuclear facilities,